TestToolkit

2.2.1 selftest-preview

TestToolkit – Test functions and utilities (Python 3)



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Chapter 1

iSyst TestToolkit Documentation

1.1 Introduction

The **TestToolkit** libraries (*TTk*) contain functions and utilities for test automation.

See https://wiki.isyst.de/wiki/TestToolkit

1.2 Overview of Packages

1.2.1 ttk_base – Base Definitions

The package `ttk_base` contains common functions as well as base definitions and abstraction layers for calibration and real time variables.
Basic access to variables will be mapped to generic `get()` and `set()` methods, regardless of the variables' original interface.

1.2.2 ttk_checks – Basic Check and Test Functions

Package `ttk_checks` provides check and test utility functions.
As convention, *"check"* functions (see `ttk_checks.basic_checks`) will return a simple boolean True/False result, while *"test"* functions (see `ttk_checks.basic_tests`) will return a formatted test result entry.
Requires package `ttk_base`.

See https://wiki.isyst.de/wiki/Testresult for more details on test results.

1.2.3 ttk_bus – Bus/Interface Test Functions

The package `ttk_bus` contains definitions and test functions for bus or interface tests (FlexRay, CAN, ...).
Requires packages `ttk_base` and `ttk_checks`

1.2.4 ttk_daq – Data Acquisition Utilities

The package `ttk_daq` contains definitions and utilities for evaluating captured data.
Requires packages `ttk_base`.

1.2.5 ttk_tools – Test Tool Interfaces

Package `ttk_tools` contains test tool interfaces e.g. for accessing CANape.
Vendor-specific tool interfaces are grouped in sub packages.
Requires package `ttk_base`.

1.3 Using TTk in a Test Project

TestToolkit packages will usually be directly added to a test project's folder structure. This way, the TTk version is fixed for a specific test project.
If an update is desired, the `ttk_*`-folders can simply be replaced by those provided with the newer distribution.

1.3.1 TTk Libraries in a Project Structure

```
Project Folder/
    Python/
        TestLibraries/
            ttk_base/
            ttk_bus/
            ttk_checks/
            ttk_daq/
            ttk_tools/

        TestPool/
            _automation_wrapper_.py
            some_test.py

    TestReports/
    TestStructure/
```

1.3.2 Example: Using TTk in a Test Script

A (very) minimal example script.

```
#*******************************************************************************
# -*- coding: utf-8 -*-
# File    : hello_test.py
# Task    : A minimal "hello world!" test script
#
# Copyright 2020 iSyst Intelligente Systeme GmbH
#
#*******************************************************************************
#******************************** Version **************************************
#*******************************************************************************
# Ver.   | Date       | Name     | Description
#-------------------------------------------------------------------------------
# 1.0.0  | 01.07.2020 | J. Doe   | initial
#*******************************************************************************
from _automation_wrapper_ import TestEnv
```

```
from ttk_checks import basic_tests

testenv = TestEnv()

try:
    # ########################################################################
    testenv.setup()

    testresult = testenv.getResults()
    hil = testenv.getHil()
    testenv.startupECU()

    testresult.append(["Hello World!", "PASSED"])
    testresult.append(
        basic_tests.checkTolerance(
            hil.CV_monitor, 14.0, rel_pos=0.05,
            descr="Check nominal supply voltage"
        )
    )
    # cleanup
    hil = None
finally:
    # ########################################################################
    testenv.breakdown()
```

Chapter 2

Namespace Index

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Chapter 5

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ttk_tools/rst/gamma_api_offline_stub.py . . . 2048
ttk_tools/rst/idaq.py . . . 2039
ttk_tools/rst/isygamma.py . . . 2048
ttk_tools/siemens/__init__.py . . . 1983
ttk_tools/siemens/simit_common.py . . . 2051
ttk_tools/siemens/simit_shm.py . . . 2051
ttk_tools/vector/__init__.py . . . 1983
ttk_tools/vector/_canapeapi.py . . . 2052
ttk_tools/vector/_canapeapi_offline_stub.py . . . 2053
ttk_tools/vector/_vflash_api.py . . . 2054
ttk_tools/vector/canapeapi.py . . . 2054
ttk_tools/vector/canapeapi_common.py . . . 2055
ttk_tools/vector/canapeapi_offline_stub.py . . . 2056
ttk_tools/vector/mdf4_reader.py . . . 2056

ttk_tools/vector/mdflib.py 2059
ttk_tools/vector/vflash_api.py 2060
ttk_tools/vector/vflash_api_common.py 2060

Chapter 6

Namespace Documentation

6.1 _basic_result_list Namespace Reference

A basic result list implementation that contains only the most commonly used features; intended for internal use in TTk test functions while avoiding external dependencies to TestSupport libs.

6.1.1 Detailed Description

A basic result list implementation that contains only the most commonly used features; intended for internal use in TTk test functions while avoiding external dependencies to TestSupport libs.

ResultList adds logging (to stdout), some sanity checks and helper functions to an otherwise plain list class in order to make it more convenient for storing test result entries.

6.2 ttk_base Namespace Reference

Namespaces

- namespace _baseutils

  General utility functions and classes.
- namespace _values_base

  Base classes for values of built-in types like long, int, float or string with additional meta information attached.
- namespace _variables_base

  Base classes for test variables (hil, cal, ...) to provide a unified interface and containers to group variables into sets.
- namespace baseutils

  Interface wrapper for general utility functions and classes in ttk_base._baseutils.
- namespace configuration

  Settings and configuration for TestToolkit packages.
- namespace dspace_rtplib

  Abstraction layer for accessing HIL variables using dSPACE RTPlib interfaces.
- namespace dspace_xilapi

  Abstraction layer for accessing HIL variables using dSPACE XIL-API/Testbench interfaces.
- namespace errors

Base classes and definitions for TestToolkit exceptions/errors.

- namespace isyst_ida

  Abstraction layer for accessing HIL variables using iSyst iDA Testware.
- namespace localization

  Base class for localization.
- namespace ni_labview

  Abstraction layer for accessing National Instruments LabView variables/controls.
- namespace pyocd

  Abstraction layer for accessing PyOCD debugger symbols/variables.
- namespace rst_gammav

  Abstraction layer for accessing RST (RST Industrie Automation GmbH) Gamma V realtime middleware process variables.
- namespace siemens_simit

  Abstraction layer for accessing variables on SIMIT SHM Coupling interfaces.
- namespace values_base

  Interface Wrapper for values with additional meta information in ttk_base._values_base.
- namespace variables_base

  Interface Wrapper for variables' base classes in ttk_base._variables_base.
- namespace vector_canape

  Abstraction layer for accessing Vector CANape device variables.

Variables

- ttk_path
- str version = ".".join(str(v) for v in sys.version_info[:3])
- str arch = "x64" if sys.maxsize > 2**32 else "x86"
- str plat
- build_info_path = os.path.join(ttk_path, "ttk_build.info")
- encoding
- errors
- tracebacklimit

6.2.1 Variable Documentation

6.2.1.1 arch

```
str ttk_base.arch = "x64" if sys.maxsize > 2**32 else "x86"
```

6.2.1.2 build_info_path

```
ttk_base.build_info_path = os.path.join(ttk_path, "ttk_build.info")
```

6.2.1.3 encoding

```
ttk_base.encoding
```

6.2.1.4 errors

```
ttk_base.errors
```

6.2.1.5 plat

```
str ttk_base.plat
```

Initial value:

```
00001 =  "win32" else \
00002                  "Linux"   if sys.platform.startswith("linux") else \
00003                   sys.platform
```

6.2.1.6 tracebacklimit

```
ttk_base.tracebacklimit
```

6.2.1.7 ttk_path

```
ttk_base.ttk_path
```

Initial value:

```
00001 =  os.path.dirname(os.path.dirname(
00002         os.path.abspath(inspect.currentframe().f_code.co_filename)
00003     ))
```

6.2.1.8 version

```
str ttk_base.version = ".".join(str(v) for v in sys.version_info[:3])
```

6.3 ttk_base._baseutils Namespace Reference

General utility functions and classes.

Classes

- class _CustomRepr

  A utility class for "marking" customized (i.e.
- class EnumInfo

  Utility/convenience class for "enum-like" value container classes.
- class HexList

  A list that returns a representation in hexadecimal notation (for easier reading).

Functions

- getSpacedHexString (value, min_bytes=1, spacer=" ")

  Get a hex string representation of Bytes in the supplied (unsigned) integer value.
- _getInteger (value)
- _bin (value, prefix='0b', min_digits=0)

  Create a binary representation of the input value.
- _hex (value, prefix='0x', min_digits=0)

  Create a hexadecimal representation of the input value.
- _dec (value, prefix='0d', min_digits=0)

  Create a integer/decimal representation of the input value.
- _oct (value, prefix='0o', min_digits=0)

  Create an octal representation of the input value.
- _pct (factor, suffix=' %', decimal_places=None)

  Create an "percentage" representation of the input factor value.
- _flt (value, decimal_places=None)

  Create a float representation of the input value.
- getCustomRepr (value, fmt=None)

  Get a custom string representation of a value.
- isCustomRepr (s)

  Check if s is formatted using a custom representation or just a default/fallback string representation (or anything else).
- bstr (obj, encoding=DEFAULT_ENCODING)

  Get a byte string representation of an object, that is, like str() but returns a raw bytes string.
- ustr (obj, encoding=DEFAULT_ENCODING)

  Get a unicode/text string representation of an object.
- uni (s)

  Enforce unicode/text strings.
- fsEncode (s)

  Get a representation of (unicode/ASCII) string s, using the current file system encoding.
- fsDecode (s)

  Get a unicode representation of byte string s, using the current file system encoding (if available).
- sleepWithFeedback (delay_ms, comment=None, step_ms=100, done_msg=None)

  Sleep for the specified delay and print some feedback/progress to console/log.

Variables

- str DEFAULT_ENCODING = "UTF-8"

  Default encoding for encoding/decoding raw byte strings as text.
- str BYTES_ENCODING = "ASCII"

  Encoding to use when adding raw byte strings to description text.
- float _FLOAT_CONVERSION_LIMIT = 0.0001

  maximum divergence to still accept a float value to be "intended" as int/long
- _FMT_KEYWORD_RE = re.compile(r"^\s*([a-z]+)\s*(\d*)\s*$")

  regex for format keyword matching, see getCustomRepr

6.3.1 Detailed Description

General utility functions and classes.

6.3.2 Function Documentation

6.3.2.1 _bin()

```
ttk_base._baseutils._bin (
              value,
              prefix = '0b',
              min_digits = 0 )  [protected]
```

Create a binary representation of the input value.

A rather recursive approach, works similar to the built-in function bin() that is available in Python 2.6+, but with some additional convenience features.

Parameters

value	value to represent as a binary string. Note that float values will be rounded towards the nearest natural number.
prefix	prefix to prepend the string (usually "0b", just like bin() would and similar to the "0x" for hex)
min_digits	a minimum bit width to enforce (filled with zeros)

Returns

a string with a binary representation of the input value.

6.3.2.2 _dec()

```
ttk_base._baseutils._dec (
              value,
              prefix = '0d',
              min_digits = 0 )  [protected]
```

Create a integer/decimal representation of the input value.

Similar to the built-in function hex(), but for base-10.

Parameters

value	value to represent as a decimal string. Note that float values will be rounded towards the nearest natural number.
prefix	prefix to prepend the string (usually empty or "0d")
min_digits	a minimum number of digits to enforce (filled with leading zeros)

Returns

a string with a decimal representation of the input value. Raises TypeError or ValueError if input value was not integer, see _getInteger.

6.3.2.3 _flt()

```
ttk_base._baseutils._flt (
```

```
                value,
                decimal_places = None )  [protected]
```

Create a float representation of the input value.

Parameters

value	(float-) value to represent as, well, a float value
decimal_places	a minimum number of fractional digits / decimal places to enforce None: use "auto" mode magic from string representation

Returns

a string with a float representation of the input factor. Raises a TypeError if value is non-numeric.

6.3.2.4 _getInteger()

```
ttk_base._baseutils._getInteger (
                value )  [protected]
```

6.3.2.5 _hex()

```
ttk_base._baseutils._hex (
                value,
                prefix = '0x',
                min_digits = 0 )  [protected]
```

Create a hexadecimal representation of the input value.

Works similar to the built-in function hex(), but generates uppercase nibbles and comes with some additional convenience features.

Parameters

value	value to represent as a hexadecimal string. Note that float values will be rounded towards the nearest natural number.
prefix	prefix to prepend the string (usually "0x")
min_digits	a minimum number of nibbles to enforce (filled with leading zeros)

Returns

a string with a hexadecimal representation of the input value. Raises TypeError or ValueError if input value was not integer, see _getInteger.

6.3.2.6 _oct()

```
ttk_base._baseutils._oct (
                value,
```

```
                prefix = '0o',
                min_digits = 0 )  [protected]
```

Create an octal representation of the input value.

Similar to the built-in function oct(), but using a prefix similar to hex/bin instead of leading zero.

Parameters

value	value to represent as an octal string. Note that float values will be rounded towards the nearest natural number.
prefix	prefix to prepend the string (usually "0o")
min_digits	a minimum number of octets to enforce (filled with leading zeros)

Returns

a string with a decimal representation of the input value. Raises TypeError or ValueError if input value was not integer, see _getInteger.

6.3.2.7 _pct()

```
ttk_base._baseutils._pct (
                factor,
                suffix = ' %',
                decimal_places = None )  [protected]
```

Create an "percentage" representation of the input factor value.

0.0 .. 1.0 => 0 .. 100 % Range is not limited, though: -3.0 .. 2.4 => -300 .. 240 %

Parameters

factor	(float-) factor to represent as percentage
suffix	suffix to add to string (usually " %")
decimal_places	a minimum number of fractional digits / decimal places to enforce None: use "auto" mode magic from string representation

Returns

a string with a percentage representation of the input factor. Raises a TypeError if value is non-numeric.

6.3.2.8 bstr()

```
ttk_base._baseutils.bstr (
                obj,
                encoding = DEFAULT_ENCODING )
```

Get a byte string representation of an object, that is, like str() but returns a raw bytes string.

Note

If characters in `obj`'s string representation cannot be mapped using the given `encoding`, a backslashed replacement sequence will be added in their stead.

Examples:

```
>> bstr("foö") # default encoding, should match source encoding
b'fo\xc3\xb6'   # two bytes for "ö" in utf-8

>> bstr("foö", "ASCII")
b'fo\\xf6'      # replacement sequence for "ö", as it is n/a in 7-bit ASCII

>> bstr("foö", "latin-1")
b'fo\xf6'       # "ö" maps to a single byte in latin-1

>> bstr(b"fo\x00\xff")
b'fo\x00'       # bytes in, bytes out

>> bstr(bytearray((102, 111, 0, 255)))
bytearray(b'fo\x00\xff') # bytearray in, bytearray out
```

Parameters

obj	object to represent as byte string
encoding	encoding to use to encode unicode/text-strings. Defaults to DEFAULT_ENCODING

Returns

a byte string (or bytearray if obj was already a bytearray)

6.3.2.9 fsDecode()

```
ttk_base._baseutils.fsDecode (
              s )
```

Get a unicode representation of byte string s, using the current file system encoding (if available).

This is mainly useful to treat file names that may contain non-ASCII characters. Note that any decoding errors will be ignored.

Parameters

s	(byte) string to decode

Returns

a unicode representation of s.

6.3.2.10 fsEncode()

```
ttk_base._baseutils.fsEncode (
              s )
```

Get a representation of (unicode/ASCII) string s, using the current file system encoding.

This is mainly useful for subprocess calls to treat command line arguments or file names that may contain non-ASCII characters.

Note

If a raw byte-string is supplied, it will first be decoded using `DEFAULT_ENCODING`, "special" characters may raise a UnicodeDecodeError.
Always supply unicode strings if special characters are necessary.

Parameters

s	(unicode) string to encode

Returns

a byte string suitable for use with file system functions.

6.3.2.11 getCustomRepr()

```
ttk_base._baseutils.getCustomRepr (
              value,
              fmt = None )
```

Get a custom string representation of a value.

Parameters

value	value to get a representation of
fmt	string format (e.g. "%.2f") or one of the predefined format keywords

Info: Predefined format keywords

- "hex" - hexadecimal representation with leading 0x and uppercase digits
  e.g. 0xC0FFEE
- "int" - integer decimal representation
  e.g. 42
- "dec" - as "int", but with leading 0d
  e.g. 0d42
- "oct" - octal representation with leading 0o
  e.g. 0o777
- "bin" - binary representation with leading 0b
  e.g. 0b101010

- "pct" - percentage representation with trailing %.
  Note that value will be interpreted as a factor,
  e.g. 0.05 will be formatted as 5 %
- "flt" - float representation with decimal places (just for convenience)

Note

Float values for hex/int/dec/oct/bin will be *rounded* towards the nearest integer number (fractional part will not just be cut-off)

Info: Format keyword options

Each keyword may be followed by a number to give additional representation hints.

For integer numbers in hex/int/dec/oct/bin, this will specify a minimum width (in nibbles, digits, octets or bits) that will be filled with leading zeroes, e.g.

- "hex8": 0x00C0FFEE
- "dec8": 0d00004711
- "bin8": 0b01000010

For pct and flt, the number will represent the number of decimal places, e.g.

- "pct2": 2.21 %
- "flt3": 2.212

6.3.2.12 getSpacedHexString()

```
ttk_base._baseutils.getSpacedHexString (
              value,
              min_bytes = 1,
              spacer = " " )
```

Get a hex string representation of Bytes in the supplied (unsigned) integer value.

Bytes will be shown in big endian order.

Parameters

value	unsigned numeric source value
min_bytes	minimum number of bytes to return. Padding with zero bytes will be added at start of string. At least one byte will always be shown.
spacer	spacer character(s) between hex-bytes

Example:

```
>> getSpacedHexString(0x123456, 4, " ")
"00 12 34 56"
```

Returns

a hex string with extra (white) space added between bytes.

6.3.2.13 isCustomRepr()

```
ttk_base._baseutils.isCustomRepr (
              s )
```

Check if s is formatted using a custom representation or just a default/fallback string representation (or anything else).

This is mainly intended as facade so the _CustomRepr class does not need to be in the public "interface" and to avoid inheritance/instance check issues due to the interface/base implementation split.

Parameters

s	string/object to check

Returns

True if s is a string containing a custom representation, otherwise False.

6.3.2.14 sleepWithFeedback()

```
ttk_base._baseutils.sleepWithFeedback (
              delay_ms,
              comment = None,
              step_ms = 100,
              done_msg = None )
```

Sleep for the specified delay and print some feedback/progress to console/log.

This is just a convenience function.

Parameters

delay_ms	delay in milliseconds
comment	comment to print at beginning, None: use default comment
step_ms	step size for feedback
done_msg	comment to print when done None: use default comment

Example:

```
>> sleepWithFeedback(1000)
Waiting 1000 ms... ......... ...done
```

6.3.2.15 uni()

```
ttk_base._baseutils.uni (
              s )
```

Enforce unicode/text strings.

Raw byte strings will be decoded according to `DEFAULT_ENCODING`. Non-byte-string values will be left as-is.

Examples:

```
uni("foo")
>> u'foo'

uni(u"foo")
>> u'foo'

uni([1, 2])
>> [1, 2]
```

Parameters

s	original string/object

Returns

s as unicode if it was a string to begin with, otherwise the unmodified object.

6.3.2.16 ustr()

```
ttk_base._baseutils.ustr (
              obj,
              encoding = DEFAULT_ENCODING )
```

Get a unicode/text string representation of an object.

Raw byte strings will be decoded using the supplied `encoding`.

Parameters

obj	object to represent as text string
encoding	encoding to use to decode byte-strings. Defaults to DEFAULT_ENCODING

Returns

a unicode text string

6.3.3 Variable Documentation

6.3.3.1 _FLOAT_CONVERSION_LIMIT

```
float ttk_base._baseutils._FLOAT_CONVERSION_LIMIT = 0.0001  [protected]
```

maximum divergence to still accept a float value to be "intended" as int/long

6.3.3.2 _FMT_KEYWORD_RE

```
ttk_base._baseutils._FMT_KEYWORD_RE = re.compile(r"^\s*([a-z]+)\s*(\d*)\s*$")  [protected]
```

regex for format keyword matching, see getCustomRepr

6.3.3.3 BYTES_ENCODING

```
str ttk_base._baseutils.BYTES_ENCODING = "ASCII"
```

Encoding to use when adding raw byte strings to description text.

As using bytes/bytearrays is a more conscious choice since Python 3, we will display raw bytes the same way byte literals are shown in Python code (only without outer quotes and b prefix). Simply decoding raw bytes with DEFAULT_ENCODING might sometimes yield unexpected characters (especially when using a multi-byte encoding like UTF-8)

6.3.3.4 DEFAULT_ENCODING

```
str ttk_base._baseutils.DEFAULT_ENCODING = "UTF-8"
```

Default encoding for encoding/decoding raw byte strings as text.

Defined here and not in ttk_base.configuration, as that module uses a few of baseutils' functions.

6.4 ttk_base._values_base Namespace Reference

Base classes for values of built-in types like long, int, float or string with additional meta information attached.

Classes

- class MetaBytearray
  Bytearray values with additional meta info.
- class MetaBytes
  Raw Byte string values with additional meta info.
- class MetaFloat
  Float values with additional meta info.
- class MetaInt
  Integer values with additional meta info.
- class MetaList
  A List with additional meta info.
- class MetaStr
  String values with additional meta info.
- class MetaTuple
  A tuple with additional meta info.
- class MetaValue
  Base class for meta attributes.

Functions

- meta (value, unit="", alias="", descr="", lookup=None, fmt=None)

  Factory function to create "meta" instances of the supplied value that have some additional meta attributes bolted on.

6.4.1 Detailed Description

Base classes for values of built-in types like long, int, float or string with additional meta information attached.

6.4.2 Function Documentation

6.4.2.1 meta()

```
ttk_base._values_base.meta (
              value,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              fmt = None )
```

Factory function to create "meta" instances of the supplied value that have some additional meta attributes bolted on.

Parameters

value	value that should get meta info added. Currently supported types are int, long, float, str and unicode
unit	meta info: value's unit, e.g. mV
alias	meta info: an alias name for the value
descr	meta info: description/comment text
lookup	meta info: lookup table for {value => "state name"} mappings
fmt	format string to use for value representation

Example:

```
>> value = meta(123, unit="mV", descr="Lorem Ipsum")
>> value.info()
'123 mV'
>> value.descr
'Lorem Ipsum'

# Please note that modifying the value afterwards might lead to
# the loss of the attached meta info (as base data types do not
# get modified in place but rather replaced with a new reference).

>> type(value)
<class 'ttk_base._values_base.MetaInt'>
>> value += 1
>> type(value)
<type 'int'>
```

Returns

an instance of a class derived from value's class, that has the additional meta attributes set.
If supplied value is None, None will be returned (without any meta information).

6.5 ttk_base._variables_base Namespace Reference

Base classes for test variables (hil, cal, ...) to provide a unified interface and containers to group variables into sets.

Classes

- class DummyVar

  A minimal dummy test variable not connected to any external interface.
- class ResettableContainer

  Simple container class that provides a non-recursive reset() method.
- class Snapshot

  Store and restore a "snapshot" of variables' values.
- class VarBase

  Base wrapper for a single variable.
- class VarContainerBase

  Base class for variables containers.

Functions

- _getArg (args, kwargs, pos, key, fallback=None)

  Extract a specific positional or keyword argument from generic args/kwargs containers.
- _voidKwArgs (kwargs, ∗keys)

  Remove keyword arguments from the supplied dict (to make sure it will not interfere with other arguments we will directly supply).
- _getModelPathAlias (model_path)

  Get a fallback name/alias based on the supplied Simulink model path.

6.5.1 Detailed Description

Base classes for test variables (hil, cal, ...) to provide a unified interface and containers to group variables into sets.

6.5.2 Function Documentation

6.5.2.1 _getArg()

```
ttk_base._variables_base._getArg (
            args,
            kwargs,
            pos,
            key,
            fallback = None )  [protected]
```

Extract a specific positional or keyword argument from generic args/kwargs containers.

Parameters

args	a list of positional arguments
kwargs	a dict of keyword arguments
pos	position of desired argument in args
key	key of desired argument in kwargs
fallback	fallback value if argument not available

Returns

the argument value or None if nothing was available.

6.5.2.2 _getModelPathAlias()

```
ttk_base._variables_base._getModelPathAlias (
              model_path )  [protected]
```

Get a fallback name/alias based on the supplied Simulink model path.

Any SIMULINK_DEFAULT_NAMES will be skipped.

Parameters

model_path	a model path string

Example:

```
_getModelPathAlias('Model Root/IOUserInterface/IO_Parameter/PowerSupply/VBAT_REMOTE[V]/Value')
=> "VBAT_REMOTE[V]"
```

Returns

an alias name (or the full model path as fallback)

6.5.2.3 _voidKwArgs()

```
ttk_base._variables_base._voidKwArgs (
              kwargs,
             * keys )  [protected]
```

Remove keyword arguments from the supplied dict (to make sure it will not interfere with other arguments we will directly supply).

Parameters

kwargs	dict (with keyword arguments)
∗*keys*	remaining arguments are key names to remove

6.6 ttk_base.baseutils Namespace Reference

Interface wrapper for general utility functions and classes in ttk_base._baseutils.

Classes

- class EnumInfo

Utility/convenience class for "enum-like" value container classes.

- class HexList

  A list that returns a representation in hexadecimal notation (for easier reading).

Functions

- getSpacedHexString (value, min_bytes=1, spacer=" ")

  Get a hex string representation of Bytes in the supplied (unsigned) integer value.
- getCustomRepr (value, fmt=None)

  Get a custom string representation of a value.
- isCustomRepr (s)

  Check if s is formatted using a custom representation or just a default/fallback string representation (or anything else).
- bstr (obj, encoding=DEFAULT_ENCODING)

  Get a byte string representation of an object, that is, like str() but returns a raw bytes string.
- ustr (obj, encoding=DEFAULT_ENCODING)

  Get a unicode/text string representation of an object.
- uni (s)

  Enforce unicode/text strings.
- fsEncode (s)

  Get a representation of (unicode/ASCII) string s, using the current file system encoding.
- fsDecode (s)

  Get a unicode representation of byte string s, using the current file system encoding (if available).
- sleepWithFeedback (delay_ms, comment=None, step_ms=100, done_msg=None)

  Sleep for the specified delay and print some feedback/progress to console/log.

6.6.1 Detailed Description

Interface wrapper for general utility functions and classes in ttk_base._baseutils.

6.6.2 Function Documentation

6.6.2.1 bstr()

```
ttk_base.baseutils.bstr (
              obj,
              encoding = DEFAULT_ENCODING )
```

Get a byte string representation of an object, that is, like str() but returns a raw bytes string.

Note

If characters in `obj`'s string representation cannot be mapped using the given `encoding`, a backslashed replacement sequence will be added in their stead.

Examples:

```
>> bstr("foö") # default encoding, should match source encoding
b'fo\xc3\xb6'   # two bytes for "ö" in utf-8

>> bstr("foö", "ASCII")
b'fo\\xf6'      # replacement sequence for "ö", as it is n/a in 7-bit ASCII

>> bstr("foö", "latin-1")
b'fo\xf6'       # "ö" maps to a single byte in latin-1

>> bstr(b"fo\x00\xff")
b'fo\x00'       # bytes in, bytes out

>> bstr(bytearray((102, 111, 0, 255)))
bytearray(b'fo\x00\xff') # bytearray in, bytearray out
```

Parameters

obj	object to represent as byte string
encoding	encoding to use to encode unicode/text-strings. Defaults to DEFAULT_ENCODING

Returns

a byte string (or bytearray if obj was already a bytearray)

6.6.2.2 fsDecode()

```
ttk_base.baseutils.fsDecode (
              s )
```

Get a unicode representation of byte string s, using the current file system encoding (if available).

This is mainly useful to treat file names that may contain non-ASCII characters. Note that any decoding errors will be ignored.

Parameters

s	(byte) string to decode

Returns

a unicode representation of s.

6.6.2.3 fsEncode()

```
ttk_base.baseutils.fsEncode (
              s )
```

Get a representation of (unicode/ASCII) string s, using the current file system encoding.

This is mainly useful for subprocess calls to treat command line arguments or file names that may contain non-ASCII characters.

Note

If a raw byte-string is supplied, it will first be decoded using the `DEFAULT_ENCODING` (see ustr()), so "special" characters may raise a UnicodeDecodeError.
Always supply unicode strings if special characters are necessary.

Parameters

s	(unicode) string to encode

Returns

a byte string suitable for use with file system functions.

6.6.2.4 getCustomRepr()

```
ttk_base.baseutils.getCustomRepr (
              value,
              fmt = None )
```

Get a custom string representation of a value.

Parameters

value	value to get a representation of
fmt	string format (e.g. "%.2f") or one of the predefined format keywords

Info: Predefined format keywords

- "hex" - hexadecimal representation with leading 0x and uppercase digits
  e.g. 0xC0FFEE
- "int" - integer decimal representation
  e.g. 42
- "dec" - as "int", but with leading 0d
  e.g. 0d42
- "oct" - octal representation with leading 0o
  e.g. 0o777
- "bin" - binary representation with leading 0b
  e.g. 0b101010
- "pct" - percentage representation with trailing %.
  Note that value will be interpreted as a factor,
  e.g. 0.05 will be formatted as 5 %
- "flt" - float representation with decimal places (just for convenience)

Note

Float values for hex/int/dec/oct/bin will be *rounded* towards the nearest integer number (fractional part will not just be cut-off)

Info: Format keyword options

Each keyword may be followed by a number to give additional representation hints.

For integer numbers in hex/int/dec/oct/bin, this will specify a minimum width (in nibbles, digits, octets or bits) that will be filled with leading zeroes, e.g.

- "hex8": 0x00C0FFEE
- "dec8": 0d00004711
- "bin8": 0b01000010

For pct and flt, the number will represent the number of decimal places, e.g.

- "pct2": 2.21 %
- "flt3": 2.212

6.6.2.5 getSpacedHexString()

```
ttk_base.baseutils.getSpacedHexString (
              value,
              min_bytes = 1,
              spacer = " " )
```

Get a hex string representation of Bytes in the supplied (unsigned) integer value.

Bytes will be shown in big endian order.

Parameters

value	unsigned numeric source value
min_bytes	minimum number of bytes to return. Padding with zero bytes will be added at start of string. At least one byte will always be shown.
spacer	spacer character(s) between hex-bytes

Example:

```
»> getSpacedHexString(0x123456, 4, " ")
"00 12 34 56"
```

Returns

a hex string with extra (white) space added between bytes.

6.6.2.6 isCustomRepr()

```
ttk_base.baseutils.isCustomRepr (
              s )
```

Check if s is formatted using a custom representation or just a default/fallback string representation (or anything else).

Parameters

s	string/object to check

Returns

True if s is a string containing a custom representation, otherwise False.

6.6.2.7 sleepWithFeedback()

```
ttk_base.baseutils.sleepWithFeedback (
              delay_ms,
              comment = None,
              step_ms = 100,
              done_msg = None )
```

Sleep for the specified delay and print some feedback/progress to console/log.

This is just a convenience function.

Parameters

delay_ms	delay in milliseconds
comment	comment to print at beginning, None: use default comment
step_ms	step size for feedback
done_msg	comment to print when done None: use default comment

Example:

```
»> sleepWithFeedback(1000)
Waiting 1000 ms... ......... ...done
```

6.6.2.8 uni()

```
ttk_base.baseutils.uni (
              s )
```

Enforce unicode/text strings.

Raw byte strings will be decoded according to `DEFAULT_ENCODING`. Non-byte-string values will be left as-is.

Examples:

```
uni("foo")
»> u'foo'

uni(u"foo")
»> u'foo'

uni([1, 2])
»> [1, 2]
```

Parameters

s	original string/object

Returns

s as unicode if it was a string to begin with, otherwise the unmodified object.

6.6.2.9 ustr()

```
ttk_base.baseutils.ustr (
              obj,
              encoding = DEFAULT_ENCODING )
```

Get a unicode/text string representation of an object.

Raw byte strings will be decoded using the supplied `encoding`.

Parameters

obj	object to represent as text string
encoding	encoding to use to decode byte-strings. Defaults to DEFAULT_ENCODING

Returns

a unicode text string

6.7 ttk_base.configuration Namespace Reference

Settings and configuration for TestToolkit packages.

Classes

- class Configuration

  A simple class that reads/stores configuration info in ini files.
- class DataParser

  A reduced "light" implementation of the original DataParser (which was in turn an alternate, less finnicky implementation of Python 2.2-era ConfigParser).
- class DataParserError

  Errors during parsing of a configuration file.
- class DongleLibDependencyError

  Base class for data parser errors.
- class DongleLibError

  Errors while loading Dongle DLLs.
- class LicenseError

  Invalid license has been encountered.
- class ParsingError

Functions

- getExpectedDongleVersion ()

  Get a string representation of the expected Dongle-DLL version for this release (see EXPECTED_DONGLE_DLL↩ _VERSION)
- findFile (file_name, start_path=None, max_parent_levels=6)

  Find a matching file by traversing the folder tree upwards, starting at start_path.
- getSettings ()

  Get the current project settings (which may be empty if no configuration file is available).
- _getTTkDebugLevel ()

  Get current TTk status output debug level.
- getL10nInstance (locale_id=None, available_locales=None)

  Get a L10N instance matching the supplied locale id.

Variables

- str DEFAULT_INI_ENCODING = "windows-1252"

  default encoding to use for configuration files (.ini)

6.7.1 Detailed Description

Settings and configuration for TestToolkit packages.

6.7.2 Function Documentation

6.7.2.1 _getTTkDebugLevel()

```
ttk_base.configuration._getTTkDebugLevel ( )  [protected]
```

Get current TTk status output debug level.

Just for convenience.

6.7.2.2 findFile()

```
ttk_base.configuration.findFile (
              file_name,
              start_path = None,
              max_parent_levels = 6 )
```

Find a matching file by traversing the folder tree upwards, starting at start_path.

Note

The starting folder does not necessarily have to exist for a file to be found, as long as somewhere "up" the tree a folder with the expected file exist.

Parameters

file_name	file name to find
start_path	path to start in, None: default to package folder
max_parent_levels	maximum number of parent folders to "go up"

Returns

a path to the first matching file or None if no matching file was found.

6.7.2.3 getExpectedDongleVersion()

```
ttk_base.configuration.getExpectedDongleVersion ( )
```

Get a string representation of the expected Dongle-DLL version for this release (see EXPECTED_DONGLE_DLL↩ _VERSION)

Returns

a formatted version string.

6.7.2.4 getL10nInstance()

```
ttk_base.configuration.getL10nInstance (
              locale_id = None,
              available_locales = None )
```

Get a L10N instance matching the supplied locale id.

Parameters

locale_id	id of locale to instantiate. If no specific id has been supplied, the ID will be taken from project settings.
available_locales	a dictionary of available locale classes (derived from L10N) as <locale name>: <LocaleClass>

Note

If a locale_id ends with a "debug" suffix, any missing entries will be logged (i.e. L10N's parameter debug_↩ missing_entries will be set to True.

Returns

a L10N instance (see ttk_base.localization.L10N).

6.7.2.5 getSettings()

```
ttk_base.configuration.getSettings ( )
```

Get the current project settings (which may be empty if no configuration file is available).

6.7.3 Variable Documentation

6.7.3.1 DEFAULT_INI_ENCODING

```
str ttk_base.configuration.DEFAULT_INI_ENCODING = "windows-1252"
```

default encoding to use for configuration files (.ini)

6.8 ttk_base.dspace_rtplib Namespace Reference

Abstraction layer for accessing HIL variables using dSPACE RTPlib interfaces.

Namespaces

- namespace _variables

  Base classes for dSPACE rtplib/rtplib2 hil variables to provide a unified interface and containers to group variables into sets.
- namespace variables

  Wrapper classes for dSPACE rtplib/rtplib2 hil variables to provide a unified interface and containers to group variables into sets.

6.8.1 Detailed Description

Abstraction layer for accessing HIL variables using dSPACE RTPlib interfaces.

6.9 ttk_base.dspace_rtplib._variables Namespace Reference

Base classes for dSPACE rtplib/rtplib2 hil variables to provide a unified interface and containers to group variables into sets.

Classes

- class Var

  Simple wrapper for dSPACE (rtplib) "HIL" application variables.
- class VarContainer

  Base class for dSPACE (rtplib) HIL Var containers.

6.9.1 Detailed Description

Base classes for dSPACE rtplib/rtplib2 hil variables to provide a unified interface and containers to group variables into sets.

6.10 ttk_base.dspace_rtplib.variables Namespace Reference

Wrapper classes for dSPACE rtplib/rtplib2 hil variables to provide a unified interface and containers to group variables into sets.

Classes

- class Var

  Simple wrapper for dSPACE (rtplib) "HIL" application variables.
- class VarContainer

  Base class for rtplib-based "HIL" variable containers.

6.10.1 Detailed Description

Wrapper classes for dSPACE rtplib/rtplib2 hil variables to provide a unified interface and containers to group variables into sets.

This serves as "interface" to the precompiled module in delivery to permit better code-analysis and code-completion in PyDev.

6.11 ttk_base.dspace_xilapi Namespace Reference

Abstraction layer for accessing HIL variables using dSPACE XIL-API/Testbench interfaces.

Namespaces

- namespace _variables

  Base classes for dSPACE XIL-API-based hil variables to provide a unified interface and containers to group variables into sets.
- namespace variables

  Wrapper classes for dSPACE XIL-API / MAPort hil variables to provide a unified interface and containers to group variables into sets.

6.11.1 Detailed Description

Abstraction layer for accessing HIL variables using dSPACE XIL-API/Testbench interfaces.

6.12 ttk_base.dspace_xilapi._variables Namespace Reference

Base classes for dSPACE XIL-API-based hil variables to provide a unified interface and containers to group variables into sets.

Classes

- class Var

  Simple wrapper for dSPACE (XIL-API) "HIL" application variables.
- class VarContainer

  Base class for dSPACE (XIL API) "HIL" variable containers.

6.12.1 Detailed Description

Base classes for dSPACE XIL-API-based hil variables to provide a unified interface and containers to group variables into sets.

6.13 ttk_base.dspace_xilapi.variables Namespace Reference

Wrapper classes for dSPACE XIL-API / MAPort hil variables to provide a unified interface and containers to group variables into sets.

Classes

- class Var

  Simple wrapper for dSPACE (XIL-API) "HIL" application variables.
- class VarContainer

  Base class for dSPACE (XIL API) "HIL" variable containers.

6.13.1 Detailed Description

Wrapper classes for dSPACE XIL-API / MAPort hil variables to provide a unified interface and containers to group variables into sets.

This serves as "interface" to the precompiled module in delivery to permit better code-analysis and code-completion in PyDev.

6.14 ttk_base.errors Namespace Reference

Base classes and definitions for TestToolkit exceptions/errors.

Classes

- class TTkErrorString

  A wrapper for error message strings so they can be distinguished from "normal" string values (like contents of ASCII-type variables in the case of calibration devices)
- class TTkException

6.14.1 Detailed Description

Base classes and definitions for TestToolkit exceptions/errors.

6.15 ttk_base.isyst_ida Namespace Reference

Abstraction layer for accessing HIL variables using iSyst iDA Testware.

Namespaces

- namespace _variables

  Base classes for iDA-Testware-based hil variables to provide a unified interface and containers to group variables into sets.
- namespace variables

  Wrapper classes for iDA-Testware-based HIL variables to provide a unified interface and containers to group variables into sets.

6.15.1 Detailed Description

Abstraction layer for accessing HIL variables using iSyst iDA Testware.

6.16 ttk_base.isyst_ida._variables Namespace Reference

Base classes for iDA-Testware-based hil variables to provide a unified interface and containers to group variables into sets.

Classes

- class Var

  Simple wrapper for iDA Testware "HIL" application variables.
- class VarContainer

  Base class for iDA Testware "HIL" variable containers.

6.16.1 Detailed Description

Base classes for iDA-Testware-based hil variables to provide a unified interface and containers to group variables into sets.

6.17 ttk_base.isyst_ida.variables Namespace Reference

Wrapper classes for iDA-Testware-based HIL variables to provide a unified interface and containers to group variables into sets.

Classes

- class Var

  Simple wrapper for iDA Testware "HIL" application variables.
- class VarContainer

  Base class for iDA Testware "HIL" variable containers.

6.17.1 Detailed Description

Wrapper classes for iDA-Testware-based HIL variables to provide a unified interface and containers to group variables into sets.

Wrapper classes for variables using SIMIT SHM Coupling signals to provide a unified interface and containers to group variables into sets.

This serves as "interface" to the precompiled module in delivery to permit better code-analysis and code-completion in PyDev.

6.18 ttk_base.localization Namespace Reference

Base class for localization.

Classes

- class L10N

  Base class for localization handling.

Functions

- showL10nDifferences (loc_a, loc_b)

  Show/print differences between localization keys in two L10N-derived instances.

6.18.1 Detailed Description

Base class for localization.

6.18.2 Function Documentation

6.18.2.1 showL10nDifferences()

```
ttk_base.localization.showL10nDifferences (
              loc_a,
              loc_b )
```

Show/print differences between localization keys in two L10N-derived instances.

Parameters

loc↩ _a	a L10N-based localization class
loc↩ _b	another L10N-based localization class

6.19 ttk_base.ni_labview Namespace Reference

Abstraction layer for accessing National Instruments LabView variables/controls.

Namespaces

- namespace _variables

  Base classes for hil variables using an interface to NI LabView to provide a unified interface and containers to group variables into sets.
- namespace variables

  Wrapper classes for hil variables using an interface to NI LabView to provide a unified interface and containers to group variables into sets.

6.19.1 Detailed Description

Abstraction layer for accessing National Instruments LabView variables/controls.

6.20 ttk_base.ni_labview._variables Namespace Reference

Base classes for hil variables using an interface to NI LabView to provide a unified interface and containers to group variables into sets.

Classes

- class Var

  Simple wrapper for NI LabView "HIL" variables/controls.
- class VarContainer

  Base class for LabView "HIL" variable containers.

6.20.1 Detailed Description

Base classes for hil variables using an interface to NI LabView to provide a unified interface and containers to group variables into sets.

6.21 ttk_base.ni_labview.variables Namespace Reference

Wrapper classes for hil variables using an interface to NI LabView to provide a unified interface and containers to group variables into sets.

Classes

- class Var

  Simple wrapper for NI LabView "HIL" variables/controls.
- class VarContainer

  Base class for LabView "HIL" variable containers.

6.21.1 Detailed Description

Wrapper classes for hil variables using an interface to NI LabView to provide a unified interface and containers to group variables into sets.

This serves as "interface" to the precompiled module in delivery to permit better code-analysis and code-completion in PyDev.

6.22 ttk_base.pyocd Namespace Reference

Abstraction layer for accessing PyOCD debugger symbols/variables.

Namespaces

- namespace _variables

  Base classes for PyOCD-based hil variables to provide a unified interface and containers to group variables into sets.
- namespace variables

6.22.1 Detailed Description

Abstraction layer for accessing PyOCD debugger symbols/variables.

6.23 ttk_base.pyocd._variables Namespace Reference

Base classes for PyOCD-based hil variables to provide a unified interface and containers to group variables into sets.

Classes

- class Var

  Simple wrapper for PyOCD variables in a attached debugging session.
- class VarContainer

  Base class for variable containers.

6.23.1 Detailed Description

Base classes for PyOCD-based hil variables to provide a unified interface and containers to group variables into sets.

Wrapper classes for PyOCD-based hil variables to provide a unified interface and containers to group variables into sets.

This serves as "interface" to the precompiled module in delivery to permit better code-analysis and code-completion in PyDev.

6.24 ttk_base.pyocd.variables Namespace Reference

Classes

- class Var

  Simple wrapper for PyOCD variables in a attached debugging session.
- class VarContainer

  Base class for variable containers.

6.25 ttk_base.rst_gammav Namespace Reference

Abstraction layer for accessing RST (RST Industrie Automation GmbH) Gamma V realtime middleware process variables.

Namespaces

- namespace _variables

  Base classes for RST Gamma V HIL variables to provide a unified interface and containers to group variables into sets.
- namespace variables

  Wrapper classes for RST Gamma V hil variables to provide a unified interface and containers to group variables into sets.

6.25.1 Detailed Description

Abstraction layer for accessing RST (RST Industrie Automation GmbH) Gamma V realtime middleware process variables.

6.26 ttk_base.rst_gammav._variables Namespace Reference

Base classes for RST Gamma V HIL variables to provide a unified interface and containers to group variables into sets.

Classes

- class Var

  Simple wrapper for Gamma V variables.
- class VarContainer

  Base class for Gamma V "GammaVar" containers.
- class VarDeferred

  Simple wrapper for Gamma V process variables with deferred initialization to speed up loading of containers with many GammaVars.

6.26.1 Detailed Description

Base classes for RST Gamma V HIL variables to provide a unified interface and containers to group variables into sets.

6.27 ttk_base.rst_gammav.variables Namespace Reference

Wrapper classes for RST Gamma V hil variables to provide a unified interface and containers to group variables into sets.

Classes

- class Var

  Simple wrapper for Gamma V variables.
- class VarContainer

  Base class for Gamma-V-based "HIL" variable containers.
- class VarDeferred

  Simple wrapper for Gamma V variables with deferred initialization to speed up loading of containers with many GammaVars.

6.27.1 Detailed Description

Wrapper classes for RST Gamma V hil variables to provide a unified interface and containers to group variables into sets.

This serves as "interface" to the precompiled module in delivery to permit better code-analysis and code-completion in PyDev.

6.28 ttk_base.siemens_simit Namespace Reference

Abstraction layer for accessing variables on SIMIT SHM Coupling interfaces.

Namespaces

- namespace _variables

  Base classes for variables using SIMIT SHM Coupling signals to provide a unified interface and containers to group variables into sets.
- namespace variables

6.28.1 Detailed Description

Abstraction layer for accessing variables on SIMIT SHM Coupling interfaces.

6.29 ttk_base.siemens_simit._variables Namespace Reference

Base classes for variables using SIMIT SHM Coupling signals to provide a unified interface and containers to group variables into sets.

Classes

- class Var

  Simple wrapper for SIMIT SHM Coupling variables.
- class VarContainer

  Base class for SIMIT SHM coupling signal variable containers.

6.29.1 Detailed Description

Base classes for variables using SIMIT SHM Coupling signals to provide a unified interface and containers to group variables into sets.

6.30 ttk_base.siemens_simit.variables Namespace Reference

Classes

- class Var

  Simple wrapper for SIMIT SHM Coupling variables.
- class VarContainer

  Base class for SIMIT SHM coupling signal variable containers.

6.31 ttk_base.values_base Namespace Reference

Interface Wrapper for values with additional meta information in ttk_base._values_base.

Functions

- meta (value, unit="", alias="", descr="", lookup=None, fmt=None)

  Factory function to create "meta" instances of the supplied value that have some additional meta attributes bolted on.

6.31.1 Detailed Description

Interface Wrapper for values with additional meta information in ttk_base._values_base.

This serves as "interface" to the precompiled module in delivery to enable code-completion in PyDev.

6.31.2 Function Documentation

6.31.2.1 meta()

```
ttk_base.values_base.meta (
              value,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              fmt = None )
```

Factory function to create "meta" instances of the supplied value that have some additional meta attributes bolted on.

Additionally, meta instances have a method info() that will return an info string (alias and current value, similar to variables from variables_base. See _values_base.MetaValue

Example: What's the point?

```
# Wrapping a value in a meta instance permits check/test functions
# to extract the associated meta data to give better feedback.
>> checkTolerance(11.3, 12, abs_pos=.8)[0]
Expected value: 12
Current value:  11.3
Absolute Tolerance:  +/- 0.8 V

from ttk_base.values_base import meta
>> checkTolerance(meta(11.3, "V"), meta(12, "V"), abs_pos=.8)[0]
Expected value: 12 V
Current value:  11.3 V
Absolute Tolerance:  +/- 0.8 V

# As variables using VarBase can also contain meta data, a simple
# check might look like this:
>> checkTolerance(cal.v_bat, meta(12, "V"), abs_pos=.8)[0]
Expected value: 12 V
Current value:  12.41 V
Absolute Tolerance:  +/- 0.8
```

Attention

Meta data is - more or less - just "eye candy" for nicer report output. It does **not** influence the actual test processing, so you could compare apples and oranges, V and mA, or mV and kV.
Just don't expect the values to be "automagically" converted to their base units for comparison purposes.

Example: Meta data does not influence checks

```
# This is just wrong (though the check will be PASSED,
# as it ignores meta info):
»> checkTolerance(meta(12, "V"), meta(12, "mV"), abs_pos=.8)[0]
Expected value: 12 mV
Current value:  12 V
Absolute Tolerance:  +/- 0.8 V
```

Meta Attributes (via MetaValue):

- .alias - alias name
- .descr - description string
- .unit - unit string
- .fmt - value formatting

If a lookup table has been supplied, it will be available as .state_lookup (same as for VarBase-derived variables).
See _values_base.MetaValue

MetaValue Methods:

- .info() - gets an info string for this value
- .hasStateLookup() - True if a lookup table is configured
- .getStateDescr(value) - get a state description text for the value
- .hasFormattedRep() - True if a non-default format is configured
- .getFormattedValue(value) - get a formatted string representation for the value

See _values_base.MetaValue
Compare interface of ttk_base._variables_base.VarBase

Parameters

value	value that should get meta info added. Currently supported types are int, long, float, str and unicode
unit	meta info: value's unit, e.g. mV
alias	meta info: an alias name for the value
descr	meta info: description/comment text
lookup	meta info: lookup table, dict with mappings {value => "state name"}
fmt	meta info: format string to use for value representation

Returns

an instance of a class derived from value's class, that has the additional meta attributes set.

6.32 ttk_base.variables_base Namespace Reference

Interface Wrapper for variables' base classes in ttk_base._variables_base.

Classes

- class DummyVar

  A minimal dummy test variable not connected to any external interface.
- class Snapshot

  Store and restore a "snapshot" of multiple variables' values.

6.32.1 Detailed Description

Interface Wrapper for variables' base classes in ttk_base._variables_base.

This serves as "interface" to the precompiled module in delivery to enable code-completion in PyDev.

Note

ttk_base._variables_base also defines VarBase, the base class for specific variable classes. This base class has no wrapped counterpart here, though, as it is only intended for internal use.

See derived classes for details:

- ttk_base.dspace_rtplib.variables.Var
- ttk_base.dspace_xilapi.variables.Var
- ttk_base.rst_gammav.variables.Var
- ttk_base.vector_canape.variables.Var

A basic implementation VarBase is included in this wrapper, which is not connected to any external interface. See DummyVar

6.33 ttk_base.vector_canape Namespace Reference

Abstraction layer for accessing Vector CANape device variables.

Namespaces

- namespace _variables

  Base classes for cal variables using Vector CANape to provide a unified interface and containers to group variables into sets.
- namespace variables

  Wrapper classes for Vector CANape cal variables to provide a unified interface and containers to group variables into sets.

6.33.1 Detailed Description

Abstraction layer for accessing Vector CANape device variables.

6.34 ttk_base.vector_canape._variables Namespace Reference

Base classes for cal variables using Vector CANape to provide a unified interface and containers to group variables into sets.

Classes

- class Var

  Simple wrapper for Vector CANape calibration variables.
- class VarContainer

  Base class for CalVar containers.

6.34.1 Detailed Description

Base classes for cal variables using Vector CANape to provide a unified interface and containers to group variables into sets.

6.35 ttk_base.vector_canape.variables Namespace Reference

Wrapper classes for Vector CANape cal variables to provide a unified interface and containers to group variables into sets.

Classes

- class Var

  Simple wrapper for Vector CANape calibration variables.
- class VarContainer

  Base class for CalVar containers.

6.35.1 Detailed Description

Wrapper classes for Vector CANape cal variables to provide a unified interface and containers to group variables into sets.

This serves as "interface" to the precompiled module in delivery to permit better code-analysis and code-completion in PyDev.

6.36 ttk_bus Namespace Reference

Test functions for bus/interface tests and abstraction layer for "bus signals".

Namespaces

- namespace _bus_signals_base

  Container classes for bus/interface signal descriptions.
- namespace _bus_signals_cangamma

  Container classes for bus/interface signal descriptions.
- namespace _bus_signals_canida

  Container classes for bus/interface signal descriptions.
- namespace _bus_signals_canmmbs

  Container classes for bus/interface signal descriptions.
- namespace _bus_signals_ethbs

  Container classes for "bus"/interface signal descriptions.
- namespace _bus_signals_frbs

  Container classes for bus/interface signal descriptions.
- namespace _bus_test_utils

  Utility functions for bus tests.
- namespace _bus_tests

  Test functions for bus/interface tests.
- namespace bus_signals_base

  Interface wrapper for bus/interface signal classes in ttk_bus._bus_signals_base.
- namespace bus_signals_cangamma

  Interface wrapper bus/interface signal classes in ttk_bus._bus_signals_cangamma.
- namespace bus_signals_canida

  Interface wrapper bus/interface signal classes in ttk_bus._bus_signals_canida.
- namespace bus_signals_canmmbs

  Interface wrapper bus/interface signal classes in ttk_bus._bus_signals_canmmbs.
- namespace bus_signals_ethbs

  Interface wrapper for "bus"/interface signal classes in ttk_bus._bus_signals_ethbs.
- namespace bus_signals_frbs

  Interface wrapper for bus/interface signal classes in ttk_bus._bus_signals_frbs.
- namespace bus_test_utils

  Interface wrapper for bus test utility functions in ttk_bus._bus_test_utils.
- namespace bus_tests

  Interface wrapper for bus/interface test functions (HIL <–bus-signal–> DUT) in ttk_bus._bus_tests.
- namespace bus_tests_loc

  Localization look-up for bus test descriptions.

6.36.1 Detailed Description

Test functions for bus/interface tests and abstraction layer for "bus signals".

6.37 ttk_bus._bus_signals_base Namespace Reference

Container classes for bus/interface signal descriptions.

Classes

- class AcBusSignal

  A RX bus signal (HIL => ECU) that models an alive counter value with additional error injections.
- class BusSignal

  Base signal container for a bus signal with a HIL-side model variable and relevant associated variables/values.
- class BusSignalContainer

  Base class for BusSignal containers.
- class CrcBusSignal
- class RxBusSignal

  A RX bus signal (HIL => ECU) that models a CRC value with additional error injections.
- class TxBusSignal

  A signal sent from the DUT to the test system (ECU Tx => HIL Rx).

6.37.1 Detailed Description

Container classes for bus/interface signal descriptions.

Convention:

Signal directions are relative to the DUT:

- Tx - transmitted by ECU => received from HIL
- Rx - received by ECU <= transmitted from HIL

6.38 ttk_bus._bus_signals_cangamma Namespace Reference

Container classes for bus/interface signal descriptions.

Classes

- class _ExtraVarsMixin

  Mixin to add a utility method to initialize "extra" helper Gamma-HilVars.
- class AcRxCanBusSignal

  Alive counter RX bus signal (HIL-Tx => ECU-Rx).
- class CanBusMessage

  CAN-Message/PDU-specific data and methods.
- class CanBusSignal

  Base wrapper for a bus signal with HIL-side BusSignals variable and relevant associated values.
- class CanBusSignalContainer

  A BusSignalContainer for CAN Bus Signals.
- class CrcRxCanBusSignal

  A CRC RX bus signal (HIL-Tx => ECU-Rx)
- class CrcTxCanBusSignal

  A CRC TX bus signal (ECU-Tx => HIL-Rx), adds CRC validity info.
- class RxCanBusMessage

CAN Message/PDU ECU-Rx (HIL-Tx => ECU-Rx).

- class RxCanBusSignal

  A CAN signal sent from the test system to the DUT (ECU Rx <= HIL Tx).
- class RxCanBusSignalBasic

  A basic CAN signal sent from the test system to the DUT (ECU Rx <= HIL Tx).
- class TxCanBusMessage

  CAN Message/PDU ECU-Tx (ECU-Tx => HIL-Rx).
- class TxCanBusSignal

  A CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).
- class TxCanBusSignalBasic

  A basic CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).

Variables

- int FALLBACK_CHANGE_DELAY_MS = 100

  Fallback value used for change_delay in event-triggered RxCanBusSignals w/o a specified change_delay, to use as a somewhat sensible default delay to permit signal propagation.
- int FALLBACK_ERROR_DELAY_MS = 200

  Fallback: Somewhat sensible default error delay used in AcCanBusSignal and CrcCanBusSignal for signals/PDUs without any specified cycle_time / debounce_time)

6.38.1 Detailed Description

Container classes for bus/interface signal descriptions.

Specific implementation for Gamma-V-based CAN blocksets

Convention:

Signal directions are relative to the DUT:

- Tx - transmitted by ECU => received from HIL
- Rx - received by ECU <= transmitted from HIL

6.38.2 Variable Documentation

6.38.2.1 FALLBACK_CHANGE_DELAY_MS

```
int ttk_bus._bus_signals_cangamma.FALLBACK_CHANGE_DELAY_MS = 100
```

Fallback value used for change_delay in event-triggered RxCanBusSignals w/o a specified change_delay, to use as a somewhat sensible default delay to permit signal propagation.

An empty change_delay may happen if neither a cycle_time nor a debounce_time has been set (and no "manual" change_delay was configured).

6.38.2.2 FALLBACK_ERROR_DELAY_MS

```
int ttk_bus._bus_signals_cangamma.FALLBACK_ERROR_DELAY_MS = 200
```

Fallback: Somewhat sensible default error delay used in AcCanBusSignal and CrcCanBusSignal for signals/PDUs without any specified cycle_time / debounce_time)

6.39 ttk_bus._bus_signals_canida Namespace Reference

Container classes for bus/interface signal descriptions.

Classes

- class _ExtraVarsMixin

  Mixin to add a utility method to initialize "extra" helper Gamma-HilVars.
- class CanBusMessage

  CAN-Message/PDU-specific data and methods.
- class CanBusSignal

  Base wrapper for a bus signal with HIL-side BusSignals variable and relevant associated values.
- class CanBusSignalContainer

  A BusSignalContainer for CAN Bus Signals.
- class RxCanBusMessage

  CAN Message/PDU ECU-Rx (HIL-Tx => ECU-Rx).
- class RxCanBusSignal

  A basic CAN signal sent from the test system to the DUT (ECU Rx <= HIL Tx).
- class TxCanBusMessage

  CAN Message/PDU ECU-Tx (ECU-Tx => HIL-Rx).
- class TxCanBusSignal

  A basic CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).

Variables

- int FALLBACK_CHANGE_DELAY_MS = 100

  Fallback value used for change_delay in event-triggered RxCanBusSignals (and RxBusMessages) w/o a specified change_delay (that is, both cycle time and change delay are 0), to use as a somewhat sensible default delay to permit signal propagation.
- int FALLBACK_ERROR_DELAY_MS = 200

  Fallback: Somewhat sensible default error delay used in AcCanBusSignal and CrcCanBusSignal for signals/PDUs without any specified cycle_time / debounce_time)

6.39.1 Detailed Description

Container classes for bus/interface signal descriptions.

Specific implementation for iDA-Blockset based CAN interfaces

Convention:

Signal directions are relative to the DUT:

- Tx - transmitted by ECU => received from HIL
- Rx - received by ECU <= transmitted from HIL

6.39.2 Variable Documentation

6.39.2.1 FALLBACK_CHANGE_DELAY_MS

```
int ttk_bus._bus_signals_canida.FALLBACK_CHANGE_DELAY_MS = 100
```

Fallback value used for change_delay in event-triggered RxCanBusSignals (and RxBusMessages) w/o a specified change_delay (that is, both cycle time and change delay are 0), to use as a somewhat sensible default delay to permit signal propagation.

An empty change_delay may happen if neither a cycle_time nor a debounce_time has been set (and no "manual" change_delay was configured).

6.39.2.2 FALLBACK_ERROR_DELAY_MS

```
int ttk_bus._bus_signals_canida.FALLBACK_ERROR_DELAY_MS = 200
```

Fallback: Somewhat sensible default error delay used in AcCanBusSignal and CrcCanBusSignal for signals/PDUs without any specified cycle_time / debounce_time)

6.40 ttk_bus._bus_signals_canmmbs Namespace Reference

Container classes for bus/interface signal descriptions.

Classes

- class _ExtraVarsMixin

  Mixin to add a utility method to initialize "extra" helper CANMM-HilVars.
- class AcCanBusSignal

  Alive counter bus signal derived from RxCanBusSignal (HIL-Tx => ECU-Rx) for use with dSPACE CAN Multi-Message blockset data structures.
- class CanBusMessage

  CAN-Message/PDU-specific data and methods.
- class CanBusSignal

  Base wrapper for a bus signal with HIL-side BusSignals variable and relevant associated values.
- class CanBusSignalContainer

  A BusSignalContainer for CAN Bus Signals.
- class CrcCanBusSignal

  A CRC RX bus signal derived from RxCanBusSignal (HIL-Tx => ECU-Rx) for use with dSPACE CAN Multi-Message blockset.
- class RxCanBusMessage

  CAN Message/PDU ECU-Rx (HIL-Tx => ECU-Rx).
- class RxCanBusSignal
- class TxCanBusMessage

  CAN Message/PDU ECU-Tx (ECU-Tx => HIL-Rx).
- class TxCanBusSignal

  A CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).

Variables

- int FALLBACK_CHANGE_DELAY_MS = 100

  Fallback value used for change_delay in event-triggered RxCanBusSignals w/o a specified change_delay, to use as a somewhat sensible default delay to permit signal propagation.
- int FALLBACK_ERROR_DELAY_MS = 200

  Fallback: Somewhat sensible default error delay used in AcCanBusSignal and CrcCanBusSignal for signals/PDUs without any specified cycle_time / debounce_time)

6.40.1 Detailed Description

Container classes for bus/interface signal descriptions.

Specific implementation for dSpace CAN Multi-Message blocksets (using "trc"-signals added to BusSystems)

Convention:

Signal directions are relative to the DUT:

- Tx - transmitted by ECU => received from HIL
- Rx - received by ECU <= transmitted from HIL

6.40.2 Variable Documentation

6.40.2.1 FALLBACK_CHANGE_DELAY_MS

```
int ttk_bus._bus_signals_canmmbs.FALLBACK_CHANGE_DELAY_MS = 100
```

Fallback value used for change_delay in event-triggered RxCanBusSignals w/o a specified change_delay, to use as a somewhat sensible default delay to permit signal propagation.

An empty change_delay may happen if neither a cycle_time nor a debounce_time has been set (and no "manual" change_delay was configured).

6.40.2.2 FALLBACK_ERROR_DELAY_MS

```
int ttk_bus._bus_signals_canmmbs.FALLBACK_ERROR_DELAY_MS = 200
```

Fallback: Somewhat sensible default error delay used in AcCanBusSignal and CrcCanBusSignal for signals/PDUs without any specified cycle_time / debounce_time)

6.41 ttk_bus._bus_signals_ethbs Namespace Reference

Container classes for "bus"/interface signal descriptions.

Classes

- class AcEthSignal

  An alive counter Rx Ethernet signal/parameter (HIL PSI => ECU CSI) for use with dSPACE Ethernet blockset.
- class CrcEthSignal

  A CRC RX Ethernet event signal/parameter (HIL-Tx => ECU-Rx) for use with dSPACE Ethernet blockset.
- class CrcEthStatus
- class EthBusSysSignal
- class EthEvent

  Ethernet Event (or Field) specific data and methods.
- class EthSignal

  Base wrapper for a bus "signal" with HIL-side Ethernet Signals variable and relevant associated values.
- class EthSignalContainer

  A container for Ethernet Signals.
- class RxEthBusSysSignal

  An Ethernet event signal/parameter/sub-parameter sent from test system to DUT (HIL PSI Tx => ECU CSI Rx).
- class RxEthEvent

  Ethernet-Event-specific data and methods - Rx (HIL-Tx => ECU-Rx).
- class RxEthSignal

  An Ethernet event signal/parameter/sub-parameter sent from test system to DUT (ECU CSI Rx <= HIL PSI Tx).
- class TxEthBusSysSignal

  An Ethernet event signal/parameter/sub-parameter sent from DUT to test system (ECU PSI Tx => HIL CSI Rx).
- class TxEthEvent

  Ethernet-Event-specific data and methods - Tx (ECU-Tx => HIL-Rx).
- class TxEthSignal

  An Ethernet event signal/parameter/sub-parameter sent from DUT to test system (ECU PSI Tx => HIL CSI Rx).

6.41.1 Detailed Description

Container classes for "bus"/interface signal descriptions.

Specific implementation for dSpace Ethernet blocksets ("Ethernet Configuration Package") using ModelRoot-signals and (for E2E) BusSystems-signals)

Convention:

Signal directions are relative to the DUT:

- Tx - transmitted by ECU => received from HIL
- Rx - received by ECU <= transmitted from HIL

- PSI - provider, Tx
- CSI - consumer, Rx

6.42 ttk_bus._bus_signals_frbs Namespace Reference

Container classes for bus/interface signal descriptions.

Classes

- class AcFrBusSignal

  An alive counter RX bus signal derived from RxFrBusSignal (HIL => ECU) for use with dSPACE FlexRay blockset PDU-based data structures.
- class CrcFrBusSignal

  A CRC RX bus signal derived from RxFrBusSignal (HIL-Tx => ECU-Rx) for use with dSPACE FlexRay blockset with PDU-based structure.
- class CrcFrBusStatus

  CRC Tx-Signal status (ECU Tx => HIL Rx) and associated switch variables for dSPACE FlexRay blockset with PDU-based modeling.
- class FrbsError

  Basic Exception for FRBS (FlexRay block set) errors.
- class FrBusSignal

  Base wrapper for a bus signal with HIL-side BusSignals variable and relevant associated values.
- class FrBusSignalContainer

  A BusSignalContainer for FlexRay Bus Signals.
- class FrPdu
- class FrRxPdu

  FlexRay ECU-Rx (HIL-Tx => ECU-Rx) PDU data.
- class FrTxPdu

  FlexRay ECU-Tx (ECU-Tx => HIL-Rx) PDU data.
- class RxFrBusSignal

  A FlexRay signal sent from test system to the DUT (ECU Rx <= HIL Tx).
- class TxFrBusSignal

  A FLexRay signal sent from DUT to the test system (ECU Tx => HIL Rx).

Variables

- int FALLBACK_CHANGE_DELAY_MS = 100

  Fallback value used for change_delay in event-triggered RxFrBusSignals w/o a specified change_delay, to use as a somewhat sensible default delay to permit signal propagation.
- int FALLBACK_ERROR_DELAY_MS = 200

  Fallback: Somewhat sensible default error delay used in AcFrBusSignal and CrcFrBusSignal for signals/PDUs without any specified cycle_time / debounce_time)

6.42.1 Detailed Description

Container classes for bus/interface signal descriptions.

Specific implementation for dSpace FlexRay blocksets using PDU-based blockset generation with signals added to Trc/BusSystems.

Convention:

Signal directions are relative to the DUT:

- Tx - transmitted by ECU => received from HIL
- Rx - received by ECU <= transmitted from HIL

6.42.2 Variable Documentation

6.42.2.1 FALLBACK_CHANGE_DELAY_MS

```
int ttk_bus._bus_signals_frbs.FALLBACK_CHANGE_DELAY_MS = 100
```

Fallback value used for change_delay in event-triggered RxFrBusSignals w/o a specified change_delay, to use as a somewhat sensible default delay to permit signal propagation.

An empty change_delay may happen if neither a cycle_time nor a debounce_time has been set (and no "manual" change_delay was configured).

6.42.2.2 FALLBACK_ERROR_DELAY_MS

```
int ttk_bus._bus_signals_frbs.FALLBACK_ERROR_DELAY_MS = 200
```

Fallback: Somewhat sensible default error delay used in AcFrBusSignal and CrcFrBusSignal for signals/PDUs without any specified cycle_time / debounce_time)

6.43 ttk_bus._bus_test_utils Namespace Reference

Utility functions for bus tests.

Functions

- getIntRangeValues (min_value=None, max_value=None, amount=5)

  Get a list of equally distributed (unique) values within min and max value boundaries.
- getRangeValues (min_value=None, max_value=None, amount=5)

  Get a list of equally distributed values within min and max value boundaries.

6.43.1 Detailed Description

Utility functions for bus tests.

6.43.2 Function Documentation

6.43.2.1 getIntRangeValues()

```
ttk_bus._bus_test_utils.getIntRangeValues (
              min_value = None,
              max_value = None,
              amount = 5 )
```

Get a list of equally distributed (unique) values within min and max value boundaries.

Parameters

min_value	lower boundary of value range
max_value	upper boundary of value range
amount	maximum amount of values to be returned (fewer values might be returned if the range cannot fit enough integers)

Returns

a list of values equally distributed between min and max values.

6.43.2.2 getRangeValues()

```
ttk_bus._bus_test_utils.getRangeValues (
              min_value = None,
              max_value = None,
              amount = 5 )
```

Get a list of equally distributed values within min and max value boundaries.

Parameters

min_value	lower boundary of value range
max_value	upper boundary of value range
amount	how many values should be returned

Returns

a list of values equally distributed between min and max values.

6.44 ttk_bus._bus_tests Namespace Reference

Test functions for bus/interface tests.

Functions

- _addToEntry (d, key, value)

  Add value to an existing key, create a new key as needed.
- testRxSignal (bus_signal_data, set_value, set_value_descr='', check_var_list=None, max_retests=0, descr='', format=None)

  Check the behaviour of a single bus signal sent from HIL system and received by the ECU.
- testTxSignal (set_var_list, bus_signal_data, check_value, check_value_descr="", rel_pos=None, abs_↩pos=None, rel_neg=None, abs_neg=None, max_retests=0, descr='', format=None)

  Check the behaviour of a single bus signal sent from the ECU and received by the HIL system.
- testSignals (set_var_list=None, check_var_list=None, min_wait_time_ms=0, descr='')

  Check the values of multiple variables/signals as a result of setting multiple other variables/signals.

- _testBasicSignalError (bus_signal_data, check_var_list_before=None, check_var_list_after=None, manual_wait_cycles=None, manual_triggers_before=0, manual_triggers_between=0, manual_trigger_delay_ms=0, log_title="testBasicSignalError", signal_type="Generic", set_error_method_name="set", clear_error_method_name="clear", descr='', format=None)

  Check ECU behaviour on errors at HIL-generated dynamic signals (like CRCs or Alive Counters) for messages sent from HIL system and received by the ECU.
- testBasicCrcError (crc_signal_data, check_var_list_before=None, check_var_list_after=None, manual_wait_cycles=None, manual_triggers_before=0, manual_triggers_between=0, manual_trigger_delay_ms=0, descr='', format=None)

  Check ECU behaviour on CRC signal errors (for messages sent from HIL system and received by the ECU).
- testBasicAcError (ac_signal_data, check_var_list_before=None, check_var_list_after=None, manual_wait_cycles=None, manual_triggers_before=0, manual_triggers_between=0, manual_trigger_delay_ms=0, descr='', format=None)

  Check ECU behaviour on alive counter signal errors (for messages sent from HIL system and received by the ECU).

6.44.1 Detailed Description

Test functions for bus/interface tests.

Requires package ttk_checks.

Conventions:

Signal directions are specified as relative to the DUT:

- Tx - transmitted from ECU => received by HIL
- Rx - received by ECU <= transmitted from HIL

See interface wrapper ttk_tools.bus_tests for more detailed docstring.

6.44.2 Function Documentation

6.44.2.1 _addToEntry()

```
ttk_bus._bus_tests._addToEntry (
            d,
            key,
            value )  [protected]
```

Add value to an existing key, create a new key as needed.

Strings will be concatenated, numbers will be added. Adding numeric values to strings (and vice-versa) will fail as usual.

Parameters

d	dictionary
key	new or existing dictionary key to add to
value	value to add

6.44.2.2 _testBasicSignalError()

```
ttk_bus._bus_tests._testBasicSignalError (
                bus_signal_data,
                check_var_list_before = None,
                check_var_list_after = None,
                manual_wait_cycles = None,
                manual_triggers_before = 0,
                manual_triggers_between = 0,
                manual_trigger_delay_ms = 0,
                log_title = "testBasicSignalError",
                signal_type = "Generic",
                set_error_method_name = "set",
                clear_error_method_name = "clear",
                descr = '',
                format = None )  [protected]
```

Check ECU behaviour on errors at HIL-generated dynamic signals (like CRCs or Alive Counters) for messages sent from HIL system and received by the ECU.

Info: This function...

- checks the "before" variables,
- calls the set error method on the bus signal (if available)
- waits a defined number of message cycles, then
- checks the "after" variables.
- calls the clear error method on the bus signal (if available)

Note

If both `check_var_list_before` and `check_var_list_after` are empty (i.e. nothing to test), the resulting verdict will also be empty (i.e. no result).

Parameters

bus_signal_data	bus signal data structure which describes the crc/ac/... signal. See bus_signals_base
check_var_list_before	variables to verify before injecting an error. A list of dictionaries with variable, rated value and tolerance See parameter example in testSignals()
check_var_list_after	variables to verify after injecting an error. A list of dictionaries with variable, rated value and tolerance See parameter example in testSignals()
manual_wait_cycles	if set, overrides the default number of cycle times to wait between error injection and variable checks
manual_triggers_before	number of manual triggers to perform on bus signal (via kickout()) prior to the "before" check. Obviously this makes the most sense for event-based signals
manual_triggers_between	number of manual triggers to perform on bus signal (via kickout()) prior to the "after" check. Again, this makes the most sense for event-based signals
manual_trigger_delay_ms	[ms] delay between trigger calls (usually the kickout() method has its own built-in minimum delay, though)
log_title	"title" to display in log output
signal_type	signal type for nicer log output, e.g. CRC or Alive Counter
set_error_method_name	name of bus_signal_data method used to set the error state
clear_error_method_name	name of bus_signal_data method used to clear the error state
descr	user specific description shown at start of result text
format	format string to control/override appearance of automatically created result text. A format override should only rarely be used.

Returns

a single test result entry (with a single verdict for all checked variables).

6.44.2.3 testBasicAcError()

```
ttk_bus._bus_tests.testBasicAcError (
              ac_signal_data,
              check_var_list_before = None,
              check_var_list_after = None,
              manual_wait_cycles = None,
              manual_triggers_before = 0,
              manual_triggers_between = 0,
              manual_trigger_delay_ms = 0,
              descr = '',
              format = None )
```

Check ECU behaviour on alive counter signal errors (for messages sent from HIL system and received by the ECU).

Info: This function...

- checks the "before" variables,
- sets the alive counter signal to an error state,
- waits a defined number of message cycles, then
- checks the "after" variables.

Note

If both `check_var_list_before` and `check_var_list_after` are empty (i.e. nothing to test), the resulting verdict will also be empty (i.e. no result).

Parameters

ac_signal_data	bus signal data structure which describes the crc signal. See bus_signals_base
check_var_list_before	variables to verify before injecting an error. A list of dictionaries with variable, rated value and tolerance See parameter example in testSignals()
check_var_list_after	variables to verify after injecting an error. A list of dictionaries with variable, rated value and tolerance See parameter example in testSignals()
manual_wait_cycles	if set, overrides the default number of cycle times to wait between crc error injection and variable checks (ac_signal_data.error_delay)
manual_triggers_before	number of manual triggers to perform on bus signal (via kickout()) prior to the "before" check. Obviously this makes the most sense for event-based signals
manual_triggers_between	number of manual triggers to perform on bus signal (via kickout()) prior to the "after" check. Again, this makes the most sense for event-based signals
manual_trigger_delay_ms	[ms] delay between trigger calls (usually the kickout() method has its own built-in minimum delay, though)
descr	user specific description shown at start of result text
format	format string to control/override appearance of automatically created result text. A format override should only rarely be used.

Returns

a single test result entry (with one verdict for all checked variables).

6.44.2.4 testBasicCrcError()

```
ttk_bus._bus_tests.testBasicCrcError (
              crc_signal_data,
              check_var_list_before = None,
              check_var_list_after = None,
              manual_wait_cycles = None,
              manual_triggers_before = 0,
              manual_triggers_between = 0,
              manual_trigger_delay_ms = 0,
              descr = '',
              format = None )
```

Check ECU behaviour on CRC signal errors (for messages sent from HIL system and received by the ECU).

Info: This function ...

- checks the "before" variables,
- sets the crc signal to an error state,
- waits a defined number of message cycles, then
- checks the "after" variables.

Note

If both `check_var_list_before` and `check_var_list_after` are empty (i.e. nothing to test), the resulting verdict will also be empty (i.e. no result).

Parameters

crc_signal_data	bus signal data structure which describes the crc signal. See bus_signals_base
check_var_list_before	variables to verify before injecting an error. A list of dictionaries with variable, rated value and tolerance See parameter example in testSignals()
check_var_list_after	variables to verify after injecting an error. A list of dictionaries with variable, rated value and tolerance See parameter example in testSignals()
manual_wait_cycles	if set, overrides the default number of cycle times to wait between crc error injection and variable checks (crc_signal_data.error_delay)
manual_triggers_before	number of manual triggers to perform on bus signal (via kickout()) prior to the "before" check. Obviously this makes the most sense for event-based signals
manual_triggers_between	number of manual triggers to perform on bus signal (via kickout()) prior to the "after" check. Again, this makes the most sense for event-based signals
manual_trigger_delay_ms	[ms] delay between trigger calls (usually the kickout() method has its own built-in minimum delay, though)
descr	user specific description shown at start of result text
format	format string to control/override appearance of automatically created result text. A format override should only rarely be used.

Returns

a single test result entry (with a single verdict for all checked variables).

6.44.2.5 testRxSignal()

```
ttk_bus._bus_tests.testRxSignal (
               bus_signal_data,
               set_value,
               set_value_descr = '',
               check_var_list = None,
               max_retests = 0,
               descr = '',
               format = None )
```

Check the behaviour of a single bus signal sent from HIL system and received by the ECU.

Info: This function ...

- sets the signal contained in bus_signal_data in the HIL model/system
- triggers sending the signal if needed (for event-triggered PDUs)
- waits for value change(s) to propagate to the ECU/DUT
- checks the value(s) inside the ECU using the supplied cal variables.

Parameters

bus_signal_data	data structure which describes the signal on the bus system. See bus_signals_base.RxBusSignal
set_value	value to set on the HIL side
set_value_descr	string which describes the value to set
check_var_list	list of dictionaries with variable, rated value and tolerance
max_retests	how many times an additional delay should be applied before re-testing failed results (0: no re-tests will be performed)
descr	user specific description shown at the start of the result text
format	format string to control/override appearance of the automatically created result text.

Example: check_var_list structure

```
check_var_list = [
    { "var":     <variable to check (CalVar, HilVar, ...)>,
      "value":   <rated/expected value>,
      "rel_pos": <relative positive tolerance (optional)>,
      "abs_pos": <absolute positive tolerance (optional)>,
      "rel_neg": <relative negative tolerance (optional)>,
      "abs_neg": <absolute negative tolerance (optional)>,
    },
    ...
]
# Note that abs_tol/rel_tol can be used as alias for abs_pos/rel_pos
# See ttk_checks.basic_checks.checkTolerance for more info on
# use of tolerance values.
```

Info: "var" and "value" in check_var_list

Both the variable to check and the expected "value" can actually be HilVar/CalVar instances, meta()-wrapped MetaValue instances or plain scalar numeric values.

This means that it is possible to compare the values of two ecu/cal variables
`{"var": cal.some_variable, "value": cal.some_other_variable}`
or between hil and ecu
`{"var": cal.foobar, "value": hil.barbaz}`
or even (admittedly, not too useful)
`{"var": 3, "value": 3.2, "abs_tol": .4}`

If you want to show units in the test report, you could wrap values in a MetaValue:
`{"var": cal.shrdlu, "value": meta(3.2, "mV"), "abs_tol": .4}`
See ttk_base.values_base.meta

If all tolerances of a check_var_list entry remain un-set (i.e. None), the values will be treated as discrete integer states for comparison. If a value does not appear to be an integer value, the test will fail with verdict ERROR (and the actual check using the un-cast value(s) will most likely fail as well).
See ttk_checks.checkutils.getInteger

Returns

a single test result entry (with a single verdict for all checked cal variables).

6.44.2.6 testSignals()

```
ttk_bus._bus_tests.testSignals (
              set_var_list = None,
              check_var_list = None,
              min_wait_time_ms = 0,
              descr = '' )
```

Check the values of multiple variables/signals as a result of setting multiple other variables/signals.

Between setting and checking, a configurable time will be waited to deal with propagation delay. This wait time defaults to the minimum practicable wait time of all involved bus signals.

This function:

- sets the value(s) to the variables in set_var_list
- triggers sending of any event-based signals included in set_var_list
- waits for value change(s) to propagate to the ECU/DUT
- checks the value(s) for variables in check_var_list

Note

While this function can perform the same tests as testRxSignal() and testTxSignal(), its flexibility somewhat limits description formatting.

Parameters

set_var_list	Variables to set: a list of dictionaries with variable and value
check_var_list	Variables to check: a list of dictionaries with variable, rated value and tolerance
min_wait_time_ms	minimum wait time before performing checks
descr	user specific description shown at the start of the result text

Example: Wait time before check (propagation delay)

```
wait_time = max(
    min_wait_time_ms,
    max(<change delay of signals to set>) +
    max(<change delay of signals to check>)
)
# Note: a bus signal's wait time defaults to 2x cycle time
```

Example: set_var_list structure

```
set_var_list = [
    { "var":   <variable to set (e.g. a CalVar, HilVar, RxBusSignal, ...)>,
      "value": <value to set>,
      "descr": <value description, optional>,
    },
    ...
]
# See ttk_base.variables_base, ttk_bus.bus_signals_base
```

Example: check_var_list structure

```
check_var_list = [
    { "var":     <variable to check (CalVar, HilVar, TxBusSignal, ...)>,
      "value":   <rated/expected value>,
      "rel_pos": <relative positive tolerance (optional)>,
      "abs_pos": <absolute positive tolerance (optional)>,
      "rel_neg": <relative negative tolerance (optional)>,
      "abs_neg": <absolute negative tolerance (optional)>,
    },
    ...
]
# Note that abs_tol/rel_tol can be used as alias for abs_pos/rel_pos
# See ttk_base.variables_base, ttk_bus.bus_signals_base
# See ttk_checks.basic_checks.checkTolerance for more info on
# use of tolerance values.
```

Info: "var" and "value" in check- and set_var_list

Both the variable to check and the expected "value" can actually be HilVar/CalVar or (Rx)BusSignal instances, meta()-wrapped MetaValue instances or plain scalar numeric values.

This means that it is possible to compare the values of two ecu/cal variables
`{"var": cal.some_variable, "value": cal.some_other_variable}`
or between hil and ecu
`{"var": cal.foobar, "value": hil.barbaz}`
`{"var": bus.barsignal, "value": cal.bazquux}`
or even (admittedly, not too useful)
`{"var": 3, "value": 3.2, "abs_tol": .4}`

If you want to show units in the test report, you could wrap values in a MetaValue:
`{"var": cal.shrdlu, "value": meta(3.2, "mV"), "abs_tol": .4}`
See ttk_base.values_base.meta

If all tolerances of a check_var_list entry remain un-set (i.e. None), the values will be treated as discrete integer states for comparison. If a value does not appear to be an integer value, the test will fail with verdict ERROR (and the actual check using the un-cast value(s) will most likely fail as well).
See ttk_checks.checkutils.getInteger

Returns

a single test result entry (with a single verdict for all checked variables).

6.44.2.7 testTxSignal()

```
ttk_bus._bus_tests.testTxSignal (
              set_var_list,
              bus_signal_data,
              check_value,
              check_value_descr = "",
              rel_pos = None,
              abs_pos = None,
              rel_neg = None,
              abs_neg = None,
              max_retests = 0,
              descr = '',
              format = None )
```

Check the behaviour of a single bus signal sent from the ECU and received by the HIL system.

Info: This function ...

- sets the value(s) inside the ECU using the supplied cal variables
- waits for value change(s) to propagate to HIL
- checks the signal of the HIL system using bus_signal_data.

Parameters

set_var_list	list of dictionaries with variable and value (see example, below)
bus_signal_data	data structure which describes the signal on the bus system. See bus_signals_base.TxBusSignal
check_value	rated/expected value that should be received on the HIL side
check_value_descr	string which describes the expected value
rel_pos	(optional) relative positive tolerance
abs_pos	(optional) absolute positive tolerance
rel_neg	(optional) relative negative tolerance
abs_neg	(optional) absolute negative tolerance
max_retests	how many times an additional delay should be applied before re-testing failed results (0: no re-tests will be performed)
descr	user specific description shown at the start of the result text
format	format string to control/override appearance of the automatically created result text.

Example: set_var_list structure

```
set_var_list = [
    { "var":   <variable to set (e.g. a CalVar, HilVar or RxBusSignal)>,
      "value": <value to set>,
      "descr": <value description, optional>,
    },
    ...
]
```

Returns

a single test result entry.

6.45 ttk_bus.bus_signals_base Namespace Reference

Interface wrapper for bus/interface signal classes in ttk_bus._bus_signals_base.

Classes

- class AcBusSignal

  A RX bus signal (HIL => ECU) that models an alive counter value with additional error injections.
- class BusSignalContainer

  Base class for BusSignal containers.
- class CrcBusSignal

  A RX bus signal (HIL => ECU) that models a CRC value with additional error injections.
- class RxBusSignal

  A signal sent from the test system to the DUT (ECU Rx <= HIL Tx).
- class TxBusSignal

  A signal sent from the DUT to the test system (ECU Tx => HIL Rx).

6.45.1 Detailed Description

Interface wrapper for bus/interface signal classes in ttk_bus._bus_signals_base.

This serves as "interface" to the precompiled module in delivery to enable code-completion in PyDev.

Convention:

Signal directions are relative to the DUT:

- Tx - transmitted by ECU => received from HIL
- Rx - received by ECU <= transmitted from HIL

6.46 ttk_bus.bus_signals_cangamma Namespace Reference

Interface wrapper bus/interface signal classes in ttk_bus._bus_signals_cangamma.

Classes

- class AcRxCanBusSignal

  Alive counter RX bus signal (HIL-Tx => ECU-Rx).
- class CanBusSignalContainer

  A BusSignalContainer for CAN Bus Signals.
- class CrcRxCanBusSignal

  A CRC RX bus signal (HIL-Tx => ECU-Rx)
- class CrcTxCanBusSignal

  A CRC TX bus signal (ECU-Tx => HIL-Rx), adds CRC validity info.
- class RxCanBusMessage

  CAN Message/PDU ECU-Rx (HIL-Tx => ECU-Rx).
- class RxCanBusSignal

  A CAN signal sent from the test system to the DUT (ECU Rx <= HIL Tx).
- class RxCanBusSignalBasic

  A basic CAN signal sent from the test system to the DUT (ECU Rx <= HIL Tx).
- class TxCanBusMessage

  CAN Message/PDU ECU-Tx (ECU-Tx => HIL-Rx).
- class TxCanBusSignal

  A CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).
- class TxCanBusSignalBasic

  A basic CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).

6.46.1 Detailed Description

Interface wrapper bus/interface signal classes in ttk_bus._bus_signals_cangamma.

Specific implementation for Gamma-V-based CAN blocksets

Convention:

Signal directions are relative to the DUT:

- Tx - transmitted by ECU => received from HIL
- Rx - received by ECU <= transmitted from HIL

6.47 ttk_bus.bus_signals_canida Namespace Reference

Interface wrapper bus/interface signal classes in ttk_bus._bus_signals_canida.

Classes

- class CanBusSignalContainer

  A BusSignalContainer for CAN Bus Signals.
- class RxCanBusMessage

  CAN Message/PDU ECU-Rx (HIL-Tx => ECU-Rx).
- class RxCanBusSignal

  A basic CAN signal sent from the test system to the DUT (ECU Rx <= HIL Tx).
- class TxCanBusMessage

  CAN Message/PDU ECU-Tx (ECU-Tx => HIL-Rx).
- class TxCanBusSignal

  A basic CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).

6.47.1 Detailed Description

Interface wrapper bus/interface signal classes in ttk_bus._bus_signals_canida.

Specific implementation for iDA-Blockset based CAN interfaces

Convention:

Signal directions are relative to the DUT:

- Tx - transmitted by ECU => received from HIL
- Rx - received by ECU <= transmitted from HIL

6.48 ttk_bus.bus_signals_canmmbs Namespace Reference

Interface wrapper bus/interface signal classes in ttk_bus._bus_signals_canmmbs.

Classes

- class AcCanBusSignal

  Alive counter bus signal derived from RxCanBusSignal (HIL-Tx => ECU-Rx) for use with dSPACE CAN Multi-↩ Message blockset data structures.
- class CanBusSignalContainer

  A BusSignalContainer for CAN Bus Signals.
- class CrcCanBusSignal

  A CRC RX bus signal derived from RxCanBusSignal (HIL-Tx => ECU-Rx) for use with dSPACE CAN Multi-Message blockset.
- class RxCanBusMessage

  CAN Message/PDU ECU-Rx (HIL-Tx => ECU-Rx).
- class RxCanBusSignal

  A CAN signal sent from the test system to the DUT (ECU Rx <= HIL Tx).
- class TxCanBusMessage

  CAN Message/PDU ECU-Tx (ECU-Tx => HIL-Rx).
- class TxCanBusSignal

  A CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).

6.48.1 Detailed Description

Interface wrapper bus/interface signal classes in ttk_bus._bus_signals_canmmbs.

Specific implementation for dSpace CAN Multi-Message blocksets (using "trc"-signals added to BusSystems)

Convention:

Signal directions are relative to the DUT:

- Tx - transmitted by ECU => received from HIL
- Rx - received by ECU <= transmitted from HIL

6.49 ttk_bus.bus_signals_ethbs Namespace Reference

Interface wrapper for "bus"/interface signal classes in ttk_bus._bus_signals_ethbs.

Classes

- class AcEthSignal

  An alive counter Rx Ethernet signal/parameter (HIL PSI => ECU CSI) for use with dSPACE Ethernet blockset.
- class CrcEthSignal

  A CRC RX Ethernet event signal/parameter (HIL-Tx => ECU-Rx) for use with dSPACE Ethernet blockset.
- class CrcEthStatus

  CRC Tx-Event/Signal status (ECU PSI Tx => HIL CSI Rx) and associated switch variables for dSPACE Ethernet blockset.
- class EthSignalContainer

  A container for Ethernet Signals.
- class RxEthBusSysSignal

  An Ethernet event signal/parameter/sub-parameter sent from test system to DUT (HIL PSI Tx => ECU CSI Rx).
- class RxEthEvent

  Ethernet-Event-specific data and methods - Rx (HIL-Tx => ECU-Rx).
- class RxEthSignal

  An Ethernet event signal/parameter/sub-parameter sent from test system to DUT (ECU CSI Rx <= HIL PSI Tx).
- class TxEthBusSysSignal

  An Ethernet event signal/parameter/sub-parameter sent from DUT to test system (ECU PSI Tx => HIL CSI Rx).
- class TxEthEvent

  Ethernet-Event-specific data and methods - Tx (ECU-Tx => HIL-Rx).
- class TxEthSignal

  An Ethernet event signal/parameter/sub-parameter sent from DUT to test system (ECU PSI Tx => HIL CSI Rx).

6.49.1 Detailed Description

Interface wrapper for "bus"/interface signal classes in ttk_bus._bus_signals_ethbs.

Specific implementation for dSpace Ethernet blocksets ("Ethernet Configuration Package") using ModelRoot-signals and (for E2E) BusSystems-signals)

Convention:

Signal directions are relative to the DUT:

- Tx - transmitted by ECU => received from HIL
- Rx - received by ECU <= transmitted from HIL

- PSI - provider, Tx
- CSI - consumer, Rx

6.50 ttk_bus.bus_signals_frbs Namespace Reference

Interface wrapper for bus/interface signal classes in ttk_bus._bus_signals_frbs.

Classes

- class AcFrBusSignal

  An alive counter RX bus signal derived from RxFrBusSignal (HIL => ECU) for use with dSPACE FlexRay blockset PDU-based data structures.
- class CrcFrBusSignal

  A CRC RX bus signal derived from RxFrBusSignal (HIL-Tx => ECU-Rx) for use with dSPACE FlexRay blockset with PDU-based structure.
- class CrcFrBusStatus

  CRC Tx-Signal status (ECU Tx => HIL Rx) and associated switch variables for dSPACE FlexRay blockset with PDU-based modeling.
- class FrBusSignalContainer

  A BusSignalContainer for FlexRay Bus Signals.
- class FrRxPdu

  FlexRay ECU-Rx (HIL-Tx => ECU-Rx) PDU data.
- class FrTxPdu

  FlexRay ECU-Tx (ECU-Tx => HIL-Rx) PDU data.
- class RxFrBusSignal

  A FlexRay signal sent from test system to the DUT (ECU Rx <= HIL Tx).
- class TxFrBusSignal

  A FLexRay signal sent from DUT to the test system (ECU Tx => HIL Rx).

6.50.1 Detailed Description

Interface wrapper for bus/interface signal classes in ttk_bus._bus_signals_frbs.

Specific implementation for dSpace FlexRay blocksets using PDU-based blockset generation with signals added to Trc/BusSystems.

Convention:

Signal directions are relative to the DUT:

- Tx - transmitted by ECU => received from HIL
- Rx - received by ECU <= transmitted from HIL

6.51 ttk_bus.bus_test_utils Namespace Reference

Interface wrapper for bus test utility functions in ttk_bus._bus_test_utils.

Functions

- getIntRangeValues (min_value=None, max_value=None, amount=5)

  Get a list of equally distributed (unique) values within min and max value boundaries.
- getRangeValues (min_value=None, max_value=None, amount=5)

  Get a list of equally distributed values within min and max value boundaries.

6.51.1 Detailed Description

Interface wrapper for bus test utility functions in ttk_bus._bus_test_utils.

6.51.2 Function Documentation

6.51.2.1 getIntRangeValues()

```
ttk_bus.bus_test_utils.getIntRangeValues (
              min_value = None,
              max_value = None,
              amount = 5 )
```

Get a list of equally distributed (unique) values within min and max value boundaries.

Parameters

min_value	lower boundary of value range
max_value	upper boundary of value range
amount	maximum amount of values to be returned (fewer values might be returned if the range cannot fit enough integers)

Returns

a list of values equally distributed between min and max values.

6.51.2.2 getRangeValues()

```
ttk_bus.bus_test_utils.getRangeValues (
              min_value = None,
              max_value = None,
              amount = 5 )
```

Get a list of equally distributed values within min and max value boundaries.

Parameters

min_value	lower boundary of value range
max_value	upper boundary of value range
amount	how many values should be returned

Returns

a list of values equally distributed between min and max values.

6.52 ttk_bus.bus_tests Namespace Reference

Interface wrapper for bus/interface test functions (HIL <–bus-signal–> DUT) in ttk_bus._bus_tests.

Functions

- testRxSignal (bus_signal_data, set_value, set_value_descr='', check_var_list=None, max_retests=0, descr='', format=None)

  Check the behaviour of a single bus signal sent from HIL system and received by the ECU.
- testTxSignal (set_var_list, bus_signal_data, check_value, check_value_descr="", rel_pos=None, abs_pos=None, rel_neg=None, abs_neg=None, max_retests=0, descr='', format=None)

  Check the behaviour of a single bus signal sent from the ECU and received by the HIL system.
- testSignals (set_var_list=None, check_var_list=None, min_wait_time_ms=0, descr='')

  Check the values of multiple variables/signals as a result of setting multiple other variables/signals.
- testBasicCrcError (crc_signal_data, check_var_list_before=None, check_var_list_after=None, manual_wait_cycles=None, manual_triggers_before=0, manual_triggers_between=0, manual_trigger_delay_ms=0, descr='', format=None)

  Check ECU behaviour on CRC signal errors (for messages sent from HIL system and received by the ECU).
- testBasicAcError (ac_signal_data, check_var_list_before=None, check_var_list_after=None, manual_wait_cycles=None, manual_triggers_before=0, manual_triggers_between=0, manual_trigger_delay_ms=0, descr='', format=None)

  Check ECU behaviour on alive counter signal errors (for messages sent from HIL system and received by the ECU).

6.52.1 Detailed Description

Interface wrapper for bus/interface test functions (HIL <–bus-signal–> DUT) in ttk_bus._bus_tests.

This serves as "interface" to the precompiled module in delivery to enable code-completion in PyDev.
Requires package ttk_checks.

Example:

```
testresult.append(
    testRxSignal(
        bus.rx_sig,
        set_value = 100,
        set_value_descr = "Some test value",
        check_var_list = [
            {"var": cal.some_var,       "value": 0,   "abs_pos": 0.01, "abs_neg": -0.1 },
            {"var": cal.some_other_var, "value": 2.1, "rel_pos": 0.05 },
            {"var": cal.status_var,     "value": 3 },

        ],
        descr = "This is the supplied testRxSignal description text"
    )
)
testresult.append(
    testTxSignal([{'var': cal.status, 'value': 1}], bus.tx_status_sig, 1 )
)
```

Conventions:

- Signal directions are specified as relative to the DUT:
  - Tx - transmitted from ECU => received by HIL
  - Rx - received by ECU <= transmitted from HIL

- Each function will take the required parameters for the test to perform plus two additional parameters for report formatting:

  - `descr` - a description string that will be placed at the start of the result description text

  - `format` - a format-string that specifies/overrides the formatting of the automatically generated part of the description.

    * Do **not** override this unless absolutely necessary.
    * If you want to change the default output formatting of a test function, you should look into changing the current localization (see ttk_bus.bus_tests_loc), so all tests will use consistent result formatting.
    * See also `http://docs.python.org/library/stdtypes.html#string-formatting`

- If an input value has a get()-method, it will be called to get the current value. Otherwise the value will be used as-is.
  - This permits simple read access to CalVar or HilVar instances.

  - See ttk_base.variables_base

- If an input value has associated meta data (like Hil/Cal Vars or basic values wrapped as MetaValues), it can show up in the automatically generated description (e.g. unit or alias names).

  - See ttk_base.variables_base, ttk_base.values_base.meta()

- Exceptions occurring "inside" a test function have to be caught; the function will return a result entry containing the error text and verdict ERROR.
  - That means it should not be necessary to wrap the test function into yet another `try ... except` block.
  - If a test function raises an (unexpected) exception anyway, please file a bug report or contact support at `support@isyst.de`

6.52.2 Function Documentation

6.52.2.1 testBasicAcError()

```
ttk_bus.bus_tests.testBasicAcError (
              ac_signal_data,
              check_var_list_before = None,
              check_var_list_after = None,
              manual_wait_cycles = None,
              manual_triggers_before = 0,
              manual_triggers_between = 0,
              manual_trigger_delay_ms = 0,
              descr = '',
              format = None )
```

Check ECU behaviour on alive counter signal errors (for messages sent from HIL system and received by the ECU).

Info: This function ...

- checks the "before" variables,
- sets the alive counter signal to an error state,
- waits a defined number of message cycles, then
- checks the "after" variables.

Parameters

ac_signal_data	AcBusSignal data structure which describes the crc signal. See bus_signals_base
check_var_list_before	variables to verify before injecting an error. A list of dictionaries with variable, rated value and tolerance See parameter example in testSignals()
check_var_list_after	variables to verify after injecting an error. A list of dictionaries with variable, rated value and tolerance See parameter example in testSignals()
manual_wait_cycles	if set, overrides the default number of cycle times to wait between crc error injection and variable checks (ac_signal_data.error_delay)
manual_triggers_before	number of manual triggers to perform on bus signal (via kickout()) prior to the "before" check. Obviously this makes the most sense for event-based signals
manual_triggers_between	number of manual triggers to perform on bus signal (via kickout()) prior to the "after" check. Again, this makes the most sense for event-based signals
manual_trigger_delay_ms	[ms] delay between trigger calls (usually the kickout() method has its own built-in minimum delay, though)
descr	user specific description shown at start of result text
format	format string to control/override appearance of automatically created result text. A format override should only rarely be used.

Returns

a single test result entry (with a single verdict for all checked variables).

6.52.2.2 testBasicCrcError()

```
ttk_bus.bus_tests.testBasicCrcError (
              crc_signal_data,
              check_var_list_before = None,
              check_var_list_after = None,
              manual_wait_cycles = None,
              manual_triggers_before = 0,
              manual_triggers_between = 0,
              manual_trigger_delay_ms = 0,
              descr = '',
              format = None )
```

Check ECU behaviour on CRC signal errors (for messages sent from HIL system and received by the ECU).

Info: This function ...

- checks the "before" variables,
- sets the crc signal to an error state,
- waits a defined number of message cycles, then
- checks the "after" variables.

Parameters

crc_signal_data	CrcBusSignal data structure which describes the crc signal. See bus_signals_base
check_var_list_before	variables to verify before injecting an error. A list of dictionaries with variable, rated value and tolerance See parameter example in testSignals()
check_var_list_after	variables to verify after injecting an error. A list of dictionaries with variable, rated value and tolerance See parameter example in testSignals()
manual_wait_cycles	if set, overrides the default number of cycle times to wait between crc error injection and variable checks (crc_signal_data.error_delay)
manual_triggers_before	number of manual triggers to perform on bus signal (via kickout()) prior to the "before" check. Obviously this makes the most sense for event-based signals
manual_triggers_between	number of manual triggers to perform on bus signal (via kickout()) prior to the "after" check. Again, this makes the most sense for event-based signals
manual_trigger_delay_ms	[ms] delay between trigger calls (usually the kickout() method has its own built-in minimum delay, though)
descr	user specific description shown at start of result text
format	format string to control/override appearance of automatically created result text. A format override should only rarely be used.

Returns

a single test result entry (with a single verdict for all checked variables).

6.52.2.3 testRxSignal()

```
ttk_bus.bus_tests.testRxSignal (
              bus_signal_data,
              set_value,
              set_value_descr = '',
              check_var_list = None,
              max_retests = 0,
              descr = '',
              format = None )
```

Check the behaviour of a single bus signal sent from HIL system and received by the ECU.

Info: This function ...

- sets the signal contained in bus_signal_data in the HIL model/system
- triggers sending the signal if needed (for event-triggered PDUs)
- waits for value change(s) to propagate to the ECU/DUT
- checks the value(s) inside the ECU using the supplied cal variables.

Parameters

bus_signal_data	data structure which describes the signal on the bus system. See bus_signals_base.RxBusSignal
set_value	value to set on the HIL side
set_value_descr	string which describes the value to set
check_var_list	list of dictionaries with variable, rated value and tolerance

Parameters

max_retests	how many times an additional delay should be applied before re-testing failed results (0: no re-tests will be performed)
descr	user specific description shown at the start of the result text
format	format string to control/override appearance of the automatically created result text. A format override should only rarely be used.

Example: check_var_list structure

```
check_var_list = [
    { "var":     <variable to check (CalVar, HilVar, ...)>,
      "value":   <rated/expected value>,
      "rel_pos": <relative positive tolerance (optional)>,
      "abs_pos": <absolute positive tolerance (optional)>,
      "rel_neg": <relative negative tolerance (optional)>,
      "abs_neg": <absolute negative tolerance (optional)>,
    },
    ...
]
Note that abs_tol/rel_tol can be used as alias for abs_pos/rel_pos
See ttk_checks.basic_checks.checkTolerance for more info on
use of tolerance values.
```

Info: "var" and "value" in check_var_list

Both the variable to check and the expected "value" can actually be HilVar/CalVar instances, meta()-wrapped MetaValue instances or plain scalar numeric values.

This means that it is possible to compare the values of two ecu/cal variables
`{"var": cal.some_variable, "value": cal.some_other_variable}`
or between hil and ecu
`{"var": cal.foobar, "value": hil.barbaz}`
or even (admittedly, not too useful)
`{"var": 3, "value": 3.2, "abs_tol": .4}`

If you want to show units in the test report, you could wrap values in a MetaValue:
`{"var": cal.shrdlu, "value": meta(3.2, "mV"), "abs_tol": .4}`
See ttk_base.values_base.meta

If all tolerances of a check_var_list entry remain un-set (i.e. None), the values will be treated as discrete integer states for comparison. If a value does not appear to be an integer value, the test will fail with verdict ERROR (and the actual check using the un-cast value(s) will most likely fail as well).
See ttk_checks.checkutils.getInteger

Returns

a single test result entry (with a single verdict for all checked cal variables).

6.52.2.4 testSignals()

```
ttk_bus.bus_tests.testSignals (
               set_var_list = None,
               check_var_list = None,
               min_wait_time_ms = 0,
               descr = '' )
```

Check the values of multiple variables/signals as a result of setting multiple other variables/signals.

Between setting and checking, a configurable time will be waited to deal with propagation delay. This wait time defaults to the minimum practicable wait time of all involved bus signals.

This function:

- sets the value(s) to the variables in set_var_list
- triggers sending of any event-based signals included in set_var_list
- waits for value change(s) to propagate to the ECU/DUT
- checks the value(s) for variables in check_var_list

Note

While this function can perform the same tests as testRxSinal() and testTxSignal(), its flexibility somewhat limits description formatting.

Parameters

set_var_list	Variables to set: a list of dictionaries with variable and value
check_var_list	Variables to check: a list of dictionaries with variable, rated value and tolerance
min_wait_time_ms	minimum wait time before performing checks
descr	user specific description shown at the start of the result text

Example: Wait time before check (propagation delay)

```
wait_time = max(
    min_wait_time_ms,
    max(<change delay of signals to set>) +
    max(<change delay of signals to check>)
)
# Note: a bus signal's wait time defaults to 2x cycle time
```

Example: set_var_list structure

```
set_var_list = [
    { "var":   <variable to set (e.g. a CalVar, HilVar, RxBusSignal, ...)>,
      "value": <value to set>,
      "descr": <value description, optional>,
    },
    ...
]
```

See ttk_base.variables_base, ttk_bus.bus_signals_base

Example: check_var_list structure

```
check_var_list = [
    { "var":     <variable to check (CalVar, HilVar, TxBusSignal, ...)>,
      "value":   <rated/expected value>,
      "rel_pos": <relative positive tolerance (optional)>,
      "abs_pos": <absolute positive tolerance (optional)>,
      "rel_neg": <relative negative tolerance (optional)>,
      "abs_neg": <absolute negative tolerance (optional)>,
    },
    ...
]
# Note that abs_tol/rel_tol can be used as alias for abs_pos/rel_pos
# See ttk_base.variables_base, ttk_bus.bus_signals_base
# See ttk_checks.basic_checks.checkTolerance for more info on
# use of tolerance values.
```

Info: "var" and "value" in check- and set_var_list

Both the variable to check and the expected "value" can actually be HilVar/CalVar or (Rx)BusSignal instances, meta()-wrapped MetaValue instances or plain scalar numeric values.

This means that it is possible to compare the values of two ecu/cal variables
`{"var": cal.some_variable, "value": cal.some_other_variable}`
or between hil and ecu
`{"var": cal.foobar, "value": hil.barbaz}`
`{"var": bus.barsignal, "value": cal.bazquux}`
or even (admittedly, not too useful)
`{"var": 3, "value": 3.2, "abs_tol": .4}`

If you want to show units in the test report, you could wrap values in a MetaValue:
`{"var": cal.shrdlu, "value": meta(3.2, "mV"), "abs_tol": .4}`
See ttk_base.values_base.meta

If all tolerances of a check_var_list entry remain un-set (i.e. None), the values will be treated as discrete integer states for comparison. If a value does not appear to be an integer value, the test will fail with verdict ERROR (and the actual check using the un-cast value(s) will most likely fail as well).
See ttk_checks.checkutils.getInteger

Returns

a single test result entry (with a single verdict for all checked variables).

6.52.2.5 testTxSignal()

```
ttk_bus.bus_tests.testTxSignal (
              set_var_list,
              bus_signal_data,
              check_value,
              check_value_descr = "",
              rel_pos = None,
              abs_pos = None,
              rel_neg = None,
              abs_neg = None,
              max_retests = 0,
              descr = '',
              format = None )
```

Check the behaviour of a single bus signal sent from the ECU and received by the HIL system.

Info: This function ...

- sets the value(s) inside the ECU using the supplied cal variables
- waits for value change(s) to propagate to HIL
- checks the signal of the HIL system using bus_signal_data.

Parameters

set_var_list	list of dictionaries with variable and value (see example, below)
bus_signal_data	data structure which describes the signal on the bus system. See bus_signals_base.TxBusSignal
check_value	rated/expected value that should be received on the HIL side
check_value_descr	string which describes the expected value
rel_pos	(optional) relative positive tolerance
abs_pos	(optional) absolute positive tolerance
rel_neg	(optional) relative negative tolerance
abs_neg	(optional) absolute negative tolerance
max_retests	how many times an additional delay should be applied before re-testing failed results (0: no re-tests will be performed)
descr	user specific description shown at the start of the result text
format	format string to control/override appearance of the automatically created result text. If a tuple of strings is supplied, it should contain three entries: (main_format>, <set_format>, <check_format>). A format override should only rarely be used.

Example: set_var_list structure

```
set_var_list = [
    { "var":   <variable to set (i.e. a CalVar, HilVar, ...)>,
      "value": <value to set>,
      "descr": <value description, optional>,
    },
    ...
]
```

Returns

a single test result entry.

6.53 ttk_bus.bus_tests_loc Namespace Reference

Localization look-up for bus test descriptions.

Classes

- class DE

  German localization.
- class EN

  English localization.

Functions

- getInstance (locale_id=None)

  Get a L10N instance matching the supplied locale id.

Variables

- dict _available_locales
- _loc = getInstance("fnord_debug")

6.53.1 Detailed Description

Localization look-up for bus test descriptions.

Conventions:

- L10N entries in `<angle-brackets>` are format identifiers that will (usually) not double as default values, but have a separate default value set. Use these in place of of short, ambiguous formats like `" [%s]"`.

- Entry names in format strings like `"Lorem: %(entry_name)s"` reference keys of a supplied value/info lookup dictionary. If no entry exists for a given name, an empty string will be inserted in its place.

- Entry names starting with `formatted_` usually contain a pre-formatted version of the same key without the prefix or some text that has had already some l10n/formatting applied (e.g. a tolerance string that may have different sub-formats, depending on tolerance type)

- The EN localization mostly contains just the keys with values set to None. A value of None uses the key itself as L10N-entry (we could just remove the keys completely and get the same effect, but this way they are easier to discover for later translation, modification or consistency checks).

6.53.2 Function Documentation

6.53.2.1 getInstance()

```
ttk_bus.bus_tests_loc.getInstance (
              locale_id = None )
```

Get a L10N instance matching the supplied locale id.

Parameters

locale↩ _id	id of locale to instantiate. If no specific id has been supplied, the ID will be taken from project settings.

Returns

a L10N instance.

6.53.3 Variable Documentation

6.53.3.1 _available_locales

```
dict ttk_bus.bus_tests_loc._available_locales  [protected]
```

Initial value:

```
00001 =  {
00002     'en':    EN,
00003     'en_us': EN,
00004     'en_gb': EN,
00005
00006     'de':    DE,
00007     'de_de': DE,
00008 }
```

6.53.3.2 _loc

```
ttk_bus.bus_tests_loc._loc = getInstance("fnord_debug")  [protected]
```

6.54 ttk_checks Namespace Reference

Check and test utility functions.

Namespaces

- namespace _adv_tests

  Advanced test/evaluation functions for test automation that directly generate test report entries.
- namespace _basic_checks

  Basic "low-level" check functions for use in test automation.
- namespace _basic_tests

  Basic test/evaluation functions for test automation that directly generate test report entries.
- namespace _checkutils

  Utilities for check functions.
- namespace _verdicts

  Defined test status verdicts.
- namespace adv_tests

  Interface wrapper for advanced test/evaluation functions in ttk_checks._adv_tests.
- namespace basic_checks

  Interface wrapper for basic check functions ("low-level") in ttk_checks._basic_checks.
- namespace basic_tests

  Interface wrapper for basic test/evaluation functions that directly generate test report entries in ttk_checks._basic_tests.
- namespace basic_tests_loc

  Localization look-up for basic test descriptions.
- namespace checkceptions

  Custom exception classes for package ttk_checks.
- namespace checkutils

  Interface wrapper for check functions utilities in ttk_checks._checkutils.
- namespace verdicts

  Interface wrapper for defined test status verdicts in ttk_checks._verdicts.

6.54.1 Detailed Description

Check and test utility functions.

6.55 ttk_checks._adv_tests Namespace Reference

Advanced test/evaluation functions for test automation that directly generate test report entries.

Functions

- _callFunction (var_list_entry)

  Call a function from the supplied entry.
- _getFuncEntryInfo (var_list_entry)

  Get a textual representation of the function in var_list_entry.
- _getToleranceFormat (abs_pos=0, rel_pos=0, abs_neg=None, rel_neg=None)

  Build a reasonably concise tolerance info format.
- _getFormattedValues (original_v, actual_v, lookup_src_var=None, fmt_src_var=None)

  Get a formatted string describing the supplied values.
- _needsTriggering (signal)

  Check whether a signal needs to be manually "triggered" for values to be passed on (if not, it should update automatically).
- _trigger (var)

  Trigger transmission of the given BusSignal/Var.
- _setVariables (set_var_list)

  Set preconditions: variables/signals.
- _checkVariables (check_var_list, show_int_note=True)

  Check a list of target values (hil/cal variables, signals).
- testVars (set_var_list=None, check_var_list=None, min_wait_time_ms=100, descr='')

  Check the values of multiple variables/signals as a result of setting multiple other variables/signals.

6.55.1 Detailed Description

Advanced test/evaluation functions for test automation that directly generate test report entries.

These functions are somewhat more complex than those in ttk_checks._basic_tests. See interface wrapper in ttk_checks.adv_tests.

6.55.2 Function Documentation

6.55.2.1 _callFunction()

```
ttk_checks._adv_tests._callFunction (
              var_list_entry )  [protected]
```

Call a function from the supplied entry.

Parameters

var_list_entry	a single entry of a set_ or check_var_list, see _setVariables and _checkVariables

Example: set_var_list and function calls

```
For special cases, the set var list can also contain function call
(which might be useful to insert additional delays or similar):

var_list_entry = {
   "func":   time.sleep,
   "args":   [1.0], # positional arguments
   "kwargs": {},    # keyword arguments
}
```

Returns

the return value of the function call (or always None if the entry contains no function call (i.e. "func" is None)). Note that any Exceptions thrown by the function will be passed on (this includes the case that the specified function reference is non-callable).

6.55.2.2 _checkVariables()

```
ttk_checks._adv_tests._checkVariables (
              check_var_list,
              show_int_note = True )  [protected]
```

Check a list of target values (hil/cal variables, signals).

Parameters

check_var_list	list of dictionaries with variable, expected value and tolerance
show_int_note	if True, a note will be logged each time a value will be interpreted as integer (see Info section below)

Example: check_var_list structure

```
# A basic entry:
check_var_list = [
    { "var":      <variable to check (CalVar, HilVar, ...)>,
      "operator": <check operator, defaults to "==">,
      "value":    <expected value(s)>
      "abs_tol":  <absolute tolerance (optional)>,
      "rel_tol":  <relative tolerance (optional)>,
    }
]
# If only abs_tol or rel_tol
# abs_tol/rel_tol aren an alias for abs_pos/rel_pos
# (when separate _pos/_neg tolerances are not required).
# If you need unsymmetrical tolerances (e.g. -1/+5), you can
# specify tolerances separately:
check_var_list = [
    { "var":      <variable to check (CalVar, HilVar, ...)>,
      "operator": <check operator, defaults to "==">,
      "value":    <expected value> or a list of expected values, of which one has to pass
      "rel_pos":  <relative positive tolerance (optional)>,
      "abs_pos":  <absolute positive tolerance (optional)>,
      "rel_neg":  <relative negative tolerance (optional)>,
      "abs_neg":  <absolute negative tolerance (optional)>,
    },
    ...
]

* See basic_checks.checkTolerance for details on tolerance parameters
```

```
* "==" is default when operator entry is omitted


See ttk_base.variables_base.CalVar
See ttk_checks.basic_checks.checkTolerance for more info on
use of tolerance values.
```

Example: extended check_var_list structure

```
check_var_list = [
    { "var":      <variable to check (CalVar, HilVar, ...)>,
      "operator": "==", # "==" or "in"
      "value":    <expected value> or a list of multiple expected values of which one has to match
      "abs_pos": <absolute tolerance (optional)>,
      "rel_pos": <relative tolerance (optional)>,
    },
    { "var":      <variable to check (CalVar, HilVar, ...)>,
      "operator": "!=", # "!=" or "not in"
      "value":    <expected value> or a list of multiple expected values, none of which must match
      "abs_pos": <absolute tolerance (optional)>,
      "rel_pos": <relative tolerance (optional)>,
    },
    { "var":      <variable to check (CalVar, HilVar, ...)>,
      "operator": ">", # "<", "<=", ">=" or ">"
      "value":    <expected value>
      # no tolerances used for "greater/less than" comparisons
    },
    ...
]
```

Example: Using a list of multiple expected values (usually for status values):

```
# Operator "==" or "in": one value has to match
check_var_list = [
    { "var":       cal.e2e_status,
      "operator": "in"
      "value":    [E2E_P04STATUS_OK, E2E_P04STATUS_OKSOMELOST],
    }
]

# Operator "!=" or "not in": no value may match:
check_var_list = [
    { "var":      cal.e2e_status,
      "operator": "not in",
      "value":    [E2E_P04STATUS_NONEWDATA, E2E_P04STATUS_ERROR,
                   E2E_P04STATUS_REPEATED, E2E_P04STATUS_WRONGSEQUENCE],
    }
]
```

Operators:

- "==" or "in":
  - Single value: var has to match value
  - Multiple values: var has to match at least one value
- "!=" or "not in":
  - Single value: var may not match value
  - Multiple values: var may not match any specified value
- ">" var has to be greater than the (scalar) value
- ">=" var has to be greater than or equal to the (scalar) value
- "<=" var has to be less than or equal to the (scalar) value
- "<" var has to be less than the (scalar) value

Info: "var" and "value" in check_var_list

Both the variable to check and the expected "value" can actually be HilVar/CalVar instances, meta()-wrapped MetaValue instances or plain scalar numeric values.

If all tolerances of a check_var_list entry remain un-set (i.e. None), the values will be treated as integer states for comparison.
If a value cannot sensibly be interpreted as as integer value, the test will fail with verdict ERROR and the actual check using the un-cast value(s) will most likely fail as well.
See ttk_checks.checkutils.getInteger

Returns

a tuple: check_verdicts - dictionary with <VERDICT>: <# of occurrences> info - an info string with (typically) one line for each checked variable

6.55.2.3 _getFormattedValues()

```
ttk_checks._adv_tests._getFormattedValues (
              original_v,
              actual_v,
              lookup_src_var = None,
              fmt_src_var = None )  [protected]
```

Get a formatted string describing the supplied values.

Note

getFormattedValues builds a string of comma-separated formatted value representations with added meta data (unit, lookup-state, alias - if available). If more than one value was given, the string will be wrapped in list-like brackets.

Parameters

original_v	list of original values/variables (potentially containing meta data like unit, alias, ...)
actual_v	list of actual values, has to be in same order and length as original_v
lookup_src_var	a Var/meta-value to provide a fallback value lookup
fmt_src_var	a Var/meta-value to provide a fallback custom value formatting

Returns

a formatted string for inclusion in result descriptions.

6.55.2.4 _getFuncEntryInfo()

```
ttk_checks._adv_tests._getFuncEntryInfo (
              var_list_entry )  [protected]
```

Get a textual representation of the function in var_list_entry.

Parameters

var_list_entry	a single entry of a set_ or check_var_list, see _setVariables and _checkVariables

Returns

a tuple function_alias, parameters_as_string

6.55.2.5 _getToleranceFormat()

```
ttk_checks._adv_tests._getToleranceFormat (
              abs_pos = 0,
              rel_pos = 0,
              abs_neg = None,
              rel_neg = None )  [protected]
```

Build a reasonably concise tolerance info format.

Format Examples:

"" (for no tolerances) or "+/- 2.0" (for symmetrical tolerances) or "+2.12/-1.12" (for the occasional asymmetrical tolerances)

Note

This is just a basic utility function that spits out different format strings, depending on the supplied tolerances. No tolerance values will be calculated

6.55.2.6 _needsTriggering()

```
ttk_checks._adv_tests._needsTriggering (
              signal )  [protected]
```

Check whether a signal needs to be manually "triggered" for values to be passed on (if not, it should update automatically).

This is determined by inspecting a member variable "cycle_time" (cycle time in ms, if available at all).

Parameters

signal	a VarBase or BusSignal instance (or any object with matching interface)

Returns

True if supplied signal is (supposedly) not being sent cyclically and requires manual triggering, otherwise False. Note that all "normal" HilVars & Co. will return False here.

6.55.2.7 _setVariables()

```
ttk_checks._adv_tests._setVariables (
              set_var_list )  [protected]
```

Set preconditions: variables/signals.

Parameters

set_var_list	list of dictionaries with variable and value (see example, below)

Example: set_var_list structure

```
set_var_list = [
    { "var":   <variable to set (e.g. a CalVar, HilVar or anything with a set() method),
      "value": <value to set>,
      "descr": <value description, optional>,
    },
    ...
]
See ttk_base.variables_base.CalVar
See ttk_base.variables_base.HilVar
```

Example: set_var_list and function calls

```
For special cases, the set var list may also contain function calls
(which might be useful to insert additional delays or similar):

set_var_list = [
    { "var":    cal.reset_var, "value": 1, descr="enable reset" },
    { "func":   time.sleep,
      "args":   [1.0], # positional arguments
      "kwargs": {},    # keyword arguments
      "descr":  "Wait for one second before clearing reset again",
    },
    { "var":   cal.reset_var, "value": 0 },
    # or a simple call like this:
    { "func":  someCustomFunction,
      "descr": "a custom function that does no need any parameters",
    },
    ...
]
If a single entry contains both "func" and "var" items, the function
gets called *before* the value will be set.
```

Returns

an info string with one line for each set variable and a list of error messages. While the error messages are also part of the info string, the separate list makes it easier to determine that errors have occurred.

6.55.2.8 _trigger()

```
ttk_checks._adv_tests._trigger (
              var )  [protected]
```

Trigger transmission of the given BusSignal/Var.

Parameters

var	Var or bus signal with a trigger kickout method

Returns

True if trigger/kickout was successfully called, False if call failed, None if no trigger/kickout method was available

6.55.2.9 testVars()

```
ttk_checks._adv_tests.testVars (
               set_var_list = None,
               check_var_list = None,
               min_wait_time_ms = 100,
               descr = '' )
```

Check the values of multiple variables/signals as a result of setting multiple other variables/signals.

Between setting and checking, a configurable time will be waited to deal with propagation delay. This wait time defaults to the minimum practicable wait time of all involved bus signals.

Info: This function ...

- sets the value(s) to the variables in set_var_list
- triggers sending of any event-based signals included in set_var_list
  - "needs triggering" is determined by evaluating `cycle_time` members of set vars (if present)
  - triggering is performed by calling a method `kickout()` or `trigger()` (whichever is available)
- waits for value change(s) to propagate to ECU/DUT
  - this depends on both the specified `min_wait_time_ms` and any `change_delay` members of entries in set or check var lists.
- checks the value(s) for variables in check_var_list

Parameters

set_var_list	Variables to set: a list of dictionaries with variable and value
check_var_list	Variables to check: a list of dictionaries with variable, expected value and tolerance
min_wait_time_ms	minimum wait time between setting variables and performing checks
descr	user specific description shown at the start of the result text

Example: Wait time before check (propagation delay)

```
wait_time = max(
    min_wait_time_ms,
    max(<change delay of signals to set>) +
    max(<change delay of signals to check>)
)
# Note: a bus signal's wait time defaults to 2x cycle time
```

Example: set_var_list structure

```
set_var_list = [
    { "var":   <variable to set (e.g. a CalVar, HilVar or anything with a set() method),
      "value": <value to set>,
      "descr": <value description, optional>,
    },
```

Example: set_var_list and function calls

```
# For special cases, the set var list may also contain function calls
# (which might be useful to insert additional delays or similar):
set_var_list = [
    { "var":    cal.reset_var, "value": 1, descr="enable reset" },
    { "func":   time.sleep,
      "args":   [1.0], # positional arguments
      "kwargs": {},    # keyword arguments
      "descr":  "Wait for one second before clearing reset again",
    },
    { "var":   cal.reset_var, "value": 0 },
    # or a simple call like this:
    { "func":  someCustomFunction,
      "descr": "a custom function that does no need any parameters",
    },
    ...
]
```

Example: check_var_list structure

```
# A basic entry:
check_var_list = [
    { "var":     <variable to check (CalVar, HilVar, ...)>,
      "operator": <check operator, defaults to "==">,
      "value":   <expected value(s)>
      "abs_tol": <absolute tolerance (optional)>,
      "rel_tol": <relative tolerance (optional)>,
    }
]
# abs_tol/rel_tol are aliases for checkTolerances's abs_pos/rel_pos
# (when separate _pos/_neg tolerances are not required).
# If you need non-symmetrical tolerances (e.g. -1/+5), you can
# specify tolerances separately:
check_var_list = [
    { "var":      <variable to check (CalVar, HilVar, ...)>,
      "operator": <check operator, defaults to "==">,
      "value":    <expected value> or a list of expected values, of which one has to pass
      "rel_pos":  <relative positive tolerance (optional)>,
      "abs_pos":  <absolute positive tolerance (optional)>,
      "rel_neg":  <relative negative tolerance (optional)>,
      "abs_neg":  <absolute negative tolerance (optional)>,
    },
    ...
]
# See basic_checks.checkTolerance for details on tolerance parameters
# "==" is default when operator entry is omitted
```

Example: extended check_var_list structure

```
check_var_list = [
    { "var":     <variable to check (CalVar, HilVar, ...)>,
      "operator": "==", # "==" or "in"
      "value":   <expected value> or a list of multiple expected values of which one has to match
      "abs_pos": <absolute tolerance (optional)>,
      "rel_pos": <relative tolerance (optional)>,
    },
    { "var":     <variable to check (CalVar, HilVar, ...)>,
      "operator": "!=", # "!=" or "not in"
      "value":   <expected value> or a list of multiple expected values, none of which must match
      "abs_pos": <absolute tolerance (optional)>,
      "rel_pos": <relative tolerance (optional)>,
    },
    { "var":     <variable to check (CalVar, HilVar, ...)>,
      "operator": ">", # "<", "<=", ">=" or ">"
      "value":   <expected value>
      # no tolerances are used for "greater/less than" comparisons
    },
    ...
]
```

Example: Using a list of multiple expected values (usually for status values):

```
# Operator "==" or "in": one value has to match
check_var_list = [
    { "var":       cal.e2e_status,
      "operator": "in"
      "value":    [E2E_P04STATUS_OK, E2E_P04STATUS_OKSOMELOST],
    }
```

```
]

# Operator "!=" or "not in": no value may match:
check_var_list = [
    { "var":      cal.e2e_status,
      "operator": "not in",
      "value":    [E2E_P04STATUS_NONEWDATA, E2E_P04STATUS_ERROR,
                   E2E_P04STATUS_REPEATED, E2E_P04STATUS_WRONGSEQUENCE],
    }
]
```

Info: "var" and "value" in check- and set_var_list

Both variable to check and the expected "value" can actually be any VarBase derived instance, meta()-wrapped MetaValue instances or plain scalar numeric values.

This means that it is possible to compare the values of two ecu/cal variables:
`{"var": cal.some_variable, "value": cal.some_other_variable}`
or between hil and ecu
`{"var": cal.foobar, "value": hil.barbaz}`
`{"var": bus.barsignal, "value": cal.bazquux}`
or even (admittedly, not too useful)
`{"var": 3, "value": 3.2, "abs_tol": .4}`

Info: Units and plain values

If none of the involved variables/values has a "unit" specified and you want to show units in the test report, you could wrap values in a MetaValue:
`{"var": cal.shrdlu, "value": meta(3.2, "mV"), "abs_tol": .4}`
See ttk_base.values_base.meta()

Status values and tolerances:

If all tolerances of a check_var_list entry remain un-set (i.e. None), the values will be treated as discrete integer states during comparison. If a value does not appear to be an integer value, the test will fail with verdict ERROR (and the actual check using the un-cast value(s) will most likely fail as well).
See ttk_checks.checkutils.getInteger()

Returns

a single test result entry [<description>, <VERDICT>] (with a single resulting verdict for all checked variables).

6.56 ttk_checks._basic_checks Namespace Reference

Basic "low-level" check functions for use in test automation.

Functions

- _checkNumeric (label, value)

  Check if value is one of the numeric types (i.e.
- checkBit (value, offset, status=1)

  Check if the bit given by the value and offset is set or cleared depending on the status.
- checkBitMask (value, mask, operator="AND", bit_length=CHECKBITMASK_DEFAULT_BIT_LENGTH)

  Check if the result of the operation given in `operator` is 'greater 0' or 'equal 0'.
- checkBitPattern (value, pattern)

  Check that a numeric value matches a certain pattern.
- checkRange (value, min_value, max_value)

  Check if value is within defined range (min_value <= value <= max_value).
- checkTolerance (current_value, rated_value, rel_pos=0, abs_pos=0, rel_neg=None, abs_neg=None)

  Check if the current value is within a specified range of the rated value.
- contains (defined_values, current_value, mode="all")

  Check if current_value is an element or sub-set of defined_values.
- compare (left_value, operator, right_value, abs_tol=0)

  Verify the conditional relationship between two operands based on the given operator.

Variables

- int CHECKBITMASK_MAX_BIT_LENGTH = 128
- int CHECKBITMASK_DEFAULT_BIT_LENGTH = 16
- tuple CHECKBITMASK_OPERATORLIST_AND = ('AND', '&')
- tuple CHECKBITMASK_OPERATORLIST_OR = ('OR', '|')
- tuple CHECKBITMASK_OPERATORLIST_XOR = ('XOR', '^')
- tuple CHECKBITMASK_OPERATORLIST_NAND = ('NAND', '!&')
- tuple CHECKBITMASK_OPERATORLIST_NOR = ('NOR', '!|')
- tuple CHECKBITMASK_OPERATORLIST_NXOR = ('NXOR', '!^', 'XNOR')
- tuple CONTAIN_MODES = ('all', 'any', 'none', 'not')
- tuple COMPARE_OPERATORS = ("=", "==", "<", ">", "<=", ">=", "!=", "<>")
- int verbosity = 1

  level of debug/log message verbosity (0: mostly silent, >0: more chatty)

6.56.1 Detailed Description

Basic "low-level" check functions for use in test automation.

See interface wrapper ttk_checks.basic_checks for more detailed docstring.

6.56.2 Function Documentation

6.56.2.1 _checkNumeric()

```
ttk_checks._basic_checks._checkNumeric (
              label,
              value )  [protected]
```

Check if value is one of the numeric types (i.e.

float or int). Raises an InputValueTypeError if value is not numeric.

Parameters

label	label/name of checked variable/parameter
value	value of checked variable/parameter

6.56.2.2 checkBit()

```
ttk_checks._basic_checks.checkBit (
              value,
              offset,
              status = 1 )
```

Check if the bit given by the value and offset is set or cleared depending on the status.

Parameters

value	Numeric value where the bit has to be checked
offset	Offset of the bit inside of value (based on 0)
status	The expected value of the bit (default 1)

Returns

a tuple (status, dictionary with intermediate values)
`status` - True if the selected bit has the desired, status, otherwise False.
`intermediate_values` - a dictionary with
''called_function': `name of this function in string format <BR>` 'value'↩
: `input value <BR>` 'offset': `input bit offset <BR>` 'bit↩
_status_detected': `actual bit status <BR>` 'bit_status_expected`` : expected bit status (from input)

6.56.2.3 checkBitMask()

```
ttk_checks._basic_checks.checkBitMask (
              value,
              mask,
              operator = "AND",
              bit_length = CHECKBITMASK_DEFAULT_BIT_LENGTH )
```

Check if the result of the operation given in `operator` is 'greater 0' or 'equal 0'.

The result is calculated with a dyadic operation. The two operands are `value` and `mask`.

Parameters

value	value to be checked (int)
mask	mask to be used for operation in operator (int or string)
operator	operator for check operation, one of ("AND", "OR", "XOR", "NAND", "NOR", "NXOR")
bit_length	bit length for value and mask

Returns

a tuple (status, dictionary with intermediate values)
`status` - True if result of operation is 'greater 0', False if it is 'equal 0'
`intermediate_values` - a dictionary with
''result_string' : result of operation in binary string format
 'result↩
_value' : result of operation as int value
 'source↩
_value' : input value
 'source_mask' : input mask
'source_operator' : input operator
 'source_bit_length' : input
bit length
 'result_verdict' : True if result value > 0

 'called_function'` : name of this check function (as string)

6.56.2.4 checkBitPattern()

```
ttk_checks._basic_checks.checkBitPattern (
            value,
            pattern )
```

Check that a numeric value matches a certain pattern.

Parameters

value	value to check. The value must match the full pattern in order to pass. Values may be numeric (int), string representations of binary/hexadecimal values ("0xABCD", "0b1010" etc.) or a list of bytes.
pattern	a string describing the desired bit pattern. Note that a pattern must always describe the full value (i.e. all bits from start to end). • A pattern starting with "0x" will be interpreted as hexadecimal • "0b" at the beginning denotes a binary pattern • "X" marks "don't care" sections (anywhere in pattern)

See examples for further details.

Examples:

```
"0xABC123"   - a hexadecimal 20-bit pattern
"123ABC"     - without a prefix, pattern defaults to hexadecimal
"0b0101101"  - a binary 6-bit pattern
"0xAABXX2"   - X designates "don't care" (4-bit-nibbles in hex
               patterns)
"0b1XX001"   - "don't care" on a bit level in binary patterns
"0xAA[8]BB"  - longer "don't care"-sections can be written as
               a repetition number in square brackets. This example
               pattern is 48 bits long with 32 don't care bits in
               the middle.
"0b10[8]00"  - as above but in a binary pattern: 12 bits with 8
               don't care bits in the middle.
"0xAA BB CC" - whitespace will be ignored for patter check, but
               plain spaces will be retained in formatted
               intermediate values
"0xab Ab aB" - patterns are not case sensitive
```

Returns

a tuple (status, dictionary with intermediate values)
`status` - True if pattern matches, otherwise False
`intermediate_values` - a dictionary with
''condensed_value': `value w/o whitespace formatting <BR>` 'condensed↩
_pattern': `pattern w/o whitespace formatting <BR>` 'formatted↩
_value': `value with whitespace formatted like pattern <BR>` 'formatted↩
_pattern': `pattern with whitespace formatting <BR>` 'mismatch↩
_position': `position where first mismatch occurred (from condensed_pattern), -1 if pattern matches. <BR>` 'called_function'` : name of this check function (as string)

6.56.2.5 checkRange()

```
ttk_checks._basic_checks.checkRange (
              value,
              min_value,
              max_value )
```

Check if value is within defined range (min_value <= value <= max_value).

Parameters

value	value to be checked (int/float)
min_value	lower limit
max_value	upper limit

Examples:

```
checkRange(0x2,-5.0, 7L)    # -> True
checkRange(0x2,  7L, 5.0)   # -> True
checkRange(  5,   7, 9)     # -> False
checkRange(  2, "a", 7)     # -> raises an exception
```

Returns

a tuple (status, dictionary with intermediate values)
`status` - True or False if value is in defined range
`intermediate_values` - a dictionary with
''value': `input value that has been checked <BR>` 'min_value': `lower Limit <BR>` 'max_value': `upper limit <BR>` 'called↩
_function'` : name of this function in string format

6.56.2.6 checkTolerance()

```
ttk_checks._basic_checks.checkTolerance (
              current_value,
              rated_value,
              rel_pos = 0,
              abs_pos = 0,
              rel_neg = None,
              abs_neg = None )
```

Check if the current value is within a specified range of the rated value.

The range is defined by an absolute and a relative tolerance. Both tolerances will be added.

Note

If a *symmetric* tolerance range is desired, only `rel_pos` and/or `abs_pos` need to be set.

Parameters

current_value	value to be checked
rated_value	expected value
rel_pos	(optional) relative positive tolerance (factor)
abs_pos	(optional) absolute positive tolerance
rel_neg	(optional) relative negative tolerance (factor)
abs_neg	(optional) absolute negative tolerance

Returns

a tuple (status, dictionary with intermediate values)
`status` - True if value is within tolerance, otherwise False
`intermediate_values` - a dictionary with
''current_value': `value to be checked <BR>` 'rated_value': `expected value to be checked against <BR>` 'rel_pos': `actual relative positive tolerance <BR>` 'abs_pos': `actual absolute positive tolerance <BR>` 'rel_neg': `actual relative negative tolerance <BR>` 'abs_neg': : `actual absolute negative tolerance <BR>` 'lower_boundary': `calculated lower boundary (from tolerances) <BR>` 'upper↩ _boundary': `calculated upper boundary (from tolerances) <BR>` 'effective↩ _abs_pos': `effective positive tolerance, absolute (based on both abs and rel pos) <BR>` 'effective_abs_neg': `effective negative tolerance, absolute (based on both abs and rel neg) <BR>` 'called_↩ function`` : name of this check function (as string)

6.56.2.7 compare()

```
ttk_checks._basic_checks.compare (
              left_value,
              operator,
              right_value,
              abs_tol = 0 )
```

Verify the conditional relationship between two operands based on the given operator.

Parameters

left_value	left operand of the relational expression
operator	the relational operator given as a string (one of "=", "==", "<", ">", "<=", ">=", "!=", "<>")
right_value	right operand of the relational expression
abs_tol	absolute tolerance for use in equality/non-equality checks of numeric types

Note

A tolerance > 0 is mandatory for equality checks (==, !=) involving float values.
See checkTolerance() for more in-depth options like relative and non-symmetric tolerances.

Info:

When comparing lists, they must contain the same number of elements or a BasicCheckException will be raised.
List compare currently only supports == and != operators, as the behavior for other operations is unspecified.

Relational operator strings:

- "=" : equal to
- "==" : equal to
- "<" : less than
- ">" : greater than
- "<=" : less than or equal to
- ">=" : greater than or equal to
- "!=" : not equal to
- "<>" : not equal to

Returns

a tuple (status, dictionary with intermediate values)
`status` - True if expression evaluates to True, otherwise False
`intermediate_values` - a dictionary with
''left_value': `left value of expression <BR>` 'right_value': `right value of expression <BR>` 'operator': `used operator <BR>` 'abs_tol': `tolerance (absolute, for numeric checks) <BR>` 'compare_mode': `kind of comparison that was performed (numeric, string, list, other) <BR>` 'called_function'` : name of this check function (as string)

6.56.2.8 contains()

```
ttk_checks._basic_checks.contains (
              defined_values,
              current_value,
              mode = "all" )
```

Check if current_value is an element or sub-set of defined_values.

The order of elements is not important.

Note

While it is - in principle - possible to compare lists of float values, it might lead to unexpected results due to rounding errors.

Parameters

defined_values	list of scalars to be searched in
current_value	function checks if this scalar (or list) is contained in 'defined_values'
mode	search mode (one of 'all', 'any', 'none', 'not', see CONTAIN_MODES)

Examples:

```
contains([1,3,5], 3, mode="any")         # -> True
contains([1,3,5], 7, mode="none")        # -> True
contains([1,3,5], [3, 45], mode="all")   # -> False
contains([1,3,5], [3, 45], mode="any")   # -> True
contains([1,3,5], [3, 45], mode="not")   # -> False
contains([1,3,5], [3, 5],  mode="all")   # -> True
```

Available Modes:

- "all": each value of 'current_value' should be in `defined_values`
- "any": at least one element of 'current_value' should be in `defined_values`
- "none"/"not": no element of 'current_value' should be in `defined_values`

Returns

a tuple (status, dictionary with intermediate values)
`status` - True or False if current_value is an element of defined_values, depending on mode
`intermediate_values` - a dictionary with
'in_list': List of all elements of current_value that are in defined_values.
 'not_in_list': List of all elements of current_value that are not in defined_values.
 'current_value': value(s) that have been checked
 'defined_values': values that have been checked against
 'mode` : check mode

6.56.3 Variable Documentation

6.56.3.1 CHECKBITMASK_DEFAULT_BIT_LENGTH

```
int ttk_checks._basic_checks.CHECKBITMASK_DEFAULT_BIT_LENGTH = 16
```

6.56.3.2 CHECKBITMASK_MAX_BIT_LENGTH

```
int ttk_checks._basic_checks.CHECKBITMASK_MAX_BIT_LENGTH = 128
```

6.56.3.3 CHECKBITMASK_OPERATORLIST_AND

```
tuple ttk_checks._basic_checks.CHECKBITMASK_OPERATORLIST_AND = ('AND', '&')
```

6.56.3.4 CHECKBITMASK_OPERATORLIST_NAND

```
tuple ttk_checks._basic_checks.CHECKBITMASK_OPERATORLIST_NAND = ('NAND', '!&')
```

6.56.3.5 CHECKBITMASK_OPERATORLIST_NOR

```
tuple ttk_checks._basic_checks.CHECKBITMASK_OPERATORLIST_NOR = ('NOR', '!|')
```

6.56.3.6 CHECKBITMASK_OPERATORLIST_NXOR

```
tuple ttk_checks._basic_checks.CHECKBITMASK_OPERATORLIST_NXOR = ('NXOR', '!^', 'XNOR')
```

6.56.3.7 CHECKBITMASK_OPERATORLIST_OR

```
tuple ttk_checks._basic_checks.CHECKBITMASK_OPERATORLIST_OR = ('OR', '|')
```

6.56.3.8 CHECKBITMASK_OPERATORLIST_XOR

```
tuple ttk_checks._basic_checks.CHECKBITMASK_OPERATORLIST_XOR = ('XOR', '^')
```

6.56.3.9 COMPARE_OPERATORS

```
tuple ttk_checks._basic_checks.COMPARE_OPERATORS = ("=", "==", "<", ">", "<=", ">=", "!=",
"<>")
```

6.56.3.10 CONTAIN_MODES

```
tuple ttk_checks._basic_checks.CONTAIN_MODES = ('all', 'any', 'none', 'not')
```

6.56.3.11 verbosity

```
int ttk_checks._basic_checks.verbosity = 1
```

level of debug/log message verbosity (0: mostly silent, >0: more chatty)

6.57 ttk_checks._basic_tests Namespace Reference

Basic test/evaluation functions for test automation that directly generate test report entries.

Functions

- _getErrorDescr (info, ex_info, ∗params)

  Get a formatted error description with added parameter information (for internal use in exception handlers).
- checkBitMask (value, mask, operator="AND", bit_length=basic_checks.CHECKBITMASK_DEFAULT_BIT_↩ LENGTH, pass_on_true=True, descr="", format=None)

  Check result of the operation given in `operator`.
- checkBitPattern (value, pattern, invert=False, descr="", format=None)

  Check that a numeric value matches a certain pattern.
- checkBit (value, offset, status=1, descr="", format=None)

  Check if the bit given by the value and offset is set or cleared depending on the status.
- checkTolerance (current_value, rated_value, rel_pos=0, abs_pos=0, rel_neg=None, abs_neg=None, invert=False, descr="", format=None)

  Check if the current value is within a specified range of the rated value.
- checkAbsTol (current_value, rated_value, abs_tol=0, invert=False, descr="")

  Check if the current value is within a specified range of the rated value.
- checkRelTol (current_value, rated_value, rel_tol=0.0, invert=False, descr="")

  Check if the current value is within a specified range of the rated value.
- checkStatus (current_status, nominal_status, equal=True, descr="", format=None)

  Check if the current status value is identical to the nominal status value.
- checkRange (value, min_value, max_value, invert=False, descr="", format=None)

  Check if value is in defined range (`min_value <= value <= max_value`).
- compare (left_value, operator, right_value, abs_tol=0, descr="", format=None)

  Verify the conditional relationship between two operands based on the given operator.
- compareCE (current_value, operator, expected_value, abs_tol=0, descr="")

  Verify the conditional relationship between current and expected value based on the given operator.
- contains (defined_values, current_value, mode="all", descr="", format=None)

  Check if current_value is an element (or sub set) of defined_values.

6.57.1 Detailed Description

Basic test/evaluation functions for test automation that directly generate test report entries.

See interface wrapper ttk_checks.basic_tests for more detailed docstring.

6.57.2 Function Documentation

6.57.2.1 _getErrorDescr()

```
ttk_checks._basic_tests._getErrorDescr (
              info,
              ex_info,
             * params )  [protected]
```

Get a formatted error description with added parameter information (for internal use in exception handlers).

Parameters

info	info text / manual description text
ex_info	exception info
∗*params*	all following arguments are parameters to add to description, add as (name, param) tuples to preserve order

Returns

a formatted description string.

6.57.2.2 checkAbsTol()

```
ttk_checks._basic_tests.checkAbsTol (
              current_value,
              rated_value,
              abs_tol = 0,
              invert = False,
              descr = "" )
```

Check if the current value is within a specified range of the rated value.

The valid range is defined by an absolute tolerance as a symmetric range around the `rated_value`.

Parameters

current_value	value to be checked
rated_value	expected value
abs_tol	absolute tolerance value, valid range will be `rated_value` +/- `abs_tol`
invert	True: inverted, check current value != rated value (+/- tolerance) False: default, check current value == rated value (+/- tolerance)
descr	user specific description shown before the single test result

Returns

a single [<description>, <verdict>] test result entry.

6.57.2.3 checkBit()

```
ttk_checks._basic_tests.checkBit (
              value,
              offset,
              status = 1,
              descr = "",
              format = None )
```

Check if the bit given by the value and offset is set or cleared depending on the status.

Parameters

value	Numeric value where the bit has to be checked
offset	Offset of the bit inside of value (based on 0)
status	The expected value of the bit (default 1)
descr	user specific description printed on the start of the single test result
format	format string to control/override appearance of the automatically created result text. A format override should only rarely be used.

Returns

a single [<description>, <verdict>] test result entry.

6.57.2.4 checkBitMask()

```
ttk_checks._basic_tests.checkBitMask (
              value,
              mask,
              operator = "AND",
              bit_length = basic_checks.CHECKBITMASK_DEFAULT_BIT_LENGTH,
              pass_on_true = True,
              descr = "",
              format = None )
```

Check result of the operation given in `operator`.

Depending on parameter `pass_on_true`, the test will be PASSED if the result is True (i.e. 'greater 0') or False (i.e. 'equal 0').

Parameters

value	value to be checked
mask	mask to be used for operation in operator
operator	operator for check operation, one of ("AND", "OR", "XOR", "NAND", "NOR", "NXOR") see *basic_checks.CHECKBITMASK*∗-constants.
bit_length	bit length for value and mask
pass_on_true	if True, test will be PASSED if result is 'greater 0', otherwise it will be PASSED if the result is 'equal 0'
descr	user specific description shown at the start of the result text
format	format string to control/override appearance of the automatically created result text. A format override should only rarely be used.

Returns

a single [<description>, <verdict>] test result entry.

6.57.2.5 checkBitPattern()

```
ttk_checks._basic_tests.checkBitPattern (
              value,
              pattern,
              invert = False,
              descr = "",
              format = None )
```

Check that a numeric value matches a certain pattern.

Parameters

value	value to check. The value must match the full pattern in order to pass. Values may be numeric (int/long), string representations of binary/hexadecimal values ("0xABCD", "0b1010" etc.) or a list of bytes.
pattern	a string describing the desired bit pattern. • a pattern must always describe the full value (i.e. all bits from start to end). • a pattern starting with "0x" will be interpreted as hexadecimal • "0b" at the beginning denotes a binary pattern • "X" marks "don't care" sections (may appear anywhere in pattern)
invert	True: inverted check: pattern must not match False: default check: pattern has to match
descr	user specific description shown at the start of the result text
format	format string to control/override appearance of the automatically created result text. A format override should only rarely be used.

Pattern String Examples:

- `0xABC123` - a hexadecimal 24-bit pattern
- `123ABC` - without a prefix, pattern defaults to hexadecimal
- `0b0101101` - a binary 6-bit pattern
- `0xAABXX2` - X designates "don't care" (4-bit-nibbles in hex patterns)
- `0b1XX001` - "don't care" on a bit level in binary patterns
- `0xAA[8]BB` - longer "don't care"-sections can be written as a repetition number in square brackets. This example pattern is 48 bits long with 32 don't care bits in the middle.
- `0b10[8]00` - as above but in a binary pattern: 12 bits with 8 don't care bits in the middle.
- `0xAA BB CC` - whitespace will be ignored for pattern check, but plain spaces will be retained in formatted intermediate values
- `0xab Ab aB` - patterns are not case sensitive

Returns

a single [<description>, <verdict>] test result entry.

6.57.2.6 checkRange()

```
ttk_checks._basic_tests.checkRange (
              value,
              min_value,
              max_value,
              invert = False,
              descr = "",
              format = None )
```

Check if value is in defined range (`min_value <= value <= max_value`).

Parameters

value	value to be checked
min_value	lower Limit
max_value	upper Limit
invert	True: inverted, check that value is outside of range False: default, check that value is inside of range
descr	user specific description printed on the start of the single test result
format	format string to control/override appearance of the automatically created result text. A format override should only rarely be used.

Returns

a single [<description>, <verdict>] test result entry.

6.57.2.7 checkRelTol()

```
ttk_checks._basic_tests.checkRelTol (
              current_value,
              rated_value,
              rel_tol = 0.0,
              invert = False,
              descr = "" )
```

Check if the current value is within a specified range of the rated value.

The valid range is defined by relative tolerance factor (as a symmetric range around the `rated_value`).

Info: Relative Tolerances

Relative tolerance values are specified as factors in relation to the supplied `rated_value`.
A range of `0.0 .. 1.0` maps to `0 .. 100 %`

Note

`checkRelTol` will treat `rel_tol` values >= 1.0 as unintended and mark the returned result as ERROR. Use the main test function `checkTolerance` if checks with tolerance values >= 100% should ever be needed.

Parameters

current_value	value to be checked
rated_value	expected value
rel_tol	relative tolerance as factor, in range 0.0..1.0
invert	True: inverted, check current value != rated value (+/- tolerance) False: default: check current value == rated value (+/- tolerance)
descr	user specific description shown before the single test result

Returns

a single [<description>, <verdict>] test result entry.

6.57.2.8 checkStatus()

```
ttk_checks._basic_tests.checkStatus (
              current_status,
              nominal_status,
              equal = True,
              descr = "",
              format = None )
```

Check if the current status value is identical to the nominal status value.

Both statuses will be interpreted as int/long values for the comparison.

Note

If one or both values cannot be represented as integer numbers, the test will fail with verdicts.ERROR.

Parameters

current_status	status to be checked
nominal_status	expected status to be checked against
equal	True: check current status == nominal status, False: check current status != nominal status
descr	user specific description printed on the start of the single test result
format	format string to control the the print out of the single test result

Returns

a single [<description>, <verdict>] test result entry.

6.57.2.9 checkTolerance()

```
ttk_checks._basic_tests.checkTolerance (
              current_value,
              rated_value,
              rel_pos = 0,
              abs_pos = 0,
              rel_neg = None,
              abs_neg = None,
              invert = False,
              descr = "",
              format = None )
```

Check if the current value is within a specified range of the rated value.

The range is defined by an absolute and a relative tolerance. Both tolerance types will be added to get the effective tolerance.

Note

If a *symmetric* tolerance range is desired, only `rel_pos` and/or `abs_pos` need to be set.

Info: Relative Tolerances

Relative tolerance values are specified as factors in relation to the supplied `rated_value`.
A range of `0.0 .. 1.0` maps to `0 .. 100 %`

Using both relative and absolute tolerances:

Internally, both tolerances are added to determine the effective boundary values:

```
lower_boundary = abs_neg + rel_neg * abs(rated_value) + rated_value
upper_boundary = abs_pos + rel_pos * abs(rated_value) + rated_value
```

`current_value` will pass the check if its value lies between the calculated boundary values:

```
lower_boundary <= current_value <= upper_boundary
```

Parameters

current_value	value to be checked
rated_value	expected value
rel_pos	(optional) relative positive tolerance as factor, typically in range 0.0..1.0
abs_pos	(optional) absolute positive tolerance
rel_neg	(optional) relative negative tolerance as factor, typically in range 0.0..-1.0
abs_neg	(optional) absolute negative tolerance
invert	True: inverted, check current value != rated value (+/- tolerance) False: default: check current value == rated value (+/- tolerance)
descr	user specific description printed on the start of the single test result
format	format string to control/override appearance of the automatically created result text. A format override should only rarely be used.

Returns

a single [<description>, <verdict>] test result entry.

6.57.2.10 compare()

```
ttk_checks._basic_tests.compare (
              left_value,
              operator,
              right_value,
              abs_tol = 0,
              descr = "",
              format = None )
```

Verify the conditional relationship between two operands based on the given operator.

Parameters

left_value	left operand of the relational expression
operator	relational operator given as a string, see below
right_value	right operand of the relational expression
abs_tol	absolute tolerance for equality/non-equality checks of numeric types
descr	user specific description printed on the start of the single test result
format	format string to control/override appearance of the automatically created result text. If a tuple of strings is supplied, it should contain two (or three) entries: `(<long_format>, <compact_format>)` or `(<long_format>, <compact_format>, <alias_length_boundary>)` where alias_length_boundary is the average alias name length at which to switch to "long" format. A format override should only rarely be used.

Info: Tolerance

A tolerance > 0 is mandatory for equality / non-equality checks involving float values.
See checkTolerance() for more in-depth options like relative and non-symmetric tolerances.

Info: Long/Compact Description Formats

If input values have no overly long aliases, a more compact output format will be used.

Compact format samples:

- 1.2 mA (short alias 1) $<$ 2.5 mA (short alias 2)
- 1.2 mA (short alias 1) == 2.5 mA (short alias 2) +/- 0.4 mA

Long format samples:

- 1.2 mA $<$ 2.5 mA
  Left Value: A rather unreasonably long alias name
  Right Value: Another quite verbose alias name
- 1.2 mA == 2.5 mA +/- 0.4 mA
  Left Value: A rather unreasonably long alias name
  Right Value: Another quite verbose alias name

Default `alias_length_boundary` for switching between formats is 32 characters.
This value can be configured in `hil_project.ini`, section `[TestToolkit]` with key/option `alias_length_boundary`.

Note

Var/Value resolving Compare resolves input values (get current value from Vars) in two variants:

- value(s) were supplied as a list/tuple:
  - each var/value will be "resolved" to its current value. See checkutils.resolveValue
- a single value was supplied: resolve it
  - if value resolved to a list, use this list for comparison
  - otherwise compare value as list of one item

Relational operator strings:

- "==" : equal to
- "<" : less than
- ">" : greater than
- "<=" : less than or equal to
- ">=" : greater than or equal to
- "!=" : not equal to

Returns

a single [<description>, <verdict>] test result entry.

6.57.2.11 compareCE()

```
ttk_checks._basic_tests.compareCE (
               current_value,
               operator,
               expected_value,
               abs_tol = 0,
               descr = "" )
```

Verify the conditional relationship between current and expected value based on the given operator.

This is a compare variant with an alternative output formatting to explicitly label input values as "current" and "expected". See also basic_tests.compare.

Parameters

current_value	left operand of the relational expression
operator	relational operator given as a string, see below
expected_value	right operand of the relational expression
abs_tol	absolute tolerance for equality/non-equality checks of numeric types
descr	user specific description printed on the start of the single test result

Example: Description Formatting

```
Comparing integer values
Current:  1 [on] (some_status_var)
Expected: == 1 [on]
```

Info: Tolerance

A tolerance > 0 is mandatory for equality / non-equality checks involving float values.

Relational operator strings:

- "==" : equal to
- "<" : less than
- ">" : greater than
- "<=" : less than or equal to
- ">=" : greater than or equal to
- "!=" : not equal to

Returns

a single [<description>, <verdict>] test result entry.

6.57.2.12 contains()

```
ttk_checks._basic_tests.contains (
              defined_values,
              current_value,
              mode = "all",
              descr = "",
              format = None )
```

Check if current_value is an element (or sub set) of defined_values.

Examples:

```
contains([1,3,5], 3, mode="any")  -> ["...", "PASSED"]
contains([1,3,5], [3, 45])        -> ["...", "FAILED"]
contains([1,3,5], 7, mode="none") -> ["...", "PASSED"]
```

Parameters

defined_values	List of scalars to be searched in
current_value	the function checks if this scalar or list is contained in 'defined_values'
mode	search mode (see below)
descr	user specific description printed on the start of the single test result
format	format string to control the the print out of the single test result. A format override should only rarely be used.

Available Modes:

- "all": each value of 'current_value' should be in `defined_values`
- "any": at least one element of 'current_value' should be in `defined_values`
- "none"/"not": no element of 'current_value' should be in `defined_values`

Returns

a single [<description>, <verdict>] test result entry.

6.58 ttk_checks._checkutils Namespace Reference

Utilities for check functions.

Classes

- class DefaultDict

  A dictionary that returns empty strings instead of raising KeyErrors.

Functions

- buildDescription (descr_format, entries)

  Build a result description from a supplied format string and entries.
- getInteger (value)

  Cast a numeric value to an integer (int/long) number, but perform sanity checks before (relevant for float input values).
- resolveValue (value)

  Get a value of the input parameter if it supports a "get" method, otherwise return it as-is.
- _getValueRepresentation (value, fallback_value=None, fallback_fmt_source=None)

  Get a string representation of the supplied value (preferably formatted if a format hint has been specified).
- _getStateDescr (value, fallback_lookup_source=None)

  Get a state description of the supplied value if a value lookup is available.
- updateMetaInfo (meta, value, key_prefix="", meta_attrs=None, setdefaults=False, fallback_value=None)

  Extract meta info from `value` (if info is available) and update the supplied meta dictionary with those values.
- updateMetaStateDescr (meta, source_var, value, key)

  Update (or create) a meta entry with a state description for the supplied value, using the lookup-table from `source_var`.
- updateMetaFormattedValue (meta, source_var, value, key="fmt", fallback_value=None)

  Update (or create) a meta entry with a formatted value representation for the supplied value, using the formatting defined in `source_var`.
- updateMetaEntries (meta, *source_vars_info)

  All-in-one update of meta data.
- bin (value, prefix='0b', min_bit_width=0)

  Create a binary representation of the input value.
- getTracebackString (limit=None)

  Get current traceback info as a string (or an empty string if no exception is currently active).

Variables

- _ = basic_tests_loc.getInstance().getText
- float FLOAT_CONVERSION_LIMIT = 0.0001

  maximum divergence to still accept a float value to be "intended" as int/long
- int FLOAT_CONVERSION_MAX_VALUE = 2**52

  IEEE-754 double-precision => 52 bits explicitly stored.

6.58.1 Detailed Description

Utilities for check functions.

See interface wrapper in ttk_checks.checkutils.

6.58.2 Function Documentation

6.58.2.1 _getStateDescr()

```
ttk_checks._checkutils._getStateDescr (
              value,
              fallback_lookup_source = None )  [protected]
```

Get a state description of the supplied value if a value lookup is available.

Parameters

value	value to get a textual state description of.
fallback_lookup_source	an instance providing a method `getStateDescr(value)` to use as optional fallback if value does not bring its own lookup data

Note

`value` is preferably meta-wrapped or VarBase-derived, plain scalar values will need a `fallback_lookup_source` as lookup "donor".

Returns

a formatted string or None if no lookup is available

6.58.2.2 _getValueRepresentation()

```
ttk_checks._checkutils._getValueRepresentation (
              value,
              fallback_value = None,
              fallback_fmt_source = None )  [protected]
```

Get a string representation of the supplied value (preferably formatted if a format hint has been specified).

Parameters

value	value/instance to get a representation of.
fallback_value	fallback value (typically of a base type like int or float) to use as representation if a getFormattedValue method is not available Defaults to `value`
fallback_fmt_source	an instance providing a method `getFormattedValue(value)` to use as optional fallback if value does not bring its own custom formatting

Note

If `value` is not an instance that provides either

- a suitable string representation of its "value" via **str** (like basic types int or float) or
- a method getFormattedValue (like all VarBase-derived variables)

the default representation would be something along the line of "`<foo.NotSoGood object at 0x000000000E8588D0>`".

If a separate value (e.g. a float value from a .get() operation) is available, it can be supplied as `fallback_value` and will be used as fallback representation for `value`.

Returns

a string representation

6.58.2.3 bin()

```
ttk_checks._checkutils.bin (
              value,
              prefix = '0b',
              min_bit_width = 0 )
```

Create a binary representation of the input value.

A rather recursive approach, works similar to the built-in function bin() that is available in Python 2.6+, but with some additional convenience features.

Parameters

value	value to represent as a binary string
prefix	prefix to prepend the string (usually "0b", just like bin() would add and similar to the "0x" for hex)
min_bit_width	a minimum bit width to enforce (filled with zeros)

Returns

a string with a binary representation of the input value.

6.58.2.4 buildDescription()

```
ttk_checks._checkutils.buildDescription (
              descr_format,
              entries )
```

Build a result description from a supplied format string and entries.

If the format is invalid, all available entries will be returned.

Parameters

descr_format	a format string (preferably referencing entries of, well, entries. If an non-existing entry is referenced, an empty string will be added in its place.
entries	a dictionary of entries to use in `format_str`

Examples:

```
>> d =  {"foo": "FOO", "baz": 12345}
>> buildDescription("Lorem: %(foo)s\nIpsum: %(bar)s%(baz)d", d)
Lorem: FOO
Ipsum: 12345
>> buildDescription("Lorem: %(foo)x\nIpsum: %(bar)s%(baz)d", d)
Invalid Format Specified: %x format: a number is required, not str
Available entries:
'baz':    12345
'foo':    FOO
```

Returns

a formatted description string

6.58.2.5 getInteger()

```
ttk_checks._checkutils.getInteger (
              value )
```

Cast a numeric value to an integer (int/long) number, but perform sanity checks before (relevant for float input values).

If `value` is already long (or int), it will simply be returned as-is.

Note

Raises an ValueConversionError if `value` does not appear to be representing an integer or if it cannot fit into a double-precision float without loss of precision.

Parameters

value	numeric value to interpret as integer number

Returns

an int (if value fits into int range), or a long value.

6.58.2.6 getTracebackString()

```
ttk_checks._checkutils.getTracebackString (
              limit = None )
```

Get current traceback info as a string (or an empty string if no exception is currently active).

6.58.2.7 resolveValue()

```
ttk_checks._checkutils.resolveValue (
              value )
```

Get a value of the input parameter if it supports a "get" method, otherwise return it as-is.

Raise a ttk_base.errors.TTkException if the get() call results in an ttk_base.errors.TTkErrorString.

Errors raised by the get() method will have to be handled by the caller, this includes "TypeError: get expected at least 1 arguments, got 0" and similar, if this get() is not one of "our" standard get() implementations (e.g. dict.get)

6.58.2.8 updateMetaEntries()

```
ttk_checks._checkutils.updateMetaEntries (
              meta,
             * source_vars_info )
```

All-in-one update of meta data.

- update all meta entries of the supplied variables (see updateMetaInfo()), then
- fill any missing entries that can be derived from other source variables. This means shared meta data like units or lookup values – checks/comparisons make not much sense if source values use different definitions.

Parameters

meta	dictionary to update, potentially with already existing meta entries
∗*source_vars_info*	all following entries must be triples of `[source_var, meta_key_prefix, current_value]`. • `current_value` may be None if no state lookup is desired. • If `current_value` is None, a VarBase or meta()-wrapped value will use its last value to provide a formatted value

6.58.2.9 updateMetaFormattedValue()

```
ttk_checks._checkutils.updateMetaFormattedValue (
              meta,
              source_var,
              value,
              key = "fmt",
              fallback_value = None )
```

Update (or create) a meta entry with a formatted value representation for the supplied value, using the formatting defined in `source_var`.

An already existing non-default formatted value representation will not be overwritten.

Parameters

meta	dictionary to update, potentially with already existing meta entries
source_var	source variable (or meta-enhanced value) with value formatting hints to update default representations
value	actual value to format (might have its own value formatting hints, though)
key	key name for state info (typically `<name>_fmt`)
fallback_value	optional fallback value (typically a a base type like int or float) to use for value representation if no suitable formatting can be found for `value`

Note

fallback_value vs. value If a "partial" variable has been supplied, e.g. a basic object with just a get() method but not much else (no getFormattedValue() and not even a custom **str** method), the fallback representation would be something like "<foo.NotSoGood object at 0x000000000E8588D0>".

By providing an optional resolved fallback_value (like used in source_vars_info), we can deal with this special case.

6.58.2.10 updateMetaInfo()

```
ttk_checks._checkutils.updateMetaInfo (
              meta,
              value,
              key_prefix = "",
              meta_attrs = None,
              setdefaults = False,
              fallback_value = None )
```

Extract meta info from `value` (if info is available) and update the supplied meta dictionary with those values.

Keys will only be present/updated if the value has matching info.

Note

Default meta_attrs:

- `unit` - unit as a string
- `descr` - description as a string
- `alias` - alias name as a string
- `state` - string description of current state (if available from lookup)
- `fmt` - value formatted using an available format string (e.g. "0x%02X")

Info: Key naming

Entries will be added with a leading `key_prefix`, so with a `key_prefix = "source_var_"` and the default `meta_attrs`, the following keys will be added/updated (as long as the `value` provides the respective members):

- "source_var_unit"
- "source_var_descr"
- "source_var_alias"
- "source_var_state"
- "source_var_fmt" plus
- "formatted_source_var_alias" (see next note)

Note

alias For `alias` an additional entry `formatted_<key_prefix>_alias` will be added that will contain the alias name formatted with the I10n-format `<alias info format>`.

value and fallback_value If `value` is not an instance that provides either

- a suitable string representation of its "value" (e.g. basic types like int or float) or
- a method getFormattedValue (like all VarBase-derived variables) the formatted representation entry ("fmt") will most likely not contain the expected value but default to something like <foo.NotSoGood object at 0x000000000E8588D0>.

If a `meta` entry matching the `key_prefix` (w/o any trailing underscores) is available, it will be used as fallback representation for `value` (as that meta entry contains the value as used in a check function)

Otherwise, if a separate value (e.g. a float value from a .get() operation) is available, it can be supplied as `fallback_value` and will be used as final fallback representation for `value`.

Parameters

meta	dictionary to update, potentially with already existing meta entries
value	value (or a VarBase derived instance) that may have additional meta info attached (unit, description, ...)
key_prefix	prefix to prepend to dictionary keys; a missing trailing underscore will be added as separator.
meta_attrs	list of custom meta attributes to extract. Defaults to `("unit", "descr", "alias", "state", "fmt")`
setdefaults	if True, already existing meta keys will not get overwritten (i.e. only default values will be set)
fallback_value	optional current/resolved value of the supplied `value` parameter. Defaults to `value`. See note.

6.58.2.11 updateMetaStateDescr()

```
ttk_checks._checkutils.updateMetaStateDescr (
              meta,
              source_var,
              value,
              key )
```

Update (or create) a meta entry with a state description for the supplied value, using the lookup-table from `source_var`.

An already existing state description will not be overwritten.

Parameters

meta	dictionary to update, potentially with already existing meta entries
source_var	source variable (or meta-enhanced value) with lookup table
value	value to lookup
key	key name for state info (typically `<name>_state`)

6.58.3 Variable Documentation

6.58.3.1 _

```
ttk_checks._checkutils._ = basic_tests_loc.getInstance().getText  [protected]
```

6.58.3.2 FLOAT_CONVERSION_LIMIT

```
float ttk_checks._checkutils.FLOAT_CONVERSION_LIMIT = 0.0001
```

maximum divergence to still accept a float value to be "intended" as int/long

6.58.3.3 FLOAT_CONVERSION_MAX_VALUE

```
int ttk_checks._checkutils.FLOAT_CONVERSION_MAX_VALUE = 2**52
```

IEEE-754 double-precision => 52 bits explicitly stored.

6.59 ttk_checks._verdicts Namespace Reference

Defined test status verdicts.

Functions

- getResultingVerdict (verdicts)

  Get the resulting verdict for a sequence of verdicts.
- getInvertedVerdict (verdict)

  Invert verdict.

Variables

- str PASSED = "PASSED"

  Verdict: Test step successfully completed.
- str FAILED = "FAILED"

  Verdict: Test step failed.
- str ERROR = "ERROR"

  Verdict: Error occurred during test execution (in environment, tool, script)
- str RETEST_PASSED = "RETEST PASSED"

  Verdict: A manual re-test of a previously failed test step passed / was successful.
- list PASS_VERDICTS = [RETEST_PASSED, PASSED]

  All verdicts that evaluate to "Passed".
- list FAIL_VERDICTS = [FAILED, ERROR]

  All verdicts that evaluate to "Failed".

6.59.1 Detailed Description

Defined test status verdicts.

See interface wrapper in ttk_checks.verdicts.

See also https://wiki.isyst.de/wiki/Testbewertungen

6.59.2 Function Documentation

6.59.2.1 getInvertedVerdict()

```
ttk_checks._verdicts.getInvertedVerdict (
              verdict )
```

Invert verdict.

Turns PASSED to FAILED and vice versa.

Note

ERROR and any info verdicts will remain untouched.

Parameters

verdict	verdict to invert

Returns

a verdict with inverted meaning.

6.59.2.2 getResultingVerdict()

```
ttk_checks._verdicts.getResultingVerdict (
              verdicts )
```

Get the resulting verdict for a sequence of verdicts.

Priority of actual pass/fail verdicts is determined by PASS_VERDICTS and FAIL_VERDICTS.

Parameters

verdicts	verdict sequence (list, tuple, dict (keys))

Returns

FAILED if at least one result is failed, ERROR if errors occurred without an explicit failed result, PASSED (or RETEST PASSED, if applicable) if passed results exist without any failed results, otherwise an empty verdict.

6.59.3 Variable Documentation

6.59.3.1 ERROR

```
str ttk_checks._verdicts.ERROR = "ERROR"
```

Verdict: Error occurred during test execution (in environment, tool, script)

6.59.3.2 FAIL_VERDICTS

```
list ttk_checks._verdicts.FAIL_VERDICTS = [FAILED, ERROR]
```

All verdicts that evaluate to "Failed".

6.59.3.3 FAILED

```
str ttk_checks._verdicts.FAILED = "FAILED"
```

Verdict: Test step failed.

6.59.3.4 PASS_VERDICTS

```
list ttk_checks._verdicts.PASS_VERDICTS = [RETEST_PASSED, PASSED]
```

All verdicts that evaluate to "Passed".

6.59.3.5 PASSED

```
str ttk_checks._verdicts.PASSED = "PASSED"
```

Verdict: Test step successfully completed.

6.59.3.6 RETEST_PASSED

```
str ttk_checks._verdicts.RETEST_PASSED = "RETEST PASSED"
```

Verdict: A manual re-test of a previously failed test step passed / was successful.

6.60 ttk_checks.adv_tests Namespace Reference

Interface wrapper for advanced test/evaluation functions in ttk_checks._adv_tests.

Functions

- testVars (set_var_list=None, check_var_list=None, min_wait_time_ms=100, descr='')

  Check the values of multiple variables/signals as a result of setting multiple other variables/signals.

6.60.1 Detailed Description

Interface wrapper for advanced test/evaluation functions in ttk_checks._adv_tests.

These functions are more complex than those in ttk_checks.basic_tests

This serves as "interface" to the precompiled module in delivery to enable code-completion in PyDev.

6.60.2 Function Documentation

6.60.2.1 testVars()

```
ttk_checks.adv_tests.testVars (
              set_var_list = None,
              check_var_list = None,
              min_wait_time_ms = 100,
              descr = '' )
```

Check the values of multiple variables/signals as a result of setting multiple other variables/signals.

Between setting and checking, a configurable time will be waited to deal with propagation delay. This wait time defaults to the minimum practicable wait time of all involved bus signals.

Info: This function ...

- sets the value(s) to the variables in set_var_list
- triggers sending of any event-based signals included in set_var_list
  - "needs triggering" is determined by evaluating `cycle_time` members of set vars (if present)
  - triggering is performed by calling a method `kickout()` or `trigger()` (whichever is available)
- waits for value change(s) to propagate to ECU/DUT
  - this depends on both the specified `min_wait_time_ms` and any `change_delay` members of entries in set or check var lists.
- checks the value(s) for variables in check_var_list

Parameters

set_var_list	Variables to set: a list of dictionaries with variable and value
check_var_list	Variables to check: a list of dictionaries with variable, expected value and tolerance
min_wait_time_ms	minimum wait time between setting variables and performing checks
descr	user specific description shown at the start of the result text

Example: Wait time before check (propagation delay)

```
wait_time = max(
    min_wait_time_ms,
    max(<change delay of signals to set>) +
    max(<change delay of signals to check>)
)
# Note: a bus signal's wait time defaults to 2x cycle time
```

Example: set_var_list structure

```
set_var_list = [
    { "var":   <variable to set (e.g. a CalVar, HilVar or anything with a set() method),
      "value": <value to set>,
      "descr": <value description, optional>,
    },
```

Example: set_var_list and function calls

```
# For special cases, the set var list may also contain function calls
# (which might be useful to insert additional delays or similar):
set_var_list = [
    { "var":    cal.reset_var, "value": 1, descr="enable reset" },
    { "func":   time.sleep,
      "args":   [1.0], # positional arguments
      "kwargs": {},    # keyword arguments
      "descr":  "Wait for one second before clearing reset again",
    },
    { "var":   cal.reset_var, "value": 0 },
    # or a simple call like this:
    { "func":  someCustomFunction,
      "descr": "a custom function that does no need any parameters",
    },
    ...
]
```

Example: check_var_list structure

```
# A basic entry:
check_var_list = [
    { "var":     <variable to check (CalVar, HilVar, ...)>,
      "operator": <check operator, defaults to "==">,
      "value":   <expected value(s)>
      "abs_tol": <absolute tolerance (optional)>,
      "rel_tol": <relative tolerance (optional)>,
    }
]
# abs_tol/rel_tol are aliases for checkTolerances's abs_pos/rel_pos
# (when separate _pos/_neg tolerances are not required).
# If you need non-symmetrical tolerances (e.g. -1/+5), you can
# specify tolerances separately:
check_var_list = [
    { "var":     <variable to check (CalVar, HilVar, ...)>,
      "operator": <check operator, defaults to "==">,
      "value":    <expected value> or a list of expected values, of which one has to pass
      "rel_pos":  <relative positive tolerance (optional)>,
      "abs_pos":  <absolute positive tolerance (optional)>,
      "rel_neg":  <relative negative tolerance (optional)>,
      "abs_neg":  <absolute negative tolerance (optional)>,
    },
    ...
]
# See basic_checks.checkTolerance for details on tolerance parameters
# "==" is default when operator entry is omitted
```

Example: extended check_var_list structure

```
check_var_list = [
    { "var":     <variable to check (CalVar, HilVar, ...)>,
      "operator": "==", # "==" or "in"
      "value":   <expected value> or a list of multiple expected values of which one has to match
      "abs_pos": <absolute tolerance (optional)>,
      "rel_pos": <relative tolerance (optional)>,
    },
    { "var":     <variable to check (CalVar, HilVar, ...)>,
      "operator": "!=", # "!=" or "not in"
```

```
      "value":   <expected value> or a list of multiple expected values, none of which must match
      "abs_pos": <absolute tolerance (optional)>,
      "rel_pos": <relative tolerance (optional)>,
    },
    { "var":     <variable to check (CalVar, HilVar, ...)>,
      "operator": ">", # "<", "<=", ">=" or ">"
      "value":   <expected value>
      # no tolerances are used for "greater/less than" comparisons
    },
    ...
]
```

Example: Using a list of multiple expected values (usually for status values):

```
# Operator "==" or "in": one value has to match
check_var_list = [
    { "var":       cal.e2e_status,
      "operator": "in"
      "value":    [E2E_P04STATUS_OK, E2E_P04STATUS_OKSOMELOST],
    }
]

# Operator "!=" or "not in": no value may match:
check_var_list = [
    { "var":      cal.e2e_status,
      "operator": "not in",
      "value":    [E2E_P04STATUS_NONEWDATA, E2E_P04STATUS_ERROR,
                   E2E_P04STATUS_REPEATED, E2E_P04STATUS_WRONGSEQUENCE],
    }
]
```

Info: "var" and "value" in check- and set_var_list

Both variable to check and the expected "value" can actually be any VarBase derived instance, meta()-wrapped MetaValue instances or plain scalar numeric values.

This means that it is possible to compare the values of two ecu/cal variables:
`{"var": cal.some_variable, "value": cal.some_other_variable}`
or between hil and ecu
`{"var": cal.foobar, "value": hil.barbaz}`
`{"var": bus.barsignal, "value": cal.bazquux}`
or even (admittedly, not too useful)
`{"var": 3, "value": 3.2, "abs_tol": .4}`

Info: Units and plain values

If none of the involved variables/values has a "unit" specified and you want to show units in the test report, you could wrap values in a MetaValue:
`{"var": cal.shrdlu, "value": meta(3.2, "mV"), "abs_tol": .4}`
See ttk_base.values_base.meta()

Status values and tolerances:

If all tolerances of a check_var_list entry remain un-set (i.e. None), the values will be treated as discrete integer states during comparison. If a value does not appear to be an integer value, the test will fail with verdict ERROR (and the actual check using the un-cast value(s) will most likely fail as well).
See ttk_checks.checkutils.getInteger()

Returns

a single test result entry [<description>, <VERDICT>] (with a single resulting verdict for all checked variables).

6.61 ttk_checks.basic_checks Namespace Reference

Interface wrapper for basic check functions ("low-level") in ttk_checks._basic_checks.

Functions

- checkBit (value, offset, status=1)

  Check if the bit given by the value and offset is set or cleared depending on the status.
- checkBitMask (value, mask, operator="AND", bit_length=CHECKBITMASK_DEFAULT_BIT_LENGTH)

  Check if the result of the operation given in 'operator' is 'greater 0' or 'equal 0'.
- checkBitPattern (value, pattern)

  Check that a numeric value matches a certain pattern.
- checkRange (value, min_value, max_value)

  Checks if value is in defined range (min_value <= value <= max_value).
- checkTolerance (current_value, rated_value, rel_pos=0, abs_pos=0, rel_neg=None, abs_neg=None)

  Check if the current value is within a specified range of the rated value.
- contains (defined_values, current_value, mode="all")

  Checks if current_value is an element or sub-set of defined_values.
- compare (left_value, operator, right_value, abs_tol=0)

  Verify the conditional relationship between two operands based on the given operator.

6.61.1 Detailed Description

Interface wrapper for basic check functions ("low-level") in ttk_checks._basic_checks.

This serves as "interface" to the precompiled module in delivery to enable code-completion in PyDev.

Example:

```
status, info = checkBitPattern(0x85, '0bxxxx0101')
```

Conventions:

Each function will return a tuple with two values:

- `status` - result check status, True/False.
  - Typically, True means that the check was successfully passed.
  - See individual function descriptions for details
- `info` - a dictionary with intermediate values calculated for the check, e.g. effective min/max boundaries calculated from relative and absolute tolerances.
  - Key names of intermediate values follow the variable naming schema from our Python Style Guide
  - See https://wiki.isyst.de/wiki/Python_Styleguide).

Intermediate values could be used to augment test result descriptions or for additional processing.
See individual function descriptions for details.

Check functions will only raise exceptions derived from the base class BasicCheckException (see module ttk_checks.checkceptions) for input value check and processing errors.
Unless something goes completely wrong, no other exceptions should get raised.

If a check function raises an (unexpected) exception anyway, please file a bug report or contact support at support@isyst.de

6.61.2 Function Documentation

6.61.2.1 checkBit()

```
ttk_checks.basic_checks.checkBit (
              value,
              offset,
              status = 1 )
```

Check if the bit given by the value and offset is set or cleared depending on the status.

Parameters

value	Numeric value where the bit has to be checked
offset	Offset of the bit inside of value (based on 0)
status	The expected value of the bit (default 1)

Returns

a tuple (status, dictionary with intermediate values)
`status` - True if the selected bit has the desired status, otherwise False.
`intermediate_values` - a dictionary with
''called_function': `name of this function in string format <BR>` 'value'↩
: `input value <BR>` 'offset': `input bit offset <BR>` 'bit↩
_status_detected': `actual bit status <BR>` 'bit_status_expected'` : expected bit status (from input)

6.61.2.2 checkBitMask()

```
ttk_checks.basic_checks.checkBitMask (
              value,
              mask,
              operator = "AND",
              bit_length = CHECKBITMASK_DEFAULT_BIT_LENGTH )
```

Check if the result of the operation given in 'operator' is 'greater 0' or 'equal 0'.

The result is calculated with a dyadic operation. The two operands are 'value' and 'mask'.

Parameters

value	value to be checked (int/long)
mask	mask to be used for operation in operator (long or string)
operator	operator for check operation, one of ("AND", "OR", "XOR", "NAND", "NOR", "NXOR")
bit_length	bit length for value and mask

Returns

a tuple (status, dictionary with intermediate values)
`status` - True if result of operation is 'greater 0', False if it is 'equal 0'
`intermediate_values` - a dictionary with
''result_string': `result of operation in binary string format <BR>` 'result↩
_value': `result of operation as long value <BR>` 'source↩
_value': `input value <BR>` 'source_mask': `input mask <BR>`
'source_operator': `input operator <BR>` 'source_bit_length': `input bit length <BR>` 'result_verdict': `True if result value > 0 <BR>` 'called_function'` : name of this check function (as string)

6.61.2.3 checkBitPattern()

```
ttk_checks.basic_checks.checkBitPattern (
              value,
              pattern )
```

Check that a numeric value matches a certain pattern.

Parameters

value	value to check. The value must match the full pattern in order to pass. Values may be numeric (int/long), string representations of binary/hexadecimal values ("0xABCD", "0b1010" etc.) or a list of bytes.
pattern	a string describing the desired bit pattern. Note that a pattern must always describe the full value (i.e. all bits from start to end). • A pattern starting with "0x" will be interpreted as hexadecimal • "0b" at the beginning denotes a binary pattern • "X" marks "don't care" sections (anywhere in pattern)

See examples for further details.

Examples:

```
"0xABC123"   - a hexadecimal 20-bit pattern
"123ABC"     - without a prefix, pattern defaults to hexadecimal
"0b0101101"  - a binary 6-bit pattern
"0xAABXX2"   - X designates "don't care" (4-bit-nibbles in hex
               patterns)
"0b1XX001"   - "don't care" on a bit level in binary patterns
"0xAA[8]BB"  - longer "don't care"-sections can be written as
               a repetition number in square brackets. This example
               pattern is 48 bits long with 32 don't care bits in
```

```
                the middle.
"0b10[8]00"   - as above but in a binary pattern: 12 bits with 8
                don't care bits in the middle.
"0xAA BB CC" - whitespace will be ignored for pattern check, but
                plain spaces will be retained in formatted
                intermediate values
"0xab Ab aB" - patterns are not case sensitive
```

Returns

a tuple (status, dictionary with intermediate values)
`status` - True if pattern matches, otherwise False
`intermediate_values` - a dictionary with
''condensed_value': `value w/o whitespace formatting <BR>` 'condensed↩
_pattern': `pattern w/o whitespace formatting <BR>` 'formatted↩
_value': `value with whitespace formatted like pattern <BR>` 'formatted↩
_pattern': `pattern with whitespace formatting <BR>` 'mismatch↩
_position': `position where first mismatch occurred (from condensed_pattern), -1 if pattern matches. <BR>` 'called_function`` : name of this check function (as string)

6.61.2.4 checkRange()

```
ttk_checks.basic_checks.checkRange (
              value,
              min_value,
              max_value )
```

Checks if value is in defined range (min_value <= value <= max_value).

Parameters

value	value to be checked (int/long/float)
min_value	lower limit
max_value	upper limit

Examples:

```
checkRange([0x2,-5.0, 7L])      -> True
checkRange([0x2,7L, 5.0])       -> True
checkRange([5, 7, 9])           -> False
checkRange([2,"a", 7])          -> raises an exception
```

Returns

a tuple (status, dictionary with intermediate values)
`status` - True or False if value is in defined range
`intermediate_values` - a dictionary with
''value': `input value that has been checked <BR>` 'min_value': `lower Limit <BR>` 'max_value': `upper limit <BR>` 'called↩
_function`` : name of this function in string format

6.61.2.5 checkTolerance()

```
ttk_checks.basic_checks.checkTolerance (
              current_value,
              rated_value,
              rel_pos = 0,
              abs_pos = 0,
              rel_neg = None,
              abs_neg = None )
```

Check if the current value is within a specified range of the rated value.

The range is defined by an absolute and a relative tolerance. Both abs and rel tolerances will be added.

Note

If a *symmetric* tolerance range is desired, only `rel_pos` and/or `abs_pos` need to be set.

Info: Relative Tolerances

Relative tolerance values are specified as factors in relation to the supplied `rated_value`.
A range of `0.0 .. 1.0` maps to `0 .. 100 %`

Parameters

current_value	value to be checked
rated_value	expected value
rel_pos	(optional) relative positive tolerance as factor, typically in range 0.0..1.0
abs_pos	(optional) absolute positive tolerance
rel_neg	(optional) relative negative tolerance as factor, typically in range 0.0..-1.0
abs_neg	(optional) absolute negative tolerance

Returns

a tuple (status, dictionary with intermediate values)
`status` - True if value is within tolerance, otherwise False
`intermediate_values` - a dictionary with
''current_value': value to be checked
 'rated_value': expected value to be checked against
 'rel_pos': actual relative positive tolerance
 'abs_pos': actual absolute positive tolerance
 'rel_neg': actual relative negative tolerance
 'abs_neg': : actual absolute negative tolerance
 'lower_boundary': calculated lower boundary (from tolerances)
 'upper_boundary': calculated upper boundary (from tolerances)
 'effective_abs_pos': effective positive tolerance, absolute (based on both abs and rel pos)
 'effective_abs_neg': effective negative tolerance, absolute (based on both abs and rel neg)
 'called_function'` : name of this check function (as string)

6.61.2.6 compare()

```
ttk_checks.basic_checks.compare (
```

```
    left_value,
    operator,
    right_value,
    abs_tol = 0 )
```

Verify the conditional relationship between two operands based on the given operator.

Parameters

left_value	left operand of the relational expression
operator	the relational operator given as a string (one of "=", "==", "<", ">", "<=", ">=", "!=", "<>")
right_value	right operand of the relational expression
abs_tol	absolute tolerance for use in equality/non-equality checks of numeric types

Note

A tolerance > 0 is mandatory for equality checks (==, !=) involving float values.
See checkTolerance() for more in-depth options like relative and non-symmetric tolerances.

Info:

When comparing lists, they must contain the same number of elements or a BasicCheckException will be raised.
List compare currently only supports == and != operators, as the behavior for other operations is unspecified.

Relational operator strings:

- "=" : equal to
- "==" : equal to
- "<" : less than
- ">" : greater than
- "<=" : less than or equal to
- ">=" : greater than or equal to
- "!=" : not equal to
- "<>" : not equal to

Returns

a tuple (status, dictionary with intermediate values)
`status` - True if expression evaluates to True, otherwise False
`intermediate_values` - a dictionary with
''left_value': `left value of expression <BR>` 'right_value': `right value of expression <BR>` 'operator': `used operator <BR>` 'abs_tol': `tolerance (absolute, for numeric checks) <BR>` 'compare↩
_mode': `kind of comparison that was performed (numeric, string, list, other) <BR>` 'called_function`` : name of this check function (as string)

6.61.2.7 contains()

```
ttk_checks.basic_checks.contains (
              defined_values,
              current_value,
              mode = "all" )
```

Checks if current_value is an element or sub-set of defined_values.

The order of elements is not important.

Note

While it is possible to compare lists of float values, it might lead to unexpected results due to rounding errors.

Parameters

defined_values	list of scalars to be searched in
current_value	function checks if this scalar (or list) is contained in 'defined_values'
mode	search mode (one of 'all', 'any', 'none', 'not', see CONTAIN_MODES)

Examples:

```
contains([1,3,5], 3, mode="any")       -> True
contains([1,3,5], 7, mode="none")      -> True
contains([1,3,5], [3, 45], mode="all") -> False
contains([1,3,5], [3, 45], mode="any") -> True
contains([1,3,5], [3, 45], mode="not") -> False
contains([1,3,5], [3, 5],  mode="all") -> True
```

Available Modes:

- "all": each value of 'current_value' should be in `defined_values`
- "any": at least one element of 'current_value' should be in `defined_values`
- "none"/"not": no element of 'current_value' should be in `defined_values`

Returns

a tuple (status, dictionary with intermediate values)
`status` - True or False if current_value is an element of defined_values, depending on mode
`intermediate_values` - a dictionary with
'in_list': List of all elements of current_value that are in defined_values.
 'not_in_list': List of all elements of current_value that are not in defined_values.
 'current_value': value(s) that have been checked
 'defined_values': values that have been checked against
 'mode` : check mode

6.62 ttk_checks.basic_tests Namespace Reference

Interface wrapper for basic test/evaluation functions that directly generate test report entries in ttk_checks._basic_tests.

Functions

- checkBitMask (value, mask, operator="AND", bit_length=CHECKBITMASK_DEFAULT_BIT_LENGTH, pass_on_true=True, descr="", format=None)

  Check result of the operation given in `operator`.
- checkBitPattern (value, pattern, invert=False, descr="", format=None)

  Check that a numeric value matches a certain pattern.
- checkBit (value, offset, status=1, descr="", format=None)

  Check if the bit given by the value and offset is set or cleared depending on the status.
- checkTolerance (current_value, rated_value, rel_pos=0, abs_pos=0, rel_neg=None, abs_neg=None, invert=False, descr="", format=None)

  Check if the current value is within a specified range of the rated value.
- checkAbsTol (current_value, rated_value, abs_tol=0, invert=False, descr="")

  Check if the current value is within a specified range of the rated value.
- checkRelTol (current_value, rated_value, rel_tol=0.0, invert=False, descr="")

  Check if the current value is within a specified range of the rated value.
- checkStatus (current_status, nominal_status, equal=True, descr="", format=None)

  Check if the current status value is identical to the nominal status value.
- checkRange (value, min_value, max_value, invert=False, descr="", format=None)

  Check if value is in defined range (`min_value <= value <= max_value`).
- compare (left_value, operator, right_value, abs_tol=0, descr="", format=None)

  Verify the conditional relationship between two operands based on the given operator.
- compareCE (current_value, operator, expected_value, abs_tol=0, descr="")

  Verify the conditional relationship between current and expected value based on the given operator.
- contains (defined_values, current_value, mode="all", descr="", format=None)

  Check if current_value is an element (or sub set) of defined_values.

6.62.1 Detailed Description

Interface wrapper for basic test/evaluation functions that directly generate test report entries in ttk_checks._basic_tests.

This serves as "interface"to the precompiled module ttk_checks._basic_tests in delivery to enable code-completion in PyDev.

Example:

```
testresult.append(
    checkBitPattern(
        cal.io_error_bits, '0bxxxx0101',
        "Both low side drivers should report errors."
    )
)
```

Conventions:

Each function will take the required parameters for the test to perform plus two additional parameters for report formatting:

- `descr` - a description string that will be placed at the start of the result description text
- `format` - a format-string that specifies/overrides the formatting of the automatically generated part of the description.

Do not override `format` unless absolutely necessary.
If you want to change the default output formatting of a test function, it would be better to look into changing the current localization (see ttk_checks.basic_tests_loc), so all tests will use consistent result formatting.
See also http://docs.python.org/library/stdtypes.html#string-formatting

If an input value has a get()-method, it will be called to get the current value. Otherwise the value will be used as-is.
This permits "CalVar" or "HilVar" instances to be directly used in test function parameters.
See ttk_base.variables_base

If an input value has associated meta data (like "Cal" or "Hil" Vars or basic values wrapped as MetaValues), it can be used in the automatically generated description (e.g. unit or alias names)
See ttk_base.variables_base, ttk_base.values_base.meta()

Each function will return exactly one result entry for a test results list, i.e. typically a list containing `["<description>", "<VERDICT>"]`
See https://wiki.isyst.de/wiki/Testresult for more info on our standard test results.

Exceptions occurring "inside" a test function have to be caught; the function will return a result entry containing the error text and verdict `ERROR`.
That means it should *not* be necessary to wrap the test function into yet another `try .. except` block.

If a test function raises an (unexpected) exception anyway, please file a bug report or contact support at support@isyst.de

6.62.2 Function Documentation

6.62.2.1 checkAbsTol()

```
ttk_checks.basic_tests.checkAbsTol (
                current_value,
                rated_value,
                abs_tol = 0,
                invert = False,
                descr = "" )
```

Check if the current value is within a specified range of the rated value.

The valid range is defined by an absolute tolerance as a symmetric range around the `rated_value`.

Parameters

current_value	value to be checked
rated_value	expected value
abs_tol	absolute tolerance value, valid range will be `rated_value` +/- `abs_tol`
invert	True: inverted, check current value != rated value (+/- tolerance) False: default, check current value == rated value (+/- tolerance)
descr	user specific description shown before the single test result

Returns

a single [<description>, <verdict>] test result entry.

6.62.2.2 checkBit()

```
ttk_checks.basic_tests.checkBit (
              value,
              offset,
              status = 1,
              descr = "",
              format = None )
```

Check if the bit given by the value and offset is set or cleared depending on the status.

Parameters

value	Numeric value where the bit has to be checked
offset	Offset of the bit inside of value (based on 0)
status	The expected value of the bit (default 1)
descr	user specific description printed on the start of the single test result
format	format string to control/override appearance of the automatically created result text. A format override should only rarely be used.

Usage:

```
testresult.append(
    checkBit(
        value  = status_bits,  # 0b00001100
        offset = 3,
        status = 1,
        descr = "Bit 3 should be set"
     )
)
testresult.append(
    checkBit(status_bits, 1, 0, descr="Bit 1 should be clear")
)
```

Sample Result Output:

```
 PASSED  Bit 3 should be set
         Offset: 3
         Value:  0b00001100 (status_bits)
         Expected state: 1
         Detected state: 1
 ----------------------------------------------------------------
 PASSED  Bit 1 should be clear
         Offset: 1
         Value:  0b00001100 (status_bits)
         Expected state: 0
         Detected state: 0
```

Returns

a single [<description>, <verdict>] test result entry.

6.62.2.3 checkBitMask()

```
ttk_checks.basic_tests.checkBitMask (
              value,
              mask,
              operator = "AND",
              bit_length = CHECKBITMASK_DEFAULT_BIT_LENGTH,
              pass_on_true = True,
              descr = "",
              format = None )
```

Check result of the operation given in `operator`.

Depending on parameter `pass_on_true`, the test will be PASSED if the result is True (i.e. 'greater 0') or False (i.e. 'equal 0').

Parameters

value	value to be checked
mask	mask to be used for operation in operator
operator	operator for check operation, one of ("AND", "OR", "XOR", "NAND", "NOR", "NXOR") see *basic_checks.CHECKBITMASK*∗-constants.
bit_length	bit length for value and mask
pass_on_true	if True, test will be PASSED if result is 'greater 0', otherwise it will be PASSED if the result is 'equal 0'
descr	user specific description shown at the start of the result text
format	format string to control/override appearance of the automatically created result text. A format override should only rarely be used.

Usage:

```
testresult.append(
    checkBitMask(
        value      = status_bits, # 0b00001100
        mask       = 0x0F,
        operator   = "AND",
        bit_length = 8,
        descr      = "Some bits of the lower nibble should be set"
    )
)
```

Sample Result Output:

```
 PASSED  Some bits of the lower nibble should be set
         Value:    12 (status_bits_alias)
         Operator: AND
         Mask:     15
         Result:   0b00001100
```

Returns

a single [<description>, <verdict>] test result entry.

6.62.2.4 checkBitPattern()

```
ttk_checks.basic_tests.checkBitPattern (
              value,
              pattern,
              invert = False,
              descr = "",
              format = None )
```

Check that a numeric value matches a certain pattern.

Parameters

value	value to check. The value must match the full pattern in order to pass. Values may be numeric (int/long), string representations of binary/hexadecimal values ("0xABCD", "0b1010" etc.) or a list of bytes.
pattern	a string describing the desired bit pattern. • a pattern must always describe the full value (i.e. all bits from start to end). • a pattern starting with "0x" will be interpreted as hexadecimal • "0b" at the beginning denotes a binary pattern • "X" marks "don't care" sections (may appear anywhere in pattern)
invert	True: inverted check: pattern must not match False: default check: pattern has to match
descr	user specific description shown at the start of the result text
format	format string to control/override appearance of the automatically created result text. A format override should only rarely be used.

Pattern String Examples:

- `0xABC123` - a hexadecimal 24-bit pattern
- `123ABC` - without a prefix, pattern defaults to hexadecimal
- `0b0101101` - a binary 6-bit pattern
- `0xAABXX2` - X designates "don't care" (4-bit-nibbles in hex patterns)
- `0b1XX001` - "don't care" on a bit level in binary patterns
- `0xAA[8]BB` - longer "don't care"-sections can be written as a repetition number in square brackets. This example pattern is 48 bits long with 32 don't care bits in the middle.
- `0b10[8]00` - as above but in a binary pattern: 12 bits with 8 don't care bits in the middle.
- `0xAA BB CC` - whitespace will be ignored for pattern check, but plain spaces will be retained in formatted intermediate values
- `0xab Ab aB` - patterns are not case sensitive

Usage:

```
testresult.append(
    checkBitPattern(
        value   = io_error_bits,
        pattern = "0bxxxx0101",
        descr   = "Both low side drivers should report errors."
    )
)
testresult.append(
    checkBitPattern(
        io_error_bits, "0b0000xxxx",
        "Both high side drivers should not report any errors."
    )
)
```

Sample Result Output:

```
 PASSED  Both low side drivers should report errors.
         Pattern: XXXX0101
         Value:   10000101 (io_error_bits)
 ------------------------------------------------------------
 FAILED  Both high side drivers should not report any errors.
         Pattern: 0000XXXX
         Value:   10000101 (io_error_bits)
```

Returns

a single [<description>, <verdict>] test result entry.

6.62.2.5 checkRange()

```
ttk_checks.basic_tests.checkRange (
              value,
              min_value,
              max_value,
              invert = False,
              descr = "",
              format = None )
```

Check if value is in defined range (`min_value <= value <= max_value`).

Parameters

value	value to be checked
min_value	lower Limit
max_value	upper Limit
invert	True: inverted, check that value is outside of range False: default, check that value is inside of range
descr	user specific description printed on the start of the single test result
format	format string to control/override appearance of the automatically created result text. A format override should only rarely be used.

Usage:

```
testresult.append(
    checkRange(
        analog_value,
        min_value = 12,
        max_value = 54,
```

```
            descr="Check that value is in range"
        )
    )
```

Sample Result Output:

```
 PASSED  Check that value is in range
         Value:     42.0 mA (analog_value)
         Min value: 12 mA
         Max value: 54 mA
```

Returns

a single [<description>, <verdict>] test result entry.

6.62.2.6 checkRelTol()

```
ttk_checks.basic_tests.checkRelTol (
              current_value,
              rated_value,
              rel_tol = 0.0,
              invert = False,
              descr = "" )
```

Check if the current value is within a specified range of the rated value.

The valid range is defined by relative tolerance factor (as a symmetric range around the `rated_value`).

Info: Relative Tolerances

Relative tolerance values are specified as factors in relation to the supplied `rated_value`.
A range of `0.0 .. 1.0` maps to `0 .. 100 %`

Note

`checkRelTol` will treat `rel_tol` values >= 1.0 as unintended and mark the returned result as ERROR. Use the main test function `checkTolerance` if checks with tolerance values >= 100% should ever be needed.

Parameters

current_value	value to be checked
rated_value	expected value
rel_tol	relative tolerance as factor, in range 0.0..1.0
invert	True: inverted, check current value != rated value (+/- tolerance) False: default: check current value == rated value (+/- tolerance)
descr	user specific description shown before the single test result

Returns

a single [<description>, <verdict>] test result entry.

6.62.2.7 checkStatus()

```
ttk_checks.basic_tests.checkStatus (
              current_status,
              nominal_status,
              equal = True,
              descr = "",
              format = None )
```

Check if the current status value is identical to the nominal status value.

Both statuses will be interpreted as int/long values for the comparison.

Note

If one or both values cannot be represented as integer numbers, the test will fail with verdicts.ERROR.

Parameters

current_status	status to be checked
nominal_status	expected status to be checked against
equal	True: check current status == nominal status, False: check current status != nominal status
descr	user specific description printed on the start of the single test result
format	format string to control the the print out of the single test result

Usage:

```
testresult.append(
    checkStatus(
        status_var, 1,
        descr="Check that status is 'on'"
     )
)
testresult.append(
    checkStatus(
        status_var, 3, equal=False,
        descr="Check that status is not 'disabled'"
     )
)
```

Sample Result Output:

```
 PASSED  Check that status is 'on'
         Current status:  1 [on] (some_status_var)
         Expected status: 1 [on]
 ------------------------------------------------------------
 PASSED  Check that status is not 'disabled'
         Current status:  1 [on] (some_status_var)
         Expected status: != 3 [disabled]
```

Returns

a single [<description>, <verdict>] test result entry.

6.62.2.8 checkTolerance()

```
ttk_checks.basic_tests.checkTolerance (
              current_value,
              rated_value,
              rel_pos = 0,
              abs_pos = 0,
              rel_neg = None,
              abs_neg = None,
              invert = False,
              descr = "",
              format = None )
```

Check if the current value is within a specified range of the rated value.

The range is defined by an absolute and a relative tolerance. Both tolerance types will be added to get the effective tolerance.

Note

If a *symmetric* tolerance range is desired, only `rel_pos` and/or `abs_pos` need to be set.

Info: Relative Tolerances

Relative tolerance values are specified as factors in relation to the supplied `rated_value`.
A range of `0.0 .. 1.0` maps to `0 .. 100 %`

Using both relative and absolute tolerances:

Internally, both tolerances are added to determine the effective boundary values:

```
lower_boundary = abs_neg + rel_neg * abs(rated_value) + rated_value
upper_boundary = abs_pos + rel_pos * abs(rated_value) + rated_value
```

`current_value` will pass the check if its value lies between the calculated boundary values:

```
lower_boundary <= current_value <= upper_boundary
```

Parameters

current_value	value to be checked
rated_value	expected value
rel_pos	(optional) relative positive tolerance as factor, typically in range 0.0..1.0
abs_pos	(optional) absolute positive tolerance
rel_neg	(optional) relative negative tolerance as factor, typically in range 0.0..-1.0
abs_neg	(optional) absolute negative tolerance
invert	True: inverted, check current value != rated value (+/- tolerance) False: default: check current value == rated value (+/- tolerance)
descr	user specific description printed on the start of the single test result
format	format string to control/override appearance of the automatically created result text. A format override should only rarely be used.

Usage:

```
testresult.append(
    checkTolerance(
        u_supply, 12.0, abs_pos = 0.8,
        descr="Check nominal supply voltage level."
     )
)
testresult.append(
    checkTolerance(
        u_supply, 12.0, abs_pos = 0.2, abs_neg = -0.8,
        descr="Check with asymmetric abs tolerance values"
    )
)
testresult.append(
    checkTolerance(
        u_supply, 12.0, rel_pos = 0.01,  rel_neg = -0.07,
        descr="Check with asymmetric rel tolerance values"
    )
)
```

Sample Result Output:

```
 PASSED  Check nominal supply voltage level.
         Expected value: 12.0 V
         Current value:  11.3 V (Supply Voltage)
         Absolute Tolerance:  +/- 0.8 V
 ---------------------------------------------------------------
 PASSED  Check with asymmetric abs tolerance values
         Expected value: 12.0 V
         Current value:  11.3 V (Supply Voltage)
         Absolute Tolerance:  +0.2 V / -0.8 V
 ---------------------------------------------------------------
 PASSED  Check with asymmetric rel tolerance values
         Expected value: 12.0 V
         Current value:  11.3 V (Supply Voltage)
         Relative Tolerance:  +0.01 / -0.07
```

Returns

a single [<description>, <verdict>] test result entry.

6.62.2.9 compare()

```
ttk_checks.basic_tests.compare (
              left_value,
              operator,
              right_value,
              abs_tol = 0,
              descr = "",
              format = None )
```

Verify the conditional relationship between two operands based on the given operator.

Parameters

left_value	left operand of the relational expression
operator	relational operator given as a string, see below
right_value	right operand of the relational expression
abs_tol	absolute tolerance for equality/non-equality checks of numeric types
descr	user specific description printed on the start of the single test result

Parameters

format	format string to control/override appearance of the automatically created result text. If a tuple of strings is supplied, it should contain two (or three) entries: (<long_format>, <compact_format>) or (<long_format>, <compact_format>, <alias_length_boundary>) where alias_length_boundary is the average alias name length at which to switch to "long" format. A format override should only rarely be used.

Info: Tolerance

A a tolerance > 0 is mandatory for equality / non-equality checks involving float values.
See checkTolerance() for more in-depth options like relative and non-symmetric tolerances.

Info: Long/Compact Description Formats

If input values have no overly long aliases, a more compact output format will be used.

Compact format samples:

- 1.2 mA (short alias 1) $<$ 2.5 mA (short alias 2)
- 1.2 mA (short alias 1) == 2.5 mA (short alias 2) +/- 0.4 mA

Long format samples:

- 1.2 mA $<$ 2.5 mA
  Left Value: A rather unreasonably long alias name
  Right Value: Another quite verbose alias name
- 1.2 mA == 2.5 mA +/- 0.4 mA
  Left Value: A rather unreasonably long alias name
  Right Value: Another quite verbose alias name

Default `alias_length_boundary` for switching between formats is 32 characters.
This value can be configured in `hil_project.ini`, section `[TestToolkit]` with key/option `alias_length_boundary`.

Relational operator strings:

- "==" : equal to
- "<" : less than
- ">" : greater than
- "<=" : less than or equal to
- ">=" : greater than or equal to
- "!=" : not equal to

Usage:

```
testresult.append(
    compare(
        left_value  = analog_value,
        operator    = "==",
        right_value = 42,
        abs_tol     = 0.01,
        descr="Comparing float values for equality requires a defined tolerance"
     )
)
testresult.append(
    compare(
        status_var, "==", 1,
        descr = "Comparing integer values"
     )
)
testresult.append(compare(status_var, "<=",         3, descr = "Comparing integer values"))
testresult.append(compare("Foo",      "==",     "Foo", descr = "Comparing strings"))
testresult.append(compare([1, 2, 3],  "!=", [2, 3, 4], descr = "Comparing lists"))
```

Sample Result Output:

```
 PASSED  Comparing float values for equality requires a defined tolerance
         42.0 mA (analog_value) == 42.0 mA +/- 0.01 mA
 ----------------------------------------------------------------
 PASSED  Comparing integer values
         1 [on] (some_status_var) == 1 [on]
 ----------------------------------------------------------------
 PASSED  Comparing integer values
         1 [on] (some_status_var) <= 3 [disabled]
 ----------------------------------------------------------------
 PASSED  Comparing strings
         Foo == Foo
 ----------------------------------------------------------------
 PASSED  Comparing lists
         [1, 2, 3] != [2, 3, 4]
```

Returns

a single [<description>, <verdict>] test result entry.

6.62.2.10 compareCE()

```
ttk_checks.basic_tests.compareCE (
              current_value,
              operator,
              expected_value,
              abs_tol = 0,
              descr = "" )
```

Verify the conditional relationship between current and expected value based on the given operator.

This is a compare variant with an alternative output formatting to explicitly label input values as "current" and "expected". See also basic_tests.compare.

Parameters

current_value	left operand of the relational expression
operator	relational operator given as a string, see below
expected_value	right operand of the relational expression
abs_tol	absolute tolerance for equality/non-equality checks of numeric types
descr	user specific description printed on the start of the single test result

Info: Tolerance

A tolerance > 0 is mandatory for equality / non-equality checks involving float values.

Relational operator strings:

- "==" : equal to
- "<" : less than
- ">" : greater than
- "<=" : less than or equal to
- ">=" : greater than or equal to
- "!=" : not equal to

Usage:

```
testresult.append(
    compareCE(
        current_value  = analog_value,
        operator       = "==",
        expected_value = 42,
        abs_tol        = 0.01,
        descr="Comparing float values for equality requires a defined tolerance"
     )
)
testresult.append(compareCE(status_var, "==",         1, descr="Comparing integer values"))
testresult.append(compareCE(status_var, "<=",         3, descr="Comparing integer values"))
testresult.append(compareCE("Foo",      "==",     "Foo", descr="Comparing strings"))
testresult.append(compareCE([1, 2, 3],  "!=", [2, 3, 4], descr="Comparing lists"))
```

Sample Result Output:

```
 PASSED  Comparing float values for equality requires a defined tolerance
         Current:  42.0 mA (analog_value)
         Expected: == 42.0 mA +/- 0.01 mA
 ----------------------------------------------------------------
 PASSED  Comparing integer values
         Current:  1 [on] (some_status_var)
         Expected: == 1 [on]
 ----------------------------------------------------------------
 PASSED  Comparing integer values
         Current:  1 [on] (some_status_var)
         Expected: <= 3 [disabled]
 ----------------------------------------------------------------
 PASSED  Comparing strings
         Current:  Foo
         Expected: == Foo
 ----------------------------------------------------------------
 PASSED  Comparing lists
         Current:  [1, 2, 3]
         Expected: != [2, 3, 4]
```

Returns

a single [<description>, <verdict>] test result entry.

6.62.2.11 contains()

```
ttk_checks.basic_tests.contains (
              defined_values,
              current_value,
              mode = "all",
              descr = "",
              format = None )
```

Check if current_value is an element (or sub set) of defined_values.

Examples:

```
contains([1,3,5], 3, mode="any")  -> ["...", "PASSED"]
contains([1,3,5], [3, 45])        -> ["...", "FAILED"]
contains([1,3,5], 7, mode="none") -> ["...", "PASSED"]
```

Parameters

defined_values	List of scalars to be searched in
current_value	the function checks if this scalar or list is contained in 'defined_values'
mode	search mode (see below)
descr	user specific description printed on the start of the single test result
format	format string to control the the print out of the single test result. A format override should only rarely be used.

Available Modes:

- "all": each value of 'current_value' should be in `defined_values`
- "any": at least one element of 'current_value' should be in `defined_values`
- "none"/"not": no element of 'current_value' should be in `defined_values`

Usage:

```
testresult.append(
    contains(
        defined_values = list_value,
        current_value = 3,
        mode = "any",
        descr = "value has to be present in list"
     )
)
testresult.append(
    contains(list_value, [1, 9], mode="any", descr="at least one value has to be defined")
)
testresult.append(
    contains(list_value, [3, 5], descr="all values have to be defined")
)
testresult.append(
    contains(list_value, [7, 9], mode="none", descr="none of the values may be contained")
)
```

Sample Result Output:

```
 PASSED  value has to be present in list
         Defined values: [1, 3, 5]
         Current value:  3
         Mode:           any
 ----------------------------------------------------------------
 PASSED  at least one value has to be defined
```

```
        Defined values: [1, 3, 5]
        Current value:  [1, 9]
        Mode:           any
-------------------------------------------------------------
PASSED  all values have to be defined
        Defined values: [1, 3, 5]
        Current value:  [3, 5]
        Mode:           all
-------------------------------------------------------------
PASSED  none of the values may be contained
        Defined values: [1, 3, 5]
        Current value:  [7, 9]
        Mode:           none
```

Returns

a single [<description>, <verdict>] test result entry.

6.63 ttk_checks.basic_tests_loc Namespace Reference

Localization look-up for basic test descriptions.

Classes

- class DE

  German localization.
- class EN

  English localization.

Functions

- getInstance (locale_id=None)

  Get a L10N instance matching the supplied locale id.

Variables

- dict _available_locales
- tuple _cached_instance = (None, None)
- _l10n = getInstance("fnord_debug")

6.63.1 Detailed Description

Localization look-up for basic test descriptions.

Conventions:

- L10N entries in `<angle-brackets>` are format identifiers that will (usually) not double as default values, but have a separate default value set. Use these in place of of short, ambiguous formats like "`[%s]`".

- Entry names in format strings like "`Lorem: %(entry_name)s`" reference keys of a supplied value/info lookup dictionary. If no entry exists for a given name, an empty string will be inserted in its place.

- Entry names starting with `formatted_` usually contain a pre-formatted version of the same key without the prefix or some text that has had already some l10n/formatting applied (e.g. a tolerance string that may have different sub-formats, depending on tolerance type)

- The EN localization mostly contains just the keys with values set to None. A value of None uses the key itself as L10N-entry (we could just remove the keys completely and get the same effect, but this way they are easier to discover for later translation, modification or consistency checks).

6.63.2 Function Documentation

6.63.2.1 getInstance()

```
ttk_checks.basic_tests_loc.getInstance (
              locale_id = None )
```

Get a L10N instance matching the supplied locale id.

Parameters

locale_id	id of locale to instantiate. If no specific id has been supplied, the ID will be taken from project settings.

Returns

a L10N instance.

6.63.3 Variable Documentation

6.63.3.1 _available_locales

```
dict ttk_checks.basic_tests_loc._available_locales  [protected]
```

Initial value:

```
00001 =  {
00002     'en':    EN,
00003     'en_us': EN,
00004     'en_gb': EN,
00005
00006     'de':    DE,
00007     'de_de': DE,
00008 }
```

6.63.3.2 _cached_instance

```
tuple ttk_checks.basic_tests_loc._cached_instance = (None, None)  [protected]
```

6.63.3.3 _l10n

```
ttk_checks.basic_tests_loc._l10n = getInstance("fnord_debug")  [protected]
```

6.64 ttk_checks.checkceptions Namespace Reference

Custom exception classes for package ttk_checks.

Classes

- class BasicCheckException

  Base Exception for basic checks.
- class InputValueTypeError

  A supplied value has a wrong format/type and cannot be converted.
- class NotSupportedError

  Exception: Feature or input is not or not yet supported.
- class PatternError

  Exception: A supplied pattern is malformed and cannot be interpreted.
- class PreconditionError
- class RangeError

  A (supplied) value is outside the supported range (but has a suitable data type otherwise).
- class TtkChecksException

  Base Exception class for ttk_checks package.
- class UnexpectedException
- class ValueConversionError

  A required precondition is not given.

6.64.1 Detailed Description

Custom exception classes for package ttk_checks.

All manually raised exceptions should inherit from the base exception TtkChecksException so they can be easily caught later.

6.65 ttk_checks.checkutils Namespace Reference

Interface wrapper for check functions utilities in ttk_checks._checkutils.

Classes

- class DefaultDict

  A dictionary that returns empty strings instead of raising KeyErrors.

Functions

- buildDescription (descr_format, entries)
- getInteger (value)

  Cast a numeric value to an integer (int/long) number, but perform sanity checks before (relevant for float input values).
- resolveValue (value)

  Get a value of the input parameter if it supports a "get" function, otherwise return it as-is.
- updateMetaInfo (meta, value, key_prefix="", meta_attrs=None, setdefaults=False, fallback_value=None)

  Extract meta info from `value` (if info is available) and update the supplied meta dictionary with those values.
- updateMetaStateDescr (meta, source_var, value, key)

  Update (or create) a meta entry with a state description for the supplied value, using the lookup-table from `source_var`.
- updateMetaFormattedValue (meta, source_var, value, key="fmt", fallback_value=None)

  Update (or create) a meta entry with a formatted value representation for the supplied value, using the formatting defined in `source_var`.
- updateMetaEntries (meta, *source_vars_info)

  All-in-one update of meta data.
- bin (value, prefix='0b', min_bit_width=0)

  Create a binary representation of the input value.
- getTracebackString (limit=None)

  Get current traceback info as a string (or an empty string if no exception is currently active).

6.65.1 Detailed Description

Interface wrapper for check functions utilities in ttk_checks._checkutils.

6.65.2 Function Documentation

6.65.2.1 bin()

```
ttk_checks.checkutils.bin (
              value,
              prefix = '0b',
              min_bit_width = 0 )
```

Create a binary representation of the input value.

A rather recursive approach, works similar to the built-in function bin() that is available in Python 2.6+, but with some additional convenience features.

Parameters

value	value to represent as a binary string
prefix	prefix to prepend the string (usually "0b", just like bin() would add and similar to the "0x" for hex)
min_bit_width	a minimum bit width to enforce (filled with zeros)

Returns

a string with a binary representation of the input value.

6.65.2.2 buildDescription()

```
ttk_checks.checkutils.buildDescription (
              descr_format,
              entries )
```

6.65.2.3 getInteger()

```
ttk_checks.checkutils.getInteger (
              value )
```

Cast a numeric value to an integer (int/long) number, but perform sanity checks before (relevant for float input values).

If `value` is already long (or int), it will simply be returned as-is.

Note

Raises an ValueConversionError if `value` does not appear to be representing an integer or if it cannot fit into a double-precision float without loss of precision.

Parameters

value	numeric value to interpret as integer number

Returns

an int (if value fits into int range), or a long value.

6.65.2.4 getTracebackString()

```
ttk_checks.checkutils.getTracebackString (
              limit = None )
```

Get current traceback info as a string (or an empty string if no exception is currently active).

6.65.2.5 resolveValue()

```
ttk_checks.checkutils.resolveValue (
              value )
```

Get a value of the input parameter if it supports a "get" function, otherwise return it as-is.

Raise a ttk_base.errors.TTkException if the get() call results in an ttk_base.errors.TTkErrorString.

6.65.2.6 updateMetaEntries()

```
ttk_checks.checkutils.updateMetaEntries (
              meta,
             * source_vars_info )
```

All-in-one update of meta data.

- update all meta entries of the supplied variables (see updateMetaInfo()), then
- fill any missing entries that can be derived from other source variables. This means shared meta data like units or lookup values – checks/comparisons make not much sense if source values use different definitions.

Parameters

meta	dictionary to update, potentially with already existing meta entries
∗*source_vars_info*	all following entries must be triples of `[source_var, meta_key_prefix, current_value]`. • `current_value` may be None if no state lookup is desired. • If `current_value` is None, a VarBase or meta()-wrapped value will use its last value to provide a formatted value

6.65.2.7 updateMetaFormattedValue()

```
ttk_checks.checkutils.updateMetaFormattedValue (
              meta,
              source_var,
              value,
              key = "fmt",
              fallback_value = None )
```

Update (or create) a meta entry with a formatted value representation for the supplied value, using the formatting defined in `source_var`.

An already existing non-default formatted value representation will not be overwritten.

Parameters

meta	dictionary to update, potentially with already existing meta entries
source_var	source variable (or meta-enhanced value) with value formatting hints to update default representations
value	actual value to format (might have its own value formatting hints, though)
key	key name for state info (typically `<name>_fmt`)
fallback_value	optional fallback value (typically a a base type like int or float) to use for value representation if no suitable formatting can be found for `value`

Note

fallback_value vs. value If a "partial" variable has been supplied, e.g. a basic object with just a get() method but not much else (no getFormattedValue() and not even a custom **str** method), the fallback representation would be something like "<foo.NotSoGood object at 0x000000000E8588D0>".

By providing an optional resolved fallback_value (like used in source_vars_info), we can deal with this special case.

6.65.2.8 updateMetaInfo()

```
ttk_checks.checkutils.updateMetaInfo (
              meta,
              value,
              key_prefix = "",
              meta_attrs = None,
              setdefaults = False,
              fallback_value = None )
```

Extract meta info from `value` (if info is available) and update the supplied meta dictionary with those values.

Keys will only be present/updated if the value has matching info.

Note

Default meta_attrs:

- `unit` - unit as a string
- `descr` - description as a string
- `alias` - alias name as a string
- `state` - string description of current state (if available from lookup)
- `fmt` - value formatted using an available format string (e.g. "0x%02X")

Info: Key naming

Entries will be added with a leading `key_prefix`, so with a `key_prefix = "source_var_"` and the default `meta_attrs`, the following keys will be added/updated (as long as the `value` provides the respective members):

- "source_var_unit"
- "source_var_descr"
- "source_var_alias"
- "source_var_state"
- "source_var_fmt" plus
- "formatted_source_var_alias" (see next note)

Note

alias For `alias` an additional entry `formatted_<key_prefix>_alias` will be added that will contain the alias name formatted with the I10n-format `<alias info format>`.

value and current_value If `value` is not an instance that provides either

- a suitable string representation of its "value" (e.g. basic types like int or float) or
- a method getFormattedValue (like all VarBase-derived variables) the formatted representation entry ("fmt") will most likely not contain the expected value but default to something like <foo.NotSoGood object at 0x000000000E8588D0>.

If a separate value (e.g. a float value from a .get() operation) is available, it can be supplied as `current_↩value` and will be used as fallback representation for `value`.

Parameters

meta	dictionary to update, potentially with already existing meta entries
value	value (or a VarBase derived instance) that may have additional meta info attached (unit, description, ...)
key_prefix	prefix to prepend to dictionary keys
meta_attrs	list of custom meta attributes to extract. Defaults to `("unit", "descr", "alias", "state", "fmt")`
setdefaults	if True, already existing meta keys will not get overwritten (i.e. only default values will be set)
fallback_value	optional current/resolved value of the supplied `value` parameter. Defaults to `value`. See note.

6.65.2.9 updateMetaStateDescr()

```
ttk_checks.checkutils.updateMetaStateDescr (
              meta,
              source_var,
              value,
              key )
```

Update (or create) a meta entry with a state description for the supplied value, using the lookup-table from `source_var`.

An already existing state description will not be overwritten.

Parameters

meta	dictionary to update, potentially with already existing meta entries
source_var	source variable (or meta-enhanced value) with lookup table
value	value to lookup
key	key name for state info (typically `<name>_state`)

6.66 ttk_checks.verdicts Namespace Reference

Interface wrapper for defined test status verdicts in ttk_checks._verdicts.

Functions

- getResultingVerdict (verdicts)

  Get the resulting verdict for a set or sequence of verdicts.
- getInvertedVerdict (verdict)

  Invert verdict.

6.66.1 Detailed Description

Interface wrapper for defined test status verdicts in ttk_checks._verdicts.

Defined test status verdicts are

- `PASSED` – ttk_checks.verdicts.PASSED
- `FAILED` – ttk_checks.verdicts.FAILED
- `ERROR` – ttk_checks.verdicts.ERROR
- `RETEST PASSED` – ttk_checks.verdicts.RETEST_PASSED

See ttk_checks._verdicts for detailed definitions.

Test Verdicts, Ordered By Priority

Verdict	Description	Verdict Class
FAILED	Test step failed	Failed
ERROR	Error during test execution (in tool, script)	Failed
RETEST PASSED	A manual re-test of a test step was passed	Passed
PASSED	Test step successfully completed	Passed
NOT TESTED	Test step currently not evaluated	n/a
NOT SPECIFIED	Test step currently has no specification	n/a
NOT PLANNED	Evaluation of this test step is not planned	n/a
INFO	Test step only contain information	n/a

Standard – and commonly used – verdicts are `PASSED`, `FAILED` and `ERROR`.

Other verdicts such as `INFO` or `TODO` (or even a blank verdict) **will** appear in the test report, but will **not** be evaluated or influence the overall test result.

See also `https://wiki.isyst.de/wiki/Testbewertungen`

6.66.2 Function Documentation

6.66.2.1 getInvertedVerdict()

```
ttk_checks.verdicts.getInvertedVerdict (
              verdict )
```

Invert verdict.

Turns PASSED to FAILED and vice versa.

Note

ERROR and any info verdicts will remain untouched.

Parameters

verdict	verdict to invert

Returns

a verdict with inverted meaning.

6.66.2.2 getResultingVerdict()

```
ttk_checks.verdicts.getResultingVerdict (
              verdicts )
```

Get the resulting verdict for a set or sequence of verdicts.

Priority of actual pass/fail verdicts is determined by PASS_VERDICTS and FAIL_VERDICTS.

Parameters

verdicts	a verdict set/sequence (list, tuple, dict (keys))

Returns

`FAILED` if at least one result is failed,
`ERROR` if errors occurred without an explicit failed result,
`PASSED` (or `RETEST PASSED`, if applicable) if passed results exist without any failed results, otherwise an empty verdict.

6.67 ttk_daq Namespace Reference

Utilities for data acquisition/signal/measurement evaluation.

Namespaces

- namespace _daq_utils

  Utility functions for recorded signal data.
- namespace _eval_signal

  Utility class for evaluation of recorded signal data.
- namespace daq_utils

  Interface wrapper for recorded signal data utility functions in ttk_daq._daq_utils.
- namespace data_sync

  Examples/utilities for daq data synchronization.
- namespace eval_signal

  Interface wrapper for recorded signal data evaluation utility class in ttk_daq._eval_signal.

6.67.1 Detailed Description

Utilities for data acquisition/signal/measurement evaluation.

6.68 ttk_daq._daq_utils Namespace Reference

Utility functions for recorded signal data.

Functions

- printDataInfo (values, indent_level=1, outer_n=2)

  Print a summary of the supplied values to stdout.
- _printDataInfo (values, indent_level=1, outer_n=2)

  alias for original protected function
- getDaqDataStruct (capture_data)

  Get a "DAQ"-like data structure from a classic dSPACE rtplib capture.Fetch() dictionary.
- roundDaqData (daq_data, digits=6)

  Round all data points in daq_data to the specified number of digits.
- getOuterTimeBoundaries (daq_data)

  Get minimum and maximum time stamps for all measurements in daq_data to contain all measurements, even if some might not have data for all time stamps (e.g.
- getInnerTimeBoundaries (daq_data)

  Get minimum and maximum time stamps for all measurements in daq_data, so that all measurements have samples between those boundaries (i.e.
- _getDerivedStepTime (daq_data, limit_data_points=1000)

  Get a suitable "average" delta time between data points in daq_data.
- shiftTimeAxes (daq_data, offset)

  Shift all time axes values in daq_data by the specified offset.
- appendDaqData (cur_daq_data, new_daq_data, only_append_to_existing=False, time_between_s=None, verbosity=0)

  Append the measurement data in new_daq_data to the data in cur_daq_data.
- trimDaqData (daq_data, verbosity=0)

  Trim measurements in a daq data structure to the area where all signals have data points (i.e.
- getNormalizedDaqData (daq_data)

  Get a data dictionary with normalized/shifted time axes, so the earliest data point(s) will start at time 0.0.
- linearInterpolate (x, x0, y0, x1, y1)

  Linear interpolation at point `x` between two points `(x0, y0)` and `(x1, y1)`.
- tableLookup (x, x_values, y_values)

  Get an interpolated y-value for a given x-value in a x/y lookup table.
- getInterpolatedValue (signal_data, time_stamp)

  Get a value at the specified time.

6.68.1 Detailed Description

Utility functions for recorded signal data.

See interface wrapper in ttk_daq.daq_utils.

Most functions expect a `daq_data` structure as input, which is essentially a dictionary containing signal-dicts with two lists:

- one list containing timestamps (in [s])
- one list containing data points / values for those timestamps.

Both lists have to contain the same number of entries for the signal to be consistent.

Example DAQ data structure:

```
daq_data = {
     <label1>: {
        'data': [<value1_0>, <value1_1>, ...],
        'time': [<time0>,  <time1>,  ...]},
     },
     <label2>: {
        'data': [<value2_0>, <value2_1>, ...],
        'time': [<time0>,  <time1>,  ...]},
    },
     <label3>: {
        'data': [<value3_0>, <value3_1>, ...],
        'time': [<time2_0>,  <time2_1>,  ...]},
    },
}
```

6.68.2 Function Documentation

6.68.2.1 _getDerivedStepTime()

```
ttk_daq._daq_utils._getDerivedStepTime (
              daq_data,
              limit_data_points = 1000 )  [protected]
```

Get a suitable "average" delta time between data points in daq_data.

This is intended to be used for appendDaqData to get a somewhat sensible time to insert between appended data sets.

Approach: Median step time will be taken of all signals. The shortest median step time across all signals will be selected (mainly to deal with signals using different step sizes)

Parameters

daq_data	a DAQ data structure
limit_data_points	maximum number of data points to process. None: process all

Returns

a delta step time. Time unit will be the same as used in daq_data, i.e. [s] for our default daq_data).

6.68.2.2 _printDataInfo()

```
ttk_daq._daq_utils._printDataInfo (
              values,
              indent_level = 1,
              outer_n = 2 )  [protected]
```

alias for original protected function

6.68.2.3 appendDaqData()

```
ttk_daq._daq_utils.appendDaqData (
              cur_daq_data,
              new_daq_data,
              only_append_to_existing = False,
              time_between_s = None,
              verbosity = 0 )
```

Append the measurement data in new_daq_data to the data in cur_daq_data.

Note

- `cur_daq_data` will be modified directly.
- `new_daq_data` will remain unmodified.
- if the same time or data axes are referenced in multiple signals, those references will be replaced by standalone copies.

Parameters

cur_daq_data	current data dictionary
new_daq_data	data dictionary with data to append
only_append_to_existing	if True, only measurements that are already in cur_daq_data will be appended to, all others in new_daq_data will be ignored.
time_between_s	time in [s] to insert between data sets (time between last point in existing data and first new point in appended data). None: determine suitable time from existing time steps
verbosity	verbosity of log output 0: only errors and "skipped signal" warnings 1: also log current signal label 2: also show summary of signal data

Example: Data dictionary format

```
{  <label>: {
    'data': [value0, value1, ...],
    'time': [time0,  time1, ...]},
}
```

6.68.2.4 getDaqDataStruct()

```
ttk_daq._daq_utils.getDaqDataStruct (
              capture_data )
```

Get a "DAQ"-like data structure from a classic dSPACE rtplib capture.Fetch() dictionary.

Note

If a multi-core data structure contains conflicting variable names (e.g. same name used in multiple cores), the core name will be used as prefix for all variables beyond the first.

Example:

```
capture_data = { # single core
    <label1>: [<value1_0>, <value1_1>, ...],
    <label2>: [<value2_0>, <value2_1>, ...],
    'xAxis' : [<time0>,  <time1>,  ...],
}
# or
capture_data = { # multi core
    <core_name>: {
        <label1>: [<value1_0>, <value1_1>, ...],
        <label2>: [<value2_0>, <value2_1>, ...],
        'xAxis' : [<time0>,  <time1>,  ...],
    },
    <core2_name>: {
        <label3>: [<value3_0>, <value3_1>, ...],
        'xAxis' : [<time2_0>,  <time2_1>,  ...],
    }
}

# maps to
daq_data = {
     <label1>: {
        'data': [<value1_0>, <value1_1>, ...],
        'time': [<time0>,  <time1>,  ...]},
     },
     <label2>: {
        'data': [<value2_0>, <value2_1>, ...],
        'time': [<time0>,  <time1>,  ...]},
    },
     <label3>: {
        'data': [<value3_0>, <value3_1>, ...],
        'time': [<time2_0>,  <time2_1>,  ...]},
    },
}
```

Note

Time Axis: Signals of the same core (with identical time axes) will reference the *same* list of timestamps. Changing timestamps of one time axis will therefore influence all associated signals.

Returns

capture signals referenced in a daq-data structure

6.68.2.5 getInnerTimeBoundaries()

```
ttk_daq._daq_utils.getInnerTimeBoundaries (
              daq_data )
```

Get minimum and maximum time stamps for all measurements in daq_data, so that all measurements have samples between those boundaries (i.e.

the overlapping section of all measurements)

Example:

```
[    S1 S1 S1 S1 S1 S1 S1 S1 S1 S1 S1 S1 S1 S1 S1 S1  ]
[ S2 S2 S2 S2 S2 S2 S2 S2 S2 S2 S2 S2 S2 S2 S2       ]
     ^- start                                ^- end
```

Returns

a tuple `(max_start_time, min_end_time)`.
If data does not overlap at all, both min/max values will be None.

6.68.2.6 getInterpolatedValue()

```
ttk_daq._daq_utils.getInterpolatedValue (
              signal_data,
              time_stamp )
```

Get a value at the specified time.

Data will be interpolated if timestamp falls between available data points.
See tableLookup() for details.

Parameters

signal_data	a dictionary with "time" and "data" lists.
time_stamp	time stamp at which to get a value

Returns

an interpolated value.

6.68.2.7 getNormalizedDaqData()

```
ttk_daq._daq_utils.getNormalizedDaqData (
              daq_data )
```

Get a data dictionary with normalized/shifted time axes, so the earliest data point(s) will start at time 0.0.

Parameters

daq_data	data dictionary with measurements

Returns

a daq data dictionary where time axes start at time 0.

6.68.2.8 getOuterTimeBoundaries()

```
ttk_daq._daq_utils.getOuterTimeBoundaries (
              daq_data )
```

Get minimum and maximum time stamps for all measurements in daq_data to contain all measurements, even if some might not have data for all time stamps (e.g.

in case acquisition started later or stopped sooner).

Example:

```
[    S1 S1 S1 S1 S1 S1 S1 S1 S1 S1 S1 S1 S1 S1 S1 S1 S1  ]
[ S2 S2 S2 S2 S2 S2 S2 S2 S2 S2 S2 S2 S2 S2 S2 S2        ]
  ^- start                                          ^- end
```

Returns

a tuple `(min_start_time, max_end_time)`

6.68.2.9 linearInterpolate()

```
ttk_daq._daq_utils.linearInterpolate (
              x,
              x0,
              y0,
              x1,
              y1 )
```

Linear interpolation at point `x` between two points `(x0, y0)` and `(x1, y1)`.

Info:

```
          y1 --+-----------+--
               |          / |
               |        /   |
          y <--+-----/      |
               |   / |      |
               | /   |      |
          y0 --+-----+-----+--
              x0     x    x1
```

```
If x is outside the specified range, the y-value will saturate
at the corresponding boundary. Precondition: x1 >= x0.
```

Returns

a float value `y`, interpolated between `y0` and `y1`.

6.68.2.10 printDataInfo()

```
ttk_daq._daq_utils.printDataInfo (
              values,
              indent_level = 1,
              outer_n = 2 )
```

Print a summary of the supplied values to stdout.

Mainly intended to be used during development / debugging as a (somewhat) compact output of DAQ data structures.

Example:

```
# Sample output using a DAQ data structure with two signals
# and the default outer_n=2
  label:
    data[20]: [100.052, 100.087, ... 100.315, 100.448]
    time[20]: [  0.000,   0.001, ...   0.018,   0.019]
  another label:
    data[20]: [ 27.162,   2.783, ...  23.078,  15.383]
    time[20]: [  0.000,   0.020, ...   0.360,   0.380]
```

Parameters

values	a scalar value or values in lists or dictionaries.
indent_level	starting indent level (as multiple of two spaces)
outer_n	print only the outer n values from start and end of data. None: all values will be printed

6.68.2.11 roundDaqData()

```
ttk_daq._daq_utils.roundDaqData (
              daq_data,
              digits = 6 )
```

Round all data points in daq_data to the specified number of digits.

Note

Data in daq_data` will be directly modified in-place.

Parameters

daq_data	data dictionary
digits	number of digits / decimal places to round to

Example: Data dictionary format

```
{  <label>: {
    'data': [value0, value1, ...],
    'time': [time0,  time1, ...]},
}
```

Returns

– (`daq_data` is modified in place)

6.68.2.12 shiftTimeAxes()

```
ttk_daq._daq_utils.shiftTimeAxes (
              daq_data,
              offset )
```

Shift all time axes values in daq_data by the specified offset.

Parameters

daq_data	a DAQ data dict with signals.
offset	offset to add to all time values

Note

- only daq_data items with "time" entries will be modified.
- if time axes are referenced multiple times (e.g. shared between signals), each unique axis will only be shifted once.
- daq_data will be modified in-place

6.68.2.13 tableLookup()

```
ttk_daq._daq_utils.tableLookup (
              x,
              x_values,
              y_values )
```

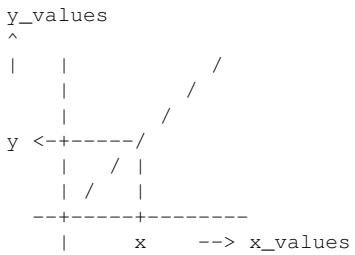
Get an interpolated y-value for a given x-value in a x/y lookup table.

Limitations:

x-values have to be monotonically increasing.
Both x- and y-value lists have to contain the same number of elements.

See linearInterpolate() for details.

Info:



Parameters

x	value on x-axis for which to get a corresponding x-value
x_values	a list of values on the x-axis
y_values	a list of values on the y-axis

Returns

an interpolated y-value for the supplied x-value. If x-value is out of lookup range, y will saturate at its boundary values.

6.68.2.14 trimDaqData()

```
ttk_daq._daq_utils.trimDaqData (
```

```
        daq_data,
        verbosity = 0 )
```

Trim measurements in a daq data structure to the area where all signals have data points (i.e.

where all signals overlap). Approach:

- determine time stamps via getInnerTimeBoundaries
- for each signal:
  - start with first item closest to start time stamp
  - stop with last item closest to stop time stamp

Note

`daq_data` will be modified in place.

Time axis references will be replaced with trimmed versions (and not modified in-place), so shared time axes (in case multiple each signals hold a reference to the same axis data) will not cause issues.

Parameters

daq_data	data dictionary with measurements
verbosity	log verbosity; 0: only error messages >0: more verbose

Example: Data dictionary format

```
{  <label>: {
    'data': [value0, value1, ...],
    'time': [time0,  time1, ...]},
}
```

6.69 ttk_daq._eval_signal Namespace Reference

Utility class for evaluation of recorded signal data.

Classes

- class EvalSignal
- class OperatorException

  Errors during custom find operator function calls.

6.69.1 Detailed Description

Utility class for evaluation of recorded signal data.

See also interface wrapper in ttk_daq.eval_signal.

6.70 ttk_daq.daq_utils Namespace Reference

Interface wrapper for recorded signal data utility functions in ttk_daq._daq_utils.

Functions

- printDataInfo (values, indent_level=1, outer_n=2)

  Print a summary of the supplied values to stdout.
- getDaqDataStruct (capture_data)

  Get a "DAQ"-like data structure from a classic dSPACE rtplib capture.Fetch() dictionary.
- roundDaqData (daq_data, digits=6)

  Round all data points in daq_data to the specified number of digits.
- getOuterTimeBoundaries (daq_data)

  Get minimum and maximum time stamps for all measurements in daq_data to contain all measurements, even if some might not have data for all time stamps (e.g.
- getInnerTimeBoundaries (daq_data)

  Get minimum and maximum time stamps for all measurements in daq_data, so that all measurements have samples between those boundaries (i.e.
- shiftTimeAxes (daq_data, offset)

  Shift all time axes values in daq_data by the specified offset.
- appendDaqData (cur_daq_data, new_daq_data, only_append_to_existing=False, time_between_s=None, verbosity=0)

  Append the measurement data in new_daq_data to the data in cur_daq_data.
- trimDaqData (daq_data, verbosity=0)

  Trim measurements in a daq data structure to the area where all signals have data points (i.e.
- getNormalizedDaqData (daq_data)

  Get a data dictionary with normalized/shifted time axes, so the earliest data point(s) will start at time 0.0.
- linearInterpolate (x, x0, y0, x1, y1)

  Linear interpolation at point `x` between two points `(x0, y0)` and `(x1, y1)`.
- tableLookup (x, x_values, y_values)

  Get an interpolated y-value for a given x-value in a x/y lookup table.
- getInterpolatedValue (signal_data, time_stamp)

  Get a value at the specified time.

6.70.1 Detailed Description

Interface wrapper for recorded signal data utility functions in ttk_daq._daq_utils.

Most functions expect a `daq_data` structure as input, which is essentially a dictionary containing signal-dicts with two lists:

- one list containing timestamps (in [s])
- one list containing data points / values for those timestamps.

Both lists have to contain the same number of entries for the signal to be consistent.

Example DAQ data structure:

```
daq_data = {
     <label1>: {
        'data': [<value1_0>, <value1_1>, ...],
        'time': [<time0>,  <time1>,  ...]},
     },
     <label2>: {
        'data': [<value2_0>, <value2_1>, ...],
        'time': [<time0>,  <time1>,  ...]},
    },
     <label3>: {
        'data': [<value3_0>, <value3_1>, ...],
        'time': [<time2_0>,  <time2_1>,  ...]},
    },
}
```

6.70.2 Function Documentation

6.70.2.1 appendDaqData()

```
ttk_daq.daq_utils.appendDaqData (
              cur_daq_data,
              new_daq_data,
              only_append_to_existing = False,
              time_between_s = None,
              verbosity = 0 )
```

Append the measurement data in new_daq_data to the data in cur_daq_data.

Note

- `cur_daq_data` will be modified directly.
- `new_daq_data` will remain unmodified.
- if the same time or data axes are referenced in multiple signals, those references will be replaced by standalone copies.

Parameters

cur_daq_data	current data dictionary
new_daq_data	data dictionary with data to append
only_append_to_existing	if True, only measurements that are already in cur_daq_data will be appended to, all others in new_daq_data will be ignored.
time_between_s	time in [s] to insert between data sets (time between last point in existing data and first new point in appended data). None: determine suitable time from existing time steps
verbosity	verbosity of log output 0: only errors and "skipped signal" warnings 1: also log current signal label 2: also show summary of signal data

Example: Data dictionary format

```
{  <label>: {
    'data': [value0, value1, ...],
    'time': [time0,  time1, ...]},
}
```

6.70.2.2 getDaqDataStruct()

```
ttk_daq.daq_utils.getDaqDataStruct (
              capture_data )
```

Get a "DAQ"-like data structure from a classic dSPACE rtplib capture.Fetch() dictionary.

Note

If a multi-core data structure contains conflicting variable names (e.g. same name used in multiple cores), the core name will be used as prefix for all variables beyond the first.

Example:

```
capture_data = { # single core
    <label1>: [<value1_0>, <value1_1>, ...],
    <label2>: [<value2_0>, <value2_1>, ...],
    'xAxis' : [<time0>,  <time1>,  ...],
}
# or
capture_data = { # multi core
    <core_name>: {
        <label1>: [<value1_0>, <value1_1>, ...],
        <label2>: [<value2_0>, <value2_1>, ...],
        'xAxis' : [<time0>,  <time1>,  ...],
    },
    <core2_name>: {
        <label3>: [<value3_0>, <value3_1>, ...],
        'xAxis' : [<time2_0>,  <time2_1>,  ...],
    }
}

# maps to
daq_data = {
     <label1>: {
        'data': [<value1_0>, <value1_1>, ...],
        'time': [<time0>,  <time1>,  ...]},
     },
     <label2>: {
        'data': [<value2_0>, <value2_1>, ...],
        'time': [<time0>,  <time1>,  ...]},
    },
     <label3>: {
        'data': [<value3_0>, <value3_1>, ...],
        'time': [<time2_0>,  <time2_1>,  ...]},
    },
}
```

Note

Time Axis: Signals of the same core (with identical time axes) will reference the *same* list of timestamps. Changing timestamps of one time axis will therefore influence all associated signals.

Returns

capture signals referenced in a daq-data structure

6.70.2.3 getInnerTimeBoundaries()

```
ttk_daq.daq_utils.getInnerTimeBoundaries (
              daq_data )
```

Get minimum and maximum time stamps for all measurements in daq_data, so that all measurements have samples between those boundaries (i.e.

the overlapping section of all measurements)

Example:

```
[    S1 S1 S1 S1 S1 S1 S1 S1 S1 S1 S1 S1 S1 S1 S1 S1  ]
[ S2 S2 S2 S2 S2 S2 S2 S2 S2 S2 S2 S2 S2 S2 S2        ]
     ^- start                                  ^- end
```

Returns

a tuple `(max_start_time, min_end_time)`.
If data does not overlap at all, both min/max values will be None.

6.70.2.4 getInterpolatedValue()

```
ttk_daq.daq_utils.getInterpolatedValue (
              signal_data,
              time_stamp )
```

Get a value at the specified time.

Data will be interpolated if timestamp falls between available data points.
See tableLookup() for details.

Parameters

signal_data	a dictionary with "time" and "data" lists.
time_stamp	time stamp at which to get a value

Returns

an interpolated value.

6.70.2.5 getNormalizedDaqData()

```
ttk_daq.daq_utils.getNormalizedDaqData (
              daq_data )
```

Get a data dictionary with normalized/shifted time axes, so the earliest data point(s) will start at time 0.0.

Parameters

daq_data	data dictionary with measurements

Returns

a daq data dictionary where time axes start at time 0.

6.70.2.6 getOuterTimeBoundaries()

```
ttk_daq.daq_utils.getOuterTimeBoundaries (
              daq_data )
```

Get minimum and maximum time stamps for all measurements in daq_data to contain all measurements, even if some might not have data for all time stamps (e.g.

in case acquisition started later or stopped sooner).

Example:

```
[    S1 S1 S1 S1 S1 S1 S1 S1 S1 S1 S1 S1 S1 S1 S1 S1  ]
[ S2 S2 S2 S2 S2 S2 S2 S2 S2 S2 S2 S2 S2 S2 S2       ]
  ^- start                                    ^- end
```

Returns

a tuple `(min_start_time, max_end_time)`

6.70.2.7 linearInterpolate()

```
ttk_daq.daq_utils.linearInterpolate (
              x,
              x0,
              y0,
              x1,
              y1 )
```

Linear interpolation at point `x` between two points `(x0, y0)` and `(x1, y1)`.

Info:

```
          y1 --+-----------+--
               |         / |
               |       /   |
          y <--+-----/     |
               |   / |     |
               | /   |     |
          y0 --+-----+-----+--
              x0     x    x1
```

```
If x is outside the specified range, the y-value will saturate
at the corresponding boundary. Precondition: x1 >= x0.
```

Returns

a float value `y`, interpolated between `y0` and `y1`.

6.70.2.8 printDataInfo()

```
ttk_daq.daq_utils.printDataInfo (
              values,
              indent_level = 1,
              outer_n = 2 )
```

Print a summary of the supplied values to stdout.

Mainly intended to be used during development / debugging as a (somewhat) compact output of DAQ data structures.

Example:

```
# Sample output using a DAQ data structure with two signals
# and the default outer_n=2
  label:
    data[20]: [100.052, 100.087, ... 100.315, 100.448]
    time[20]: [  0.000,   0.001, ...   0.018,   0.019]
  another label:
    data[20]: [ 27.162,   2.783, ...  23.078,  15.383]
    time[20]: [  0.000,   0.020, ...   0.360,   0.380]
```

Parameters

values	a scalar value or values in lists or dictionaries.
indent_level	starting indent level (as multiple of two spaces)
outer_n	print only the outer n values from start and end of data. None: all values will be printed

6.70.2.9 roundDaqData()

```
ttk_daq.daq_utils.roundDaqData (
              daq_data,
              digits = 6 )
```

Round all data points in daq_data to the specified number of digits.

Note

Data in daq_data` will be directly modified in-place.

Parameters

daq_data	data dictionary
digits	number of digits / decimal places to round to

Example: Data dictionary format

```
{  <label>: {
    'data': [value0, value1, ...],
    'time': [time0,  time1, ...]},
}
```

Returns

– (`daq_data` is modified in place)

6.70.2.10 shiftTimeAxes()

```
ttk_daq.daq_utils.shiftTimeAxes (
              daq_data,
              offset )
```

Shift all time axes values in daq_data by the specified offset.

Parameters

daq_data	a DAQ data dict with signals.
offset	offset to add to all time values

Note

- only daq_data items with "time" entries will be modified.
- if time axes are referenced multiple times (e.g. shared between signals), each unique axis will only be shifted once.
- daq_data will be modified in-place

6.70.2.11 tableLookup()

```
ttk_daq.daq_utils.tableLookup (
              x,
              x_values,
              y_values )
```

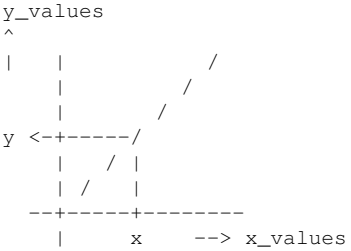
Get an interpolated y-value for a given x-value in a x/y lookup table.

Limitations:

x-values have to be monotonically increasing.
Both x- and y-value lists have to contain the same number of elements.

See linearInterpolate() for details.

Info:



Parameters

x	value on x-axis for which to get a corresponding x-value
x_values	a list of values on the x-axis
y_values	a list of values on the y-axis

Returns

an interpolated y-value for the supplied x-value. If x-value is out of lookup range, y will saturate at its boundary values.

6.70.2.12 trimDaqData()

```
ttk_daq.daq_utils.trimDaqData (
              daq_data,
              verbosity = 0 )
```

Trim measurements in a daq data structure to the area where all signals have data points (i.e.

where all signals overlap).

Note

`daq_data` will be modified in place.

Time axis references will be replaced with trimmed versions (and not modified in-place), so shared time axes (in case multiple each signals hold a reference to the same axis data) will not cause issues.

Parameters

daq_data	data dictionary with measurements
verbosity	log verbosity; 0: only error messages >0: more verbose

Example: Data dictionary format

```
{  <label>: {
    'data': [value0, value1, ...],
    'time': [time0,  time1, ...]},
}
```

6.71 ttk_daq.data_sync Namespace Reference

Examples/utilities for daq data synchronization.

Functions

- synchronize (daq_data_a, sync_signal_a, sync_value_a, daq_data_b, sync_signal_b, sync_value_b, interpolate=True, offset=0)

  Synchronize timestamps of two sets of DAQ measurements (a and b) by using a synchronization signal value change recorded in both.

6.71.1 Detailed Description

Examples/utilities for daq data synchronization.

Note

As synchronization requirements vary with available HW and SW features, the provided functions are intended as a starting point for developing individual sync functions suitable for specific project needs.

6.71.2 Function Documentation

6.71.2.1 synchronize()

```
ttk_daq.data_sync.synchronize (
              daq_data_a,
              sync_signal_a,
              sync_value_a,
              daq_data_b,
              sync_signal_b,
              sync_value_b,
              interpolate = True,
              offset = 0 )
```

Synchronize timestamps of two sets of DAQ measurements (a and b) by using a synchronization signal value change recorded in both.

Note

- Time axes in `daq_data_a` will be updated in-place, if a sync time was successfully detected.
- `daq_data_b` will remain unmodified.
- it is assumed that signals/measurements in a daq-data container are already synchronous to all other signals in that container.

Parameters

daq_data_a	daq_data dict structure of first set of measurements
sync_signal_a	name/label of sync signal in `daq_data_a`
sync_value_a	sync time will be found if value exceed this threshold for the first time
daq_data_b	daq_data data structure of second set of measurements
sync_signal_b	name/label of sync signal in `daq_data_b`
sync_value_b	sync time will be found if value exceed this threshold for the first time
interpolate	True: interpolate time of sync edge between the first matching and the previous data point (might be helpful for analog values and slower measurements) False: use timestamp of first match (usually preferable if sync signals are digital signals with discrete states)
offset	additional time axis offset [s] data a vs. data b.

Note

Offset example If `daq_data_a`'s source is a HIL/RT capture and `daq_data_b`'s source is a Calibration tool measurement, `offset` might be set to a positive value to offset processing and/or filter time in ECU (e.g. a 2.5ms delay due to low pass filter on ADC input plus 5ms cycle time of ADC read task)

Example:

```
# given recorded HIL data in `hil_data' (e.g. from a real time logger)
# and recorded calibration data (e.g. via CANape) `cal_data'
# as well als `hil' and `cal' variable containers:
synchronize(
    # hil capture
    daq_data_a      = hil_data,
    sync_signal_a   = hil.vbat,
    sync_value_a    = 11.5,

    # daq measurement
    daq_data_b      = cal_data,
    sync_signal_b   = cal.adc_vbat,
    sync_value_b    = 11.5,

    interpolate     = True,
    # 5ms offset to compensate e.g. for input filters and (worst-case) task timing
    offset          = 0.005
)
# merge synchronized data
synched = {}
synched.update(hil_data)
synched.update(cal_data)

# trim off excess data points to get a cleaner area if you want to create a plot
daq_utils.trimDaqData(synched)
```

Returns

True if synchronization was successful, otherwise False.
Time axis of `daq_data_a` will be modified in-place and synch'ed to the axis of `daq_data_b`.

closure; detect timestamp where data first reaches threshold value

6.72 ttk_daq.eval_signal Namespace Reference

Interface wrapper for recorded signal data evaluation utility class in ttk_daq._eval_signal.

Classes

- class EvalSignal

  Class for evaluation of recorded signal data.

6.72.1 Detailed Description

Interface wrapper for recorded signal data evaluation utility class in ttk_daq._eval_signal.

6.73 ttk_selftest Namespace Reference

Test functions for HIL system self test and abstraction layer for "HIL devices".

Namespaces

- namespace _adjust_exception

  Exception class definitions.
- namespace _adjustment_calculation

  Adjustment functions for self test at hil systems.
- namespace _basic_result_list
- namespace _selftest_tests

  Test functions for self test of hil systems.
- namespace _xml_handling

  Read and Write methods for self test at hil systems.
- namespace adjustment_calculation

  Interface wrapper for adjustment calculation class.
- namespace devices_base

  Interface wrapper for HIL device descriptions.
- namespace hil_devices

  Access to hil devices (dSPACE and GammaV).
- namespace selftest_tests

  Wrapper for self test test functions This serves as "interface" to the precompiled module in delivery to enable code-completion in PyDev.

6.73.1 Detailed Description

Test functions for HIL system self test and abstraction layer for "HIL devices".

6.74 ttk_selftest._adjust_exception Namespace Reference

Exception class definitions.

Classes

- class AdjustError

  Base class to raise adjustment Errors.

6.74.1 Detailed Description

Exception class definitions.

6.75 ttk_selftest._adjustment_calculation Namespace Reference

Adjustment functions for self test at hil systems.

Classes

- class AdjustmentCalculation

  Class for adjustment one hil channel which is currently in state self test and adjustment.

Variables

- str _NOTTESTED = "NOT TESTED"
- str _PASSED = "PASSED"
- str _FAILED = "FAILED"
- str _MAX = "max"
- str _MIN = "min"
- delta = TestVar.DummyVar(context=None, identifier="", default=1.00)
- bravo = TestVar.DummyVar(context=None, identifier="", default=0.00)
- a = AdjustmentCalculation(hilvar_gain=delta, hilvar_offset=bravo)
- expected
- measured

6.75.1 Detailed Description

Adjustment functions for self test at hil systems.

standard libraries ##

6.75.2 Variable Documentation

6.75.2.1 _FAILED

```
str ttk_selftest._adjustment_calculation._FAILED = "FAILED"  [protected]
```

6.75.2.2 _MAX

```
str ttk_selftest._adjustment_calculation._MAX = "max"  [protected]
```

6.75.2.3 _MIN

```
str ttk_selftest._adjustment_calculation._MIN = "min"  [protected]
```

6.75.2.4 _NOTTESTED

```
str ttk_selftest._adjustment_calculation._NOTTESTED = "NOT TESTED"  [protected]
```

6.75.2.5 _PASSED

```
str ttk_selftest._adjustment_calculation._PASSED = "PASSED"  [protected]
```

6.75.2.6 a

```
ttk_selftest._adjustment_calculation.a = AdjustmentCalculation(hilvar_gain=delta, hilvar_↩
offset=bravo)
```

6.75.2.7 bravo

```
ttk_selftest._adjustment_calculation.bravo = TestVar.DummyVar(context=None, identifier="",
default=0.00)
```

6.75.2.8 delta

```
ttk_selftest._adjustment_calculation.delta = TestVar.DummyVar(context=None, identifier="",
default=1.00)
```

6.75.2.9 expected

```
ttk_selftest._adjustment_calculation.expected
```

6.75.2.10 measured

```
ttk_selftest._adjustment_calculation.measured
```

6.76 ttk_selftest._basic_result_list Namespace Reference

Classes

- class ResultList

  Add logging, some sanity checks and helper functions to a plain list class in order to make it more convenient for storing test result entries.
- class SubResult

  Context manager for Sub-Results.

Functions

- _parseDescription (description)

  Parse a description text to detect "special" keywords.
- _removeDoubledBeginnings (descriptions)

  Remove first lines of description texts that are already present in the (directly) preceding description.
- printDescriptionBlock (descr, verdict='', min_indent=7)

  Print a formatted description and verdict (if available)

Variables

- str TESTCASE_ID_MARKER = "[[TCID]]"

  default marker for testcase-IDs (pre-wrapped with brackets)
- str COMMENT_MARKER = "[[COMMENT]]"

  default marker for comments (pre-wrapped with brackets)
- str DEFAULT_DIVIDER = "\n----------\n"

  default divider to use with getCombinedResult
- str DEFAULT_FAILED_PREFIX = "> "

  default prefix marker for failed sub result entries
- dict COMMENT_KEYWORDS = {'COMMENT', 'CMNT'}
- dict TESTCASE_ID_KEYWORDS = {'TCID', 'TC-ID', 'TC'}
- dict OBJECT_KEYWORDS = {'IMG', 'OBJ'}
- dict DELTATIME_KEYWORDS = {"DELTATIME"}
- _keyword_regex = re.compile(r'\[\[([-\w]+)\]\]\s*(.*?)\s*$', re.DOTALL|re.ASCII)

6.76.1 Function Documentation

6.76.1.1 _parseDescription()

```
ttk_selftest._basic_result_list._parseDescription (
              description ) [protected]
```

Parse a description text to detect "special" keywords.

Parameters

description	description string to parse

Returns

a tuple (<keyword>, <data>), keyword will be None if no keyword was found.

6.76.1.2 _removeDoubledBeginnings()

```
ttk_selftest._basic_result_list._removeDoubledBeginnings (
              descriptions ) [protected]
```

Remove first lines of description texts that are already present in the (directly) preceding description.

This may help to reduce clutter when combining sub-results later.

Note that if a description is identical to its preceding description, it will be kept untouched (and not reduced to an empty string).

Parameters

descriptions	list of description strings

Returns

- (descriptions will be modified in-place)

6.76.1.3 printDescriptionBlock()

```
ttk_selftest._basic_result_list.printDescriptionBlock (
              descr,
              verdict = '',
              min_indent = 7 )
```

Print a formatted description and verdict (if available)

Parameters

descr	description to print
verdict	verdict to print left of description
min_indent	minimum indent to use for description (for shorter/empty verdicts)

6.76.2 Variable Documentation

6.76.2.1 _keyword_regex

```
ttk_selftest._basic_result_list._keyword_regex = re.compile(r'\[\[([-\w]+)\]\]\s*(.*?)\s*$',
re.DOTALL|re.ASCII)  [protected]
```

6.76.2.2 COMMENT_KEYWORDS

```
dict ttk_selftest._basic_result_list.COMMENT_KEYWORDS = {'COMMENT', 'CMNT'}
```

6.76.2.3 COMMENT_MARKER

```
str ttk_selftest._basic_result_list.COMMENT_MARKER = "[[COMMENT]]"
```

default marker for comments (pre-wrapped with brackets)

6.76.2.4 DEFAULT_DIVIDER

```
str ttk_selftest._basic_result_list.DEFAULT_DIVIDER = "\n------------------------\n"
```

default divider to use with getCombinedResult

6.76.2.5 DEFAULT_FAILED_PREFIX

```
str ttk_selftest._basic_result_list.DEFAULT_FAILED_PREFIX = "> "
```

default prefix marker for failed sub result entries

6.76.2.6 DELTATIME_KEYWORDS

```
dict ttk_selftest._basic_result_list.DELTATIME_KEYWORDS = {"DELTATIME"}
```

6.76.2.7 OBJECT_KEYWORDS

```
dict ttk_selftest._basic_result_list.OBJECT_KEYWORDS = {'IMG', 'OBJ'}
```

6.76.2.8 TESTCASE_ID_KEYWORDS

```
dict ttk_selftest._basic_result_list.TESTCASE_ID_KEYWORDS = {'TCID', 'TC-ID', 'TC'}
```

6.76.2.9 TESTCASE_ID_MARKER

```
str ttk_selftest._basic_result_list.TESTCASE_ID_MARKER = "[[TCID]]"
```

default marker for testcase-IDs (pre-wrapped with brackets)

6.77 ttk_selftest._selftest_tests Namespace Reference

Test functions for self test of hil systems.

Classes

- class NegWrapper

  Class used to get values multiplied by -1.

Functions

- mean (it)
- _setFiu (fiu_channel, OpenLoad=True, EI1=True, EI2=False)

  Function to set the FIU channel.
- _resetFiu (fiu_channel)

  Function to reset the whole FIU channel.
- _setFiuCanFr (fiu_channel, High_OpenLoad=True, High_EI1=True, High_EI2=False, Low_OpenLoad=True, Low_EI1=False, Low_EI2=True)

  Function to set the FIU_CAN_FR channel.
- _resetFiuCanFr (fiu_channel)

  Function to reset the whole FIU_CAN_FR channel.
- _getCommonVoltageRange (ch1, ch2, offset_voltage=0)

  Function to get common voltage range of two analog channels.
- _setPreconditions (preconditions)

  Function to set the preconditions for the test of a hil device.
- _resetPreconditions (preconditions)

  Function to reset the preconditions for the test of a hil device.

- _setPowerSupply (power_supply, voltage, current_limit)

  Function to set the power supply output.
- _resetPowerSupply (power_supply)

  Function to reset the power supply output.
- _checkAndSetPowerSupply (testenv, power_supply, voltage, current_limit)

  Function to set the power supply output, if the test result of the power supply is PASSED.
- _resetAndWait (device, stimulation_var, min_wait_time_ms, timestamp_start=None)

  Function to reset the stimulation and pause the test execution.
- _addValuesToSelftestDataset (device, expected_value, measured_value)

  Function to add a measurement snapshot consisting of expected value and measured value to the dataset dictionary of the device.
- testGND (testenv, gnd, threshold=10000, ignore_results=False)

  Function to test ground connections.
- testAnalogOutput (testenv, channel, ref_channel, num_testvalues=5, gnd_offset=0.5)

  Function to test an analog output channel.
- testAnalogVoltageInput (testenv, device, num_testvalues=5)

  Function to test an analog voltage input channel.
- testAnalogCurrentInput (testenv, device, num_testvalues=5)

  Function to test an analog current input channel.
- testPowerSupplyVoltage (testenv, power_supply, num_testvalues=5)

  Function to test power supply voltage control and measurement.
- testPowerSupplyCurrent (testenv, power_supply, max_test_current=None, num_testvalues=5)

  Function to test power supply current control and measurement.
- testPowerSupply (testenv, power_supply, max_test_current=None, num_testvalues=5)

  Function to test Power Supply voltage and current.
- testPwmInput (testenv, device, num_testvalues=5)

  Function to test a PWM input channel.
- testLoad (testenv, load, num_testvalues=5)

  Function to test a load.
- testPwmOutput (testenv, channel, num_testvalues=5)

  Function to test an PWM output channel.
- testCan (testenv, channel)

  Function to test a CAN channel.
- testDigitalInput (testenv, device)

  Function to test a digital input channel.
- testDigitalOutput (testenv, channel)

  Function to test a digital output channel.
- testiDigiRes (testenv, idigires)

  Function to test an iDigiRes.
- resetFiu (hil_devices)

  Function to reset the FIU.

6.77.1 Detailed Description

Test functions for self test of hil systems.

6.77.2 Function Documentation

6.77.2.1 _addValuesToSelftestDataset()

```
ttk_selftest._selftest_tests._addValuesToSelftestDataset (
              device,
              expected_value,
              measured_value )  [protected]
```

Function to add a measurement snapshot consisting of expected value and measured value to the dataset dictionary of the device.

Parameters

device	device/channel that is tested
expected_value	expected value of the snapshot
measured_value	measured value of the snapshot

Returns

-

6.77.2.2 _checkAndSetPowerSupply()

```
ttk_selftest._selftest_tests._checkAndSetPowerSupply (
              testenv,
              power_supply,
              voltage,
              current_limit )  [protected]
```

Function to set the power supply output, if the test result of the power supply is PASSED.

This includes the defined preconditions of the supplied power supply. If the power supply has not been tested before, the power supply tests will be performed. The test result will be added to the test result list of the test environment.

Parameters

testenv	test environment
power_supply	power supply to be configured
voltage	target voltage of power supply
current_limit	current limitation of the power supply

6.77.2.3 _getCommonVoltageRange()

```
ttk_selftest._selftest_tests._getCommonVoltageRange (
              ch1,
              ch2,
              offset_voltage = 0 )  [protected]
```

Function to get common voltage range of two analog channels.

Parameters

ch1	first analog channel
ch2	second analog channel
offset_voltage	offset voltage that will be subtracted from the maximum values of the channels (e.g. for test of analog outputs) defaults to 0V

Returns

An array consisting of first element: result verdict second element: result string third element: lower common voltage limit fourth element: upper common voltage limit

6.77.2.4 _resetAndWait()

```
ttk_selftest._selftest_tests._resetAndWait (
              device,
              stimulation_var,
              min_wait_time_ms,
              timestamp_start = None )  [protected]
```

Function to reset the stimulation and pause the test execution.

Thus loads will be protected from damage due to over load and inductive loads will be unloaded to prevent voltage peaks that might cause damages.

Parameters

device	load device that is tested
stimulation_var	variables used for load stimulation
min_wait_time_ms	minimum waiting time in ms
timestamp_start	timestamp of the test start defaults to None

Returns

a single test result entry

6.77.2.5 _resetFiu()

```
ttk_selftest._selftest_tests._resetFiu (
              fiu_channel )  [protected]
```

Function to reset the whole FIU channel.

Parameters

fiu_channel	channel in the FIU or GND device to be reset

Returns

a single test result entry (with verdict info)

6.77.2.6 _resetFiuCanFr()

```
ttk_selftest._selftest_tests._resetFiuCanFr (
              fiu_channel )  [protected]
```

Function to reset the whole FIU_CAN_FR channel.

Parameters

fiu_channel	channel in the FIU to be reset

Returns

a single test result entry (with verdict info)

6.77.2.7 _resetPowerSupply()

```
ttk_selftest._selftest_tests._resetPowerSupply (
              power_supply )  [protected]
```

Function to reset the power supply output.

This includes the defined preconditions of the supplied power supply.

Parameters

power_supply	power supply to be configured

Returns

a single test result entry

6.77.2.8 _resetPreconditions()

```
ttk_selftest._selftest_tests._resetPreconditions (
              preconditions )  [protected]
```

Function to reset the preconditions for the test of a hil device.

Parameters

preconditions	preconditions to be reset: a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". For the reset, the key "value" will be ignored as the reset()-function will be called on the device. If the device needs triggering, the trigger method will be called. Alternatively, functions can be called. In this case, the keys "func" or "reset_func" and optional arguments have to be part of the dictionary. The function specified with "func" will be called when the preconditions are set and reset, while the functions specified with "reset_func" will only be called when preconditions are set.

Note

preconditions will be reset in reverse order!

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
# For special cases, the preconditions may also contain function calls
# (which might be useful to insert additional delays or similar):
preconditions = [
    { "device": Voltage, "value": 1},
    { "func":   time.sleep,
      "args":   [1.0], # positional arguments
      "kwargs": {},    # keyword arguments
    },
    {"device": ClampSwitch, "value": 1},
    # or a simple call like this:
    { "func":  someCustomFunction,
    },
    ...
]
```

Returns

a single test result entry (with verdict info)

6.77.2.9 _setFiu()

```
ttk_selftest._selftest_tests._setFiu (
                fiu_channel,
                OpenLoad = True,
                EI1 = True,
                EI2 = False )  [protected]
```

Function to set the FIU channel.

Parameters

fiu_channel	channel in the FIU or GND device to be set
OpenLoad	(opt) flag to set open load
EI1	(opt) flag to connect channel to EI1
EI2	(opt) flag to connect channel to EI2

Returns

a single test result entry (with verdict info)

6.77.2.10 _setFiuCanFr()

```
ttk_selftest._selftest_tests._setFiuCanFr (
              fiu_channel,
              High_OpenLoad = True,
              High_EI1 = True,
              High_EI2 = False,
              Low_OpenLoad = True,
              Low_EI1 = False,
              Low_EI2 = True )  [protected]
```

Function to set the FIU_CAN_FR channel.

Parameters

fiu_channel	channel in the FIU to be set
High_OpenLoad	(opt) flag to set open load of bus signal high defaults to True
High_EI1	(opt) flag to connect high bus signal to EI1 defaults to True
High_EI2	(opt) flag to connect high bus signal to EI2 defaults to False
Low_OpenLoad	(opt) flag to set open load of bus signal low defaults to True
Low_EI1	(opt) flag to connect low bus signal to EI1 defaults to False
Low_EI2	(opt) flag to connect low bus signal to EI2 defaults to True

Returns

a single test result entry (with verdict info)

6.77.2.11 _setPowerSupply()

```
ttk_selftest._selftest_tests._setPowerSupply (
              power_supply,
              voltage,
              current_limit )  [protected]
```

Function to set the power supply output.

This includes the defined preconditions of the supplied power supply.

Parameters

power_supply	power supply to be configured
voltage	target voltage of power supply
current_limit	current limitation of the power supply

Returns

a single test result entry

6.77.2.12 _setPreconditions()

```
ttk_selftest._selftest_tests._setPreconditions (
              preconditions )  [protected]
```

Function to set the preconditions for the test of a hil device.

Parameters

preconditions	preconditions to be set: a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". If the device needs triggering, the trigger method will be called. Alternatively, functions can be called. In this case, the keys "func" or "set_func" and optional arguments have to be part of the dictionary. The function specified with "func" will be called when the preconditions are set and reset, while the functions specified with "set_func" will only be called when preconditions are set.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
# For special cases, the preconditions may also contain function calls
# (which might be useful to insert additional delays or similar):
preconditions = [
    { "device": Voltage, "value": 1},
    { "func":   time.sleep,
      "args":   [1.0], # positional arguments
      "kwargs": {},    # keyword arguments
    },
    {"device": ClampSwitch, "value": 1},
    # or a simple call like this:
    { "func":  someCustomFunction,
    },
    ...
]
```

Returns

a single test result entry (with verdict info)

6.77.2.13 mean()

```
ttk_selftest._selftest_tests.mean (
              it )
```

6.77.2.14 resetFiu()

```
ttk_selftest._selftest_tests.resetFiu (
              hil_devices )
```

Function to reset the FIU.

This affects the iFIUCards and iFIU_CAN_FRs, whose channels will be set to the no error state.

Parameters

hil_devices	devices of the HIL system

Returns

a single test result entry.

6.77.2.15 testAnalogCurrentInput()

```
ttk_selftest._selftest_tests.testAnalogCurrentInput (
              testenv,
              device,
              num_testvalues = 5 )
```

Function to test an analog current input channel.

Parameters

testenv	test environment
device	load device whose current measurement shall be tested
num_testvalues	(opt) number of different values to be tested equally split within common range of the channel and the range of the iSRC

Returns

the testresult parameter extended by test result entries.

6.77.2.16 testAnalogOutput()

```
ttk_selftest._selftest_tests.testAnalogOutput (
              testenv,
              channel,
              ref_channel,
              num_testvalues = 5,
              gnd_offset = 0.5 )
```

Function to test an analog output channel.

Parameters

testenv	test environment
channel	analog output that shall be tested
ref_channel	iSystemReferenceCard device to be used as reference
num_testvalues	(opt) number of different values to be tested equally split within common range of both analog channels defaults to 5 values
gnd_offset	(opt) voltage to be applied to channel ground if it is not equal to the ground of the PSU that is connected to the iSRC valid range: 0-2V, defaults to 0.5V

Returns

the testresult parameter extended by test result entries.

6.77.2.17 testAnalogVoltageInput()

```
ttk_selftest._selftest_tests.testAnalogVoltageInput (
              testenv,
              device,
              num_testvalues = 5 )
```

Function to test an analog voltage input channel.

Parameters

testenv	test environment
device	analog input channel to be tested or load device whose voltage measurement shall be tested
num_testvalues	(opt) number of different values to be tested equally split within common range of the channel and the range of the iSRC (only relevant for load devices)

Returns

the testresult parameter extended by test result entries.

6.77.2.18 testCan()

```
ttk_selftest._selftest_tests.testCan (
              testenv,
              channel )
```

Function to test a CAN channel.

Parameters

testenv	test environment
channel	CAN channel that shall be tested

Returns

the testresult parameter extended by test result entries.

6.77.2.19 testDigitalInput()

```
ttk_selftest._selftest_tests.testDigitalInput (
              testenv,
              device )
```

Function to test a digital input channel.

Parameters

testenv	test environment
device	digital input channel to be tested or load device whose digital input channel(s) shall be tested

Returns

the testresult parameter extended by test result entries.

6.77.2.20 testDigitalOutput()

```
ttk_selftest._selftest_tests.testDigitalOutput (
              testenv,
              channel )
```

Function to test a digital output channel.

Parameters

testenv	test environment
channel	digital output that shall be tested

Returns

the testresult parameter extended by test result entries.

6.77.2.21 testGND()

```
ttk_selftest._selftest_tests.testGND (
              testenv,
              gnd,
              threshold = 10000,
              ignore_results = False )
```

Function to test ground connections.

The test is passed if the gnd has no connection to any other defined ground.

Parameters

testenv	test environment
gnd	gnd device which shall be tested
threshold	(opt) minimum required resistance between two ground devices in Ohm, defaults to 10k

Note

The measured resistance depends on the system configuration. Therefore the threshold has to be chosen individually for each system! ignore_results - If set to True, the test will be performed ignoring any already

available results. If set to False, the available results will be taken into account. For example, the test of gnd1 to gnd2 will not be performed anymore if the test gnd2 to gnd1 has already been performed before.

Returns

the testresult parameter extended by test result entries.

6.77.2.22 testiDigiRes()

```
ttk_selftest._selftest_tests.testiDigiRes (
              testenv,
              idigires )
```

Function to test an iDigiRes.

Parameters

testenv	test environment
idigires	iDigiRes that shall be tested

Returns

the testresult parameter extended by test result entries.

6.77.2.23 testLoad()

```
ttk_selftest._selftest_tests.testLoad (
              testenv,
              load,
              num_testvalues = 5 )
```

Function to test a load.

Parameters

testenv	test environment
load	load that shall be tested
num_testvalues	(opt) number of different values to be tested equally split within common range of loads specification and iSRC

Returns

the testresult parameter extended by test result entries.

6.77.2.24 testPowerSupply()

```
ttk_selftest._selftest_tests.testPowerSupply (
              testenv,
```

```
                power_supply,
                max_test_current = None,
                num_testvalues = 5 )
```

Function to test Power Supply voltage and current.

Parameters

testenv	test environment
power_supply	power supply that shall be tested
max_test_current	(opt) current limit during current test defaults to 18A
num_testvalues	(opt) number of different values to be tested equally split within common range of the power supply and the iSRC

Returns

the testresult parameter extended by test result entries.

6.77.2.25 testPowerSupplyCurrent()

```
ttk_selftest._selftest_tests.testPowerSupplyCurrent (
                testenv,
                power_supply,
                max_test_current = None,
                num_testvalues = 5 )
```

Function to test power supply current control and measurement.

Parameters

testenv	test environment
power_supply	power supply that shall be tested
max_test_current	(opt) current limit during current test defaults to 18A
num_testvalues	(opt) number of different values to be tested equally split within common range of the power supply and the iSRC

Returns

the testresult parameter extended by test result entries.

6.77.2.26 testPowerSupplyVoltage()

```
ttk_selftest._selftest_tests.testPowerSupplyVoltage (
                testenv,
                power_supply,
                num_testvalues = 5 )
```

Function to test power supply voltage control and measurement.

Parameters

testenv	test environment
power_supply	power supply that shall be tested
num_testvalues	(opt) number of different values to be tested equally split within common range of the power supply and the iSRC

Returns

the testresult parameter extended by test result entries.

6.77.2.27 testPwmInput()

```
ttk_selftest._selftest_tests.testPwmInput (
              testenv,
              device,
              num_testvalues = 5 )
```

Function to test a PWM input channel.

Parameters

testenv	test environment
device	PWM input channel to be tested or load device whose PWM input channel(s) shall be tested
num_testvalues	(opt) number of different values to be tested equally split within common range of both frequency and duty cycle

Returns

the testresult parameter extended by test result entries.

6.77.2.28 testPwmOutput()

```
ttk_selftest._selftest_tests.testPwmOutput (
              testenv,
              channel,
              num_testvalues = 5 )
```

Function to test an PWM output channel.

Parameters

testenv	test environment
channel	PWM output that shall be tested
num_testvalues	(opt) number of different values to be tested equally split within common range of both frequency and duty cycle

Returns

the testresult parameter extended by test result entries.

6.78 ttk_selftest._xml_handling Namespace Reference

Read and Write methods for self test at hil systems.

Functions

- writeDefaultXml (filepath, filename, prjname="ProjectName", hilinfo={}, boardinfo={})
  Write a default xml file in specified structure.

Variables

- dict a = {1: "0215-02686", 2: "0215-3386", 3: "0555-02686"}
- current_path = os.path.realpath(inspect.currentframe().f_code.co_filename)
- ttk_selftest_path = os.path.dirname(current_path)
- lib_path = os.path.dirname(ttk_selftest_path)
- python_path = os.path.dirname(lib_path)
- project_path = os.path.dirname(python_path)
- file_path = os.path.join(project_path, "Model", "_Standardmodel")
- str file_name = "FMC4008A_Test.xml"
- filepath
- filename
- prjname
- hilinfo
- boardinfo

6.78.1 Detailed Description

Read and Write methods for self test at hil systems.

standard libraries ##

6.78.2 Function Documentation

6.78.2.1 writeDefaultXml()

```
ttk_selftest._xml_handling.writeDefaultXml (
              filepath,
              filename,
              prjname = "ProjectName",
              hilinfo = {},
              boardinfo = {} )
```

Write a default xml file in specified structure.

Attention

In case of file is defined, it will be overwritten.

Parameters

path	path xml file will generated in
name	name of xml file which will generated
prjname	project name to specify tag attribute
hilinfo	information about number of hil's and serial number type: <dict>-type structure: {HIL-Number: Serial-Number, ...} HIL-Number: <int>-type Serial-Number: <string>-type

example: {1:"0125-2222", 2:"0125-2244", 3:"0125-4444"}

```
boardinfo - information about used card and her configuration
board name should be have a unambiguous specification
type: <dict>-type
structure: {<Card-Name>: {"Port": <Port Number>,
                          "Type": <Analog or Digital>,
                          "Subtype": <>,
                          "Direction": {
                              "In": <number of input pins> (<int>-type),
                              "Out": <number of output pins> (<int>-type),
                              }
                          }, ...}
```

example: {"TPMC530_A": {

```
"Port": "0",
"Type": "Analog",
"Subtype": "abc",
"Direction": {
    "In": 16,
    "Out": 8,
    }}, "TPMC530_B": ...}
```

6.78.3 Variable Documentation

6.78.3.1 a

```
dict ttk_selftest._xml_handling.a = {1:  "0215-02686", 2:  "0215-3386", 3:  "0555-02686"}
```

6.78.3.2 boardinfo

```
ttk_selftest._xml_handling.boardinfo
```

6.78.3.3 current_path

```
ttk_selftest._xml_handling.current_path = os.path.realpath(inspect.currentframe().f_code.co_↩
filename)
```

6.78.3.4 file_name

```
str ttk_selftest._xml_handling.file_name = "FMC4008A_Test.xml"
```

6.78.3.5 file_path

```
ttk_selftest._xml_handling.file_path = os.path.join(project_path, "Model", "_Standardmodel")
```

6.78.3.6 filename

```
ttk_selftest._xml_handling.filename
```

6.78.3.7 filepath

```
ttk_selftest._xml_handling.filepath
```

6.78.3.8 hilinfo

```
ttk_selftest._xml_handling.hilinfo
```

6.78.3.9 lib_path

```
ttk_selftest._xml_handling.lib_path = os.path.dirname(ttk_selftest_path)
```

6.78.3.10 prjname

```
ttk_selftest._xml_handling.prjname
```

6.78.3.11 project_path

```
ttk_selftest._xml_handling.project_path = os.path.dirname(python_path)
```

6.78.3.12 python_path

```
ttk_selftest._xml_handling.python_path = os.path.dirname(lib_path)
```

6.78.3.13 ttk_selftest_path

```
ttk_selftest._xml_handling.ttk_selftest_path = os.path.dirname(current_path)
```

6.79 ttk_selftest.adjustment_calculation Namespace Reference

Interface wrapper for adjustment calculation class.

Classes

- class AdjustmentCalculation

  Class for adjustment one hil channel which is currently in state self test and adjustment.

6.79.1 Detailed Description

Interface wrapper for adjustment calculation class.

6.80 ttk_selftest.devices_base Namespace Reference

Interface wrapper for HIL device descriptions.

6.80.1 Detailed Description

Interface wrapper for HIL device descriptions.

Imports

6.81 ttk_selftest.hil_devices Namespace Reference

Access to hil devices (dSPACE and GammaV).

Namespaces

- namespace _analog_devices

  Container classes for analog device descriptions.
- namespace _bus_devices

  Container classes for bus device descriptions.
- namespace _components

  Container classes for several hil component descriptions.
- namespace _devices_base

  Container classes for HIL device descriptions.
- namespace _digital_devices

  Container classes for digital device descriptions.
- namespace _fiu_devices

  Container classes for fiu device descriptions.
- namespace _selftest_devices

  Container classes for selftest device descriptions.
- namespace analog_devices

  Interface wrapper for analog device descriptions.
- namespace bus_devices

  Interface wrapper for bus device descriptions.
- namespace components

  Wrapper classes for several hil component descriptions.
- namespace devices_base
- namespace digital_devices

  Interface wrapper for digital device descriptions.
- namespace fiu_devices

  Interface wrapper for fiu device descriptions.
- namespace selftest_devices

  Interface wrapper for selftest device descriptions.

6.81.1 Detailed Description

Access to hil devices (dSPACE and GammaV).

6.82 ttk_selftest.hil_devices._analog_devices Namespace Reference

Container classes for analog device descriptions.

Classes

- class AnalogChannel

  Base analog channel container for a testable analog channel and relevant associated variables/values.
- class AnalogInput

  Base analog input container for a testable analog input and relevant associated variables/values.
- class AnalogOutput

  Base analog output container for a testable analog output and relevant associated variables/values.

6.82.1 Detailed Description

Container classes for analog device descriptions.

Imports

6.83 ttk_selftest.hil_devices._bus_devices Namespace Reference

Container classes for bus device descriptions.

Classes

- class CanChannel

  Base CAN channel container for a testable CAN channel and relevant associated variables/values.

6.83.1 Detailed Description

Container classes for bus device descriptions.

Imports

6.84 ttk_selftest.hil_devices._components Namespace Reference

Container classes for several hil component descriptions.

Classes

- class iDigiRes

  Base iDigiRes container for a testable iDigiRes device and relevant associated variables/values.
- class Load

  Base load container for a testable load and relevant associated variables/values.
- class PowerSupply

  Base power supply container for a testable power supply and relevant associated variables/values.

6.84.1 Detailed Description

Container classes for several hil component descriptions.

Imports

6.85 ttk_selftest.hil_devices._devices_base Namespace Reference

Container classes for HIL device descriptions.

Classes

- class DeviceContainer

  Base class for HIL device containers.
- class GND

  Base ground signal container for a testable ground signal and relevant associated variables/values.
- class HilDevice

  Base HIL device container for a testable HIL device and relevant associated variables/values.

Functions

- isGamma (context)

  Function to detect if the system context is a GammaV environment.
- isDspace (context)

  Function to detect if the system context is a dSPACE environment.
- isIdaTestware (context)

  Function to detect if the system context is an iDA Testware environment.
- hilVarFactory (context, identifier, ∗args, ∗∗kwargs)

  Factory for "HilVars", decides actual implementation to use based on supplied context (i.e.
- _buildCanMessagePath (context, base_path, msg_name, direction)

  Function to build a CAN message path, decides actual implementation to use based on supplied context and message direction based on supplied direction.
- _buildCanSignalPath (context, base_path, msg_name, sig_name, direction)

  Function to build a CAN signal path, decides actual implementation to use based on supplied context and signal direction based on supplied direction.
- _readHilVarFactory (context, identifier, variable=None, ∗args, ∗∗kwargs)

  Factory for "ReadHilVars" (HilVars that will be read only e.g.
- _writeHilVarFactory (context, identifier, variable=None, ∗args, ∗∗kwargs)

  Factory for "WriteHilVars" (HilVars that will be read/write e.g.

- _rxCanBusMessageFactory (context, base_path, msg_name, ∗args, ∗∗kwargs)

  Factory for "RxCanBusMessages", decides actual implementation to use based on supplied context.
- _txCanBusMessageFactory (context, base_path, msg_name, ∗args, ∗∗kwargs)

  Factory for "TxCanBusMessages", decides actual implementation to use based on supplied context.
- _rxCanBusSignalFactory (context, base_path, msg_name, sig_name, ∗args, ∗∗kwargs)

  Factory for "RxCanBusSignals", decides actual implementation to use based on supplied context.
- _txCanBusSignalFactory (context, base_path, msg_name, sig_name, ∗args, ∗∗kwargs)

  Factory for "TxCanBusSignals", decides actual implementation to use based on supplied context.

Variables

- str DSPACE_SEPARATOR = "/"
- XilTestbench = OfflineXilTestbench
- str GAMMA_SEPARATOR = "."
- gaapi = _GaApiStub
- str IDA_SEPARATOR = "."
- IdaTestware = OfflineIdaTestware

6.85.1 Detailed Description

Container classes for HIL device descriptions.

Imports

6.85.2 Function Documentation

6.85.2.1 _buildCanMessagePath()

```
ttk_selftest.hil_devices._devices_base._buildCanMessagePath (
              context,
              base_path,
              msg_name,
              direction )  [protected]
```

Function to build a CAN message path, decides actual implementation to use based on supplied context and message direction based on supplied direction.

Parameters

context	context for variable (either an rtplib.Appl, xil_api.XilTestbench instance or gaapi reference)
base_path	base path of the CAN bus e.g. "BusSystems/CAN/<CAN_NAME>" for dSPACE systems
msg_name	name of the CAN message
direction	direction of the CAN message, can be either "RX" for messages sent from HIL to DUT or "TX" for messages sent from DUT to HIL

Returns

the CAN message path as string

6.85.2.2 _buildCanSignalPath()

```
ttk_selftest.hil_devices._devices_base._buildCanSignalPath (
              context,
              base_path,
              msg_name,
              sig_name,
              direction )  [protected]
```

Function to build a CAN signal path, decides actual implementation to use based on supplied context and signal direction based on supplied direction.

Parameters

context	context for variable (either an rtplib.Appl, xil_api.XilTestbench instance or gaapi reference)
base_path	base path of the CAN bus e.g. "BusSystems/CAN/<CAN_NAME>" for dSPACE systems
msg_name	name of the CAN message
sig_name	name of the CAN signal
direction	direction of the CAN signal, can be either "RX" for signals sent from HIL to DUT or "TX" for signals sent from DUT to HIL

Returns

the CAN signal path as string

6.85.2.3 _readHilVarFactory()

```
ttk_selftest.hil_devices._devices_base._readHilVarFactory (
              context,
              identifier,
              variable = None,
             * args,
             ** kwargs )  [protected]
```

Factory for "ReadHilVars" (HilVars that will be read only e.g.

input signal values), decides actual implementation to use based on supplied context. Furthermore, the necessary suffix will be added to the identifier according to the supplied context.

Parameters

context	context for variable (either an rtplib.Appl, xil_api.XilTestbench instance or gaapi reference)
identifier	full identifier or base path for the variable
variable	(opt) variable name that will be added to identifier separated by context specific separator defaults to None
∗*args*	additional positional arguments to be passed on to Var implementation
∗∗*kwargs*	additional keyword arguments to be passed on to Var implementation

Returns

an Var instance, either based on rtplib, xil_api or gaapi

6.85.2.4 _rxCanBusMessageFactory()

```
ttk_selftest.hil_devices._devices_base._rxCanBusMessageFactory (
              context,
              base_path,
              msg_name,
             * args,
             ** kwargs )  [protected]
```

Factory for "RxCanBusMessages", decides actual implementation to use based on supplied context.

Parameters

context	context for CanBusMessage (either an rtplib.Appl, xil_api.XilTestbench instance or gaapi reference)
base_path	base path of the CAN bus e.g. "BusSystems/CAN/<CAN_NAME>" for dSPACE systems
msg_name	name of the CAN message
∗*args*	additional positional arguments to be passed on to Var implementation
∗∗*kwargs*	additional keyword arguments to be passed on to Var implementation

Returns

an RxCanBusMessage instance, either based on dSPACE or Gamma implementation.

6.85.2.5 _rxCanBusSignalFactory()

```
ttk_selftest.hil_devices._devices_base._rxCanBusSignalFactory (
              context,
              base_path,
              msg_name,
              sig_name,
             * args,
             ** kwargs )  [protected]
```

Factory for "RxCanBusSignals", decides actual implementation to use based on supplied context.

Parameters

context	context for variable (either an rtplib.Appl, xil_api.XilTestbench instance or gaapi reference)
base_path	base path of the CAN bus e.g. "BusSystems/CAN/<CAN_NAME>" for dSPACE systems
msg_name	name of the CAN message
sig_name	name of the CAN signal
∗*args*	additional positional arguments to be passed on to Var implementation
∗∗*kwargs*	additional keyword arguments to be passed on to Var implementation

Returns

an RxCanBusSignal instance, either based on dSPACE or Gamma implementation.

6.85.2.6 _txCanBusMessageFactory()

```
ttk_selftest.hil_devices._devices_base._txCanBusMessageFactory (
              context,
              base_path,
              msg_name,
             * args,
             ** kwargs )  [protected]
```

Factory for "TxCanBusMessages", decides actual implementation to use based on supplied context.

Parameters

context	context for CanBusMessage (either an rtplib.Appl, xil_api.XilTestbench instance or gaapi reference)
base_path	base path of the CAN bus e.g. "BusSystems/CAN/<CAN_NAME>" for dSPACE systems
msg_name	name of the CAN message
∗*args*	additional positional arguments to be passed on to Var implementation
∗∗*kwargs*	additional keyword arguments to be passed on to Var implementation

Returns

an TxCanBusMessage instance, either based on dSPACE or Gamma implementation.

6.85.2.7 _txCanBusSignalFactory()

```
ttk_selftest.hil_devices._devices_base._txCanBusSignalFactory (
              context,
              base_path,
              msg_name,
              sig_name,
             * args,
             ** kwargs )  [protected]
```

Factory for "TxCanBusSignals", decides actual implementation to use based on supplied context.

Parameters

context	context for variable (either an rtplib.Appl, xil_api.XilTestbench instance or gaapi reference)
base_path	base path of the CAN bus e.g. "BusSystems/CAN/<CAN_NAME>" for dSPACE systems
msg_name	name of the CAN message
sig_name	name of the CAN signal
∗*args*	additional positional arguments to be passed on to Var implementation
∗∗*kwargs*	additional keyword arguments to be passed on to Var implementation

Returns

an TxCanBusSignal instance, either based on dSPACE or Gamma implementation.

6.85.2.8 _writeHilVarFactory()

```
ttk_selftest.hil_devices._devices_base._writeHilVarFactory (
               context,
               identifier,
               variable = None,
              * args,
              ** kwargs )  [protected]
```

Factory for "WriteHilVars" (HilVars that will be read/write e.g.

output signal values or gain/offset values), decides actual implementation to use based on supplied context. Furthermore, the necessary suffix will be added to the identifier according to the supplied context.

Parameters

context	context for variable (either an rtplib.Appl, xil_api.XilTestbench instance or gaapi reference)
identifier	full identifier or base path for the variable
variable	(opt) variable name that will be added to identifier separated by context specific separator defaults to None
∗*args*	additional positional arguments to be passed on to Var implementation
∗∗*kwargs*	additional keyword arguments to be passed on to Var implementation

Returns

an Var instance, either based on rtplib, xil_api or gaapi

6.85.2.9 hilVarFactory()

```
ttk_selftest.hil_devices._devices_base.hilVarFactory (
               context,
               identifier,
              * args,
              ** kwargs )
```

Factory for "HilVars", decides actual implementation to use based on supplied context (i.e.

rtplib-, xil-testbench-based or Gamma instance).

Parameters

context	context for variable (either an rtplib.Appl or xil_api.XilTestbench or gamma_api.Gamma instance)
identifier	identifier/model path for variable
∗*args*	additional positional arguments to be passed on to Var implementation
∗∗*kwargs*	additional keyword arguments to be passed on to Var implementation

Returns

an Var instance, either based on rtplib or xil_api or gamma_api

6.85.2.10 isDspace()

```
ttk_selftest.hil_devices._devices_base.isDspace (
              context )
```

Function to detect if the system context is a dSPACE environment.

Parameters

context	system context (either an rtplib.Appl or xil_api.XilTestbench or gamma_api.Gamma or iDA instance)

Returns

True if the context is a dSPACE instance or False if not

6.85.2.11 isGamma()

```
ttk_selftest.hil_devices._devices_base.isGamma (
              context )
```

Function to detect if the system context is a GammaV environment.

Parameters

context	system context (either an rtplib.Appl or xil_api.XilTestbench or gamma_api.Gamma or iDA instance)

Returns

True if the context is a Gamma instance or False if not

6.85.2.12 isIdaTestware()

```
ttk_selftest.hil_devices._devices_base.isIdaTestware (
              context )
```

Function to detect if the system context is an iDA Testware environment.

Parameters

context	system context (IdaTestware, "real" or offline stub)

Returns

True if the context is an IdaTestware instance, otherwise False

6.85.3 Variable Documentation

6.85.3.1 DSPACE_SEPARATOR

```
str ttk_selftest.hil_devices._devices_base.DSPACE_SEPARATOR = "/"
```

6.85.3.2 gaapi

```
ttk_selftest.hil_devices._devices_base.gaapi = _GaApiStub
```

6.85.3.3 GAMMA_SEPARATOR

```
str ttk_selftest.hil_devices._devices_base.GAMMA_SEPARATOR = "."
```

6.85.3.4 IDA_SEPARATOR

```
str ttk_selftest.hil_devices._devices_base.IDA_SEPARATOR = "."
```

6.85.3.5 IdaTestware

```
ttk_selftest.hil_devices._devices_base.IdaTestware = OfflineIdaTestware
```

6.85.3.6 XilTestbench

```
ttk_selftest.hil_devices._devices_base.XilTestbench = OfflineXilTestbench
```

6.86 ttk_selftest.hil_devices._digital_devices Namespace Reference

Container classes for digital device descriptions.

Classes

- class C2DOutput

  Base iCAN2DIG output container for a testable iCAN2DIG output and relevant associated variables/values.
- class DigitalChannel

  Base digital channel container for a testable digital channel and relevant associated variables/values.
- class DigitalInput

  Base digital input container for a testable digital input and relevant associated variables/values.
- class DigitalOutput

  Base digital output container for a testable digital output and relevant associated variables/values.
- class PwmChannel

  Base PWM channel container for a testable PWM channel and relevant associated variables/values.
- class PwmInput

  Base PWM input container for a testable PWM input and relevant associated variables/values.
- class PwmOutput

  Base PWM output container for a testable PWM output and relevant associated variables/values.
- class Switch

  Base switch container for a testable switch and relevant associated variables/values.

Functions

- _isCanRxBusSignal (identifier)

  Function to identify CAN RX bus signals (e.g.

6.86.1 Detailed Description

Container classes for digital device descriptions.

Imports

6.86.2 Function Documentation

6.86.2.1 _isCanRxBusSignal()

```
ttk_selftest.hil_devices._digital_devices._isCanRxBusSignal (
              identifier )  [protected]
```

Function to identify CAN RX bus signals (e.g.

dSpaceRxCanBusSignal or gammaRxCanBusSignal)

Parameters

identifier	variable/signal/identifier to be checked

Returns

True if identifier is CAN Rx bus signal False if identifier is not CAN Rx bus signal

6.87 ttk_selftest.hil_devices._fiu_devices Namespace Reference

Container classes for fiu device descriptions.

Classes

- class FiuCanFrChannel

  Base FIU CAN FR channel container.
- class FiuChannel

  Base FIU channel container.

Functions

- _checkValueAndRange (label, value, upper_limit)

  Function to check if value is one of the integer types (i.e.

6.87.1 Detailed Description

Container classes for fiu device descriptions.

Imports

6.87.2 Function Documentation

6.87.2.1 _checkValueAndRange()

```
ttk_selftest.hil_devices._fiu_devices._checkValueAndRange (
              label,
              value,
              upper_limit )  [protected]
```

Function to check if value is one of the integer types (i.e.

long or int) and within valid range (positive and smaller/equal to upper limit). Raises an InputValueTypeError if value is not integer type. Raises an RangeError if value is not in valid range.

Parameters

label	label/name of checked variable/parameter
value	value of checked variable/parameter
upper_limit	upper limit for checked variable/parameter

Note

function has been derived from _checkNumeric of package ttk_checks._basic_checks

6.88 ttk_selftest.hil_devices._selftest_devices Namespace Reference

Container classes for selftest device descriptions.

Classes

- class _SrcMode

  Base class for modes of the iSRC.
- class SRC

  Base iSystemReferenceCard (iSRC) container.

6.88.1 Detailed Description

Container classes for selftest device descriptions.

Imports

6.89 ttk_selftest.hil_devices.analog_devices Namespace Reference

Interface wrapper for analog device descriptions.

Classes

- class AnalogChannel

  Base analog channel container for a testable analog channel and relevant associated variables/values.
- class AnalogInput

  Base analog input container for a testable analog input and relevant associated variables/values.
- class AnalogOutput

  Base analog output container for a testable analog output and relevant associated variables/values.

6.89.1 Detailed Description

Interface wrapper for analog device descriptions.

Imports

6.90 ttk_selftest.hil_devices.bus_devices Namespace Reference

Interface wrapper for bus device descriptions.

Classes

- class CanChannel

  Base CAN channel container for a testable CAN channel and relevant associated variables/values.

6.90.1 Detailed Description

Interface wrapper for bus device descriptions.

Imports

6.91 ttk_selftest.hil_devices.components Namespace Reference

Wrapper classes for several hil component descriptions.

Classes

- class iDigiRes

  Base iDigiRes container for a testable iDigiRes device and relevant associated variables/values.
- class Load

  Base load container for a testable load and relevant associated variables/values.
- class PowerSupply

  Base power supply container for a testable power supply and relevant associated variables/values.

6.91.1 Detailed Description

Wrapper classes for several hil component descriptions.

Imports

6.92 ttk_selftest.hil_devices.devices_base Namespace Reference

Classes

- class DeviceContainer

  Base class for HIL device containers.
- class GND

  Base ground signal container for a testable ground signal and relevant associated variables/values.
- class HilDevice

  Base HIL device container for a testable HIL device and relevant associated variables/values.

6.93 ttk_selftest.hil_devices.digital_devices Namespace Reference

Interface wrapper for digital device descriptions.

Classes

- class C2DOutput

  Base iCAN2DIG output container for a testable iCAN2DIG output and relevant associated variables/values.
- class DigitalChannel

  Base digital channel container for a testable digital channel and relevant associated variables/values.
- class DigitalInput

  Base digital input container for a testable digital input and relevant associated variables/values.
- class DigitalOutput

  Base digital output container for a testable digital output and relevant associated variables/values.
- class PwmChannel

  Base PWM channel container for a testable PWM channel and relevant associated variables/values.
- class PwmInput

  Base PWM input container for a testable PWM input and relevant associated variables/values.
- class PwmOutput

  Base PWM output container for a testable PWM output and relevant associated variables/values.
- class Switch

  Base switch container for a testable switch and relevant associated variables/values.

6.93.1 Detailed Description

Interface wrapper for digital device descriptions.

Imports

6.94 ttk_selftest.hil_devices.fiu_devices Namespace Reference

Interface wrapper for fiu device descriptions.

Classes

- class FiuCanFrChannel

  Base FIU CAN FR channel container.
- class FiuChannel

  Base FIU channel container.

6.94.1 Detailed Description

Interface wrapper for fiu device descriptions.

6.95 ttk_selftest.hil_devices.selftest_devices Namespace Reference

Interface wrapper for selftest device descriptions.

Classes

- class SRC

  Base iSystemReferenceCard (iSRC) container.

6.95.1 Detailed Description

Interface wrapper for selftest device descriptions.

Imports

6.96 ttk_selftest.selftest_tests Namespace Reference

Wrapper for self test test functions This serves as "interface" to the precompiled module in delivery to enable code-completion in PyDev.

Functions

- testGND (testenv, gnd, threshold=10000, ignore_results=False)

  Function to test ground connections.
- testAnalogOutput (testenv, channel, ref_channel, num_testvalues=5, gnd_offset=0.5)

  Function to test an analog output channel.
- testAnalogVoltageInput (testenv, device, num_testvalues=5)

  Function to test an analog voltage input channel.
- testAnalogCurrentInput (testenv, device, num_testvalues=5)

  Function to test an analog current input channel.
- testPowerSupplyVoltage (testenv, power_supply, num_testvalues=5)

  Function to test power supply voltage control and measurement.
- testPowerSupplyCurrent (testenv, power_supply, max_test_current=None, num_testvalues=5)

  Function to test power supply current control and measurement.
- testPowerSupply (testenv, power_supply, max_test_current=None, num_testvalues=5)

  Function to test Power Supply voltage and current.
- testPwmInput (testenv, device, num_testvalues=5)

  Function to test a PWM input channel.
- testLoad (testenv, load, num_testvalues=5)

  Function to test a load.
- testPwmOutput (testenv, channel, num_testvalues=5)

  Function to test an PWM output channel.
- testCan (testenv, channel)

  Function to test a CAN channel.
- testDigitalInput (testenv, device)

  Function to test a digital input channel.
- testDigitalOutput (testenv, channel)

  Function to test a digital output channel.
- testiDigiRes (testenv, idigires)

  Function to test an iDigiRes.
- resetFiu (hil_devices)

  Function to reset the FIU.

6.96.1 Detailed Description

Wrapper for self test test functions This serves as "interface" to the precompiled module in delivery to enable code-completion in PyDev.

6.96.2 Function Documentation

6.96.2.1 resetFiu()

```
ttk_selftest.selftest_tests.resetFiu (
              hil_devices )
```

Function to reset the FIU.

This affects the iFIUCards and iFIU_CAN_FRs, whose channels will be set to the no error state.

Parameters

hil_devices	devices of the HIL system

Returns

a single test result entry.

6.96.2.2 testAnalogCurrentInput()

```
ttk_selftest.selftest_tests.testAnalogCurrentInput (
              testenv,
              device,
              num_testvalues = 5 )
```

Function to test an analog current input channel.

Parameters

testenv	test environment
device	load device whose current measurement shall be tested
num_testvalues	(opt) number of different values to be tested equally split within common range of the channel and the range of the iSRC

Returns

the testresult parameter extended by test result entries.

6.96.2.3 testAnalogOutput()

```
ttk_selftest.selftest_tests.testAnalogOutput (
              testenv,
```

```
                channel,
                ref_channel,
                num_testvalues = 5,
                gnd_offset = 0.5 )
```

Function to test an analog output channel.

Parameters

testenv	test environment
channel	analog output that shall be tested
ref_channel	iSystemReferenceCard device to be used as reference
num_testvalues	(opt) number of different values to be tested equally split within common range of both analog channels defaults to 5 values
gnd_offset	(opt) voltage to be applied to channel ground if it is not equal to the ground of the PSU that is connected to the iSRC valid range: 0-2V, defaults to 0.5V

Returns

the testresult parameter extended by test result entries.

6.96.2.4 testAnalogVoltageInput()

```
ttk_selftest.selftest_tests.testAnalogVoltageInput (
                testenv,
                device,
                num_testvalues = 5 )
```

Function to test an analog voltage input channel.

Parameters

testenv	test environment
device	analog input channel to be tested or load device whose voltage measurement shall be tested
num_testvalues	(opt) number of different values to be tested equally split within common range of the channel and the range of the iSRC (only relevant for load devices)

Returns

the testresult parameter extended by test result entries.

6.96.2.5 testCan()

```
ttk_selftest.selftest_tests.testCan (
                testenv,
                channel )
```

Function to test a CAN channel.

Parameters

testenv	test environment
channel	CAN channel that shall be tested

Returns

the testresult parameter extended by test result entries.

6.96.2.6 testDigitalInput()

```
ttk_selftest.selftest_tests.testDigitalInput (
              testenv,
              device )
```

Function to test a digital input channel.

Parameters

testenv	test environment
device	digital input channel to be tested or load device whose digital input channel(s) shall be tested

Returns

the testresult parameter extended by test result entries.

6.96.2.7 testDigitalOutput()

```
ttk_selftest.selftest_tests.testDigitalOutput (
              testenv,
              channel )
```

Function to test a digital output channel.

Parameters

testenv	test environment
channel	digital output that shall be tested

Returns

the testresult parameter extended by test result entries.

6.96.2.8 testGND()

```
ttk_selftest.selftest_tests.testGND (
              testenv,
```

```
                gnd,
                threshold = 10000,
                ignore_results = False )
```

Function to test ground connections.

The test is passed if the gnd has no connection to any other defined ground.

Parameters

testenv	test environment
gnd	gnd device which shall be tested
threshold	(opt) minimum required resistance between two ground devices in Ohm, defaults to 10k

Note

The measured resistance depends on the system configuration. Therefore the threshold has to be chosen individually for each system! ignore_results - If set to True, the test will be performed ignoring any already available results. If set to False, the available results will be taken into account. For example, the test of gnd1 to gnd2 will not be performed anymore if the test gnd2 to gnd1 has already been performed before.

Returns

the testresult parameter extended by test result entries.

6.96.2.9 testiDigiRes()

```
ttk_selftest.selftest_tests.testiDigiRes (
                testenv,
                idigires )
```

Function to test an iDigiRes.

Parameters

testenv	test environment
idigires	iDigiRes that shall be tested

Returns

the testresult parameter extended by test result entries.

6.96.2.10 testLoad()

```
ttk_selftest.selftest_tests.testLoad (
                testenv,
                load,
                num_testvalues = 5 )
```

Function to test a load.

Parameters

testenv	test environment
load	load that shall be tested
num_testvalues	(opt) number of different values to be tested equally split within common range of loads specification and iSRC

Returns

the testresult parameter extended by test result entries.

6.96.2.11 testPowerSupply()

```
ttk_selftest.selftest_tests.testPowerSupply (
              testenv,
              power_supply,
              max_test_current = None,
              num_testvalues = 5 )
```

Function to test Power Supply voltage and current.

Parameters

testenv	test environment
power_supply	power supply that shall be tested
max_test_current	(opt) current limit during current test defaults to 18A
num_testvalues	(opt) number of different values to be tested equally split within common range of the power supply and the iSRC

Returns

the testresult parameter extended by test result entries.

6.96.2.12 testPowerSupplyCurrent()

```
ttk_selftest.selftest_tests.testPowerSupplyCurrent (
              testenv,
              power_supply,
              max_test_current = None,
              num_testvalues = 5 )
```

Function to test power supply current control and measurement.

Parameters

testenv	test environment
power_supply	power supply that shall be tested
max_test_current	(opt) current limit during current test defaults to 18A
num_testvalues	(opt) number of different values to be tested equally split within common range of the power supply and the iSRC

Returns

the testresult parameter extended by test result entries.

6.96.2.13 testPowerSupplyVoltage()

```
ttk_selftest.selftest_tests.testPowerSupplyVoltage (
              testenv,
              power_supply,
              num_testvalues = 5 )
```

Function to test power supply voltage control and measurement.

Parameters

testenv	test environment
power_supply	power supply that shall be tested
num_testvalues	(opt) number of different values to be tested equally split within common range of the power supply and the iSRC

Returns

the testresult parameter extended by test result entries.

6.96.2.14 testPwmInput()

```
ttk_selftest.selftest_tests.testPwmInput (
              testenv,
              device,
              num_testvalues = 5 )
```

Function to test a PWM input channel.

Parameters

testenv	test environment
device	PWM input channel to be tested or load device whose PWM input channel(s) shall be tested
num_testvalues	(opt) number of different values to be tested equally split within common range of both frequency and duty cycle

Returns

the testresult parameter extended by test result entries.

6.96.2.15 testPwmOutput()

```
ttk_selftest.selftest_tests.testPwmOutput (
              testenv,
```

```
    channel,
    num_testvalues = 5 )
```

Function to test an PWM output channel.

Parameters

testenv	test environment
channel	PWM output that shall be tested
num_testvalues	(opt) number of different values to be tested equally split within common range of both frequency and duty cycle

Returns

the testresult parameter extended by test result entries.

6.97 ttk_tools Namespace Reference

Modules for tool access.

Namespaces

- namespace _process_utils

  Utilities for handling win32 (and x64) processes.
- namespace _win32_utils

  Utility functions for use on Windows systems.
- namespace can_hal

  Access to iSyst Hardware Abstraction Layer for CAN interfaces.
- namespace dspace

  Access to dSPACE (dSPACE GmbH) tools and interfaces.
- namespace isyst

  Access to iSyst Intelligente Systeme GmbH tools and interfaces.
- namespace ni

  Access to NI (National Instruments) tools and interfaces, primarily LabView.
- namespace offline_data_provider

  Data provider for dummy/offline/mock implementations of tool access modules (like ccp/xcp or real time platform/model data).
- namespace process_utils

  Interface to utilities for handling win32 (and x64) processes in ttk_tools._process_utils.
- namespace pyocd

  Wrapper for using PyOCD (https://pyocd.io/) as a data source for HIL tests.
- namespace rst

  Access to RST (RST Industrie Automation GmbH) tools and interfaces.
- namespace siemens

  Access to SIEMENS automation technology tools and interfaces.
- namespace vector

  Access to Vector (Vector Informatik GmbH) tools and interfaces.
- namespace win32_utils

  Interface wrapper for Windows systems utility functions, see ttk_tools._win32_utils.

6.97.1 Detailed Description

Modules for tool access.

6.98 ttk_tools._process_utils Namespace Reference

Utilities for handling win32 (and x64) processes.

Functions

- killProcess (target_names, verbosity=1)

  Kill all processes that match the given target_name(s) (using kernel32).
- getCurrentProcessNames (normalize_case=True)

  Get image names of currently active processes (that is, the executable file names).
- getCurrentProcessPaths (normalize_case=True, file_names_only=False)

  Get image names/paths of currently active processes (that is, names or paths of the executable files).
- isProcessActive (process_name, ignore_case=True)

  Check whether at least one process using the supplied executable name is currently running.
- killProcess2 (image_name)

  Kill all processes that match the given image_name using the windows "taskkill" command running in a sub process.
- getCurrentPIDs ()

  Get a list of PIDs of running processes, similar to the output of win32process.EnumProcesses() (which was not yet available in older Python versions).
- killByPID (pid)

  Kill the process with the supplied ID (as long as the user has sufficient access rights).

Variables

- int PROCESS_TERMINATE = 0x0001
- int PROCESS_VM_READ = 0x0010
- int PROCESS_QUERY_INFORMATION = 0x0400
- int PROCESS_QUERY_LIMITED_INFORMATION = 0x1000

6.98.1 Detailed Description

Utilities for handling win32 (and x64) processes.

See also interface wrapper in ttk_tools.process_utils.

It appears that "handling" processes mostly seems to boil down to "killing" them, though.

6.98.2 Function Documentation

6.98.2.1 getCurrentPIDs()

```
ttk_tools._process_utils.getCurrentPIDs ( )
```

Get a list of PIDs of running processes, similar to the output of win32process.EnumProcesses() (which was not yet available in older Python versions).

Directly queries psapi.EnumProcesses.

Returns

a list of PID values.

6.98.2.2 getCurrentProcessNames()

```
ttk_tools._process_utils.getCurrentProcessNames (
              normalize_case = True )
```

Get image names of currently active processes (that is, the executable file names).

Parameters

normalize_case	cast process names to lowercase (as windows is not case sensitive); defaults to True

Returns

a dictionary {<process_name>: <number of active processes>, ...}

6.98.2.3 getCurrentProcessPaths()

```
ttk_tools._process_utils.getCurrentProcessPaths (
              normalize_case = True,
              file_names_only = False )
```

Get image names/paths of currently active processes (that is, names or paths of the executable files).

Limitation:

The used QueryFullProcessImageName is only supported since Windows Vista

Parameters

normalize_case	cast process names/paths to lowercase (as windows is not case sensitive); defaults to True
file_names_only	True: return only base names (e.g. "notepad.exe") False: return full file paths

Returns

a dictionary {<process name or path>: <number of active processes>, ...}

6.98.2.4 isProcessActive()

```
ttk_tools._process_utils.isProcessActive (
            process_name,
            ignore_case = True )
```

Check whether at least one process using the supplied executable name is currently running.

Parameters

process_name	name of process to check
ignore_case	True: ignore case of process_name

Returns

True if a matching process is running, otherwise False.

6.98.2.5 killByPID()

```
ttk_tools._process_utils.killByPID (
            pid )
```

Kill the process with the supplied ID (as long as the user has sufficient access rights).

Parameters

pid	process ID of the process to be killed

Returns

the return status of `TerminateProcess` (non-zero on success)

6.98.2.6 killProcess()

```
ttk_tools._process_utils.killProcess (
            target_names,
            verbosity = 1 )
```

Kill all processes that match the given target_name(s) (using kernel32).

Note that target may be a single name or list of names.

Parameters

target_names	a single name or a list of names of processes to be killed. If a process name contains unicode characters, it should be supplied as unicode string.
verbosity	verbosity of status output 0: (mostly) silent 1: show names/pids of killed processes 2: also show process access warnings

Returns

the number of killed processes.

6.98.2.7 killProcess2()

```
ttk_tools._process_utils.killProcess2 (
              image_name )
```

Kill all processes that match the given image_name using the windows "taskkill" command running in a sub process.

Parameters

image_name	the name of the process to be killed

6.98.3 Variable Documentation

6.98.3.1 PROCESS_QUERY_INFORMATION

```
int ttk_tools._process_utils.PROCESS_QUERY_INFORMATION = 0x0400
```

6.98.3.2 PROCESS_QUERY_LIMITED_INFORMATION

```
int ttk_tools._process_utils.PROCESS_QUERY_LIMITED_INFORMATION = 0x1000
```

6.98.3.3 PROCESS_TERMINATE

```
int ttk_tools._process_utils.PROCESS_TERMINATE = 0x0001
```

6.98.3.4 PROCESS_VM_READ

```
int ttk_tools._process_utils.PROCESS_VM_READ = 0x0010
```

6.99 ttk_tools._win32_utils Namespace Reference

Utility functions for use on Windows systems.

Functions

- disableAppCrashDialog ()

  Disable displaying of application/program crash dialog for the current process ("XYZ has stopped working" / "XYZ funktioniert nicht mehr").
- waitPumpMessages (delay)

  Wait for (approximately) delay [s] and pump windows messages/events
  while waiting, which is a "good thing" to do if a COM connection is active in the current process/thread.
- _getRegKeys (path)

  Get a `_winreg.HKEY_*`-constant matching the supplied `start_key`.
- getRegPath (path, name, wow64_32_key=None, verbosity=1)

  Get a value from an entry in the windows registry.
- setRegPath (path, name, value, value_type=None, wow64_32_key=None, verbosity=1)

  Set a value to an entry in the windows registry.

Variables

- _stop_event = win32event.CreateEvent(None, 0, 0, None)

  alibi-event to wait for with waitPumpMessages() Note: this event could be triggered via win32event.SetEvent(stop↩ _event) to stop a running waitPumpMessages() early

6.99.1 Detailed Description

Utility functions for use on Windows systems.

See also interface wrapper in ttk_tools.win32_utils.

6.99.2 Function Documentation

6.99.2.1 _getRegKeys()

```
ttk_tools._win32_utils._getRegKeys (
              path )  [protected]
```

Get a `_winreg.HKEY_*`-constant matching the supplied `start_key`.

Parameters

path	a registry path starting with one of the HKEY entries, either as full HKEY or abbreviated, e.g. "HKEY_LOCAL_MACHINE\\SOFTWARE\\Microsoft" or "HKLM\\SOFTWARE\\Microsoft"

Returns

a tuple: matching *winreg.HKEY*∗-constant plus the remaining sub-key

6.99.2.2 disableAppCrashDialog()

```
ttk_tools._win32_utils.disableAppCrashDialog ( )
```

Disable displaying of application/program crash dialog for the current process ("XYZ has stopped working" / "XYZ funktioniert nicht mehr").

This tells Windows Error Reporting to skip displaying the dialog. See `SEM_NOGPFAULTERRORBOX`.

6.99.2.3 getRegPath()

```
ttk_tools._win32_utils.getRegPath (
              path,
              name,
              wow64_32_key = None,
              verbosity = 1 )
```

Get a value from an entry in the windows registry.

Parameters

path	a registry path starting with one of the HKEY entries, e.g. "HKEY_LOCAL_MACHINE\\SOFTWARE\\Microsoft" or "HKLM\\SOFTWARE\\Microsoft"
name	name of entry below path
wow64_32_key	None: use default key view depending on current bitness True: access 32-bit keys from 32-bit and 64-bit applications False: access 64-bit keys from 32-bit and 64-bit applications
verbosity	verbosity of error messages 0: silent, 1: print error to stdout, 2: also print exception info

Note

Registry paths may start with both full or abbreviated HKEY values, e.g.

- `HKEY_LOCAL_MACHINE\\...` or abbreviated `HKLM\\...`
- `HKEY_CLASSES_ROOT\\...` or abbreviated `HKCR\\...`
- `HKEY_CURRENT_USER\\...` or abbreviated `HKCU\\...`

Returns

the contents of the specified key/name or None if access failed.

6.99.2.4 setRegPath()

```
ttk_tools._win32_utils.setRegPath (
              path,
              name,
              value,
              value_type = None,
              wow64_32_key = None,
              verbosity = 1 )
```

Set a value to an entry in the windows registry.

Parameters

path	a registry path starting with one of the HKEY entries, e.g. "HKEY_LOCAL_MACHINE\\SOFTWARE\\Microsoft" or "HKLM\\SOFTWARE\\Microsoft". See note at getRegPath().
name	name of entry below path
value	value to set
value_type	type of value to set, should be one of the defined `_winreg.REG_*` values or None to "auto detect": • string values => `REG_SZ` • integer values => `REG_DWORD` (note: max 32 bit) • float values => `REG_DWORD` (will be rounded towards the nearest integer)
wow64_32_key	None: use default key view depending on current bitness True: access 32-bit keys from 32-bit and 64-bit applications False: access 64-bit keys from 32-bit and 64-bit applications
verbosity	verbosity of error messages 0: silent, 1: print error to stdout, 2: also print exception info

Returns

True if value was set, otherwise False.

6.99.2.5 waitPumpMessages()

```
ttk_tools._win32_utils.waitPumpMessages (
              delay )
```

Wait for (approximately) delay [s] and pump windows messages/events
while waiting, which is a "good thing" to do if a COM connection is active in the current process/thread.

Note

The wait time is rather approximate, and may be quite a bit longer (like about +500ms on occasion) if the system is busy and lots of events are fired.
The usable resolution appears to be about 200ms.

Parameters

delay	wait delay in seconds

6.99.3 Variable Documentation

6.99.3.1 _stop_event

```
ttk_tools._win32_utils._stop_event = win32event.CreateEvent(None, 0, 0, None)  [protected]
```

alibi-event to wait for with waitPumpMessages() Note: this event could be triggered via win32event.SetEvent(stop↩ _event) to stop a running waitPumpMessages() early

6.100 ttk_tools.can_hal Namespace Reference

Access to iSyst Hardware Abstraction Layer for CAN interfaces.

Namespaces

- namespace _hal

  Access to CAN HAL (via CAN_HAL_32.dll / CAN_HAL_64.dll).
- namespace _hal_offline_stub

  Offline stub for CAN HAL access in ttk_tools.can_hal.hal.
- namespace common

  Common CAN HAL definitions, classes and utility functions.
- namespace hal
- namespace hal_offline_stub
- namespace hal_structures

  C data structures for use with CAN_HAL DLL.

6.100.1 Detailed Description

Access to iSyst Hardware Abstraction Layer for CAN interfaces.

6.101 ttk_tools.can_hal._hal Namespace Reference

Access to CAN HAL (via CAN_HAL_32.dll / CAN_HAL_64.dll).

Classes

- class CanHal

  Low(-ish)-Level interface to CAN_HAL_(32/64).dll.
- class CanHalDevice

  CAN device connected to a CanHal instance.

Functions

- _errCheck (result, func, args)

  Default "errcheck" callable for API calls that return a tHalError value.
- _settypes (lib, name, argtypes=None, restype=None)

  Assign "restype", "errcheck" and "argtypes" attributes to DLL functions.

Variables

- _base_path = os.path.abspath(inspect.currentframe().f_code.co_filename)

6.101.1 Detailed Description

Access to CAN HAL (via CAN_HAL_32.dll / CAN_HAL_64.dll).

Interface wrapper for CAN HAL access offline stub.

Interface wrapper for CAN HAL access (via CAN_HAL_32.dll / CAN_HAL_64.dll) in ttk_tools.can_hal._hal.

CAN HAL currently supports CAN interfaces by:

- PEAK-System Technik
- Vector Informatik
- SYS TEC electronic

See also interface wrapper in ttk_tools.can_hal.hal

Note

HAL DLLs are currently only available on Windows

CAN HAL currently supports CAN interfaces by:

- PEAK-System Technik
- Vector Informatik
- SYS TEC electronic

Note

Windows device drivers for CAN devices have to be installed separately.

6.101.2 Function Documentation

6.101.2.1 _errCheck()

```
ttk_tools.can_hal._hal._errCheck (
              result,
              func,
              args )  [protected]
```

Default "errcheck" callable for API calls that return a tHalError value.

See https://docs.python.org/3/library/ctypes.html#ctypes._FuncPtr.errcheck
See HalErrors

Info:

- No error and "message buffer empty" will just return the status code
- Read errors will raise CanHalReadError
- Write errors will raise CanHalWriteError
- All other error codes will be raised as generic CanHalApiError

6.101.2.2 _settypes()

```
ttk_tools.can_hal._hal._settypes (
              lib,
              name,
              argtypes = None,
              restype = None )  [protected]
```

Assign "restype", "errcheck" and "argtypes" attributes to DLL functions.

If no restype is specified, a default ErrorID and errcheck callback will be assigned.

Parameters

lib	loaded DLL
name	name of lib function
argtypes	list of c-types to specify the argument types
restype	c-type to assign as result type None: use default ErrorID as restype and _errCheck callback

6.101.3 Variable Documentation

6.101.3.1 _base_path

```
ttk_tools.can_hal._hal._base_path = os.path.abspath(inspect.currentframe().f_code.co_filename)
[protected]
```

6.102 ttk_tools.can_hal._hal_offline_stub Namespace Reference

Offline stub for CAN HAL access in ttk_tools.can_hal.hal.

Classes

- class CanHal
  Low(-ish)-Level interface to CAN_HAL_(32/64).dll.
- class CanHalDevice
  CAN device connected to a CanHal instance.

Variables

- _base_path = os.path.abspath(inspect.currentframe().f_code.co_filename)

6.102.1 Detailed Description

Offline stub for CAN HAL access in ttk_tools.can_hal.hal.

6.102.2 Variable Documentation

6.102.2.1 _base_path

```
ttk_tools.can_hal._hal_offline_stub._base_path = os.path.abspath(inspect.currentframe().f_↩
code.co_filename)  [protected]
```

6.103 ttk_tools.can_hal.common Namespace Reference

Common CAN HAL definitions, classes and utility functions.

Classes

- class CanBitrate
  Connection speed settings (CanHal.h, tCanHalBautrate), bits per s.
- class CanBusState
  CAN bus states (CanHal.h, tCanHalCanBusState).
- class CanHalApiError
- class CanHalError
  Base class for CAN HAL exceptions.
- class CanHalReadError
  Error during read access.
- class CanHalWriteError
  Device Type (CanHAl.h, tCanHalDeviceType).
- class ConfigFlag
  Flags used during CAN configuration (CanHal.h, tCanHalCanConfig).
- class DeviceType
- class HalErrors
  HAL Error Codes (HalErrors.h, enum tHalError).
- class MessageStatus
  CAN Message Status (CanHal.h, tCanHalMessageStatus).

Variables

- ms = MessageStatus.RxMessage | MessageStatus.RemoteRequest
- int error_code = 292

6.103.1 Detailed Description

Common CAN HAL definitions, classes and utility functions.

6.103.2 Variable Documentation

6.103.2.1 error_code

```
int ttk_tools.can_hal.common.error_code = 292
```

6.103.2.2 ms

```
ttk_tools.can_hal.common.ms = MessageStatus.RxMessage | MessageStatus.RemoteRequest
```

6.104 ttk_tools.can_hal.hal Namespace Reference

Classes

- class CanHal

  Low(-ish)-Level interface to CAN_HAL_(32/64).dll.
- class CanHalDevice

  CAN device connected to a CanHal instance.
- class CanMessage

  CAN message data (CanHal.h, tCanHalCanMessage).

6.105 ttk_tools.can_hal.hal_offline_stub Namespace Reference

Classes

- class CanHal

  Low(-ish)-Level interface to CAN_HAL_(32/64).dll.
- class CanHalDevice

  CAN device connected to a CanHal instance.
- class CanMessage

  CAN message data (CanHal.h, tCanHalCanMessage).

6.106 ttk_tools.can_hal.hal_structures Namespace Reference

C data structures for use with CAN_HAL DLL.

Classes

- class _CanId
  An int-wrapper that uses a hexadecimal representation.
- class _EnumMapMixin
  Mix-in for ctypes.Structure.
- class CanDeviceConfig
  CAN device configuration (CanHal.h, tCanHalDeviceConfig).
- class CanHalVersion
  tCanHalVersion (CanHal.h)
- class CanMessage
  CAN message data (CanHal.h, tCanHalCanMessage).
- class DeviceListEntry
  A single device list entry (CanHal.h, tCanHalDeviceList).
- class Timestamp
  Message time stamps (CanHal.h, tCanHalTimestamp), values in µs.

Variables

- CanDeviceHdl = ctypes.c_ulong
  Handle for connected devices: typedef unsigned long tHalCanDeviceHdl;.
- int CanPayload = ctypes.c_ubyte * 8
  data type for CAN payload data (8 bytes, just for convenience)

6.106.1 Detailed Description

C data structures for use with CAN_HAL DLL.

Used both in online implementation and offline stub.

6.106.2 Variable Documentation

6.106.2.1 CanDeviceHdl

```
ttk_tools.can_hal.hal_structures.CanDeviceHdl = ctypes.c_ulong
```

Handle for connected devices: typedef unsigned long tHalCanDeviceHdl;.

6.106.2.2 CanPayload

```
int ttk_tools.can_hal.hal_structures.CanPayload = ctypes.c_ubyte * 8
```

data type for CAN payload data (8 bytes, just for convenience)

6.107 ttk_tools.dspace Namespace Reference

Access to dSPACE (dSPACE GmbH) tools and interfaces.

Namespaces

- namespace _xil_api

  Interface to the dSPACE XIL-2.x-Testbench implementation.
- namespace _xil_api_offline_stub

  A dummy/stub implementation of the default ttk_tools.dspace.xil_api for debugging without available hardware or license keys.
- namespace dspace_utils

  Utilities for classic dSPACE rtplib-based real-time platform access.
- namespace rtplib_offline_stub

  A dummy/stub implementation of the dSPACE rtplib to enable offline debugging without available real time hardware or license keys.
- namespace sdmlib_offline_stub

  A dummy/stub implementation of the classic dSPACE sdmlib Stimulus Engine to enable offline debugging without available real-time hardware or license keys.
- namespace xil_api

  Wrapper for interface to the dSPACE XIL-2.x-Testbench implementation in ttk_tools.dspace._xil_api.
- namespace xil_api_common

  Common shared definitions and error classes for xil_api.
- namespace xil_api_offline_stub

  Wrapper for a dummy/stub implementation of the default ttk_tools.dspace.xil_api for debugging without available hardware or license keys.
- namespace xil_net

  Import helper module as workaround for current import hook compatibility issues between packages Cython and pythonnet.

6.107.1 Detailed Description

Access to dSPACE (dSPACE GmbH) tools and interfaces.

6.108 ttk_tools.dspace._xil_api Namespace Reference

Interface to the dSPACE XIL-2.x-Testbench implementation.

Classes

- class Capture

  Capture / Data Acquisition.
- class Stimulus

  Stimulus / Signal Generator.
- class StimulusSignalDescription

  Signal value description for stimulus / signal generator.
- class XilTestbench

  XilTestbench access.

Variables

- dict _SCALAR_DATA_TYPES
- dict _VECTOR_DATA_TYPES
- dict _MATRIX_DATA_TYPES

6.108.1 Detailed Description

Interface to the dSPACE XIL-2.x-Testbench implementation.

See also interface wrapper in ttk_tools.dspace.xil_api.

All .NET bindings are wrapped to provide a plain (and somewhat less obnoxious) Python interface.

Note

Requires "Python for .NET" (pythonnet) for .NET access
(which is provided in dSPACE releases that have XIL-API support).

Current dSPACE XIL-2.x-Testbench implementations might only be available for x64.

6.108.2 Variable Documentation

6.108.2.1 _MATRIX_DATA_TYPES

```
dict ttk_tools.dspace._xil_api._MATRIX_DATA_TYPES  [protected]
```

Initial value:

```
00001 =  {
00002     DataType.eINT_MATRIX,
00003     DataType.eUINT_MATRIX,
00004     DataType.eFLOAT_MATRIX,
00005     DataType.eSTRING_MATRIX,
00006     DataType.eBOOLEAN_MATRIX,
00007 }
```

6.108.2.2 _SCALAR_DATA_TYPES

```
dict ttk_tools.dspace._xil_api._SCALAR_DATA_TYPES  [protected]
```

Initial value:

```
00001 =  {
00002     DataType.eINT,
00003     DataType.eUINT,
00004     DataType.eFLOAT,
00005     DataType.eSTRING,
00006     DataType.eBOOLEAN,
00007 }
```

6.108.2.3 _VECTOR_DATA_TYPES

```
dict ttk_tools.dspace._xil_api._VECTOR_DATA_TYPES  [protected]
```

Initial value:

```
00001 =   {
00002     DataType.eINT_VECTOR,
00003     DataType.eUINT_VECTOR,
00004     DataType.eFLOAT_VECTOR,
00005     DataType.eSTRING_VECTOR,
00006     DataType.eBOOLEAN_VECTOR,
00007 }
```

6.109 ttk_tools.dspace._xil_api_offline_stub Namespace Reference

A dummy/stub implementation of the default ttk_tools.dspace.xil_api for debugging without available hardware or license keys.

Classes

- class _MaPortStub

  Minimal stub for a MAPort-ish implementation.
- class _TestbenchStub

  Minimal (very minimal) stub for XIL-API testbench implementation.
- class Capture

  Capture / Data Acquisition Offline Stub.
- class DotNetException

  Stub for .NET System.Exception.
- class MAPortError

  XIL Model-Access-Port Error Stub.
- class Stimulus

  Offline stub for Stimulus / Signal Generator.
- class StimulusSignalDescription

  Stub for signal value description for stimulus / signal generator.
- class TestbenchPortException
- class TimeoutException

  XIL TestbenchPortException Stub.
- class XilTestbench

  XilTestbench access offline stub.

6.109.1 Detailed Description

A dummy/stub implementation of the default ttk_tools.dspace.xil_api for debugging without available hardware or license keys.

As relevant "constants"/definitions for xil_api will be imported from their .NET implementation, this module will provide stubs for them as well.

XilTestbech offline stubs use the "port_name" parameter as context for connecting to the default offline data provider.

Example: Provide initial data

```
dp = offline_data_provider.getDefaultProvider()
ctx = "MAPort" # xil_api

# hil var values
dp.setData(ctx, 'Model Root/Analog_In/KL30_Rueckmessung/Out1', 13.98) # CV_monitor
dp.setData(ctx, 'Model Root/Analog_In/CCmon/Out1',              0.48) # CC_monitor
dp.setData(ctx, 'Model Root/Analog_Out/motor_temp/Value',      20.00) # motor_temp
```

6.110 ttk_tools.dspace.dspace_utils Namespace Reference

Utilities for classic dSPACE rtplib-based real-time platform access.

Functions

- hilVarFactory (context, identifier, ∗args, ∗∗kwargs)

  Factory for "HilVars", decides actual implementation to use based on supplied context (i.e.

Variables

- XilTestbench = OfflineXilTestbench

6.110.1 Detailed Description

Utilities for classic dSPACE rtplib-based real-time platform access.

6.110.2 Function Documentation

6.110.2.1 hilVarFactory()

```
ttk_tools.dspace.dspace_utils.hilVarFactory (
              context,
              identifier,
             * args,
             ** kwargs )
```

Factory for "HilVars", decides actual implementation to use based on supplied context (i.e.

rtplib- or xil-testbench-based).

Parameters

context	context for variable (either an rtplib.Appl or xil_api.XilTestbench instance)
identifier	identifier/model path for variable
∗*args*	additional positional arguments to be passed on to Var implementation
∗∗*kwargs*	additional keyword arguments to be passed on to Var implementation

Returns

an Var instance, either based on rtplib or or xil_api/testbench

6.110.3 Variable Documentation

6.110.3.1 XilTestbench

```
ttk_tools.dspace.dspace_utils.XilTestbench = OfflineXilTestbench
```

6.111 ttk_tools.dspace.rtplib_offline_stub Namespace Reference

A dummy/stub implementation of the dSPACE rtplib to enable offline debugging without available real time hardware or license keys.

Classes

- class _RTPApplicationDummy

  Dummy implementation of `rtplib.Appl()` compatible objects for offline use.
- class _RTPCaptureDummy

  A dummy implementation of `rtplib.Capture()` compatible objects for offline use.
- class _RTPVariableDummy

  Dummy implementation of `rtplib.Appl().Variable()` compatible objects for offline use.
- class rtplibError

  Exception of the module rtplib.

Functions

- ApplCheck (Object)
- VariableCheck (Object)

  Check if an Python object has the type Variable.
- CaptureCheck (Object)

  Check if an Python object has the type Capture.
- Appl (VariableFile, SystemName, SystemType)

  Create an Appl class object.

Variables

- str rtplibVersion = "0.0.0.0"
- int CaptureNone = 0
- int CaptureTriggerArmed = 1
- int CaptureInProgress = 2
- int CaptureDone = 3
- int CaptureAborted = 4
- int CaptureWaitingForTriggerArmed = 5

6.111.1 Detailed Description

A dummy/stub implementation of the dSPACE rtplib to enable offline debugging without available real time hardware or license keys.

Mainly retained for use in demo projects. The actual rtplib/rtplib2 is deprecated and not available for Python 3.

6.111.2 Function Documentation

6.111.2.1 Appl()

```
ttk_tools.dspace.rtplib_offline_stub.Appl (
              VariableFile,
              SystemName,
              SystemType )
```

Create an Appl class object.

Parameters

VariableFile	path to variable file (∗.sdf or ∗.trc)
SystemName	name of rtp system, e.g. "ds1005"
SystemType	type of rtp system, e.g. "PPC"

Returns

an rtp application instance

6.111.2.2 ApplCheck()

```
ttk_tools.dspace.rtplib_offline_stub.ApplCheck (
              Object )
```

6.111.2.3 CaptureCheck()

```
ttk_tools.dspace.rtplib_offline_stub.CaptureCheck (
              Object )
```

Check if an Python object has the type Capture.

6.111.2.4 VariableCheck()

```
ttk_tools.dspace.rtplib_offline_stub.VariableCheck (
              Object )
```

Check if an Python object has the type Variable.

6.111.3 Variable Documentation

6.111.3.1 CaptureAborted

```
int ttk_tools.dspace.rtplib_offline_stub.CaptureAborted = 4
```

6.111.3.2 CaptureDone

```
int ttk_tools.dspace.rtplib_offline_stub.CaptureDone = 3
```

6.111.3.3 CaptureInProgress

```
int ttk_tools.dspace.rtplib_offline_stub.CaptureInProgress = 2
```

6.111.3.4 CaptureNone

```
int ttk_tools.dspace.rtplib_offline_stub.CaptureNone = 0
```

6.111.3.5 CaptureTriggerArmed

```
int ttk_tools.dspace.rtplib_offline_stub.CaptureTriggerArmed = 1
```

6.111.3.6 CaptureWaitingForTriggerArmed

```
int ttk_tools.dspace.rtplib_offline_stub.CaptureWaitingForTriggerArmed = 5
```

6.111.3.7 rtplibVersion

```
str ttk_tools.dspace.rtplib_offline_stub.rtplibVersion = "0.0.0.0"
```

6.112 ttk_tools.dspace.sdmlib_offline_stub Namespace Reference

A dummy/stub implementation of the classic dSPACE sdmlib Stimulus Engine to enable offline debugging without available real-time hardware or license keys.

Classes

- class BUFFER

  Instances of the class BUFFER are used to hold values of Stimulus Engine channels for later use.
- class Constants

  Constants-Enum.
- class sdmlibError

  sdmlib-exceptions
- class SIGNAL

Functions

- SG_INIT (TrcFile, BoardName=None, MPSubsystem=None)

  Initialize the global stimulus object with an application.
- SG_CLOSE ()

  Shut down the Signal Generator if it is running in a separate thread.
- SG_GET_STATE ()

  Get the current state of the stimulus engine.
- SG_GET_TIME ()

  Get the current time of the Stimulus Engine.
- SG_SET_STATE (NewState)

  Set the state of the stimulus engine.
- SG_STOP ()

  Stop the Stimulus Engine on the real-time platform.

- SG_RUN ()

  Start the Stimulus Engine on the real-time platform.
- SG_PAUSE ()

  Pause the stimulus engine on the real-time platform.
- ELSE ()

  Mark the beginning of the else branch of an if statement.
- ENDIF ()

  Mark the end of an IF statement.
- ENDLOOP ()

  Mark the end of a repeated code block.
- EVAL (eval_time_length, save_per_period=30, save_time_step=None, mat_file_name=None)

  Calculate the specified signal forms offline on the host PC and save the results in a binary MATLAB-compatible file.
- GOTO (Label)

  Go to the specified label.
- IF (result, timeout=None)

  Define a condition which is evaluated by the Stimulus Engine.
- LABEL (Label)

  Set a label for GOTO() calls.
- LOAD_SEQ ()

  Download the instruction table.
- LOOP (Number)

  Specify the number of repetitions of a repeated code block.
- MAKE_SEQ (seq_file_name=None)

  Create the sequence control table (SEQ) for the signal generation.
- MsgSleep (Interval)

  Suspends the execution of the current thread for at least the specified interval.
- STOP ()

  This function is used to finish the stimulus description.
- TH_START ()

  Start the data replay mechanism.
- TH_STOP ()

  Stop the data replay mechanism.
- TT (Delay)

  Specify the time delay between two Stimulus Engine actions.
- UNLOAD_SEQ ()

  Unload the instruction table.
- ASCIItoMat (ASCII_file, mat_file)

  Convert ASCII data to a binary MAT file.

Variables

- str _version_ = '0.0.0.0'

6.112.1 Detailed Description

A dummy/stub implementation of the classic dSPACE sdmlib Stimulus Engine to enable offline debugging without available real-time hardware or license keys.

Mainly retained for use in demo projects. The actual sdmlib is deprecated and not available for Python 3.

6.112.2 Function Documentation

6.112.2.1 ASCIItoMat()

```
ttk_tools.dspace.sdmlib_offline_stub.ASCIItoMat (
              ASCII_file,
              mat_file )
```

Convert ASCII data to a binary MAT file.

Parameters

ASCII_file	Name (path) of the source ASCII file
mat_file	Name (path) of the target MAT file

6.112.2.2 ELSE()

```
ttk_tools.dspace.sdmlib_offline_stub.ELSE ( )
```

Mark the beginning of the else branch of an if statement.

6.112.2.3 ENDIF()

```
ttk_tools.dspace.sdmlib_offline_stub.ENDIF ( )
```

Mark the end of an IF statement.

6.112.2.4 ENDLOOP()

```
ttk_tools.dspace.sdmlib_offline_stub.ENDLOOP ( )
```

Mark the end of a repeated code block.

6.112.2.5 EVAL()

```
ttk_tools.dspace.sdmlib_offline_stub.EVAL (
              eval_time_length,
              save_per_period = 30,
              save_time_step = None,
              mat_file_name = None )
```

Calculate the specified signal forms offline on the host PC and save the results in a binary MATLAB-compatible file.

Parameters

eval_time_length	Time during which signals are evaluated [s]
save_per_period	number of values (int) to be saved for nonlinear functions in a period. Defaults to 30.
save_time_step	step size used to save nonlinear signals if the parameter that describes the time behavior is a buffer or another signal. The default is five times the sampling period.
mat_file_name	User-defined name/path of the evaluation file. If this parameter is omitted, the model name with extension "_eval.mat" is used for the evaluation file.

6.112.2.6 GOTO()

```
ttk_tools.dspace.sdmlib_offline_stub.GOTO (
              Label )
```

Go to the specified label.

Parameters

Label	name of label to branch to

6.112.2.7 IF()

```
ttk_tools.dspace.sdmlib_offline_stub.IF (
              result,
              timeout = None )
```

Define a condition which is evaluated by the Stimulus Engine.

Parameters

result	condition to evaluate
timeout	duration during which the comparison expression should be checked.

6.112.2.8 LABEL()

```
ttk_tools.dspace.sdmlib_offline_stub.LABEL (
              Label )
```

Set a label for GOTO() calls.

Parameters

Label	name of label to set.

6.112.2.9 LOAD_SEQ()

```
ttk_tools.dspace.sdmlib_offline_stub.LOAD_SEQ ( )
```

Download the instruction table.

6.112.2.10 LOOP()

```
ttk_tools.dspace.sdmlib_offline_stub.LOOP (
              Number )
```

Specify the number of repetitions of a repeated code block.

Parameters

Number	Number of repetitions

6.112.2.11 MAKE_SEQ()

```
ttk_tools.dspace.sdmlib_offline_stub.MAKE_SEQ (
              seq_file_name = None )
```

Create the sequence control table (SEQ) for the signal generation.

The sequence control table contains all the information for the specified signal behavior, such as waveforms and parameters, in a coded form.

Parameters

seq_file_name	user-defined name of the SEQ file. Defaults to the model name with the extension SEQ.

6.112.2.12 MsgSleep()

```
ttk_tools.dspace.sdmlib_offline_stub.MsgSleep (
              Interval )
```

Suspends the execution of the current thread for at least the specified interval.

Windows message of the current process will be handled with PumpWaitingMessages. It uses no or less processor time while waiting.

6.112.2.13 SG_CLOSE()

```
ttk_tools.dspace.sdmlib_offline_stub.SG_CLOSE ( )
```

Shut down the Signal Generator if it is running in a separate thread.

A call to this function is not necessary if the Signal Generator is running in the main thread. It is used to leave a function when it is running in a separate thread.

6.112.2.14 SG_GET_STATE()

```
ttk_tools.dspace.sdmlib_offline_stub.SG_GET_STATE ( )
```

Get the current state of the stimulus engine.

Possible states of the stimulus engine are:
0: stopped
1: running
2: paused

Returns

the current state of the dSPACE stimulus engine.

6.112.2.15 SG_GET_TIME()

```
ttk_tools.dspace.sdmlib_offline_stub.SG_GET_TIME ( )
```

Get the current time of the Stimulus Engine.

Returns

the current time as a float value.

6.112.2.16 SG_INIT()

```
ttk_tools.dspace.sdmlib_offline_stub.SG_INIT (
              TrcFile,
              BoardName = None,
              MPSubsystem = None )
```

Initialize the global stimulus object with an application.

Parameters

TrcFile	name of the trc- or sdf-file of the application
BoardName	name of the target platform/system or None if no system should be used
MPSubsystem	name of the subsystem, if a multi-processor system is used (or None)

6.112.2.17 SG_PAUSE()

```
ttk_tools.dspace.sdmlib_offline_stub.SG_PAUSE ( )
```

Pause the stimulus engine on the real-time platform.

6.112.2.18 SG_RUN()

```
ttk_tools.dspace.sdmlib_offline_stub.SG_RUN ( )
```

Start the Stimulus Engine on the real-time platform.

6.112.2.19 SG_SET_STATE()

```
ttk_tools.dspace.sdmlib_offline_stub.SG_SET_STATE (
              NewState )
```

Set the state of the stimulus engine.

Parameters

NewState	The new state to be set. Possible states of the stimulus engine are: 0: stopped 1: running 2: paused

6.112.2.20 SG_STOP()

```
ttk_tools.dspace.sdmlib_offline_stub.SG_STOP ( )
```

Stop the Stimulus Engine on the real-time platform.

6.112.2.21 STOP()

```
ttk_tools.dspace.sdmlib_offline_stub.STOP ( )
```

This function is used to finish the stimulus description.

6.112.2.22 TH_START()

```
ttk_tools.dspace.sdmlib_offline_stub.TH_START ( )
```

Start the data replay mechanism.

6.112.2.23 TH_STOP()

```
ttk_tools.dspace.sdmlib_offline_stub.TH_STOP ( )
```

Stop the data replay mechanism.

6.112.2.24 TT()

```
ttk_tools.dspace.sdmlib_offline_stub.TT (
              Delay )
```

Specify the time delay between two Stimulus Engine actions.

```
Delay: time delay in seconds until the next Stimulus Engine
       instructions occurs.
```

6.112.2.25 UNLOAD_SEQ()

```
ttk_tools.dspace.sdmlib_offline_stub.UNLOAD_SEQ ( )
```

Unload the instruction table.

6.112.3 Variable Documentation

6.112.3.1 _version_

```
str ttk_tools.dspace.sdmlib_offline_stub._version_ = '0.0.0.0'  [protected]
```

6.113 ttk_tools.dspace.xil_api Namespace Reference

Wrapper for interface to the dSPACE XIL-2.x-Testbench implementation in ttk_tools.dspace._xil_api.

Classes

- class Capture
  Capture / Data Acquisition.
- class Stimulus
  Stimulus / Signal Generator.
- class StimulusSignalDescription
  Signal value description for stimulus / signal generator.
- class XilTestbench
  XilTestbench access.

6.113.1 Detailed Description

Wrapper for interface to the dSPACE XIL-2.x-Testbench implementation in ttk_tools.dspace._xil_api.

All .NET bindings are wrapped to provide a plain (and somewhat less obnoxious) Python interface.

This serves as "interface" to the precompiled module in delivery to enable code-completion in PyDev.

Note

Requires "Python for .NET" (pythonnet) for .NET access
(which is provided in dSPACE releases that have XIL-API support).

Current dSPACE XIL-2.x-Testbench implementations might only be available for x64.

6.114 ttk_tools.dspace.xil_api_common Namespace Reference

Common shared definitions and error classes for xil_api.

Classes

- class CaptureState

  Data Capture States.
- class DataType

  Data Types.
- class InterpolationTypes

  Data Interpolation Types.
- class MAPortState

  Model Access Port States.
- class SignalGeneratorState

  Signal Generator States.
- class StimulusError

  Errors during Stimulus setup/execution.
- class TestbenchError

  Base class for "other" testbench errors that are not handled by XIL TestbenchPortExceptions.

Functions

- writeMAPortConfigXML (sdf_file_path, platform_name="DS1006", xml_file_path=None)

6.114.1 Detailed Description

Common shared definitions and error classes for xil_api.

6.114.2 Function Documentation

6.114.2.1 writeMAPortConfigXML()

```
ttk_tools.dspace.xil_api_common.writeMAPortConfigXML (
              sdf_file_path,
              platform_name = "DS1006",
              xml_file_path = None )
```

6.115 ttk_tools.dspace.xil_api_offline_stub Namespace Reference

Wrapper for a dummy/stub implementation of the default ttk_tools.dspace.xil_api for debugging without available hardware or license keys.

Classes

- class Capture

  Capture / Data Acquisition Offline Stub.
- class Stimulus

  Offline stub for Stimulus / Signal Generator.
- class StimulusSignalDescription

  Stub for signal value description for stimulus / signal generator.
- class XilTestbench

  XilTestbench access offline stub.

6.115.1 Detailed Description

Wrapper for a dummy/stub implementation of the default ttk_tools.dspace.xil_api for debugging without available hardware or license keys.

This serves as "interface" to the precompiled module in delivery to enable code-completion in PyDev.

XilTestbech offline stubs use the "port_name" parameter as context for connecting to the default offline data provider.

Example: Provide initial data

```
dp = offline_data_provider.getDefaultProvider()
ctx = "MAPort" # xil_api

# hil var values
dp.setData(ctx, 'Model Root/Analog_In/KL30_Rueckmessung/Out1', 13.98) # CV_monitor
dp.setData(ctx, 'Model Root/Analog_In/CCmon/Out1',              0.48) # CC_monitor
dp.setData(ctx, 'Model Root/Analog_Out/motor_temp/Value',      20.00) # motor_temp
```

6.116 ttk_tools.dspace.xil_net Namespace Reference

Import helper module as workaround for current import hook compatibility issues between packages Cython and pythonnet.

Variables

- str default_xil_version = "2.2.0"

  fallback/default version
- available_xil_versions = _detectXilVersions(verbose=False)

  detected versions in ascending order (if any)
- preferred_xil_version

  used specified version (from current hil_project.ini)
- list candidates = [default_xil_version]
- list xil_version = candidates[-1]

6.116.1 Detailed Description

Import helper module as workaround for current import hook compatibility issues between packages Cython and pythonnet.

See https://github.com/pythonnet/pythonnet/issues/941 https://github.com/cython/cython/issues/3084

Note

If a specific XIL API version should be used (and auto detection is not sufficient), a version can be defined in project settings

Example:

```
# define a specific XIL API version to use in hil_project.ini

[TestToolkit]
# currently supported versions: 2.0.1, 2.1.0, 2.2.0
xil_api_version = 2.1.0
```

6.116.2 Variable Documentation

6.116.2.1 available_xil_versions

```
ttk_tools.dspace.xil_net.available_xil_versions = _detectXilVersions(verbose=False)
```

detected versions in ascending order (if any)

6.116.2.2 candidates

```
list ttk_tools.dspace.xil_net.candidates = [default_xil_version]
```

6.116.2.3 default_xil_version

```
str ttk_tools.dspace.xil_net.default_xil_version = "2.2.0"
```

fallback/default version

6.116.2.4 preferred_xil_version

```
ttk_tools.dspace.xil_net.preferred_xil_version
```

Initial value:

```
00001 =  configuration.getSettings().get(
00002     "TestToolkit", "xil_api_version", ""
00003 )
```

used specified version (from current hil_project.ini)

6.116.2.5 xil_version

```
list ttk_tools.dspace.xil_net.xil_version = candidates[-1]
```

6.117 ttk_tools.isyst Namespace Reference

Access to iSyst Intelligente Systeme GmbH tools and interfaces.

Namespaces

- namespace ida_common

  Common iDA Testware / middleware definitions, classes and utility functions.
- namespace ida_testware

  Access to iDA Testware interfaces.
- namespace ida_testware_offline_stub
- namespace idaq

  I/O for "idaq" data acquisition format.
- namespace mwapp_client

  iDA Testware data access via gRPC "MwApp" interface
- namespace mwapp_client_offline_stub

  Offline stub for iDA Testware data access via gRPC "MwApp" interface.
- namespace mwrec_client

  Access to iDA Recorder (data acquisition) via gRPC "MwRec" interface.
- namespace mwrec_client_offline_stub

  Offline stub for access to iDA Recorder (data acquisition)

6.117.1 Detailed Description

Access to iSyst Intelligente Systeme GmbH tools and interfaces.

6.118 ttk_tools.isyst.ida_common Namespace Reference

Common iDA Testware / middleware definitions, classes and utility functions.

Classes

- class MetaKind
  MetaKind as defined in protobuf message MetaId.
- class MwAppError
- class MwAppStatus
  Application RPC status info (internal success/error).
- class MwDataType
  Data types for middleware values, compare mwvalue.h: dataType_t.
- class MwError
  Generic iDA Testware middleware interface error.
- class MwRecError
  Errors during data recording.
- class RecCommandStatus
  Recorder command RPC status info (internal success/error).
- class RecState
  Recorder state machine state.

Functions

- _enumName (EnumClass, value)
  Get value's name of an enum, or "<undefined>" for undefined values.
- _hexRepr (value, nibbles=0)
  Get a hexadecimal representation of value, fallback to default representation for non-integer values.

6.118.1 Detailed Description

Common iDA Testware / middleware definitions, classes and utility functions.

6.118.2 Function Documentation

6.118.2.1 _enumName()

```
ttk_tools.isyst.ida_common._enumName (
              EnumClass,
              value )  [protected]
```

Get value's name of an enum, or "<undefined>" for undefined values.

6.118.2.2 _hexRepr()

```
ttk_tools.isyst.ida_common._hexRepr (
              value,
              nibbles = 0 )  [protected]
```

Get a hexadecimal representation of value, fallback to default representation for non-integer values.

Parameters

value	value to format as hex
nibbles	minimum number of nibbles to show

Returns

a string

6.119 ttk_tools.isyst.ida_testware Namespace Reference

Access to iDA Testware interfaces.

Classes

- class IdaRecorder

  Data acquisition using an iDA Recorder application.
- class IdaTestware

  Access to iDA Testware MwApp interface for reading/writing variables.

6.119.1 Detailed Description

Access to iDA Testware interfaces.

Access to stubbed/mocked iDA Testware interfaces.

Note

Interfaces require package grpcio for gRPC access.

Example:

```
HIL_IP = "192.168.56.8"

with IdaRecorder(addr=HIL_IP) as recorder: # using the default port
    recorder.setup(["Triangle.uint32", "Sine.float64", "Counter"])
    # recorder.setup will raise an MwRecError if setup failed

    recorder.start()
    time.sleep(3.0)
    recorder.stop()

    daq_data = recorder.getData()
    idaq.printDataInfo(daq_data, outer_n=8)

with IdaTestware(addr=HIL_IP) as ida: # using the default port
    mwvar = ida.getMwVar("foo")
    print(mwvar.getValue())

Instead of with-context handlers, the iDA interfaces are often configured
as part of a TestEnv instance and accessible via getters, e.g.
testenv.getHilRecorder().
```

Example: Variable abstraction and Var containers

```
# In test scripts, Variables will usually be accessed via a (hil) var container:

# hil_variables.py
from ttk_base.isyst_ida.variables import Var as HilVar
from ttk_base.isyst_ida.variables import VarContainer as HilVarContainer

class HilVars(HilVarContainer):
    def __init__(self, ida):
        super().__init__(ida)
        self.CV_vorgabe = HilVar(ida, 'Analog_Out.CVprog', unit="V", fmt="flt2", resettable=False)
        self.CC_vorgabe = HilVar(ida, 'Analog_Out.CCprog', unit="A", fmt="flt3")
        self.CV_monitor = HilVar(ida, 'Analog_In.KL30_Rueckmessung', unit="V", fmt="flt2")
        self.CC_monitor = HilVar(ida, 'Analog_In.CCmon',             unit="A", fmt="flt3")
        self.KL30       = HilVar(ida, 'Digital_Out.KL30', lookup={0: "off", 1: "on"}, fmt="int",
resettable=False)
        self.KL15       = HilVar(ida, 'Digital_Out.KL15', lookup={0: "off", 1: "on"}, fmt="int",
resettable=False)
        self.motor_temp = HilVar(ida, 'Analog_Out.MotorTemp', unit="°C", descr="Motor temperature")

# Usually, an instance of the container can later (in a test script) be accessed via
hil = testenv.getHil()
print(hil.CC_monitor.get())
print(hil.CC_monitor.info())
```

iDA Testware offline stubs use the connection address as context for connecting to the default offline data provider (for offline implementations of MwAppRpc/MwRecRpc).

Example: Provide initial data

```
addr = "127.0.0.1"

dp = offline_data_provider.getDefaultProvider()

# initialize default data (uninitialized entries will default to float 0.0)
dp.setData(addr, "foo", 113)
dp.setData(addr, "bar", 456)
dp.setData(addr, "baz", 789)
dp.setData(addr, "qux", 11.11)
dp.setData(addr, "etaoin[0]", 1)
dp.setData(addr, "etaoin[1]", 2)
dp.setData(addr, "etaoin[2]", 3)

dp.setData(addr, 'Analog_In.KL30_Rueckmessung', 13.98) # CV_monitor
dp.setData(addr, 'Analog_In.CCmon',              0.48) # CC_monitor
dp.setData(addr, 'Analog_Out.MotorTemp',        20.00) # motor_temp

with IdaRecorder(addr) as recorder: # using the default port
    recorder.setup(["Triangle.uint32", "Sine.float64", "Counter"])
    # recorder.setup will raise an MwRecError if setup failed

    recorder.start()
    time.sleep(3.0)
    recorder.stop()

    daq_data = recorder.getData()
    idaq.printDataInfo(daq_data, outer_n=8)

with IdaTestware(addr) as ida: # using the default port
    mwvar = ida.getMwVar("foo")
    print(mwvar.getValue())

Instead of with-context handlers, the iDA interfaces are often configured
as part of a TestEnv instance and accessible via getters, e.g.
testenv.getHilRecorder().
```

Example: Variable abstraction and Var containers

```
# In test scripts, Variables will usually be accessed via a (hil) var container:

# hil_variables.py
from ttk_base.isyst_ida.variables import Var as HilVar
from ttk_base.isyst_ida.variables import VarContainer as HilVarContainer

class HilVars(HilVarContainer):
    def __init__(self, ida):
        super().__init__(ida)
        self.CV_vorgabe = HilVar(ida, 'Analog_Out.CVprog', unit="V", fmt="flt2", resettable=False)
        self.CC_vorgabe = HilVar(ida, 'Analog_Out.CCprog', unit="A", fmt="flt3")
```

```
        self.CV_monitor = HilVar(ida, 'Analog_In.KL30_Rueckmessung', unit="V", fmt="flt2")
        self.CC_monitor = HilVar(ida, 'Analog_In.CCmon',             unit="A", fmt="flt3")
        self.KL30       = HilVar(ida, 'Digital_Out.KL30', lookup={0: "off", 1: "on"}, fmt="int",
resettable=False)
        self.KL15       = HilVar(ida, 'Digital_Out.KL15', lookup={0: "off", 1: "on"}, fmt="int",
resettable=False)
        self.motor_temp = HilVar(ida, 'Analog_Out.MotorTemp', unit="°C", descr="Motor temperature")

# Usually, an instance of the container can later (in a test script) be accessed via
hil = testenv.getHil()
print(hil.CC_monitor.get())
print(hil.CC_monitor.info())
```

6.120 ttk_tools.isyst.ida_testware_offline_stub Namespace Reference

Classes

- class IdaRecorder

  Data acquisition using an iDA Recorder application.
- class IdaTestware

  Access to iDA Testware MwApp interface for reading/writing variables.

6.121 ttk_tools.isyst.idaq Namespace Reference

I/O for "idaq" data acquisition format.

Classes

- class DaqDataError

  Errors during DAQ data decoding.

Functions

- _bstr (s)
- _ustr (b)

  Get a text string using the default encoding.
- _getByteSizeString (size)

  Get a string representation of the supplied size in Bytes.
- _getSamplesPath ()

  Get path to a folder /samples on same level as a parent folder /src.
- readDaqData (file_handle)

  Read DAQ data in "idaq" format.
- writeDaqData (daq_data, file_handle, comment=None, verbosity=1)

  Write Read DAQ data in "idaq" format.
- printDataInfo (values, indent_level=1, outer_n=2)

  Print a summary of the supplied values to stdout.

Variables

- str FILE_FORMAT = "1.2"

  marker for supported file format version
- str ENCODING = "latin-1"

  text encoding to use for non-binary parts (header, meta data)
- dict data_type_to_format_mapping

  Mapping of data type names to Python struct format codes.

6.121.1 Detailed Description

I/O for "idaq" data acquisition format.

6.121.2 Function Documentation

6.121.2.1 _bstr()

```
ttk_tools.isyst.idaq._bstr (
              s )  [protected]
```

6.121.2.2 _getByteSizeString()

```
ttk_tools.isyst.idaq._getByteSizeString (
              size )  [protected]
```

Get a string representation of the supplied size in Bytes.

Parameters

size	a size in Bytes

Returns

a formatted string with size and suitable SI unit.

6.121.2.3 _getSamplesPath()

```
ttk_tools.isyst.idaq._getSamplesPath ( )  [protected]
```

Get path to a folder /samples on same level as a parent folder /src.

Returns

a path to /samples or to the current script's folder, if no /samples folder was detected.

6.121.2.4 _ustr()

```
ttk_tools.isyst.idaq._ustr (
              b )  [protected]
```

Get a text string using the default encoding.

Encoding errors (un-encodable characters) will be replaced. Compare ttk_base.baseutils.ustr.

6.121.2.5 printDataInfo()

```
ttk_tools.isyst.idaq.printDataInfo (
              values,
              indent_level = 1,
              outer_n = 2 )
```

Print a summary of the supplied values to stdout.

Mainly intended to be used during development / debugging as a (somewhat) compact output of DAQ data structures.

Example:

```
# Sample output using a DAQ data structure with two signals
# and the default outer_n=2
  label:
    data[20]: [100.052, 100.087, ... 100.315, 100.448]
    time[20]: [  0.000,   0.001, ...   0.018,   0.019]
  another label:
    data[20]: [ 27.162,   2.783, ...  23.078,  15.383]
    time[20]: [  0.000,   0.020, ...   0.360,   0.380]
```

Parameters

values	a scalar value or values in lists or dictionaries.
indent_level	starting indent level (as multiple of two spaces)
outer_n	print only the outer n values from start and end of data. None: all values will be printed

6.121.2.6 readDaqData()

```
ttk_tools.isyst.idaq.readDaqData (
              file_handle )
```

Read DAQ data in "idaq" format.

Parameters

file_handle	opened file handle (binary mode) with DAQ data

Returns

a daq data structure with {
<label>: {
"time": [<time_s>, <time_s>, ...],
"data": [<value>, <value>, ...],
"type": "<datatype-string>"
},
} entries.

6.121.2.7 writeDaqData()

```
ttk_tools.isyst.idaq.writeDaqData (
              daq_data,
              file_handle,
              comment = None,
              verbosity = 1 )
```

Write Read DAQ data in "idaq" format.

Parameters

file_handle	file handle opened for binary writing

daq_data	a daq data structure with {
	<label>: {
	"time": [<time_s>, <time_s>, ...],
	"data": [<value>, <value>, ...],
	"type": "<datatype-string>" # optional },
	} entries.

comment	string comment to add to [FILE] section
verbosity	verbosity of log info 0: silent 1: + info messages 2: + show labels

Note

"type" entry in daq_data The IDAQ format uses fixed data types and Python data types may map to to different data types. The `"type"` entry can be used to select a suitable data type for storing the values in `"data"`. See data_type_to_format_mapping for available data types.
Without a defined `"type"`, values will default to type "float64".

"time" axis value IDAQ format only support a single share time axis, so all signals will have to share the same time values

6.121.3 Variable Documentation

6.121.3.1 data_type_to_format_mapping

```
dict ttk_tools.isyst.idaq.data_type_to_format_mapping
```

Initial value:
```
00001 =  {
```

```
00002     "int8":         b"b",
00003     "sint8":        b"b", # permit sint<#> as alias for int<#>
00004     "uint8":        b"B",
00005
00006     "int16":        b"h",
00007     "sint16":       b"h",
00008     "uint16":       b"H",
00009
00010     "int32":        b"i",
00011     "sint32":       b"i",
00012     "uint32":       b"I",
00013
00014     "int64":        b"q",
00015     "sint64":       b"q",
00016     "uint64":       b"Q",
00017
00018     "float32":      b"f",
00019     "float64":      b"d",
00020 }
```

Mapping of data type names to Python struct format codes.

```
        Format    C Type            Python type          Bytes
```

int8 b signed char integer 1 uint8 B unsigned char integer 1 int16 h short integer 2 uint16 H unsigned short integer 2 int32 i int integer 4 uint32 I unsigned int integer 4 (int32) l long integer 4 (uint32) L unsigned long integer 4 int64 q long long integer 8 uint64 Q unsigned long long integer 8 float32 f float float 4 float64 d double float 8

Compare https://docs.python.org/2/library/struct.html => float32/float 64 use IEEE 754 binary32 or binary64 format

6.121.3.2 ENCODING

```
str ttk_tools.isyst.idaq.ENCODING = "latin-1"
```

text encoding to use for non-binary parts (header, meta data)

6.121.3.3 FILE_FORMAT

```
str ttk_tools.isyst.idaq.FILE_FORMAT = "1.2"
```

marker for supported file format version

6.122 ttk_tools.isyst.mwapp_client Namespace Reference

iDA Testware data access via gRPC "MwApp" interface

Classes

- class _VarMetaCache

  Cache for iDA-Variable meta data.
- class MwAppRpc

  Simple wrapper for MwApp RPCs.
- class MwVar

  Container for a single MW-Variable.

6.122.1 Detailed Description

iDA Testware data access via gRPC "MwApp" interface

Note

Requires package grpcio-tools (or separately packages grpcio and protobuf) for gRPC access.

6.123 ttk_tools.isyst.mwapp_client_offline_stub Namespace Reference

Offline stub for iDA Testware data access via gRPC "MwApp" interface.

Classes

- class _VarMetaCache

  Cache for iDA-variable meta data.
- class MwAppRpc

  Simple wrapper for MwApp RPCs (offline stub/mock implementation)
- class MwVar

  Container for a single MW-Variable.

Variables

- str _STUB_VERSION = "1.3.0"

6.123.1 Detailed Description

Offline stub for iDA Testware data access via gRPC "MwApp" interface.

A dummy/stub implementation of the default ttk_tools.isyst.mwapp_client for debugging without available hardware/connection.

MwAppRpc offline stubs use the connection "target" as context for connecting to the default offline data provider.

Example: Provide initial data

```
        addr, port = "127.0.0.1", 50051
        dp = offline_data_provider.getDefaultProvider()

        # initialize default data (uninitialized entries will default to float 0.0)
        dp.setData(addr, "foo", 113)
        dp.setData(addr, "bar", 456)
        dp.setData(addr, "baz", 789)
        dp.setData(addr, "qux", 11.11)
        dp.setData(addr, "etaoin[0]", 1)
        dp.setData(addr, "etaoin[1]", 2)
        dp.setData(addr, "etaoin[2]", 3)

        dp.setData(addr, "shrdlu", b"\x00\x01\x02\x03")
        dp.setData(addr, "lorem.ipsum", "dolor sit amet")

        dp.setData(addr, 'Analog_In.KL30_Rueckmessung', 13.98) # CV_monitor
        dp.setData(addr, 'Analog_In.CCmon',              0.48) # CC_monitor
        dp.setData(addr, 'Analog_Out.MotorTemp',        20.00) # motor_temp

        # and later (direct):
        with MwAppRpc(target=f"{addr}:{port}") as mwapp:
            mwvar = mwapp.getMwVar("foo")
            print(mwvar.getValue())

        # or via the default interface (in ttk_tools.isyst.ida_testware):
        with IdaTestware(addr, port) as ida:
            mwvar = ida.getMwVar("foo")
            print(mwvar.getValue())

    In test projects, a `MwAppRpc` instance will usually be created in the project's
    TestEnv `setup()` via the IdaTestware interface class.
```

Example: in *automation_wrapper*.py

```
        # in class TestEnv
        def setup(self):
            ”’ Test Setup: initialize test environment. ”’
            super().setup()
            self.ida = ida_testware.IdaTestware(addr=TARGET_IP)
            # [...]

    Variable abstraction will internally access `MwVar` instances via a `getMwVar`
    method (available both via `MwAppRpc` and the `IdaTestware` interface)
```

Example: in hil_variables.py

```
    from ttk_base.isyst_ida.variables import Var
    from ttk_base.isyst_ida.variables import VarContainer
    class HilVars(VarContainer):
        def __init__(self, ida):
        super().__init__(ida)
        self.varname = Var(ida, "variable.name.string")
        # [...]
```

6.123.2 Variable Documentation

6.123.2.1 _STUB_VERSION

```
str ttk_tools.isyst.mwapp_client_offline_stub._STUB_VERSION = "1.3.0"  [protected]
```

6.124 ttk_tools.isyst.mwrec_client Namespace Reference

Access to iDA Recorder (data acquisition) via gRPC "MwRec" interface.

Classes

- class MwRecRpc

  Simple wrapper for MwRec RPCs.

6.124.1 Detailed Description

Access to iDA Recorder (data acquisition) via gRPC "MwRec" interface.

Note

Requires package grpcio-tools (or separately packages grpcio and protobuf) for gRPC access.

6.125 ttk_tools.isyst.mwrec_client_offline_stub Namespace Reference

Offline stub for access to iDA Recorder (data acquisition)

Classes

- class MwRecRpc

  Simple wrapper for MwRec RPCs (offline stub/mock implementation)

Variables

- str _STUB_VERSION = "1.0.0"

6.125.1 Detailed Description

Offline stub for access to iDA Recorder (data acquisition)

MwRecRpc offline stubs use the connection address as context for connecting to the default offline data provider.

Example: Provide initial data

```
addr, port = "127.0.0.1", 50055 # default iDA-Recorder port is 50055

dp = offline_data_provider.getDefaultProvider()

# initialize default data (uninitialized entries will default to float 0.0)
dp.setData(addr, 'Analog_In.KL30_Rueckmessung', 13.98) # CV_monitor
dp.setData(addr, 'Analog_In.CCmon',              0.48) # CC_monitor
dp.setData(addr, 'Analog_Out.MotorTemp',        20.00) # motor_temp

# and later (direct):
with MwAppRpc(target=f"{addr}:{port}") as mwrec:
    mwrec.setupVars(
        var_names=["scheduler.Primary.runtime", "tasklevel.Recorder_runtime"]
    )
    state = mwrec.getState(verbose=True))
    if mwrec.getState(verbose=True) not in (RecState.ready, RecState.stopped):
        print(f"--> not in a valid state to start recording: {state.name}")
    else:
        mwrec.startRecording()
        time.sleep(3.0)
        mwrec.stopRecording()

        mwrec.getState(verbose=True)
        daq_data = mwrec.getData()
        idaq.printDataInfo(daq_data, outer_n=8)

# or via the default interface (in ttk_tools.isyst.ida_testware):
with IdaRecorder(addr, port) as recorder:
    recorder.setup(["Triangle.uint32", "Sine.float64", "Counter"])
    # recorder.setup will raise an MwRecError if setup failed

    recorder.start()
    time.sleep(3.0)
    recorder.stop()

    daq_data = recorder.getData()
    idaq.printDataInfo(daq_data, outer_n=8)
```

6.125.2 Variable Documentation

6.125.2.1 _STUB_VERSION

```
str ttk_tools.isyst.mwrec_client_offline_stub._STUB_VERSION = "1.0.0"  [protected]
```

6.126 ttk_tools.ni Namespace Reference

Access to NI (National Instruments) tools and interfaces, primarily LabView.

Namespaces

- namespace _labview

  Access to LabView VIs via COM server.
- namespace labview

  Interface wrapper for LabView VI COM access in ttk_tools.ni._labview.

6.126.1 Detailed Description

Access to NI (National Instruments) tools and interfaces, primarily LabView.

6.127 ttk_tools.ni._labview Namespace Reference

Access to LabView VIs via COM server.

Classes

- class LabView

  Access to LabView VI controls/variables via COM interface.

6.127.1 Detailed Description

Access to LabView VIs via COM server.

6.128 ttk_tools.ni.labview Namespace Reference

Interface wrapper for LabView VI COM access in ttk_tools.ni._labview.

Classes

- class LabView

  Access to LabView VI controls/variables via COM interface.

6.128.1 Detailed Description

Interface wrapper for LabView VI COM access in ttk_tools.ni._labview.

This module serves as "interface" to the precompiled module in delivery to enable code-completion in PyDev.

6.129 ttk_tools.offline_data_provider Namespace Reference

Data provider for dummy/offline/mock implementations of tool access modules (like ccp/xcp or real time platform/model data).

Classes

- class OfflineDataProvider

  Provider of dummy data for use in offline debugging.

Functions

- getDefaultProvider ()

  Get global default provider.
- setDefaultProvider (data_provider)

  Set a new global default provider.

Variables

- float DEFAULT_VALUE = 0.0

  initial/default value for new "variables"

6.129.1 Detailed Description

Data provider for dummy/offline/mock implementations of tool access modules (like ccp/xcp or real time platform/model data).

This servers as a source of read/writable dummy data for simulated interfaces in "offline" mode.

6.129.2 Function Documentation

6.129.2.1 getDefaultProvider()

```
ttk_tools.offline_data_provider.getDefaultProvider ( )
```

Get global default provider.

6.129.2.2 setDefaultProvider()

```
ttk_tools.offline_data_provider.setDefaultProvider (
              data_provider )
```

Set a new global default provider.

6.129.3 Variable Documentation

6.129.3.1 DEFAULT_VALUE

```
float ttk_tools.offline_data_provider.DEFAULT_VALUE = 0.0
```

initial/default value for new "variables"

6.130 ttk_tools.process_utils Namespace Reference

Interface to utilities for handling win32 (and x64) processes in ttk_tools._process_utils.

Functions

- killProcess (target_names, verbosity=1)

  Kill all processes that match the given target_name(s) (using kernel32).
- getCurrentProcessNames (normalize_case=True)

  Get image names of currently active processes (that is, the executable file names).
- getCurrentProcessPaths (normalize_case=True, file_names_only=False)

  Get image names/paths of currently active processes (that is, names or paths of the executable files).
- isProcessActive (process_name, ignore_case=True)

  Check whether at least one process using the supplied executable name is currently running.
- killProcess2 (image_name)

  Kill all processes that match the given image_name using the windows "taskkill" command running in a sub process.
- getCurrentPIDs ()

  Get a list of PIDs of running processes, similar to the output of win32process.EnumProcesses() (which was not yet available in older Python versions).
- killByPID (pid)

  Kill the process with the supplied ID (as long as the user has sufficient access rights).

6.130.1 Detailed Description

Interface to utilities for handling win32 (and x64) processes in ttk_tools._process_utils.

It appears that "handling" processes mostly seems to boil down to "killing" them, though.

6.130.2 Function Documentation

6.130.2.1 getCurrentPIDs()

```
ttk_tools.process_utils.getCurrentPIDs ( )
```

Get a list of PIDs of running processes, similar to the output of win32process.EnumProcesses() (which was not yet available in older Python versions).

Returns

a list of PID values.

6.130.2.2 getCurrentProcessNames()

```
ttk_tools.process_utils.getCurrentProcessNames (
              normalize_case = True )
```

Get image names of currently active processes (that is, the executable file names).

Parameters

normalize_case	cast process names to lowercase (as windows is not case sensitive); defaults to True

Returns

a dictionary {<process_name>: <number of active processes>, ...}

6.130.2.3 getCurrentProcessPaths()

```
ttk_tools.process_utils.getCurrentProcessPaths (
              normalize_case = True,
              file_names_only = False )
```

Get image names/paths of currently active processes (that is, names or paths of the executable files).

Limitation:

The used QueryFullProcessImageName is only supported since Windows Vista

Parameters

normalize_case	cast process names/paths to lowercase (as windows is not case sensitive); defaults to True
file_names_only	True: return only base names (e.g. "notepad.exe") False: return full file paths

Returns

a dictionary {<process name or path>: <number of active processes>, ...}

6.130.2.4 isProcessActive()

```
ttk_tools.process_utils.isProcessActive (
              process_name,
              ignore_case = True )
```

Check whether at least one process using the supplied executable name is currently running.

Parameters

process_name	name of process to check
ignore_case	True: ignore case of process_name

Returns

True if a matching process is running, otherwise False.

6.130.2.5 killByPID()

```
ttk_tools.process_utils.killByPID (
              pid )
```

Kill the process with the supplied ID (as long as the user has sufficient access rights).

Parameters

pid	process ID of the process to be killed

Returns

the return status of `TerminateProcess` (non-zero on success)

6.130.2.6 killProcess()

```
ttk_tools.process_utils.killProcess (
              target_names,
              verbosity = 1 )
```

Kill all processes that match the given target_name(s) (using kernel32).

Note that target may be a single name or list of names.

Parameters

target_names	a single name or a list of names of processes to be killed. If a process name contains unicode characters, it should be supplied as unicode string.
verbosity	verbosity of status output 0: (mostly) silent 1: show names/pids of killed processes 2: also show process access warnings

Returns

the number of killed processes.

6.130.2.7 killProcess2()

```
ttk_tools.process_utils.killProcess2 (
              image_name )
```

Kill all processes that match the given image_name using the windows "taskkill" command running in a sub process.

Parameters

image_name	the name of the process to be killed

6.131 ttk_tools.pyocd Namespace Reference

Wrapper for using PyOCD (https://pyocd.io/) as a data source for HIL tests.

Namespaces

- namespace _debugger

  Access to PyOCD session, provides attaching to running controllers and read/write of variables via debug symbols.
- namespace debugger

  Access to PyOCD session, provides attaching to running controllers and read/write of variables via debug symbols.

6.131.1 Detailed Description

Wrapper for using PyOCD (https://pyocd.io/) as a data source for HIL tests.

6.132 ttk_tools.pyocd._debugger Namespace Reference

Access to PyOCD session, provides attaching to running controllers and read/write of variables via debug symbols.

Classes

- class Debugger
- class SymbolNotFound

  No matching symbol was found in ELF file.

Variables

- level

6.132.1 Detailed Description

Access to PyOCD session, provides attaching to running controllers and read/write of variables via debug symbols.

Requirements:

- installed pyocd package:
  pip install pyocd

Note

- some dependencies might be missing:
  - ModuleNotFoundError: No module named 'importlib_metadata' pip install importlib-metadata
- debugger target might not be supported in the base release, e.g. "Target type stm32f407vgtx not recognized" See https://github.com/pyocd/pyOCD/blob/master/docs/target_support.md for how to install additional target support.
  - additional target packs can be installed:
    pyocd pack install stm32f407vgtx
    pyocd pack install stm32h743zitx
- ST-Link-Firmware has to be recent-ish
  - V2.J37.M26 seems to work

6.132.2 Variable Documentation

6.132.2.1 level

`ttk_tools.pyocd._debugger.level`

6.133 ttk_tools.pyocd.debugger Namespace Reference

Access to PyOCD session, provides attaching to running controllers and read/write of variables via debug symbols.

Classes

- class Debugger

  Wrapper for a PyOCD session and utility methods.

6.133.1 Detailed Description

Access to PyOCD session, provides attaching to running controllers and read/write of variables via debug symbols.

Requirements:

- installed pyocd package:
  pip install pyocd

Note

- some dependencies might be missing:
  - ModuleNotFoundError: No module named 'importlib_metadata' pip install importlib-metadata
- debugger target might not be supported in the base release, e.g. "Target type stm32f407vgtx not recognized" See https://github.com/pyocd/pyOCD/blob/master/docs/target_support.md for how to install additional target support.
  - additional target packs can be installed:
    pyocd pack install stm32f407vgtx
    pyocd pack install stm32h743zitx
- ST-Link-Firmware has to be recent-ish
  - V2.J37.M26 seems to work

6.134 ttk_tools.rst Namespace Reference

Access to RST (RST Industrie Automation GmbH) tools and interfaces.

Namespaces

- namespace _gamma_api

  Access to GammaV Service and related functionality.
- namespace _gamma_api_offline_stub

  A stub implementation of ttk_tools.rst.gamma_api for debugging without a installed GammaV middleware.
- namespace gamma_api

  Interface wrapper for GammaV Service Access in ttk_tools.rst._gamma_api.
- namespace gamma_api_common

  Common shared definitions and error classes for gamma_api.
- namespace gamma_api_offline_stub

  Interface wrapper for a stub implementation of gamma_api in ttk_tools.rst._gamma_api_offline_stub.
- namespace idaq
- namespace isygamma

  Access to Gamma V DLLs.

6.134.1 Detailed Description

Access to RST (RST Industrie Automation GmbH) tools and interfaces.

6.135 ttk_tools.rst._gamma_api Namespace Reference

Access to GammaV Service and related functionality.

Classes

- class DataAcquisition

  Data Acquisition with GammaV Logger Task.
- class Gamma

  GammaService-connection (application).

Functions

- _getIdArgs (ident)

  Get an identifier (or "address") argument as numeric index or as a name for use with gacmd calls.
- loadDaqDataFile (file_path)

  Read DAQ data from a file in "idaq" format.

Variables

- tuple is_win32 = (sys.platform == "win32")

  True if current platform is detected as Windows, otherwise False.
- tuple is_linux = (sys.platform.startswith("linux"))

  True if current platform is detected as Linux, otherwise False.
- str GAMMA_PROCESS_NAME = "gaservice.exe"

  name of gamma service executable (depends on used platform)
- str GAMMA_CMD_PROCESS_NAME = "gacmd.exe"

  name of gamma command executable (depends on used platform)
- int _GAMMA_SERVICE_START_DELAY_MS = 2000

  delay in [ms] to wait after starting a "new" Gamma V Service
- int _GAMMA_SERVICE_UPDATE_DELAY_MS = 2000

  delay in [ms] to wait after attaching to a "new" Gamma V Service (to give it time to update the variables list)
- bool _DEFAULT_SHOW_GAMMA_CALLS = False

  default value for `Gamma._show_gamma_calls`, set to True to print gamma process calls to stdout for debugging purposes.

6.135.1 Detailed Description

Access to GammaV Service and related functionality.

See also interface wrapper in ttk_tools.rst.gamma_api.

Note

Requires an installed Gamma V middleware (x64, matching the architecture of the used Python Interpreter).

6.135.2 Function Documentation

6.135.2.1 _getIdArgs()

```
ttk_tools.rst._gamma_api._getIdArgs (
              ident )  [protected]
```

Get an identifier (or "address") argument as numeric index or as a name for use with gacmd calls.

Parameters

ident	identifier/address for gamma PV/scheduler/schedulerSlot etc.

Returns

a tuple (identifier_type, identifier) with an address type depending on supplied identifier format and an identifier/address string.
A missing SystemName prefix will be added (as "default" system)

6.135.2.2 loadDaqDataFile()

```
ttk_tools.rst._gamma_api.loadDaqDataFile (
              file_path )
```

Read DAQ data from a file in "idaq" format.

Convenience function. See idaq.readDataData().

Parameters

file_path	path to DAQ data file

Returns

a daq data structure with {
<label>: {
"time": [<time_s>, <time_s>, ...],
"data": [<value>, <value>, ...],
"type": "<datatype-string>"
},
} entries.

6.135.3 Variable Documentation

6.135.3.1 _DEFAULT_SHOW_GAMMA_CALLS

```
bool ttk_tools.rst._gamma_api._DEFAULT_SHOW_GAMMA_CALLS = False  [protected]
```

default value for `Gamma._show_gamma_calls`, set to True to print gamma process calls to stdout for debugging purposes.

6.135.3.2 _GAMMA_SERVICE_START_DELAY_MS

```
int ttk_tools.rst._gamma_api._GAMMA_SERVICE_START_DELAY_MS = 2000  [protected]
```

delay in [ms] to wait after starting a "new" Gamma V Service

6.135.3.3 _GAMMA_SERVICE_UPDATE_DELAY_MS

```
int ttk_tools.rst._gamma_api._GAMMA_SERVICE_UPDATE_DELAY_MS = 2000  [protected]
```

delay in [ms] to wait after attaching to a "new" Gamma V Service (to give it time to update the variables list)

6.135.3.4 GAMMA_CMD_PROCESS_NAME

```
str ttk_tools.rst._gamma_api.GAMMA_CMD_PROCESS_NAME = "gacmd.exe"
```

name of gamma command executable (depends on used platform)

6.135.3.5 GAMMA_PROCESS_NAME

```
str ttk_tools.rst._gamma_api.GAMMA_PROCESS_NAME = "gaservice.exe"
```

name of gamma service executable (depends on used platform)

6.135.3.6 is_linux

```
tuple ttk_tools.rst._gamma_api.is_linux = (sys.platform.startswith("linux"))
```

True if current platform is detected as Linux, otherwise False.

6.135.3.7 is_win32

```
tuple ttk_tools.rst._gamma_api.is_win32 = (sys.platform == "win32")
```

True if current platform is detected as Windows, otherwise False.

6.136 ttk_tools.rst._gamma_api_offline_stub Namespace Reference

A stub implementation of ttk_tools.rst.gamma_api for debugging without a installed GammaV middleware.

Classes

- class _GaApiStub
  Minimal stub for "gaapi" (which is normally a module)
- class _PVStub
  Basic stub for class gaapi.PV (or rather, isygamma.PV).
- class DataAcquisition
  Data Acquisition with GammaV Logger Task.
- class Gamma
  GammaService-connection (application) stub.

Functions

- loadDaqDataFile (file_path)
  Read DAQ data from a file in "idaq" format.

Variables

- tuple is_win32 = (sys.platform == "win32")
  True if current platform is detected as Windows, otherwise False.
- tuple is_linux = (sys.platform.startswith("linux"))
  True if current platform is detected as Linux, otherwise False.
- str GAMMA_PROCESS_NAME = "gaservice.exe"
  name of gamma service executable (depends on used platform)
- str GAMMA_CMD_PROCESS_NAME = "gacmd.exe"
  name of gamma command executable (depends on used platform)
- str _OFFLINE_DATA_CONTEXT = "gamma"
- gaapi = _GaApiStub()

6.136.1 Detailed Description

A stub implementation of ttk_tools.rst.gamma_api for debugging without a installed GammaV middleware.

Gamma offline stubs use the fixed string "gamma" as context for connecting to the default offline data provider.

Example: Provide initial data

```
dp = offline_data_provider.getDefaultProvider()
ctx = "gamma"

# hil var values
dp.setData(ctx, 'iSyst:IOUserInterface.IO_Display.Konstanter.CV_mon', 13.98) # CV_monitor
dp.setData(ctx, 'iSyst:IOUserInterface.IO_Display.Konstanter.CC_mon',  0.48) # CC_monitor
dp.setData(ctx, 'iSyst:IOUserInterface.IO_Parameter.Analog.NTC_1',    20.00) # motor_temp
```

6.136.2 Function Documentation

6.136.2.1 loadDaqDataFile()

```
ttk_tools.rst._gamma_api_offline_stub.loadDaqDataFile (
              file_path )
```

Read DAQ data from a file in "idaq" format.

Convenience function. See idaq.readDataData().

Parameters

file_path	path to DAQ data file

Returns

a daq data structure with {
<label>: {
"time": [<time_s>, <time_s>, ...],
"data": [<value>, <value>, ...],
"type": "<datatype-string>"
},
} entries.

6.136.3 Variable Documentation

6.136.3.1 _OFFLINE_DATA_CONTEXT

```
str ttk_tools.rst._gamma_api_offline_stub._OFFLINE_DATA_CONTEXT = "gamma"  [protected]
```

6.136.3.2 gaapi

```
ttk_tools.rst._gamma_api_offline_stub.gaapi = _GaApiStub()
```

6.136.3.3 GAMMA_CMD_PROCESS_NAME

```
str ttk_tools.rst._gamma_api_offline_stub.GAMMA_CMD_PROCESS_NAME = "gacmd.exe"
```

name of gamma command executable (depends on used platform)

6.136.3.4 GAMMA_PROCESS_NAME

```
str ttk_tools.rst._gamma_api_offline_stub.GAMMA_PROCESS_NAME = "gaservice.exe"
```

name of gamma service executable (depends on used platform)

6.136.3.5 is_linux

```
tuple ttk_tools.rst._gamma_api_offline_stub.is_linux = (sys.platform.startswith("linux"))
```

True if current platform is detected as Linux, otherwise False.

6.136.3.6 is_win32

```
tuple ttk_tools.rst._gamma_api_offline_stub.is_win32 = (sys.platform == "win32")
```

True if current platform is detected as Windows, otherwise False.

6.137 ttk_tools.rst.gamma_api Namespace Reference

Interface wrapper for GammaV Service Access in ttk_tools.rst._gamma_api.

Classes

- class DataAcquisition

  Data Acquisition with GammaV Logger Task.
- class Gamma

  GammaService-connection (application).

Functions

- loadDaqDataFile (file_path)

  Read DAQ data from a file in "idaq" format.

6.137.1 Detailed Description

Interface wrapper for GammaV Service Access in ttk_tools.rst._gamma_api.

Note

Requires an installed Gamma V middleware (x64, matching the architecture of the used Python Interpreter).
This module serves as "interface" to the precompiled module in delivery to enable code-completion in PyDev.

6.137.2 Function Documentation

6.137.2.1 loadDaqDataFile()

```
ttk_tools.rst.gamma_api.loadDaqDataFile (
              file_path )
```

Read DAQ data from a file in "idaq" format.

Convenience function. See idaq.readDataData().

Parameters

file_path	path to DAQ data file

Returns

a daq data structure with {
<label>: {
"time": [<time_s>, <time_s>, ...],
"data": [<value>, <value>, ...],
"type": "<datatype-string>"
},
} entries.

6.138 ttk_tools.rst.gamma_api_common Namespace Reference

Common shared definitions and error classes for gamma_api.

Classes

- class BrowsingType
  Wrapper for enum gc_BrowsingType (BROWSING_TYPE_∗).
- class DaqMeasurementStatus
  State machine states for DAQ task.
- class DataType
  Gamma Data Types (gc_DataType).
- class ErrorCodes
  Gamma error codes (default return codes of API functions).

- class GammaApiError
- class GammaDaqError

  Base Error for GammaV DAQ / data logger functionality.
- class GammaDaqStateError
- class GammaError

  Base Error for GammaV connection.
- class LOG_LEVEL

  Log levels.
- class MESSAGE_MODE

  Message modes.
- class PVFieldType

  gc_PVFieldType - Field types of numeric PVs (gc_PV_FIELD_∗ entries).
- class PVProperty

  Properties for PV operation/access mode.
- class PVType

  ga_PVType

Variables

- str DEFAULT_LOGFILE_NAME = "Gamma_Logfile.txt"

  default name for log files
- int DEFAULT_LISTEN_PORT = 0

  default listen port of the gamma service (see gamma service configuration file)
- int DEFAULT_ATTACH_TIMEOUT = 5000

  default timeout in [ms] to wait for the gamma service during attach

6.138.1 Detailed Description

Common shared definitions and error classes for gamma_api.

6.138.2 Variable Documentation

6.138.2.1 DEFAULT_ATTACH_TIMEOUT

```
int ttk_tools.rst.gamma_api_common.DEFAULT_ATTACH_TIMEOUT = 5000
```

default timeout in [ms] to wait for the gamma service during attach

6.138.2.2 DEFAULT_LISTEN_PORT

```
int ttk_tools.rst.gamma_api_common.DEFAULT_LISTEN_PORT = 0
```

default listen port of the gamma service (see gamma service configuration file)

6.138.2.3 DEFAULT_LOGFILE_NAME

```
str ttk_tools.rst.gamma_api_common.DEFAULT_LOGFILE_NAME = "Gamma_Logfile.txt"
```

default name for log files

6.139 ttk_tools.rst.gamma_api_offline_stub Namespace Reference

Interface wrapper for a stub implementation of gamma_api in ttk_tools.rst._gamma_api_offline_stub.

Classes

- class DataAcquisition

  Data Acquisition with GammaV Logger Task.
- class Gamma

  GammaService-connection (application) stub.

Functions

- loadDaqDataFile (file_path)

  Read DAQ data from a file in "idaq" format.

6.139.1 Detailed Description

Interface wrapper for a stub implementation of gamma_api in ttk_tools.rst._gamma_api_offline_stub.

Intended for debugging without a installed Gamma V middleware.

This module serves as "interface" to the precompiled module in delivery to enable code-completion in PyDev.

Gamma offline stubs use the fixed string "gamma" as context for connecting to the default offline data provider.

Example: Provide initial data

```
dp = offline_data_provider.getDefaultProvider()
ctx = "gamma"

# hil var values
dp.setData(ctx, 'iSyst:IOUserInterface.IO_Display.Konstanter.CV_mon', 13.98) # CV_monitor
dp.setData(ctx, 'iSyst:IOUserInterface.IO_Display.Konstanter.CC_mon',  0.48) # CC_monitor
dp.setData(ctx, 'iSyst:IOUserInterface.IO_Parameter.Analog.NTC_1',    20.00) # motor_temp
```

6.139.2 Function Documentation

6.139.2.1 loadDaqDataFile()

```
ttk_tools.rst.gamma_api_offline_stub.loadDaqDataFile (
              file_path )
```

Read DAQ data from a file in "idaq" format.

Convenience function. See idaq.readDataData().

Parameters

file_path	path to DAQ data file

Returns

a daq data structure with {
<label>: {
"time": [<time_s>, <time_s>, ...],
"data": [<value>, <value>, ...],
"type": "<datatype-string>"
},
} entries.

6.140 ttk_tools.rst.idaq Namespace Reference

Classes

- class DaqDataError

  Errors during DAQ data decoding.

Functions

- _bstr (s)
- _ustr (b)

  Get a text string using the default CANape encoding.
- _getByteSizeString (size)

  Get a string representation of the supplied size in Bytes.
- _getSamplesPath ()

  Get path to a folder /samples on same level as a parent folder /src.
- readDaqData (file_handle)

  Read DAQ data in "idaq" format.
- writeDaqData (daq_data, file_handle, comment=None, verbosity=1)

  Write Read DAQ data in "idaq" format.

Variables

- str FILE_FORMAT = "1.2"

  marker for supported file format version
- str ENCODING = "latin-1"

  text encoding to use for non-binary parts (header, meta data)
- dict data_type_to_format_mapping

  Mapping of data type names to Python struct format codes.

6.140.1 Function Documentation

6.140.1.1 _bstr()

```
ttk_tools.rst.idaq._bstr (
              s ) [protected]
```

6.140.1.2 _getByteSizeString()

```
ttk_tools.rst.idaq._getByteSizeString (
              size )  [protected]
```

Get a string representation of the supplied size in Bytes.

Parameters

size	a size in Bytes

Returns

a formatted string with size and suitable SI unit.

6.140.1.3 _getSamplesPath()

```
ttk_tools.rst.idaq._getSamplesPath ( )  [protected]
```

Get path to a folder /samples on same level as a parent folder /src.

Returns

a path to /samples or to the current script's folder, if no /samples folder was detected.

6.140.1.4 _ustr()

```
ttk_tools.rst.idaq._ustr (
              b )  [protected]
```

Get a text string using the default CANape encoding.

Intended for use w/ canapapi.dll functions that return byte strings. Encoding errors (un-encodable characters) will be replaced. Compare ttk_base.baseutils.ustr.

6.140.1.5 readDaqData()

```
ttk_tools.rst.idaq.readDaqData (
              file_handle )
```

Read DAQ data in "idaq" format.

Parameters

file_handle	opened file handle (binary mode) with DAQ data

Returns

a daq data structure with {
<label>: {
"time": [<time_s>, <time_s>, ...],
"data": [<value>, <value>, ...],
"type": "<datatype-string>"
},
} entries.

6.140.1.6 writeDaqData()

```
ttk_tools.rst.idaq.writeDaqData (
              daq_data,
              file_handle,
              comment = None,
              verbosity = 1 )
```

Write Read DAQ data in "idaq" format.

Parameters

file_handle	file handle opened for binary writing

daq_data	a daq data structure with {
	<label>: {
	"time": [<time_s>, <time_s>, ...],
	"data": [<value>, <value>, ...],
	"type": "<datatype-string>" # optional },
	} entries.

comment	string comment to add to [FILE] section
verbosity	verbosity of log info 0: silent 1: + info messages 2: + show labels

Note

"type" entry in daq_data The IDAQ format uses fixed data types and Python data types may map to to different data types. The `"type"` entry can be used to select a suitable data type for storing the values in `"data"`. See data_type_to_format_mapping for available data types.
Without a defined `"type"`, values will default to type "float64".

"time" axis value IDAQ format only support a single share time axis, so all signals will have to share the same time values

6.140.2 Variable Documentation

6.140.2.1 data_type_to_format_mapping

```
dict ttk_tools.rst.idaq.data_type_to_format_mapping
```

Initial value:
```
00001 =  {
```

```
00002     "int8":         b"b",
00003     "sint8":        b"b", # permit sint<#> as alias for int<#>
00004     "uint8":        b"B",
00005
00006     "int16":        b"h",
00007     "sint16":       b"h",
00008     "uint16":       b"H",
00009
00010     "int32":        b"i",
00011     "sint32":       b"i",
00012     "uint32":       b"I",
00013
00014     "int64":        b"q",
00015     "sint64":       b"q",
00016     "uint64":       b"Q",
00017
00018     "float32":      b"f",
00019     "float64":      b"d",
00020 }
```

Mapping of data type names to Python struct format codes.

```
        Format    C Type            Python type           Bytes
```

int8 b signed char integer 1 uint8 B unsigned char integer 1 int16 h short integer 2 uint16 H unsigned short integer 2 int32 i int integer 4 uint32 I unsigned int integer 4 (int32) l long integer 4 (uint32) L unsigned long integer 4 int64 q long long integer 8 uint64 Q unsigned long long integer 8 float32 f float float 4 float64 d double float 8

Compare https://docs.python.org/2/library/struct.html => float32/float 64 use IEEE 754 binary32 or binary64 format

6.140.2.2 ENCODING

```
str ttk_tools.rst.idaq.ENCODING = "latin-1"
```

text encoding to use for non-binary parts (header, meta data)

6.140.2.3 FILE_FORMAT

```
str ttk_tools.rst.idaq.FILE_FORMAT = "1.2"
```

marker for supported file format version

6.141 ttk_tools.rst.isygamma Namespace Reference

Access to Gamma V DLLs.

Classes

- class _GaAPI

  Wrapper for gaapi dynamic library.
- class _GaCommon

  Wrapper for gacommon dynamic library.
- class Addresses

  Gamma addresses by nesting level.
- class BrowserInfo

  typedef struct gc_BrowserInfo.
- class Level1

  Address Level 1 elements (system). More...
- class Level2

  Address Level 2 elements (node/io/task/action/scheduler) More...
- class Level3

  Address Level 3 elements (memory/io-channel/scheduler-slot). More...
- class Level4

  Address Level 4 elements (group). More...
- class Level5

  Address Level 5 elements (pv-array/fifo). More...
- class NumericValue
- class PV

  Gamma PV.
- class PVHandle

  Opaque PV handle.
- class VersionNumber

  gc_VersionNumber

Functions

- _errCheck (result, func, args)

  Default "errcheck" callable for API calls that return a Gamma ErrorID.
- _settypes (lib, name, argtypes=None, restype=None)

  Assign "restype", "errcheck" and "argtypes" attributes to DLL functions.
- _bstr (s)

  Get a raw bytes string using the default encoding.
- _ustr (b)

  Get a text string using the default encoding.

Variables

- tuple IS_WIN32 = (sys.platform == "win32")
- str API_ENCODING = "latin-1"
- LengthInfo = namedtuple('LengthInfo', ['cur_len', 'max_len'])
- VersionInfo = namedtuple('VersionInfo', ['major', 'minor', 'release', 'build'])
- gaapi = _GaAPI()

  instance of wrapped gaapi DLL
- gacommon = _GaCommon()

  instance of gacommon DLL

6.141.1 Detailed Description

Access to Gamma V DLLs.

Note

Requires an installed Gamma V middleware (x64, matching the architecture of the used Python Interpreter)

The environment variable RST_GAMMA_DIR will be used to locate Gamma API libraries. The Gamma V installer should have created this variable.

6.141.2 Class Documentation

6.141.2.1 class ttk_tools::rst::isygamma::Level1

Address Level 1 elements (system).

6.141.2.2 class ttk_tools::rst::isygamma::Level2

Address Level 2 elements (node/io/task/action/scheduler)

6.141.2.3 class ttk_tools::rst::isygamma::Level3

Address Level 3 elements (memory/io-channel/scheduler-slot).

6.141.2.4 class ttk_tools::rst::isygamma::Level4

Address Level 4 elements (group).

6.141.2.5 class ttk_tools::rst::isygamma::Level5

Address Level 5 elements (pv-array/fifo).

6.141.3 Function Documentation

6.141.3.1 _bstr()

```
ttk_tools.rst.isygamma._bstr (
              s ) [protected]
```

Get a raw bytes string using the default encoding.

Intended for use w/ dll functions that expect byte strings. Encoding errors (un-encodable characters) will be ignored. Compare ttk_base.baseutils.bstr.

6.141.3.2 _errCheck()

```
ttk_tools.rst.isygamma._errCheck (
              result,
              func,
              args )  [protected]
```

Default "errcheck" callable for API calls that return a Gamma ErrorID.

See https://docs.python.org/3/library/ctypes.html#ctypes._FuncPtr.errcheck
See ErrorCodes

6.141.3.3 _settypes()

```
ttk_tools.rst.isygamma._settypes (
              lib,
              name,
              argtypes = None,
              restype = None )  [protected]
```

Assign "restype", "errcheck" and "argtypes" attributes to DLL functions.

If no restype is specified, a default ErrorID and errcheck callback will be assigned.

Parameters

lib	loaded DLL
name	name of lib function
argtypes	list of c-types to specify the argument types
restype	c-type to assign as result type None: use default ErrorID as restype and _errCheck callback

6.141.3.4 _ustr()

```
ttk_tools.rst.isygamma._ustr (
              b )  [protected]
```

Get a text string using the default encoding.

Intended for use w/ dll functions that return byte strings. Encoding errors (un-encodable characters) will be replaced. Compare ttk_base.baseutils.ustr.

6.141.4 Variable Documentation

6.141.4.1 API_ENCODING

```
str ttk_tools.rst.isygamma.API_ENCODING = "latin-1"
```

6.141.4.2 gaapi

```
ttk_tools.rst.isygamma.gaapi = _GaAPI()
```

instance of wrapped gaapi DLL

6.141.4.3 gacommon

```
ttk_tools.rst.isygamma.gacommon = _GaCommon()
```

instance of gacommon DLL

6.141.4.4 IS_WIN32

```
tuple ttk_tools.rst.isygamma.IS_WIN32 = (sys.platform == "win32")
```

6.141.4.5 LengthInfo

```
ttk_tools.rst.isygamma.LengthInfo = namedtuple('LengthInfo', ['cur_len', 'max_len'])
```

6.141.4.6 VersionInfo

```
ttk_tools.rst.isygamma.VersionInfo = namedtuple('VersionInfo', ['major', 'minor', 'release',
'build'])
```

6.142 ttk_tools.siemens Namespace Reference

Access to SIEMENS automation technology tools and interfaces.

Namespaces

- namespace simit_common

  Common definitions for SIMIT signal data access via SHM coupling.
- namespace simit_shm

  Access SIMIT signal data via SHM coupling.

6.142.1 Detailed Description

Access to SIEMENS automation technology tools and interfaces.

6.143 ttk_tools.siemens.simit_common Namespace Reference

Common definitions for SIMIT signal data access via SHM coupling.

Classes

- class DataType

  Data types from "Tabelle 2-3 Typkennung der Signale".
- class Direction

  "E/A-Kennung, d.

6.143.1 Detailed Description

Common definitions for SIMIT signal data access via SHM coupling.

6.144 ttk_tools.siemens.simit_shm Namespace Reference

Access SIMIT signal data via SHM coupling.

Classes

- class SimitShm

  Wrapper for SIMIT SHM access.
- class SimitSignal

  A SIMIT signal configured in SHM Coupling.

Variables

- str shm_name = "SIMIT_SHM"
- shm = None

6.144.1 Detailed Description

Access SIMIT signal data via SHM coupling.

Note

- Windows only
- Requires pywin32 extensions for Windows SHM/Mutex access

6.144.2 Variable Documentation

6.144.2.1 shm

```
ttk_tools.siemens.simit_shm.shm = None
```

6.144.2.2 shm_name

```
str ttk_tools.siemens.simit_shm.shm_name = "SIMIT_SHM"
```

6.145 ttk_tools.vector Namespace Reference

Access to Vector (Vector Informatik GmbH) tools and interfaces.

Namespaces

- namespace _canapeapi

  Access to CANape devices via CANapeAPI.
- namespace _canapeapi_offline_stub

  A dummy/stub implementation of the default ttk_tools.vector.canapeapi for debugging without access to vector software, hardware or license keys.
- namespace _vflash_api

  Access to vFlash API functions.
- namespace canapeapi

  Wrapper for Access to CANape devices via CANapeAPI.
- namespace canapeapi_common

  Common utility functions and classes for CANape access (used in canapeapi.py).
- namespace canapeapi_offline_stub

  Interface to a stub implementation of the default canapeapi.py for debugging without access to vector software, hardware or license keys.
- namespace mdf4_reader
- namespace mdflib

  MDF-file I/O via Vector MdfLib DLL (for MDF version 3).
- namespace vflash_api

  Interface to vFlash API function access in ttk_tools.vector._vflash_api.
- namespace vflash_api_common

  Common definitions, utility functions and classes for vFlash API access.

6.145.1 Detailed Description

Access to Vector (Vector Informatik GmbH) tools and interfaces.

6.146 ttk_tools.vector._canapeapi Namespace Reference

Access to CANape devices via CANapeAPI.

Classes

- class Appversion

  CANape API application version info structure.
- class BaseDevice

  Base class for CANape devices with generic/common/basic functionality.
- class CANapeASAP3

  CANape ASAP3-connection (application).
- class CANapeDevice

  CANape device class (ASAP3-devices: CCP, XCP).
- class CANapeDiag

Diagnostics device class (UDS, ...).

- class DataAcquisition

  Data acquisition for multiple CANape devices.
- class DiagNumericParameter

  Numeric values for diagnostic parameters.
- class Scripting

  Access to extended scripting functions added in CANape 9.
- class TCalibrationObjectValue

  Calibration Object Value data structure as used in CANapeAPI.
- class TConverterInfo

  Basic information about available MDF converters.
- class TTaskInfo

  ECU Task data structure as used in CANapeAPI.
- class Version_t

  CANape API version info structure.

Functions

- _bstr (s)

  Get a raw bytes string using the default CANape encoding.
- _ustr (b)

  Get a text string using the default CANape encoding.
- _printCurrentExceptionInfo (header=None)

  Print information about the currently active exception (mainly for debugging purposes).
- _getCanapeRegPath ()

  Get the current installation path of CANape from the windows registry (HKLM\SOFTWARE\VECTOR\CANape).
- _setCanapeRegPath (path)

  Select the active CANape installation by setting the path in the windows registry (HKLM\SOFTWARE\VECTOR\CANape).

Variables

- str CANAPE_REG_KEY = r"HKEY_LOCAL_MACHINE\SOFTWARE\VECTOR\CANape"

  Registry key for CANape installations.

6.146.1 Detailed Description

Access to CANape devices via CANapeAPI.

6.146.2 Function Documentation

6.146.2.1 _bstr()

```
ttk_tools.vector._canapeapi._bstr (
              s )  [protected]
```

Get a raw bytes string using the default CANape encoding.

Intended for use w/ canapapi.dll functions that expect byte strings. Encoding errors (un-encodable characters) will be ignored. Compare ttk_base.baseutils.bstr.

6.146.2.2 _getCanapeRegPath()

```
ttk_tools.vector._canapeapi._getCanapeRegPath ( )  [protected]
```

Get the current installation path of CANape from the windows registry (HKLM\SOFTWARE\VECTOR\CANape).

See CANAPE_REG_KEY.

Returns

the current installation path setting, or None if access failed.

6.146.2.3 _printCurrentExceptionInfo()

```
ttk_tools.vector._canapeapi._printCurrentExceptionInfo (
              header = None )  [protected]
```

Print information about the currently active exception (mainly for debugging purposes).

Parameters

header	an optional header line to print

6.146.2.4 _setCanapeRegPath()

```
ttk_tools.vector._canapeapi._setCanapeRegPath (
              path )  [protected]
```

Select the active CANape installation by setting the path in the windows registry (HKLM\SOFTWARE\VECTOR\CANape).

See CANAPE_REG_KEY.

Note

Multiple CANape installations can exist in parallel, change the installation path in the windows registry to select the active installation.

Parameters

path	absolute path to the CANape installation dir

Returns

True if value was set, otherwise False

6.146.2.5 _ustr()

```
ttk_tools.vector._canapeapi._ustr (
              b )  [protected]
```

Get a text string using the default CANape encoding.

Intended for use w/ canapapi.dll functions that return byte strings. Encoding errors (un-encodable characters) will be replaced. Compare ttk_base.baseutils.ustr.

6.146.3 Variable Documentation

6.146.3.1 CANAPE_REG_KEY

```
str ttk_tools.vector._canapeapi.CANAPE_REG_KEY = r"HKEY_LOCAL_MACHINE\SOFTWARE\VECTOR\CANape"
```

Registry key for CANape installations.

Note that on 64bit systems, this automatically maps to the actual HKEY_LOCAL_MACHINE\SOFTWARE\Wow6432↩ Node\Vector\CANape due to Windows shenanigans.

6.147 ttk_tools.vector._canapeapi_offline_stub Namespace Reference

A dummy/stub implementation of the default ttk_tools.vector.canapeapi for debugging without access to vector software, hardware or license keys.

Classes

- class BaseDevice

  Base class (stub) for CANape devices with generic/common/basic functionality.
- class CANapeASAP3

  Stub for CANape ASAP3-connection (application).
- class CANapeDevice

  CANape device class (ASAP3-devices: CCP, XCP).
- class CANapeDiag

  Diagnostics device stub class (UDS, ...).
- class CANapeOfflineStubNotSupported

  Error class for features not supported in the stub implementation.
- class DataAcquisition

  Data acquisition (stub) for multiple CANape devices.
- class Scripting

  Access to extended scripting functions.

Variables

- tuple STUB_VERSION = (4, 7, 1)

6.147.1 Detailed Description

A dummy/stub implementation of the default ttk_tools.vector.canapeapi for debugging without access to vector software, hardware or license keys.

CANapeAPI offline stubs use a device's configured module_name as context for connecting to the default offline data provider (see CANapeDevice).

Example: Provide initial data

```
dp = offline_data_provider.getDefaultProvider()

# cal var values, XCP device/module "DEMO_ECU"
dp.setData("DEMO_ECU", "EMotor_Temp_Ist", 20.0)
dp.setData("DEMO_ECU", "EMotor_N_Ist",    1000)

# can var values, CAN device/module "CAN"
dp.setData("CAN",      "GE_Oiltemp",      24.4)
```

6.147.2 Variable Documentation

6.147.2.1 STUB_VERSION

```
tuple ttk_tools.vector._canapeapi_offline_stub.STUB_VERSION = (4, 7, 1)
```

6.148 ttk_tools.vector._vflash_api Namespace Reference

Access to vFlash API functions.

Classes

- class VFlashAPI

  Python Wrapper for accessing vFlash API functions.

Functions

- _bstr (s)

  Get a raw bytes string using the default encoding.

6.148.1 Detailed Description

Access to vFlash API functions.

6.148.2 Function Documentation

6.148.2.1 _bstr()

```
ttk_tools.vector._vflash_api._bstr (
              s )  [protected]
```

Get a raw bytes string using the default encoding.

Intended for dll functions that expect byte strings. Encoding errors (un-encodable characters) will be ignored. Compare ttk_base.baseutils.bstr.

6.149 ttk_tools.vector.canapeapi Namespace Reference

Wrapper for Access to CANape devices via CANapeAPI.

Classes

- class CANapeASAP3

  CANape ASAP3-connection (application).
- class CANapeDevice

  CANape device class (ASAP3-devices: CCP, XCP).
- class CANapeDiag

  Diagnostics device class (UDS, ...).
- class DataAcquisition

  Data acquisition for multiple CANape devices.
- class Scripting

  Access to extended scripting functions added in CANape 9.

6.149.1 Detailed Description

Wrapper for Access to CANape devices via CANapeAPI.

This serves as "interface" to the precompiled module in delivery to enable code-completion in PyDev.

See ttk_tools.vector._canapeapi for base implementation.

6.150 ttk_tools.vector.canapeapi_common Namespace Reference

Common utility functions and classes for CANape access (used in canapeapi.py).

Classes

- class CANapeAccessError

  Cal-object access errors for CANape connection.
- class CANapeDaqError

  Data Acquisition errors for CANape connection.
- class CANapeDiagError

  Errors for CANape diagnostic connections.
- class CANapeDiagTimeoutError

  Timeout occurred during a diag request.
- class CANapeError

  Base error for CANape connection.
- class CANapeErrorString

  Container for diagnostic response values that may have multiple value representations.
- class CANapeScriptError

  A wrapper for error message strings so they can be distinguished from "normal" string values (like contents of ASCII-type variables).
- class DATA_TYPE

Data type constants (`TYPE_*`) from enum `TAsap3DataType`

- class DEV

  Device channel constants (DEV_*) from CANapAPI.h.
- class DIAG_PARAM_TYPE

  Diagnostic response parameter types (`Param*`) from enum `EnParamType`.
- class DIAG_SERVICE

  Diagnostic service states (`DIAG_SERVICE_*`) from enum `eServiceStates`.
- class DiagParameterName

  Container for diagnostic response name that may contain sub-parameter names for "complex parameters".
- class DiagParameterValue
- class DRIVER

  Driver type constants (`ASAP3_DRIVER_*`) from enum `tDriverType`.
- class MEASUREMENT_STATE

  Measurement state constants (`eT_MEASUREMENT_*`) from enum `tMeasurementState` (new in CANape 9).
- class REPR

  Measurement data format representation constants from enum `TFormat` (original constants have no specific prefix)
- class SCRIPT_STATUS

  Script execution status values from enum `TScriptStatus`.
- class TYPE_SWITCH

  Device online switch state constants (`TYPE_SWITCH_*`) from enum `TAsap3ECUState`.
- class VALUE_TYPE

  Value type constants from enum `TValueType` (original constants have no specific prefix)

6.150.1 Detailed Description

Common utility functions and classes for CANape access (used in canapeapi.py).

6.151 ttk_tools.vector.canapeapi_offline_stub Namespace Reference

Interface to a stub implementation of the default canapeapi.py for debugging without access to vector software, hardware or license keys.

Classes

- class CANapeASAP3

  CANape ASAP3-connection (application).
- class CANapeDevice

  CANape device class (ASAP3-devices: CCP, XCP).
- class CANapeDiag

  Diagnostics device class (UDS, ...).
- class DataAcquisition

  Data acquisition (stub) for multiple CANape devices.
- class Scripting

  Access to extended scripting functions added in CANape 9.

6.151.1 Detailed Description

Interface to a stub implementation of the default canapeapi.py for debugging without access to vector software, hardware or license keys.

See ttk_tools.vector._canapeapi_offline_stub for base implementation.

CANapeAPI offline stubs use a device's configured module_name as context for connecting to the default offline data provider (see CANapeDevice).

Example: Provide initial data

```
dp = offline_data_provider.getDefaultProvider()

# cal var values, XCP device/module "DEMO_ECU"
dp.setData("DEMO_ECU", "EMotor_Temp_Ist", 20.0)
dp.setData("DEMO_ECU", "EMotor_N_Ist",    1000)

# can var values, CAN device/module "CAN"
dp.setData("CAN",      "GE_Oiltemp",      24.4)
```

6.152 ttk_tools.vector.mdf4_reader Namespace Reference

Classes

- class Block

  A generic block with header, links and data sections (for all blocks except for special case IDBLOCK)
- class BlockHeader

  String encoding used in an MDF file is UTF-8 (applies to TXBLOCK and MDBLOCK data)
- class CaBlock

  Channel Array Block.
- class CaFlags

  Container for decoded uint32 ca_flags.
- class CANopenDate

  Container for channel data type "CANopen date".
- class CANopenTime

  Container for channel data type "CANopen time".
- class CaStorage

  Channel storage (ca_storage) values: how element values are stored.
- class CaTypes

  Channel array type (ca_type) values: array semantics.
- class CcBlock

  Channel Conversion Block.
- class CcFlags

  Container for decoded uint16 cc_flags.
- class CcTypes

  Conversion type / formula identifier (cc_type) values.
- class CgBlock

  Channel Group Block.
- class CgFlags

  Container for decoded cg_flags.
- class CnBlock

  Channel Block.

- class CnDataTypes

  Channel data type (cn_data_type) values.
- class CnFlags

  Container for decoded uint32 cn_flags.
- class CnSyncTypes

  Channel synchronization type (cn_sync_type) values.
- class CnTypes

  Channel type (cn_type) values.
- class DataRecords

  Iterator for fixed and variable length signal data records, returns sliced records from raw byte payload data.
- class DgBlock

  Data Group Block.
- class DlBlock

  Data List Block.
- class DlFlags

  Container for decoded uint8 dl_flags.
- class DtBlock

  Data Block.
- class DzZipType

  Zip algorithm type (dz_zip_type) used to compress data stored in dz_data.
- class EnumInfo

  Placeholder for ttk_base.baseutils.EnumInfo.
- class EnumInfoMeta
- class FhBlock

  File History Block.
- class Flags

  An integer/long as container for decoded bit values.
- class FloatWithRepr

  A float value that has an associated (textual) representation.
- class HdBlock

  Header Block, unique.
- class HlBlock

  Header List Block.
- class HlFlags

  Container for decoded uint8 hl_flags.
- class IdBlock

  File Identification Block.
- class IntWithEnumRepr

  An integer value that shows an EnumInfo repr of itself (if a match is found)
- class IntWithRepr

  An integer value that has an associated (textual) representation.
- class Link

  LINK file position/offset as a long value with custom representation.
- class Mdf4Reader

  Reader for MDF4 files; read data will be stored to member vars.
- class MdfDataError

  Feature not (yet) supported)
- class MdfError
- class MdfNotYetSupported

  Meta class for EnumInfo; adds a mechanism for custom value lookups.
- class MdfStructureError

Data or data types are inconsistent.

- class MdfVersionError

  Detected MDF version is unsupported.

- class MdTxBlock

  Text block with optional meta data.

- class SdBlock

  Signal Data Block.

- class SiBlock

  Source Information Block.

- class SiBusTypes

  Source bus type (si_bus_type), additional classification of used bus (should be 0 for si_type >= 3).

- class SiFlags

  Container for decoded uint8 si_flags.

- class SiTypes

  Source type (si_type), additional source classification.

Functions

- printMembers (inst, header=None, indent="")
- parseXmlSnippet (xml)
- getNodeText (node)

  Get text content from a DOM node.

- getTagText (parent_node, tag_name="TX", joiner="\n")

  Get text from all elements with the given tag name.

- decodeZeroTerminatedString (raw_bytes, encoding="utf-8")

  Get a string representation from a zero-terminated byte string buffer.

- getLinkedBlocks (file_handle, block_class, first_pos, next_pos_attr_name)

  Get/resolve a linked list of MDF-Blocks where each block contains a LINK / start position of the following block.

- _fixup2digitYear (yy_year)

  Get "full" year value for the supplied two-digit year that is closest to the current year.

- getUnzippedData (raw_data)

  Uncompress block data from a DZBLOCK to get the "original" data.

- getData (file_handle, pos=None)

  Get (potentially merged) data payload.

- getSignalData (file_handle, pos=None)

  Get (potentially merged) signal data payload (i.e.

- clearCache ()

  MDF Spec 4.1.0, Section 5.4.4 Linking of Blocks: "Note that there are block types for which there are no unique links.

- getCachedBlock (file_handle, pos, block_class, fallback=None, optional=True)

  Get a block from file handle (see Block) and cache it.

- getCachedText (file_handle, pos, fallback=None)

  Get contents of an (optional) text block (TX or MD) link (i.e.

- printLineHeader (text, indent="", width=80)

  Print a "header" line for log output.

- readMdf4 (file_path, verbosity=0)

  Read an MDF4 file.

- _printDataInfo (values, indent_level=1, print_all=False, outer_n=3, label_min_width=16)

  Print a summary of the supplied values to stdout.

Variables

- _current_year = time.localtime()[0]
- _current_century = _current_year - (_current_year % 100)
- dict _block_cache = {}
- _handle_ref = None
- examples_path = os.path.join("..", "samples")
- file_path = os.path.join(examples_path, "Simple", "Vector_CANape.MF4")

6.152.1 Function Documentation

6.152.1.1 _fixup2digitYear()

```
ttk_tools.vector.mdf4_reader._fixup2digitYear (
              yy_year )  [protected]
```

Get "full" year value for the supplied two-digit year that is closest to the current year.

Parameters

yy_year	two-digit year value 0..99

Returns

a fill year (for the foreseeable future: a 4-digit value)

6.152.1.2 _printDataInfo()

```
ttk_tools.vector.mdf4_reader._printDataInfo (
              values,
              indent_level = 1,
              print_all = False,
              outer_n = 3,
              label_min_width = 16 )  [protected]
```

Print a summary of the supplied values to stdout.

Used internally as (somewhat) verbose output of daq data.

Parameters

values	a scalar value or values in lists or dictionaries
indent_level	starting indent level
print_all	True: list values will not be summarized
outer_n	if `print_all` is False, only the outer n values from start and end of data will be printed.
label_min_width	minimum width for data label display formatting

6.152.1.3 clearCache()

```
ttk_tools.vector.mdf4_reader.clearCache ( )
```

MDF Spec 4.1.0, Section 5.4.4 Linking of Blocks: "Note that there are block types for which there are no unique links.

These blocks are called "sharable" blocks. Currently, TX, MD, SI and CC blocks are sharable blocks. Sharing may be used to reduce duplication of identical blocks (e.g. several channels sharing the same CC block or SI block)."

Clear currently cached blocks

6.152.1.4 decodeZeroTerminatedString()

```
ttk_tools.vector.mdf4_reader.decodeZeroTerminatedString (
              raw_bytes,
              encoding = "utf-8" )
```

Get a string representation from a zero-terminated byte string buffer.

Additionally, MDF4/Windows line endings (\r\n) will be normalized to just \n.

Parameters

raw_bytes	raw byte string
encoding	optional encoding with which to decode the raw string

Returns

a string cut-off at the first zero byte, unicode if an encoding was specified, otherwise a byte string.

6.152.1.5 getCachedBlock()

```
ttk_tools.vector.mdf4_reader.getCachedBlock (
              file_handle,
              pos,
              block_class,
              fallback = None,
              optional = True )
```

Get a block from file handle (see Block) and cache it.

Later calls for the same block/position will return the cached block instance.

Note

For optional=True, basic_blocks.getOptionalBlock behaves similarly (but without caching)

Parameters

file_handle	opened file handle
pos	position ("LINK") in file to start of block, see Block
block_class	class of block to be read
fallback	fallback value to use for optional blocks if link cannot be resolved
optional	True: block is optional, False: block is required (link/pos may not be NIL/0)

Returns

a Block-derived instance or the fallback value.

6.152.1.6 getCachedText()

```
ttk_tools.vector.mdf4_reader.getCachedText (
              file_handle,
              pos,
              fallback = None )
```

Get contents of an (optional) text block (TX or MD) link (i.e.

just plain text, not the TxMdBlock instance). Just for convenience. Block access will be cached. If link position cannot be resolved (NIL/0), the fallback value will be returned. See basic_blocks.getText for un-cached version.

Parameters

file_handle	opened file handle
pos	position ("LINK") in file to start of TXBLOCK
fallback	fallback value to return if block is not available

Returns

a string or the given fallback value.

6.152.1.7 getData()

```
ttk_tools.vector.mdf4_reader.getData (
              file_handle,
              pos = None )
```

Get (potentially merged) data payload.

Data source may be

- a single DtBlock
- DlBlocks + linked DtBlocks
- HlBlock + linked DlBlocks + linked DzBlocks/DtBlocks

Returns

a byte string.

6.152.1.8 getLinkedBlocks()

```
ttk_tools.vector.mdf4_reader.getLinkedBlocks (
              file_handle,
              block_class,
              first_pos,
              next_pos_attr_name )
```

Get/resolve a linked list of MDF-Blocks where each block contains a LINK / start position of the following block.

Parameters

file_handle	handle to MDF file
block_class	a Block-derived class to use to resolve/read each linked block.
first_pos	position of first Block to retrieve
next_pos_attr_name	name of attribute containing the next LINK position/offset in block_class

Example: Resolving linked file history blocks in a HDBLOCK

```
file_history = getLinkedBlocks(
    file_handle, FhBlock, self.hd_fh_first, "fh_fh_next"
)
```

Returns

a list of block_class instances in defined order.

6.152.1.9 getNodeText()

```
ttk_tools.vector.mdf4_reader.getNodeText (
              node )
```

Get text content from a DOM node.

6.152.1.10 getSignalData()

```
ttk_tools.vector.mdf4_reader.getSignalData (
              file_handle,
              pos = None )
```

Get (potentially merged) signal data payload (i.e.

raw data for for VLSD). This is quite similar to getData(), as DlBlocks (and potentially HLBLocks) may again be involved to split data across multiple chunks.

Data source may be

- a single SdBlock
- DlBlocks + linked SdBlocks
- HlBlock + linked DlBlocks + linked DzBlocks/SdBlocks

Returns

a byte string.

6.152.1.11 getTagText()

```
ttk_tools.vector.mdf4_reader.getTagText (
              parent_node,
              tag_name = "TX",
              joiner = "\n" )
```

Get text from all elements with the given tag name.

Text will be joined if more than one tag exists. Any leading/trailing whitespace will be removed.

Parameters

parent_node	parent minidom-node
tag_name	name of tag with text content to extract
joiner	character/string which which to join text of multiple tags

Returns

a string (might be empty if no matching tag was found or tag content was empty)

6.152.1.12 getUnzippedData()

```
ttk_tools.vector.mdf4_reader.getUnzippedData (
              raw_data )
```

Uncompress block data from a DZBLOCK to get the "original" data.

Parameters

raw_data	byte string of DZBLOCK payload data

Returns

a tuple: (dz_org_block_type, data).

6.152.1.13 parseXmlSnippet()

```
ttk_tools.vector.mdf4_reader.parseXmlSnippet (
              xml )
```

6.152.1.14 printLineHeader()

```
ttk_tools.vector.mdf4_reader.printLineHeader (
              text,
              indent = "",
              width = 80 )
```

Print a "header" line for log output.

Parameters

text	header text
indent	indent (typically a string of spaces) to place before header
width	width of header line

Example:

```
printLineHeader("foo", width=42)
>> # foo ####################################
```

6.152.1.15 printMembers()

```
ttk_tools.vector.mdf4_reader.printMembers (
              inst,
              header = None,
              indent = "" )
```

6.152.1.16 readMdf4()

```
ttk_tools.vector.mdf4_reader.readMdf4 (
              file_path,
              verbosity = 0 )
```

Read an MDF4 file.

Parameters

file_path	path to MDF4 file
verbosity	level of log verbosity. 0: (mostly) silent

Returns

a DAQ data structure, S

6.152.2 Variable Documentation

6.152.2.1 _block_cache

```
dict ttk_tools.vector.mdf4_reader._block_cache = {}  [protected]
```

6.152.2.2 _current_century

```
ttk_tools.vector.mdf4_reader._current_century = _current_year - (_current_year % 100)  [protected]
```

6.152.2.3 _current_year

```
ttk_tools.vector.mdf4_reader._current_year = time.localtime()[0]  [protected]
```

6.152.2.4 _handle_ref

```
ttk_tools.vector.mdf4_reader._handle_ref = None  [protected]
```

6.152.2.5 examples_path

```
ttk_tools.vector.mdf4_reader.examples_path = os.path.join("..", "samples")
```

6.152.2.6 file_path

```
ttk_tools.vector.mdf4_reader.file_path = os.path.join(examples_path, "Simple", "Vector_↩
CANape.MF4")
```

6.153 ttk_tools.vector.mdflib Namespace Reference

MDF-file I/O via Vector MdfLib DLL (for MDF version 3).

Classes

- class MDF_CHN
  MDF Channel Types (MDF_CHN_∗-constants)
- class MDF_ERROR
  MDF Error Values (E_MDF_∗-constants)
- class MDF_MODULE
  MDF Module/Block Types (MDF_MODULE_∗-constants)
- class MDF_OPEN
  Values for MDF_OPEN_∗-constants.
- class MDF_SEEK
  Values for MDF_SEEK_∗-constants.
- class MDF_STATUS
  MDF status values (MDF_∗-constants).
- class MdfLibError
  Base Error for MdfLib access.

Functions

- setMdfLibPath (dll_path)

  Set the path to the MDFLib DLL.
- getMdfLibHandle ()

  Get a handle to the MdfLib DLL.
- getMdfError (error_code)

  Get a correct interpretation of MDFERROR return values, which are 16-bit signed int values.
- getCurrentError (verbosity=1)

  Get info on the most recent error condition from the MDFLib DLL.
- _handleMdfErrorStatus (mdf_error, context="", ignore_non_severe_errors=False)

  Raise a MdfLibError if decoded MDF-Error indicates an error state.
- readMdf (file_path, verbosity=1, signal_names=None)

  Read channel data from the supplied MDF-File.

Variables

- _lib_path = None
- _lib = None
- int _IS_X64 = sys.maxsize > 2**32

6.153.1 Detailed Description

MDF-file I/O via Vector MdfLib DLL (for MDF version 3).

Actually, right now only "I" is implemented, i.e. read data from MDF files.

6.153.2 Function Documentation

6.153.2.1 _handleMdfErrorStatus()

```
ttk_tools.vector.mdflib._handleMdfErrorStatus (
              mdf_error,
              context = "",
              ignore_non_severe_errors = False )  [protected]
```

Raise a MdfLibError if decoded MDF-Error indicates an error state.

Parameters

mdf_error	mdf-lib error code
context	info string that describes where the error occurred
ignore_non_severe_errors	if True, only severe errors will be raised as MdfLibError

6.153.2.2 getCurrentError()

```
ttk_tools.vector.mdflib.getCurrentError (
              verbosity = 1 )
```

Get info on the most recent error condition from the MDFLib DLL.

This seems to work **only** for severe/critical errors (i.e. error codes < 0), though.

Parameters

verbosity	> 0: log info to stdout if an error is active

Returns

a tuple (error_code, error_text).

6.153.2.3 getMdfError()

```
ttk_tools.vector.mdflib.getMdfError (
              error_code )
```

Get a correct interpretation of MDFERROR return values, which are 16-bit signed int values.

Parameters

error_code	signed 16-bit error value

Returns

a tuple (error_code, error_message).

6.153.2.4 getMdfLibHandle()

```
ttk_tools.vector.mdflib.getMdfLibHandle ( )
```

Get a handle to the MdfLib DLL.

Returns

a DLL handle.

6.153.2.5 readMdf()

```
ttk_tools.vector.mdflib.readMdf (
              file_path,
              verbosity = 1,
              signal_names = None )
```

Read channel data from the supplied MDF-File.

Parameters

file_path	path to .mdf-file
verbosity	level of debug feedback 0: only error/warning messages 1: + channel info 2: + extended info
signal_names	list of signal (mdf: channel) names to read from MDF. None: read all available signals

Example: Data Dictionary Structure:

```
mdf_data[<signal name>] = {
    'name':    <signal name>,
    'context': <signal device/message context>,
    'group':   <channel group number in mdf-data>,
    'data':    <list of data entries>,
    'time':    <list of time entries>,
    'descr':   <channel description>,
    'text':    <channel comment text>,
}
```

Note

Channel groups will be "flattened" to simplify access, so signals are no longer grouped by channels .

To handle signals with the same name appearing in different groups (not a very common case), further signals of the same name will only be added with their `context.` prefix (or, if context is not available, with a prefix of `group<number>.`).

Returns

a data dictionary with data for each found signal (see data structure example, above).

Signals will be added as both `signal_name` and `context.signal_name`, e.g. "u8ecuErrorState" and "asap.u8ecuErrorState" (for a cal signal) or "EngTrqMin" and "EM2_RQ_ENG.EngTrqMin" (for a can signal).

6.153.2.6 setMdfLibPath()

```
ttk_tools.vector.mdflib.setMdfLibPath (
               dll_path )
```

Set the path to the MDFLib DLL.

Parameters

dll_path	path to DLL-file or to its containing folder, in which case the first "mfdlib∗.dll" will be chosen, or None, which will enable the default windows dll search path.

Note

A copy of `MdfLibms8.dll` will be available in CANape's `Exec` folder. A copy of `MdfLibms8_x64.dll` will be available in CANape's `Exec64` folder (CANape 17+) Also, `MdfLibSort_x64.dll` will work as a substitute.

6.153.3 Variable Documentation

6.153.3.1 _IS_X64

```
int ttk_tools.vector.mdflib._IS_X64 = sys.maxsize > 2**32  [protected]
```

6.153.3.2 _lib

```
ttk_tools.vector.mdflib._lib = None  [protected]
```

6.153.3.3 _lib_path

```
ttk_tools.vector.mdflib._lib_path = None  [protected]
```

6.154 ttk_tools.vector.vflash_api Namespace Reference

Interface to vFlash API function access in ttk_tools.vector._vflash_api.

Classes

- class VFlashAPI
  Python Wrapper for accessing vFlash API functions.

6.154.1 Detailed Description

Interface to vFlash API function access in ttk_tools.vector._vflash_api.

6.155 ttk_tools.vector.vflash_api_common Namespace Reference

Common definitions, utility functions and classes for vFlash API access.

Classes

- class VFlashDLLError
  Flashing ended with an error state.
- class VFlashError
  Base Error for vFlash API.
- class VFlashResult
  Flash programming result values.
- class VFlashResultError
- class VFlashStatus
  Flash programming status values.

Variables

- int FLASH_TIMEOUT = 2000

  timeout for flash process in [s]

6.155.1 Detailed Description

Common definitions, utility functions and classes for vFlash API access.

6.155.2 Variable Documentation

6.155.2.1 FLASH_TIMEOUT

```
int ttk_tools.vector.vflash_api_common.FLASH_TIMEOUT = 2000
```

timeout for flash process in [s]

6.156 ttk_tools.win32_utils Namespace Reference

Interface wrapper for Windows systems utility functions, see ttk_tools._win32_utils.

Functions

- disableAppCrashDialog ()

  Disable displaying of application/program crash dialog for the current process ("XYZ has stopped working" / "XYZ funktioniert nicht mehr").
- waitPumpMessages (delay)

  Wait for (approximately) delay [s] and pump windows messages/events
  while waiting, which is a "good thing" to do if a COM connection is active in the current process/thread.
- getRegPath (path, name, wow64_32_key=None, verbosity=1)

  Get a value from an entry in the windows registry.
- setRegPath (path, name, value, value_type=None, wow64_32_key=None, verbosity=1)

  Set a value to an entry in the windows registry.

6.156.1 Detailed Description

Interface wrapper for Windows systems utility functions, see ttk_tools._win32_utils.

6.156.2 Function Documentation

6.156.2.1 disableAppCrashDialog()

```
ttk_tools.win32_utils.disableAppCrashDialog ( )
```

Disable displaying of application/program crash dialog for the current process ("XYZ has stopped working" / "XYZ funktioniert nicht mehr").

This tells Windows Error Reporting to skip displaying the dialog. See SEM_NOGPFAULTERRORBOX.

6.156.2.2 getRegPath()

```
ttk_tools.win32_utils.getRegPath (
              path,
              name,
              wow64_32_key = None,
              verbosity = 1 )
```

Get a value from an entry in the windows registry.

Parameters

path	a registry path starting with one of the HKEY entries, e.g. "HKEY_LOCAL_MACHINE\\SOFTWARE\\Microsoft" or "HKLM\\SOFTWARE\\Microsoft"
name	name of entry below path
wow64_32_key	None: use default key view depending on current bitness True: access 32-bit keys from 32-bit and 64-bit applications False: access 64-bit keys from 32-bit and 64-bit applications
verbosity	verbosity of error messages 0: silent, 1: print error to stdout, 2: also print exception info

Note

Registry paths may start with both full or abbreviated HKEY values, e.g.

- `HKEY_LOCAL_MACHINE\\...` or abbreviated `HKLM\\...`
- `HKEY_CLASSES_ROOT\\...` or abbreviated `HKCR\\...`
- `HKEY_CURRENT_USER\\...` or abbreviated `HKCU\\...`

Returns

the contents of the specified key/name or None if access failed.

6.156.2.3 setRegPath()

```
ttk_tools.win32_utils.setRegPath (
              path,
              name,
              value,
              value_type = None,
              wow64_32_key = None,
              verbosity = 1 )
```

Set a value to an entry in the windows registry.

Parameters

path	a registry path starting with one of the HKEY entries, e.g. "HKEY_LOCAL_MACHINE\\SOFTWARE\\Microsoft" or "HKLM\\SOFTWARE\\Microsoft". See note at getRegPath().
name	name of entry below path
value	value to set
value_type	type of value to set, should be one of the defined `_winreg.REG_*` values or None to "auto detect": • string values => `REG_SZ` • integer values => `REG_DWORD` (note: max 32 bit) • float values => `REG_DWORD` (will be rounded towards the nearest integer)
wow64_32_key	None: use default key view depending on current bitness True: access 32-bit keys from 32-bit and 64-bit applications False: access 64-bit keys from 32-bit and 64-bit applications
verbosity	verbosity of error messages 0: silent, 1: print error to stdout, 2: also print exception info

Returns

True if value was set, otherwise False.

6.156.2.4 waitPumpMessages()

```
ttk_tools.win32_utils.waitPumpMessages (
              delay )
```

Wait for (approximately) delay [s] and pump windows messages/events
while waiting, which is a "good thing" to do if a COM connection is active in the current process/thread.

Note

The wait time is rather approximate, and may be quite a bit longer (like about +500ms on occasion) if the system is busy and lots of events are fired.
The usable resolution appears to be about 200ms.

Parameters

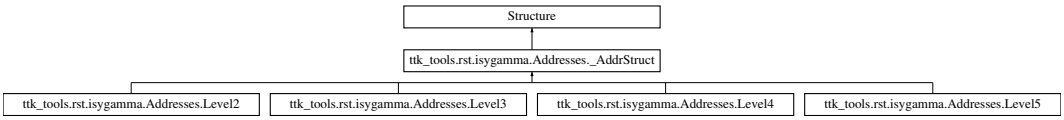
delay	wait delay in seconds

Chapter 7

Class Documentation

7.1 ttk_tools.rst.isygamma.Addresses._AddrStruct Class Reference

Inheritance diagram for ttk_tools.rst.isygamma.Addresses._AddrStruct:



Public Member Functions

- __str__ (self)

Protected Attributes

- _fields_

7.1.1 Member Function Documentation

7.1.1.1 __str__()

```
ttk_tools.rst.isygamma.Addresses._AddrStruct.__str__ (
              self )
```

7.1.2 Member Data Documentation

7.1.2.1 _fields_

```
ttk_tools.rst.isygamma.Addresses._AddrStruct._fields_  [protected]
```

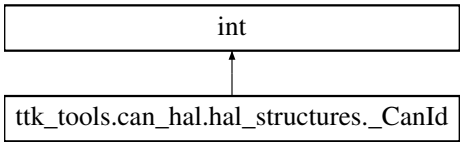
The documentation for this class was generated from the following file:

- ttk_tools/rst/isygamma.py (v1.6)

7.2 ttk_tools.can_hal.hal_structures._CanId Class Reference

An int-wrapper that uses a hexadecimal representation.

Inheritance diagram for ttk_tools.can_hal.hal_structures._CanId:



Public Member Functions

- __str__ (self)
- __repr__ (self)

7.2.1 Detailed Description

An int-wrapper that uses a hexadecimal representation.

Just to a get "nicer" default CAN-ID value display.

7.2.2 Member Function Documentation

7.2.2.1 __repr__()

```
ttk_tools.can_hal.hal_structures._CanId.__repr__ (
              self )
```

7.2.2.2 __str__()

```
ttk_tools.can_hal.hal_structures._CanId.__str__ (
              self )
```

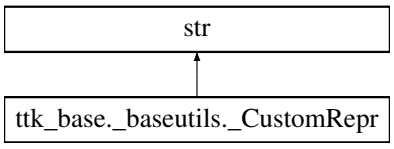
The documentation for this class was generated from the following file:

- ttk_tools/can_hal/hal_structures.py (v1.0.0)

7.3 ttk_base._baseutils._CustomRepr Class Reference

A utility class for "marking" customized (i.e.

Inheritance diagram for ttk_base._baseutils._CustomRepr:



7.3.1 Detailed Description

A utility class for "marking" customized (i.e.

non-default) representation strings generated in VarBase-derived variables and meta-wrapped values.

In test functions, default representations will be overwritten/recreated if another related variable has a customized format defined (and therefore a non-default representation).

See isCustomRepr, below.

Interpret a numeric value as an integer (int/long) value, but perform sanity checks before (relevant for float input values).
If `value` is already long (or int), it will simply be returned as-is.

Note

Raises

- TypeError if `value` is non-numeric
- ValueError if a float value does not appear to be representing an integer

This is quite similar to ttk_check.checkutils.getInteger, but

- only checks if value is not too far off to be integer(-ish)
- raises only basic Type/ValueErrors with less verbose exception messages.

IEEE-754 double-precision can store 52 bits explicitly, so integer values larger than 2**52 cannot be represented precisely, e.g. `int(float(2**54 + 2)) != 2**54 + 2`.

If value gets closer to 2**52, the loss of precision will cause us to detect more values as "integer", as errors creep into the range of whole numbers. float.is_integer() (Python 2.6+) has the same issue.

Parameters

value	numeric value to interpret as integer number

Returns

an int value.

The documentation for this class was generated from the following file:

- ttk_base/_baseutils.py (v2.3.2)

7.4 ttk_tools.can_hal._hal_offline_stub.CanHal._DeviceStub Class Reference

Public Member Functions

- __init__ (self, dev_type=DeviceType.PeakUSB, dev_id=0x00)

Public Attributes

- device_list_entry
- rx_queue
- bus_state

7.4.1 Constructor & Destructor Documentation

7.4.1.1 __init__()

```
ttk_tools.can_hal._hal_offline_stub.CanHal._DeviceStub.__init__ (
              self,
              dev_type = DeviceType.PeakUSB,
              dev_id = 0x00 )
```

7.4.2 Member Data Documentation

7.4.2.1 bus_state

```
ttk_tools.can_hal._hal_offline_stub.CanHal._DeviceStub.bus_state
```

7.4.2.2 device_list_entry

```
ttk_tools.can_hal._hal_offline_stub.CanHal._DeviceStub.device_list_entry
```

7.4.2.3 rx_queue

```
ttk_tools.can_hal._hal_offline_stub.CanHal._DeviceStub.rx_queue
```

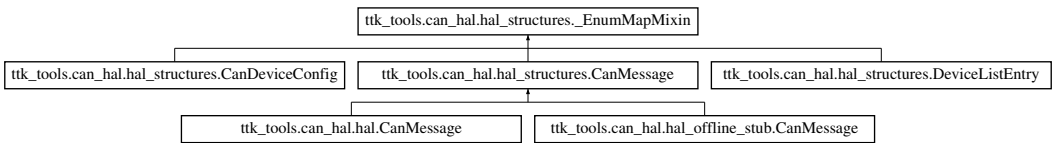
The documentation for this class was generated from the following file:

- ttk_tools/can_hal/_hal_offline_stub.py (v1.1.1)

7.5 ttk_tools.can_hal.hal_structures._EnumMapMixin Class Reference

Mix-in for ctypes.Structure.

Inheritance diagram for ttk_tools.can_hal.hal_structures._EnumMapMixin:



Public Member Functions

- __getattribute__ (self, name)

7.5.1 Detailed Description

Mix-in for ctypes.Structure.

Maps scalar values to enums (usually IntEnum-derived classes). Mapping of structure elements happens via a class member `_map` containing `{"name from _fields_ member": EnumClass}` entries.

7.5.2 Member Function Documentation

7.5.2.1 __getattribute__()

```
ttk_tools.can_hal.hal_structures._EnumMapMixin.__getattribute__ (
            self,
            name )
```

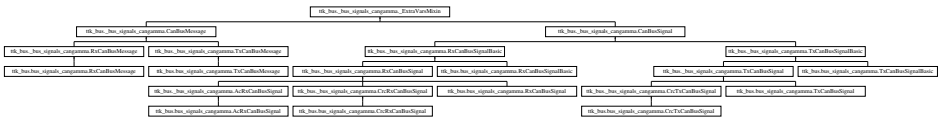
The documentation for this class was generated from the following file:

- ttk_tools/can_hal/hal_structures.py (v1.0.0)

7.6 ttk_bus._bus_signals_cangamma._ExtraVarsMixin Class Reference

Mixin to add a utility method to initialize "extra" helper Gamma-HilVars.

Inheritance diagram for ttk_bus._bus_signals_cangamma._ExtraVarsMixin:



Public Attributes

- context

Protected Member Functions

- _initExtraVar (self, varname, suffix, unit=None, lookup=None, fmt=None, silent=False)

  Add an "extra" info or access PV, based on the supplied base identifier identifier.

7.6.1 Detailed Description

Mixin to add a utility method to initialize "extra" helper Gamma-HilVars.

Moved to a separate class to avoid code duplication.

7.6.2 Member Function Documentation

7.6.2.1 _initExtraVar()

```
ttk_bus._bus_signals_cangamma._ExtraVarsMixin._initExtraVar (
              self,
              varname,
              suffix,
              unit = None,
              lookup = None,
              fmt = None,
              silent = False )  [protected]
```

Add an "extra" info or access PV, based on the supplied base identifier identifier.

This gets used in "Message" and "Signal" implementations

Expects the following members to be available:

- self.base_identifier
- self.context

Uses values from a member _default_values (if available) to set default values to created HilVar instances

Parameters

varname	name of member var to initialize
suffix	suffix of variable (to complete base_identifier)
unit	unit for variable (optional)
lookup	lookup-table for variable (optional)
fmt	formatting hints (optional)
silent	if True, no error will be printed/logged if variable was not available (use only for optional variables)

Returns

- (<varname> member variable will be set directly)

7.6.3 Member Data Documentation

7.6.3.1 context

```
ttk_bus._bus_signals_cangamma._ExtraVarsMixin.context
```

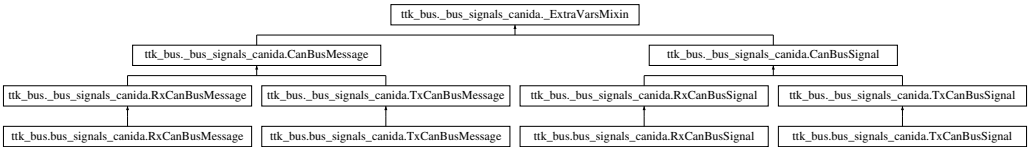
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_cangamma.py (v1.12)

7.7 ttk_bus._bus_signals_canida._ExtraVarsMixin Class Reference

Mixin to add a utility method to initialize "extra" helper Gamma-HilVars.

Inheritance diagram for ttk_bus._bus_signals_canida._ExtraVarsMixin:



Public Attributes

- context

Protected Member Functions

- _initExtraVar (self, varname, suffix, unit="", lookup=None, fmt=None, silent=False)

  Add an "extra" info or access PV, based on the supplied base identifier identifier.

7.7.1 Detailed Description

Mixin to add a utility method to initialize "extra" helper Gamma-HilVars.

Moved to a separate class to avoid code duplication.

7.7.2 Member Function Documentation

7.7.2.1 _initExtraVar()

```
ttk_bus._bus_signals_canida._ExtraVarsMixin._initExtraVar (
              self,
              varname,
              suffix,
              unit = "",
              lookup = None,
              fmt = None,
              silent = False )  [protected]
```

Add an "extra" info or access PV, based on the supplied base identifier identifier.

This gets used in "Message" and "Signal" implementations

Expects the following members to be available:

- self.base_identifier
- self.context

Uses values from a member _default_values (if available) to set default values to created HilVar instances

Parameters

varname	name of member var to initialize
suffix	suffix of variable (to complete base_identifier)
unit	unit for variable (optional)
lookup	lookup-table for variable (optional)
fmt	formatting hints (optional)
silent	if True, no error will be printed/logged if variable was not available (use only for optional variables)

7.7.3 Member Data Documentation

7.7.3.1 context

`ttk_bus._bus_signals_canida._ExtraVarsMixin.context`

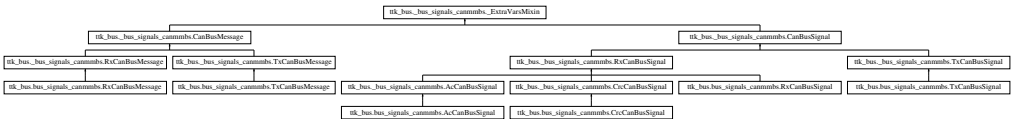
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_canida.py (v1.0.2)

7.8 ttk_bus._bus_signals_canmmbs._ExtraVarsMixin Class Reference

Mixin to add a utility method to initialize "extra" helper CANMM-HilVars.

Inheritance diagram for ttk_bus._bus_signals_canmmbs._ExtraVarsMixin:



Public Attributes

- context
- alias

Protected Member Functions

- _initExtraVar (self, varname, suffix, unit=None, lookup=None, fmt=None, silent=False)

  Add an "extra" info or access PV, based on the supplied base identifier identifier.

7.8.1 Detailed Description

Mixin to add a utility method to initialize "extra" helper CANMM-HilVars.

Moved to a separate class to avoid code duplication.

7.8.2 Member Function Documentation

7.8.2.1 _initExtraVar()

```
ttk_bus._bus_signals_canmmbs._ExtraVarsMixin._initExtraVar (
            self,
            varname,
            suffix,
            unit = None,
            lookup = None,
            fmt = None,
            silent = False )  [protected]
```

Add an "extra" info or access PV, based on the supplied base identifier identifier.

This gets used in "Message" and "Signal" implementations

Expects the following members to be available:

- self.base_path (base path to prefix)
- self.context

Uses values from a member _default_values (if available) to set default values to created HilVar instances

Parameters

varname	name of member var to initialize
suffix	suffix of variable (to complete base_identifier)
unit	unit for variable (optional)
lookup	lookup-table for variable (optional)
fmt	formatting hints (optional)
silent	if True, no error will be printed/logged if variable was not available (use only for optional variables)

Returns

- (<varname> member variable will be set directly)

7.8.3 Member Data Documentation

7.8.3.1 alias

```
ttk_bus._bus_signals_canmmbs._ExtraVarsMixin.alias
```

7.8.3.2 context

```
ttk_bus._bus_signals_canmmbs._ExtraVarsMixin.context
```

The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_canmmbs.py (v2.3)

7.9 ttk_tools.rst.isygamma._GaAPI Class Reference

Wrapper for gaapi dynamic library.

Public Member Functions

- __init__ (self)
- getPVs (self)

  Get PV addresses available on the currently connected system(s).
- getVersion (self)

  Get gaapi DLL version.
- PV (self, address, properties=PVProperty.ACC_NON_BLOCKING)

  Factory function.

Protected Member Functions

- _getInfoByIndices (self, browsing_type, system=0, index1=0, index2=0, index3=0, index4=0)

  Get enumerated information for the given BrowsingType.

Static Protected Attributes

- str _lib_name = "gaapi.dll" if IS_WIN32 else "libgaapi.so.1"
- _lib = None

7.9.1 Detailed Description

Wrapper for gaapi dynamic library.

7.9.2 Constructor & Destructor Documentation

7.9.2.1 __init__()

```
ttk_tools.rst.isygamma._GaAPI.__init__ (
              self )
```

7.9.3 Member Function Documentation

7.9.3.1 _getInfoByIndices()

```
ttk_tools.rst.isygamma._GaAPI._getInfoByIndices (
              self,
              browsing_type,
              system = 0,
              index1 = 0,
              index2 = 0,
              index3 = 0,
              index4 = 0 )  [protected]
```

Get enumerated information for the given BrowsingType.

Parameters

browsing_type	a BrowsingType to, well, browse. See BrowsingType
system	system index
index1	interpretation depends on used browsing_type
index2	interpretation depends on used browsing_type
index3	interpretation depends on used browsing_type
index4	interpretation depends on used browsing_type

Returns

a sequence of _BrowserInfo structures

7.9.3.2 getPVs()

```
ttk_tools.rst.isygamma._GaAPI.getPVs (
              self )
```

Get PV addresses available on the currently connected system(s).

Example: PV info dicts

```
# Each info dict contains entries similar to this:
pv_info = {
    "address":     'iSyst:DAQ.Memory.Group.DIO_IN_00', # full PV address as string (again)
    "name":        'DIO_IN_00',     # just the PV (array) name
    "data_type":   'UINT_8',        # data type as string, see gacommon.DataType
    "num_elements": 1,              # number of elements of the PV (array)
}
```

Note

For data type BLOB there will be an additional entry "length" that contains the BLOB's length.

The PV (or actually: PV-Array) name will be the unique name below the "group" level (i.e. address level 4), and may still contain further dots for sub-structured PVs

Returns

a dict with { <pv_address>: {pv_info} } mappings.

7.9.3.3 getVersion()

```
ttk_tools.rst.isygamma._GaAPI.getVersion (
              self )
```

Get gaapi DLL version.

Returns

a VersionInfo named tuple.

7.9.3.4 PV()

```
ttk_tools.rst.isygamma._GaAPI.PV (
              self,
              address,
              properties = PVProperty.ACC_NON_BLOCKING )
```

Factory function.

Parameters

address	PV "address" as string
properties	PV properties to set, see PVProperty

Returns

a new isygamma.PV instance with this GaAPI.

7.9.4 Member Data Documentation

7.9.4.1 _lib

```
ttk_tools.rst.isygamma._GaAPI._lib = None  [static], [protected]
```

7.9.4.2 _lib_name

```
str ttk_tools.rst.isygamma._GaAPI._lib_name = "gaapi.dll" if IS_WIN32 else "libgaapi.so.1"
[static], [protected]
```

The documentation for this class was generated from the following file:

- ttk_tools/rst/isygamma.py (v1.6)

7.10 ttk_tools.rst._gamma_api_offline_stub._GaApiStub Class Reference

Minimal stub for "gaapi" (which is normally a module)

Public Member Functions

- __init__ (self)
- getVersion (self)
- PV (self, address, properties=0)

  basic stub that provides an interface similar to class gaapi.PV

7.10.1 Detailed Description

Minimal stub for "gaapi" (which is normally a module)

7.10.2 Constructor & Destructor Documentation

7.10.2.1 __init__()

```
ttk_tools.rst._gamma_api_offline_stub._GaApiStub.__init__ (
            self )
```

7.10.3 Member Function Documentation

7.10.3.1 getVersion()

```
ttk_tools.rst._gamma_api_offline_stub._GaApiStub.getVersion (
              self )
```

7.10.3.2 PV()

```
ttk_tools.rst._gamma_api_offline_stub._GaApiStub.PV (
              self,
              address,
              properties = 0 )
```

basic stub that provides an interface similar to class gaapi.PV

The documentation for this class was generated from the following file:

- ttk_tools/rst/_gamma_api_offline_stub.py (v1.7)

7.11 ttk_tools.rst.isygamma._GaCommon Class Reference

Wrapper for gacommon dynamic library.

Public Member Functions

- __init__ (self)
- attachToService (self, port=0, timeout_ms=0)

  Attach to Gamma service.
- detachFromService (self)

  Detach from Gamma service.
- getErrorDescr (self, error_id)

  Get a textual description for the supplied error_id.
- getVersion (self)

  Get gacommon DLL version.

Static Protected Attributes

- str _lib_name = "gacommon.dll" if IS_WIN32 else "libgacommon.so.2"
- _lib = None

7.11.1 Detailed Description

Wrapper for gacommon dynamic library.

7.11.2 Constructor & Destructor Documentation

7.11.2.1 __init__()

```
ttk_tools.rst.isygamma._GaCommon.__init__ (
              self )
```

7.11.3 Member Function Documentation

7.11.3.1 attachToService()

```
ttk_tools.rst.isygamma._GaCommon.attachToService (
              self,
              port = 0,
              timeout_ms = 0 )
```

Attach to Gamma service.

Parameters

port	listen port of the Gamma service.
timeout_ms	attach timeout in [ms]

7.11.3.2 detachFromService()

```
ttk_tools.rst.isygamma._GaCommon.detachFromService (
              self )
```

Detach from Gamma service.

After a detach, all still opened PVs will be invalid.

7.11.3.3 getErrorDescr()

```
ttk_tools.rst.isygamma._GaCommon.getErrorDescr (
              self,
              error_id )
```

Get a textual description for the supplied error_id.

7.11.3.4 getVersion()

```
ttk_tools.rst.isygamma._GaCommon.getVersion (
              self )
```

Get gacommon DLL version.

Returns

a VersionInfo named tuple.

7.11.4 Member Data Documentation

7.11.4.1 _lib

```
ttk_tools.rst.isygamma._GaCommon._lib = None  [static], [protected]
```

7.11.4.2 _lib_name

```
str ttk_tools.rst.isygamma._GaCommon._lib_name = "gacommon.dll" if IS_WIN32 else "libgacommon.↩
so.2"  [static], [protected]
```

The documentation for this class was generated from the following file:

- ttk_tools/rst/isygamma.py (v1.6)

7.12 ttk_tools.dspace._xil_api_offline_stub._MaPortStub Class Reference

Minimal stub for a MAPort-ish implementation.

Public Member Functions

- __init__ (self, port_name)
- read (self, model_path)
- write (self, model_path, value)

Public Attributes

- State
  port state, our port stubs always start with a running model...

Protected Attributes

- _context
  context for accessing offline date provider
- _data_provider
  offline data provider

7.12.1 Detailed Description

Minimal stub for a MAPort-ish implementation.

Note that this is not a "true" stub but merely a somewhat similar implementation (interfaces differ)

7.12.2 Constructor & Destructor Documentation

7.12.2.1 __init__()

```
ttk_tools.dspace._xil_api_offline_stub._MaPortStub.__init__ (
            self,
            port_name )
```

Parameters

port_name	name of MAPort, used as context for accessing offline data

7.12.3 Member Function Documentation

7.12.3.1 read()

```
ttk_tools.dspace._xil_api_offline_stub._MaPortStub.read (
              self,
              model_path )
```

7.12.3.2 write()

```
ttk_tools.dspace._xil_api_offline_stub._MaPortStub.write (
              self,
              model_path,
              value )
```

7.12.4 Member Data Documentation

7.12.4.1 _context

```
ttk_tools.dspace._xil_api_offline_stub._MaPortStub._context  [protected]
```

context for accessing offline date provider

7.12.4.2 _data_provider

```
ttk_tools.dspace._xil_api_offline_stub._MaPortStub._data_provider  [protected]
```

offline data provider

7.12.4.3 State

```
ttk_tools.dspace._xil_api_offline_stub._MaPortStub.State
```

port state, our port stubs always start with a running model...

The documentation for this class was generated from the following file:

- ttk_tools/dspace/_xil_api_offline_stub.py (v2.1.2)

7.13 ttk_tools.rst._gamma_api_offline_stub._PVStub Class Reference

Basic stub for class gaapi.PV (or rather, isygamma.PV).

Public Member Functions

- __init__ (self, address, properties=PVProperty.ACC_NON_BLOCKING)
- __enter__ (self)

  Context handler: Enter.
- __exit__ (self, _exc_type, _exc_value, _tb)

  Context handler: Exit.
- __str__ (self)
- value (self)

  Get current value of the PV.
- value (self, value)

  Set current value of the PV.
- address (self)

  PV address as supplied in init.
- data_type (self)

  Data Type of PV.
- pv_type (self)

  Type/kind of PV.
- length (self)

  current and max length of a PV with a "complex" data type.

7.13.1 Detailed Description

Basic stub for class gaapi.PV (or rather, isygamma.PV).

7.13.2 Constructor & Destructor Documentation

7.13.2.1 __init__()

```
ttk_tools.rst._gamma_api_offline_stub._PVStub.__init__ (
              self,
              address,
              properties = PVProperty.ACC_NON_BLOCKING )
```

Parameters

address	address/identifier of the PV.
properties	properties to be used for the PV.

7.13.3 Member Function Documentation

7.13.3.1 __enter__()

```
ttk_tools.rst._gamma_api_offline_stub._PVStub.__enter__ (
              self )
```

Context handler: Enter.

7.13.3.2 __exit__()

```
ttk_tools.rst._gamma_api_offline_stub._PVStub.__exit__ (
              self,
              _exc_type,
              _exc_value,
              _tb )
```

Context handler: Exit.

7.13.3.3 __str__()

```
ttk_tools.rst._gamma_api_offline_stub._PVStub.__str__ (
              self )
```

7.13.3.4 address()

```
ttk_tools.rst._gamma_api_offline_stub._PVStub.address (
              self )
```

PV address as supplied in init.

7.13.3.5 data_type()

```
ttk_tools.rst._gamma_api_offline_stub._PVStub.data_type (
              self )
```

Data Type of PV.

7.13.3.6 length()

```
ttk_tools.rst._gamma_api_offline_stub._PVStub.length (
              self )
```

current and max length of a PV with a "complex" data type.

7.13.3.7 pv_type()

```
ttk_tools.rst._gamma_api_offline_stub._PVStub.pv_type (
              self )
```

Type/kind of PV.

7.13.3.8 value() [1/2]

```
ttk_tools.rst._gamma_api_offline_stub._PVStub.value (
              self )
```

Get current value of the PV.

7.13.3.9 value() [2/2]

```
ttk_tools.rst._gamma_api_offline_stub._PVStub.value (
              self,
              value )
```

Set current value of the PV.

The documentation for this class was generated from the following file:

- ttk_tools/rst/_gamma_api_offline_stub.py (v1.7)

7.14 ttk_tools.dspace.rtplib_offline_stub._RTPApplicationDummy Class Reference

Dummy implementation of `rtplib.Appl()` compatible objects for offline use.

Public Member Functions

- __init__ (self, VariableFile, SystemName, SystemType)
  Create an Appl class object.
- GetSamplingPeriod (self, MPSubsystem=None)
  Get the base sampling rate of the application on the real-time processor.
- GetVarNames (self)
  Create a list with all names which are specified within the variable file.
- Read (self, TargetType, TargetAddr, ProcIndex, NoOfValues=1)
  Read an amount of data from an address on the real-time processor.
- Write (self, TargetType, TargetAddr, Value, ProcIndex)
  Write a value to an address on the real-time processor.
- SetEditMode (self)
  Activate the edit mode for Variable and Capture objects.
- SetAnimationMode (self)
  Activate the animation mode for Variable and Capture objects.
- Variable (self, VariableName)
  Create an Variable class object.
- Capture (self, variables)
  Create an object of the class Capture.

Public Attributes

- trace_file
- system_name
- system_type

7.14.1 Detailed Description

Dummy implementation of `rtplib.Appl()` compatible objects for offline use.

7.14.2 Constructor & Destructor Documentation

7.14.2.1 __init__()

```
ttk_tools.dspace.rtplib_offline_stub._RTPApplicationDummy.__init__ (
              self,
              VariableFile,
              SystemName,
              SystemType )
```

Create an Appl class object.

Parameters

VariableFile	path to variable file (∗.sdf or ∗.trc)
SystemName	name of rtp system, e.g. "ds1005"
SystemType	type of rtp system, e.g. "PPC"

7.14.3 Member Function Documentation

7.14.3.1 Capture()

```
ttk_tools.dspace.rtplib_offline_stub._RTPApplicationDummy.Capture (
              self,
              variables )
```

Create an object of the class Capture.

Parameters

variables	a list of variables (i.e. a list of model paths)

Returns

a Capture instance containing the specified variables.

7.14.3.2 GetSamplingPeriod()

```
ttk_tools.dspace.rtplib_offline_stub._RTPApplicationDummy.GetSamplingPeriod (
              self,
              MPSubsystem = None )
```

Get the base sampling rate of the application on the real-time processor.

Parameters

MPSubsystem	which multiprocessor subsystem to query (or None)

Returns

a sampling period.

7.14.3.3 GetVarNames()

```
ttk_tools.dspace.rtplib_offline_stub._RTPApplicationDummy.GetVarNames (
              self )
```

Create a list with all names which are specified within the variable file.

Returns

a list of variable names.

7.14.3.4 Read()

```
ttk_tools.dspace.rtplib_offline_stub._RTPApplicationDummy.Read (
              self,
              TargetType,
              TargetAddr,
              ProcIndex,
              NoOfValues = 1 )
```

Read an amount of data from an address on the real-time processor.

7.14.3.5 SetAnimationMode()

```
ttk_tools.dspace.rtplib_offline_stub._RTPApplicationDummy.SetAnimationMode (
              self )
```

Activate the animation mode for Variable and Capture objects.

7.14.3.6 SetEditMode()

```
ttk_tools.dspace.rtplib_offline_stub._RTPApplicationDummy.SetEditMode (
              self )
```

Activate the edit mode for Variable and Capture objects.

7.14.3.7 Variable()

```
ttk_tools.dspace.rtplib_offline_stub._RTPApplicationDummy.Variable (
              self,
              VariableName )
```

Create an Variable class object.

Parameters

VariableName	model path of variable to create

Returns

an instance of Variable.

7.14.3.8 Write()

```
ttk_tools.dspace.rtplib_offline_stub._RTPApplicationDummy.Write (
              self,
              TargetType,
              TargetAddr,
              Value,
              ProcIndex )
```

Write a value to an address on the real-time processor.

7.14.4 Member Data Documentation

7.14.4.1 system_name

```
ttk_tools.dspace.rtplib_offline_stub._RTPApplicationDummy.system_name
```

7.14.4.2 system_type

```
ttk_tools.dspace.rtplib_offline_stub._RTPApplicationDummy.system_type
```

7.14.4.3 trace_file

```
ttk_tools.dspace.rtplib_offline_stub._RTPApplicationDummy.trace_file
```

The documentation for this class was generated from the following file:

- ttk_tools/dspace/rtplib_offline_stub.py (v2.8)

7.15 ttk_tools.dspace.rtplib_offline_stub._RTPCaptureDummy Class Reference

A dummy implementation of `rtplib.Capture()` compatible objects for offline use.

Public Member Functions

- __init__ (self, variables, context)
- StartCapturing (self)

  Start a new capture.
- StartStreamToDisk (self, IdfFileName)

  Enable the data acquisition mode stream to disk.
- StopStreamToDisk (self)

  Stop the data stream to disk and close the data file.
- Fetch (self, Timeout=None, MaxNumber=None)

  Fetch data from application buffers to Python workspace.
- CaptureState (self)

  Retrieve the current capture state.
- DefineTrigger (self, variable, level, edge, initial=0)

  Specify a variable on which to trigger.
- TriggerOff (self)

  Disable the trigger.
- TriggerOn (self)

  Enable the trigger.
- SetFrame (self, Downsampling, Delay, Interval)

  Set time frame for capturing application data.
- SetFramePoints (self, Downsampling, Delay, Interval)

  Specify the time frame for capturing application.

Public Attributes

- cycle_time

Protected Attributes

- _variables
- _context
- _data_provider
- _data
- _downsampling
- _delay
- _duration

7.15.1 Detailed Description

A dummy implementation of `rtplib.Capture()` compatible objects for offline use.

7.15.2 Constructor & Destructor Documentation

7.15.2.1 __init__()

```
ttk_tools.dspace.rtplib_offline_stub._RTPCaptureDummy.__init__ (
              self,
              variables,
              context )
```

7.15.3 Member Function Documentation

7.15.3.1 CaptureState()

```
ttk_tools.dspace.rtplib_offline_stub._RTPCaptureDummy.CaptureState (
              self )
```

Retrieve the current capture state.

See also

CaptureNone
CaptureTriggerArmed
CaptureInProgress
CaptureDone
CaptureAborted
CaptureWaitingForTriggerArmed

7.15.3.2 DefineTrigger()

```
ttk_tools.dspace.rtplib_offline_stub._RTPCaptureDummy.DefineTrigger (
              self,
              variable,
              level,
              edge,
              initial = 0 )
```

Specify a variable on which to trigger.

7.15.3.3 Fetch()

```
ttk_tools.dspace.rtplib_offline_stub._RTPCaptureDummy.Fetch (
              self,
              Timeout = None,
              MaxNumber = None )
```

Fetch data from application buffers to Python workspace.

Parameters

Timeout	timeout for fetch operation (None: no timeout)
MaxNumber	maximum number of next entries to fetch (None: fetch all available)

Returns

a Capture data dictionary (one entry for each captured variable (list of data points) and an entry "xAxis" (list of time stamps).

7.15.3.4 SetFrame()

```
ttk_tools.dspace.rtplib_offline_stub._RTPCaptureDummy.SetFrame (
              self,
              Downsampling,
              Delay,
              Interval )
```

Set time frame for capturing application data.

Parameters

Downsampling	define data downsampling factor (n model cycles = 1 capture cycle)
Delay	delay before capture starts [s]
Interval	capture duration [s]

Returns

a tuple with current (Downsampling, Delay, Interval)

7.15.3.5 SetFramePoints()

```
ttk_tools.dspace.rtplib_offline_stub._RTPCaptureDummy.SetFramePoints (
              self,
              Downsampling,
              Delay,
              Interval )
```

Specify the time frame for capturing application.

7.15.3.6 StartCapturing()

```
ttk_tools.dspace.rtplib_offline_stub._RTPCaptureDummy.StartCapturing (
              self )
```

Start a new capture.

7.15.3.7 StartStreamToDisk()

```
ttk_tools.dspace.rtplib_offline_stub._RTPCaptureDummy.StartStreamToDisk (
              self,
              IdfFileName )
```

Enable the data acquisition mode stream to disk.

7.15.3.8 StopStreamToDisk()

```
ttk_tools.dspace.rtplib_offline_stub._RTPCaptureDummy.StopStreamToDisk (
              self )
```

Stop the data stream to disk and close the data file.

7.15.3.9 TriggerOff()

```
ttk_tools.dspace.rtplib_offline_stub._RTPCaptureDummy.TriggerOff (
              self )
```

Disable the trigger.

7.15.3.10 TriggerOn()

```
ttk_tools.dspace.rtplib_offline_stub._RTPCaptureDummy.TriggerOn (
              self )
```

Enable the trigger.

7.15.4 Member Data Documentation

7.15.4.1 _context

```
ttk_tools.dspace.rtplib_offline_stub._RTPCaptureDummy._context  [protected]
```

7.15.4.2 _data

```
ttk_tools.dspace.rtplib_offline_stub._RTPCaptureDummy._data  [protected]
```

7.15.4.3 _data_provider

```
ttk_tools.dspace.rtplib_offline_stub._RTPCaptureDummy._data_provider  [protected]
```

7.15.4.4 _delay

```
ttk_tools.dspace.rtplib_offline_stub._RTPCaptureDummy._delay  [protected]
```

7.15.4.5 _downsampling

ttk_tools.dspace.rtplib_offline_stub._RTPCaptureDummy._downsampling [protected]

7.15.4.6 _duration

ttk_tools.dspace.rtplib_offline_stub._RTPCaptureDummy._duration [protected]

7.15.4.7 _variables

ttk_tools.dspace.rtplib_offline_stub._RTPCaptureDummy._variables [protected]

7.15.4.8 cycle_time

ttk_tools.dspace.rtplib_offline_stub._RTPCaptureDummy.cycle_time

The documentation for this class was generated from the following file:

- ttk_tools/dspace/rtplib_offline_stub.py (v2.8)

7.16 ttk_tools.dspace.rtplib_offline_stub._RTPVariableDummy Class Reference

Dummy implementation of `rtplib.Appl().Variable()` compatible objects for offline use.

Public Member Functions

- __init__ (self, variable_name, context)
- Read (self)

  Read an amount of data from the application.
- Write (self, Data)

  Write a value to a variable of the application.
- GetProperties (self)

  Obtain variable properties.
- GetProcessorIndex (self)

  Get the index of the processor on which the variable resides.

Protected Attributes

- _name
- _context
- _data_provider

7.16.1 Detailed Description

Dummy implementation of `rtplib.Appl().Variable()` compatible objects for offline use.

7.16.2 Constructor & Destructor Documentation

7.16.2.1 __init__()

```
ttk_tools.dspace.rtplib_offline_stub._RTPVariableDummy.__init__ (
              self,
              variable_name,
              context )
```

7.16.3 Member Function Documentation

7.16.3.1 GetProcessorIndex()

```
ttk_tools.dspace.rtplib_offline_stub._RTPVariableDummy.GetProcessorIndex (
              self )
```

Get the index of the processor on which the variable resides.

7.16.3.2 GetProperties()

```
ttk_tools.dspace.rtplib_offline_stub._RTPVariableDummy.GetProperties (
              self )
```

Obtain variable properties.

Returns

a tuple (VarType, VarAddr, VarOffset, VarLength)

7.16.3.3 Read()

```
ttk_tools.dspace.rtplib_offline_stub._RTPVariableDummy.Read (
              self )
```

Read an amount of data from the application.

7.16.3.4 Write()

```
ttk_tools.dspace.rtplib_offline_stub._RTPVariableDummy.Write (
              self,
              Data )
```

Write a value to a variable of the application.

7.16.4 Member Data Documentation

7.16.4.1 _context

ttk_tools.dspace.rtplib_offline_stub._RTPVariableDummy._context [protected]

7.16.4.2 _data_provider

ttk_tools.dspace.rtplib_offline_stub._RTPVariableDummy._data_provider [protected]

7.16.4.3 _name

ttk_tools.dspace.rtplib_offline_stub._RTPVariableDummy._name [protected]

The documentation for this class was generated from the following file:

- ttk_tools/dspace/rtplib_offline_stub.py (v2.8)

7.17 ttk_selftest.hil_devices._selftest_devices._SrcMode Class Reference

Base class for modes of the iSRC.

Public Member Functions

- __init__ (self, mode_id, mode_type, alias, msg=None, vol_check=None, cur_check=None, vol_set=None, limit=None, ref_value=None, factor_vol_to_si=1, factor_cur_to_si=1)

  iSystemReferenceCard mode initialization.

Public Attributes

- mode_id
- mode_type
- alias
- msg
- vol_check
- cur_check
- vol_set
- limit
- ref_value
- factor_vol_to_si
- factor_cur_to_si

7.17.1 Detailed Description

Base class for modes of the iSRC.

7.17.2 Constructor & Destructor Documentation

7.17.2.1 __init__()

```
ttk_selftest.hil_devices._selftest_devices._SrcMode.__init__ (
              self,
              mode_id,
              mode_type,
              alias,
              msg = None,
              vol_check = None,
              cur_check = None,
              vol_set = None,
              limit = None,
              ref_value = None,
              factor_vol_to_si = 1,
              factor_cur_to_si = 1 )
```

iSystemReferenceCard mode initialization.

Parameters

id	unique mode id
type	type of the mode
alias	alias of the mode
msg	(opt) CAN message for measurement/generation defaults to None
vol_check	(opt) variable/signal for voltage measurement defaults to None
cur_check	(opt) variable/signal for current measurement defaults to None
vol_set	(opt) variable/signal for voltage generation defaults to None
limit	(opt) current limit of measurement defaults to None
ref_value	(opt) reference voltage defaults to None
factor_vol_to↩ _si	(opt) factor for the conversion of the voltage measurement value to the si unit (V) defaults to 1
factor_cur_to↩ _si	(opt) factor for the conversion of the current measurement value to the si unit (A) defaults to 1

7.17.3 Member Data Documentation

7.17.3.1 alias

```
ttk_selftest.hil_devices._selftest_devices._SrcMode.alias
```

7.17.3.2 cur_check

```
ttk_selftest.hil_devices._selftest_devices._SrcMode.cur_check
```

7.17.3.3 factor_cur_to_si

```
ttk_selftest.hil_devices._selftest_devices._SrcMode.factor_cur_to_si
```

7.17.3.4 factor_vol_to_si

ttk_selftest.hil_devices._selftest_devices._SrcMode.factor_vol_to_si

7.17.3.5 limit

ttk_selftest.hil_devices._selftest_devices._SrcMode.limit

7.17.3.6 mode_id

ttk_selftest.hil_devices._selftest_devices._SrcMode.mode_id

7.17.3.7 mode_type

ttk_selftest.hil_devices._selftest_devices._SrcMode.mode_type

7.17.3.8 msg

ttk_selftest.hil_devices._selftest_devices._SrcMode.msg

7.17.3.9 ref_value

ttk_selftest.hil_devices._selftest_devices._SrcMode.ref_value

7.17.3.10 vol_check

ttk_selftest.hil_devices._selftest_devices._SrcMode.vol_check

7.17.3.11 vol_set

ttk_selftest.hil_devices._selftest_devices._SrcMode.vol_set

The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/_selftest_devices.py (v2.1.1)

7.18 ttk_tools.dspace._xil_api_offline_stub._TestbenchStub Class Reference

Minimal (very minimal) stub for XIL-API testbench implementation.

Public Member Functions

- __init__ (self, vendor_name, product_name, product_version)

Public Attributes

- VendorName
- ProductName
- ProductVersion

7.18.1 Detailed Description

Minimal (very minimal) stub for XIL-API testbench implementation.

7.18.2 Constructor & Destructor Documentation

7.18.2.1 __init__()

```
ttk_tools.dspace._xil_api_offline_stub._TestbenchStub.__init__ (
              self,
              vendor_name,
              product_name,
              product_version )
```

7.18.3 Member Data Documentation

7.18.3.1 ProductName

```
ttk_tools.dspace._xil_api_offline_stub._TestbenchStub.ProductName
```

7.18.3.2 ProductVersion

```
ttk_tools.dspace._xil_api_offline_stub._TestbenchStub.ProductVersion
```

7.18.3.3 VendorName

```
ttk_tools.dspace._xil_api_offline_stub._TestbenchStub.VendorName
```

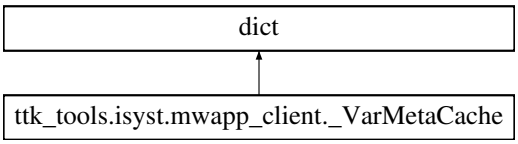
The documentation for this class was generated from the following file:

- ttk_tools/dspace/_xil_api_offline_stub.py (v2.1.2)

7.19 ttk_tools.isyst.mwapp_client._VarMetaCache Class Reference

Cache for iDA-Variable meta data.

Inheritance diagram for ttk_tools.isyst.mwapp_client._VarMetaCache:



Classes

- class VarMeta

  Simple container for iDA-Variable meta data.

Public Member Functions

- __init__ (self, mwapp_stub)
- getEntry (self, var_id)

  Get a VarMeta entry, which might be empty if var_id is not found.
- getDisplayName (self, var_id, fallback="<unknown var-id {var_id}>")

  Get display name of MW-Variable from cached meta data.
- getDataType (self, var_id)
- getSize (self, var_id)
- getUnit (self, var_id)
- getFmt (self, var_id)
- getDescr (self, var_id)
- getDefaultValue (self, var_id)
- getMinLimit (self, var_id)
- getMaxLimit (self, var_id)

Public Attributes

- stub
- fallback

Protected Member Functions

- _preloadData (self)

  Retrieve "essential" meta data for all available variables and place them in the cache.
- _ensureFullVarMeta (self, var_id)

  Ensure that all meta entries are loaded for a Var-ID.

7.19.1 Detailed Description

Cache for iDA-Variable meta data.

Maps Var-IDs to VarMeta data instances. Uses bulkGetMeta RPC to retrieve meta entries.

7.19.2 Constructor & Destructor Documentation

7.19.2.1 __init__()

```
ttk_tools.isyst.mwapp_client._VarMetaCache.__init__ (
              self,
              mwapp_stub )
```

Parameters

mwapp_stub	MwApp grpc-stub (RPCs, not an offline_stub)

7.19.3 Member Function Documentation

7.19.3.1 _ensureFullVarMeta()

```
ttk_tools.isyst.mwapp_client._VarMetaCache._ensureFullVarMeta (
              self,
              var_id )  [protected]
```

Ensure that all meta entries are loaded for a Var-ID.

Loads remaining entries not already filled during _preloadData.

Parameters

var↩ _id	Var-ID of MW-Variable

7.19.3.2 _preloadData()

```
ttk_tools.isyst.mwapp_client._VarMetaCache._preloadData (
              self )  [protected]
```

Retrieve "essential" meta data for all available variables and place them in the cache.

- var display name (for name <-> id mapping)
- data type
- size (dimension/length) also:
- unit (nice to have as fallback in Var abstraction)
- display_format (nice to have as fallback for fmt in Var abstraction)

7.19.3.3 getDataType()

```
ttk_tools.isyst.mwapp_client._VarMetaCache.getDataType (
              self,
              var_id )
```

7.19.3.4 getDefaultValue()

```
ttk_tools.isyst.mwapp_client._VarMetaCache.getDefaultValue (
              self,
              var_id )
```

7.19.3.5 getDescr()

```
ttk_tools.isyst.mwapp_client._VarMetaCache.getDescr (
              self,
              var_id )
```

7.19.3.6 getDisplayName()

```
ttk_tools.isyst.mwapp_client._VarMetaCache.getDisplayName (
              self,
              var_id,
              fallback = "<unknown var-id {var_id}>" )
```

Get display name of MW-Variable from cached meta data.

Parameters

var_id	Var-ID of MW-Variable
fallback	fallback value for not found variables. If a string is supplied, it may contain `var_id` as format key

Returns

a MwDataType or fallback if `var_id` was not found.

7.19.3.7 getEntry()

```
ttk_tools.isyst.mwapp_client._VarMetaCache.getEntry (
              self,
              var_id )
```

Get a VarMeta entry, which might be empty if var_id is not found.

Parameters

var↩ _id	MW-Variable-ID

Returns

a VarMeta entry.

7.19.3.8 getFmt()

```
ttk_tools.isyst.mwapp_client._VarMetaCache.getFmt (
              self,
              var_id )
```

7.19.3.9 getMaxLimit()

```
ttk_tools.isyst.mwapp_client._VarMetaCache.getMaxLimit (
              self,
              var_id )
```

7.19.3.10 getMinLimit()

```
ttk_tools.isyst.mwapp_client._VarMetaCache.getMinLimit (
              self,
              var_id )
```

7.19.3.11 getSize()

```
ttk_tools.isyst.mwapp_client._VarMetaCache.getSize (
              self,
              var_id )
```

7.19.3.12 getUnit()

```
ttk_tools.isyst.mwapp_client._VarMetaCache.getUnit (
              self,
              var_id )
```

7.19.4 Member Data Documentation

7.19.4.1 fallback

```
ttk_tools.isyst.mwapp_client._VarMetaCache.fallback
```

7.19.4.2 stub

```
ttk_tools.isyst.mwapp_client._VarMetaCache.stub
```

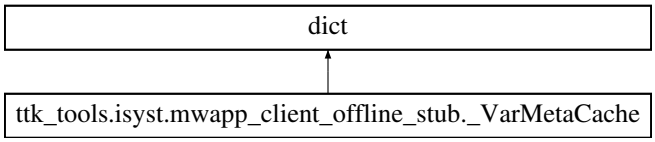
The documentation for this class was generated from the following file:

- ttk_tools/isyst/mwapp_client.py (v1.2.4)

7.20 ttk_tools.isyst.mwapp_client_offline_stub._VarMetaCache Class Reference

Cache for iDA-variable meta data.

Inheritance diagram for ttk_tools.isyst.mwapp_client_offline_stub._VarMetaCache:



Classes

- class VarMeta

  Simple container for iDA-Variable meta data.

Public Member Functions

- getEntry (self, var_id)

  Get a VarMeta entry, which might be empty if var_id is not found.
- getDisplayName (self, var_id, fallback="<unknown var-id {var_id}>")

  Get display name of MW-Variable from cached meta data.
- getDataType (self, var_id)
- getSize (self, var_id)
- getUnit (self, var_id)
- getFmt (self, var_id)
- getDescr (self, var_id)
- getDefaultValue (self, var_id)
- getMinLimit (self, var_id)
- getMaxLimit (self, var_id)

Protected Member Functions

- _addEntry (self, var_id, var_name)

  Add new entry to cache.

7.20.1 Detailed Description

Cache for iDA-variable meta data.

Maps Var-IDs to VarMeta data instances.

7.20.2 Member Function Documentation

7.20.2.1 _addEntry()

```
ttk_tools.isyst.mwapp_client_offline_stub._VarMetaCache._addEntry (
              self,
              var_id,
              var_name )  [protected]
```

Add new entry to cache.

Only used in stup to populate cache on-the-fly

7.20.2.2 getDataType()

```
ttk_tools.isyst.mwapp_client_offline_stub._VarMetaCache.getDataType (
              self,
              var_id )
```

7.20.2.3 getDefaultValue()

```
ttk_tools.isyst.mwapp_client_offline_stub._VarMetaCache.getDefaultValue (
              self,
              var_id )
```

7.20.2.4 getDescr()

```
ttk_tools.isyst.mwapp_client_offline_stub._VarMetaCache.getDescr (
              self,
              var_id )
```

7.20.2.5 getDisplayName()

```
ttk_tools.isyst.mwapp_client_offline_stub._VarMetaCache.getDisplayName (
              self,
              var_id,
              fallback = "<unknown var-id {var_id}>" )
```

Get display name of MW-Variable from cached meta data.

Parameters

var_id	Var-ID of MW-Variable
fallback	fallback value for not found variables. If a string is supplied, it may contain `var_id` as format key

Returns

a MwDataType or fallback if `var_id` was not found.

7.20.2.6 getEntry()

```
ttk_tools.isyst.mwapp_client_offline_stub._VarMetaCache.getEntry (
              self,
              var_id )
```

Get a VarMeta entry, which might be empty if var_id is not found.

Parameters

var_id	MW-Variable-ID

Returns

a VarMeta entry.

7.20.2.7 getFmt()

```
ttk_tools.isyst.mwapp_client_offline_stub._VarMetaCache.getFmt (
              self,
              var_id )
```

7.20.2.8 getMaxLimit()

```
ttk_tools.isyst.mwapp_client_offline_stub._VarMetaCache.getMaxLimit (
              self,
              var_id )
```

7.20.2.9 getMinLimit()

```
ttk_tools.isyst.mwapp_client_offline_stub._VarMetaCache.getMinLimit (
              self,
              var_id )
```

7.20.2.10 getSize()

```
ttk_tools.isyst.mwapp_client_offline_stub._VarMetaCache.getSize (
              self,
              var_id )
```

7.20.2.11 getUnit()

```
ttk_tools.isyst.mwapp_client_offline_stub._VarMetaCache.getUnit (
              self,
              var_id )
```

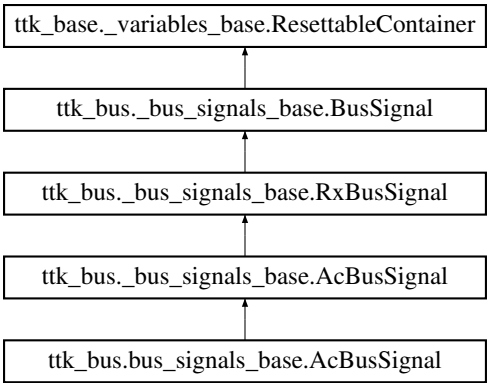
The documentation for this class was generated from the following file:

- ttk_tools/isyst/mwapp_client_offline_stub.py (v1.3.0)

7.21 ttk_bus._bus_signals_base.AcBusSignal Class Reference

A RX bus signal (HIL => ECU) that models an alive counter value with additional error injections.

Inheritance diagram for ttk_bus._bus_signals_base.AcBusSignal:



Public Member Functions

- __init__ (self, context, identifier, cycle_time=0, debounce_time=0, signal_var=None, alias="", descr="", error_delay=None, change_delay=None, timeout=None, recovery_time=None)

  Alive counter bus signal initialization.
- setBasicAcError (self)

  Switch to basic alive counter error state (which is usually "counter stopped")
- clearBasicAcError (self)

  Resume normal alive counter operation (which is typically "continue counting").

Public Attributes

- error_delay

  delay to wait after a "basic" alive counter error injection [ms]

7.21.1 Detailed Description

A RX bus signal (HIL => ECU) that models an alive counter value with additional error injections.

This is the generic base class for AC signals, blockset/implementation specific AC signal classes will inherit from this.

Attention

This is just a base class, using it directly will not accomplish much. At the very least, setBasicAcError and clearBasicAcError need to be overridden depending on the target interface.

See bus_signals_frbs or bus_signals_canmmbs for existing implementations.

7.21.2 Constructor & Destructor Documentation

7.21.2.1 __init__()

```
ttk_bus._bus_signals_base.AcBusSignal.__init__ (
              self,
              context,
              identifier,
              cycle_time = 0,
              debounce_time = 0,
              signal_var = None,
              alias = "",
              descr = "",
              error_delay = None,
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Alive counter bus signal initialization.

Parameters

context	parent context for instantiating HIL variables. This gets mainly used in derived classes
identifier	base signal identifier/name/path in the HIL model pointing to the alive counter signal value. Derived classes typically use this identifier as starting point for locating additional helper variables.
cycle_time	cycle time in [ms]
debounce_time	minimum debounce delay between events in [ms]
signal_var	(opt) an already initialized instance of Var (or most likely one of its derived classes) for the main signal value.
alias	(opt) bus signal alias/name, defaults to signal_var.alias.
descr	(opt) signal description (e.g. may get used in report entries)
error_delay	(opt) delay to wait until a basic alive counter error injection should be detected [ms], defaults to 10x max(cycle_time, debounce_time)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)

Note

The parameters `change_delay`, `timeout` and `recovery_time` are just here for completeness' sake, only `cycle_time` and `error_delay` are used during typical Alive Counter checks.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

Reimplemented in ttk_bus.bus_signals_base.AcBusSignal.

7.21.3 Member Function Documentation

7.21.3.1 clearBasicAcError()

```
ttk_bus._bus_signals_base.AcBusSignal.clearBasicAcError (
              self )
```

Resume normal alive counter operation (which is typically "continue counting").

Reimplemented in ttk_bus.bus_signals_base.AcBusSignal.

7.21.3.2 setBasicAcError()

```
ttk_bus._bus_signals_base.AcBusSignal.setBasicAcError (
              self )
```

Switch to basic alive counter error state (which is usually "counter stopped")

Reimplemented in ttk_bus.bus_signals_base.AcBusSignal.

7.21.4 Member Data Documentation

7.21.4.1 error_delay

```
ttk_bus._bus_signals_base.AcBusSignal.error_delay
```

delay to wait after a "basic" alive counter error injection [ms]

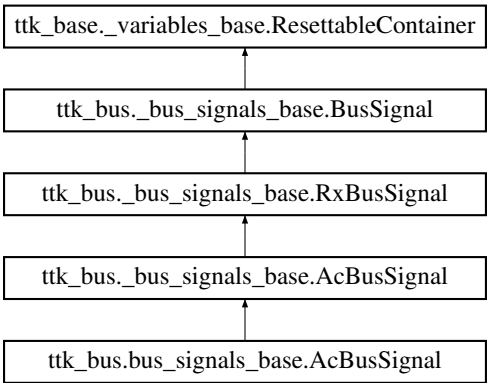
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_base.py (v3.0.1)

7.22 ttk_bus.bus_signals_base.AcBusSignal Class Reference

A RX bus signal (HIL => ECU) that models an alive counter value with additional error injections.

Inheritance diagram for ttk_bus.bus_signals_base.AcBusSignal:



Public Member Functions

- __init__ (self, context, identifier, cycle_time=0, debounce_time=0, signal_var=None, alias="", descr="", error_delay=None, change_delay=None, timeout=None, recovery_time=None)

  Alive counter bus signal initialization.
- setBasicAcError (self)

  Switch to basic alive counter error state (which is usually "counter stopped")
- clearBasicAcError (self)

  Resume normal alive counter operation (which is typically "continue counting").
- info (self)

  Get a current info string about the HIL signal variable.
- get (self)

  Get the current value of the HIL signal variable (typically not that useful for ECU-Rx/HIL-Tx AC-signals, though).
- set (self, value)

  Set a value to the HIL signal variable.
- getState (self)

  Get a textual state for the HIL signal variable's current data.
- setState (self, state)

  Set a textual state to the HIL signal variable.
- hasStateLookup (self)

  Return True if a state/value lookup table is defined for the HIL signal variable.
- getStateDescr (self, value=None, fallback=None)

  Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).
- hasFormattedRepr (self)

  Return True if a formatting for value representation is defined for the HIL signal variable.
- getFormattedValue (self, value=None)

  Get a formatted representation for the given value using the defined format string of the the HIL signal variable.
- reset (self, *args, **kwargs)

  Call reset() on all contained variables/signals that support it.

Public Attributes

- error_delay

  delay to wait after a "basic" crc error injection[ms]

7.22.1 Detailed Description

A RX bus signal (HIL => ECU) that models an alive counter value with additional error injections.

This is the generic base class for AC signals, blockset/implementation specific AC signal classes will inherit from this.

Attention

This is just a base class, using it directly will not accomplish much. At the very least, setBasicAcError and clearBasicAcError need to be overridden depending on the target interface.

See bus_signals_frbs or bus_signals_canmmbs for existing implementations.

7.22.2 Constructor & Destructor Documentation

7.22.2.1 __init__()

```
ttk_bus.bus_signals_base.AcBusSignal.__init__ (
              self,
              context,
              identifier,
              cycle_time = 0,
              debounce_time = 0,
              signal_var = None,
              alias = "",
              descr = "",
              error_delay = None,
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Alive counter bus signal initialization.

Parameters

context	parent context for instantiating HIL variables. This gets mainly used in derived classes
identifier	base signal identifier/name/path in the HIL model pointing to the alive counter signal value. Derived classes typically use this identifier as starting point for locating additional helper variables.
cycle_time	cycle time in [ms]
debounce_time	minimum debounce delay between events in [ms]
signal_var	(opt) an already initialized instance of Var (or most likely one of its derived classes) for the main signal value.
alias	(opt) bus signal alias/name, defaults to signal_var.alias.
descr	(opt) signal description (e.g. may get used in report entries)
error_delay	(opt) delay to wait until a basic alive counter error injection should be detected [ms], defaults to 10x max(cycle_time, debounce_time)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)

Note

The parameters `change_delay`, `timeout` and `recovery_time` are just here for completeness' sake, only `cycle_time` and `error_delay` are used during typical Alive Counter checks.

Reimplemented from ttk_bus._bus_signals_base.AcBusSignal.

7.22.3 Member Function Documentation

7.22.3.1 clearBasicAcError()

```
ttk_bus.bus_signals_base.AcBusSignal.clearBasicAcError (
              self )
```

Resume normal alive counter operation (which is typically "continue counting").

Reimplemented from ttk_bus._bus_signals_base.AcBusSignal.

7.22.3.2 get()

```
ttk_bus.bus_signals_base.AcBusSignal.get (
              self )
```

Get the current value of the HIL signal variable (typically not that useful for ECU-Rx/HIL-Tx AC-signals, though).

Returns

current value.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.22.3.3 getFormattedValue()

```
ttk_bus.bus_signals_base.AcBusSignal.getFormattedValue (
              self,
              value = None )
```

Get a formatted representation for the given value using the defined format string of the the HIL signal variable.

An empty (or non-defined) format string will use the value's default string representation.

Parameters

value	value for which to get a string representation. If None, the last value (read or written) will be used instead.

Returns

a formatted string representation.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.22.3.4 getState()

```
ttk_bus.bus_signals_base.AcBusSignal.getState (
              self )
```

Get a textual state for the HIL signal variable's current data.

(typically not that useful for ECU-Rx/HIL-Tx AC-signals, though).

Returns

the current state's description

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.22.3.5 getStateDescr()

```
ttk_bus.bus_signals_base.AcBusSignal.getStateDescr (
              self,
              value = None,
              fallback = None )
```

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the last read value will be used instead.
fallback	fallback value to return if no match was found. Default: None

Returns

the state text if a match was found, otherwise the specified fallback value.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.22.3.6 hasFormattedRepr()

```
ttk_bus.bus_signals_base.AcBusSignal.hasFormattedRepr (
              self )
```

Return True if a formatting for value representation is defined for the HIL signal variable.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.22.3.7 hasStateLookup()

```
ttk_bus.bus_signals_base.AcBusSignal.hasStateLookup (
              self )
```

Return True if a state/value lookup table is defined for the HIL signal variable.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.22.3.8 info()

```
ttk_bus.bus_signals_base.AcBusSignal.info (
              self )
```

Get a current info string about the HIL signal variable.

Returns

an info string.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.22.3.9 reset()

```
ttk_bus.bus_signals_base.AcBusSignal.reset (
              self,
             * args,
             ** kwargs )
```

Call reset() on all contained variables/signals that support it.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.ResettableContainer.

7.22.3.10 set()

```
ttk_bus.bus_signals_base.AcBusSignal.set (
              self,
              value )
```

Set a value to the HIL signal variable.

Parameters

value	new value to set

Returns

the previous value.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.22.3.11 setBasicAcError()

```
ttk_bus.bus_signals_base.AcBusSignal.setBasicAcError (
              self )
```

Switch to basic alive counter error state (which is usually "counter stopped")

Reimplemented from ttk_bus._bus_signals_base.AcBusSignal.

7.22.3.12 setState()

```
ttk_bus.bus_signals_base.AcBusSignal.setState (
              self,
              state )
```

Set a textual state to the HIL signal variable.

This uses the lookup values supplied in constructor.

Parameters

state	named state to set

Returns

the previous state's description

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.22.4 Member Data Documentation

7.22.4.1 error_delay

```
ttk_bus.bus_signals_base.AcBusSignal.error_delay
```

delay to wait after a "basic" crc error injection[ms]

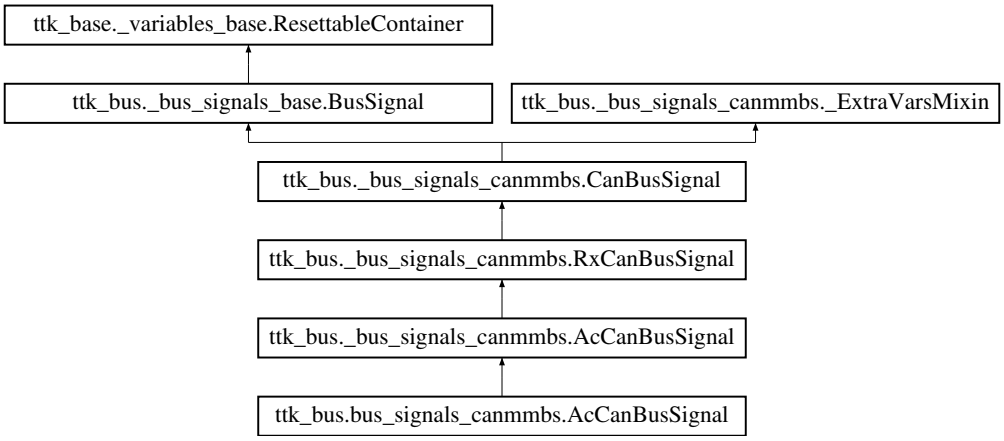
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_base.py (v3.1.0)

7.23 ttk_bus._bus_signals_canmmbs.AcCanBusSignal Class Reference

Alive counter bus signal derived from RxCanBusSignal (HIL-Tx => ECU-Rx) for use with dSPACE CAN Multi-↩ Message blockset data structures.

Inheritance diagram for ttk_bus._bus_signals_canmmbs.AcCanBusSignal:



Public Member Functions

- __init__ (self, context, signal_path, message=None, cycle_time=None, debounce_time=None, alias="", descr="", lookup=None, error_delay=None, change_delay=None, timeout=None, recovery_time=None)

  Alive counter bus signal (HIL-Tx => ECU-Rx) initialization for dSPACE CAN Multi-Message blockset.
- setBasicAcError (self)

  Stop and hold Alive Counter at the current count.
- clearBasicAcError (self)

  (Re-)enable normal Alive Counter behaviour (i.e.

Public Attributes

- error_delay

  delay to wait after a "basic" AC error injection [ms]
- ac_switch_var

  Alive Counter mode switch (uses same switch as other input manipulation options).
- ac_tmpsteplen_var

  "HilVar" for "temporary step length", holds counter for n ∗ cycle time (will reset automatically to default step length)
- ac_offset_var

  "HilVar" for additional alive counter offset (offset will simply be added to current counter value)

Static Public Attributes

- int cfg_basic_error_tmpsteplen = 2000

  Configuration settings for new instances:
  Temporary step length used to "freeze" counter in setBasicAcError (arbitrary value, but should be selected large enough to hold counter for a full check duration, typically > error_delay)
- dict cfg_custom_defaults

  Configuration settings for new instances:
  Custom "hard" default value(s) to use for internal real time system variables.

Protected Attributes

- _basic_error_tmpsteplen

  temporary step length used to "freeze" counter in setBasicAcError See self.cfg_basic_error_tmpsteplen

7.23.1 Detailed Description

Alive counter bus signal derived from RxCanBusSignal (HIL-Tx => ECU-Rx) for use with dSPACE CAN Multi-↩ Message blockset data structures.

See RxCanBusSignal for inherited methods.

7.23.2 Constructor & Destructor Documentation

7.23.2.1 __init__()

```
ttk_bus._bus_signals_canmmbs.AcCanBusSignal.__init__ (
              self,
              context,
              signal_path,
              message = None,
              cycle_time = None,
              debounce_time = None,
              alias = "",
              descr = "",
              lookup = None,
              error_delay = None,
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Alive counter bus signal (HIL-Tx => ECU-Rx) initialization for dSPACE CAN Multi-Message blockset.

Parameters

context	rtplib application instance
signal_path	base path to signal below BusSystems (`BusSystems/.../<message>/TX/<signal>`)
message	reference to associated RxCanBusMessage instance
cycle_time	(opt) cycle time in [ms], overrides settings in Message
debounce_time	(opt) minimum debounce delay between events in [ms], overrides settings in Message
alias	(opt) bus signal alias/name
descr	(opt) signal description
lookup	(opt) lookup table for values, see VarBase
error_delay	(opt) delay to wait until a simple crc error should be detected [ms], defaults to 10x cycle time
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x cycle_time
timeout	(opt) detection time for timeout errors [ms], defaults to 2x cycle_time
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x cycle_time

Reimplemented from ttk_bus._bus_signals_canmmbs.RxCanBusSignal.

Reimplemented in ttk_bus.bus_signals_canmmbs.AcCanBusSignal.

7.23.3 Member Function Documentation

7.23.3.1 clearBasicAcError()

```
ttk_bus._bus_signals_canmmbs.AcCanBusSignal.clearBasicAcError (
              self )
```

(Re-)enable normal Alive Counter behaviour (i.e.

continue counting).

Reimplemented in ttk_bus.bus_signals_canmmbs.AcCanBusSignal.

7.23.3.2 setBasicAcError()

```
ttk_bus._bus_signals_canmmbs.AcCanBusSignal.setBasicAcError (
              self )
```

Stop and hold Alive Counter at the current count.

Reimplemented in ttk_bus.bus_signals_canmmbs.AcCanBusSignal.

7.23.4 Member Data Documentation

7.23.4.1 _basic_error_tmpsteplen

```
ttk_bus._bus_signals_canmmbs.AcCanBusSignal._basic_error_tmpsteplen  [protected]
```

temporary step length used to "freeze" counter in setBasicAcError See self.cfg_basic_error_tmpsteplen

7.23.4.2 ac_offset_var

```
ttk_bus._bus_signals_canmmbs.AcCanBusSignal.ac_offset_var
```

"HilVar" for additional alive counter offset (offset will simply be added to current counter value)

7.23.4.3 ac_switch_var

```
ttk_bus._bus_signals_canmmbs.AcCanBusSignal.ac_switch_var
```

Alive Counter mode switch (uses same switch as other input manipulation options).

See self.switch_var

7.23.4.4 ac_tmpsteplen_var

```
ttk_bus._bus_signals_canmmbs.AcCanBusSignal.ac_tmpsteplen_var
```

"HilVar" for "temporary step length", holds counter for n ∗ cycle time (will reset automatically to default step length)

7.23.4.5 cfg_basic_error_tmpsteplen

```
int ttk_bus._bus_signals_canmmbs.AcCanBusSignal.cfg_basic_error_tmpsteplen = 2000  [static]
```

Configuration settings for new instances:
Temporary step length used to "freeze" counter in setBasicAcError (arbitrary value, but should be selected large enough to hold counter for a full check duration, typically > error_delay)

7.23.4.6 cfg_custom_defaults

```
dict ttk_bus._bus_signals_canmmbs.AcCanBusSignal.cfg_custom_defaults  [static]
```

Initial value:

```
=  {
        "ac_tmpsteplen_var": None, #
        "ac_offset_var":     None, #

        "switch_var":        None, # ac_switch_var is just an alias for switch_var
        "dyncount_var":      None, # for parent RxCanBusSignal
        "dynvalue_var":      None, # for parent RxCanBusSignal
    }
```

Configuration settings for new instances:
Custom "hard" default value(s) to use for internal real time system variables.

See RxFrBusSignal for further details.

Mapping: <variable name as string>: <default value>

Typical default values:

- source switches for alive counters usually default to "running counter" See RxFrBusSignal for further details.
- Temporary Step Length: auto-resets to default step length, which is usually 1
- Counter Offset, additive offset to counter value defaults to 0

7.23.4.7 error_delay

```
ttk_bus._bus_signals_canmmbs.AcCanBusSignal.error_delay
```

delay to wait after a "basic" AC error injection [ms]

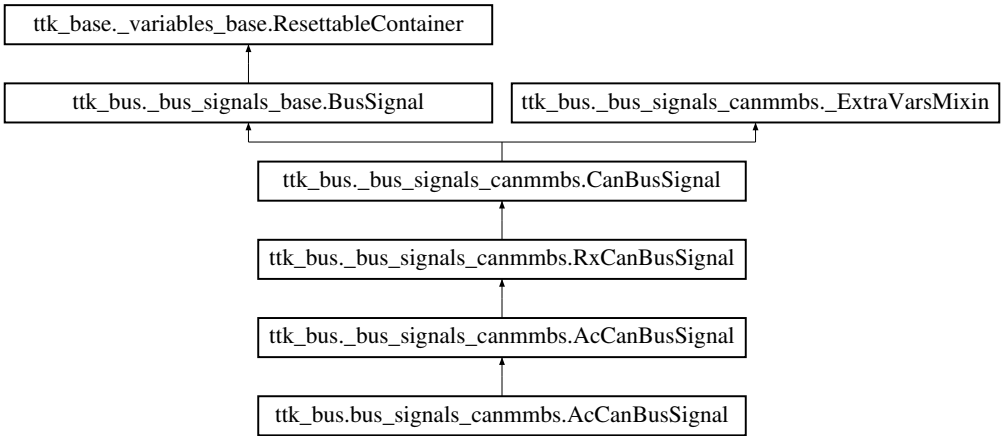
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_canmmbs.py (v2.3)

7.24 ttk_bus.bus_signals_canmmbs.AcCanBusSignal Class Reference

Alive counter bus signal derived from RxCanBusSignal (HIL-Tx => ECU-Rx) for use with dSPACE CAN Multi-↩ Message blockset data structures.

Inheritance diagram for ttk_bus.bus_signals_canmmbs.AcCanBusSignal:



Public Member Functions

- __init__ (self, context, signal_path, message=None, cycle_time=None, debounce_time=None, alias="", descr="", lookup=None, error_delay=None, change_delay=None, timeout=None, recovery_time=None)

  Alive counter bus signal (HIL-Tx => ECU-Rx) initialization for dSPACE CAN Multi-Message blockset.
- setBasicAcError (self)

  Stop and hold Alive Counter at the current count.
- clearBasicAcError (self)

  (Re-)enable normal Alive Counter behaviour (i.e.
- enableHilTx (self)

  Enable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).
- disableHilTx (self)

  Disable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).
- kickout (self)

  Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent CAN-Message/PDU).
- get (self)

  Get the current value of the HIL signal variable (typically not that useful for ECU-Rx/HIL-Tx AC-signals, though).
- set (self, value)

  Set a value to the HIL signal variable.
- getState (self)

  Get a textual state for the HIL signal variable's current data (typically not that useful for ECU-Rx/HIL-Tx AC-signals, though).
- setState (self, state)

  Set a textual state to the HIL signal variable.
- getStateDescr (self, value=None, fallback=None)

  Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).
- info (self)

  Get a current info string about the HIL signal variable.
- reset (self, *args, **kwargs)

  Reset all resettable variables of this bus signal.

Public Attributes

- error_delay
  delay to wait after a "simple" ac error injection [ms]
- ac_offset_var
  HilVar for additional alive counter offset (offset will simply be added to current counter value)
- ac_tmpsteplen_var
  HilVar for "temporary step length", holds counter for n ∗ cycle time (will reset automatically to default step length)
- ac_switch_var
  Alive Counter mode switch (uses same switch as other input manipulation options).
- base_path
  Signal model path as base path for deriving sub-element paths.
- message
  reference to associated (Rx-)CanBusMessage instance
- signal_var
  signal value variable, get/info methods of this TxCanBusSignal will access this variable (a HilVar instance).
- context
  parent context (an RTE application instance to resolve signal/variable paths)
- unit
  unit of physical signal value (as string, mainly used for report entries)
- alias
  bus signal alias name
- descr
  bus signal description text
- cycle_time
  cycle time of Message/PDU in [ms]
- debounce_time
  minimum debounce delay for event-triggered Messages/PDUs in [ms]
- change_delay
  delay to wait after a value change [ms]
- timeout
  detection time for timeout errors [ms]
- recovery_time
  time for recovery/"wiedergut" checks [ms]

Static Public Attributes

- int cfg_basic_error_tmpsteplen = 2000
  Configuration settings for new instances:
  Temporary step length used to "freeze" counter in setBasicAcError (arbitrary value, but should be selected large enough to hold counter for a full check duration, typically > error_delay)
- dict cfg_custom_defaults
  Configuration settings for new instances:
  Custom "hard" default value(s) to use for internal real time system variables.

7.24.1 Detailed Description

Alive counter bus signal derived from RxCanBusSignal (HIL-Tx => ECU-Rx) for use with dSPACE CAN Multi-↩ Message blockset data structures.

See RxCanBusSignal for inherited methods.

7.24.2 Constructor & Destructor Documentation

7.24.2.1 __init__()

```
ttk_bus.bus_signals_canmmbs.AcCanBusSignal.__init__ (
                self,
                context,
                signal_path,
                message = None,
                cycle_time = None,
                debounce_time = None,
                alias = "",
                descr = "",
                lookup = None,
                error_delay = None,
                change_delay = None,
                timeout = None,
                recovery_time = None )
```

Alive counter bus signal (HIL-Tx => ECU-Rx) initialization for dSPACE CAN Multi-Message blockset.

Parameters

context	rtplib application instance
signal_path	base path to signal below BusSystems (BusSystems/.../<message>/TX/<signal>)
message	reference to associated RxCanBusMessage instance
cycle_time	(opt) cycle time in [ms], overrides settings in Message
debounce_time	(opt) minimum debounce delay between events in [ms], overrides settings in Message
alias	(opt) bus signal alias/name
descr	(opt) signal description
lookup	(opt) lookup table for values, see VarBase
error_delay	(opt) delay to wait until a simple crc error should be detected [ms], defaults to 10x cycle time
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x cycle_time
timeout	(opt) detection time for timeout errors [ms], defaults to 2x cycle_time
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x cycle_time

Reimplemented from ttk_bus._bus_signals_canmmbs.AcCanBusSignal.

7.24.3 Member Function Documentation

7.24.3.1 clearBasicAcError()

```
ttk_bus.bus_signals_canmmbs.AcCanBusSignal.clearBasicAcError (
                self )
```

(Re-)enable normal Alive Counter behaviour (i.e.

continue counting).

Reimplemented from ttk_bus._bus_signals_canmmbs.AcCanBusSignal.

7.24.3.2 disableHilTx()

```
ttk_bus.bus_signals_canmmbs.AcCanBusSignal.disableHilTx (
              self )
```

Disable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).

Note

This will disable transmission of *all* signals of the parent Message/PDU.

Reimplemented from ttk_bus._bus_signals_canmmbs.RxCanBusSignal.

7.24.3.3 enableHilTx()

```
ttk_bus.bus_signals_canmmbs.AcCanBusSignal.enableHilTx (
              self )
```

Enable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).

Note

This will enable transmission of *all* signals of the parent Message/PDU.

Reimplemented from ttk_bus._bus_signals_canmmbs.RxCanBusSignal.

7.24.3.4 get()

```
ttk_bus.bus_signals_canmmbs.AcCanBusSignal.get (
              self )
```

Get the current value of the HIL signal variable (typically not that useful for ECU-Rx/HIL-Tx AC-signals, though).

Returns

the current value from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.24.3.5 getState()

```
ttk_bus.bus_signals_canmmbs.AcCanBusSignal.getState (
              self )
```

Get a textual state for the HIL signal variable's current data (typically not that useful for ECU-Rx/HIL-Tx AC-signals, though).

Returns

the current state's description.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.24.3.6 getStateDescr()

```
ttk_bus.bus_signals_canmmbs.AcCanBusSignal.getStateDescr (
              self,
              value = None,
              fallback = None )
```

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the last read value will be used instead.
fallback	fallback value to return if no match was found. Default: None

Returns

the state text if a match was found, otherwise the specified fallback value.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.24.3.7 info()

```
ttk_bus.bus_signals_canmmbs.AcCanBusSignal.info (
              self )
```

Get a current info string about the HIL signal variable.

Returns

an info string.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.24.3.8 kickout()

```
ttk_bus.bus_signals_canmmbs.AcCanBusSignal.kickout (
              self )
```

Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent CAN-Message/PDU).

Note

All other signals of the parent Message/PDU will also be sent.

Reimplemented from ttk_bus._bus_signals_canmmbs.RxCanBusSignal.

7.24.3.9 reset()

```
ttk_bus.bus_signals_canmmbs.AcCanBusSignal.reset (
              self,
             * args,
             ** kwargs )
```

Reset all resettable variables of this bus signal.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.ResettableContainer.

7.24.3.10 set()

```
ttk_bus.bus_signals_canmmbs.AcCanBusSignal.set (
              self,
              value )
```

Set a value to the HIL signal variable.

Parameters

value	value to set

Returns

the previous value from signal_var.

Reimplemented from ttk_bus._bus_signals_canmmbs.RxCanBusSignal.

7.24.3.11 setBasicAcError()

```
ttk_bus.bus_signals_canmmbs.AcCanBusSignal.setBasicAcError (
              self )
```

Stop and hold Alive Counter at the current count.

Reimplemented from ttk_bus._bus_signals_canmmbs.AcCanBusSignal.

7.24.3.12 setState()

```
ttk_bus.bus_signals_canmmbs.AcCanBusSignal.setState (
              self,
              state )
```

Set a textual state to the HIL signal variable.

This uses the lookup values supplied in constructor.

Parameters

state	named state to set

Returns

the previous state's description.

Reimplemented from ttk_bus._bus_signals_canmmbs.RxCanBusSignal.

7.24.4 Member Data Documentation

7.24.4.1 ac_offset_var

```
ttk_bus.bus_signals_canmmbs.AcCanBusSignal.ac_offset_var
```

HilVar for additional alive counter offset (offset will simply be added to current counter value)

7.24.4.2 ac_switch_var

```
ttk_bus.bus_signals_canmmbs.AcCanBusSignal.ac_switch_var
```

Alive Counter mode switch (uses same switch as other input manipulation options).

See self.switch_var

7.24.4.3 ac_tmpsteplen_var

```
ttk_bus.bus_signals_canmmbs.AcCanBusSignal.ac_tmpsteplen_var
```

HilVar for "temporary step length", holds counter for n $*$ cycle time (will reset automatically to default step length)

7.24.4.4 alias

```
ttk_bus.bus_signals_canmmbs.AcCanBusSignal.alias
```

bus signal alias name

7.24.4.5 base_path

```
ttk_bus.bus_signals_canmmbs.AcCanBusSignal.base_path
```

Signal model path as base path for deriving sub-element paths.

For CANMM signals, related variables follow the form <path_to_signal>_suffix

7.24.4.6 cfg_basic_error_tmpsteplen

```
int ttk_bus.bus_signals_canmmbs.AcCanBusSignal.cfg_basic_error_tmpsteplen = 2000  [static]
```

Configuration settings for new instances:
Temporary step length used to "freeze" counter in setBasicAcError (arbitrary value, but should be selected large enough to hold counter for a full check duration, typically > error_delay)

7.24.4.7 cfg_custom_defaults

```
dict ttk_bus.bus_signals_canmmbs.AcCanBusSignal.cfg_custom_defaults  [static]
```

Initial value:

```
=  {
        "ac_tmpsteplen_var": None, #
        "ac_offset_var":     None, #

        "switch_var":        None, # ac_switch_var is just an alias for switch_var
        "dyncount_var":      None, # for parent RxCanBusSignal
        "dynvalue_var":      None, # for parent RxCanBusSignal
    }
```

Configuration settings for new instances:
Custom "hard" default value(s) to use for internal real time system variables.

See RxFrBusSignal for further details.

Mapping: <variable name as string>: <default value>

Typical default values:

- source switches for alive counters usually default to "running counter" See RxFrBusSignal for further details.
- Temporary Step Length: auto-resets to default step length, which is usually 1
- Counter Offset, additive offset to counter value defaults to 0

7.24.4.8 change_delay

ttk_bus.bus_signals_canmmbs.AcCanBusSignal.change_delay

delay to wait after a value change [ms]

7.24.4.9 context

ttk_bus.bus_signals_canmmbs.AcCanBusSignal.context

parent context (an RTE application instance to resolve signal/variable paths)

7.24.4.10 cycle_time

ttk_bus.bus_signals_canmmbs.AcCanBusSignal.cycle_time

cycle time of Message/PDU in [ms]

7.24.4.11 debounce_time

ttk_bus.bus_signals_canmmbs.AcCanBusSignal.debounce_time

minimum debounce delay for event-triggered Messages/PDUs in [ms]

7.24.4.12 descr

ttk_bus.bus_signals_canmmbs.AcCanBusSignal.descr

bus signal description text

7.24.4.13 error_delay

ttk_bus.bus_signals_canmmbs.AcCanBusSignal.error_delay

delay to wait after a "simple" ac error injection [ms]

delay to wait after a "basic" AC error injection [ms]

7.24.4.14 message

ttk_bus.bus_signals_canmmbs.AcCanBusSignal.message

reference to associated (Rx-)CanBusMessage instance

7.24.4.15 recovery_time

`ttk_bus.bus_signals_canmmbs.AcCanBusSignal.recovery_time`

time for recovery/"wiedergut" checks [ms]

7.24.4.16 signal_var

`ttk_bus.bus_signals_canmmbs.AcCanBusSignal.signal_var`

signal value variable, get/info methods of this TxCanBusSignal will access this variable (a HilVar instance).

7.24.4.17 timeout

`ttk_bus.bus_signals_canmmbs.AcCanBusSignal.timeout`

detection time for timeout errors [ms]

7.24.4.18 unit

`ttk_bus.bus_signals_canmmbs.AcCanBusSignal.unit`

unit of physical signal value (as string, mainly used for report entries)

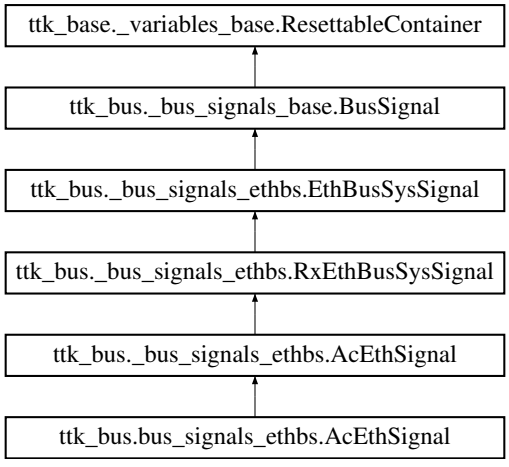
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_canmmbs.py (v1.8)

7.25 ttk_bus._bus_signals_ethbs.AcEthSignal Class Reference

An alive counter Rx Ethernet signal/parameter (HIL PSI => ECU CSI) for use with dSPACE Ethernet blockset.

Inheritance diagram for ttk_bus._bus_signals_ethbs.AcEthSignal:



Public Member Functions

- __init__ (self, context, base_path, event=None, cycle_time=None, debounce_time=None, alias="", descr="", error_delay=None, change_delay=None, timeout=None, recovery_time=None, **kwargs)

  Alive counter signal initialization for dSPACE Ethernet blockset.
- setBasicAcError (self)

  Stop and hold Alive Counter at the current count.
- clearBasicAcError (self)

  (Re-)enable normal Alive Counter behaviour (continue counting).

Public Attributes

- error_delay

  delay [ms] to wait until a basic Alive Counter error (like "AC stopped") should be detected.
- ac_runtime_behavior_var

  Counter Runtime Behavior variable (a HilVar instance)
- ac_offset_var

  Counter Offset variable.
- ac_switch_var

  ac_switch_var is only defined for backwards-compatibility to CANMMBS variant Runtime Behavior entry is used as AC enable/disable switch

Static Public Attributes

- int fallback_error_delay = 20

  fallback error delay in [ms] to use for signals/events without any specified or derived error_delay

7.25.1 Detailed Description

An alive counter Rx Ethernet signal/parameter (HIL PSI => ECU CSI) for use with dSPACE Ethernet blockset.

Note

This class expects a BusSystems path structure.

7.25.2 Constructor & Destructor Documentation

7.25.2.1 __init__()

```
ttk_bus._bus_signals_ethbs.AcEthSignal.__init__ (
              self,
              context,
              base_path,
              event = None,
              cycle_time = None,
              debounce_time = None,
              alias = "",
              descr = "",
              error_delay = None,
              change_delay = None,
              timeout = None,
              recovery_time = None,
             ** kwargs )
```

Alive counter signal initialization for dSPACE Ethernet blockset.

Example: Alive Counter Signal Path Structure:

```
BusSystems/Ethernet/FRR/PSIs/ACME.ENVIRONMENTALMODEL/ObjectFusionFront_1/Events/ObjectPosition/E2E
Protection/counter/Counter/Calculated Value
BusSystems/Ethernet/FRR/PSIs/ACME.ENVIRONMENTALMODEL/ObjectFusionFront_1/Events/ObjectPosition/E2E
Protection/counter/Counter/Enable
BusSystems/Ethernet/FRR/PSIs/ACME.ENVIRONMENTALMODEL/ObjectFusionFront_1/Events/ObjectPosition/E2E
Protection/counter/Counter/Offset
BusSystems/Ethernet/FRR/PSIs/ACME.ENVIRONMENTALMODEL/ObjectFusionFront_1/Events/ObjectPosition/E2E
Protection/counter/Counter/Runtime Behavior
BusSystems/Ethernet/FRR/PSIs/ACME.ENVIRONMENTALMODEL/ObjectFusionFront_1/Events/ObjectPosition/E2E
Protection/counter/Source Switch
BusSystems/Ethernet/FRR/PSIs/ACME.ENVIRONMENTALMODEL/ObjectFusionFront_1/Events/ObjectPosition/E2E
Protection/counter/Value
# or
BusSystems/Ethernet/FRR/PSIs/ACME.DASS/RadarData_1/Events/LaneInfo/E2E
Protection/counter/Counter/Calculated Value
BusSystems/Ethernet/FRR/PSIs/ACME.DASS/RadarData_1/Events/LaneInfo/E2E
Protection/counter/Counter/Enable
BusSystems/Ethernet/FRR/PSIs/ACME.DASS/RadarData_1/Events/LaneInfo/E2E
Protection/counter/Counter/Offset
BusSystems/Ethernet/FRR/PSIs/ACME.DASS/RadarData_1/Events/LaneInfo/E2E
Protection/counter/Counter/Runtime Behavior
BusSystems/Ethernet/FRR/PSIs/ACME.DASS/RadarData_1/Events/LaneInfo/E2E Protection/counter/Source Switch
BusSystems/Ethernet/FRR/PSIs/ACME.DASS/RadarData_1/Events/LaneInfo/E2E Protection/counter/Tx
Inspect/Available SL Value
BusSystems/Ethernet/FRR/PSIs/ACME.DASS/RadarData_1/Events/LaneInfo/E2E Protection/counter/Tx Inspect/Tx
Value
BusSystems/Ethernet/FRR/PSIs/ACME.DASS/RadarData_1/Events/LaneInfo/E2E Protection/counter/Value
```

Suitable base_paths / examples:

BusSystems/Ethernet/FRR/PSIs/ACME.DASS/RadarData/Events/LaneInfo/E2E Protection/counter/↩ Counter/Enable BusSystems/Ethernet/FRR/PSIs/ACME.DASS/RadarData/Events/LaneInfo/E2E Protection/counter/↩ Value => anything below .../Counter/* or the .../Value entry

Parameters

context	rtplib Application or XilTestbench instance
base_path	base model path (below BusSystems) from which all required paths will be derived.
event	reference to associated EthEvent instance
cycle_time	(opt) cycle time in [ms], overrides settings in EthEvent
debounce_time	(opt) minimum debounce delay time between events in [ms], overrides settings in EthEvent
alias	(opt) bus signal alias/name
descr	(opt) signal description
error_delay	(opt) delay to wait until a simple alive counter error should be detected [ms], defaults to 10x cycle time
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x cycle_time
timeout	(opt) detection time for timeout errors [ms], defaults to 2x cycle_time
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x cycle_time
***kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.

Reimplemented in ttk_bus.bus_signals_ethbs.AcEthSignal.

7.25.3 Member Function Documentation

7.25.3.1 clearBasicAcError()

```
ttk_bus._bus_signals_ethbs.AcEthSignal.clearBasicAcError (
              self )
```

(Re-)enable normal Alive Counter behaviour (continue counting).

Reimplemented in ttk_bus.bus_signals_ethbs.AcEthSignal.

7.25.3.2 setBasicAcError()

```
ttk_bus._bus_signals_ethbs.AcEthSignal.setBasicAcError (
              self )
```

Stop and hold Alive Counter at the current count.

Reimplemented in ttk_bus.bus_signals_ethbs.AcEthSignal.

7.25.4 Member Data Documentation

7.25.4.1 ac_offset_var

```
ttk_bus._bus_signals_ethbs.AcEthSignal.ac_offset_var
```

Counter Offset variable.

Permits to add an offset to the value of the alive counter at run time.

7.25.4.2 ac_runtime_behavior_var

```
ttk_bus._bus_signals_ethbs.AcEthSignal.ac_runtime_behavior_var
```

Counter Runtime Behavior variable (a HilVar instance)

- 0: "The alive counter runs independently of the selected source for the signal (see switch_var)"
- 1: counter running
  "The alive counter runs only if its value is really sent (Switch is 2 or 12). If it is switched to another source than the alive counter, for example, a value from the Simulink model, the alive counter stops."
- 2: counter stopped (at current value)
  "The alive counter stops at the current value."

7.25.4.3 ac_switch_var

```
ttk_bus._bus_signals_ethbs.AcEthSignal.ac_switch_var
```

ac_switch_var is only defined for backwards-compatibility to CANMMBS variant Runtime Behavior entry is used as AC enable/disable switch

7.25.4.4 error_delay

```
ttk_bus._bus_signals_ethbs.AcEthSignal.error_delay
```

delay [ms] to wait until a basic Alive Counter error (like "AC stopped") should be detected.

This will be used in test functions.

7.25.4.5 fallback_error_delay

```
int ttk_bus._bus_signals_ethbs.AcEthSignal.fallback_error_delay = 20  [static]
```

fallback error delay in [ms] to use for signals/events without any specified or derived error_delay

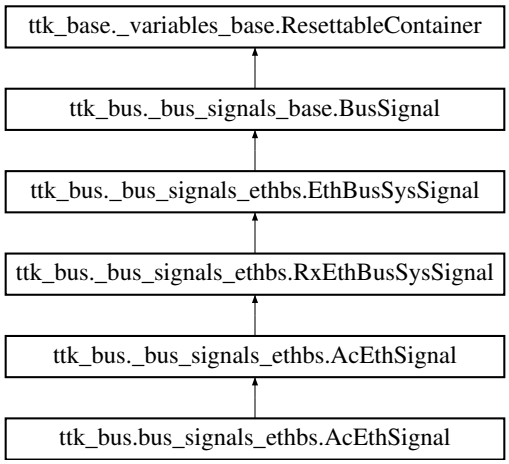
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_ethbs.py (v2.5)

7.26 ttk_bus.bus_signals_ethbs.AcEthSignal Class Reference

An alive counter Rx Ethernet signal/parameter (HIL PSI => ECU CSI) for use with dSPACE Ethernet blockset.

Inheritance diagram for ttk_bus.bus_signals_ethbs.AcEthSignal:



Public Member Functions

- __init__ (self, context, base_path, event=None, cycle_time=None, debounce_time=None, alias="", descr="", error_delay=None, change_delay=None, timeout=None, recovery_time=None, **kwargs)

  Alive counter signal initialization for dSPACE Ethernet blockset.
- setBasicAcError (self)

  Stop and hold Alive Counter at the current count.
- clearBasicAcError (self)

  (Re-)enable normal Alive Counter behaviour (continue counting).
- enableHilTx (self)

Enable transmission of signal HIL => DUT (if the assigned EthEvent supports it).

- disableHilTx (self)

  Disable transmission of signal HIL => DUT (if the assigned EthEvent supports it).
- kickout (self)

  Trigger sending of the Ethernet Event / Field.
- get (self)

  Get the current value of the HIL signal variable (typically not that useful for ECU-Rx/HIL-Tx AC-signals, though).
- set (self, value)

  Set a value to the HIL signal variable.
- getState (self)

  Get a textual state for the HIL signal variable's current data.
- setState (self, state)

  Set a textual state to the HIL signal variable.
- getStateDescr (self, value=None, fallback=None)

  Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).
- info (self)

  Get a current info string about the HIL signal variable.
- reset (self, ∗args, ∗∗kwargs)

  Reset all resettable variables of this bus signal.

Public Attributes

- error_delay

  delay [ms] to wait until a basic Alive Counter error (like "AC stopped") should be detected.
- ac_runtime_behavior_var

  Counter Runtime Behavior variable (a HilVar instance)
- ac_offset_var

  Counter Offset variable.
- ac_switch_var

  ac_switch_var is only defined for backwards-compatibility to CANMMBS variant Runtime Behavior entry is used as AC enable/disable switch
- context

  parent context (an RTE application instance to resolve signal/variable paths)
- signal_path

  base path to signal below BusSystems
- event

  reference to associated EthEvent instance
- unit

  unit of physical signal value (as string, mainly used for report entries)
- alias

  bus signal alias name
- descr

  signal description text
- lookup

  lookup-table for mapping discrete signal states to textual descriptions
- cycle_time

  cycle time of signal in [ms]
- debounce_time

  minimum debounce delay for triggered signals in [ms]
- change_delay

delay to wait after a value change [ms]

- timeout

  detection time for timeout errors [ms]

- recovery_time

  time for recovery/"wiedergut" checks [ms]

7.26.1 Detailed Description

An alive counter Rx Ethernet signal/parameter (HIL PSI => ECU CSI) for use with dSPACE Ethernet blockset.

Note

This class expects a BusSystems path structure.

7.26.2 Constructor & Destructor Documentation

7.26.2.1 __init__()

```
ttk_bus.bus_signals_ethbs.AcEthSignal.__init__ (
              self,
              context,
              base_path,
              event = None,
              cycle_time = None,
              debounce_time = None,
              alias = "",
              descr = "",
              error_delay = None,
              change_delay = None,
              timeout = None,
              recovery_time = None,
             ** kwargs )
```

Alive counter signal initialization for dSPACE Ethernet blockset.

Example: Alive Counter Signal Path Structure:

```
BusSystems/Ethernet/FRR/PSIs/ACME.ENVIRONMENTALMODEL/ObjectFusionFront_1/Events/ObjectPosition/E2E
Protection/counter/Counter/Calculated Value
BusSystems/Ethernet/FRR/PSIs/ACME.ENVIRONMENTALMODEL/ObjectFusionFront_1/Events/ObjectPosition/E2E
Protection/counter/Counter/Enable
BusSystems/Ethernet/FRR/PSIs/ACME.ENVIRONMENTALMODEL/ObjectFusionFront_1/Events/ObjectPosition/E2E
Protection/counter/Counter/Offset
BusSystems/Ethernet/FRR/PSIs/ACME.ENVIRONMENTALMODEL/ObjectFusionFront_1/Events/ObjectPosition/E2E
Protection/counter/Counter/Runtime Behavior
BusSystems/Ethernet/FRR/PSIs/ACME.ENVIRONMENTALMODEL/ObjectFusionFront_1/Events/ObjectPosition/E2E
Protection/counter/Source Switch
BusSystems/Ethernet/FRR/PSIs/ACME.ENVIRONMENTALMODEL/ObjectFusionFront_1/Events/ObjectPosition/E2E
Protection/counter/Value
# or
BusSystems/Ethernet/FRR/PSIs/ACME.DASS/RadarData_1/Events/LaneInfo/E2E
Protection/counter/Counter/Calculated Value
BusSystems/Ethernet/FRR/PSIs/ACME.DASS/RadarData_1/Events/LaneInfo/E2E
Protection/counter/Counter/Enable
BusSystems/Ethernet/FRR/PSIs/ACME.DASS/RadarData_1/Events/LaneInfo/E2E
Protection/counter/Counter/Offset
BusSystems/Ethernet/FRR/PSIs/ACME.DASS/RadarData_1/Events/LaneInfo/E2E
Protection/counter/Counter/Runtime Behavior
BusSystems/Ethernet/FRR/PSIs/ACME.DASS/RadarData_1/Events/LaneInfo/E2E Protection/counter/Source Switch
BusSystems/Ethernet/FRR/PSIs/ACME.DASS/RadarData_1/Events/LaneInfo/E2E Protection/counter/Tx
Inspect/Available SL Value
BusSystems/Ethernet/FRR/PSIs/ACME.DASS/RadarData_1/Events/LaneInfo/E2E Protection/counter/Tx Inspect/Tx
Value
BusSystems/Ethernet/FRR/PSIs/ACME.DASS/RadarData_1/Events/LaneInfo/E2E Protection/counter/Value
```

Suitable base_paths / examples:

BusSystems/Ethernet/FRR/PSIs/ACME.DASS/RadarData/Events/LaneInfo/E2E Protection/counter/↩ Counter/Enable BusSystems/Ethernet/FRR/PSIs/ACME.DASS/RadarData/Events/LaneInfo/E2E Protection/counter/↩ Value => anything below .../Counter/* or the .../Value entry

Parameters

context	rtplib Application or XilTestbench instance
base_path	base model path (below BusSystems) from which all required paths will be derived.
event	reference to associated EthEvent instance
cycle_time	(opt) cycle time in [ms], overrides settings in EthEvent
debounce_time	(opt) minimum debounce delay time between events in [ms], overrides settings in EthEvent
alias	(opt) bus signal alias/name
descr	(opt) signal description
error_delay	(opt) delay to wait until a simple alive counter error should be detected [ms], defaults to 10x cycle time
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x cycle_time
timeout	(opt) detection time for timeout errors [ms], defaults to 2x cycle_time
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x cycle_time
***kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_bus._bus_signals_ethbs.AcEthSignal.

7.26.3 Member Function Documentation

7.26.3.1 clearBasicAcError()

```
ttk_bus.bus_signals_ethbs.AcEthSignal.clearBasicAcError (
              self )
```

(Re-)enable normal Alive Counter behaviour (continue counting).

Reimplemented from ttk_bus._bus_signals_ethbs.AcEthSignal.

7.26.3.2 disableHilTx()

```
ttk_bus.bus_signals_ethbs.AcEthSignal.disableHilTx (
              self )
```

Disable transmission of signal HIL => DUT (if the assigned EthEvent supports it).

Note

This disables transmission of all signals in the parent Event.

Reimplemented from ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.

7.26.3.3 enableHilTx()

```
ttk_bus.bus_signals_ethbs.AcEthSignal.enableHilTx (
              self )
```

Enable transmission of signal HIL => DUT (if the assigned EthEvent supports it).

Note

This enables transmission of all signals in the parent Event.

Reimplemented from ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.

7.26.3.4 get()

```
ttk_bus.bus_signals_ethbs.AcEthSignal.get (
              self )
```

Get the current value of the HIL signal variable (typically not that useful for ECU-Rx/HIL-Tx AC-signals, though).

Returns

the current value from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.26.3.5 getState()

```
ttk_bus.bus_signals_ethbs.AcEthSignal.getState (
              self )
```

Get a textual state for the HIL signal variable's current data.

(typically not that useful for ECU-Rx/HIL-Tx AC-signals, though).

Returns

the current state's description.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.26.3.6 getStateDescr()

```
ttk_bus.bus_signals_ethbs.AcEthSignal.getStateDescr (
              self,
              value = None,
              fallback = None )
```

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the last read value will be used instead.
fallback	fallback value to return if no match was found. Default: None

Returns

the state text if a match was found, otherwise the specified fallback value.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.26.3.7 info()

```
ttk_bus.bus_signals_ethbs.AcEthSignal.info (
              self )
```

Get a current info string about the HIL signal variable.

Returns

an info string.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.26.3.8 kickout()

```
ttk_bus.bus_signals_ethbs.AcEthSignal.kickout (
              self )
```

Trigger sending of the Ethernet Event / Field.

Note

All other signals of the parent EthEvent also will be sent.

Reimplemented from ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.

7.26.3.9 reset()

```
ttk_bus.bus_signals_ethbs.AcEthSignal.reset (
              self,
             * args,
             ** kwargs )
```

Reset all resettable variables of this bus signal.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.ResettableContainer.

7.26.3.10 set()

```
ttk_bus.bus_signals_ethbs.AcEthSignal.set (
              self,
              value )
```

Set a value to the HIL signal variable.

Parameters

value	value to be set

Returns

the original value before the set operation occurred.

Reimplemented from ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.

7.26.3.11 setBasicAcError()

```
ttk_bus.bus_signals_ethbs.AcEthSignal.setBasicAcError (
              self )
```

Stop and hold Alive Counter at the current count.

Reimplemented from ttk_bus._bus_signals_ethbs.AcEthSignal.

7.26.3.12 setState()

```
ttk_bus.bus_signals_ethbs.AcEthSignal.setState (
              self,
              state )
```

Set a textual state to the HIL signal variable.

This uses the lookup values supplied in constructor.

Parameters

state	named state to set

Returns

the previous state's description.

Reimplemented from ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.

7.26.4 Member Data Documentation

7.26.4.1 ac_offset_var

`ttk_bus.bus_signals_ethbs.AcEthSignal.ac_offset_var`

Counter Offset variable.

Permits to add an offset to the value of the alive counter at run time.

7.26.4.2 ac_runtime_behavior_var

`ttk_bus.bus_signals_ethbs.AcEthSignal.ac_runtime_behavior_var`

Counter Runtime Behavior variable (a HilVar instance)

- 0: "The alive counter runs independently of the selected source for the signal (see switch_var)"
- 1: counter running
  "The alive counter runs only if its value is really sent (Switch is 2 or 12). If it is switched to another source than the alive counter, for example, a value from the Simulink model, the alive counter stops."
- 2: counter stopped (at current value)
  "The alive counter stops at the current value."

7.26.4.3 ac_switch_var

`ttk_bus.bus_signals_ethbs.AcEthSignal.ac_switch_var`

ac_switch_var is only defined for backwards-compatibility to CANMMBS variant Runtime Behavior entry is used as AC enable/disable switch

7.26.4.4 alias

`ttk_bus.bus_signals_ethbs.AcEthSignal.alias`

bus signal alias name

7.26.4.5 change_delay

`ttk_bus.bus_signals_ethbs.AcEthSignal.change_delay`

delay to wait after a value change [ms]

7.26.4.6 context

ttk_bus.bus_signals_ethbs.AcEthSignal.context

parent context (an RTE application instance to resolve signal/variable paths)

7.26.4.7 cycle_time

ttk_bus.bus_signals_ethbs.AcEthSignal.cycle_time

cycle time of signal in [ms]

7.26.4.8 debounce_time

ttk_bus.bus_signals_ethbs.AcEthSignal.debounce_time

minimum debounce delay for triggered signals in [ms]

7.26.4.9 descr

ttk_bus.bus_signals_ethbs.AcEthSignal.descr

signal description text

7.26.4.10 error_delay

ttk_bus.bus_signals_ethbs.AcEthSignal.error_delay

delay [ms] to wait until a basic Alive Counter error (like "AC stopped") should be detected.

This will be used in test functions.

7.26.4.11 event

ttk_bus.bus_signals_ethbs.AcEthSignal.event

reference to associated EthEvent instance

7.26.4.12 lookup

ttk_bus.bus_signals_ethbs.AcEthSignal.lookup

lookup-table for mapping discrete signal states to textual descriptions

7.26.4.13 recovery_time

`ttk_bus.bus_signals_ethbs.AcEthSignal.recovery_time`

time for recovery/"wiedergut" checks [ms]

7.26.4.14 signal_path

`ttk_bus.bus_signals_ethbs.AcEthSignal.signal_path`

base path to signal below BusSystems

7.26.4.15 timeout

`ttk_bus.bus_signals_ethbs.AcEthSignal.timeout`

detection time for timeout errors [ms]

7.26.4.16 unit

`ttk_bus.bus_signals_ethbs.AcEthSignal.unit`

unit of physical signal value (as string, mainly used for report entries)

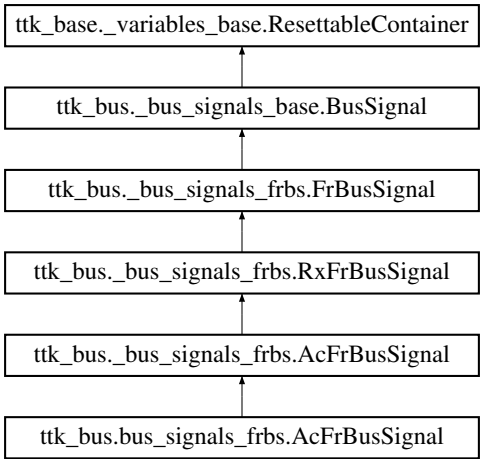
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_ethbs.py (v2.5)

7.27 ttk_bus._bus_signals_frbs.AcFrBusSignal Class Reference

An alive counter RX bus signal derived from RxFrBusSignal (HIL => ECU) for use with dSPACE FlexRay blockset PDU-based data structures.

Inheritance diagram for ttk_bus._bus_signals_frbs.AcFrBusSignal:



Public Member Functions

- __init__ (self, context, base_model_path, pdu=None, cycle_time=None, debounce_time=None, alias="", descr="", error_delay=None, change_delay=None, timeout=None, recovery_time=None)

  Alive counter bus signal initialization for dSPACE FlexRay blockset.
- setBasicAcError (self)

  Stop and hold Alive Counter at the current count.
- clearBasicAcError (self)

  (Re-)enable normal Alive Counter behaviour (continue counting).

Public Attributes

- switch_var

  Source Switch variable for static counter value.
- error_delay

  delay [ms] to wait until a basic Alive Counter error (like "AC stopped") should be detected.
- ac_runtime_behavior_var

  Counter Runtime Behavior variable (a "HilVar" instance).
- ac_offset_var

  Counter Offset variable: offset to current counter value (a "HilVar" instance).
- ac_countdown_value_var

  Countdown Value: countdown timer for dynamic manipulation (a "HilVar" instance).
- ac_switch_var

  ac_switch_var only defined for backwards-compatibility to CANMMBS variant.

Static Public Attributes

- dict cfg_custom_defaults

  Configuration settings for new instances: Custom "hard" default value(s) to use for internal real time system variables.

7.27.1 Detailed Description

An alive counter RX bus signal derived from RxFrBusSignal (HIL => ECU) for use with dSPACE FlexRay blockset PDU-based data structures.

See RxFrBusSignal for inherited methods.

7.27.2 Constructor & Destructor Documentation

7.27.2.1 __init__()

```
ttk_bus._bus_signals_frbs.AcFrBusSignal.__init__ (
              self,
              context,
              base_model_path,
              pdu = None,
              cycle_time = None,
              debounce_time = None,
              alias = "",
              descr = "",
              error_delay = None,
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Alive counter bus signal initialization for dSPACE FlexRay blockset.

Example: Alive Counter Signal Path Structure:

```
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/ALIV_PDU_NAME_OUT/Value
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/ALIV_PDU_NAME_OUT/Dynamic Value
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/ALIV_PDU_NAME_OUT/Validity
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/ALIV_PDU_NAME_OUT/Dynamic Validity
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/ALIV_PDU_NAME_OUT/Countdown Value
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/ALIV_PDU_NAME_OUT/Source Switch
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/ALIV_PDU_NAME_OUT/AliveCounter/Value   #
Note: read-only
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/ALIV_PDU_NAME_OUT/AliveCounter/Runtime
Behavior
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/ALIV_PDU_NAME_OUT/AliveCounter/Offset

# base_model_path == "Value" path:
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/ALIV_PDU_NAME_OUT/Value
```

Parameters

context	rtplib application instance
base_model_path	base model path from which all required paths will be derived, use /Value of counter signal.
pdu	reference to associated FrPdu instance
cycle_time	(opt) cycle time in [ms], overrides settings in PDU
debounce_time	(opt) minimum debounce delay time in [ms], overrides settings in PDU
alias	(opt) bus signal alias/name
descr	(opt) signal description
error_delay	(opt) delay to wait until a simple alive counter error should be detected [ms], defaults to 10x max(cycle_time, debounce_time)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_frbs.RxFrBusSignal.

Reimplemented in ttk_bus.bus_signals_frbs.AcFrBusSignal.

7.27.3 Member Function Documentation

7.27.3.1 clearBasicAcError()

```
ttk_bus._bus_signals_frbs.AcFrBusSignal.clearBasicAcError (
              self )
```

(Re-)enable normal Alive Counter behaviour (continue counting).

Reimplemented in ttk_bus.bus_signals_frbs.AcFrBusSignal.

7.27.3.2 setBasicAcError()

```
ttk_bus._bus_signals_frbs.AcFrBusSignal.setBasicAcError (
              self )
```

Stop and hold Alive Counter at the current count.

Reimplemented in ttk_bus.bus_signals_frbs.AcFrBusSignal.

7.27.4 Member Data Documentation

7.27.4.1 ac_countdown_value_var

`ttk_bus._bus_signals_frbs.AcFrBusSignal.ac_countdown_value_var`

Countdown Value: countdown timer for dynamic manipulation (a "HilVar" instance).

> "Time span for dynamic signal manipulation. It specifies how often the dynamic values are sent."

7.27.4.2 ac_offset_var

`ttk_bus._bus_signals_frbs.AcFrBusSignal.ac_offset_var`

Counter Offset variable: offset to current counter value (a "HilVar" instance).

> "Allows you to add an offset to the value of the alive counter at run time."

7.27.4.3 ac_runtime_behavior_var

`ttk_bus._bus_signals_frbs.AcFrBusSignal.ac_runtime_behavior_var`

Counter Runtime Behavior variable (a "HilVar" instance).

- 0: "The alive counter runs independently of the selected source for the signal (Source Switch variable)"
- 1: counter running
  "The alive counter runs only if its value is really sent (Switch is 2 or 12). If it is switched to another source than the alive counter, for example, a value from the Simulink model, the alive counter stops."
- 2: counter stopped (at current value)
  "The alive counter stops at the current value."

7.27.4.4 ac_switch_var

`ttk_bus._bus_signals_frbs.AcFrBusSignal.ac_switch_var`

ac_switch_var only defined for backwards-compatibility to CANMMBS variant.

Runtime Behavior entry is used as AC enable/disable switch

7.27.4.5 cfg_custom_defaults

```
dict ttk_bus._bus_signals_frbs.AcFrBusSignal.cfg_custom_defaults  [static]
```

Initial value:

```
=  {
        "ac_runtime_behavior_var":  None,
        "ac_offset_var":            None,
        "ac_countdown_value_var":   None,
        "switch_var":               None, # default for "value" source switch (see note above)
    }
```

Configuration settings for new instances: Custom "hard" default value(s) to use for internal real time system variables.

See RxFrBusSignal for further details.

Mapping: <variable name as string>: <default value>

Typical default values:

- source switches for alive counters usually default to "running counter"
  - 0: static value from Simulink Model
  - 1: static value from BusSystems
  - 2: running counter from BusSystems (usually the default)
  - 12: counter is stopped for time span specified by Countdown Value, last value of alive counter is sent. When Countdown Value reaches 0, Source Switch is switched back to last Switch value.
- Counter Runtime Behavior, usually defaults to "running"
  - 0: counter runs independently of the selected source
  - 1: counter running
  - 2: counter stopped (at current value)
- Counter Offset, additive offset defaults to 0
- Countdown Value (countdown timer for dynamic manipulation), default should be 0 but value auto decrements down to 0 anyway.

7.27.4.6 error_delay

```
ttk_bus._bus_signals_frbs.AcFrBusSignal.error_delay
```

delay [ms] to wait until a basic Alive Counter error (like "AC stopped") should be detected.

This will be used in test functions.

7.27.4.7 switch_var

`ttk_bus._bus_signals_frbs.AcFrBusSignal.switch_var`

Source Switch variable for static counter value.

- 0: static value from Simulink Model
- 1: static value from BusSystems
- 2: running counter from BusSystems
- 12: counter is stopped for time span specified by Countdown Value, last value of alive counter is sent. When Countdown Value reaches 0, Source Switch is switched back to the last Switch value.

See dSPACE: FlexRayConfigFeatures.pdf "Using an Alive Counter in the Simulation"

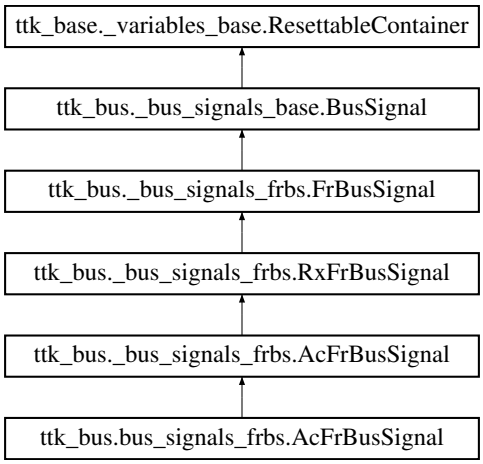
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_frbs.py (v2.5)

7.28 ttk_bus.bus_signals_frbs.AcFrBusSignal Class Reference

An alive counter RX bus signal derived from RxFrBusSignal (HIL => ECU) for use with dSPACE FlexRay blockset PDU-based data structures.

Inheritance diagram for ttk_bus.bus_signals_frbs.AcFrBusSignal:



Public Member Functions

- __init__ (self, context, base_model_path, pdu=None, cycle_time=None, debounce_time=None, alias="", descr="", error_delay=None, change_delay=None, timeout=None, recovery_time=None)

  Alive counter bus signal initialization for dSPACE FlexRay blockset.
- setBasicAcError (self)

  Stop and hold Alive Counter at the current count.
- clearBasicAcError (self)

  (Re-)enable normal Alive Counter behaviour (continue counting).
- enableHilTx (self)

  (Re-)enable sending of PDU, depending on selected disable_mode (see constructor).
- disableHilTx (self)

  Disable sending of PDU, depending on selected disable_mode (see constructor).
- kickout (self)

  Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent PDU).
- get (self)

  Get the current value of the HIL signal variable (typically not that useful for ECU-Rx/HIL-Tx AC-signals, though).
- set (self, value)

  Set a value to the HIL Alive Counter signal variable below BusSystems (and switch the signal source to "BusSystems", static value).
- getState (self)

  Get a textual state for the HIL signal variable's current data (typically not that useful for ECU-Rx/HIL-Tx AC-signals, though).
- setState (self, state)

  Set a textual state to the HIL signal variable.
- getStateDescr (self, value=None, fallback=None)

  Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).
- info (self)

  Get a current info string about the HIL signal variable.
- reset (self, *args, **kwargs)

  Reset all resettable variables of this bus signal.

Public Attributes

- error_delay

  delay [ms] to wait until a basic Alive Counter error (like "AC stopped") should be detected.
- ac_runtime_behavior_var

  Counter Runtime Behavior variable (a "HilVar" instance).
- ac_offset_var

  Counter Offset variable: offset to current counter value (a "HilVar" instance).
- ac_countdown_value_var

  Countdown Value: countdown timer for dynamic manipulation (a "HilVar" instance).
- switch_var

  Source Switch variable for static counter value (a "HilVar" instance).
- value_path

  model path of value variable
- base_path

  base model path (base path for deriving sub-element paths)
- pdu

  reference to parent FrRxPdu
- context

context (reference to assigned RTE application instance)

- signal_var

  main Alive Counter bus signal value variable on HIL-side (a "HilVar" instance)
- alias

  Alive Counter signal alias name (from signal_var)
- unit

  signal unit (from signal_var; usually empty for Alive Counters)
- descr

  Alive Counter signal description text.
- cycle_time

  cycle time of signal in [ms] (that is, of the associated message or PDU)
- debounce_time

  minimum debounce delay for event-triggered signals in [ms]
- change_delay

  delay to wait after a value change [ms]
- timeout

  detection time for timeout errors [ms]
- recovery_time

  time for recovery/"wiedergut" checks [ms]

Static Public Attributes

- dict cfg_custom_defaults

  Configuration settings for new instances: Custom "hard" default value(s) to use for internal real time system variables.

7.28.1 Detailed Description

An alive counter RX bus signal derived from RxFrBusSignal (HIL => ECU) for use with dSPACE FlexRay blockset PDU-based data structures.

See RxFrBusSignal for inherited methods.

7.28.2 Constructor & Destructor Documentation

7.28.2.1 __init__()

```
ttk_bus.bus_signals_frbs.AcFrBusSignal.__init__ (
              self,
              context,
              base_model_path,
              pdu = None,
              cycle_time = None,
              debounce_time = None,
              alias = "",
              descr = "",
              error_delay = None,
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Alive counter bus signal initialization for dSPACE FlexRay blockset.

Example: Alive Counter Signal Path Structure:

```
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/ALIV_PDU_NAME_OUT/Value
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/ALIV_PDU_NAME_OUT/Dynamic Value
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/ALIV_PDU_NAME_OUT/Validity
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/ALIV_PDU_NAME_OUT/Dynamic Validity
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/ALIV_PDU_NAME_OUT/Countdown Value
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/ALIV_PDU_NAME_OUT/Source Switch
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/ALIV_PDU_NAME_OUT/AliveCounter/Value   #
Note: read-only
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/ALIV_PDU_NAME_OUT/AliveCounter/Runtime
Behavior
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/ALIV_PDU_NAME_OUT/AliveCounter/Offset

# base_model_path == "Value" path:
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/ALIV_PDU_NAME_OUT/Value
```

Parameters

context	rtplib application instance
base_model_path	base model path from which all required paths will be derived, use /Value of counter signal.
pdu	reference to associated FrPdu instance
cycle_time	(opt) cycle time in [ms], overrides settings in PDU
debounce_time	(opt) minimum debounce delay time in [ms], overrides settings in PDU
alias	(opt) bus signal alias/name
descr	(opt) signal description
error_delay	(opt) delay to wait until a simple alive counter error should be detected [ms], defaults to 10x max(cycle_time, debounce_time)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_frbs.AcFrBusSignal.

7.28.3 Member Function Documentation

7.28.3.1 clearBasicAcError()

```
ttk_bus.bus_signals_frbs.AcFrBusSignal.clearBasicAcError (
              self )
```

(Re-)enable normal Alive Counter behaviour (continue counting).

Reimplemented from ttk_bus._bus_signals_frbs.AcFrBusSignal.

7.28.3.2 disableHilTx()

```
ttk_bus.bus_signals_frbs.AcFrBusSignal.disableHilTx (
              self )
```

Disable sending of PDU, depending on selected disable_mode (see constructor).

This overrides current settings from Simulink Model, use reset() to restore initial behavior.

Note

This disables transmission of all signals of the parent PDU.

Reimplemented from ttk_bus._bus_signals_frbs.RxFrBusSignal.

7.28.3.3 enableHilTx()

```
ttk_bus.bus_signals_frbs.AcFrBusSignal.enableHilTx (
              self )
```

(Re-)enable sending of PDU, depending on selected disable_mode (see constructor).

This overrides current settings from Simulink Model, use reset() to restore initial behavior.

Note

This enables transmission of all signals of the parent PDU.

Reimplemented from ttk_bus._bus_signals_frbs.RxFrBusSignal.

7.28.3.4 get()

```
ttk_bus.bus_signals_frbs.AcFrBusSignal.get (
              self )
```

Get the current value of the HIL signal variable (typically not that useful for ECU-Rx/HIL-Tx AC-signals, though).

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.28.3.5 getState()

```
ttk_bus.bus_signals_frbs.AcFrBusSignal.getState (
              self )
```

Get a textual state for the HIL signal variable's current data (typically not that useful for ECU-Rx/HIL-Tx AC-signals, though).

Returns

the current state's description from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.28.3.6 getStateDescr()

```
ttk_bus.bus_signals_frbs.AcFrBusSignal.getStateDescr (
              self,
              value = None,
              fallback = None )
```

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the last read value will be used instead.
fallback	fallback value to return if no match was found. Default: None

Returns

the state text if a match was found, otherwise the specified fallback value.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.28.3.7 info()

```
ttk_bus.bus_signals_frbs.AcFrBusSignal.info (
              self )
```

Get a current info string about the HIL signal variable.

Returns

a formatted info string.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.28.3.8 kickout()

```
ttk_bus.bus_signals_frbs.AcFrBusSignal.kickout (
              self )
```

Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent PDU).

Note

This means all other signals of the parent PDU will be sent as well.

Reimplemented from ttk_bus._bus_signals_frbs.RxFrBusSignal.

7.28.3.9 reset()

```
ttk_bus.bus_signals_frbs.AcFrBusSignal.reset (
              self,
             * args,
             ** kwargs )
```

Reset all resettable variables of this bus signal.

Note

This will also switch the Source Switch back to its default state (which is usually `2: running counter from BusSystems`).

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.ResettableContainer.

7.28.3.10 set()

```
ttk_bus.bus_signals_frbs.AcFrBusSignal.set (
              self,
              value )
```

Set a value to the HIL Alive Counter signal variable below BusSystems (and switch the signal source to "Bus↩ Systems", static value).

If a value is set to an Alive Counter signal, the Alive Counter value will stay at this static value until reset() gets called. See possible AcFrBusSignal.switch_var values for Alive Counter signals.

Parameters

value	static value to set

Returns

the previous value from signal_var.

Reimplemented from ttk_bus._bus_signals_frbs.RxFrBusSignal.

7.28.3.11 setBasicAcError()

```
ttk_bus.bus_signals_frbs.AcFrBusSignal.setBasicAcError (
              self )
```

Stop and hold Alive Counter at the current count.

Reimplemented from ttk_bus._bus_signals_frbs.AcFrBusSignal.

7.28.3.12 setState()

```
ttk_bus.bus_signals_frbs.AcFrBusSignal.setState (
              self,
              state )
```

Set a textual state to the HIL signal variable.

This uses the lookup values supplied in constructor.

Parameters

state	named state to set

Returns

the previous state's description.

Reimplemented from ttk_bus._bus_signals_frbs.RxFrBusSignal.

7.28.4 Member Data Documentation

7.28.4.1 ac_countdown_value_var

`ttk_bus.bus_signals_frbs.AcFrBusSignal.ac_countdown_value_var`

Countdown Value: countdown timer for dynamic manipulation (a "HilVar" instance).

"Time span for dynamic signal manipulation. It specifies how often the dynamic values are sent."

7.28.4.2 ac_offset_var

`ttk_bus.bus_signals_frbs.AcFrBusSignal.ac_offset_var`

Counter Offset variable: offset to current counter value (a "HilVar" instance).

"Allows you to add an offset to the value of the alive counter at run time."

7.28.4.3 ac_runtime_behavior_var

`ttk_bus.bus_signals_frbs.AcFrBusSignal.ac_runtime_behavior_var`

Counter Runtime Behavior variable (a "HilVar" instance).

- 0: "The alive counter runs independently of the selected source for the signal (Source Switch variable)"
- 1: counter running
  "The alive counter runs only if its value is really sent (Switch is 2 or 12). If it is switched to another source than the alive counter, for example, a value from the Simulink model, the alive counter stops."
- 2: counter stopped (at current value)
  "The alive counter stops at the current value."

7.28.4.4 alias

`ttk_bus.bus_signals_frbs.AcFrBusSignal.alias`

Alive Counter signal alias name (from signal_var)

7.28.4.5 base_path

```
ttk_bus.bus_signals_frbs.AcFrBusSignal.base_path
```

base model path (base path for deriving sub-element paths)

7.28.4.6 cfg_custom_defaults

```
dict ttk_bus.bus_signals_frbs.AcFrBusSignal.cfg_custom_defaults  [static]
```

Initial value:

```
=  {
        "ac_runtime_behavior_var":  None,
        "ac_offset_var":            None,
        "ac_countdown_value_var":   None,
        "switch_var":               None, # default for "value" source switch (see note above)
    }
```

Configuration settings for new instances: Custom "hard" default value(s) to use for internal real time system variables.

See RxFrBusSignal for further details.

Mapping: <variable name as string>: <default value>

Typical default values:

- source switches for alive counters usually default to "running counter"
  - 0: static value from Simulink Model
  - 1: static value from BusSystems
  - 2: running counter from BusSystems (usually the default)
  - 12: counter is stopped for time span specified by Countdown Value, last value of alive counter is sent. When Countdown Value reaches 0, Source Switch is switched back to last Switch value.
- Counter Runtime Behavior, usually defaults to "running"
  - 0: counter runs independently of the selected source
  - 1: counter running
  - 2: counter stopped (at current value)
- Counter Offset, additive offset defaults to 0
- Countdown Value (countdown timer for dynamic manipulation), default should be 0 but value auto decrements down to 0 anyway.

7.28.4.7 change_delay

```
ttk_bus.bus_signals_frbs.AcFrBusSignal.change_delay
```

delay to wait after a value change [ms]

7.28.4.8 context

ttk_bus.bus_signals_frbs.AcFrBusSignal.context

context (reference to assigned RTE application instance)

7.28.4.9 cycle_time

ttk_bus.bus_signals_frbs.AcFrBusSignal.cycle_time

cycle time of signal in [ms] (that is, of the associated message or PDU)

7.28.4.10 debounce_time

ttk_bus.bus_signals_frbs.AcFrBusSignal.debounce_time

minimum debounce delay for event-triggered signals in [ms]

7.28.4.11 descr

ttk_bus.bus_signals_frbs.AcFrBusSignal.descr

Alive Counter signal description text.

7.28.4.12 error_delay

ttk_bus.bus_signals_frbs.AcFrBusSignal.error_delay

delay [ms] to wait until a basic Alive Counter error (like "AC stopped") should be detected.

This will be used in test functions.

7.28.4.13 pdu

ttk_bus.bus_signals_frbs.AcFrBusSignal.pdu

reference to parent FrRxPdu

7.28.4.14 recovery_time

ttk_bus.bus_signals_frbs.AcFrBusSignal.recovery_time

time for recovery/"wiedergut" checks [ms]

7.28.4.15 signal_var

`ttk_bus.bus_signals_frbs.AcFrBusSignal.signal_var`

main Alive Counter bus signal value variable on HIL-side (a "HilVar" instance)

7.28.4.16 switch_var

`ttk_bus.bus_signals_frbs.AcFrBusSignal.switch_var`

Source Switch variable for static counter value (a "HilVar" instance).

- 0: static value from Simulink Model
- 1: static value from BusSystems
- 2: running counter from BusSystems
- 12: counter is stopped for time span specified by Countdown Value, last value of alive counter is sent. When Countdown Value reaches 0, Source Switch is switched back to the last Switch value.

See dSPACE: FlexRayConfigFeatures.pdf "Using an Alive Counter in the Simulation"

7.28.4.17 timeout

`ttk_bus.bus_signals_frbs.AcFrBusSignal.timeout`

detection time for timeout errors [ms]

7.28.4.18 unit

`ttk_bus.bus_signals_frbs.AcFrBusSignal.unit`

signal unit (from signal_var; usually empty for Alive Counters)

7.28.4.19 value_path

`ttk_bus.bus_signals_frbs.AcFrBusSignal.value_path`

model path of value variable

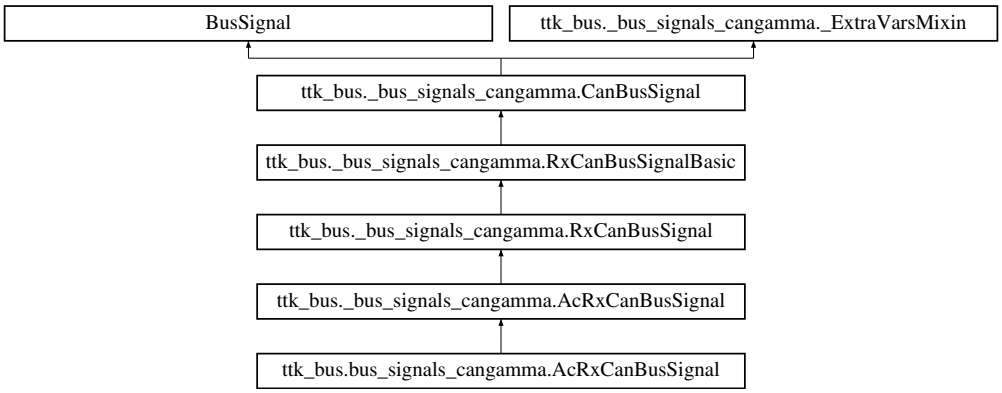
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_frbs.py (v1.8)

7.29 ttk_bus._bus_signals_cangamma.AcRxCanBusSignal Class Reference

Alive counter RX bus signal (HIL-Tx => ECU-Rx).

Inheritance diagram for ttk_bus._bus_signals_cangamma.AcRxCanBusSignal:



Public Member Functions

- __init__ (self, context, signal_identifier, message=None, cycle_time=None, debounce_time=None, alias="", descr="", lookup=None, error_delay=None, change_delay=None, timeout=None, recovery_time=None)

  Alive counter bus signal (HIL-Tx => ECU-Rx) initialization for Gamma-V based CAN signal manipulation options.
- setBasicAcError (self)

  Stop and hold Alive Counter at the current count.
- clearBasicAcError (self)

  (Re-)enable normal Alive Counter behaviour (i.e.

Public Attributes

- error_delay

  delay to wait after a "basic" AC error injection [ms]
- offset_var

  Variable for additional alive counter offset (offset will simply be added to current counter value)

Static Public Attributes

- dict cfg_custom_defaults

  Configuration settings for new instances:
  Custom "hard" default value(s) to use for internal real time system variables.

7.29.1 Detailed Description

Alive counter RX bus signal (HIL-Tx => ECU-Rx).

7.29.2 Constructor & Destructor Documentation

7.29.2.1 __init__()

```
ttk_bus._bus_signals_cangamma.AcRxCanBusSignal.__init__ (
              self,
              context,
              signal_identifier,
              message = None,
              cycle_time = None,
              debounce_time = None,
              alias = "",
              descr = "",
              lookup = None,
              error_delay = None,
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Alive counter bus signal (HIL-Tx => ECU-Rx) initialization for Gamma-V based CAN signal manipulation options.

Parameters

context	parent gaapi context
signal_identifier	identifier for alive counter signal PV
message	reference to associated RxCanBusMessage instance
cycle_time	(opt) cycle time in [ms], overrides settings in Message
debounce_time	(opt) minimum debounce delay between events in [ms], overrides settings in Message
alias	(opt) bus signal alias/name
descr	(opt) signal description
lookup	(opt) lookup table for values, see VarBase
error_delay	(opt) delay to wait until a simple crc error should be detected [ms], defaults to 10x cycle time
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x cycle_time
timeout	(opt) detection time for timeout errors [ms], defaults to 2x cycle_time
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x cycle_time

Reimplemented from ttk_bus._bus_signals_cangamma.RxCanBusSignal.

Reimplemented in ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.

7.29.3 Member Function Documentation

7.29.3.1 clearBasicAcError()

```
ttk_bus._bus_signals_cangamma.AcRxCanBusSignal.clearBasicAcError (
              self )
```

(Re-)enable normal Alive Counter behaviour (i.e.

continue counting).

Reimplemented in ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.

7.29.3.2 setBasicAcError()

```
ttk_bus._bus_signals_cangamma.AcRxCanBusSignal.setBasicAcError (
              self )
```

Stop and hold Alive Counter at the current count.

Reimplemented in ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.

7.29.4 Member Data Documentation

7.29.4.1 cfg_custom_defaults

```
dict ttk_bus._bus_signals_cangamma.AcRxCanBusSignal.cfg_custom_defaults  [static]
```

Initial value:

```
=  {
        "offset_var":        None, # offset to add to current counter value
        "mode_var":          None, # for parent RxCanBusSignal
        "dyn_count_var":     None, # for parent RxCanBusSignal
        "dyn_value_phy_var": None, # for parent RxCanBusSignal
    }
```

Configuration settings for new instances:
Custom "hard" default value(s) to use for internal real time system variables.

See RxFrBusSignal for further details.

Mapping: <variable name as string>: <default value>

Typical default values:

- mode switches for alive counters usually default to "running counter" See RxFrBusSignal for further details.
- Counter Offset, additive offset to counter value defaults to 0

7.29.4.2 error_delay

```
ttk_bus._bus_signals_cangamma.AcRxCanBusSignal.error_delay
```

delay to wait after a "basic" AC error injection [ms]

7.29.4.3 offset_var

```
ttk_bus._bus_signals_cangamma.AcRxCanBusSignal.offset_var
```

Variable for additional alive counter offset (offset will simply be added to current counter value)

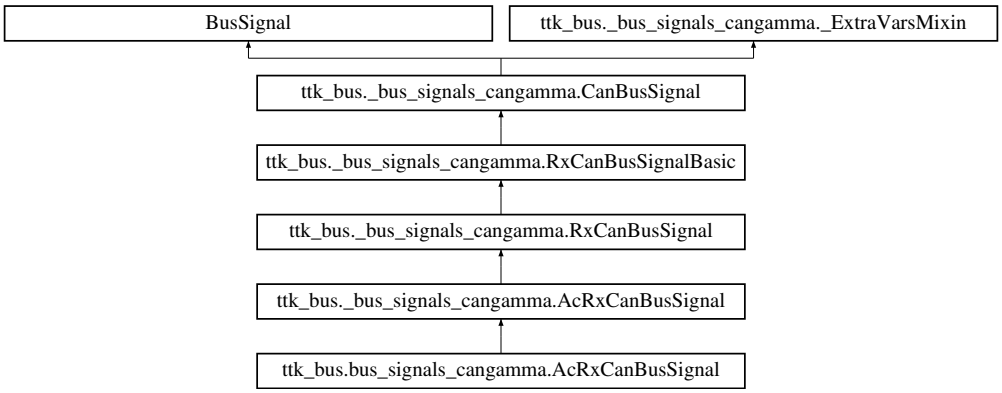
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_cangamma.py (v1.12)

7.30 ttk_bus.bus_signals_cangamma.AcRxCanBusSignal Class Reference

Alive counter RX bus signal (HIL-Tx => ECU-Rx).

Inheritance diagram for ttk_bus.bus_signals_cangamma.AcRxCanBusSignal:



Public Member Functions

- __init__ (self, context, signal_identifier, message=None, cycle_time=None, debounce_time=None, alias="", descr="", lookup=None, error_delay=None, change_delay=None, timeout=None, recovery_time=None)

  Alive counter bus signal (HIL-Tx => ECU-Rx) initialization for Gamma-V based CAN signal manipulation options.
- setBasicAcError (self)

  Stop and hold Alive Counter at the current count.
- clearBasicAcError (self)

  (Re-)enable normal Alive Counter behaviour (i.e.
- enableHilTx (self)

  Enable transmission of signal HIL => DUT.
- disableHilTx (self)

  Disable transmission of signal HIL => DUT.
- kickout (self)

  Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent CAN-Message/PDU).
- get (self)

  Get the current value of the HIL signal variable (typically not that useful for ECU-Rx/HIL-Tx AC-signals, though).
- set (self, value)

  Set a value to the HIL signal variable.
- getState (self)

  Get a textual state for the HIL signal variable's current data (typically not that useful for ECU-Rx/HIL-Tx AC-signals, though).
- setState (self, state)

  Set a textual state to the HIL signal variable.
- getStateDescr (self, value=None, fallback=None)

  Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).
- info (self)

  Get a current info string about the HIL signal variable.
- reset (self, *args, **kwargs)

  Reset all resettable variables of this bus signal.

Public Attributes

- error_delay
  delay to wait after a "simple" alive counter error injection [ms]
- offset_var
  Variable with additional alive counter offset (offset will simply be added to current counter value)
- dyn_count_var
  Dynamic Countdown value.
- dyn_value_var
  Dynamic Value: value active while dyn_count_var decrements (depending on selected mode).
- dyn_pattern_var
  Dynamic Pattern value.
- static_value_phy_var
  Physical static override value (for mode 0x01)
- static_value_raw_var
  Raw static override value (for mode 0x02)
- mode_var
  Mode switch variable for signal manipulation (see RxCanBusSignal for full details)
- base_identifier
  Signal PV address/identifier as base identifier for deriving sub-elements.
- message
  reference to associated (Rx-)CanBusMessage instance
- alias
  bus signal alias name
- descr
  bus signal description text
- cycle_time
  cycle time of Message/PDU in [ms]
- debounce_time
  minimum debounce delay for event-triggered Messages/PDUs in [ms]
- change_delay
  delay to wait after a value change [ms]
- timeout
  detection time for timeout errors [ms]
- recovery_time
  time for recovery/"wiedergut" checks [ms]

Static Public Attributes

- dict cfg_custom_defaults
  Configuration settings for new instances:
  Custom "hard" default value(s) to use for internal real time system variables.

7.30.1 Detailed Description

Alive counter RX bus signal (HIL-Tx => ECU-Rx).

7.30.2 Constructor & Destructor Documentation

7.30.2.1 __init__()

```
ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.__init__ (
               self,
               context,
               signal_identifier,
               message = None,
               cycle_time = None,
               debounce_time = None,
               alias = "",
               descr = "",
               lookup = None,
               error_delay = None,
               change_delay = None,
               timeout = None,
               recovery_time = None )
```

Alive counter bus signal (HIL-Tx => ECU-Rx) initialization for Gamma-V based CAN signal manipulation options.

Parameters

context	parent gaapi context
signal_identifier	identifier for alive counter signal PV
message	reference to associated RxCanBusMessage instance
cycle_time	(opt) cycle time in [ms], overrides settings in Message
debounce_time	(opt) minimum debounce delay between events in [ms], overrides settings in Message
alias	(opt) bus signal alias/name
descr	(opt) signal description
lookup	(opt) lookup table for values, see VarBase
error_delay	(opt) delay to wait until a simple crc error should be detected [ms], defaults to 10x cycle time
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x cycle_time
timeout	(opt) detection time for timeout errors [ms], defaults to 2x cycle_time
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x cycle_time

Reimplemented from ttk_bus._bus_signals_cangamma.AcRxCanBusSignal.

7.30.3 Member Function Documentation

7.30.3.1 clearBasicAcError()

```
ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.clearBasicAcError (
               self )
```

(Re-)enable normal Alive Counter behaviour (i.e.

continue counting).

Reimplemented from ttk_bus._bus_signals_cangamma.AcRxCanBusSignal.

7.30.3.2 disableHilTx()

```
ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.disableHilTx (
              self )
```

Disable transmission of signal HIL => DUT.

Note

This will disable transmission of *all* signals of the parent Message/PDU.

Reimplemented from ttk_bus._bus_signals_cangamma.RxCanBusSignalBasic.

7.30.3.3 enableHilTx()

```
ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.enableHilTx (
              self )
```

Enable transmission of signal HIL => DUT.

Note

This will enable transmission of *all* signals of the parent Message/PDU.

Reimplemented from ttk_bus._bus_signals_cangamma.RxCanBusSignalBasic.

7.30.3.4 get()

```
ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.get (
              self )
```

Get the current value of the HIL signal variable (typically not that useful for ECU-Rx/HIL-Tx AC-signals, though).

Returns

the current value from `signal_var`.

7.30.3.5 getState()

```
ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.getState (
              self )
```

Get a textual state for the HIL signal variable's current data (typically not that useful for ECU-Rx/HIL-Tx AC-signals, though).

Returns

the current state's description.

7.30.3.6 getStateDescr()

```
ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.getStateDescr (
              self,
              value = None,
              fallback = None )
```

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the last read value will be used instead.
fallback	fallback value to return if no match was found. Default: None

Returns

the state text if a match was found, otherwise the specified fallback value.

7.30.3.7 info()

```
ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.info (
              self )
```

Get a current info string about the HIL signal variable.

Returns

an info string.

7.30.3.8 kickout()

```
ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.kickout (
              self )
```

Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent CAN-Message/PDU).

Note

All other signals of the parent Message/PDU will also be sent.

Reimplemented from ttk_bus._bus_signals_cangamma.RxCanBusSignalBasic.

7.30.3.9 reset()

```
ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.reset (
              self,
             * args,
             ** kwargs )
```

Reset all resettable variables of this bus signal.

Parameters

∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

7.30.3.10 set()

```
ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.set (
              self,
              value )
```

Set a value to the HIL signal variable.

Parameters

value	value to set

Returns

the previous value from `signal_var`.

Reimplemented from ttk_bus._bus_signals_cangamma.RxCanBusSignal.

7.30.3.11 setBasicAcError()

```
ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.setBasicAcError (
              self )
```

Stop and hold Alive Counter at the current count.

Reimplemented from ttk_bus._bus_signals_cangamma.AcRxCanBusSignal.

7.30.3.12 setState()

```
ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.setState (
              self,
              state )
```

Set a textual state to the HIL signal variable.

This uses the lookup values supplied in constructor.

Parameters

state	named state to set

Returns

the previous state's description.

Reimplemented from ttk_bus._bus_signals_cangamma.RxCanBusSignal.

7.30.4 Member Data Documentation

7.30.4.1 alias

ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.alias

bus signal alias name

7.30.4.2 base_identifier

ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.base_identifier

Signal PV address/identifier as base identifier for deriving sub-elements.

For CAN/Gamma signals, related variables follow the form <base_identifier>.<suffix>

7.30.4.3 cfg_custom_defaults

dict ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.cfg_custom_defaults [static]

Initial value:

```
=  {
        "offset_var":        None, # offset to add to current counter value
        "mode_var":          None, # for parent RxCanBusSignal
        "dyn_count_var":     None, # for parent RxCanBusSignal
        "dyn_value_phy_var": None, # for parent RxCanBusSignal
    }
```

Configuration settings for new instances:
Custom "hard" default value(s) to use for internal real time system variables.

See RxFrBusSignal for further details.

Mapping: <variable name as string>: <default value>

Typical default values:

- mode switches for alive counters usually default to "running counter" See RxFrBusSignal for further details.
- Counter Offset, additive offset to counter value defaults to 0

7.30.4.4 change_delay

ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.change_delay

delay to wait after a value change [ms]

7.30.4.5 cycle_time

ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.cycle_time

cycle time of Message/PDU in [ms]

7.30.4.6 debounce_time

ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.debounce_time

minimum debounce delay for event-triggered Messages/PDUs in [ms]

7.30.4.7 descr

ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.descr

bus signal description text

7.30.4.8 dyn_count_var

ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.dyn_count_var

Dynamic Countdown value.

7.30.4.9 dyn_pattern_var

ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.dyn_pattern_var

Dynamic Pattern value.

7.30.4.10 dyn_value_var

ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.dyn_value_var

Dynamic Value: value active while dyn_count_var decrements (depending on selected mode).

7.30.4.11 error_delay

ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.error_delay

delay to wait after a "simple" alive counter error injection [ms]

delay to wait after a "basic" AC error injection [ms]

7.30.4.12 message

ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.message

reference to associated (Rx-)CanBusMessage instance

7.30.4.13 mode_var

`ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.mode_var`

Mode switch variable for signal manipulation (see RxCanBusSignal for full details)

For alive-counter signals:

- 0x10: alive-counter, counting
- 0x11: alive-counter, stopped (static value)
- 0x12: alive-counter, stopped while .dyn_count decrements, then continues counting from last count (mode 0x10)
- 0x13: alive-counter, stopped while .dyn_count decrements, continues counting in background (then mode 0x10)
- 0x14: alive-counter, add .offset to value
- 0x15: alive-counter stopped/started according to .dyn_pattern
- 0x16: alive-counter stopped/started according to .dyn_pattern (continues counting while stopped)

7.30.4.14 offset_var

`ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.offset_var`

Variable with additional alive counter offset (offset will simply be added to current counter value)

7.30.4.15 recovery_time

`ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.recovery_time`

time for recovery/"wiedergut" checks [ms]

7.30.4.16 static_value_phy_var

`ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.static_value_phy_var`

Physical static override value (for mode 0x01)

7.30.4.17 static_value_raw_var

`ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.static_value_raw_var`

Raw static override value (for mode 0x02)

7.30.4.18 timeout

`ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.timeout`

detection time for timeout errors [ms]

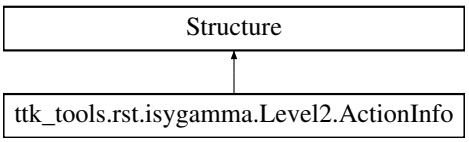
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_cangamma.py (v1.3)

7.31 ttk_tools.rst.isygamma.Level2.ActionInfo Class Reference

typedef struct gc_ActionInfo

Inheritance diagram for ttk_tools.rst.isygamma.Level2.ActionInfo:



7.31.1 Detailed Description

typedef struct gc_ActionInfo

The documentation for this class was generated from the following file:

- ttk_tools/rst/isygamma.py (v1.6)

7.32 ttk_tools.rst.isygamma.Addresses Class Reference

Gamma addresses by nesting level.

Classes

- class _AddrStruct
- class Level2
- class Level3
- class Level4
- class Level5

Static Public Attributes

- Level1 = c_int32

7.32.1 Detailed Description

Gamma addresses by nesting level.

7.32.2 Member Data Documentation

7.32.2.1 Level1

```
ttk_tools.rst.isygamma.Addresses.Level1 = c_int32  [static]
```

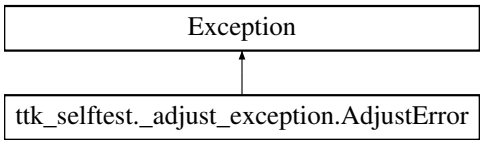
The documentation for this class was generated from the following file:

- ttk_tools/rst/isygamma.py (v1.6)

7.33 ttk_selftest._adjust_exception.AdjustError Class Reference

Base class to raise adjustment Errors.

Inheritance diagram for ttk_selftest._adjust_exception.AdjustError:



Public Member Functions

- __init__ (self, m_call, msg="")

7.33.1 Detailed Description

Base class to raise adjustment Errors.

Exception to use for the special handling during the adjustment calculation.

7.33.2 Constructor & Destructor Documentation

7.33.2.1 __init__()

```
ttk_selftest._adjust_exception.AdjustError.__init__ (
              self,
              m_call,
              msg = "" )
```

Parameters

m_call	Method name which throw the exception.
msg	exception message

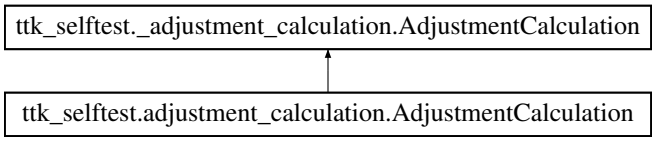
The documentation for this class was generated from the following file:

- ttk_selftest/_adjust_exception.py (v2.0.0)

7.34 ttk_selftest._adjustment_calculation.AdjustmentCalculation Class Reference

Class for adjustment one hil channel which is currently in state self test and adjustment.

Inheritance diagram for ttk_selftest._adjustment_calculation.AdjustmentCalculation:



Public Member Functions

- __init__ (self, hilvar_gain, hilvar_offset, limit_gain=None, limit_offset=None, precision=6)

  Initialize the instance and prepare the self adjustment environment.
- calibData (self, expected, measured, verbosity=True)

  Calibration Method - The method describe the complete hil adjustment calculation:
- getCurrentGain (self)

  Get back the current gain which is specified in running simulation model at current test system.
- getCurrentOffset (self)

  Get back the current offset which is specified in running simulation model at current test system.
- getNewGain (self)

  Get back the calculated gain.
- getNewOffset (self)

  Get back the new offset.
- getCalibVerdict (self)

  Get back the qualification string from the check the new parameter against the specified ranges.
- getCalibTimeStamp (self)

  Get back the time stamp of running calibration.
- getLimitGainMax (self)

  Get back the maximum specified limitation value of gain.
- getLimitGainMin (self)

  Get back the minimum specified limitation value of gain.
- getLimitOffsetMax (self)

  Get back the maximum specified limitation value of offset.
- getLimitOffsetMin (self)

  Get back the minimum specified limitation value of offset.

- getPrecision (self)

  Get back the specified number of decimal places, which is used to round the new gain and offset value before get back.
- getStandardVariance (self)

  Return the standard variance of linear regression back.
- getGainVariance (self)

  Return the gain variance of linear regression back.
- getOffsetVariance (self)

  Return the offset variance of linear regression back.

Protected Member Functions

- _dataCompensation (self)

  Compensate the measured values from self test against the old gain and offset specification currently in use in the simulation model.
- _calculateAvgValue (self, val_list)

  Calculate the average value of a list of values.
- _denominatorValue (self)

  The method will be used in different other methods to calculate the same denominator like definition: sum(x^2) = sum((x(i)-avg(x))^2) / number of counting x: expected values.
- _enumeratorValue (self)

  The method will be used in different other methods to calculate the same enumerator like definition: sum(xy) = sum(x(i)-avg(x))$*$(y(i)-avg(y)) / number of counting x: expected values y: measured values.
- _calculateGainValue (self)

  Calculate the gain value by using the linear regression definition and the measured and expected values.
- _calculateOffsetValue (self)

  Calculate the offset value by using the linear regression definition and the measured and expected values.
- _calculateCalibData (self)

  Calculate the new Gain and Offset values after linear regression is run.
- _checkLimitation (self)

  Check the new gain and offset are in range of specified limitation.
- _calculateStandardVariance (self)

  During linear regression variance will be happen.
- _calculateGainVariance (self)

  During linear regression variance will be happen.
- _calculateOffsetVariance (self)

  During linear regression variance will be happen.

Protected Attributes

- _current_gain

  current gain parameter is used in model
- _current_offset

  current offset parameter is used in model
- _limit_gain_max

  specify the limitation of the calculated new gain is only allowed to have maximum
- _limit_gain_min

  specify the limitation of the calculated new gain is only allowed to have minimum
- _limit_offset_max

  specify the limitation of the calculated new offset is only allowed to have maximum

- _limit_offset_min
  specify the limitation of the calculated new offset is only allowed to have minimum
- _precision
  define the precision of return values to convert in correct format
- _new_gain
  calculated gain parameter
- _new_offset
  calculated offset parameter
- _standard_variance
  calculated standard variance parameter
- _gain_variance
  calculated gain variance parameter
- _offset_variance
  calculated offset variance parameter
- _verdict
  result parameter of limitation check
- _timestamp
  timestamp parameter of calibration was run
- _expect_data
  expected data list parameter for input data
- _measure_data
  measured data list parameter for input data
- _measure_compensate
  measured data list parameter for compensate data
- _len_data_element
  length parameter for input data
- _avg_expect
  average value parameter of expected input data
- _avg_measure
  average value parameter of compensate measured input data
- _lin_reg_gain
  linear regression calculated gain parameter
- _lin_reg_offset
  linear regression calculated offset parameter
- _lin_reg_enumerator
  calculated enumerator value parameter
- _lin_reg_denominator
  calculated denominator value parameter

7.34.1 Detailed Description

Class for adjustment one hil channel which is currently in state self test and adjustment.

7.34.2 Constructor & Destructor Documentation

7.34.2.1 __init__()

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation.__init__ (
                self,
                hilvar_gain,
                hilvar_offset,
                limit_gain = None,
                limit_offset = None,
                precision = 6 )
```

Initialize the instance and prepare the self adjustment environment.

Parameters

hilvar_gain	variable definition with a get() and set() method to read current specified gain for channel is specified to run the adjustment calculation.
hilvar_offset	variable definition with a get() and set() method to read current specified offset for channel is specified to run the adjustment calculation.
limit_gain	(opt) dict with struct {"max":<value>, "min":<value>} Limitation of calculated new gain. In use to check the calculation and in case of incorrect set the calculation to false. • "max": maximum limitation (> min) • "min": minimum limitation (< max) • Type: Type of "Float", "Long" and "Int" acceptable
limit_offset	(opt) dict with struct {"max":<value>, "min":<value>} Limitation of calculated new offset. In use to check the calculation and in case of incorrect set the calculation to false. • "max": maximum limitation (> min) • "min": minimum limitation (< max) • Type: Type of "Float", "Long" and "Int" acceptable
precision	(opt) Specify the number of decimal places the new gain and offset value should round after the calculation. Type: "Integer"

Returns

In case of sanity check is failed throw an Exception. Else return None.

Reimplemented in ttk_selftest.adjustment_calculation.AdjustmentCalculation.

7.34.3 Member Function Documentation

7.34.3.1 _calculateAvgValue()

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation._calculateAvgValue (
                self,
                val_list )  [protected]
```

Calculate the average value of a list of values.

Will be used very often during the linear regression.

Parameters

val_list	list of values to calculate the average e.g.: [0.002, 1.001, 1.999, ...]

Returns

The calculated average value

7.34.3.2 _calculateCalibData()

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation._calculateCalibData (
              self )  [protected]
```

Calculate the new Gain and Offset values after linear regression is run.

Definition: new gain = 1 / linear regression gain new offset = - linear regression offset / linear regression gain

Info:

Method use the parameter

- self._lin_reg_gain
- self._lin_reg_offset Method will overwrite the parameter
- self._new_gain
- self._new_offset

7.34.3.3 _calculateGainValue()

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation._calculateGainValue (
              self )  [protected]
```

Calculate the gain value by using the linear regression definition and the measured and expected values.

Definition: b = sum(xy) / sum(x^2) with:

- sum(xy) = sum(x(i)$*$avg(x))$*$(y(i)$*$avg(y)) / number of counting
- sum(x^2) = sum((x(i)$*$avg(x))^2) / number of counting

x: expected values y: measured values b: calculated gain value

Info:

Method use the parameter

- self._lin_reg_denominator
- self._lin_reg_enumerator Method will overwrite the parameter
- self._lin_reg_gain

7.34.3.4 _calculateGainVariance()

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation._calculateGainVariance (
            self )  [protected]
```

During linear regression variance will be happen.

After the linear regression it is necessary to define the gain variance to have a quality number of linear regression of gain. The value will be calculate with the definition: variance(b) = variance(y) $*$ sqrt((sum(x^2)/sum((x-avg)^2)/number of counting) with: sum(x^2) = x(i)^2 sum((x-avg)^2)= (x(i) - avg(x))^2 x: expected values

Info:

Method use the parameter

- self._expect_data
- self._len_data_element
- self._standard_variance
- self._lin_reg_denominator Method will overwrite the parameter
- self._gain_variance

7.34.3.5 _calculateOffsetValue()

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation._calculateOffsetValue (
            self )  [protected]
```

Calculate the offset value by using the linear regression definition and the measured and expected values.

Definition: a = avg(y) - b $*$ avg(x)

x: expected values y: measured values a: calculated offset value b: calculated gain value

Info:

Method use the parameter

- self._avg_expect
- self._avg_measure
- self._lin_reg_gain Method will overwrite the parameter
- self._lin_reg_offset

7.34.3.6 _calculateOffsetVariance()

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation._calculateOffsetVariance (
            self )  [protected]
```

During linear regression variance will be happen.

After the linear regression it is necessary to define the offset variance to have a quality number of linear regression of offset. The value will be calculate with the definition: variance(a) = variance(y) $*$ sqrt(1/sum((x-avg)^2)) with: sum((x-avg)^2)= (x(i) - avg(x))^2 x: expected values

Info:

Method use the parameter

- self._standard_variance
- self._lin_reg_denominator Method will overwrite the parameter
- self._offset_variance

7.34.3.7 _calculateStandardVariance()

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation._calculateStandardVariance (
              self )  [protected]
```

During linear regression variance will be happen.

After the linear regression it is necessary to define also the standard variance to have a quality number of linear regression. The value will be calculate with the definition: variance(y) = sqrt(sum(xy) / (number of counting - 2)) with: sum(xy) = (y(i) - (a $*$ x(i) - b))^2 a: linear regression offset b: linear regression gain x: expected values y: measured values

Info:

Method use the parameter

- self._expect_data
- self._lin_reg_gain
- self._lin_reg_offset
- self._len_data_element
- self._measure_compensate Method will overwrite the parameter
- self._standard_variance

7.34.3.8 _checkLimitation()

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation._checkLimitation (
              self )  [protected]
```

Check the new gain and offset are in range of specified limitation.

In case one or both are not in range set the verdict to failed.

- Passed: Both values are in range
- Failed: One or both are not in range

  Info:

  Method use the parameter

- self._new_gain
- self._new_offset Method will overwrite the parameter
- self._verdict

7.34.3.9 _dataCompensation()

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation._dataCompensation (
              self )  [protected]
```

Compensate the measured values from self test against the old gain and offset specification currently in use in the simulation model.

Info:

Method use the parameter

- self._measure_data
- self._current_gain
- self._current_offset
- self._len_data_element Method will overwrite the parameter
- self._measure_compensate

7.34.3.10 _denominatorValue()

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation._denominatorValue (
              self )  [protected]
```

The method will be used in different other methods to calculate the same denominator like definition: sum(x^2) = sum((x(i)-avg(x))^2) / number of counting x: expected values.

Info:

Method use the parameter

- self._avg_expect
- self._expect_data Method will overwrite the parameter
- self._lin_reg_denominator

7.34.3.11 _enumeratorValue()

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation._enumeratorValue (
              self )  [protected]
```

The method will be used in different other methods to calculate the same enumerator like definition: sum(xy) = sum(x(i)-avg(x))$*$(y(i)-avg(y)) / number of counting x: expected values y: measured values.

Info:

Method use the parameter

- self._avg_expect
- self._avg_measure
- self._expect_data
- self._len_data_element
- self._measure_compensate Method will overwrite the parameter
- self._lin_reg_enumerator

7.34.3.12 calibData()

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation.calibData (
              self,
              expected,
              measured,
              verbosity = True )
```

Calibration Method - The method describe the complete hil adjustment calculation:

- Check parameter are correct.
- Compensate possible old gain and offset values different to the default in model.
- Calculate the linear regression.
- Calculate the new calibrated data (gain and offset).
- Check the new calibrated data against the limitation.

Parameters

expected	list of expected data
measured	list of measured data verbosity- (opt) state to define the depth of logging False: No print out True: Print the call of methods...

Attention

Expected and measured data should have the same value order.

Returns

None

Reimplemented in ttk_selftest.adjustment_calculation.AdjustmentCalculation.

7.34.3.13 getCalibTimeStamp()

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation.getCalibTimeStamp (
              self )
```

Get back the time stamp of running calibration.

Returns

The timestamp of calibration is run. If None the calibration is not running correctly. Value can be convert like: time.strftime("%Y-%m-%dT%H:%M:%S", time.localtime(getCalibTimeStamp()))

Reimplemented in ttk_selftest.adjustment_calculation.AdjustmentCalculation.

7.34.3.14 getCalibVerdict()

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation.getCalibVerdict (
              self )
```

Get back the qualification string from the check the new parameter against the specified ranges.

Returns

PASSED or FAILED or NOT TESTED

- PASSED: Check of new parameters against defined ranges pass
- FAILED: Check of new parameters against defined ranges fails
- NOTTESTED: Check is not running

Reimplemented in ttk_selftest.adjustment_calculation.AdjustmentCalculation.

7.34.3.15 getCurrentGain()

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation.getCurrentGain (
              self )
```

Get back the current gain which is specified in running simulation model at current test system.

The value will be read out by use the specified variable.

Returns

The current gain read out from used hil variable.

Reimplemented in ttk_selftest.adjustment_calculation.AdjustmentCalculation.

7.34.3.16 getCurrentOffset()

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation.getCurrentOffset (
              self )
```

Get back the current offset which is specified in running simulation model at current test system.

The value will be read out by use the specified variable.

Returns

The current offset read out from used hil variable.

Reimplemented in ttk_selftest.adjustment_calculation.AdjustmentCalculation.

7.34.3.17 getGainVariance()

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation.getGainVariance (
              self )
```

Return the gain variance of linear regression back.

Will be generated during calibration values will be calculated.

Returns

The calculated gain variance to use after calibration or None in case calibration is not running correctly.

Reimplemented in ttk_selftest.adjustment_calculation.AdjustmentCalculation.

7.34.3.18 getLimitGainMax()

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation.getLimitGainMax (
              self )
```

Get back the maximum specified limitation value of gain.

Returns

The maximum limitation value of the gain.

Reimplemented in ttk_selftest.adjustment_calculation.AdjustmentCalculation.

7.34.3.19 getLimitGainMin()

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation.getLimitGainMin (
              self )
```

Get back the minimum specified limitation value of gain.

Returns

The minimum limitation value of the gain.

Reimplemented in ttk_selftest.adjustment_calculation.AdjustmentCalculation.

7.34.3.20 getLimitOffsetMax()

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation.getLimitOffsetMax (
              self )
```

Get back the maximum specified limitation value of offset.

Returns

The maximum limitation value of the offset.

Reimplemented in ttk_selftest.adjustment_calculation.AdjustmentCalculation.

7.34.3.21 getLimitOffsetMin()

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation.getLimitOffsetMin (
              self )
```

Get back the minimum specified limitation value of offset.

Returns

The minimum limitation value of the offset.

Reimplemented in ttk_selftest.adjustment_calculation.AdjustmentCalculation.

7.34.3.22 getNewGain()

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation.getNewGain (
              self )
```

Get back the calculated gain.

(linear regression)

Returns

The new/calculated gain to use after calibration or None in case calibration is not running correctly.

Reimplemented in ttk_selftest.adjustment_calculation.AdjustmentCalculation.

7.34.3.23 getNewOffset()

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation.getNewOffset (
              self )
```

Get back the new offset.

(linear regression)

Returns

The new/calculated offset to use after calibration or None in case calibration is not running correctly.

Reimplemented in ttk_selftest.adjustment_calculation.AdjustmentCalculation.

7.34.3.24 getOffsetVariance()

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation.getOffsetVariance (
              self )
```

Return the offset variance of linear regression back.

Will be generated during calibration values will be calculated.

Returns

The calculated offset variance to use after calibration or None in case calibration is not running correctly.

Reimplemented in ttk_selftest.adjustment_calculation.AdjustmentCalculation.

7.34.3.25 getPrecision()

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation.getPrecision (
              self )
```

Get back the specified number of decimal places, which is used to round the new gain and offset value before get back.

Returns

Number of decimal places should be use to round the values.

Reimplemented in ttk_selftest.adjustment_calculation.AdjustmentCalculation.

7.34.3.26 getStandardVariance()

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation.getStandardVariance (
              self )
```

Return the standard variance of linear regression back.

Will be generated during calibration values will be calculated.

Returns

The calculated standard variance to use after calibration or None in case calibration is not running correctly.

Reimplemented in ttk_selftest.adjustment_calculation.AdjustmentCalculation.

7.34.4 Member Data Documentation

7.34.4.1 _avg_expect

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation._avg_expect  [protected]
```

average value parameter of expected input data

7.34.4.2 _avg_measure

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation._avg_measure  [protected]
```

average value parameter of compensate measured input data

7.34.4.3 _current_gain

```
ttk_selftest._adjustment_calculation.AdjustmentCalculation._current_gain  [protected]
```

current gain parameter is used in model

7.34.4.4 _current_offset

ttk_selftest._adjustment_calculation.AdjustmentCalculation._current_offset [protected]

current offset parameter is used in model

7.34.4.5 _expect_data

ttk_selftest._adjustment_calculation.AdjustmentCalculation._expect_data [protected]

expected data list parameter for input data

7.34.4.6 _gain_variance

ttk_selftest._adjustment_calculation.AdjustmentCalculation._gain_variance [protected]

calculated gain variance parameter

7.34.4.7 _len_data_element

ttk_selftest._adjustment_calculation.AdjustmentCalculation._len_data_element [protected]

length parameter for input data

7.34.4.8 _limit_gain_max

ttk_selftest._adjustment_calculation.AdjustmentCalculation._limit_gain_max [protected]

specify the limitation of the calculated new gain is only allowed to have maximum

7.34.4.9 _limit_gain_min

ttk_selftest._adjustment_calculation.AdjustmentCalculation._limit_gain_min [protected]

specify the limitation of the calculated new gain is only allowed to have minimum

7.34.4.10 _limit_offset_max

ttk_selftest._adjustment_calculation.AdjustmentCalculation._limit_offset_max [protected]

specify the limitation of the calculated new offset is only allowed to have maximum

7.34.4.11 _limit_offset_min

ttk_selftest._adjustment_calculation.AdjustmentCalculation._limit_offset_min [protected]

specify the limitation of the calculated new offset is only allowed to have minimum

7.34.4.12 _lin_reg_denominator

ttk_selftest._adjustment_calculation.AdjustmentCalculation._lin_reg_denominator [protected]

calculated denominator value parameter

7.34.4.13 _lin_reg_enumerator

ttk_selftest._adjustment_calculation.AdjustmentCalculation._lin_reg_enumerator [protected]

calculated enumerator value parameter

7.34.4.14 _lin_reg_gain

ttk_selftest._adjustment_calculation.AdjustmentCalculation._lin_reg_gain [protected]

linear regression calculated gain parameter

7.34.4.15 _lin_reg_offset

ttk_selftest._adjustment_calculation.AdjustmentCalculation._lin_reg_offset [protected]

linear regression calculated offset parameter

7.34.4.16 _measure_compensate

ttk_selftest._adjustment_calculation.AdjustmentCalculation._measure_compensate [protected]

measured data list parameter for compensate data

7.34.4.17 _measure_data

ttk_selftest._adjustment_calculation.AdjustmentCalculation._measure_data [protected]

measured data list parameter for input data

7.34.4.18 _new_gain

ttk_selftest._adjustment_calculation.AdjustmentCalculation._new_gain [protected]

calculated gain parameter

7.34.4.19 _new_offset

ttk_selftest._adjustment_calculation.AdjustmentCalculation._new_offset [protected]

calculated offset parameter

7.34.4.20 _offset_variance

ttk_selftest._adjustment_calculation.AdjustmentCalculation._offset_variance [protected]

calculated offset variance parameter

7.34.4.21 _precision

ttk_selftest._adjustment_calculation.AdjustmentCalculation._precision [protected]

define the precision of return values to convert in correct format

7.34.4.22 _standard_variance

ttk_selftest._adjustment_calculation.AdjustmentCalculation._standard_variance [protected]

calculated standard variance parameter

7.34.4.23 _timestamp

ttk_selftest._adjustment_calculation.AdjustmentCalculation._timestamp [protected]

timestamp parameter of calibration was run

7.34.4.24 _verdict

ttk_selftest._adjustment_calculation.AdjustmentCalculation._verdict [protected]

result parameter of limitation check

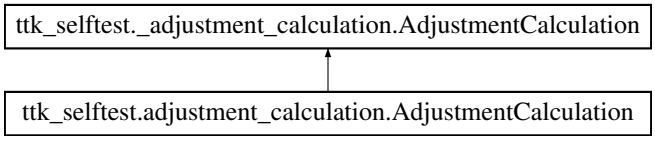
The documentation for this class was generated from the following file:

- ttk_selftest/_adjustment_calculation.py (v2.0.1)

7.35 ttk_selftest.adjustment_calculation.AdjustmentCalculation Class Reference

Class for adjustment one hil channel which is currently in state self test and adjustment.

Inheritance diagram for ttk_selftest.adjustment_calculation.AdjustmentCalculation:



Public Member Functions

- __init__ (self, hilvar_gain, hilvar_offset, limit_gain=None, limit_offset=None, precision=6)

  Initialize the instance and prepare the self adjustment environment.
- calibData (self, expected, measured, verbosity=True)

  Calibration Method - The method describe the complete hil adjustment calculation:
- getCurrentGain (self)

  Get back the current gain which is specified in running simulation model at current test system.
- getCurrentOffset (self)

  Get back the current offset which is specified in running simulation model at current test system.
- getNewGain (self)

  Get back the calculated gain (linear regression).
- getNewOffset (self)

  Get back the new offset (linear regression).
- getCalibVerdict (self)

  Get back the qualification string from the check the new parameter against the specified ranges.
- getCalibTimeStamp (self)

  Get back the time stamp of running calibration.
- getLimitGainMax (self)

  Get back the maximum specified limitation value of gain.
- getLimitGainMin (self)

  Get back the minimum specified limitation value of gain.
- getLimitOffsetMax (self)

  Get back the maximum specified limitation value of offset.
- getLimitOffsetMin (self)

  Get back the minimum specified limitation value of offset.
- getPrecision (self)

  Get back the specified number of decimal places, which is used to round the new gain and offset value before get back.
- getStandardVariance (self)

  Return the standard variance of linear regression back.
- getGainVariance (self)

  Return the gain variance of linear regression back.
- getOffsetVariance (self)

  Return the offset variance of linear regression back.

7.35.1 Detailed Description

Class for adjustment one hil channel which is currently in state self test and adjustment.

7.35.2 Constructor & Destructor Documentation

7.35.2.1 __init__()

```
ttk_selftest.adjustment_calculation.AdjustmentCalculation.__init__ (
              self,
              hilvar_gain,
              hilvar_offset,
              limit_gain = None,
              limit_offset = None,
              precision = 6 )
```

Initialize the instance and prepare the self adjustment environment.

Parameters

hilvar_gain	variable definition with a get() and set() method to read current specified gain for channel is specified to run the adjustment calculation.
hilvar_offset	variable definition with a get() and set() method to read current specified offset for channel is specified to run the adjustment calculation.
limit_gain	(opt) dict with struct {"max":<value>, "min":<value>} Limitation of calculated new gain. In use to check the calculation and in case of incorrect set the calculation to false. • "max": maximum limitation (> min) • "min": minimum limitation (< max) • Type: Type of "Float", "Long" and "Int" acceptable
limit_offset	(opt) dict with struct {"max":<value>, "min":<value>} Limitation of calculated new offset. In use to check the calculation and in case of incorrect set the calculation to false. • "max": maximum limitation (> min) • "min": minimum limitation (< max) • Type: Type of "Float", "Long" and "Int" acceptable
precision	(opt) Specify the number of decimal places the new gain and offset value should round after the calculation. Type: "Integer"

Raises Exception in case of sanity check fails

Reimplemented from ttk_selftest._adjustment_calculation.AdjustmentCalculation.

7.35.3 Member Function Documentation

7.35.3.1 calibData()

```
ttk_selftest.adjustment_calculation.AdjustmentCalculation.calibData (
               self,
               expected,
               measured,
               verbosity = True )
```

Calibration Method - The method describe the complete hil adjustment calculation:

- Check parameter are correct
- compensate possible old gain and offset values different to the default in model
- calculate the linear regression
- calculate the new calibrated data (gain and offset)
- check the new calibrated data against the limitation

Parameters

expected	list of expected data
measured	list of measured data verbosity- (opt) state to define the depth of logging False: No print out True: Print the call of methods...

Attention

expected and measured data should have the same value order.

Reimplemented from ttk_selftest._adjustment_calculation.AdjustmentCalculation.

7.35.3.2 getCalibTimeStamp()

```
ttk_selftest.adjustment_calculation.AdjustmentCalculation.getCalibTimeStamp (
              self )
```

Get back the time stamp of running calibration.

Returns

timestamp of calibration run or None if calibration is not running correctly.

Note

Value can be converted like: time.strftime("%Y-%m-%dT%H:%M:%S", time.localtime(getCalibTimeStamp()))

Reimplemented from ttk_selftest._adjustment_calculation.AdjustmentCalculation.

7.35.3.3 getCalibVerdict()

```
ttk_selftest.adjustment_calculation.AdjustmentCalculation.getCalibVerdict (
              self )
```

Get back the qualification string from the check the new parameter against the specified ranges.

Returns

a verdict string, "PASSED", "FAILED" or "NOT TESTED"

- PASSED: Check of new parameters against defined ranges passed
- FAILED: Check of new parameters against defined ranges failed
- NOTTESTED: Check is not running

Reimplemented from ttk_selftest._adjustment_calculation.AdjustmentCalculation.

7.35.3.4 getCurrentGain()

```
ttk_selftest.adjustment_calculation.AdjustmentCalculation.getCurrentGain (
              self )
```

Get back the current gain which is specified in running simulation model at current test system.

The value will be read out by use the specified variable.

Returns

current gain value read from used hil variable.

Reimplemented from ttk_selftest._adjustment_calculation.AdjustmentCalculation.

7.35.3.5 getCurrentOffset()

```
ttk_selftest.adjustment_calculation.AdjustmentCalculation.getCurrentOffset (
              self )
```

Get back the current offset which is specified in running simulation model at current test system.

The value will be read out by use the specified variable.

Returns

current offset value read from used hil variable.

Reimplemented from ttk_selftest._adjustment_calculation.AdjustmentCalculation.

7.35.3.6 getGainVariance()

```
ttk_selftest.adjustment_calculation.AdjustmentCalculation.getGainVariance (
              self )
```

Return the gain variance of linear regression back.

Will be generated during calibration values calculated.

Returns

the calculated gain variance to use after calibration or None in case calibration is not running correctly.

Reimplemented from ttk_selftest._adjustment_calculation.AdjustmentCalculation.

7.35.3.7 getLimitGainMax()

```
ttk_selftest.adjustment_calculation.AdjustmentCalculation.getLimitGainMax (
              self )
```

Get back the maximum specified limitation value of gain.

Returns

maximum limitation value of the gain.

Reimplemented from ttk_selftest._adjustment_calculation.AdjustmentCalculation.

7.35.3.8 getLimitGainMin()

```
ttk_selftest.adjustment_calculation.AdjustmentCalculation.getLimitGainMin (
              self )
```

Get back the minimum specified limitation value of gain.

Returns

minimum limitation value of the gain.

Reimplemented from ttk_selftest._adjustment_calculation.AdjustmentCalculation.

7.35.3.9 getLimitOffsetMax()

```
ttk_selftest.adjustment_calculation.AdjustmentCalculation.getLimitOffsetMax (
              self )
```

Get back the maximum specified limitation value of offset.

Returns

maximum limitation value of the offset.

Reimplemented from ttk_selftest._adjustment_calculation.AdjustmentCalculation.

7.35.3.10 getLimitOffsetMin()

```
ttk_selftest.adjustment_calculation.AdjustmentCalculation.getLimitOffsetMin (
              self )
```

Get back the minimum specified limitation value of offset.

Returns

minimum limitation value of the offset.

Reimplemented from ttk_selftest._adjustment_calculation.AdjustmentCalculation.

7.35.3.11 getNewGain()

```
ttk_selftest.adjustment_calculation.AdjustmentCalculation.getNewGain (
              self )
```

Get back the calculated gain (linear regression).

Returns

new/calculated gain value to use after calibration or None in case calibration is not running correctly.

Reimplemented from ttk_selftest._adjustment_calculation.AdjustmentCalculation.

7.35.3.12 getNewOffset()

```
ttk_selftest.adjustment_calculation.AdjustmentCalculation.getNewOffset (
              self )
```

Get back the new offset (linear regression).

Returns

new/calculated offset to use after calibration or None in case calibration is not running correctly.

Reimplemented from ttk_selftest._adjustment_calculation.AdjustmentCalculation.

7.35.3.13 getOffsetVariance()

```
ttk_selftest.adjustment_calculation.AdjustmentCalculation.getOffsetVariance (
              self )
```

Return the offset variance of linear regression back.

Will be generated during calibration value calculation.

Returns

the calculated offset variance to use after calibration or None in case calibration is not running correctly.

Reimplemented from ttk_selftest._adjustment_calculation.AdjustmentCalculation.

7.35.3.14 getPrecision()

```
ttk_selftest.adjustment_calculation.AdjustmentCalculation.getPrecision (
              self )
```

Get back the specified number of decimal places, which is used to round the new gain and offset value before get back.

Returns

number of decimal places that should be used to round values.

Reimplemented from ttk_selftest._adjustment_calculation.AdjustmentCalculation.

7.35.3.15 getStandardVariance()

```
ttk_selftest.adjustment_calculation.AdjustmentCalculation.getStandardVariance (
              self )
```

Return the standard variance of linear regression back.

Will be generated during calibration value calculation.

Returns

the calculated standard variance to use after calibration or None in case calibration is not running correctly.

Reimplemented from ttk_selftest._adjustment_calculation.AdjustmentCalculation.

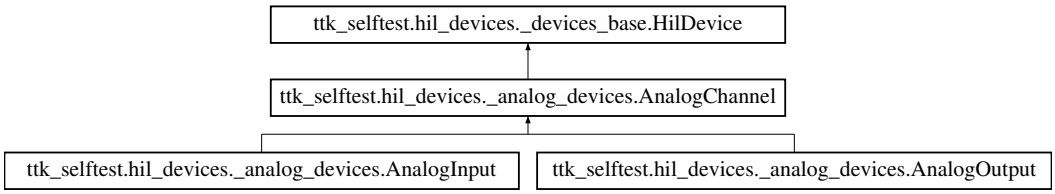
The documentation for this class was generated from the following file:

- ttk_selftest/adjustment_calculation.py (v2.0.1)

7.36 ttk_selftest.hil_devices._analog_devices.AnalogChannel Class Reference

Base analog channel container for a testable analog channel and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices._analog_devices.AnalogChannel:



Public Member Functions

- __init__ (self, context, base_path, sig_channel, ref_channel, max_value, min_value=0, abs_tol=0.05, rel_tol=0.01, preconditions=None, wait_time_ms=None, unit="", alias="", descr="")

  Analog channel initialization.
- info (self)

  Get an info string for this analog channel: alias, FIU channels, minimum and maximum value plus description, if available.
- getFormattedValue (self, value=None)

  Get a formatted representation of the SelfTestVariable for the given value using the defined format string.

Public Attributes

- base_path
- selftest_var
- offset_var
- factor_var
- max_value
- min_value
- abs_tol
- rel_tol
- ref_channel
- selftest_dataset
- unit

7.36.1 Detailed Description

Base analog channel container for a testable analog channel and relevant associated variables/values.

7.36.2 Constructor & Destructor Documentation

7.36.2.1 __init__()

```
ttk_selftest.hil_devices._analog_devices.AnalogChannel.__init__ (
               self,
               context,
               base_path,
               sig_channel,
               ref_channel,
               max_value,
               min_value = 0,
               abs_tol = 0.05,
               rel_tol = 0.01,
               preconditions = None,
               wait_time_ms = None,
               unit = "",
               alias = "",
               descr = "" )
```

Analog channel initialization.

Parameters

context	parent context
base_path	base path to analog channel subsystem
sig_channel	fiu channel of the analog channel equals positive path of the analog signal
ref_channel	reference channel of the analog channel equals ground path of analog signal can be either GND device or FiuChannel
max_value	maximum allowed physical value of the channel at ECU connector
min_value	(opt) minimum allowed physical value of the channel at the ECU connector defaults to 0
abs_tol	(opt) absolute tolerance, defaults to 0.05
rel_tol	(opt) relative tolerance, defaults to 0.01 = 1%
preconditions	(opt) preconditions that have to be set before the device can be checked. defaults to None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The functions specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
# For special cases, the preconditions may also contain function calls
# (which might be useful to insert additional delays or similar):
preconditions = [
    { "device": Voltage, "value": 1},
    { "func":   time.sleep,
```

```
                "args":   [1.0], # positional arguments
                "kwargs": {},    # keyword arguments
            },
            {"device": ClampSwitch, "value": 1},
            # or a simple call like this:
            { "func":  someCustomFunction,
            },
            ...
        ]
    wait_time_ms  - (opt) waiting time in milliseconds between
    channel stimulation and channel check
    defaults to 100ms
    unit          - (opt) unit of the physical value
    defaults to "V"
    alias         - (opt) channel alias/name, defaults to the
    last (non-default) entry in identifier
    (i.e. signal model path)
    descr         - (opt) channel description (e.g. used in report entries)
```

Reimplemented from ttk_selftest.hil_devices._devices_base.HilDevice.

Reimplemented in ttk_selftest.hil_devices._analog_devices.AnalogOutput, and ttk_selftest.hil_devices._analog_devices.AnalogInput.

7.36.3 Member Function Documentation

7.36.3.1 getFormattedValue()

```
ttk_selftest.hil_devices._analog_devices.AnalogChannel.getFormattedValue (
                self,
                value = None )
```

Get a formatted representation of the SelfTestVariable for the given value using the defined format string.

An empty (or non-defined) format string will use the value's default string representation.

Parameters

value	value for which to get a string representation. If None, the last value (read or written) will be used instead (see self.last_value).

Returns

a formatted string representation.

Reimplemented in ttk_selftest.hil_devices._analog_devices.AnalogInput.

7.36.3.2 info()

```
ttk_selftest.hil_devices._analog_devices.AnalogChannel.info (
                self )
```

Get an info string for this analog channel: alias, FIU channels, minimum and maximum value plus description, if available.

Returns

a formatted string.

7.36.4 Member Data Documentation

7.36.4.1 abs_tol

ttk_selftest.hil_devices._analog_devices.AnalogChannel.abs_tol

7.36.4.2 base_path

ttk_selftest.hil_devices._analog_devices.AnalogChannel.base_path

7.36.4.3 factor_var

ttk_selftest.hil_devices._analog_devices.AnalogChannel.factor_var

7.36.4.4 max_value

ttk_selftest.hil_devices._analog_devices.AnalogChannel.max_value

7.36.4.5 min_value

ttk_selftest.hil_devices._analog_devices.AnalogChannel.min_value

7.36.4.6 offset_var

ttk_selftest.hil_devices._analog_devices.AnalogChannel.offset_var

7.36.4.7 ref_channel

ttk_selftest.hil_devices._analog_devices.AnalogChannel.ref_channel

7.36.4.8 rel_tol

ttk_selftest.hil_devices._analog_devices.AnalogChannel.rel_tol

7.36.4.9 selftest_dataset

ttk_selftest.hil_devices._analog_devices.AnalogChannel.selftest_dataset

7.36.4.10 selftest_var

ttk_selftest.hil_devices._analog_devices.AnalogChannel.selftest_var

7.36.4.11 unit

`ttk_selftest.hil_devices._analog_devices.AnalogChannel.unit`

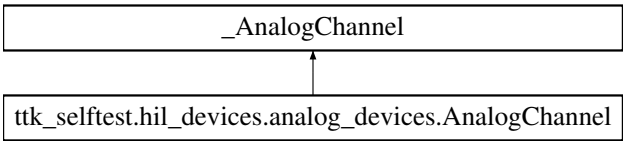
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/_analog_devices.py (v2.0.1)

7.37 ttk_selftest.hil_devices.analog_devices.AnalogChannel Class Reference

Base analog channel container for a testable analog channel and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices.analog_devices.AnalogChannel:



Public Member Functions

- __init__ (self, context, base_path, sig_channel, ref_channel, max_value, min_value=0, abs_tol=0.05, rel_tol=0.01, preconditions=None, wait_time_ms=None, unit="", alias="", descr="")

  Analog channel initialization.
- info (self)

  Get an info string for this analog channel: alias, FIU channels, minimum and maximum value plus description, if available.
- getFormattedValue (self, value=None)

  Get a formatted representation of the SelfTestVariable for the given value using the defined format string.

7.37.1 Detailed Description

Base analog channel container for a testable analog channel and relevant associated variables/values.

7.37.2 Constructor & Destructor Documentation

7.37.2.1 __init__()

```
ttk_selftest.hil_devices.analog_devices.AnalogChannel.__init__ (
               self,
               context,
               base_path,
               sig_channel,
               ref_channel,
               max_value,
               min_value = 0,
               abs_tol = 0.05,
               rel_tol = 0.01,
               preconditions = None,
               wait_time_ms = None,
               unit = "",
               alias = "",
               descr = "" )
```

Analog channel initialization.

Parameters

context	parent context
base_path	base path to analog channel subsystem
sig_channel	fiu channel of the analog channel equals positive path of the analog signal
ref_channel	reference channel of the analog channel equals ground path of analog signal can be either GND device or FiuChannel
max_value	maximum allowed physical value of the channel at ECU connector
min_value	(opt) minimum allowed physical value of the channel at the ECU connector defaults to 0
abs_tol	(opt) absolute tolerance, defaults to 0.05
rel_tol	(opt) relative tolerance, defaults to 0.01 = 1%
preconditions	(opt) preconditions that have to be set before the device can be checked. defaults to None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The functions specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
# For special cases, the preconditions may also contain function calls
# (which might be useful to insert additional delays or similar):
preconditions = [
    { "device": Voltage, "value": 1},
    { "func":   time.sleep,
```

```
                "args":   [1.0], # positional arguments
                "kwargs": {},    # keyword arguments
            },
            {"device": ClampSwitch, "value": 1},
            # or a simple call like this:
            { "func":  someCustomFunction,
            },
            ...
        ]
    wait_time_ms  - (opt) waiting time in milliseconds between
    channel stimulation and channel check
    defaults to 100ms
    unit          - (opt) unit of the physical value
    defaults to "V"
    alias         - (opt) channel alias/name, defaults to the
    last (non-default) entry in identifier
    (i.e. signal model path)
    descr         - (opt) channel description (e.g. used in report entries)
```

7.37.3 Member Function Documentation

7.37.3.1 getFormattedValue()

```
ttk_selftest.hil_devices.analog_devices.AnalogChannel.getFormattedValue (
                self,
                value = None )
```

Get a formatted representation of the SelfTestVariable for the given value using the defined format string.

An empty (or non-defined) format string will use the value's default string representation.

Parameters

value	value for which to get a string representation. If None, the last value (read or written) will be used instead (see self.last_value).

Returns

a formatted string representation.

7.37.3.2 info()

```
ttk_selftest.hil_devices.analog_devices.AnalogChannel.info (
                self )
```

Get an info string for this analog channel: alias, FIU channels, minimum and maximum value plus description, if available.

Returns

a formatted string.

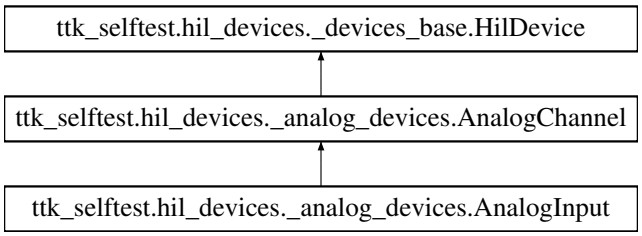
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/analog_devices.py (v2.0.0)

7.38 ttk_selftest.hil_devices._analog_devices.AnalogInput Class Reference

Base analog input container for a testable analog input and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices._analog_devices.AnalogInput:



Public Member Functions

- __init__ (self, context, base_path, sig_channel, ref_channel, max_value, min_value=0, use_filter=False, abs_tol=0.05, rel_tol=0.01, preconditions=None, wait_time_ms=None, unit="", alias="", descr="")

  Analog input initialization.
- get (self)

  Function to get the current value of the SelfTestVariable depended of the filter setting.
- getFormattedValue (self, value=None)

  Get a formatted representation of the filtered or unfiltered SelfTestVariable for the given value using the defined format string.
- getLastValue (self)

  Get the value of the last get() operation on the filtered or unfiltered value.

Public Attributes

- use_filter
- selftest_var
- selftest_var_filter
- unit

7.38.1 Detailed Description

Base analog input container for a testable analog input and relevant associated variables/values.

7.38.2 Constructor & Destructor Documentation

7.38.2.1 __init__()

```
ttk_selftest.hil_devices._analog_devices.AnalogInput.__init__ (
                self,
                context,
                base_path,
                sig_channel,
                ref_channel,
                max_value,
                min_value = 0,
                use_filter = False,
                abs_tol = 0.05,
                rel_tol = 0.01,
                preconditions = None,
                wait_time_ms = None,
                unit = "",
                alias = "",
                descr = "" )
```

Analog input initialization.

Parameters

context	parent context
base_path	base path to analog channel subsystem
sig_channel	fiu channel of the analog channel equals positive path of the analog signal
ref_channel	reference channel of the analog channel equals ground path of analog signal can be either GND device or FiuChannel
max_value	maximum allowed physical value of the channel at ECU connector
min_value	(opt) minimum allowed physical value of the channel at the ECU connector default is 0
use_filter	(opt) enable/disable usage of filtered value False: use (normal) unfiltered value True: use filtered value defaults to unfiltered value (False)
abs_tol	(opt) absolute tolerance, defaults to 0.05
rel_tol	(opt) relative tolerance, defaults to 0.01 = 1%
preconditions	(opt) preconditions that have to be set before the device can be checked. defaults to None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The functions specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
        # For special cases, the preconditions may also contain function calls
        # (which might be useful to insert additional delays or similar):
        preconditions = [
            { "device": Voltage, "value": 1},
            { "func":   time.sleep,
              "args":   [1.0], # positional arguments
              "kwargs": {},    # keyword arguments
            },
            {"device": ClampSwitch, "value": 1},
            # or a simple call like this:
            { "func":  someCustomFunction,
            },
            ...
        ]
    wait_time_ms  - (opt) waiting time in milliseconds between
    channel stimulation and channel check
    defaults to 100ms
    unit          - (opt) unit of the physical value
    defaults to "V"
    alias         - (opt) channel alias/name, defaults to the
    last (non-default) entry in identifier
    (i.e. signal model path)
    descr         - (opt) channel description (e.g. used in report entries)
```

Reimplemented from ttk_selftest.hil_devices._analog_devices.AnalogChannel.

7.38.3 Member Function Documentation

7.38.3.1 get()

```
ttk_selftest.hil_devices._analog_devices.AnalogInput.get (
              self )
```

Function to get the current value of the SelfTestVariable depended of the filter setting.

```
None
```

Returns

the current value of the filtered or unfiltered SelfTestVariable, depended on the setting of the use_filter parameter of the object.

7.38.3.2 getFormattedValue()

```
ttk_selftest.hil_devices._analog_devices.AnalogInput.getFormattedValue (
              self,
              value = None )
```

Get a formatted representation of the filtered or unfiltered SelfTestVariable for the given value using the defined format string.

An empty (or non-defined) format string will use the value's default string representation.

Parameters

value	value for which to get a string representation. If None, the last value (read or written) will be used instead (see self.last_value).

Returns

a formatted string representation.

Reimplemented from ttk_selftest.hil_devices._analog_devices.AnalogChannel.

7.38.3.3 getLastValue()

```
ttk_selftest.hil_devices._analog_devices.AnalogInput.getLastValue (
              self )
```

Get the value of the last get() operation on the filtered or unfiltered value.

```
None
```

Returns

the value of the last get() operation of the of the filtered or unfiltered SelfTestVariable, depended on the setting of the use_filter parameter of the object.

7.38.4 Member Data Documentation

7.38.4.1 selftest_var

```
ttk_selftest.hil_devices._analog_devices.AnalogInput.selftest_var
```

7.38.4.2 selftest_var_filter

```
ttk_selftest.hil_devices._analog_devices.AnalogInput.selftest_var_filter
```

7.38.4.3 unit

```
ttk_selftest.hil_devices._analog_devices.AnalogInput.unit
```

7.38.4.4 use_filter

```
ttk_selftest.hil_devices._analog_devices.AnalogInput.use_filter
```

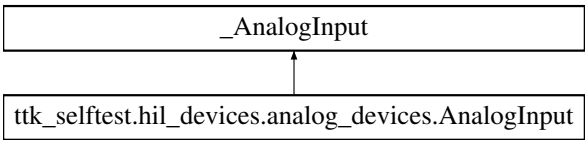
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/_analog_devices.py (v2.0.1)

7.39 ttk_selftest.hil_devices.analog_devices.AnalogInput Class Reference

Base analog input container for a testable analog input and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices.analog_devices.AnalogInput:



Public Member Functions

- __init__ (self, context, base_path, sig_channel, ref_channel, max_value, min_value=0, use_filter=False, abs_tol=0.05, rel_tol=0.01, preconditions=None, wait_time_ms=None, unit="", alias="", descr="")

  Analog input initialization.
- get (self)

  Function to get the current value of the SelfTestVariable depended of the filter setting.

7.39.1 Detailed Description

Base analog input container for a testable analog input and relevant associated variables/values.

7.39.2 Constructor & Destructor Documentation

7.39.2.1 __init__()

```
ttk_selftest.hil_devices.analog_devices.AnalogInput.__init__ (
              self,
              context,
              base_path,
              sig_channel,
              ref_channel,
              max_value,
              min_value = 0,
              use_filter = False,
              abs_tol = 0.05,
              rel_tol = 0.01,
              preconditions = None,
              wait_time_ms = None,
              unit = "",
              alias = "",
              descr = "" )
```

Analog input initialization.

Parameters

context	parent context
base_path	base path to analog channel subsystem
sig_channel	fiu channel of the analog channel equals positive path of the analog signal
ref_channel	reference channel of the analog channel equals ground path of analog signal can be either GND device or FiuChannel
max_value	maximum allowed physical value of the channel at ECU connector
min_value	(opt) minimum allowed physical value of the channel at the ECU connector default is 0
use_filter	(opt) enable/disable usage of filtered value False: use (normal) unfiltered value True: use filtered value defaults to unfiltered value (False)
abs_tol	(opt) absolute tolerance, defaults to 0.05
rel_tol	(opt) relative tolerance, defaults to 0.01 = 1%
preconditions	(opt) preconditions that have to be set before the device can be checked. defaults to None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The functions specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
    # For special cases, the preconditions may also contain function calls
    # (which might be useful to insert additional delays or similar):
    preconditions = [
        { "device": Voltage, "value": 1},
        { "func":   time.sleep,
          "args":   [1.0], # positional arguments
          "kwargs": {},    # keyword arguments
        },
        {"device": ClampSwitch, "value": 1},
        # or a simple call like this:
        { "func":  someCustomFunction,
        },
        ...
    ]
wait_time_ms  - (opt) waiting time in milliseconds between
channel stimulation and channel check
defaults to 100ms
unit          - (opt) unit of the physical value
defaults to "V"
alias         - (opt) channel alias/name, defaults to the
last (non-default) entry in identifier
(i.e. signal model path)
descr         - (opt) channel description (e.g. used in report entries)
```

7.39.3 Member Function Documentation

7.39.3.1 get()

```
ttk_selftest.hil_devices.analog_devices.AnalogInput.get (
            self )
```

Function to get the current value of the SelfTestVariable depended of the filter setting.

```
None
```

Returns

the current value of the filtered or unfiltered SelfTestVariable, depended on the setting of the use_filter parameter of the object.

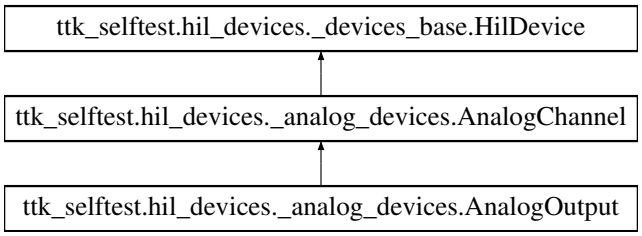
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/analog_devices.py (v2.0.0)

7.40 ttk_selftest.hil_devices._analog_devices.AnalogOutput Class Reference

Base analog output container for a testable analog output and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices._analog_devices.AnalogOutput:



Public Member Functions

- __init__ (self, context, base_path, sig_channel, ref_channel, max_value, min_value=0, abs_tol=0.05, rel_tol=0.01, preconditions=None, wait_time_ms=None, unit="", alias="", descr="")

  Analog output initialization.
- set (self, value, ∗args, ∗∗kwargs)

  Function to set the SelfTestVariable of the channel.
- reset (self, verbosity=1)

  Function to reset the SelfTestVariable of the channel to its initial value.

Public Attributes

- selftest_var

7.40.1 Detailed Description

Base analog output container for a testable analog output and relevant associated variables/values.

7.40.2 Constructor & Destructor Documentation

7.40.2.1 __init__()

```
ttk_selftest.hil_devices._analog_devices.AnalogOutput.__init__ (
                self,
                context,
                base_path,
                sig_channel,
                ref_channel,
                max_value,
                min_value = 0,
                abs_tol = 0.05,
                rel_tol = 0.01,
                preconditions = None,
                wait_time_ms = None,
                unit = "",
                alias = "",
                descr = "" )
```

Analog output initialization.

Parameters

context	parent context
base_path	base path to analog channel subsystem
sig_channel	fiu channel of the analog channel equals positive path of the analog signal
ref_channel	reference channel of the analog channel equals ground path of analog signal can be either GND device or FiuChannel
max_value	maximum allowed physical value of the channel in V at ECU connector
min_value	(opt) minimum allowed physical value of the channel at the ECU connector default is 0
abs_tol	(opt) absolute tolerance, defaults to 0.05
rel_tol	(opt) relative tolerance, defaults to 0.01 = 1%
preconditions	(opt) preconditions that have to be set before the device can be checked. defaults to None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The functions specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
# For special cases, the preconditions may also contain function calls
# (which might be useful to insert additional delays or similar):
preconditions = [
    { "device": Voltage, "value": 1},
    { "func":   time.sleep,
```

```
                "args":   [1.0], # positional arguments
                "kwargs": {},    # keyword arguments
            },
            {"device": ClampSwitch, "value": 1},
            # or a simple call like this:
            { "func":  someCustomFunction,
            },
            ...
        ]
    wait_time_ms  - (opt) waiting time in milliseconds between
    channel stimulation and channel check
    defaults to 100ms
    unit          - (opt) unit of the physical value
    defaults to "V"
    alias         - (opt) channel alias/name, defaults to the
    last (non-default) entry in identifier
    (i.e. signal model path)
    descr         - (opt) channel description (e.g. used in report entries)
```

Reimplemented from ttk_selftest.hil_devices._analog_devices.AnalogChannel.

7.40.3 Member Function Documentation

7.40.3.1 reset()

```
ttk_selftest.hil_devices._analog_devices.AnalogOutput.reset (
              self,
              verbosity = 1 )
```

Function to reset the SelfTestVariable of the channel to its initial value.

Parameters

verbosity	verbosity of log output, 0: silent, 1: show calls to non-resettable variables 2: show all reset calls

Returns

the original value (which may be None if the variable has not been set before)

7.40.3.2 set()

```
ttk_selftest.hil_devices._analog_devices.AnalogOutput.set (
              self,
              value,
             * args,
             ** kwargs )
```

Function to set the SelfTestVariable of the channel.

Parameters

value	value to set
∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Note

Any additional parameters will be forwarded to the write implementation of the variable instance

Returns

the original value before the set operation occurred.

7.40.4 Member Data Documentation

7.40.4.1 selftest_var

`ttk_selftest.hil_devices._analog_devices.AnalogOutput.selftest_var`

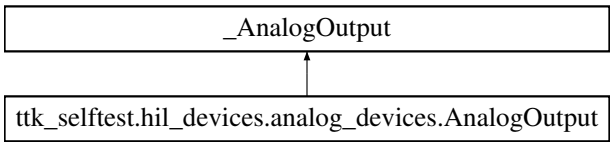
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/_analog_devices.py (v2.0.1)

7.41 ttk_selftest.hil_devices.analog_devices.AnalogOutput Class Reference

Base analog output container for a testable analog output and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices.analog_devices.AnalogOutput:



Public Member Functions

- __init__ (self, context, base_path, sig_channel, ref_channel, max_value, min_value=0, abs_tol=0.05, rel_tol=0.01, preconditions=None, wait_time_ms=None, unit="", alias="", descr="")

  Analog output initialization.
- set (self, value, *args, **kwargs)

  Function to set the SelfTestVariable of the channel.
- reset (self, verbosity=1)

  Function to reset the SelfTestVariable of the channel to its initial value.

7.41.1 Detailed Description

Base analog output container for a testable analog output and relevant associated variables/values.

7.41.2 Constructor & Destructor Documentation

7.41.2.1 __init__()

```
ttk_selftest.hil_devices.analog_devices.AnalogOutput.__init__ (
                self,
                context,
                base_path,
                sig_channel,
                ref_channel,
                max_value,
                min_value = 0,
                abs_tol = 0.05,
                rel_tol = 0.01,
                preconditions = None,
                wait_time_ms = None,
                unit = "",
                alias = "",
                descr = "" )
```

Analog output initialization.

Parameters

context	parent context
base_path	base path to analog channel subsystem
sig_channel	fiu channel of the analog channel equals positive path of the analog signal
ref_channel	reference channel of the analog channel equals ground path of analog signal can be either GND device or FiuChannel
max_value	maximum allowed physical value of the channel in V at ECU connector
min_value	(opt) minimum allowed physical value of the channel at the ECU connector default is 0
abs_tol	(opt) absolute tolerance, defaults to 0.05
rel_tol	(opt) relative tolerance, defaults to 0.01 = 1%
preconditions	(opt) preconditions that have to be set before the device can be checked. defaults to None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The functions specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
# For special cases, the preconditions may also contain function calls
# (which might be useful to insert additional delays or similar):
preconditions = [
    { "device": Voltage, "value": 1},
    { "func":   time.sleep,
```

```
                "args":   [1.0], # positional arguments
                "kwargs": {},    # keyword arguments
            },
            {"device": ClampSwitch, "value": 1},
            # or a simple call like this:
            { "func":  someCustomFunction,
            },
            ...
        ]
    wait_time_ms  - (opt) waiting time in milliseconds between
    channel stimulation and channel check
    defaults to 100ms
    unit          - (opt) unit of the physical value
    defaults to "V"
    alias         - (opt) channel alias/name, defaults to the
    last (non-default) entry in identifier
    (i.e. signal model path)
    descr         - (opt) channel description (e.g. used in report entries)
```

7.41.3 Member Function Documentation

7.41.3.1 reset()

```
ttk_selftest.hil_devices.analog_devices.AnalogOutput.reset (
                self,
                verbosity = 1 )
```

Function to reset the SelfTestVariable of the channel to its initial value.

Parameters

verbosity	verbosity of log output, 0: silent, 1: show calls to non-resettable variables 2: show all reset calls

Returns

the original value (which may be None if the variable has not been set before)

7.41.3.2 set()

```
ttk_selftest.hil_devices.analog_devices.AnalogOutput.set (
                self,
                value,
               * args,
               ** kwargs )
```

Function to set the SelfTestVariable of the channel.

Parameters

value	value to set
∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Note

Any additional parameters will be forwarded to the write implementation of the variable instance

Returns

the original value before the set operation occurred.

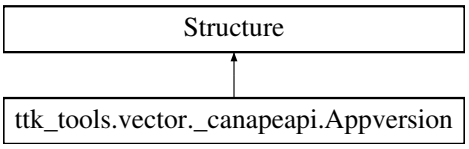
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/analog_devices.py (v2.0.0)

7.42 ttk_tools.vector._canapeapi.Appversion Class Reference

CANape API application version info structure.

Inheritance diagram for ttk_tools.vector._canapeapi.Appversion:



Static Protected Attributes

- int _pack_ = 1
- list _fields_

7.42.1 Detailed Description

CANape API application version info structure.

7.42.2 Member Data Documentation

7.42.2.1 _fields_

```
list ttk_tools.vector._canapeapi.Appversion._fields_  [static], [protected]
```

Initial value:

```
=  [
        ('MainVersion',   c_int),         # Main version of the current application
        ('SubVersion',    c_int),         # Subversion version of the current application
        ('ServicePack',   c_int),         # Installed service pack
        ('Application',   c_char * 30),   # Application name (CANape or CANdito)
    ]
```

7.42.2.2 _pack_

```
int ttk_tools.vector._canapeapi.Appversion._pack_ = 1  [static], [protected]
```

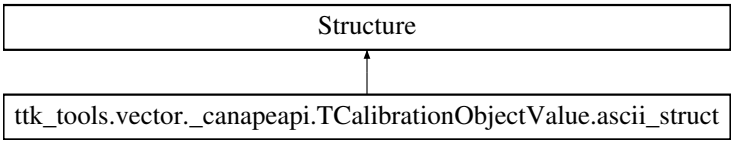
The documentation for this class was generated from the following file:

- ttk_tools/vector/_canapeapi.py (v7.3.3)

7.43 ttk_tools.vector._canapeapi.TCalibrationObjectValue.ascii_struct Class Reference

ASCII/string data.

Inheritance diagram for ttk_tools.vector._canapeapi.TCalibrationObjectValue.ascii_struct:



Static Protected Attributes

- int _pack_ = 1
- list _fields_

7.43.1 Detailed Description

ASCII/string data.

7.43.2 Member Data Documentation

7.43.2.1 _fields_

```
list ttk_tools.vector._canapeapi.TCalibrationObjectValue.ascii_struct._fields_  [static],
[protected]
```

Initial value:

```
=  [
            ("type",        c_int),
            ("len",         c_short),
            ("ascii",       POINTER(c_char)),
        ]
```

7.43.2.2 _pack_

```
int ttk_tools.vector._canapeapi.TCalibrationObjectValue.ascii_struct._pack_ = 1  [static],
[protected]
```

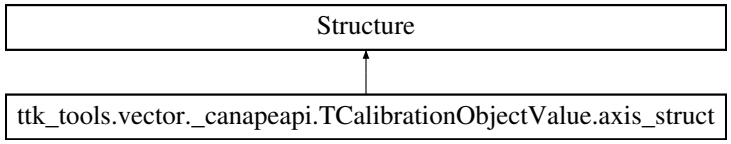
The documentation for this class was generated from the following file:

- ttk_tools/vector/_canapeapi.py (v7.3.3)

7.44 ttk_tools.vector._canapeapi.TCalibrationObjectValue.axis_struct Class Reference

axis definitions for maps or curves.

Inheritance diagram for ttk_tools.vector._canapeapi.TCalibrationObjectValue.axis_struct:



Static Protected Attributes

- int _pack_ = 1
- list _fields_

7.44.1 Detailed Description

axis definitions for maps or curves.

7.44.2 Member Data Documentation

7.44.2.1 _fields_

```
list ttk_tools.vector._canapeapi.TCalibrationObjectValue.axis_struct._fields_  [static], [protected]
```

Initial value:

```
=  [
            ("type",        c_int),
            ("dimension",   c_short),
            ("axis",        POINTER(c_double)),
        ]
```

7.44.2.2 _pack_

```
int ttk_tools.vector._canapeapi.TCalibrationObjectValue.axis_struct._pack_ = 1  [static],
[protected]
```

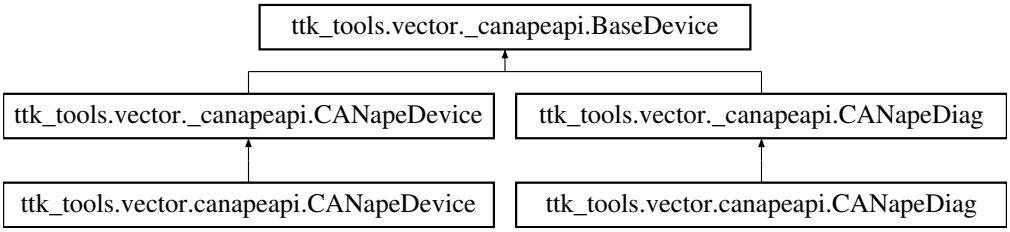
The documentation for this class was generated from the following file:

- ttk_tools/vector/_canapeapi.py (v7.3.3)

7.45 ttk_tools.vector._canapeapi.BaseDevice Class Reference

Base class for CANape devices with generic/common/basic functionality.

Inheritance diagram for ttk_tools.vector._canapeapi.BaseDevice:



Public Member Functions

- __init__ (self, canape_asap3, db_filename="hil.a2l", comm_channel=DEV.CAN1, driver_type=DRIVER.XCP, module_name="ECU", connect=True, enable_cache=None)

  Initialize a generic CANape Device connection.
- release (self, verbosity=2)

  Manually release the module/device handle in CANape.
- __del__ (self)

  De-initialize a generic CANape Device connection.
- __str__ (self)

  Get a string representation of the device.
- getModuleName (self, verbose=False)

  Get the current module name for the active module (device) handle.
- getModuleHandle (self)

  Get the current module handle.
- getCommunicationType (self, verbose=False)

  Get the current communication/driver type for the active handle.
- runScript (self, script_command, is_file_name=False)

  Execute a CAPL script or script command via CANapeAPI.
- isOnline (self)

  Get online state of current device.
- goOnline (self, retry_timeout=10000)

  Switch the CANape device connection to online mode.
- goOffline (self, retry_timeout=10000)

  Switch the CANape device connection to offline mode.
- getECUTasks (self)

  Get a dictionary containing available measurement tasks (as defined in the ECU).

Public Attributes

- verbose

  Verbosity of debug/error messages.

Static Public Attributes

- cnf_override_createmodule_version = None

  config: override used Asap3CreateModule variant.

Protected Member Functions

- _setECUState (self, state=TYPE_SWITCH.ONLINE, retry_timeout=0)

  Switch the CANape device "online" state, that is, enable or disable the current connection (ccp/xcp/...).
- _getECUState (self)

  Get current online/offline state of the CANape device.

Protected Attributes

- _comm_channel
- _driver_type
- _db_filename
- _module_name
- _asap3
- _module_handle
- _api_log_level
- _logApiCallStart
- _logApiCallEnd

7.45.1 Detailed Description

Base class for CANape devices with generic/common/basic functionality.

7.45.2 Constructor & Destructor Documentation

7.45.2.1 __init__()

```
ttk_tools.vector._canapeapi.BaseDevice.__init__ (
              self,
              canape_asap3,
              db_filename = "hil.a2l",
              comm_channel = DEV.CAN1,
              driver_type = DRIVER.XCP,
              module_name = "ECU",
              connect = True,
              enable_cache = None )
```

Initialize a generic CANape Device connection.

Parameters

canape_asap3	a CANapeASAP3 instance (application interface)
db_filename	filename of an appropriate database (.a2l, .dbc, ...)
comm_channel	communication channel (usually the CAN channel #), see canapeapi_common.DEV for device channel constants
driver_type	device driver to use, see canapeapi_common.DRIVER for driver constants
module_name	name of device/module in CANape's device list
connect	Device connection behavior: True: immediately connect DUT after device has been created. Device will start in state "online" False: create device but remain in state "offline" if DUT cannot be connected at this time.
enable_cache	None: ignore parameter, use default True: enable caching False: disable caching Available since CANape 10, see Asap3CreateModule3

Note

Parameter `connect` Setting `connect` to False permits attaching/creating a device that is currently not active (e.g. DUT is still shut down), that is, the device-init will not fail right away with error message "No response from ECU (attach Asap2 failed)".

An **inactive** DUT can be successfully attached and start out in state "offline". See isOnline() and goOnline(). However, if the DUT is **active** after all, the device's online state may end up in state "online" even if `connect` has been set to False.

Reimplemented in ttk_tools.vector._canapeapi.CANapeDevice, ttk_tools.vector.canapeapi.CANapeDevice, ttk_tools.vector._canapeapi.CANapeDiag, and ttk_tools.vector.canapeapi.CANapeDiag.

7.45.2.2 __del__()

```
ttk_tools.vector._canapeapi.BaseDevice.__del__ (
              self )
```

De-initialize a generic CANape Device connection.

7.45.3 Member Function Documentation

7.45.3.1 __str__()

```
ttk_tools.vector._canapeapi.BaseDevice.__str__ (
              self )
```

Get a string representation of the device.

Returns

an info string.

Reimplemented in ttk_tools.vector._canapeapi.CANapeDevice, and ttk_tools.vector._canapeapi.CANapeDiag.

7.45.3.2 _getECUState()

```
ttk_tools.vector._canapeapi.BaseDevice._getECUState (
              self )  [protected]
```

Get current online/offline state of the CANape device.

Returns

a `TAsap3ECUState`, i.e. 0 if device is currently online or 1 if it is offline. See canapeapi_common.TYPE↩ _SWITCH.

7.45.3.3 _setECUState()

```
ttk_tools.vector._canapeapi.BaseDevice._setECUState (
              self,
              state = TYPE_SWITCH.ONLINE,
              retry_timeout = 0 )  [protected]
```

Switch the CANape device "online" state, that is, enable or disable the current connection (ccp/xcp/...).

Parameters

state	0: switch online, 1: switch offline (a little bit counter- intuitive, direct complains to ASAM) see TYPE_SWITCH.ONLINE, TYPE_SWITCH.OFFLINE
retry_timeout	timeout in ms; if specified, failed attempts will be retried until timeout is reached.

Returns

a list of status messages, one entry per failed retry attempt

7.45.3.4 getCommunicationType()

```
ttk_tools.vector._canapeapi.BaseDevice.getCommunicationType (
              self,
              verbose = False )
```

Get the current communication/driver type for the active handle.

7.45.3.5 getECUTasks()

```
ttk_tools.vector._canapeapi.BaseDevice.getECUTasks (
              self )
```

Get a dictionary containing available measurement tasks (as defined in the ECU).

Returns

a task dictionary with entries like:
<task name/descr.>: {
"id": <internal task id>,
"cycle_time": <cycle time in ms, 0 if n/a>
}

Reimplemented in ttk_tools.vector.canapeapi.CANapeDevice.

7.45.3.6 getModuleHandle()

```
ttk_tools.vector._canapeapi.BaseDevice.getModuleHandle (
              self )
```

Get the current module handle.

7.45.3.7 getModuleName()

```
ttk_tools.vector._canapeapi.BaseDevice.getModuleName (
              self,
              verbose = False )
```

Get the current module name for the active module (device) handle.

Parameters

verbose	if True, the name will also be printed to stdout

Returns

the name of the module/device.

7.45.3.8 goOffline()

```
ttk_tools.vector._canapeapi.BaseDevice.goOffline (
              self,
              retry_timeout = 10000 )
```

Switch the CANape device connection to offline mode.

Parameters

retry_timeout	[ms], if > 0, repeated attempts to set the ecu state will be made until timeout occurs

Returns

a list of testresult entries containing info about retried attempts.

Reimplemented in ttk_tools.vector._canapeapi.CANapeDiag, ttk_tools.vector.canapeapi.CANapeDevice, and ttk_tools.vector.canapeapi.CANapeDiag.

7.45.3.9 goOnline()

```
ttk_tools.vector._canapeapi.BaseDevice.goOnline (
              self,
              retry_timeout = 10000 )
```

Switch the CANape device connection to online mode.

Parameters

retry_timeout	[ms], if > 0, repeated attempts to set the ecu state will be made until timeout occurs

Returns

a list of testresult entries containing info about retried attempts.

Reimplemented in ttk_tools.vector._canapeapi.CANapeDiag, ttk_tools.vector.canapeapi.CANapeDevice, and ttk_tools.vector.canapeapi.CANapeDiag.

7.45.3.10 isOnline()

```
ttk_tools.vector._canapeapi.BaseDevice.isOnline (
              self )
```

Get online state of current device.

Returns

True if the device is online, otherwise False.

Reimplemented in ttk_tools.vector.canapeapi.CANapeDevice, and ttk_tools.vector.canapeapi.CANapeDiag.

7.45.3.11 release()

```
ttk_tools.vector._canapeapi.BaseDevice.release (
              self,
              verbosity = 2 )
```

Manually release the module/device handle in CANape.

This will normally be called during **del**, but sometimes it might be advantageous to release the handle a defined time (and not just implicitly when the destructor gets called eventually).

Parameters

verbosity	verbosity of stdout/log output: -1: do not show anything if handle is already gone (for calls in destructor) 0: errors only 1: Asap3Exit status output 2: also show "nothing to do" notice if handle is already gone

Returns

True if successful (or already released), otherwise False.

Reimplemented in ttk_tools.vector.canapeapi.CANapeDevice, and ttk_tools.vector.canapeapi.CANapeDiag.

7.45.3.12 runScript()

```
ttk_tools.vector._canapeapi.BaseDevice.runScript (
              self,
              script_command,
              is_file_name = False )
```

Execute a CAPL script or script command via CANapeAPI.

Note

This uses the "classic" `Asap3ExecuteScript` API function which does not provide any script status or return values.

New extended script functions are available since CANape version 9. The classic approach is sufficient for simple cases, though.

Parameters

script_command	CAPL command or name of CAPL script file to execute. A CAPL command can also be the name of a custom "project function" that has been defined in the current workspace.
is_file_name	True if `script_command` contains the name of a script file to be run, False if it directly contains a CAPL command.

Example:

```
# multiple commands can be added in sequence
device.runScript('Speak("Hello World"); Write("Done.");')

# some CAPL commands/methods have to be called on an existing
# device object
device.runScript('%s.SaveDatabase(0);'%(device.getModuleName())

# default location of script files is the folder of the current
# CANape workspace
device.runScript('capl_script.scr', is_file_name=True)
```

Note

The executed script or script command might still be running when this function returns.

Returns

True if script execution was successful (though not necessarily completed), otherwise False.

Reimplemented in ttk_tools.vector.canapeapi.CANapeDevice, and ttk_tools.vector.canapeapi.CANapeDiag.

7.45.4 Member Data Documentation

7.45.4.1 _api_log_level

`ttk_tools.vector._canapeapi.BaseDevice._api_log_level [protected]`

7.45.4.2 _asap3

`ttk_tools.vector._canapeapi.BaseDevice._asap3 [protected]`

7.45.4.3 _comm_channel

`ttk_tools.vector._canapeapi.BaseDevice._comm_channel [protected]`

7.45.4.4 _db_filename

`ttk_tools.vector._canapeapi.BaseDevice._db_filename [protected]`

7.45.4.5 _driver_type

`ttk_tools.vector._canapeapi.BaseDevice._driver_type [protected]`

7.45.4.6 _logApiCallEnd

`ttk_tools.vector._canapeapi.BaseDevice._logApiCallEnd [protected]`

7.45.4.7 _logApiCallStart

`ttk_tools.vector._canapeapi.BaseDevice._logApiCallStart [protected]`

7.45.4.8 _module_handle

`ttk_tools.vector._canapeapi.BaseDevice._module_handle [protected]`

7.45.4.9 _module_name

`ttk_tools.vector._canapeapi.BaseDevice._module_name [protected]`

7.45.4.10 cnf_override_createmodule_version

`ttk_tools.vector._canapeapi.BaseDevice.cnf_override_createmodule_version = None [static]`

config: override used Asap3CreateModule variant.

By default the highest available variant will be used:

- Asap3CreateModule - adds moduleName and driverType (available at least since CANape 6)
- Asap3CreateModule2 - adds flag `goOnline` (available at least since CANape 6, marked as "For internal use only" until CANape 9)
- Asap3CreateModule3 - adds flag `enablecache` (available since CANape 10)

(before Asap3CreateModule there was a function Asap3AttachAsap2, but it only supported CCP)

Setting this member will force a specific version to be used instead:

- 1: use Asap3CreateModule (connect and enable_cache not available)
- 2: use Asap3CreateModule2 (enable_cache not available)
- 3: use Asap3CreateModule3
- default: use "best" available function

7.45.4.11 verbose

`ttk_tools.vector._canapeapi.BaseDevice.verbose`

Verbosity of debug/error messages.

True: show messages
False: skip (most) messages

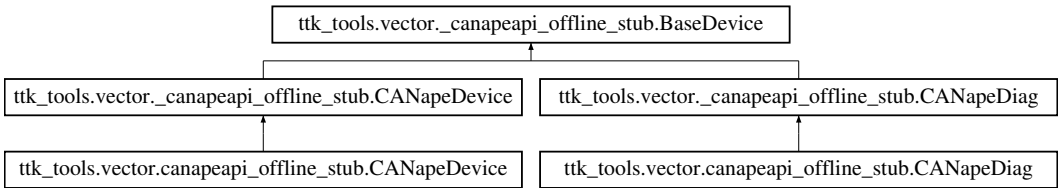
The documentation for this class was generated from the following file:

- ttk_tools/vector/_canapeapi.py (v7.3.3)

7.46 ttk_tools.vector._canapeapi_offline_stub.BaseDevice Class Reference

Base class (stub) for CANape devices with generic/common/basic functionality.

Inheritance diagram for ttk_tools.vector._canapeapi_offline_stub.BaseDevice:



Public Member Functions

- __init__ (self, canape_asap3, db_filename="hil.a2l", comm_channel=DEV.CAN1, driver_type=DRIVER.XCP, module_name="ECU", connect=True, enable_cache=None)

  Initialize a generic CANape Device connection.
- release (self, verbosity=2)

  Manually release the module/device handle in CANape.
- __del__ (self)

  De-initialize a generic CANape Device connection.
- __str__ (self)

  Get a string representation of the device.
- getModuleName (self, verbose=False)

  Get the current module name for the active module (device) handle.
- getModuleHandle (self)

  Get the current module handle.
- getCommunicationType (self, verbose=False)

  Get the current communication/driver type for the active handle.
- runScript (self, script_command, is_file_name=False)

  Execute a CAPL script or script command via CANapeAPI.
- isOnline (self)

  Get online state of current device.
- goOnline (self, retry_timeout=10000)

  Switch the CANape device connection to online mode.
- goOffline (self, retry_timeout=10000)

  Switch the CANape device connection to offline mode.
- getECUTasks (self)

  Get a dictionary containing available measurement tasks (as defined in the ECU).

Public Attributes

- verbose

  Verbosity of debug/error messages.

Static Public Attributes

- cnf_override_createmodule_version = None

  config: override used Asap3CreateModule variant.

Protected Attributes

- _comm_channel
- _driver_type
- _db_filename
- _module_name
- _asap3
- _module_handle
- _device_online
- _context

7.46.1 Detailed Description

Base class (stub) for CANape devices with generic/common/basic functionality.

7.46.2 Constructor & Destructor Documentation

7.46.2.1 __init__()

```
ttk_tools.vector._canapeapi_offline_stub.BaseDevice.__init__ (
              self,
              canape_asap3,
              db_filename = "hil.a2l",
              comm_channel = DEV.CAN1,
              driver_type = DRIVER.XCP,
              module_name = "ECU",
              connect = True,
              enable_cache = None )
```

Initialize a generic CANape Device connection.

Parameters

canape_asap3	a CANapeASAP3 instance (application interface)
db_filename	filename of an appropriate database (.a2l, .dbc, ...)
comm_channel	communication channel (usually the CAN channel #), see canapeapi_common.DEV for device channel constants
driver_type	device driver to use, see canapeapi_common.DRIVER for driver constants
module_name	name of device/module in CANape's device list
connect	Device connection behavior: True: immediately connect DUT after device has been created. Device will start in state "online" False: create device but remain in state "offline" if DUT cannot be connected at this time.
enable_cache	None: ignore parameter, use default True: enable caching False: disable caching Available since CANape 10, see Asap3CreateModule3

Note

Parameter `connect` Setting `connect` to False permits attaching/creating a device that is currently not active (e.g. DUT is still shut down), that is, the device-init will not fail right away with error message "No response from ECU (attach Asap2 failed)".

An **inactive** DUT can be successfully attached and start out in state "offline". See isOnline() and goOnline(). However, if the DUT is **active** after all, the device's online state may end up in state "online" even if `connect` has been set to False.

Reimplemented in ttk_tools.vector._canapeapi_offline_stub.CANapeDevice, ttk_tools.vector.canapeapi_offline_stub.CANapeDevice, ttk_tools.vector._canapeapi_offline_stub.CANapeDiag, and ttk_tools.vector.canapeapi_offline_stub.CANapeDiag.

7.46.2.2 __del__()

```
ttk_tools.vector._canapeapi_offline_stub.BaseDevice.__del__ (
              self )
```

De-initialize a generic CANape Device connection.

7.46.3 Member Function Documentation

7.46.3.1 __str__()

```
ttk_tools.vector._canapeapi_offline_stub.BaseDevice.__str__ (
              self )
```

Get a string representation of the device.

Returns

an info string.

Reimplemented in ttk_tools.vector._canapeapi_offline_stub.CANapeDevice, and ttk_tools.vector._canapeapi_offline_stub.CANapeDia

7.46.3.2 getCommunicationType()

```
ttk_tools.vector._canapeapi_offline_stub.BaseDevice.getCommunicationType (
              self,
              verbose = False )
```

Get the current communication/driver type for the active handle.

7.46.3.3 getECUTasks()

```
ttk_tools.vector._canapeapi_offline_stub.BaseDevice.getECUTasks (
              self )
```

Get a dictionary containing available measurement tasks (as defined in the ECU).

Returns

a task dictionary with entries like:
<task name/descr.>: {
"id": <internal task id>,
"cycle_time": <cycle time in ms, 0 if n/a>
}

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.CANapeDevice.

7.46.3.4 getModuleHandle()

```
ttk_tools.vector._canapeapi_offline_stub.BaseDevice.getModuleHandle (
              self )
```

Get the current module handle.

7.46.3.5 getModuleName()

```
ttk_tools.vector._canapeapi_offline_stub.BaseDevice.getModuleName (
              self,
              verbose = False )
```

Get the current module name for the active module (device) handle.

Parameters

verbose	if True, the name will also be printed to stdout

Returns

the name of the module/device.

7.46.3.6 goOffline()

```
ttk_tools.vector._canapeapi_offline_stub.BaseDevice.goOffline (
              self,
              retry_timeout = 10000 )
```

Switch the CANape device connection to offline mode.

Parameters

retry_timeout	[ms], if > 0, repeated attempts to set the ecu state will be made until timeout occurs

Returns

a list of testresult entries containing info about retried attempts.

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.CANapeDevice, and ttk_tools.vector.canapeapi_offline_stub.CANapeDiag.

7.46.3.7 goOnline()

```
ttk_tools.vector._canapeapi_offline_stub.BaseDevice.goOnline (
              self,
              retry_timeout = 10000 )
```

Switch the CANape device connection to online mode.

Parameters

retry_timeout	[ms], if > 0, repeated attempts to set the ecu state will be made until timeout occurs

Returns

a list of testresult entries containing info about retried attempts.

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.CANapeDevice, and ttk_tools.vector.canapeapi_offline_stub.CANapeDiag.

7.46.3.8 isOnline()

```
ttk_tools.vector._canapeapi_offline_stub.BaseDevice.isOnline (
              self )
```

Get online state of current device.

Returns

True if the device is online, otherwise False.

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.CANapeDevice, and ttk_tools.vector.canapeapi_offline_stub.CANapeDiag.

7.46.3.9 release()

```
ttk_tools.vector._canapeapi_offline_stub.BaseDevice.release (
              self,
              verbosity = 2 )
```

Manually release the module/device handle in CANape.

This will normally be called during **del**, but sometimes it might be advantageous to release the handle a defined time (and not just implicitly when the destructor gets called eventually).

Parameters

verbosity	verbosity of stdout/log output: -1: do not show anything if handle is already gone (for calls in destructor) 0: errors only 1: Asap3Exit status output 2: also show "nothing to do" notice if handle is already gone

Returns

True if successful (or already released), otherwise False.

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.CANapeDevice, and ttk_tools.vector.canapeapi_offline_stub.CANapeDiag.

7.46.3.10 runScript()

```
ttk_tools.vector._canapeapi_offline_stub.BaseDevice.runScript (
              self,
              script_command,
              is_file_name = False )
```

Execute a CAPL script or script command via CANapeAPI.

Note

This uses the "classic" `Asap3ExecuteScript` API function which does not provide any script status or return values.

New extended script functions are available since CANape version 9. The classic approach is sufficient for simple cases, though.

Parameters

script_command	CAPL command or name of CAPL script file to execute. A CAPL command can also be the name of a custom "project function" that has been defined in the current workspace.
is_file_name	True if `script_command` contains the name of a script file to be run, False if it directly contains a CAPL command.

Example:

```
# multiple commands can be added in sequence
device.runScript('Speak("Hello World"); Write("Done.");')

# some CAPL commands/methods have to be called on an existing
# device object
device.runScript('%s.SaveDatabase(0);'%(device.getModuleName())

# default location of script files is the folder of the current
# CANape workspace
device.runScript('capl_script.scr', is_file_name=True)
```

Note

The executed script or script command might still be running when this function returns.

Returns

True if script execution was successful (though not necessarily completed), otherwise False. Stub implementation will always return True.

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.CANapeDevice, and ttk_tools.vector.canapeapi_offline_stub.CANapeDiag.

7.46.4 Member Data Documentation

7.46.4.1 _asap3

ttk_tools.vector._canapeapi_offline_stub.BaseDevice._asap3 [protected]

7.46.4.2 _comm_channel

ttk_tools.vector._canapeapi_offline_stub.BaseDevice._comm_channel [protected]

7.46.4.3 _context

ttk_tools.vector._canapeapi_offline_stub.BaseDevice._context [protected]

7.46.4.4 _db_filename

ttk_tools.vector._canapeapi_offline_stub.BaseDevice._db_filename [protected]

7.46.4.5 _device_online

ttk_tools.vector._canapeapi_offline_stub.BaseDevice._device_online [protected]

7.46.4.6 _driver_type

ttk_tools.vector._canapeapi_offline_stub.BaseDevice._driver_type [protected]

7.46.4.7 _module_handle

ttk_tools.vector._canapeapi_offline_stub.BaseDevice._module_handle [protected]

7.46.4.8 _module_name

ttk_tools.vector._canapeapi_offline_stub.BaseDevice._module_name [protected]

7.46.4.9 cnf_override_createmodule_version

```
ttk_tools.vector._canapeapi_offline_stub.BaseDevice.cnf_override_createmodule_version = None
[static]
```

config: override used Asap3CreateModule variant.

By default the highest available variant will be used:

- Asap3CreateModule - adds moduleName and driverType (available at least since CANape 6)
- Asap3CreateModule2 - adds flag `goOnline` (available at least since CANape 6, marked as "For internal use only" until CANape 9)
- Asap3CreateModule3 - adds flag `enablecache` (available since CANape 10)

(before Asap3CreateModule there was a function Asap3AttachAsap2, but it only supported CCP)

Setting this member will force a specific version to be used instead:

- 1: use Asap3CreateModule (connect and enable_cache not available)
- 2: use Asap3CreateModule2 (enable_cache not available)
- 3: use Asap3CreateModule3
- default: use "best" available function

7.46.4.10 verbose

```
ttk_tools.vector._canapeapi_offline_stub.BaseDevice.verbose
```

Verbosity of debug/error messages.

True: show messages
False: skip (most) messages

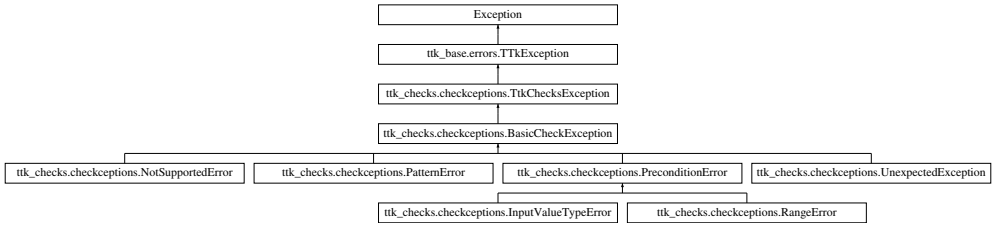
The documentation for this class was generated from the following file:

- ttk_tools/vector/_canapeapi_offline_stub.py (v4.7.1)

7.47 ttk_checks.checkceptions.BasicCheckException Class Reference

Base Exception for basic checks.

Inheritance diagram for ttk_checks.checkceptions.BasicCheckException:



7.47.1 Detailed Description

Base Exception for basic checks.

A value conversion failed or is not plausible.

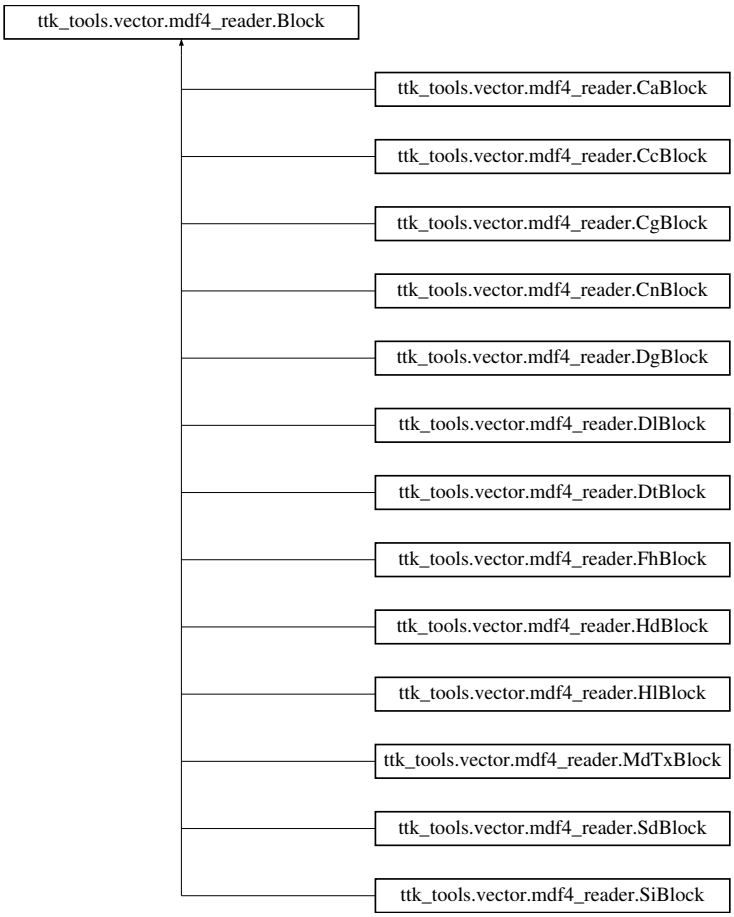
The documentation for this class was generated from the following file:

- ttk_checks/checkceptions.py (v2.0)

7.48 ttk_tools.vector.mdf4_reader.Block Class Reference

A generic block with header, links and data sections (for all blocks except for special case IDBLOCK)

Inheritance diagram for ttk_tools.vector.mdf4_reader.Block:



Public Member Functions

- __init__ (self, file_handle, pos=None, expected_id=None, expected_links=None)

  Read a block from file_handle, starting at the current or supplied position.
- freeRawDataBuffer (self)

Protected Attributes

- _orig_pos
- _header
- _links

  A list of LINKs in order of appearance.
- _data

7.48.1 Detailed Description

A generic block with header, links and data sections (for all blocks except for special case IDBLOCK)

7.48.2 Constructor & Destructor Documentation

7.48.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.Block.__init__ (
              self,
              file_handle,
              pos = None,
              expected_id = None,
              expected_links = None )
```

Read a block from file_handle, starting at the current or supplied position.

Note: File pointer position will be modified

Parameters

file_handle	open file handle from which to read raw data
pos	file position to seek() before reading. None: use current position
expected_id	expected ID if this block, see BlockHeader
expected_links	expected number of LINK entries for this block or None to skip sanity check

Info:

Block contents will be placed in member variables: _header - a BlockHeader instance _links - a list of LINK offsets/positions (in order of appearance) _data - raw payload data as a byte string

_orig_pos - original start position of block (in case we have to have to match LINKs later)

Reimplemented in ttk_tools.vector.mdf4_reader.MdTxBlock, ttk_tools.vector.mdf4_reader.HdBlock, ttk_tools.vector.mdf4_reader.FhBl
ttk_tools.vector.mdf4_reader.DgBlock, ttk_tools.vector.mdf4_reader.CgBlock, ttk_tools.vector.mdf4_reader.CnBlock,
ttk_tools.vector.mdf4_reader.CaBlock, ttk_tools.vector.mdf4_reader.DtBlock, ttk_tools.vector.mdf4_reader.SdBlock,
ttk_tools.vector.mdf4_reader.DlBlock, ttk_tools.vector.mdf4_reader.HlBlock, ttk_tools.vector.mdf4_reader.SiBlock,
and ttk_tools.vector.mdf4_reader.CcBlock.

7.48.3 Member Function Documentation

7.48.3.1 freeRawDataBuffer()

```
ttk_tools.vector.mdf4_reader.Block.freeRawDataBuffer (
              self )
```

7.48.4 Member Data Documentation

7.48.4.1 _data

```
ttk_tools.vector.mdf4_reader.Block._data  [protected]
```

7.48.4.2 _header

```
ttk_tools.vector.mdf4_reader.Block._header  [protected]
```

7.48.4.3 _links

```
ttk_tools.vector.mdf4_reader.Block._links  [protected]
```

A list of LINKs in order of appearance.

7.48.4.4 _orig_pos

```
ttk_tools.vector.mdf4_reader.Block._orig_pos  [protected]
```

The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.49 ttk_tools.vector.mdf4_reader.BlockHeader Class Reference

String encoding used in an MDF file is UTF-8 (applies to TXBLOCK and MDBLOCK data)

Public Member Functions

- __init__ (self, data, expected_id=None)

  Unpack the supplied 24-byte block header.
- getRawData (self)
- __repr__ (self)
- getDataLength (self)

  Get byte-length of data in block.

Public Attributes

- id
- length
- link_count

Static Public Attributes

- fmt = struct.Struct("< 4s 4x Q Q")

7.49.1 Detailed Description

String encoding used in an MDF file is UTF-8 (applies to TXBLOCK and MDBLOCK data)

- For strings in IDBLOCK and for the block identifiers, always single byte character (SBC) encoding is used (ISO-8859-1 Latin character set).
  - For string signal values stored in the DTBLOCK, RDBLOCK or SDBLOCK, different encodings can be specified by the channel data type or the MIME type.

  Byte order in an MDF file is Little Endian (Intel Byte order)
  - All values (i.e. integers, floats) have to be interpreted accordingly. The only exception is that signal values stored in DTBLOCK, RDBLOCK or SDBLOCK, or that attachments stored as embedded data in ATBLOCK could use a different Byte order. In this case, the Byte order specified by the channel data type or by the MIME type overrules the default Byte order.

  The start address of a block must always be a multiple of 8 Bytes (8-Byte alignment).
  - As a consequence, if the length of a block is not a multiple of 8, there will be a gap between this and the next block. However, this can only happen for blocks which do not automatically have a length that is a multiple of 8 Byte, e.g. for DTBLOCK, RDBLOCK, SDBLOCK, ATBLOCK, TXBLOCK or MDBLOCK.
  - Note that for records in DTBLOCK or RDBLOCK or for the VLSD values in SDBLOCK only a 1-Byte alignment is used.
  - To achieve 8-Byte alignment of block structures, reserved fields are inserted. Writing an MDF file, all reserved Bytes must be filled with zeros.

Common header for all blocks (except IDBLOCK), contains information about type, length and number of links in the block's attached link section. The header has a fixed length of 24 Bytes.

7.49.2 Constructor & Destructor Documentation

7.49.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.BlockHeader.__init__ (
              self,
              data,
              expected_id = None )
```

Unpack the supplied 24-byte block header.

Parameters

data	24 bytes of block header data
expected_id	Expected 4-byte block id for sanity check or a list/tuple of multiple expected ids. None: skip check

7.49.3 Member Function Documentation

7.49.3.1 __repr__()

```
ttk_tools.vector.mdf4_reader.BlockHeader.__repr__ (
              self )
```

7.49.3.2 getDataLength()

```
ttk_tools.vector.mdf4_reader.BlockHeader.getDataLength (
              self )
```

Get byte-length of data in block.

7.49.3.3 getRawData()

```
ttk_tools.vector.mdf4_reader.BlockHeader.getRawData (
              self )
```

7.49.4 Member Data Documentation

7.49.4.1 fmt

```
ttk_tools.vector.mdf4_reader.BlockHeader.fmt = struct.Struct("< 4s 4x Q Q")  [static]
```

7.49.4.2 id

```
ttk_tools.vector.mdf4_reader.BlockHeader.id
```

7.49.4.3 length

```
ttk_tools.vector.mdf4_reader.BlockHeader.length
```

7.49.4.4 link_count

```
ttk_tools.vector.mdf4_reader.BlockHeader.link_count
```

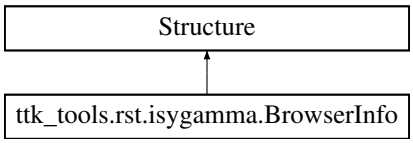
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.50 ttk_tools.rst.isygamma.BrowserInfo Class Reference

typedef struct gc_BrowserInfo.

Inheritance diagram for ttk_tools.rst.isygamma.BrowserInfo:



Classes

- class BrowsingInfoUnion
  container for the various "info" flavors.

Static Protected Attributes

- _fields_

7.50.1 Detailed Description

typedef struct gc_BrowserInfo.

See GaAPI.getPVs

7.50.2 Member Data Documentation

7.50.2.1 _fields_

`ttk_tools.rst.isygamma.BrowserInfo._fields_ [static], [protected]`

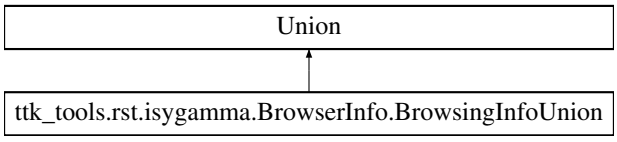
The documentation for this class was generated from the following file:

- ttk_tools/rst/isygamma.py (v1.6)

7.51 ttk_tools.rst.isygamma.BrowserInfo.BrowsingInfoUnion Class Reference

container for the various "info" flavors.

Inheritance diagram for ttk_tools.rst.isygamma.BrowserInfo.BrowsingInfoUnion:



Static Protected Attributes

- list _fields_

7.51.1 Detailed Description

container for the various "info" flavors.

7.51.2 Member Data Documentation

7.51.2.1 _fields_

```
list ttk_tools.rst.isygamma.BrowserInfo.BrowsingInfoUnion._fields_  [static], [protected]
```

Initial value:

```
=  [
            ("system",        Level1.SystemInfo),
            ("node",          Level2.NodeInfo),
            ("action",        Level2.ActionInfo),         # (not used here)
            ("loop",          Level2.LoopInfo),           # (not used here)
            ("task",          Level2.TaskInfo),           # (not used here)
            ("io",            Level2.IOInfo),             # (not used here)
            ("scheduler",     Level2.SchedulerInfo),      # (not used here)
            ("memory",        Level3.MemoryInfo),
            ("ioChannel",     Level3.IOChannelInfo),      # (not used here)
            ("schedulerSlot", Level3.SchedulerSlotInfo),  # (not used here)
            ("group",         Level4.GroupInfo),
            ("pvArray",       Level5.PVArrayInfo),
            ("fifo",          Level5.FifoInfo),           # (not used here)
            ("string",        c_char * 256),
            # => remaining elements added to make sure we'll end up at the             #    correct maximum
      size, though _TaskInfo's size would probably
            #    be sufficient.
        ]
```

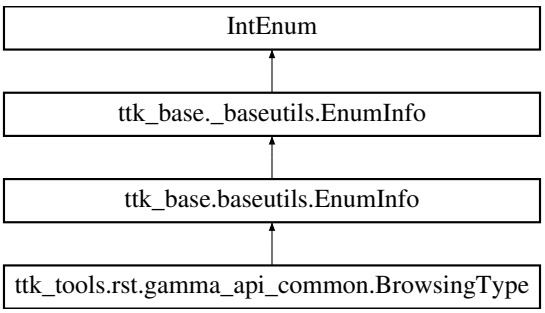
The documentation for this class was generated from the following file:

- ttk_tools/rst/isygamma.py (v1.6)

7.52 ttk_tools.rst.gamma_api_common.BrowsingType Class Reference

Wrapper for enum gc_BrowsingType (BROWSING_TYPE_∗).

Inheritance diagram for ttk_tools.rst.gamma_api_common.BrowsingType:



Static Public Attributes

- int SYSTEM = 0
- int NODE = 1
- int ACTION = 2
- int LOOP = 3
- int TASK = 4
- int IO = 5
- int SCHEDULER = 6
- int MEMORY = 7
- int IO_INPUT = 8
- int IO_OUTPUT = 9
- int SCHEDULER_SLOT = 10
- int GROUP = 11
- int PV_ARRAY = 12
- int FIFO = 13
- int NODE_ALIAS = 14
- int LOOP_ALIAS = 15
- int ACTION_ALIAS = 16
- int TASK_ALIAS = 17
- int SCHEDULER_ALIAS = 18
- int PV_ARRAY_ALIAS = 19
- int FIFO_ALIAS = 20
- int GROUP_ALIAS = 21
- int MEMORY_ALIAS = 22
- int SCHEDULER_SLOT_ALIAS = 23

7.52.1 Detailed Description

Wrapper for enum gc_BrowsingType (BROWSING_TYPE_∗).

See GaAPI.getPVs.

7.52.2 Member Data Documentation

7.52.2.1 ACTION

```
int ttk_tools.rst.gamma_api_common.BrowsingType.ACTION = 2  [static]
```

7.52.2.2 ACTION_ALIAS

```
int ttk_tools.rst.gamma_api_common.BrowsingType.ACTION_ALIAS = 16  [static]
```

7.52.2.3 FIFO

```
int ttk_tools.rst.gamma_api_common.BrowsingType.FIFO = 13  [static]
```

7.52.2.4 FIFO_ALIAS

```
int ttk_tools.rst.gamma_api_common.BrowsingType.FIFO_ALIAS = 20  [static]
```

7.52.2.5 GROUP

```
int ttk_tools.rst.gamma_api_common.BrowsingType.GROUP = 11  [static]
```

7.52.2.6 GROUP_ALIAS

```
int ttk_tools.rst.gamma_api_common.BrowsingType.GROUP_ALIAS = 21  [static]
```

7.52.2.7 IO

```
int ttk_tools.rst.gamma_api_common.BrowsingType.IO = 5  [static]
```

7.52.2.8 IO_INPUT

```
int ttk_tools.rst.gamma_api_common.BrowsingType.IO_INPUT = 8  [static]
```

7.52.2.9 IO_OUTPUT

```
int ttk_tools.rst.gamma_api_common.BrowsingType.IO_OUTPUT = 9  [static]
```

7.52.2.10 LOOP

```
int ttk_tools.rst.gamma_api_common.BrowsingType.LOOP = 3  [static]
```

7.52.2.11 LOOP_ALIAS

```
int ttk_tools.rst.gamma_api_common.BrowsingType.LOOP_ALIAS = 15  [static]
```

7.52.2.12 MEMORY

```
int ttk_tools.rst.gamma_api_common.BrowsingType.MEMORY = 7  [static]
```

7.52.2.13 MEMORY_ALIAS

```
int ttk_tools.rst.gamma_api_common.BrowsingType.MEMORY_ALIAS = 22  [static]
```

7.52.2.14 NODE

```
int ttk_tools.rst.gamma_api_common.BrowsingType.NODE = 1  [static]
```

7.52.2.15 NODE_ALIAS

```
int ttk_tools.rst.gamma_api_common.BrowsingType.NODE_ALIAS = 14  [static]
```

7.52.2.16 PV_ARRAY

```
int ttk_tools.rst.gamma_api_common.BrowsingType.PV_ARRAY = 12  [static]
```

7.52.2.17 PV_ARRAY_ALIAS

```
int ttk_tools.rst.gamma_api_common.BrowsingType.PV_ARRAY_ALIAS = 19  [static]
```

7.52.2.18 SCHEDULER

```
int ttk_tools.rst.gamma_api_common.BrowsingType.SCHEDULER = 6  [static]
```

7.52.2.19 SCHEDULER_ALIAS

```
int ttk_tools.rst.gamma_api_common.BrowsingType.SCHEDULER_ALIAS = 18  [static]
```

7.52.2.20 SCHEDULER_SLOT

```
int ttk_tools.rst.gamma_api_common.BrowsingType.SCHEDULER_SLOT = 10  [static]
```

7.52.2.21 SCHEDULER_SLOT_ALIAS

```
int ttk_tools.rst.gamma_api_common.BrowsingType.SCHEDULER_SLOT_ALIAS = 23  [static]
```

7.52.2.22 SYSTEM

```
int ttk_tools.rst.gamma_api_common.BrowsingType.SYSTEM = 0  [static]
```

7.52.2.23 TASK

```
int ttk_tools.rst.gamma_api_common.BrowsingType.TASK = 4  [static]
```

7.52.2.24 TASK_ALIAS

```
int ttk_tools.rst.gamma_api_common.BrowsingType.TASK_ALIAS = 17  [static]
```

The documentation for this class was generated from the following file:

- ttk_tools/rst/gamma_api_common.py (v1.5)

7.53 ttk_tools.dspace.sdmlib_offline_stub.BUFFER Class Reference

Instances of the class BUFFER are used to hold values of Stimulus Engine channels for later use.

Public Member Functions

- __init__ (self, initial_value=0)
  Define a buffer in the declaration section of a signal description.
- SET (self, value)
  Define a buffer with the current value of a SIGNAL object.

7.53.1 Detailed Description

Instances of the class BUFFER are used to hold values of Stimulus Engine channels for later use.

7.53.2 Constructor & Destructor Documentation

7.53.2.1 __init__()

```
ttk_tools.dspace.sdmlib_offline_stub.BUFFER.__init__ (
              self,
              initial_value = 0 )
```

Define a buffer in the declaration section of a signal description.

Parameters

initial_value	initial buffer value

7.53.3 Member Function Documentation

7.53.3.1 SET()

```
ttk_tools.dspace.sdmlib_offline_stub.BUFFER.SET (
              self,
              value )
```

Define a buffer with the current value of a SIGNAL object.

Parameters

value	SIGNAL object whose current value is stored in the buffer

The documentation for this class was generated from the following file:

- ttk_tools/dspace/sdmlib_offline_stub.py (v1.4)

7.54 ttk_bus._bus_signals_base.BusSignal Class Reference

Base signal container for a bus signal with a HIL-side model variable and relevant associated variables/values.

Inheritance diagram for ttk_bus._bus_signals_base.BusSignal:



Public Member Functions

- __init__ (self, context, identifier, cycle_time=0, debounce_time=0, signal_var=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  Bus signal initialization.
- __str__ (self)

  Return the actual signal var's string/identifier to make Capture setup more convenient.
- info (self)

  Get a current info string about the HIL signal variable.
- get (self)

  Get the current value of the HIL signal variable.
- set (self, value)

  Set a value to the HIL signal variable.
- getState (self)

  Get a textual state for the HIL signal variable's current data.
- setState (self, state)

  Set a textual state to the HIL signal variable.
- hasStateLookup (self)

  Return True if a state/value lookup table is defined for the HIL signal variable.
- getStateDescr (self, value=None, fallback=None)

  Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).
- hasFormattedRepr (self)

  Return True if a formatting for value representation is defined for the HIL signal variable.
- getFormattedValue (self, value=None)

  Get a formatted representation for the given value using the defined format string of the the HIL signal variable.

Public Attributes

- context
  context for "HilVar" variables
- signal_var
  Main bus signal variable on HIL-side.
- alias
  alias/name for bus signal.
- unit
  value unit for bus signal.
- descr
  bus signal description text
- cycle_time
  cycle time of signal in [ms] (that is, of the associated message or PDU)
- debounce_time
  minimum debounce delay for event-triggered signals in [ms]
- change_delay
  delay to wait after a value change [ms]
- timeout
  detection time for timeout errors [ms]
- recovery_time
  time for recovery/"wiedergut" checks [ms]

7.54.1 Detailed Description

Base signal container for a bus signal with a HIL-side model variable and relevant associated variables/values.

7.54.2 Constructor & Destructor Documentation

7.54.2.1 __init__()

```
ttk_bus._bus_signals_base.BusSignal.__init__ (
              self,
              context,
              identifier,
              cycle_time = 0,
              debounce_time = 0,
              signal_var = None,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Bus signal initialization.

Parameters

context	parent context for instantiating HIL variables. This gets mainly used in derived classes

Parameters

identifier	base signal identifier/name/path in the HIL model (typically this is read-only for Tx-signals and writable for Rx-signals). Derived classes typically use this identifier as starting point for deriving additional helper variables.
cycle_time	cycle time in [ms], 0 if event-triggered
debounce_time	minimum debounce delay time between events in [ms]
signal_var	(opt) an already initialized instance of Var (or most likely one of its derived classes) for the main signal value. If not supplied, a DummyVar will be used/ created as fall back
unit	(opt) unit of physical signal value (as string, mainly used for report entries), defaults to signal_var.unit.
alias	(opt) bus signal alias/name, defaults to signal_var.alias.
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions, defaults to signal_var.unit.
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_base._variables_base.ResettableContainer.

Reimplemented in ttk_bus._bus_signals_frbs.CrcFrBusSignal, ttk_bus._bus_signals_frbs.AcFrBusSignal, ttk_bus.bus_signals_frbs.Cr
ttk_bus.bus_signals_frbs.AcFrBusSignal, ttk_bus._bus_signals_ethbs.CrcEthSignal, ttk_bus._bus_signals_ethbs.AcEthSignal,
ttk_bus.bus_signals_ethbs.CrcEthSignal, ttk_bus.bus_signals_ethbs.AcEthSignal, ttk_bus._bus_signals_ethbs.RxEthBusSysSignal,
ttk_bus.bus_signals_ethbs.RxEthBusSysSignal, ttk_bus._bus_signals_frbs.FrBusSignal, ttk_bus._bus_signals_frbs.TxFrBusSignal,
ttk_bus._bus_signals_frbs.RxFrBusSignal, ttk_bus.bus_signals_frbs.TxFrBusSignal, ttk_bus.bus_signals_frbs.RxFrBusSignal,
ttk_bus._bus_signals_base.CrcBusSignal, ttk_bus._bus_signals_base.AcBusSignal, ttk_bus.bus_signals_base.CrcBusSignal,
ttk_bus.bus_signals_base.AcBusSignal, ttk_bus.bus_signals_base.TxBusSignal, ttk_bus.bus_signals_base.RxBusSignal,
ttk_bus._bus_signals_ethbs.EthBusSysSignal, ttk_bus._bus_signals_ethbs.EthSignal, ttk_bus._bus_signals_ethbs.RxEthSignal,
ttk_bus.bus_signals_ethbs.RxEthSignal, ttk_bus._bus_signals_canmmbs.CrcCanBusSignal, ttk_bus._bus_signals_canmmbs.AcCanE
ttk_bus.bus_signals_canmmbs.CrcCanBusSignal, ttk_bus.bus_signals_canmmbs.AcCanBusSignal, ttk_bus._bus_signals_canmmbs.
ttk_bus._bus_signals_canmmbs.RxCanBusSignal, ttk_bus.bus_signals_canmmbs.TxCanBusSignal, and ttk_bus.bus_signals_canmm

7.54.3 Member Function Documentation

7.54.3.1 __str__()

```
ttk_bus._bus_signals_base.BusSignal.__str__ (
            self )
```

Return the actual signal var's string/identifier to make Capture setup more convenient.

7.54.3.2 get()

```
ttk_bus._bus_signals_base.BusSignal.get (
              self )
```

Get the current value of the HIL signal variable.

Returns

the current value.

Reimplemented in ttk_bus.bus_signals_base.TxBusSignal, ttk_bus.bus_signals_base.RxBusSignal, ttk_bus.bus_signals_base.CrcBu
ttk_bus.bus_signals_base.AcBusSignal, ttk_bus.bus_signals_canmmbs.TxCanBusSignal, ttk_bus.bus_signals_canmmbs.RxCanBus
ttk_bus.bus_signals_canmmbs.CrcCanBusSignal, ttk_bus.bus_signals_canmmbs.AcCanBusSignal, ttk_bus.bus_signals_ethbs.RxEt
ttk_bus.bus_signals_ethbs.TxEthSignal, ttk_bus.bus_signals_ethbs.RxEthBusSysSignal, ttk_bus.bus_signals_ethbs.TxEthBusSysSig
ttk_bus.bus_signals_ethbs.CrcEthSignal, ttk_bus.bus_signals_ethbs.AcEthSignal, ttk_bus.bus_signals_frbs.TxFrBusSignal,
ttk_bus.bus_signals_frbs.RxFrBusSignal, ttk_bus.bus_signals_frbs.CrcFrBusSignal, and ttk_bus.bus_signals_frbs.AcFrBusSignal.

7.54.3.3 getFormattedValue()

```
ttk_bus._bus_signals_base.BusSignal.getFormattedValue (
              self,
              value = None )
```

Get a formatted representation for the given value using the defined format string of the the HIL signal variable.

An empty (or non-defined) format string will use the value's default string representation.

Parameters

value	value for which to get a string representation. If None, the last value (read or written) will be used instead.

Returns

a formatted string representation.

Reimplemented in ttk_bus.bus_signals_base.TxBusSignal, ttk_bus.bus_signals_base.RxBusSignal, ttk_bus.bus_signals_base.CrcBu
and ttk_bus.bus_signals_base.AcBusSignal.

7.54.3.4 getState()

```
ttk_bus._bus_signals_base.BusSignal.getState (
              self )
```

Get a textual state for the HIL signal variable's current data.

This uses the lookup values supplied in constructor. Fallback (if no matching state was found) will be the unmapped numeric value.

Returns

current state's description.

Reimplemented in ttk_bus.bus_signals_base.TxBusSignal, ttk_bus.bus_signals_base.RxBusSignal, ttk_bus.bus_signals_base.CrcBu
ttk_bus.bus_signals_base.AcBusSignal, ttk_bus.bus_signals_canmmbs.TxCanBusSignal, ttk_bus.bus_signals_canmmbs.RxCanBus
ttk_bus.bus_signals_canmmbs.CrcCanBusSignal, ttk_bus.bus_signals_canmmbs.AcCanBusSignal, ttk_bus.bus_signals_ethbs.RxEth
ttk_bus.bus_signals_ethbs.TxEthSignal, ttk_bus.bus_signals_ethbs.RxEthBusSysSignal, ttk_bus.bus_signals_ethbs.TxEthBusSysSig
ttk_bus.bus_signals_ethbs.CrcEthSignal, ttk_bus.bus_signals_ethbs.AcEthSignal, ttk_bus.bus_signals_frbs.TxFrBusSignal,
ttk_bus.bus_signals_frbs.RxFrBusSignal, ttk_bus.bus_signals_frbs.CrcFrBusSignal, and ttk_bus.bus_signals_frbs.AcFrBusSignal.

7.54.3.5 getStateDescr()

```
ttk_bus._bus_signals_base.BusSignal.getStateDescr (
            self,
            value = None,
            fallback = None )
```

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the last read value will be used instead.
fallback	fallback value to return if no match was found. Default: None

Returns

the state text if a match was found, otherwise the specified fallback value.

Reimplemented in ttk_bus.bus_signals_base.TxBusSignal, ttk_bus.bus_signals_base.RxBusSignal, ttk_bus.bus_signals_base.CrcBu
ttk_bus.bus_signals_base.AcBusSignal, ttk_bus.bus_signals_canmmbs.TxCanBusSignal, ttk_bus.bus_signals_canmmbs.RxCanBus
ttk_bus.bus_signals_canmmbs.CrcCanBusSignal, ttk_bus.bus_signals_canmmbs.AcCanBusSignal, ttk_bus.bus_signals_ethbs.RxEth
ttk_bus.bus_signals_ethbs.TxEthSignal, ttk_bus.bus_signals_ethbs.RxEthBusSysSignal, ttk_bus.bus_signals_ethbs.TxEthBusSysSig
ttk_bus.bus_signals_ethbs.CrcEthSignal, ttk_bus.bus_signals_ethbs.AcEthSignal, ttk_bus.bus_signals_frbs.TxFrBusSignal,
ttk_bus.bus_signals_frbs.RxFrBusSignal, ttk_bus.bus_signals_frbs.CrcFrBusSignal, and ttk_bus.bus_signals_frbs.AcFrBusSignal.

7.54.3.6 hasFormattedRepr()

```
ttk_bus._bus_signals_base.BusSignal.hasFormattedRepr (
            self )
```

Return True if a formatting for value representation is defined for the HIL signal variable.

Reimplemented in ttk_bus.bus_signals_base.TxBusSignal, ttk_bus.bus_signals_base.RxBusSignal, ttk_bus.bus_signals_base.CrcBu
and ttk_bus.bus_signals_base.AcBusSignal.

7.54.3.7 hasStateLookup()

```
ttk_bus._bus_signals_base.BusSignal.hasStateLookup (
            self )
```

Return True if a state/value lookup table is defined for the HIL signal variable.

Reimplemented in ttk_bus.bus_signals_base.TxBusSignal, ttk_bus.bus_signals_base.RxBusSignal, ttk_bus.bus_signals_base.CrcBu
and ttk_bus.bus_signals_base.AcBusSignal.

7.54.3.8 info()

```
ttk_bus._bus_signals_base.BusSignal.info (
            self )
```

Get a current info string about the HIL signal variable.

Reimplemented in ttk_bus.bus_signals_base.TxBusSignal, ttk_bus.bus_signals_base.RxBusSignal, ttk_bus.bus_signals_base.CrcBu
ttk_bus.bus_signals_base.AcBusSignal, ttk_bus.bus_signals_canmmbs.TxCanBusSignal, ttk_bus.bus_signals_canmmbs.RxCanBus
ttk_bus.bus_signals_canmmbs.CrcCanBusSignal, ttk_bus.bus_signals_canmmbs.AcCanBusSignal, ttk_bus.bus_signals_ethbs.RxEt
ttk_bus.bus_signals_ethbs.TxEthSignal, ttk_bus.bus_signals_ethbs.RxEthBusSysSignal, ttk_bus.bus_signals_ethbs.TxEthBusSysSig
ttk_bus.bus_signals_ethbs.CrcEthSignal, ttk_bus.bus_signals_ethbs.AcEthSignal, ttk_bus.bus_signals_frbs.TxFrBusSignal,
ttk_bus.bus_signals_frbs.RxFrBusSignal, ttk_bus.bus_signals_frbs.CrcFrBusSignal, and ttk_bus.bus_signals_frbs.AcFrBusSignal.

7.54.3.9 set()

```
ttk_bus._bus_signals_base.BusSignal.set (
            self,
            value )
```

Set a value to the HIL signal variable.

Parameters

value	new value to be set

Returns

the original value before the set operation occurred.

Reimplemented in ttk_bus._bus_signals_canmmbs.RxCanBusSignal, ttk_bus._bus_signals_ethbs.RxEthBusSysSignal,
ttk_bus._bus_signals_frbs.RxFrBusSignal, ttk_bus.bus_signals_base.TxBusSignal, ttk_bus.bus_signals_base.RxBusSignal,
ttk_bus.bus_signals_base.CrcBusSignal, ttk_bus.bus_signals_base.AcBusSignal, ttk_bus.bus_signals_canmmbs.TxCanBusSignal,
ttk_bus.bus_signals_canmmbs.RxCanBusSignal, ttk_bus.bus_signals_canmmbs.CrcCanBusSignal, ttk_bus.bus_signals_canmmbs.A
ttk_bus.bus_signals_ethbs.RxEthSignal, ttk_bus.bus_signals_ethbs.TxEthSignal, ttk_bus.bus_signals_ethbs.RxEthBusSysSignal,
ttk_bus.bus_signals_ethbs.TxEthBusSysSignal, ttk_bus.bus_signals_ethbs.CrcEthSignal, ttk_bus.bus_signals_ethbs.AcEthSignal,
ttk_bus.bus_signals_frbs.TxFrBusSignal, ttk_bus.bus_signals_frbs.RxFrBusSignal, ttk_bus.bus_signals_frbs.CrcFrBusSignal,
and ttk_bus.bus_signals_frbs.AcFrBusSignal.

7.54.3.10 setState()

```
ttk_bus._bus_signals_base.BusSignal.setState (
            self,
            state )
```

Set a textual state to the HIL signal variable.

This uses the lookup values supplied in constructor.

Parameters

state	named state to set

Returns

the previous state's description.

Reimplemented in ttk_bus._bus_signals_canmmbs.RxCanBusSignal, ttk_bus._bus_signals_ethbs.RxEthBusSysSignal, ttk_bus._bus_signals_frbs.RxFrBusSignal, ttk_bus.bus_signals_base.TxBusSignal, ttk_bus.bus_signals_base.RxBusSignal, ttk_bus.bus_signals_base.CrcBusSignal, ttk_bus.bus_signals_base.AcBusSignal, ttk_bus.bus_signals_canmmbs.TxCanBusSignal, ttk_bus.bus_signals_canmmbs.RxCanBusSignal, ttk_bus.bus_signals_canmmbs.CrcCanBusSignal, ttk_bus.bus_signals_canmmbs.A ttk_bus.bus_signals_ethbs.RxEthSignal, ttk_bus.bus_signals_ethbs.TxEthSignal, ttk_bus.bus_signals_ethbs.RxEthBusSysSignal, ttk_bus.bus_signals_ethbs.TxEthBusSysSignal, ttk_bus.bus_signals_ethbs.CrcEthSignal, ttk_bus.bus_signals_ethbs.AcEthSignal, ttk_bus.bus_signals_frbs.TxFrBusSignal, ttk_bus.bus_signals_frbs.RxFrBusSignal, ttk_bus.bus_signals_frbs.CrcFrBusSignal, and ttk_bus.bus_signals_frbs.AcFrBusSignal.

7.54.4 Member Data Documentation

7.54.4.1 alias

`ttk_bus._bus_signals_base.BusSignal.alias`

alias/name for bus signal.

Defaults to associated signal variable's alias entry (and thus inherits context-dependent alias fallback from identifier)

7.54.4.2 change_delay

`ttk_bus._bus_signals_base.BusSignal.change_delay`

delay to wait after a value change [ms]

7.54.4.3 context

`ttk_bus._bus_signals_base.BusSignal.context`

context for "HilVar" variables

7.54.4.4 cycle_time

`ttk_bus._bus_signals_base.BusSignal.cycle_time`

cycle time of signal in [ms] (that is, of the associated message or PDU)

7.54.4.5 debounce_time

`ttk_bus._bus_signals_base.BusSignal.debounce_time`

minimum debounce delay for event-triggered signals in [ms]

7.54.4.6 descr

`ttk_bus._bus_signals_base.BusSignal.descr`

bus signal description text

7.54.4.7 recovery_time

`ttk_bus._bus_signals_base.BusSignal.recovery_time`

time for recovery/"wiedergut" checks [ms]

7.54.4.8 signal_var

`ttk_bus._bus_signals_base.BusSignal.signal_var`

Main bus signal variable on HIL-side.

7.54.4.9 timeout

`ttk_bus._bus_signals_base.BusSignal.timeout`

detection time for timeout errors [ms]

7.54.4.10 unit

`ttk_bus._bus_signals_base.BusSignal.unit`

value unit for bus signal.

Defaults to associated signal variable's unit entry.

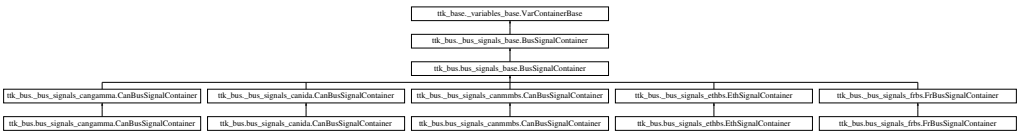
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_base.py (v3.0.1)

7.55 ttk_bus._bus_signals_base.BusSignalContainer Class Reference

Base class for BusSignal containers.

Inheritance diagram for ttk_bus._bus_signals_base.BusSignalContainer:



Public Member Functions

- __init__ (self, context)
  Override to add signals.

7.55.1 Detailed Description

Base class for BusSignal containers.

7.55.2 Constructor & Destructor Documentation

7.55.2.1 __init__()

```
ttk_bus._bus_signals_base.BusSignalContainer.__init__ (
              self,
              context )
```

Override to add signals.

Parameters

context	context used for contained BusSignals

Reimplemented in ttk_bus.bus_signals_base.BusSignalContainer, ttk_bus.bus_signals_cangamma.CanBusSignalContainer, ttk_bus.bus_signals_canida.CanBusSignalContainer, ttk_bus.bus_signals_canmmbs.CanBusSignalContainer, ttk_bus.bus_signals_ethbs.EthSignalContainer, and ttk_bus.bus_signals_frbs.FrBusSignalContainer.

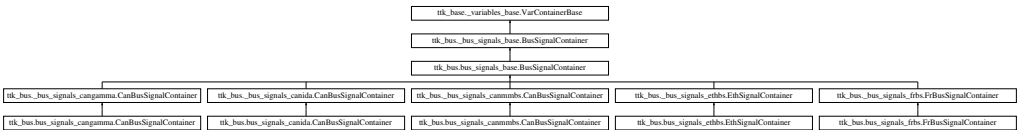
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_base.py (v3.0.1)

7.56 ttk_bus.bus_signals_base.BusSignalContainer Class Reference

Base class for BusSignal containers.

Inheritance diagram for ttk_bus.bus_signals_base.BusSignalContainer:



Public Member Functions

- __init__ (self, context)
  Override to add signals.
- resetAll (self, verbosity=1, recursion_depth=4)
  Call reset() on all contained BusSignals that support it.

7.56.1 Detailed Description

Base class for BusSignal containers.

7.56.2 Constructor & Destructor Documentation

7.56.2.1 __init__()

```
ttk_bus.bus_signals_base.BusSignalContainer.__init__ (
               self,
               context )
```

Override to add signals.

Parameters

context	context used for contained BusSignals

Reimplemented from ttk_bus._bus_signals_base.BusSignalContainer.

Reimplemented in ttk_bus.bus_signals_cangamma.CanBusSignalContainer, ttk_bus.bus_signals_canida.CanBusSignalContainer, ttk_bus.bus_signals_canmmbs.CanBusSignalContainer, ttk_bus.bus_signals_ethbs.EthSignalContainer, and ttk_bus.bus_signals_frbs.FrBusSignalContainer.

7.56.3 Member Function Documentation

7.56.3.1 resetAll()

```
ttk_bus.bus_signals_base.BusSignalContainer.resetAll (
               self,
               verbosity = 1,
               recursion_depth = 4 )
```

Call reset() on all contained BusSignals that support it.

Parameters

verbosity	verbosity of log output: 0: silent except for errors 1: only minimal status info (begin/end) 2: also show processed variable names
recursion_depth	how "deep" to recurse into lists/dictionary structures when looking for variables to reset.

Reimplemented from ttk_base._variables_base.VarContainerBase.

Reimplemented in ttk_bus.bus_signals_cangamma.CanBusSignalContainer, ttk_bus.bus_signals_canida.CanBusSignalContainer, ttk_bus.bus_signals_canmmbs.CanBusSignalContainer, ttk_bus.bus_signals_ethbs.EthSignalContainer, and ttk_bus.bus_signals_frbs.FrBusSignalContainer.

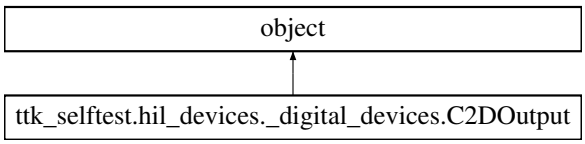
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_base.py (v3.1.0)

7.57 ttk_selftest.hil_devices._digital_devices.C2DOutput Class Reference

Base iCAN2DIG output container for a testable iCAN2DIG output and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices._digital_devices.C2DOutput:



Public Member Functions

- __init__ (self, context, base_path, num_c2d_card, num_output, alias="")

  iCAN2DIG output initialization.
- setOutput (self)

  Function to set the output of the iCAN2DIG.
- resetOutput (self)

  Function to reset the output of the iCAN2DIG.

Public Attributes

- c2d_msg_alias
- c2d_vars_alias
- alias
- msg
- output

7.57.1 Detailed Description

Base iCAN2DIG output container for a testable iCAN2DIG output and relevant associated variables/values.

7.57.2 Constructor & Destructor Documentation

7.57.2.1 __init__()

```
ttk_selftest.hil_devices._digital_devices.C2DOutput.__init__ (
              self,
              context,
              base_path,
              num_c2d_card,
              num_output,
              alias = "" )
```

iCAN2DIG output initialization.

Parameters

context	parent context
base_path	base path of the iCAN2DIG in the model (all required variables will be derived from this path)
num_c2d_card	number of the iCAN2DIG
num_output	output number of the specified iCAN2DIG
alias	(opt) output alias/name defaults to the iCAN2DIG and output number

7.57.3 Member Function Documentation

7.57.3.1 resetOutput()

```
ttk_selftest.hil_devices._digital_devices.C2DOutput.resetOutput (
              self )
```

Function to reset the output of the iCAN2DIG.

```
None
```

Returns

None

7.57.3.2 setOutput()

```
ttk_selftest.hil_devices._digital_devices.C2DOutput.setOutput (
              self )
```

Function to set the output of the iCAN2DIG.

```
None
```

Returns

None

7.57.4 Member Data Documentation

7.57.4.1 alias

```
ttk_selftest.hil_devices._digital_devices.C2DOutput.alias
```

7.57.4.2 c2d_msg_alias

```
ttk_selftest.hil_devices._digital_devices.C2DOutput.c2d_msg_alias
```

7.57.4.3 c2d_vars_alias

```
ttk_selftest.hil_devices._digital_devices.C2DOutput.c2d_vars_alias
```

7.57.4.4 msg

```
ttk_selftest.hil_devices._digital_devices.C2DOutput.msg
```

7.57.4.5 output

```
ttk_selftest.hil_devices._digital_devices.C2DOutput.output
```

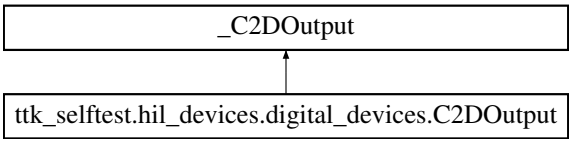
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/_digital_devices.py (v2.0.0)

7.58 ttk_selftest.hil_devices.digital_devices.C2DOutput Class Reference

Base iCAN2DIG output container for a testable iCAN2DIG output and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices.digital_devices.C2DOutput:



Public Member Functions

- __init__ (self, context, base_path, num_c2d_card, num_output, alias="")
  iCAN2DIG output initialization.
- setOutput (self)
  Function to set the output of the iCAN2DIG.
- resetOutput (self)
  Function to reset the output of the iCAN2DIG.

7.58.1 Detailed Description

Base iCAN2DIG output container for a testable iCAN2DIG output and relevant associated variables/values.

7.58.2 Constructor & Destructor Documentation

7.58.2.1 __init__()

```
ttk_selftest.hil_devices.digital_devices.C2DOutput.__init__ (
              self,
              context,
              base_path,
              num_c2d_card,
              num_output,
              alias = "" )
```

iCAN2DIG output initialization.

Parameters

context	parent context
base_path	base path of the iCAN2DIG in the model (all required variables will be derived from this path)
num_c2d_card	number of the iCAN2DIG
num_output	output number of the specified iCAN2DIG
alias	(opt) output alias/name defaults to the iCAN2DIG and output number

7.58.3 Member Function Documentation

7.58.3.1 resetOutput()

```
ttk_selftest.hil_devices.digital_devices.C2DOutput.resetOutput (
              self )
```

Function to reset the output of the iCAN2DIG.

```
None
```

Returns

None

7.58.3.2 setOutput()

```
ttk_selftest.hil_devices.digital_devices.C2DOutput.setOutput (
              self )
```

Function to set the output of the iCAN2DIG.

```
None
```

Returns

None

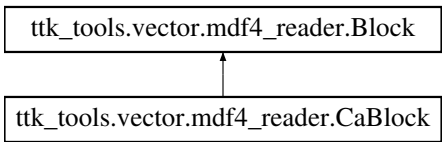
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/digital_devices.py (v2.0.0)

7.59 ttk_tools.vector.mdf4_reader.CaBlock Class Reference

Channel Array Block.

Inheritance diagram for ttk_tools.vector.mdf4_reader.CaBlock:



Public Member Functions

- __init__ (self, file_handle, pos=None)

  Read left-open interval from file_handle, starting at the current or supplied position.
- __repr__ (self)

Public Attributes

- ca_ndim
- ca_byte_offset_base
- ca_inval_bit_pos_base
- ca_type
- ca_storage
- ca_flags
- ca_dim_size
- total_elements
- ca_axis_value
- ca_cycle_count
- ca_composition

  Array of composed elements: Pointer to a CNBLOCK for array of structures, or to a CABLOCK for array of arrays (can be NIL).
- ca_data

  List of links to either a data block (DTBLOCK or DZBLOCK for this block type) or a data list block (DLBLOCK of data blocks or its HLBLOCK) for each element.
- ca_dynamic_size

  References to channels for size signal of each dimension (can be NIL).
- ca_input_quantity

  Reference to channels for input quantity signal for each dimension (can be NIL).
- ca_output_quantity
- ca_comparison_quantity

  Reference to channel for comparison quantity (can be NIL).
- ca_cc_axis_conversion

  Pointer to a conversion rule (CCBLOCK) for the axis of each dimension.
- ca_axis
- composition
- name

7.59.1 Detailed Description

Channel Array Block.

A CABLOCK is used to model an array.

7.59.2 Constructor & Destructor Documentation

7.59.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.CaBlock.__init__ (
              self,
              file_handle,
              pos = None )
```

Read left-open interval from file_handle, starting at the current or supplied position.

Reimplemented from ttk_tools.vector.mdf4_reader.Block.

7.59.3 Member Function Documentation

7.59.3.1 __repr__()

```
ttk_tools.vector.mdf4_reader.CaBlock.__repr__ (
              self )
```

7.59.4 Member Data Documentation

7.59.4.1 ca_axis

```
ttk_tools.vector.mdf4_reader.CaBlock.ca_axis
```

7.59.4.2 ca_axis_value

```
ttk_tools.vector.mdf4_reader.CaBlock.ca_axis_value
```

7.59.4.3 ca_byte_offset_base

```
ttk_tools.vector.mdf4_reader.CaBlock.ca_byte_offset_base
```

7.59.4.4 ca_cc_axis_conversion

ttk_tools.vector.mdf4_reader.CaBlock.ca_cc_axis_conversion

Pointer to a conversion rule (CCBLOCK) for the axis of each dimension.

If a link NIL a 1:1 conversion must be used for this axis. If the "fixed axis" flag (Bit 5) is set, the conversion must be applied to the fixed axis values of the respective axis/dimension (ca_axis_value list stores the raw values as REAL). If the link to the CCBLOCK is NIL already the physical values are stored in the ca_axis_value list. If the "fixed axes" flag (Bit 5) is not set, the conversion must be applied to the raw values of the respective axis channel, i.e. it overrules the conversion specified for the axis channel, even if the ca_axis_conversion link is NIL! Note: ca_axis_conversion may reference the same CCBLOCK as referenced by the respective axis channel ("sharing" of CCBLOCK).

7.59.4.5 ca_comparison_quantity

ttk_tools.vector.mdf4_reader.CaBlock.ca_comparison_quantity

Reference to channel for comparison quantity (can be NIL).

The reference is a link triple with pointer to parent DGBLOCK, parent CGBLOCK and CNBLOCK for the channel (either all three links are assigned or NIL). Since the comparison quantity signal and the array signal must be synchronized, their channel groups must contain at least one common master channel type. The comparison quantity should have the same physical unit as the array elements.

7.59.4.6 ca_composition

ttk_tools.vector.mdf4_reader.CaBlock.ca_composition

Array of composed elements: Pointer to a CNBLOCK for array of structures, or to a CABLOCK for array of arrays (can be NIL).

If a CABLOCK is referenced, it must use the "CN template" storage type (ca_storage = 0).

7.59.4.7 ca_cycle_count

ttk_tools.vector.mdf4_reader.CaBlock.ca_cycle_count

7.59.4.8 ca_data

ttk_tools.vector.mdf4_reader.CaBlock.ca_data

List of links to either a data block (DTBLOCK or DZBLOCK for this block type) or a data list block (DLBLOCK of data blocks or its HLBLOCK) for each element.

A link in this list may only be NIL if the cycle count of the respective element is 0: ca_data[k] = NIL => ca_cycle_↩ count[k] = 0 If ca_data[k] is NIL, this should be interpreted that the respective array element was not recorded. If the element was recorded, but no single sample occurred for it, this should be indicated by ca_cycle_count[k] = 0, but ca_data[k] referencing an "empty" DTBLOCK (i.e. only DT block header with 24 bytes). Thus a simple distinction between "not recorded elements" and "recorded elements without samples" is possible. The links are stored line-oriented, i.e. element k uses ca_data[k] The size of the list must be equal to prod(N(d)), i.e. to the product of the number of elements per dimension N(d) over all dimensions D. Note: link ca_data[0] must be equal to dg_data link of the parent DGBLOCK.

7.59.4.9 ca_dim_size

ttk_tools.vector.mdf4_reader.CaBlock.ca_dim_size

7.59.4.10 ca_dynamic_size

ttk_tools.vector.mdf4_reader.CaBlock.ca_dynamic_size

References to channels for size signal of each dimension (can be NIL).

Each reference is a link triple with pointer to parent DGBLOCK, parent CGBLOCK and CNBLOCK for the channel (either all three links are assigned or NIL). Thus the links have the following order:

- DGBLOCK for size signal of dimension 1
- CGBLOCK for size signal of dimension 1
- CNBLOCK for size signal of dimension 1 ...
- DGBLOCK for size signal of dimension D
- CGBLOCK for size signal of dimension D
- CNBLOCK for size signal of dimension D The size signal can be used to model arrays whose number of elements per dimension can vary over time. If a size signal is specified for a dimension, the number of elements for this dimension at some point in time is equal to the value of the size signal at this time (i.e. for time-synchronized signals, the size signal value with highest time stamp less or equal to current time stamp). If the size signal has no recorded signal value for this time (yet), assume 0 as size. Since each referenced channel defines the current size of a dimension, its physical values must be Integer values >= 0 and <= N(d), i.e. the values must not exceed the maximum number of elements for the respective dimension. Usually, a size signal should have some Integer data type (cn_data_type <= 3) without conversion rule and without unit. Since the size signal and the array signal must be synchronized, their channel groups must contain at least one common master channel type. Note: the "positions" of the array elements are fixed as defined for the maximum allowed array size. They will not change when the array is reduced by a size signal value less than N(d). However, the "removed" array elements must not be considered. For example, assume a max 2x3 matrix. For row-oriented CN template storage, the elements would be stored in the record in following order: a11, a12, a13, a21, a22, a23 Assume that the size of the second dimension is reduced from 3 to 2. Then the values of elements a13 and a23 will be undefined and should not be used. However, the other elements stay at the same position (byte offset): a11, a12, [a13], a21, a22, [a23]

7.59.4.11 ca_flags

ttk_tools.vector.mdf4_reader.CaBlock.ca_flags

7.59.4.12 ca_input_quantity

ttk_tools.vector.mdf4_reader.CaBlock.ca_input_quantity

Reference to channels for input quantity signal for each dimension (can be NIL).

Each reference is a link triple with pointer to parent DGBLOCK, parent CGBLOCK and CNBLOCK for the channel (either all three links are assigned or NIL). Thus the links have the following order:

- DGBLOCK for size signal of dimension 1
- CGBLOCK for size signal of dimension 1
- CNBLOCK for size signal of dimension 1 ...
- DGBLOCK for size signal of dimension D
- CGBLOCK for size signal of dimension D
- CNBLOCK for size signal of dimension D Since the input quantity signal and the array signal must be synchronized, their channel groups must contain at least one common master channel type. The input quantity often is denoted as "working point" because it is used for look-up of a value in the array. For array type "look-up", the value of the input quantity will be used for looking up a value in the scaling axis of the respective dimension (either axis channel or fixed axis values listed in ca_axis_value list). For array type "scaling axis", the value of the input quantity will be used for looking up a value in the axis itself. The input quantity should have the same physical unit as the axis it is applied to.

7.59.4.13 ca_inval_bit_pos_base

ttk_tools.vector.mdf4_reader.CaBlock.ca_inval_bit_pos_base

7.59.4.14 ca_ndim

ttk_tools.vector.mdf4_reader.CaBlock.ca_ndim

7.59.4.15 ca_output_quantity

ttk_tools.vector.mdf4_reader.CaBlock.ca_output_quantity

7.59.4.16 ca_storage

ttk_tools.vector.mdf4_reader.CaBlock.ca_storage

7.59.4.17 ca_type

ttk_tools.vector.mdf4_reader.CaBlock.ca_type

7.59.4.18 composition

`ttk_tools.vector.mdf4_reader.CaBlock.composition`

7.59.4.19 name

`ttk_tools.vector.mdf4_reader.CaBlock.name`

7.59.4.20 total_elements

`ttk_tools.vector.mdf4_reader.CaBlock.total_elements`

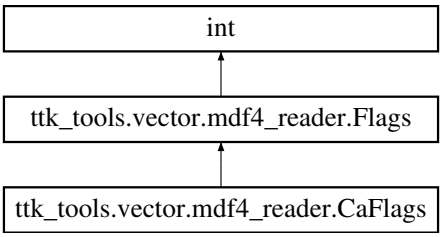
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.60 ttk_tools.vector.mdf4_reader.CaFlags Class Reference

Container for decoded uint32 ca_flags.

Inheritance diagram for ttk_tools.vector.mdf4_reader.CaFlags:



Public Member Functions

- __init__ (self, flags)

Public Attributes

- dynamic_size
  Bit 0: dynamic size flag.
- input_quantity
  Bit 1: input quantity flag.
- output_quantity
  Bit 2: output quantity flag.
- comparison_quantity
  Bit 3: comparison quantity flag.
- axis
  Bit 4: axis flag.
- fixed_axis
  Bit 5: fixed axis flag .
- inverse_layout
  Bit 6: inverse layout flag If set, the record layout is "column oriented" instead of "row oriented".
- left_open_interval
  Bit 7: left-open interval flag.

7.60.1 Detailed Description

Container for decoded uint32 ca_flags.

Only intended for read-only access, changes in named flags will not modify original "flags" value.

7.60.2 Constructor & Destructor Documentation

7.60.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.CaFlags.__init__ (
            self,
            flags )
```

Reimplemented from ttk_tools.vector.mdf4_reader.Flags.

7.60.3 Member Data Documentation

7.60.3.1 axis

```
ttk_tools.vector.mdf4_reader.CaFlags.axis
```

Bit 4: axis flag.

If set, a scaling axis is given for each dimension of the array, either as fixed or as dynamic axis, depending on "fixed axis" flag (bit 5). Can only be set for array types "look-up", "scaling axis" and "classification result".

7.60.3.2 comparison_quantity

```
ttk_tools.vector.mdf4_reader.CaFlags.comparison_quantity
```

Bit 3: comparison quantity flag.

If set, a channel for the comparison quantity is specified by ca_comparison_quantity. Can only be set for array type "look-up".

7.60.3.3 dynamic_size

```
ttk_tools.vector.mdf4_reader.CaFlags.dynamic_size
```

Bit 0: dynamic size flag.

If set, the number of scaling points for the array is not fixed but can vary over time. Can only be set for array types "look-up" and "scaling axis".

7.60.3.4 fixed_axis

ttk_tools.vector.mdf4_reader.CaFlags.fixed_axis

Bit 5: fixed axis flag .

If set, the scaling axis is fixed and the axis points are stored as raw values in ca_axis_value list. If not set, the scaling axis may vary over time and the axis points are stored as channel referenced in ca_axis for each dimension. Only relevant if "axis" flag (bit 4) is set. Can only not be set for array types "look-up" and "scaling axis".

7.60.3.5 input_quantity

ttk_tools.vector.mdf4_reader.CaFlags.input_quantity

Bit 1: input quantity flag.

If set, a channel for the input quantity is specified for each dimension by ca_input_quantity. Can only be set for array types "look-up" and "scaling axis".

7.60.3.6 inverse_layout

ttk_tools.vector.mdf4_reader.CaFlags.inverse_layout

Bit 6: inverse layout flag If set, the record layout is "column oriented" instead of "row oriented".

Only relevant for "CN template" storage type and for number of dimensions > 1.

7.60.3.7 left_open_interval

ttk_tools.vector.mdf4_reader.CaFlags.left_open_interval

Bit 7: left-open interval flag.

If set, the interval ranges for the class interval axes are left-open and right-closed, i.e.]a, b] = {x | a $<$ x $<=$ b}. If not set, the interval ranges for the class interval axes are left-closed and right-open, i.e. [a, b[= {x | a $<=$ x $<$ b}.

Note that opening or closing is irrelevant for infinite end points (there can only be -INF for the left border of the very first interval, or +INF for right border of the very last interval). Only relevant for array type "interval axes". Valid since MDF 4.1.0, should not be set for earlier versions.

7.60.3.8 output_quantity

ttk_tools.vector.mdf4_reader.CaFlags.output_quantity

Bit 2: output quantity flag.

If set, a channel for the output quantity is specified by ca_output_quantity. Can only be set for array type "look-up".

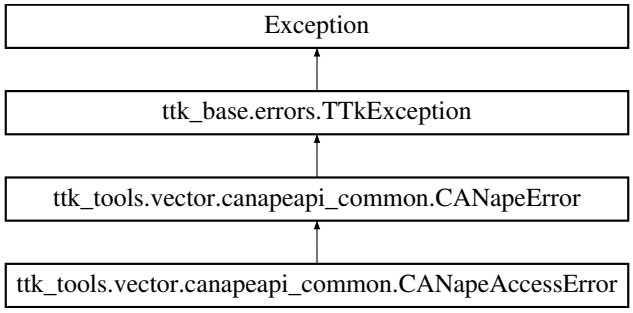
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.61 ttk_tools.vector.canapeapi_common.CANapeAccessError Class Reference

Cal-object access errors for CANape connection.

Inheritance diagram for ttk_tools.vector.canapeapi_common.CANapeAccessError:



Public Member Functions

- __init__ (self, varname='', msg='', log_msg=False, api_msg=None, error_code=None)

7.61.1 Detailed Description

Cal-object access errors for CANape connection.

7.61.2 Constructor & Destructor Documentation

7.61.2.1 __init__()

```
ttk_tools.vector.canapeapi_common.CANapeAccessError.__init__ (
              self,
              msg = '',
              log_msg = '',
              api_msg = False,
              error_code = None,
              error_code = None )
```

Parameters

msg	error message
log_msg	True: log message to stdout, False: no log output
api_msg	CANape-provided "last error" message, if available
error_code	error code from CANapeAPI, if available.

Reimplemented from ttk_tools.vector.canapeapi_common.CANapeError.

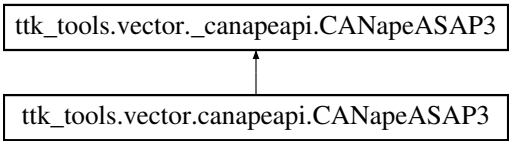
The documentation for this class was generated from the following file:

- ttk_tools/vector/canapeapi_common.py (v3.1.2)

7.62 ttk_tools.vector._canapeapi.CANapeASAP3 Class Reference

CANape ASAP3-connection (application).

Inheritance diagram for ttk_tools.vector._canapeapi.CANapeASAP3:



Public Member Functions

- __init__ (self, working_dir, clear_device_list=False, debug_mode=True, keep_canape_alive=False, canape_path=None, init_timeout=100000, init_retries=3, api_log_level=0)

  Initialize a CANape connection (that is, a connection to the base application).
- exit (self, verbosity=2)

  Manually shut down ASAP3 connection to CANape.
- __del__ (self)

  Destructor/Cleanup.
- getHandle (self)

  Get the current ASAP3-handle for use in API calls.
- getVersion (self, verbose=False)

  Get the current CANape/API version.
- __str__ (self)

  Get a string representation of the ASAP3 interface instance.
- getDebugWindowText (self)

  Get current contents of CANape's write/debug window.
- getLastError (self, also_return_code=False)

  Get an error text/description for the most recent CANape error.
- convertMDF2MAT (self, mdf_filename, mat_filename=None)

  Convert an existing .MDF file to .MAT (Matlab) format.
- getAvailableConverters (self, verbosity=0)

  Get available options for data file conversion.
- convertData (self, converter_id, source_file_name, dest_file_name=None, overwrite=True)

  Convert data between different formats.

Public Attributes

- verbose

  Verbosity of debug/error messages.
- api_log_level

  Logging of CANape-API-DLL calls.
- canape_path

  path to current CANape installation directory
- api

  CANape-API (DLL) handle.
- sample_size

Static Public Attributes

- int fifo_size = 512

  FIFO_SIZE: Count of measurements to fit in the FIFO (so using a single 10ms task for data acquisition, FIFO_SIZE ∗ 10ms worth of data can be stored).
- int sample_size = 128

  SAMPLE_SIZE: number of signals/variables/channels that can be assigned to each task for measuring (default: 128, see Application Note AN-AMC-1-103)
- TAsap3Hdl = c_void_p
- TModulHdl = c_ushort
- TScriptHdl = c_ulong
- TAsap3DiagHdl = c_ulong
- restype
- optional
- TAsap3ECUState = c_int
- TFormat = c_int
- tMeasurementState = c_int
- TAsap3DataType = c_int
- eServiceStates = c_int
- RespID_t = c_long
- TScriptStatus = c_int

Protected Member Functions

- _logApiCallStart (self, function_name, ∗∗kwargs)

  Log start of API calls to stdout.
- _logApiCallEnd (self, function_name, retval=None, ∗∗kwargs)

  Log end of API calls to stdout.
- _updateCANapePath (self, path=None)

  Update the CANape installation path setting in the windows registry (HKLM\SOFTWARE\VECTOR\CANape) to select the active installation.
- _restoreCANapePath (self)

  Restore the original/previous CANape installation path setting in the registry.
- _getApiHandle (self)

  Get a handle to the CANapeAPI (canapapi.dll).
- _getLastErrorException (self, exc_class, message="", ∗∗kwargs)

  Get a CANapeError-derived Exception instance pre-initialized with a custom message, the current API error message and the associated error code.

Protected Attributes

- _init_time

  how long init took in [s]
- _init_attempts

  how many init attempts were made
- _max_init_attempts

  maximum number of attempts to perform Asap3Init
- _init_attempts_before_recovery_attempt

  number of init attempts before trying to kill an unresponsive CANape process
- _working_dir

  path to working directory
- _clear_device_list

0: retain list of configured devices (from current workspace)
1: clear list of configured devices

- _debug_mode

  0: start CANape minimized,
  1: show CANape window

- _keep_canape_alive

  0: close CANape upon exit (slower),
  1: keep process alive, just disconnect (faster)

- _response_timeout
- _asap_handle

  handle to ASAP3 instance.

- _buffer_size

  size of FIFO-buffer in bytes

- _prev_canape_path

7.62.1 Detailed Description

CANape ASAP3-connection (application).

7.62.2 Constructor & Destructor Documentation

7.62.2.1 __init__()

```
ttk_tools.vector._canapeapi.CANapeASAP3.__init__ (
              self,
              working_dir,
              clear_device_list = False,
              debug_mode = True,
              keep_canape_alive = False,
              canape_path = None,
              init_timeout = 100000,
              init_retries = 3,
              api_log_level = 0 )
```

Initialize a CANape connection (that is, a connection to the base application).

Parameters

working_dir	working directory containing `CANape.ini`
clear_device_list	if device list gets **not** cleared and later a module name matches a device in the list, the existing device (i.e. its settings) will be assigned.
debug_mode	if True, the CANape window will be visible to show debug messages (i.e. the "Write Window")
keep_canape_alive	if True, the CANape instance will stay active between calls (thus drastically decreasing startup time)
canape_path	path to CANape installation directory, set to None to use the current path from registry (e.g. for single CANape installations)
init_timeout	[ms] timeout for connection init and responses (not too short, or timeout will occur while device database(s) are being read)
init_retries	how often the ASAP3 connection init should be retried in case of failure
api_log_level	log level for CANape-API calls: 0: silent 1: log before/after execution of API call to stdout 2: additionally log API call parameters (if applicable/sensible)

Reimplemented in ttk_tools.vector.canapeapi.CANapeASAP3.

7.62.2.2 __del__()

```
ttk_tools.vector._canapeapi.CANapeASAP3.__del__ (
              self )
```

Destructor/Cleanup.

7.62.3 Member Function Documentation

7.62.3.1 __str__()

```
ttk_tools.vector._canapeapi.CANapeASAP3.__str__ (
              self )
```

Get a string representation of the ASAP3 interface instance.

Returns

an info string.

7.62.3.2 _getApiHandle()

```
ttk_tools.vector._canapeapi.CANapeASAP3._getApiHandle (
              self )  [protected]
```

Get a handle to the CANapeAPI (canapapi.dll).

Handle will only be initialized the first time this method is called, subsequent calls will just return the existing handle.

Returns

a DLL-handle.

7.62.3.3 _getLastErrorException()

```
ttk_tools.vector._canapeapi.CANapeASAP3._getLastErrorException (
              self,
              exc_class,
              message = "",
             ** kwargs )  [protected]
```

Get a CANapeError-derived Exception instance pre-initialized with a custom message, the current API error message and the associated error code.

See getLastError()

Parameters

exc_class	CANapeError class to instantiate
message	custom message / error description
∗∗*kwargs*	additional keyword arguments to be passed on to exc_class' constructor (if any).

Returns

an initialized ready-to-be-raised instance (raise it at the appropriate code location to make stack traces more helpful)

7.62.3.4 _logApiCallEnd()

```
ttk_tools.vector._canapeapi.CANapeASAP3._logApiCallEnd (
              self,
              function_name,
              retval = None,
             ** kwargs )  [protected]
```

Log end of API calls to stdout.

Call this immediately after the API call returns.

Parameters

function_name	name of called API function
retval	return value of API function
∗∗*kwargs*	remaining keyword arguments (catch-all)

7.62.3.5 _logApiCallStart()

```
ttk_tools.vector._canapeapi.CANapeASAP3._logApiCallStart (
              self,
              function_name,
             ** kwargs )  [protected]
```

Log start of API calls to stdout.

Call this immediately before performing the API call.

Parameters

function_name	name of called API function (add relevant API call args as keyword arguments)
∗∗*kwargs*	remaining keyword arguments (catch-all)

7.62.3.6 _restoreCANapePath()

```
ttk_tools.vector._canapeapi.CANapeASAP3._restoreCANapePath (
```

```
                self ) [protected]
```

Restore the original/previous CANape installation path setting in the registry.

7.62.3.7 _updateCANapePath()

```
ttk_tools.vector._canapeapi.CANapeASAP3._updateCANapePath (
                self,
                path = None ) [protected]
```

Update the CANape installation path setting in the windows registry (HKLM\SOFTWARE\VECTOR\CANape) to select the active installation.

Parameters

path	path to CANape installation folder, or None to keep the current setting

Returns

the current path to the CANape installation folder.

7.62.3.8 convertData()

```
ttk_tools.vector._canapeapi.CANapeASAP3.convertData (
                self,
                converter_id,
                source_file_name,
                dest_file_name = None,
                overwrite = True )
```

Convert data between different formats.

Note

Support for Asap3MDFConvert was added in CANape 13.

Parameters

converter_id	id/name of converter to use, e.g. "MDF2MAT". See getAvailableConverters() for, well, available converters.
source_file_name	name/path of input/source file
dest_file_name	name/path of output/destination file None: CANape will choose name of output file, which typically means same path but a different file extension.
overwrite	True: overwrite existing files, False: do not overwrite existing files (may raise a CANapeError)

Reimplemented in ttk_tools.vector.canapeapi.CANapeASAP3.

7.62.3.9 convertMDF2MAT()

```
ttk_tools.vector._canapeapi.CANapeASAP3.convertMDF2MAT (
              self,
              mdf_filename,
              mat_filename = None )
```

Convert an existing .MDF file to .MAT (Matlab) format.

Attention

The used API function (Asap3MatlabConversion) has been deprecated starting with CANape 13.

Note

Some errors might trigger a modal CANape dialog containing an error message.
In most cases it might be a better idea to use mdflib.readMdf() to access mdf data...

Parameters

mdf_filename	file name/path to source MDF file
mat_filename	file name/path to destination MAT file (will be created)

Returns

True if conversion was successful, otherwise False.

7.62.3.10 exit()

```
ttk_tools.vector._canapeapi.CANapeASAP3.exit (
              self,
              verbosity = 2 )
```

Manually shut down ASAP3 connection to CANape.

If keep_canape_alive was set to False in constructor, the CANape application will also be closed.

This will normally be called during the destructor call, but it might sometimes be advantageous to exit the connection at a defined time (and not just implicitly when the destructor gets called eventually).

Parameters

verbosity	verbosity of stdout/log output: -1: do not show anything if handle is already gone (for calls in CANapeASAP3.__del__) 0: errors only 1: Asap3Exit status output 2: also show "nothing to do" notice if handle is already gone

Returns

True if successful (or already closed), otherwise False

Reimplemented in ttk_tools.vector.canapeapi.CANapeASAP3.

7.62.3.11 getAvailableConverters()

```
ttk_tools.vector._canapeapi.CANapeASAP3.getAvailableConverters (
              self,
              verbosity = 0 )
```

Get available options for data file conversion.

Note

Support for Asap3MDFConverterInfo was added in CANape 13.

Parameters

verbosity	0: silent, 1: log converters to stdout

Returns

a dictionary with entries for each converter option:
<converter id> : {
'id'" <converter id>,
''name': <converter name>,
 'comment'`: <converter comment>,
}
or None if API does not support this feature (available starting with CANape 13).

Reimplemented in ttk_tools.vector.canapeapi.CANapeASAP3.

7.62.3.12 getDebugWindowText()

```
ttk_tools.vector._canapeapi.CANapeASAP3.getDebugWindowText (
              self )
```

Get current contents of CANape's write/debug window.

Returns

a contents as a string. If access to debug window contents failed, returned string will contain an error message.

Reimplemented in ttk_tools.vector.canapeapi.CANapeASAP3.

7.62.3.13 getHandle()

```
ttk_tools.vector._canapeapi.CANapeASAP3.getHandle (
              self )
```

Get the current ASAP3-handle for use in API calls.

7.62.3.14 getLastError()

```
ttk_tools.vector._canapeapi.CANapeASAP3.getLastError (
              self,
              also_return_code = False )
```

Get an error text/description for the most recent CANape error.

Parameters

also_return_code	if True, the CANape error code will also be returned

Returns

the string "No Error" if, well, no error occurred, otherwise an error description supplied by CANape.

If parameter `also_return_code` is set, the result will be a tuple containing the error info: (<description>, <code>)

Reimplemented in ttk_tools.vector.canapeapi.CANapeASAP3.

7.62.3.15 getVersion()

```
ttk_tools.vector._canapeapi.CANapeASAP3.getVersion (
              self,
              verbose = False )
```

Get the current CANape/API version.

Parameters

verbose	if True, version info will also be printed to stdout

Returns

a dictionary with version information.

Reimplemented in ttk_tools.vector.canapeapi.CANapeASAP3.

7.62.4 Member Data Documentation

7.62.4.1 _asap_handle

```
ttk_tools.vector._canapeapi.CANapeASAP3._asap_handle  [protected]
```

handle to ASAP3 instance.

7.62.4.2 _buffer_size

ttk_tools.vector._canapeapi.CANapeASAP3._buffer_size [protected]

size of FIFO-buffer in bytes

7.62.4.3 _clear_device_list

ttk_tools.vector._canapeapi.CANapeASAP3._clear_device_list [protected]

0: retain list of configured devices (from current workspace)
1: clear list of configured devices

7.62.4.4 _debug_mode

ttk_tools.vector._canapeapi.CANapeASAP3._debug_mode [protected]

0: start CANape minimized,
1: show CANape window

7.62.4.5 _init_attempts

ttk_tools.vector._canapeapi.CANapeASAP3._init_attempts [protected]

how many init attempts were made

7.62.4.6 _init_attempts_before_recovery_attempt

ttk_tools.vector._canapeapi.CANapeASAP3._init_attempts_before_recovery_attempt [protected]

number of init attempts before trying to kill an unresponsive CANape process

7.62.4.7 _init_time

ttk_tools.vector._canapeapi.CANapeASAP3._init_time [protected]

how long init took in [s]

7.62.4.8 _keep_canape_alive

ttk_tools.vector._canapeapi.CANapeASAP3._keep_canape_alive [protected]

0: close CANape upon exit (slower),
1: keep process alive, just disconnect (faster)

7.62.4.9 _max_init_attempts

ttk_tools.vector._canapeapi.CANapeASAP3._max_init_attempts [protected]

maximum number of attempts to perform Asap3Init

7.62.4.10 _prev_canape_path

ttk_tools.vector._canapeapi.CANapeASAP3._prev_canape_path [protected]

7.62.4.11 _response_timeout

ttk_tools.vector._canapeapi.CANapeASAP3._response_timeout [protected]

7.62.4.12 _working_dir

ttk_tools.vector._canapeapi.CANapeASAP3._working_dir [protected]

path to working directory

7.62.4.13 api

ttk_tools.vector._canapeapi.CANapeASAP3.api

CANape-API (DLL) handle.

7.62.4.14 api_log_level

ttk_tools.vector._canapeapi.CANapeASAP3.api_log_level

Logging of CANape-API-DLL calls.

0: silent 1: log before/after execution of API call to stdout 2: additionally log API call parameters (if applicable/sensible)

7.62.4.15 canape_path

ttk_tools.vector._canapeapi.CANapeASAP3.canape_path

path to current CANape installation directory

7.62.4.16 eServiceStates

ttk_tools.vector._canapeapi.CANapeASAP3.eServiceStates = c_int [static]

7.62.4.17 fifo_size

```
int ttk_tools.vector._canapeapi.CANapeASAP3.fifo_size = 512  [static]
```

FIFO_SIZE: Count of measurements to fit in the FIFO (so using a single 10ms task for data acquisition, FIFO_SIZE ∗ 10ms worth of data can be stored).

If more than one task is active, the FIFO buffer gets divided evenly between tasks (default: 512, see Application Note AN-AMC-1-103).

7.62.4.18 optional

```
ttk_tools.vector._canapeapi.CANapeASAP3.optional  [static]
```

7.62.4.19 RespID_t

```
ttk_tools.vector._canapeapi.CANapeASAP3.RespID_t = c_long  [static]
```

7.62.4.20 restype

```
ttk_tools.vector._canapeapi.CANapeASAP3.restype  [static]
```

7.62.4.21 sample_size [1/2]

```
int ttk_tools.vector._canapeapi.CANapeASAP3.sample_size = 128  [static]
```

SAMPLE_SIZE: number of signals/variables/channels that can be assigned to each task for measuring (default: 128, see Application Note AN-AMC-1-103)

7.62.4.22 sample_size [2/2]

```
ttk_tools.vector._canapeapi.CANapeASAP3.sample_size
```

7.62.4.23 TAsap3DataType

```
ttk_tools.vector._canapeapi.CANapeASAP3.TAsap3DataType = c_int  [static]
```

7.62.4.24 TAsap3DiagHdl

```
ttk_tools.vector._canapeapi.CANapeASAP3.TAsap3DiagHdl = c_ulong  [static]
```

7.62.4.25 TAsap3ECUState

```
ttk_tools.vector._canapeapi.CANapeASAP3.TAsap3ECUState = c_int  [static]
```

7.62.4.26 TAsap3Hdl

```
ttk_tools.vector._canapeapi.CANapeASAP3.TAsap3Hdl = c_void_p  [static]
```

7.62.4.27 TFormat

```
ttk_tools.vector._canapeapi.CANapeASAP3.TFormat = c_int  [static]
```

7.62.4.28 tMeasurementState

```
ttk_tools.vector._canapeapi.CANapeASAP3.tMeasurementState = c_int  [static]
```

7.62.4.29 TModulHdl

```
ttk_tools.vector._canapeapi.CANapeASAP3.TModulHdl = c_ushort  [static]
```

7.62.4.30 TScriptHdl

```
ttk_tools.vector._canapeapi.CANapeASAP3.TScriptHdl = c_ulong  [static]
```

7.62.4.31 TScriptStatus

```
ttk_tools.vector._canapeapi.CANapeASAP3.TScriptStatus = c_int  [static]
```

7.62.4.32 verbose

```
ttk_tools.vector._canapeapi.CANapeASAP3.verbose
```

Verbosity of debug/error messages.

True: show messages
False: skip (most) messages

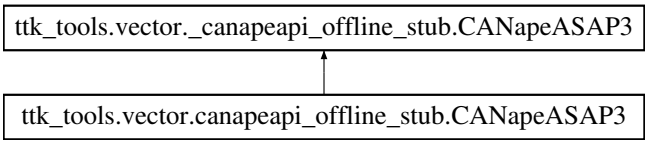
The documentation for this class was generated from the following file:

- ttk_tools/vector/_canapeapi.py (v7.3.3)

7.63 ttk_tools.vector._canapeapi_offline_stub.CANapeASAP3 Class Reference

Stub for CANape ASAP3-connection (application).

Inheritance diagram for ttk_tools.vector._canapeapi_offline_stub.CANapeASAP3:



Public Member Functions

- __init__ (self, working_dir, clear_device_list=False, debug_mode=True, keep_canape_alive=False, canape↩ _path=None, init_timeout=100000, init_retries=3, api_log_level=0)

  Initialize a CANape connection (that is, a connection to the base application).
- exit (self, verbosity=2)

  Manually shut down ASAP3 connection to CANape.
- __del__ (self)

  Destructor/Cleanup.
- getHandle (self)

  Get the current ASAP3-handle for use in API calls.
- getVersion (self, verbose=False)

  Get the current CANape/API version.
- __str__ (self)

  Get a string representation of the ASAP3 interface instance.
- getDebugWindowText (self)

  Get current contents of CANape's write/debug window.
- getLastError (self, also_return_code=False)

  Get an error text/description for the most recent CANape error.
- convertMDF2MAT (self, mdf_filename, mat_filename=None)

  Convert an existing .MDF file to .MAT (Matlab) format.
- getAvailableConverters (self, verbosity=0)

  Get available options for data file conversion.
- convertData (self, converter_id, source_file_name, dest_file_name=None, overwrite=True)

  Convert data between different formats.

Static Public Attributes

- int fifo_size = 512

  FIFO_SIZE: Count of measurements to fit in the FIFO (so using a single 10ms task for data acquisition, FIFO_SIZE ∗ 10ms worth of data can be stored).
- int sample_size = 128

  SAMPLE_SIZE: number of signals/variables/channels that can be assigned to each task for measuring (default: 128, see Application Note AN-AMC-1-103)

Protected Attributes

- _asap_handle
- _working_dir

7.63.1 Detailed Description

Stub for CANape ASAP3-connection (application).

7.63.2 Constructor & Destructor Documentation

7.63.2.1 __init__()

```
ttk_tools.vector._canapeapi_offline_stub.CANapeASAP3.__init__ (
              self,
              working_dir,
              clear_device_list = False,
              debug_mode = True,
              keep_canape_alive = False,
              canape_path = None,
              init_timeout = 100000,
              init_retries = 3,
              api_log_level = 0 )
```

Initialize a CANape connection (that is, a connection to the base application).

Parameters

working_dir	working directory containing `CANape.ini`
clear_device_list	if device list gets **not** cleared and later a module name matches a device in the list, the existing device (i.e. its settings) will be assigned.
debug_mode	if True, the CANape window will be visible to show debug messages (i.e. the "Write Window")
keep_canape_alive	if True, the CANape instance will stay active between calls (thus drastically decreasing startup time)
canape_path	path to CANape installation directory, set to None to use the current path from registry (e.g. for single CANape installations)
init_timeout	[ms] timeout for connection init and responses (not too short, or timeout will occur while device database(s) are being read)
init_retries	how often the ASAP3 connection init should be retried in case of failure
api_log_level	log level for CANape-API calls (not relevant in stub)

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.CANapeASAP3.

7.63.2.2 __del__()

```
ttk_tools.vector._canapeapi_offline_stub.CANapeASAP3.__del__ (
              self )
```

Destructor/Cleanup.

7.63.3 Member Function Documentation

7.63.3.1 __str__()

```
ttk_tools.vector._canapeapi_offline_stub.CANapeASAP3.__str__ (
              self )
```

Get a string representation of the ASAP3 interface instance.

Returns

an info string.

7.63.3.2 convertData()

```
ttk_tools.vector._canapeapi_offline_stub.CANapeASAP3.convertData (
              self,
              converter_id,
              source_file_name,
              dest_file_name = None,
              overwrite = True )
```

Convert data between different formats.

Note

Support for Asap3MDFConvert was added in CANape 13.

Parameters

converter_id	id/name of converter to use, e.g. "MDF2MAT". See getAvailableConverters() for, well, available converters.
source_file_name	name/path of input/source file
dest_file_name	name/path of output/destination file None: CANape will choose name of output file, which typically means same path but a different file extension.
overwrite	True: overwrite existing files, False: do not overwrite existing files (may raise a CANapeError)

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.CANapeASAP3.

7.63.3.3 convertMDF2MAT()

```
ttk_tools.vector._canapeapi_offline_stub.CANapeASAP3.convertMDF2MAT (
              self,
              mdf_filename,
              mat_filename = None )
```

Convert an existing .MDF file to .MAT (Matlab) format.

Attention

The used API function (Asap3MatlabConversion) has been deprecated starting with CANape 13.

Note

Some errors might trigger a modal CANape dialog containing an error message.
In most cases it might be a better idea to use mdflib.readMdf() to access mdf data...

Parameters

mdf_filename	file name/path to source MDF file
mat_filename	file name/path to destination MAT file (will be created)

Returns

True if conversion was successful, otherwise False.

7.63.3.4 exit()

```
ttk_tools.vector._canapeapi_offline_stub.CANapeASAP3.exit (
              self,
              verbosity = 2 )
```

Manually shut down ASAP3 connection to CANape.

If keep_canape_alive was set to False in constructor, the CANape application will also be closed.

This will normally be called during the destructor call, but it might sometimes be advantageous to exit the connection at a defined time (and not just implicitly when the destructor gets called eventually).

Parameters

verbosity	verbosity of stdout/log output: -1: do not show anything if handle is already gone (for calls in CANapeASAP3.__del__) 0: errors only 1: Asap3Exit status output 2: also show "nothing to do" notice if handle is already gone

Returns

True if successful (or already closed), otherwise False

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.CANapeASAP3.

7.63.3.5 getAvailableConverters()

```
ttk_tools.vector._canapeapi_offline_stub.CANapeASAP3.getAvailableConverters (
              self,
              verbosity = 0 )
```

Get available options for data file conversion.

Note

Support for Asap3MDFConverterInfo was added in CANape 13.

Parameters

verbosity	0: silent, 1: log converters to stdout

Returns

a dictionary with entries for each converter option:
<converter id> : {
'id"' <converter id>,
''name'`: <converter name>, <BR> 'comment'`: <converter comment>,
}
or None if API does not support this feature (available starting with CANape 13).

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.CANapeASAP3.

7.63.3.6 getDebugWindowText()

```
ttk_tools.vector._canapeapi_offline_stub.CANapeASAP3.getDebugWindowText (
            self )
```

Get current contents of CANape's write/debug window.

Returns

a contents as a string. If access to debug window contents failed, returned string will contain an error message.

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.CANapeASAP3.

7.63.3.7 getHandle()

```
ttk_tools.vector._canapeapi_offline_stub.CANapeASAP3.getHandle (
            self )
```

Get the current ASAP3-handle for use in API calls.

7.63.3.8 getLastError()

```
ttk_tools.vector._canapeapi_offline_stub.CANapeASAP3.getLastError (
            self,
            also_return_code = False )
```

Get an error text/description for the most recent CANape error.

Parameters

also_return_code	if True, the CANape error code will also be returned

Returns

the string "No Error" if, well, no error occurred, otherwise an error description supplied by CANape.

If parameter `also_return_code` is set, the result will be a tuple containing the error info: (<description>, <code>)

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.CANapeASAP3.

7.63.3.9 getVersion()

```
ttk_tools.vector._canapeapi_offline_stub.CANapeASAP3.getVersion (
              self,
              verbose = False )
```

Get the current CANape/API version.

Parameters

verbose	if True, version info will also be printed to stdout

Returns

a dictionary with version information.

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.CANapeASAP3.

7.63.4 Member Data Documentation

7.63.4.1 _asap_handle

```
ttk_tools.vector._canapeapi_offline_stub.CANapeASAP3._asap_handle  [protected]
```

7.63.4.2 _working_dir

```
ttk_tools.vector._canapeapi_offline_stub.CANapeASAP3._working_dir  [protected]
```

7.63.4.3 fifo_size

```
int ttk_tools.vector._canapeapi_offline_stub.CANapeASAP3.fifo_size = 512  [static]
```

FIFO_SIZE: Count of measurements to fit in the FIFO (so using a single 10ms task for data acquisition, FIFO_SIZE $*$ 10ms worth of data can be stored).

If more than one task is active, the FIFO buffer gets divided evenly between tasks (default: 512, see Application Note AN-AMC-1-103).

7.63.4.4 sample_size

```
int ttk_tools.vector._canapeapi_offline_stub.CANapeASAP3.sample_size = 128  [static]
```

SAMPLE_SIZE: number of signals/variables/channels that can be assigned to each task for measuring (default: 128, see Application Note AN-AMC-1-103)

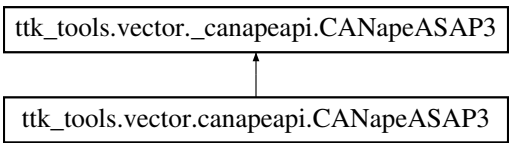
The documentation for this class was generated from the following file:

- ttk_tools/vector/_canapeapi_offline_stub.py (v4.7.1)

7.64 ttk_tools.vector.canapeapi.CANapeASAP3 Class Reference

CANape ASAP3-connection (application).

Inheritance diagram for ttk_tools.vector.canapeapi.CANapeASAP3:



Public Member Functions

- __init__ (self, working_dir, clear_device_list=False, debug_mode=True, keep_canape_alive=False, canape_path=None, init_timeout=100000, init_retries=3, api_log_level=0)

  Initialize a CANape connection (that is, a connection to the base application).
- exit (self, verbosity=2)

  Manually shut down ASAP3 connection to CANape.
- getVersion (self, verbose=False)

  Get the current CANape/API version.
- getDebugWindowText (self)

  Get current contents of CANape's write/debug window.
- getLastError (self, also_return_code=False)

  Get an error text/description for the most recent CANape error.
- getAvailableConverters (self, verbosity=0)

  Get available options for data file conversion.
- convertData (self, converter_id, source_file_name, dest_file_name=None, overwrite=True)

  Convert data between different formats.

7.64.1 Detailed Description

CANape ASAP3-connection (application).

7.64.2 Constructor & Destructor Documentation

7.64.2.1 __init__()

```
ttk_tools.vector.canapeapi.CANapeASAP3.__init__ (
              self,
              working_dir,
              clear_device_list = False,
              debug_mode = True,
              keep_canape_alive = False,
              canape_path = None,
              init_timeout = 100000,
              init_retries = 3,
              api_log_level = 0 )
```

Initialize a CANape connection (that is, a connection to the base application).

Parameters

working_dir	working directory containing `CANape.ini`
clear_device_list	if device list gets **not** cleared and later a module name matches a device in the list, the existing device (i.e. its settings) will be assigned.
debug_mode	if True, the CANape window will be visible to show debug messages (i.e. the "Write Window")
keep_canape_alive	if True, the CANape instance will stay active between calls (thus drastically decreasing startup time)
canape_path	path to CANape installation directory, set to None to use the current path from registry (e.g. for single CANape installations)
init_timeout	[ms] timeout for connection init and responses (not too short, or timeout will occur while device database(s) are being read)
init_retries	how often the ASAP3 connection init should be retried in case of failure
api_log_level	log level for CANape-API calls: 0: silent 1: log before/after execution of API call to stdout 2: additionally log API call parameters (if applicable/sensible)

Reimplemented from ttk_tools.vector._canapeapi.CANapeASAP3.

7.64.3 Member Function Documentation

7.64.3.1 convertData()

```
ttk_tools.vector.canapeapi.CANapeASAP3.convertData (
              self,
              converter_id,
              source_file_name,
              dest_file_name = None,
              overwrite = True )
```

Convert data between different formats.

Note

Support for Asap3MDFConvert was added in CANape 13.

Parameters

converter_id	id/name of converter to use, e.g. "MDF2MAT". See getAvailableConverters() for, well, available converters.
source_file_name	name/path of input/source file
dest_file_name	name/path of output/destination file None: CANape will choose name of output file, which typically means same path but a different file extension.
overwrite	True: overwrite existing files, False: do not overwrite existing files (may raise a CANapeError)

Reimplemented from ttk_tools.vector._canapeapi.CANapeASAP3.

7.64.3.2 exit()

```
ttk_tools.vector.canapeapi.CANapeASAP3.exit (
              self,
              verbosity = 2 )
```

Manually shut down ASAP3 connection to CANape.

If keep_canape_alive was set to False in constructor, the CANape application will also be closed.

This will normally be called during the destructor call, but it might sometimes be advantageous to exit the connection at a defined time (and not just implicitly when the destructor gets called eventually).

Parameters

verbosity	verbosity of stdout/log output: -1: do not show anything if handle is already gone (for calls in CANapeASAP3.__del__) 0: errors only 1: Asap3Exit status output 2: also show "nothing to do" notice if handle is already gone

Returns

True if successful (or already closed), otherwise False

Reimplemented from ttk_tools.vector._canapeapi.CANapeASAP3.

7.64.3.3 getAvailableConverters()

```
ttk_tools.vector.canapeapi.CANapeASAP3.getAvailableConverters (
              self,
              verbosity = 0 )
```

Get available options for data file conversion.

Note

Support for Asap3MDFConverterInfo was added in CANape 13.

Parameters

verbosity	0: silent, 1: log converters to stdout

Returns

a dictionary with entries for each converter option:
<converter id> : {
'id"' <converter id>,
''name': <converter name>,
 'comment'`: <converter comment>,

}
or None if API does not support this feature (available starting with CANape 13).

Reimplemented from ttk_tools.vector._canapeapi.CANapeASAP3.

7.64.3.4 getDebugWindowText()

```
ttk_tools.vector.canapeapi.CANapeASAP3.getDebugWindowText (
              self )
```

Get current contents of CANape's write/debug window.

Returns

a contents as a string. If access to debug window contents failed, returned string will contain an error message.

Reimplemented from ttk_tools.vector._canapeapi.CANapeASAP3.

7.64.3.5 getLastError()

```
ttk_tools.vector.canapeapi.CANapeASAP3.getLastError (
              self,
              also_return_code = False )
```

Get an error text/description for the most recent CANape error.

Parameters

also_return_code	if True, the CANape error code will also be returned

Returns

the string "No Error" if, well, no error occurred, otherwise an error description supplied by CANape.

If parameter `also_return_code` is set, the result will be a tuple containing the error info: (<description>, <code>)

Reimplemented from ttk_tools.vector._canapeapi.CANapeASAP3.

7.64.3.6 getVersion()

```
ttk_tools.vector.canapeapi.CANapeASAP3.getVersion (
              self,
              verbose = False )
```

Get the current CANape/API version.

Parameters

verbose	if True, version info will also be printed to stdout

Returns

a dictionary with version information.

Reimplemented from ttk_tools.vector._canapeapi.CANapeASAP3.

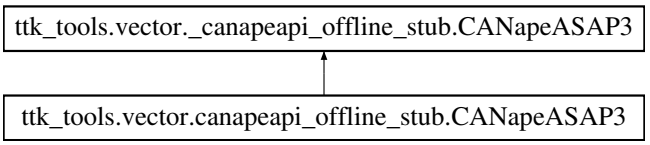
The documentation for this class was generated from the following file:

- ttk_tools/vector/canapeapi.py (v3.1.1)

7.65 ttk_tools.vector.canapeapi_offline_stub.CANapeASAP3 Class Reference

CANape ASAP3-connection (application).

Inheritance diagram for ttk_tools.vector.canapeapi_offline_stub.CANapeASAP3:



Public Member Functions

- __init__ (self, working_dir, clear_device_list=False, debug_mode=True, keep_canape_alive=False, canape↩ _path=None, init_timeout=100000, init_retries=3, api_log_level=0)

  Initialize a CANape connection (that is, a connection to the base application).
- exit (self, verbosity=2)

  Manually shut down ASAP3 connection to CANape.
- getVersion (self, verbose=False)

  Get the current CANape/API version.
- getDebugWindowText (self)

  Get current contents of CANape's write/debug window.
- getLastError (self, also_return_code=False)

  Get an error text/description for the most recent CANape error.
- getAvailableConverters (self, verbosity=0)

  Get available options for data file conversion.
- convertData (self, converter_id, source_file_name, dest_file_name=None, overwrite=True)

  Convert data between different formats.

7.65.1 Detailed Description

CANape ASAP3-connection (application).

7.65.2 Constructor & Destructor Documentation

7.65.2.1 __init__()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeASAP3.__init__ (
              self,
              working_dir,
              clear_device_list = False,
              debug_mode = True,
              keep_canape_alive = False,
              canape_path = None,
              init_timeout = 100000,
              init_retries = 3,
              api_log_level = 0 )
```

Initialize a CANape connection (that is, a connection to the base application).

Parameters

working_dir	working directory containing `CANape.ini`
clear_device_list	if device list gets **not** cleared and later a module name matches a device in the list, the existing device (i.e. its settings) will be assigned.
debug_mode	if True, the CANape window will be visible to show debug messages (i.e. the "Write Window")
keep_canape_alive	if True, the CANape instance will stay active between calls (thus drastically decreasing startup time)
canape_path	path to CANape installation directory, set to None to use the current path from registry (e.g. for single CANape installations)
init_timeout	[ms] timeout for connection init and responses (not too short, or timeout will occur while device database(s) are being read)
init_retries	how often the ASAP3 connection init should be retried in case of failure
api_log_level	log level for CANape-API calls (not relevant in stub)

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.CANapeASAP3.

7.65.3 Member Function Documentation

7.65.3.1 convertData()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeASAP3.convertData (
              self,
              converter_id,
              source_file_name,
              dest_file_name = None,
              overwrite = True )
```

Convert data between different formats.

Note

Support for Asap3MDFConvert was added in CANape 13.

Parameters

converter_id	id/name of converter to use, e.g. "MDF2MAT". See getAvailableConverters() for, well, available converters.
source_file_name	name/path of input/source file
dest_file_name	name/path of output/destination file None: CANape will choose name of output file, which typically means same path but a different file extension.
overwrite	True: overwrite existing files, False: do not overwrite existing files (may raise a CANapeError)

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.CANapeASAP3.

7.65.3.2 exit()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeASAP3.exit (
              self,
              verbosity = 2 )
```

Manually shut down ASAP3 connection to CANape.

If keep_canape_alive was set to False in constructor, the CANape application will also be closed.

This will normally be called during the destructor call, but it might sometimes be advantageous to exit the connection at a defined time (and not just implicitly when the destructor gets called eventually).

Parameters

verbosity	verbosity of stdout/log output: -1: do not show anything if handle is already gone (for calls in CANapeASAP3.__del__) 0: errors only 1: Asap3Exit status output 2: also show "nothing to do" notice if handle is already gone

Returns

True if successful (or already closed), otherwise False

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.CANapeASAP3.

7.65.3.3 getAvailableConverters()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeASAP3.getAvailableConverters (
              self,
              verbosity = 0 )
```

Get available options for data file conversion.

Note

Support for Asap3MDFConverterInfo was added in CANape 13.

Parameters

verbosity	0: silent, 1: log converters to stdout

Returns

a dictionary with entries for each converter option:
<converter id> : {
'id"' <converter id>,
''name'`: <converter name>, <BR> 'comment'`: <converter comment>,
}
or None if API does not support this feature (available starting with CANape 13).

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.CANapeASAP3.

7.65.3.4 getDebugWindowText()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeASAP3.getDebugWindowText (
            self )
```

Get current contents of CANape's write/debug window.

Returns

a contents as a string. If access to debug window contents failed, returned string will contain an error message.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.CANapeASAP3.

7.65.3.5 getLastError()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeASAP3.getLastError (
            self,
            also_return_code = False )
```

Get an error text/description for the most recent CANape error.

Parameters

also_return_code	if True, the CANape error code will also be returned

Returns

the string "No Error" if, well, no error occurred, otherwise an error description supplied by CANape.

If parameter `also_return_code` is set, the result will be a tuple containing the error info: (<description>, <code>)

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.CANapeASAP3.

7.65.3.6 getVersion()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeASAP3.getVersion (
              self,
              verbose = False )
```

Get the current CANape/API version.

Parameters

verbose	if True, version info will also be printed to stdout

Returns

a dictionary with version information.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.CANapeASAP3.

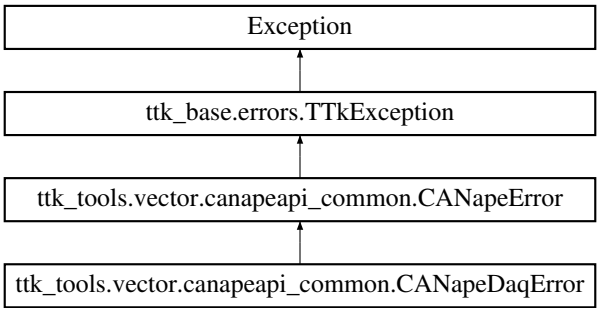
The documentation for this class was generated from the following file:

- ttk_tools/vector/canapeapi_offline_stub.py (v4.1.0)

7.66 ttk_tools.vector.canapeapi_common.CANapeDaqError Class Reference

Data Acquisition errors for CANape connection.

Inheritance diagram for ttk_tools.vector.canapeapi_common.CANapeDaqError:



Public Member Functions

- __init__ (self, msg='', log_msg=False, api_msg=None, error_code=None)

7.66.1 Detailed Description

Data Acquisition errors for CANape connection.

7.66.2 Constructor & Destructor Documentation

7.66.2.1 __init__()

```
ttk_tools.vector.canapeapi_common.CANapeDaqError.__init__ (
              self,
              msg = '',
              log_msg = False,
              api_msg = None,
              error_code = None )
```

Parameters

msg	error message
log_msg	True: log message to stdout, False: no log output
api_msg	CANape-provided "last error" message, if available
error_code	error code from CANapeAPI, if available.

Reimplemented from ttk_tools.vector.canapeapi_common.CANapeError.

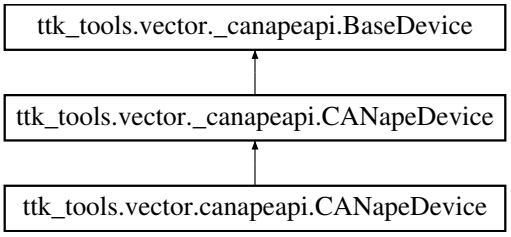
The documentation for this class was generated from the following file:

- ttk_tools/vector/canapeapi_common.py (v3.1.2)

7.67 ttk_tools.vector._canapeapi.CANapeDevice Class Reference

CANape device class (ASAP3-devices: CCP, XCP).

Inheritance diagram for ttk_tools.vector._canapeapi.CANapeDevice:



Public Member Functions

- __init__ (self, canape_asap3, db_filename="hil.a2l", comm_channel=DEV.CAN1, driver_type=DRIVER.XCP, module_name="ECU", connect=True, enable_cache=None, obj_access_retries=4, raise_get_exceptions=True, raise_set_exceptions=True)

  Initializes a standard CANape device (XCP, CAN, ...) connection.
- __str__ (self)

  Get a string representation of the device.
- getECUVar (self, varname, representation=REPR.PHYS, extended_mode=False, silent=False)

  Get the current value of a single ECU variable (single read access via polling).

- setECUVar (self, varname, data, representation=REPR.PHYS, extended_mode=False, return_original=True)

  Set data of a single ECU variable.
- getECUVarInfoBasic (self, varname, representation=REPR.PHYS)

  Get information about an ECU var/object.
- getECUVarByAddress (self, varname, length)

  Read `length` bytes from beginning at the address of an object defined by a given name.
- readBytes (self, address, length, addr_ext=0, verbosity=1)

  Read `length` bytes, starting at the supplied address.
- writeBytes (self, address, data, addr_ext=0, verbosity=1)

  Write the supplied bytes to ECU, starting at address.

Public Attributes

- obj_access_retries

  number of retries to perform when object access fails
- ascii_encoding

  encoding to use for decoding values of VALUE_TYPE.ASCII.

Static Public Attributes

- handle
- module
- status

Protected Member Functions

- _readCalibrationObject (self, varname, representation=REPR.PHYS)

  Simple wrapper for CANapeAPI.Asap3ReadCalibrationObject().
- _writeCalibrationObject (self, varname, calobj, representation=REPR.PHYS)

  Simple wrapper for CANapeAPI.Asap3WriteCalibrationObject().
- _readByAddress (self, addr, size, addr_ext=0, verbosity=1)

  A simple wrapper for CANapeAPI.Asap3ReadByAddress().
- _writeByAddress (self, addr, data, addr_ext=0, verbosity=1)

  A simple wrapper for CANapeAPI.Asap3WriteByAddress().
- _getCalObjectData (self, calobj, extended_mode=False)

  Retrieve data from an calibration object as a simple Python type (as float, a list of floats or a string, according to type of calibration object).
- _getFilledDoubleArray (self, desired_length, data, varname='')

  Create an array of c_double values, pre-filled with supplied data.

Protected Attributes

- _raise_get_exceptions

  Migration setting for getECUVar behavior True: Raise Exceptions for failed get/read attempts False: Return a CANapeErrorString.
- _raise_set_exceptions

  Migration setting for setECUVar behavior True: Raise Exceptions for failed set/write attempts False: Return a CANapeErrorString.
- _driver_type
- _comm_channel
- _module_name
- _db_filename
- _module_handle

7.67.1 Detailed Description

CANape device class (ASAP3-devices: CCP, XCP).

Also supports Bus-devices (like CAN/FlexRay), although getECUVar() will not return correct values, as those devices do not support polling. DAQ will work just fine, though.

7.67.2 Constructor & Destructor Documentation

7.67.2.1 __init__()

```
ttk_tools.vector._canapeapi.CANapeDevice.__init__ (
              self,
              canape_asap3,
              db_filename = "hil.a2l",
              comm_channel = DEV.CAN1,
              driver_type = DRIVER.XCP,
              module_name = "ECU",
              connect = True,
              enable_cache = None,
              obj_access_retries = 4,
              raise_get_exceptions = True,
              raise_set_exceptions = True )
```

Initializes a standard CANape device (XCP, CAN, ...) connection.

Parameters

canape_asap3	a CANapeASAP3 instance (application interface)
db_filename	filename of .a2l database (or .dbc for CAN, fibex/.xml for FlexRay, ...)
comm_channel	communication channel (see canapeapi_common.DEV constants)
driver_type	driver type (see canapeapi_common.DRIVER constants)
module_name	name of device/module in CANape
connect	Device connection behavior: True: immediately connect DUT after device has been created. Device will start in state "online" False: create device but remain in state "offline" if DUT cannot be connected at this time.
enable_cache	None: ignore parameter, use default True: enable caching False: disable caching Available since CANape 10, see Asap3CreateModule3
obj_access_retries	number of retries to perform when object access fails
raise_get_exceptions	Error handling for get/read access (see getECUVar()): True: Raise exceptions False: Return error strings (classic behavior)
raise_set_exceptions	Error handling for set/write access (see setECUVar()): True: Raise exceptions False: Return error strings (classic behavior)

Note

Parameter `connect` Setting `connect` to False permits attaching/creating a device that is currently not active (e.g. DUT is still shut down), that is, the device-init will not fail right away with error message "No response from ECU (attach Asap2 failed)".

An **inactive** DUT can be successfully attached and start out in state "offline". See isOnline() and goOnline(). However, if the DUT is **active** after all, the device's online state may end up in state "online" even if `connect` has been set to False.

Reimplemented from ttk_tools.vector._canapeapi.BaseDevice.

Reimplemented in ttk_tools.vector.canapeapi.CANapeDevice.

7.67.3 Member Function Documentation

7.67.3.1 __str__()

```
ttk_tools.vector._canapeapi.CANapeDevice.__str__ (
              self )
```

Get a string representation of the device.

Returns

an info string.

Reimplemented from ttk_tools.vector._canapeapi.BaseDevice.

7.67.3.2 _getCalObjectData()

```
ttk_tools.vector._canapeapi.CANapeDevice._getCalObjectData (
              self,
              calobj,
              extended_mode = False )  [protected]
```

Retrieve data from an calibration object as a simple Python type (as float, a list of floats or a string, according to type of calibration object).

Parameters

calobj	calibration object
extended_mode	if true, existing axis data for CURVE and MAP objects will also be returned

Example (simple mode): VALUE: a float value CURVE: [values] # (n float values) MAP: [values] # (x∗y float values) AXIS_PTS: [values] # (n float values) ASCII: a string

Example (extended mode): VALUE: a float value CURVE: ([axis values], [data values]) MAP: ([x-axis values], [y-axis values], [data values]) AXIS_PTS: [values] ASCII: a string

Note

If a `CURVE` or `MAP` entry has no assigned axis/axes (i.e. no referenced `AXIS_PTS` entries), CANape will return all axis values as 0.0

Returns

the calibration object data.
If an unknown calibration object type was encountered, a CANapeError will be raised (which should not happen unless the API changed between versions).

7.67.3.3 _getFilledDoubleArray()

```
ttk_tools.vector._canapeapi.CANapeDevice._getFilledDoubleArray (
              self,
              desired_length,
              data,
              varname = '' )  [protected]
```

Create an array of c_double values, pre-filled with supplied data.

Parameters

desired_length	desired length of new c_double array
data	a list of new values (or a scalar numeric value to be set to each entry)
varname	variable name (optional, for more informative warning messages)

Returns

a c_double array object

7.67.3.4 _readByAddress()

```
ttk_tools.vector._canapeapi.CANapeDevice._readByAddress (
              self,
              addr,
              size,
              addr_ext = 0,
              verbosity = 1 )  [protected]
```

A simple wrapper for CANapeAPI.Asap3ReadByAddress().

Parameters

addr	address to read from (preferably a c_ulong)
size	size of data to read (bytes)
addr_ext	address extension, only used with some special multiprocessor ECUs(leave at 0 if not required)
verbosity	verbosity of log output (0: only errors, 1: status info)

Returns

a string buffer (see ctypes.create_string_buffer) containing the read bytes

7.67.3.5 _readCalibrationObject()

```
ttk_tools.vector._canapeapi.CANapeDevice._readCalibrationObject (
              self,
              varname,
              representation = REPR.PHYS )  [protected]
```

Simple wrapper for CANapeAPI.Asap3ReadCalibrationObject().

7.67.3.6 _writeByAddress()

```
ttk_tools.vector._canapeapi.CANapeDevice._writeByAddress (
              self,
              addr,
              data,
              addr_ext = 0,
              verbosity = 1 )  [protected]
```

A simple wrapper for CANapeAPI.Asap3WriteByAddress().

Parameters

addr	start address to write to (preferably a c_ulong)
data	data to write (bytes), either as an initialized ctypes buffer (see ctypes.create_string_buffer) or just a list of byte values
addr_ext	address extension, only used with some special multiprocessor ECUs (leave at 0 if not required)
verbosity	verbosity of log output (0: only errors, 1: status info)

Returns

the write status from API (i.e. 0 on success)

7.67.3.7 _writeCalibrationObject()

```
ttk_tools.vector._canapeapi.CANapeDevice._writeCalibrationObject (
              self,
              varname,
              calobj,
              representation = REPR.PHYS )  [protected]
```

Simple wrapper for CANapeAPI.Asap3WriteCalibrationObject().

7.67.3.8 getECUVar()

```
ttk_tools.vector._canapeapi.CANapeDevice.getECUVar (
              self,
              varname,
              representation = REPR.PHYS,
              extended_mode = False,
              silent = False )
```

Get the current value of a single ECU variable (single read access via polling).

Parameters

varname	name of variable (signal/measurement/calibration)
representation	data representation, can be either REPR.HEX (ecu internal value) or REPR.PHYS (physical representation of data). See canapeapi_common.REPR.
extended_mode	if True, existing axis data for `CURVE` and `MAP` objects will also be returned. See canapeapi_common.VALUE_TYPE
silent	True: suppress debug error messages

Example: Data format in "simple mode" depending on value types

```
VALUE:     a float value
CURVE:     [values]          # a list of n float values
MAP:       [values]          # a list of x*y float values
AXIS_PTS:  [values]          # a list of n float values
ASCII:     a string
VAL_BLK:   [values]          # a list of x*y float values
```

Example: Data format in "extended mode" depending on value types

```
VALUE:     a float value
CURVE:     ([axis values], [data values])
MAP:       ([x-axis values], [y-axis values], [data values])
AXIS_PTS:  [values]
ASCII:     a string
VAL_BLK:   [values]          # a list of x*y float values

# Note: If a CURVE or MAP entry has no assigned axis/axes
#       (i.e. it has no referenced AXIS_PTS entries),
#       CANape will return all axis values as 0.0
```

Returns

the current data found for the variable (calibration object).

In case of access errors, either a CANapeErrorString will be returned or a CANapeAccessError will be raised, depending on the setting of raise_get_exceptions, see CANapeDevice.__init__.

For other error cases (e.g. an invalid data type was encountered), a CANapeError will be raised (or CANape↩ ErrorString returned; again, see raise_get_exceptions)

Reimplemented in ttk_tools.vector.canapeapi.CANapeDevice.

7.67.3.9 getECUVarByAddress()

```
ttk_tools.vector._canapeapi.CANapeDevice.getECUVarByAddress (
              self,
              varname,
              length )
```

Read `length` bytes from beginning at the address of an object defined by a given name.

Parameters

varname	name of variable at the start address
length	number of bytes to read

Returns

a raw bytes string.

Reimplemented in ttk_tools.vector.canapeapi.CANapeDevice.

7.67.3.10 getECUVarInfoBasic()

```
ttk_tools.vector._canapeapi.CANapeDevice.getECUVarInfoBasic (
              self,
              varname,
              representation = REPR.PHYS )
```

Get information about an ECU var/object.

Parameters

varname	name of variable
representation	data representation, can be either REPR.HEX (ecu internal value) or REPR.PHYS (physical representation of data). See canapeapi_common.REPR

Returns

a dict with some basic variable information:
‘'type’: data type (see canapeapi_common.DATA_TYPE) as string 'address': memory address
 'minimum': minimum value
 'maximum': maximum value
 'increment'`: increment value

Reimplemented in ttk_tools.vector.canapeapi.CANapeDevice.

7.67.3.11 readBytes()

```
ttk_tools.vector._canapeapi.CANapeDevice.readBytes (
              self,
              address,
              length,
              addr_ext = 0,
              verbosity = 1 )
```

Read `length` bytes, starting at the supplied address.

Parameters

address	start address in ECU to read from (int)
length	number of bytes to read
addr_ext	address extension, only used with some special multiprocessor ECUs (leave at 0 if not required)
verbosity	verbosity of log output 0: only errors 1: status info

Example:

```
buf = canape.readBytes(0x123456, 8)
print buf                   # buf is a bytes string, e.g. b'foo\x01bar.'
print list(buf)             # list of bytes (as int values)
print HexList(buf)          # list of bytes with "nicer" hex display
print binascii.hexlify(buf) # hex string
```

Returns

a raw bytes string.

Reimplemented in ttk_tools.vector.canapeapi.CANapeDevice.

7.67.3.12 setECUVar()

```
ttk_tools.vector._canapeapi.CANapeDevice.setECUVar (
              self,
              varname,
              data,
              representation = REPR.PHYS,
              extended_mode = False,
              return_original = True )
```

Set data of a single ECU variable.

Parameters

varname	name of variable (signal/measurement/calibration)
data	data to set (either a scalar value or a list of values)
representation	data representation, can be either REPR.HEX (ecu internal value) or REPR.PHYS (physical representation of data). See canapeapi_common.REPR
extended_mode	see description and examples at getECUVar()
return_original	True: return original value before "set" False: return current value after "set"

Returns

the previous/original value if `return_original` is True (useful to reset changes later), otherwise the current value (as read back from ECU, useful to check if the new value has been successfully set)

In case of errors, either a CANapeErrorString will be returned or a CANapeAccessError will be raised, depending on the setting of raise_set_exceptions, see CANapeDevice.__init__.

Reimplemented in ttk_tools.vector.canapeapi.CANapeDevice.

7.67.3.13 writeBytes()

```
ttk_tools.vector._canapeapi.CANapeDevice.writeBytes (
              self,
              address,
              data,
              addr_ext = 0,
              verbosity = 1 )
```

Write the supplied bytes to ECU, starting at address.

Parameters

address	start address in ECU to read from (preferably already a `c_ulong`)
data	an already initialized ctypes string buffer OR a list of byte values to write OR a hex-string (a leading "0x" will be ignored) Note that a missing nibble in the hex string will be zero-padded to get a full byte
addr_ext	address extension, only used with some special multiprocessor ECUs (leave at 0 if not required)
verbosity	verbosity of log output 0: only errors 1: status info

Examples:

```
canape.writeBytes(0x123456, "0xAABBCC")
canape.writeBytes(0x123456, "AA BB CC")
canape.writeBytes(0x123456, "ABBCC") # will be padded to 0ABBCC
canape.writeBytes(0x123456, [0xAA, 0xBB, 0xCC])
canape.writeBytes(0x123456, [170, 187, 204])
```

Returns

the write status from low-level API (i.e. 0 on success)

Reimplemented in ttk_tools.vector.canapeapi.CANapeDevice.

7.67.4 Member Data Documentation

7.67.4.1 _comm_channel

```
ttk_tools.vector._canapeapi.CANapeDevice._comm_channel  [protected]
```

7.67.4.2 _db_filename

```
ttk_tools.vector._canapeapi.CANapeDevice._db_filename  [protected]
```

7.67.4.3 _driver_type

ttk_tools.vector._canapeapi.CANapeDevice._driver_type [protected]

7.67.4.4 _module_handle

ttk_tools.vector._canapeapi.CANapeDevice._module_handle [protected]

7.67.4.5 _module_name

ttk_tools.vector._canapeapi.CANapeDevice._module_name [protected]

7.67.4.6 _raise_get_exceptions

ttk_tools.vector._canapeapi.CANapeDevice._raise_get_exceptions [protected]

Migration setting for getECUVar behavior True: Raise Exceptions for failed get/read attempts False: Return a CANapeErrorString.

7.67.4.7 _raise_set_exceptions

ttk_tools.vector._canapeapi.CANapeDevice._raise_set_exceptions [protected]

Migration setting for setECUVar behavior True: Raise Exceptions for failed set/write attempts False: Return a CANapeErrorString.

7.67.4.8 ascii_encoding

ttk_tools.vector._canapeapi.CANapeDevice.ascii_encoding

encoding to use for decoding values of VALUE_TYPE.ASCII.

If set to None, values will be returned as raw byte strings

7.67.4.9 handle

ttk_tools.vector._canapeapi.CANapeDevice.handle [static]

7.67.4.10 module

ttk_tools.vector._canapeapi.CANapeDevice.module [static]

7.67.4.11 obj_access_retries

```
ttk_tools.vector._canapeapi.CANapeDevice.obj_access_retries
```

number of retries to perform when object access fails

7.67.4.12 status

```
ttk_tools.vector._canapeapi.CANapeDevice.status  [static]
```

Initial value:

```
=  self._asap3.api.Asap3ReleaseModule(
                    self._asap3.getHandle(),
                    self._module_handle
                )
```

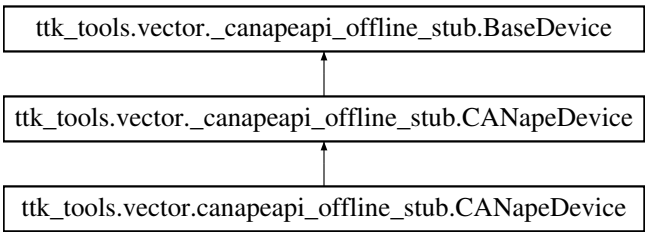
The documentation for this class was generated from the following file:

- ttk_tools/vector/_canapeapi.py (v7.3.3)

7.68 ttk_tools.vector._canapeapi_offline_stub.CANapeDevice Class Reference

CANape device class (ASAP3-devices: CCP, XCP).

Inheritance diagram for ttk_tools.vector._canapeapi_offline_stub.CANapeDevice:



Public Member Functions

- __init__ (self, canape_asap3, db_filename="hil.a2l", comm_channel=DEV.CAN1, driver_type=DRIVER.XCP, module_name="ECU", connect=True, enable_cache=None, obj_access_retries=4, raise_get_exceptions=True, raise_set_exceptions=True)

  Initializes a standard CANape device (XCP, CAN, ...) connection.
- __str__ (self)

  Get a string representation of the device.
- getECUVar (self, varname, representation=REPR.PHYS, extended_mode=False, silent=False)

  Get the current value of a single ECU variable (single read access via polling).
- setECUVar (self, varname, data, representation=REPR.PHYS, extended_mode=False, return_original=True)

  Set data of a single ECU variable.
- getECUVarInfoBasic (self, varname, representation=REPR.PHYS)

  Get information about an ECU var/object.
- getECUVarByAddress (self, varname, length)

  Read `length` bytes from beginning at the address of an object defined by a given name.
- readBytes (self, address, length, addr_ext=0, verbosity=1)

  Read `length` bytes, starting at the supplied address.
- writeBytes (self, address, data, addr_ext=0, verbosity=1)

  Write the supplied bytes to ECU, starting at address.

Public Attributes

- obj_access_retries

  number of retries to perform when object access fails
- ascii_encoding

  encoding to use for decoding values of VALUE_TYPE.ASCII If set to None, values will be returned as raw byte strings

Protected Attributes

- _data_provider

  data provider for stub implementation
- _raise_get_exceptions

  Migration setting for getECUVar behavior True: Raise Exceptions for failed get/read attempts False: Return a CANapeErrorString.
- _raise_set_exceptions

  Migration setting for setECUVar behavior True: Raise Exceptions for failed set/write attempts False: Return a CANapeErrorString.
- _driver_type
- _comm_channel
- _module_name
- _db_filename
- _context

7.68.1 Detailed Description

CANape device class (ASAP3-devices: CCP, XCP).

Also supports Bus-devices (like CAN/FlexRay), although getECUVar() will not return correct values, as those devices do not support polling. DAQ will work just fine, though.

7.68.2 Constructor & Destructor Documentation

7.68.2.1 __init__()

```
ttk_tools.vector._canapeapi_offline_stub.CANapeDevice.__init__ (
              self,
              canape_asap3,
              db_filename = "hil.a2l",
              comm_channel = DEV.CAN1,
              driver_type = DRIVER.XCP,
              module_name = "ECU",
              connect = True,
              enable_cache = None,
              obj_access_retries = 4,
              raise_get_exceptions = True,
              raise_set_exceptions = True )
```

Initializes a standard CANape device (XCP, CAN, ...) connection.

Parameters

canape_asap3	a CANapeASAP3 instance (application interface)
db_filename	filename of .a2l database (or .dbc for CAN, fibex/.xml for FlexRay, ...)
comm_channel	communication channel (see canapeapi_common.DEV constants)
driver_type	driver type (see canapeapi_common.DRIVER constants)
module_name	name of device/module in CANape
connect	Device connection behavior: True: immediately connect DUT after device has been created. Device will start in state "online" False: create device but remain in state "offline" if DUT cannot be connected at this time.
enable_cache	None: ignore parameter, use default True: enable caching False: disable caching Available since CANape 10, see Asap3CreateModule3
obj_access_retries	number of retries to perform when object access fails
raise_get_exceptions	Error handling for get/read access (see getECUVar()): True: Raise exceptions False: Return error strings (classic behavior)
raise_set_exceptions	Error handling for set/write access (see setECUVar()): True: Raise exceptions False: Return error strings (classic behavior)

Note

Parameter `connect` Setting `connect` to False permits attaching/creating a device that is currently not active (e.g. DUT is still shut down), that is, the device-init will not fail right away with error message "No response from ECU (attach Asap2 failed)".

An **inactive** DUT can be successfully attached and start out in state "offline". See isOnline() and goOnline(). However, if the DUT is **active** after all, the device's online state may end up in state "online" even if `connect` has been set to False.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.BaseDevice.

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.CANapeDevice.

7.68.3 Member Function Documentation

7.68.3.1 __str__()

```
ttk_tools.vector._canapeapi_offline_stub.CANapeDevice.__str__ (
            self )
```

Get a string representation of the device.

Returns

an info string.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.BaseDevice.

7.68.3.2 getECUVar()

```
ttk_tools.vector._canapeapi_offline_stub.CANapeDevice.getECUVar (
              self,
              varname,
              representation = REPR.PHYS,
              extended_mode = False,
              silent = False )
```

Get the current value of a single ECU variable (single read access via polling).

Parameters

varname	name of variable (signal/measurement/calibration)
representation	data representation, can be either REPR.HEX (ecu internal value) or REPR.PHYS (physical representation of data). See canapeapi_common.REPR.
extended_mode	if True, existing axis data for `CURVE` and `MAP` objects will also be returned. See canapeapi_common.VALUE_TYPE
silent	True: suppress debug error messages

Example: Data format in "simple mode" depending on value types

```
VALUE:     a float value
CURVE:     [values]          # a list of n float values
MAP:       [values]          # a list of x*y float values
AXIS_PTS:  [values]          # a list of n float values
ASCII:     a string
VAL_BLK:   [values]          # a list of x*y float values
```

Example: Data format in "extended mode" depending on value types

```
VALUE:     a float value
CURVE:     ([axis values], [data values])
MAP:       ([x-axis values], [y-axis values], [data values])
AXIS_PTS:  [values]
ASCII:     a string
VAL_BLK:   [values]          # a list of x*y float values

# Note: If a CURVE or MAP entry has no assigned axis/axes
#       (i.e. it has no referenced AXIS_PTS entries),
#       CANape will return all axis values as 0.0
```

Returns

the current data found for the variable (calibration object).

In case of access errors, either a CANapeErrorString will be returned or a CANapeAccessError will be raised, depending on the setting of raise_get_exceptions, see CANapeDevice.__init__.

For other error cases (e.g. an invalid data type was encountered), a CANapeError will be raised (or CANape↩ ErrorString returned; again, see raise_get_exceptions)

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.CANapeDevice.

7.68.3.3 getECUVarByAddress()

```
ttk_tools.vector._canapeapi_offline_stub.CANapeDevice.getECUVarByAddress (
              self,
              varname,
              length )
```

Read `length` bytes from beginning at the address of an object defined by a given name.

Parameters

varname	name of variable at the start address
length	number of bytes to read

Returns

a raw bytes string.

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.CANapeDevice.

7.68.3.4 getECUVarInfoBasic()

```
ttk_tools.vector._canapeapi_offline_stub.CANapeDevice.getECUVarInfoBasic (
              self,
              varname,
              representation = REPR.PHYS )
```

Get information about an ECU var/object.

Parameters

varname	name of variable
representation	data representation, can be either REPR.HEX (ecu internal value) or REPR.PHYS (physical representation of data). See canapeapi_common.REPR

Returns

a dict with some basic variable information:
‘'type’: data type (see canapeapi_common.DATA_TYPE) as string 'address': memory address
 'minimum': minimum value
 'maximum': maximum value
 'increment'`: increment value

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.CANapeDevice.

7.68.3.5 readBytes()

```
ttk_tools.vector._canapeapi_offline_stub.CANapeDevice.readBytes (
              self,
              address,
              length,
              addr_ext = 0,
              verbosity = 1 )
```

Read `length` bytes, starting at the supplied address.

Parameters

address	start address in ECU to read from (preferably already a `c_ulong`)
length	number of bytes to read
addr_ext	address extension, only used with some special multiprocessor ECUs (leave at 0 if not required)
verbosity	verbosity of log output 0: only errors 1: status info

Example:

```
buf = canape.readBytes(0x123456, 16)
print [ord(b) for b in buf] # list of bytes (as int values)
print HexList(buf)          # list of bytes with "nicer" hex display
print binascii.hexlify(buf) # hex string
```

Returns

a raw bytes string.

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.CANapeDevice.

7.68.3.6 setECUVar()

```
ttk_tools.vector._canapeapi_offline_stub.CANapeDevice.setECUVar (
              self,
              varname,
              data,
              representation = REPR.PHYS,
              extended_mode = False,
              return_original = True )
```

Set data of a single ECU variable.

Parameters

varname	name of variable (signal/measurement/calibration)
data	data to set (either a scalar value or a list of values)
representation	data representation, can be either REPR.HEX (ecu internal value) or REPR.PHYS (physical representation of data). See canapeapi_common.REPR
extended_mode	see description and examples at getECUVar()
return_original	True: return original value before "set" False: return current value after "set"

Returns

the previous/original value if `return_original` is True (useful to reset changes later), otherwise the current value (as read back from ECU, useful to check if the new value has been successfully set)

In case of errors, either a CANapeErrorString will be returned or a CANapeAccessError will be raised, depending on the setting of raise_set_exceptions, see CANapeDevice.__init__.

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.CANapeDevice.

7.68.3.7 writeBytes()

```
ttk_tools.vector._canapeapi_offline_stub.CANapeDevice.writeBytes (
              self,
              address,
              data,
              addr_ext = 0,
              verbosity = 1 )
```

Write the supplied bytes to ECU, starting at address.

Parameters

address	start address in ECU to read from (preferably already a `c_ulong`)
data	an already initialized ctypes string buffer OR a list of byte values to write OR a hex-string (a leading "0x" will be ignored) Note that a missing nibble in the hex string will be zero-padded to get a full byte
addr_ext	address extension, only used with some special multiprocessor ECUs (leave at 0 if not required)
verbosity	verbosity of log output 0: only errors 1: status info

Examples:

```
canape.writeBytes(0x123456, "0xAABBCC")
canape.writeBytes(0x123456, "AA BB CC")
canape.writeBytes(0x123456, "ABBCC") # will be padded to 0ABBCC
canape.writeBytes(0x123456, [0xAA, 0xBB, 0xCC])
canape.writeBytes(0x123456, [170, 187, 204])
```

Returns

the write status from low-level API (i.e. 0 on success)

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.CANapeDevice.

7.68.4 Member Data Documentation

7.68.4.1 _comm_channel

```
ttk_tools.vector._canapeapi_offline_stub.CANapeDevice._comm_channel  [protected]
```

7.68.4.2 _context

```
ttk_tools.vector._canapeapi_offline_stub.CANapeDevice._context  [protected]
```

7.68.4.3 _data_provider

ttk_tools.vector._canapeapi_offline_stub.CANapeDevice._data_provider [protected]

data provider for stub implementation

7.68.4.4 _db_filename

ttk_tools.vector._canapeapi_offline_stub.CANapeDevice._db_filename [protected]

7.68.4.5 _driver_type

ttk_tools.vector._canapeapi_offline_stub.CANapeDevice._driver_type [protected]

7.68.4.6 _module_name

ttk_tools.vector._canapeapi_offline_stub.CANapeDevice._module_name [protected]

7.68.4.7 _raise_get_exceptions

ttk_tools.vector._canapeapi_offline_stub.CANapeDevice._raise_get_exceptions [protected]

Migration setting for getECUVar behavior True: Raise Exceptions for failed get/read attempts False: Return a CANapeErrorString.

7.68.4.8 _raise_set_exceptions

ttk_tools.vector._canapeapi_offline_stub.CANapeDevice._raise_set_exceptions [protected]

Migration setting for setECUVar behavior True: Raise Exceptions for failed set/write attempts False: Return a CANapeErrorString.

7.68.4.9 ascii_encoding

ttk_tools.vector._canapeapi_offline_stub.CANapeDevice.ascii_encoding

encoding to use for decoding values of VALUE_TYPE.ASCII If set to None, values will be returned as raw byte strings

7.68.4.10 obj_access_retries

ttk_tools.vector._canapeapi_offline_stub.CANapeDevice.obj_access_retries

number of retries to perform when object access fails

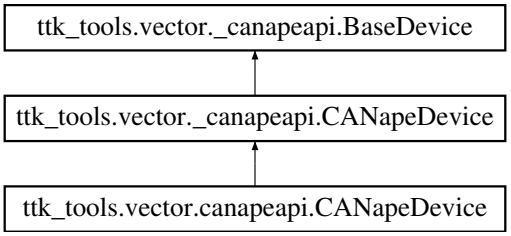
The documentation for this class was generated from the following file:

- ttk_tools/vector/_canapeapi_offline_stub.py (v4.7.1)

7.69 ttk_tools.vector.canapeapi.CANapeDevice Class Reference

CANape device class (ASAP3-devices: CCP, XCP).

Inheritance diagram for ttk_tools.vector.canapeapi.CANapeDevice:



Public Member Functions

- __init__ (self, canape_asap3, db_filename="hil.a2l", comm_channel=DEV.CAN1, driver_type=DRIVER.XCP, module_name="ECU", connect=True, enable_cache=None, obj_access_retries=4, raise_get_exceptions=True, raise_set_exceptions=True)

  Initializes a standard CANape device (XCP, CAN, ...) connection.
- getECUVar (self, varname, representation=REPR.PHYS, extended_mode=False, silent=False)

  Get the current value of a single ECU variable (single read access via polling).
- setECUVar (self, varname, data, representation=REPR.PHYS, extended_mode=False, return_original=True)

  Set data of a single ECU variable.
- getECUVarInfoBasic (self, varname, representation=REPR.PHYS)

  Get information about an ECU var/object.
- getECUVarByAddress (self, varname, length)

  Read `length` bytes from beginning at the address of an object defined by a given name.
- readBytes (self, address, length, addr_ext=0, verbosity=1)

  Read `length` bytes, starting at the supplied address.
- writeBytes (self, address, data, addr_ext=0, verbosity=1)

  Write the supplied bytes to ECU, starting at address.
- release (self, verbosity=2)

  Manually release the module/device handle in CANape.
- runScript (self, script_command, is_file_name=False)

  Execute a CAPL script or script command via CANapeAPI.
- isOnline (self)

  Get online state of current device.
- goOnline (self, retry_timeout=10000)

  Switch the CANape device connection to online mode.
- goOffline (self, retry_timeout=10000)

  Switch the CANape device connection to offline mode.
- getECUTasks (self)

  Get a dictionary containing available measurement tasks (as defined in the ECU)

7.69.1 Detailed Description

CANape device class (ASAP3-devices: CCP, XCP).

Also supports Bus-devices (like CAN/FlexRay), although getECUVar() will not return correct values, as those devices do not support polling. DAQ will work just fine, though.

7.69.2 Constructor & Destructor Documentation

7.69.2.1 __init__()

```
ttk_tools.vector.canapeapi.CANapeDevice.__init__ (
              self,
              canape_asap3,
              db_filename = "hil.a2l",
              comm_channel = DEV.CAN1,
              driver_type = DRIVER.XCP,
              module_name = "ECU",
              connect = True,
              enable_cache = None,
              obj_access_retries = 4,
              raise_get_exceptions = True,
              raise_set_exceptions = True )
```

Initializes a standard CANape device (XCP, CAN, ...) connection.

Parameters

canape_asap3	a CANapeASAP3 instance (application interface)
db_filename	filename of .a2l database (or .dbc for CAN, fibex/.xml for FlexRay, ...)
comm_channel	communication channel (see canapeapi_common.DEV constants)
driver_type	driver type (see canapeapi_common.DRIVER constants)
module_name	name of device/module in CANape
connect	Device connection behavior: True: immediately connect DUT after device has been created. Device will start in state "online" False: create device but remain in state "offline" if DUT cannot be connected at this time.
enable_cache	None: ignore parameter, use default True: enable caching False: disable caching Available since CANape 10, see Asap3CreateModule3
obj_access_retries	number of retries to perform when object access fails
raise_get_exceptions	Error handling for get/read access (see getECUVar()): True: Raise exceptions False: Return error strings (classic behavior)
raise_set_exceptions	Error handling for set/write access (see setECUVar()): True: Raise exceptions False: Return error strings (classic behavior)

Note

Parameter `connect` Setting `connect` to False permits attaching/creating a device that is currently not active (e.g. DUT is still shut down), that is, the device-init will not fail right away with error message "No response from ECU (attach Asap2 failed)".

An **inactive** DUT can be successfully attached and start out in state "offline". See isOnline() and goOnline(). However, if the DUT is **active** after all, the device's online state may end up in state "online" even if `connect` has been set to False.

Reimplemented from ttk_tools.vector._canapeapi.CANapeDevice.

7.69.3 Member Function Documentation

7.69.3.1 getECUTasks()

```
ttk_tools.vector.canapeapi.CANapeDevice.getECUTasks (
              self )
```

Get a dictionary containing available measurement tasks (as defined in the ECU)

Returns

a task dictionary with entries like:
<task name/descr.>: {
"id": <internal task id>,
"cycle_time": <cycle time in ms, 0 if n/a>
}

Reimplemented from ttk_tools.vector._canapeapi.BaseDevice.

7.69.3.2 getECUVar()

```
ttk_tools.vector.canapeapi.CANapeDevice.getECUVar (
              self,
              varname,
              representation = REPR.PHYS,
              extended_mode = False,
              silent = False )
```

Get the current value of a single ECU variable (single read access via polling).

Parameters

varname	name of variable (signal/measurement/calibration)
representation	data representation, can be either REPR.HEX (ecu internal value) or REPR.PHYS (physical representation of data). See canapeapi_common.REPR.
extended_mode	if True, existing axis data for `CURVE` and `MAP` objects will also be returned. See canapeapi_common.VALUE_TYPE
silent	True: suppress debug error messages

Example: Data format in "simple mode" depending on value types

```
VALUE:     a float value
CURVE:     [values]         # a list of n float values
MAP:       [values]         # a list of x*y float values
AXIS_PTS:  [values]         # a list of n float values
ASCII:     a string
```

Example: Data format in "extended mode" depending on value types

```
VALUE:     a float value
CURVE:     ([axis values], [data values])
MAP:       ([x-axis values], [y-axis values], [data values])
AXIS_PTS:  [values]
ASCII:     a string

# Note: If a CURVE or MAP entry has no assigned axis/axes
#       (i.e. it has no referenced AXIS_PTS entries),
#       CANape will return all axis values as 0.0
```

Returns

the current data found for the variable (calibration object).

In case of access errors, either a CANapeErrorString will be returned or a CANapeAccessError will be raised, depending on the setting of raise_get_exceptions, see CANapeDevice.__init__.

For other error cases (e.g. an invalid data type was encountered), a CANapeError will be raised (or CANape↩ ErrorString returned; again, see raise_get_exceptions)

Reimplemented from ttk_tools.vector._canapeapi.CANapeDevice.

7.69.3.3 getECUVarByAddress()

```
ttk_tools.vector.canapeapi.CANapeDevice.getECUVarByAddress (
              self,
              varname,
              length )
```

Read `length` bytes from beginning at the address of an object defined by a given name.

Parameters

varname	name of variable at the start address
length	number of bytes to read

Returns

a raw bytes string.

Reimplemented from ttk_tools.vector._canapeapi.CANapeDevice.

7.69.3.4 getECUVarInfoBasic()

```
ttk_tools.vector.canapeapi.CANapeDevice.getECUVarInfoBasic (
              self,
              varname,
              representation = REPR.PHYS )
```

Get information about an ECU var/object.

Parameters

varname	name of variable
representation	data representation, can be either REPR.HEX (ecu internal value) or REPR.PHYS (physical representation of data). See canapeapi_common.REPR

Returns

a dict with some basic variable information:
"type": data type (see canapeapi_common.DATA_TYPE) as string 'address': memory address
 'minimum': minimum value
 'maximum': maximum value
 'increment`: increment value

Reimplemented from ttk_tools.vector._canapeapi.CANapeDevice.

7.69.3.5 goOffline()

```
ttk_tools.vector.canapeapi.CANapeDevice.goOffline (
              self,
              retry_timeout = 10000 )
```

Switch the CANape device connection to offline mode.

Parameters

retry_timeout	[ms], if > 0, repeated attempts to set the ecu state will be made until timeout occurs

Returns

a list of testresult entries containing info about retried attempts.

Reimplemented from ttk_tools.vector._canapeapi.BaseDevice.

7.69.3.6 goOnline()

```
ttk_tools.vector.canapeapi.CANapeDevice.goOnline (
              self,
              retry_timeout = 10000 )
```

Switch the CANape device connection to online mode.

Parameters

retry_timeout	[ms], if > 0, repeated attempts to set the ECU state will be made until timeout occurs

Returns

a list of testresult entries containing info about retried attempts

Reimplemented from ttk_tools.vector._canapeapi.BaseDevice.

7.69.3.7 isOnline()

```
ttk_tools.vector.canapeapi.CANapeDevice.isOnline (
              self )
```

Get online state of current device.

Returns

True if the device is online, otherwise False.

Reimplemented from ttk_tools.vector._canapeapi.BaseDevice.

7.69.3.8 readBytes()

```
ttk_tools.vector.canapeapi.CANapeDevice.readBytes (
              self,
              address,
              length,
              addr_ext = 0,
              verbosity = 1 )
```

Read `length` bytes, starting at the supplied address.

Parameters

address	start address in ECU to read from (int)
length	number of bytes to read
addr_ext	address extension, only used with some special multiprocessor ECUs (leave at 0 if not required)
verbosity	verbosity of log output 0: only errors 1: status info

Example:

```
buf = canape.readBytes(0x123456, 8)
print buf                   # buf is a bytes string, e.g. b'foo\x01bar.'
print list(buf)             # list of bytes (as int values)
print HexList(buf)          # list of bytes with "nicer" hex display
print binascii.hexlify(buf) # hex string
```

Returns

a raw bytes string.

Reimplemented from ttk_tools.vector._canapeapi.CANapeDevice.

7.69.3.9 release()

```
ttk_tools.vector.canapeapi.CANapeDevice.release (
              self,
              verbosity = 2 )
```

Manually release the module/device handle in CANape.

This will normally be called during **del**, but sometimes it might be advantageous to release the handle a defined time (and not just implicitly when the destructor gets called eventually).

Parameters

verbosity	verbosity of stdout/log output: -1: do not show anything if handle is already gone (for calls in **del**) 0: errors only 1: Asap3Exit status output 2: also show "nothing to do" notice if handle is already gone

Returns

True if successful (or already released), otherwise False.

Reimplemented from ttk_tools.vector._canapeapi.BaseDevice.

7.69.3.10 runScript()

```
ttk_tools.vector.canapeapi.CANapeDevice.runScript (
              self,
              script_command,
              is_file_name = False )
```

Execute a CAPL script or script command via CANapeAPI.

Note

This uses the "classic" `Asap3ExecuteScript` API function which does not provide any script status or return values.

New extended script functions are available since CANape version 9. The classic approach is sufficient for simple cases, though.

Parameters

script_command	CAPL command or name of CAPL script file to execute. A CAPL command can also be the name of a custom "project function" that has been defined in the current workspace.
is_file_name	True if `script_command` contains the name of a script file to be run, False if it directly contains a CAPL command.

Example:

```
# multiple commands can be added in sequence
device.runScript('Speak("Hello World"); Write("Done.");')

# some CAPL commands/methods have to be called on an existing
# device object
device.runScript('%s.SaveDatabase(0);'%(device.getModuleName())

# default location of script files is the folder of the current
# CANape workspace
device.runScript('capl_script.scr', is_file_name=True)
```

Note

The executed script or script command might still be running when this function returns.

Returns

True if script execution was successful (though not necessarily completed), otherwise False.

Reimplemented from ttk_tools.vector._canapeapi.BaseDevice.

7.69.3.11 setECUVar()

```
ttk_tools.vector.canapeapi.CANapeDevice.setECUVar (
                self,
                varname,
                data,
                representation = REPR.PHYS,
                extended_mode = False,
                return_original = True )
```

Set data of a single ECU variable.

Parameters

varname	name of variable (signal/measurement/calibration)
data	data to set (either a scalar value or a list of values)
representation	data representation, can be either REPR.HEX (ecu internal value) or REPR.PHYS (physical representation of data). See canapeapi_common.REPR
extended_mode	see description and examples at getECUVar()
return_original	True: return original value before "set" False: return current value after "set"

Returns

the previous/original value if `return_original` is True (useful to reset changes later), otherwise the current value (as read back from ECU, useful to check if the new value has been successfully set)

In case of errors, either a CANapeErrorString will be returned or a CANapeAccessError will be raised, depending on the setting of raise_set_exceptions, see CANapeDevice.__init__.

Reimplemented from ttk_tools.vector._canapeapi.CANapeDevice.

7.69.3.12 writeBytes()

```
ttk_tools.vector.canapeapi.CANapeDevice.writeBytes (
                self,
                address,
                data,
                addr_ext = 0,
                verbosity = 1 )
```

Write the supplied bytes to ECU, starting at address.

Parameters

address	start address in ECU to read from (preferably already a `c_ulong`)
data	an already initialized ctypes string buffer OR a list of byte values to write OR a hex-string (a leading "0x" will be ignored) Note that a missing nibble in the hex string will be zero-padded to get a full byte
addr_ext	address extension, only used with some special multiprocessor ECUs (leave at 0 if not required)
verbosity	verbosity of log output 0: only errors 1: status info

Examples:

```
canape.writeBytes(0x123456, "0xAABBCC")
canape.writeBytes(0x123456, "AA BB CC")
canape.writeBytes(0x123456, "ABBCC") # will be padded to 0ABBCC
canape.writeBytes(0x123456, [0xAA, 0xBB, 0xCC])
canape.writeBytes(0x123456, [170, 187, 204])
```

Returns

the write status from low-level API (i.e. 0 on success)

Reimplemented from ttk_tools.vector._canapeapi.CANapeDevice.

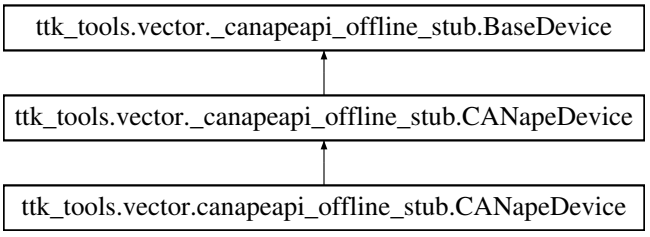
The documentation for this class was generated from the following file:

- ttk_tools/vector/canapeapi.py (v3.1.1)

7.70 ttk_tools.vector.canapeapi_offline_stub.CANapeDevice Class Reference

CANape device class (ASAP3-devices: CCP, XCP).

Inheritance diagram for ttk_tools.vector.canapeapi_offline_stub.CANapeDevice:



Public Member Functions

- __init__ (self, canape_asap3, db_filename="hil.a2l", comm_channel=DEV.CAN1, driver_type=DRIVER.XCP, module_name="ECU", connect=True, enable_cache=None, obj_access_retries=4, raise_get_exceptions=True, raise_set_exceptions=True)

  Initializes a standard CANape device (XCP, CAN, ...) connection.
- getECUVar (self, varname, representation=REPR.PHYS, extended_mode=False, silent=False)

  Get the current value of a single ECU variable (single read access via polling).
- setECUVar (self, varname, data, representation=REPR.PHYS, extended_mode=False, return_original=True)

  Set data of a single ECU variable.
- getECUVarInfoBasic (self, varname, representation=REPR.PHYS)

  Get information about an ECU var/object.
- getECUVarByAddress (self, varname, length)

  Read `length` bytes from beginning at the address of an object defined by a given name.
- readBytes (self, address, length, addr_ext=0, verbosity=1)

  Read `length` bytes, starting at the supplied address.
- writeBytes (self, address, data, addr_ext=0, verbosity=1)

  Write the supplied bytes to ECU, starting at address.
- release (self, verbosity=2)

  Manually release the module/device handle in CANape.
- runScript (self, script_command, is_file_name=False)

  Execute a CAPL script or script command via CANapeAPI.
- isOnline (self)

  Get online state of current device.
- goOnline (self, retry_timeout=10000)

  Switch the CANape device connection to online mode.
- goOffline (self, retry_timeout=10000)

  Switch the CANape device connection to offline mode.
- getECUTasks (self)

  Get a dictionary containing available measurement tasks (as defined in the ECU)

7.70.1 Detailed Description

CANape device class (ASAP3-devices: CCP, XCP).

Also supports Bus-devices (like CAN/FlexRay), although getECUVar() will not return correct values, as those devices do not support polling. DAQ will work just fine, though.

7.70.2 Constructor & Destructor Documentation

7.70.2.1 __init__()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeDevice.__init__ (
              self,
              canape_asap3,
              db_filename = "hil.a2l",
              comm_channel = DEV.CAN1,
              driver_type = DRIVER.XCP,
              module_name = "ECU",
              connect = True,
              enable_cache = None,
              obj_access_retries = 4,
              raise_get_exceptions = True,
              raise_set_exceptions = True )
```

Initializes a standard CANape device (XCP, CAN, ...) connection.

Parameters

canape_asap3	a CANapeASAP3 instance (application interface)
db_filename	filename of .a2l database (or .dbc for CAN, fibex/.xml for FlexRay, ...)
comm_channel	communication channel (see canapeapi_common.DEV constants)
driver_type	driver type (see canapeapi_common.DRIVER constants)
module_name	name of device/module in CANape
connect	Device connection behavior: True: immediately connect DUT after device has been created. Device will start in state "online" False: create device but remain in state "offline" if DUT cannot be connected at this time.
enable_cache	None: ignore parameter, use default True: enable caching False: disable caching Available since CANape 10, see Asap3CreateModule3
obj_access_retries	number of retries to perform when object access fails
raise_get_exceptions	Error handling for get/read access (see getECUVar()): True: Raise exceptions False: Return error strings (classic behavior)
raise_set_exceptions	Error handling for set/write access (see setECUVar()): True: Raise exceptions False: Return error strings (classic behavior)

Note

Parameter `connect` Setting `connect` to False permits attaching/creating a device that is currently not active (e.g. DUT is still shut down), that is, the device-init will not fail right away with error message "No response from ECU (attach Asap2 failed)".

An **inactive** DUT can be successfully attached and start out in state "offline". See isOnline() and goOnline(). However, if the DUT is **active** after all, the device's online state may end up in state "online" even if `connect` has been set to False.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.CANapeDevice.

7.70.3 Member Function Documentation

7.70.3.1 getECUTasks()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeDevice.getECUTasks (
              self )
```

Get a dictionary containing available measurement tasks (as defined in the ECU)

Returns

a task dictionary with entries like:
<task name/descr.>: {
"id": <internal task id>,
"cycle_time": <cycle time in ms, 0 if n/a>
}

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.BaseDevice.

7.70.3.2 getECUVar()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeDevice.getECUVar (
              self,
              varname,
              representation = REPR.PHYS,
              extended_mode = False,
              silent = False )
```

Get the current value of a single ECU variable (single read access via polling).

Parameters

varname	name of variable (signal/measurement/calibration)
representation	data representation, can be either REPR.HEX (ecu internal value) or REPR.PHYS (physical representation of data). See canapeapi_common.REPR.
extended_mode	if True, existing axis data for `CURVE` and `MAP` objects will also be returned. See canapeapi_common.VALUE_TYPE
silent	True: suppress debug error messages

Example: Data format in "simple mode" depending on value types

```
VALUE:     a float value
CURVE:     [values]         # a list of n float values
MAP:       [values]         # a list of x*y float values
AXIS_PTS:  [values]         # a list of n float values
ASCII:     a string
```

Example: Data format in "extended mode" depending on value types

```
VALUE:     a float value
CURVE:     ([axis values], [data values])
MAP:       ([x-axis values], [y-axis values], [data values])
AXIS_PTS:  [values]
ASCII:     a string

# Note: If a CURVE or MAP entry has no assigned axis/axes
#       (i.e. it has no referenced AXIS_PTS entries),
#       CANape will return all axis values as 0.0
```

Returns

the current data found for the variable (calibration object).

In case of access errors, either a CANapeErrorString will be returned or a CANapeAccessError will be raised, depending on the setting of raise_get_exceptions, see CANapeDevice.__init__.

For other error cases (e.g. an invalid data type was encountered), a CANapeError will be raised (or CANape↩ ErrorString returned; again, see raise_get_exceptions)

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.CANapeDevice.

7.70.3.3 getECUVarByAddress()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeDevice.getECUVarByAddress (
              self,
              varname,
              length )
```

Read `length` bytes from beginning at the address of an object defined by a given name.

Parameters

varname	name of variable at the start address
length	number of bytes to read

Returns

a raw bytes string.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.CANapeDevice.

7.70.3.4 getECUVarInfoBasic()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeDevice.getECUVarInfoBasic (
              self,
              varname,
              representation = REPR.PHYS )
```

Get information about an ECU var/object.

Parameters

varname	name of variable
representation	data representation, can be either REPR.HEX (ecu internal value) or REPR.PHYS (physical representation of data). See canapeapi_common.REPR

Returns

a dict with some basic variable information:
"type": data type (see canapeapi_common.DATA_TYPE)
 'address': memory address
 'minimum': minimum value
 'maximum': maximum value
 'increment`: increment value

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.CANapeDevice.

7.70.3.5 goOffline()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeDevice.goOffline (
              self,
              retry_timeout = 10000 )
```

Switch the CANape device connection to offline mode.

Parameters

retry_timeout	[ms], if $>$ 0, repeated attempts to set the ecu state will be made until timeout occurs

Returns

a list of testresult entries containing info about retried attempts.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.BaseDevice.

7.70.3.6 goOnline()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeDevice.goOnline (
              self,
              retry_timeout = 10000 )
```

Switch the CANape device connection to online mode.

Parameters

retry_timeout	[ms], if $>$ 0, repeated attempts to set the ECU state will be made until timeout occurs

Returns

a list of testresult entries containing info about retried attempts

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.BaseDevice.

7.70.3.7 isOnline()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeDevice.isOnline (
              self )
```

Get online state of current device.

Returns

True if the device is online, otherwise False.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.BaseDevice.

7.70.3.8 readBytes()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeDevice.readBytes (
              self,
              address,
              length,
              addr_ext = 0,
              verbosity = 1 )
```

Read `length` bytes, starting at the supplied address.

Parameters

address	start address in ECU to read from (preferably already a `c_ulong`)
length	number of bytes to read
addr_ext	address extension, only used with some special multiprocessor ECUs (leave at 0 if not required)
verbosity	verbosity of log output 0: only errors 1: status info

Example:

```
buf = canape.readBytes(0x123456, 16)
print [ord(b) for b in buf] # list of bytes (as int values)
print HexList(buf)          # list of bytes with "nicer" hex display
print binascii.hexlify(buf) # hex string
```

Returns

a raw bytes string.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.CANapeDevice.

7.70.3.9 release()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeDevice.release (
              self,
              verbosity = 2 )
```

Manually release the module/device handle in CANape.

This will normally be called during **del**, but sometimes it might be advantageous to release the handle a defined time (and not just implicitly when the destructor gets called eventually).

Parameters

verbosity	verbosity of stdout/log output: -1: do not show anything if handle is already gone (for calls in **del**) 0: errors only 1: Asap3Exit status output 2: also show "nothing to do" notice if handle is already gone

Returns

True if successful (or already released), otherwise False.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.BaseDevice.

7.70.3.10 runScript()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeDevice.runScript (
              self,
```

```
                script_command,
                is_file_name = False )
```

Execute a CAPL script or script command via CANapeAPI.

Note

This uses the "classic" `Asap3ExecuteScript` API function which does not provide any script status or return values.

New extended script functions are available since CANape version 9. The classic approach is sufficient for simple cases, though.

Parameters

script_command	CAPL command or name of CAPL script file to execute. A CAPL command can also be the name of a custom "project function" that has been defined in the current workspace.
is_file_name	True if `script_command` contains the name of a script file to be run, False if it directly contains a CAPL command.

Example:

```
# multiple commands can be added in sequence
device.runScript('Speak("Hello World"); Write("Done.");')

# some CAPL commands/methods have to be called on an existing
# device object
device.runScript('%s.SaveDatabase(0);'%(device.getModuleName())

# default location of script files is the folder of the current
# CANape workspace
device.runScript('capl_script.scr', is_file_name=True)
```

Note

The executed script or script command might still be running when this function returns.

Returns

True if script execution was successful (though not necessarily completed), otherwise False.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.BaseDevice.

7.70.3.11 setECUVar()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeDevice.setECUVar (
                self,
                varname,
                data,
                representation = REPR.PHYS,
                extended_mode = False,
                return_original = True )
```

Set data of a single ECU variable.

Parameters

varname	name of variable (signal/measurement/calibration)
data	data to set (either a scalar value or a list of values)
representation	data representation, can be either REPR.HEX (ecu internal value) or REPR.PHYS (physical representation of data). See canapeapi_common.REPR
extended_mode	see description and examples at getECUVar()
return_original	True: return original value before "set" False: return current value after "set"

Returns

the previous/original value if `return_original` is True (useful to reset changes later), otherwise the current value (as read back from ECU, useful to check if the new value has been successfully set)

In case of errors, either a CANapeErrorString will be returned or a CANapeAccessError will be raised, depending on the setting of raise_set_exceptions, see CANapeDevice.__init__.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.CANapeDevice.

7.70.3.12 writeBytes()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeDevice.writeBytes (
              self,
              address,
              data,
              addr_ext = 0,
              verbosity = 1 )
```

Write the supplied bytes to ECU, starting at address.

Parameters

address	start address in ECU to read from (preferably already a `c_ulong`)
data	an already initialized ctypes string buffer OR a list of byte values to write OR a hex-string (a leading "0x" will be ignored) Note that a missing nibble in the hex string will be zero-padded to get a full byte
addr_ext	address extension, only used with some special multiprocessor ECUs (leave at 0 if not required)
verbosity	verbosity of log output 0: only errors 1: status info

Examples:

```
canape.writeBytes(0x123456, "0xAABBCC")
canape.writeBytes(0x123456, "AA BB CC")
canape.writeBytes(0x123456, "ABBCC") # will be padded to 0ABBCC
canape.writeBytes(0x123456, [0xAA, 0xBB, 0xCC])
canape.writeBytes(0x123456, [170, 187, 204])
```

Returns

the write status from low-level API (i.e. 0 on success)

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.CANapeDevice.

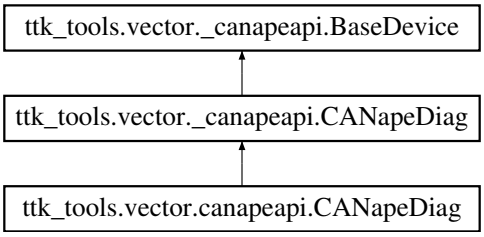
The documentation for this class was generated from the following file:

- ttk_tools/vector/canapeapi_offline_stub.py (v4.1.0)

7.71 ttk_tools.vector._canapeapi.CANapeDiag Class Reference

Diagnostics device class (UDS, ...).

Inheritance diagram for ttk_tools.vector._canapeapi.CANapeDiag:



Public Member Functions

- __init__ (self, canape_asap3, db_filename="hil.cdd", comm_channel=DEV.CAN1, driver_type=DRIVER.CANDELA, module_name="DIAG", connect=True, enable_cache=None, request_timeout=30000, enable_tp=None, restore_tp=True)

  Initialize a CANape Diagnostics Device connection.
- __str__ (self)

  Get a string representation of the device.
- isTesterPresentEnabled (self, ignore_errors=True, verbosity=1)

  Get current state of automatic sending of Tester Present (TP).
- enableTesterPresent (self, verbosity=1)

  Enable automatic sending of Tester Present (TP) service.
- disableTesterPresent (self, verbosity=1)

  Disable automatic sending of Tester Present (TP) service.
- goOnline (self, retry_timeout=10000)

  Switch the CANape diag device connection to online mode.
- goOffline (self, retry_timeout=10000)

  Switch the CANape diag device connection to offline mode.
- sendHexRequest (self, request)

  Send a raw service request as a series of bytes (usually as hex).
- sendSymbolicRequest (self, request, request_params=None, response_params=None, skip_empty_↩ iterations=False)

  Send a symbolic service request (name/path as defined in CDD/ODX-file).
- sendHexRequestSym (self, request, response_params=None)

  Send a raw service request as a series of bytes (usually as hex), but attempt to decode the response using parameter names/qualifiers of the matching symbolic request.

Public Attributes

- diag_verbosity

  verbosity level of diag requests 0: mostly silent 1: basic info 2: log diag response data 3: log missing data representations when decoding symbolic parameters, show currently used diag handle
- request_timeout

  timeout for diag requests [ms]
- request_verbosity

  verbosity of log output during Asap3DiagExecute 0: mostly silent 1: compact progress 2: show current execution status
- restore_tp

  restore previous TesterPresent state after goOnline() (as CANape seems to revert to TP setting of workspace)

Protected Member Functions

- _logInfo (self, min_verbosity, msg)

  Log messages to stdout if min_verbosity is >= the current diag_verbosity.
- _Asap3DiagIsTesterPresentEnabled (self)

  Get current "Tester Present sending enabled" state.
- _Asap3DiagEnableTesterPresent (self, enable=True)

  Enable/disable automatic sending of the "Tester Present" service.
- _Asap3DiagCreateRawRequest (self, service_bytes)

  Create a raw request.
- _Asap3DiagCreateSymbolicRequest (self, service_name)

  Create a symbolic request.
- _Asap3DiagGetServiceState (self, request_handle)

  Query the state of a service/request.
- _Asap3DiagGetResponseStream (self, request_handle, response_number)

  Get the raw response byte stream.
- _Asap3DiagReleaseService (self, request_handle)

  Release and remove a used service request handle.
- _Asap3DiagExecute (self, request_handle, max_attempts=1)

  Execute/send an already defined request.
- _Asap3DiagSetRawParameter (self, request_handle, qualifier, byte_values)

  Set a raw parameter value (as a list of bytes).
- _Asap3DiagSetStringParameter (self, request_handle, qualifier, value)

  Set a string parameter value.
- _Asap3DiagSetNumericParameter (self, request_handle, qualifier, value)

  Set a numeric (int, float) parameter value.
- _Asap3DiagGetStringResponseParameter (self, request_handle, qualifier, response_number)

  Get a string parameter value from the indexed response.
- _Asap3DiagGetNumericResponseParameter (self, request_handle, qualifier, response_number)

  Get a numeric parameter value from the indexed response.
- _Asap3DiagGetRawResponseParameter (self, request_handle, qualifier, response_number)

  Get a raw parameter value from the indexed response.
- _Asap3DiagIsPositiveResponse (self, request_handle, response_number)

  Check whether the response at index `response_number` is a positive response.
- _Asap3DiagIsComplexResponseParameter (self, request_handle, param_qualifier, response_number)

  Check whether the response parameter is "complex" (that is, if it contains named sub-parameters).
- _Asap3DiagGetComplexIterationCount (self, request_handle, param_qualifier, response_number)

  Get the number of iterations for the supplied complex parameter.

- _Asap3DiagGetComplexStringResponseParameter (self, request_handle, qualifier, response_number, subparam_qualifier, iteration_idx)

  Get a string parameter value for a complex sub-parameter from the indexed response.
- _Asap3DiagGetComplexNumericResponseParameter (self, request_handle, qualifier, response_number, subparam_qualifier, iteration_idx)

  Get a numeric parameter value for a complex sub-parameter from the indexed response.
- _Asap3DiagGetComplexRawResponseParameter (self, request_handle, qualifier, response_number, subparam_qualifier, iteration_idx)

  Get a raw parameter value for a complex sub-parameter from the indexed response.
- _decodeResponseParameters (self, request_handle, response_params=None, response_index=0, skip_↩ empty_iterations=False)

  Extract/decode symbolic response parameters, that is, parameter values in raw, string and numeric representations (as available).

Protected Attributes

- _prev_tp_enabled_state

  internal "previous" TesterPresent state before going offline [unused in stub]
- _driver_type
- _comm_channel
- _module_name
- _module_handle

7.71.1 Detailed Description

Diagnostics device class (UDS, ...).

7.71.2 Constructor & Destructor Documentation

7.71.2.1 __init__()

```
ttk_tools.vector._canapeapi.CANapeDiag.__init__ (
              self,
              canape_asap3,
              db_filename = "hil.cdd",
              comm_channel = DEV.CAN1,
              driver_type = DRIVER.CANDELA,
              module_name = "DIAG",
              connect = True,
              enable_cache = None,
              request_timeout = 30000,
              enable_tp = None,
              restore_tp = True )
```

Initialize a CANape Diagnostics Device connection.

Parameters

canape_asap3	a CANapeASAP3 instance (application interface)
db_filename	filename of .cdd/.odx database (or similar)
comm_channel	communication channel (see canapeapi_common.DEV constants)

Parameters

driver_type	driver type (see canapeapi_common.DRIVER constants)
module_name	name of device/module in CANape
connect	Device connection behavior: True: immediately connect DUT after device has been created. Device will start in state "online" False: create device but remain in state "offline" if DUT cannot be connected at this time.
enable_cache	None: ignore parameter, use default True: enable caching False: disable caching Available since CANape 10, see Asap3CreateModule3
request_timeout	timeout for diag requests [ms]
enable_tp	auto-set Tester Present status: None: leave as configured in workspace True: Enable automatic sending False: Disable automatic sending
restore_tp	TesterPresent status after going online (again): True: Restore TP status, so it will be the same as before going offline. False: Do nothing, TP status after going online will depend on current CANape configuration.

Reimplemented from ttk_tools.vector._canapeapi.BaseDevice.

Reimplemented in ttk_tools.vector.canapeapi.CANapeDiag.

7.71.3 Member Function Documentation

7.71.3.1 __str__()

```
ttk_tools.vector._canapeapi.CANapeDiag.__str__ (
            self )
```

Get a string representation of the device.

Returns

an info string.

Reimplemented from ttk_tools.vector._canapeapi.BaseDevice.

7.71.3.2 _Asap3DiagCreateRawRequest()

```
ttk_tools.vector._canapeapi.CANapeDiag._Asap3DiagCreateRawRequest (
            self,
            service_bytes )  [protected]
```

Create a raw request.

Parameters

service_bytes	list of bytes containing the request

Returns

a diag service handle for the raw request

7.71.3.3 _Asap3DiagCreateSymbolicRequest()

```
ttk_tools.vector._canapeapi.CANapeDiag._Asap3DiagCreateSymbolicRequest (
              self,
              service_name )  [protected]
```

Create a symbolic request.

Parameters

service_name	symbolic name of the service

Returns

a diag service handle for the symbolic request

7.71.3.4 _Asap3DiagEnableTesterPresent()

```
ttk_tools.vector._canapeapi.CANapeDiag._Asap3DiagEnableTesterPresent (
              self,
              enable = True )  [protected]
```

Enable/disable automatic sending of the "Tester Present" service.

Available: CANape 11+

Parameters

enable	True: enable sending of TP False: disable TP

7.71.3.5 _Asap3DiagExecute()

```
ttk_tools.vector._canapeapi.CANapeDiag._Asap3DiagExecute (
              self,
              request_handle,
              max_attempts = 1 )  [protected]
```

Execute/send an already defined request.

Parameters

request_handle	a valid request handle as provided by _Asap3DiagCreateRawRequest() or _Asap3DiagCreateSymbolicRequest()
max_attempts	number of attempts to make when trying to send the request

Returns

the number of responses

7.71.3.6 _Asap3DiagGetComplexIterationCount()

```
ttk_tools.vector._canapeapi.CANapeDiag._Asap3DiagGetComplexIterationCount (
              self,
              request_handle,
              param_qualifier,
              response_number )  [protected]
```

Get the number of iterations for the supplied complex parameter.

Parameters

request_handle	a currently valid request handle
param_qualifier	qualifier/name of parameter to check
response_number	index of response to check

Returns

the number of possible iterations

7.71.3.7 _Asap3DiagGetComplexNumericResponseParameter()

```
ttk_tools.vector._canapeapi.CANapeDiag._Asap3DiagGetComplexNumericResponseParameter (
              self,
              request_handle,
              qualifier,
              response_number,
              subparam_qualifier,
              iteration_idx )  [protected]
```

Get a numeric parameter value for a complex sub-parameter from the indexed response.

Parameters

request_handle	a currently valid request handle
qualifier	parameter name/qualifier for which to get the sub-parameter value
response_number	index of response to check
subparam_qualifier	sub-parameter name/qualifier for which to get the value
iteration_idx	current iteration (sub parameter count), one complex parameter may have multiple sets of sub parameter values

Returns

an int/float value, depending on numerical type of the parameter, or None if no valid type has been returned or the sub-parameter has no numeric representations

7.71.3.8 _Asap3DiagGetComplexRawResponseParameter()

```
ttk_tools.vector._canapeapi.CANapeDiag._Asap3DiagGetComplexRawResponseParameter (
              self,
              request_handle,
              qualifier,
              response_number,
              subparam_qualifier,
              iteration_idx )  [protected]
```

Get a raw parameter value for a complex sub-parameter from the indexed response.

Parameters

request_handle	a currently valid request handle
qualifier	parameter name/qualifier for which to get the sub-parameter value
response_number	index of response to check
subparam_qualifier	sub-parameter name/qualifier for which to get the value
iteration_idx	current iteration (sub parameter count), one complex parameter may have multiple sets of sub parameter values

Returns

a raw value (list of bytes) for the specified sub-parameter or None if the sub-parameter can not – for whatever reason – be represented as raw bytes.

7.71.3.9 _Asap3DiagGetComplexStringResponseParameter()

```
ttk_tools.vector._canapeapi.CANapeDiag._Asap3DiagGetComplexStringResponseParameter (
              self,
              request_handle,
              qualifier,
              response_number,
              subparam_qualifier,
              iteration_idx )  [protected]
```

Get a string parameter value for a complex sub-parameter from the indexed response.

Parameters

request_handle	a currently valid request handle
qualifier	parameter name/qualifier for which to get the sub-parameter value
response_number	index of response to check
subparam_qualifier	sub-parameter name/qualifier for which to get the value
iteration_idx	current iteration (sub parameter count), one complex parameter may have multiple sets of sub parameter values

Returns

a string value for the specified sub-parameter or None if the sub-parameter has no string representation.

7.71.3.10 _Asap3DiagGetNumericResponseParameter()

```
ttk_tools.vector._canapeapi.CANapeDiag._Asap3DiagGetNumericResponseParameter (
              self,
              request_handle,
              qualifier,
              response_number )  [protected]
```

Get a numeric parameter value from the indexed response.

Parameters

request_handle	a currently valid request handle
qualifier	parameter name/qualifier for which to get the value
response_number	index of response to check

Returns

an int/float value, depending on numerical type of the parameter, or None if no valid type has been returned or the parameter has no numeric representation.

7.71.3.11 _Asap3DiagGetRawResponseParameter()

```
ttk_tools.vector._canapeapi.CANapeDiag._Asap3DiagGetRawResponseParameter (
              self,
              request_handle,
              qualifier,
              response_number )  [protected]
```

Get a raw parameter value from the indexed response.

Parameters

request_handle	a currently valid request handle
qualifier	parameter name/qualifier for which to get the value
response_number	index of response to check

Returns

a list of byte values.

7.71.3.12 _Asap3DiagGetResponseStream()

```
ttk_tools.vector._canapeapi.CANapeDiag._Asap3DiagGetResponseStream (
              self,
```

```
            request_handle,
            response_number )  [protected]
```

Get the raw response byte stream.

Parameters

request_handle	a valid request handle as provided by _Asap3DiagCreateRawRequest() or _Asap3DiagCreateSymbolicRequest()
response_number	index of response that should be retrieved

Returns

a raw response byte stream as a list of bytes

7.71.3.13 _Asap3DiagGetServiceState()

```
ttk_tools.vector._canapeapi.CANapeDiag._Asap3DiagGetServiceState (
            self,
            request_handle )  [protected]
```

Query the state of a service/request.

Parameters

request_handle	a valid request handle as provided by _Asap3DiagCreateRawRequest() or _Asap3DiagCreateSymbolicRequest()

Returns

a numeric service state (created, running, finished or timeout), see canapeapi_common.DIAG_SERVICE states.

7.71.3.14 _Asap3DiagGetStringResponseParameter()

```
ttk_tools.vector._canapeapi.CANapeDiag._Asap3DiagGetStringResponseParameter (
            self,
            request_handle,
            qualifier,
            response_number )  [protected]
```

Get a string parameter value from the indexed response.

Parameters

request_handle	a currently valid request handle
qualifier	parameter name/qualifier for which to get the value
response_number	index of response to check

Returns

a string value for the specified parameter

7.71.3.15 _Asap3DiagIsComplexResponseParameter()

```
ttk_tools.vector._canapeapi.CANapeDiag._Asap3DiagIsComplexResponseParameter (
              self,
              request_handle,
              param_qualifier,
              response_number )  [protected]
```

Check whether the response parameter is "complex" (that is, if it contains named sub-parameters).

Parameters

request_handle	a currently valid request handle
param_qualifier	qualifier/name of parameter to check
response_number	index of response to check

Returns

True if response parameter is "complex", otherwise False

7.71.3.16 _Asap3DiagIsPositiveResponse()

```
ttk_tools.vector._canapeapi.CANapeDiag._Asap3DiagIsPositiveResponse (
              self,
              request_handle,
              response_number )  [protected]
```

Check whether the response at index `response_number` is a positive response.

Parameters

request_handle	a currently valid request handle
response_number	index of response to check

Returns

True if response is positive, otherwise False.

7.71.3.17 _Asap3DiagIsTesterPresentEnabled()

```
ttk_tools.vector._canapeapi.CANapeDiag._Asap3DiagIsTesterPresentEnabled (
              self )  [protected]
```

Get current "Tester Present sending enabled" state.

Available: CANape 11+

7.71.3.18 _Asap3DiagReleaseService()

```
ttk_tools.vector._canapeapi.CANapeDiag._Asap3DiagReleaseService (
              self,
              request_handle )  [protected]
```

Release and remove a used service request handle.

Parameters

request_handle	a valid request handle as provided by _Asap3DiagCreateRawRequest() or _Asap3DiagCreateSymbolicRequest()

Returns

True if handle was successfully released, otherwise False.

7.71.3.19 _Asap3DiagSetNumericParameter()

```
ttk_tools.vector._canapeapi.CANapeDiag._Asap3DiagSetNumericParameter (
              self,
              request_handle,
              qualifier,
              value )  [protected]
```

Set a numeric (int, float) parameter value.

Parameters

request_handle	a valid request handle as provided by _Asap3DiagCreateRawRequest() or _Asap3DiagCreateSymbolicRequest()
qualifier	name/qualifier of the parameter to set
value	parameter value as int or float

7.71.3.20 _Asap3DiagSetRawParameter()

```
ttk_tools.vector._canapeapi.CANapeDiag._Asap3DiagSetRawParameter (
              self,
              request_handle,
              qualifier,
              byte_values )  [protected]
```

Set a raw parameter value (as a list of bytes).

Parameters

request_handle	a valid request handle as provided by _Asap3DiagCreateRawRequest() or _Asap3DiagCreateSymbolicRequest()
qualifier	name/qualifier of the parameter to set
byte_values	parameter value as raw byte data (list of bytes or bytes/bytearray)

7.71.3.21 _Asap3DiagSetStringParameter()

```
ttk_tools.vector._canapeapi.CANapeDiag._Asap3DiagSetStringParameter (
              self,
              request_handle,
              qualifier,
              value )  [protected]
```

Set a string parameter value.

Parameters

request_handle	a valid request handle as provided by _Asap3DiagCreateRawRequest() or _Asap3DiagCreateSymbolicRequest()
qualifier	name/qualifier of the parameter to set
value	parameter value as string

7.71.3.22 _decodeResponseParameters()

```
ttk_tools.vector._canapeapi.CANapeDiag._decodeResponseParameters (
              self,
              request_handle,
              response_params = None,
              response_index = 0,
              skip_empty_iterations = False )  [protected]
```

Extract/decode symbolic response parameters, that is, parameter values in raw, string and numeric representations (as available).

Note

A basic string representation should always be available, same for "raw" byte representation. However, numeric representations tend to be only available for numeric parameters types.
See canapeapi_common.DiagParameterValue for info on accessing available representations.
See canapeapi_common.DiagParameterName for setting "sub-names" required for decoding complex parameters.

Parameters

request_handle	handle to diagnostic service/request, compare _Asap3DiagCreateRawRequest, _Asap3DiagCreateSymbolicRequest or _Asap3DiagExecute
response_params	list of parameter qualifiers/names that should be extracted from the response
response_index	index of response for which to decode parameters (this is typically 0, as there tends to be only a single response available)
skip_empty_iterations	for complex parameters: skip iterations if they contain no data at all

Returns

a tuple (decoded_params, failed_params).
decoded_params: parameter names mapped to DiagParameterValue instances
failed_params: parameter names mapped to error info for all failed parameters.

Note that additional decoding of sub-parameters ("complex parameters") might fail even if basic decoding ("simple parameter") was successful for a parameter.

7.71.3.23 _logInfo()

```
ttk_tools.vector._canapeapi.CANapeDiag._logInfo (
              self,
              min_verbosity,
              msg )  [protected]
```

Log messages to stdout if min_verbosity is >= the current diag_verbosity.

7.71.3.24 disableTesterPresent()

```
ttk_tools.vector._canapeapi.CANapeDiag.disableTesterPresent (
              self,
              verbosity = 1 )
```

Disable automatic sending of Tester Present (TP) service.

Note

Available since CANape 11.

Parameters

verbosity	0: silent 1: log current TP status

Reimplemented in ttk_tools.vector.canapeapi.CANapeDiag.

7.71.3.25 enableTesterPresent()

```
ttk_tools.vector._canapeapi.CANapeDiag.enableTesterPresent (
              self,
              verbosity = 1 )
```

Enable automatic sending of Tester Present (TP) service.

Note

Available since CANape 11.

Parameters

verbosity	0: silent 1: log current TP status

Reimplemented in ttk_tools.vector.canapeapi.CANapeDiag.

7.71.3.26 goOffline()

```
ttk_tools.vector._canapeapi.CANapeDiag.goOffline (
              self,
              retry_timeout = 10000 )
```

Switch the CANape diag device connection to offline mode.

Parameters

retry_timeout	[ms], if > 0, repeated attempts to set the ECU state will be made until timeout occurs.

Returns

a list of testresult entries containing info about retried attempts.

Reimplemented from ttk_tools.vector._canapeapi.BaseDevice.

Reimplemented in ttk_tools.vector.canapeapi.CANapeDiag.

7.71.3.27 goOnline()

```
ttk_tools.vector._canapeapi.CANapeDiag.goOnline (
              self,
              retry_timeout = 10000 )
```

Switch the CANape diag device connection to online mode.

Parameters

retry_timeout	[ms], if > 0, repeated attempts to set the ECU state will be made until timeout occurs

Returns

a list of testresult entries containing info about retried attempts.

Reimplemented from ttk_tools.vector._canapeapi.BaseDevice.

Reimplemented in ttk_tools.vector.canapeapi.CANapeDiag.

7.71.3.28 isTesterPresentEnabled()

```
ttk_tools.vector._canapeapi.CANapeDiag.isTesterPresentEnabled (
              self,
              ignore_errors = True,
              verbosity = 1 )
```

Get current state of automatic sending of Tester Present (TP).

Note

Available since CANape 11.

Parameters

ignore_errors	if True, access errors will be ignored and return value will be None.
verbosity	0: silent 1: log current TP status

Returns

True if Tester Present sending is currently active,
False if TP sending is inactive or
None if API function access failed and ignore_errors is set (e.g. older CANape releases).

Reimplemented in ttk_tools.vector.canapeapi.CANapeDiag.

7.71.3.29 sendHexRequest()

```
ttk_tools.vector._canapeapi.CANapeDiag.sendHexRequest (
              self,
              request )
```

Send a raw service request as a series of bytes (usually as hex).

Parameters

request	a request as list of bytes

Example:

```
asap3 = CANapeASAP3(working_dir=r"path/to/canape_working_dir")
diag  = CANapeDiag(asap3, module_name="device_name")
diag.goOnline()
response = diag.sendHexRequest([0x10, 0x03])
print "response:", response
diag.goOffline()
```

Tester Present issues:

In order to get diagnostics via CANapeAPI to work, cyclic sending of "Tester Present" may have to be deactivated in CANape config (no, really).

Device Configuration -> [Diagnostics device] -> Tab Communication -> uncheck "Send Tester Present" -> Save

See also disableTesterPresent() for CANape 11 or later.

Returns

a list of responses [[response 1], [response 2], ...]. Typically, there will only be a single response, though.

Reimplemented in ttk_tools.vector.canapeapi.CANapeDiag.

7.71.3.30 sendHexRequestSym()

```
ttk_tools.vector._canapeapi.CANapeDiag.sendHexRequestSym (
              self,
              request,
              response_params = None )
```

Send a raw service request as a series of bytes (usually as hex), but attempt to decode the response using parameter names/qualifiers of the matching symbolic request.

Parameters

request	a request as list of bytes
response_params	a list of parameter names/qualifiers to decode, compare sendSymbolicRequest

Example:

```
# using a symbolic request
responses = canape_diag.sendSymbolicRequest(
    request = "EcuIdentification_Read",
    request_params  = None,
    response_params = ["RecordDataIdentifier", "Part_Number"],
)
# dto. using a raw request
responses = canape_diag.sendHexRequestSym(
    request = [0x22, 0x00, 0x9F],
    response_params = ["RecordDataIdentifier", "Part_Number"],
)
```

Returns

a list of response dicts [{response 1}, {response 2}, ...] i.e. one dict per response (which usually means just one).

Each response dict contains: ''raw_response': raw response as a list of bytes
 'pos_response': True if response was positive, otherwise False
 'params'↩ : a dictionary with name: value entries for each parameter name from theresponse_paramslist.
 @returns If a parameter name forresponse_params` could not be decoded, its value will be set to None.

Reimplemented in ttk_tools.vector.canapeapi.CANapeDiag.

7.71.3.31 sendSymbolicRequest()

```
ttk_tools.vector._canapeapi.CANapeDiag.sendSymbolicRequest (
              self,
              request,
              request_params = None,
              response_params = None,
              skip_empty_iterations = False )
```

Send a symbolic service request (name/path as defined in CDD/ODX-file).

Parameters

request	a request as service name/path (from CDD/ODX)
request_params	dict with parameters for the request
response_params	list of parameters that should be extracted from the response
skip_empty_iterations	for complex parameters: skip iterations if they contain no data at all

Example:

```
# RoutineCtrl/SR015_Idle_Speed_Modification/Start
response = diag.sendSymbolicRequest(
    'SR015_Idle_Speed_Modification_Start',
    {'engine_speed_request': 1200}
)
print(f"{response[0]['pos_response'] = })

# Stored Data / Sine Write
response = diag.sendSymbolicRequest(
    'Sine_Write', # qualifier
    request_params = {"sine_ampl": 0.800, "sine_period": 0.160, "sine_value": 1.0}
)
print(f"{response[0]['pos_response'] = })

# Stored Data / Sine Read
response_params = ["sine_ampl", "sine_period", "sine_value"],
responses = canape_diag.sendSymbolicRequest(
    'Sine_Read', # qualifier,
    response_params = response_params,
)
print(f"{response[0]['pos_response'] = })
for param in response_params:
    diag_value = response[0]['response_params'][param]
    print(f"{param}: default/str: {diag_value}, numeric: {diag_value.numeric}")

response = diag.sendSymbolicRequest(
    'Windows_Read',
    None,
    ['Window_Front_Left', 'Window_Front_Right'],
)
print(f"{response[0]['pos_response'] = })
print(f"{response[0]['response_params']['Window_Front_Left'] = })
```

Note

For complex response parameters, supply appropriate instances of `DiagParameterName` (with set sub-parameter names) as entries in `response_params` in order to extract/access the complex sub-responses. See canapeapi_common.DiagParameterName, canapeapi_common.DiagParameterValue

Returns

a list of response dicts [{response 1}, {response 2}, ...] i.e. one dict per response (which usually means just one).

Each response dict contains: ''raw_response': raw response as a list of bytes
 'pos_response': True if response was positive, otherwise False
 'params'↩ : a dictionary with name: value entries for each parameter name from theresponse_paramslist.
 @returns If a parameter name forresponse_params` could not be decoded, its value will be set to None.

Reimplemented in ttk_tools.vector.canapeapi.CANapeDiag.

7.71.4 Member Data Documentation

7.71.4.1 _comm_channel

ttk_tools.vector._canapeapi.CANapeDiag._comm_channel [protected]

7.71.4.2 _driver_type

ttk_tools.vector._canapeapi.CANapeDiag._driver_type [protected]

7.71.4.3 _module_handle

ttk_tools.vector._canapeapi.CANapeDiag._module_handle [protected]

7.71.4.4 _module_name

ttk_tools.vector._canapeapi.CANapeDiag._module_name [protected]

7.71.4.5 _prev_tp_enabled_state

ttk_tools.vector._canapeapi.CANapeDiag._prev_tp_enabled_state [protected]

internal "previous" TesterPresent state before going offline [unused in stub]

7.71.4.6 diag_verbosity

ttk_tools.vector._canapeapi.CANapeDiag.diag_verbosity

verbosity level of diag requests 0: mostly silent 1: basic info 2: log diag response data 3: log missing data representations when decoding symbolic parameters, show currently used diag handle

7.71.4.7 request_timeout

ttk_tools.vector._canapeapi.CANapeDiag.request_timeout

timeout for diag requests [ms]

7.71.4.8 request_verbosity

ttk_tools.vector._canapeapi.CANapeDiag.request_verbosity

verbosity of log output during Asap3DiagExecute 0: mostly silent 1: compact progress 2: show current execution status

7.71.4.9 restore_tp

ttk_tools.vector._canapeapi.CANapeDiag.restore_tp

restore previous TesterPresent state after goOnline() (as CANape seems to revert to TP setting of workspace)

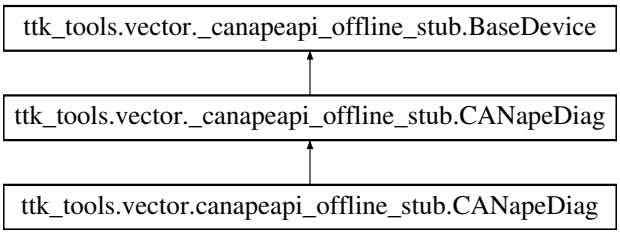
The documentation for this class was generated from the following file:

- ttk_tools/vector/_canapeapi.py (v7.3.3)

7.72 ttk_tools.vector._canapeapi_offline_stub.CANapeDiag Class Reference

Diagnostics device stub class (UDS, ...).

Inheritance diagram for ttk_tools.vector._canapeapi_offline_stub.CANapeDiag:



Public Member Functions

- __init__ (self, canape_asap3, db_filename="hil.cdd", comm_channel=DEV.CAN1, driver_type=DRIVER.CANDELA, module_name="DIAG", connect=True, enable_cache=None, request_timeout=30000, enable_tp=None, restore_tp=True)

  Initialize a CANape Diagnostics Device connection.
- __str__ (self)

  Get a string representation of the device.
- isTesterPresentEnabled (self, ignore_errors=True, verbosity=1)

  Get current state of automatic sending of Tester Present (TP).
- enableTesterPresent (self, verbosity=1)

  Enable automatic sending of Tester Present (TP) service.
- disableTesterPresent (self, verbosity=1)

  Disable automatic sending of Tester Present (TP) service.
- sendHexRequest (self, request)

  Send a raw service request as a series of bytes (usually as hex).
- sendSymbolicRequest (self, request, request_params=None, response_params=None, skip_empty_↩ iterations=False)

  Send a symbolic service request (name/path as defined in CDD/ODX-file).
- sendHexRequestSym (self, request, response_params=None)

  Send a raw service request as a series of bytes (usually as hex), but attempt to decode the response using parameter names/qualifiers of the matching symbolic request.

Public Attributes

- request_timeout

  timeout for diag requests [ms]
- restore_tp

  restore previous TesterPresent state after goOnline() [unused in stub]

Protected Attributes

- _prev_tp_enabled_state

  internal "previous" TesterPresent state before going offline [unused in stub]
- _driver_type
- _comm_channel
- _module_name

7.72.1 Detailed Description

Diagnostics device stub class (UDS, ...).

7.72.2 Constructor & Destructor Documentation

7.72.2.1 __init__()

```
ttk_tools.vector._canapeapi_offline_stub.CANapeDiag.__init__ (
                self,
                canape_asap3,
                db_filename = "hil.cdd",
                comm_channel = DEV.CAN1,
                driver_type = DRIVER.CANDELA,
                module_name = "DIAG",
                connect = True,
                enable_cache = None,
                request_timeout = 30000,
                enable_tp = None,
                restore_tp = True )
```

Initialize a CANape Diagnostics Device connection.

Parameters

canape_asap3	a CANapeASAP3 instance (application interface)
db_filename	filename of .cdd/.odx database (or similar)
comm_channel	communication channel (see canapeapi_common.DEV constants)
driver_type	driver type (see canapeapi_common.DRIVER constants)
module_name	name of device/module in CANape
connect	Device connection behavior: True: immediately connect DUT after device has been created. Device will start in state "online" False: create device but remain in state "offline" if DUT cannot be connected at this time.
enable_cache	None: ignore parameter, use default True: enable caching False: disable caching Available since CANape 10, see Asap3CreateModule3
request_timeout	timeout for diag requests [ms]
enable_tp	auto-set Tester Present status: None: leave as configured in workspace True: Enable automatic sending False: Disable automatic sending
restore_tp	TesterPresent status after going online (again): True: Restore TP status, so it will be the same as before going offline. False: Do nothing, TP status after going online will depend on current CANape configuration.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.BaseDevice.

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.CANapeDiag.

7.72.3 Member Function Documentation

7.72.3.1 __str__()

```
ttk_tools.vector._canapeapi_offline_stub.CANapeDiag.__str__ (
              self )
```

Get a string representation of the device.

Returns

an info string.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.BaseDevice.

7.72.3.2 disableTesterPresent()

```
ttk_tools.vector._canapeapi_offline_stub.CANapeDiag.disableTesterPresent (
              self,
              verbosity = 1 )
```

Disable automatic sending of Tester Present (TP) service.

Note

Available since CANape 11.

Parameters

verbosity	0: silent 1: log current TP status

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.CANapeDiag.

7.72.3.3 enableTesterPresent()

```
ttk_tools.vector._canapeapi_offline_stub.CANapeDiag.enableTesterPresent (
              self,
              verbosity = 1 )
```

Enable automatic sending of Tester Present (TP) service.

Note

Available since CANape 11.

Parameters

verbosity	0: silent 1: log current TP status

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.CANapeDiag.

7.72.3.4 isTesterPresentEnabled()

```
ttk_tools.vector._canapeapi_offline_stub.CANapeDiag.isTesterPresentEnabled (
              self,
              ignore_errors = True,
              verbosity = 1 )
```

Get current state of automatic sending of Tester Present (TP).

Note

Available since CANape 11.

Parameters

ignore_errors	if True, access errors will be ignored and return value will be None.
verbosity	0: silent 1: log current TP status

Returns

True if Tester Present sending is currently active,
False if TP sending is inactive or
None if API function access failed and ignore_errors is set (e.g. older CANape releases).

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.CANapeDiag.

7.72.3.5 sendHexRequest()

```
ttk_tools.vector._canapeapi_offline_stub.CANapeDiag.sendHexRequest (
              self,
              request )
```

Send a raw service request as a series of bytes (usually as hex).

Parameters

request	a request as list of bytes

Example:

```
asap3 = CANapeASAP3(working_dir=r"path/to/canape_working_dir")
diag  = CANapeDiag(asap3, module_name="device_name")
diag.goOnline()
response = diag.sendHexRequest([0x10, 0x03])
print "response:", response
diag.goOffline()
```

Tester Present issues:

In order to get diagnostics via CANapeAPI to work, cyclic sending of "Tester Present" may have to be deactivated in CANape config (no, really).

Device Configuration -> [Diagnostics device] -> Tab Communication -> uncheck "Send Tester Present" -> Save

See also disableTesterPresent() for CANape 11 or later.

Returns

a list of responses [[response 1], [response 2], ...]. Typically, there will only be a single response, though.

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.CANapeDiag.

7.72.3.6 sendHexRequestSym()

```
ttk_tools.vector._canapeapi_offline_stub.CANapeDiag.sendHexRequestSym (
              self,
              request,
              response_params = None )
```

Send a raw service request as a series of bytes (usually as hex), but attempt to decode the response using parameter names/qualifiers of the matching symbolic request.

Parameters

request	a request as list of bytes
response_params	a list of parameter names/qualifiers to decode, compare sendSymbolicRequest

Example:

```
# using a symbolic request
responses = canape_diag.sendSymbolicRequest(
    request = "EcuIdentification_Read",
    request_params  = None,
    response_params = ["RecordDataIdentifier", "Part_Number"],
)
# dto. using a raw request
responses = canape_diag.sendHexRequestSym(
    request = [0x22, 0x00, 0x9F],
    response_params = ["RecordDataIdentifier", "Part_Number"],
)
```

Returns

a list of response dicts [{response 1}, {response 2}, ...] i.e. one dict per response (which usually means just one).

Each response dict contains: ‘‘raw_response’: raw response as a list of bytes
 'pos_response': True if response was positive, otherwise False
 'params'↩ : a dictionary with name: value entries for each parameter name from theresponse_paramslist.
 @returns If a parameter name forresponse_params` could not be decoded, its value will be set to None.

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.CANapeDiag.

7.72.3.7 sendSymbolicRequest()

```
ttk_tools.vector._canapeapi_offline_stub.CANapeDiag.sendSymbolicRequest (
              self,
              request,
              request_params = None,
              response_params = None,
              skip_empty_iterations = False )
```

Send a symbolic service request (name/path as defined in CDD/ODX-file).

Parameters

request	a request as service name/path (from CDD/ODX)
request_params	dict with parameters for the request
response_params	list of parameters that should be extracted from the response
skip_empty_iterations	for complex parameters: skip iterations if they contain no data at all

Example:

```
# RoutineCtrl/SR015_Idle_Speed_Modification/Start
response = diag.sendSymbolicRequest(
    'SR015_Idle_Speed_Modification_Start',
    {'engine_speed_request': 1200}
)
print(f"{response[0]['pos_response'] = })

# Stored Data / Sine Write
response = diag.sendSymbolicRequest(
    'Sine_Write', # qualifier
    request_params = {"sine_ampl": 0.800, "sine_period": 0.160, "sine_value": 1.0}
)
print(f"{response[0]['pos_response'] = })

# Stored Data / Sine Read
response_params = ["sine_ampl", "sine_period", "sine_value"],
responses = canape_diag.sendSymbolicRequest(
    'Sine_Read', # qualifier,
    response_params = response_params,
)
print(f"{response[0]['pos_response'] = })
for param in response_params:
    diag_value = response[0]['response_params'][param]
    print(f"{param}: default/str: {diag_value}, numeric: {diag_value.numeric}")

response = diag.sendSymbolicRequest(
    'Windows_Read',
    None,
    ['Window_Front_Left', 'Window_Front_Right'],
)
print(f"{response[0]['pos_response'] = })
print(f"{response[0]['response_params']['Window_Front_Left'] = })
```

Note

For complex response parameters, supply appropriate instances of `DiagParameterName` (with set sub-parameter names) as entries in `response_params` in order to extract/access the complex sub-responses. See canapeapi_common.DiagParameterName, canapeapi_common.DiagParameterValue

Returns

a list of response dicts [{response 1}, {response 2}, ...] i.e. one dict per response (which usually means just one).

Each response dict contains: 'raw_response': raw response as a list of bytes
 'pos_response': True if response was positive, otherwise False
 'params'↩ : a dictionary with name: value entries for each parameter name from theresponse_paramslist.
 @returns If a parameter name forresponse_params` could not be decoded, its value will be set to None.

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.CANapeDiag.

7.72.4 Member Data Documentation

7.72.4.1 _comm_channel

ttk_tools.vector._canapeapi_offline_stub.CANapeDiag._comm_channel [protected]

7.72.4.2 _driver_type

ttk_tools.vector._canapeapi_offline_stub.CANapeDiag._driver_type [protected]

7.72.4.3 _module_name

ttk_tools.vector._canapeapi_offline_stub.CANapeDiag._module_name [protected]

7.72.4.4 _prev_tp_enabled_state

ttk_tools.vector._canapeapi_offline_stub.CANapeDiag._prev_tp_enabled_state [protected]

internal "previous" TesterPresent state before going offline [unused in stub]

7.72.4.5 request_timeout

ttk_tools.vector._canapeapi_offline_stub.CANapeDiag.request_timeout

timeout for diag requests [ms]

7.72.4.6 restore_tp

`ttk_tools.vector._canapeapi_offline_stub.CANapeDiag.restore_tp`

restore previous TesterPresent state after goOnline() [unused in stub]

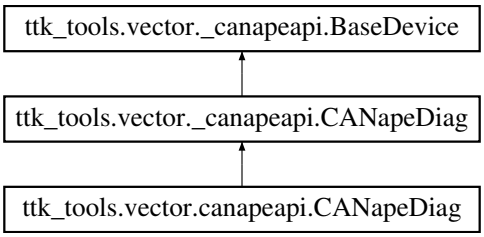
The documentation for this class was generated from the following file:

- ttk_tools/vector/_canapeapi_offline_stub.py (v4.7.1)

7.73 ttk_tools.vector.canapeapi.CANapeDiag Class Reference

Diagnostics device class (UDS, ...).

Inheritance diagram for ttk_tools.vector.canapeapi.CANapeDiag:



Public Member Functions

- __init__ (self, canape_asap3, db_filename="hil.cdd", comm_channel=DEV.CAN1, driver_type=DRIVER.CANDELA, module_name="DIAG", connect=True, enable_cache=None, request_timeout=30000, enable_tp=None, restore_tp=True)

  Initialize a CANape Diagnostics Device connection.
- isTesterPresentEnabled (self, ignore_errors=True, verbosity=1)

  Get current state of automatic sending of Tester Present (TP).
- enableTesterPresent (self, verbosity=1)

  Enable automatic sending of Tester Present (TP) service.
- disableTesterPresent (self, verbosity=1)

  Disable automatic sending of Tester Present (TP) service.
- release (self, verbosity=2)

  Manually release the module/device handle in CANape.
- runScript (self, script_command, is_file_name=False)

  Execute a CAPL script or script command via CANapeAPI.
- isOnline (self)

  Get online state of current device.
- goOnline (self, retry_timeout=10000)

  Switch the CANape diag device connection to online mode.
- goOffline (self, retry_timeout=10000)

  Switch the CANape diag device connection to offline mode.
- sendHexRequest (self, request)

  Send a raw service request as a series of bytes (usually as hex).
- sendSymbolicRequest (self, request, request_params=None, response_params=None, skip_empty_↩ iterations=False)

  Send a symbolic service request (name/path as defined in CDD/ODX-file).
- sendHexRequestSym (self, request, response_params=None)

  Send a raw service request as a series of bytes (usually as hex), but attempt to decode the response using parameter names/qualifiers of the matching symbolic request.

7.73.1 Detailed Description

Diagnostics device class (UDS, ...).

7.73.2 Constructor & Destructor Documentation

7.73.2.1 __init__()

```
ttk_tools.vector.canapeapi.CANapeDiag.__init__ (
                self,
                canape_asap3,
                db_filename = "hil.cdd",
                comm_channel = DEV.CAN1,
                driver_type = DRIVER.CANDELA,
                module_name = "DIAG",
                connect = True,
                enable_cache = None,
                request_timeout = 30000,
                enable_tp = None,
                restore_tp = True )
```

Initialize a CANape Diagnostics Device connection.

Parameters

canape_asap3	a CANapeASAP3 instance (application interface)
db_filename	filename of .cdd/.odx database (or similar)
comm_channel	communication channel (see canapeapi_common.DEV constants)
driver_type	driver type (see canapeapi_common.DRIVER constants)
module_name	name of device/module in CANape
connect	Device connection behavior: True: immediately connect DUT after device has been created. Device will start in state "online" False: create device but remain in state "offline" if DUT cannot be connected at this time.
enable_cache	None: ignore parameter, use default True: enable caching False: disable caching Available since CANape 10, see Asap3CreateModule3
request_timeout	timeout for diag requests [ms]
enable_tp	auto-set Tester Present status: None: leave as configured in workspace True: Enable automatic sending False: Disable automatic sending
restore_tp	TesterPresent status after going online (again): True: Restore TP status, so it will be the same as before going offline. False: Do nothing, TP status after going online will depend on current CANape configuration.

Reimplemented from ttk_tools.vector._canapeapi.CANapeDiag.

7.73.3 Member Function Documentation

7.73.3.1 disableTesterPresent()

```
ttk_tools.vector.canapeapi.CANapeDiag.disableTesterPresent (
              self,
              verbosity = 1 )
```

Disable automatic sending of Tester Present (TP) service.

Note

Available since CANape 11.

Parameters

verbosity	0: silent 1: log current TP status

Reimplemented from ttk_tools.vector._canapeapi.CANapeDiag.

7.73.3.2 enableTesterPresent()

```
ttk_tools.vector.canapeapi.CANapeDiag.enableTesterPresent (
              self,
              verbosity = 1 )
```

Enable automatic sending of Tester Present (TP) service.

Note

Available since CANape 11.

Parameters

verbosity	0: silent 1: log current TP status

Reimplemented from ttk_tools.vector._canapeapi.CANapeDiag.

7.73.3.3 goOffline()

```
ttk_tools.vector.canapeapi.CANapeDiag.goOffline (
              self,
              retry_timeout = 10000 )
```

Switch the CANape diag device connection to offline mode.

Parameters

retry_timeout	[ms], if > 0, repeated attempts to set the ECU state will be made until timeout occurs.

Returns

a list of testresult entries containing info about retried attempts.

Reimplemented from ttk_tools.vector._canapeapi.CANapeDiag.

7.73.3.4 goOnline()

```
ttk_tools.vector.canapeapi.CANapeDiag.goOnline (
              self,
              retry_timeout = 10000 )
```

Switch the CANape diag device connection to online mode.

Parameters

retry_timeout	[ms], if > 0, repeated attempts to set the ECU state will be made until timeout occurs

Returns

a list of testresult entries containing info about retried attempts.

Reimplemented from ttk_tools.vector._canapeapi.CANapeDiag.

7.73.3.5 isOnline()

```
ttk_tools.vector.canapeapi.CANapeDiag.isOnline (
              self )
```

Get online state of current device.

Returns

True if the device is online, otherwise False.

Reimplemented from ttk_tools.vector._canapeapi.BaseDevice.

7.73.3.6 isTesterPresentEnabled()

```
ttk_tools.vector.canapeapi.CANapeDiag.isTesterPresentEnabled (
              self,
              ignore_errors = True,
              verbosity = 1 )
```

Get current state of automatic sending of Tester Present (TP).

Note

Available since CANape 11.

Parameters

ignore_errors	if True, access errors will be ignored and return value will be None.
verbosity	0: silent 1: log current TP status

Returns

True if Tester Present sending is currently active,
False if TP sending is inactive or
None if API function access failed and ignore_errors is set (e.g. older CANape releases).

Reimplemented from ttk_tools.vector._canapeapi.CANapeDiag.

7.73.3.7 release()

```
ttk_tools.vector.canapeapi.CANapeDiag.release (
              self,
              verbosity = 2 )
```

Manually release the module/device handle in CANape.

This will normally be called during **del**, but sometimes it might be advantageous to release the handle a defined time (and not just implicitly when the destructor gets called eventually).

Parameters

verbosity	verbosity of stdout/log output: -1: do not show anything if handle is already gone (for calls in **del**) 0: errors only 1: Asap3Exit status output 2: also show "nothing to do" notice if handle is already gone

Returns

True if successful (or already released), otherwise False.

Reimplemented from ttk_tools.vector._canapeapi.BaseDevice.

7.73.3.8 runScript()

```
ttk_tools.vector.canapeapi.CANapeDiag.runScript (
              self,
              script_command,
              is_file_name = False )
```

Execute a CAPL script or script command via CANapeAPI.

Note

This uses the "classic" `Asap3ExecuteScript` API function which does not provide any script status or return values.

New extended script functions are available since CANape version 9. The classic approach is sufficient for simple cases, though.

Parameters

script_command	CAPL command or name of CAPL script file to execute. A CAPL command can also be the name of a custom "project function" that has been defined in the current workspace.
is_file_name	True if `script_command` contains the name of a script file to be run, False if it directly contains a CAPL command.

Example:

```
# multiple commands can be added in sequence
device.runScript('Speak("Hello World"); Write("Done.");')

# some CAPL commands/methods have to be called on an existing
# device object
device.runScript('%s.SaveDatabase(0);'%(device.getModuleName())

# default location of script files is the folder of the current
# CANape workspace
device.runScript('capl_script.scr', is_file_name=True)
```

Note

The executed script or script command might still be running when this function returns.

Returns

True if script execution was successful (though not necessarily completed), otherwise False.

Reimplemented from ttk_tools.vector._canapeapi.BaseDevice.

7.73.3.9 sendHexRequest()

```
ttk_tools.vector.canapeapi.CANapeDiag.sendHexRequest (
              self,
              request )
```

Send a raw service request as a series of bytes (usually as hex).

Parameters

request	a request as list of bytes

Example:

```
asap3 = CANapeASAP3(working_dir=r"path/to/canape_working_dir")
diag  = CANapeDiag(asap3, module_name="device_name")
diag.goOnline()
response = diag.sendHexRequest([0x10, 0x03])
print "response:", response
diag.goOffline()
```

Tester Present issues:

In order to get diagnostics via CANapeAPI to work, cyclic sending of "Tester Present" may have to be deactivated in CANape config (no, really).

Device Configuration -> [Diagnostics device] -> Tab Communication -> uncheck "Send Tester Present" -> Save

See also disableTesterPresent() for CANape 11 or later.

Returns

a list of responses [[response 1], [response 2], ...]. Typically, there will only be a single response, though.

Reimplemented from ttk_tools.vector._canapeapi.CANapeDiag.

7.73.3.10 sendHexRequestSym()

```
ttk_tools.vector.canapeapi.CANapeDiag.sendHexRequestSym (
              self,
              request,
              response_params = None )
```

Send a raw service request as a series of bytes (usually as hex), but attempt to decode the response using parameter names/qualifiers of the matching symbolic request.

Parameters

request	a request as list of bytes
response_params	a list of parameter names/qualifiers to decode, compare sendSymbolicRequest

Example:

```
# using a symbolic request
responses = canape_diag.sendSymbolicRequest(
    request = "EcuIdentification_Read",
    request_params  = None,
    response_params = ["RecordDataIdentifier", "Part_Number"],
)
# dto. using a raw request
responses = canape_diag.sendHexRequestSym(
    request = [0x22, 0x00, 0x9F],
    response_params = ["RecordDataIdentifier", "Part_Number"],
)
```

Returns

a list of response dicts [{response 1}, {response 2}, ...] i.e. one dict per response (which usually means just one).

Each response dict contains: ''raw_response': raw response as a list of bytes
 'pos_response': True if response was positive, otherwise False
 'params'↩ : a dictionary with name: value entries for each parameter name from theresponse_paramslist.
 @returns If a parameter name forresponse_params` could not be decoded, its value will be set to None.

Reimplemented from ttk_tools.vector._canapeapi.CANapeDiag.

7.73.3.11 sendSymbolicRequest()

```
ttk_tools.vector.canapeapi.CANapeDiag.sendSymbolicRequest (
                self,
                request,
                request_params = None,
                response_params = None,
                skip_empty_iterations = False )
```

Send a symbolic service request (name/path as defined in CDD/ODX-file).

Parameters

request	a request as service name/path (from CDD/ODX)
request_params	dict with parameters for the request
response_params	list of parameters that should be extracted from the response
skip_empty_iterations	for complex parameters: skip iterations if they contain no data at all

Example:

```
# RoutineCtrl/SR015_Idle_Speed_Modification/Start
response = diag.sendSymbolicRequest(
    'SR015_Idle_Speed_Modification_Start',
    {'engine_speed_request': 1200}
)
print(f"{response[0]['pos_response'] = })

# Stored Data / Sine Write
response = diag.sendSymbolicRequest(
    'Sine_Write', # qualifier
    request_params = {"sine_ampl": 0.800, "sine_period": 0.160, "sine_value": 1.0}
)
print(f"{response[0]['pos_response'] = })

# Stored Data / Sine Read
response_params = ["sine_ampl", "sine_period", "sine_value"],
responses = canape_diag.sendSymbolicRequest(
    'Sine_Read', # qualifier,
    response_params = response_params,
)
print(f"{response[0]['pos_response'] = })
for param in response_params:
    diag_value = response[0]['response_params'][param]
    print(f"{param}: default/str: {diag_value}, numeric: {diag_value.numeric}")

response = diag.sendSymbolicRequest(
    'Windows_Read',
    None,
    ['Window_Front_Left', 'Window_Front_Right'],
)
print(f"{response[0]['pos_response'] = })
print(f"{response[0]['response_params']['Window_Front_Left'] = })
```

Note

For complex response parameters, supply appropriate instances of `DiagParameterName` (with set sub-parameter names) as entries in `response_params` in order to extract/access the complex sub-responses. See canapeapi_common.DiagParameterName, canapeapi_common.DiagParameterValue

Returns

a list of response dicts [{response 1}, {response 2}, ...] i.e. one dict per response (which usually means just one).

Each response dict contains: ‘‘raw_response’: raw response as a list of bytes
 'pos_response': True if response was positive, otherwise False
 'params'↩ : a dictionary with name: value entries for each parameter name from theresponse_paramslist.
 @returns If a parameter name forresponse_params` could not be decoded, its value will be set to None.

Reimplemented from ttk_tools.vector._canapeapi.CANapeDiag.

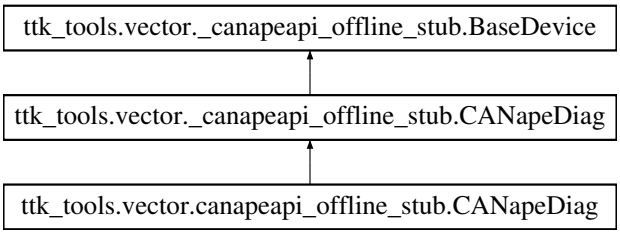
The documentation for this class was generated from the following file:

- ttk_tools/vector/canapeapi.py (v3.1.1)

7.74 ttk_tools.vector.canapeapi_offline_stub.CANapeDiag Class Reference

Diagnostics device class (UDS, ...).

Inheritance diagram for ttk_tools.vector.canapeapi_offline_stub.CANapeDiag:



Public Member Functions

- __init__ (self, canape_asap3, db_filename="hil.cdd", comm_channel=DEV.CAN1, driver_type=DRIVER.CANDELA, module_name="DIAG", connect=True, enable_cache=None, request_timeout=30000, enable_tp=None, restore_tp=True)

  Initialize a CANape Diagnostics Device connection.
- isTesterPresentEnabled (self, ignore_errors=True, verbosity=1)

  Get current state of automatic sending of Tester Present (TP).
- enableTesterPresent (self, verbosity=1)

  Enable automatic sending of Tester Present (TP) service.
- disableTesterPresent (self, verbosity=1)

  Disable automatic sending of Tester Present (TP) service.
- release (self, verbosity=2)

  Manually release the module/device handle in CANape.
- runScript (self, script_command, is_file_name=False)

  Execute a CAPL script or script command via CANapeAPI.
- isOnline (self)

  Get online state of current device.

- goOnline (self, retry_timeout=10000)

  Switch the CANape diag device connection to online mode.
- goOffline (self, retry_timeout=10000)

  Switch the CANape diag device connection to offline mode.
- sendHexRequest (self, request)

  Send a raw service request as a series of bytes (usually as hex).
- sendSymbolicRequest (self, request, request_params=None, response_params=None, skip_empty_↩ iterations=False)

  Send a symbolic service request (name/path as defined in CDD/ODX-file).
- sendHexRequestSym (self, request, response_params=None)

  Send a raw service request as a series of bytes (usually as hex), but attempt to decode the response using parameter names/qualifiers of the matching symbolic request.

7.74.1 Detailed Description

Diagnostics device class (UDS, ...).

7.74.2 Constructor & Destructor Documentation

7.74.2.1 __init__()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeDiag.__init__ (
              self,
              canape_asap3,
              db_filename = "hil.cdd",
              comm_channel = DEV.CAN1,
              driver_type = DRIVER.CANDELA,
              module_name = "DIAG",
              connect = True,
              enable_cache = None,
              request_timeout = 30000,
              enable_tp = None,
              restore_tp = True )
```

Initialize a CANape Diagnostics Device connection.

Parameters

canape_asap3	a CANapeASAP3 instance (application interface)
db_filename	filename of .cdd/.odx database (or similar)
comm_channel	communication channel (see canapeapi_common.DEV constants)
driver_type	driver type (see canapeapi_common.DRIVER constants)
module_name	name of device/module in CANape
connect	Device connection behavior: True: immediately connect DUT after device has been created. Device will start in state "online" False: create device but remain in state "offline" if DUT cannot be connected at this time.
enable_cache	None: ignore parameter, use default True: enable caching False: disable caching Available since CANape 10, see Asap3CreateModule3

Parameters

request_timeout	timeout for diag requests [ms]
enable_tp	auto-set Tester Present status: None: leave as configured in workspace True: Enable automatic sending False: Disable automatic sending
restore_tp	TesterPresent status after going online (again): True: Restore TP status, so it will be the same as before going offline. False: Do nothing, TP status after going online will depend on current CANape configuration.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.CANapeDiag.

7.74.3 Member Function Documentation

7.74.3.1 disableTesterPresent()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeDiag.disableTesterPresent (
              self,
              verbosity = 1 )
```

Disable automatic sending of Tester Present (TP) service.

Note

Available since CANape 11.

Parameters

verbosity	0: silent 1: log current TP status

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.CANapeDiag.

7.74.3.2 enableTesterPresent()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeDiag.enableTesterPresent (
              self,
              verbosity = 1 )
```

Enable automatic sending of Tester Present (TP) service.

Note

Available since CANape 11.

Parameters

verbosity	0: silent 1: log current TP status

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.CANapeDiag.

7.74.3.3 goOffline()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeDiag.goOffline (
              self,
              retry_timeout = 10000 )
```

Switch the CANape diag device connection to offline mode.

Parameters

retry_timeout	[ms], if > 0, repeated attempts to set the ECU state will be made until timeout occurs.

Returns

a list of testresult entries containing info about retried attempts.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.BaseDevice.

7.74.3.4 goOnline()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeDiag.goOnline (
              self,
              retry_timeout = 10000 )
```

Switch the CANape diag device connection to online mode.

Parameters

retry_timeout	[ms], if > 0, repeated attempts to set the ECU state will be made until timeout occurs

Returns

a list of testresult entries containing info about retried attempts.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.BaseDevice.

7.74.3.5 isOnline()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeDiag.isOnline (
              self )
```

Get online state of current device.

Returns

True if the device is online, otherwise False.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.BaseDevice.

7.74.3.6 isTesterPresentEnabled()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeDiag.isTesterPresentEnabled (
              self,
              ignore_errors = True,
              verbosity = 1 )
```

Get current state of automatic sending of Tester Present (TP).

Note

Available since CANape 11.

Parameters

ignore_errors	if True, access errors will be ignored and return value will be None.
verbosity	0: silent 1: log current TP status

Returns

True if Tester Present sending is currently active,
False if TP sending is inactive or (if ignore_errors is set)
None if API function access failed (e.g. older CANape releases).

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.CANapeDiag.

7.74.3.7 release()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeDiag.release (
              self,
              verbosity = 2 )
```

Manually release the module/device handle in CANape.

This will normally be called during **del**, but sometimes it might be advantageous to release the handle a defined time (and not just implicitly when the destructor gets called eventually).

Parameters

verbosity	verbosity of stdout/log output: -1: do not show anything if handle is already gone (for calls in **del**) 0: errors only 1: Asap3Exit status output 2: also show "nothing to do" notice if handle is already gone

Returns

True if successful (or already released), otherwise False.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.BaseDevice.

7.74.3.8 runScript()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeDiag.runScript (
              self,
              script_command,
              is_file_name = False )
```

Execute a CAPL script or script command via CANapeAPI.

Note

This uses the "classic" `Asap3ExecuteScript` API function which does not provide any script status or return values.

New extended script functions are available since CANape version 9. The classic approach is sufficient for simple cases, though.

Parameters

script_command	CAPL command or name of CAPL script file to execute. A CAPL command can also be the name of a custom "project function" that has been defined in the current workspace.
is_file_name	True if `script_command` contains the name of a script file to be run, False if it directly contains a CAPL command.

Example:

```
# multiple commands can be added in sequence
device.runScript('Speak("Hello World"); Write("Done.");')

# some CAPL commands/methods have to be called on an existing
# device object
device.runScript('%s.SaveDatabase(0);'%(device.getModuleName())

# default location of script files is the folder of the current
# CANape workspace
device.runScript('capl_script.scr', is_file_name=True)
```

Note

The executed script or script command might still be running when this function returns.

Returns

True if script execution was successful (though not necessarily completed), otherwise False.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.BaseDevice.

7.74.3.9 sendHexRequest()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeDiag.sendHexRequest (
              self,
              request )
```

Send a raw service request as a series of bytes (usually as hex).

Parameters

request	a request as list of bytes

Example:

```
asap3 = CANapeASAP3(working_dir=r"path/to/canape_working_dir")
diag  = CANapeDiag(asap3, module_name="device_name")
diag.goOnline()
response = diag.sendHexRequest([0x10, 0x03])
print "response:", response
diag.goOffline()
```

Tester Present issues:

In order to get diagnostics via CANapeAPI to work, cyclic sending of "Tester Present" may have to be deactivated in CANape config (no, really).

Device Configuration -> [Diagnostics device] -> Tab Communication -> uncheck "Send Tester Present" -> Save

See also disableTesterPresent() for CANape 11 or later.

Returns

a list of responses [[response 1], [response 2], ...]. Typically, there will only be a single response, though.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.CANapeDiag.

7.74.3.10 sendHexRequestSym()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeDiag.sendHexRequestSym (
              self,
              request,
              response_params = None )
```

Send a raw service request as a series of bytes (usually as hex), but attempt to decode the response using parameter names/qualifiers of the matching symbolic request.

Parameters

request	a request as list of bytes
response_params	a list of parameter names/qualifiers to decode, compare sendSymbolicRequest

Example:

```
# using a symbolic request
responses = canape_diag.sendSymbolicRequest(
    request = "EcuIdentification_Read",
    request_params  = None,
    response_params = ["RecordDataIdentifier", "Part_Number"],
)
# dto. using a raw request
responses = canape_diag.sendHexRequestSym(
    request = [0x22, 0x00, 0x9F],
    response_params = ["RecordDataIdentifier", "Part_Number"],
)
```

Returns

a list of response dicts [{response 1}, {response 2}, ...] i.e. one dict per response (which usually means just one).

Each response dict contains: ‘‘raw_response’: raw response as a list of bytes
 'pos_response': True if response was positive, otherwise False
 'params'↩ : a dictionary with name: value entries for each parameter name from theresponse_paramslist.
 @returns If a parameter name forresponse_params` could not be decoded, its value will be set to None.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.CANapeDiag.

7.74.3.11 sendSymbolicRequest()

```
ttk_tools.vector.canapeapi_offline_stub.CANapeDiag.sendSymbolicRequest (
              self,
              request,
              request_params = None,
              response_params = None,
              skip_empty_iterations = False )
```

Send a symbolic service request (name/path as defined in CDD/ODX-file).

Parameters

request	a request as service name/path (from CDD/ODX)
request_params	dict with parameters for the request
response_params	list of parameters that should be extracted from the response
skip_empty_iterations	for complex parameters: skip iterations if they contain no data at all

Example:

```
# RoutineCtrl/SR015_Idle_Speed_Modification/Start
response = diag.sendSymbolicRequest(
    'SR015_Idle_Speed_Modification_Start',
```

```
    {'engine_speed_request': 1200}
)
print(f"{response[0]['pos_response'] = })

# Stored Data / Sine Write
response = diag.sendSymbolicRequest(
    'Sine_Write', # qualifier
    request_params = {"sine_ampl": 0.800, "sine_period": 0.160, "sine_value": 1.0}
)
print(f"{response[0]['pos_response'] = })

# Stored Data / Sine Read
response_params = ["sine_ampl", "sine_period", "sine_value"],
responses = canape_diag.sendSymbolicRequest(
    'Sine_Read', # qualifier,
    response_params = response_params,
)
print(f"{response[0]['pos_response'] = })
for param in response_params:
    diag_value = response[0]['response_params'][param]
    print(f"{param}: default/str: {diag_value}, numeric: {diag_value.numeric}")

response = diag.sendSymbolicRequest(
    'Windows_Read',
    None,
    ['Window_Front_Left', 'Window_Front_Right'],
)
print(f"{response[0]['pos_response'] = })
print(f"{response[0]['response_params']['Window_Front_Left'] = })
```

Note

For complex response parameters, supply appropriate instances of `DiagParameterName` (with set sub-parameter names) as entries in `response_params` in order to extract/access the complex sub-responses. See canapeapi_common.DiagParameterName, canapeapi_common.DiagParameterValue

Returns

a list of response dicts [{response 1}, {response 2}, ...] i.e. one dict per response (which usually means just one).

Each response dict contains: ''raw_response': raw response as a list of bytes
 'pos_response': True if response was positive, otherwise False
 'params'↩ : a dictionary with name: value entries for each parameter name from theresponse_paramslist.
 @returns If a parameter name forresponse_params` could not be decoded, its value will be set to None.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.CANapeDiag.

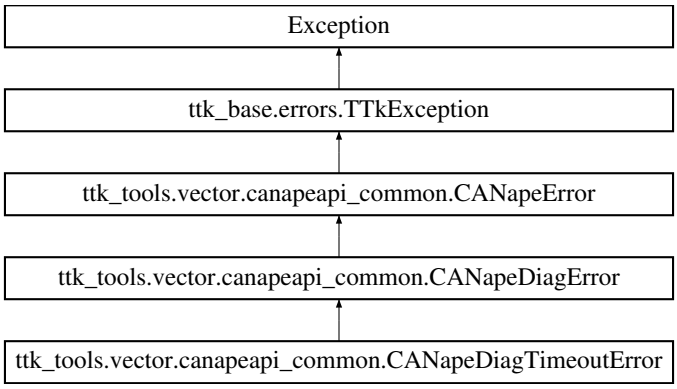
The documentation for this class was generated from the following file:

- ttk_tools/vector/canapeapi_offline_stub.py (v4.1.0)

7.75 ttk_tools.vector.canapeapi_common.CANapeDiagError Class Reference

Errors for CANape diagnostic connections.

Inheritance diagram for ttk_tools.vector.canapeapi_common.CANapeDiagError:



Public Member Functions

- __init__ (self, msg='', log_msg=False, api_msg=None, error_code=None)

7.75.1 Detailed Description

Errors for CANape diagnostic connections.

7.75.2 Constructor & Destructor Documentation

7.75.2.1 __init__()

```
ttk_tools.vector.canapeapi_common.CANapeDiagError.__init__ (
              self,
              msg = '',
              log_msg = False,
              api_msg = None,
              error_code = None )
```

Parameters

msg	error message
log_msg	True: log message to stdout, False: no log output
api_msg	CANape-provided "last error" message, if available
error_code	error code from CANapeAPI, if available.

Reimplemented from ttk_tools.vector.canapeapi_common.CANapeError.

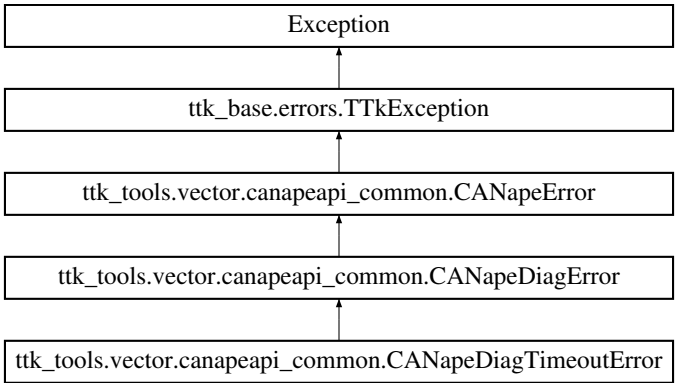
The documentation for this class was generated from the following file:

- ttk_tools/vector/canapeapi_common.py (v3.1.2)

7.76 ttk_tools.vector.canapeapi_common.CANapeDiagTimeoutError Class Reference

Timeout occurred during a diag request.

Inheritance diagram for ttk_tools.vector.canapeapi_common.CANapeDiagTimeoutError:



7.76.1 Detailed Description

Timeout occurred during a diag request.

Errors during script execution

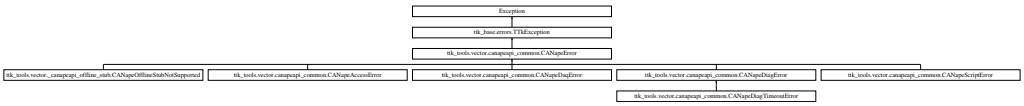
The documentation for this class was generated from the following file:

- ttk_tools/vector/canapeapi_common.py (v3.1.2)

7.77 ttk_tools.vector.canapeapi_common.CANapeError Class Reference

Base error for CANape connection.

Inheritance diagram for ttk_tools.vector.canapeapi_common.CANapeError:



Public Member Functions

- __init__ (self, msg='', log_msg=True, api_msg=None, error_code=None)

Public Attributes

- error_code

  error code from API function call, if available
- api_msg

  error description from APU ("last error"), if available

7.77.1 Detailed Description

Base error for CANape connection.

7.77.2 Constructor & Destructor Documentation

7.77.2.1 __init__()

```
ttk_tools.vector.canapeapi_common.CANapeError.__init__ (
              self,
              msg = '',
              log_msg = True,
              api_msg = None,
              error_code = None )
```

Parameters

msg	error message
log_msg	True: log message to stdout, False: no log output
api_msg	CANape-provided "last error" message, if available
error_code	error code from CANapeAPI, if available.

Reimplemented in ttk_tools.vector._canapeapi_offline_stub.CANapeOfflineStubNotSupported, ttk_tools.vector.canapeapi_common.C
ttk_tools.vector.canapeapi_common.CANapeDiagError, and ttk_tools.vector.canapeapi_common.CANapeAccessError.

7.77.3 Member Data Documentation

7.77.3.1 api_msg

```
ttk_tools.vector.canapeapi_common.CANapeError.api_msg
```

error description from APU ("last error"), if available

7.77.3.2 error_code

```
ttk_tools.vector.canapeapi_common.CANapeError.error_code
```

error code from API function call, if available

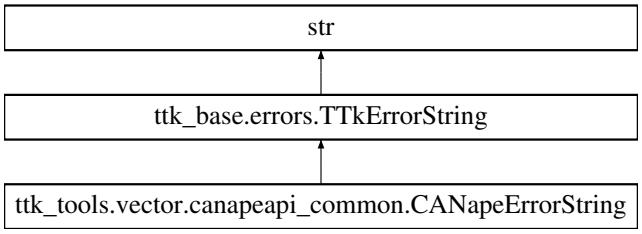
The documentation for this class was generated from the following file:

- ttk_tools/vector/canapeapi_common.py (v3.1.2)

7.78 ttk_tools.vector.canapeapi_common.CANapeErrorString Class Reference

Container for diagnostic response values that may have multiple value representations.

Inheritance diagram for ttk_tools.vector.canapeapi_common.CANapeErrorString:



7.78.1 Detailed Description

Container for diagnostic response values that may have multiple value representations.

Base representation is a "string", others might be "raw", "numeric" and "complex" parameters that contain a list of sub-parameters as before: DiagParameterValue - value as basic string representation (which should always be available) DiagParameterValue.raw - raw value representation DiagParameterValue.numeric - numeric value representation DiagParameterValue.complex - complex value representation (a list of sub-responses with sub-value dicts)

A response parameter may have multiple representations (as configured in .cdd):

- string: default representation, always available (may be empty).
- numeric: raw data interpreted as a numeric value, using a configured conversion. May be int or float.
- raw: raw bytes (usually as list of byte values)
- complex: complex parameters will contain a list of sub-responses. Common use case is fault memory access

String data may be anything. Common occurrences are

- raw data bytes interpreted as characters
  e.g. EcuIdentification_Read: Part_Number with value='Vector #12345', raw=[0x56, 0x65, 0x63, 0x74, ...]
- enum keys/labels (like "open"/"closed", "true"/"false")
  e.g. DoorStatus_Read: DoorContactFrontLeft with value='closed', raw=[0x00], numeric=None, complex=None
- raw data as hex string
  e.g. EcuIdentification_Read: RecordDataIdentifier with value='0x9F', raw=[0x00, 0x9F], numeric=159)
- numeric data as string
  e.g. Stored Data/Sine_Read: sine_period with value='0.16', raw=[0x02], numeric=0.1568627450980392

Numeric data is only available if a conversion has been configured in .cdd and sometimes it is missing even if it would seem to make sense
e.g. for bool values, like in FaultMemory_ReadAllIdentified DtcStatusbyte.TestFailedSinceLastClear: value='true', raw=[0x01], numeric=None, complex=None

Raw data is usually available for normal (non-complex) parameters and contains the raw/undecoded bytes of the parameter value. Note that raw representations are available for complex parameters (for whatever reason).

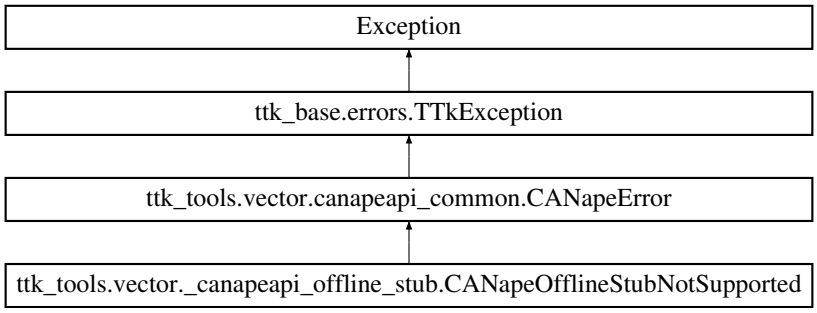
The documentation for this class was generated from the following file:

- ttk_tools/vector/canapeapi_common.py (v3.1.2)

7.79 ttk_tools.vector._canapeapi_offline_stub.CANapeOfflineStubNotSupported Class Reference

Error class for features not supported in the stub implementation.

Inheritance diagram for ttk_tools.vector._canapeapi_offline_stub.CANapeOfflineStubNotSupported:



Public Member Functions

- __init__ (self, msg='')

7.79.1 Detailed Description

Error class for features not supported in the stub implementation.

7.79.2 Constructor & Destructor Documentation

7.79.2.1 __init__()

```
ttk_tools.vector._canapeapi_offline_stub.CANapeOfflineStubNotSupported.__init__ (
              self,
              msg = '' )
```

Parameters

msg	error message

Reimplemented from ttk_tools.vector.canapeapi_common.CANapeError.

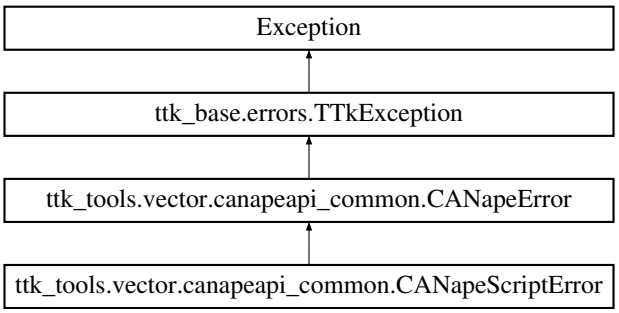
The documentation for this class was generated from the following file:

- ttk_tools/vector/_canapeapi_offline_stub.py (v4.7.1)

7.80 ttk_tools.vector.canapeapi_common.CANapeScriptError Class Reference

A wrapper for error message strings so they can be distinguished from "normal" string values (like contents of ASCII-type variables).

Inheritance diagram for ttk_tools.vector.canapeapi_common.CANapeScriptError:



7.80.1 Detailed Description

A wrapper for error message strings so they can be distinguished from "normal" string values (like contents of ASCII-type variables).

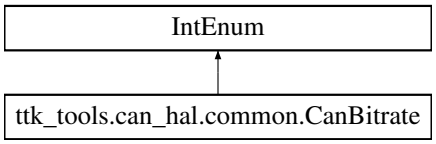
The documentation for this class was generated from the following file:

- ttk_tools/vector/canapeapi_common.py (v3.1.2)

7.81 ttk_tools.can_hal.common.CanBitrate Class Reference

Connection speed settings (CanHal.h, tCanHalBautrate), bits per s.

Inheritance diagram for ttk_tools.can_hal.common.CanBitrate:



Static Public Attributes

- int Bitrate1MBit = 0x00
- int Bitrate800kBit = 0x01
- int Bitrate500kBit = 0x02
- int Bitrate250kBit = 0x03
- int Bitrate125kBit = 0x04
- int Bitrate100kBit = 0x05
- int Bitrate50kBit = 0x06
- int Bitrate20kBit = 0x07
- int Bitrate10kBit = 0x08

7.81.1 Detailed Description

Connection speed settings (CanHal.h, tCanHalBautrate), bits per s.

7.81.2 Member Data Documentation

7.81.2.1 Bitrate100kBit

```
int ttk_tools.can_hal.common.CanBitrate.Bitrate100kBit = 0x05  [static]
```

7.81.2.2 Bitrate10kBit

```
int ttk_tools.can_hal.common.CanBitrate.Bitrate10kBit = 0x08  [static]
```

7.81.2.3 Bitrate125kBit

```
int ttk_tools.can_hal.common.CanBitrate.Bitrate125kBit = 0x04  [static]
```

7.81.2.4 Bitrate1MBit

```
int ttk_tools.can_hal.common.CanBitrate.Bitrate1MBit = 0x00  [static]
```

7.81.2.5 Bitrate20kBit

```
int ttk_tools.can_hal.common.CanBitrate.Bitrate20kBit = 0x07  [static]
```

7.81.2.6 Bitrate250kBit

```
int ttk_tools.can_hal.common.CanBitrate.Bitrate250kBit = 0x03  [static]
```

7.81.2.7 Bitrate500kBit

```
int ttk_tools.can_hal.common.CanBitrate.Bitrate500kBit = 0x02  [static]
```

7.81.2.8 Bitrate50kBit

```
int ttk_tools.can_hal.common.CanBitrate.Bitrate50kBit = 0x06  [static]
```

7.81.2.9 Bitrate800kBit

```
int ttk_tools.can_hal.common.CanBitrate.Bitrate800kBit = 0x01  [static]
```

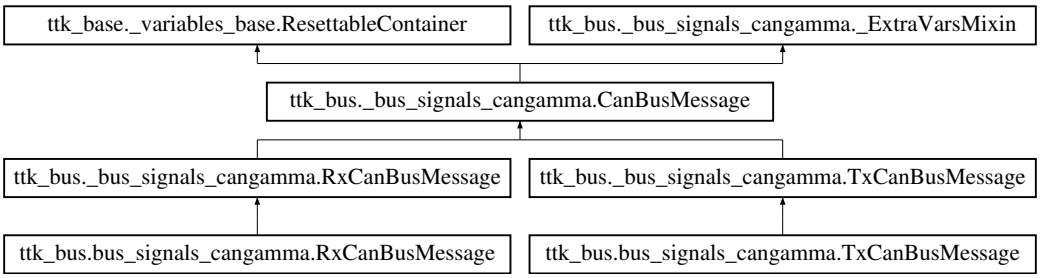
The documentation for this class was generated from the following file:

- ttk_tools/can_hal/common.py (v1.2)

7.82 ttk_bus._bus_signals_cangamma.CanBusMessage Class Reference

CAN-Message/PDU-specific data and methods.

Inheritance diagram for ttk_bus._bus_signals_cangamma.CanBusMessage:



Public Member Functions

- __init__ (self, context, base_identifier, cycle_time=0, debounce_time=0, alias="", descr='', change_delay=None, timeout=None, recovery_time=None)

  CAN Message/PDU data.
- __repr__ (self)

Public Attributes

- alias

  Alias name.
- descr

  Description.
- context

  Context (reference to current gaapi)
- base_identifier

  base identifier of Message/PDU helper variables
- cycle_time

  cycle time of Message/PDU in [ms]
- debounce_time

  minimum debounce delay for event-triggered PDUs in [ms]
- change_delay

  delay to wait after a value change [ms]
- timeout

  detection time for timeout errors [ms]
- recovery_time

  time for recovery/"wiedergut" checks [ms]
- length_var

  Message (Rx or Tx) length.
- time_var

  Message (Rx or Tx) timestamp.

7.82.1 Detailed Description

CAN-Message/PDU-specific data and methods.

7.82.2 Constructor & Destructor Documentation

7.82.2.1 __init__()

```
ttk_bus._bus_signals_cangamma.CanBusMessage.__init__ (
                self,
                context,
                base_identifier,
                cycle_time = 0,
                debounce_time = 0,
                alias = "",
                descr = '',
                change_delay = None,
                timeout = None,
                recovery_time = None )
```

CAN Message/PDU data.

Parameters

context	parent gamma_api context
base_identifier	base identifier to PDU/message Helper entries will be derived from t e.g. kickout or status variables (though this is mostly relevant to messages sent to DUT, see RxCanBusMessage)
cycle_time	cycle time in [ms]. Event-triggered messages have a cycle time of 0 (similar to DBC value)
debounce_time	minimum debounce delay between events in [ms]
alias	Message alias/name
descr	(opt) Message description (may be used in report entries)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_base._variables_base.ResettableContainer.

Reimplemented in ttk_bus._bus_signals_cangamma.TxCanBusMessage, ttk_bus.bus_signals_cangamma.TxCanBusMessage, ttk_bus._bus_signals_cangamma.RxCanBusMessage, and ttk_bus.bus_signals_cangamma.RxCanBusMessage.

7.82.3 Member Function Documentation

7.82.3.1 __repr__()

```
ttk_bus._bus_signals_cangamma.CanBusMessage.__repr__ (
                self )
```

7.82.4 Member Data Documentation

7.82.4.1 alias

ttk_bus._bus_signals_cangamma.CanBusMessage.alias

Alias name.

7.82.4.2 base_identifier

ttk_bus._bus_signals_cangamma.CanBusMessage.base_identifier

base identifier of Message/PDU helper variables

message vars:
<node>.<memory>.<group>.<message-id>.<var-name>
signal vars:
<node>.<memory>.<group>.<message-id>.signals.<signal-name>.<var-name>

Examples:
BusSystems.can3_iFIU_iSRC.HIL_TX.N0FIU_SetError_Card1.signals.Path_1_EI_1.↩
value_phy
BusSystems.can3_iFIU_iSRC.HIL_RX.N0FIU_Status_Card19.signals.State_Path_3_↩
break.value_phy

7.82.4.3 change_delay

ttk_bus._bus_signals_cangamma.CanBusMessage.change_delay

delay to wait after a value change [ms]

7.82.4.4 context

ttk_bus._bus_signals_cangamma.CanBusMessage.context

Context (reference to current gaapi)

7.82.4.5 cycle_time

ttk_bus._bus_signals_cangamma.CanBusMessage.cycle_time

cycle time of Message/PDU in [ms]

7.82.4.6 debounce_time

ttk_bus._bus_signals_cangamma.CanBusMessage.debounce_time

minimum debounce delay for event-triggered PDUs in [ms]

7.82.4.7 descr

`ttk_bus._bus_signals_cangamma.CanBusMessage.descr`

Description.

7.82.4.8 length_var

`ttk_bus._bus_signals_cangamma.CanBusMessage.length_var`

Message (Rx or Tx) length.

7.82.4.9 recovery_time

`ttk_bus._bus_signals_cangamma.CanBusMessage.recovery_time`

time for recovery/"wiedergut" checks [ms]

7.82.4.10 time_var

`ttk_bus._bus_signals_cangamma.CanBusMessage.time_var`

Message (Rx or Tx) timestamp.

7.82.4.11 timeout

`ttk_bus._bus_signals_cangamma.CanBusMessage.timeout`

detection time for timeout errors [ms]

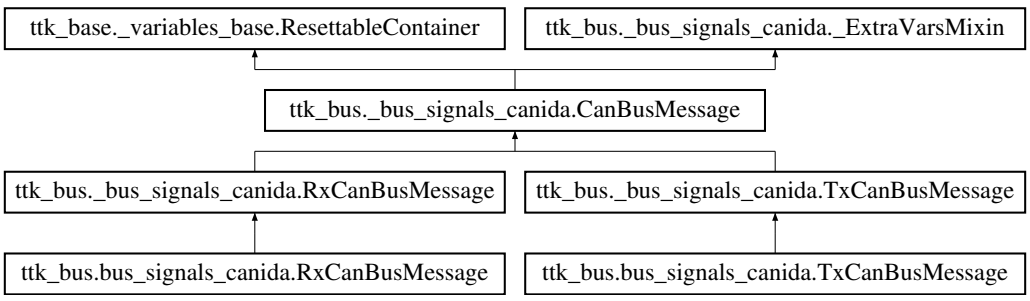
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_cangamma.py (v1.12)

7.83 ttk_bus._bus_signals_canida.CanBusMessage Class Reference

CAN-Message/PDU-specific data and methods.

Inheritance diagram for ttk_bus._bus_signals_canida.CanBusMessage:



Public Member Functions

- __init__ (self, context, base_identifier, cycle_time=0, debounce_time=0, alias="", descr='', change_delay=None, timeout=None, recovery_time=None)

  CAN Message/PDU data.
- __repr__ (self)

Public Attributes

- alias

  Alias name.
- descr

  Description.
- context

  Context (reference to current iDA connection)
- base_identifier

  base identifier of Message/PDU helper variables Note that the static "CAN"-prefix in variable names is mainly a way to group all CAN variables (for multiple CAN interfaces) together in tree views.
- cycle_time

  cycle time of Message/PDU in [ms]
- debounce_time

  minimum debounce delay for event-triggered PDUs in [ms]
- change_delay

  delay to wait after a value change [ms]
- timeout

  detection time for timeout errors [ms]
- recovery_time

  time for recovery/"wiedergut" checks [ms]
- length_var

  Message (Rx or Tx) length.
- time_var

  Message (Rx or Tx) timestamp.

7.83.1 Detailed Description

CAN-Message/PDU-specific data and methods.

7.83.2 Constructor & Destructor Documentation

7.83.2.1 __init__()

```
ttk_bus._bus_signals_canida.CanBusMessage.__init__ (
              self,
              context,
              base_identifier,
              cycle_time = 0,
              debounce_time = 0,
              alias = "",
              descr = '',
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

CAN Message/PDU data.

Parameters

context	parent iDA context
base_identifier	base var name/identifier to PDU/message Helper entries will be derived from this name. e.g. kickout or status variables (though this is mostly relevant to messages sent to DUT, see RxCanBusMessage) e.g. "CAN.can3_iFIU_iSRC.HIL_TX.N0FIU_SetError_Card1"
cycle_time	cycle time in [ms]. Event-triggered messages have a cycle time of 0 (similar to DBC value)
debounce_time	minimum debounce delay between events in [ms]
alias	Message alias/name
descr	(opt) Message description (may be used in report entries)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_base._variables_base.ResettableContainer.

Reimplemented in ttk_bus._bus_signals_canida.TxCanBusMessage, ttk_bus._bus_signals_canida.RxCanBusMessage, ttk_bus.bus_signals_canida.TxCanBusMessage, and ttk_bus.bus_signals_canida.RxCanBusMessage.

7.83.3 Member Function Documentation

7.83.3.1 __repr__()

```
ttk_bus._bus_signals_canida.CanBusMessage.__repr__ (
              self )
```

7.83.4 Member Data Documentation

7.83.4.1 alias

```
ttk_bus._bus_signals_canida.CanBusMessage.alias
```

Alias name.

7.83.4.2 base_identifier

```
ttk_bus._bus_signals_canida.CanBusMessage.base_identifier
```

base identifier of Message/PDU helper variables Note that the static "CAN"-prefix in variable names is mainly a way to group all CAN variables (for multiple CAN interfaces) together in tree views.

message vars:
base (hil-tx): `CAN.<can-name>.HIL_TX.<message-id>`
base (hil-rx): `CAN.<can-name>.HIL_RX.<message-id>`
vars: `CAN.<can-name>.HIL_<TX/RX>.<message-id>{.<var-name>}`

signal vars:
base (hil-tx): CAN.<can-name>.HIL_TX.<message-id>.signals.<signal-name>
base (hil-rx): CAN.<can-name>.HIL_RX.<message-id>.signals.<signal-name>
vars: CAN.<can-name>.HIL_<TX/RX>.<message-id>.signals.<signal-name>{.<var-name>}

Examples:
CAN.can3_iFIU_iSRC.HIL_TX.N0FIU_SetError_Card1.signals.Path_1_EI_1.value
CAN.can3_iFIU_iSRC.HIL_RX.N0FIU_Status_Card19.signals.State_Path_3_break.↩
value

7.83.4.3 change_delay

ttk_bus._bus_signals_canida.CanBusMessage.change_delay

delay to wait after a value change [ms]

7.83.4.4 context

ttk_bus._bus_signals_canida.CanBusMessage.context

Context (reference to current iDA connection)

7.83.4.5 cycle_time

ttk_bus._bus_signals_canida.CanBusMessage.cycle_time

cycle time of Message/PDU in [ms]

7.83.4.6 debounce_time

ttk_bus._bus_signals_canida.CanBusMessage.debounce_time

minimum debounce delay for event-triggered PDUs in [ms]

7.83.4.7 descr

ttk_bus._bus_signals_canida.CanBusMessage.descr

Description.

7.83.4.8 length_var

ttk_bus._bus_signals_canida.CanBusMessage.length_var

Message (Rx or Tx) length.

7.83.4.9 recovery_time

`ttk_bus._bus_signals_canida.CanBusMessage.recovery_time`

time for recovery/"wiedergut" checks [ms]

7.83.4.10 time_var

`ttk_bus._bus_signals_canida.CanBusMessage.time_var`

Message (Rx or Tx) timestamp.

7.83.4.11 timeout

`ttk_bus._bus_signals_canida.CanBusMessage.timeout`

detection time for timeout errors [ms]

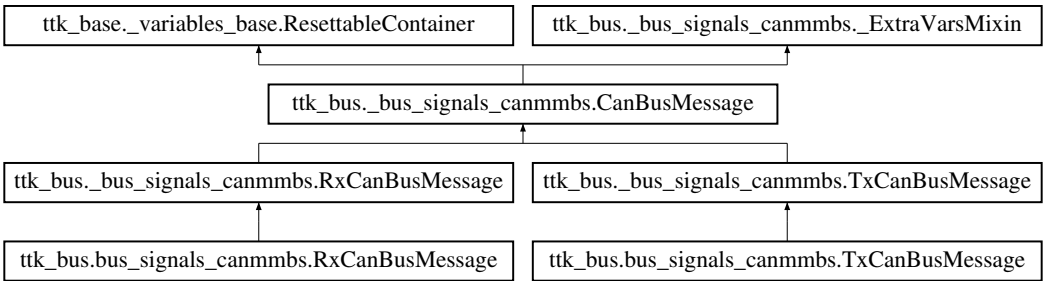
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_canida.py (v1.0.2)

7.84 ttk_bus._bus_signals_canmmbs.CanBusMessage Class Reference

CAN-Message/PDU-specific data and methods.

Inheritance diagram for ttk_bus._bus_signals_canmmbs.CanBusMessage:



Public Member Functions

- __init__ (self, context, bus_systems_path, cycle_time=0, debounce_time=0, alias="", descr='', change_delay=None, timeout=None, recovery_time=None)

  CAN Message/PDU data.
- __str__ (self)

  Return Message/PDU alias.

Public Attributes

- alias
  alias name of message/pdu
- descr
  description of message/pdu
- context
  Context (reference to assigned RTE application instance)
- bus_systems_path
  base BusSystems path to Message/PDU helper variables below BusSystems e.g.
- base_path
  base path for derived variables (just for convenience, mirrors cleaned bus_systems_path)
- cycle_time
  cycle time of Message/PDU in [ms]
- debounce_time
  minimum debounce delay for event-triggered PDUs in [ms]
- change_delay
  delay to wait after a value change [ms]
- timeout
  detection time for timeout errors [ms]
- recovery_time
  time for recovery/"wiedergut" checks [ms]
- status_var
  message (Rx or Tx) status

7.84.1 Detailed Description

CAN-Message/PDU-specific data and methods.

7.84.2 Constructor & Destructor Documentation

7.84.2.1 __init__()

```
ttk_bus._bus_signals_canmmbs.CanBusMessage.__init__ (
              self,
              context,
              bus_systems_path,
              cycle_time = 0,
              debounce_time = 0,
              alias = "",
              descr = '',
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

CAN Message/PDU data.

Parameters

context	parent context (an RTE application instance to resolve signal/variable paths)

Parameters

bus_systems_path	base path to PDU/message below BusSystems, use "BusSystems/CAN/<CAN_NAME>/<MSG_NAME>/TX/<MSG_NAME>" or "BusSystems/CAN/<CAN_NAME>/<MSG_NAME>/TX/<MSG_NAME>_Enable" Helper entries will be derived from this path, e.g. kickout or status variables (though this is mostly relevant to messages sent to DUT, see RxCanBusMessage)
cycle_time	cycle time in [ms]. Event-triggered messages have a cycle time of 0 (similar to DBC value)
debounce_time	minimum debounce delay between events in [ms]
alias	Message alias/name
descr	(opt) Message description (may be used in report entries)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_base._variables_base.ResettableContainer.

Reimplemented in ttk_bus._bus_signals_canmmbs.TxCanBusMessage, ttk_bus.bus_signals_canmmbs.TxCanBusMessage, ttk_bus._bus_signals_canmmbs.RxCanBusMessage, and ttk_bus.bus_signals_canmmbs.RxCanBusMessage.

7.84.3 Member Function Documentation

7.84.3.1 __str__()

```
ttk_bus._bus_signals_canmmbs.CanBusMessage.__str__ (
              self )
```

Return Message/PDU alias.

7.84.4 Member Data Documentation

7.84.4.1 alias

```
ttk_bus._bus_signals_canmmbs.CanBusMessage.alias
```

alias name of message/pdu

7.84.4.2 base_path

```
ttk_bus._bus_signals_canmmbs.CanBusMessage.base_path
```

base path for derived variables (just for convenience, mirrors cleaned bus_systems_path)

7.84.4.3 bus_systems_path

ttk_bus._bus_signals_canmmbs.CanBusMessage.bus_systems_path

base BusSystems path to Message/PDU helper variables below BusSystems
e.g.

"BusSystems/CAN/<CAN_NAME>/<MSG_NAME>/TX/<MSG_NAME>"

7.84.4.4 change_delay

ttk_bus._bus_signals_canmmbs.CanBusMessage.change_delay

delay to wait after a value change [ms]

7.84.4.5 context

ttk_bus._bus_signals_canmmbs.CanBusMessage.context

Context (reference to assigned RTE application instance)

7.84.4.6 cycle_time

ttk_bus._bus_signals_canmmbs.CanBusMessage.cycle_time

cycle time of Message/PDU in [ms]

7.84.4.7 debounce_time

ttk_bus._bus_signals_canmmbs.CanBusMessage.debounce_time

minimum debounce delay for event-triggered PDUs in [ms]

7.84.4.8 descr

ttk_bus._bus_signals_canmmbs.CanBusMessage.descr

description of message/pdu

7.84.4.9 recovery_time

ttk_bus._bus_signals_canmmbs.CanBusMessage.recovery_time

time for recovery/"wiedergut" checks [ms]

7.84.4.10 status_var

`ttk_bus._bus_signals_canmmbs.CanBusMessage.status_var`

message (Rx or Tx) status

7.84.4.11 timeout

`ttk_bus._bus_signals_canmmbs.CanBusMessage.timeout`

detection time for timeout errors [ms]

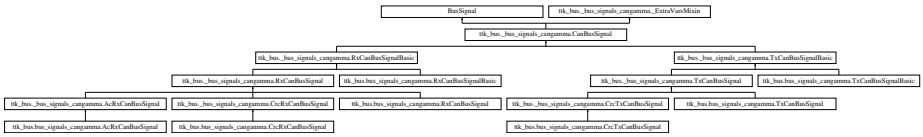
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_canmmbs.py (v2.3)

7.85 ttk_bus._bus_signals_cangamma.CanBusSignal Class Reference

Base wrapper for a bus signal with HIL-side BusSignals variable and relevant associated values.

Inheritance diagram for ttk_bus._bus_signals_cangamma.CanBusSignal:



Public Member Functions

- __init__ (self, context, signal_identifier, message=None, cycle_time=None, debounce_time=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  Bus signal initialization.
- pdu (self)

  Alias for message.
- msg (self)

  Alias for message.
- trigger (self)

  Alias for kickout.

Public Attributes

- base_identifier

  Signal model path as base path for deriving any helper PVs.
- message

  Reference to parent message/PDU.
- cycle_time
- debounce_time
- change_delay

7.85.1 Detailed Description

Base wrapper for a bus signal with HIL-side BusSignals variable and relevant associated values.

7.85.2 Constructor & Destructor Documentation

7.85.2.1 __init__()

```
ttk_bus._bus_signals_cangamma.CanBusSignal.__init__ (
                self,
                context,
                signal_identifier,
                message = None,
                cycle_time = None,
                debounce_time = None,
                unit = "",
                alias = "",
                descr = "",
                lookup = None,
                change_delay = None,
                timeout = None,
                recovery_time = None )
```

Bus signal initialization.

Parameters

context	parent gamma_api context
signal_identifier	identifier for signal value PV.
message	reference to associated CanBusMessage instance
cycle_time	(opt) cycle time in [ms], overrides settings in message
debounce_time	(opt) minimum debounce delay between events in [ms], overrides settings in Message
unit	(opt) unit of physical signal value (as string, mainly used for report entries)
alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

Reimplemented in ttk_bus._bus_signals_cangamma.CrcTxCanBusSignal, ttk_bus.bus_signals_cangamma.CrcTxCanBusSignal, ttk_bus._bus_signals_cangamma.CrcRxCanBusSignal, ttk_bus._bus_signals_cangamma.AcRxCanBusSignal, ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal, ttk_bus.bus_signals_cangamma.AcRxCanBusSignal, ttk_bus._bus_signals_cangamma.TxCanBusSignalBasic, ttk_bus._bus_signals_cangamma.TxCanBusSignal, ttk_bus._bus_signals_cangamma.RxCanBusSignalBasic, ttk_bus._bus_signals_cangamma.RxCanBusSignal, ttk_bus.bus_signals_cangamma.TxCanBusSignalBasic, ttk_bus.bus_signals_cangamma.TxCanBusSignal, ttk_bus.bus_signals_cangamma.RxCanBusSignalBasic, and ttk_bus.bus_signals_cangamma.RxCanBusSignal.

7.85.3 Member Function Documentation

7.85.3.1 msg()

```
ttk_bus._bus_signals_cangamma.CanBusSignal.msg (
              self )
```

Alias for message.

7.85.3.2 pdu()

```
ttk_bus._bus_signals_cangamma.CanBusSignal.pdu (
              self )
```

Alias for message.

7.85.3.3 trigger()

```
ttk_bus._bus_signals_cangamma.CanBusSignal.trigger (
              self )
```

Alias for kickout.

7.85.4 Member Data Documentation

7.85.4.1 base_identifier

```
ttk_bus._bus_signals_cangamma.CanBusSignal.base_identifier
```

Signal model path as base path for deriving any helper PVs.

7.85.4.2 change_delay

```
ttk_bus._bus_signals_cangamma.CanBusSignal.change_delay
```

7.85.4.3 cycle_time

```
ttk_bus._bus_signals_cangamma.CanBusSignal.cycle_time
```

7.85.4.4 debounce_time

```
ttk_bus._bus_signals_cangamma.CanBusSignal.debounce_time
```

7.85.4.5 message

`ttk_bus._bus_signals_cangamma.CanBusSignal.message`

Reference to parent message/PDU.

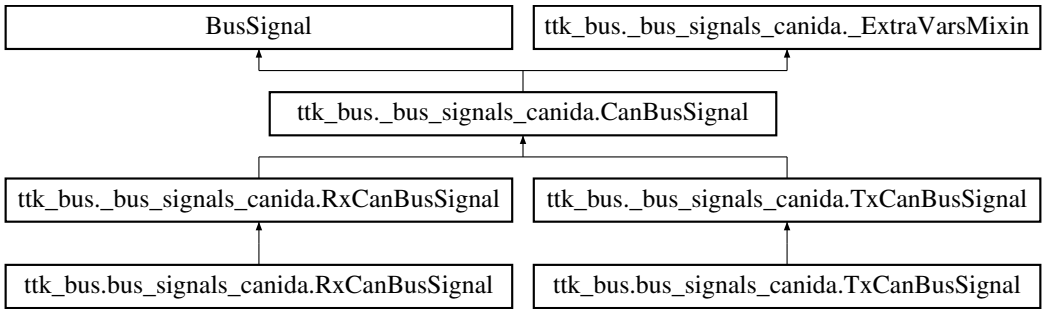
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_cangamma.py (v1.12)

7.86 ttk_bus._bus_signals_canida.CanBusSignal Class Reference

Base wrapper for a bus signal with HIL-side BusSignals variable and relevant associated values.

Inheritance diagram for ttk_bus._bus_signals_canida.CanBusSignal:



Public Member Functions

- __init__ (self, context, signal_identifier, message=None, cycle_time=None, debounce_time=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  Bus signal initialization.
- pdu (self)

  Alias for message.
- msg (self)

  Alias for message.
- trigger (self)

  Alias for kickout.

Public Attributes

- base_identifier

  Signal model path as base path for deriving any helper PVs.
- message

  Reference to parent message/PDU.
- cycle_time
- debounce_time
- change_delay

7.86.1 Detailed Description

Base wrapper for a bus signal with HIL-side BusSignals variable and relevant associated values.

7.86.2 Constructor & Destructor Documentation

7.86.2.1 __init__()

```
ttk_bus._bus_signals_canida.CanBusSignal.__init__ (
                self,
                context,
                signal_identifier,
                message = None,
                cycle_time = None,
                debounce_time = None,
                unit = "",
                alias = "",
                descr = "",
                lookup = None,
                change_delay = None,
                timeout = None,
                recovery_time = None )
```

Bus signal initialization.

Note

`signal_identifier` var name Use `.value` entry (or optionally `.value_phy`/`.value_raw`, if available), e.g.
`"CAN.CanName.HIL_TX.MessageName.signals.SignalName.value"`
or just the base signal identifier to default to use `.value`:
`"CAN.CanName.HIL_TX.MessageName.signals.SignalName"

Parameters

context	parent iDA context
signal_identifier	identifier for signal value var.
message	reference to associated CanBusMessage instance
cycle_time	(opt) cycle time in [ms], overrides settings in message
debounce_time	(opt) minimum debounce delay between events in [ms], overrides settings in Message
unit	(opt) unit of physical signal value (as string, mainly used for report entries)
alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

Reimplemented in ttk_bus._bus_signals_canida.TxCanBusSignal, ttk_bus._bus_signals_canida.RxCanBusSignal,

ttk_bus.bus_signals_canida.TxCanBusSignal, and ttk_bus.bus_signals_canida.RxCanBusSignal.

7.86.3 Member Function Documentation

7.86.3.1 msg()

```
ttk_bus._bus_signals_canida.CanBusSignal.msg (
              self )
```

Alias for message.

7.86.3.2 pdu()

```
ttk_bus._bus_signals_canida.CanBusSignal.pdu (
              self )
```

Alias for message.

7.86.3.3 trigger()

```
ttk_bus._bus_signals_canida.CanBusSignal.trigger (
              self )
```

Alias for kickout.

7.86.4 Member Data Documentation

7.86.4.1 base_identifier

```
ttk_bus._bus_signals_canida.CanBusSignal.base_identifier
```

Signal model path as base path for deriving any helper PVs.

7.86.4.2 change_delay

```
ttk_bus._bus_signals_canida.CanBusSignal.change_delay
```

7.86.4.3 cycle_time

```
ttk_bus._bus_signals_canida.CanBusSignal.cycle_time
```

7.86.4.4 debounce_time

```
ttk_bus._bus_signals_canida.CanBusSignal.debounce_time
```

7.86.4.5 message

`ttk_bus._bus_signals_canida.CanBusSignal.message`

Reference to parent message/PDU.

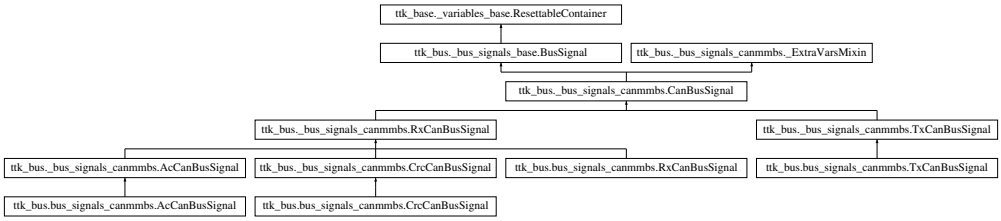
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_canida.py (v1.0.2)

7.87 ttk_bus._bus_signals_canmmbs.CanBusSignal Class Reference

Base wrapper for a bus signal with HIL-side BusSignals variable and relevant associated values.

Inheritance diagram for ttk_bus._bus_signals_canmmbs.CanBusSignal:



Public Member Functions

- __init__ (self, context, signal_path, message=None, cycle_time=None, debounce_time=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  Bus signal initialization.

Public Attributes

- base_path

  Signal model path as base path for deriving sub-element paths (for CANMM signals, related variables follow the form <path_to_signal>_suffix)
- message

  Reference to parent message/PDU.

7.87.1 Detailed Description

Base wrapper for a bus signal with HIL-side BusSignals variable and relevant associated values.

7.87.2 Constructor & Destructor Documentation

7.87.2.1 __init__()

```
ttk_bus._bus_signals_canmmbs.CanBusSignal.__init__ (
                self,
                context,
                signal_path,
                message = None,
                cycle_time = None,
                debounce_time = None,
                unit = "",
                alias = "",
                descr = "",
                lookup = None,
                change_delay = None,
                timeout = None,
                recovery_time = None )
```

Bus signal initialization.

Parameters

context	parent context (an RTE application instance to resolve signal/variable paths)
signal_path	base path to signal value below BusSystems
message	reference to associated CanBusMessage instance
cycle_time	(opt) cycle time in [ms], overrides settings in Message
debounce_time	(opt) minimum debounce delay between events in [ms], overrides settings in Message
unit	(opt) unit of physical signal value (as string, mainly used for report entries)
alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

Reimplemented in ttk_bus._bus_signals_canmmbs.CrcCanBusSignal, ttk_bus._bus_signals_canmmbs.AcCanBusSignal, ttk_bus.bus_signals_canmmbs.CrcCanBusSignal, ttk_bus.bus_signals_canmmbs.AcCanBusSignal, ttk_bus._bus_signals_canmmbs.TxCanBusSignal, ttk_bus.bus_signals_canmmbs.TxCanBusSignal, and ttk_bus.bus_signals_canmmbs.RxCanBusSignal.

7.87.3 Member Data Documentation

7.87.3.1 base_path

```
ttk_bus._bus_signals_canmmbs.CanBusSignal.base_path
```

Signal model path as base path for deriving sub-element paths (for CANMM signals, related variables follow the form <path_to_signal>_suffix)

7.87.3.2 message

```
ttk_bus._bus_signals_canmmbs.CanBusSignal.message
```

Reference to parent message/PDU.

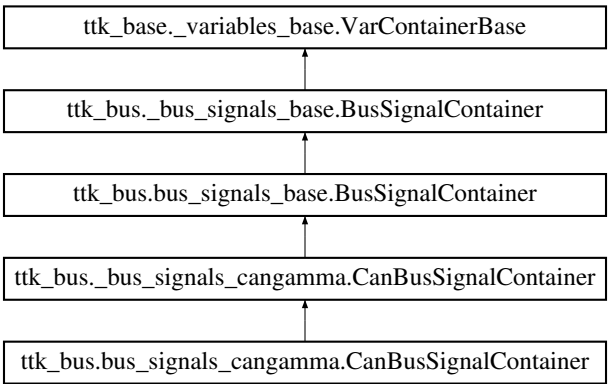
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_canmmbs.py (v2.3)

7.88 ttk_bus._bus_signals_cangamma.CanBusSignalContainer Class Reference

A BusSignalContainer for CAN Bus Signals.

Inheritance diagram for ttk_bus._bus_signals_cangamma.CanBusSignalContainer:



Public Member Functions

- iterSignals (self, msg)

  Iterate over all signals that reference the specified CAN message/PDU.

7.88.1 Detailed Description

A BusSignalContainer for CAN Bus Signals.

7.88.2 Member Function Documentation

7.88.2.1 iterSignals()

```
ttk_bus._bus_signals_cangamma.CanBusSignalContainer.iterSignals (
              self,
              msg )
```

Iterate over all signals that reference the specified CAN message/PDU.

Parameters

msg	a CAN message/PDU instance

Usage:

```
for sig in bus.iterSignals(bus.tx_msg_FNORD):
    print sig
```

Returns

an iterator object.

Reimplemented in ttk_bus.bus_signals_cangamma.CanBusSignalContainer.

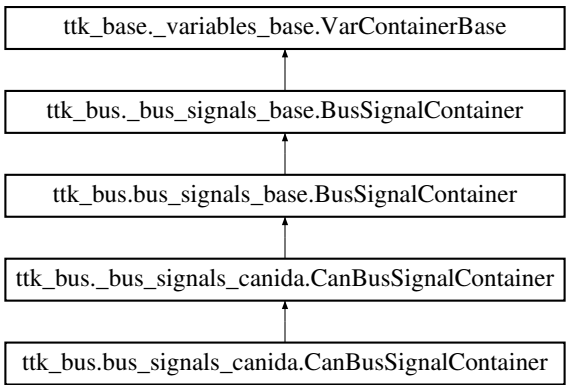
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_cangamma.py (v1.12)

7.89 ttk_bus._bus_signals_canida.CanBusSignalContainer Class Reference

A BusSignalContainer for CAN Bus Signals.

Inheritance diagram for ttk_bus._bus_signals_canida.CanBusSignalContainer:



Public Member Functions

- iterSignals (self, msg)

  Iterate over all signals that reference the specified CAN message/PDU.

7.89.1 Detailed Description

A BusSignalContainer for CAN Bus Signals.

7.89.2 Member Function Documentation

7.89.2.1 iterSignals()

```
ttk_bus._bus_signals_canida.CanBusSignalContainer.iterSignals (
            self,
            msg )
```

Iterate over all signals that reference the specified CAN message/PDU.

Parameters

msg	a CAN message/PDU instance

Usage:

```
for sig in bus.iterSignals(bus.tx_msg_FNORD):
    print sig
```

Returns

an iterator object.

Reimplemented in ttk_bus.bus_signals_canida.CanBusSignalContainer.

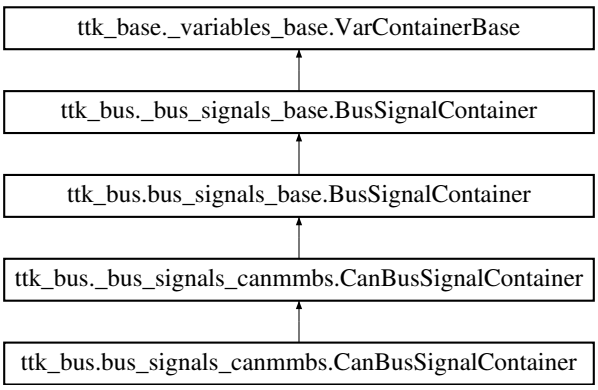
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_canida.py (v1.0.2)

7.90 ttk_bus._bus_signals_canmmbs.CanBusSignalContainer Class Reference

A BusSignalContainer for CAN Bus Signals.

Inheritance diagram for ttk_bus._bus_signals_canmmbs.CanBusSignalContainer:



Public Member Functions

- iterSignals (self, msg)

  Iterate over all signals that reference the specified CAN message/PDU.

7.90.1 Detailed Description

A BusSignalContainer for CAN Bus Signals.

7.90.2 Member Function Documentation

7.90.2.1 iterSignals()

```
ttk_bus._bus_signals_canmmbs.CanBusSignalContainer.iterSignals (
            self,
            msg )
```

Iterate over all signals that reference the specified CAN message/PDU.

Parameters

msg	a CAN message/pdu instance

Usage:

```
for sig in bus.iterSignals(bus.tx_msg_FNORD):
    print sig
```

Returns

an iterator object.

Reimplemented in ttk_bus.bus_signals_canmmbs.CanBusSignalContainer.

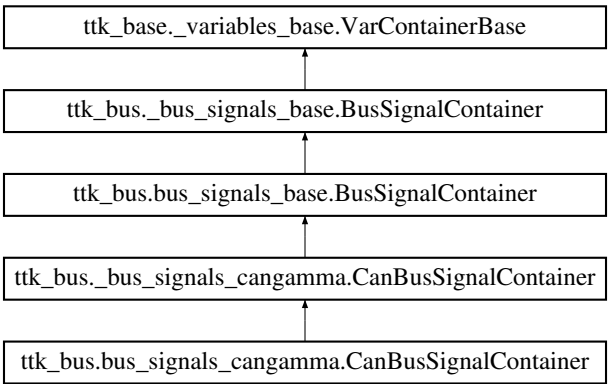
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_canmmbs.py (v2.3)

7.91 ttk_bus.bus_signals_cangamma.CanBusSignalContainer Class Reference

A BusSignalContainer for CAN Bus Signals.

Inheritance diagram for ttk_bus.bus_signals_cangamma.CanBusSignalContainer:



Public Member Functions

- __init__ (self, context)

  Override to add signals.
- resetAll (self, verbosity=1, recursion_depth=4)

  Call reset() on all contained BusSignals that support it.
- iterSignals (self, msg)

  Iterate over all signals that reference the specified CAN message/PDU.

7.91.1 Detailed Description

A BusSignalContainer for CAN Bus Signals.

7.91.2 Constructor & Destructor Documentation

7.91.2.1 __init__()

```
ttk_bus.bus_signals_cangamma.CanBusSignalContainer.__init__ (
               self,
               context )
```

Override to add signals.

Parameters

context	context used for contained BusSignals

Reimplemented from ttk_bus.bus_signals_base.BusSignalContainer.

7.91.3 Member Function Documentation

7.91.3.1 iterSignals()

```
ttk_bus.bus_signals_cangamma.CanBusSignalContainer.iterSignals (
               self,
               msg )
```

Iterate over all signals that reference the specified CAN message/PDU.

Parameters

msg	a CAN message/PDU instance

Usage:

```
for sig in bus.iterSignals(bus.tx_msg_FNORD):
    print sig
```

Returns

an iterator object.

Reimplemented from ttk_bus._bus_signals_cangamma.CanBusSignalContainer.

7.91.3.2 resetAll()

```
ttk_bus.bus_signals_cangamma.CanBusSignalContainer.resetAll (
               self,
               verbosity = 1,
               recursion_depth = 4 )
```

Call reset() on all contained BusSignals that support it.

Parameters

verbosity	verbosity of log output: 0: silent except for errors 1: only minimal status info (begin/end) 2: also show processed variable names
recursion_depth	how "deep" to recurse into lists/dictionary structures when looking for variables to reset.

Reimplemented from ttk_bus.bus_signals_base.BusSignalContainer.

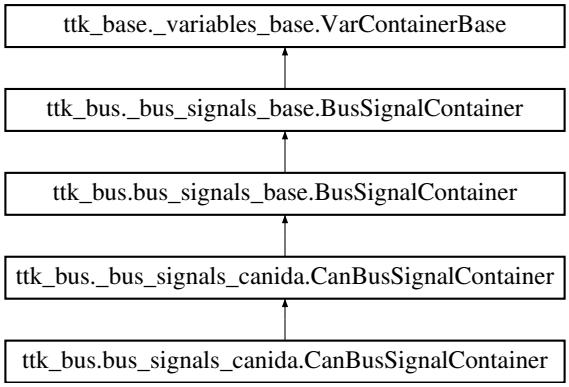
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_cangamma.py (v1.3)

7.92 ttk_bus.bus_signals_canida.CanBusSignalContainer Class Reference

A BusSignalContainer for CAN Bus Signals.

Inheritance diagram for ttk_bus.bus_signals_canida.CanBusSignalContainer:



Public Member Functions

- __init__ (self, context)

  Override to add signals.
- resetAll (self, verbosity=1, recursion_depth=4)

  Call reset() on all contained BusSignals that support it.
- iterSignals (self, msg)

  Iterate over all signals that reference the specified CAN message/PDU.

7.92.1 Detailed Description

A BusSignalContainer for CAN Bus Signals.

7.92.2 Constructor & Destructor Documentation

7.92.2.1 __init__()

```
ttk_bus.bus_signals_canida.CanBusSignalContainer.__init__ (
              self,
              context )
```

Override to add signals.

Parameters

context	context used for contained BusSignals

Reimplemented from ttk_bus.bus_signals_base.BusSignalContainer.

7.92.3 Member Function Documentation

7.92.3.1 iterSignals()

```
ttk_bus.bus_signals_canida.CanBusSignalContainer.iterSignals (
              self,
              msg )
```

Iterate over all signals that reference the specified CAN message/PDU.

Parameters

msg	a CAN message/PDU instance

Usage:

```
for sig in bus.iterSignals(bus.tx_msg_FNORD):
    print sig
```

Returns

an iterator object.

Reimplemented from ttk_bus._bus_signals_canida.CanBusSignalContainer.

7.92.3.2 resetAll()

```
ttk_bus.bus_signals_canida.CanBusSignalContainer.resetAll (
               self,
               verbosity = 1,
               recursion_depth = 4 )
```

Call reset() on all contained BusSignals that support it.

Parameters

verbosity	verbosity of log output: 0: silent except for errors 1: only minimal status info (begin/end) 2: also show processed variable names
recursion_depth	how "deep" to recurse into lists/dictionary structures when looking for variables to reset.

Reimplemented from ttk_bus.bus_signals_base.BusSignalContainer.

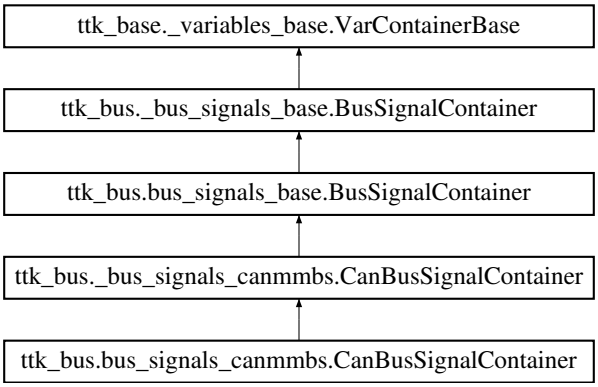
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_canida.py (v1.0.0)

7.93 ttk_bus.bus_signals_canmmbs.CanBusSignalContainer Class Reference

A BusSignalContainer for CAN Bus Signals.

Inheritance diagram for ttk_bus.bus_signals_canmmbs.CanBusSignalContainer:



Public Member Functions

- __init__ (self, context)

  Override to add signals.
- resetAll (self, verbosity=1, recursion_depth=4)

  Call reset() on all contained BusSignals that support it.
- iterSignals (self, msg)

  Iterate over all signals that reference the specified CAN message/PDU.

7.93.1 Detailed Description

A BusSignalContainer for CAN Bus Signals.

7.93.2 Constructor & Destructor Documentation

7.93.2.1 __init__()

```
ttk_bus.bus_signals_canmmbs.CanBusSignalContainer.__init__ (
              self,
              context )
```

Override to add signals.

Parameters

context	context used for contained BusSignals

Reimplemented from ttk_bus.bus_signals_base.BusSignalContainer.

7.93.3 Member Function Documentation

7.93.3.1 iterSignals()

```
ttk_bus.bus_signals_canmmbs.CanBusSignalContainer.iterSignals (
              self,
              msg )
```

Iterate over all signals that reference the specified CAN message/PDU.

Parameters

msg	a CAN message/pdu instance

Usage:

```
for sig in bus.iterSignals(bus.tx_msg_FNORD):
    print sig
```

Returns

an iterator object.

Reimplemented from ttk_bus._bus_signals_canmmbs.CanBusSignalContainer.

7.93.3.2 resetAll()

```
ttk_bus.bus_signals_canmmbs.CanBusSignalContainer.resetAll (
               self,
               verbosity = 1,
               recursion_depth = 4 )
```

Call reset() on all contained BusSignals that support it.

Parameters

verbosity	verbosity of log output: 0: silent except for errors 1: only minimal status info (begin/end) 2: also show processed variable names
recursion_depth	how "deep" to recurse into lists/dictionary structures when looking for variables to reset.

Reimplemented from ttk_bus.bus_signals_base.BusSignalContainer.

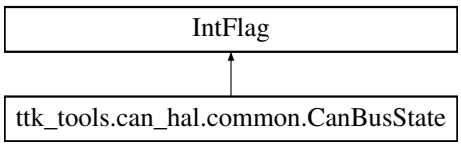
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_canmmbs.py (v1.8)

7.94 ttk_tools.can_hal.common.CanBusState Class Reference

CAN bus states (CanHal.h, tCanHalCanBusState).

Inheritance diagram for ttk_tools.can_hal.common.CanBusState:



Static Public Attributes

- int NoError = 0b000000000000
- int ErrorPassive = 0b000000000001
- int ErrorBusOff = 0b000000000010
- int WarningLimitSet = 0b000000000100
- int Overrun = 0b000000001000
- int StuffError = 0b000000010000
- int FormatError = 0b000000100000
- int AcknowledgeError = 0b000001000000
- int CrcError = 0b000010000000
- int ErrorFrame = 0b000100000000
- int RxBufferFull = 0b001000000000
- int TxBufferFull = 0b010000000000
- int HardwareError = 0b100000000000

7.94.1 Detailed Description

CAN bus states (CanHal.h, tCanHalCanBusState).

7.94.2 Member Data Documentation

7.94.2.1 AcknowledgeError

```
int ttk_tools.can_hal.common.CanBusState.AcknowledgeError = 0b000001000000  [static]
```

7.94.2.2 CrcError

```
int ttk_tools.can_hal.common.CanBusState.CrcError = 0b000010000000  [static]
```

7.94.2.3 ErrorBusOff

```
int ttk_tools.can_hal.common.CanBusState.ErrorBusOff = 0b000000000010  [static]
```

7.94.2.4 ErrorFrame

```
int ttk_tools.can_hal.common.CanBusState.ErrorFrame = 0b000100000000  [static]
```

7.94.2.5 ErrorPassive

```
int ttk_tools.can_hal.common.CanBusState.ErrorPassive = 0b000000000001  [static]
```

7.94.2.6 FormatError

```
int ttk_tools.can_hal.common.CanBusState.FormatError = 0b000000100000  [static]
```

7.94.2.7 HardwareError

```
int ttk_tools.can_hal.common.CanBusState.HardwareError = 0b100000000000  [static]
```

7.94.2.8 NoError

```
int ttk_tools.can_hal.common.CanBusState.NoError = 0b000000000000  [static]
```

7.94.2.9 Overrun

```
int ttk_tools.can_hal.common.CanBusState.Overrun = 0b000000001000  [static]
```

7.94.2.10 RxBufferFull

```
int ttk_tools.can_hal.common.CanBusState.RxBufferFull = 0b001000000000  [static]
```

7.94.2.11 StuffError

```
int ttk_tools.can_hal.common.CanBusState.StuffError = 0b000000010000  [static]
```

7.94.2.12 TxBufferFull

```
int ttk_tools.can_hal.common.CanBusState.TxBufferFull = 0b010000000000  [static]
```

7.94.2.13 WarningLimitSet

```
int ttk_tools.can_hal.common.CanBusState.WarningLimitSet = 0b000000000100  [static]
```

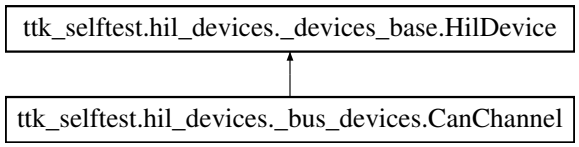
The documentation for this class was generated from the following file:

- ttk_tools/can_hal/common.py (v1.2)

7.95 ttk_selftest.hil_devices._bus_devices.CanChannel Class Reference

Base CAN channel container for a testable CAN channel and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices._bus_devices.CanChannel:



Public Member Functions

- __init__ (self, context, base_path, test_msg_name, test_msg_id, test_msg_length, fiu_channel, bitrate, termination, preconditions=None, wait_time_ms=None, alias="", descr="")

  CAN channel initialization.
- info (self)

  Get an info string for this CAN channel: alias, FIU channel, bitrate, and termination plus description, if available.

Public Attributes

- base_path
- bitrate
- termination
- test_msg
- test_msg_id
- test_msg_length
- alias

7.95.1 Detailed Description

Base CAN channel container for a testable CAN channel and relevant associated variables/values.

7.95.2 Constructor & Destructor Documentation

7.95.2.1 __init__()

```
ttk_selftest.hil_devices._bus_devices.CanChannel.__init__ (
                self,
                context,
                base_path,
                test_msg_name,
                test_msg_id,
                test_msg_length,
                fiu_channel,
                bitrate,
                termination,
                preconditions = None,
                wait_time_ms = None,
                alias = "",
                descr = "" )
```

CAN channel initialization.

Parameters

context	parent context
base_path	base path to the CAN channel in the model
test_msg_name	name of the CAN message that will be used for the test of the channel
test_msg_id	ID of the CAN message to be used for the test of the channel
test_msg_length	length in byte of the CAN message to be used for the test of the channel
fiu_channel	fiu channel of the CAN channel
bitrate	bitrate of the CAN channel in kBit/s
termination	flag to specify the hardware termination of the channel TRUE: Channel is already fully terminated FALSE: Channel is not fully terminated
preconditions	(opt) preconditions that have to be set before the device can be checked. defaults to None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The functions specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
        # For special cases, the preconditions may also contain function calls
        # (which might be useful to insert additional delays or similar):
        preconditions = [
            { "device": Voltage, "value": 1},
            { "func":   time.sleep,
              "args":   [1.0], # positional arguments
              "kwargs": {},    # keyword arguments
            },
            {"device": ClampSwitch, "value": 1},
            # or a simple call like this:
            { "func":  someCustomFunction,
            },
            ...
        ]
    wait_time_ms      - (opt) waiting time in milliseconds between
    channel stimulation and channel check
    defaults to 100ms
    alias             - (opt) channel alias/name, defaults to the
    base path of the CAN channel in the model
    descr             - (opt) channel description (e.g. used in report entries)
```

Reimplemented from ttk_selftest.hil_devices._devices_base.HilDevice.

7.95.3 Member Function Documentation

7.95.3.1 info()

```
ttk_selftest.hil_devices._bus_devices.CanChannel.info (
            self )
```

Get an info string for this CAN channel: alias, FIU channel, bitrate, and termination plus description, if available.

Returns

a formatted string.

7.95.4 Member Data Documentation

7.95.4.1 alias

```
ttk_selftest.hil_devices._bus_devices.CanChannel.alias
```

7.95.4.2 base_path

```
ttk_selftest.hil_devices._bus_devices.CanChannel.base_path
```

7.95.4.3 bitrate

```
ttk_selftest.hil_devices._bus_devices.CanChannel.bitrate
```

7.95.4.4 termination

ttk_selftest.hil_devices._bus_devices.CanChannel.termination

7.95.4.5 test_msg

ttk_selftest.hil_devices._bus_devices.CanChannel.test_msg

7.95.4.6 test_msg_id

ttk_selftest.hil_devices._bus_devices.CanChannel.test_msg_id

7.95.4.7 test_msg_length

ttk_selftest.hil_devices._bus_devices.CanChannel.test_msg_length

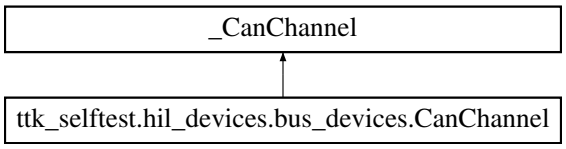
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/_bus_devices.py (v2.0.0)

7.96 ttk_selftest.hil_devices.bus_devices.CanChannel Class Reference

Base CAN channel container for a testable CAN channel and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices.bus_devices.CanChannel:



Public Member Functions

- __init__ (self, context, base_path, test_msg_name, test_msg_id, test_msg_length, fiu_channel, bitrate, termination, preconditions=None, wait_time_ms=None, alias="", descr="")

  CAN channel initialization.
- info (self)

  Get an info string for this CAN channel: alias, FIU channel, bitrate, and termination plus description, if available.

7.96.1 Detailed Description

Base CAN channel container for a testable CAN channel and relevant associated variables/values.

7.96.2 Constructor & Destructor Documentation

7.96.2.1 __init__()

```
ttk_selftest.hil_devices.bus_devices.CanChannel.__init__ (
                self,
                context,
                base_path,
                test_msg_name,
                test_msg_id,
                test_msg_length,
                fiu_channel,
                bitrate,
                termination,
                preconditions = None,
                wait_time_ms = None,
                alias = "",
                descr = "" )
```

CAN channel initialization.

Parameters

context	parent context
base_path	base path to the CAN channel in the model
test_msg_name	name of the CAN message that will be used for the test of the channel
test_msg_id	ID of the CAN message to be used for the test of the channel
test_msg_length	length in byte of the CAN message to be used for the test of the channel
fiu_channel	fiu channel of the CAN channel
bitrate	bitrate of the CAN channel in kBit/s
termination	flag to specify the hardware termination of the channel TRUE: Channel is already fully terminated FALSE: Channel is not fully terminated
preconditions	(opt) preconditions that have to be set before the device can be checked. defaults to None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The functions specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
# For special cases, the preconditions may also contain function calls
# (which might be useful to insert additional delays or similar):
preconditions = [
    { "device": Voltage, "value": 1},
    { "func":   time.sleep,
      "args":   [1.0], # positional arguments
```

```
            "kwargs": {},    # keyword arguments
        },
        {"device": ClampSwitch, "value": 1},
        # or a simple call like this:
        { "func":  someCustomFunction,
        },
        ...
    ]
wait_time_ms      - (opt) waiting time in milliseconds between
channel stimulation and channel check
defaults to 100ms
alias             - (opt) channel alias/name, defaults to the
base path of the CAN channel in the model
descr             - (opt) channel description (e.g. used in report entries)
```

7.96.3 Member Function Documentation

7.96.3.1 info()

```
ttk_selftest.hil_devices.bus_devices.CanChannel.info (
            self )
```

Get an info string for this CAN channel: alias, FIU channel, bitrate, and termination plus description, if available.

Returns

a formatted string.

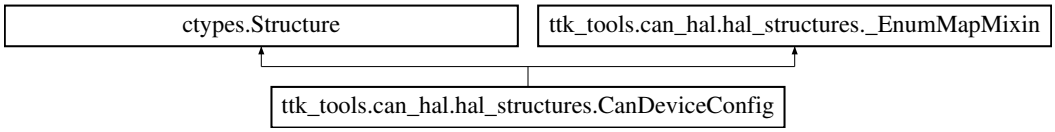
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/bus_devices.py (v2.0.0)

7.97 ttk_tools.can_hal.hal_structures.CanDeviceConfig Class Reference

CAN device configuration (CanHal.h, tCanHalDeviceConfig).

Inheritance diagram for ttk_tools.can_hal.hal_structures.CanDeviceConfig:



Public Member Functions

- setDevice (self, device_list_entry)

  Sets members DeviceType/DeviceNumber/Channel from a supplied DeviceListEntry.

Public Attributes

- DeviceType
- DeviceNumber
- Channel

Static Protected Attributes

- list _fields_
- dict _map

7.97.1 Detailed Description

CAN device configuration (CanHal.h, tCanHalDeviceConfig).

7.97.2 Member Function Documentation

7.97.2.1 setDevice()

```
ttk_tools.can_hal.hal_structures.CanDeviceConfig.setDevice (
            self,
            device_list_entry )
```

Sets members DeviceType/DeviceNumber/Channel from a supplied DeviceListEntry.

Parameters

device_list_entry	a DeviceListEntry instance

7.97.3 Member Data Documentation

7.97.3.1 _fields_

```
list ttk_tools.can_hal.hal_structures.CanDeviceConfig._fields_  [static], [protected]
```

Initial value:

```
=  [
        ("DeviceType",          ctypes.c_int),        # tCanHalDeviceType   m_DeviceType
        ("DeviceNumber",        ctypes.c_uint),       # unsigned int        m_uiDeviceNumber
        ("Channel",             ctypes.c_uint),       # unsigned int        m_uiChannel
        ("SerialNumber",        ctypes.c_long),       # unsigned long       m_ulSerialNumber
        ("Bitrate",             ctypes.c_int),        # tCanHalBautrate     m_uiBaudNumber, see CanBitrate

        ("ReceiveBufferSize",   ctypes.c_uint),       # unsigned int        m_uiReceiveBufferSize

        ("TransmitBufferSize",  ctypes.c_uint),       # unsigned int        m_uiTransmittBufferSize

        ("ConfigurationFlags",  ctypes.c_uint),       # tCanHalCanConfig    m_ConfigurationFlags

        ("Timeout",             ctypes.c_longlong),   # unsigned long int   m_uliTimeout

        ("NumberOfRepeats",     ctypes.c_uint)        # unsigned int        m_uiNumberOfRepeats
    ]
```

7.97.3.2 _map

```
dict ttk_tools.can_hal.hal_structures.CanDeviceConfig._map  [static], [protected]
```

Initial value:

```
=  {
        "DeviceType":         DeviceType,
        "Bitrate":            CanBitrate,
        "ConfigurationFlags": ConfigFlag
    }
```

7.97.3.3 Channel

```
ttk_tools.can_hal.hal_structures.CanDeviceConfig.Channel
```

7.97.3.4 DeviceNumber

```
ttk_tools.can_hal.hal_structures.CanDeviceConfig.DeviceNumber
```

7.97.3.5 DeviceType

```
ttk_tools.can_hal.hal_structures.CanDeviceConfig.DeviceType
```

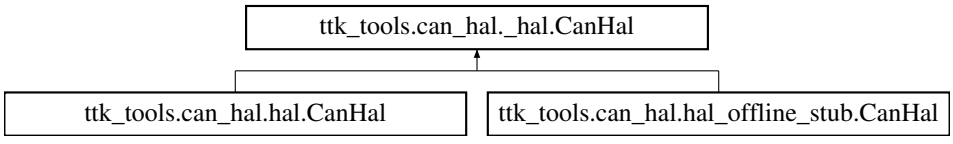
The documentation for this class was generated from the following file:

- ttk_tools/can_hal/hal_structures.py (v1.0.0)

7.98 ttk_tools.can_hal._hal.CanHal Class Reference

Low(-ish)-Level interface to CAN_HAL_(32/64).dll.

Inheritance diagram for ttk_tools.can_hal._hal.CanHal:



Public Member Functions

- __init__ (self)
- init (self)

  CAN HAL init.
- deinit (self)

  CAN HAL deinit (also deinitializes any current devices).
- initDevice (self, device_index, bit_rate_setting=CanBitrate.Bitrate1MBit, rx_buffer_size=10, tx_buffer_size=10, config_flags=None)

  Initialize a connected CAN device.
- deinitDevice (self, device_hdl)

  De-initialize a CAN device.
- getDeviceList (self)

  Get a list of available CAN devices.
- getDeviceBusState (self, device_hdl)

  Get state of CAN bus the given device.
- getVersion (self)

  Get version of HAL.
- sendMessage (self, can_message, device_hdl)

  Send a CAN message to the given device handle.
- receiveMessage (self, device_hdl)

  Receive a CAN message from the given device handle.

Public Attributes

- lib

Static Public Attributes

- int max_number_of_devices = 32

  buffer size for getDeviceList; maximum number of device list entries that will be read.

Protected Attributes

- _initialized
- _initialized_devices

7.98.1 Detailed Description

Low(-ish)-Level interface to CAN_HAL_(32/64).dll.

7.98.2 Constructor & Destructor Documentation

7.98.2.1 __init__()

```
ttk_tools.can_hal._hal.CanHal.__init__ (
            self )
```

7.98.3 Member Function Documentation

7.98.3.1 deinit()

```
ttk_tools.can_hal._hal.CanHal.deinit (
              self )
```

CAN HAL deinit (also deinitializes any current devices).

Reimplemented in ttk_tools.can_hal.hal.CanHal, and ttk_tools.can_hal.hal_offline_stub.CanHal.

7.98.3.2 deinitDevice()

```
ttk_tools.can_hal._hal.CanHal.deinitDevice (
              self,
              device_hdl )
```

De-initialize a CAN device.

Parameters

device_hdl	can device handle to deinit

Reimplemented in ttk_tools.can_hal.hal.CanHal, and ttk_tools.can_hal.hal_offline_stub.CanHal.

7.98.3.3 getDeviceBusState()

```
ttk_tools.can_hal._hal.CanHal.getDeviceBusState (
              self,
              device_hdl )
```

Get state of CAN bus the given device.

Parameters

device_hdl	handle of an initialized CAN device, see initDevice.

Returns

a CanBusState value.

Reimplemented in ttk_tools.can_hal.hal.CanHal, and ttk_tools.can_hal.hal_offline_stub.CanHal.

7.98.3.4 getDeviceList()

```
ttk_tools.can_hal._hal.CanHal.getDeviceList (
              self )
```

Get a list of available CAN devices.

Returns

a list of DeviceListEntry items.

Reimplemented in ttk_tools.can_hal.hal.CanHal, and ttk_tools.can_hal.hal_offline_stub.CanHal.

7.98.3.5 getVersion()

```
ttk_tools.can_hal._hal.CanHal.getVersion (
              self )
```

Get version of HAL.

Returns

a tuple: (major, minor, patch) version

Reimplemented in ttk_tools.can_hal.hal.CanHal, and ttk_tools.can_hal.hal_offline_stub.CanHal.

7.98.3.6 init()

```
ttk_tools.can_hal._hal.CanHal.init (
              self )
```

CAN HAL init.

Note

gets implicitly called at end of `__init__`

Reimplemented in ttk_tools.can_hal.hal.CanHal, and ttk_tools.can_hal.hal_offline_stub.CanHal.

7.98.3.7 initDevice()

```
ttk_tools.can_hal._hal.CanHal.initDevice (
              self,
              device_index,
              bit_rate_setting = CanBitrate.Bitrate1MBit,
              rx_buffer_size = 10,
              tx_buffer_size = 10,
              config_flags = None )
```

Initialize a connected CAN device.

Parameters

device_index	index of connected device, see getDeviceList
bit_rate_setting	connection speed, see CanBitrate
rx_buffer_size	receive buffer size
tx_buffer_size	transmit buffer size
config_flags	optional ConfigFlag instance with set flags. None: Use defaults

Note

config_flags So far, it does not seem that those flags are actually used in CAN_HAL

Returns

a CAN HAL device handle.

Reimplemented in ttk_tools.can_hal.hal.CanHal, and ttk_tools.can_hal.hal_offline_stub.CanHal.

7.98.3.8 receiveMessage()

```
ttk_tools.can_hal._hal.CanHal.receiveMessage (
              self,
              device_hdl )
```

Receive a CAN message from the given device handle.

Parameters

device_hdl	handle of an initialized CAN device, see initDevice.

Returns

a CanMessage instance or None if no message was waiting in receive buffer Raises a CanHalReadError for, well, read errors or a generic CanHalApiError for anything else.

Reimplemented in ttk_tools.can_hal.hal.CanHal, and ttk_tools.can_hal.hal_offline_stub.CanHal.

7.98.3.9 sendMessage()

```
ttk_tools.can_hal._hal.CanHal.sendMessage (
              self,
              can_message,
              device_hdl )
```

Send a CAN message to the given device handle.

Parameters

can_message	a filled CanMessage instance
device_hdl	handle of an initialized CAN device, see initDevice.

Raises a CanHalWriteError for, well, read errors or a generic CanHalApiError for anything else.

Reimplemented in ttk_tools.can_hal.hal.CanHal, and ttk_tools.can_hal.hal_offline_stub.CanHal.

7.98.4 Member Data Documentation

7.98.4.1 _initialized

```
ttk_tools.can_hal._hal.CanHal._initialized  [protected]
```

7.98.4.2 _initialized_devices

```
ttk_tools.can_hal._hal.CanHal._initialized_devices  [protected]
```

7.98.4.3 lib

```
ttk_tools.can_hal._hal.CanHal.lib
```

7.98.4.4 max_number_of_devices

```
int ttk_tools.can_hal._hal.CanHal.max_number_of_devices = 32  [static]
```

buffer size for getDeviceList; maximum number of device list entries that will be read.

The documentation for this class was generated from the following file:

- ttk_tools/can_hal/_hal.py (v1.3.1)

7.99 ttk_tools.can_hal._hal_offline_stub.CanHal Class Reference

Low(-ish)-Level interface to CAN_HAL_(32/64).dll.

Classes

- class _DeviceStub

Public Member Functions

- __init__ (self)
- init (self)

  CAN HAL init.
- deinit (self)

  CAN HAL deinit (also deinitializes any current devices).
- initDevice (self, device_index, bit_rate_setting=CanBitrate.Bitrate1MBit, rx_buffer_size=10, tx_buffer_size=10, config_flags=None)

  Initialize a connected CAN device.
- deinitDevice (self, device_hdl)

  De-initialize a CAN device.
- getDeviceList (self)

  Get a list of available CAN devices.
- getDeviceBusState (self, device_hdl)

  Get state of CAN bus the given device.
- getVersion (self)

  Get version of HAL.
- sendMessage (self, can_message, device_hdl)

  Send a CAN message to the given device handle.
- receiveMessage (self, device_hdl)

  Receive a CAN message from the given device handle.

Public Attributes

- lib

Protected Attributes

- _devices
- _initialized
- _initialized_devices

7.99.1 Detailed Description

Low(-ish)-Level interface to CAN_HAL_(32/64).dll.

7.99.2 Constructor & Destructor Documentation

7.99.2.1 __init__()

```
ttk_tools.can_hal._hal_offline_stub.CanHal.__init__ (
              self )
```

7.99.3 Member Function Documentation

7.99.3.1 deinit()

```
ttk_tools.can_hal._hal_offline_stub.CanHal.deinit (
              self )
```

CAN HAL deinit (also deinitializes any current devices).

7.99.3.2 deinitDevice()

```
ttk_tools.can_hal._hal_offline_stub.CanHal.deinitDevice (
              self,
              device_hdl )
```

De-initialize a CAN device.

Parameters

device_hdl	can device handle to deinit

7.99.3.3 getDeviceBusState()

```
ttk_tools.can_hal._hal_offline_stub.CanHal.getDeviceBusState (
```

```
                self,
                device_hdl )
```

Get state of CAN bus the given device.

Parameters

device_hdl	handle of an initialized CAN device, see initDevice.

Returns

a CanBusState value.

7.99.3.4 getDeviceList()

```
ttk_tools.can_hal._hal_offline_stub.CanHal.getDeviceList (
                self )
```

Get a list of available CAN devices.

Returns

a list of DeviceListEntry items.

7.99.3.5 getVersion()

```
ttk_tools.can_hal._hal_offline_stub.CanHal.getVersion (
                self )
```

Get version of HAL.

Returns

a tuple: (major, minor, patch) version

7.99.3.6 init()

```
ttk_tools.can_hal._hal_offline_stub.CanHal.init (
                self )
```

CAN HAL init.

Note

gets implicitly called at end of `__init__`

7.99.3.7 initDevice()

```
ttk_tools.can_hal._hal_offline_stub.CanHal.initDevice (
                self,
                device_index,
                bit_rate_setting = CanBitrate.Bitrate1MBit,
                rx_buffer_size = 10,
                tx_buffer_size = 10,
                config_flags = None )
```

Initialize a connected CAN device.

Parameters

device_index	index of connected device, see getDeviceList
bit_rate_setting	connection speed, see CanBitrate
rx_buffer_size	receive buffer size
tx_buffer_size	transmit buffer size
config_flags	optional ConfigFlag instance with set flags. None: Use defaults

Note

config_flags So far, it does not seem that those flags are actually used in CAN_HAL

Returns

a CAN HAL device handle.

7.99.3.8 receiveMessage()

```
ttk_tools.can_hal._hal_offline_stub.CanHal.receiveMessage (
              self,
              device_hdl )
```

Receive a CAN message from the given device handle.

Parameters

device_hdl	handle of an initialized CAN device, see initDevice.

Returns

a CanMessage instance or None if no message was waiting in receive buffer Raises a CanHalReadError for, well, read errors or a generic CanHalApiError for anything else.

7.99.3.9 sendMessage()

```
ttk_tools.can_hal._hal_offline_stub.CanHal.sendMessage (
              self,
              can_message,
              device_hdl )
```

Send a CAN message to the given device handle.

Parameters

can_message	a filled CanMessage instance
device_hdl	handle of an initialized CAN device, see initDevice.

Raises a CanHalWriteError for, well, read errors or a generic CanHalApiError for anything else.

7.99.4 Member Data Documentation

7.99.4.1 _devices

`ttk_tools.can_hal._hal_offline_stub.CanHal._devices [protected]`

7.99.4.2 _initialized

`ttk_tools.can_hal._hal_offline_stub.CanHal._initialized [protected]`

7.99.4.3 _initialized_devices

`ttk_tools.can_hal._hal_offline_stub.CanHal._initialized_devices [protected]`

7.99.4.4 lib

`ttk_tools.can_hal._hal_offline_stub.CanHal.lib`

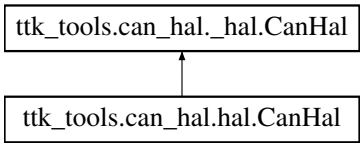
The documentation for this class was generated from the following file:

- ttk_tools/can_hal/_hal_offline_stub.py (v1.1.1)

7.100 ttk_tools.can_hal.hal.CanHal Class Reference

Low(-ish)-Level interface to CAN_HAL_(32/64).dll.

Inheritance diagram for ttk_tools.can_hal.hal.CanHal:



Public Member Functions

- init (self)

  CAN HAL init.
- deinit (self)

  CAN HAL deinit (also deinitializes any current devices).
- initDevice (self, device_index, bit_rate_setting=CanBitrate.Bitrate1MBit, rx_buffer_size=10, tx_buffer_size=10, config_flags=None)

  Initialize a connected CAN device.
- deinitDevice (self, device_hdl)

  De-initialize a CAN device.
- getDeviceList (self)

  Get a list of available CAN devices.
- getDeviceBusState (self, device_hdl)

  Get state of CAN bus the given device.
- getVersion (self)

  Get version of HAL.
- sendMessage (self, can_message, device_hdl)

  Send a CAN message to the given device handle.
- receiveMessage (self, device_hdl)

  Receive a CAN message from the given device handle.

7.100.1 Detailed Description

Low(-ish)-Level interface to CAN_HAL_(32/64).dll.

7.100.2 Member Function Documentation

7.100.2.1 deinit()

```
ttk_tools.can_hal.hal.CanHal.deinit (
              self )
```

CAN HAL deinit (also deinitializes any current devices).

Reimplemented from ttk_tools.can_hal._hal.CanHal.

7.100.2.2 deinitDevice()

```
ttk_tools.can_hal.hal.CanHal.deinitDevice (
              self,
              device_hdl )
```

De-initialize a CAN device.

Parameters

device_hdl	can device handle to deinit

Reimplemented from ttk_tools.can_hal._hal.CanHal.

7.100.2.3 getDeviceBusState()

```
ttk_tools.can_hal.hal.CanHal.getDeviceBusState (
              self,
              device_hdl )
```

Get state of CAN bus the given device.

Parameters

device_hdl	handle of an initialized CAN device, see initDevice.

Returns

a CanBusState value.

Reimplemented from ttk_tools.can_hal._hal.CanHal.

7.100.2.4 getDeviceList()

```
ttk_tools.can_hal.hal.CanHal.getDeviceList (
              self )
```

Get a list of available CAN devices.

Returns

a list of DeviceListEntry items.

Reimplemented from ttk_tools.can_hal._hal.CanHal.

7.100.2.5 getVersion()

```
ttk_tools.can_hal.hal.CanHal.getVersion (
              self )
```

Get version of HAL.

Returns

a tuple: (major, minor, patch) version

Reimplemented from ttk_tools.can_hal._hal.CanHal.

7.100.2.6 init()

```
ttk_tools.can_hal.hal.CanHal.init (
              self )
```

CAN HAL init.

Note

gets implicitly called at end of `__init__`

Reimplemented from ttk_tools.can_hal._hal.CanHal.

7.100.2.7 initDevice()

```
ttk_tools.can_hal.hal.CanHal.initDevice (
              self,
              device_index,
              bit_rate_setting = CanBitrate.Bitrate1MBit,
              rx_buffer_size = 10,
              tx_buffer_size = 10,
              config_flags = None )
```

Initialize a connected CAN device.

Parameters

device_index	index of connected device, see getDeviceList
bit_rate_setting	connection speed, see CanBitrate
rx_buffer_size	receive buffer size
tx_buffer_size	transmit buffer size
config_flags	optional ConfigFlag instance with set flags. None: Use defaults

Note

config_flags So far, it does not seem that those flags are actually used in CAN_HAL

Returns

a CAN HAL device handle.

Reimplemented from ttk_tools.can_hal._hal.CanHal.

7.100.2.8 receiveMessage()

```
ttk_tools.can_hal.hal.CanHal.receiveMessage (
              self,
              device_hdl )
```

Receive a CAN message from the given device handle.

Parameters

device_hdl	handle of an initialized CAN device, see initDevice.

Returns

a CanMessage instance or None if no message was waiting in receive buffer Raises a CanHalReadError for, well, read errors or a generic CanHalApiError for anything else.

Reimplemented from ttk_tools.can_hal._hal.CanHal.

7.100.2.9 sendMessage()

```
ttk_tools.can_hal.hal.CanHal.sendMessage (
              self,
              can_message,
              device_hdl )
```

Send a CAN message to the given device handle.

Parameters

can_message	a filled CanMessage instance
device_hdl	handle of an initialized CAN device, see initDevice.

Raises a CanHalWriteError for, well, read errors or a generic CanHalApiError for anything else.

Reimplemented from ttk_tools.can_hal._hal.CanHal.

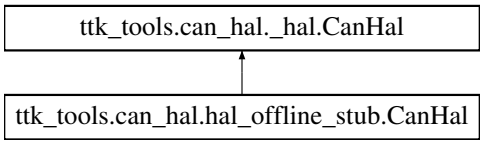
The documentation for this class was generated from the following file:

- ttk_tools/can_hal/hal.py (v1.0.2)

7.101 ttk_tools.can_hal.hal_offline_stub.CanHal Class Reference

Low(-ish)-Level interface to CAN_HAL_(32/64).dll.

Inheritance diagram for ttk_tools.can_hal.hal_offline_stub.CanHal:



Public Member Functions

- init (self)

  CAN HAL init.
- deinit (self)

  CAN HAL deinit (also deinitializes any current devices).
- initDevice (self, device_index, bit_rate_setting=CanBitrate.Bitrate1MBit, rx_buffer_size=10, tx_buffer_size=10, config_flags=None)

  Initialize a connected CAN device.
- deinitDevice (self, device_hdl)

  De-initialize a CAN device.
- getDeviceList (self)

  Get a list of available CAN devices.
- getDeviceBusState (self, device_hdl)

  Get state of CAN bus the given device.
- getVersion (self)

  Get version of HAL.
- sendMessage (self, can_message, device_hdl)

  Send a CAN message to the given device handle.
- receiveMessage (self, device_hdl)

  Receive a CAN message from the given device handle.

7.101.1 Detailed Description

Low(-ish)-Level interface to CAN_HAL_(32/64).dll.

7.101.2 Member Function Documentation

7.101.2.1 deinit()

```
ttk_tools.can_hal.hal_offline_stub.CanHal.deinit (
              self )
```

CAN HAL deinit (also deinitializes any current devices).

Reimplemented from ttk_tools.can_hal._hal.CanHal.

7.101.2.2 deinitDevice()

```
ttk_tools.can_hal.hal_offline_stub.CanHal.deinitDevice (
              self,
              device_hdl )
```

De-initialize a CAN device.

Parameters

device_hdl	can device handle to deinit

Reimplemented from ttk_tools.can_hal._hal.CanHal.

7.101.2.3 getDeviceBusState()

```
ttk_tools.can_hal.hal_offline_stub.CanHal.getDeviceBusState (
              self,
              device_hdl )
```

Get state of CAN bus the given device.

Parameters

device_hdl	handle of an initialized CAN device, see initDevice.

Returns

a CanBusState value.

Reimplemented from ttk_tools.can_hal._hal.CanHal.

7.101.2.4 getDeviceList()

```
ttk_tools.can_hal.hal_offline_stub.CanHal.getDeviceList (
              self )
```

Get a list of available CAN devices.

Returns

a list of DeviceListEntry items.

Reimplemented from ttk_tools.can_hal._hal.CanHal.

7.101.2.5 getVersion()

```
ttk_tools.can_hal.hal_offline_stub.CanHal.getVersion (
              self )
```

Get version of HAL.

Returns

a tuple: (major, minor, patch) version

Reimplemented from ttk_tools.can_hal._hal.CanHal.

7.101.2.6 init()

```
ttk_tools.can_hal.hal_offline_stub.CanHal.init (
              self )
```

CAN HAL init.

Note

gets implicitly called at end of `__init__`

Reimplemented from ttk_tools.can_hal._hal.CanHal.

7.101.2.7 initDevice()

```
ttk_tools.can_hal.hal_offline_stub.CanHal.initDevice (
              self,
              device_index,
              bit_rate_setting = CanBitrate.Bitrate1MBit,
              rx_buffer_size = 10,
              tx_buffer_size = 10,
              config_flags = None )
```

Initialize a connected CAN device.

Parameters

device_index	index of connected device, see getDeviceList
bit_rate_setting	connection speed, see CanBitrate
rx_buffer_size	receive buffer size
tx_buffer_size	transmit buffer size
config_flags	optional ConfigFlag instance with set flags. None: Use defaults

Note

config_flags So far, it does not seem that those flags are actually used in CAN_HAL

Returns

a CAN HAL device handle.

Reimplemented from ttk_tools.can_hal._hal.CanHal.

7.101.2.8 receiveMessage()

```
ttk_tools.can_hal.hal_offline_stub.CanHal.receiveMessage (
              self,
              device_hdl )
```

Receive a CAN message from the given device handle.

Parameters

device_hdl	handle of an initialized CAN device, see initDevice.

Returns

a CanMessage instance or None if no message was waiting in receive buffer Raises a CanHalReadError for, well, read errors or a generic CanHalApiError for anything else.

Reimplemented from ttk_tools.can_hal._hal.CanHal.

7.101.2.9 sendMessage()

```
ttk_tools.can_hal.hal_offline_stub.CanHal.sendMessage (
              self,
              can_message,
              device_hdl )
```

Send a CAN message to the given device handle.

Parameters

can_message	a filled CanMessage instance
device_hdl	handle of an initialized CAN device, see initDevice.

Raises a CanHalWriteError for, well, read errors or a generic CanHalApiError for anything else.

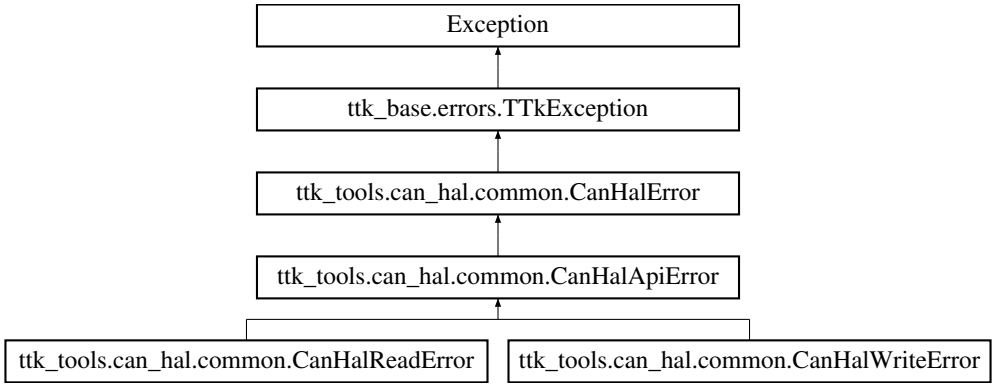
Reimplemented from ttk_tools.can_hal._hal.CanHal.

The documentation for this class was generated from the following file:

- ttk_tools/can_hal/hal_offline_stub.py (v1.0.2)

7.102 ttk_tools.can_hal.common.CanHalApiError Class Reference

Inheritance diagram for ttk_tools.can_hal.common.CanHalApiError:



Public Member Functions

- __init__ (self, message, error_code, func_name=None)

Public Attributes

- error_code
- func_name

7.102.1 Constructor & Destructor Documentation

7.102.1.1 __init__()

```
ttk_tools.can_hal.common.CanHalApiError.__init__ (
              self,
              message,
              error_code,
              func_name = None )
```

Parameters

message	error message
error_code	error code returned by API function, see HalErrors
func_name	name of API function that caused this error (if available)

7.102.2 Member Data Documentation

7.102.2.1 error_code

```
ttk_tools.can_hal.common.CanHalApiError.error_code
```

7.102.2.2 func_name

```
ttk_tools.can_hal.common.CanHalApiError.func_name
```

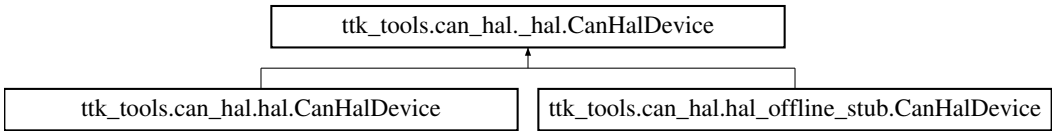
The documentation for this class was generated from the following file:

- ttk_tools/can_hal/common.py (v1.2)

7.103 ttk_tools.can_hal._hal.CanHalDevice Class Reference

CAN device connected to a CanHal instance.

Inheritance diagram for ttk_tools.can_hal._hal.CanHalDevice:



Public Member Functions

- __init__ (self, can_hal, device_index=0, bit_rate_setting=CanBitrate.Bitrate1MBit, rx_buffer_size=10, tx_↩ buffer_size=10, config_flags=None)
- deinit (self)

  Deinitialize this device on HAL.
- getBusState (self)

  Get state of CAN bus for this device.
- sendMessage (self, can_id, data, length=8, ext_id=None, feedback=False)

  Send a CAN message.
- sendCanMessage (self, can_message)

  Send a preconfigured CanMessage.
- receiveMessage (self, timeout_s=2, step_s=0.100, feedback=False)

  Receive a message (get a pending message from receive buffer).

Public Attributes

- hal
- device_hdl

7.103.1 Detailed Description

CAN device connected to a CanHal instance.

7.103.2 Constructor & Destructor Documentation

7.103.2.1 __init__()

```
ttk_tools.can_hal._hal.CanHalDevice.__init__ (
              self,
              can_hal,
              device_index = 0,
              bit_rate_setting = CanBitrate.Bitrate1MBit,
              rx_buffer_size = 10,
              tx_buffer_size = 10,
              config_flags = None )
```

Parameters

can_hal	parent CanHal instance
device_index	index of can device in CanHal, defaults to first device
bit_rate_setting	connection speed, see CanBitrate
rx_buffer_size	receive buffer size
tx_buffer_size	transmit buffer size
config_flags	optional ConfigFlag instance with set flags. None: Use defaults. See CanHal.initDevice.

Reimplemented in ttk_tools.can_hal.hal.CanHalDevice, and ttk_tools.can_hal.hal_offline_stub.CanHalDevice.

7.103.3 Member Function Documentation

7.103.3.1 deinit()

```
ttk_tools.can_hal._hal.CanHalDevice.deinit (
              self )
```

Deinitialize this device on HAL.

Reimplemented in ttk_tools.can_hal.hal.CanHalDevice, and ttk_tools.can_hal.hal_offline_stub.CanHalDevice.

7.103.3.2 getBusState()

```
ttk_tools.can_hal._hal.CanHalDevice.getBusState (
              self )
```

Get state of CAN bus for this device.

Returns

a CanBusState value.

Reimplemented in ttk_tools.can_hal.hal.CanHalDevice, and ttk_tools.can_hal.hal_offline_stub.CanHalDevice.

7.103.3.3 receiveMessage()

```
ttk_tools.can_hal._hal.CanHalDevice.receiveMessage (
              self,
              timeout_s = 2,
              step_s = 0.100,
              feedback = False )
```

Receive a message (get a pending message from receive buffer).

Parameters

timeout_s	maximum time to wait for a message to appear in [s]. 0: return immediately after checking once.
step_s	wait time between checks for new messages in [s]
feedback	True: Print progress/status info to stdout

Returns

a CanMessage instance, or None if buffer was empty

Reimplemented in ttk_tools.can_hal.hal.CanHalDevice, and ttk_tools.can_hal.hal_offline_stub.CanHalDevice.

7.103.3.4 sendCanMessage()

```
ttk_tools.can_hal._hal.CanHalDevice.sendCanMessage (
              self,
              can_message )
```

Send a preconfigured CanMessage.

Parameters

can_message	an already configured CanMessage struct instance.

Reimplemented in ttk_tools.can_hal.hal.CanHalDevice, and ttk_tools.can_hal.hal_offline_stub.CanHalDevice.

7.103.3.5 sendMessage()

```
ttk_tools.can_hal._hal.CanHalDevice.sendMessage (
              self,
              can_id,
              data,
              length = 8,
              ext_id = None,
              feedback = False )
```

Send a CAN message.

Parameters

can_id	ID of message, 11 bits (max 0x7FF) for standard, 29 bits (max 0x1FFFFFFF) for extended identifiers.
data	payload data, either • as a byte/bytearray string or • as a list/tuple of byte values
length	length (data length code) of message data in bytes [0..8]
ext_id	True: set "extended identifier" bit to `CanMessage.status` None: set bit automatically if `can_id` is larger than 11 bits. See CanMessage
feedback	True: Print message info before sending to stdout

Reimplemented in ttk_tools.can_hal.hal.CanHalDevice, and ttk_tools.can_hal.hal_offline_stub.CanHalDevice.

7.103.4 Member Data Documentation

7.103.4.1 device_hdl

```
ttk_tools.can_hal._hal.CanHalDevice.device_hdl
```

7.103.4.2 hal

`ttk_tools.can_hal._hal.CanHalDevice.hal`

The documentation for this class was generated from the following file:

- ttk_tools/can_hal/_hal.py (v1.3.1)

7.104 ttk_tools.can_hal._hal_offline_stub.CanHalDevice Class Reference

CAN device connected to a CanHal instance.

Public Member Functions

- __init__ (self, can_hal, device_index=0, bit_rate_setting=CanBitrate.Bitrate1MBit, rx_buffer_size=10, tx_↩ buffer_size=10, config_flags=None)
- deinit (self)

  Deinitialize this device on HAL.
- getBusState (self)

  Get state of CAN bus for this device.
- sendMessage (self, can_id, data, length=8, ext_id=None, feedback=False)

  Send a CAN message.
- sendCanMessage (self, can_message)

  Send a preconfigured CanMessage.
- receiveMessage (self, timeout_s=2, step_s=0.100, feedback=False)

  Receive a message (get a pending message from receive buffer).

Public Attributes

- hal
- device_hdl

7.104.1 Detailed Description

CAN device connected to a CanHal instance.

7.104.2 Constructor & Destructor Documentation

7.104.2.1 __init__()

```
ttk_tools.can_hal._hal_offline_stub.CanHalDevice.__init__ (
              self,
              can_hal,
              device_index = 0,
              bit_rate_setting = CanBitrate.Bitrate1MBit,
              rx_buffer_size = 10,
              tx_buffer_size = 10,
              config_flags = None )
```

Parameters

can_hal	parent CanHal instance
device_index	index of can device in CanHal, defaults to first device
bit_rate_setting	connection speed, see CanBitrate
rx_buffer_size	receive buffer size
tx_buffer_size	transmit buffer size
config_flags	optional ConfigFlag instance with set flags. None: Use defaults. See CanHal.initDevice.

7.104.3 Member Function Documentation

7.104.3.1 deinit()

```
ttk_tools.can_hal._hal_offline_stub.CanHalDevice.deinit (
              self )
```

Deinitialize this device on HAL.

7.104.3.2 getBusState()

```
ttk_tools.can_hal._hal_offline_stub.CanHalDevice.getBusState (
              self )
```

Get state of CAN bus for this device.

Returns

a CanBusState value.

7.104.3.3 receiveMessage()

```
ttk_tools.can_hal._hal_offline_stub.CanHalDevice.receiveMessage (
              self,
              timeout_s = 2,
              step_s = 0.100,
              feedback = False )
```

Receive a message (get a pending message from receive buffer).

Parameters

timeout_s	maximum time to wait for a message to appear in [s]. 0: return immediately after checking once.
step_s	wait time between checks for new messages in [s]
feedback	True: Print progress/status info to stdout

Returns

a CanMessage instance, or None if buffer was empty

7.104.3.4 sendCanMessage()

```
ttk_tools.can_hal._hal_offline_stub.CanHalDevice.sendCanMessage (
              self,
              can_message )
```

Send a preconfigured CanMessage.

Parameters

can_message	an already configured CanMessage struct instance.

7.104.3.5 sendMessage()

```
ttk_tools.can_hal._hal_offline_stub.CanHalDevice.sendMessage (
              self,
              can_id,
              data,
              length = 8,
              ext_id = None,
              feedback = False )
```

Send a CAN message.

Parameters

can_id	ID of message, 11 bits (max 0x7FF) for standard, 29 bits (max 0x1FFFFFFF) for extended identifiers.
data	payload data, either • as a byte/bytearray string or • as a list/tuple of byte values
length	length (data length code) of message data in bytes [0..8]
ext_id	True: set "extended identifier" bit to `CanMessage.status` None: set bit automatically if `can_id` is larger than 11 bits. See CanMessage
feedback	True: Print message info before sending to stdout

7.104.4 Member Data Documentation

7.104.4.1 device_hdl

```
ttk_tools.can_hal._hal_offline_stub.CanHalDevice.device_hdl
```

7.104.4.2 hal

```
ttk_tools.can_hal._hal_offline_stub.CanHalDevice.hal
```

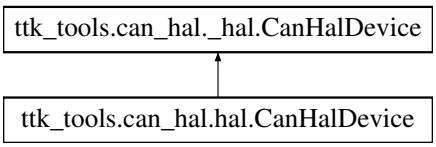
The documentation for this class was generated from the following file:

- ttk_tools/can_hal/_hal_offline_stub.py (v1.1.1)

7.105 ttk_tools.can_hal.hal.CanHalDevice Class Reference

CAN device connected to a CanHal instance.

Inheritance diagram for ttk_tools.can_hal.hal.CanHalDevice:



Public Member Functions

- __init__ (self, can_hal, device_index=0, bit_rate_setting=CanBitrate.Bitrate1MBit, rx_buffer_size=10, tx_↩ buffer_size=10, config_flags=None)
- deinit (self)

  Deinitialize this device on HAL.
- getBusState (self)

  Get state of CAN bus for this device.
- sendMessage (self, can_id, data, length=8, ext_id=None, feedback=False)

  Send a CAN message.
- sendCanMessage (self, can_message)

  Send a preconfigured CanMessage.
- receiveMessage (self, timeout_s=2, step_s=0.100, feedback=False)

  Receive a message (get a pending message from receive buffer).

7.105.1 Detailed Description

CAN device connected to a CanHal instance.

7.105.2 Constructor & Destructor Documentation

7.105.2.1 __init__()

```
ttk_tools.can_hal.hal.CanHalDevice.__init__ (
              self,
              can_hal,
              device_index = 0,
              bit_rate_setting = CanBitrate.Bitrate1MBit,
              rx_buffer_size = 10,
              tx_buffer_size = 10,
              config_flags = None )
```

Parameters

can_hal	parent CanHal instance
device_index	index of can device in CanHal, defaults to first device
bit_rate_setting	connection speed, see CanBitrate
rx_buffer_size	receive buffer size
tx_buffer_size	transmit buffer size
config_flags	optional ConfigFlag instance with set flags. None: Use defaults. See CanHal.initDevice.

Reimplemented from ttk_tools.can_hal._hal.CanHalDevice.

7.105.3 Member Function Documentation

7.105.3.1 deinit()

```
ttk_tools.can_hal.hal.CanHalDevice.deinit (
              self )
```

Deinitialize this device on HAL.

Reimplemented from ttk_tools.can_hal._hal.CanHalDevice.

7.105.3.2 getBusState()

```
ttk_tools.can_hal.hal.CanHalDevice.getBusState (
              self )
```

Get state of CAN bus for this device.

Returns

a CanBusState value.

Reimplemented from ttk_tools.can_hal._hal.CanHalDevice.

7.105.3.3 receiveMessage()

```
ttk_tools.can_hal.hal.CanHalDevice.receiveMessage (
              self,
              timeout_s = 2,
              step_s = 0.100,
              feedback = False )
```

Receive a message (get a pending message from receive buffer).

Parameters

timeout_s	maximum time to wait for a message to appear in [s]. 0: return immediately after checking once.
step_s	wait time between checks for new messages in [s]
feedback	True: Print progress/status info to stdout

Returns

a CanMessage instance, or None if buffer was empty

Reimplemented from ttk_tools.can_hal._hal.CanHalDevice.

7.105.3.4 sendCanMessage()

```
ttk_tools.can_hal.hal.CanHalDevice.sendCanMessage (
              self,
              can_message )
```

Send a preconfigured CanMessage.

Parameters

can_message	an already configured CanMessage struct instance.

Reimplemented from ttk_tools.can_hal._hal.CanHalDevice.

7.105.3.5 sendMessage()

```
ttk_tools.can_hal.hal.CanHalDevice.sendMessage (
              self,
              can_id,
              data,
              length = 8,
              ext_id = None,
              feedback = False )
```

Send a CAN message.

Parameters

can_id	ID of message, 11 bits (max 0x7FF) for standard, 29 bits (max 0x1FFFFFFF) for extended identifiers.
data	payload data, either • as a byte/bytearray string or • as a list/tuple of byte values

Parameters

length	length (data length code) of message data in bytes [0..8]
ext_id	True: set "extended identifier" bit to `CanMessage.status` None: set bit automatically if `can_id` is larger than 11 bits. See CanMessage
feedback	True: Print message info before sending to stdout

Reimplemented from ttk_tools.can_hal._hal.CanHalDevice.

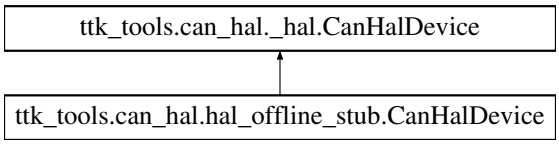
The documentation for this class was generated from the following file:

- ttk_tools/can_hal/hal.py (v1.0.2)

7.106 ttk_tools.can_hal.hal_offline_stub.CanHalDevice Class Reference

CAN device connected to a CanHal instance.

Inheritance diagram for ttk_tools.can_hal.hal_offline_stub.CanHalDevice:



Public Member Functions

- __init__ (self, can_hal, device_index=0, bit_rate_setting=CanBitrate.Bitrate1MBit, rx_buffer_size=10, tx_↩ buffer_size=10, config_flags=None)
- deinit (self)

  Deinitialize this device on HAL.
- getBusState (self)

  Get state of CAN bus for this device.
- sendMessage (self, can_id, data, length=8, ext_id=None, feedback=False)

  Send a CAN message.
- sendCanMessage (self, can_message)

  Send a preconfigured CanMessage.
- receiveMessage (self, timeout_s=2, step_s=0.100, feedback=False)

  Receive a message (get a pending message from receive buffer).

7.106.1 Detailed Description

CAN device connected to a CanHal instance.

7.106.2 Constructor & Destructor Documentation

7.106.2.1 __init__()

```
ttk_tools.can_hal.hal_offline_stub.CanHalDevice.__init__ (
              self,
              can_hal,
              device_index = 0,
              bit_rate_setting = CanBitrate.Bitrate1MBit,
              rx_buffer_size = 10,
              tx_buffer_size = 10,
              config_flags = None )
```

Parameters

can_hal	parent CanHal instance
device_index	index of can device in CanHal, defaults to first device
bit_rate_setting	connection speed, see CanBitrate
rx_buffer_size	receive buffer size
tx_buffer_size	transmit buffer size
config_flags	optional ConfigFlag instance with set flags. None: Use defaults. See CanHal.initDevice.

Reimplemented from ttk_tools.can_hal._hal.CanHalDevice.

7.106.3 Member Function Documentation

7.106.3.1 deinit()

```
ttk_tools.can_hal.hal_offline_stub.CanHalDevice.deinit (
              self )
```

Deinitialize this device on HAL.

Reimplemented from ttk_tools.can_hal._hal.CanHalDevice.

7.106.3.2 getBusState()

```
ttk_tools.can_hal.hal_offline_stub.CanHalDevice.getBusState (
              self )
```

Get state of CAN bus for this device.

Returns

a CanBusState value.

Reimplemented from ttk_tools.can_hal._hal.CanHalDevice.

7.106.3.3 receiveMessage()

```
ttk_tools.can_hal.hal_offline_stub.CanHalDevice.receiveMessage (
              self,
              timeout_s = 2,
              step_s = 0.100,
              feedback = False )
```

Receive a message (get a pending message from receive buffer).

Parameters

timeout_s	maximum time to wait for a message to appear in [s]. 0: return immediately after checking once.
step_s	wait time between checks for new messages in [s]
feedback	True: Print progress/status info to stdout

Returns

a CanMessage instance, or None if buffer was empty

Reimplemented from ttk_tools.can_hal._hal.CanHalDevice.

7.106.3.4 sendCanMessage()

```
ttk_tools.can_hal.hal_offline_stub.CanHalDevice.sendCanMessage (
              self,
              can_message )
```

Send a preconfigured CanMessage.

Parameters

can_message	an already configured CanMessage struct instance.

Reimplemented from ttk_tools.can_hal._hal.CanHalDevice.

7.106.3.5 sendMessage()

```
ttk_tools.can_hal.hal_offline_stub.CanHalDevice.sendMessage (
              self,
              can_id,
              data,
              length = 8,
              ext_id = None,
              feedback = False )
```

Send a CAN message.

Parameters

can_id	ID of message, 11 bits (max 0x7FF) for standard, 29 bits (max 0x1FFFFFFF) for extended identifiers.
data	payload data, either • as a byte/bytearray string or • as a list/tuple of byte values
length	length (data length code) of message data in bytes [0..8]
ext_id	True: set "extended identifier" bit to `CanMessage.status` None: set bit automatically if `can_id` is larger than 11 bits. See CanMessage
feedback	True: Print message info before sending to stdout

Reimplemented from ttk_tools.can_hal._hal.CanHalDevice.

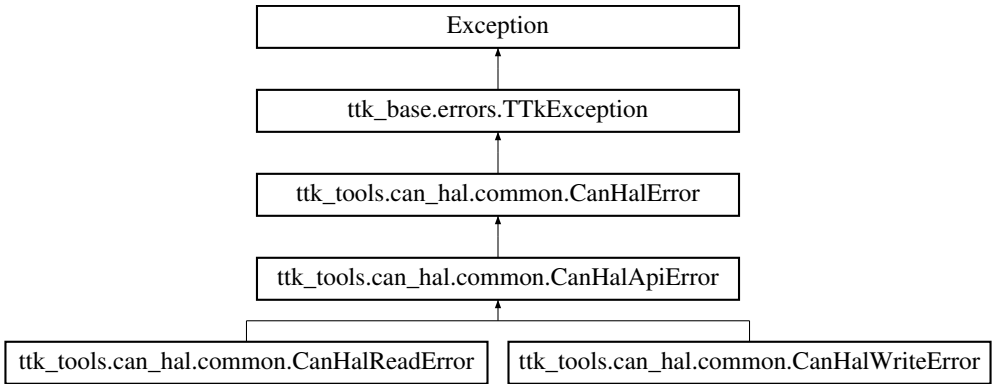
The documentation for this class was generated from the following file:

- ttk_tools/can_hal/hal_offline_stub.py (v1.0.2)

7.107 ttk_tools.can_hal.common.CanHalError Class Reference

Base class for CAN HAL exceptions.

Inheritance diagram for ttk_tools.can_hal.common.CanHalError:



7.107.1 Detailed Description

Base class for CAN HAL exceptions.

Errors during HAL API access (wraps error codes).

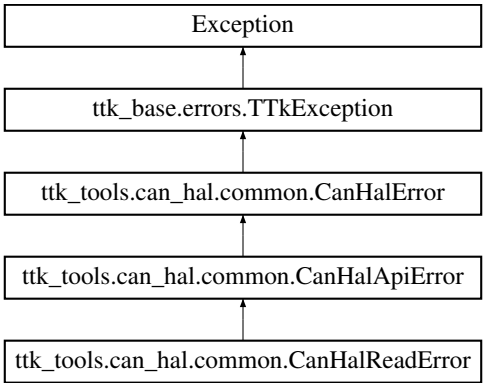
The documentation for this class was generated from the following file:

- ttk_tools/can_hal/common.py (v1.2)

7.108 ttk_tools.can_hal.common.CanHalReadError Class Reference

Error during read access.

Inheritance diagram for ttk_tools.can_hal.common.CanHalReadError:



7.108.1 Detailed Description

Error during read access.

This class is used to represent read errors of all devices (each device class has its own "read" error code.)

Error during read access. This class is used to represent write errors of all devices (each device class has its own "write" error code.)

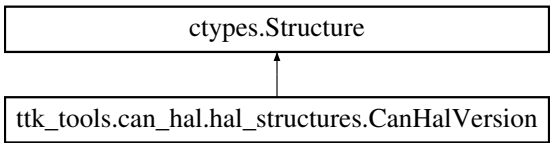
The documentation for this class was generated from the following file:

- ttk_tools/can_hal/common.py (v1.2)

7.109 ttk_tools.can_hal.hal_structures.CanHalVersion Class Reference

tCanHalVersion (CanHal.h)

Inheritance diagram for ttk_tools.can_hal.hal_structures.CanHalVersion:



Public Member Functions

- getVersion (self)

  Get version info as a named tuple with major, minor and patch entries.
- __repr__ (self)
- __str__ (self)

Public Attributes

- MajorVersion
- MinorVersion

Static Protected Attributes

- list _fields_
- _HalVersion = namedtuple("HalVersion", ["major", "minor", "patch"])

7.109.1 Detailed Description

tCanHalVersion (CanHal.h)

7.109.2 Member Function Documentation

7.109.2.1 __repr__()

```
ttk_tools.can_hal.hal_structures.CanHalVersion.__repr__ (
              self )
```

7.109.2.2 __str__()

```
ttk_tools.can_hal.hal_structures.CanHalVersion.__str__ (
              self )
```

7.109.2.3 getVersion()

```
ttk_tools.can_hal.hal_structures.CanHalVersion.getVersion (
              self )
```

Get version info as a named tuple with major, minor and patch entries.

7.109.3 Member Data Documentation

7.109.3.1 _fields_

```
list ttk_tools.can_hal.hal_structures.CanHalVersion._fields_  [static], [protected]
```

Initial value:

```
=  [
        ("MajorVersion", ctypes.c_uint), # unsigned int m_CanHalMajorVersion
        ("MinorVersion", ctypes.c_uint), # unsigned int m_CanHalMinorVersion
        ("PatchLevel",   ctypes.c_uint), # unsigned int m_CanHalPatchLevel
    ]
```

7.109.3.2 _HalVersion

```
ttk_tools.can_hal.hal_structures.CanHalVersion._HalVersion = namedtuple("HalVersion", ["major",
"minor", "patch"])  [static], [protected]
```

7.109.3.3 MajorVersion

```
ttk_tools.can_hal.hal_structures.CanHalVersion.MajorVersion
```

7.109.3.4 MinorVersion

```
ttk_tools.can_hal.hal_structures.CanHalVersion.MinorVersion
```

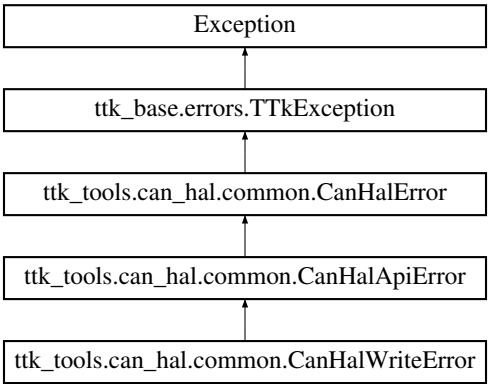
The documentation for this class was generated from the following file:

- ttk_tools/can_hal/hal_structures.py (v1.0.0)

7.110 ttk_tools.can_hal.common.CanHalWriteError Class Reference

Device Type (CanHAl.h, tCanHalDeviceType).

Inheritance diagram for ttk_tools.can_hal.common.CanHalWriteError:



7.110.1 Detailed Description

Device Type (CanHAl.h, tCanHalDeviceType).

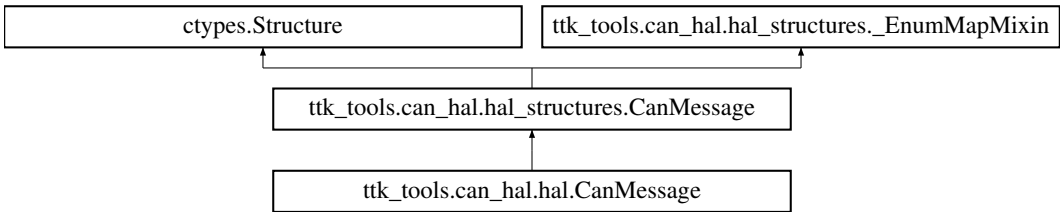
The documentation for this class was generated from the following file:

- ttk_tools/can_hal/common.py (v1.2)

7.111 ttk_tools.can_hal.hal.CanMessage Class Reference

CAN message data (CanHal.h, tCanHalCanMessage).

Inheritance diagram for ttk_tools.can_hal.hal.CanMessage:



Public Member Functions

- __init__ (self, can_id=None, dlc=None, data=None, tx=None, ext_id=None)
  Create an optionally pre-initialized CanMessage struct.
- getData (self)
  Get payload data as a list of byte values.
- getBytes (self)
  Get payload data as raw bytes string.
- getSystemTimestamp (self)
  Get system time stamp of a received message in µs.
- getHardwareTimestamp (self)
  Get hardware-based time stamp of a received message in µs.

Public Attributes

- can_id
- length
- payload
- timestamp
- status

7.111.1 Detailed Description

CAN message data (CanHal.h, tCanHalCanMessage).

7.111.2 Constructor & Destructor Documentation

7.111.2.1 __init__()

```
ttk_tools.can_hal.hal.CanMessage.__init__ (
              self,
              can_id = None,
              dlc = None,
              data = None,
              tx = None,
              ext_id = None )
```

Create an optionally pre-initialized CanMessage struct.

Parameters

can_id	ID of message, 11 bits (max 0x7FF) for standard, 29 bits (max 0x1FFFFFFF) for extended identifiers.
dlc	data length code; length of message data in bytes [0..8]
data	payload data, either • as a byte/bytearray string or • as a list/tuple of byte values
tx	True for Tx messages, False for Rx messages. Sets corresponding bits to CanMessage.status. None: set no bits (in case an empty message is sufficient)
ext_id	True: set "extended identifier" bit to `CanMessage.status` None: set bit automatically if `can_id` is larger than 11 bits.

Note

Extended Identifiers If a `can_id` > 0x7FF is supplied and `ext_id` is False, the supplied id will be truncated to 11 bits, as standard CAN frames only contain the first 11-bit "Identifier A" header.

Reimplemented from ttk_tools.can_hal.hal_structures.CanMessage.

7.111.3 Member Function Documentation

7.111.3.1 getBytes()

```
ttk_tools.can_hal.hal.CanMessage.getBytes (
              self )
```

Get payload data as raw bytes string.

Number of values depends on message length/dlc, see getData().

Reimplemented from ttk_tools.can_hal.hal_structures.CanMessage.

7.111.3.2 getData()

```
ttk_tools.can_hal.hal.CanMessage.getData (
              self )
```

Get payload data as a list of byte values.

Number of values depends on message length/dlc.

Note

Remote Requests RemoteRequest messages (RTR bit set in header) will not return any data. The length/DLC field indicates data length of the *requested* message (not the transmitted one)

Reimplemented from ttk_tools.can_hal.hal_structures.CanMessage.

7.111.3.3 getHardwareTimestamp()

```
ttk_tools.can_hal.hal.CanMessage.getHardwareTimestamp (
              self )
```

Get hardware-based time stamp of a received message in µs.

Time stamp will be generated by CAN driver, any reference to an external time base is driver/device dependent.

Returns

a time stamp in µs.

Reimplemented from ttk_tools.can_hal.hal_structures.CanMessage.

7.111.3.4 getSystemTimestamp()

```
ttk_tools.can_hal.hal.CanMessage.getSystemTimestamp (
              self )
```

Get system time stamp of a received message in µs.

Time stamp is based on Window's QueryPerformanceCounter value at time when message gets read from message queue.

Note

Performance counter value is not synchronized to an external time reference, so the time stamps are only useful to check intervals between messages.

Returns

a time stamp in µs.

Reimplemented from ttk_tools.can_hal.hal_structures.CanMessage.

7.111.4 Member Data Documentation

7.111.4.1 can_id

```
ttk_tools.can_hal.hal.CanMessage.can_id
```

7.111.4.2 length

```
ttk_tools.can_hal.hal.CanMessage.length
```

7.111.4.3 payload

```
ttk_tools.can_hal.hal.CanMessage.payload
```

7.111.4.4 status

`ttk_tools.can_hal.hal.CanMessage.status`

7.111.4.5 timestamp

`ttk_tools.can_hal.hal.CanMessage.timestamp`

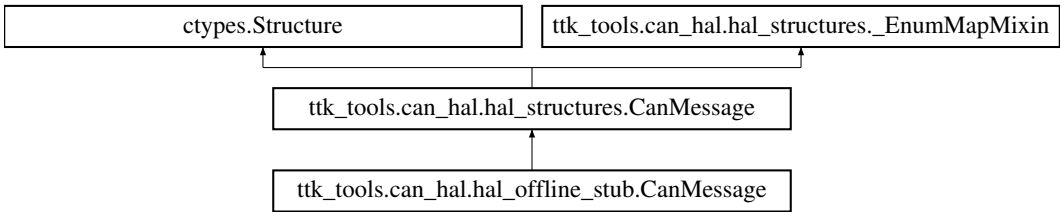
The documentation for this class was generated from the following file:

- ttk_tools/can_hal/hal.py (v1.0.2)

7.112 ttk_tools.can_hal.hal_offline_stub.CanMessage Class Reference

CAN message data (CanHal.h, tCanHalCanMessage).

Inheritance diagram for ttk_tools.can_hal.hal_offline_stub.CanMessage:



Public Member Functions

- __init__ (self, can_id=None, dlc=None, data=None, tx=None, ext_id=None)
  Create an optionally pre-initialized CanMessage struct.
- getData (self)
  Get payload data as a list of byte values.
- getBytes (self)
  Get payload data as raw bytes string.
- getSystemTimestamp (self)
  Get system time stamp of a received message in µs.
- getHardwareTimestamp (self)
  Get hardware-based time stamp of a received message in µs.

Public Attributes

- can_id
- length
- payload
- timestamp
- status

7.112.1 Detailed Description

CAN message data (CanHal.h, tCanHalCanMessage).

7.112.2 Constructor & Destructor Documentation

7.112.2.1 __init__()

```
ttk_tools.can_hal.hal_offline_stub.CanMessage.__init__ (
                self,
                can_id = None,
                dlc = None,
                data = None,
                tx = None,
                ext_id = None )
```

Create an optionally pre-initialized CanMessage struct.

Parameters

can_id	ID of message, 11 bits (max 0x7FF) for standard, 29 bits (max 0x1FFFFFFF) for extended identifiers.
dlc	data length code; length of message data in bytes [0..8]
data	payload data, either • as a byte/bytearray string or • as a list/tuple of byte values
tx	True for Tx messages, False for Rx messages. Sets corresponding bits to CanMessage.status. None: set no bits (in case an empty message is sufficient)
ext_id	True: set "extended identifier" bit to `CanMessage.status` None: set bit automatically if `can_id` is larger than 11 bits.

Note

Extended Identifiers If a `can_id` > 0x7FF is supplied and `ext_id` is False, the supplied id will be truncated to 11 bits, as standard CAN frames only contain the first 11-bit "Identifier A" header.

Reimplemented from ttk_tools.can_hal.hal_structures.CanMessage.

7.112.3 Member Function Documentation

7.112.3.1 getBytes()

```
ttk_tools.can_hal.hal_offline_stub.CanMessage.getBytes (
                self )
```

Get payload data as raw bytes string.

Number of values depends on message length/dlc, see getData().

Reimplemented from ttk_tools.can_hal.hal_structures.CanMessage.

7.112.3.2 getData()

```
ttk_tools.can_hal.hal_offline_stub.CanMessage.getData (
              self )
```

Get payload data as a list of byte values.

Number of values depends on message length/dlc.

Note

Remote Requests RemoteRequest messages (RTR bit set in header) will not return any data. The length/DLC field indicates data length of the *requested* message (not the transmitted one)

Reimplemented from ttk_tools.can_hal.hal_structures.CanMessage.

7.112.3.3 getHardwareTimestamp()

```
ttk_tools.can_hal.hal_offline_stub.CanMessage.getHardwareTimestamp (
              self )
```

Get hardware-based time stamp of a received message in µs.

Time stamp will be generated by CAN driver, any reference to an external time base is driver/device dependent.

Returns

a time stamp in µs.

Reimplemented from ttk_tools.can_hal.hal_structures.CanMessage.

7.112.3.4 getSystemTimestamp()

```
ttk_tools.can_hal.hal_offline_stub.CanMessage.getSystemTimestamp (
              self )
```

Get system time stamp of a received message in µs.

Time stamp is based on Window's QueryPerformanceCounter value at time when message gets read from message queue.

Note

Performance counter value is not synchronized to an external time reference, so the time stamps are only useful to check intervals between messages.

Returns

a time stamp in µs.

Reimplemented from ttk_tools.can_hal.hal_structures.CanMessage.

7.112.4 Member Data Documentation

7.112.4.1 can_id

ttk_tools.can_hal.hal_offline_stub.CanMessage.can_id

7.112.4.2 length

ttk_tools.can_hal.hal_offline_stub.CanMessage.length

7.112.4.3 payload

ttk_tools.can_hal.hal_offline_stub.CanMessage.payload

7.112.4.4 status

ttk_tools.can_hal.hal_offline_stub.CanMessage.status

7.112.4.5 timestamp

ttk_tools.can_hal.hal_offline_stub.CanMessage.timestamp

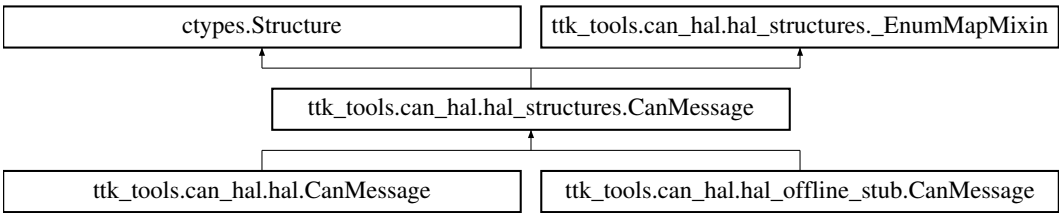
The documentation for this class was generated from the following file:

- ttk_tools/can_hal/hal_offline_stub.py (v1.0.2)

7.113 ttk_tools.can_hal.hal_structures.CanMessage Class Reference

CAN message data (CanHal.h, tCanHalCanMessage).

Inheritance diagram for ttk_tools.can_hal.hal_structures.CanMessage:



Public Member Functions

- __init__ (self, can_id=None, dlc=None, data=None, tx=None, ext_id=None)

  Create an optionally pre-initialized CanMessage struct.
- getData (self)

  Get payload data as a list of byte values.
- getBytes (self)

  Get payload data as raw bytes string.
- getSystemTimestamp (self)

  Get system time stamp in µs.
- getHardwareTimestamp (self)

  Get hardware-based time stamp in µs.
- __repr__ (self)
- __str__ (self)

Public Attributes

- can_id
- length
- payload

Static Protected Attributes

- list _fields_
- dict _map

7.113.1 Detailed Description

CAN message data (CanHal.h, tCanHalCanMessage).

7.113.2 Constructor & Destructor Documentation

7.113.2.1 __init__()

```
ttk_tools.can_hal.hal_structures.CanMessage.__init__ (
              self,
              can_id = None,
              dlc = None,
              data = None,
              tx = None,
              ext_id = None )
```

Create an optionally pre-initialized CanMessage struct.

Parameters

can↩ _id	ID of message, 11 bits (max 0x7FF) for standard, 29 bits (max 0x1FFFFFFF) for extended identifiers.
dlc	data length code; length of message data in bytes [0..8]

Parameters

data	payload data, either • as a byte/bytearray string or • as a list/tuple of byte values
tx	True for Tx messages, False for Rx messages. Sets corresponding bits to CanMessage.status. None: set no bits (in case an empty message is sufficient)
ext_id	True: set "extended identifier" bit to `CanMessage.status` None: set bit automatically if `can_id` is larger than 11 bits.

Note

Extended Identifiers If a `can_id` > 0x7FF is supplied and `ext_id` is False, the supplied id will be truncated to 11 bits, as standard CAN frames only contain the first 11-bit "Identifier A" header.

Reimplemented in ttk_tools.can_hal.hal.CanMessage, and ttk_tools.can_hal.hal_offline_stub.CanMessage.

7.113.3 Member Function Documentation

7.113.3.1 __repr__()

```
ttk_tools.can_hal.hal_structures.CanMessage.__repr__ (
              self )
```

7.113.3.2 __str__()

```
ttk_tools.can_hal.hal_structures.CanMessage.__str__ (
              self )
```

7.113.3.3 getBytes()

```
ttk_tools.can_hal.hal_structures.CanMessage.getBytes (
              self )
```

Get payload data as raw bytes string.

Number of values depends on message length/dlc.

Reimplemented in ttk_tools.can_hal.hal.CanMessage, and ttk_tools.can_hal.hal_offline_stub.CanMessage.

7.113.3.4 getData()

```
ttk_tools.can_hal.hal_structures.CanMessage.getData (
              self )
```

Get payload data as a list of byte values.

Number of values depends on message length/dlc.

Note

RemoteRequest RemoteRequest messages (RTR bit set in header) will not return any data. The length/DLC field indicates data length of the *requested* message (not the transmitted one)

Reimplemented in ttk_tools.can_hal.hal.CanMessage, and ttk_tools.can_hal.hal_offline_stub.CanMessage.

7.113.3.5 getHardwareTimestamp()

```
ttk_tools.can_hal.hal_structures.CanMessage.getHardwareTimestamp (
              self )
```

Get hardware-based time stamp in µs.

Time stamp is generated by CAN driver, reference to external time (if any) is driver/device dependent.

Returns

a time stamp in µs

Reimplemented in ttk_tools.can_hal.hal.CanMessage, and ttk_tools.can_hal.hal_offline_stub.CanMessage.

7.113.3.6 getSystemTimestamp()

```
ttk_tools.can_hal.hal_structures.CanMessage.getSystemTimestamp (
              self )
```

Get system time stamp in µs.

Time stamp is based on Window's QueryPerformanceCounter value at time when message gets read from message queue.

Note

Performance counter value is not synchronized to an external time reference, so the time stamps are only useful to check intervals between messages.

Returns

a time stamp in µs

Reimplemented in ttk_tools.can_hal.hal.CanMessage, and ttk_tools.can_hal.hal_offline_stub.CanMessage.

7.113.4 Member Data Documentation

7.113.4.1 _fields_

list ttk_tools.can_hal.hal_structures.CanMessage._fields_ [static], [protected]

Initial value:

```
=  [
        ("can_id",      ctypes.c_uint),      # unsigned int         m_uiCanId
        ("length",      ctypes.c_ubyte),     # unsigned char        m_ucLength
        ("payload",     CanPayload),         # unsigned char        m_ucData
        ("timestamp",   Timestamp),          # tCanHalTimestamp     m_Timestamp
        ("status",      ctypes.c_uint)       # tCanHalMessageStatus m_CanMessageStatus
    ]
```

7.113.4.2 _map

dict ttk_tools.can_hal.hal_structures.CanMessage._map [static], [protected]

Initial value:

```
=  {
        "status": MessageStatus,
        "can_id": _CanId,         # not really an Enum, but works just as well
    }
```

7.113.4.3 can_id

ttk_tools.can_hal.hal_structures.CanMessage.can_id

7.113.4.4 length

ttk_tools.can_hal.hal_structures.CanMessage.length

7.113.4.5 payload

ttk_tools.can_hal.hal_structures.CanMessage.payload

The documentation for this class was generated from the following file:

- ttk_tools/can_hal/hal_structures.py (v1.0.0)

7.114 ttk_tools.vector.mdf4_reader.CANopenDate Class Reference

Container for channel data type "CANopen date".

Public Member Functions

- __init__ (self, raw_data)
- time (self)

  value as seconds since epoch
- struct (self)

  value as a tuple in time.struct_time format.
- __str__ (self)
- __repr__ (self)

Public Attributes

- ms
- min
- hour
- day
- month
- year
- is_dst
- day_of_month
- day_of_week
- struct

Static Public Attributes

- fmt = struct.Struct("< H B B B B B")

  Data Structure:

7.114.1 Detailed Description

Container for channel data type "CANopen date".

7.114.2 Constructor & Destructor Documentation

7.114.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.CANopenDate.__init__ (
            self,
            raw_data )
```

7.114.3 Member Function Documentation

7.114.3.1 __repr__()

```
ttk_tools.vector.mdf4_reader.CANopenDate.__repr__ (
            self )
```

7.114.3.2 __str__()

```
ttk_tools.vector.mdf4_reader.CANopenDate.__str__ (
              self )
```

7.114.3.3 struct()

```
ttk_tools.vector.mdf4_reader.CANopenDate.struct (
              self )
```

value as a tuple in time.struct_time format.

Note that CANopen's 2-digit year will be moved to the closest century.

7.114.3.4 time()

```
ttk_tools.vector.mdf4_reader.CANopenDate.time (
              self )
```

value as seconds since epoch

7.114.4 Member Data Documentation

7.114.4.1 day

```
ttk_tools.vector.mdf4_reader.CANopenDate.day
```

7.114.4.2 day_of_month

```
ttk_tools.vector.mdf4_reader.CANopenDate.day_of_month
```

7.114.4.3 day_of_week

```
ttk_tools.vector.mdf4_reader.CANopenDate.day_of_week
```

7.114.4.4 fmt

```
ttk_tools.vector.mdf4_reader.CANopenDate.fmt = struct.Struct("< H B B B B B")  [static]
```

Data Structure:

Type Identifier Description ~~~

7.114.4.5 hour

ttk_tools.vector.mdf4_reader.CANopenDate.hour

7.114.4.6 is_dst

ttk_tools.vector.mdf4_reader.CANopenDate.is_dst

7.114.4.7 min

ttk_tools.vector.mdf4_reader.CANopenDate.min

7.114.4.8 month

ttk_tools.vector.mdf4_reader.CANopenDate.month

7.114.4.9 ms

ttk_tools.vector.mdf4_reader.CANopenDate.ms

7.114.4.10 struct

ttk_tools.vector.mdf4_reader.CANopenDate.struct

7.114.4.11 year

ttk_tools.vector.mdf4_reader.CANopenDate.year

The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.115 ttk_tools.vector.mdf4_reader.CANopenTime Class Reference

Container for channel data type "CANopen time".

Public Member Functions

- __init__ (self, raw_data)
- getDHMS (self)

  Get value as days/hours/minutes/seconds since 1984-01-01T00:00:00.
- __str__ (self)
- __repr__ (self)

Public Attributes

- ms
- days

Static Public Attributes

- fmt = struct.Struct("< L H")

  Data Structure:

7.115.1 Detailed Description

Container for channel data type "CANopen time".

7.115.2 Constructor & Destructor Documentation

7.115.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.CANopenTime.__init__ (
              self,
              raw_data )
```

7.115.3 Member Function Documentation

7.115.3.1 __repr__()

```
ttk_tools.vector.mdf4_reader.CANopenTime.__repr__ (
              self )
```

7.115.3.2 __str__()

```
ttk_tools.vector.mdf4_reader.CANopenTime.__str__ (
              self )
```

7.115.3.3 getDHMS()

```
ttk_tools.vector.mdf4_reader.CANopenTime.getDHMS (
              self )
```

Get value as days/hours/minutes/seconds since 1984-01-01T00:00:00.

Returns

a tuple (days, hours, minutes, seconds>.

7.115.4 Member Data Documentation

7.115.4.1 days

`ttk_tools.vector.mdf4_reader.CANopenTime.days`

7.115.4.2 fmt

`ttk_tools.vector.mdf4_reader.CANopenTime.fmt = struct.Struct("< L H")  [static]`

Data Structure:

Type Identifier Description ~~~

Public Attributes

- parent
  reference to parent XilTestbench instance
- testbench
  reference to actual testbench api
- capture
  actual Capture instance, will be configured during setup()
- variables
  list of variables / model_paths to capture (configured during setup())
- task_name
  capture task name (configured during setup())
- mdf_file_path
  file path to MDF-file in stream-to-disk mode (configured during setup())
- result_writer
  result writer instance to use for writing captured data (configured during setup())

Protected Member Functions

- _captureResultToDaqData (self, capture_result)
- _showMdfInfo (self, prefix="")
  Show some info about current MDF file (name, file size...).

7.116.1 Detailed Description

Capture / Data Acquisition.

7.116.2 Constructor & Destructor Documentation

7.116.2.1 __init__()

```
ttk_tools.dspace._xil_api.Capture.__init__ (
              self,
              xil_testbench )
```

Capture/DAQ.

Parameters

xil_testbench	parent XilTestbench instance

Reimplemented in ttk_tools.dspace.xil_api.Capture.

7.116.2.2 __del__()

```
ttk_tools.dspace._xil_api.Capture.__del__ (
              self )
```

7.116.3 Member Function Documentation

7.116.3.1 _captureResultToDaqData()

```
ttk_tools.dspace._xil_api.Capture._captureResultToDaqData (
              self,
              capture_result )  [protected]
```

7.116.3.2 _showMdfInfo()

```
ttk_tools.dspace._xil_api.Capture._showMdfInfo (
              self,
              prefix = "" )  [protected]
```

Show some info about current MDF file (name, file size...).

Only sensible in stream-to-disk mode, otherwise there will be no file configured.

Parameters

prefix	prefix text for printed error messages

7.116.3.3 dispose()

```
ttk_tools.dspace._xil_api.Capture.dispose (
              self )
```

Dispose of resources allocated by this Capture instance.

Note

After this method has been called, further access to capture data will fail.

Reimplemented in ttk_tools.dspace.xil_api.Capture.

7.116.3.4 fetch()

```
ttk_tools.dspace._xil_api.Capture.fetch (
              self )
```

Fetch data while capture is running.

Only works if data is written to memory (and specifically not in stream-to-disk mode).

Only works in states `eACTIVATED`, `eRUNNING`, `eFINISHED` (and not in `eCONFIGURED`, so it will not work once a capture has been stopped.)

Returns

data captured since the last call to fetch.

Reimplemented in ttk_tools.dspace.xil_api.Capture.

7.116.3.5 getData()

```
ttk_tools.dspace._xil_api.Capture.getData (
               self,
               verbosity = 1 )
```

Get a DAQ data structure containing the captured data.

Example: DAQ data structure:

```
daq_data = {
    "<signal name>": {
        "time": [ t1, t2, t3, ..., tn ], # timestamps
        "data": [ d1, d2, d3, ..., dn ], # data points
    }
}
```

Parameters

verbosity	0: silent 1: show MDF file info (only for stream-to-disk mode)

Returns

a dictionary with entries for each captured signal (see example).

Reimplemented in ttk_tools.dspace.xil_api.Capture.

7.116.3.6 setup()

```
ttk_tools.dspace._xil_api.Capture.setup (
               self,
               variables,
               mdf_file_path = None,
               downsampling = None,
               task_name = "Periodic Task 1" )
```

Setup data capture.

Task Name:

Default capture/measurement task_name is

- "HostService" for PHS bus systems and
- "Periodic Task 1" for VEOS/SCALEXIO systems.

See TRC file (<model_name>.trc) for available/configured tasks, e.g.
sampling_period[0]
{
value: 0.001
alias: "HostService"
increment: 0.001
unit: "s"
}

Parameters

variables	list of variables/paths to capture
mdf_file_path	path to MDF-file to use for stream-to-disk mode. None: keep captured data in memory.
downsampling	downsampling factor (1:n, one sample captured for every n capture cycles). Setting None uses the capture service's default setting (which should be 1:1).
task_name	name of capture task to use (see note)

Reimplemented in ttk_tools.dspace.xil_api.Capture.

7.116.3.7 start()

```
ttk_tools.dspace._xil_api.Capture.start (
              self )
```

Start capturing.

Attention

stream-to-disk:
start() will raise a (somewhat generic) `TestbenchPortException` ("Could not start capturing.") if the configured MDF-file is locked by another application.

Reimplemented in ttk_tools.dspace.xil_api.Capture.

7.116.3.8 stop()

```
ttk_tools.dspace._xil_api.Capture.stop (
              self )
```

Stop capture process.

Reimplemented in ttk_tools.dspace.xil_api.Capture.

7.116.4 Member Data Documentation

7.116.4.1 capture

```
ttk_tools.dspace._xil_api.Capture.capture
```

actual Capture instance, will be configured during setup()

7.116.4.2 mdf_file_path

```
ttk_tools.dspace._xil_api.Capture.mdf_file_path
```

file path to MDF-file in stream-to-disk mode (configured during setup())

7.116.4.3 parent

`ttk_tools.dspace._xil_api.Capture.parent`

reference to parent XilTestbench instance

7.116.4.4 result_writer

`ttk_tools.dspace._xil_api.Capture.result_writer`

result writer instance to use for writing captured data (configured during setup())

7.116.4.5 task_name

`ttk_tools.dspace._xil_api.Capture.task_name`

capture task name (configured during setup())

7.116.4.6 testbench

`ttk_tools.dspace._xil_api.Capture.testbench`

reference to actual testbench api

7.116.4.7 variables

`ttk_tools.dspace._xil_api.Capture.variables`

list of variables / model_paths to capture (configured during setup())

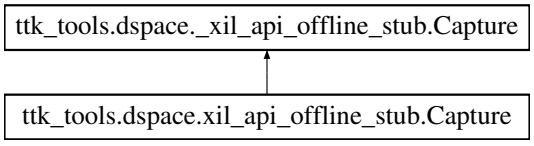
The documentation for this class was generated from the following file:

- ttk_tools/dspace/_xil_api.py (v2.3.1)

7.117 ttk_tools.dspace._xil_api_offline_stub.Capture Class Reference

Capture / Data Acquisition Offline Stub.

Inheritance diagram for ttk_tools.dspace._xil_api_offline_stub.Capture:



Public Member Functions

- __init__ (self, xil_testbench)

  Capture/DAQ stub.
- setup (self, variables, mdf_file_path=None, downsampling=None, task_name="Periodic Task 1")

  Setup data capture.
- dispose (self)

  Dispose of resources allocated by this Capture instance.
- start (self)

  Start capturing.
- stop (self)

  Stop capture process.
- fetch (self)

  Fetch data while capture is running.
- getData (self, verbosity=1)

  Get a DAQ data structure containing the captured data.

Public Attributes

- parent

  reference to parent XilTestbench instance
- testbench

  reference to actual testbench api
- capture

  actual Capture instance, will be configured during setup()
- variables

  list of variables / model_paths to capture (configured during setup())
- task_name

  capture task name (configured during setup())
- mdf_file_path

  file path to MDF-file in stream-to-disk mode (configured during setup())

Protected Member Functions

- _getDummyDaqData (self, duration, offset=0.0)

  Offline Stub: Get some dummy data.

Protected Attributes

- _cycle_time
- _downsampling
- _mdf_file_path
- _start_time
- _last_fetch_time
- _stop_time
- _state

7.117.1 Detailed Description

Capture / Data Acquisition Offline Stub.

7.117.2 Constructor & Destructor Documentation

7.117.2.1 __init__()

```
ttk_tools.dspace._xil_api_offline_stub.Capture.__init__ (
              self,
              xil_testbench )
```

Capture/DAQ stub.

Parameters

xil_testbench	parent XilTestbench instance

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.Capture.

7.117.3 Member Function Documentation

7.117.3.1 _getDummyDaqData()

```
ttk_tools.dspace._xil_api_offline_stub.Capture._getDummyDaqData (
              self,
              duration,
              offset = 0.0 )  [protected]
```

Offline Stub: Get some dummy data.

Uses a time stamp step size derived from self._cycle_time and self._downsampling

Parameters

duration	duration in [s] to get data for
offset	offset of first time stamp

Returns

a DAQ data structure with data for each variable in self.variables.

7.117.3.2 dispose()

```
ttk_tools.dspace._xil_api_offline_stub.Capture.dispose (
              self )
```

Dispose of resources allocated by this Capture instance.

Note

After this has been called further access to capture data will fail (... if not in the stub implementation).

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.Capture.

7.117.3.3 fetch()

```
ttk_tools.dspace._xil_api_offline_stub.Capture.fetch (
              self )
```

Fetch data while capture is running.

Only works if data is written to memory (and specifically not in stream-to-disk mode).

Only works in states `eACTIVATED`, `eRUNNING`, `eFINISHED` (and not in `eCONFIGURED`, so it will not work once a capture has been stopped.)

Returns

data captured since the last call to fetch.

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.Capture.

7.117.3.4 getData()

```
ttk_tools.dspace._xil_api_offline_stub.Capture.getData (
              self,
              verbosity = 1 )
```

Get a DAQ data structure containing the captured data.

Example: DAQ data structure:

```
daq_data = {
    "<signal name>": {
        "time": [ t1, t2, t3, ..., tn ], # timestamps
        "data": [ d1, d2, d3, ..., dn ], # data points
    }
}
```

Parameters

verbosity	0: silent 1: show MDF file info (only for stream-to-disk mode)

Returns

a dictionary with entries for each captured signal (see example)

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.Capture.

7.117.3.5 setup()

```
ttk_tools.dspace._xil_api_offline_stub.Capture.setup (
              self,
              variables,
```

```
                mdf_file_path = None,
                downsampling = None,
                task_name = "Periodic Task 1" )
```

Setup data capture.

Task Name:

Default capture/measurement task_name is

- "HostService" for PHS bus systems and
- "Periodic Task 1" for VEOS/SCALEXIO systems.

See TRC file (`<model_name>.trc`) for available/configured tasks, e.g.
sampling_period[0]
{
value: 0.001
alias: "HostService"
increment: 0.001
unit: "s"
}

Parameters

variables	list of variables/paths to capture
mdf_file_path	path to MDF-file to use for stream-to-disk mode. None: keep captured data in memory.
downsampling	downsampling factor (1:n, one sample captured for every n capture cycles). Setting None uses the capture service's default setting (which should be 1:1).
task_name	name of capture task to use (see note)

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.Capture.

7.117.3.6 start()

```
ttk_tools.dspace._xil_api_offline_stub.Capture.start (
                self )
```

Start capturing.

Note

stream-to-disk: start() will raise a (somewhat generic) `TestbenchPortException` ("Could not start capturing.") if the configured MDF-file is locked by another application.
The stub implementation will not really care about this, though.

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.Capture.

7.117.3.7 stop()

```
ttk_tools.dspace._xil_api_offline_stub.Capture.stop (
             self )
```

Stop capture process.

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.Capture.

7.117.4 Member Data Documentation

7.117.4.1 _cycle_time

```
ttk_tools.dspace._xil_api_offline_stub.Capture._cycle_time  [protected]
```

7.117.4.2 _downsampling

```
ttk_tools.dspace._xil_api_offline_stub.Capture._downsampling  [protected]
```

7.117.4.3 _last_fetch_time

```
ttk_tools.dspace._xil_api_offline_stub.Capture._last_fetch_time  [protected]
```

7.117.4.4 _mdf_file_path

```
ttk_tools.dspace._xil_api_offline_stub.Capture._mdf_file_path  [protected]
```

7.117.4.5 _start_time

```
ttk_tools.dspace._xil_api_offline_stub.Capture._start_time  [protected]
```

7.117.4.6 _state

```
ttk_tools.dspace._xil_api_offline_stub.Capture._state  [protected]
```

7.117.4.7 _stop_time

```
ttk_tools.dspace._xil_api_offline_stub.Capture._stop_time  [protected]
```

7.117.4.8 capture

```
ttk_tools.dspace._xil_api_offline_stub.Capture.capture
```

actual Capture instance, will be configured during setup()

7.117.4.9 mdf_file_path

`ttk_tools.dspace._xil_api_offline_stub.Capture.mdf_file_path`

file path to MDF-file in stream-to-disk mode (configured during setup())

7.117.4.10 parent

`ttk_tools.dspace._xil_api_offline_stub.Capture.parent`

reference to parent XilTestbench instance

7.117.4.11 task_name

`ttk_tools.dspace._xil_api_offline_stub.Capture.task_name`

capture task name (configured during setup())

7.117.4.12 testbench

`ttk_tools.dspace._xil_api_offline_stub.Capture.testbench`

reference to actual testbench api

7.117.4.13 variables

`ttk_tools.dspace._xil_api_offline_stub.Capture.variables`

list of variables / model_paths to capture (configured during setup())

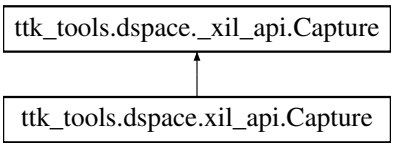
The documentation for this class was generated from the following file:

- ttk_tools/dspace/_xil_api_offline_stub.py (v2.1.2)

7.118 ttk_tools.dspace.xil_api.Capture Class Reference

Capture / Data Acquisition.

Inheritance diagram for ttk_tools.dspace.xil_api.Capture:



Public Member Functions

- __init__ (self, xil_testbench)
  Capture/DAQ.
- setup (self, variables, mdf_file_path=None, downsampling=None, task_name="Periodic Task 1")
  Setup data capture.
- dispose (self)
  Dispose of resources allocated by this Capture instance.
- start (self)
  Start capturing.
- stop (self)
  Stop capture process.
- fetch (self)
  Fetch data while capture is running.
- getData (self, verbosity=1)
  Get a DAQ data structure containing the captured data.

Public Attributes

- parent
  reference to parent XilTestbench instance
- testbench
  reference to actual testbench api
- capture
  actual Capture instance, will be configured during setup()
- variables
  list of variables / model_paths to capture (configured during setup())
- task_name
  capture task name (configured during setup())
- mdf_file_path
  file path to MDF-file in stream-to-disk mode (configured during setup())
- result_writer
  result writer instance to use for writing captured data (configured during setup())

7.118.1 Detailed Description

Capture / Data Acquisition.

7.118.2 Constructor & Destructor Documentation

7.118.2.1 __init__()

```
ttk_tools.dspace.xil_api.Capture.__init__ (
            self,
            xil_testbench )
```

Capture/DAQ.

Parameters

xil_testbench	parent XilTestbench instance

Reimplemented from ttk_tools.dspace._xil_api.Capture.

7.118.3 Member Function Documentation

7.118.3.1 dispose()

```
ttk_tools.dspace.xil_api.Capture.dispose (
              self )
```

Dispose of resources allocated by this Capture instance.

Note

After this method has been called, further access to capture data will fail.

Reimplemented from ttk_tools.dspace._xil_api.Capture.

7.118.3.2 fetch()

```
ttk_tools.dspace.xil_api.Capture.fetch (
              self )
```

Fetch data while capture is running.

Only works if data is written to memory (and specifically not in stream-to-disk mode).

Only works in states `eACTIVATED`, `eRUNNING`, `eFINISHED` (and not in `eCONFIGURED`, so it will not work once a capture has been stopped.)

Returns

data captured since the last call to fetch.

Reimplemented from ttk_tools.dspace._xil_api.Capture.

7.118.3.3 getData()

```
ttk_tools.dspace.xil_api.Capture.getData (
              self,
              verbosity = 1 )
```

Get a DAQ data structure containing the captured data.

Example: DAQ data structure:

```
daq_data = {
    "<signal name>": {
        "time": [ t1, t2, t3, ..., tn ], # timestamps
        "data": [ d1, d2, d3, ..., dn ], # data points
    }
}
```

Parameters

verbosity	0: silent 1: show MDF file info (only for stream-to-disk mode)

Returns

a dictionary with entries for each captured signal (see example).

Reimplemented from ttk_tools.dspace._xil_api.Capture.

7.118.3.4 setup()

```
ttk_tools.dspace.xil_api.Capture.setup (
              self,
              variables,
              mdf_file_path = None,
              downsampling = None,
              task_name = "Periodic Task 1" )
```

Setup data capture.

Task Name:

Default capture/measurement task_name is

- "HostService" for PHS bus systems and
- "Periodic Task 1" for VEOS/SCALEXIO systems.

See TRC file (<model_name>.trc) for available/configured tasks, e.g.
sampling_period[0]
{
value: 0.001
alias: "HostService"
increment: 0.001
unit: "s"
}

Parameters

variables	list of variables/paths to capture
mdf_file_path	path to MDF-file to use for stream-to-disk mode. None: keep captured data in memory.
downsampling	downsampling factor (1:n, one sample captured for every n capture cycles). Setting None uses the capture service's default setting (which should be 1:1).
task_name	name of capture task to use (see note)

Reimplemented from ttk_tools.dspace._xil_api.Capture.

7.118.3.5 start()

```
ttk_tools.dspace.xil_api.Capture.start (
              self )
```

Start capturing.

Attention

stream-to-disk:
start() will raise a (somewhat generic) `TestbenchPortException` ("Could not start capturing.") if the configured MDF-file is locked by another application.

Reimplemented from ttk_tools.dspace._xil_api.Capture.

7.118.3.6 stop()

```
ttk_tools.dspace.xil_api.Capture.stop (
              self )
```

Stop capture process.

Reimplemented from ttk_tools.dspace._xil_api.Capture.

7.118.4 Member Data Documentation

7.118.4.1 capture

```
ttk_tools.dspace.xil_api.Capture.capture
```

actual Capture instance, will be configured during setup()

7.118.4.2 mdf_file_path

```
ttk_tools.dspace.xil_api.Capture.mdf_file_path
```

file path to MDF-file in stream-to-disk mode (configured during setup())

7.118.4.3 parent

```
ttk_tools.dspace.xil_api.Capture.parent
```

reference to parent XilTestbench instance

7.118.4.4 result_writer

```
ttk_tools.dspace.xil_api.Capture.result_writer
```

result writer instance to use for writing captured data (configured during setup())

7.118.4.5 task_name

`ttk_tools.dspace.xil_api.Capture.task_name`

capture task name (configured during setup())

7.118.4.6 testbench

`ttk_tools.dspace.xil_api.Capture.testbench`

reference to actual testbench api

7.118.4.7 variables

`ttk_tools.dspace.xil_api.Capture.variables`

list of variables / model_paths to capture (configured during setup())

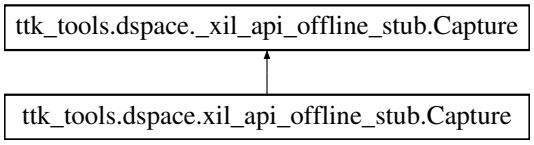
The documentation for this class was generated from the following file:

- ttk_tools/dspace/xil_api.py (v2.0.1)

7.119 ttk_tools.dspace.xil_api_offline_stub.Capture Class Reference

Capture / Data Acquisition Offline Stub.

Inheritance diagram for ttk_tools.dspace.xil_api_offline_stub.Capture:



Public Member Functions

- __init__ (self, xil_testbench)
  Capture/DAQ stub.
- setup (self, variables, mdf_file_path=None, downsampling=None, task_name="Periodic Task 1")
  Setup data capture.
- dispose (self)
  Dispose of resources allocated by this Capture instance.
- start (self)
  Start capturing.
- stop (self)
  Stop capture process.
- fetch (self)
  Fetch data while capture is running.
- getData (self, verbosity=1)
  Get a DAQ data structure containing the captured data.

Public Attributes

- parent
  reference to parent XilTestbench instance
- testbench
  reference to actual testbench api
- capture
  actual Capture instance, will be configured during setup()
- variables
  list of variables / model_paths to capture (configured during setup())
- task_name
  capture task name (configured during setup())
- mdf_file_path
  file path to MDF-file in stream-to-disk mode (configured during setup())

7.119.1 Detailed Description

Capture / Data Acquisition Offline Stub.

7.119.2 Constructor & Destructor Documentation

7.119.2.1 __init__()

```
ttk_tools.dspace.xil_api_offline_stub.Capture.__init__ (
              self,
              xil_testbench )
```

Capture/DAQ stub.

Parameters

xil_testbench	parent XilTestbench instance

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.Capture.

7.119.3 Member Function Documentation

7.119.3.1 dispose()

```
ttk_tools.dspace.xil_api_offline_stub.Capture.dispose (
              self )
```

Dispose of resources allocated by this Capture instance.

Note

After this has been called further access to capture data will fail (... if not in the stub implementation).

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.Capture.

7.119.3.2 fetch()

```
ttk_tools.dspace.xil_api_offline_stub.Capture.fetch (
              self )
```

Fetch data while capture is running.

Only works if data is written to memory (and specifically not in stream-to-disk mode).

Only works in states `eACTIVATED`, `eRUNNING`, `eFINISHED` (and not in `eCONFIGURED`, so it will not work once a capture has been stopped.)

Returns

data captured since the last call to fetch.

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.Capture.

7.119.3.3 getData()

```
ttk_tools.dspace.xil_api_offline_stub.Capture.getData (
              self,
              verbosity = 1 )
```

Get a DAQ data structure containing the captured data.

Example: DAQ data structure:

```
daq_data = {
    "<signal name>": {
        "time": [ t1, t2, t3, ..., tn ], # timestamps
        "data": [ d1, d2, d3, ..., dn ], # data points
    }
}
```

Parameters

verbosity	0: silent 1: show MDF file info (only for stream-to-disk mode)

Returns

a dictionary with entries for each captured signal (see example)

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.Capture.

7.119.3.4 setup()

```
ttk_tools.dspace.xil_api_offline_stub.Capture.setup (
              self,
              variables,
```

```
                mdf_file_path = None,
                downsampling = None,
                task_name = "Periodic Task 1" )
```

Setup data capture.

Task Name:

Default capture/measurement task_name is

- "HostService" for PHS bus systems and
- "Periodic Task 1" for VEOS/SCALEXIO systems.

See TRC file (`<model_name>.trc`) for available/configured tasks, e.g.
sampling_period[0]
{
value: 0.001
alias: "HostService"
increment: 0.001
unit: "s"
}

Parameters

variables	list of variables/paths to capture
mdf_file_path	path to MDF-file to use for stream-to-disk mode. None: keep captured data in memory.
downsampling	downsampling factor (1:n, one sample captured for every n capture cycles). Setting None uses the capture service's default setting (which should be 1:1).
task_name	name of capture task to use (see note)

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.Capture.

7.119.3.5 start()

```
ttk_tools.dspace.xil_api_offline_stub.Capture.start (
                self )
```

Start capturing.

Note

stream-to-disk: start() will raise a (somewhat generic) `TestbenchPortException` ("Could not start capturing.") if the configured MDF-file is locked by another application.
The stub implementation will not really care about this, though.

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.Capture.

7.119.3.6 stop()

```
ttk_tools.dspace.xil_api_offline_stub.Capture.stop (
              self )
```

Stop capture process.

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.Capture.

7.119.4 Member Data Documentation

7.119.4.1 capture

```
ttk_tools.dspace.xil_api_offline_stub.Capture.capture
```

actual Capture instance, will be configured during setup()

7.119.4.2 mdf_file_path

```
ttk_tools.dspace.xil_api_offline_stub.Capture.mdf_file_path
```

file path to MDF-file in stream-to-disk mode (configured during setup())

7.119.4.3 parent

```
ttk_tools.dspace.xil_api_offline_stub.Capture.parent
```

reference to parent XilTestbench instance

7.119.4.4 task_name

```
ttk_tools.dspace.xil_api_offline_stub.Capture.task_name
```

capture task name (configured during setup())

7.119.4.5 testbench

```
ttk_tools.dspace.xil_api_offline_stub.Capture.testbench
```

reference to actual testbench api

7.119.4.6 variables

```
ttk_tools.dspace.xil_api_offline_stub.Capture.variables
```

list of variables / model_paths to capture (configured during setup())

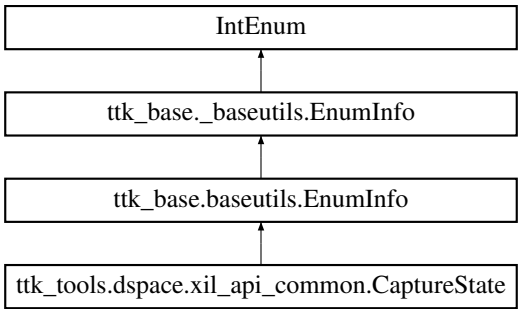
The documentation for this class was generated from the following file:

- ttk_tools/dspace/xil_api_offline_stub.py (v2.0.1)

7.120 ttk_tools.dspace.xil_api_common.CaptureState Class Reference

Data Capture States.

Inheritance diagram for ttk_tools.dspace.xil_api_common.CaptureState:



Static Public Attributes

- int eCONFIGURED = 0x00
  capture created/setup (and is stopped)
- int eACTIVATED = 0x01
  waiting for trigger (if any)
- int eRUNNING = 0x02
  capture running
- int eFINISHED = 0x03
  stop trigger reached (if any)

7.120.1 Detailed Description

Data Capture States.

See ASAM.XIL.Interfaces.Testbench.Common.Capturing.Enum.CaptureState

7.120.2 Member Data Documentation

7.120.2.1 eACTIVATED

```
int ttk_tools.dspace.xil_api_common.CaptureState.eACTIVATED = 0x01  [static]
```

waiting for trigger (if any)

7.120.2.2 eCONFIGURED

```
int ttk_tools.dspace.xil_api_common.CaptureState.eCONFIGURED = 0x00  [static]
```

capture created/setup (and is stopped)

7.120.2.3 eFINISHED

```
int ttk_tools.dspace.xil_api_common.CaptureState.eFINISHED = 0x03  [static]
```

stop trigger reached (if any)

7.120.2.4 eRUNNING

```
int ttk_tools.dspace.xil_api_common.CaptureState.eRUNNING = 0x02  [static]
```

capture running

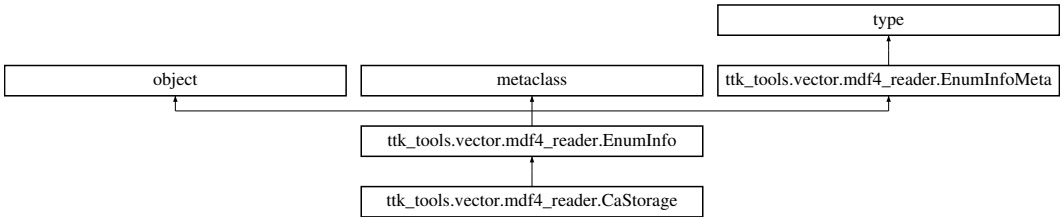
The documentation for this class was generated from the following file:

- ttk_tools/dspace/xil_api_common.py (v1.5.0)

7.121 ttk_tools.vector.mdf4_reader.CaStorage Class Reference

Channel storage (ca_storage) values: how element values are stored.

Inheritance diagram for ttk_tools.vector.mdf4_reader.CaStorage:



Static Public Attributes

- int CN_TEMPLATE = 0
- int CG_TEMPLATE = 1
- int DG_TEMPLATE = 2

7.121.1 Detailed Description

Channel storage (ca_storage) values: how element values are stored.

7.121.2 Member Data Documentation

7.121.2.1 CG_TEMPLATE

```
int ttk_tools.vector.mdf4_reader.CaStorage.CG_TEMPLATE = 1  [static]
```

7.121.2.2 CN_TEMPLATE

```
int ttk_tools.vector.mdf4_reader.CaStorage.CN_TEMPLATE = 0  [static]
```

7.121.2.3 DG_TEMPLATE

```
int ttk_tools.vector.mdf4_reader.CaStorage.DG_TEMPLATE = 2  [static]
```

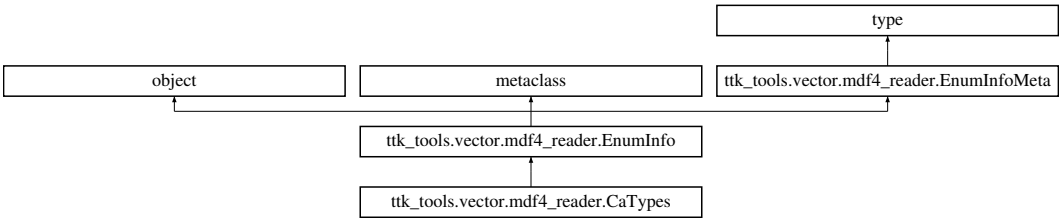
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.122 ttk_tools.vector.mdf4_reader.CaTypes Class Reference

Channel array type (ca_type) values: array semantics.

Inheritance diagram for ttk_tools.vector.mdf4_reader.CaTypes:



Static Public Attributes

- int ARRAY = 0
- int SCALING_AXIS = 1
- int LOOK_UP = 2
- int INTERVAL_AXIS = 3
- int CLASSIFICATION_RESULT = 4

7.122.1 Detailed Description

Channel array type (ca_type) values: array semantics.

7.122.2 Member Data Documentation

7.122.2.1 ARRAY

```
int ttk_tools.vector.mdf4_reader.CaTypes.ARRAY = 0  [static]
```

7.122.2.2 CLASSIFICATION_RESULT

```
int ttk_tools.vector.mdf4_reader.CaTypes.CLASSIFICATION_RESULT = 4  [static]
```

7.122.2.3 INTERVAL_AXIS

```
int ttk_tools.vector.mdf4_reader.CaTypes.INTERVAL_AXIS = 3  [static]
```

7.122.2.4 LOOK_UP

```
int ttk_tools.vector.mdf4_reader.CaTypes.LOOK_UP = 2  [static]
```

7.122.2.5 SCALING_AXIS

```
int ttk_tools.vector.mdf4_reader.CaTypes.SCALING_AXIS = 1  [static]
```

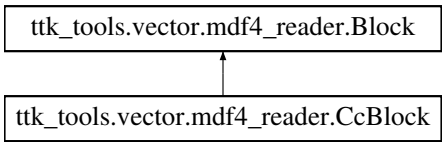
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.123 ttk_tools.vector.mdf4_reader.CcBlock Class Reference

Channel Conversion Block.

Inheritance diagram for ttk_tools.vector.mdf4_reader.CcBlock:



Public Member Functions

- __init__ (self, file_handle, pos=None)

  Read CCBLOCK and any associated MDBLOCK from file_handle, starting at the current or supplied position.
- convert (self, value)

  Convert value according to conversion type.
- __repr__ (self)

Public Attributes

- cc_precision
- cc_phy_range_min
- cc_phy_range_max
- cc_val
- cc_flags
- cc_type
- cc_tx_name

  Link to TXBLOCK with name (identifier) of conversion (can be NIL).
- cc_md_unit

  Link to TXBLOCK with name (identifier) of conversion (can be NIL).
- cc_md_comment

  Link to TXBLOCK with name (identifier) of conversion (can be NIL).
- cc_cc_inverse

  Link to TXBLOCK with name (identifier) of conversion (can be NIL).
- cc_ref

  List of additional links to TXBLOCKs with strings or to CCBLOCKs with partial conversion rules.
- name
- unit
- comment
- convert
- cc_ref_data

  list of resolved cc_ref blocks/text data
- cc_val_lut

7.123.1 Detailed Description

Channel Conversion Block.

Data records can be used to store raw values (often also denoted as implementation values or internal values). The CCBLOCK serves to specify a conversion formula to convert those raw values to physical values with a physical unit.

The result of a conversion will always be either

- a floating-point value (REAL => float) or
- a character string (UTF-8 => unicode).

7.123.2 Constructor & Destructor Documentation

7.123.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.CcBlock.__init__ (
              self,
              file_handle,
              pos = None )
```

Read CCBLOCK and any associated MDBLOCK from file_handle, starting at the current or supplied position.

Note

File pointer position will be modified

Reimplemented from ttk_tools.vector.mdf4_reader.Block.

7.123.3 Member Function Documentation

7.123.3.1 __repr__()

```
ttk_tools.vector.mdf4_reader.CcBlock.__repr__ (
              self )
```

7.123.3.2 convert()

```
ttk_tools.vector.mdf4_reader.CcBlock.convert (
              self,
              value )
```

Convert value according to conversion type.

Default 1:1 "conversion", actual method implementation may be overridden during **init**

7.123.4 Member Data Documentation

7.123.4.1 cc_cc_inverse

```
ttk_tools.vector.mdf4_reader.CcBlock.cc_cc_inverse
```

Link to TXBLOCK with name (identifier) of conversion (can be NIL).

Link to TXBLOCK/MDBLOCK with physical unit of signal data after conversion. (can be NIL). Unit only applies if no unit defined in CNBLOCK. Otherwise the unit of the channel overwrites the conversion unit. An MDBLOCK can be used to additionally reference the A-HDO unit definition. Note: for channels with cn_sync_type > 0, the unit is already defined, thus a reference to an A-HDO definition should be omitted to avoid redundancy.

Link to TXBLOCK/MDBLOCK with comment of conversion and additional information (can be NIL)

Link to CCBLOCK for inverse formula (can be NIL, must be NIL for CCBLOCK of the inverse formula (no cyclic reference allowed).

7.123.4.2 cc_flags

```
ttk_tools.vector.mdf4_reader.CcBlock.cc_flags
```

7.123.4.3 cc_md_comment

```
ttk_tools.vector.mdf4_reader.CcBlock.cc_md_comment
```

Link to TXBLOCK with name (identifier) of conversion (can be NIL).

Link to TXBLOCK/MDBLOCK with physical unit of signal data after conversion. (can be NIL). Unit only applies if no unit defined in CNBLOCK. Otherwise the unit of the channel overwrites the conversion unit. An MDBLOCK can be used to additionally reference the A-HDO unit definition. Note: for channels with cn_sync_type > 0, the unit is already defined, thus a reference to an A-HDO definition should be omitted to avoid redundancy.

Link to TXBLOCK/MDBLOCK with comment of conversion and additional information (can be NIL)

7.123.4.4 cc_md_unit

ttk_tools.vector.mdf4_reader.CcBlock.cc_md_unit

Link to TXBLOCK with name (identifier) of conversion (can be NIL).

Link to TXBLOCK/MDBLOCK with physical unit of signal data after conversion. (can be NIL). Unit only applies if no unit defined in CNBLOCK. Otherwise the unit of the channel overwrites the conversion unit. An MDBLOCK can be used to additionally reference the A-HDO unit definition. Note: for channels with cn_sync_type > 0, the unit is already defined, thus a reference to an A-HDO definition should be omitted to avoid redundancy.

7.123.4.5 cc_phy_range_max

ttk_tools.vector.mdf4_reader.CcBlock.cc_phy_range_max

7.123.4.6 cc_phy_range_min

ttk_tools.vector.mdf4_reader.CcBlock.cc_phy_range_min

7.123.4.7 cc_precision

ttk_tools.vector.mdf4_reader.CcBlock.cc_precision

7.123.4.8 cc_ref

ttk_tools.vector.mdf4_reader.CcBlock.cc_ref

List of additional links to TXBLOCKs with strings or to CCBLOCKs with partial conversion rules.

Length of list is given by cc_ref_count. The list can be empty. Details are explained in formula-specific block supplement.

7.123.4.9 cc_ref_data

ttk_tools.vector.mdf4_reader.CcBlock.cc_ref_data

list of resolved cc_ref blocks/text data

7.123.4.10 cc_tx_name

ttk_tools.vector.mdf4_reader.CcBlock.cc_tx_name

Link to TXBLOCK with name (identifier) of conversion (can be NIL).

7.123.4.11 cc_type

ttk_tools.vector.mdf4_reader.CcBlock.cc_type

7.123.4.12 cc_val

ttk_tools.vector.mdf4_reader.CcBlock.cc_val

7.123.4.13 cc_val_lut

ttk_tools.vector.mdf4_reader.CcBlock.cc_val_lut

7.123.4.14 comment

ttk_tools.vector.mdf4_reader.CcBlock.comment

7.123.4.15 convert

ttk_tools.vector.mdf4_reader.CcBlock.convert

7.123.4.16 name

ttk_tools.vector.mdf4_reader.CcBlock.name

7.123.4.17 unit

ttk_tools.vector.mdf4_reader.CcBlock.unit

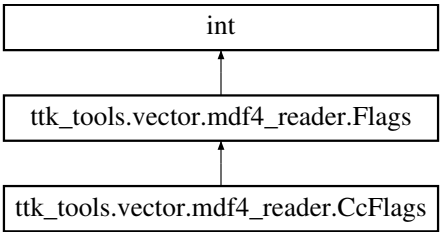
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.124 ttk_tools.vector.mdf4_reader.CcFlags Class Reference

Container for decoded uint16 cc_flags.

Inheritance diagram for ttk_tools.vector.mdf4_reader.CcFlags:



Public Member Functions

- __init__ (self, flags)

Public Attributes

- precision_valid
  Bit 0: Precision valid flag.
- phys_value_range_valid
  Bit 1: Physical value range valid flag.
- status_string
  Bit 2: Status string flag.

7.124.1 Detailed Description

Container for decoded uint16 cc_flags.

Only intended for read-only access, changes in named flags will not modify original "flags" value.

7.124.2 Constructor & Destructor Documentation

7.124.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.CcFlags.__init__ (
            self,
            flags )
```

Reimplemented from ttk_tools.vector.mdf4_reader.Flags.

7.124.3 Member Data Documentation

7.124.3.1 phys_value_range_valid

```
ttk_tools.vector.mdf4_reader.CcFlags.phys_value_range_valid
```

Bit 1: Physical value range valid flag.

If set, both the minimum and the maximum physical value that occurred after conversion for this signal within the samples recorded in this file are known and stored incc_phy_range_min and cc_phy_range_max.
Otherwise the two fields are not valid. Note: physical value ranges can only be expressed for conversions which return a numeric value (REAL). For conversions returning a string value or for the inverse conversion rule the flag must not be set.

7.124.3.2 precision_valid

```
ttk_tools.vector.mdf4_reader.CcFlags.precision_valid
```

Bit 0: Precision valid flag.

If set, the precision value for display of floating point values specified in cc_precision is valid

7.124.3.3 status_string

`ttk_tools.vector.mdf4_reader.CcFlags.status_string`

Bit 2: Status string flag.

This flag indicates for conversion types 7 and 8 (value/value range to text/scale conversion tabular look-up) that the normal table entries are status strings (only reference to TXBLOCK), and the actual conversion rule is given in CCBLOCK referenced by default value.
This also implies special handling of limits, see [MCD-2 MC] keyword STATUS_STRING_REF. Can only be set for 7 <= cc_type <= 8.

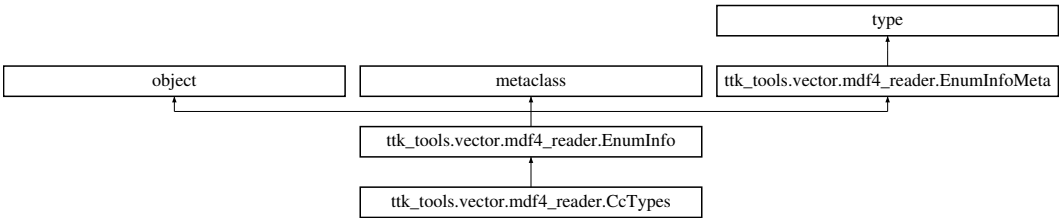
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.125 ttk_tools.vector.mdf4_reader.CcTypes Class Reference

Conversion type / formula identifier (cc_type) values.

Inheritance diagram for ttk_tools.vector.mdf4_reader.CcTypes:



Static Public Attributes

- int NONE = 0
- int LINEAR = 1
- int RATIONAL = 2
- int ALGEBRAIC = 3
- int TAB_VAL2VAL_IN = 4
- int TAB_VAL2VAL = 5
- int TAB_RNG2VAL = 6
- int TAB_VAL2TXT = 7
- int TAB_RNG2TXT = 8
- int TAB_TXT2VAL = 9
- int TAB_TXT2TXT = 10

7.125.1 Detailed Description

Conversion type / formula identifier (cc_type) values.

7.125.2 Member Data Documentation

7.125.2.1 ALGEBRAIC

```
int ttk_tools.vector.mdf4_reader.CcTypes.ALGEBRAIC = 3  [static]
```

7.125.2.2 LINEAR

```
int ttk_tools.vector.mdf4_reader.CcTypes.LINEAR = 1  [static]
```

7.125.2.3 NONE

```
int ttk_tools.vector.mdf4_reader.CcTypes.NONE = 0  [static]
```

7.125.2.4 RATIONAL

```
int ttk_tools.vector.mdf4_reader.CcTypes.RATIONAL = 2  [static]
```

7.125.2.5 TAB_RNG2TXT

```
int ttk_tools.vector.mdf4_reader.CcTypes.TAB_RNG2TXT = 8  [static]
```

7.125.2.6 TAB_RNG2VAL

```
int ttk_tools.vector.mdf4_reader.CcTypes.TAB_RNG2VAL = 6  [static]
```

7.125.2.7 TAB_TXT2TXT

```
int ttk_tools.vector.mdf4_reader.CcTypes.TAB_TXT2TXT = 10  [static]
```

7.125.2.8 TAB_TXT2VAL

```
int ttk_tools.vector.mdf4_reader.CcTypes.TAB_TXT2VAL = 9  [static]
```

7.125.2.9 TAB_VAL2TXT

```
int ttk_tools.vector.mdf4_reader.CcTypes.TAB_VAL2TXT = 7  [static]
```

7.125.2.10 TAB_VAL2VAL

```
int ttk_tools.vector.mdf4_reader.CcTypes.TAB_VAL2VAL = 5  [static]
```

7.125.2.11 TAB_VAL2VAL_IN

```
int ttk_tools.vector.mdf4_reader.CcTypes.TAB_VAL2VAL_IN = 4  [static]
```

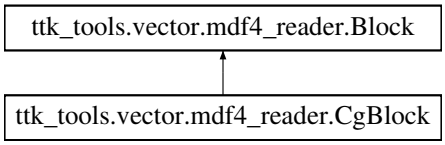
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.126 ttk_tools.vector.mdf4_reader.CgBlock Class Reference

Channel Group Block.

Inheritance diagram for ttk_tools.vector.mdf4_reader.CgBlock:



Public Member Functions

- __init__ (self, file_handle, pos=None)

  Read CGBLOCK and any associated MDBLOCK from file_handle, starting at the current position.
- vlsd_total_data_size (self)

  For VLSD channel groups: Total size of VLSD values in bytes (not including cg_cycle_count ∗ 4 bytes for uint32 length header value).
- __repr__ (self)

  Pseudo-repr.

Public Attributes

- cg_cg_next

  Pointer to next channel group block (CGBLOCK) (can be NIL)
- cg_cn_first

  Pointer to next channel group block (CGBLOCK) (can be NIL)
- cg_tx_acq_name

  Pointer to next channel group block (CGBLOCK) (can be NIL)
- cg_si_acq_source

  Pointer to next channel group block (CGBLOCK) (can be NIL)
- cg_sr_first

  Pointer to next channel group block (CGBLOCK) (can be NIL)
- cg_md_comment

  Pointer to next channel group block (CGBLOCK) (can be NIL)
- cg_record_id
- cg_cycle_count
- cg_path_separator
- cg_data_bytes
- cg_inval_bytes
- cg_flags
- acq_name
- acq_source
- comment
- channels

7.126.1 Detailed Description

Channel Group Block.

The CGBLOCK contains a collection of channels which are stored in one record, i.e. which have equal sampling.

Note

Special Case VLSD: The only exception is a "VLSD channel group" for variable length signal data (VLSD). It has no channel collection and can only occur in an unsorted data group.
A VLSD channel group describes signal values of variable length which are stored as variable length records in a normal DTBLOCK.

7.126.2 Constructor & Destructor Documentation

7.126.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.CgBlock.__init__ (
              self,
              file_handle,
              pos = None )
```

Read CGBLOCK and any associated MDBLOCK from file_handle, starting at the current position.

Note

File pointer position will be modified

Reimplemented from ttk_tools.vector.mdf4_reader.Block.

7.126.3 Member Function Documentation

7.126.3.1 __repr__()

```
ttk_tools.vector.mdf4_reader.CgBlock.__repr__ (
              self )
```

Pseudo-repr.

7.126.3.2 vlsd_total_data_size()

```
ttk_tools.vector.mdf4_reader.CgBlock.vlsd_total_data_size (
              self )
```

For VLSD channel groups: Total size of VLSD values in bytes (not including cg_cycle_count $*$ 4 bytes for uint32 length header value).

Returns

a size in bytes or None if this is not a VLSD channel group.

7.126.4 Member Data Documentation

7.126.4.1 acq_name

ttk_tools.vector.mdf4_reader.CgBlock.acq_name

7.126.4.2 acq_source

ttk_tools.vector.mdf4_reader.CgBlock.acq_source

7.126.4.3 cg_cg_next

ttk_tools.vector.mdf4_reader.CgBlock.cg_cg_next

Pointer to next channel group block (CGBLOCK) (can be NIL)

7.126.4.4 cg_cn_first

ttk_tools.vector.mdf4_reader.CgBlock.cg_cn_first

Pointer to next channel group block (CGBLOCK) (can be NIL)

Pointer to first channel block (CNBLOCK) (can be NIL, must be NIL for VLSD CGBLOCK, i.e. if "VLSD channel group" flag (bit 0) is set)

7.126.4.5 cg_cycle_count

ttk_tools.vector.mdf4_reader.CgBlock.cg_cycle_count

7.126.4.6 cg_data_bytes

ttk_tools.vector.mdf4_reader.CgBlock.cg_data_bytes

7.126.4.7 cg_flags

ttk_tools.vector.mdf4_reader.CgBlock.cg_flags

7.126.4.8 cg_inval_bytes

ttk_tools.vector.mdf4_reader.CgBlock.cg_inval_bytes

7.126.4.9 cg_md_comment

ttk_tools.vector.mdf4_reader.CgBlock.cg_md_comment

Pointer to next channel group block (CGBLOCK) (can be NIL)

Pointer to first channel block (CNBLOCK) (can be NIL, must be NIL for VLSD CGBLOCK, i.e. if "VLSD channel group" flag (bit 0) is set)

Pointer to acquisition name (TXBLOCK) (can be NI, must be NIL for VLSD CGBLOCK9

Pointer to acquisition source (SIBLOCK) (can be NIL, must be NIL for VLSD CGBLOCK) See also rules for uniqueness in Chapter 5.4.3 Identification of Channels.

Pointer to first sample reduction block (SRBLOCK) (can be NIL, must be NIL for VLSD CGBLOCK)

Pointer to comment and additional information (TXBLOCK or MDBLOCK) (can be NIL, must be NIL for VLSD CGBLOCK)

7.126.4.10 cg_path_separator

ttk_tools.vector.mdf4_reader.CgBlock.cg_path_separator

7.126.4.11 cg_record_id

ttk_tools.vector.mdf4_reader.CgBlock.cg_record_id

7.126.4.12 cg_si_acq_source

ttk_tools.vector.mdf4_reader.CgBlock.cg_si_acq_source

Pointer to next channel group block (CGBLOCK) (can be NIL)

Pointer to first channel block (CNBLOCK) (can be NIL, must be NIL for VLSD CGBLOCK, i.e. if "VLSD channel group" flag (bit 0) is set)

Pointer to acquisition name (TXBLOCK) (can be NI, must be NIL for VLSD CGBLOCK9

Pointer to acquisition source (SIBLOCK) (can be NIL, must be NIL for VLSD CGBLOCK) See also rules for uniqueness in Chapter 5.4.3 Identification of Channels.

7.126.4.13 cg_sr_first

ttk_tools.vector.mdf4_reader.CgBlock.cg_sr_first

Pointer to next channel group block (CGBLOCK) (can be NIL)

Pointer to first channel block (CNBLOCK) (can be NIL, must be NIL for VLSD CGBLOCK, i.e. if "VLSD channel group" flag (bit 0) is set)

Pointer to acquisition name (TXBLOCK) (can be NI, must be NIL for VLSD CGBLOCK9

Pointer to acquisition source (SIBLOCK) (can be NIL, must be NIL for VLSD CGBLOCK) See also rules for uniqueness in Chapter 5.4.3 Identification of Channels.

Pointer to first sample reduction block (SRBLOCK) (can be NIL, must be NIL for VLSD CGBLOCK)

7.126.4.14 cg_tx_acq_name

`ttk_tools.vector.mdf4_reader.CgBlock.cg_tx_acq_name`

Pointer to next channel group block (CGBLOCK) (can be NIL)

Pointer to first channel block (CNBLOCK) (can be NIL, must be NIL for VLSD CGBLOCK, i.e. if "VLSD channel group" flag (bit 0) is set)

Pointer to acquisition name (TXBLOCK) (can be NI, must be NIL for VLSD CGBLOCK9

7.126.4.15 channels

`ttk_tools.vector.mdf4_reader.CgBlock.channels`

7.126.4.16 comment

`ttk_tools.vector.mdf4_reader.CgBlock.comment`

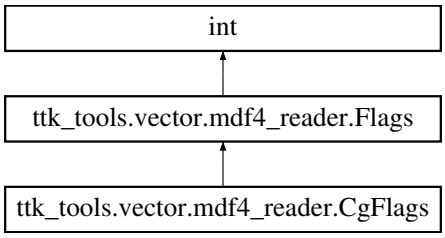
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.127 ttk_tools.vector.mdf4_reader.CgFlags Class Reference

Container for decoded cg_flags.

Inheritance diagram for ttk_tools.vector.mdf4_reader.CgFlags:



Public Member Functions

- __init__ (self, flags)

Public Attributes

- vlsd_channel_group

  Bit 0: VLSD channel group flag.
- bus_event_channel_group

  Bit 1: Bus event channel group flag.
- plain_bus_event_channel_group

7.127.1 Detailed Description

Container for decoded cg_flags.

Only intended for read-only access, changes in named flags will not modify original "flags" value.

7.127.2 Constructor & Destructor Documentation

7.127.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.CgFlags.__init__ (
            self,
            flags )
```

Reimplemented from ttk_tools.vector.mdf4_reader.Flags.

7.127.3 Member Data Documentation

7.127.3.1 bus_event_channel_group

```
ttk_tools.vector.mdf4_reader.CgFlags.bus_event_channel_group
```

Bit 1: Bus event channel group flag.

If set, this channel group contains information about a bus event, i.e. it contains a structure channel with bit 10 (bus event flag) set in cn_flags.
For details please refer to [MDF-BL].
Valid since MDF 4.1.0, should not be set for earlier versions

7.127.3.2 plain_bus_event_channel_group

```
ttk_tools.vector.mdf4_reader.CgFlags.plain_bus_event_channel_group
```

7.127.3.3 vlsd_channel_group

```
ttk_tools.vector.mdf4_reader.CgFlags.vlsd_channel_group
```

Bit 0: VLSD channel group flag.

If set, this is a "variable length signal data" (VLSD) channel group.

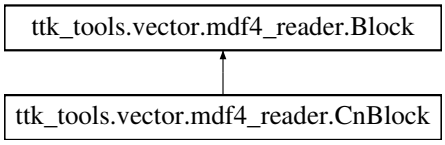
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.128 ttk_tools.vector.mdf4_reader.CnBlock Class Reference

Channel Block.

Inheritance diagram for ttk_tools.vector.mdf4_reader.CnBlock:



Public Member Functions

- __init__ (self, file_handle, pos=None)

  Read CNBLOCK and any associated MDBLOCK from file_handle, starting at the current or supplied position.
- decodeSingleValue (self, raw)

  Decode a single value from raw data to the configured data type.
- decode (self, data_record, data_record_index=0)

  Decode relevant raw bytes from data_record to get a single data point for this channel.
- __repr__ (self)

Public Attributes

- cn_type
- cn_sync_type
- cn_data_type
- cn_bit_offset
- cn_byte_offset
- cn_bit_count
- cn_inval_bit_pos
- cn_precision
- cn_val_range_min
- cn_val_range_max
- cn_limit_min
- cn_limit_max
- cn_limit_ext_min
- cn_limit_ext_max
- cn_flags
- cg_inval_bytes
- cn_cn_next

  Pointer to next channel block (CNBLOCK) (can be NIL) Note: this will be used in parent CGBLOCK to resolve all assigned channels.
- cn_composition

  Pointer to next channel block (CNBLOCK) (can be NIL) Note: this will be used in parent CGBLOCK to resolve all assigned channels.
- cn_tx_name

  Pointer to next channel block (CNBLOCK) (can be NIL) Note: this will be used in parent CGBLOCK to resolve all assigned channels.
- cn_si_source

Pointer to next channel block (CNBLOCK) (can be NIL) Note: this will be used in parent CGBLOCK to resolve all assigned channels.

- cn_cc_conversion

  Pointer to next channel block (CNBLOCK) (can be NIL) Note: this will be used in parent CGBLOCK to resolve all assigned channels.
- cn_data

  Pointer to next channel block (CNBLOCK) (can be NIL) Note: this will be used in parent CGBLOCK to resolve all assigned channels.
- cn_md_unit

  Pointer to next channel block (CNBLOCK) (can be NIL) Note: this will be used in parent CGBLOCK to resolve all assigned channels.
- cn_md_comment

  Pointer to next channel block (CNBLOCK) (can be NIL) Note: this will be used in parent CGBLOCK to resolve all assigned channels.
- cn_at_references

  List of attachments for this channel (references to ATBLOCKs in global linked list of ATBLOCKs).
- cn_default_x
- name
- composition
- conversion

  Conversion formula (CCBLOCK, from cn_cc_conversion).
- unit
- comment

  additional information about the channel self.comment = getOptionalBlock(file_handle, self.cn_md_comment, Md↩ TxBlock)
- source

  Channel source (SIBLOCK, from cn_si_source).
- bytes_used

  total number of full bytes within value will be located.
- bit_mask

  mask for used bits, aligned to bit 0 (i.e.
- data

  decoded data of this channel (will be filled later)

Static Public Attributes

- dict fmt_lookup

  lookup for simple value unpacking formats (for byte-aligned values)

Protected Attributes

- _has_array_composition

  just for convenience:
- _has_struct_composition
- _sd_data

  Signal Data (raw bytes)
- _sd_data_cg_link

  Link to VLSD channel group block (CGBLOCK), if any.
- _is_unsorted_vlsd

  unsorted: variable length data spliced into data group's DTBLOCK data
- _is_sorted_vlsd

sorted: variable length data in this channel's SDBLOCK data

- _struct_fmt
- _is_byte_aligned

  data is aligned to n full bytes
- _is_big_endian

  data is in big endian byte order
- _is_virtual

  data is calculated from record index
- _is_master

  channel is a "master" channel (e.g.

7.128.1 Detailed Description

Channel Block.

A CNBLOCK describes a channel, i.e. it contains information about the recorded signal and how its signal values are stored in the MDF file.

7.128.2 Constructor & Destructor Documentation

7.128.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.CnBlock.__init__ (
              self,
              file_handle,
              pos = None )
```

Read CNBLOCK and any associated MDBLOCK from file_handle, starting at the current or supplied position.

Note

File pointer position will be modified

Reimplemented from ttk_tools.vector.mdf4_reader.Block.

7.128.3 Member Function Documentation

7.128.3.1 __repr__()

```
ttk_tools.vector.mdf4_reader.CnBlock.__repr__ (
              self )
```

7.128.3.2 decode()

```
ttk_tools.vector.mdf4_reader.CnBlock.decode (
              self,
              data_record,
              data_record_index = 0 )
```

Decode relevant raw bytes from data_record to get a single data point for this channel.

Parameters

data_record	reference to full raw data record (but just the data, no record_id/header or invalidation bits)
data_record_index	sequential index of data_record (only relevant for decoding virtual data like virtual master channels)

7.128.3.3 decodeSingleValue()

```
ttk_tools.vector.mdf4_reader.CnBlock.decodeSingleValue (
              self,
              raw )
```

Decode a single value from raw data to the configured data type.

Note

Split from decode() to better deal with arrays.

Parameters

raw	buffer of raw bytes, has to be the exact size for the channel's configured data type

Returns

a single value

7.128.4 Member Data Documentation

7.128.4.1 _has_array_composition

```
ttk_tools.vector.mdf4_reader.CnBlock._has_array_composition  [protected]
```

just for convenience:

7.128.4.2 _has_struct_composition

```
ttk_tools.vector.mdf4_reader.CnBlock._has_struct_composition  [protected]
```

7.128.4.3 _is_big_endian

```
ttk_tools.vector.mdf4_reader.CnBlock._is_big_endian  [protected]
```

data is in big endian byte order

7.128.4.4 _is_byte_aligned

ttk_tools.vector.mdf4_reader.CnBlock._is_byte_aligned [protected]

data is aligned to n full bytes

7.128.4.5 _is_master

ttk_tools.vector.mdf4_reader.CnBlock._is_master [protected]

channel is a "master" channel (e.g.

time channel)

7.128.4.6 _is_sorted_vlsd

ttk_tools.vector.mdf4_reader.CnBlock._is_sorted_vlsd [protected]

sorted: variable length data in this channel's SDBLOCK data

7.128.4.7 _is_unsorted_vlsd

ttk_tools.vector.mdf4_reader.CnBlock._is_unsorted_vlsd [protected]

unsorted: variable length data spliced into data group's DTBLOCK data

7.128.4.8 _is_virtual

ttk_tools.vector.mdf4_reader.CnBlock._is_virtual [protected]

data is calculated from record index

7.128.4.9 _sd_data

ttk_tools.vector.mdf4_reader.CnBlock._sd_data [protected]

Signal Data (raw bytes)

7.128.4.10 _sd_data_cg_link

ttk_tools.vector.mdf4_reader.CnBlock._sd_data_cg_link [protected]

Link to VLSD channel group block (CGBLOCK), if any.

7.128.4.11 _struct_fmt

ttk_tools.vector.mdf4_reader.CnBlock._struct_fmt [protected]

7.128.4.12 bit_mask

ttk_tools.vector.mdf4_reader.CnBlock.bit_mask

mask for used bits, aligned to bit 0 (i.e.

shift bits before if needed)

7.128.4.13 bytes_used

ttk_tools.vector.mdf4_reader.CnBlock.bytes_used

total number of full bytes within value will be located.

Note: for bit_offset == 0 and bit_count % 8 == 0, this will simply whittle down to bytes_used = bit_count // 8

7.128.4.14 cg_inval_bytes

ttk_tools.vector.mdf4_reader.CnBlock.cg_inval_bytes

7.128.4.15 cn_at_references

ttk_tools.vector.mdf4_reader.CnBlock.cn_at_references

List of attachments for this channel (references to ATBLOCKs in global linked list of ATBLOCKs).

The length of the list is given by cn_attachment_count. It can be empty (cn_attachment_count = 0), i.e. there are no attachments for this channel. Valid since MDF 4.1.0

7.128.4.16 cn_bit_count

ttk_tools.vector.mdf4_reader.CnBlock.cn_bit_count

7.128.4.17 cn_bit_offset

ttk_tools.vector.mdf4_reader.CnBlock.cn_bit_offset

7.128.4.18 cn_byte_offset

ttk_tools.vector.mdf4_reader.CnBlock.cn_byte_offset

7.128.4.19 cn_cc_conversion

ttk_tools.vector.mdf4_reader.CnBlock.cn_cc_conversion

Pointer to next channel block (CNBLOCK) (can be NIL) Note: this will be used in parent CGBLOCK to resolve all assigned channels.

Composition of channels: Pointer to channel array block (CABLOCK) or channel block (CNBLOCK) (can be NIL).

Pointer to TXBLOCK with name (identification) of channel. Name must conform to MDF naming rules. The combination of name and source name and path (both from SIBLOCK) must be unique within all channels of this channel group. Note: Alternative names (e.g. display name) can be stored in MDBLOCK of cn_md_comment

Pointer to channel source (SIBLOCK) (can be NIL) Must be NIL for component channels (members of a structure or array elements) because they all must have the same source and thus simply use the SIBLOCK of their parent CNBLOCK (direct child of CGBLOCK).

Pointer to the conversion formula (CCBLOCK) (can be NIL, must be NIL for complex channel data types, i.e. for cn_data_type >= 10). If the pointer is NIL, this means that a 1:1 conversion is used (phys = int).

7.128.4.20 cn_cn_next

ttk_tools.vector.mdf4_reader.CnBlock.cn_cn_next

Pointer to next channel block (CNBLOCK) (can be NIL) Note: this will be used in parent CGBLOCK to resolve all assigned channels.

7.128.4.21 cn_composition

ttk_tools.vector.mdf4_reader.CnBlock.cn_composition

Pointer to next channel block (CNBLOCK) (can be NIL) Note: this will be used in parent CGBLOCK to resolve all assigned channels.

Composition of channels: Pointer to channel array block (CABLOCK) or channel block (CNBLOCK) (can be NIL).

7.128.4.22 cn_data

ttk_tools.vector.mdf4_reader.CnBlock.cn_data

Pointer to next channel block (CNBLOCK) (can be NIL) Note: this will be used in parent CGBLOCK to resolve all assigned channels.

Composition of channels: Pointer to channel array block (CABLOCK) or channel block (CNBLOCK) (can be NIL).

Pointer to TXBLOCK with name (identification) of channel. Name must conform to MDF naming rules. The combination of name and source name and path (both from SIBLOCK) must be unique within all channels of this channel group. Note: Alternative names (e.g. display name) can be stored in MDBLOCK of cn_md_comment

Pointer to channel source (SIBLOCK) (can be NIL) Must be NIL for component channels (members of a structure or array elements) because they all must have the same source and thus simply use the SIBLOCK of their parent CNBLOCK (direct child of CGBLOCK).

Pointer to the conversion formula (CCBLOCK) (can be NIL, must be NIL for complex channel data types, i.e. for cn_data_type >= 10). If the pointer is NIL, this means that a 1:1 conversion is used (phys = int).

Pointer to channel type specific signal data

- For variable length data channel (cn_type = 1): unique link to signal data block (SDBLOCK or DZBLOCK for this block type) or data list block (DLBLOCK of signal data blocks or its HLBLOCK) or, only for unsorted data groups, referencing link to a VLSD channel group block (CGBLOCK). Can only be NIL if SDBLOCK would be empty.
- For synchronization channel (cn_type = 4): referencing link to attachment block (ATBLOCK) in global linked list of ATBLOCKs starting at hd_at_first. Cannot be NIL.
- For maximum length data channel (cn_type = 5): referencing link to channel block (CNBLOCK) in same channel group. Cannot be NIL.

Must be NIL for all other channel types. See respective channel type for further details.

7.128.4.23 cn_data_type

ttk_tools.vector.mdf4_reader.CnBlock.cn_data_type

7.128.4.24 cn_default_x

ttk_tools.vector.mdf4_reader.CnBlock.cn_default_x

7.128.4.25 cn_flags

ttk_tools.vector.mdf4_reader.CnBlock.cn_flags

7.128.4.26 cn_inval_bit_pos

ttk_tools.vector.mdf4_reader.CnBlock.cn_inval_bit_pos

7.128.4.27 cn_limit_ext_max

ttk_tools.vector.mdf4_reader.CnBlock.cn_limit_ext_max

7.128.4.28 cn_limit_ext_min

ttk_tools.vector.mdf4_reader.CnBlock.cn_limit_ext_min

7.128.4.29 cn_limit_max

ttk_tools.vector.mdf4_reader.CnBlock.cn_limit_max

7.128.4.30 cn_limit_min

ttk_tools.vector.mdf4_reader.CnBlock.cn_limit_min

7.128.4.31 cn_md_comment

ttk_tools.vector.mdf4_reader.CnBlock.cn_md_comment

Pointer to next channel block (CNBLOCK) (can be NIL) Note: this will be used in parent CGBLOCK to resolve all assigned channels.

Composition of channels: Pointer to channel array block (CABLOCK) or channel block (CNBLOCK) (can be NIL).

Pointer to TXBLOCK with name (identification) of channel. Name must conform to MDF naming rules. The combination of name and source name and path (both from SIBLOCK) must be unique within all channels of this channel group. Note: Alternative names (e.g. display name) can be stored in MDBLOCK of cn_md_comment

Pointer to channel source (SIBLOCK) (can be NIL) Must be NIL for component channels (members of a structure or array elements) because they all must have the same source and thus simply use the SIBLOCK of their parent CNBLOCK (direct child of CGBLOCK).

Pointer to the conversion formula (CCBLOCK) (can be NIL, must be NIL for complex channel data types, i.e. for cn_data_type >= 10). If the pointer is NIL, this means that a 1:1 conversion is used (phys = int).

Pointer to channel type specific signal data

- For variable length data channel (cn_type = 1): unique link to signal data block (SDBLOCK or DZBLOCK for this block type) or data list block (DLBLOCK of signal data blocks or its HLBLOCK) or, only for unsorted data groups, referencing link to a VLSD channel group block (CGBLOCK). Can only be NIL if SDBLOCK would be empty.
- For synchronization channel (cn_type = 4): referencing link to attachment block (ATBLOCK) in global linked list of ATBLOCKs starting at hd_at_first. Cannot be NIL.
- For maximum length data channel (cn_type = 5): referencing link to channel block (CNBLOCK) in same channel group. Cannot be NIL.

Must be NIL for all other channel types. See respective channel type for further details.

Pointer to TXBLOCK/MDBLOCK with designation for physical unit of signal data (after conversion) or (only for channel data types "MIME sample" and "MIME stream") to MIME context-type text. (can be NIL).

- The unit can be used if no conversion rule is specified or to overwrite the unit specified for the conversion rule (e.g. if a conversion rule is shared between channels). If the link is NIL, then the unit from the conversion rule must be used. If the content is an empty string, no unit should be displayed. If an MDBLOCK is used, in addition the A-HDO unit definition can be stored, see Table 39. Note: for (virtual) master and synchronization channels the A-HDO definition should be omitted to avoid redundancy. Here the unit is already specified by cn_sync_type of the channel.
- In case of channel data types "MIME sample" and "MIME stream", the text of the unit must be the content-type text of a MIME type which specifies the content of the values of the channel (either fixed length in record or variable length in SDBLOCK). The MIME content-type string must be written in lowercase, and it must apply to the same rules as defined for at_tx_mimetype in ATBLOCKs.

Pointer to TXBLOCK/MDBLOCK with comment and additional information about the channel (can be NIL)

7.128.4.32 cn_md_unit

ttk_tools.vector.mdf4_reader.CnBlock.cn_md_unit

Pointer to next channel block (CNBLOCK) (can be NIL) Note: this will be used in parent CGBLOCK to resolve all assigned channels.

Composition of channels: Pointer to channel array block (CABLOCK) or channel block (CNBLOCK) (can be NIL).

Pointer to TXBLOCK with name (identification) of channel. Name must conform to MDF naming rules. The combination of name and source name and path (both from SIBLOCK) must be unique within all channels of this channel group. Note: Alternative names (e.g. display name) can be stored in MDBLOCK of cn_md_comment

Pointer to channel source (SIBLOCK) (can be NIL) Must be NIL for component channels (members of a structure or array elements) because they all must have the same source and thus simply use the SIBLOCK of their parent CNBLOCK (direct child of CGBLOCK).

Pointer to the conversion formula (CCBLOCK) (can be NIL, must be NIL for complex channel data types, i.e. for cn_data_type >= 10). If the pointer is NIL, this means that a 1:1 conversion is used (phys = int).

Pointer to channel type specific signal data

- For variable length data channel (cn_type = 1): unique link to signal data block (SDBLOCK or DZBLOCK for this block type) or data list block (DLBLOCK of signal data blocks or its HLBLOCK) or, only for unsorted data groups, referencing link to a VLSD channel group block (CGBLOCK). Can only be NIL if SDBLOCK would be empty.
- For synchronization channel (cn_type = 4): referencing link to attachment block (ATBLOCK) in global linked list of ATBLOCKs starting at hd_at_first. Cannot be NIL.
- For maximum length data channel (cn_type = 5): referencing link to channel block (CNBLOCK) in same channel group. Cannot be NIL.

Must be NIL for all other channel types. See respective channel type for further details.

Pointer to TXBLOCK/MDBLOCK with designation for physical unit of signal data (after conversion) or (only for channel data types "MIME sample" and "MIME stream") to MIME context-type text. (can be NIL).

- The unit can be used if no conversion rule is specified or to overwrite the unit specified for the conversion rule (e.g. if a conversion rule is shared between channels). If the link is NIL, then the unit from the conversion rule must be used. If the content is an empty string, no unit should be displayed. If an MDBLOCK is used, in addition the A-HDO unit definition can be stored, see Table 39. Note: for (virtual) master and synchronization channels the A-HDO definition should be omitted to avoid redundancy. Here the unit is already specified by cn_sync_type of the channel.
- In case of channel data types "MIME sample" and "MIME stream", the text of the unit must be the content-type text of a MIME type which specifies the content of the values of the channel (either fixed length in record or variable length in SDBLOCK). The MIME content-type string must be written in lowercase, and it must apply to the same rules as defined for at_tx_mimetype in ATBLOCKs.

7.128.4.33 cn_precision

ttk_tools.vector.mdf4_reader.CnBlock.cn_precision

7.128.4.34 cn_si_source

ttk_tools.vector.mdf4_reader.CnBlock.cn_si_source

Pointer to next channel block (CNBLOCK) (can be NIL) Note: this will be used in parent CGBLOCK to resolve all assigned channels.

Composition of channels: Pointer to channel array block (CABLOCK) or channel block (CNBLOCK) (can be NIL).

Pointer to TXBLOCK with name (identification) of channel. Name must conform to MDF naming rules. The combination of name and source name and path (both from SIBLOCK) must be unique within all channels of this channel group. Note: Alternative names (e.g. display name) can be stored in MDBLOCK of cn_md_comment

Pointer to channel source (SIBLOCK) (can be NIL) Must be NIL for component channels (members of a structure or array elements) because they all must have the same source and thus simply use the SIBLOCK of their parent CNBLOCK (direct child of CGBLOCK).

7.128.4.35 cn_sync_type

ttk_tools.vector.mdf4_reader.CnBlock.cn_sync_type

7.128.4.36 cn_tx_name

ttk_tools.vector.mdf4_reader.CnBlock.cn_tx_name

Pointer to next channel block (CNBLOCK) (can be NIL) Note: this will be used in parent CGBLOCK to resolve all assigned channels.

Composition of channels: Pointer to channel array block (CABLOCK) or channel block (CNBLOCK) (can be NIL).

Pointer to TXBLOCK with name (identification) of channel. Name must conform to MDF naming rules. The combination of name and source name and path (both from SIBLOCK) must be unique within all channels of this channel group. Note: Alternative names (e.g. display name) can be stored in MDBLOCK of cn_md_comment

7.128.4.37 cn_type

ttk_tools.vector.mdf4_reader.CnBlock.cn_type

7.128.4.38 cn_val_range_max

ttk_tools.vector.mdf4_reader.CnBlock.cn_val_range_max

7.128.4.39 cn_val_range_min

ttk_tools.vector.mdf4_reader.CnBlock.cn_val_range_min

7.128.4.40 comment

ttk_tools.vector.mdf4_reader.CnBlock.comment

additional information about the channel self.comment = getOptionalBlock(file_handle, self.cn_md_comment, Md↩ TxBlock)

7.128.4.41 composition

ttk_tools.vector.mdf4_reader.CnBlock.composition

7.128.4.42 conversion

ttk_tools.vector.mdf4_reader.CnBlock.conversion

Conversion formula (CCBLOCK, from cn_cc_conversion).

Conversion formula can be NIL, must be NIL for complex channel data types, i.e. for cn_data_type >= 10). self.conversion = getOptionalBlock(file_handle, self.cn_cc_conversion, CcBlock)

7.128.4.43 data

ttk_tools.vector.mdf4_reader.CnBlock.data

decoded data of this channel (will be filled later)

7.128.4.44 fmt_lookup

dict ttk_tools.vector.mdf4_reader.CnBlock.fmt_lookup [static]

lookup for simple value unpacking formats (for byte-aligned values)

7.128.4.45 name

ttk_tools.vector.mdf4_reader.CnBlock.name

7.128.4.46 source

ttk_tools.vector.mdf4_reader.CnBlock.source

Channel source (SIBLOCK, from cn_si_source).

Channel source can be NIL, must be NIL for component channels (members of a structure or array elements) because they all must have the same source and thus simply use the SIBLOCK of their parent CNBLOCK (direct child of CGBLOCK).

7.128.4.47 unit

`ttk_tools.vector.mdf4_reader.CnBlock.unit`

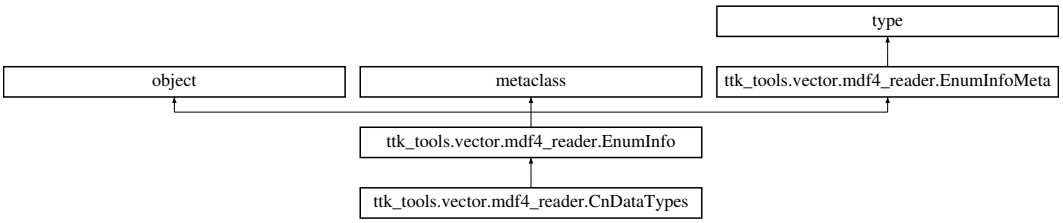
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.129 ttk_tools.vector.mdf4_reader.CnDataTypes Class Reference

Channel data type (cn_data_type) values.

Inheritance diagram for ttk_tools.vector.mdf4_reader.CnDataTypes:



Static Public Attributes

- int UINT_LE = 0
- int UINT_BE = 1
- int SINT_LE = 2
- int SINT_BE = 3
- int IEEE754_LE = 4
- int IEEE754_BE = 5
- int STRING_SBC = 6
- int STRING_UTF8 = 7
- int STRING_UTF16_LE = 8
- int STRING_UTF16_BE = 9
- int BYTE_ARRAY = 10
- int MIME_SAMPLE = 11
- int MIME_STREAM = 12
- int CANOPEN_DATE = 13
- int CANOPEN_TIME = 14
- signed_types = set((SINT_LE, SINT_BE))
- unsigned_types = set((UINT_LE, UINT_BE))
- float_types = set((IEEE754_LE, IEEE754_BE))
- string_types = set((STRING_SBC, STRING_UTF8, STRING_UTF16_LE, STRING_UTF16_BE))
- complex_types = set((BYTE_ARRAY, MIME_SAMPLE, MIME_STREAM))
- canopen_types = set((CANOPEN_DATE, CANOPEN_TIME))
- tuple numeric_types = (signed_types | unsigned_types | float_types)
- big_endian_types = set((SINT_BE, UINT_BE, IEEE754_BE, STRING_UTF16_BE))

7.129.1 Detailed Description

Channel data type (cn_data_type) values.

7.129.2 Member Data Documentation

7.129.2.1 big_endian_types

ttk_tools.vector.mdf4_reader.CnDataTypes.big_endian_types = set((SINT_BE, UINT_BE, IEEE754_BE, STRING_UTF16_BE)) [static]

7.129.2.2 BYTE_ARRAY

int ttk_tools.vector.mdf4_reader.CnDataTypes.BYTE_ARRAY = 10 [static]

7.129.2.3 CANOPEN_DATE

int ttk_tools.vector.mdf4_reader.CnDataTypes.CANOPEN_DATE = 13 [static]

7.129.2.4 CANOPEN_TIME

int ttk_tools.vector.mdf4_reader.CnDataTypes.CANOPEN_TIME = 14 [static]

7.129.2.5 canopen_types

ttk_tools.vector.mdf4_reader.CnDataTypes.canopen_types = set((CANOPEN_DATE, CANOPEN_TIME)) [static]

7.129.2.6 complex_types

ttk_tools.vector.mdf4_reader.CnDataTypes.complex_types = set((BYTE_ARRAY, MIME_SAMPLE, MIME_STREAM)) [static]

7.129.2.7 float_types

ttk_tools.vector.mdf4_reader.CnDataTypes.float_types = set((IEEE754_LE, IEEE754_BE)) [static]

7.129.2.8 IEEE754_BE

int ttk_tools.vector.mdf4_reader.CnDataTypes.IEEE754_BE = 5 [static]

7.129.2.9 IEEE754_LE

int ttk_tools.vector.mdf4_reader.CnDataTypes.IEEE754_LE = 4 [static]

7.129.2.10 MIME_SAMPLE

int ttk_tools.vector.mdf4_reader.CnDataTypes.MIME_SAMPLE = 11 [static]

7.129.2.11 MIME_STREAM

int ttk_tools.vector.mdf4_reader.CnDataTypes.MIME_STREAM = 12 [static]

7.129.2.12 numeric_types

tuple ttk_tools.vector.mdf4_reader.CnDataTypes.numeric_types = (signed_types | unsigned_types | float_types) [static]

7.129.2.13 signed_types

ttk_tools.vector.mdf4_reader.CnDataTypes.signed_types = set((SINT_LE, SINT_BE)) [static]

7.129.2.14 SINT_BE

int ttk_tools.vector.mdf4_reader.CnDataTypes.SINT_BE = 3 [static]

7.129.2.15 SINT_LE

int ttk_tools.vector.mdf4_reader.CnDataTypes.SINT_LE = 2 [static]

7.129.2.16 STRING_SBC

int ttk_tools.vector.mdf4_reader.CnDataTypes.STRING_SBC = 6 [static]

7.129.2.17 string_types

ttk_tools.vector.mdf4_reader.CnDataTypes.string_types = set((STRING_SBC, STRING_UTF8, STRING_UTF16_LE, STRING_UTF16_BE)) [static]

7.129.2.18 STRING_UTF16_BE

int ttk_tools.vector.mdf4_reader.CnDataTypes.STRING_UTF16_BE = 9 [static]

7.129.2.19 STRING_UTF16_LE

int ttk_tools.vector.mdf4_reader.CnDataTypes.STRING_UTF16_LE = 8 [static]

7.129.2.20 STRING_UTF8

```
int ttk_tools.vector.mdf4_reader.CnDataTypes.STRING_UTF8 = 7  [static]
```

7.129.2.21 UINT_BE

```
int ttk_tools.vector.mdf4_reader.CnDataTypes.UINT_BE = 1  [static]
```

7.129.2.22 UINT_LE

```
int ttk_tools.vector.mdf4_reader.CnDataTypes.UINT_LE = 0  [static]
```

7.129.2.23 unsigned_types

```
ttk_tools.vector.mdf4_reader.CnDataTypes.unsigned_types = set((UINT_LE, UINT_BE))  [static]
```

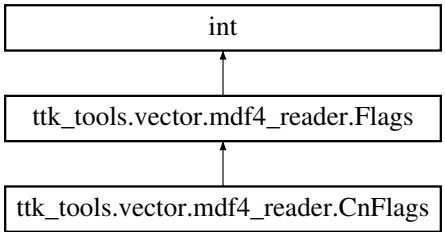
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.130 ttk_tools.vector.mdf4_reader.CnFlags Class Reference

Container for decoded uint32 cn_flags.

Inheritance diagram for ttk_tools.vector.mdf4_reader.CnFlags:



Public Member Functions

- __init__ (self, flags)

Public Attributes

- all_values_invalid
  Bit 0: All values invalid flag.
- invalidation_bit_valid
  Bit 1: Invalidation bit valid flag.
- precision_valid
  Bit 2: Precision valid flag.
- value_range_valid
  Bit 3: Value range valid flag.
- limit_range_valid
  Bit 4: Limit range valid flag.
- extended_limit_range_valid
  Bit 5: Extended limit range valid flag.
- discrete_value
  Bit 6: Discrete value flag.
- calibration
  Bit 7: Calibration flag.
- calculated
  Bit 8: Calculated flag.
- virtual
  Bit 9: Virtual flag.
- bus_event
  Bit 10: Bus event flag.
- monotonous
  Bit 11: Monotonous flag.
- default_x_axis
  Bit 12: Default X axis flag.

7.130.1 Detailed Description

Container for decoded uint32 cn_flags.

Only intended for read-only access, changes in named flags will not modify original "flags" value.

7.130.2 Constructor & Destructor Documentation

7.130.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.CnFlags.__init__ (
            self,
            flags )
```

Reimplemented from ttk_tools.vector.mdf4_reader.Flags.

7.130.3 Member Data Documentation

7.130.3.1 all_values_invalid

ttk_tools.vector.mdf4_reader.CnFlags.all_values_invalid

Bit 0: All values invalid flag.

If set, all values of this channel are invalid. If in addition an invalidation bit is used (bit 1 set), then the value of the invalidation bit must be set (high) for every value of this channel.
Must not be set for a master channel (channel types 2 and 3).

7.130.3.2 bus_event

ttk_tools.vector.mdf4_reader.CnFlags.bus_event

Bit 10: Bus event flag.

If set, this channel contains information about a bus event. For details please refer to [MDF-BL]. Valid since MDF 4.1.0, should not be set for earlier versions

7.130.3.3 calculated

ttk_tools.vector.mdf4_reader.CnFlags.calculated

Bit 8: Calculated flag.

If set, the values of this channel have been calculated from other channel inputs. (see [MCD-2 MC] keywords VIRTUAL and DEPENDENT_CHARACTERISTIC) In MDBLOCK for cn_md_comment the used input signals and the calculation formula can be documented.

7.130.3.4 calibration

ttk_tools.vector.mdf4_reader.CnFlags.calibration

Bit 7: Calibration flag.

If set, the signal values of this channel correspond to a calibration object, otherwise to a measurement object (see [MCD-2 MC] keywords MEASUREMENT and CHARACTERISTIC)

7.130.3.5 default_x_axis

ttk_tools.vector.mdf4_reader.CnFlags.default_x_axis

Bit 12: Default X axis flag.

If set, a channel to be preferably used as X axis is specified by cn_default_x. This is only a recommendation, a tool may choose to use a different X axis.
Valid since MDF 4.1.0, should not be set for earlier versions

7.130.3.6 discrete_value

ttk_tools.vector.mdf4_reader.CnFlags.discrete_value

Bit 6: Discrete value flag.

If set, the signal values of this channel are discrete and must not be interpolated. (see [MCD-2 MC] keyword DISCRETE)

7.130.3.7 extended_limit_range_valid

ttk_tools.vector.mdf4_reader.CnFlags.extended_limit_range_valid

Bit 5: Extended limit range valid flag.

If set, the extended limits of the signal value are known and stored in cn_limit_ext_min and cn_limit_ext_max. Otherwise the two fields are not valid. (see [MCD-2 MC] keyword EXTENDED_LIMITS)
The extended limit range must be larger or equal to the limit range (if valid). Values outside the extended limit range indicate an error during measurement acquisition. The extended limit range can be used for any limits, e.g. physical ranges from analog sensors.
See also remarks for "limit range valid" flag (bit 4).

7.130.3.8 invalidation_bit_valid

ttk_tools.vector.mdf4_reader.CnFlags.invalidation_bit_valid

Bit 1: Invalidation bit valid flag.

If set, this channel uses an invalidation bit (position specified by cn_inval_bit_pos).
Must not be set if cg_inval_bytes is zero.
Must not be set for a master channel (channel types 2 and 3).

7.130.3.9 limit_range_valid

ttk_tools.vector.mdf4_reader.CnFlags.limit_range_valid

Bit 4: Limit range valid flag.

If set, the limits of the signal value are known and stored in cn_limit_min and cn_limit_max. Otherwise the two fields are not valid. (see [MCD-2 MC] keywords LowerLimit/UpperLimit for MEASUREMENT and CHARACTERISTIC)
This limit range defines the range of plausible values for this channel and may be used for display a warning. Note: the (extended) limit range can only be expressed for a channel whose conversion rule returns a numeric value (REAL) or which has a numeric channel data type (cn_data_type $<=$ 5).
In all other cases, the flag must not be set.
Depending on the type of conversion, the limit values are interpreted as physical or raw values. If the conversion rule results in numeric values, the limits must be interpreted as physical values, otherwise (e.g. for verbal / scale conversion) as raw values.

7.130.3.10 monotonous

ttk_tools.vector.mdf4_reader.CnFlags.monotonous

Bit 11: Monotonous flag.

If set, this channel contains only strictly monotonously increasing/decreasing values.
The flag is optional.
Valid since MDF 4.1.0, should not be set for earlier versions

7.130.3.11 precision_valid

ttk_tools.vector.mdf4_reader.CnFlags.precision_valid

Bit 2: Precision valid flag.

If set, the precision value for display of floating point values specified in cn_precision is valid and overrules a possibly specified precision value of the conversion rule (cc_precision).

7.130.3.12 value_range_valid

ttk_tools.vector.mdf4_reader.CnFlags.value_range_valid

Bit 3: Value range valid flag.

If set, both the minimum and the maximum raw value that occurred for this signal within the samples recorded in this file are known and stored in cn_val_range_min and cn_val_range_max. Otherwise the two fields are not valid. Note: raw value range can only be expressed for numeric channel data types (cn_data_type <= 5). For all other data types, the flag must not be set.

7.130.3.13 virtual

ttk_tools.vector.mdf4_reader.CnFlags.virtual

Bit 9: Virtual flag.

If set, this channel is virtual, i.e. it is simulated by the recording tool. (see [MCD-2 MC] keywords VIRTUAL and VIRTUAL_CHARACTERISTIC) Note: for a virtual measurement according to MCD-2 MC both the "Virtual" flag (bit 9) and the "Calculated" flag (bit 8) should be set.

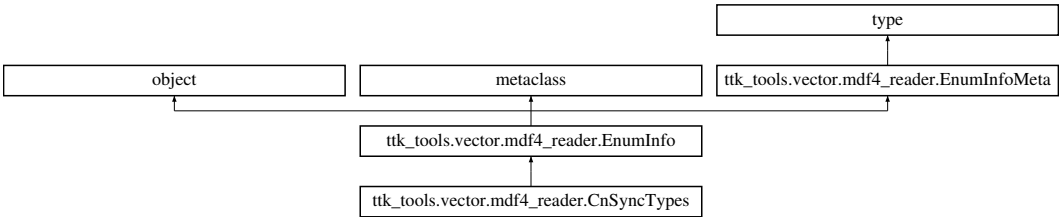
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.131 ttk_tools.vector.mdf4_reader.CnSyncTypes Class Reference

Channel synchronization type (cn_sync_type) values.

Inheritance diagram for ttk_tools.vector.mdf4_reader.CnSyncTypes:



Static Public Attributes

- int NONE = 0
- int TIME = 1
- int ANGLE = 2
- int DISTANCE = 3
- int INDEX = 4

7.131.1 Detailed Description

Channel synchronization type (cn_sync_type) values.

7.131.2 Member Data Documentation

7.131.2.1 ANGLE

```
int ttk_tools.vector.mdf4_reader.CnSyncTypes.ANGLE = 2  [static]
```

7.131.2.2 DISTANCE

```
int ttk_tools.vector.mdf4_reader.CnSyncTypes.DISTANCE = 3  [static]
```

7.131.2.3 INDEX

```
int ttk_tools.vector.mdf4_reader.CnSyncTypes.INDEX = 4  [static]
```

7.131.2.4 NONE

```
int ttk_tools.vector.mdf4_reader.CnSyncTypes.NONE = 0  [static]
```

7.131.2.5 TIME

```
int ttk_tools.vector.mdf4_reader.CnSyncTypes.TIME = 1  [static]
```

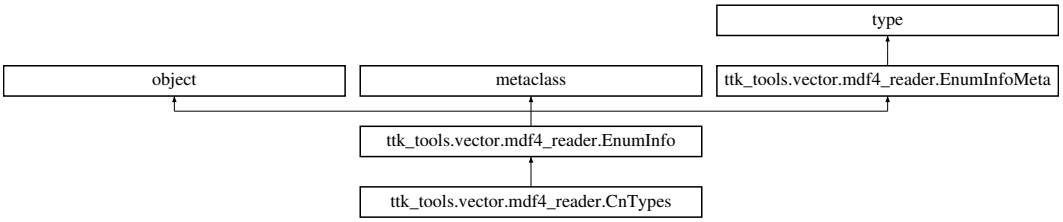
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.132 ttk_tools.vector.mdf4_reader.CnTypes Class Reference

Channel type (cn_type) values.

Inheritance diagram for ttk_tools.vector.mdf4_reader.CnTypes:



Static Public Attributes

- int FIXED_LENGTH = 0
- int VARIABLE_LENGTH = 1
- int MASTER = 2
- int VIRTUAL_MASTER = 3
- int SYNC = 4
- int MAXIMUM_LENGTH = 5
- int VIRTUAL_DATA = 6

7.132.1 Detailed Description

Channel type (cn_type) values.

7.132.2 Member Data Documentation

7.132.2.1 FIXED_LENGTH

```
int ttk_tools.vector.mdf4_reader.CnTypes.FIXED_LENGTH = 0  [static]
```

7.132.2.2 MASTER

```
int ttk_tools.vector.mdf4_reader.CnTypes.MASTER = 2  [static]
```

7.132.2.3 MAXIMUM_LENGTH

```
int ttk_tools.vector.mdf4_reader.CnTypes.MAXIMUM_LENGTH = 5  [static]
```

7.132.2.4 SYNC

```
int ttk_tools.vector.mdf4_reader.CnTypes.SYNC = 4  [static]
```

7.132.2.5 VARIABLE_LENGTH

```
int ttk_tools.vector.mdf4_reader.CnTypes.VARIABLE_LENGTH = 1  [static]
```

7.132.2.6 VIRTUAL_DATA

```
int ttk_tools.vector.mdf4_reader.CnTypes.VIRTUAL_DATA = 6  [static]
```

7.132.2.7 VIRTUAL_MASTER

```
int ttk_tools.vector.mdf4_reader.CnTypes.VIRTUAL_MASTER = 3  [static]
```

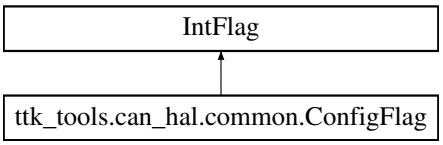
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.133 ttk_tools.can_hal.common.ConfigFlag Class Reference

Flags used during CAN configuration (CanHal.h, tCanHalCanConfig).

Inheritance diagram for ttk_tools.can_hal.common.ConfigFlag:



Static Public Attributes

- int NoConfig = 0b00000000
- int ConfigAckOn = 0b00000001
  enable acknowledge for CAN-interface
- int ConfigReceiveOn = 0b00000010
  enable receiving of messages
- int ConfigSendOn = 0b00000100
  enable sending of messages
- int ConfigBusStatisticsOn = 0b00001000
  enable evaluation of bus statistics
- int ConfigErrorframeCheckOn = 0b00010000
  enable evaluation of error frames

7.133.1 Detailed Description

Flags used during CAN configuration (CanHal.h, tCanHalCanConfig).

7.133.2 Member Data Documentation

7.133.2.1 ConfigAckOn

```
int ttk_tools.can_hal.common.ConfigFlag.ConfigAckOn = 0b00000001  [static]
```

enable acknowledge for CAN-interface

7.133.2.2 ConfigBusStatisticsOn

```
int ttk_tools.can_hal.common.ConfigFlag.ConfigBusStatisticsOn = 0b00001000  [static]
```

enable evaluation of bus statistics

7.133.2.3 ConfigErrorframeCheckOn

```
int ttk_tools.can_hal.common.ConfigFlag.ConfigErrorframeCheckOn = 0b00010000  [static]
```

enable evaluation of error frames

7.133.2.4 ConfigReceiveOn

```
int ttk_tools.can_hal.common.ConfigFlag.ConfigReceiveOn = 0b00000010  [static]
```

enable receiving of messages

7.133.2.5 ConfigSendOn

```
int ttk_tools.can_hal.common.ConfigFlag.ConfigSendOn = 0b00000100  [static]
```

enable sending of messages

7.133.2.6 NoConfig

```
int ttk_tools.can_hal.common.ConfigFlag.NoConfig = 0b00000000  [static]
```

The documentation for this class was generated from the following file:

- ttk_tools/can_hal/common.py (v1.2)

7.134 ttk_base.configuration.Configuration Class Reference

A simple class that reads/stores configuration info in ini files.

Public Member Functions

- __init__ (self, config_file_name, defaults_file_name=None, override_file_name=None, file_encoding=DEFAULT_INI_ENCODIN verbosity=0)

  Configuration initialization.
- get (self, section, key, default='')

  Get a config entry as string value.
- getParsedString (self, section, key, default='')

  Get a config entry interpreted as parsed/evaluated string.
- getInt (self, section, key, default=0)

  Get a config entry interpreted as integer value.
- getFloat (self, section, key, default=0.0)

  Get a config entry interpreted as float value.
- getBool (self, section, key, default=False)

  Get value for key/section interpreted as boolean value.
- getCSValues (self, section, key, default=None, separator=',', remove_empty_values=True)

  Get the value for key/section interpreted as a list of comma- separated strings (or rather "separator"-separated strings, but this makes for a rather un-intuitive method name ;)
- getAbsPath (self, section, key, default='', base_path=None)

  Get the value for key/section interpreted as an absolute path.
- getSection (self, section=DEFAULT_SECTION)

  Get a dictionary containing the key:value contents of the specified section.
- set (self, section, key, value=None)

  Set a config entry to custom config data.
- clear (self, section, key)

  Clear/remove a single config entry from custom data (master data will remain unaffected)
- store (self)

  Store current "custom" configuration back to disk.
- __str__ (self)

  Get an information string with currently loaded config files.

Public Attributes

- custom_data_file_path

  file path to custom/user settings file
- master_data_file_path

  file path to master/default settings file
- override_data_file_path

  file path to override settings file
- verbosity

  verbosity level
- file_encoding

  file encoding for config files
- master_data

  main config with basic settings, will not change during runtime

- custom_data

  custom user data overrides data in master_data, may get saved
- override_data

  "override" data, well, overrides data in master_data and custom_data, will not change during runtime
- config_path

  folder path of current configuration (for getAbsPath())

7.134.1 Detailed Description

A simple class that reads/stores configuration info in ini files.

Two (or three) files are used, one "master" file with default values and one "user" file that can contain user specific overrides (and that can be stored back to disk). Optionally a third file can be used which will contain key that will override user settings.

7.134.2 Constructor & Destructor Documentation

7.134.2.1 __init__()

```
ttk_base.configuration.Configuration.__init__ (
              self,
              config_file_name,
              defaults_file_name = None,
              override_file_name = None,
              file_encoding = DEFAULT_INI_ENCODING,
              verbosity = 0 )
```

Configuration initialization.

Parameters

config_file_name	file name/path of configuration file ("custom data", i.e. user settings, can be stored)
defaults_file_name	file name/path of configuration file ("master data", i.e. default settings)
override_file_name	file name/path of configuration file ("override data", overrides settings in other files)
file_encoding	file encoding to use for all config files
verbosity	0: (mostly) silent, 1+: more chatty

7.134.3 Member Function Documentation

7.134.3.1 __str__()

```
ttk_base.configuration.Configuration.__str__ (
              self )
```

Get an information string with currently loaded config files.

7.134.3.2 clear()

```
ttk_base.configuration.Configuration.clear (
              self,
              section,
              key )
```

Clear/remove a single config entry from custom data (master data will remain unaffected)

Parameters

section	configuration section
key	configuration key within section

Returns

the previous data or None if no value was set in custom data.

7.134.3.3 get()

```
ttk_base.configuration.Configuration.get (
              self,
              section,
              key,
              default = '' )
```

Get a config entry as string value.

Parameters

section	configuration section
key	configuration key within section
default	value to return if no value is set

Returns

the value for the supplied section/key, or the default value if key was not yet set.

7.134.3.4 getAbsPath()

```
ttk_base.configuration.Configuration.getAbsPath (
              self,
              section,
              key,
              default = '',
              base_path = None )
```

Get the value for key/section interpreted as an absolute path.

It the config value is a relative path (i.e. not starting with a drive letter or slash), it will be interpreted as relative to the location of the the config file.

For relative paths, config file location will be derived from `self.override_data_file_path`, `self.custom_data_file_path` or `self.master_data_file_path` (in order of availability) or the current working folder as fallback. See `self.config_path`.

Example:

```
# With a config file located at D:/Temp/cfg.ini containing:
# [STUFF]
# key = foo/bar/baz.log

>> getAbsPath("STUFF", "key")
D:/Temp/foo/bar/baz.log
```

Parameters

section	configuration section of key
key	configuration key to lookup within section
default	default path
base_path	manual base path override for relative paths, will be used instead of self.config_path

Note

default Supplied default value will be interpreted as a path or part of a path, so providing `default=None` for a missing key will return a path to a folder "None" below the base_path.

Returns

an absolute file system path.

7.134.3.5 getBool()

```
ttk_base.configuration.Configuration.getBool (
              self,
              section,
              key,
              default = False )
```

Get value for key/section interpreted as boolean value.

Parameters

section	configuration section
key	configuration key within section
default	value to return if no value is set

Returns

the boolean value for the supplied section/key, or the default value if key was not yet set (or not interpretable as boolean, see Configuration.BOOL_TRUE_STRINGS).

7.134.3.6 getCSValues()

```
ttk_base.configuration.Configuration.getCSValues (
              self,
              section,
              key,
              default = None,
              separator = ',',
              remove_empty_values = True )
```

Get the value for key/section interpreted as a list of comma- separated strings (or rather "separator"-separated strings, but this makes for a rather un-intuitive method name ;)

Parameters

section	configuration section of key
key	configuration key to lookup within section
default	default value (may be already a list)
separator	separator char, default to comma
remove_empty_values	True: empty list entries will be removed

Returns

a list of string values

7.134.3.7 getFloat()

```
ttk_base.configuration.Configuration.getFloat (
              self,
              section,
              key,
              default = 0.0 )
```

Get a config entry interpreted as float value.

Parameters

section	configuration section
key	configuration key within section
default	value to return if no value is set

Returns

the float value for the supplied section/key, or the default value if key was not yet set (or not interpretable as an float value).

7.134.3.8 getInt()

```
ttk_base.configuration.Configuration.getInt (
              self,
```

```
                section,
                key,
                default = 0 )
```

Get a config entry interpreted as integer value.

Parameters

section	configuration section
key	configuration key within section
default	value to return if no value is set

Returns

the integer value for the supplied section/key, or the default value if key was not yet set (or not interpretable as an integer value).

7.134.3.9 getParsedString()

```
ttk_base.configuration.Configuration.getParsedString (
                self,
                section,
                key,
                default = '' )
```

Get a config entry interpreted as parsed/evaluated string.

This permits escaping of special characters.
If a value starts and ends with string delimiters, it will be evaluated as a Python string.

Parameters

section	configuration section
key	configuration key within section
default	value to return if no value is set

Returns

the value for the supplied section/key, or the default value if key was not yet set. If the value could not be parsed, it will be returned in its unparsed state.

7.134.3.10 getSection()

```
ttk_base.configuration.Configuration.getSection (
                self,
                section = DEFAULT_SECTION )
```

Get a dictionary containing the key:value contents of the specified section.

Parameters

section	name of configuration section

Returns

the section data as dictionary

7.134.3.11 set()

```
ttk_base.configuration.Configuration.set (
              self,
              section,
              key,
              value = None )
```

Set a config entry to custom config data.

This will override an existing entry found in master data.

Note

If the entry exists in "override data", it will still be set to user data but get() will still return the overridden value.

Parameters

section	configuration section
key	configuration key within section
value	new value to set

7.134.3.12 store()

```
ttk_base.configuration.Configuration.store (
              self )
```

Store current "custom" configuration back to disk.

7.134.4 Member Data Documentation

7.134.4.1 config_path

```
ttk_base.configuration.Configuration.config_path
```

folder path of current configuration (for getAbsPath())

7.134.4.2 custom_data

ttk_base.configuration.Configuration.custom_data

custom user data overrides data in master_data, may get saved

7.134.4.3 custom_data_file_path

ttk_base.configuration.Configuration.custom_data_file_path

file path to custom/user settings file

7.134.4.4 file_encoding

ttk_base.configuration.Configuration.file_encoding

file encoding for config files

7.134.4.5 master_data

ttk_base.configuration.Configuration.master_data

main config with basic settings, will not change during runtime

7.134.4.6 master_data_file_path

ttk_base.configuration.Configuration.master_data_file_path

file path to master/default settings file

7.134.4.7 override_data

ttk_base.configuration.Configuration.override_data

"override" data, well, overrides data in master_data and custom_data, will not change during runtime

7.134.4.8 override_data_file_path

ttk_base.configuration.Configuration.override_data_file_path

file path to override settings file

7.134.4.9 verbosity

```
ttk_base.configuration.Configuration.verbosity
```

verbosity level

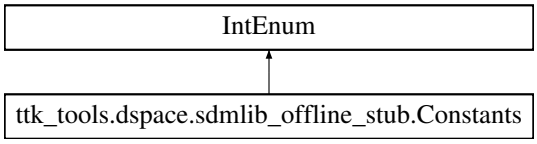
The documentation for this class was generated from the following file:

- ttk_base/configuration.py (v3.8.1)

7.135 ttk_tools.dspace.sdmlib_offline_stub.Constants Class Reference

Constants-Enum.

Inheritance diagram for ttk_tools.dspace.sdmlib_offline_stub.Constants:



Static Public Attributes

- int TH_STATUS_STOP = 0
- int TH_STATUS_START = 1
- int TH_STATUS_TERMINATE = 2

7.135.1 Detailed Description

Constants-Enum.

7.135.2 Member Data Documentation

7.135.2.1 TH_STATUS_START

```
int ttk_tools.dspace.sdmlib_offline_stub.Constants.TH_STATUS_START = 1  [static]
```

7.135.2.2 TH_STATUS_STOP

```
int ttk_tools.dspace.sdmlib_offline_stub.Constants.TH_STATUS_STOP = 0  [static]
```

7.135.2.3 TH_STATUS_TERMINATE

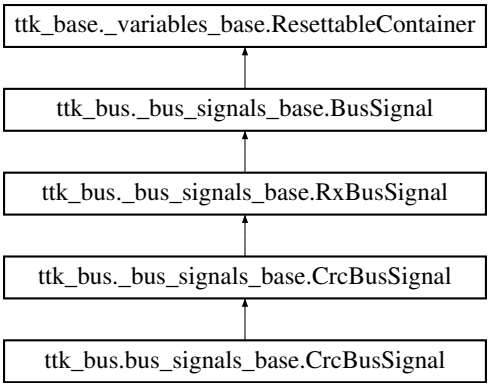
```
int ttk_tools.dspace.sdmlib_offline_stub.Constants.TH_STATUS_TERMINATE = 2  [static]
```

The documentation for this class was generated from the following file:

- ttk_tools/dspace/sdmlib_offline_stub.py (v1.4)

7.136 ttk_bus._bus_signals_base.CrcBusSignal Class Reference

Inheritance diagram for ttk_bus._bus_signals_base.CrcBusSignal:



Public Member Functions

- __init__ (self, context, identifier, cycle_time=0, debounce_time=0, signal_var=None, alias="", descr="", error_delay=None, change_delay=None, timeout=None, recovery_time=None)

  CRC bus signal initialization.
- setBasicCrcError (self)

  Switch to basic CRC error state.
- clearBasicCrcError (self)

  Resume normal CRC calculation.

Public Attributes

- error_delay

  delay to wait after a "basic" crc error injection[ms]

7.136.1 Constructor & Destructor Documentation

7.136.1.1 __init__()

```
ttk_bus._bus_signals_base.CrcBusSignal.__init__ (
                self,
                context,
                identifier,
                cycle_time = 0,
                debounce_time = 0,
                signal_var = None,
                alias = "",
                descr = "",
                error_delay = None,
                change_delay = None,
                timeout = None,
                recovery_time = None )
```

CRC bus signal initialization.

Parameters

context	parent context for instantiating HIL variables. This gets mainly used in derived classes
identifier	base signal identifier/name/path in the HIL model pointing to the CRC signal value. Derived classes typically use this identifier as starting point for locating additional helper variables.
cycle_time	cycle time in [ms]
debounce_time	minimum debounce delay between events in [ms]
signal_var	(opt) an already initialized instance of Var (or most likely one of its derived classes) for the main signal value.
alias	(opt) bus signal alias/name, defaults to signal_var.alias.
descr	(opt) signal description (e.g. may get used in report entries)
error_delay	(opt) delay to wait until a basic CRC error injection should be detected [ms], defaults to 10x max(cycle_time, debounce_time)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)

Note

The parameters `change_delay`, `timeout` and `recovery_time` are just here for completeness' sake, only `cycle_time` and `error_delay` are used during typical CRC checks.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

Reimplemented in ttk_bus.bus_signals_base.CrcBusSignal.

7.136.2 Member Function Documentation

7.136.2.1 clearBasicCrcError()

```
ttk_bus._bus_signals_base.CrcBusSignal.clearBasicCrcError (
                self )
```

Resume normal CRC calculation.

Reimplemented in ttk_bus.bus_signals_base.CrcBusSignal.

7.136.2.2 setBasicCrcError()

```
ttk_bus._bus_signals_base.CrcBusSignal.setBasicCrcError (
              self )
```

Switch to basic CRC error state.

Reimplemented in ttk_bus.bus_signals_base.CrcBusSignal.

7.136.3 Member Data Documentation

7.136.3.1 error_delay

```
ttk_bus._bus_signals_base.CrcBusSignal.error_delay
```

delay to wait after a "basic" crc error injection[ms]

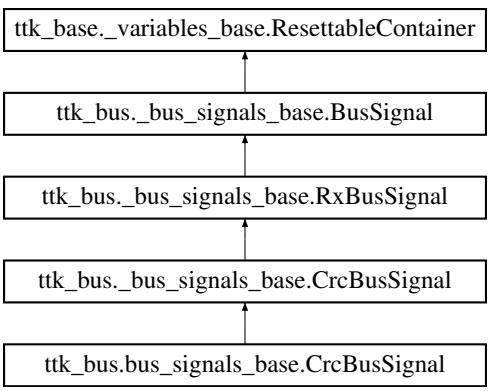
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_base.py (v3.0.1)

7.137 ttk_bus.bus_signals_base.CrcBusSignal Class Reference

A RX bus signal (HIL => ECU) that models a CRC value with additional error injections.

Inheritance diagram for ttk_bus.bus_signals_base.CrcBusSignal:



Public Member Functions

- __init__ (self, context, identifier, cycle_time=0, debounce_time=0, signal_var=None, alias="", descr="", error_delay=None, change_delay=None, timeout=None, recovery_time=None)

  CRC bus signal initialization.
- setBasicCrcError (self)

  Switch to basic CRC error state.
- clearBasicCrcError (self)

  Resume normal CRC calculation.
- info (self)

  Get a current info string about the HIL signal variable.
- get (self)

  Get the current value of the HIL signal variable (typically not that useful for ECU-Rx/HIL-Tx CRC-signals, though).
- set (self, value)

  Set a value to the HIL signal variable.
- getState (self)

  Get a textual state for the HIL signal variable's current data.
- setState (self, state)

  Set a textual state to the HIL signal variable.
- hasStateLookup (self)

  Return True if a state/value lookup table is defined for the HIL signal variable.
- getStateDescr (self, value=None, fallback=None)

  Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).
- hasFormattedRepr (self)

  Return True if a formatting for value representation is defined for the HIL signal variable.
- getFormattedValue (self, value=None)

  Get a formatted representation for the given value using the defined format string of the the HIL signal variable.
- reset (self, *args, **kwargs)

  Call reset() on all contained variables/signals that support it.

Public Attributes

- error_delay

  delay to wait after a "basic" crc error injection[ms]

7.137.1 Detailed Description

A RX bus signal (HIL => ECU) that models a CRC value with additional error injections.

This is the generic base class for CRC signals, blockset/implementation specific CRC signal classes will inherit from this.

Attention

This is just a base class, using it directly will not accomplish much. At the very least, setBasicCrcError and clearBasicCrcError need to be overridden depending on the target interface.

See bus_signals_frbs or bus_signals_canmmbs for existing implementations.

7.137.2 Constructor & Destructor Documentation

7.137.2.1 __init__()

```
ttk_bus.bus_signals_base.CrcBusSignal.__init__ (
                self,
                context,
                identifier,
                cycle_time = 0,
                debounce_time = 0,
                signal_var = None,
                alias = "",
                descr = "",
                error_delay = None,
                change_delay = None,
                timeout = None,
                recovery_time = None )
```

CRC bus signal initialization.

Parameters

context	parent context for instantiating HIL variables. This gets mainly used in derived classes
identifier	base signal identifier/name/path in the HIL model pointing to the CRC signal value. Derived classes typically use this identifier as starting point for locating additional helper variables.
cycle_time	cycle time in [ms]
debounce_time	minimum debounce delay between events in [ms]
signal_var	(opt) an already initialized instance of Var (or most likely one of its derived classes) for the main signal value.
alias	(opt) bus signal alias/name, defaults to signal_var.alias.
descr	(opt) signal description (e.g. may get used in report entries)
error_delay	(opt) delay to wait until a basic CRC error injection should be detected [ms], defaults to 10x max(cycle_time, debounce_time)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)

Note

The parameters `change_delay`, `timeout` and `recovery_time` are just here for completeness' sake, only `cycle_time` and `error_delay` are used during typical CRC checks.

Reimplemented from ttk_bus._bus_signals_base.CrcBusSignal.

7.137.3 Member Function Documentation

7.137.3.1 clearBasicCrcError()

```
ttk_bus.bus_signals_base.CrcBusSignal.clearBasicCrcError (
                self )
```

Resume normal CRC calculation.

Reimplemented from ttk_bus._bus_signals_base.CrcBusSignal.

7.137.3.2 get()

```
ttk_bus.bus_signals_base.CrcBusSignal.get (
              self )
```

Get the current value of the HIL signal variable (typically not that useful for ECU-Rx/HIL-Tx CRC-signals, though).

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.137.3.3 getFormattedValue()

```
ttk_bus.bus_signals_base.CrcBusSignal.getFormattedValue (
              self,
              value = None )
```

Get a formatted representation for the given value using the defined format string of the the HIL signal variable.

An empty (or non-defined) format string will use the value's default string representation.

Parameters

value	value for which to get a string representation. If None, the last value (read or written) will be used instead.

Returns

a formatted string representation.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.137.3.4 getState()

```
ttk_bus.bus_signals_base.CrcBusSignal.getState (
              self )
```

Get a textual state for the HIL signal variable's current data.

(typically not that useful for ECU-Rx/HIL-Tx CRC-signals, though).

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.137.3.5 getStateDescr()

```
ttk_bus.bus_signals_base.CrcBusSignal.getStateDescr (
              self,
              value = None,
              fallback = None )
```

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the last read value will be used instead.
fallback	fallback value to return if no match was found. Default: None

Returns

the state text if a match was found, otherwise the specified fallback value.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.137.3.6 hasFormattedRepr()

```
ttk_bus.bus_signals_base.CrcBusSignal.hasFormattedRepr (
              self )
```

Return True if a formatting for value representation is defined for the HIL signal variable.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.137.3.7 hasStateLookup()

```
ttk_bus.bus_signals_base.CrcBusSignal.hasStateLookup (
              self )
```

Return True if a state/value lookup table is defined for the HIL signal variable.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.137.3.8 info()

```
ttk_bus.bus_signals_base.CrcBusSignal.info (
              self )
```

Get a current info string about the HIL signal variable.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.137.3.9 reset()

```
ttk_bus.bus_signals_base.CrcBusSignal.reset (
              self,
             * args,
             ** kwargs )
```

Call reset() on all contained variables/signals that support it.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.ResettableContainer.

7.137.3.10 set()

```
ttk_bus.bus_signals_base.CrcBusSignal.set (
              self,
              value )
```

Set a value to the HIL signal variable.

Parameters

value	new value to set

Returns

the previous value.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.137.3.11 setBasicCrcError()

```
ttk_bus.bus_signals_base.CrcBusSignal.setBasicCrcError (
              self )
```

Switch to basic CRC error state.

Reimplemented from ttk_bus._bus_signals_base.CrcBusSignal.

7.137.3.12 setState()

```
ttk_bus.bus_signals_base.CrcBusSignal.setState (
              self,
              state )
```

Set a textual state to the HIL signal variable.

This uses the lookup values supplied in constructor.

Parameters

state	named state to set

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.137.4 Member Data Documentation

7.137.4.1 error_delay

`ttk_bus.bus_signals_base.CrcBusSignal.error_delay`

delay to wait after a "basic" crc error injection[ms]

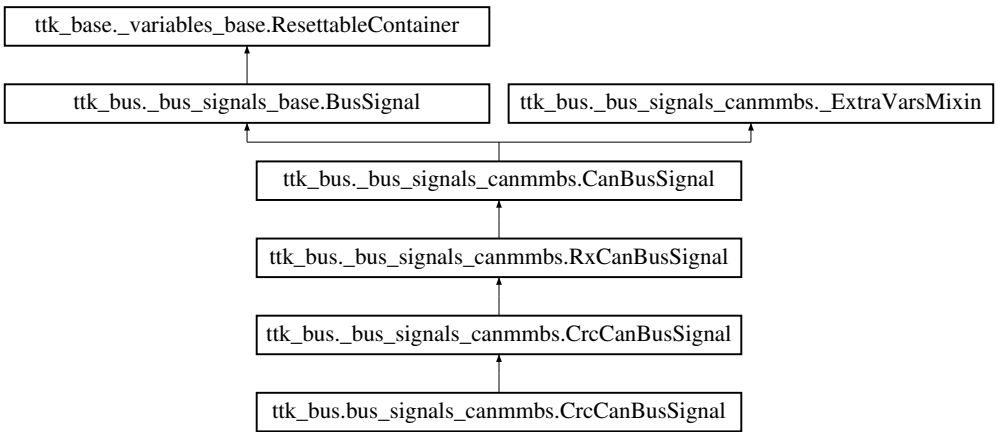
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_base.py (v3.1.0)

7.138 ttk_bus._bus_signals_canmmbs.CrcCanBusSignal Class Reference

A CRC RX bus signal derived from RxCanBusSignal (HIL-Tx => ECU-Rx) for use with dSPACE CAN Multi-Message blockset.

Inheritance diagram for ttk_bus._bus_signals_canmmbs.CrcCanBusSignal:



Public Member Functions

- __init__ (self, context, signal_path, message=None, cycle_time=None, debounce_time=None, alias="", descr="", lookup=None, error_delay=None, change_delay=None, timeout=None, recovery_time=None)

  CRC bus signal (HIL-Tx => ECU-Rx) initialization for dSPACE CAN Multi-Message blockset.
- setBasicCrcError (self)

  Switch CRC calculation to an algorithm producing incorrect CRCs.
- clearBasicCrcError (self)

  Switch CRC calculation back to an algorithm producing correct CRCs.

Public Attributes

- error_delay

  delay to wait after a "basic" CRC error injection [ms]
- crc_type_var

  CRC Type variable, selects active CRC algorithm (a "HilVar" instance)
- crc_enable_var

  CRC Enable variable, enables/disables automatic CRC calculation (a "HilVar" instance)
- crc_dyntype_var

  CRC Dynamic Type variable, selects active CRC algorithm (a "HilVar" instance), only available if configured in CANMM and enabled in class config.
- crc_dyncount_var

  CRC Dynamic Countdown variable, enables dynamic CRC type while counting down to 0 at each message transmission (a "HilVar" instance), only available if configured in CANMM and enabled with in class config.
- crc_normal_type

  CRC type for "correct" CRC algorithm (normal mode)
- crc_error_type

  CRC type for "incorrect" CRC algorithm (for error injection)

Static Public Attributes

- int cfg_crc_normal_type = 1

  Configuration setting for new instances:
  CRC type for "correct" CRC algorithm (normal mode)
- int cfg_crc_error_type = 2

  Configuration setting for new instances:
  CRC type for "incorrect" CRC algorithm (for error injection, for example calculated CRC + 1).
- bool cfg_init_crc_dynamic_manipulation = False

  Configuration setting for new instances:
  Try to initialize (additional) variables that are only available if "dynamic crc" manipulation has been selected during RTICANMM configuration for a signal.
- dict cfg_custom_defaults

  Configuration setting for new instances:
  Custom "hard" default value(s) to use for internal real time system variables.

7.138.1 Detailed Description

A CRC RX bus signal derived from RxCanBusSignal (HIL-Tx => ECU-Rx) for use with dSPACE CAN Multi-Message blockset.

See RxCanBusSignal for inherited methods.

7.138.2 Constructor & Destructor Documentation

7.138.2.1 __init__()

```
ttk_bus._bus_signals_canmmbs.CrcCanBusSignal.__init__ (
                self,
                context,
                signal_path,
                message = None,
                cycle_time = None,
                debounce_time = None,
                alias = "",
                descr = "",
                lookup = None,
                error_delay = None,
                change_delay = None,
                timeout = None,
                recovery_time = None )
```

CRC bus signal (HIL-Tx => ECU-Rx) initialization for dSPACE CAN Multi-Message blockset.

Example: CRC Signal Path Structure:

```
# (static) crc signal value
BusSystems/CAN/FOO_CAN/QUX_MESSAGE/TX/CRC_QUX_MESSAGE
# crc calculation enable
BusSystems/CAN/FOO_CAN/QUX_MESSAGE/TX/QUX_MESSAGE_crc
# selected crc algorithm
BusSystems/CAN/FOO_CAN/QUX_MESSAGE/TX/QUX_MESSAGE_crc_algorithm
# variables for dynamic crc manipulation
BusSystems/CAN/FOO_CAN/QUX_MESSAGE/TX/QUX_MESSAGE_dyn_algorithm
BusSystems/CAN/FOO_CAN/QUX_MESSAGE/TX/QUX_MESSAGE_dyn_countdown
```

Parameters

context	rtplib application instance
signal_path	base path to signal below BusSystems (BusSystems/.../<message>/TX/<signal>)
message	reference to associated RxCanBusMessage instance
cycle_time	(opt) cycle time in [ms], overrides settings in Message
debounce_time	(opt) minimum debounce delay between events in [ms], overrides settings in Message
alias	(opt) bus signal alias/name
descr	(opt) signal description
lookup	(opt) lookup table for values, see VarBase
error_delay	(opt) delay to wait until a simple crc error should be detected [ms], defaults to 10x cycle time
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x cycle_time
timeout	(opt) detection time for timeout errors [ms], defaults to 2x cycle_time
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x cycle_time

Reimplemented from ttk_bus._bus_signals_canmmbs.RxCanBusSignal.

Reimplemented in ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.

7.138.3 Member Function Documentation

7.138.3.1 clearBasicCrcError()

```
ttk_bus._bus_signals_canmmbs.CrcCanBusSignal.clearBasicCrcError (
              self )
```

Switch CRC calculation back to an algorithm producing correct CRCs.

See CrcCanBusSignal.crc_normal_type.

Reimplemented in ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.

7.138.3.2 setBasicCrcError()

```
ttk_bus._bus_signals_canmmbs.CrcCanBusSignal.setBasicCrcError (
              self )
```

Switch CRC calculation to an algorithm producing incorrect CRCs.

See CrcCanBusSignal.crc_error_type.

Reimplemented in ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.

7.138.4 Member Data Documentation

7.138.4.1 cfg_crc_error_type

```
int ttk_bus._bus_signals_canmmbs.CrcCanBusSignal.cfg_crc_error_type = 2  [static]
```

Configuration setting for new instances:
CRC type for "incorrect" CRC algorithm (for error injection, for example calculated CRC + 1).

Note: Simply switching CRC calculation off (e.g. CRC = 0x00) may lead to detection failures, as a static CRC value might actually be correct for the current data.

7.138.4.2 cfg_crc_normal_type

```
int ttk_bus._bus_signals_canmmbs.CrcCanBusSignal.cfg_crc_normal_type = 1  [static]
```

Configuration setting for new instances:
CRC type for "correct" CRC algorithm (normal mode)

7.138.4.3 cfg_custom_defaults

```
dict ttk_bus._bus_signals_canmmbs.CrcCanBusSignal.cfg_custom_defaults  [static]
```

Initial value:

```
=  {
        "crc_enable_var": None,
        "crc_type_var":   None,

        "switch_var":     None,   # for parent RxCanBusSignal
        "dyncount_var":   None,   # for parent RxCanBusSignal
        "dynvalue_var":   None,   # for parent RxCanBusSignal
    }
```

Configuration setting for new instances:
Custom "hard" default value(s) to use for internal real time system variables.

Mapping: <variable name as string>: <default value>

A value of None (or a missing variable name) enables normal/automatic reset behavior.
See RxCanBusSignal for further details.

Typical default values:

- CRC enable (automatic CRC calculation) usually defaults to "enabled"
  - 0: CRC calculation disabled
  - 1: CRC calculation enabled
- CRC type (active CRC algorithm) should default to "correct CRC" (which is typically algorithm 1)

7.138.4.4 cfg_init_crc_dynamic_manipulation

```
bool ttk_bus._bus_signals_canmmbs.CrcCanBusSignal.cfg_init_crc_dynamic_manipulation = False
[static]
```

Configuration setting for new instances:
Try to initialize (additional) variables that are only available if "dynamic crc" manipulation has been selected during RTICANMM configuration for a signal.

Additional variables:

- .crc_dyntype_var
- .crc_dyncount_var

7.138.4.5 crc_dyncount_var

```
ttk_bus._bus_signals_canmmbs.CrcCanBusSignal.crc_dyncount_var
```

CRC Dynamic Countdown variable, enables dynamic CRC type while counting down to 0 at each message transmission (a "HilVar" instance), only available if configured in CANMM and enabled with in class config.

See cfg_init_crc_dynamic_manipulation

7.138.4.6 crc_dyntype_var

`ttk_bus._bus_signals_canmmbs.CrcCanBusSignal.crc_dyntype_var`

CRC Dynamic Type variable, selects active CRC algorithm (a "HilVar" instance), only available if configured in CANMM and enabled in class config.

See cfg_init_crc_dynamic_manipulation

7.138.4.7 crc_enable_var

`ttk_bus._bus_signals_canmmbs.CrcCanBusSignal.crc_enable_var`

CRC Enable variable, enables/disables automatic CRC calculation (a "HilVar" instance)

- 0: CRC calculation disabled
- 1: CRC calculation enabled

7.138.4.8 crc_error_type

`ttk_bus._bus_signals_canmmbs.CrcCanBusSignal.crc_error_type`

CRC type for "incorrect" CRC algorithm (for error injection)

7.138.4.9 crc_normal_type

`ttk_bus._bus_signals_canmmbs.CrcCanBusSignal.crc_normal_type`

CRC type for "correct" CRC algorithm (normal mode)

7.138.4.10 crc_type_var

`ttk_bus._bus_signals_canmmbs.CrcCanBusSignal.crc_type_var`

CRC Type variable, selects active CRC algorithm (a "HilVar" instance)

7.138.4.11 error_delay

`ttk_bus._bus_signals_canmmbs.CrcCanBusSignal.error_delay`

delay to wait after a "basic" CRC error injection [ms]

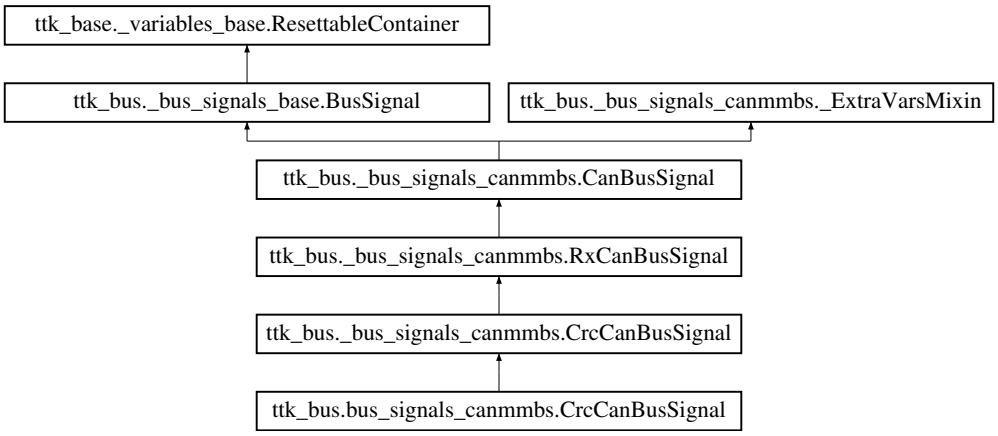
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_canmmbs.py (v2.3)

7.139 ttk_bus.bus_signals_canmmbs.CrcCanBusSignal Class Reference

A CRC RX bus signal derived from RxCanBusSignal (HIL-Tx => ECU-Rx) for use with dSPACE CAN Multi-Message blockset.

Inheritance diagram for ttk_bus.bus_signals_canmmbs.CrcCanBusSignal:



Public Member Functions

- __init__ (self, context, signal_path, message=None, cycle_time=None, debounce_time=None, alias="", descr="", lookup=None, error_delay=None, change_delay=None, timeout=None, recovery_time=None)

  CRC bus signal (HIL-Tx => ECU-Rx) initialization for dSPACE CAN Multi-Message blockset.
- setBasicCrcError (self)

  Switch CRC calculation to an algorithm producing incorrect CRCs.
- clearBasicCrcError (self)

  Switch CRC calculation back to an algorithm producing correct CRCs.
- enableHilTx (self)

  Enable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).
- disableHilTx (self)

  Disable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).
- kickout (self)

  Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent CAN-Message/PDU).
- get (self)

  Get the current value of the HIL signal variable (typically not that useful for ECU-Rx/HIL-Tx CRC-Signals, though).
- set (self, value)

  Set a value to the HIL signal variable.
- getState (self)

  Get a textual state for the HIL signal variable's current data (typically not that useful for ECU-Rx/HIL-Tx CRC-Signals, though).
- setState (self, state)

  Set a textual state to the HIL signal variable.
- getStateDescr (self, value=None, fallback=None)

  Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).
- info (self)

  Get a current info string about the HIL signal variable.
- reset (self, *args, **kwargs)

  Reset all resettable variables of this bus signal.

Public Attributes

- crc_normal_type

  CRC type for "correct" CRC algorithm (normal mode)
- crc_error_type

  CRC type for "incorrect" CRC algorithm (for error injection)
- error_delay

  delay [ms] to wait until a basic CRC error (like "CRC invalid") should be detected.
- crc_type_var

  CRC Type variable, selects active CRC algorithm (a "HilVar" instance)
- crc_enable_var

  CRC Enable variable, enables/disables automatic CRC calculation (a "HilVar" instance)
- crc_dyntype_var

  CRC Dynamic Type variable, selects active CRC algorithm (a "HilVar" instance), only available if configured in CANMM and enabled in class config.
- crc_dyncount_var

  CRC Dynamic Countdown variable, enables dynamic CRC type while counting down to 0 at each message transmission (a "HilVar" instance), only available if configured in CANMM and enabled with in class config.
- switch_var

  Switch variable for signal manipulation.
- dyncount_var

  Dynamic Countdown Value.
- dynvalue_var

  Dynamic Value: value that is active while dyncount_var decrements.
- base_path

  Signal model path as base path for deriving sub-element paths.
- message

  reference to associated (Rx-)CanBusMessage instance
- signal_var

  Signal value variable.
- context

  parent context (an RTE application instance to resolve signal/variable paths)
- unit

  unit of physical signal value (as string, mainly used for report entries)
- alias

  bus signal alias name
- descr

  bus signal description text
- cycle_time

  cycle time of Message/PDU in [ms]
- debounce_time

  minimum debounce delay for event-triggered Messages/PDUs in [ms]
- change_delay

  delay to wait after a value change [ms]
- timeout

  detection time for timeout errors [ms]
- recovery_time

  time for recovery/"wiedergut" checks [ms]

Static Public Attributes

- int cfg_crc_normal_type = 1

  Configuration setting for new instances:
  CRC type for "correct" CRC algorithm (normal mode)
- int cfg_crc_error_type = 2

  Configuration setting for new instances:
  CRC type for "incorrect" CRC algorithm (for error injection, for example calculated CRC + 1).
- bool cfg_init_crc_dynamic_manipulation = False

  Configuration setting for new instances:
  Try to initialize (additional) variables that are only available if "dynamic crc" manipulation has been selected during RTICANMM configuration for a signal.
- dict cfg_custom_defaults

  Configuration setting for new instances:
  Custom "hard" default value(s) to use for internal real time system variables.

7.139.1 Detailed Description

A CRC RX bus signal derived from RxCanBusSignal (HIL-Tx => ECU-Rx) for use with dSPACE CAN Multi-Message blockset.

See RxCanBusSignal for inherited methods.

7.139.2 Constructor & Destructor Documentation

7.139.2.1 __init__()

```
ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.__init__ (
              self,
              context,
              signal_path,
              message = None,
              cycle_time = None,
              debounce_time = None,
              alias = "",
              descr = "",
              lookup = None,
              error_delay = None,
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

CRC bus signal (HIL-Tx => ECU-Rx) initialization for dSPACE CAN Multi-Message blockset.

Example: CRC Signal Path Structure:

```
# (static) crc signal value
BusSystems/CAN/FOO_CAN/QUX_MESSAGE/TX/CRC_QUX_MESSAGE
# crc calculation enable
BusSystems/CAN/FOO_CAN/QUX_MESSAGE/TX/QUX_MESSAGE_crc
# selected crc algorithm
BusSystems/CAN/FOO_CAN/QUX_MESSAGE/TX/QUX_MESSAGE_crc_algorithm
# variables for dynamic crc manipulation
BusSystems/CAN/FOO_CAN/QUX_MESSAGE/TX/QUX_MESSAGE_dyn_algorithm
BusSystems/CAN/FOO_CAN/QUX_MESSAGE/TX/QUX_MESSAGE_dyn_countdown
```

Parameters

context	rtplib application instance
signal_path	base path to signal below BusSystems (`BusSystems/.../<message>/TX/<signal>`)
message	reference to associated RxCanBusMessage instance
cycle_time	(opt) cycle time in [ms], overrides settings in Message
debounce_time	(opt) minimum debounce delay between events in [ms], overrides settings in Message
alias	(opt) bus signal alias/name
descr	(opt) signal description
lookup	(opt) lookup table for values, see VarBase
error_delay	(opt) delay to wait until a simple crc error should be detected [ms], defaults to 10x cycle time
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x cycle_time
timeout	(opt) detection time for timeout errors [ms], defaults to 2x cycle_time
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x cycle_time

Reimplemented from ttk_bus._bus_signals_canmmbs.CrcCanBusSignal.

7.139.3 Member Function Documentation

7.139.3.1 clearBasicCrcError()

```
ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.clearBasicCrcError (
              self )
```

Switch CRC calculation back to an algorithm producing correct CRCs.

See CrcCanBusSignal.crc_normal_type.

Reimplemented from ttk_bus._bus_signals_canmmbs.CrcCanBusSignal.

7.139.3.2 disableHilTx()

```
ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.disableHilTx (
              self )
```

Disable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).

Note

This will disable transmission of *all* signals of the parent Message/PDU.

Reimplemented from ttk_bus._bus_signals_canmmbs.RxCanBusSignal.

7.139.3.3 enableHilTx()

```
ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.enableHilTx (
              self )
```

Enable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).

Note

This will enable transmission of *all* signals of the parent Message/PDU.

Reimplemented from ttk_bus._bus_signals_canmmbs.RxCanBusSignal.

7.139.3.4 get()

```
ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.get (
              self )
```

Get the current value of the HIL signal variable (typically not that useful for ECU-Rx/HIL-Tx CRC-Signals, though).

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.139.3.5 getState()

```
ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.getState (
              self )
```

Get a textual state for the HIL signal variable's current data (typically not that useful for ECU-Rx/HIL-Tx CRC-Signals, though).

Returns

the current state's description from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.139.3.6 getStateDescr()

```
ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.getStateDescr (
              self,
              value = None,
              fallback = None )
```

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the last read value will be used instead.
fallback	fallback value to return if no match was found. Default: None

Returns

the state text if a match was found, otherwise the specified fallback value.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.139.3.7 info()

```
ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.info (
              self )
```

Get a current info string about the HIL signal variable.

Returns

an info string.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.139.3.8 kickout()

```
ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.kickout (
              self )
```

Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent CAN-Message/PDU).

Note

All other signals of the parent Message/PDU will also be sent.

Reimplemented from ttk_bus._bus_signals_canmmbs.RxCanBusSignal.

7.139.3.9 reset()

```
ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.reset (
              self,
             * args,
             ** kwargs )
```

Reset all resettable variables of this bus signal.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.ResettableContainer.

7.139.3.10 set()

```
ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.set (
              self,
              value )
```

Set a value to the HIL signal variable.

Parameters

value	value to set

Returns

the previous value from signal_var.

Reimplemented from ttk_bus._bus_signals_canmmbs.RxCanBusSignal.

7.139.3.11 setBasicCrcError()

```
ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.setBasicCrcError (
              self )
```

Switch CRC calculation to an algorithm producing incorrect CRCs.

See CrcCanBusSignal.crc_error_type.

Reimplemented from ttk_bus._bus_signals_canmmbs.CrcCanBusSignal.

7.139.3.12 setState()

```
ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.setState (
              self,
              state )
```

Set a textual state to the HIL signal variable.

This uses the lookup values supplied in constructor.

Parameters

state	named state to set

Returns

the previous state's description.

Reimplemented from ttk_bus._bus_signals_canmmbs.RxCanBusSignal.

7.139.4 Member Data Documentation

7.139.4.1 alias

```
ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.alias
```

bus signal alias name

7.139.4.2 base_path

```
ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.base_path
```

Signal model path as base path for deriving sub-element paths.

For CANMM signals, related variables follow the form <path_to_signal>_suffix

7.139.4.3 cfg_crc_error_type

```
int ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.cfg_crc_error_type = 2  [static]
```

Configuration setting for new instances:
CRC type for "incorrect" CRC algorithm (for error injection, for example calculated CRC + 1).

Note: Simply switching CRC calculation off (e.g. CRC = 0x00) may lead to detection failures, as a static CRC value might actually be correct for the current data.

7.139.4.4 cfg_crc_normal_type

```
int ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.cfg_crc_normal_type = 1  [static]
```

Configuration setting for new instances:
CRC type for "correct" CRC algorithm (normal mode)

7.139.4.5 cfg_custom_defaults

```
dict ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.cfg_custom_defaults  [static]
```

Initial value:

```
=  {
        "crc_enable_var": None,
        "crc_type_var":   None,

        "switch_var":     None,   # for parent RxCanBusSignal
        "dyncount_var":   None,   # for parent RxCanBusSignal
        "dynvalue_var":   None,   # for parent RxCanBusSignal
    }
```

Configuration setting for new instances:
Custom "hard" default value(s) to use for internal real time system variables.

Mapping: <variable name as string>: <default value>

A value of None (or a missing variable name) enables normal/automatic reset behavior.
See RxCanBusSignal for further details.

Typical default values:

- CRC enable (automatic CRC calculation) usually defaults to "enabled"
  - 0: CRC calculation disabled
  - 1: CRC calculation enabled
- CRC type (active CRC algorithm) should default to "correct CRC" (which is typically algorithm 1)

7.139.4.6 cfg_init_crc_dynamic_manipulation

```
bool ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.cfg_init_crc_dynamic_manipulation = False
[static]
```

Configuration setting for new instances:
Try to initialize (additional) variables that are only available if "dynamic crc" manipulation has been selected during RTICANMM configuration for a signal.

Additional variables:

- .crc_dyntype_var
- .crc_dyncount_var

7.139.4.7 change_delay

```
ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.change_delay
```

delay to wait after a value change [ms]

7.139.4.8 context

```
ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.context
```

parent context (an RTE application instance to resolve signal/variable paths)

7.139.4.9 crc_dyncount_var

```
ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.crc_dyncount_var
```

CRC Dynamic Countdown variable, enables dynamic CRC type while counting down to 0 at each message transmission (a "HilVar" instance), only available if configured in CANMM and enabled with in class config.

See cfg_init_crc_dynamic_manipulation

7.139.4.10 crc_dyntype_var

```
ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.crc_dyntype_var
```

CRC Dynamic Type variable, selects active CRC algorithm (a "HilVar" instance), only available if configured in CANMM and enabled in class config.

See cfg_init_crc_dynamic_manipulation

7.139.4.11 crc_enable_var

ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.crc_enable_var

CRC Enable variable, enables/disables automatic CRC calculation (a "HilVar" instance)

- 0: CRC calculation disabled
- 1: CRC calculation enabled

7.139.4.12 crc_error_type

ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.crc_error_type

CRC type for "incorrect" CRC algorithm (for error injection)

7.139.4.13 crc_normal_type

ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.crc_normal_type

CRC type for "correct" CRC algorithm (normal mode)

7.139.4.14 crc_type_var

ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.crc_type_var

CRC Type variable, selects active CRC algorithm (a "HilVar" instance)

7.139.4.15 cycle_time

ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.cycle_time

cycle time of Message/PDU in [ms]

7.139.4.16 debounce_time

ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.debounce_time

minimum debounce delay for event-triggered Messages/PDUs in [ms]

7.139.4.17 descr

ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.descr

bus signal description text

7.139.4.18 dyncount_var

ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.dyncount_var

Dynamic Countdown Value.

Only available if signal manipulation is configured in CANMM and enabled in class config.
See self.cfg_init_signal_manipulation

7.139.4.19 dynvalue_var

ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.dynvalue_var

Dynamic Value: value that is active while dyncount_var decrements.

Only available if signal manipulation is configured in CANMM and enabled in class config.
See self.cfg_init_signal_manipulation

7.139.4.20 error_delay

ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.error_delay

delay [ms] to wait until a basic CRC error (like "CRC invalid") should be detected.

delay to wait after a "basic" CRC error injection [ms]

This will be used in test functions.

7.139.4.21 message

ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.message

reference to associated (Rx-)CanBusMessage instance

7.139.4.22 recovery_time

ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.recovery_time

time for recovery/"wiedergut" checks [ms]

7.139.4.23 signal_var

ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.signal_var

Signal value variable.

get/info methods of this CanBusSignal will access this variable (a HilVar instance).

7.139.4.24 switch_var

`ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.switch_var`

Switch variable for signal manipulation.

Only available if more than one signal manipulation option is configured in CANMM.

Values:

- 0: model input (if configured/available)
- 1: constant from Trc/BusSystems
- 2: counter running (for counter signals, default switch setting for enabled counters)
- 3: constant + counter (for counter signals)
- 4: toggle value (for toggle signals)
- 5: parity value (for parity signals)
- 6: ∼parity value (for parity signals)
- 7: error value (if an error value is configured)
- 8: dynamic value (if dynamic value manipulation is configured)

7.139.4.25 timeout

`ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.timeout`

detection time for timeout errors [ms]

7.139.4.26 unit

`ttk_bus.bus_signals_canmmbs.CrcCanBusSignal.unit`

unit of physical signal value (as string, mainly used for report entries)

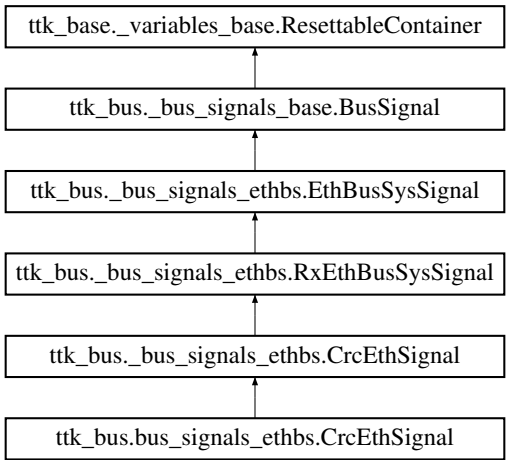
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_canmmbs.py (v1.8)

7.140 ttk_bus._bus_signals_ethbs.CrcEthSignal Class Reference

A CRC RX Ethernet event signal/parameter (HIL-Tx => ECU-Rx) for use with dSPACE Ethernet blockset.

Inheritance diagram for ttk_bus._bus_signals_ethbs.CrcEthSignal:



Public Member Functions

- __init__ (self, context, base_path, event=None, cycle_time=None, debounce_time=None, alias="", descr="", error_delay=None, change_delay=None, timeout=None, recovery_time=None, **kwargs)

  CRC Ethernet signal initialization (with error injection, via BusSystems) for dSPACE Ethernet blockset.
- setBasicCrcError (self)

  Switch CRC calculation to an algorithm producing incorrect CRCs.
- clearBasicCrcError (self)

  Clear a currently active CRC error injection.

Public Attributes

- crc_type_var

  CRC Type variable, selects active CRC algorithm (a "HilVar" instance)
- crc_enable_var

  CRC Enable variable, enables/disables automatic CRC calculation (a "HilVar" instance)
- error_delay

  delay to wait after a "basic" CRC error injection [ms]
- crc_normal_type

  CRC Type for normal mode (correct CRC calculation).
- crc_error_type

  CRC Type for error mode (incorrect CRC calculation, for example calculated CRC + 1).

Static Public Attributes

- int fallback_error_delay = 20

  fallback error delay in [ms] to use for signals/events without any specified or derived error_delay

7.140.1 Detailed Description

A CRC RX Ethernet event signal/parameter (HIL-Tx => ECU-Rx) for use with dSPACE Ethernet blockset.

Note

This class expects a BusSystems path structure.

7.140.2 Constructor & Destructor Documentation

7.140.2.1 __init__()

```
ttk_bus._bus_signals_ethbs.CrcEthSignal.__init__ (
              self,
              context,
              base_path,
              event = None,
              cycle_time = None,
              debounce_time = None,
              alias = "",
              descr = "",
              error_delay = None,
              change_delay = None,
              timeout = None,
              recovery_time = None,
             ** kwargs )
```

CRC Ethernet signal initialization (with error injection, via BusSystems) for dSPACE Ethernet blockset.

Example: BusSystems CRC Signal Path Structure

```
BusSystems/Ethernet/FRR/PSIs/ACME.ENVIRONMENTALMODEL/BrakeAssistFrontRadar_1/Events/BrakeAssistFrontRadar/E2E
Protection/cRC/CRC/Enable
BusSystems/Ethernet/FRR/PSIs/ACME.ENVIRONMENTALMODEL/BrakeAssistFrontRadar_1/Events/BrakeAssistFrontRadar/E2E
Protection/cRC/CRC/Type
BusSystems/Ethernet/FRR/PSIs/ACME.ENVIRONMENTALMODEL/BrakeAssistFrontRadar_1/Events/BrakeAssistFrontRadar/E2E
Protection/cRC/Source Switch
BusSystems/Ethernet/FRR/PSIs/ACME.ENVIRONMENTALMODEL/BrakeAssistFrontRadar_1/Events/BrakeAssistFrontRadar/E2E
Protection/cRC/Value
```

Suitable base_paths / examples:

BusSystems/Ethernet/FRR/PSIs/ACME.ENVMDL/BrakeAssistFR/Events/BrakeAssistFR/E2E Protection/c↩
RC/CRC/Enable
BusSystems/Ethernet/FRR/PSIs/ACME.ENVMDL/BrakeAssistFR/Events/BrakeAssistFR/E2E Protection/c↩
RC/Value
=> anything below .../CRC/* or the .../Value entry

Parameters

context	parent context (an RTE application or XilTestbench instance to resolve signal/variable paths)
base_path	base model path below BusSystems from which all required paths will be derived (use /CRC/Enable of actual crc signal)
event	reference to associated EthEvent instance
cycle_time	(opt) cycle time in [ms], overrides settings in EthEvent

Parameters

debounce_time	(opt) minimum debounce delay time between events in [ms], overrides settings in EthEvent
alias	(opt) bus signal alias/name
descr	(opt) signal description
error_delay	(opt) delay to wait until a simple crc error should be detected [ms], defaults to 10x cycle time
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x cycle_time
timeout	(opt) detection time for timeout errors [ms], defaults to 2x cycle_time
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x cycle_time
∗∗*kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.

Reimplemented in ttk_bus.bus_signals_ethbs.CrcEthSignal.

7.140.3 Member Function Documentation

7.140.3.1 clearBasicCrcError()

```
ttk_bus._bus_signals_ethbs.CrcEthSignal.clearBasicCrcError (
              self )
```

Clear a currently active CRC error injection.

Reimplemented in ttk_bus.bus_signals_ethbs.CrcEthSignal.

7.140.3.2 setBasicCrcError()

```
ttk_bus._bus_signals_ethbs.CrcEthSignal.setBasicCrcError (
              self )
```

Switch CRC calculation to an algorithm producing incorrect CRCs.

Reimplemented in ttk_bus.bus_signals_ethbs.CrcEthSignal.

7.140.4 Member Data Documentation

7.140.4.1 crc_enable_var

```
ttk_bus._bus_signals_ethbs.CrcEthSignal.crc_enable_var
```

CRC Enable variable, enables/disables automatic CRC calculation (a "HilVar" instance)

- 0: CRC calculation disabled
- 1: CRC calculation enabled

7.140.4.2 crc_error_type

`ttk_bus._bus_signals_ethbs.CrcEthSignal.crc_error_type`

CRC Type for error mode (incorrect CRC calculation, for example calculated CRC + 1).

See EthernetConfigurationPackageApplicationNote.pdf, PSI Group, E2E Protection group:

1: An invalid checksum is calculated.
The checksum calculated according to AUTOSAR E2E Protection Profile 4 CRC-32 is incremented by 1 to ensure an invalid checksum.

7.140.4.3 crc_normal_type

`ttk_bus._bus_signals_ethbs.CrcEthSignal.crc_normal_type`

CRC Type for normal mode (correct CRC calculation).

See EthernetConfigurationPackageApplicationNote.pdf, PSI Group, E2E Protection group:

0: The checksum is calculated according to AUTOSAR E2E Protection Profile 4.

7.140.4.4 crc_type_var

`ttk_bus._bus_signals_ethbs.CrcEthSignal.crc_type_var`

CRC Type variable, selects active CRC algorithm (a "HilVar" instance)

7.140.4.5 error_delay

`ttk_bus._bus_signals_ethbs.CrcEthSignal.error_delay`

delay to wait after a "basic" CRC error injection [ms]

7.140.4.6 fallback_error_delay

`int ttk_bus._bus_signals_ethbs.CrcEthSignal.fallback_error_delay = 20 [static]`

fallback error delay in [ms] to use for signals/events without any specified or derived error_delay

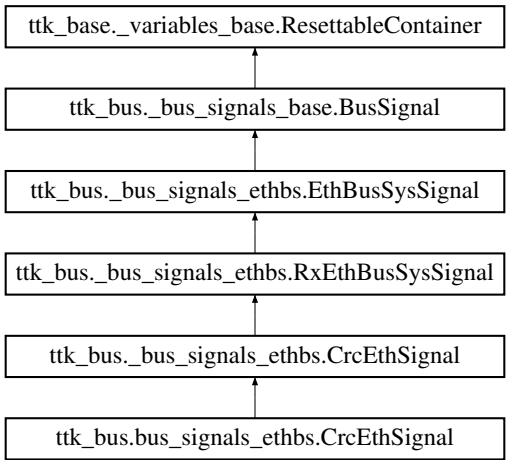
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_ethbs.py (v2.5)

7.141 ttk_bus.bus_signals_ethbs.CrcEthSignal Class Reference

A CRC RX Ethernet event signal/parameter (HIL-Tx => ECU-Rx) for use with dSPACE Ethernet blockset.

Inheritance diagram for ttk_bus.bus_signals_ethbs.CrcEthSignal:



Public Member Functions

- __init__ (self, context, base_path, event=None, cycle_time=None, debounce_time=None, alias="", descr="", error_delay=None, change_delay=None, timeout=None, recovery_time=None, **kwargs)

  CRC Ethernet signal initialization (with error injection, via BusSystems) for dSPACE Ethernet blockset.
- setBasicCrcError (self)

  Switch CRC calculation to an algorithm producing incorrect CRCs.
- clearBasicCrcError (self)

  Clear a currently active CRC error injection.
- enableHilTx (self)

  Enable transmission of signal HIL => DUT (if the assigned EthEvent supports it).
- disableHilTx (self)

  Disable transmission of signal HIL => DUT (if the assigned EthEvent supports it).
- kickout (self)

  Trigger sending of the Ethernet Event / Field.
- get (self)

  Get the current value of the HIL signal variable (typically not that useful for ECU-Rx/HIL-Tx AC-signals, though).
- set (self, value)

  Set a value to the HIL signal variable.
- getState (self)

  Get a textual state for the HIL signal variable's current data.
- setState (self, state)

  Set a textual state to the HIL signal variable.
- getStateDescr (self, value=None, fallback=None)

  Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).
- info (self)

  Get a current info string about the HIL signal variable.
- reset (self, *args, **kwargs)

  Reset all resettable variables of this bus signal.

Public Attributes

- crc_normal_type
  CRC type for "correct" CRC algorithm (normal mode)
- crc_error_type
  CRC type for "incorrect" CRC algorithm (for error injection)
- error_delay
  delay [ms] to wait until a basic CRC error (like "CRC invalid") should be detected.
- crc_type_var
  CRC Type variable, selects active CRC algorithm (a "HilVar" instance)
- crc_enable_var
  CRC Enable variable, enables/disables automatic CRC calculation (a "HilVar" instance)
- context
  parent context (an RTE application instance to resolve signal/variable paths)
- signal_path
  base path to signal below BusSystems
- event
  reference to associated EthEvent instance
- unit
  unit of physical signal value (as string, mainly used for report entries)
- alias
  bus signal alias name
- descr
  signal description text
- lookup
  lookup-table for mapping discrete signal states to textual descriptions
- cycle_time
  cycle time of signal in [ms]
- debounce_time
  minimum debounce delay for triggered signals in [ms]
- change_delay
  delay to wait after a value change [ms]
- timeout
  detection time for timeout errors [ms]
- recovery_time
  time for recovery/"wiedergut" checks [ms]

7.141.1 Detailed Description

A CRC RX Ethernet event signal/parameter (HIL-Tx => ECU-Rx) for use with dSPACE Ethernet blockset.

Note

This class expects a BusSystems path structure.

7.141.2 Constructor & Destructor Documentation

7.141.2.1 __init__()

```
ttk_bus.bus_signals_ethbs.CrcEthSignal.__init__ (
              self,
              context,
              base_path,
              event = None,
              cycle_time = None,
              debounce_time = None,
              alias = "",
              descr = "",
              error_delay = None,
              change_delay = None,
              timeout = None,
              recovery_time = None,
             ** kwargs )
```

CRC Ethernet signal initialization (with error injection, via BusSystems) for dSPACE Ethernet blockset.

Example: BusSystems CRC Signal Path Structure

```
BusSystems/Ethernet/FRR/PSIs/ACME.ENVIRONMENTALMODEL/BrakeAssistFrontRadar_1/Events/BrakeAssistFrontRadar/E2E
Protection/cRC/CRC/Enable
BusSystems/Ethernet/FRR/PSIs/ACME.ENVIRONMENTALMODEL/BrakeAssistFrontRadar_1/Events/BrakeAssistFrontRadar/E2E
Protection/cRC/CRC/Type
BusSystems/Ethernet/FRR/PSIs/ACME.ENVIRONMENTALMODEL/BrakeAssistFrontRadar_1/Events/BrakeAssistFrontRadar/E2E
Protection/cRC/Source Switch
BusSystems/Ethernet/FRR/PSIs/ACME.ENVIRONMENTALMODEL/BrakeAssistFrontRadar_1/Events/BrakeAssistFrontRadar/E2E
Protection/cRC/Value
```

Suitable base_paths / examples:

BusSystems/Ethernet/FRR/PSIs/ACME.ENVMDL/BrakeAssistFR/Events/BrakeAssistFR/E2E Protection/c↩
RC/CRC/Enable
BusSystems/Ethernet/FRR/PSIs/ACME.ENVMDL/BrakeAssistFR/Events/BrakeAssistFR/E2E Protection/c↩
RC/Value
=> anything below .../CRC/* or the .../Value entry

Parameters

context	parent context (an RTE application or XilTestbench instance to resolve signal/variable paths)
base_path	base model path below BusSystems from which all required paths will be derived (use /CRC/Enable of actual crc signal)
event	reference to associated EthEvent instance
cycle_time	(opt) cycle time in [ms], overrides settings in EthEvent
debounce_time	(opt) minimum debounce delay time between events in [ms], overrides settings in EthEvent
alias	(opt) bus signal alias/name
descr	(opt) signal description
error_delay	(opt) delay to wait until a simple crc error should be detected [ms], defaults to 10x cycle time
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x cycle_time
timeout	(opt) detection time for timeout errors [ms], defaults to 2x cycle_time
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x cycle_time
∗∗*kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_bus._bus_signals_ethbs.CrcEthSignal.

7.141.3 Member Function Documentation

7.141.3.1 clearBasicCrcError()

```
ttk_bus.bus_signals_ethbs.CrcEthSignal.clearBasicCrcError (
              self )
```

Clear a currently active CRC error injection.

Reimplemented from ttk_bus._bus_signals_ethbs.CrcEthSignal.

7.141.3.2 disableHilTx()

```
ttk_bus.bus_signals_ethbs.CrcEthSignal.disableHilTx (
              self )
```

Disable transmission of signal HIL => DUT (if the assigned EthEvent supports it).

Note

This disables transmission of all signals in the parent Event.

Reimplemented from ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.

7.141.3.3 enableHilTx()

```
ttk_bus.bus_signals_ethbs.CrcEthSignal.enableHilTx (
              self )
```

Enable transmission of signal HIL => DUT (if the assigned EthEvent supports it).

Note

This enables transmission of all signals in the parent Event.

Reimplemented from ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.

7.141.3.4 get()

```
ttk_bus.bus_signals_ethbs.CrcEthSignal.get (
              self )
```

Get the current value of the HIL signal variable (typically not that useful for ECU-Rx/HIL-Tx AC-signals, though).

Returns

the current value from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.141.3.5 getState()

```
ttk_bus.bus_signals_ethbs.CrcEthSignal.getState (
              self )
```

Get a textual state for the HIL signal variable's current data.

(typically not that useful for ECU-Rx/HIL-Tx AC-signals, though).

Returns

the current state's description.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.141.3.6 getStateDescr()

```
ttk_bus.bus_signals_ethbs.CrcEthSignal.getStateDescr (
              self,
              value = None,
              fallback = None )
```

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the last read value will be used instead.
fallback	fallback value to return if no match was found. Default: None

Returns

the state text if a match was found, otherwise the specified fallback value.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.141.3.7 info()

```
ttk_bus.bus_signals_ethbs.CrcEthSignal.info (
              self )
```

Get a current info string about the HIL signal variable.

Returns

an info string.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.141.3.8 kickout()

```
ttk_bus.bus_signals_ethbs.CrcEthSignal.kickout (
              self )
```

Trigger sending of the Ethernet Event / Field.

Note

All other signals of the parent EthEvent also will be sent.

Reimplemented from ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.

7.141.3.9 reset()

```
ttk_bus.bus_signals_ethbs.CrcEthSignal.reset (
              self,
             * args,
             ** kwargs )
```

Reset all resettable variables of this bus signal.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.ResettableContainer.

7.141.3.10 set()

```
ttk_bus.bus_signals_ethbs.CrcEthSignal.set (
              self,
              value )
```

Set a value to the HIL signal variable.

Parameters

value	value to set

Returns

the previous value from signal_var.

Reimplemented from ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.

7.141.3.11 setBasicCrcError()

```
ttk_bus.bus_signals_ethbs.CrcEthSignal.setBasicCrcError (
              self )
```

Switch CRC calculation to an algorithm producing incorrect CRCs.

Reimplemented from ttk_bus._bus_signals_ethbs.CrcEthSignal.

7.141.3.12 setState()

```
ttk_bus.bus_signals_ethbs.CrcEthSignal.setState (
              self,
              state )
```

Set a textual state to the HIL signal variable.

This uses the lookup values supplied in constructor.

Parameters

state	named state to set

Returns

the previous state's description.

Reimplemented from ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.

7.141.4 Member Data Documentation

7.141.4.1 alias

```
ttk_bus.bus_signals_ethbs.CrcEthSignal.alias
```

bus signal alias name

7.141.4.2 change_delay

```
ttk_bus.bus_signals_ethbs.CrcEthSignal.change_delay
```

delay to wait after a value change [ms]

7.141.4.3 context

```
ttk_bus.bus_signals_ethbs.CrcEthSignal.context
```

parent context (an RTE application instance to resolve signal/variable paths)

7.141.4.4 crc_enable_var

`ttk_bus.bus_signals_ethbs.CrcEthSignal.crc_enable_var`

CRC Enable variable, enables/disables automatic CRC calculation (a "HilVar" instance)

- 0: CRC calculation disabled
- 1: CRC calculation enabled

7.141.4.5 crc_error_type

`ttk_bus.bus_signals_ethbs.CrcEthSignal.crc_error_type`

CRC type for "incorrect" CRC algorithm (for error injection)

7.141.4.6 crc_normal_type

`ttk_bus.bus_signals_ethbs.CrcEthSignal.crc_normal_type`

CRC type for "correct" CRC algorithm (normal mode)

7.141.4.7 crc_type_var

`ttk_bus.bus_signals_ethbs.CrcEthSignal.crc_type_var`

CRC Type variable, selects active CRC algorithm (a "HilVar" instance)

7.141.4.8 cycle_time

`ttk_bus.bus_signals_ethbs.CrcEthSignal.cycle_time`

cycle time of signal in [ms]

7.141.4.9 debounce_time

`ttk_bus.bus_signals_ethbs.CrcEthSignal.debounce_time`

minimum debounce delay for triggered signals in [ms]

7.141.4.10 descr

`ttk_bus.bus_signals_ethbs.CrcEthSignal.descr`

signal description text

7.141.4.11 error_delay

ttk_bus.bus_signals_ethbs.CrcEthSignal.error_delay

delay [ms] to wait until a basic CRC error (like "CRC invalid") should be detected.

This will be used in test functions.

7.141.4.12 event

ttk_bus.bus_signals_ethbs.CrcEthSignal.event

reference to associated EthEvent instance

7.141.4.13 lookup

ttk_bus.bus_signals_ethbs.CrcEthSignal.lookup

lookup-table for mapping discrete signal states to textual descriptions

7.141.4.14 recovery_time

ttk_bus.bus_signals_ethbs.CrcEthSignal.recovery_time

time for recovery/"wiedergut" checks [ms]

7.141.4.15 signal_path

ttk_bus.bus_signals_ethbs.CrcEthSignal.signal_path

base path to signal below BusSystems

7.141.4.16 timeout

ttk_bus.bus_signals_ethbs.CrcEthSignal.timeout

detection time for timeout errors [ms]

7.141.4.17 unit

ttk_bus.bus_signals_ethbs.CrcEthSignal.unit

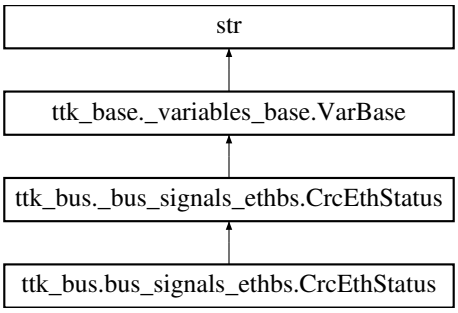
unit of physical signal value (as string, mainly used for report entries)

The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_ethbs.py (v2.5)

7.142 ttk_bus._bus_signals_ethbs.CrcEthStatus Class Reference

Inheritance diagram for ttk_bus._bus_signals_ethbs.CrcEthStatus:



Public Member Functions

- __new__ (cls, context, crc_data_path, ∗args, ∗∗kwargs)

  See VarBase.__new__.
- __init__ (self, context, crc_data_path, alias="", descr="")

  HIL-Rx CRC-check status variables.
- enableCrcCalc (self)

  Enable CRC calculation/evaluation regardless of setting in model.
- get (self)

  Get current CRC status value.
- reset (self)

  Reset CRC calculation settings (switch back to calculation settings from "model").

Public Attributes

- status_var

  Status variable for CRC check ("CRC Status", read-only)
- enable_var

  CRC calculation/check enable variable ("CRC Enable"):
- switch_var

  Source Switch variable for CRC check enable:

7.142.1 Constructor & Destructor Documentation

7.142.1.1 __init__()

```
ttk_bus._bus_signals_ethbs.CrcEthStatus.__init__ (
              self,
              context,
              crc_data_path,
              alias = "",
              descr = "" )
```

HIL-Rx CRC-check status variables.

CRC check status values:

- 0: CRC valid
- 1: CRC invalid
- 2: CRC check is disabled

Example: BusSystems Path Structure

```
# CRC enable variable:
BusSystems/Ethernet/FRR/CSIs/ACME.ENVIRONMENTALMODEL/RoadDescription_3/Events/RoadForesight/E2E
Protection/cRC/CRC/Enable
# CRC status variable:
BusSystems/Ethernet/FRR/CSIs/ACME.ENVIRONMENTALMODEL/RoadDescription_3/Events/RoadForesight/E2E
Protection/cRC/CRC/Status
# CRC type variable:
BusSystems/Ethernet/FRR/CSIs/ACME.ENVIRONMENTALMODEL/RoadDescription_3/Events/RoadForesight/E2E
Protection/cRC/CRC/Type
# CRC signal variable:
BusSystems/Ethernet/FRR/CSIs/ACME.ENVIRONMENTALMODEL/RoadDescription_3/Events/RoadForesight/E2E
Protection/cRC/Value
```

Parameters

context	parent context (an RTE application or XilTestbench instance to resolve signal/variable paths)
crc_data_path	model path to /CRC/Enable branch of the event group (or directly to the /CRC/Status variable)
alias	(opt) CRC signal alias/name
descr	(opt) CRC signal description

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_bus.bus_signals_ethbs.CrcEthStatus.

7.142.2 Member Function Documentation

7.142.2.1 __new__()

```
ttk_bus._bus_signals_ethbs.CrcEthStatus.__new__ (
              cls,
              context,
              crc_data_path,
             * args,
             ** kwargs )
```

See VarBase.__new__.

Parameters

context	parent context, see **init**
crc_data_path	a model path, see **init**
∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.VarBase.

7.142.2.2 enableCrcCalc()

```
ttk_bus._bus_signals_ethbs.CrcEthStatus.enableCrcCalc (
              self )
```

Enable CRC calculation/evaluation regardless of setting in model.

Reimplemented in ttk_bus.bus_signals_ethbs.CrcEthStatus.

7.142.2.3 get()

```
ttk_bus._bus_signals_ethbs.CrcEthStatus.get (
              self )
```

Get current CRC status value.

Possible values are

- 0: CRC value is valid
- 1: CRC value is invalid
- 2: CRC check is disabled

Returns

the current status value.

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_bus.bus_signals_ethbs.CrcEthStatus.

7.142.2.4 reset()

```
ttk_bus._bus_signals_ethbs.CrcEthStatus.reset (
              self )
```

Reset CRC calculation settings (switch back to calculation settings from "model").

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_bus.bus_signals_ethbs.CrcEthStatus.

7.142.3 Member Data Documentation

7.142.3.1 enable_var

```
ttk_bus._bus_signals_ethbs.CrcEthStatus.enable_var
```

CRC calculation/check enable variable ("CRC Enable"):

- 0: calculation/check disabled
- 1: calculation/check enabled

7.142.3.2 status_var

`ttk_bus._bus_signals_ethbs.CrcEthStatus.status_var`

Status variable for CRC check ("CRC Status", read-only)

- 0: CRC value is valid
- 1: CRC value is invalid
- 2: CRC check is disabled

7.142.3.3 switch_var

`ttk_bus._bus_signals_ethbs.CrcEthStatus.switch_var`

Source Switch variable for CRC check enable:

- 0: Simulink Model
- 1: BusSystems

This is unused for CRC-Enable (but kept for consistency with FR & CAN equivalents). => ECP E2E BuySystems structure does not use/have a switch var (as E2E info is n/a in SL/Model)

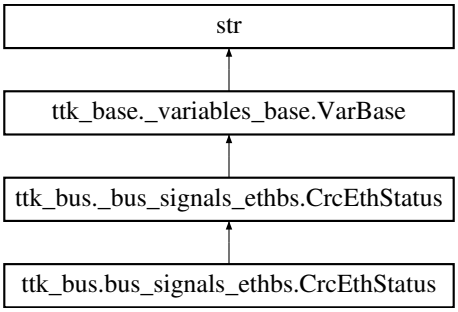
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_ethbs.py (v2.5)

7.143 ttk_bus.bus_signals_ethbs.CrcEthStatus Class Reference

CRC Tx-Event/Signal status (ECU PSI Tx => HIL CSI Rx) and associated switch variables for dSPACE Ethernet blockset.

Inheritance diagram for ttk_bus.bus_signals_ethbs.CrcEthStatus:



Public Member Functions

- __init__ (self, context, crc_data_path, alias="", descr="")

  HIL-Rx CRC-check status variables.
- enableCrcCalc (self)

  Enable CRC calculation/evaluation regardless of setting in model.
- get (self)

  Get current CRC status value.
- getState (self)

  Get a textual state description for the current CRC status value.
- reset (self)

  Reset CRC calculation settings (switch back to calculation settings from "model").

Public Attributes

- enable_var

  CRC calculation/check enable variable ("CRC Enable"):
- switch_var

  Source Switch variable for CRC check enable:

7.143.1 Detailed Description

CRC Tx-Event/Signal status (ECU PSI Tx => HIL CSI Rx) and associated switch variables for dSPACE Ethernet blockset.

Note

This class expects a BusSystems path structure.

This behaves essentially as a basic "HilVar" containing the CRC status plus additional sub-variables.

7.143.2 Constructor & Destructor Documentation

7.143.2.1 __init__()

```
ttk_bus.bus_signals_ethbs.CrcEthStatus.__init__ (
              self,
              context,
              crc_data_path,
              alias = "",
              descr = "" )
```

HIL-Rx CRC-check status variables.

CRC check status values:

- 0: CRC valid
- 1: CRC invalid
- 2: CRC check is disabled

Example: BusSystems Path Structure

```
# CRC enable variable:
BusSystems/Ethernet/FRR/CSIs/ACME.ENVIRONMENTALMODEL/RoadDescription_3/Events/RoadForesight/E2E
Protection/cRC/CRC/Enable
# CRC status variable:
BusSystems/Ethernet/FRR/CSIs/ACME.ENVIRONMENTALMODEL/RoadDescription_3/Events/RoadForesight/E2E
Protection/cRC/CRC/Status
# CRC type variable:
BusSystems/Ethernet/FRR/CSIs/ACME.ENVIRONMENTALMODEL/RoadDescription_3/Events/RoadForesight/E2E
Protection/cRC/CRC/Type
# CRC signal variable:
BusSystems/Ethernet/FRR/CSIs/ACME.ENVIRONMENTALMODEL/RoadDescription_3/Events/RoadForesight/E2E
Protection/cRC/Value
```

Parameters

context	parent context (an RTE application or XilTestbench instance to resolve signal/variable paths)
crc_data_path	model path to /CRC/Enable branch of the event group (or directly to the /CRC/Status variable)
alias	(opt) CRC signal alias/name
descr	(opt) CRC signal description

Reimplemented from ttk_bus._bus_signals_ethbs.CrcEthStatus.

7.143.3 Member Function Documentation

7.143.3.1 enableCrcCalc()

```
ttk_bus.bus_signals_ethbs.CrcEthStatus.enableCrcCalc (
              self )
```

Enable CRC calculation/evaluation regardless of setting in model.

Reimplemented from ttk_bus._bus_signals_ethbs.CrcEthStatus.

7.143.3.2 get()

```
ttk_bus.bus_signals_ethbs.CrcEthStatus.get (
              self )
```

Get current CRC status value.

Possible values are

- 0: CRC value is valid
- 1: CRC value is invalid
- 2: CRC check is disabled

Returns

the current status value.

Reimplemented from ttk_bus._bus_signals_ethbs.CrcEthStatus.

7.143.3.3 getState()

```
ttk_bus.bus_signals_ethbs.CrcEthStatus.getState (
              self )
```

Get a textual state description for the current CRC status value.

Returns

the current state representation.

Reimplemented from ttk_base._variables_base.VarBase.

7.143.3.4 reset()

```
ttk_bus.bus_signals_ethbs.CrcEthStatus.reset (
              self )
```

Reset CRC calculation settings (switch back to calculation settings from "model").

Reimplemented from ttk_bus._bus_signals_ethbs.CrcEthStatus.

7.143.4 Member Data Documentation

7.143.4.1 enable_var

```
ttk_bus.bus_signals_ethbs.CrcEthStatus.enable_var
```

CRC calculation/check enable variable ("CRC Enable"):

- 0: calculation/check disabled
- 1: calculation/check enabled

7.143.4.2 switch_var

```
ttk_bus.bus_signals_ethbs.CrcEthStatus.switch_var
```

Source Switch variable for CRC check enable:

- 0: Simulink Model
- 1: BusSystems

This is unused for CRC-Enable (but kept for consistency with FR & CAN equivalents). => ECP E2E BuySystems structure does not use/have a switch var (as E2E info is n/a in SL/Model)

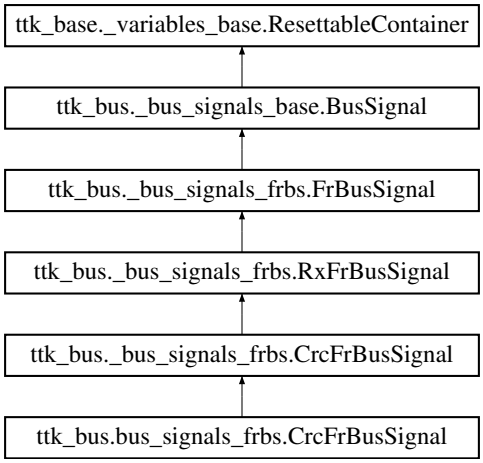
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_ethbs.py (v2.5)

7.144 ttk_bus._bus_signals_frbs.CrcFrBusSignal Class Reference

A CRC RX bus signal derived from RxFrBusSignal (HIL-Tx => ECU-Rx) for use with dSPACE FlexRay blockset with PDU-based structure.

Inheritance diagram for ttk_bus._bus_signals_frbs.CrcFrBusSignal:



Public Member Functions

- __init__ (self, context, base_model_path, pdu=None, cycle_time=None, debounce_time=None, alias="", descr="", error_delay=None, change_delay=None, timeout=None, recovery_time=None)

  CRC bus signal initialization for dSPACE FlexRay blockset.
- setBasicCrcError (self)

  Switch CRC calculation to an algorithm producing incorrect CRCs.
- clearBasicCrcError (self)

  Switch CRC calculation to an algorithm producing correct CRCs (which should be the default case).

Public Attributes

- crc_switch_var

  Source Switch variable for CRC Type/Enable (settings below /CRC Data/) (a "HilVar" instance)
- crc_type_var

  CRC Type variable, selects active CRC algorithm (a "HilVar" instance)
- crc_enable_var

  CRC Enable variable, enables/disables automatic CRC calculation (a "HilVar" instance)
- crc_normal_type

  CRC type for "correct" CRC algorithm (normal mode).
- crc_error_type

  CRC type for "incorrect" CRC algorithm (for error injection).
- error_delay

  delay [ms] to wait until a basic CRC error (like "CRC invalid") should be detected.

Static Public Attributes

- int cfg_crc_normal_type = 1

  Configuration setting for new instances: CRC type for "correct" CRC algorithm (normal mode).
- int cfg_crc_error_type = 2

  Configuration setting for new instances: CRC type for "incorrect" CRC algorithm (for error injection).
- dict cfg_custom_defaults

  Configuration settings for new instances:
  Custom "hard" default value(s) to use for internal real time system variables.

7.144.1 Detailed Description

A CRC RX bus signal derived from RxFrBusSignal (HIL-Tx => ECU-Rx) for use with dSPACE FlexRay blockset with PDU-based structure.

See RxFrBusSignal for inherited methods.

7.144.2 Constructor & Destructor Documentation

7.144.2.1 __init__()

```
ttk_bus._bus_signals_frbs.CrcFrBusSignal.__init__ (
              self,
              context,
              base_model_path,
              pdu = None,
              cycle_time = None,
              debounce_time = None,
              alias = "",
              descr = "",
              error_delay = None,
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

CRC bus signal initialization for dSPACE FlexRay blockset.

See RxFrBusSignal.

Note

CRC Types / Switch Codes: CRC Types for normal mode (correct CRC calculation) and error mode (incorrect CRC calculation, e.g. calculated CRC + 1) are defined in members crc_normal_type and crc_error_type.

Note that simply switching the CRC calculation off (e.g. CRC = 0x00) may lead to detection failures, as a static CRC value might actually be correct for the current data.

If you need different switch codes in your project, consider setting CrcFrBusSignal.cfg_crc_normal_type and/or CrcFrBusSignal.cfg_crc_error_type before instantiating signals or sub-classing CrcFrBusSignal and overriding the crc types.

Example: BusSystems CRC Signal Path Structure

```
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/CRC_PDU_NAME_OUT/Value
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/CRC_PDU_NAME_OUT/Dynamic Value
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/CRC_PDU_NAME_OUT/Countdown Value
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/CRC_PDU_NAME_OUT/Source Switch
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/CRC Data/CRC Enable
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/CRC Data/Type
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/CRC Data/Source Switch

# base_model_path == "Value" path:
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/CRC_PDU_NAME_OUT/Value
```

Parameters

context	rtplib application instance
base_model_path	base model path from which all required paths will be derived (use /Value of actual crc signal)
pdu	reference to associated FrRxPdu instance
cycle_time	(opt) cycle time in [ms], overrides settings in PDU
debounce_time	(opt) minimum debounce delay time in [ms], overrides settings in PDU
alias	(opt) bus signal alias/name
descr	(opt) signal description
error_delay	(opt) delay to wait until a simple CRC error should be detected [ms], defaults to 10x max(cycle_time, debounce_time)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_frbs.RxFrBusSignal.

Reimplemented in ttk_bus.bus_signals_frbs.CrcFrBusSignal.

7.144.3 Member Function Documentation

7.144.3.1 clearBasicCrcError()

```
ttk_bus._bus_signals_frbs.CrcFrBusSignal.clearBasicCrcError (
              self )
```

Switch CRC calculation to an algorithm producing correct CRCs (which should be the default case).

See CrcFrBusSignal.crc_normal_type.

Note

Use reset() to switch back to CRC Type from Simulink/Model.

Reimplemented in ttk_bus.bus_signals_frbs.CrcFrBusSignal.

7.144.3.2 setBasicCrcError()

```
ttk_bus._bus_signals_frbs.CrcFrBusSignal.setBasicCrcError (
              self )
```

Switch CRC calculation to an algorithm producing incorrect CRCs.

See CrcFrBusSignal.crc_error_type.

Note

If you want to use a different `crc_error_type` for a few test cases and re-use test functions that call this method for simple error injection, you could reconfigure the active `crc_error_type` on-the-fly.

Example: Temporarily reconfiguring crc_error_type

```
# use error type 4711 for a single test
orig_error_type = crc_signal.crc_error_type
crc_signal.crc_error_type = 4711
testBasicCrcError(crc_signal, ...)
crc_signal.crc_error_type = orig_error_type
```

Reimplemented in ttk_bus.bus_signals_frbs.CrcFrBusSignal.

7.144.4 Member Data Documentation

7.144.4.1 cfg_crc_error_type

```
int ttk_bus._bus_signals_frbs.CrcFrBusSignal.cfg_crc_error_type = 2  [static]
```

Configuration setting for new instances: CRC type for "incorrect" CRC algorithm (for error injection).

7.144.4.2 cfg_crc_normal_type

```
int ttk_bus._bus_signals_frbs.CrcFrBusSignal.cfg_crc_normal_type = 1  [static]
```

Configuration setting for new instances: CRC type for "correct" CRC algorithm (normal mode).

7.144.4.3 cfg_custom_defaults

```
dict ttk_bus._bus_signals_frbs.CrcFrBusSignal.cfg_custom_defaults  [static]
```

Initial value:

```
=  {
        "crc_type_var":     None,
        "crc_enable_var":   None,
        "crc_switch_var":   None, # default for crc enable/type switch
        "switch_var":       None, # default for static "value" switch (from base RxFrBusSignal)
    }
```

Configuration settings for new instances:
Custom "hard" default value(s) to use for internal real time system variables.

See RxFrBusSignal for further details.

Mapping: <variable name as string>: <default value>

Typical default values:

- source switches usually default to "source: simulink model"
  - 0: source: Simulink Model / TRC
  - 1: source: BusSystems
- CRC enable (automatic CRC calculation) usually defaults to "enabled"
  - 0: CRC calculation disabled
  - 1: CRC calculation enabled
- CRC type (active CRC algorithm) should default to "correct CRC" (which is typically algorithm 1)

7.144.4.4 crc_enable_var

`ttk_bus._bus_signals_frbs.CrcFrBusSignal.crc_enable_var`

CRC Enable variable, enables/disables automatic CRC calculation (a "HilVar" instance)

- 0: CRC calculation disabled
- 1: CRC calculation enabled

7.144.4.5 crc_error_type

`ttk_bus._bus_signals_frbs.CrcFrBusSignal.crc_error_type`

CRC type for "incorrect" CRC algorithm (for error injection).

Initial value gets copied from class attribute, see CrcFrBusSignal.cfg_crc_error_type

7.144.4.6 crc_normal_type

`ttk_bus._bus_signals_frbs.CrcFrBusSignal.crc_normal_type`

CRC type for "correct" CRC algorithm (normal mode).

Initial value gets copied from class attribute, see CrcFrBusSignal.cfg_crc_normal_type

7.144.4.7 crc_switch_var

`ttk_bus._bus_signals_frbs.CrcFrBusSignal.crc_switch_var`

Source Switch variable for CRC Type/Enable (settings below /CRC Data/) (a "HilVar" instance)

- 0: Simulink Model
- 1: BusSystems

7.144.4.8 crc_type_var

`ttk_bus._bus_signals_frbs.CrcFrBusSignal.crc_type_var`

CRC Type variable, selects active CRC algorithm (a "HilVar" instance)

7.144.4.9 error_delay

`ttk_bus._bus_signals_frbs.CrcFrBusSignal.error_delay`

delay [ms] to wait until a basic CRC error (like "CRC invalid") should be detected.

This will be used in test functions.

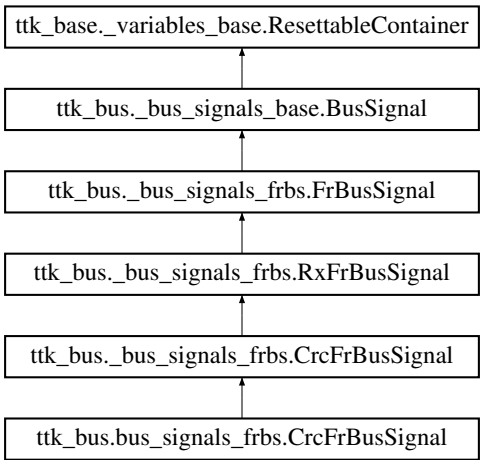
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_frbs.py (v2.5)

7.145 ttk_bus.bus_signals_frbs.CrcFrBusSignal Class Reference

A CRC RX bus signal derived from RxFrBusSignal (HIL-Tx => ECU-Rx) for use with dSPACE FlexRay blockset with PDU-based structure.

Inheritance diagram for ttk_bus.bus_signals_frbs.CrcFrBusSignal:



Public Member Functions

- __init__ (self, context, base_model_path, pdu=None, cycle_time=None, debounce_time=None, alias="", descr="", error_delay=None, change_delay=None, timeout=None, recovery_time=None)

  CRC bus signal initialization for dSPACE FlexRay blockset.
- setBasicCrcError (self)

  Switch CRC calculation to an algorithm producing incorrect CRCs.
- clearBasicCrcError (self)

  Switch CRC calculation to an algorithm producing correct CRCs (which should be the default case).
- enableHilTx (self)

  (Re-)enable sending of PDU, depending on selected disable_mode (see constructor).
- disableHilTx (self)

  Disable sending of PDU, depending on selected disable_mode (see constructor).
- kickout (self)

  Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent PDU).
- get (self)

Get the current value of the HIL signal variable (typically not that useful for ECU-Rx/HIL-Tx CRC-signals, though).

- set (self, value)

  Set a value to the HIL signal variable.
- getState (self)

  Get a textual state for the HIL signal variable's current data (typically not that useful for ECU-Rx/HIL-Tx CRC-signals, though).
- setState (self, state)

  Set a textual state to the HIL signal variable.
- getStateDescr (self, value=None, fallback=None)

  Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).
- info (self)

  Get a current info string about the HIL signal variable.
- reset (self, ∗args, ∗∗kwargs)

  Reset all resettable variables of this bus signal.

Public Attributes

- crc_switch_var

  Source Switch variable for CRC Type/Enable (settings below /CRC Data/) (a "HilVar" instance)
- crc_type_var

  CRC Type variable, selects active CRC algorithm (a "HilVar" instance)
- crc_enable_var

  CRC Enable variable, enables/disables automatic CRC calculation (a "HilVar" instance)
- crc_normal_type

  CRC type for "correct" CRC algorithm (normal mode).
- crc_error_type

  CRC type for "incorrect" CRC algorithm (for error injection).
- error_delay

  delay [ms] to wait until a basic CRC error (like "CRC invalid") should be detected.
- switch_var

  Source Switch variable for static value (if automatic CRC calculation is disabled) (a "HilVar" instance)
- value_path

  model path of value variable
- base_path

  base model path (base path for deriving sub-element paths)
- pdu

  reference to parent FrRxPdu
- context

  context (reference to assigned RTE application instance)
- signal_var

  main bus signal variable on HIL-side, (a "HilVar" instance)
- alias

  signal alias name (from signal_var)
- unit

  signal unit (from signal_var)
- descr

  signal description text
- cycle_time

  cycle time of signal in [ms] (that is, of the associated message or PDU)
- debounce_time

minimum debounce delay for event-triggered signals in [ms]

- change_delay

  delay to wait after a value change [ms]
- timeout

  detection time for timeout errors [ms]
- recovery_time

  time for recovery/"wiedergut" checks [ms]

Static Public Attributes

- int cfg_crc_normal_type = 1

  Configuration setting for new instances: CRC type for "correct" CRC algorithm (normal mode).
- int cfg_crc_error_type = 2

  Configuration setting for new instances: CRC type for "incorrect" CRC algorithm (for error injection).
- dict cfg_custom_defaults

  Configuration settings for new instances:
  Custom "hard" default value(s) to use for internal real time system variables.

7.145.1 Detailed Description

A CRC RX bus signal derived from RxFrBusSignal (HIL-Tx => ECU-Rx) for use with dSPACE FlexRay blockset with PDU-based structure.

See RxFrBusSignal for inherited methods.

7.145.2 Constructor & Destructor Documentation

7.145.2.1 __init__()

```
ttk_bus.bus_signals_frbs.CrcFrBusSignal.__init__ (
              self,
              context,
              base_model_path,
              pdu = None,
              cycle_time = None,
              debounce_time = None,
              alias = "",
              descr = "",
              error_delay = None,
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

CRC bus signal initialization for dSPACE FlexRay blockset.

See RxFrBusSignal.

Note

CRC Types / Switch Codes: CRC Types for normal mode (correct CRC calculation) and error mode (incorrect CRC calculation, e.g. calculated CRC + 1) are defined in members crc_normal_type and crc_error_type.

Note that simply switching the CRC calculation off (e.g. CRC = 0x00) may lead to detection failures, as a static CRC value might actually be correct for the current data.

If you need different switch codes in your project, consider setting CrcFrBusSignal.cfg_crc_normal_type and/or CrcFrBusSignal.cfg_crc_error_type before instantiating signals or sub-classing CrcFrBusSignal and overriding the crc types.

Example: BusSystems CRC Signal Path Structure

```
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/CRC_PDU_NAME_OUT/Value
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/CRC_PDU_NAME_OUT/Dynamic Value
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/CRC_PDU_NAME_OUT/Countdown Value
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/CRC_PDU_NAME_OUT/Source Switch
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/CRC Data/CRC Enable
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/CRC Data/Type
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/CRC Data/Source Switch

# base_model_path == "Value" path:
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/CRC_PDU_NAME_OUT/Value
```

Parameters

context	rtplib application instance
base_model_path	base model path from which all required paths will be derived (use /Value of actual crc signal)
pdu	reference to associated FrRxPdu instance
cycle_time	(opt) cycle time in [ms], overrides settings in PDU
debounce_time	(opt) minimum debounce delay time in [ms], overrides settings in PDU
alias	(opt) bus signal alias/name
descr	(opt) signal description
error_delay	(opt) delay to wait until a simple CRC error should be detected [ms], defaults to 10x max(cycle_time, debounce_time)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_frbs.CrcFrBusSignal.

7.145.3 Member Function Documentation

7.145.3.1 clearBasicCrcError()

```
ttk_bus.bus_signals_frbs.CrcFrBusSignal.clearBasicCrcError (
              self )
```

Switch CRC calculation to an algorithm producing correct CRCs (which should be the default case).

See CrcFrBusSignal.crc_normal_type.

Note

Use reset() to switch back to CRC Type from Simulink/Model.

Reimplemented from ttk_bus._bus_signals_frbs.CrcFrBusSignal.

7.145.3.2 disableHilTx()

```
ttk_bus.bus_signals_frbs.CrcFrBusSignal.disableHilTx (
              self )
```

Disable sending of PDU, depending on selected disable_mode (see constructor).

This overrides current settings from Simulink Model, use reset() to restore initial behavior.

Note

This disables transmission of all signals of the parent PDU.

Reimplemented from ttk_bus._bus_signals_frbs.RxFrBusSignal.

7.145.3.3 enableHilTx()

```
ttk_bus.bus_signals_frbs.CrcFrBusSignal.enableHilTx (
              self )
```

(Re-)enable sending of PDU, depending on selected disable_mode (see constructor).

This overrides current settings from Simulink Model, use reset() to restore initial behavior.

Note

This enables transmission of all signals of the parent PDU.

Reimplemented from ttk_bus._bus_signals_frbs.RxFrBusSignal.

7.145.3.4 get()

```
ttk_bus.bus_signals_frbs.CrcFrBusSignal.get (
              self )
```

Get the current value of the HIL signal variable (typically not that useful for ECU-Rx/HIL-Tx CRC-signals, though).

Returns

the current value from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.145.3.5 getState()

```
ttk_bus.bus_signals_frbs.CrcFrBusSignal.getState (
              self )
```

Get a textual state for the HIL signal variable's current data (typically not that useful for ECU-Rx/HIL-Tx CRC-signals, though).

Returns

the current state representation from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.145.3.6 getStateDescr()

```
ttk_bus.bus_signals_frbs.CrcFrBusSignal.getStateDescr (
              self,
              value = None,
              fallback = None )
```

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the last read value will be used instead.
fallback	fallback value to return if no match was found. Default: None

Returns

the state text if a match was found, otherwise the specified fallback value.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.145.3.7 info()

```
ttk_bus.bus_signals_frbs.CrcFrBusSignal.info (
              self )
```

Get a current info string about the HIL signal variable.

Returns

an info string.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.145.3.8 kickout()

```
ttk_bus.bus_signals_frbs.CrcFrBusSignal.kickout (
              self )
```

Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent PDU).

Note

This means all other signals of the parent PDU will be sent as well.

Reimplemented from ttk_bus._bus_signals_frbs.RxFrBusSignal.

7.145.3.9 reset()

```
ttk_bus.bus_signals_frbs.CrcFrBusSignal.reset (
              self,
             * args,
             ** kwargs )
```

Reset all resettable variables of this bus signal.

Note

This will also switch the Source Switch back to its default state (which is usually `2: running counter from BusSystems`).

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.ResettableContainer.

7.145.3.10 set()

```
ttk_bus.bus_signals_frbs.CrcFrBusSignal.set (
              self,
              value )
```

Set a value to the HIL signal variable.

Parameters

value	value to set

Returns

the previous value from signal_var

Reimplemented from ttk_bus._bus_signals_frbs.RxFrBusSignal.

7.145.3.11 setBasicCrcError()

```
ttk_bus.bus_signals_frbs.CrcFrBusSignal.setBasicCrcError (
              self )
```

Switch CRC calculation to an algorithm producing incorrect CRCs.

See CrcFrBusSignal.crc_error_type.

Note

If you want to use a different `crc_error_type` for a few test cases and re-use test functions that call this method for simple error injection, you could reconfigure the active `crc_error_type` on-the-fly.

Example: Temporarily reconfiguring crc_error_type

```
# use error type 4711 for a single test
orig_error_type = crc_signal.crc_error_type
crc_signal.crc_error_type = 4711
testBasicCrcError(crc_signal, ...)
crc_signal.crc_error_type = orig_error_type
```

Reimplemented from ttk_bus._bus_signals_frbs.CrcFrBusSignal.

7.145.3.12 setState()

```
ttk_bus.bus_signals_frbs.CrcFrBusSignal.setState (
              self,
              state )
```

Set a textual state to the HIL signal variable.

This uses the lookup values supplied in constructor

Parameters

state	named state to set

Returns

the previous state representation from signal_var.

Reimplemented from ttk_bus._bus_signals_frbs.RxFrBusSignal.

7.145.4 Member Data Documentation

7.145.4.1 alias

```
ttk_bus.bus_signals_frbs.CrcFrBusSignal.alias
```

signal alias name (from signal_var)

7.145.4.2 base_path

```
ttk_bus.bus_signals_frbs.CrcFrBusSignal.base_path
```

base model path (base path for deriving sub-element paths)

7.145.4.3 cfg_crc_error_type

```
int ttk_bus.bus_signals_frbs.CrcFrBusSignal.cfg_crc_error_type = 2  [static]
```

Configuration setting for new instances: CRC type for "incorrect" CRC algorithm (for error injection).

7.145.4.4 cfg_crc_normal_type

```
int ttk_bus.bus_signals_frbs.CrcFrBusSignal.cfg_crc_normal_type = 1  [static]
```

Configuration setting for new instances: CRC type for "correct" CRC algorithm (normal mode).

7.145.4.5 cfg_custom_defaults

```
dict ttk_bus.bus_signals_frbs.CrcFrBusSignal.cfg_custom_defaults  [static]
```

Initial value:

```
=  {
        "crc_type_var":     None,
        "crc_enable_var":   None,
        "crc_switch_var":   None, # default for crc enable/type switch
        "switch_var":       None, # default for static "value" switch (from base RxFrBusSignal)
    }
```

Configuration settings for new instances:
Custom "hard" default value(s) to use for internal real time system variables.

See RxFrBusSignal for further details.

Mapping: <variable name as string>: <default value>

Typical default values:

- source switches usually default to "source: simulink model"
  - 0: source: Simulink Model / TRC
  - 1: source: BusSystems
- CRC enable (automatic CRC calculation) usually defaults to "enabled"
  - 0: CRC calculation disabled
  - 1: CRC calculation enabled
- CRC type (active CRC algorithm) should default to "correct CRC" (which is typically algorithm 1)

7.145.4.6 change_delay

`ttk_bus.bus_signals_frbs.CrcFrBusSignal.change_delay`

delay to wait after a value change [ms]

7.145.4.7 context

`ttk_bus.bus_signals_frbs.CrcFrBusSignal.context`

context (reference to assigned RTE application instance)

7.145.4.8 crc_enable_var

`ttk_bus.bus_signals_frbs.CrcFrBusSignal.crc_enable_var`

CRC Enable variable, enables/disables automatic CRC calculation (a "HilVar" instance)

- 0: CRC calculation disabled
- 1: CRC calculation enabled

7.145.4.9 crc_error_type

`ttk_bus.bus_signals_frbs.CrcFrBusSignal.crc_error_type`

CRC type for "incorrect" CRC algorithm (for error injection).

Initial value gets copied from class attribute, see CrcFrBusSignal.cfg_crc_error_type

7.145.4.10 crc_normal_type

`ttk_bus.bus_signals_frbs.CrcFrBusSignal.crc_normal_type`

CRC type for "correct" CRC algorithm (normal mode).

Initial value gets copied from class attribute, see CrcFrBusSignal.cfg_crc_normal_type

7.145.4.11 crc_switch_var

`ttk_bus.bus_signals_frbs.CrcFrBusSignal.crc_switch_var`

Source Switch variable for CRC Type/Enable (settings below /CRC Data/) (a "HilVar" instance)

- 0: Simulink Model
- 1: BusSystems

7.145.4.12 crc_type_var

`ttk_bus.bus_signals_frbs.CrcFrBusSignal.crc_type_var`

CRC Type variable, selects active CRC algorithm (a "HilVar" instance)

7.145.4.13 cycle_time

`ttk_bus.bus_signals_frbs.CrcFrBusSignal.cycle_time`

cycle time of signal in [ms] (that is, of the associated message or PDU)

7.145.4.14 debounce_time

`ttk_bus.bus_signals_frbs.CrcFrBusSignal.debounce_time`

minimum debounce delay for event-triggered signals in [ms]

7.145.4.15 descr

`ttk_bus.bus_signals_frbs.CrcFrBusSignal.descr`

signal description text

7.145.4.16 error_delay

`ttk_bus.bus_signals_frbs.CrcFrBusSignal.error_delay`

delay [ms] to wait until a basic CRC error (like "CRC invalid") should be detected.

This will be used in test functions.

7.145.4.17 pdu

`ttk_bus.bus_signals_frbs.CrcFrBusSignal.pdu`

reference to parent FrRxPdu

7.145.4.18 recovery_time

`ttk_bus.bus_signals_frbs.CrcFrBusSignal.recovery_time`

time for recovery/"wiedergut" checks [ms]

7.145.4.19 signal_var

ttk_bus.bus_signals_frbs.CrcFrBusSignal.signal_var

main bus signal variable on HIL-side, (a "HilVar" instance)

7.145.4.20 switch_var

ttk_bus.bus_signals_frbs.CrcFrBusSignal.switch_var

Source Switch variable for static value (if automatic CRC calculation is disabled) (a "HilVar" instance)

- 0: Simulink Model
- 1: BusSystems

7.145.4.21 timeout

ttk_bus.bus_signals_frbs.CrcFrBusSignal.timeout

detection time for timeout errors [ms]

7.145.4.22 unit

ttk_bus.bus_signals_frbs.CrcFrBusSignal.unit

signal unit (from signal_var)

7.145.4.23 value_path

ttk_bus.bus_signals_frbs.CrcFrBusSignal.value_path

model path of value variable

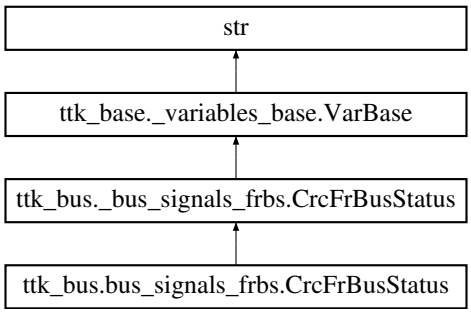
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_frbs.py (v1.8)

7.146 ttk_bus._bus_signals_frbs.CrcFrBusStatus Class Reference

CRC Tx-Signal status (ECU Tx => HIL Rx) and associated switch variables for dSPACE FlexRay blockset with PDU-based modeling.

Inheritance diagram for ttk_bus._bus_signals_frbs.CrcFrBusStatus:



Public Member Functions

- __init__ (self, context, crc_data_path, alias="", descr="")

  Constructor for HIL-Rx CRC-check status variables.
- enableCrcCalc (self)

  Enable CRC calculation/evaluation regardless of current setting in simulink model.
- get (self)

  Get the current CRC status value from FlexRay blockset.
- reset (self)

  Reset CRC check/calculation settings (default: switch back to current setting in Simulink model).

Public Attributes

- switch_var

  Source Switch variable for CRC check enable.
- enable_var

  CRC calculation/check enable variable ("CRC Enable")
- status_var

  Status variable for CRC check (read-only)

Static Public Attributes

- dict cfg_custom_defaults

  Configuration settings for new instances: Custom "hard" default value(s) to use for internal real time system variables (for example: switch settings)

Protected Attributes

- _default_values

7.146.1 Detailed Description

CRC Tx-Signal status (ECU Tx => HIL Rx) and associated switch variables for dSPACE FlexRay blockset with PDU-based modeling.

Note

This is essentially a basic "HilVar" containing the CRC status plus additional sub-variables.

7.146.2 Constructor & Destructor Documentation

7.146.2.1 __init__()

```
ttk_bus._bus_signals_frbs.CrcFrBusStatus.__init__ (
               self,
               context,
               crc_data_path,
               alias = "",
               descr = "" )
```

Constructor for HIL-Rx CRC-check status variables.

CRC check status values:

- 0: CRC valid
- 1: CRC invalid
- 2: CRC check is disabled

Example: BusSystems Path Structure

```
# CRC signal variable:
BusSystems/FlexRay/Monitoring/ECU_NAME/Channel A/PDU_NAME/Signals/CRC_PDU_NAME_IN/Value
# CRC status variable:
BusSystems/FlexRay/Monitoring/ECU_NAME/Channel A/PDU_NAME/CRC Data/Status
```

Parameters

context	parent context (an RTE application instance to resolve signal/variable paths)
crc_data_path	model path to /CRC Data/Status variable
alias	(opt) CRC signal alias/name
descr	(opt) CRC signal description

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_bus.bus_signals_frbs.CrcFrBusStatus.

7.146.3 Member Function Documentation

7.146.3.1 enableCrcCalc()

```
ttk_bus._bus_signals_frbs.CrcFrBusStatus.enableCrcCalc (
               self )
```

Enable CRC calculation/evaluation regardless of current setting in simulink model.

Use reset() to clear this override again.

Reimplemented in ttk_bus.bus_signals_frbs.CrcFrBusStatus.

7.146.3.2 get()

```
ttk_bus._bus_signals_frbs.CrcFrBusStatus.get (
              self )
```

Get the current CRC status value from FlexRay blockset.

CRC check status values are:

- 0: crc valid
- 1: crc invalid
- 2: check disabled

Returns

the current value from status var.

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_bus.bus_signals_frbs.CrcFrBusStatus.

7.146.3.3 reset()

```
ttk_bus._bus_signals_frbs.CrcFrBusStatus.reset (
              self )
```

Reset CRC check/calculation settings (default: switch back to current setting in Simulink model).

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_bus.bus_signals_frbs.CrcFrBusStatus.

7.146.4 Member Data Documentation

7.146.4.1 _default_values

```
ttk_bus._bus_signals_frbs.CrcFrBusStatus._default_values  [protected]
```

7.146.4.2 cfg_custom_defaults

```
dict ttk_bus._bus_signals_frbs.CrcFrBusStatus.cfg_custom_defaults  [static]
```

Initial value:

```
=  {
        "switch_var":   None,
        "enable_var":   None,
    }
```

Configuration settings for new instances: Custom "hard" default value(s) to use for internal real time system variables (for example: switch settings)

Mapping: <variable name as string>: <default value>

See description at FrRxPdu for additional details.

Typical default values:

- source switches usually default to "source: simulink model"
  - 0: source: simulink model / trc
  - 1: source: bus system
- CRC calculation/check enable variable: "disabled" is a safe default (enable in test as needed) as enabled checks w/ faulty CRCs will freeze remaining signal data at the "last known good" value.
  - 0: calculation/check disabled
  - 1: calculation/check enabled

7.146.4.3 enable_var

```
ttk_bus._bus_signals_frbs.CrcFrBusStatus.enable_var
```

CRC calculation/check enable variable ("CRC Enable")

- 0: calculation/check disabled
- 1: calculation/check enabled

7.146.4.4 status_var

```
ttk_bus._bus_signals_frbs.CrcFrBusStatus.status_var
```

Status variable for CRC check (read-only)

- 0: CRC value is valid
- 1: CRC value is invalid
- 2: CRC check is disabled

7.146.4.5 switch_var

`ttk_bus._bus_signals_frbs.CrcFrBusStatus.switch_var`

Source Switch variable for CRC check enable.

- 0: Simulink Model
- 1: BusSystems

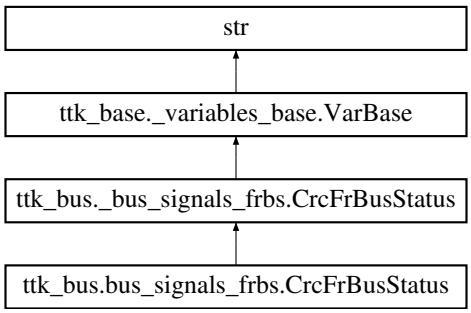
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_frbs.py (v2.5)

7.147 ttk_bus.bus_signals_frbs.CrcFrBusStatus Class Reference

CRC Tx-Signal status (ECU Tx => HIL Rx) and associated switch variables for dSPACE FlexRay blockset with PDU-based modeling.

Inheritance diagram for ttk_bus.bus_signals_frbs.CrcFrBusStatus:



Public Member Functions

- __init__ (self, context, crc_data_path, alias="", descr="")

  Constructor for HIL-Rx CRC-check status variables.
- enableCrcCalc (self)

  Enable CRC calculation/evaluation regardless of current setting in simulink model.
- get (self)

  Get the current CRC status value from FlexRay blockset.
- getState (self)

  Get a textual state for the current CRC status value from FlexRay blockset.
- reset (self)

  Reset CRC check/calculation settings (default: switch back to current setting in Simulink model).

Public Attributes

- switch_var

  Source Switch variable for CRC check enable.
- enable_var

  CRC calculation/check enable variable ("CRC Enable")
- status_var

  Status variable for CRC check (read-only)

Static Public Attributes

- dict cfg_custom_defaults

  Configuration settings for new instances: Custom "hard" default value(s) to use for internal real time system variables (for example: switch settings)

7.147.1 Detailed Description

CRC Tx-Signal status (ECU Tx => HIL Rx) and associated switch variables for dSPACE FlexRay blockset with PDU-based modeling.

Note

This is essentially a basic "HilVar" containing the CRC status plus additional sub-variables.

7.147.2 Constructor & Destructor Documentation

7.147.2.1 __init__()

```
ttk_bus.bus_signals_frbs.CrcFrBusStatus.__init__ (
              self,
              context,
              crc_data_path,
              alias = "",
              descr = "" )
```

Constructor for HIL-Rx CRC-check status variables.

CRC check status values:

- 0: CRC valid
- 1: CRC invalid
- 2: CRC check is disabled

Example: BusSystems Path Structure

```
# CRC signal variable:
BusSystems/FlexRay/Monitoring/ECU_NAME/Channel A/PDU_NAME/Signals/CRC_PDU_NAME_IN/Value
# CRC status variable:
BusSystems/FlexRay/Monitoring/ECU_NAME/Channel A/PDU_NAME/CRC Data/Status
```

Parameters

context	parent context (an RTE application instance to resolve signal/variable paths)
crc_data_path	model path to /CRC Data/Status variable
alias	(opt) CRC signal alias/name
descr	(opt) CRC signal description

Reimplemented from ttk_bus._bus_signals_frbs.CrcFrBusStatus.

7.147.3 Member Function Documentation

7.147.3.1 enableCrcCalc()

```
ttk_bus.bus_signals_frbs.CrcFrBusStatus.enableCrcCalc (
              self )
```

Enable CRC calculation/evaluation regardless of current setting in simulink model.

Use reset() to clear this override again.

Reimplemented from ttk_bus._bus_signals_frbs.CrcFrBusStatus.

7.147.3.2 get()

```
ttk_bus.bus_signals_frbs.CrcFrBusStatus.get (
              self )
```

Get the current CRC status value from FlexRay blockset.

CRC check status values are:

- 0: crc valid
- 1: crc invalid
- 2: check disabled

Returns

the current value from status var.

Reimplemented from ttk_bus._bus_signals_frbs.CrcFrBusStatus.

7.147.3.3 getState()

```
ttk_bus.bus_signals_frbs.CrcFrBusStatus.getState (
              self )
```

Get a textual state for the current CRC status value from FlexRay blockset.

Returns

the current state representation.

Reimplemented from ttk_base._variables_base.VarBase.

7.147.3.4 reset()

```
ttk_bus.bus_signals_frbs.CrcFrBusStatus.reset (
              self )
```

Reset CRC check/calculation settings (default: switch back to current setting in Simulink model).

Reimplemented from ttk_bus._bus_signals_frbs.CrcFrBusStatus.

7.147.4 Member Data Documentation

7.147.4.1 cfg_custom_defaults

```
dict ttk_bus.bus_signals_frbs.CrcFrBusStatus.cfg_custom_defaults  [static]
```

Initial value:

```
=  {
        "switch_var":   None,
        "enable_var":   None,
    }
```

Configuration settings for new instances: Custom "hard" default value(s) to use for internal real time system variables (for example: switch settings)

Mapping: <variable name as string>: <default value>

See description at FrRxPdu for additional details.

Typical default values:

- source switches usually default to "source: simulink model"
  - 0: source: simulink model / trc
  - 1: source: bus system
- CRC calculation/check enable variable: "disabled" is a safe default (enable in test as needed) as enabled checks w/ faulty CRCs will freeze remaining signal data at the "last known good" value.
  - 0: calculation/check disabled
  - 1: calculation/check enabled

7.147.4.2 enable_var

```
ttk_bus.bus_signals_frbs.CrcFrBusStatus.enable_var
```

CRC calculation/check enable variable ("CRC Enable")

- 0: calculation/check disabled
- 1: calculation/check enabled

7.147.4.3 status_var

`ttk_bus.bus_signals_frbs.CrcFrBusStatus.status_var`

Status variable for CRC check (read-only)

- 0: CRC value is valid
- 1: CRC value is invalid
- 2: CRC check is disabled

7.147.4.4 switch_var

`ttk_bus.bus_signals_frbs.CrcFrBusStatus.switch_var`

Source Switch variable for CRC check enable.

- 0: Simulink Model
- 1: BusSystems

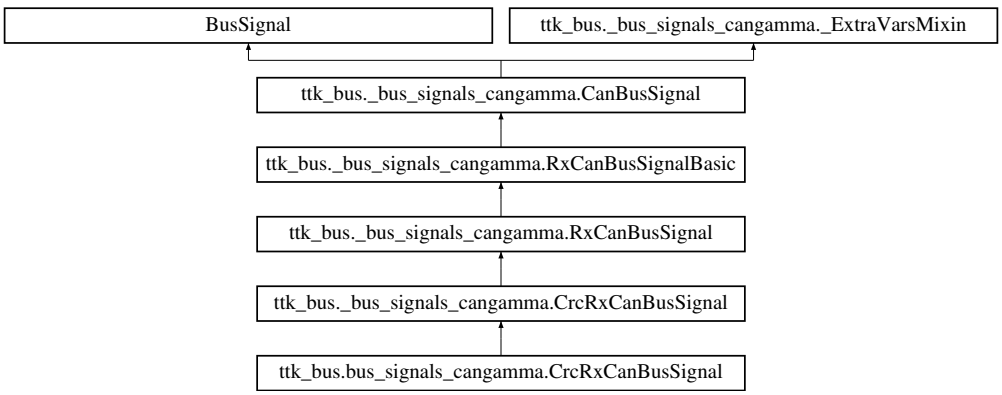
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_frbs.py (v1.8)

7.148 ttk_bus._bus_signals_cangamma.CrcRxCanBusSignal Class Reference

A CRC RX bus signal (HIL-Tx => ECU-Rx)

Inheritance diagram for ttk_bus._bus_signals_cangamma.CrcRxCanBusSignal:



Public Member Functions

- __init__ (self, context, signal_identifier, message=None, cycle_time=None, debounce_time=None, alias="", descr="", lookup=None, error_delay=None, change_delay=None, timeout=None, recovery_time=None)

  CRC bus signal (HIL-Tx => ECU-Rx) initialization for Gamma-V based CAN signal manipulation options.
- setBasicCrcError (self)

  Switch CRC calculation to an algorithm producing incorrect CRCs.
- clearBasicCrcError (self)

  Switch CRC calculation back to an algorithm producing correct CRCs.

Public Attributes

- error_delay

  delay to wait after a "basic" CRC error injection [ms]
- algo_var

  CRC algorithm type variable, selects active/default CRC algorithm.
- dyn_algo_var

  CRC dynamic algorithm type variable, selects active CRC algorithm.
- crc_normal_type

  CRC type for "correct" CRC algorithm (normal mode)
- crc_error_type

  CRC type for "incorrect" CRC algorithm (for error injection)

Static Public Attributes

- int cfg_crc_normal_type = 1

  Configuration setting for new instances:
  CRC type for "correct" CRC algorithm (normal mode)
- int cfg_crc_error_type = 2

  Configuration setting for new instances:
  CRC type for "incorrect" CRC algorithm (for error injection, for example calculated CRC + 1).
- dict cfg_custom_defaults

  Configuration setting for new instances:
  Custom "hard" default value(s) to use for internal real time system variables.

7.148.1 Detailed Description

A CRC RX bus signal (HIL-Tx => ECU-Rx)

7.148.2 Constructor & Destructor Documentation

7.148.2.1 __init__()

```
ttk_bus._bus_signals_cangamma.CrcRxCanBusSignal.__init__ (
                self,
                context,
                signal_identifier,
                message = None,
                cycle_time = None,
                debounce_time = None,
                alias = "",
                descr = "",
                lookup = None,
                error_delay = None,
                change_delay = None,
                timeout = None,
                recovery_time = None )
```

CRC bus signal (HIL-Tx => ECU-Rx) initialization for Gamma-V based CAN signal manipulation options.

Parameters

context	parent gaapi context
signal_identifier	identifier for signal PV
message	reference to parent RxCanBusMessage instance
cycle_time	(opt) cycle time in [ms], overrides settings in Message
debounce_time	(opt) minimum debounce delay between events in [ms], overrides settings in Message
alias	(opt) bus signal alias/name
descr	(opt) signal description
lookup	(opt) lookup table for values, see VarBase
error_delay	(opt) delay to wait until a simple crc error should be detected [ms], defaults to 10x cycle time
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x cycle_time
timeout	(opt) detection time for timeout errors [ms], defaults to 2x cycle_time
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x cycle_time

Reimplemented from ttk_bus._bus_signals_cangamma.RxCanBusSignal.

Reimplemented in ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.

7.148.3 Member Function Documentation

7.148.3.1 clearBasicCrcError()

```
ttk_bus._bus_signals_cangamma.CrcRxCanBusSignal.clearBasicCrcError (
                self )
```

Switch CRC calculation back to an algorithm producing correct CRCs.

See CrcRxCanBusSignal.crc_normal_type.

Reimplemented in ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.

7.148.3.2 setBasicCrcError()

```
ttk_bus._bus_signals_cangamma.CrcRxCanBusSignal.setBasicCrcError (
              self )
```

Switch CRC calculation to an algorithm producing incorrect CRCs.

See CrcRxCanBusSignal.crc_error_type.

Reimplemented in ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.

7.148.4 Member Data Documentation

7.148.4.1 algo_var

```
ttk_bus._bus_signals_cangamma.CrcRxCanBusSignal.algo_var
```

CRC algorithm type variable, selects active/default CRC algorithm.

7.148.4.2 cfg_crc_error_type

```
int ttk_bus._bus_signals_cangamma.CrcRxCanBusSignal.cfg_crc_error_type = 2  [static]
```

Configuration setting for new instances:
CRC type for "incorrect" CRC algorithm (for error injection, for example calculated CRC + 1).

Note: Simply switching CRC calculation off (e.g. CRC = 0x00) may lead to detection failures, as a static CRC value might actually be correct for the current data.

7.148.4.3 cfg_crc_normal_type

```
int ttk_bus._bus_signals_cangamma.CrcRxCanBusSignal.cfg_crc_normal_type = 1  [static]
```

Configuration setting for new instances:
CRC type for "correct" CRC algorithm (normal mode)

7.148.4.4 cfg_custom_defaults

```
dict ttk_bus._bus_signals_cangamma.CrcRxCanBusSignal.cfg_custom_defaults  [static]
```

Initial value:

```
=  {
        "algo_var":          None,   #
        "dyn_algo_var":      None,   #
        "mode_var":          None,   # for parent RxCanBusSignal
        "dyn_count_var":     None,   # for parent RxCanBusSignal
        "dyn_value_phy_var": None,   # for parent RxCanBusSignal
    }
```

Configuration setting for new instances:
Custom "hard" default value(s) to use for internal real time system variables.

Mapping: <variable name as string>: <default value>

A value of None (or a missing variable name) enables normal/automatic reset behavior.
See RxCanBusSignal for further details.

Typical default values:

- mode_var usually defaults to "CRC calculation active/enabled"
  - 0x22: CRC calculation enabled
- algo_var: CRC type (active CRC algorithm) should default to "correct CRC" (which is typically algorithm 1)
- dyn_algo_var: dynamic algorithm override

7.148.4.5 crc_error_type

```
ttk_bus._bus_signals_cangamma.CrcRxCanBusSignal.crc_error_type
```

CRC type for "incorrect" CRC algorithm (for error injection)

7.148.4.6 crc_normal_type

```
ttk_bus._bus_signals_cangamma.CrcRxCanBusSignal.crc_normal_type
```

CRC type for "correct" CRC algorithm (normal mode)

7.148.4.7 dyn_algo_var

```
ttk_bus._bus_signals_cangamma.CrcRxCanBusSignal.dyn_algo_var
```

CRC dynamic algorithm type variable, selects active CRC algorithm.

7.148.4.8 error_delay

`ttk_bus._bus_signals_cangamma.CrcRxCanBusSignal.error_delay`

delay to wait after a "basic" CRC error injection [ms]

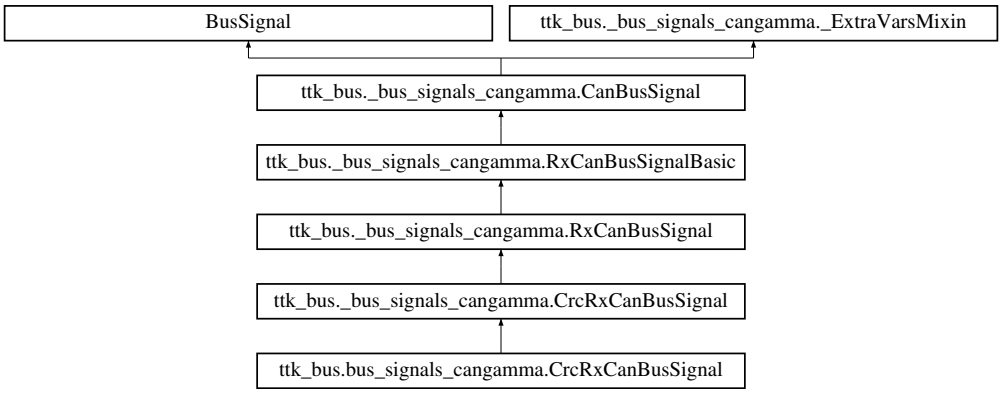
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_cangamma.py (v1.12)

7.149 ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal Class Reference

A CRC RX bus signal (HIL-Tx => ECU-Rx)

Inheritance diagram for ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal:



Public Member Functions

- __init__ (self, context, signal_identifier, message=None, cycle_time=None, debounce_time=None, alias="", descr="", lookup=None, error_delay=None, change_delay=None, timeout=None, recovery_time=None)

  CRC bus signal (HIL-Tx => ECU-Rx) initialization for Gamma-V based CAN signal manipulation options.
- setBasicCrcError (self)

  Switch CRC calculation to an algorithm producing incorrect CRCs.
- clearBasicCrcError (self)

  Switch CRC calculation back to an algorithm producing correct CRCs.
- enableHilTx (self)

  Enable transmission of signal HIL => DUT.
- disableHilTx (self)

  Disable transmission of signal HIL => DUT.
- kickout (self)

  Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent CAN-Message/PDU).
- get (self)

  Get the current value of the HIL signal variable (typically not that useful for ECU-Rx/HIL-Tx CRC-Signals, though).
- set (self, value)

  Set a value to the HIL signal variable.

- getState (self)

  Get a textual state for the HIL signal variable's current data (typically not that useful for ECU-Rx/HIL-Tx CRC-Signals, though).
- setState (self, state)

  Set a textual state to the HIL signal variable.
- getStateDescr (self, value=None, fallback=None)

  Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).
- info (self)

  Get a current info string about the HIL signal variable.
- reset (self, ∗args, ∗∗kwargs)

  Reset all resettable variables of this bus signal.

Public Attributes

- crc_normal_type

  CRC algorithm type for "correct" CRC (normal mode)
- crc_error_type

  CRC algorithm type for "incorrect" CRC (for error injection)
- algo_var

  CRC algorithm type variable, selects active/default CRC algorithm.
- dyn_algo_var

  CRC dynamic algorithm type variable, selects active CRC algorithm during dynamic error injection.
- dyn_count_var

  Dynamic Countdown value.
- dyn_value_var

  Dynamic Value: value that is active while dyn_count_var decrements.
- dyn_pattern_var

  Dynamic Pattern value.
- static_value_phy_var

  Physical static override value (for mode 0x01)
- static_value_raw_var

  Raw static override value (for mode 0x02)
- mode_var

  Mode switch variable for signal manipulation.
- base_identifier

  Signal PV address/identifier as base identifier for deriving sub-elements.
- message

  reference to associated (Rx-)CanBusMessage instance
- alias

  bus signal alias name
- descr

  bus signal description text
- cycle_time

  cycle time of Message/PDU in [ms]
- debounce_time

  minimum debounce delay for event-triggered Messages/PDUs in [ms]
- change_delay

  delay to wait after a value change [ms]
- timeout

  detection time for timeout errors [ms]
- recovery_time

  time for recovery/"wiedergut" checks [ms]
- error_delay

  delay to wait after a "basic" CRC error injection [ms]

Static Public Attributes

- int cfg_crc_normal_type = 1

  Configuration setting for new instances:
  CRC type for "correct" CRC algorithm (normal mode)
- int cfg_crc_error_type = 2

  Configuration setting for new instances:
  CRC type for "incorrect" CRC algorithm (for error injection, for example calculated CRC + 1).
- dict cfg_custom_defaults

  Configuration setting for new instances:
  Custom "hard" default value(s) to use for internal real time system variables.

7.149.1 Detailed Description

A CRC RX bus signal (HIL-Tx => ECU-Rx)

7.149.2 Constructor & Destructor Documentation

7.149.2.1 __init__()

```
ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.__init__ (
              self,
              context,
              signal_identifier,
              message = None,
              cycle_time = None,
              debounce_time = None,
              alias = "",
              descr = "",
              lookup = None,
              error_delay = None,
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

CRC bus signal (HIL-Tx => ECU-Rx) initialization for Gamma-V based CAN signal manipulation options.

Parameters

context	parent gaapi context
signal_identifier	identifier for signal PV
message	reference to parent RxCanBusMessage instance
cycle_time	(opt) cycle time in [ms], overrides settings in Message
debounce_time	(opt) minimum debounce delay between events in [ms], overrides settings in Message
alias	(opt) bus signal alias/name
descr	(opt) signal description
lookup	(opt) lookup table for values, see VarBase
error_delay	(opt) delay to wait until a simple crc error should be detected [ms], defaults to 10x cycle time
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x cycle_time
timeout	(opt) detection time for timeout errors [ms], defaults to 2x cycle_time
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x cycle_time

Reimplemented from ttk_bus._bus_signals_cangamma.CrcRxCanBusSignal.

7.149.3 Member Function Documentation

7.149.3.1 clearBasicCrcError()

```
ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.clearBasicCrcError (
              self )
```

Switch CRC calculation back to an algorithm producing correct CRCs.

See CrcRxCanBusSignal.crc_normal_type.

Reimplemented from ttk_bus._bus_signals_cangamma.CrcRxCanBusSignal.

7.149.3.2 disableHilTx()

```
ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.disableHilTx (
              self )
```

Disable transmission of signal HIL => DUT.

Note

This will disable transmission of *all* signals of the parent Message/PDU.

Reimplemented from ttk_bus._bus_signals_cangamma.RxCanBusSignalBasic.

7.149.3.3 enableHilTx()

```
ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.enableHilTx (
              self )
```

Enable transmission of signal HIL => DUT.

Note

This will enable transmission of *all* signals of the parent Message/PDU.

Reimplemented from ttk_bus._bus_signals_cangamma.RxCanBusSignalBasic.

7.149.3.4 get()

```
ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.get (
              self )
```

Get the current value of the HIL signal variable (typically not that useful for ECU-Rx/HIL-Tx CRC-Signals, though).

7.149.3.5 getState()

```
ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.getState (
              self )
```

Get a textual state for the HIL signal variable's current data (typically not that useful for ECU-Rx/HIL-Tx CRC-Signals, though).

Returns

the current state's description from `signal_var`.

7.149.3.6 getStateDescr()

```
ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.getStateDescr (
              self,
              value = None,
              fallback = None )
```

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the last read value will be used instead.
fallback	fallback value to return if no match was found. Default: None

Returns

the state text if a match was found, otherwise the specified fallback value.

7.149.3.7 info()

```
ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.info (
              self )
```

Get a current info string about the HIL signal variable.

Returns

an info string.

7.149.3.8 kickout()

```
ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.kickout (
              self )
```

Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent CAN-Message/PDU).

Note

All other signals of the parent Message/PDU will also be sent.

Reimplemented from ttk_bus._bus_signals_cangamma.RxCanBusSignalBasic.

7.149.3.9 reset()

```
ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.reset (
              self,
             * args,
             ** kwargs )
```

Reset all resettable variables of this bus signal.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

7.149.3.10 set()

```
ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.set (
              self,
              value )
```

Set a value to the HIL signal variable.

Parameters

value	value to set

Returns

the previous value from `signal_var`.

Reimplemented from ttk_bus._bus_signals_cangamma.RxCanBusSignal.

7.149.3.11 setBasicCrcError()

```
ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.setBasicCrcError (
              self )
```

Switch CRC calculation to an algorithm producing incorrect CRCs.

See CrcRxCanBusSignal.crc_error_type.

Reimplemented from ttk_bus._bus_signals_cangamma.CrcRxCanBusSignal.

7.149.3.12 setState()

```
ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.setState (
              self,
              state )
```

Set a textual state to the HIL signal variable.

This uses the lookup values supplied in constructor.

Parameters

state	named state to set

Returns

the previous state's description.

Reimplemented from ttk_bus._bus_signals_cangamma.RxCanBusSignal.

7.149.4 Member Data Documentation

7.149.4.1 algo_var

`ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.algo_var`

CRC algorithm type variable, selects active/default CRC algorithm.

7.149.4.2 alias

`ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.alias`

bus signal alias name

7.149.4.3 base_identifier

`ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.base_identifier`

Signal PV address/identifier as base identifier for deriving sub-elements.

For CAN/Gamma signals, related variables follow the form `<base_identifier>.<suffix>`

7.149.4.4 cfg_crc_error_type

`int ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.cfg_crc_error_type = 2  [static]`

Configuration setting for new instances:
CRC type for "incorrect" CRC algorithm (for error injection, for example calculated CRC + 1).

Note: Simply switching CRC calculation off (e.g. CRC = 0x00) may lead to detection failures, as a static CRC value might actually be correct for the current data.

7.149.4.5 cfg_crc_normal_type

```
int ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.cfg_crc_normal_type = 1  [static]
```

Configuration setting for new instances:
CRC type for "correct" CRC algorithm (normal mode)

7.149.4.6 cfg_custom_defaults

```
dict ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.cfg_custom_defaults  [static]
```

Initial value:

```
=  {
        "algo_var":          None,   #
        "dyn_algo_var":      None,   #
        "mode_var":          None,   # for parent RxCanBusSignal
        "dyn_count_var":     None,   # for parent RxCanBusSignal
        "dyn_value_phy_var": None,   # for parent RxCanBusSignal
    }
```

Configuration setting for new instances:
Custom "hard" default value(s) to use for internal real time system variables.

Mapping: <variable name as string>: <default value>

A value of None (or a missing variable name) enables normal/automatic reset behavior.
See RxCanBusSignal for further details.

Typical default values:

- mode_var usually defaults to "CRC calculation active/enabled"
  - 0x22: CRC calculation enabled
- algo_var: CRC type (active CRC algorithm) should default to "correct CRC" (which is typically algorithm 1)
- dyn_algo_var: dynamic algorithm override

7.149.4.7 change_delay

```
ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.change_delay
```

delay to wait after a value change [ms]

7.149.4.8 crc_error_type

```
ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.crc_error_type
```

CRC algorithm type for "incorrect" CRC (for error injection)

7.149.4.9 crc_normal_type

```
ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.crc_normal_type
```

CRC algorithm type for "correct" CRC (normal mode)

7.149.4.10 cycle_time

ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.cycle_time

cycle time of Message/PDU in [ms]

7.149.4.11 debounce_time

ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.debounce_time

minimum debounce delay for event-triggered Messages/PDUs in [ms]

7.149.4.12 descr

ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.descr

bus signal description text

7.149.4.13 dyn_algo_var

ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.dyn_algo_var

CRC dynamic algorithm type variable, selects active CRC algorithm during dynamic error injection.

7.149.4.14 dyn_count_var

ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.dyn_count_var

Dynamic Countdown value.

7.149.4.15 dyn_pattern_var

ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.dyn_pattern_var

Dynamic Pattern value.

7.149.4.16 dyn_value_var

ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.dyn_value_var

Dynamic Value: value that is active while dyn_count_var decrements.

7.149.4.17 error_delay

ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.error_delay

delay to wait after a "basic" CRC error injection [ms]

7.149.4.18 message

`ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.message`

reference to associated (Rx-)CanBusMessage instance

7.149.4.19 mode_var

`ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.mode_var`

Mode switch variable for signal manipulation.

(see RxCanBusSignal for full details)

For CRC signals:

- 0x20: CRC calculated with default/correct algorithm (.algo)
- 0x21: CRC calculated with algorithm set in .dyn_algo
- 0x22: CRC calculated with algorithm from .dyn_algo while .dyn_count decrements, then mode 0x20
- 0x23: CRC calculated with .algo/.dyn_algo according to .dyn_pattern

7.149.4.20 recovery_time

`ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.recovery_time`

time for recovery/"wiedergut" checks [ms]

7.149.4.21 static_value_phy_var

`ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.static_value_phy_var`

Physical static override value (for mode 0x01)

7.149.4.22 static_value_raw_var

`ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.static_value_raw_var`

Raw static override value (for mode 0x02)

7.149.4.23 timeout

`ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal.timeout`

detection time for timeout errors [ms]

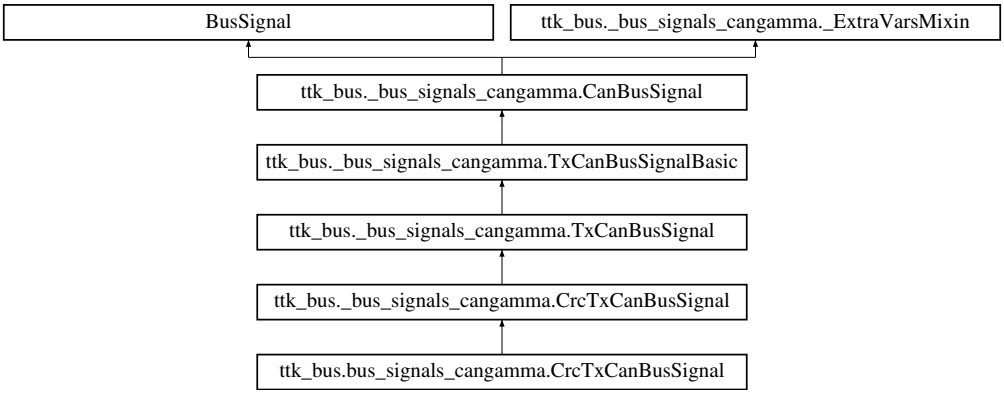
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_cangamma.py (v1.3)

7.150 ttk_bus._bus_signals_cangamma.CrcTxCanBusSignal Class Reference

A CRC TX bus signal (ECU-Tx => HIL-Rx), adds CRC validity info.

Inheritance diagram for ttk_bus._bus_signals_cangamma.CrcTxCanBusSignal:



Public Member Functions

- __init__ (self, context, signal_identifier, message=None, cycle_time=None, debounce_time=None, alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  CRC bus signal (ECU-Tx => HIL-Rx) initialization for Gamma-V based CAN signal manipulation options.
- getCrcStatus (self)

Public Attributes

- algo_var

  CRC algorithm type variable, selects CRC algorithm to use for CRC validation.
- crc_valid_var

  CRC check status.

Static Public Attributes

- dict cfg_custom_defaults

  Configuration setting for new instances:
  Custom "hard" default value(s) to use for internal real time system variables.

7.150.1 Detailed Description

A CRC TX bus signal (ECU-Tx => HIL-Rx), adds CRC validity info.

7.150.2 Constructor & Destructor Documentation

7.150.2.1 __init__()

```
ttk_bus._bus_signals_cangamma.CrcTxCanBusSignal.__init__ (
               self,
               context,
               signal_identifier,
               message = None,
               cycle_time = None,
               debounce_time = None,
               alias = "",
               descr = "",
               lookup = None,
               change_delay = None,
               timeout = None,
               recovery_time = None )
```

CRC bus signal (ECU-Tx => HIL-Rx) initialization for Gamma-V based CAN signal manipulation options.

Parameters

context	parent gaapi context
signal_identifier	identifier for CRC signal PV
message	reference to associated RxCanBusMessage instance
cycle_time	(opt) cycle time in [ms], overrides settings in Message
debounce_time	(opt) minimum debounce delay between events in [ms], overrides settings in Message
alias	(opt) bus signal alias/name
descr	(opt) signal description
lookup	(opt) lookup table for values, see VarBase
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x cycle_time
timeout	(opt) detection time for timeout errors [ms], defaults to 2x cycle_time
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x cycle_time

Reimplemented from ttk_bus._bus_signals_cangamma.TxCanBusSignal.

Reimplemented in ttk_bus.bus_signals_cangamma.CrcTxCanBusSignal.

7.150.3 Member Function Documentation

7.150.3.1 getCrcStatus()

```
ttk_bus._bus_signals_cangamma.CrcTxCanBusSignal.getCrcStatus (
               self )
```

Reimplemented in ttk_bus.bus_signals_cangamma.CrcTxCanBusSignal.

7.150.4 Member Data Documentation

7.150.4.1 algo_var

```
ttk_bus._bus_signals_cangamma.CrcTxCanBusSignal.algo_var
```

CRC algorithm type variable, selects CRC algorithm to use for CRC validation.

7.150.4.2 cfg_custom_defaults

```
dict ttk_bus._bus_signals_cangamma.CrcTxCanBusSignal.cfg_custom_defaults  [static]
```

Initial value:

```
=  {
        "algo_var":       None,    #
    }
```

Configuration setting for new instances:
Custom "hard" default value(s) to use for internal real time system variables.

Mapping: <variable name as string>: <default value>

A value of None (or a missing variable name) enables normal/automatic reset behavior.
See RxCanBusSignal for further details.

Typical default values:

- algo_var: CRC type (active CRC algorithm) should default to "correct CRC" (which is typically algorithm 1)

7.150.4.3 crc_valid_var

```
ttk_bus._bus_signals_cangamma.CrcTxCanBusSignal.crc_valid_var
```

CRC check status.

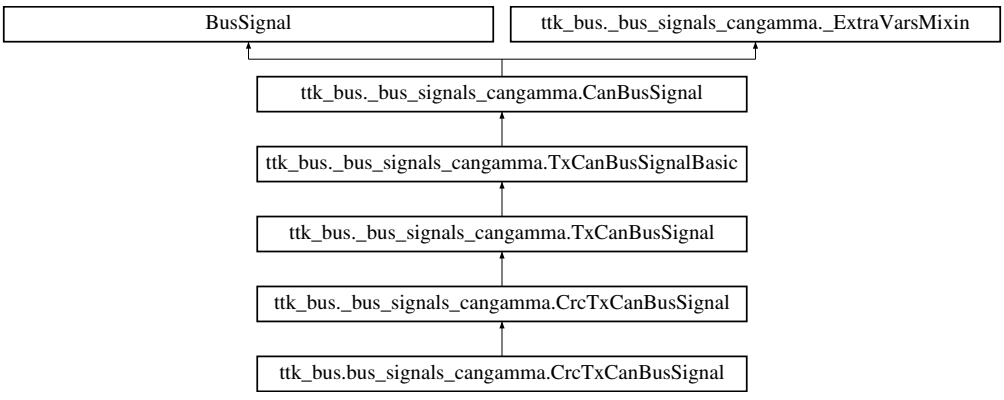
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_cangamma.py (v1.12)

7.151 ttk_bus.bus_signals_cangamma.CrcTxCanBusSignal Class Reference

A CRC TX bus signal (ECU-Tx => HIL-Rx), adds CRC validity info.

Inheritance diagram for ttk_bus.bus_signals_cangamma.CrcTxCanBusSignal:



Public Member Functions

- __init__ (self, context, signal_identifier, message=None, cycle_time=None, debounce_time=None, alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  CRC bus signal (ECU-Tx => HIL-Rx) initialization for Gamma-V based CAN signal manipulation options.
- getCrcStatus (self)

  Get status result of last CRC validation.
- info (self)

  Get a current info string about the HIL signal variable, see `signal_var`
- get (self)

  Get the current value of the HIL CRC signal variable, see `signal_var`*.*
- set (self, value)

  Set a value to the HIL signal variable, see `signal_var`
- getState (self)

  Get a textual state for the HIL signal variable's current data, see `signal_var`*.*
- setState (self, state)

  Set a textual state to the HIL signal variable, see `signal_var`*.*
- getStateDescr (self, value=None, fallback=None)

  Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).
- reset (self, *args, **kwargs)

  Reset all resettable variables of this bus signal.

Public Attributes

- algo_var

  CRC algorithm type variable, selects active CRC algorithm to use for CRC validation.
- crc_valid_var

  CRC check status variable.

Static Public Attributes

- dict cfg_custom_defaults

  Configuration setting for new instances:
  Custom "hard" default value(s) to use for internal real time system variables.

7.151.1 Detailed Description

A CRC TX bus signal (ECU-Tx => HIL-Rx), adds CRC validity info.

7.151.2 Constructor & Destructor Documentation

7.151.2.1 __init__()

```
ttk_bus.bus_signals_cangamma.CrcTxCanBusSignal.__init__ (
                self,
                context,
                signal_identifier,
                message = None,
                cycle_time = None,
                debounce_time = None,
                alias = "",
                descr = "",
                lookup = None,
                change_delay = None,
                timeout = None,
                recovery_time = None )
```

CRC bus signal (ECU-Tx => HIL-Rx) initialization for Gamma-V based CAN signal manipulation options.

Parameters

context	parent gaapi context
signal_identifier	identifier for CRC signal PV
message	reference to associated RxCanBusMessage instance
cycle_time	(opt) cycle time in [ms], overrides settings in Message
debounce_time	(opt) minimum debounce delay between events in [ms], overrides settings in Message
alias	(opt) bus signal alias/name
descr	(opt) signal description
lookup	(opt) lookup table for values, see VarBase
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x cycle_time
timeout	(opt) detection time for timeout errors [ms], defaults to 2x cycle_time
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x cycle_time

Reimplemented from ttk_bus._bus_signals_cangamma.CrcTxCanBusSignal.

7.151.3 Member Function Documentation

7.151.3.1 get()

```
ttk_bus.bus_signals_cangamma.CrcTxCanBusSignal.get (
                self )
```

Get the current value of the HIL CRC signal variable, see `signal_var`.

Returns

the current value from `signal_var`.

7.151.3.2 getCrcStatus()

```
ttk_bus.bus_signals_cangamma.CrcTxCanBusSignal.getCrcStatus (
              self )
```

Get status result of last CRC validation.

Reimplemented from ttk_bus._bus_signals_cangamma.CrcTxCanBusSignal.

7.151.3.3 getState()

```
ttk_bus.bus_signals_cangamma.CrcTxCanBusSignal.getState (
              self )
```

Get a textual state for the HIL signal variable's current data, see `signal_var`.

This uses the lookup values supplied in constructor. Fallback (if no matching state was found) will be the unmapped numeric value.

Returns

the current state representation from `signal_var`.

7.151.3.4 getStateDescr()

```
ttk_bus.bus_signals_cangamma.CrcTxCanBusSignal.getStateDescr (
              self,
              value = None,
              fallback = None )
```

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the last read value will be used instead.
fallback	fallback value to return if no match was found. Default: None

Returns

the state text if a match was found, otherwise the specified fallback value.

7.151.3.5 info()

```
ttk_bus.bus_signals_cangamma.CrcTxCanBusSignal.info (
              self )
```

Get a current info string about the HIL signal variable, see `signal_var`

Returns

the current info string from `signal_var`.

7.151.3.6 reset()

```
ttk_bus.bus_signals_cangamma.CrcTxCanBusSignal.reset (
              self,
             * args,
             ** kwargs )
```

Reset all resettable variables of this bus signal.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

7.151.3.7 set()

```
ttk_bus.bus_signals_cangamma.CrcTxCanBusSignal.set (
              self,
              value )
```

Set a value to the HIL signal variable, see `signal_var`

Attention

For TxCanBusSignals, this will only overwrite the current contents of the HIl receive buffer of the signal (which is typically not too useful as it will be overwritten on next update with data from CAN plugin)

Parameters

value	value to set

Returns

the previous value from `signal_var`.

7.151.3.8 setState()

```
ttk_bus.bus_signals_cangamma.CrcTxCanBusSignal.setState (
              self,
              state )
```

Set a textual state to the HIL signal variable, see `signal_var`.

This uses the lookup values supplied in constructor.

Attention

For TxFrBusSignals this will only overwrite the current contents of the HIl receive buffer of the signal (which typically is not too useful).

Parameters

state	named state to set

Returns

the previous state representation from `signal_var`.

7.151.4 Member Data Documentation

7.151.4.1 algo_var

`ttk_bus.bus_signals_cangamma.CrcTxCanBusSignal.algo_var`

CRC algorithm type variable, selects active CRC algorithm to use for CRC validation.

7.151.4.2 cfg_custom_defaults

`dict ttk_bus.bus_signals_cangamma.CrcTxCanBusSignal.cfg_custom_defaults [static]`

Initial value:

```
= {
        "algo_var":       None,   #
    }
```

Configuration setting for new instances:
Custom "hard" default value(s) to use for internal real time system variables.

Mapping: <variable name as string>: <default value>

A value of None (or a missing variable name) enables normal/automatic reset behavior.
See RxCanBusSignal for further details.

Typical default values:

- algo_var: CRC type (active CRC algorithm) should default to "correct CRC" (which is typically algorithm 1)

7.151.4.3 crc_valid_var

`ttk_bus.bus_signals_cangamma.CrcTxCanBusSignal.crc_valid_var`

CRC check status variable.

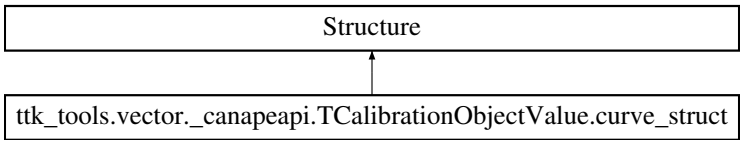
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_cangamma.py (v1.3)

7.152 ttk_tools.vector._canapeapi.TCalibrationObjectValue.curve_struct Class Reference

2D table data.

Inheritance diagram for ttk_tools.vector._canapeapi.TCalibrationObjectValue.curve_struct:



Static Protected Attributes

- int _pack_ = 1
- list _fields_

7.152.1 Detailed Description

2D table data.

7.152.2 Member Data Documentation

7.152.2.1 _fields_

```
list ttk_tools.vector._canapeapi.TCalibrationObjectValue.curve_struct._fields_  [static],
[protected]
```

Initial value:

```
=  [
            ("type",        c_int),
            ("dimension",   c_short),
            ("axis",        POINTER(c_double)),
            ("values",      POINTER(c_double)),
        ]
```

7.152.2.2 _pack_

```
int ttk_tools.vector._canapeapi.TCalibrationObjectValue.curve_struct._pack_ = 1  [static],
[protected]
```

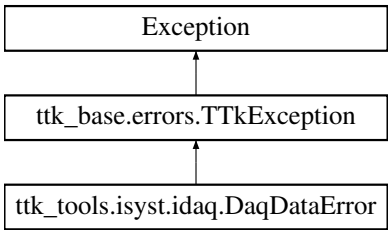
The documentation for this class was generated from the following file:

- ttk_tools/vector/_canapeapi.py (v7.3.3)

7.153 ttk_tools.isyst.idaq.DaqDataError Class Reference

Errors during DAQ data decoding.

Inheritance diagram for ttk_tools.isyst.idaq.DaqDataError:



7.153.1 Detailed Description

Errors during DAQ data decoding.

Get a raw bytes string using the default encoding. Encoding errors (un-encodable characters) will be ignored. Compare ttk_base.baseutils.bstr.

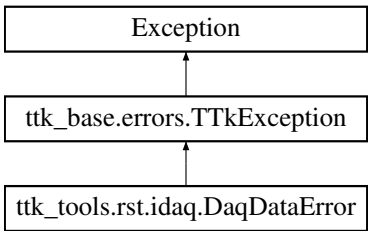
The documentation for this class was generated from the following file:

- ttk_tools/isyst/idaq.py (v1.2.1)

7.154 ttk_tools.rst.idaq.DaqDataError Class Reference

Errors during DAQ data decoding.

Inheritance diagram for ttk_tools.rst.idaq.DaqDataError:



7.154.1 Detailed Description

Errors during DAQ data decoding.

Get a raw bytes string using the default encoding. Encoding errors (un-encodable characters) will be ignored. Compare ttk_base.baseutils.bstr.

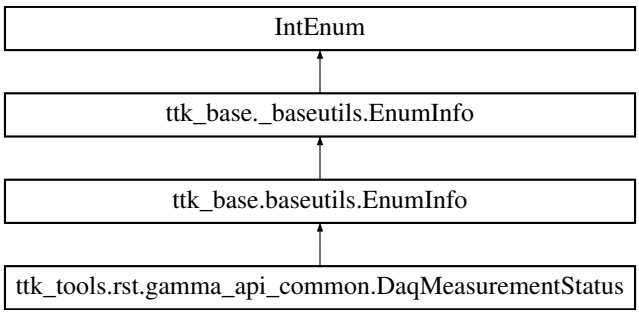
The documentation for this class was generated from the following file:

- ttk_tools/rst/idaq.py (v1.2.0)

7.155 ttk_tools.rst.gamma_api_common.DaqMeasurementStatus Class Reference

State machine states for DAQ task.

Inheritance diagram for ttk_tools.rst.gamma_api_common.DaqMeasurementStatus:



Static Public Attributes

- int M_INIT = 0
- int M_WAITING_FOR_CONFIG = 1
- int M_CREATE_THREAD_GET_CONFIG = 2
- int M_WAITING_FOR_CONFIGDATA = 3
- int M_CREATE_THREAD_VALIDATE_CONFIG = 4
- int M_SETUP_CONFIGURATION = 5
- int M_WAITING_FOR_RUN_INFO = 6
- int M_WAITING_FOR_RUN = 7
- int M_RUNNING_INFO = 8
- int M_RUNNING = 9
- int M_RUNNING_STOPPED_INFO = 10
- int M_RUNNING_STOPPED = 11
- int M_DEINIT = 12

7.155.1 Detailed Description

State machine states for DAQ task.

7.155.2 Member Data Documentation

7.155.2.1 M_CREATE_THREAD_GET_CONFIG

```
int ttk_tools.rst.gamma_api_common.DaqMeasurementStatus.M_CREATE_THREAD_GET_CONFIG = 2  [static]
```

7.155.2.2 M_CREATE_THREAD_VALIDATE_CONFIG

```
int ttk_tools.rst.gamma_api_common.DaqMeasurementStatus.M_CREATE_THREAD_VALIDATE_CONFIG = 4
[static]
```

7.155.2.3 M_DEINIT

```
int ttk_tools.rst.gamma_api_common.DaqMeasurementStatus.M_DEINIT = 12  [static]
```

7.155.2.4 M_INIT

```
int ttk_tools.rst.gamma_api_common.DaqMeasurementStatus.M_INIT = 0  [static]
```

7.155.2.5 M_RUNNING

```
int ttk_tools.rst.gamma_api_common.DaqMeasurementStatus.M_RUNNING = 9  [static]
```

7.155.2.6 M_RUNNING_INFO

```
int ttk_tools.rst.gamma_api_common.DaqMeasurementStatus.M_RUNNING_INFO = 8  [static]
```

7.155.2.7 M_RUNNING_STOPPED

```
int ttk_tools.rst.gamma_api_common.DaqMeasurementStatus.M_RUNNING_STOPPED = 11  [static]
```

7.155.2.8 M_RUNNING_STOPPED_INFO

```
int ttk_tools.rst.gamma_api_common.DaqMeasurementStatus.M_RUNNING_STOPPED_INFO = 10  [static]
```

7.155.2.9 M_SETUP_CONFIGURATION

```
int ttk_tools.rst.gamma_api_common.DaqMeasurementStatus.M_SETUP_CONFIGURATION = 5  [static]
```

7.155.2.10 M_WAITING_FOR_CONFIG

```
int ttk_tools.rst.gamma_api_common.DaqMeasurementStatus.M_WAITING_FOR_CONFIG = 1  [static]
```

7.155.2.11 M_WAITING_FOR_CONFIGDATA

```
int ttk_tools.rst.gamma_api_common.DaqMeasurementStatus.M_WAITING_FOR_CONFIGDATA = 3  [static]
```

7.155.2.12 M_WAITING_FOR_RUN

```
int ttk_tools.rst.gamma_api_common.DaqMeasurementStatus.M_WAITING_FOR_RUN = 7  [static]
```

7.155.2.13 M_WAITING_FOR_RUN_INFO

```
int ttk_tools.rst.gamma_api_common.DaqMeasurementStatus.M_WAITING_FOR_RUN_INFO = 6  [static]
```

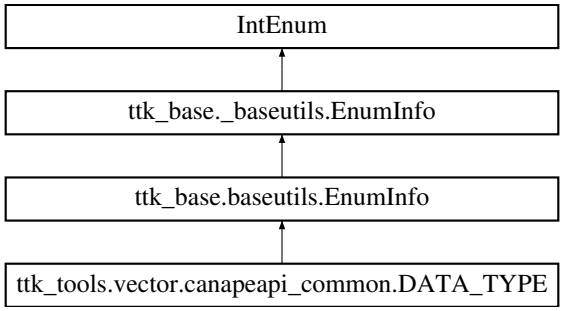
The documentation for this class was generated from the following file:

- ttk_tools/rst/gamma_api_common.py (v1.5)

7.156 ttk_tools.vector.canapeapi_common.DATA_TYPE Class Reference

Data type constants (`TYPE_*`) from enum `TAsap3DataType`

Inheritance diagram for ttk_tools.vector.canapeapi_common.DATA_TYPE:



Static Public Attributes

- int UNKNOWN = 0
- int INT = 1
- int FLOAT = 2
- int DOUBLE = 3
- int SIGNED = 4
- int UNSIGNED = 5
- int STRING = 6
- int INT64 = 7
- int UINT64 = 8
- int UWORD = 9
- int WORD = 10
- int UINT = 11
- int UBYTE = 12
- int SBYTE = 13
- int FLOAT16 = 14

7.156.1 Detailed Description

Data type constants (`TYPE_*`) from enum `TAsap3DataType`

7.156.2 Member Data Documentation

7.156.2.1 DOUBLE

```
int ttk_tools.vector.canapeapi_common.DATA_TYPE.DOUBLE = 3  [static]
```

7.156.2.2 FLOAT

```
int ttk_tools.vector.canapeapi_common.DATA_TYPE.FLOAT = 2  [static]
```

7.156.2.3 FLOAT16

```
int ttk_tools.vector.canapeapi_common.DATA_TYPE.FLOAT16 = 14  [static]
```

7.156.2.4 INT

```
int ttk_tools.vector.canapeapi_common.DATA_TYPE.INT = 1  [static]
```

7.156.2.5 INT64

```
int ttk_tools.vector.canapeapi_common.DATA_TYPE.INT64 = 7  [static]
```

7.156.2.6 SBYTE

```
int ttk_tools.vector.canapeapi_common.DATA_TYPE.SBYTE = 13  [static]
```

7.156.2.7 SIGNED

```
int ttk_tools.vector.canapeapi_common.DATA_TYPE.SIGNED = 4  [static]
```

7.156.2.8 STRING

```
int ttk_tools.vector.canapeapi_common.DATA_TYPE.STRING = 6  [static]
```

7.156.2.9 UBYTE

```
int ttk_tools.vector.canapeapi_common.DATA_TYPE.UBYTE = 12  [static]
```

7.156.2.10 UINT

```
int ttk_tools.vector.canapeapi_common.DATA_TYPE.UINT = 11  [static]
```

7.156.2.11 UINT64

```
int ttk_tools.vector.canapeapi_common.DATA_TYPE.UINT64 = 8  [static]
```

7.156.2.12 UNKNOWN

```
int ttk_tools.vector.canapeapi_common.DATA_TYPE.UNKNOWN = 0  [static]
```

7.156.2.13 UNSIGNED

```
int ttk_tools.vector.canapeapi_common.DATA_TYPE.UNSIGNED = 5  [static]
```

7.156.2.14 UWORD

```
int ttk_tools.vector.canapeapi_common.DATA_TYPE.UWORD = 9  [static]
```

7.156.2.15 WORD

```
int ttk_tools.vector.canapeapi_common.DATA_TYPE.WORD = 10  [static]
```

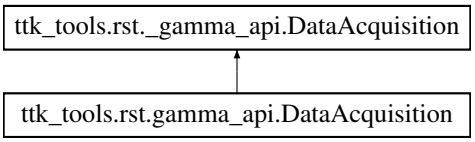
The documentation for this class was generated from the following file:

- ttk_tools/vector/canapeapi_common.py (v3.1.2)

7.157 ttk_tools.rst._gamma_api.DataAcquisition Class Reference

Data Acquisition with GammaV Logger Task.

Inheritance diagram for ttk_tools.rst._gamma_api.DataAcquisition:



Public Member Functions

- __init__ (self, gamma, system_name="iSyst", ftp_host=None)

  GammaV Data Acquisition.
- setup (self, variables)

  Setup data acquisition (configure what variables to measure).
- start (self)

  Start a configured data acquisition.
- stop (self)

  Stop a currently running data acquisition.
- getData (self)

  Get data from last measurement run.
- getRemoteDataFilePath (self)

  Get the current data file name (and relative path) to file on remote Gamma system's FTP server.
- getDaqDataFile (self, file_path)

  Retrieve contents of remote data file.
- setDataStorageFolder (self, folder_path=None)

  Set a storage folder for retrieved ∗.idaq files will to store there.

Public Attributes

- gamma

  context for accessing PVs (a reference to gaapi)
- data_storage_folder

  If data_storage_folder is set to a folder path, retrieved ∗.idaq files will be stored there, otherwise a temporary file will be used.
- verbosity

  level of verbosity.
- pv_capture_config_flag
- pv_capture_run_flag
- pv_capture_config_vars
- pv_file_name
- pv_logger_status
- pv_logger_status_code
- pv_retval
- pv_retval_descr
- ftp_host

  FTP host for DAQ data transfer (an IP address string)
- ftp_user

  FTP user for DAQ data transfer.
- ftp_passwd

  FTP password for DAQ data transfer.

Static Public Attributes

- str pvaddr_capture_config_flag = "iSyRTA.measurement.config.CaptureConfigFlag"

  Default PV addresses for access/utility variables.
- str pvaddr_capture_run_flag = "iSyRTA.measurement.config.CaptureRunFlag"
- str pvaddr_capture_config_vars = "iSyRTA.measurement.config.ConfigVars"
- str pvaddr_retval = "iSyRTA.measurement.config.GL_CommandRetVal"
- str pvaddr_retval_descr = "iSyRTA.measurement.config.GL_RetValDescription"
- str pvaddr_file_name = "iSyRTA.measurement.config.GL_FileName"
- str pvaddr_system_ip = "iSyRTA.measurement.config.GL_IPAddress"
- str pvaddr_system_free_space = "iSyRTA.measurement.config.GL_HDDSpace"
- str pvaddr_logger_status = "iSyRTA.measurement.config.GL_LoggerStatus"
- str pvaddr_logger_status_code = "iSyRTA.measurement.config.GL_LoggerStatusID"
- str pvaddr_task_status = "iSyRTA.measurement.config.GL_TaskStatus"

Protected Member Functions

- _getCurrentStatus (self)

  Get the current status of the logger state machine.
- _waitForStatus (self, expected_status, timeout_s=10, step_s=0.200)

  Wait for state machine state to reach the given state.

7.157.1 Detailed Description

Data Acquisition with GammaV Logger Task.

7.157.2 Constructor & Destructor Documentation

7.157.2.1 __init__()

```
ttk_tools.rst._gamma_api.DataAcquisition.__init__ (
              self,
              gamma,
              system_name = "iSyst",
              ftp_host = None )
```

GammaV Data Acquisition.

Parameters

gamma	context for accessing PVs (i.e. a reference to gaapi for Gamma PVs, see Gamma.getReference())
system_name	name of remote Gamma system to access
ftp_host	host name / ip of remote system to retrieve captured data from. None: automatically retrieve setting from a support PV

Reimplemented in ttk_tools.rst.gamma_api.DataAcquisition.

7.157.3 Member Function Documentation

7.157.3.1 _getCurrentStatus()

```
ttk_tools.rst._gamma_api.DataAcquisition._getCurrentStatus (
              self )  [protected]
```

Get the current status of the logger state machine.

7.157.3.2 _waitForStatus()

```
ttk_tools.rst._gamma_api.DataAcquisition._waitForStatus (
              self,
              expected_status,
              timeout_s = 10,
              step_s = 0.200 )  [protected]
```

Wait for state machine state to reach the given state.

A GammaDaqStateError will be raised if the expected expected_status was not reached in time.

Parameters

expected_status	expected_status/state to reach (numeric, see DaqMeasurementStatus)
timeout_s	timeout in seconds
step_s	step size / delay between expected_status checks

7.157.3.3 getDaqDataFile()

```
ttk_tools.rst._gamma_api.DataAcquisition.getDaqDataFile (
              self,
              file_path )
```

Retrieve contents of remote data file.

Parameters

file_path	name of file to retrieve from Gamma server

Returns

an open file handle in mode r+b, file pointer position will be at last written position.

Reimplemented in ttk_tools.rst.gamma_api.DataAcquisition.

7.157.3.4 getData()

```
ttk_tools.rst._gamma_api.DataAcquisition.getData (
              self )
```

Get data from last measurement run.

Data acquisition should be stopped before retrieving data (and should have been started at least once before).

Returns

a daq data structure with {
<label>: {
"time": [<time_s>, <time_s>, ...],
"data": [<value>, <value>, ...],
"type": "<datatype-string>"
},
} entries.

Reimplemented in ttk_tools.rst.gamma_api.DataAcquisition.

7.157.3.5 getRemoteDataFilePath()

```
ttk_tools.rst._gamma_api.DataAcquisition.getRemoteDataFilePath (
              self )
```

Get the current data file name (and relative path) to file on remote Gamma system's FTP server.

Reimplemented in ttk_tools.rst.gamma_api.DataAcquisition.

7.157.3.6 setDataStorageFolder()

```
ttk_tools.rst._gamma_api.DataAcquisition.setDataStorageFolder (
              self,
              folder_path = None )
```

Set a storage folder for retrieved ∗.idaq files will to store there.

If set to None, a disposable temporary file will be used for data exchange.

Parameters

folder_path	path to a folder or None to use temp files

Reimplemented in ttk_tools.rst.gamma_api.DataAcquisition.

7.157.3.7 setup()

```
ttk_tools.rst._gamma_api.DataAcquisition.setup (
              self,
              variables )
```

Setup data acquisition (configure what variables to measure).

Note

setup() may only be performed while data acquisition is not already running.

Parameters

variables	list/iterable of gamma variables to measure

Usage:

```
# direct setup with PV addresses/identifiers:
daq.setup([
    "iSyst:Data1.Data1.Data1.Data1",
    "iSyst:Data2.Data2.Data2.Data2"
])
# or, when using a VarContainer:
daq.setup([hil.some_var, hil.some_other_var])
```

Reimplemented in ttk_tools.rst.gamma_api.DataAcquisition.

7.157.3.8 start()

```
ttk_tools.rst._gamma_api.DataAcquisition.start (
              self )
```

Start a configured data acquisition.

Reimplemented in ttk_tools.rst.gamma_api.DataAcquisition.

7.157.3.9 stop()

```
ttk_tools.rst._gamma_api.DataAcquisition.stop (
              self )
```

Stop a currently running data acquisition.

Reimplemented in ttk_tools.rst.gamma_api.DataAcquisition.

7.157.4 Member Data Documentation

7.157.4.1 data_storage_folder

```
ttk_tools.rst._gamma_api.DataAcquisition.data_storage_folder
```

If data_storage_folder is set to a folder path, retrieved ∗.idaq files will be stored there, otherwise a temporary file will be used.

See setDataStoragefolder().

7.157.4.2 ftp_host

```
ttk_tools.rst._gamma_api.DataAcquisition.ftp_host
```

FTP host for DAQ data transfer (an IP address string)

7.157.4.3 ftp_passwd

```
ttk_tools.rst._gamma_api.DataAcquisition.ftp_passwd
```

FTP password for DAQ data transfer.

7.157.4.4 ftp_user

```
ttk_tools.rst._gamma_api.DataAcquisition.ftp_user
```

FTP user for DAQ data transfer.

Empty user/password: login anonymously

7.157.4.5 gamma

```
ttk_tools.rst._gamma_api.DataAcquisition.gamma
```

context for accessing PVs (a reference to gaapi)

7.157.4.6 pv_capture_config_flag

ttk_tools.rst._gamma_api.DataAcquisition.pv_capture_config_flag

7.157.4.7 pv_capture_config_vars

ttk_tools.rst._gamma_api.DataAcquisition.pv_capture_config_vars

7.157.4.8 pv_capture_run_flag

ttk_tools.rst._gamma_api.DataAcquisition.pv_capture_run_flag

7.157.4.9 pv_file_name

ttk_tools.rst._gamma_api.DataAcquisition.pv_file_name

7.157.4.10 pv_logger_status

ttk_tools.rst._gamma_api.DataAcquisition.pv_logger_status

7.157.4.11 pv_logger_status_code

ttk_tools.rst._gamma_api.DataAcquisition.pv_logger_status_code

7.157.4.12 pv_retval

ttk_tools.rst._gamma_api.DataAcquisition.pv_retval

7.157.4.13 pv_retval_descr

ttk_tools.rst._gamma_api.DataAcquisition.pv_retval_descr

7.157.4.14 pvaddr_capture_config_flag

str ttk_tools.rst._gamma_api.DataAcquisition.pvaddr_capture_config_flag = "iSyRTA.measurement.↩ config.CaptureConfigFlag" [static]

Default PV addresses for access/utility variables.

7.157.4.15 pvaddr_capture_config_vars

str ttk_tools.rst._gamma_api.DataAcquisition.pvaddr_capture_config_vars = "iSyRTA.measurement.↩ config.ConfigVars" [static]

7.157.4.16 pvaddr_capture_run_flag

str ttk_tools.rst._gamma_api.DataAcquisition.pvaddr_capture_run_flag = "iSyRTA.measurement.↩
config.CaptureRunFlag" [static]

7.157.4.17 pvaddr_file_name

str ttk_tools.rst._gamma_api.DataAcquisition.pvaddr_file_name = "iSyRTA.measurement.config.↩
GL_FileName" [static]

7.157.4.18 pvaddr_logger_status

str ttk_tools.rst._gamma_api.DataAcquisition.pvaddr_logger_status = "iSyRTA.measurement.↩
config.GL_LoggerStatus" [static]

7.157.4.19 pvaddr_logger_status_code

str ttk_tools.rst._gamma_api.DataAcquisition.pvaddr_logger_status_code = "iSyRTA.measurement.↩
config.GL_LoggerStatusID" [static]

7.157.4.20 pvaddr_retval

str ttk_tools.rst._gamma_api.DataAcquisition.pvaddr_retval = "iSyRTA.measurement.config.GL_↩
CommandRetVal" [static]

7.157.4.21 pvaddr_retval_descr

str ttk_tools.rst._gamma_api.DataAcquisition.pvaddr_retval_descr = "iSyRTA.measurement.↩
config.GL_RetValDescription" [static]

7.157.4.22 pvaddr_system_free_space

str ttk_tools.rst._gamma_api.DataAcquisition.pvaddr_system_free_space = "iSyRTA.measurement.↩
config.GL_HDDSpace" [static]

7.157.4.23 pvaddr_system_ip

str ttk_tools.rst._gamma_api.DataAcquisition.pvaddr_system_ip = "iSyRTA.measurement.config.↩
GL_IPAddress" [static]

7.157.4.24 pvaddr_task_status

str ttk_tools.rst._gamma_api.DataAcquisition.pvaddr_task_status = "iSyRTA.measurement.config.↩
GL_TaskStatus" [static]

7.157.4.25 verbosity

`ttk_tools.rst._gamma_api.DataAcquisition.verbosity`

level of verbosity.

Mainly used in _waitForStatus: 0: silent
1: Waiting for expected_status FOO... . . . .done.
2: show current expected_status code instead of "."s
3: show current expected_status name instead of "."s
4: also list available idaq files on the remote system during getDaqDataFile

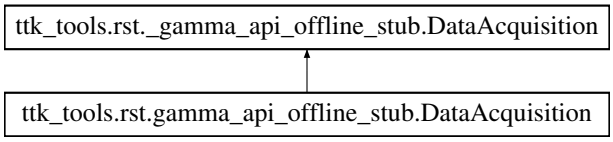
The documentation for this class was generated from the following file:

- ttk_tools/rst/_gamma_api.py (v3.4.1)

7.158 ttk_tools.rst._gamma_api_offline_stub.DataAcquisition Class Reference

Data Acquisition with GammaV Logger Task.

Inheritance diagram for ttk_tools.rst._gamma_api_offline_stub.DataAcquisition:



Public Member Functions

- __init__ (self, gamma, system_name="iSyst", ftp_host=None)
  GammaV Data Acquisition stub.
- setup (self, variables)
  Setup data acquisition (configure what variables to measure).
- start (self)
  Start a configured data acquisition.
- stop (self)
  Stop a currently running data acquisition.
- getData (self)
  Get data from last measurement run.
- getRemoteDataFilePath (self)
  Get the current data file name (and relative path) to file on remote Gamma system's FTP server.
- getDaqDataFile (self, file_path)
  Retrieve contents of remote data file.
- setDataStorageFolder (self, folder_path=None)
  Set a storage folder for retrieved ∗.idaq files will to store there.

Public Attributes

- gamma
- data_storage_folder

  If data_storage_folder is set to a folder path, retrieved ∗.idaq files will be stored there, otherwise a temporary file will be used.
- verbosity

  level of verbosity.
- pv_capture_config_flag
- pv_capture_run_flag
- pv_capture_config_vars
- pv_file_name
- pv_logger_status
- pv_retval
- pv_retval_descr
- ftp_host

  FTP host for DAQ data transfer (an IP address string)
- ftp_sub_folder
- ftp_user

  FTP user for DAQ data transfer.
- ftp_passwd

  FTP password for DAQ data transfer.

Static Public Attributes

- str pvaddr_capture_config_flag = "iSyRTA.measurement.config.CaptureConfigFlag"

  Default PV addresses for access/utility variables.
- str pvaddr_capture_run_flag = "iSyRTA.measurement.config.CaptureRunFlag"
- str pvaddr_capture_config_vars = "iSyRTA.measurement.config.ConfigVars"
- str pvaddr_retval = "iSyRTA.measurement.config.GL_CommandRetVal"
- str pvaddr_retval_descr = "iSyRTA.measurement.config.GL_RetValDescription"
- str pvaddr_file_name = "iSyRTA.measurement.config.GL_FileName"
- str pvaddr_system_ip = "iSyRTA.measurement.config.GL_IPAddress"
- str pvaddr_system_free_space = "iSyRTA.measurement.config.GL_HDDSpace"
- str pvaddr_logger_status = "iSyRTA.measurement.config.GL_LoggerStatus"
- str pvaddr_logger_status_code = "iSyRTA.measurement.config.GL_LoggerStatusID"
- str pvaddr_task_status = "iSyRTA.measurement.config.GL_TaskStatus"

Protected Member Functions

- _getCurrentStatus (self)

  Get the current status of the logger state machine.
- _waitForStatus (self, expected_status, timeout_s=10, step_s=0.200)

  Wait for state machine state to reach the given state.
- _getDummyDaqData (self, duration, offset=0.0)

  Offline Stub: Get some dummy data.

Protected Attributes

- _cycle_time
- _downsampling
- _mdf_file_path
- _start_time
- _last_fetch_time
- _stop_time
- _variables
- _state

7.158.1 Detailed Description

Data Acquisition with GammaV Logger Task.

7.158.2 Constructor & Destructor Documentation

7.158.2.1 __init__()

```
ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.__init__ (
              self,
              gamma,
              system_name = "iSyst",
              ftp_host = None )
```

GammaV Data Acquisition stub.

Parameters

gamma	context for accessing PVs (i.e. a reference to gaapi for Gamma PVs, see Gamma.getReference())
system_name	name of remote Gamma system to access
ftp_host	host name / ip of remote system to retrieve captured data from. None: automatically retrieve setting from a support PV

instantiate a PV with a prefixed system name

Reimplemented in ttk_tools.rst.gamma_api_offline_stub.DataAcquisition.

7.158.3 Member Function Documentation

7.158.3.1 _getCurrentStatus()

```
ttk_tools.rst._gamma_api_offline_stub.DataAcquisition._getCurrentStatus (
              self )  [protected]
```

Get the current status of the logger state machine.

7.158.3.2 _getDummyDaqData()

```
ttk_tools.rst._gamma_api_offline_stub.DataAcquisition._getDummyDaqData (
              self,
              duration,
              offset = 0.0 )  [protected]
```

Offline Stub: Get some dummy data.

Uses a time stamp step size derived from self._cycle_time and self._downsampling

Parameters

duration	duration in [s] to get data for
offset	offset of first time stamp

Returns

a DAQ data structure with data for each variable in self.variables.

7.158.3.3 _waitForStatus()

```
ttk_tools.rst._gamma_api_offline_stub.DataAcquisition._waitForStatus (
              self,
              expected_status,
              timeout_s = 10,
              step_s = 0.200 )  [protected]
```

Wait for state machine state to reach the given state.

A GammaDaqStateError will be raised if the expected expected_status was not reached in time.

Parameters

expected_status	expected_status/state to reach (numeric, see DaqMeasurementStatus)
timeout_s	timeout in seconds
step_s	step size / delay between expected_status checks

7.158.3.4 getDaqDataFile()

```
ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.getDaqDataFile (
              self,
              file_path )
```

Retrieve contents of remote data file.

Parameters

file_path	name of file to retrieve from Gamma server

Returns

an open file handle in mode r+b, file pointer position will be at last written position.

Reimplemented in ttk_tools.rst.gamma_api_offline_stub.DataAcquisition.

7.158.3.5 getData()

```
ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.getData (
              self )
```

Get data from last measurement run.

Data acquisition should be stopped before retrieving data (and should have been started at least once before).

Returns

a daq data structure with {
<label>: {
"time": [<time_s>, <time_s>, ...],
"data": [<value>, <value>, ...],
"type": "<datatype-string>"
},
} entries.

Reimplemented in ttk_tools.rst.gamma_api_offline_stub.DataAcquisition.

7.158.3.6 getRemoteDataFilePath()

```
ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.getRemoteDataFilePath (
              self )
```

Get the current data file name (and relative path) to file on remote Gamma system's FTP server.

Reimplemented in ttk_tools.rst.gamma_api_offline_stub.DataAcquisition.

7.158.3.7 setDataStorageFolder()

```
ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.setDataStorageFolder (
              self,
              folder_path = None )
```

Set a storage folder for retrieved ∗.idaq files will to store there.

If set to None, a disposable temporary file will be used for data exchange.

Parameters

folder_path	path to a folder or None to use temp files

Reimplemented in ttk_tools.rst.gamma_api_offline_stub.DataAcquisition.

7.158.3.8 setup()

```
ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.setup (
              self,
              variables )
```

Setup data acquisition (configure what variables to measure).

Note

setup() may only be performed while data acquisition is not already running.

Parameters

variables	list/iterable of gamma variables to measure

Usage:

```
# direct setup with PV addresses/identifiers:
daq.setup([
    "iSyst:Data1.Data1.Data1.Data1",
    "iSyst:Data2.Data2.Data2.Data2"
])
# or, when using a VarContainer:
daq.setup([hil.some_var, hil.some_other_var])
```

Reimplemented in ttk_tools.rst.gamma_api_offline_stub.DataAcquisition.

7.158.3.9 start()

```
ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.start (
              self )
```

Start a configured data acquisition.

Reimplemented in ttk_tools.rst.gamma_api_offline_stub.DataAcquisition.

7.158.3.10 stop()

```
ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.stop (
              self )
```

Stop a currently running data acquisition.

Reimplemented in ttk_tools.rst.gamma_api_offline_stub.DataAcquisition.

7.158.4 Member Data Documentation

7.158.4.1 _cycle_time

```
ttk_tools.rst._gamma_api_offline_stub.DataAcquisition._cycle_time  [protected]
```

7.158.4.2 _downsampling

ttk_tools.rst._gamma_api_offline_stub.DataAcquisition._downsampling [protected]

7.158.4.3 _last_fetch_time

ttk_tools.rst._gamma_api_offline_stub.DataAcquisition._last_fetch_time [protected]

7.158.4.4 _mdf_file_path

ttk_tools.rst._gamma_api_offline_stub.DataAcquisition._mdf_file_path [protected]

7.158.4.5 _start_time

ttk_tools.rst._gamma_api_offline_stub.DataAcquisition._start_time [protected]

7.158.4.6 _state

ttk_tools.rst._gamma_api_offline_stub.DataAcquisition._state [protected]

7.158.4.7 _stop_time

ttk_tools.rst._gamma_api_offline_stub.DataAcquisition._stop_time [protected]

7.158.4.8 _variables

ttk_tools.rst._gamma_api_offline_stub.DataAcquisition._variables [protected]

7.158.4.9 data_storage_folder

ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.data_storage_folder

If data_storage_folder is set to a folder path, retrieved $*$.idaq files will be stored there, otherwise a temporary file will be used.

This is mainly intended for debugging purposes.

7.158.4.10 ftp_host

ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.ftp_host

FTP host for DAQ data transfer (an IP address string)

7.158.4.11 ftp_passwd

ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.ftp_passwd

FTP password for DAQ data transfer.

7.158.4.12 ftp_sub_folder

ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.ftp_sub_folder

7.158.4.13 ftp_user

ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.ftp_user

FTP user for DAQ data transfer.

Empty user/password: login anonymously

7.158.4.14 gamma

ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.gamma

7.158.4.15 pv_capture_config_flag

ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.pv_capture_config_flag

7.158.4.16 pv_capture_config_vars

ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.pv_capture_config_vars

7.158.4.17 pv_capture_run_flag

ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.pv_capture_run_flag

7.158.4.18 pv_file_name

ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.pv_file_name

7.158.4.19 pv_logger_status

ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.pv_logger_status

7.158.4.20 pv_retval

ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.pv_retval

7.158.4.21 pv_retval_descr

ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.pv_retval_descr

7.158.4.22 pvaddr_capture_config_flag

str ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.pvaddr_capture_config_flag = "iSy↩
RTA.measurement.config.CaptureConfigFlag" [static]

Default PV addresses for access/utility variables.

7.158.4.23 pvaddr_capture_config_vars

str ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.pvaddr_capture_config_vars = "iSy↩
RTA.measurement.config.ConfigVars" [static]

7.158.4.24 pvaddr_capture_run_flag

str ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.pvaddr_capture_run_flag = "iSyRTA.↩
measurement.config.CaptureRunFlag" [static]

7.158.4.25 pvaddr_file_name

str ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.pvaddr_file_name = "iSyRTA.measurement.↩
config.GL_FileName" [static]

7.158.4.26 pvaddr_logger_status

str ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.pvaddr_logger_status = "iSyRTA.↩
measurement.config.GL_LoggerStatus" [static]

7.158.4.27 pvaddr_logger_status_code

str ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.pvaddr_logger_status_code = "iSy↩
RTA.measurement.config.GL_LoggerStatusID" [static]

7.158.4.28 pvaddr_retval

str ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.pvaddr_retval = "iSyRTA.measurement.↩
config.GL_CommandRetVal" [static]

7.158.4.29 pvaddr_retval_descr

```
str ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.pvaddr_retval_descr = "iSyRTA.↩
measurement.config.GL_RetValDescription"  [static]
```

7.158.4.30 pvaddr_system_free_space

```
str ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.pvaddr_system_free_space = "iSy↩
RTA.measurement.config.GL_HDDSpace"  [static]
```

7.158.4.31 pvaddr_system_ip

```
str ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.pvaddr_system_ip = "iSyRTA.measurement.↩
config.GL_IPAddress"  [static]
```

7.158.4.32 pvaddr_task_status

```
str ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.pvaddr_task_status = "iSyRTA.↩
measurement.config.GL_TaskStatus"  [static]
```

7.158.4.33 verbosity

```
ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.verbosity
```

level of verbosity.

Mainly used in _waitForStatus: 0: silent 1: Waiting for expected_status FOO... . . . .done. 2: show current expected_status code instead of "."s 3: show current expected_status name instead of "."s

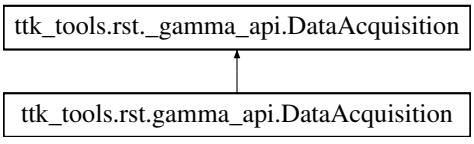
The documentation for this class was generated from the following file:

- ttk_tools/rst/_gamma_api_offline_stub.py (v1.7)

7.159 ttk_tools.rst.gamma_api.DataAcquisition Class Reference

Data Acquisition with GammaV Logger Task.

Inheritance diagram for ttk_tools.rst.gamma_api.DataAcquisition:



Public Member Functions

- __init__ (self, gamma, system_name="iSyst", ftp_host=None)

  GammaV Data Acquisition.
- setup (self, variables)

  Setup data acquisition (configure what variables to measure).
- start (self)

  Start a configured data acquisition.
- stop (self)

  Stop a currently running data acquisition.
- getData (self)

  Get data from last measurement run.
- getRemoteDataFilePath (self)

  Get the current data file name (and relative path) to file on remote Gamma system's FTP server.
- getDaqDataFile (self, file_path)

  Retrieve contents of remote data file.
- setDataStorageFolder (self, folder_path=None)

  Set a storage folder for retrieved ∗.idaq files will to store there.

Static Public Attributes

- str pvaddr_capture_config_flag = "iSyRTA.measurement.config.CaptureConfigFlag"

  Default PV addresses for access/utility variables.
- str pvaddr_capture_run_flag = "iSyRTA.measurement.config.CaptureRunFlag"
- str pvaddr_capture_config_vars = "iSyRTA.measurement.config.ConfigVars"
- str pvaddr_retval = "iSyRTA.measurement.config.GL_CommandRetVal"
- str pvaddr_retval_descr = "iSyRTA.measurement.config.GL_RetValDescription"
- str pvaddr_file_name = "iSyRTA.measurement.config.GL_FileName"
- str pvaddr_system_ip = "iSyRTA.measurement.config.GL_IPAddress"
- str pvaddr_system_free_space = "iSyRTA.measurement.config.GL_HDDSpace"
- str pvaddr_logger_status = "iSyRTA.measurement.config.GL_LoggerStatus"
- str pvaddr_logger_status_code = "iSyRTA.measurement.config.GL_LoggerStatusID"
- str pvaddr_task_status = "iSyRTA.measurement.config.GL_TaskStatus"

7.159.1 Detailed Description

Data Acquisition with GammaV Logger Task.

7.159.2 Constructor & Destructor Documentation

7.159.2.1 __init__()

```
ttk_tools.rst.gamma_api.DataAcquisition.__init__ (
              self,
              gamma,
              system_name = "iSyst",
              ftp_host = None )
```

GammaV Data Acquisition.

Parameters

gamma	context for accessing PVs (i.e. a reference to gaapi for Gamma PVs, see Gamma.getReference())
system_name	name of remote Gamma system to access
ftp_host	host name / ip of remote system to retrieve captured data from. None: automatically retrieve setting from a support PV

Reimplemented from ttk_tools.rst._gamma_api.DataAcquisition.

7.159.3 Member Function Documentation

7.159.3.1 getDaqDataFile()

```
ttk_tools.rst.gamma_api.DataAcquisition.getDaqDataFile (
              self,
              file_path )
```

Retrieve contents of remote data file.

Parameters

file_path	name of file to retrieve from Gamma server

Returns

an open file handle in mode r+b, file pointer position will be at last written position.

Reimplemented from ttk_tools.rst._gamma_api.DataAcquisition.

7.159.3.2 getData()

```
ttk_tools.rst.gamma_api.DataAcquisition.getData (
              self )
```

Get data from last measurement run.

Data acquisition should be stopped before retrieving data (and should have been started at least once before).

Returns

a daq data structure with {
<label>: {
"time": [<time_s>, <time_s>, ...],
"data": [<value>, <value>, ...],
"type": "<datatype-string>"
},
} entries.

Reimplemented from ttk_tools.rst._gamma_api.DataAcquisition.

7.159.3.3 getRemoteDataFilePath()

```
ttk_tools.rst.gamma_api.DataAcquisition.getRemoteDataFilePath (
              self )
```

Get the current data file name (and relative path) to file on remote Gamma system's FTP server.

Reimplemented from ttk_tools.rst._gamma_api.DataAcquisition.

7.159.3.4 setDataStorageFolder()

```
ttk_tools.rst.gamma_api.DataAcquisition.setDataStorageFolder (
              self,
              folder_path = None )
```

Set a storage folder for retrieved ∗.idaq files will to store there.

If set to None, a disposable temporary file will be used for data exchange.

Parameters

folder_path	path to a folder or None to use temp files

Reimplemented from ttk_tools.rst._gamma_api.DataAcquisition.

7.159.3.5 setup()

```
ttk_tools.rst.gamma_api.DataAcquisition.setup (
              self,
              variables )
```

Setup data acquisition (configure what variables to measure).

Note

setup() may only be performed while data acquisition is not already running.

Parameters

variables	list/iterable of gamma variables to measure

Usage:

```
# direct setup with PV addresses/identifiers:
daq.setup([
    "iSyst:Data1.Data1.Data1.Data1",
    "iSyst:Data2.Data2.Data2.Data2"
])
# or, when using a VarContainer:
daq.setup([hil.some_var, hil.some_other_var])
```

Reimplemented from ttk_tools.rst._gamma_api.DataAcquisition.

7.159.3.6 start()

```
ttk_tools.rst.gamma_api.DataAcquisition.start (
              self )
```

Start a configured data acquisition.

Reimplemented from ttk_tools.rst._gamma_api.DataAcquisition.

7.159.3.7 stop()

```
ttk_tools.rst.gamma_api.DataAcquisition.stop (
              self )
```

Stop a currently running data acquisition.

Reimplemented from ttk_tools.rst._gamma_api.DataAcquisition.

7.159.4 Member Data Documentation

7.159.4.1 pvaddr_capture_config_flag

```
str ttk_tools.rst.gamma_api.DataAcquisition.pvaddr_capture_config_flag = "iSyRTA.measurement.↩
config.CaptureConfigFlag"  [static]
```

Default PV addresses for access/utility variables.

7.159.4.2 pvaddr_capture_config_vars

```
str ttk_tools.rst.gamma_api.DataAcquisition.pvaddr_capture_config_vars = "iSyRTA.measurement.↩
config.ConfigVars"  [static]
```

7.159.4.3 pvaddr_capture_run_flag

```
str ttk_tools.rst.gamma_api.DataAcquisition.pvaddr_capture_run_flag = "iSyRTA.measurement.↩
config.CaptureRunFlag"  [static]
```

7.159.4.4 pvaddr_file_name

```
str ttk_tools.rst.gamma_api.DataAcquisition.pvaddr_file_name = "iSyRTA.measurement.config.GL↩
_FileName"  [static]
```

7.159.4.5 pvaddr_logger_status

```
str ttk_tools.rst.gamma_api.DataAcquisition.pvaddr_logger_status = "iSyRTA.measurement.↩
config.GL_LoggerStatus"  [static]
```

7.159.4.6 pvaddr_logger_status_code

```
str ttk_tools.rst.gamma_api.DataAcquisition.pvaddr_logger_status_code = "iSyRTA.measurement.↩
config.GL_LoggerStatusID"  [static]
```

7.159.4.7 pvaddr_retval

```
str ttk_tools.rst.gamma_api.DataAcquisition.pvaddr_retval = "iSyRTA.measurement.config.GL_↩
CommandRetVal"  [static]
```

7.159.4.8 pvaddr_retval_descr

```
str ttk_tools.rst.gamma_api.DataAcquisition.pvaddr_retval_descr = "iSyRTA.measurement.config.↩
GL_RetValDescription"  [static]
```

7.159.4.9 pvaddr_system_free_space

```
str ttk_tools.rst.gamma_api.DataAcquisition.pvaddr_system_free_space = "iSyRTA.measurement.↩
config.GL_HDDSpace"  [static]
```

7.159.4.10 pvaddr_system_ip

```
str ttk_tools.rst.gamma_api.DataAcquisition.pvaddr_system_ip = "iSyRTA.measurement.config.GL↩
_IPAddress"  [static]
```

7.159.4.11 pvaddr_task_status

```
str ttk_tools.rst.gamma_api.DataAcquisition.pvaddr_task_status = "iSyRTA.measurement.config.↩
GL_TaskStatus"  [static]
```

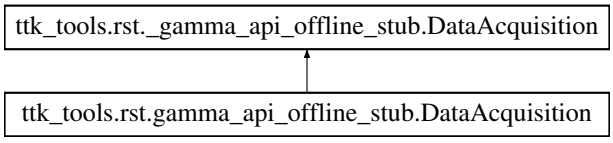
The documentation for this class was generated from the following file:

- ttk_tools/rst/gamma_api.py (v3.0)

7.160 ttk_tools.rst.gamma_api_offline_stub.DataAcquisition Class Reference

Data Acquisition with GammaV Logger Task.

Inheritance diagram for ttk_tools.rst.gamma_api_offline_stub.DataAcquisition:



Public Member Functions

- __init__ (self, gamma, system_name="iSyst", ftp_host=None)

  GammaV Data Acquisition stub.
- setup (self, variables)

  Setup data acquisition (configure what variables to measure).
- start (self)

  Start a configured data acquisition.
- stop (self)

  Stop a currently running data acquisition.
- getData (self)

  Get data from last measurement run.
- getRemoteDataFilePath (self)

  Get the current data file name (and relative path) to file on remote Gamma system's FTP server.
- getDaqDataFile (self, file_path)

  Retrieve contents of remote data file.
- setDataStorageFolder (self, folder_path=None)

  Set a storage folder for retrieved ∗.idaq files will to store there.

Static Public Attributes

- str pvaddr_capture_config_flag = "iSyRTA.measurement.config.CaptureConfigFlag"

  Default PV addresses for access/utility variables.
- str pvaddr_capture_run_flag = "iSyRTA.measurement.config.CaptureRunFlag"
- str pvaddr_capture_config_vars = "iSyRTA.measurement.config.ConfigVars"
- str pvaddr_retval = "iSyRTA.measurement.config.GL_CommandRetVal"
- str pvaddr_retval_descr = "iSyRTA.measurement.config.GL_RetValDescription"
- str pvaddr_file_name = "iSyRTA.measurement.config.GL_FileName"
- str pvaddr_system_ip = "iSyRTA.measurement.config.GL_IPAddress"
- str pvaddr_system_free_space = "iSyRTA.measurement.config.GL_HDDSpace"
- str pvaddr_logger_status = "iSyRTA.measurement.config.GL_LoggerStatus"
- str pvaddr_logger_status_code = "iSyRTA.measurement.config.GL_LoggerStatusID"
- str pvaddr_task_status = "iSyRTA.measurement.config.GL_TaskStatus"

7.160.1 Detailed Description

Data Acquisition with GammaV Logger Task.

7.160.2 Constructor & Destructor Documentation

7.160.2.1 __init__()

```
ttk_tools.rst.gamma_api_offline_stub.DataAcquisition.__init__ (
              self,
              gamma,
              system_name = "iSyst",
              ftp_host = None )
```

GammaV Data Acquisition stub.

Parameters

gamma	context for accessing PVs (i.e. a reference to gaapi for Gamma PVs, see Gamma.getReference())
system_name	name of remote Gamma system to access
ftp_host	host name / ip of remote system to retrieve captured data from. None: automatically retrieve setting from a support PV

Reimplemented from ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.

7.160.3 Member Function Documentation

7.160.3.1 getDaqDataFile()

```
ttk_tools.rst.gamma_api_offline_stub.DataAcquisition.getDaqDataFile (
              self,
              file_path )
```

Retrieve contents of remote data file.

Parameters

file_path	name of file to retrieve from Gamma server

Returns

an open file handle in mode r+b, file pointer position will be at last written position.

Reimplemented from ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.

7.160.3.2 getData()

```
ttk_tools.rst.gamma_api_offline_stub.DataAcquisition.getData (
              self )
```

Get data from last measurement run.

Data acquisition should be stopped before retrieving data (and should have been started at least once before).

Returns

a daq data structure with {
<label>: {
"time": [<time_s>, <time_s>, ...],
"data": [<value>, <value>, ...],
"type": "<datatype-string>"
},
} entries.

Reimplemented from ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.

7.160.3.3 getRemoteDataFilePath()

```
ttk_tools.rst.gamma_api_offline_stub.DataAcquisition.getRemoteDataFilePath (
              self )
```

Get the current data file name (and relative path) to file on remote Gamma system's FTP server.

Reimplemented from ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.

7.160.3.4 setDataStorageFolder()

```
ttk_tools.rst.gamma_api_offline_stub.DataAcquisition.setDataStorageFolder (
              self,
              folder_path = None )
```

Set a storage folder for retrieved ∗.idaq files will to store there.

If set to None, a disposable temporary file will be used for data exchange.

Parameters

folder_path	path to a folder or None to use temp files

Reimplemented from ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.

7.160.3.5 setup()

```
ttk_tools.rst.gamma_api_offline_stub.DataAcquisition.setup (
              self,
              variables )
```

Setup data acquisition (configure what variables to measure).

Note

setup() may only be performed while data acquisition is not already running.

Parameters

variables	list/iterable of gamma variables to measure

Usage:

```
# direct setup with PV addresses/identifiers:
daq.setup([
    "iSyst:Data1.Data1.Data1.Data1",
    "iSyst:Data2.Data2.Data2.Data2"
])
# or, when using a VarContainer:
daq.setup([hil.some_var, hil.some_other_var])
```

Reimplemented from ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.

7.160.3.6 start()

```
ttk_tools.rst.gamma_api_offline_stub.DataAcquisition.start (
              self )
```

Start a configured data acquisition.

Reimplemented from ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.

7.160.3.7 stop()

```
ttk_tools.rst.gamma_api_offline_stub.DataAcquisition.stop (
              self )
```

Stop a currently running data acquisition.

Reimplemented from ttk_tools.rst._gamma_api_offline_stub.DataAcquisition.

7.160.4 Member Data Documentation

7.160.4.1 pvaddr_capture_config_flag

```
str ttk_tools.rst.gamma_api_offline_stub.DataAcquisition.pvaddr_capture_config_flag = "iSy↩
RTA.measurement.config.CaptureConfigFlag"  [static]
```

Default PV addresses for access/utility variables.

7.160.4.2 pvaddr_capture_config_vars

```
str ttk_tools.rst.gamma_api_offline_stub.DataAcquisition.pvaddr_capture_config_vars = "iSy↩
RTA.measurement.config.ConfigVars"  [static]
```

7.160.4.3 pvaddr_capture_run_flag

```
str ttk_tools.rst.gamma_api_offline_stub.DataAcquisition.pvaddr_capture_run_flag = "iSyRTA.↩
measurement.config.CaptureRunFlag"  [static]
```

7.160.4.4 pvaddr_file_name

```
str ttk_tools.rst.gamma_api_offline_stub.DataAcquisition.pvaddr_file_name = "iSyRTA.measurement.↩
config.GL_FileName"  [static]
```

7.160.4.5 pvaddr_logger_status

```
str ttk_tools.rst.gamma_api_offline_stub.DataAcquisition.pvaddr_logger_status = "iSyRTA.↩
measurement.config.GL_LoggerStatus"  [static]
```

7.160.4.6 pvaddr_logger_status_code

```
str ttk_tools.rst.gamma_api_offline_stub.DataAcquisition.pvaddr_logger_status_code = "iSy↩
RTA.measurement.config.GL_LoggerStatusID"  [static]
```

7.160.4.7 pvaddr_retval

```
str ttk_tools.rst.gamma_api_offline_stub.DataAcquisition.pvaddr_retval = "iSyRTA.measurement.↩
config.GL_CommandRetVal"  [static]
```

7.160.4.8 pvaddr_retval_descr

```
str ttk_tools.rst.gamma_api_offline_stub.DataAcquisition.pvaddr_retval_descr = "iSyRTA.↩
measurement.config.GL_RetValDescription"  [static]
```

7.160.4.9 pvaddr_system_free_space

```
str ttk_tools.rst.gamma_api_offline_stub.DataAcquisition.pvaddr_system_free_space = "iSyRTA.↩
measurement.config.GL_HDDSpace"  [static]
```

7.160.4.10 pvaddr_system_ip

```
str ttk_tools.rst.gamma_api_offline_stub.DataAcquisition.pvaddr_system_ip = "iSyRTA.measurement.↩
config.GL_IPAddress"  [static]
```

7.160.4.11 pvaddr_task_status

```
str ttk_tools.rst.gamma_api_offline_stub.DataAcquisition.pvaddr_task_status = "iSyRTA.measurement.↩
config.GL_TaskStatus"  [static]
```

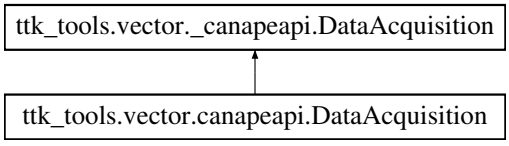
The documentation for this class was generated from the following file:

- ttk_tools/rst/gamma_api_offline_stub.py (v1.5)

7.161 ttk_tools.vector._canapeapi.DataAcquisition Class Reference

Data acquisition for multiple CANape devices.

Inheritance diagram for ttk_tools.vector._canapeapi.DataAcquisition:



Public Member Functions

- __init__ (self, canape_asap3, *devices)

  Setup data acquisition devices.
- setup (self, measurement_vars, mdf_file_path=None, polling_rate=20)

  Setup data acquisition.
- start (self)

  Start the currently configured data acquisition.
- stop (self)

  Stop a currently running data acquisition.
- getData (self, verbosity=0)

  Get data of the last measurement from the measurement data file.
- getStatus (self, verbosity=1)

  Get status of current DAQ/measurement, see canapeapi_common.MEASUREMENT_STATE.

Public Attributes

- mdf_file_path

  file path of current mdf file.
- default_mdf_filename

  default file name for MDF files ("daq.mdf").
- default_mdf_folder

  default output folder for MDF files.
- polling_rate

  default/fallback polling-rate in [ms] when using DAQ with polling (which is a rather unusual thing to do).
- extra_delay_before_mdf_read

  [s] extra delay before MDF file is read.
- read_data_attempts_timeout

  [s] timeout value for MDF-read attempts.
- read_data_attempts_delay

  [s] delay between MDF-read attempts.

Protected Member Functions

- _readMdf4Data (self, mdf_file_path, verbosity=0)

  Read data from a file in MDF4 format.
- _readMdf3Data (self, mdf_file_path, verbosity=0)

  Read data from a file in MDF3 format (using mdflib DLL).
- _getMeasurementState (self)

  Get state of current measurement.

Protected Attributes

- _devices
- _asap3
- _daq_configured
- _daq_running
- _api_log_level
- _logApiCallStart
- _logApiCallEnd

7.161.1 Detailed Description

Data acquisition for multiple CANape devices.

Data will be stored as MDF files ("streamed to disk") and later read back after the measurement is completed.

Note

In some CANape versions, CANape will retain a file lock on the current MDF-file after a measurement has been run (and stopped).
So multiple start() / stop() cycles after a single setup() may not work in all releases.

Workaround:
Explicitly perform another DAQ setup before each new measurement (works even with the same mdf file path).

Note

In some versions (probably starting with CANape 13), CANape seems to continue writing to the MDF-file for some time after data acquisition has been successfully stopped.
This may cause the current MDF-data to be interpreted as "unsorted" but the call to `mdfSort` will fail with an undocumented error code (1022). As this appears to be a race condition, the actual error symptoms might vary.

Workaround:
see DataAcquisition.extra_delay_before_mdf_read
see DataAcquisition.read_data_attempts_timeout
see DataAcquisition.read_data_attempts_delay

7.161.2 Constructor & Destructor Documentation

7.161.2.1 __init__()

```
ttk_tools.vector._canapeapi.DataAcquisition.__init__ (
              self,
              canape_asap3,
             * devices )
```

Setup data acquisition devices.

Parameters

canape_asap3	ASAP3-handle to main application server
∗*devices*	CANape devices from which to acquire data [1..n]. See CANapeDevice

Usage:

```
# with two CANapeDevice instances, one for XCP and one for CAN:
daq = DataAcquisition(asap3, canape_xcp, canape_can)
```

Reimplemented in ttk_tools.vector.canapeapi.DataAcquisition.

7.161.3 Member Function Documentation

7.161.3.1 _getMeasurementState()

```
ttk_tools.vector._canapeapi.DataAcquisition._getMeasurementState (
              self )  [protected]
```

Get state of current measurement.

Returns

a MEASUREMENT_STATE value.

7.161.3.2 _readMdf3Data()

```
ttk_tools.vector._canapeapi.DataAcquisition._readMdf3Data (
              self,
              mdf_file_path,
              verbosity = 0 )  [protected]
```

Read data from a file in MDF3 format (using mdflib DLL).

Parameters

mdf_file_path	path to mdf file
verbosity	verbosity of log messages

Returns

a DAQ data dict.

7.161.3.3 _readMdf4Data()

```
ttk_tools.vector._canapeapi.DataAcquisition._readMdf4Data (
              self,
              mdf_file_path,
              verbosity = 0 )  [protected]
```

Read data from a file in MDF4 format.

Parameters

mdf_file_path	path to mdf file
verbosity	verbosity of log messages

Returns

a DAQ data dict.

7.161.3.4 getData()

```
ttk_tools.vector._canapeapi.DataAcquisition.getData (
              self,
              verbosity = 0 )
```

Get data of the last measurement from the measurement data file.

Parameters

verbosity	verbosity level of mdflib status messages. 0: off. See mdflib.readMdf() for more details.

Example: Basic Data Structure:

```
daq_data[<canape var name>] = {
    'data':    <list of data values>,
    'time':    <list of corresponding time stamps>,
}
```

Returns

a data dictionary with entries for each recorded variable, see Basic Data Structure, above.
See mdflib.readMdf() for a more detailed description.

Reimplemented in ttk_tools.vector.canapeapi.DataAcquisition.

7.161.3.5 getStatus()

```
ttk_tools.vector._canapeapi.DataAcquisition.getStatus (
              self,
              verbosity = 1 )
```

Get status of current DAQ/measurement, see canapeapi_common.MEASUREMENT_STATE.

Parameters

verbosity	$>$ 0: status text will be printed to stdout

Returns

an numeric status value or None if status is not available (function was added in CANape 9).

Reimplemented in ttk_tools.vector.canapeapi.DataAcquisition.

7.161.3.6 setup()

```
ttk_tools.vector._canapeapi.DataAcquisition.setup (
              self,
              measurement_vars,
```

```
        mdf_file_path = None,
        polling_rate = 20 )
```

Setup data acquisition.

Parameters

measurement_vars	list of variables to measure.
mdf_file_path	path to mdf-file into which to store results. If left empty, a default "daq.mdf" file will be used/created in the CANape working directory. See DataAcquisition.default_mdf_filename.
polling_rate	[ms] polling rate to use for task-type "polling". Use polling only as fallback if there are no other/better measurement tasks available.

Format of measurement_vars:

`measurement_vars` with list entries:
`[[var, task, device, repr], ...]`

`measurement_vars` with dictionary entries:
`[{"var": cal.nii, "task": "10ms", "device": "ECU", "repr": REPR.↩ PHY}, ...]`

Available entries

- `var` - name/identifier of variable (this can simply be a "CalVar" instance, as VarBase-derived variables have their identifier as default representation)
- `task` - name of DAQ-task, see CANapeDevice.getECUTasks() for available tasks.
- `device` - index of device (order as supplied during DataAcquisition.__init__), defaults to first device (device index 0)
  OR
  device/module name as supplied during device's constructor call.
- `repr` - data representation to use for measured variable, REPR.HEX or REPR.PHYS (defaults to PHYS). See canapeapi_common.REPR

Note that at least `var` and `task` entries have to be present, `device` and `repr` are optional.

Example: DAQ setup and execution

```
meas_vars = [
    # <variable/identifier>, <measurement task>, <device index>
    [cal.emotor_temp,       "10ms", 0],
    [cal.emotor_n_ist,      "10ms", 0],
    [cal.speicherdruck,     "10ms", 0],
    ["GE_Oiltemp",          "CAN",  1],
]

daq.setup(meas_vars)

# start measurement/recorder
daq.start()

# do stuff here...

# stop measurement/recorder
daq.stop()

# get measured data
daq_data = daq.getData()
```

Reimplemented in ttk_tools.vector.canapeapi.DataAcquisition.

7.161.3.7 start()

```
ttk_tools.vector._canapeapi.DataAcquisition.start (
              self )
```

Start the currently configured data acquisition.

Returns

True if data acquisition has been started or is already running.

Reimplemented in ttk_tools.vector.canapeapi.DataAcquisition.

7.161.3.8 stop()

```
ttk_tools.vector._canapeapi.DataAcquisition.stop (
              self )
```

Stop a currently running data acquisition.

Returns

True if data acquisition has been stopped or was not running to begin with.

Reimplemented in ttk_tools.vector.canapeapi.DataAcquisition.

7.161.4 Member Data Documentation

7.161.4.1 _api_log_level

```
ttk_tools.vector._canapeapi.DataAcquisition._api_log_level  [protected]
```

7.161.4.2 _asap3

```
ttk_tools.vector._canapeapi.DataAcquisition._asap3  [protected]
```

7.161.4.3 _daq_configured

```
ttk_tools.vector._canapeapi.DataAcquisition._daq_configured  [protected]
```

7.161.4.4 _daq_running

```
ttk_tools.vector._canapeapi.DataAcquisition._daq_running  [protected]
```

7.161.4.5 _devices

ttk_tools.vector._canapeapi.DataAcquisition._devices [protected]

7.161.4.6 _logApiCallEnd

ttk_tools.vector._canapeapi.DataAcquisition._logApiCallEnd [protected]

7.161.4.7 _logApiCallStart

ttk_tools.vector._canapeapi.DataAcquisition._logApiCallStart [protected]

7.161.4.8 default_mdf_filename

ttk_tools.vector._canapeapi.DataAcquisition.default_mdf_filename

default file name for MDF files ("daq.mdf").

This will be used if no explicit mdf_file_path will be supplied during setup().

7.161.4.9 default_mdf_folder

ttk_tools.vector._canapeapi.DataAcquisition.default_mdf_folder

default output folder for MDF files.

Defaults to the current CANape working folder.

7.161.4.10 extra_delay_before_mdf_read

ttk_tools.vector._canapeapi.DataAcquisition.extra_delay_before_mdf_read

[s] extra delay before MDF file is read.

This might be useful for some CANape releases (e.g. at least some builds of CANNpe13), where the MDF fill is still being processed when stop() returns.

7.161.4.11 mdf_file_path

ttk_tools.vector._canapeapi.DataAcquisition.mdf_file_path

file path of current mdf file.

This value will be set during setup().

7.161.4.12 polling_rate

`ttk_tools.vector._canapeapi.DataAcquisition.polling_rate`

default/fallback polling-rate in [ms] when using DAQ with polling (which is a rather unusual thing to do).

This value will be updated during calls to setup().

7.161.4.13 read_data_attempts_delay

`ttk_tools.vector._canapeapi.DataAcquisition.read_data_attempts_delay`

[s] delay between MDF-read attempts.

See DataAcquisition.read_data_attempts_timeout

7.161.4.14 read_data_attempts_timeout

`ttk_tools.vector._canapeapi.DataAcquisition.read_data_attempts_timeout`

[s] timeout value for MDF-read attempts.

If a MDF file is not yet completely written, mdfLib will raise errors that will "go away" once the file is finished/consistent.

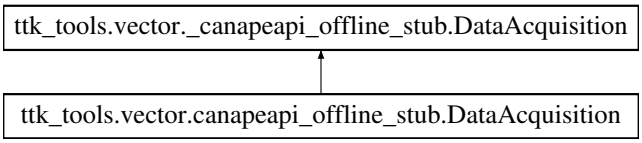
The documentation for this class was generated from the following file:

- ttk_tools/vector/_canapeapi.py (v7.3.3)

7.162 ttk_tools.vector._canapeapi_offline_stub.DataAcquisition Class Reference

Data acquisition (stub) for multiple CANape devices.

Inheritance diagram for ttk_tools.vector._canapeapi_offline_stub.DataAcquisition:



Public Member Functions

- __init__ (self, canape_asap3, *devices)

  Setup data acquisition devices.
- setup (self, measurement_vars, mdf_file_path=None, polling_rate=20)

  Setup data acquisition.
- start (self)

  Start the currently configured data acquisition.
- stop (self)

  Stop a currently running data acquisition.
- getData (self, verbosity=0)

  Get data of the last measurement from the measurement data file.
- getStatus (self, verbosity=1)

  Get status of current DAQ/measurement, see canapeapi_common.MEASUREMENT_STATE.

Public Attributes

- mdf_file_path

  file path of current mdf file.
- default_mdf_filename

  default file name for MDF files ("daq.mdf") [unused in stub].
- default_mdf_folder

  default output folder for MDF files [unused in stub].
- polling_rate

  default/fallback polling-rate in [ms] when using DAQ with polling (which is a rather unusual thing to do).
- extra_delay_before_mdf_read

  [s] extra delay before MDF file is read [unused in stub].
- read_data_attempts_timeout

  [s] timeout value for MDF-read attempts [unused in stub] If a MDF file is not yet completely written, mdfLib will raise errors that will "go away" once the file is finished/consistent.
- read_data_attempts_delay

  [s] delay between MDF-read attempts [unused in stub] See DataAcquisition.read_data_attempts_timeout

Protected Attributes

- _devices
- _asap3
- _daq_configured
- _daq_running
- _data_provider
- _variables
- _start_time
- _stop_time

7.162.1 Detailed Description

Data acquisition (stub) for multiple CANape devices.

Normally, data will be stored as MDF files ("streamed to disk") and later read back after the measurement is completed.

This stub will currently only provide generated dummy data for the measured time range (current value of measured stub variable ∗ (time range / step size) values).
See also DataAcquisition.polling_rate

Note

In some CANape versions, CANape will retain a file lock on the current MDF-file after a measurement has been run (and stopped).
So multiple start() / stop() cycles after a single setup() may not work in all releases.

Workaround:
Explicitly perform another DAQ setup before each new measurement (works even with the same mdf file path).

Note

In some versions (probably starting with CANape 13), CANape seems to continue writing to the MDF-file for some time after data acquisition has been successfully stopped.
This may cause the current MDF-data to be interpreted as "unsorted" but the call to `mdfSort` will fail with an undocumented error code (1022). As this appears to be a race condition, the actual error symptoms might vary.

Workaround:
see DataAcquisition.extra_delay_before_mdf_read
see DataAcquisition.read_data_attempts_timeout
see DataAcquisition.read_data_attempts_delay

7.162.2 Constructor & Destructor Documentation

7.162.2.1 __init__()

```
ttk_tools.vector._canapeapi_offline_stub.DataAcquisition.__init__ (
              self,
              canape_asap3,
             * devices )
```

Setup data acquisition devices.

Parameters

canape_asap3	ASAP3-handle to main application server
∗*devices*	CANape devices from which to acquire data [1..n]. See CANapeDevice

Usage:

```
# with two CANapeDevice instances, one for XCP and one for CAN:
daq = DataAcquisition(asap3, canape_xcp, canape_can)
```

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.DataAcquisition.

7.162.3 Member Function Documentation

7.162.3.1 getData()

```
ttk_tools.vector._canapeapi_offline_stub.DataAcquisition.getData (
              self,
              verbosity = 0 )
```

Get data of the last measurement from the measurement data file.

Parameters

verbosity	verbosity level of mdflib status messages. 0: off. See mdflib.readMdf() for more details.

Example: Basic Data Structure:

```
daq_data[<canape var name>] = {
    'data':    <list of data values>,
    'time':    <list of corresponding time stamps>,
}
```

Returns

a data dictionary with entries for each recorded variable, see Basic Data Structure, above.
See mdflib.readMdf() for a more detailed description.

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.DataAcquisition.

7.162.3.2 getStatus()

```
ttk_tools.vector._canapeapi_offline_stub.DataAcquisition.getStatus (
              self,
              verbosity = 1 )
```

Get status of current DAQ/measurement, see canapeapi_common.MEASUREMENT_STATE.

Parameters

verbosity	$>$ 0: status text will be printed to stdout

Returns

an numeric status value or None if status is not available (function was added in CANape 9).

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.DataAcquisition.

7.162.3.3 setup()

```
ttk_tools.vector._canapeapi_offline_stub.DataAcquisition.setup (
              self,
              measurement_vars,
              mdf_file_path = None,
              polling_rate = 20 )
```

Setup data acquisition.

Parameters

measurement_vars	list of variables to measure.
mdf_file_path	path to mdf-file into which to store results. If left empty, a default "daq.mdf" file will be used/created in the CANape working directory. See DataAcquisition.default_mdf_filename.
polling_rate	[ms] polling rate to use for task-type "polling". Use polling only as fallback if there are no other/better measurement tasks available.

Format of measurement_vars:

`measurement_vars` with list entries:

```
[[var, task, device, repr], ...]
```

`measurement_vars` with dictionary entries:

```
[{"var": cal.nii, "task": "10ms", "device": "ECU", "repr": REPR.↩
PHY}, ...]
```

Available entries

- `var` - name/identifier of variable (this can simply be a "CalVar" instance, as VarBase-derived variables have their identifier as default representation)
- `task` - name of DAQ-task, see CANapeDevice.getECUTasks() for available tasks.
- `device` - index of device (order as supplied during DataAcquisition.__init__), defaults to first device (device index 0)
  OR
  device/module name as supplied during device's constructor call.

- `repr` - data representation to use for measured variable, REPR.HEX or REPR.PHYS (defaults to PHYS). See canapeapi_common.REPR

Note that at least `var` and `task` entries have to be present, `device` and `repr` are optional.

Example: DAQ setup and execution

```
meas_vars = [
    # <variable/identifier>, <measurement task>, <device index>
    [cal.emotor_temp,       "10ms", 0],
    [cal.emotor_n_ist,      "10ms", 0],
    [cal.speicherdruck,     "10ms", 0],
    ["GE_Oiltemp",          "CAN",  1],
]

daq.setup(meas_vars)

# start measurement/recorder
daq.start()

# do stuff here...

# stop measurement/recorder
daq.stop()

# get measured data
daq_data = daq.getData()
```

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.DataAcquisition.

7.162.3.4 start()

```
ttk_tools.vector._canapeapi_offline_stub.DataAcquisition.start (
              self )
```

Start the currently configured data acquisition.

Returns

True if data acquisition has been started or is already running.

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.DataAcquisition.

7.162.3.5 stop()

```
ttk_tools.vector._canapeapi_offline_stub.DataAcquisition.stop (
              self )
```

Stop a currently running data acquisition.

Returns

True if data acquisition has been stopped or was not running to begin with.

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.DataAcquisition.

7.162.4 Member Data Documentation

7.162.4.1 _asap3

ttk_tools.vector._canapeapi_offline_stub.DataAcquisition._asap3 [protected]

7.162.4.2 _daq_configured

ttk_tools.vector._canapeapi_offline_stub.DataAcquisition._daq_configured [protected]

7.162.4.3 _daq_running

ttk_tools.vector._canapeapi_offline_stub.DataAcquisition._daq_running [protected]

7.162.4.4 _data_provider

ttk_tools.vector._canapeapi_offline_stub.DataAcquisition._data_provider [protected]

7.162.4.5 _devices

ttk_tools.vector._canapeapi_offline_stub.DataAcquisition._devices [protected]

7.162.4.6 _start_time

ttk_tools.vector._canapeapi_offline_stub.DataAcquisition._start_time [protected]

7.162.4.7 _stop_time

ttk_tools.vector._canapeapi_offline_stub.DataAcquisition._stop_time [protected]

7.162.4.8 _variables

ttk_tools.vector._canapeapi_offline_stub.DataAcquisition._variables [protected]

7.162.4.9 default_mdf_filename

ttk_tools.vector._canapeapi_offline_stub.DataAcquisition.default_mdf_filename

default file name for MDF files ("daq.mdf") [unused in stub].

This will be used if no explicit mdf_file_path will be supplied during setup().

7.162.4.10 default_mdf_folder

`ttk_tools.vector._canapeapi_offline_stub.DataAcquisition.default_mdf_folder`

default output folder for MDF files [unused in stub].

Defaults to the current CANape working folder.

7.162.4.11 extra_delay_before_mdf_read

`ttk_tools.vector._canapeapi_offline_stub.DataAcquisition.extra_delay_before_mdf_read`

[s] extra delay before MDF file is read [unused in stub].

This might be useful for some CANape releases (e.g. at least some builds of CANNpe13), where the MDF fill is still being processed when stop() returns.

7.162.4.12 mdf_file_path

`ttk_tools.vector._canapeapi_offline_stub.DataAcquisition.mdf_file_path`

file path of current mdf file.

This value will be set during setup().

7.162.4.13 polling_rate

`ttk_tools.vector._canapeapi_offline_stub.DataAcquisition.polling_rate`

default/fallback polling-rate in [ms] when using DAQ with polling (which is a rather unusual thing to do).

This value will be updated during calls to setup().

Note: This stub implementation will use the `polling_rate` value as step size for generating dummy measurement data (as any actual measurement task information is unavailable in stub)

7.162.4.14 read_data_attempts_delay

`ttk_tools.vector._canapeapi_offline_stub.DataAcquisition.read_data_attempts_delay`

[s] delay between MDF-read attempts [unused in stub] See DataAcquisition.read_data_attempts_timeout

7.162.4.15 read_data_attempts_timeout

`ttk_tools.vector._canapeapi_offline_stub.DataAcquisition.read_data_attempts_timeout`

[s] timeout value for MDF-read attempts [unused in stub] If a MDF file is not yet completely written, mdfLib will raise errors that will "go away" once the file is finished/consistent.

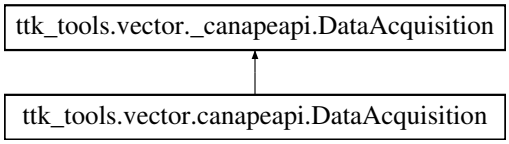
The documentation for this class was generated from the following file:

- ttk_tools/vector/_canapeapi_offline_stub.py (v4.7.1)

7.163 ttk_tools.vector.canapeapi.DataAcquisition Class Reference

Data acquisition for multiple CANape devices.

Inheritance diagram for ttk_tools.vector.canapeapi.DataAcquisition:



Public Member Functions

- __init__ (self, canape_asap3, *devices)

  Setup data acquisition devices.
- setup (self, measurement_vars, mdf_file_path=None, polling_rate=20)

  Setup data acquisition.
- start (self)

  Start the currently configured data acquisition.
- stop (self)

  Stop a currently running data acquisition.
- getData (self, verbosity=0)

  Get data of the last measurement from the measurement data file.
- getStatus (self, verbosity=1)

  Get status of current DAQ/measurement, see canapeapi_common.MEASUREMENT_STATE.

Public Attributes

- extra_delay_before_mdf_read

  [s] extra delay before MDF file is read.
- read_data_attempts_timeout

  [s] timeout value for MDF-read attempts If a MDF file is not yet completely written, mdfLib will raise errors that will "go away" once the file is finished/consistent.
- read_data_attempts_delay

  [s] delay between MDF-read attempts [unused in stub] See read_data_attempts_timeout

7.163.1 Detailed Description

Data acquisition for multiple CANape devices.

Data will be stored as MDF files ("streamed to disk") and later read back after the measurement is completed.

Note

In some CANape versions, CANape will retain a file lock on the current MDF-file after a measurement has been run (and stopped).
So multiple start() / stop() cycles after a single setup() may not work in all releases.

Workaround:
Explicitly perform another DAQ setup before each new measurement (works even with the same mdf file path).

Note

In some versions (probably starting with CANape 13), CANape seems to continue writing to the MDF-file for some time after data acquisition has been successfully stopped.
This may cause the current MDF-data to be interpreted as "unsorted" but the call to `mdfSort` will fail with an undocumented error code (1022). As this appears to be a race condition, the actual error symptoms might vary.

Workaround:
see DataAcquisition.extra_delay_before_mdf_read
see DataAcquisition.read_data_attempts_timeout
see DataAcquisition.read_data_attempts_delay

7.163.2 Constructor & Destructor Documentation

7.163.2.1 __init__()

```
ttk_tools.vector.canapeapi.DataAcquisition.__init__ (
              self,
              canape_asap3,
             * devices )
```

Setup data acquisition devices.

Parameters

canape_asap3	ASAP3-handle to main application server
∗*devices*	CANape devices from which to acquire data [1..n]. See CANapeDevice

Usage:

```
# with two CANapeDevice instances, one for XCP and one for CAN:
daq = DataAcquisition(asap3, canape_xcp, canape_can)
```

Reimplemented from ttk_tools.vector._canapeapi.DataAcquisition.

7.163.3 Member Function Documentation

7.163.3.1 getData()

```
ttk_tools.vector.canapeapi.DataAcquisition.getData (
              self,
              verbosity = 0 )
```

Get data of the last measurement from the measurement data file.

Parameters

verbosity	verbosity level of mdflib status messages. 0: off. See mdflib.readMdf() for more details.

Example: Basic Data Structure:

```
daq_data[<canape var name>] = {
    'data':    <list of data values>,
    'time':    <list of corresponding time stamps>,
}
```

Returns

a data dictionary with entries for each recorded variable, see Basic Data Structure, above.
See mdflib.readMdf() for a more detailed description.

Reimplemented from ttk_tools.vector._canapeapi.DataAcquisition.

7.163.3.2 getStatus()

```
ttk_tools.vector.canapeapi.DataAcquisition.getStatus (
              self,
              verbosity = 1 )
```

Get status of current DAQ/measurement, see canapeapi_common.MEASUREMENT_STATE.

Parameters

verbosity	$>$ 0: status text will be printed to stdout

Returns

an numeric status value or None if status is not available (function was added in CANape 9).

Reimplemented from ttk_tools.vector._canapeapi.DataAcquisition.

7.163.3.3 setup()

```
ttk_tools.vector.canapeapi.DataAcquisition.setup (
              self,
              measurement_vars,
              mdf_file_path = None,
              polling_rate = 20 )
```

Setup data acquisition.

Parameters

measurement_vars	list of variables to measure.
mdf_file_path	path to mdf-file into which to store results. If left empty, a default "daq.mdf" file will be used/created in the CANape working directory. See DataAcquisition.default_mdf_filename.
polling_rate	[ms] polling rate to use for task-type "polling". Use polling only as fallback if there are no other/better measurement tasks available.

Format of measurement_vars:

`measurement_vars` with list entries:
`[[var, task, device, repr], ...]`

`measurement_vars` with dictionary entries:
`[{"var": cal.nii, "task": "10ms", "device": "ECU", "repr": REPR.↩ PHY}, ...]`

Available entries

- `var` - name/identifier of variable (this can simply be a "CalVar" instance, as VarBase-derived variables have their identifier as default representation)
- `task` - name of DAQ-task, see CANapeDevice.getECUTasks() for available tasks.
- `device` - index of device (order as supplied during DataAcquisition.__init__), defaults to first device (device index 0)
  OR
  device/module name as supplied during device's constructor call.
- `repr` - data representation to use for measured variable, REPR.HEX or REPR.PHYS (defaults to PHYS). See canapeapi_common.REPR

Note that at least `var` and `task` entries have to be present, `device` and `repr` are optional.

Example: DAQ setup and execution

```
meas_vars = [
    # <variable/identifier>, <measurement task>, <device index>
    [cal.emotor_temp,       "10ms", 0],
    [cal.emotor_n_ist,      "10ms", 0],
    [cal.speicherdruck,     "10ms", 0],
    ["GE_Oiltemp",          "CAN",  1],
]

daq.setup(meas_vars)

# start measurement/recorder
daq.start()

# do stuff here...

# stop measurement/recorder
daq.stop()

# get measured data
daq_data = daq.getData()
```

Reimplemented from ttk_tools.vector._canapeapi.DataAcquisition.

7.163.3.4 start()

```
ttk_tools.vector.canapeapi.DataAcquisition.start (
              self )
```

Start the currently configured data acquisition.

Returns

True if data acquisition has been started or is already running.

Reimplemented from ttk_tools.vector._canapeapi.DataAcquisition.

7.163.3.5 stop()

```
ttk_tools.vector.canapeapi.DataAcquisition.stop (
              self )
```

Stop a currently running data acquisition.

Returns

True if data acquisition has been stopped or was not running to begin with.

Reimplemented from ttk_tools.vector._canapeapi.DataAcquisition.

7.163.4 Member Data Documentation

7.163.4.1 extra_delay_before_mdf_read

```
ttk_tools.vector.canapeapi.DataAcquisition.extra_delay_before_mdf_read
```

[s] extra delay before MDF file is read.

This might be useful for some CANape releases (e.g. at least some builds of CANNpe13), where the MDF fill is still being processed when stop() returns

7.163.4.2 read_data_attempts_delay

```
ttk_tools.vector.canapeapi.DataAcquisition.read_data_attempts_delay
```

[s] delay between MDF-read attempts [unused in stub] See read_data_attempts_timeout

7.163.4.3 read_data_attempts_timeout

```
ttk_tools.vector.canapeapi.DataAcquisition.read_data_attempts_timeout
```

[s] timeout value for MDF-read attempts If a MDF file is not yet completely written, mdfLib will raise errors that will "go away" once the file is finished/consistent.

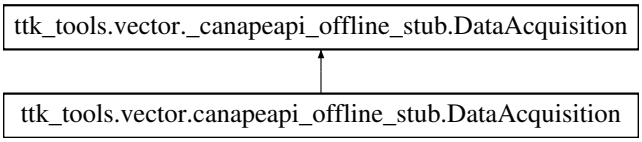
The documentation for this class was generated from the following file:

- ttk_tools/vector/canapeapi.py (v3.1.1)

7.164 ttk_tools.vector.canapeapi_offline_stub.DataAcquisition Class Reference

Data acquisition (stub) for multiple CANape devices.

Inheritance diagram for ttk_tools.vector.canapeapi_offline_stub.DataAcquisition:



Public Member Functions

- __init__ (self, canape_asap3, *devices)

  Setup data acquisition devices.
- setup (self, measurement_vars, mdf_file_path=None, polling_rate=20)

  Setup data acquisition.
- start (self)

  Start the currently configured data acquisition.
- stop (self)

  Stop a currently running data acquisition.
- getData (self, verbosity=0)

  Get data of the last measurement from the measurement data file.
- getStatus (self, verbosity=1)

  Get status of current DAQ/measurement, see canapeapi_common.MEASUREMENT_STATE.

Public Attributes

- mdf_file_path

  file path of current mdf file.
- default_mdf_filename

  default file name for MDF files ("daq.mdf") [unused in stub].
- default_mdf_folder

  default output folder for MDF files [unused in stub].
- polling_rate

  default/fallback polling-rate in [ms] when using DAQ with polling (which is a rather unusual thing to do).
- extra_delay_before_mdf_read

  [s] extra delay before MDF file is read [unused in stub].
- read_data_attempts_timeout

  [s] timeout value for MDF-read attempts [unused in stub] If a MDF file is not yet completely written, mdfLib will raise errors that will "go away" once the file is finished/consistent.
- read_data_attempts_delay

  [s] delay between MDF-read attempts [unused in stub] See DataAcquisition.read_data_attempts_timeout

7.164.1 Detailed Description

Data acquisition (stub) for multiple CANape devices.

Normally, data will be stored as MDF files ("streamed to disk") and later read back after the measurement is completed.

This stub will currently only provide generated dummy data for the measured time range (current value of measured stub variable $*$ (time range / step size) values).
See also DataAcquisition.polling_rate

Note

In some CANape versions, CANape will retain a file lock on the current MDF-file after a measurement has been run (and stopped).
So multiple start() / stop() cycles after a single setup() may not work in all releases.

Workaround:
Explicitly perform another DAQ setup before each new measurement (works even with the same mdf file path).

Note

In some versions (probably starting with CANape 13), CANape seems to continue writing to the MDF-file for some time after data acquisition has been successfully stopped.
This may cause the current MDF-data to be interpreted as "unsorted" but the call to `mdfSort` will fail with an undocumented error code (1022). As this appears to be a race condition, the actual error symptoms might vary.

Workaround:
see DataAcquisition.extra_delay_before_mdf_read
see DataAcquisition.read_data_attempts_timeout
see DataAcquisition.read_data_attempts_delay

7.164.2 Constructor & Destructor Documentation

7.164.2.1 __init__()

```
ttk_tools.vector.canapeapi_offline_stub.DataAcquisition.__init__ (
              self,
              canape_asap3,
             * devices )
```

Setup data acquisition devices.

Parameters

canape_asap3	ASAP3-handle to main application server
$*$*devices*	CANape devices from which to acquire data [1..n]. See CANapeDevice

Usage:

```
# with two CANapeDevice instances, one for XCP and one for CAN:
daq = DataAcquisition(asap3, canape_xcp, canape_can)
```

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.DataAcquisition.

7.164.3 Member Function Documentation

7.164.3.1 getData()

```
ttk_tools.vector.canapeapi_offline_stub.DataAcquisition.getData (
              self,
              verbosity = 0 )
```

Get data of the last measurement from the measurement data file.

Parameters

verbosity	verbosity level of mdflib status messages. 0: off. See mdflib.readMdf() for more details.

Example: Basic Data Structure:

```
daq_data[<canape var name>] = {
    'data':    <list of data values>,
    'time':    <list of corresponding time stamps>,
}
```

Returns

a data dictionary with entries for each recorded variable. See basic example, above. See mdflib.readMdf() for a more detailed description.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.DataAcquisition.

7.164.3.2 getStatus()

```
ttk_tools.vector.canapeapi_offline_stub.DataAcquisition.getStatus (
              self,
              verbosity = 1 )
```

Get status of current DAQ/measurement, see canapeapi_common.MEASUREMENT_STATE.

Parameters

verbosity	$>$ 0: status text will be printed to stdout

Returns

an numeric status value or None if status is not available (function was added in CANape 9).

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.DataAcquisition.

7.164.3.3 setup()

```
ttk_tools.vector.canapeapi_offline_stub.DataAcquisition.setup (
              self,
              measurement_vars,
              mdf_file_path = None,
              polling_rate = 20 )
```

Setup data acquisition.

Parameters

measurement_vars	list of variables to measure.
mdf_file_path	path to mdf-file into which to store results. If left empty, a default "daq.mdf" file will be used/created in the CANape working directory. See DataAcquisition.default_mdf_filename.
polling_rate	[ms] polling rate to use for task-type "polling". Use polling only as fallback if there are no other/better measurement tasks available.

Format of measurement_vars:

`measurement_vars` with list entries:

```
[[var, task, device, repr], ...]
```

`measurement_vars` with dictionary entries:

```
[{"var": cal.nii, "task": "10ms", "device": "ECU", "repr": REPR.↩
PHY}, ...]
```

Available entries

- `var` - name/identifier of variable (this can simply be a "CalVar" instance, as VarBase-derived variables have their identifier as default representation)
- `task` - name of DAQ-task, see CANapeDevice.getECUTasks() for available tasks.
- `device` - index of device (order as supplied during DataAcquisition.__init__), defaults to first device (device index 0)
  OR
  device/module name as supplied during device's constructor call.

- `repr` - data representation to use for measured variable, REPR.HEX or REPR.PHYS (defaults to PHYS). See canapeapi_common.REPR

Note that at least `var` and `task` entries have to be present, `device` and `repr` are optional.

Example: DAQ setup and execution

```
meas_vars = [
    # <variable/identifier>, <measurement task>, <device index>
    [cal.emotor_temp,       "10ms", 0],
    [cal.emotor_n_ist,      "10ms", 0],
    [cal.speicherdruck,     "10ms", 0],
    ["GE_Oiltemp",          "CAN",  1],
]

daq.setup(meas_vars)

# start measurement/recorder
daq.start()

# do stuff here...

# stop measurement/recorder
daq.stop()

# get measured data
daq_data = daq.getData()
```

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.DataAcquisition.

7.164.3.4 start()

```
ttk_tools.vector.canapeapi_offline_stub.DataAcquisition.start (
              self )
```

Start the currently configured data acquisition.

Returns

True if data acquisition has been started or is already running.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.DataAcquisition.

7.164.3.5 stop()

```
ttk_tools.vector.canapeapi_offline_stub.DataAcquisition.stop (
              self )
```

Stop a currently running data acquisition.

Returns

True if data acquisition has been stopped or was not running to begin with.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.DataAcquisition.

7.164.4 Member Data Documentation

7.164.4.1 default_mdf_filename

ttk_tools.vector.canapeapi_offline_stub.DataAcquisition.default_mdf_filename

default file name for MDF files ("daq.mdf") [unused in stub].

This will be used if no explicit mdf_file_path will be supplied during setup().

7.164.4.2 default_mdf_folder

ttk_tools.vector.canapeapi_offline_stub.DataAcquisition.default_mdf_folder

default output folder for MDF files [unused in stub].

Defaults to the current CANape working folder.

7.164.4.3 extra_delay_before_mdf_read

ttk_tools.vector.canapeapi_offline_stub.DataAcquisition.extra_delay_before_mdf_read

[s] extra delay before MDF file is read [unused in stub].

This might be useful for some CANape releases (e.g. at least some builds of CANNpe13), where the MDF fill is still being processed when stop() returns.

7.164.4.4 mdf_file_path

ttk_tools.vector.canapeapi_offline_stub.DataAcquisition.mdf_file_path

file path of current mdf file.

This value will be set during setup().

7.164.4.5 polling_rate

ttk_tools.vector.canapeapi_offline_stub.DataAcquisition.polling_rate

default/fallback polling-rate in [ms] when using DAQ with polling (which is a rather unusual thing to do).

This value will be updated during calls to setup().

Note: This stub implementation will use the `polling_rate` value as step size for generating dummy measurement data (as any actual measurement task information is unavailable in stub)

7.164.4.6 read_data_attempts_delay

`ttk_tools.vector.canapeapi_offline_stub.DataAcquisition.read_data_attempts_delay`

[s] delay between MDF-read attempts [unused in stub] See DataAcquisition.read_data_attempts_timeout

7.164.4.7 read_data_attempts_timeout

`ttk_tools.vector.canapeapi_offline_stub.DataAcquisition.read_data_attempts_timeout`

[s] timeout value for MDF-read attempts [unused in stub] If a MDF file is not yet completely written, mdfLib will raise errors that will "go away" once the file is finished/consistent.

The documentation for this class was generated from the following file:

- ttk_tools/vector/canapeapi_offline_stub.py (v4.1.0)

7.165 ttk_base.configuration.DataParser Class Reference

A reduced "light" implementation of the original DataParser (which was in turn an alternate, less finnicky implementation of Python 2.2-era ConfigParser).

Public Member Functions

- __init__ (self, default_section_name='')
- add_section (self, section)
  Create a new section in the configuration unless it already exists.
- has_section (self, section)
  Indicate whether the named section is present in the configuration.
- has_key (self, section, key)
  Check for the existence of a given key in a given section.
- sections (self)
  Returns a list of section names.
- options (self, section=None)
  Get available option names (i.e.
- get (self, section, key, default=None)
  Get a value for a given section and key.
- getInt (self, section, key, fallback=0)
- getFloat (self, section, key, fallback=0.0)
- getBoolean (self, section, key, fallback=False)
- set (self, section, key, value)
  Set an option (a key/value pair in a defined section).
- setDefault (self, section, key, default)
  Set a default option (a key/value pair in a defined section).
- removeEntry (self, section, key)
  Remove an option (key/value in a section).
- removeSection (self, section)
  Remove the specified data section.
- getDict (self, section=None, auto_create=False)
  Get a reference to the actual dictionary in the specified section.
- write (self, fp)
  Write an .ini-format representation of the configuration state.
- readFile (self, filenames, encoding=DEFAULT_INI_ENCODING)
  Read and parse a single file name/path or a list of file names/paths.
- read (self, fp, fpname=None)
  Read and parse a structured data/config file.

Protected Member Functions

- _normalizeKey (self, key)

  Return a normalized representation of the supplied dictionary key.
- _escapeKey (self, key)

  Escape separator characters (":", "=") from key strings for writing.

7.165.1 Detailed Description

A reduced "light" implementation of the original DataParser (which was in turn an alternate, less finnicky implementation of Python 2.2-era ConfigParser).

This reduced version contains only the methods relevant for use with our Configuration class and provides sensible fallbacks when encountering missing or duplicate sections

Note

Differences to RawConfigParser: The current RawConfigParser implementation does much of what our Data↩ Parser was created for. Some differences remain:

- Allows to use an anonymous/empty section
  - This section will be active right at start of file
  - Can be entered again as section [] (section name defaults to an empty string)
- There is no "DEFAULT" section to contain default values for entries in all other sections.
  - Instead, get() has a fallback value to use for unset options
  - DataParser's "default" section is the anonymous section, but that behaves like any normal section
- More lenient error handling;
  - missing sections will read as empty or be created as needed
  - missing entries (or entries in missing sections) will read as unset (and return fallback values)
- comments may only start at the beginning of a line to avoid issues with multiline values
- both section names and entries/options are case sensitive
- Written ini files will align values within a section to start at the same column (for better readability)
- Empty lines in multiline values will be retained (similar to Py3's RawConfigParser)

7.165.2 Constructor & Destructor Documentation

7.165.2.1 __init__()

```
ttk_base.configuration.DataParser.__init__ (
            self,
            default_section_name = '' )
```

Parameters

default_section_name	name of default section (i.e. anything without a defined [SECTION] name).

7.165.3 Member Function Documentation

7.165.3.1 _escapeKey()

```
ttk_base.configuration.DataParser._escapeKey (
              self,
              key )  [protected]
```

Escape separator characters (":", "=") from key strings for writing.

Parameters

key	key name to escape

Returns

the escaped key name

7.165.3.2 _normalizeKey()

```
ttk_base.configuration.DataParser._normalizeKey (
              self,
              key )  [protected]
```

Return a normalized representation of the supplied dictionary key.

Override this method if you want to force all keys to lowercase or similar.

7.165.3.3 add_section()

```
ttk_base.configuration.DataParser.add_section (
              self,
              section )
```

Create a new section in the configuration unless it already exists.

Parameters

section	name of new section

7.165.3.4 get()

```
ttk_base.configuration.DataParser.get (
              self,
              section,
              key,
              default = None )
```

Get a value for a given section and key.

Parameters

section	name of section
key	name of key ("option") to get
default	default value to return if key is not yet defined

Returns

the value for key, or the default value as-is if key is undefined.

7.165.3.5 getBoolean()

```
ttk_base.configuration.DataParser.getBoolean (
              self,
              section,
              key,
              fallback = False )
```

7.165.3.6 getDict()

```
ttk_base.configuration.DataParser.getDict (
              self,
              section = None,
              auto_create = False )
```

Get a reference to the actual dictionary in the specified section.

Note

If `auto_create` is selected and the specified section does not exist, it will be created on the fly. Use only if section data should be modified (i.e. during set())

Parameters

section	name of section
auto_create	True: section will be created if missing False: no section will be created, only an anonymous empty dict will be returned

Returns

a section's data dict.

7.165.3.7 getFloat()

```
ttk_base.configuration.DataParser.getFloat (
              self,
              section,
              key,
              fallback = 0.0 )
```

7.165.3.8 getInt()

```
ttk_base.configuration.DataParser.getInt (
              self,
              section,
              key,
              fallback = 0 )
```

7.165.3.9 has_key()

```
ttk_base.configuration.DataParser.has_key (
              self,
              section,
              key )
```

Check for the existence of a given key in a given section.

Parameters

section	name of section to check in
key	name of key to check

Returns

True if key exists in section, otherwise False

7.165.3.10 has_section()

```
ttk_base.configuration.DataParser.has_section (
              self,
              section )
```

Indicate whether the named section is present in the configuration.

Parameters

section	name of section

Returns

True if named section exists, otherwise False

7.165.3.11 options()

```
ttk_base.configuration.DataParser.options (
              self,
              section = None )
```

Get available option names (i.e.

entries) for the given section.

Parameters

section	name of section

Returns

a list of option names

7.165.3.12 read()

```
ttk_base.configuration.DataParser.read (
              self,
              fp,
              fpname = None )
```

Read and parse a structured data/config file.

Parse a data file structured like windows '.ini' files (though keys may contain whitespace and value interpolation will not be performed).

Sections in the file contain a section title (in square brackets) line followed by lines containing "key: value" pairs for the current section. Section titles and key names have to start at the beginning of a line.

Value continuations are represented by follow-up lines with leading whitespace. Note that while a newline char between value and continuation will be retained, whitespace chars at the start and end of values will be trimmed.

Blank lines, lines beginning with '#' or ';' and just about everything else will be ignored, though a ParsingError exception will be raised if malformed lines have been encountered.

Parameters

fp	file-like object opened for reading
fpname	name of file (just for parsing error messages) will default to fp.name or '<???>' if neither is available

Raises a ParsingError if malformed lines have been encountered.

7.165.3.13 readFile()

```
ttk_base.configuration.DataParser.readFile (
              self,
              filenames,
              encoding = DEFAULT_INI_ENCODING )
```

Read and parse a single file name/path or a list of file names/paths.

Files that cannot be opened are silently ignored; this is designed so that you can specify a list of potential configuration file locations (e.g. current directory, user's home directory, system wide directory), and all existing configuration files in the list will be read. A single filename may also be given.

Parameters

filenames	name/path of an ini file (or multiple as list)
encoding	file encoding to use for reading the ini file(s)

Returns

a list of successfully read filenames/-paths.

7.165.3.14 removeEntry()

```
ttk_base.configuration.DataParser.removeEntry (
              self,
              section,
              key )
```

Remove an option (key/value in a section).

Parameters

section	name of section
key	name of key to remove

Returns

True if the specified key existed and has been removed, otherwise False.

7.165.3.15 removeSection()

```
ttk_base.configuration.DataParser.removeSection (
              self,
              section )
```

Remove the specified data section.

Parameters

section	name of section to remove

Returns

True if the specified section existed and has been removed, otherwise False.

7.165.3.16 sections()

```
ttk_base.configuration.DataParser.sections (
              self )
```

Returns a list of section names.

7.165.3.17 set()

```
ttk_base.configuration.DataParser.set (
              self,
              section,
              key,
              value )
```

Set an option (a key/value pair in a defined section).

- If key already exists, it will be overwritten.
- If the supplied section does not yet exist, it will be created on the fly.

Parameters

section	name of section
key	name of key
value	value to set

7.165.3.18 setDefault()

```
ttk_base.configuration.DataParser.setDefault (
              self,
              section,
              key,
              default )
```

Set a default option (a key/value pair in a defined section).

If key already exists, it will remain unchanged.

Parameters

section	name of section
key	name of key
default	default value to set

7.165.3.19 write()

```
ttk_base.configuration.DataParser.write (
              self,
              fp )
```

Write an .ini-format representation of the configuration state.

Parameters

fp	an already opened file-like object with write access.

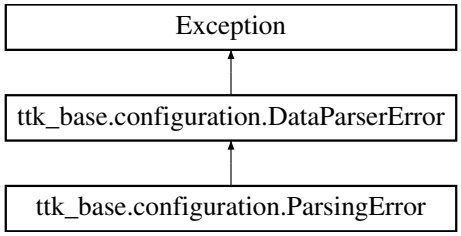
The documentation for this class was generated from the following file:

- ttk_base/configuration.py (v3.8.1)

7.166 ttk_base.configuration.DataParserError Class Reference

Errors during parsing of a configuration file.

Inheritance diagram for ttk_base.configuration.DataParserError:



7.166.1 Detailed Description

Errors during parsing of a configuration file.

The documentation for this class was generated from the following file:

- ttk_base/configuration.py (v3.8.1)

7.167 ttk_tools.vector.mdf4_reader.DataRecords Class Reference

Iterator for fixed and variable length signal data records, returns sliced records from raw byte payload data.

Public Member Functions

- __init__ (self, data_group)
- __iter__ (self)
- __next__ (self)

  Get data for next iteration.

Public Attributes

- data_group
- data
- rec_id_bytes

  header bytes for record id
- offset
- total_size
- record_index
- rec_id_fmt

7.167.1 Detailed Description

Iterator for fixed and variable length signal data records, returns sliced records from raw byte payload data.

7.167.2 Constructor & Destructor Documentation

7.167.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.DataRecords.__init__ (
            self,
            data_group )
```

Parameters

data_group	reference to parent data group

7.167.3 Member Function Documentation

7.167.3.1 __iter__()

```
ttk_tools.vector.mdf4_reader.DataRecords.__iter__ (
            self )
```

7.167.3.2 __next__()

```
ttk_tools.vector.mdf4_reader.DataRecords.__next__ (
            self )
```

Get data for next iteration.

Returns

a tuple with four elements (<record_index>, <channel_group>, <data>, <inval_bits>). record_index - incrementing count/index of data records (to be used for calculating virtual channel data) channel_group - parent channel group of the current record data - raw payload data inval_bits - invalidation bits for payload data

Header or inval elements may be None if they are not available.

7.167.4 Member Data Documentation

7.167.4.1 data

```
ttk_tools.vector.mdf4_reader.DataRecords.data
```

7.167.4.2 data_group

ttk_tools.vector.mdf4_reader.DataRecords.data_group

7.167.4.3 offset

ttk_tools.vector.mdf4_reader.DataRecords.offset

7.167.4.4 rec_id_bytes

ttk_tools.vector.mdf4_reader.DataRecords.rec_id_bytes

header bytes for record id

7.167.4.5 rec_id_fmt

ttk_tools.vector.mdf4_reader.DataRecords.rec_id_fmt

7.167.4.6 record_index

ttk_tools.vector.mdf4_reader.DataRecords.record_index

7.167.4.7 total_size

ttk_tools.vector.mdf4_reader.DataRecords.total_size

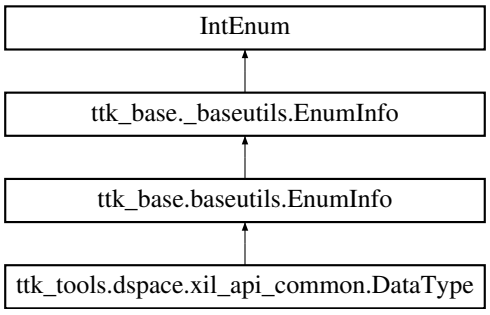
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.168 ttk_tools.dspace.xil_api_common.DataType Class Reference

Data Types.

Inheritance diagram for ttk_tools.dspace.xil_api_common.DataType:



Static Public Attributes

- int eNO_TYPE = 0x00

  No explicit data type.
- int eBOOLEAN = 0x01

  A_BOOLEAN (simple data type)
- int eINT = 0x02

  A_INT64 (simple data type)
- int eUINT = 0x03

  A_UINT64 (simple data type)
- int eFLOAT = 0x04

  A_FLOAT64 (simple data type)
- int eSTRING = 0x05

  A_UNICODE2STRING (simple data type)
- int eINT_VECTOR = 0x06

  integer vector (complex data type)
- int eFLOAT_VECTOR = 0x07

  float vector (complex data type)
- int eSTRING_VECTOR = 0x08

  string vector (complex data type)
- int eBOOLEAN_VECTOR = 0x09

  boolean vector (complex data type)
- int eINT_MATRIX = 0x0A

  integer matrix (complex data type)
- int eFLOAT_MATRIX = 0x0B

  float matrix (complex data type)
- int eSTRING_MATRIX = 0x0C

  string matrix (complex data type)
- int eBOOLEAN_MATRIX = 0x0D

  boolean matrix (complex data type)
- int eMAP = 0x0E

  map (complex data type)
- int eCURVE = 0x0F

  curve (complex data type)
- int eXYVALUE = 0x10

  x-y value (complex data type)
- int eUINT_VECTOR = 0x11

  unsigned integer vector (complex data type)
- int eUINT_MATRIX = 0x12

  unsigned integer matrix (complex data type)
- int eSIGNALVALUE = 0x13

  signal value (complex data type)
- int eSIGNALGROUPVALUE = 0x14

  signal group value (complex data type)

7.168.1 Detailed Description

Data Types.

See ASAM.XIL.Interfaces.Testbench.Common.ValueContainer.Enum.DataType

7.168.2 Member Data Documentation

7.168.2.1 eBOOLEAN

```
int ttk_tools.dspace.xil_api_common.DataType.eBOOLEAN = 0x01  [static]
```

A_BOOLEAN (simple data type)

7.168.2.2 eBOOLEAN_MATRIX

```
int ttk_tools.dspace.xil_api_common.DataType.eBOOLEAN_MATRIX = 0x0D  [static]
```

boolean matrix (complex data type)

7.168.2.3 eBOOLEAN_VECTOR

```
int ttk_tools.dspace.xil_api_common.DataType.eBOOLEAN_VECTOR = 0x09  [static]
```

boolean vector (complex data type)

7.168.2.4 eCURVE

```
int ttk_tools.dspace.xil_api_common.DataType.eCURVE = 0x0F  [static]
```

curve (complex data type)

7.168.2.5 eFLOAT

```
int ttk_tools.dspace.xil_api_common.DataType.eFLOAT = 0x04  [static]
```

A_FLOAT64 (simple data type)

7.168.2.6 eFLOAT_MATRIX

```
int ttk_tools.dspace.xil_api_common.DataType.eFLOAT_MATRIX = 0x0B  [static]
```

float matrix (complex data type)

7.168.2.7 eFLOAT_VECTOR

```
int ttk_tools.dspace.xil_api_common.DataType.eFLOAT_VECTOR = 0x07  [static]
```

float vector (complex data type)

7.168.2.8 eINT

```
int ttk_tools.dspace.xil_api_common.DataType.eINT = 0x02  [static]
```

A_INT64 (simple data type)

7.168.2.9 eINT_MATRIX

```
int ttk_tools.dspace.xil_api_common.DataType.eINT_MATRIX = 0x0A  [static]
```

integer matrix (complex data type)

7.168.2.10 eINT_VECTOR

```
int ttk_tools.dspace.xil_api_common.DataType.eINT_VECTOR = 0x06  [static]
```

integer vector (complex data type)

7.168.2.11 eMAP

```
int ttk_tools.dspace.xil_api_common.DataType.eMAP = 0x0E  [static]
```

map (complex data type)

7.168.2.12 eNO_TYPE

```
int ttk_tools.dspace.xil_api_common.DataType.eNO_TYPE = 0x00  [static]
```

No explicit data type.

7.168.2.13 eSIGNALGROUPVALUE

```
int ttk_tools.dspace.xil_api_common.DataType.eSIGNALGROUPVALUE = 0x14  [static]
```

signal group value (complex data type)

7.168.2.14 eSIGNALVALUE

```
int ttk_tools.dspace.xil_api_common.DataType.eSIGNALVALUE = 0x13  [static]
```

signal value (complex data type)

7.168.2.15 eSTRING

```
int ttk_tools.dspace.xil_api_common.DataType.eSTRING = 0x05  [static]
```

A_UNICODE2STRING (simple data type)

7.168.2.16 eSTRING_MATRIX

```
int ttk_tools.dspace.xil_api_common.DataType.eSTRING_MATRIX = 0x0C  [static]
```

string matrix (complex data type)

7.168.2.17 eSTRING_VECTOR

```
int ttk_tools.dspace.xil_api_common.DataType.eSTRING_VECTOR = 0x08  [static]
```

string vector (complex data type)

7.168.2.18 eUINT

```
int ttk_tools.dspace.xil_api_common.DataType.eUINT = 0x03  [static]
```

A_UINT64 (simple data type)

7.168.2.19 eUINT_MATRIX

```
int ttk_tools.dspace.xil_api_common.DataType.eUINT_MATRIX = 0x12  [static]
```

unsigned integer matrix (complex data type)

7.168.2.20 eUINT_VECTOR

```
int ttk_tools.dspace.xil_api_common.DataType.eUINT_VECTOR = 0x11  [static]
```

unsigned integer vector (complex data type)

7.168.2.21 eXYVALUE

```
int ttk_tools.dspace.xil_api_common.DataType.eXYVALUE = 0x10  [static]
```

x-y value (complex data type)

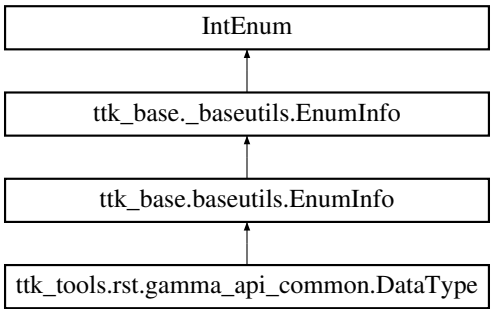
The documentation for this class was generated from the following file:

- ttk_tools/dspace/xil_api_common.py (v1.5.0)

7.169 ttk_tools.rst.gamma_api_common.DataType Class Reference

Gamma Data Types (gc_DataType).

Inheritance diagram for ttk_tools.rst.gamma_api_common.DataType:



Public Member Functions

- isValid (cls, data_type)
- isNumeric (cls, data_type)
- isBlobby (cls, data_type)

Public Attributes

- STRING
- BLOB

Static Public Attributes

- int SINT_8 = 0x00
- int UINT_8 = 0x01
- int SINT_16 = 0x02
- int UINT_16 = 0x03
- int SINT_32 = 0x04
- int UINT_32 = 0x05
- int SINT_64 = 0x06
- int UINT_64 = 0x07
- int IEEE_32 = 0x08
- int IEEE_64 = 0x09
- int STRING = 0x0A
- int BLOB = 0x0B
- int INVALID = 0xFF

7.169.1 Detailed Description

Gamma Data Types (gc_DataType).

7.169.2 Member Function Documentation

7.169.2.1 isBlobby()

```
ttk_tools.rst.gamma_api_common.DataType.isBlobby (
              cls,
              data_type )
```

7.169.2.2 isNumeric()

```
ttk_tools.rst.gamma_api_common.DataType.isNumeric (
              cls,
              data_type )
```

7.169.2.3 isValid()

```
ttk_tools.rst.gamma_api_common.DataType.isValid (
              cls,
              data_type )
```

7.169.3 Member Data Documentation

7.169.3.1 BLOB [1/2]

```
int ttk_tools.rst.gamma_api_common.DataType.BLOB = 0x0B  [static]
```

7.169.3.2 BLOB [2/2]

```
ttk_tools.rst.gamma_api_common.DataType.BLOB
```

7.169.3.3 IEEE_32

```
int ttk_tools.rst.gamma_api_common.DataType.IEEE_32 = 0x08  [static]
```

7.169.3.4 IEEE_64

```
int ttk_tools.rst.gamma_api_common.DataType.IEEE_64 = 0x09  [static]
```

7.169.3.5 INVALID

```
int ttk_tools.rst.gamma_api_common.DataType.INVALID = 0xFF  [static]
```

7.169.3.6 SINT_16

```
int ttk_tools.rst.gamma_api_common.DataType.SINT_16 = 0x02  [static]
```

7.169.3.7 SINT_32

```
int ttk_tools.rst.gamma_api_common.DataType.SINT_32 = 0x04  [static]
```

7.169.3.8 SINT_64

```
int ttk_tools.rst.gamma_api_common.DataType.SINT_64 = 0x06  [static]
```

7.169.3.9 SINT_8

```
int ttk_tools.rst.gamma_api_common.DataType.SINT_8 = 0x00  [static]
```

7.169.3.10 STRING [1/2]

```
int ttk_tools.rst.gamma_api_common.DataType.STRING = 0x0A  [static]
```

7.169.3.11 STRING [2/2]

```
ttk_tools.rst.gamma_api_common.DataType.STRING
```

7.169.3.12 UINT_16

```
int ttk_tools.rst.gamma_api_common.DataType.UINT_16 = 0x03  [static]
```

7.169.3.13 UINT_32

```
int ttk_tools.rst.gamma_api_common.DataType.UINT_32 = 0x05  [static]
```

7.169.3.14 UINT_64

```
int ttk_tools.rst.gamma_api_common.DataType.UINT_64 = 0x07  [static]
```

7.169.3.15 UINT_8

```
int ttk_tools.rst.gamma_api_common.DataType.UINT_8 = 0x01  [static]
```

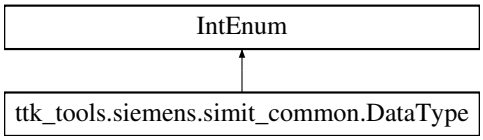
The documentation for this class was generated from the following file:

- ttk_tools/rst/gamma_api_common.py (v1.5)

7.170 ttk_tools.siemens.simit_common.DataType Class Reference

Data types from "Tabelle 2-3 Typkennung der Signale".

Inheritance diagram for ttk_tools.siemens.simit_common.DataType:



Public Member Functions

- is_bool (self)
- byte_size (self)

  data type size in bytes
- struct_fmt (self)

  struct format for binary packing/unpacking.

Static Public Attributes

- int bool_b0 = 0
- int bool_b1 = 1
- int bool_b2 = 2
- int bool_b3 = 3
- int bool_b4 = 4
- int bool_b5 = 5
- int bool_b6 = 6
- int bool_b7 = 7
- int byte = 8
- int word = 9
- int int = 10
- int dword = 11
- int dint = 12
- int real = 13

7.170.1 Detailed Description

Data types from "Tabelle 2-3 Typkennung der Signale".

7.170.2 Member Function Documentation

7.170.2.1 byte_size()

```
ttk_tools.siemens.simit_common.DataType.byte_size (
            self )
```

data type size in bytes

7.170.2.2 is_bool()

```
ttk_tools.siemens.simit_common.DataType.is_bool (
              self )
```

7.170.2.3 struct_fmt()

```
ttk_tools.siemens.simit_common.DataType.struct_fmt (
              self )
```

struct format for binary packing/unpacking.

Bool/bit types default to the containing byte.

7.170.3 Member Data Documentation

7.170.3.1 bool_b0

```
int ttk_tools.siemens.simit_common.DataType.bool_b0 = 0  [static]
```

7.170.3.2 bool_b1

```
int ttk_tools.siemens.simit_common.DataType.bool_b1 = 1  [static]
```

7.170.3.3 bool_b2

```
int ttk_tools.siemens.simit_common.DataType.bool_b2 = 2  [static]
```

7.170.3.4 bool_b3

```
int ttk_tools.siemens.simit_common.DataType.bool_b3 = 3  [static]
```

7.170.3.5 bool_b4

```
int ttk_tools.siemens.simit_common.DataType.bool_b4 = 4  [static]
```

7.170.3.6 bool_b5

```
int ttk_tools.siemens.simit_common.DataType.bool_b5 = 5  [static]
```

7.170.3.7 bool_b6

```
int ttk_tools.siemens.simit_common.DataType.bool_b6 = 6  [static]
```

7.170.3.8 bool_b7

```
int ttk_tools.siemens.simit_common.DataType.bool_b7 = 7  [static]
```

7.170.3.9 byte

```
int ttk_tools.siemens.simit_common.DataType.byte = 8  [static]
```

7.170.3.10 dint

```
int ttk_tools.siemens.simit_common.DataType.dint = 12  [static]
```

7.170.3.11 dword

```
int ttk_tools.siemens.simit_common.DataType.dword = 11  [static]
```

7.170.3.12 int

```
int ttk_tools.siemens.simit_common.DataType.int = 10  [static]
```

7.170.3.13 real

```
int ttk_tools.siemens.simit_common.DataType.real = 13  [static]
```

7.170.3.14 word

```
int ttk_tools.siemens.simit_common.DataType.word = 9  [static]
```

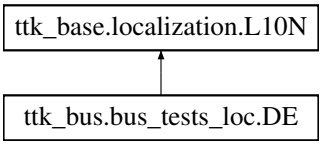
The documentation for this class was generated from the following file:

- ttk_tools/siemens/simit_common.py (v1.0.0)

7.171 ttk_bus.bus_tests_loc.DE Class Reference

German localization.

Inheritance diagram for ttk_bus.bus_tests_loc.DE:



Static Protected Attributes

- dict _localization

7.171.1 Detailed Description

German localization.

7.171.2 Member Data Documentation

7.171.2.1 _localization

```
dict ttk_bus.bus_tests_loc.DE._localization  [static], [protected]
```

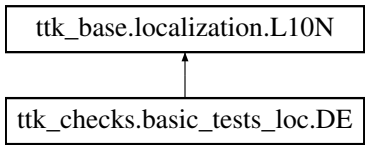
The documentation for this class was generated from the following file:

- ttk_bus/bus_tests_loc.py (v1.6)

7.172 ttk_checks.basic_tests_loc.DE Class Reference

German localization.

Inheritance diagram for ttk_checks.basic_tests_loc.DE:



Static Protected Attributes

- dict _localization

7.172.1 Detailed Description

German localization.

7.172.2 Member Data Documentation

7.172.2.1 _localization

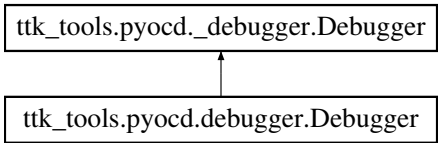
```
dict ttk_checks.basic_tests_loc.DE._localization  [static], [protected]
```

The documentation for this class was generated from the following file:

- ttk_checks/basic_tests_loc.py (v1.20.1)

7.173 ttk_tools.pyocd._debugger.Debugger Class Reference

Inheritance diagram for ttk_tools.pyocd._debugger.Debugger:



Public Member Functions

- __init__ (self, elf_file_path)
- __del__ (self)
- connect (self)

  Connect to target (open session)
- disconnect (self)

  Disconnect from target (close session).
- getAddress (self, symbol)

  Get memory address of an object on the ELF symbol table.
- read_uint8 (self, addr)
- read_uint16 (self, addr)
- read_uint32 (self, addr)
- read_uint64 (self, addr)
- read_sint8 (self, addr)
- read_sint16 (self, addr)
- read_sint32 (self, addr)
- read_sint64 (self, addr)
- read_float32 (self, addr)
- read_float64 (self, addr)
- write_uint8 (self, addr, value)
- write_uint16 (self, addr, value)
- write_uint32 (self, addr, value)
- write_uint64 (self, addr, value)
- write_sint8 (self, addr, value)
- write_sint16 (self, addr, value)
- write_sint32 (self, addr, value)
- write_sint64 (self, addr, value)
- write_float32 (self, addr, value)
- write_float64 (self, addr, value)

Public Attributes

- session

  main session instance
- target

  Target instance of session.
- symbol_provider

Protected Attributes

- _elf_file_path

7.173.1 Constructor & Destructor Documentation

7.173.1.1 __init__()

```
ttk_tools.pyocd._debugger.Debugger.__init__ (
              self,
              elf_file_path )
```

Note

Currently connects to the first detected probe

Parameters

elf_file_path	path to ELF file with debug symbols

Reimplemented in ttk_tools.pyocd.debugger.Debugger.

7.173.1.2 __del__()

```
ttk_tools.pyocd._debugger.Debugger.__del__ (
              self )
```

7.173.2 Member Function Documentation

7.173.2.1 connect()

```
ttk_tools.pyocd._debugger.Debugger.connect (
              self )
```

Connect to target (open session)

Reimplemented in ttk_tools.pyocd.debugger.Debugger.

7.173.2.2 disconnect()

```
ttk_tools.pyocd._debugger.Debugger.disconnect (
              self )
```

Disconnect from target (close session).

Reimplemented in ttk_tools.pyocd.debugger.Debugger.

7.173.2.3 getAddress()

```
ttk_tools.pyocd._debugger.Debugger.getAddress (
              self,
              symbol )
```

Get memory address of an object on the ELF symbol table.

Parameters

symbol	symbol name string

Returns

a memory address (or raises a SymbolNotFound exception)

Reimplemented in ttk_tools.pyocd.debugger.Debugger.

7.173.2.4 read_float32()

```
ttk_tools.pyocd._debugger.Debugger.read_float32 (
              self,
              addr )
```

Reimplemented in ttk_tools.pyocd.debugger.Debugger.

7.173.2.5 read_float64()

```
ttk_tools.pyocd._debugger.Debugger.read_float64 (
              self,
              addr )
```

Reimplemented in ttk_tools.pyocd.debugger.Debugger.

7.173.2.6 read_sint16()

```
ttk_tools.pyocd._debugger.Debugger.read_sint16 (
              self,
              addr )
```

Reimplemented in ttk_tools.pyocd.debugger.Debugger.

7.173.2.7 read_sint32()

```
ttk_tools.pyocd._debugger.Debugger.read_sint32 (
              self,
              addr )
```

Reimplemented in ttk_tools.pyocd.debugger.Debugger.

7.173.2.8 read_sint64()

```
ttk_tools.pyocd._debugger.Debugger.read_sint64 (
              self,
              addr )
```

Reimplemented in ttk_tools.pyocd.debugger.Debugger.

7.173.2.9 read_sint8()

```
ttk_tools.pyocd._debugger.Debugger.read_sint8 (
              self,
              addr )
```

Reimplemented in ttk_tools.pyocd.debugger.Debugger.

7.173.2.10 read_uint16()

```
ttk_tools.pyocd._debugger.Debugger.read_uint16 (
              self,
              addr )
```

Reimplemented in ttk_tools.pyocd.debugger.Debugger.

7.173.2.11 read_uint32()

```
ttk_tools.pyocd._debugger.Debugger.read_uint32 (
              self,
              addr )
```

Reimplemented in ttk_tools.pyocd.debugger.Debugger.

7.173.2.12 read_uint64()

```
ttk_tools.pyocd._debugger.Debugger.read_uint64 (
              self,
              addr )
```

Reimplemented in ttk_tools.pyocd.debugger.Debugger.

7.173.2.13 read_uint8()

```
ttk_tools.pyocd._debugger.Debugger.read_uint8 (
              self,
              addr )
```

Reimplemented in ttk_tools.pyocd.debugger.Debugger.

7.173.2.14 write_float32()

```
ttk_tools.pyocd._debugger.Debugger.write_float32 (
              self,
              addr,
              value )
```

Reimplemented in ttk_tools.pyocd.debugger.Debugger.

7.173.2.15 write_float64()

```
ttk_tools.pyocd._debugger.Debugger.write_float64 (
              self,
              addr,
              value )
```

Reimplemented in ttk_tools.pyocd.debugger.Debugger.

7.173.2.16 write_sint16()

```
ttk_tools.pyocd._debugger.Debugger.write_sint16 (
              self,
              addr,
              value )
```

Reimplemented in ttk_tools.pyocd.debugger.Debugger.

7.173.2.17 write_sint32()

```
ttk_tools.pyocd._debugger.Debugger.write_sint32 (
              self,
              addr,
              value )
```

Reimplemented in ttk_tools.pyocd.debugger.Debugger.

7.173.2.18 write_sint64()

```
ttk_tools.pyocd._debugger.Debugger.write_sint64 (
              self,
              addr,
              value )
```

Reimplemented in ttk_tools.pyocd.debugger.Debugger.

7.173.2.19 write_sint8()

```
ttk_tools.pyocd._debugger.Debugger.write_sint8 (
              self,
              addr,
              value )
```

Reimplemented in ttk_tools.pyocd.debugger.Debugger.

7.173.2.20 write_uint16()

```
ttk_tools.pyocd._debugger.Debugger.write_uint16 (
              self,
              addr,
              value )
```

Reimplemented in ttk_tools.pyocd.debugger.Debugger.

7.173.2.21 write_uint32()

```
ttk_tools.pyocd._debugger.Debugger.write_uint32 (
              self,
              addr,
              value )
```

Reimplemented in ttk_tools.pyocd.debugger.Debugger.

7.173.2.22 write_uint64()

```
ttk_tools.pyocd._debugger.Debugger.write_uint64 (
              self,
              addr,
              value )
```

Reimplemented in ttk_tools.pyocd.debugger.Debugger.

7.173.2.23 write_uint8()

```
ttk_tools.pyocd._debugger.Debugger.write_uint8 (
              self,
              addr,
              value )
```

Reimplemented in ttk_tools.pyocd.debugger.Debugger.

7.173.3 Member Data Documentation

7.173.3.1 _elf_file_path

```
ttk_tools.pyocd._debugger.Debugger._elf_file_path  [protected]
```

7.173.3.2 session

```
ttk_tools.pyocd._debugger.Debugger.session
```

main session instance

7.173.3.3 symbol_provider

```
ttk_tools.pyocd._debugger.Debugger.symbol_provider
```

7.173.3.4 target

`ttk_tools.pyocd._debugger.Debugger.target`

Target instance of session.

Usage Examples:

- target.reset_and_halt()
- target.step()
- target.resume()
- target.halt()
- target.reset()

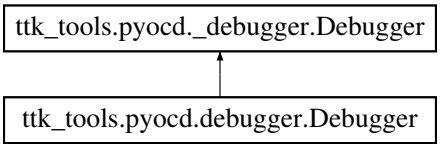
The documentation for this class was generated from the following file:

- ttk_tools/pyocd/_debugger.py (v1.2.0)

7.174 ttk_tools.pyocd.debugger.Debugger Class Reference

Wrapper for a PyOCD session and utility methods.

Inheritance diagram for ttk_tools.pyocd.debugger.Debugger:



Public Member Functions

- __init__ (self, elf_file_path)
- connect (self)

  Connect to target (open session)
- disconnect (self)

  Disconnect from target (close session).
- getAddress (self, symbol)

  Get memory address of an object on the ELF symbol table.
- read_uint8 (self, addr)

  Read uint8 value at `addr`.
- read_uint16 (self, addr)

  Read uint16 value at `addr`.
- read_uint32 (self, addr)

  Read uint32 value at `addr`.
- read_uint64 (self, addr)

  Read uint64 value at `addr`.

- read_sint8 (self, addr)
  Read sint8 value at `addr`.
- read_sint16 (self, addr)
  Read sint16 value at `addr`.
- read_sint32 (self, addr)
  Read sint32 value at `addr`.
- read_sint64 (self, addr)
  Read sint64 value at `addr`.
- read_float32 (self, addr)
  Read 32bit float value ("single") at `addr`.
- read_float64 (self, addr)
  Read 64bit float value ("double") at `addr`.
- write_uint8 (self, addr, value)
  Write uint8 `value` *to* `addr`.
- write_uint16 (self, addr, value)
  Write uint16 `value` *to* `addr`.
- write_uint32 (self, addr, value)
  Write uint32 `value` *to* `addr`.
- write_uint64 (self, addr, value)
  Write uint64 `value` *to* `addr`.
- write_sint8 (self, addr, value)
  Write sint8 `value` *to* `addr`.
- write_sint16 (self, addr, value)
  Write sint16 `value` *to* `addr`.
- write_sint32 (self, addr, value)
  Write sint32 `value` *to* `addr`.
- write_sint64 (self, addr, value)
  Write sint64 `value` *to* `addr`.
- write_float32 (self, addr, value)
  Write 32bit float `value` *("single") to* `addr`.
- write_float64 (self, addr, value)
  Write 64bit float `value` *("double") to* `addr`.

7.174.1 Detailed Description

Wrapper for a PyOCD session and utility methods.

Debugger session will be connected to running system in "attach" mode.

7.174.2 Constructor & Destructor Documentation

7.174.2.1 __init__()

```
ttk_tools.pyocd.debugger.Debugger.__init__ (
            self,
            elf_file_path )
```

Note

Currently connects to the first detected probe

Parameters

elf_file_path	path to ELF file with debug symbols

Reimplemented from ttk_tools.pyocd._debugger.Debugger.

7.174.3 Member Function Documentation

7.174.3.1 connect()

```
ttk_tools.pyocd.debugger.Debugger.connect (
              self )
```

Connect to target (open session)

Reimplemented from ttk_tools.pyocd._debugger.Debugger.

7.174.3.2 disconnect()

```
ttk_tools.pyocd.debugger.Debugger.disconnect (
              self )
```

Disconnect from target (close session).

Reimplemented from ttk_tools.pyocd._debugger.Debugger.

7.174.3.3 getAddress()

```
ttk_tools.pyocd.debugger.Debugger.getAddress (
              self,
              symbol )
```

Get memory address of an object on the ELF symbol table.

Parameters

symbol	symbol name string

Note

PyOCD's symbol provider can only resolve base addresses (start address of arrays, structs, ...)

Returns

a memory address (or raises a SymbolNotFound exception)

Reimplemented from ttk_tools.pyocd._debugger.Debugger.

7.174.3.4 read_float32()

```
ttk_tools.pyocd.debugger.Debugger.read_float32 (
              self,
              addr )
```

Read 32bit float value ("single") at addr.

Reimplemented from ttk_tools.pyocd._debugger.Debugger.

7.174.3.5 read_float64()

```
ttk_tools.pyocd.debugger.Debugger.read_float64 (
              self,
              addr )
```

Read 64bit float value ("double") at addr.

Not supported on 32bit SOCs.

Reimplemented from ttk_tools.pyocd._debugger.Debugger.

7.174.3.6 read_sint16()

```
ttk_tools.pyocd.debugger.Debugger.read_sint16 (
              self,
              addr )
```

Read sint16 value at addr.

Reimplemented from ttk_tools.pyocd._debugger.Debugger.

7.174.3.7 read_sint32()

```
ttk_tools.pyocd.debugger.Debugger.read_sint32 (
              self,
              addr )
```

Read sint32 value at addr.

Reimplemented from ttk_tools.pyocd._debugger.Debugger.

7.174.3.8 read_sint64()

```
ttk_tools.pyocd.debugger.Debugger.read_sint64 (
              self,
              addr )
```

Read sint64 value at addr.

Not supported on 32bit SOCs.

Reimplemented from ttk_tools.pyocd._debugger.Debugger.

7.174.3.9 read_sint8()

```
ttk_tools.pyocd.debugger.Debugger.read_sint8 (
              self,
              addr )
```

Read sint8 value at `addr`.

Reimplemented from ttk_tools.pyocd._debugger.Debugger.

7.174.3.10 read_uint16()

```
ttk_tools.pyocd.debugger.Debugger.read_uint16 (
              self,
              addr )
```

Read uint16 value at `addr`.

Reimplemented from ttk_tools.pyocd._debugger.Debugger.

7.174.3.11 read_uint32()

```
ttk_tools.pyocd.debugger.Debugger.read_uint32 (
              self,
              addr )
```

Read uint32 value at `addr`.

Reimplemented from ttk_tools.pyocd._debugger.Debugger.

7.174.3.12 read_uint64()

```
ttk_tools.pyocd.debugger.Debugger.read_uint64 (
              self,
              addr )
```

Read uint64 value at `addr`.

Not supported on 32bit SOCs.

Reimplemented from ttk_tools.pyocd._debugger.Debugger.

7.174.3.13 read_uint8()

```
ttk_tools.pyocd.debugger.Debugger.read_uint8 (
              self,
              addr )
```

Read uint8 value at `addr`.

Reimplemented from ttk_tools.pyocd._debugger.Debugger.

7.174.3.14 write_float32()

```
ttk_tools.pyocd.debugger.Debugger.write_float32 (
                self,
                addr,
                value )
```

Write 32bit float `value` ("single") to `addr`.

Note

Not supported on 32bit SOCs.

Reimplemented from ttk_tools.pyocd._debugger.Debugger.

7.174.3.15 write_float64()

```
ttk_tools.pyocd.debugger.Debugger.write_float64 (
                self,
                addr,
                value )
```

Write 64bit float `value` ("double") to `addr`.

Not supported on 32bit SOCs.

Reimplemented from ttk_tools.pyocd._debugger.Debugger.

7.174.3.16 write_sint16()

```
ttk_tools.pyocd.debugger.Debugger.write_sint16 (
                self,
                addr,
                value )
```

Write sint16 `value` to `addr`.

Reimplemented from ttk_tools.pyocd._debugger.Debugger.

7.174.3.17 write_sint32()

```
ttk_tools.pyocd.debugger.Debugger.write_sint32 (
                self,
                addr,
                value )
```

Write sint32 `value` to `addr`.

Reimplemented from ttk_tools.pyocd._debugger.Debugger.

7.174.3.18 write_sint64()

```
ttk_tools.pyocd.debugger.Debugger.write_sint64 (
                self,
                addr,
                value )
```

Write sint64 `value` to `addr`.

Not supported on 32bit SOCs.

Reimplemented from ttk_tools.pyocd._debugger.Debugger.

7.174.3.19 write_sint8()

```
ttk_tools.pyocd.debugger.Debugger.write_sint8 (
                self,
                addr,
                value )
```

Write sint8 `value` to `addr`.

Reimplemented from ttk_tools.pyocd._debugger.Debugger.

7.174.3.20 write_uint16()

```
ttk_tools.pyocd.debugger.Debugger.write_uint16 (
                self,
                addr,
                value )
```

Write uint16 `value` to `addr`.

Reimplemented from ttk_tools.pyocd._debugger.Debugger.

7.174.3.21 write_uint32()

```
ttk_tools.pyocd.debugger.Debugger.write_uint32 (
                self,
                addr,
                value )
```

Write uint32 `value` to `addr`.

Reimplemented from ttk_tools.pyocd._debugger.Debugger.

7.174.3.22 write_uint64()

```
ttk_tools.pyocd.debugger.Debugger.write_uint64 (
              self,
              addr,
              value )
```

Write uint64 `value` to `addr`.

Not supported on 32bit SOCs.

Reimplemented from ttk_tools.pyocd._debugger.Debugger.

7.174.3.23 write_uint8()

```
ttk_tools.pyocd.debugger.Debugger.write_uint8 (
              self,
              addr,
              value )
```

Write uint8 `value` to `addr`.

Reimplemented from ttk_tools.pyocd._debugger.Debugger.

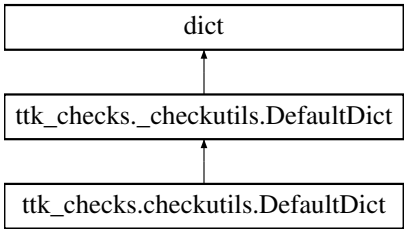
The documentation for this class was generated from the following file:

- ttk_tools/pyocd/debugger.py (v1.0.0)

7.175 ttk_checks._checkutils.DefaultDict Class Reference

A dictionary that returns empty strings instead of raising KeyErrors.

Inheritance diagram for ttk_checks._checkutils.DefaultDict:



Public Member Functions

- __getitem__ (self, key)

7.175.1 Detailed Description

A dictionary that returns empty strings instead of raising KeyErrors.

7.175.2 Member Function Documentation

7.175.2.1 __getitem__()

```
ttk_checks._checkutils.DefaultDict.__getitem__ (
            self,
            key )
```

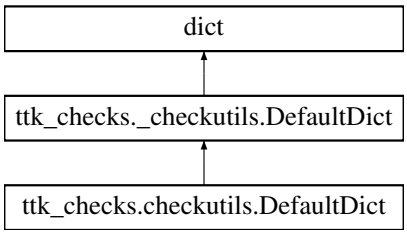
The documentation for this class was generated from the following file:

- ttk_checks/_checkutils.py (v2.1)

7.176 ttk_checks.checkutils.DefaultDict Class Reference

A dictionary that returns empty strings instead of raising KeyErrors.

Inheritance diagram for ttk_checks.checkutils.DefaultDict:



7.176.1 Detailed Description

A dictionary that returns empty strings instead of raising KeyErrors.

Build a result description from a supplied format string and entries. If the format is invalid, all available entries will be returned.

Parameters

descr_format	a format string (preferably referencing entries of, well, entries. If an non-existing entry is referenced, an empty string will be added in its place.
entries	a dictionary of entries to use in `format_str`

Examples:

```
>> d =  {"foo": "FOO", "baz": 12345}
>> buildDescription("Lorem: %(foo)s\nIpsum: %(bar)s%(baz)d", d)
Lorem: FOO
Ipsum: 12345
>> buildDescription("Lorem: %(foo)x\nIpsum: %(bar)s%(baz)d", d)
Invalid Format Specified: %x format: a number is required, not str
Available entries:
'baz':    12345
'foo':    FOO
```

Returns

a formatted description string

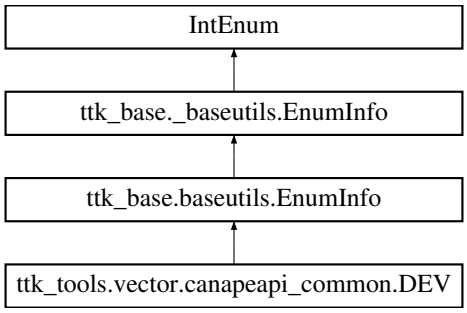
The documentation for this class was generated from the following file:

- ttk_checks/checkutils.py (v2.0)

7.177 ttk_tools.vector.canapeapi_common.DEV Class Reference

Device channel constants (DEV_*) from CANapAPI.h.

Inheritance diagram for ttk_tools.vector.canapeapi_common.DEV:



Static Public Attributes

- int CAN1 = 1
- int CAN2 = 2
- int CAN3 = 3
- int CAN4 = 4
- int CAN5 = 5
- int CAN6 = 6
- int CAN7 = 7
- int CAN8 = 8
- int CAN20 = 20
- int FLX1 = 31
- int FLX2 = 32
- int FLX3 = 33
- int FLX4 = 34
- int FLX5 = 35
- int FLX6 = 36
- int FLX7 = 37
- int FLX8 = 38
- int LIN1 = 61
- int LIN2 = 62
- int LIN3 = 63
- int LIN4 = 64
- int LIN5 = 65
- int LIN6 = 66
- int LIN7 = 67
- int LIN8 = 68

- int VX_CAN1 = 81
- int VX_CAN2 = 82
- int VX_CAN3 = 83
- int VX_CAN4 = 84
- int VX_TCP = 85
- int VX_UDP = 86
- int SXI1 = 91
- int SXI2 = 92
- int SXI3 = 93
- int SXI4 = 94
- int SXI5 = 95
- int SXI6 = 96
- int SXI7 = 97
- int SXI8 = 98
- int USB = 110
- int TCP = 255
- int UDP = 256
- int USERDEFINED = 261

  "User Defined Interface" (gets used for FlexRay connections in CANape releases pre-9.0)
- int VX_ETHERNET1 = 271
- int VX_ETHERNET2 = 272
- int DAIO_DLL = 280

7.177.1 Detailed Description

Device channel constants (DEV_∗) from CANapAPI.h.

7.177.2 Member Data Documentation

7.177.2.1 CAN1

```
int ttk_tools.vector.canapeapi_common.DEV.CAN1 = 1  [static]
```

7.177.2.2 CAN2

```
int ttk_tools.vector.canapeapi_common.DEV.CAN2 = 2  [static]
```

7.177.2.3 CAN20

```
int ttk_tools.vector.canapeapi_common.DEV.CAN20 = 20  [static]
```

7.177.2.4 CAN3

```
int ttk_tools.vector.canapeapi_common.DEV.CAN3 = 3  [static]
```

7.177.2.5 CAN4

```
int ttk_tools.vector.canapeapi_common.DEV.CAN4 = 4  [static]
```

7.177.2.6 CAN5

```
int ttk_tools.vector.canapeapi_common.DEV.CAN5 = 5  [static]
```

7.177.2.7 CAN6

```
int ttk_tools.vector.canapeapi_common.DEV.CAN6 = 6  [static]
```

7.177.2.8 CAN7

```
int ttk_tools.vector.canapeapi_common.DEV.CAN7 = 7  [static]
```

7.177.2.9 CAN8

```
int ttk_tools.vector.canapeapi_common.DEV.CAN8 = 8  [static]
```

7.177.2.10 DAIO_DLL

```
int ttk_tools.vector.canapeapi_common.DEV.DAIO_DLL = 280  [static]
```

7.177.2.11 FLX1

```
int ttk_tools.vector.canapeapi_common.DEV.FLX1 = 31  [static]
```

7.177.2.12 FLX2

```
int ttk_tools.vector.canapeapi_common.DEV.FLX2 = 32  [static]
```

7.177.2.13 FLX3

```
int ttk_tools.vector.canapeapi_common.DEV.FLX3 = 33  [static]
```

7.177.2.14 FLX4

```
int ttk_tools.vector.canapeapi_common.DEV.FLX4 = 34  [static]
```

7.177.2.15 FLX5

```
int ttk_tools.vector.canapeapi_common.DEV.FLX5 = 35  [static]
```

7.177.2.16 FLX6

```
int ttk_tools.vector.canapeapi_common.DEV.FLX6 = 36  [static]
```

7.177.2.17 FLX7

```
int ttk_tools.vector.canapeapi_common.DEV.FLX7 = 37  [static]
```

7.177.2.18 FLX8

```
int ttk_tools.vector.canapeapi_common.DEV.FLX8 = 38  [static]
```

7.177.2.19 LIN1

```
int ttk_tools.vector.canapeapi_common.DEV.LIN1 = 61  [static]
```

7.177.2.20 LIN2

```
int ttk_tools.vector.canapeapi_common.DEV.LIN2 = 62  [static]
```

7.177.2.21 LIN3

```
int ttk_tools.vector.canapeapi_common.DEV.LIN3 = 63  [static]
```

7.177.2.22 LIN4

```
int ttk_tools.vector.canapeapi_common.DEV.LIN4 = 64  [static]
```

7.177.2.23 LIN5

```
int ttk_tools.vector.canapeapi_common.DEV.LIN5 = 65  [static]
```

7.177.2.24 LIN6

```
int ttk_tools.vector.canapeapi_common.DEV.LIN6 = 66  [static]
```

7.177.2.25 LIN7

```
int ttk_tools.vector.canapeapi_common.DEV.LIN7 = 67  [static]
```

7.177.2.26 LIN8

```
int ttk_tools.vector.canapeapi_common.DEV.LIN8 = 68  [static]
```

7.177.2.27 SXI1

```
int ttk_tools.vector.canapeapi_common.DEV.SXI1 = 91  [static]
```

7.177.2.28 SXI2

```
int ttk_tools.vector.canapeapi_common.DEV.SXI2 = 92  [static]
```

7.177.2.29 SXI3

```
int ttk_tools.vector.canapeapi_common.DEV.SXI3 = 93  [static]
```

7.177.2.30 SXI4

```
int ttk_tools.vector.canapeapi_common.DEV.SXI4 = 94  [static]
```

7.177.2.31 SXI5

```
int ttk_tools.vector.canapeapi_common.DEV.SXI5 = 95  [static]
```

7.177.2.32 SXI6

```
int ttk_tools.vector.canapeapi_common.DEV.SXI6 = 96  [static]
```

7.177.2.33 SXI7

```
int ttk_tools.vector.canapeapi_common.DEV.SXI7 = 97  [static]
```

7.177.2.34 SXI8

```
int ttk_tools.vector.canapeapi_common.DEV.SXI8 = 98  [static]
```

7.177.2.35 TCP

```
int ttk_tools.vector.canapeapi_common.DEV.TCP = 255  [static]
```

7.177.2.36 UDP

```
int ttk_tools.vector.canapeapi_common.DEV.UDP = 256  [static]
```

7.177.2.37 USB

```
int ttk_tools.vector.canapeapi_common.DEV.USB = 110  [static]
```

7.177.2.38 USERDEFINED

```
int ttk_tools.vector.canapeapi_common.DEV.USERDEFINED = 261  [static]
```

"User Defined Interface" (gets used for FlexRay connections in CANape releases pre-9.0)

7.177.2.39 VX_CAN1

```
int ttk_tools.vector.canapeapi_common.DEV.VX_CAN1 = 81  [static]
```

7.177.2.40 VX_CAN2

```
int ttk_tools.vector.canapeapi_common.DEV.VX_CAN2 = 82  [static]
```

7.177.2.41 VX_CAN3

```
int ttk_tools.vector.canapeapi_common.DEV.VX_CAN3 = 83  [static]
```

7.177.2.42 VX_CAN4

```
int ttk_tools.vector.canapeapi_common.DEV.VX_CAN4 = 84  [static]
```

7.177.2.43 VX_ETHERNET1

```
int ttk_tools.vector.canapeapi_common.DEV.VX_ETHERNET1 = 271  [static]
```

7.177.2.44 VX_ETHERNET2

```
int ttk_tools.vector.canapeapi_common.DEV.VX_ETHERNET2 = 272  [static]
```

7.177.2.45 VX_TCP

```
int ttk_tools.vector.canapeapi_common.DEV.VX_TCP = 85  [static]
```

7.177.2.46 VX_UDP

```
int ttk_tools.vector.canapeapi_common.DEV.VX_UDP = 86  [static]
```

The documentation for this class was generated from the following file:

- ttk_tools/vector/canapeapi_common.py (v3.1.2)

7.178 ttk_selftest.hil_devices._devices_base.DeviceContainer Class Reference

Base class for HIL device containers.

Public Member Functions

- __init__ (self, context)
- iterDevices (self, device_type)

  Iterate over all devices of type "device_type".
- __enter__ (self)

  Start Selftest as context manager.
- __exit__ (self, exc_type, exc_value, tb)

  End context manager.

7.178.1 Detailed Description

Base class for HIL device containers.

7.178.2 Constructor & Destructor Documentation

7.178.2.1 __init__()

```
ttk_selftest.hil_devices._devices_base.DeviceContainer.__init__ (
            self,
            context )
```

Parameters

context	parent context

7.178.3 Member Function Documentation

7.178.3.1 __enter__()

```
ttk_selftest.hil_devices._devices_base.DeviceContainer.__enter__ (
               self )
```

Start Selftest as context manager.

Enables .selftest_enable on enter and disables it again at exit.

Example:

```
# with testenv.hil_devices being an instance of the project's
# configured DeviceContainer class:
with testenv.hil_devices as hil_devices:
    for power_supply in hil_devices.iterDevices(PowerSupply):
       selftest_tests.testPowerSupply(
           testenv,
           power_supply,
           max_test_current=18,
           num_testvalues=6
        )
```

7.178.3.2 __exit__()

```
ttk_selftest.hil_devices._devices_base.DeviceContainer.__exit__ (
               self,
               exc_type,
               exc_value,
               tb )
```

End context manager.

Disables .selftest_enable again.

7.178.3.3 iterDevices()

```
ttk_selftest.hil_devices._devices_base.DeviceContainer.iterDevices (
               self,
               device_type )
```

Iterate over all devices of type "device_type".

Parameters

device_type	type specification of the requested devices

Usage:

```
for dev in dev.iterDevices(AnalogOutput):
    print dev
```

Returns

an iterator object.

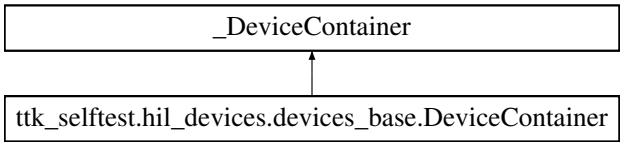
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/_devices_base.py (v2.2.0)

7.179 ttk_selftest.hil_devices.devices_base.DeviceContainer Class Reference

Base class for HIL device containers.

Inheritance diagram for ttk_selftest.hil_devices.devices_base.DeviceContainer:



Public Member Functions

- __init__ (self, context)
- iterDevices (self, device_type)

  Iterate over all devices of type "device_type".

7.179.1 Detailed Description

Base class for HIL device containers.

7.179.2 Constructor & Destructor Documentation

7.179.2.1 __init__()

```
ttk_selftest.hil_devices.devices_base.DeviceContainer.__init__ (
            self,
            context )
```

Parameters

context	parent context

7.179.3 Member Function Documentation

7.179.3.1 iterDevices()

```
ttk_selftest.hil_devices.devices_base.DeviceContainer.iterDevices (
                self,
                device_type )
```

Iterate over all devices of type "device_type".

Parameters

device_type	type specification of the requested devices

Usage:

```
for dev in dev.iterDevices(AnalogOutput):
    print dev
```

Returns

an iterator object.

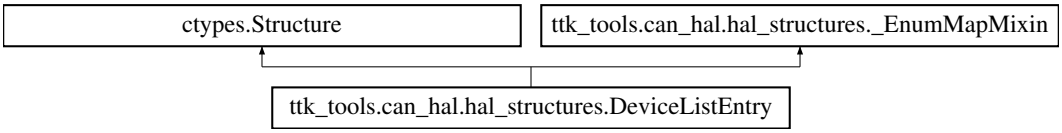
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/devices_base.py (v2.0.0)

7.180 ttk_tools.can_hal.hal_structures.DeviceListEntry Class Reference

A single device list entry (CanHal.h, tCanHalDeviceList).

Inheritance diagram for ttk_tools.can_hal.hal_structures.DeviceListEntry:



Public Member Functions

- is_free (self)
- __repr__ (self)

Static Protected Attributes

- list _fields_
- dict _map = {"DeviceType": DeviceType}

7.180.1 Detailed Description

A single device list entry (CanHal.h, tCanHalDeviceList).

7.180.2 Member Function Documentation

7.180.2.1 __repr__()

```
ttk_tools.can_hal.hal_structures.DeviceListEntry.__repr__ (
              self )
```

7.180.2.2 is_free()

```
ttk_tools.can_hal.hal_structures.DeviceListEntry.is_free (
              self )
```

7.180.3 Member Data Documentation

7.180.3.1 _fields_

```
list ttk_tools.can_hal.hal_structures.DeviceListEntry._fields_  [static], [protected]
```

Initial value:

```
=  [
        ("DeviceType",      ctypes.c_int),      # tCanHalDeviceType   m_DeviceType

        ("SubModuleHandle", CanDeviceHdl),      # tHalCanDeviceHdl    m_SubModuleHandle
        ("SerialNumber",    ctypes.c_ulong),    # unsigned long       m_ulSerialNumber
        ("DeviceNumber",    ctypes.c_uint),     # unsigned int        m_uiDeviceNumber
        ("DeviceChannel",   ctypes.c_uint),     # unsigned int        m_uiDeviceChannel

        ("DeviceUsed",      ctypes.c_bool)      # BOOL                m_bDeviceUsed
    ]
```

7.180.3.2 _map

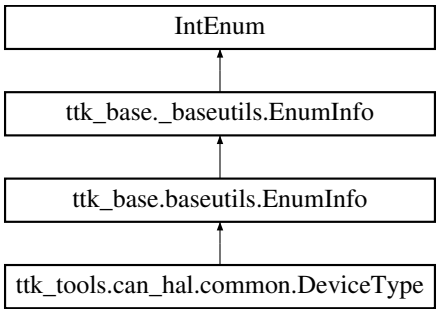
```
dict ttk_tools.can_hal.hal_structures.DeviceListEntry._map = {"DeviceType":  DeviceType}  [static],
[protected]
```

The documentation for this class was generated from the following file:

- ttk_tools/can_hal/hal_structures.py (v1.0.0)

7.181 ttk_tools.can_hal.common.DeviceType Class Reference

Inheritance diagram for ttk_tools.can_hal.common.DeviceType:



Static Public Attributes

- int NoDevice = 0x0000
- int PeakPCI = 0x0001
- int PeakUSB = 0x0002
- int PeakLAN = 0x0003
- int PeakPCCARD = 0x0004
- int VectorCaseX = 0x0020
- int VectorAc2PCI = 0x0021
- int VectorCaseY = 0x0022
- int VectorCardXL = 0x0023
- int VectorCaseXL = 0x0024
- int VectorCaseXL_Log = 0x0025
- int VectorCanBoardXL = 0x0026
- int VectorCanBoardXL_PXI = 0x0027
- int VectorVN2600_2610 = 0x0028
- int VectorVN3300 = 0x002A
- int VectorVN3600 = 0x002B
- int VectorVN7600 = 0x002C
- int VectorCardXLE = 0x002D
- int VectorVN8900 = 0x002E
- int VectorVN8950 = 0x002F
- int VectorVN2640 = 0x0030
- int VectorVN1610 = 0x0031
- int VectorVN1630 = 0x0032
- int VectorVN1640 = 0x0033
- int VectorVN8970 = 0x0034
- int VectorVN1611 = 0x0035
- int VectorVN5610 = 0x0036
- int VectorVN7570 = 0x0037
- int SystecUsb = 0x0060

7.181.1 Member Data Documentation

7.181.1.1 NoDevice

```
int ttk_tools.can_hal.common.DeviceType.NoDevice = 0x0000  [static]
```

7.181.1.2 PeakLAN

```
int ttk_tools.can_hal.common.DeviceType.PeakLAN = 0x0003  [static]
```

7.181.1.3 PeakPCCARD

```
int ttk_tools.can_hal.common.DeviceType.PeakPCCARD = 0x0004  [static]
```

7.181.1.4 PeakPCI

```
int ttk_tools.can_hal.common.DeviceType.PeakPCI = 0x0001  [static]
```

7.181.1.5 PeakUSB

```
int ttk_tools.can_hal.common.DeviceType.PeakUSB = 0x0002  [static]
```

7.181.1.6 SystecUsb

```
int ttk_tools.can_hal.common.DeviceType.SystecUsb = 0x0060  [static]
```

7.181.1.7 VectorAc2PCI

```
int ttk_tools.can_hal.common.DeviceType.VectorAc2PCI = 0x0021  [static]
```

7.181.1.8 VectorCanBoardXL

```
int ttk_tools.can_hal.common.DeviceType.VectorCanBoardXL = 0x0026  [static]
```

7.181.1.9 VectorCanBoardXL_PXI

```
int ttk_tools.can_hal.common.DeviceType.VectorCanBoardXL_PXI = 0x0027  [static]
```

7.181.1.10 VectorCardXL

```
int ttk_tools.can_hal.common.DeviceType.VectorCardXL = 0x0023  [static]
```

7.181.1.11 VectorCardXLE

```
int ttk_tools.can_hal.common.DeviceType.VectorCardXLE = 0x002D  [static]
```

7.181.1.12 VectorCaseX

```
int ttk_tools.can_hal.common.DeviceType.VectorCaseX = 0x0020  [static]
```

7.181.1.13 VectorCaseXL

```
int ttk_tools.can_hal.common.DeviceType.VectorCaseXL = 0x0024  [static]
```

7.181.1.14 VectorCaseXL_Log

```
int ttk_tools.can_hal.common.DeviceType.VectorCaseXL_Log = 0x0025  [static]
```

7.181.1.15 VectorCaseY

```
int ttk_tools.can_hal.common.DeviceType.VectorCaseY = 0x0022  [static]
```

7.181.1.16 VectorVN1610

```
int ttk_tools.can_hal.common.DeviceType.VectorVN1610 = 0x0031  [static]
```

7.181.1.17 VectorVN1611

```
int ttk_tools.can_hal.common.DeviceType.VectorVN1611 = 0x0035  [static]
```

7.181.1.18 VectorVN1630

```
int ttk_tools.can_hal.common.DeviceType.VectorVN1630 = 0x0032  [static]
```

7.181.1.19 VectorVN1640

```
int ttk_tools.can_hal.common.DeviceType.VectorVN1640 = 0x0033  [static]
```

7.181.1.20 VectorVN2600_2610

```
int ttk_tools.can_hal.common.DeviceType.VectorVN2600_2610 = 0x0028  [static]
```

7.181.1.21 VectorVN2640

```
int ttk_tools.can_hal.common.DeviceType.VectorVN2640 = 0x0030  [static]
```

7.181.1.22 VectorVN3300

```
int ttk_tools.can_hal.common.DeviceType.VectorVN3300 = 0x002A  [static]
```

7.181.1.23 VectorVN3600

```
int ttk_tools.can_hal.common.DeviceType.VectorVN3600 = 0x002B  [static]
```

7.181.1.24 VectorVN5610

```
int ttk_tools.can_hal.common.DeviceType.VectorVN5610 = 0x0036  [static]
```

7.181.1.25 VectorVN7570

```
int ttk_tools.can_hal.common.DeviceType.VectorVN7570 = 0x0037  [static]
```

7.181.1.26 VectorVN7600

```
int ttk_tools.can_hal.common.DeviceType.VectorVN7600 = 0x002C  [static]
```

7.181.1.27 VectorVN8900

```
int ttk_tools.can_hal.common.DeviceType.VectorVN8900 = 0x002E  [static]
```

7.181.1.28 VectorVN8950

```
int ttk_tools.can_hal.common.DeviceType.VectorVN8950 = 0x002F  [static]
```

7.181.1.29 VectorVN8970

```
int ttk_tools.can_hal.common.DeviceType.VectorVN8970 = 0x0034  [static]
```

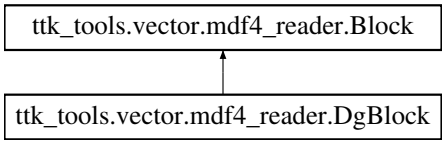
The documentation for this class was generated from the following file:

- ttk_tools/can_hal/common.py (v1.2)

7.182 ttk_tools.vector.mdf4_reader.DgBlock Class Reference

Data Group Block.

Inheritance diagram for ttk_tools.vector.mdf4_reader.DgBlock:



Public Member Functions

- __init__ (self, file_handle, pos=None)

  Read DGBLOCK and any associated MDBLOCK from file_handle, starting at the current position.
- updateVLSDChannelReferences (self)

  Update VLSD CGBLOCKs so they contain back-references to the VLSD channels.

Public Attributes

- dg_dg_next

  Pointer to next data group block (DGBLOCK) (can be NIL)
- dg_cg_first

  Pointer to next data group block (DGBLOCK) (can be NIL)
- dg_data

  Pointer to next data group block (DGBLOCK) (can be NIL)
- dg_md_comment

  Pointer to next data group block (DGBLOCK) (can be NIL)
- dg_rec_id_size
- comment

  Comment (defaults to None)
- channel_groups

  Linked channel groups (if any)
- channel_groups_by_rec_id

  mapping: {cg_record_id: <CgBlock instance>} for all channel groups.
- data

  Data container for this channel group.

7.182.1 Detailed Description

Data Group Block.

The DGBLOCK gathers information and links related to its data block. Thus the branch in the tree of MDF blocks that is opened by the DGBLOCK contains all information necessary to understand and decode the data block referenced by the DGBLOCK.

A DGBLOCK can contain several channel groups. In this case the data group (and thus the MDF file) is "unsorted". If there is only a single channel group in the DGBLOCK, the data group is "sorted".

7.182.2 Constructor & Destructor Documentation

7.182.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.DgBlock.__init__ (
            self,
            file_handle,
            pos = None )
```

Read DGBLOCK and any associated MDBLOCK from file_handle, starting at the current position.

Note

File pointer position will be modified

Reimplemented from ttk_tools.vector.mdf4_reader.Block.

7.182.3 Member Function Documentation

7.182.3.1 updateVLSDChannelReferences()

```
ttk_tools.vector.mdf4_reader.DgBlock.updateVLSDChannelReferences (
              self )
```

Update VLSD CGBLOCKs so they contain back-references to the VLSD channels.

Note

Variable signal length channels may reference a "special" VLSD channel group block instead of having a signal data block with variable length data. VLSD channel groups are not really "groups", they contain no channels but rather serve as stand-in for one channel with variable length entries (looks like they re-used CGs, as they can be referenced by "record ids" from DTBLOCK data)

7.182.4 Member Data Documentation

7.182.4.1 channel_groups

```
ttk_tools.vector.mdf4_reader.DgBlock.channel_groups
```

Linked channel groups (if any)

7.182.4.2 channel_groups_by_rec_id

```
ttk_tools.vector.mdf4_reader.DgBlock.channel_groups_by_rec_id
```

mapping: {cg_record_id: <CgBlock instance>} for all channel groups.

Will be used later when iterating over payload data (but will only be relevant for variable-length channel groups)

7.182.4.3 comment

```
ttk_tools.vector.mdf4_reader.DgBlock.comment
```

Comment (defaults to None)

7.182.4.4 data

```
ttk_tools.vector.mdf4_reader.DgBlock.data
```

Data container for this channel group.

7.182.4.5 dg_cg_first

ttk_tools.vector.mdf4_reader.DgBlock.dg_cg_first

Pointer to next data group block (DGBLOCK) (can be NIL)

Pointer to first channel group block (CGBLOCK) (can be NIL)

7.182.4.6 dg_data

ttk_tools.vector.mdf4_reader.DgBlock.dg_data

Pointer to next data group block (DGBLOCK) (can be NIL)

Pointer to first channel group block (CGBLOCK) (can be NIL)

Pointer to data block (DTBLOCK or DZBLOCK for this block type) or data list block (DLBLOCK of data blocks or its HLBLOCK) (can be NIL)

7.182.4.7 dg_dg_next

ttk_tools.vector.mdf4_reader.DgBlock.dg_dg_next

Pointer to next data group block (DGBLOCK) (can be NIL)

7.182.4.8 dg_md_comment

ttk_tools.vector.mdf4_reader.DgBlock.dg_md_comment

Pointer to next data group block (DGBLOCK) (can be NIL)

Pointer to first channel group block (CGBLOCK) (can be NIL)

Pointer to data block (DTBLOCK or DZBLOCK for this block type) or data list block (DLBLOCK of data blocks or its HLBLOCK) (can be NIL)

Pointer to comment and additional information (TXBLOCK or MDBLOCK) (can be NIL)

7.182.4.9 dg_rec_id_size

ttk_tools.vector.mdf4_reader.DgBlock.dg_rec_id_size

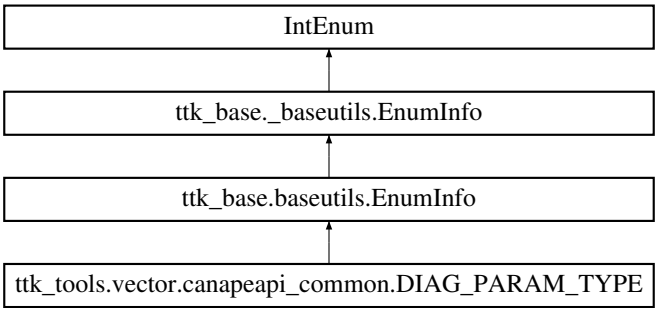
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.183 ttk_tools.vector.canapeapi_common.DIAG_PARAM_TYPE Class Reference

Diagnostic response parameter types (`Param*`) from enum `EnParamType`.

Inheritance diagram for ttk_tools.vector.canapeapi_common.DIAG_PARAM_TYPE:



Static Public Attributes

- int SIGNED = 1
- int DOUBLE = 2
- int BCD = 3
- int UNSIGNED = 4
- int FLOAT = 5
- int AUTODETECT = 6

7.183.1 Detailed Description

Diagnostic response parameter types (`Param*`) from enum `EnParamType`.

7.183.2 Member Data Documentation

7.183.2.1 AUTODETECT

```
int ttk_tools.vector.canapeapi_common.DIAG_PARAM_TYPE.AUTODETECT = 6  [static]
```

7.183.2.2 BCD

```
int ttk_tools.vector.canapeapi_common.DIAG_PARAM_TYPE.BCD = 3  [static]
```

7.183.2.3 DOUBLE

```
int ttk_tools.vector.canapeapi_common.DIAG_PARAM_TYPE.DOUBLE = 2  [static]
```

7.183.2.4 FLOAT

```
int ttk_tools.vector.canapeapi_common.DIAG_PARAM_TYPE.FLOAT = 5  [static]
```

7.183.2.5 SIGNED

```
int ttk_tools.vector.canapeapi_common.DIAG_PARAM_TYPE.SIGNED = 1  [static]
```

7.183.2.6 UNSIGNED

```
int ttk_tools.vector.canapeapi_common.DIAG_PARAM_TYPE.UNSIGNED = 4  [static]
```

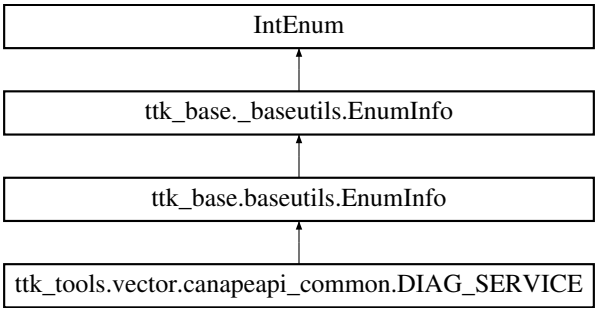
The documentation for this class was generated from the following file:

- ttk_tools/vector/canapeapi_common.py (v3.1.2)

7.184 ttk_tools.vector.canapeapi_common.DIAG_SERVICE Class Reference

Diagnostic service states (`DIAG_SERVICE_*`) from enum `eServiceStates`.

Inheritance diagram for ttk_tools.vector.canapeapi_common.DIAG_SERVICE:



Static Public Attributes

- int CREATED = 10

  Diagnostic Service created but not yet sent.
- int RUNNING = 20

  Diagnostic Service created and sent, currently running -> no response received.
- int FINISHED = 30

  Diagnostic Service finished running -> response received.
- int TIMEOUT = 40

  Diagnostic Service was created and sent, but a timeout occurred while CANape was waiting for response.

7.184.1 Detailed Description

Diagnostic service states (`DIAG_SERVICE_*`) from enum `eServiceStates`.

Note

Service states have explicitly been defined in ascending order:
`CREATED` => `RUNNING` => `FINISHED` => `TIMEOUT`

7.184.2 Member Data Documentation

7.184.2.1 CREATED

```
int ttk_tools.vector.canapeapi_common.DIAG_SERVICE.CREATED = 10  [static]
```

Diagnostic Service created but not yet sent.

7.184.2.2 FINISHED

```
int ttk_tools.vector.canapeapi_common.DIAG_SERVICE.FINISHED = 30  [static]
```

Diagnostic Service finished running -> response received.

7.184.2.3 RUNNING

```
int ttk_tools.vector.canapeapi_common.DIAG_SERVICE.RUNNING = 20  [static]
```

Diagnostic Service created and sent, currently running -> no response received.

7.184.2.4 TIMEOUT

```
int ttk_tools.vector.canapeapi_common.DIAG_SERVICE.TIMEOUT = 40  [static]
```

Diagnostic Service was created and sent, but a timeout occurred while CANape was waiting for response.

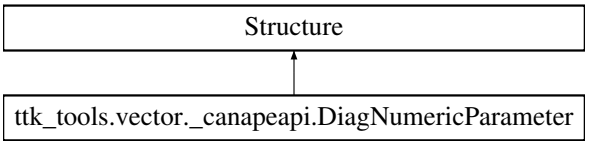
The documentation for this class was generated from the following file:

- ttk_tools/vector/canapeapi_common.py (v3.1.2)

7.185 ttk_tools.vector._canapeapi.DiagNumericParameter Class Reference

Numeric values for diagnostic parameters.

Inheritance diagram for ttk_tools.vector._canapeapi.DiagNumericParameter:



Classes

- class PValues

  Different value representations.

Public Member Functions

- getValue (self)

  Get un-wrapped contained value.
- setValue (self, value)

  Set numeric value.

Public Attributes

- DiagNumeric

Static Protected Attributes

- _pack_
- list _fields_

7.185.1 Detailed Description

Numeric values for diagnostic parameters.

7.185.2 Member Function Documentation

7.185.2.1 getValue()

```
ttk_tools.vector._canapeapi.DiagNumericParameter.getValue (
              self )
```

Get un-wrapped contained value.

Returns

the value as numeric (int or float) value, or None if DiagNumeric contains an undefined or unsupported type (see DIAG_PARAM_TYPE).

7.185.2.2 setValue()

```
ttk_tools.vector._canapeapi.DiagNumericParameter.setValue (
              self,
              value )
```

Set numeric value.

Data type of value will be used to determine DiagNumeric type.

Note

DiagNumericParameter only supports 32bit integers

7.185.3 Member Data Documentation

7.185.3.1 _fields_

```
list ttk_tools.vector._canapeapi.DiagNumericParameter._fields_  [static], [protected]
```

Initial value:

```
= [
        ("DiagNumeric", c_int), # see EnParamType enum / DIAG_PARAM_TYPE
        ("Values",      PValues)
    ]
```

7.185.3.2 _pack_

```
ttk_tools.vector._canapeapi.DiagNumericParameter._pack_  [static], [protected]
```

7.185.3.3 DiagNumeric

```
ttk_tools.vector._canapeapi.DiagNumericParameter.DiagNumeric
```

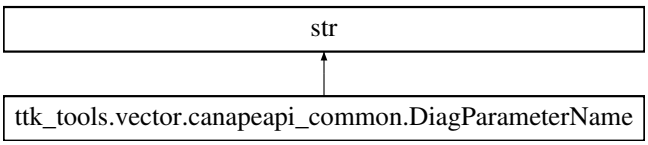
The documentation for this class was generated from the following file:

- ttk_tools/vector/_canapeapi.py (v7.3.3)

7.186 ttk_tools.vector.canapeapi_common.DiagParameterName Class Reference

Container for diagnostic response name that may contain sub-parameter names for "complex parameters".

Inheritance diagram for ttk_tools.vector.canapeapi_common.DiagParameterName:



Public Member Functions

- __new__ (cls, name, ∗args, ∗∗kwargs)

  Create a custom instance of the immutable str type.
- __init__ (self, name, sub_names=None)

  Create a new diagnostics parameter name that may have sub-parameter names.
- __repr__ (self)

Public Attributes

- sub_names

  list of sub-parameter names

7.186.1 Detailed Description

Container for diagnostic response name that may contain sub-parameter names for "complex parameters".

7.186.2 Constructor & Destructor Documentation

7.186.2.1 __init__()

```
ttk_tools.vector.canapeapi_common.DiagParameterName.__init__ (
              self,
              name,
              sub_names = None )
```

Create a new diagnostics parameter name that may have sub-parameter names.

See CANapeDiag.sendSymbolicRequest.

Note. For "normal" (non-complex) parameters, a name as string will usually suffice.

Parameters

name	name of the parameter
sub_names	for complex parameters: a list of sub-parameter names (i.e. for sub-parameters contained in this complex parameter)

Example:

```
param = DiagParameterName(
    name      = "main_parameter_name",
    sub_names = ["sub_param_name_1", "sub_param_name_2"]
)
```

7.186.3 Member Function Documentation

7.186.3.1 __new__()

```
ttk_tools.vector.canapeapi_common.DiagParameterName.__new__ (
              cls,
              name,
             * args,
             ** kwargs )
```

Create a custom instance of the immutable str type.

Parameters

name	name of parameter, see **init**
∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

7.186.3.2 __repr__()

```
ttk_tools.vector.canapeapi_common.DiagParameterName.__repr__ (
            self )
```

7.186.4 Member Data Documentation

7.186.4.1 sub_names

```
ttk_tools.vector.canapeapi_common.DiagParameterName.sub_names
```

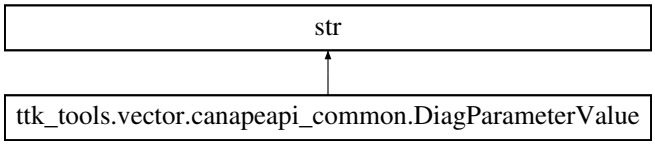
list of sub-parameter names

The documentation for this class was generated from the following file:

- ttk_tools/vector/canapeapi_common.py (v3.1.2)

7.187 ttk_tools.vector.canapeapi_common.DiagParameterValue Class Reference

Inheritance diagram for ttk_tools.vector.canapeapi_common.DiagParameterValue:



Public Member Functions

- __new__ (cls, value, *args, **kwargs)
  Create a custom instance of the immutable str type.
- __init__ (self, value, raw_val=None, numeric_val=None, complex_val=None)
  Create a new diagnostics parameter value.
- value (self)
  Return default string value representation.
- raw_as_int (self)
  Return raw bytes interpreted as an integer.
- __repr__ (self)

Public Attributes

- raw
  raw value representation.
- numeric
  numeric value representation
- complex
  complex value (list of sub-responses)

7.187.1 Constructor & Destructor Documentation

7.187.1.1 __init__()

```
ttk_tools.vector.canapeapi_common.DiagParameterValue.__init__ (
              self,
              value,
              raw_val = None,
              numeric_val = None,
              complex_val = None )
```

Create a new diagnostics parameter value.

It should normally not be necessary to create those values directly. However, instances of DiagParameterValue will be returned by diagnostic requests to give access to all available value representations.

Parameters

value	value as default string representation
raw_val	raw value representation (e.g. bytes)
numeric_val	numeric value representation
complex_val	a list of sub-responses containing a dictionary with entries for each requested sub-parameter

7.187.2 Member Function Documentation

7.187.2.1 __new__()

```
ttk_tools.vector.canapeapi_common.DiagParameterValue.__new__ (
              cls,
              value,
             * args,
             ** kwargs )
```

Create a custom instance of the immutable str type.

Parameters

value	value as string representation, see **init**
∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

7.187.2.2 __repr__()

```
ttk_tools.vector.canapeapi_common.DiagParameterValue.__repr__ (
              self )
```

7.187.2.3 raw_as_int()

```
ttk_tools.vector.canapeapi_common.DiagParameterValue.raw_as_int (
              self )
```

Return raw bytes interpreted as an integer.

This may is sometimes be useful if a parameter has only a string description but no numeric representation (e.g. DTCs, where raw data corresponds to the actual diagnostic trouble code and the string data contains a configured error text)

Example:

```
>> DiagParameterValue("This is a error description text",
                      raw_val=[0x00, 0x12, 0x34]).raw_as_int
4660 # i.e. 0x001234

>> DiagParameterValue("lorem ipsum", b"\x01\x04").raw_as_int
260 # i.e. 0x0104
```

7.187.2.4 value()

```
ttk_tools.vector.canapeapi_common.DiagParameterValue.value (
              self )
```

Return default string value representation.

Note

Just for completeness, so all representations can be accessed by an attribute (.value, .raw, .numeric, .complex)

7.187.3 Member Data Documentation

7.187.3.1 complex

```
ttk_tools.vector.canapeapi_common.DiagParameterValue.complex
```

complex value (list of sub-responses)

7.187.3.2 numeric

```
ttk_tools.vector.canapeapi_common.DiagParameterValue.numeric
```

numeric value representation

7.187.3.3 raw

```
ttk_tools.vector.canapeapi_common.DiagParameterValue.raw
```

raw value representation.

Note that complex responses do not seem to have a raw value repr

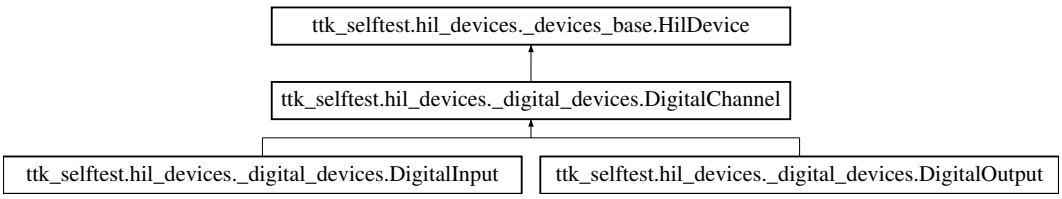
The documentation for this class was generated from the following file:

- ttk_tools/vector/canapeapi_common.py (v3.1.2)

7.188 ttk_selftest.hil_devices._digital_devices.DigitalChannel Class Reference

Base digital channel container for a testable digital channel and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices._digital_devices.DigitalChannel:



Public Member Functions

- __init__ (self, base_path, sig_channel, max_voltage, preconditions=None, wait_time_ms=None, alias="", descr="")

  Digital channel initialization.
- info (self)

  Get an info string for this digital channel: alias, FIU channel and maximum voltage plus description, if available.

Public Attributes

- base_path
- selftest_var
- max_voltage
- lut_state

7.188.1 Detailed Description

Base digital channel container for a testable digital channel and relevant associated variables/values.

7.188.2 Constructor & Destructor Documentation

7.188.2.1 __init__()

```
ttk_selftest.hil_devices._digital_devices.DigitalChannel.__init__ (
              self,
              base_path,
              sig_channel,
              max_voltage,
              preconditions = None,
              wait_time_ms = None,
              alias = "",
              descr = "" )
```

Digital channel initialization.

Parameters

base_path	base path to digital channel subsystem
sig_channel	fiu channel of the digital channel
max_voltage	maximum physical voltage in V of the channel
preconditions	(opt) preconditions that have to be set before the device can be checked. default is None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The function specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
  # For special cases, the preconditions may also contain function calls
  # (which might be useful to insert additional delays or similar):
  preconditions = [
      { "device": Voltage, "value": 1},
      { "func":   time.sleep,
        "args":   [1.0], # positional arguments
        "kwargs": {},    # keyword arguments
      },
      {"device": ClampSwitch, "value": 1},
      # or a simple call like this:
      { "func":  someCustomFunction,
      },
      ...
  ]
wait_time_ms      - (opt) waiting time in milliseconds between
channel stimulation and channel check
defaults to 100ms
alias             - (opt) channel alias/name, defaults to the
last (non-default) entry in identifier
(i.e. signal model path)
descr             - (opt) channel description (e.g. used in report entries)
```

Reimplemented from ttk_selftest.hil_devices._devices_base.HilDevice.

Reimplemented in ttk_selftest.hil_devices._digital_devices.DigitalOutput, and ttk_selftest.hil_devices._digital_devices.DigitalInput.

7.188.3 Member Function Documentation

7.188.3.1 info()

```
ttk_selftest.hil_devices._digital_devices.DigitalChannel.info (
            self )
```

Get an info string for this digital channel: alias, FIU channel and maximum voltage plus description, if available.

Returns

a formatted string.

Reimplemented in ttk_selftest.hil_devices._digital_devices.DigitalInput, and ttk_selftest.hil_devices._digital_devices.DigitalOutput.

7.188.4 Member Data Documentation

7.188.4.1 base_path

`ttk_selftest.hil_devices._digital_devices.DigitalChannel.base_path`

7.188.4.2 lut_state

`ttk_selftest.hil_devices._digital_devices.DigitalChannel.lut_state`

7.188.4.3 max_voltage

`ttk_selftest.hil_devices._digital_devices.DigitalChannel.max_voltage`

7.188.4.4 selftest_var

`ttk_selftest.hil_devices._digital_devices.DigitalChannel.selftest_var`

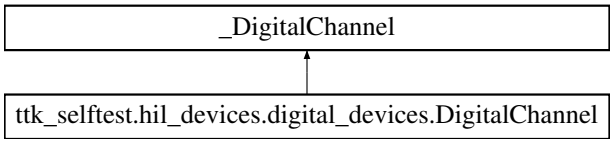
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/_digital_devices.py (v2.0.0)

7.189 ttk_selftest.hil_devices.digital_devices.DigitalChannel Class Reference

Base digital channel container for a testable digital channel and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices.digital_devices.DigitalChannel:



Public Member Functions

- __init__ (self, base_path, sig_channel, max_voltage, preconditions=None, wait_time_ms=None, alias="", descr="")

  Digital channel initialization.
- info (self)

  Get an info string for this digital channel: alias, FIU channel and maximum voltage plus description, if available.

7.189.1 Detailed Description

Base digital channel container for a testable digital channel and relevant associated variables/values.

7.189.2 Constructor & Destructor Documentation

7.189.2.1 __init__()

```
ttk_selftest.hil_devices.digital_devices.DigitalChannel.__init__ (
                self,
                base_path,
                sig_channel,
                max_voltage,
                preconditions = None,
                wait_time_ms = None,
                alias = "",
                descr = "" )
```

Digital channel initialization.

Parameters

base_path	base path to digital channel subsystem
sig_channel	fiu channel of the digital channel
max_voltage	maximum physical voltage in V of the channel
preconditions	(opt) preconditions that have to be set before the device can be checked. default is None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The function specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
  # For special cases, the preconditions may also contain function calls
  # (which might be useful to insert additional delays or similar):
  preconditions = [
      { "device": Voltage, "value": 1},
      { "func":   time.sleep,
        "args":   [1.0], # positional arguments
        "kwargs": {},    # keyword arguments
      },
      {"device": ClampSwitch, "value": 1},
      # or a simple call like this:
      { "func":  someCustomFunction,
      },
      ...
  ]
wait_time_ms      - (opt) waiting time in milliseconds between
```

```
channel stimulation and channel check
defaults to 100ms
alias             - (opt) channel alias/name, defaults to the
last (non-default) entry in identifier
(i.e. signal model path)
descr             - (opt) channel description (e.g. used in report entries)
```

7.189.3 Member Function Documentation

7.189.3.1 info()

```
ttk_selftest.hil_devices.digital_devices.DigitalChannel.info (
              self )
```

Get an info string for this digital channel: alias, FIU channel and maximum voltage plus description, if available.

Returns

a formatted string.

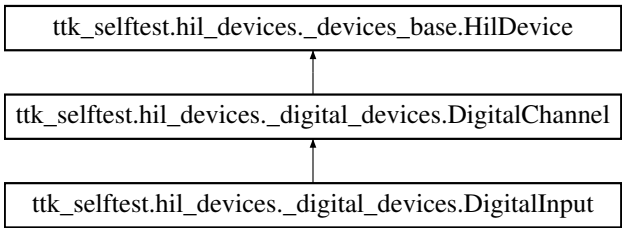
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/digital_devices.py (v2.0.0)

7.190 ttk_selftest.hil_devices._digital_devices.DigitalInput Class Reference

Base digital input container for a testable digital input and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices._digital_devices.DigitalInput:



Public Member Functions

- __init__ (self, context, base_path, sig_channel, max_voltage, threshold_voltage=None, preconditions=None, wait_time_ms=None, alias="", descr="")

  Digital input initialization.
- info (self)

  Get an info string for this digital input channel: alias, FIU channel, maximum voltage and threshold voltage plus description, if available.

Public Attributes

- threshold_voltage
- selftest_var
- max_voltage

7.190.1 Detailed Description

Base digital input container for a testable digital input and relevant associated variables/values.

7.190.2 Constructor & Destructor Documentation

7.190.2.1 __init__()

```
ttk_selftest.hil_devices._digital_devices.DigitalInput.__init__ (
               self,
               context,
               base_path,
               sig_channel,
               max_voltage,
               threshold_voltage = None,
               preconditions = None,
               wait_time_ms = None,
               alias = "",
               descr = "" )
```

Digital input initialization.

Parameters

context	parent context
base_path	base path to digital input subsystem
sig_channel	fiu channel of the digital input channel
max_voltage	maximum physical voltage in V of the channel
threshold_voltage	(opt) voltage threshold in V for high/low detection defaults to 2.5V (due to the threshold voltage of the digital signal conditioning card DSK)
preconditions	(opt) preconditions that have to be set before the device can be checked. default is None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The function specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
    # For special cases, the preconditions may also contain function calls
    # (which might be useful to insert additional delays or similar):
    preconditions = [
        { "device": Voltage, "value": 1},
        { "func":   time.sleep,
          "args":   [1.0], # positional arguments
          "kwargs": {},    # keyword arguments
        },
        {"device": ClampSwitch, "value": 1},
        # or a simple call like this:
        { "func":  someCustomFunction,
        },
        ...
    ]
  wait_time_ms      - (opt) waiting time in milliseconds between
  channel stimulation and channel check
  defaults to 100ms
  alias             - (opt) channel alias/name, defaults to the
  last (non-default) entry in identifier
  (i.e. signal model path)
  descr             - (opt) channel description (e.g. used in report entries)
```

Reimplemented from ttk_selftest.hil_devices._digital_devices.DigitalChannel.

7.190.3 Member Function Documentation

7.190.3.1 info()

```
ttk_selftest.hil_devices._digital_devices.DigitalInput.info (
              self )
```

Get an info string for this digital input channel: alias, FIU channel, maximum voltage and threshold voltage plus description, if available.

Returns

a formatted string.

Reimplemented from ttk_selftest.hil_devices._digital_devices.DigitalChannel.

7.190.4 Member Data Documentation

7.190.4.1 max_voltage

```
ttk_selftest.hil_devices._digital_devices.DigitalInput.max_voltage
```

7.190.4.2 selftest_var

```
ttk_selftest.hil_devices._digital_devices.DigitalInput.selftest_var
```

7.190.4.3 threshold_voltage

```
ttk_selftest.hil_devices._digital_devices.DigitalInput.threshold_voltage
```

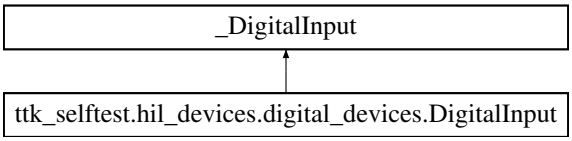
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/_digital_devices.py (v2.0.0)

7.191 ttk_selftest.hil_devices.digital_devices.DigitalInput Class Reference

Base digital input container for a testable digital input and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices.digital_devices.DigitalInput:



Public Member Functions

- __init__ (self, context, base_path, sig_channel, max_voltage, threshold_voltage=None, preconditions=None, wait_time_ms=None, alias="", descr="")

  Digital input initialization.
- info (self)

  Get an info string for this digital input channel: alias, FIU channel, maximum voltage and threshold voltage plus description, if available.

7.191.1 Detailed Description

Base digital input container for a testable digital input and relevant associated variables/values.

7.191.2 Constructor & Destructor Documentation

7.191.2.1 __init__()

```
ttk_selftest.hil_devices.digital_devices.DigitalInput.__init__ (
            self,
            context,
            base_path,
            sig_channel,
            max_voltage,
            threshold_voltage = None,
            preconditions = None,
            wait_time_ms = None,
            alias = "",
            descr = "" )
```

Digital input initialization.

Parameters

context	parent context
base_path	base path to digital input subsystem
sig_channel	fiu channel of the digital input channel
max_voltage	maximum physical voltage in V of the channel
threshold_voltage	(opt) voltage threshold in V for high/low detection defaults to 2.5V (due to the threshold voltage of the digital signal conditioning card DSK)
preconditions	(opt) preconditions that have to be set before the device can be checked. default is None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The function specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
  # For special cases, the preconditions may also contain function calls
  # (which might be useful to insert additional delays or similar):
  preconditions = [
      { "device": Voltage, "value": 1},
      { "func":   time.sleep,
        "args":   [1.0], # positional arguments
        "kwargs": {},    # keyword arguments
      },
      {"device": ClampSwitch, "value": 1},
      # or a simple call like this:
      { "func":  someCustomFunction,
      },
      ...
  ]
wait_time_ms      - (opt) waiting time in milliseconds between
channel stimulation and channel check
defaults to 100ms
alias             - (opt) channel alias/name, defaults to the
last (non-default) entry in identifier
(i.e. signal model path)
descr             - (opt) channel description (e.g. used in report entries)
```

7.191.3 Member Function Documentation

7.191.3.1 info()

```
ttk_selftest.hil_devices.digital_devices.DigitalInput.info (
              self )
```

Get an info string for this digital input channel: alias, FIU channel, maximum voltage and threshold voltage plus description, if available.

Returns

a formatted string.

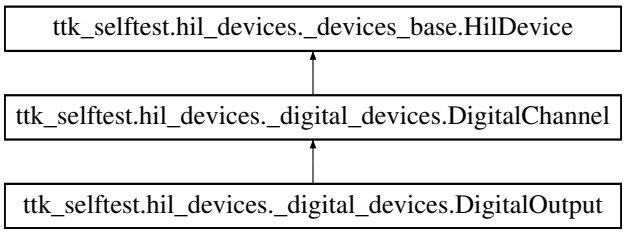
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/digital_devices.py (v2.0.0)

7.192 ttk_selftest.hil_devices._digital_devices.DigitalOutput Class Reference

Base digital output container for a testable digital output and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices._digital_devices.DigitalOutput:



Public Member Functions

- __init__ (self, context, base_path, sig_channel, max_voltage, max_low_voltage=None, min_high_voltage=None, preconditions=None, wait_time_ms=None, alias="", descr="")

  Digital output initialization.
- info (self)

  Get an info string for this digital output channel: alias, FIU channel, maximum voltage, maximum voltage for state "Low" and minimum voltage for state "High" plus description, if available.

Public Attributes

- max_low_voltage
- min_high_voltage
- selftest_var
- alias
- max_voltage

7.192.1 Detailed Description

Base digital output container for a testable digital output and relevant associated variables/values.

7.192.2 Constructor & Destructor Documentation

7.192.2.1 __init__()

```
ttk_selftest.hil_devices._digital_devices.DigitalOutput.__init__ (
                self,
                context,
                base_path,
                sig_channel,
                max_voltage,
                max_low_voltage = None,
                min_high_voltage = None,
                preconditions = None,
                wait_time_ms = None,
                alias = "",
                descr = "" )
```

Digital output initialization.

Parameters

context	parent context
base_path	base path to digital output subsystem
sig_channel	fiu channel of the digital output channel
max_voltage	maximum physical voltage in V of the channel
max_low_voltage	(opt) maximum allowed voltage in V of state "Low" defaults to 0.5V
min_high_voltage	(opt) minimum allowed voltage in V of state "High" defaults to 90% of max_voltage
preconditions	(opt) preconditions that have to be set before the device can be checked. default is None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The function specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
# For special cases, the preconditions may also contain function calls
# (which might be useful to insert additional delays or similar):
preconditions = [
    { "device": Voltage, "value": 1},
    { "func":   time.sleep,
      "args":   [1.0], # positional arguments
      "kwargs": {},    # keyword arguments
    },
    {"device": ClampSwitch, "value": 1},
    # or a simple call like this:
    { "func":  someCustomFunction,
    },
    ...
```

```
  ]
wait_time_ms      - (opt) waiting time in milliseconds between
channel stimulation and channel check
defaults to 100ms
alias             - (opt) channel alias/name, defaults to the
last (non-default) entry in identifier
(i.e. signal model path)
descr             - (opt) channel description (e.g. used in report entries)
```

Reimplemented from ttk_selftest.hil_devices._digital_devices.DigitalChannel.

7.192.3 Member Function Documentation

7.192.3.1 info()

```
ttk_selftest.hil_devices._digital_devices.DigitalOutput.info (
              self )
```

Get an info string for this digital output channel: alias, FIU channel, maximum voltage, maximum voltage for state "Low" and minimum voltage for state "High" plus description, if available.

Returns

a formatted string.

Reimplemented from ttk_selftest.hil_devices._digital_devices.DigitalChannel.

7.192.4 Member Data Documentation

7.192.4.1 alias

```
ttk_selftest.hil_devices._digital_devices.DigitalOutput.alias
```

7.192.4.2 max_low_voltage

```
ttk_selftest.hil_devices._digital_devices.DigitalOutput.max_low_voltage
```

7.192.4.3 max_voltage

```
ttk_selftest.hil_devices._digital_devices.DigitalOutput.max_voltage
```

7.192.4.4 min_high_voltage

```
ttk_selftest.hil_devices._digital_devices.DigitalOutput.min_high_voltage
```

7.192.4.5 selftest_var

```
ttk_selftest.hil_devices._digital_devices.DigitalOutput.selftest_var
```

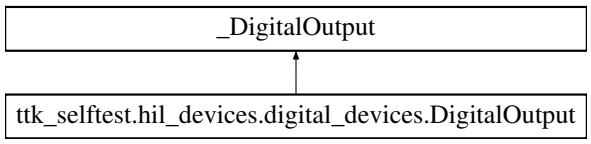
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/_digital_devices.py (v2.0.0)

7.193 ttk_selftest.hil_devices.digital_devices.DigitalOutput Class Reference

Base digital output container for a testable digital output and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices.digital_devices.DigitalOutput:



Public Member Functions

- __init__ (self, context, base_path, sig_channel, max_voltage, max_low_voltage=None, min_high_voltage=None, preconditions=None, wait_time_ms=None, alias="", descr="")

  Digital output initialization.
- info (self)

  Get an info string for this digital output channel: alias, FIU channel, maximum voltage, maximum voltage for state "Low" and minimum voltage for state "High" plus description, if available.

7.193.1 Detailed Description

Base digital output container for a testable digital output and relevant associated variables/values.

7.193.2 Constructor & Destructor Documentation

7.193.2.1 __init__()

```
ttk_selftest.hil_devices.digital_devices.DigitalOutput.__init__ (
              self,
              context,
              base_path,
              sig_channel,
              max_voltage,
              max_low_voltage = None,
              min_high_voltage = None,
              preconditions = None,
              wait_time_ms = None,
              alias = "",
              descr = "" )
```

Digital output initialization.

Parameters

context	parent context
base_path	base path to digital output subsystem
sig_channel	fiu channel of the digital output channel
max_voltage	maximum physical voltage in V of the channel
max_low_voltage	(opt) maximum allowed voltage in V of state "Low" defaults to 0.5V
min_high_voltage	(opt) minimum allowed voltage in V of state "High" defaults to 90% of max_voltage
preconditions	(opt) preconditions that have to be set before the device can be checked. default is None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The function specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
  # For special cases, the preconditions may also contain function calls
  # (which might be useful to insert additional delays or similar):
  preconditions = [
      { "device": Voltage, "value": 1},
      { "func":   time.sleep,
        "args":   [1.0], # positional arguments
        "kwargs": {},    # keyword arguments
      },
      {"device": ClampSwitch, "value": 1},
      # or a simple call like this:
      { "func":  someCustomFunction,
      },
      ...
  ]
wait_time_ms      - (opt) waiting time in milliseconds between
channel stimulation and channel check
defaults to 100ms
alias             - (opt) channel alias/name, defaults to the
last (non-default) entry in identifier
(i.e. signal model path)
descr             - (opt) channel description (e.g. used in report entries)
```

7.193.3 Member Function Documentation

7.193.3.1 info()

```
ttk_selftest.hil_devices.digital_devices.DigitalOutput.info (
              self )
```

Get an info string for this digital output channel: alias, FIU channel, maximum voltage, maximum voltage for state "Low" and minimum voltage for state "High" plus description, if available.

Returns

a formatted string.

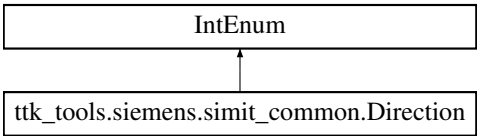
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/digital_devices.py (v2.0.0)

7.194 ttk_tools.siemens.simit_common.Direction Class Reference

"E/A-Kennung, d.

Inheritance diagram for ttk_tools.siemens.simit_common.Direction:



Static Public Attributes

- int shm_to_simit = 0
- int simit_to_shm = 1

7.194.1 Detailed Description

"E/A-Kennung, d.

h. die Kennung, ob das Signal ein Eingangssignal (Kennwert 0) oder Ausgangssignal (Kennwert 1) ist"

7.194.2 Member Data Documentation

7.194.2.1 shm_to_simit

```
int ttk_tools.siemens.simit_common.Direction.shm_to_simit = 0  [static]
```

7.194.2.2 simit_to_shm

```
int ttk_tools.siemens.simit_common.Direction.simit_to_shm = 1  [static]
```

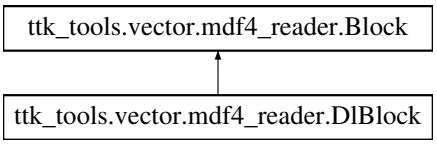
The documentation for this class was generated from the following file:

- ttk_tools/siemens/simit_common.py (v1.0.0)

7.195 ttk_tools.vector.mdf4_reader.DlBlock Class Reference

Data List Block.

Inheritance diagram for ttk_tools.vector.mdf4_reader.DlBlock:



Public Member Functions

- __init__ (self, file_handle, pos=None)

  Read DLBLOCK from file_handle, starting at the current or supplied position.

Public Attributes

- dl_count
- dl_flags
- dl_equal_length
- dl_offset
- dl_dl_next

  Pointer to next data list block (DLBLOCK) (can be NIL).
- dl_data

  Pointers to the data blocks (DTBLOCK, SDBLOCK or RDBLOCK or a DZBLOCK of the respective block type).

7.195.1 Detailed Description

Data List Block.

The DLBLOCK references

- a list of data blocks (DTBLOCK) or
- a list of signal data blocks (SDBLOCK) or
- a list of reduction data blocks (RDBLOCK). This list of blocks is equivalent to using a single (signal/reduction) data block and can be used to avoid a huge data block by splitting it into smaller parts.

Each single block referenced in this list also can be replaced by a zipped data block (DZBLOCK) for the respective block type. There is no change to the DLBLOCK when using a DZBLOCK instead of the original block type with uncompressed data section: A DZBLOCK always must be treated as if it was its uncompressed equivalent.

In order to optimize read access of a specific record / signal value in the list of DTBLOCKs/SDBLOCKs/RDBLOCKs, a list of "start offsets" for the data sections of the referenced blocks is maintained. If the (uncompressed) data sections of all referenced blocks would be concatenated, the start offset of a block defines the position of its data section in this concatenation.

For the special case that all (uncompressed) data sections of the referenced blocks have the same length, a calculation can be used instead of a search in the list of start offsets. To indicate this possibility, an extra element specifies the equal data section length.

7.195.2 Constructor & Destructor Documentation

7.195.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.DlBlock.__init__ (
              self,
              file_handle,
              pos = None )
```

Read DLBLOCK from file_handle, starting at the current or supplied position.

Reimplemented from ttk_tools.vector.mdf4_reader.Block.

7.195.3 Member Data Documentation

7.195.3.1 dl_count

```
ttk_tools.vector.mdf4_reader.DlBlock.dl_count
```

7.195.3.2 dl_data

```
ttk_tools.vector.mdf4_reader.DlBlock.dl_data
```

Pointers to the data blocks (DTBLOCK, SDBLOCK or RDBLOCK or a DZBLOCK of the respective block type).

None of the links in the list can be NIL.

It is not allowed to mix the data block types: all links must uniformly reference the same data block type (DTBLOCK/SDBLOCK/RDBLOCK or an equivalent DZBLOCK). Also all DLBLOCKs in the linked list of DLBLOCKs must reference the same data block type

Note: a mixture between DZBLOCKs and uncompressed data blocks is allowed. However, if a DZBLOCK is in the list, the complete linked list of DLBLOCKs must be preceded by a HLBLOCK. See also explanation of Header List Block HLBLOCK .

7.195.3.3 dl_dl_next

```
ttk_tools.vector.mdf4_reader.DlBlock.dl_dl_next
```

Pointer to next data list block (DLBLOCK) (can be NIL).

7.195.3.4 dl_equal_length

```
ttk_tools.vector.mdf4_reader.DlBlock.dl_equal_length
```

7.195.3.5 dl_flags

```
ttk_tools.vector.mdf4_reader.DlBlock.dl_flags
```

7.195.3.6 dl_offset

```
ttk_tools.vector.mdf4_reader.DlBlock.dl_offset
```

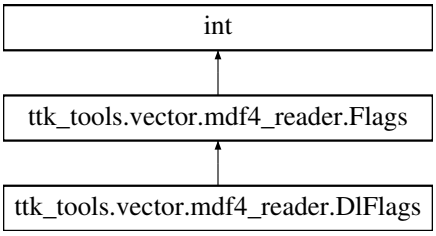
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.196 ttk_tools.vector.mdf4_reader.DlFlags Class Reference

Container for decoded uint8 dl_flags.

Inheritance diagram for ttk_tools.vector.mdf4_reader.DlFlags:



Public Member Functions

- __init__ (self, flags)

Public Attributes

- equal_length

  Bit 0: Equal length flag If set, each DLBLOCK in the linked list has the same number of referenced blocks (dl_count) and the (uncompressed) data sections of the blocks referenced by dl_data have a common length given by dl_equal_length.

7.196.1 Detailed Description

Container for decoded uint8 dl_flags.

Only intended for read-only access, changes in named flags will not modify original "flags" value.

7.196.2 Constructor & Destructor Documentation

7.196.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.DlFlags.__init__ (
            self,
            flags )
```

Reimplemented from ttk_tools.vector.mdf4_reader.Flags.

7.196.3 Member Data Documentation

7.196.3.1 equal_length

`ttk_tools.vector.mdf4_reader.DlFlags.equal_length`

Bit 0: Equal length flag If set, each DLBLOCK in the linked list has the same number of referenced blocks (dl_↩ count) and the (uncompressed) data sections of the blocks referenced by dl_data have a common length given by dl_equal_length.

The only exception is that for the last DLBLOCK in the list (dl_dl_next = NIL), its number of referenced blocks dl_count can be <= dl_count of the previous DLBLOCK, and the data section length of its last referenced block (dl_data[dl_count-1]) can be <= dl_equal_length. If not set, the number of referenced blocks dl_count may be different for each DLBLOCK in the linked list, and the data section lengths of the referenced blocks may be different and a table of offsets is given in dl_offset. Note: The value of the "equal length" flag must be equal for all DLBLOCKs in the linked list.

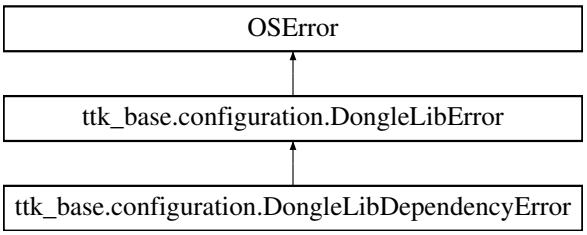
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.197 ttk_base.configuration.DongleLibDependencyError Class Reference

Base class for data parser errors.

Inheritance diagram for ttk_base.configuration.DongleLibDependencyError:



7.197.1 Detailed Description

Base class for data parser errors.

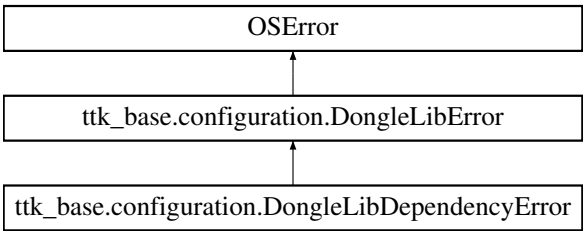
The documentation for this class was generated from the following file:

- ttk_base/configuration.py (v3.8.1)

7.198 ttk_base.configuration.DongleLibError Class Reference

Errors while loading Dongle DLLs.

Inheritance diagram for ttk_base.configuration.DongleLibError:



7.198.1 Detailed Description

Errors while loading Dongle DLLs.

DLL is present but failed to load; most likely a matching Visual-C Redistributable is missing. Detecting missing dependencies in Visual-C-built binaries is a bit less straight-forward than it could be, as is checking for installed redistributables.

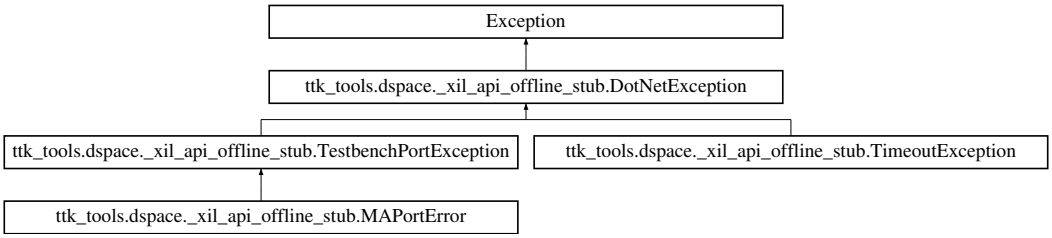
The documentation for this class was generated from the following file:

- ttk_base/configuration.py (v3.8.1)

7.199 ttk_tools.dspace._xil_api_offline_stub.DotNetException Class Reference

Stub for .NET System.Exception.

Inheritance diagram for ttk_tools.dspace._xil_api_offline_stub.DotNetException:



7.199.1 Detailed Description

Stub for .NET System.Exception.

Non-stubbed hierarchy:

Exception -> (.NET) System.Exception

Stub for .NET System.TimeoutException (this gets raised when MAPortFactory times out).

Non-stubbed hierarchy:

Exception -> (.NET) System.DotNetException -> System.TimeoutException

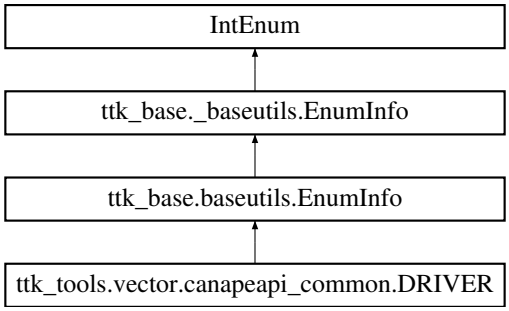
The documentation for this class was generated from the following file:

- ttk_tools/dspace/_xil_api_offline_stub.py (v2.1.2)

7.200 ttk_tools.vector.canapeapi_common.DRIVER Class Reference

Driver type constants (`ASAP3_DRIVER_*`) from enum `tDriverType`.

Inheritance diagram for ttk_tools.vector.canapeapi_common.DRIVER:



Static Public Attributes

- int UNKNOWN = 0

  unknown driver type
- int CCP = 1

  CCP: CAN Calibration Protocol Driver.
- int XCP = 2

  XCP: Universal Measurement and Calibration Protocol Driver.
- int CAN = 20

  CAN: Controller Area Network Driver.
- int KWPCAN = 30

  Keyword protocol 2000 on CAN via ISO/TF2.
- int HEXEDIT = 40

  Pure offline driver.
- int ANALOG = 50

  Analog measurement data (e.g.
- int CANOPEN = 60

  CANopen Driver.
- int CANDELA = 70

  CANdela Diagnostic Driver.
- int ENVIRONMENT = 80

  Environment (access to global variables)
- int LIN = 90

  LIN: Local Interconnect Network.
- int FLX = 100

  FlexRay Driver.
- int FUNC = 110

  Functional Diagnostic Driver.
- int NIDAQMX = 120

  NI DAQ Driver 'National Instruments'.
- int XCP_RAMSCOPE = 130

  XCP Driver for RAMscope.
- int SYSTEM = 140

  System Driver.

- int ETH = 150

  Ethernet driver (CANape 14++)
- int DAIO_SYSTEM = 160

  DAIO system driver (CANape 14++)
- int KWP2000 = 255

  Keyword protocol 2000 on K-Line.

7.200.1 Detailed Description

Driver type constants (`ASAP3_DRIVER_*`) from enum `tDriverType`.

7.200.2 Member Data Documentation

7.200.2.1 ANALOG

```
int ttk_tools.vector.canapeapi_common.DRIVER.ANALOG = 50  [static]
```

Analog measurement data (e.g.

'National Instruments' PCMCIA-card)

7.200.2.2 CAN

```
int ttk_tools.vector.canapeapi_common.DRIVER.CAN = 20  [static]
```

CAN: Controller Area Network Driver.

7.200.2.3 CANDELA

```
int ttk_tools.vector.canapeapi_common.DRIVER.CANDELA = 70  [static]
```

CANdela Diagnostic Driver.

7.200.2.4 CANOPEN

```
int ttk_tools.vector.canapeapi_common.DRIVER.CANOPEN = 60  [static]
```

CANopen Driver.

7.200.2.5 CCP

```
int ttk_tools.vector.canapeapi_common.DRIVER.CCP = 1  [static]
```

CCP: CAN Calibration Protocol Driver.

7.200.2.6 DAIO_SYSTEM

```
int ttk_tools.vector.canapeapi_common.DRIVER.DAIO_SYSTEM = 160  [static]
```

DAIO system driver (CANape 14++)

7.200.2.7 ENVIRONMENT

```
int ttk_tools.vector.canapeapi_common.DRIVER.ENVIRONMENT = 80  [static]
```

Environment (access to global variables)

7.200.2.8 ETH

```
int ttk_tools.vector.canapeapi_common.DRIVER.ETH = 150  [static]
```

Ethernet driver (CANape 14++)

7.200.2.9 FLX

```
int ttk_tools.vector.canapeapi_common.DRIVER.FLX = 100  [static]
```

FlexRay Driver.

7.200.2.10 FUNC

```
int ttk_tools.vector.canapeapi_common.DRIVER.FUNC = 110  [static]
```

Functional Diagnostic Driver.

7.200.2.11 HEXEDIT

```
int ttk_tools.vector.canapeapi_common.DRIVER.HEXEDIT = 40  [static]
```

Pure offline driver.

7.200.2.12 KWP2000

```
int ttk_tools.vector.canapeapi_common.DRIVER.KWP2000 = 255  [static]
```

Keyword protocol 2000 on K-Line.

7.200.2.13 KWPCAN

```
int ttk_tools.vector.canapeapi_common.DRIVER.KWPCAN = 30  [static]
```

Keyword protocol 2000 on CAN via ISO/TF2.

7.200.2.14 LIN

```
int ttk_tools.vector.canapeapi_common.DRIVER.LIN = 90  [static]
```

LIN: Local Interconnect Network.

7.200.2.15 NIDAQMX

```
int ttk_tools.vector.canapeapi_common.DRIVER.NIDAQMX = 120  [static]
```

NI DAQ Driver 'National Instruments'.

7.200.2.16 SYSTEM

```
int ttk_tools.vector.canapeapi_common.DRIVER.SYSTEM = 140  [static]
```

System Driver.

7.200.2.17 UNKNOWN

```
int ttk_tools.vector.canapeapi_common.DRIVER.UNKNOWN = 0  [static]
```

unknown driver type

7.200.2.18 XCP

```
int ttk_tools.vector.canapeapi_common.DRIVER.XCP = 2  [static]
```

XCP: Universal Measurement and Calibration Protocol Driver.

7.200.2.19 XCP_RAMSCOPE

```
int ttk_tools.vector.canapeapi_common.DRIVER.XCP_RAMSCOPE = 130  [static]
```

XCP Driver for RAMscope.

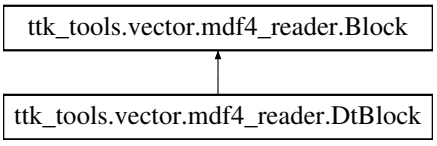
The documentation for this class was generated from the following file:

- ttk_tools/vector/canapeapi_common.py (v3.1.2)

7.201 ttk_tools.vector.mdf4_reader.DtBlock Class Reference

Data Block.

Inheritance diagram for ttk_tools.vector.mdf4_reader.DtBlock:



Public Member Functions

- __init__ (self, file_handle, pos=None)

  Read DTBLOCK from file_handle, starting at the current or supplied position.

Protected Attributes

- _data

7.201.1 Detailed Description

Data Block.

The data section of a DTBLOCK contains a sequence of records. It contains records of all channel groups assigned to its parent DGBLOCK.

7.201.2 Constructor & Destructor Documentation

7.201.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.DtBlock.__init__ (
              self,
              file_handle,
              pos = None )
```

Read DTBLOCK from file_handle, starting at the current or supplied position.

Also handles DTBLOCKs with compressed data (DZBLOCKs).

Reimplemented from ttk_tools.vector.mdf4_reader.Block.

7.201.3 Member Data Documentation

7.201.3.1 _data

```
ttk_tools.vector.mdf4_reader.DtBlock._data  [protected]
```

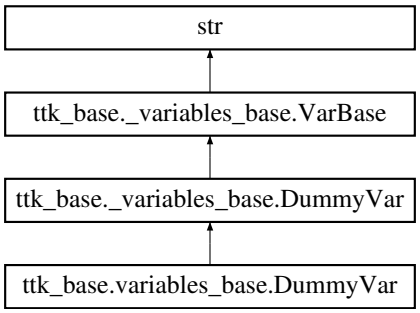
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.202 ttk_base._variables_base.DummyVar Class Reference

A minimal dummy test variable not connected to any external interface.

Inheritance diagram for ttk_base._variables_base.DummyVar:



Public Member Functions

- __init__ (self, context, identifier, unit="", alias="", descr="", lookup=None, default=None, resettable=True, fmt=None)

  Variable initialization.
- get (self, *args, **kwargs)

  Get the current value.
- set (self, value, *args, **kwargs)

  Set a new value.

Public Attributes

- alias
- last_value

  initial "dummy" value
- orig_value

7.202.1 Detailed Description

A minimal dummy test variable not connected to any external interface.

This is a simple and direct implementation of _variables_base.VarBase with get()/set() methods adapted to write to a member variable.

7.202.2 Constructor & Destructor Documentation

7.202.2.1 __init__()

```
ttk_base._variables_base.DummyVar.__init__ (
              self,
              context,
              identifier,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              default = None,
              resettable = True,
              fmt = None )
```

Variable initialization.

Parameters

context	parent context that handles tool references This is usually a reference to a variables container or to a tool/api instance, but can be anything for a "DummyVar"
identifier	variable identifier/name in the calibration tool
unit	(optional) unit string for the variable
alias	(optional) alias name, defaults to identifier
descr	description for the variable
lookup	lookup dictionary for states {value => "state name"}
default	manually defined default value for the variable. Will be used as "reset" value instead of the original value before the first set operation if set to anything but None.
resettable	enable/disable reset() for this variable. If set to False, a call to reset() will be ignored.
fmt	format string or keyword to use for custom value representation, e.g. "%.2f" or "hex"

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_base.variables_base.DummyVar.

7.202.3 Member Function Documentation

7.202.3.1 get()

```
ttk_base._variables_base.DummyVar.get (
              self,
             * args,
             ** kwargs )
```

Get the current value.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_base.variables_base.DummyVar.

7.202.3.2 set()

```
ttk_base._variables_base.DummyVar.set (
              self,
              value,
             * args,
             ** kwargs )
```

Set a new value.

Parameters

value	value to set
∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Returns

the original value before the set operation occurred.

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_base.variables_base.DummyVar.

7.202.4 Member Data Documentation

7.202.4.1 alias

`ttk_base._variables_base.DummyVar.alias`

7.202.4.2 last_value

`ttk_base._variables_base.DummyVar.last_value`

initial "dummy" value

7.202.4.3 orig_value

`ttk_base._variables_base.DummyVar.orig_value`

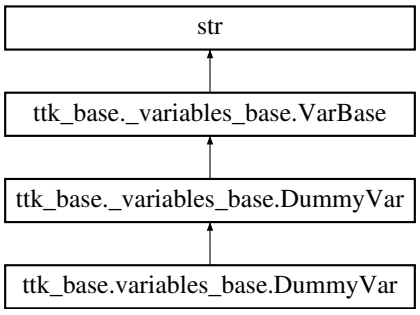
The documentation for this class was generated from the following file:

- ttk_base/_variables_base.py (v2.8.1)

7.203 ttk_base.variables_base.DummyVar Class Reference

A minimal dummy test variable not connected to any external interface.

Inheritance diagram for ttk_base.variables_base.DummyVar:



Public Member Functions

- __init__ (self, context, identifier, unit="", alias="", descr="", lookup=None, default=None, resettable=True, fmt=None)

  Variable initialization.
- get (self, ∗args, ∗∗kwargs)

  Get the current value.
- set (self, value, ∗args, ∗∗kwargs)

  Set a new value.
- getState (self)

  Get a textual state for the variable's current data.
- setState (self, state)

  Set a textual state to the variable.
- reset (self, verbosity=1)

  Reset the variable to its initial value before the first set() call.
- info (self)

  Get an info string for this variable (alias and current value plus unit and lookup value, if available).

7.203.1 Detailed Description

A minimal dummy test variable not connected to any external interface.

This is a simple (and direct) implementation of _variables_base.VarBase with get() and set() methods adapted to write to a member variable instead of accessing an external API.

7.203.2 Constructor & Destructor Documentation

7.203.2.1 __init__()

```
ttk_base.variables_base.DummyVar.__init__ (
              self,
              context,
              identifier,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              default = None,
              resettable = True,
              fmt = None )
```

Variable initialization.

Parameters

context	parent context that handles tool references This is usually a reference to a variables container or to a tool/api instance, but can be anything for a "DummyVar"
identifier	variable identifier/name in the calibration tool
unit	(optional) unit string for the variable
alias	(optional) alias name, defaults to varname
descr	description for the variable
lookup	lookup dictionary for states {value => "state name"}
default	manually defined default value for the variable. Will be used as "reset" value instead of the original value before the first set operation if set to anything but None.
resettable	enable/disable reset() for this variable. If set to False, a call to reset() will be ignored.
fmt	format string or keyword to use for value representation, e.g. "%.2f" or "hex"

Reimplemented from ttk_base._variables_base.DummyVar.

7.203.3 Member Function Documentation

7.203.3.1 get()

```
ttk_base.variables_base.DummyVar.get (
              self,
             * args,
             ** kwargs )
```

Get the current value.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Returns

the current value.

Reimplemented from ttk_base._variables_base.DummyVar.

7.203.3.2 getState()

```
ttk_base.variables_base.DummyVar.getState (
              self )
```

Get a textual state for the variable's current data.

This uses the lookup values supplied in constructor. Fallback (if no matching state was found) will be the unmapped numeric value.

Returns

the description for the value.

Reimplemented from ttk_base._variables_base.VarBase.

7.203.3.3 info()

```
ttk_base.variables_base.DummyVar.info (
              self )
```

Get an info string for this variable (alias and current value plus unit and lookup value, if available).

Example:

```
>> print voltage_var.info()
Var Name: 12 V
>> print status_var.info()
Status Var Name: 1 (valid)
```

Returns

a formatted string.

Reimplemented from ttk_base._variables_base.VarBase.

7.203.3.4 reset()

```
ttk_base.variables_base.DummyVar.reset (
              self,
              verbosity = 1 )
```

Reset the variable to its initial value before the first set() call.

Parameters

verbosity	verbosity of log output 0: silent 1: show calls to non-resettable variables 2: show all reset calls

Reimplemented from ttk_base._variables_base.VarBase.

7.203.3.5 set()

```
ttk_base.variables_base.DummyVar.set (
              self,
              value,
             * args,
             ** kwargs )
```

Set a new value.

Parameters

value	value to set
∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Returns

the original value before the set operation occurred.

Reimplemented from ttk_base._variables_base.DummyVar.

7.203.3.6 setState()

```
ttk_base.variables_base.DummyVar.setState (
              self,
              state )
```

Set a textual state to the variable.

This uses the lookup values supplied in constructor. Fallback (if no matching state text was found) will be the unmapped input state.

Parameters

state	named state to set

Returns

the description for the previously set value.

Reimplemented from ttk_base._variables_base.VarBase.

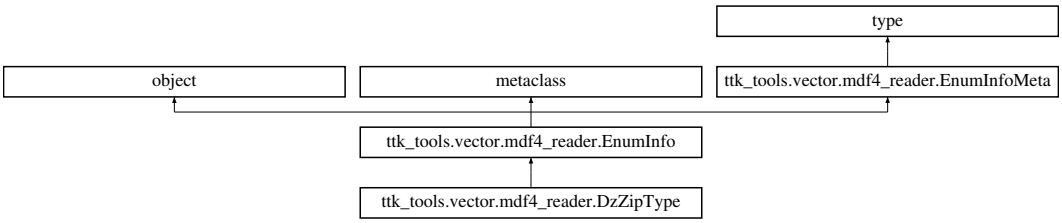
The documentation for this class was generated from the following file:

- ttk_base/variables_base.py (v2.2)

7.204 ttk_tools.vector.mdf4_reader.DzZipType Class Reference

Zip algorithm type (dz_zip_type) used to compress data stored in dz_data.

Inheritance diagram for ttk_tools.vector.mdf4_reader.DzZipType:



Static Public Attributes

- int DEFLATE = 0
- int TRANSP_DEFLATE = 1

7.204.1 Detailed Description

Zip algorithm type (dz_zip_type) used to compress data stored in dz_data.

7.204.2 Member Data Documentation

7.204.2.1 DEFLATE

```
int ttk_tools.vector.mdf4_reader.DzZipType.DEFLATE = 0  [static]
```

7.204.2.2 TRANSP_DEFLATE

```
int ttk_tools.vector.mdf4_reader.DzZipType.TRANSP_DEFLATE = 1  [static]
```

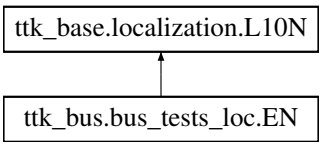
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.205 ttk_bus.bus_tests_loc.EN Class Reference

English localization.

Inheritance diagram for ttk_bus.bus_tests_loc.EN:



Static Protected Attributes

- dict _localization

7.205.1 Detailed Description

English localization.

7.205.2 Member Data Documentation

7.205.2.1 _localization

```
dict ttk_bus.bus_tests_loc.EN._localization  [static], [protected]
```

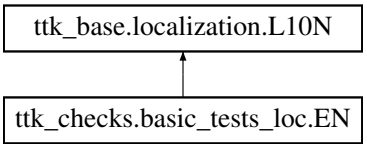
The documentation for this class was generated from the following file:

- ttk_bus/bus_tests_loc.py (v1.6)

7.206 ttk_checks.basic_tests_loc.EN Class Reference

English localization.

Inheritance diagram for ttk_checks.basic_tests_loc.EN:



Static Protected Attributes

- dict _localization

7.206.1 Detailed Description

English localization.

7.206.2 Member Data Documentation

7.206.2.1 _localization

```
dict ttk_checks.basic_tests_loc.EN._localization  [static], [protected]
```

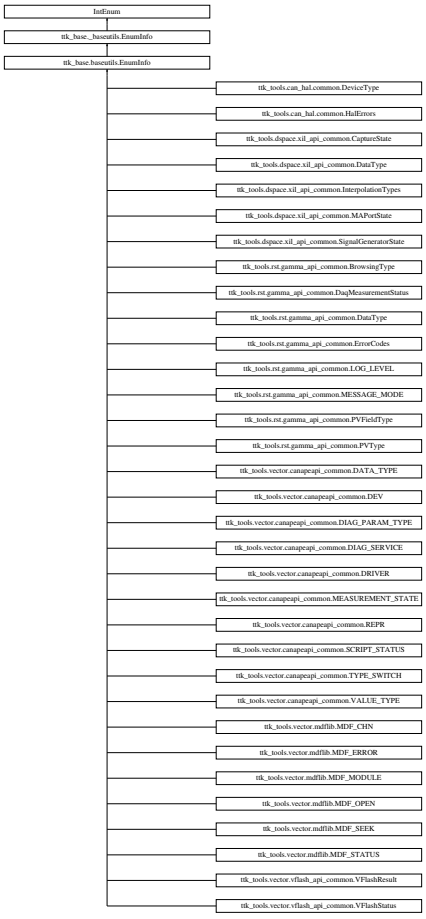
The documentation for this class was generated from the following file:

- ttk_checks/basic_tests_loc.py (v1.20.1)

7.207 ttk_base._baseutils.EnumInfo Class Reference

Utility/convenience class for "enum-like" value container classes.

Inheritance diagram for ttk_base._baseutils.EnumInfo:



Public Member Functions

- getInfo (cls, value, fallback="<undefined: %(value)s>")

  Get a textual info for the supplied value (if one is available).
- getName (cls, value, fallback="<undefined: %(value)s>")

  Get a textual representation for the supplied value from matching "constant" in class members.
- values (cls)

  Get all defined values as a list.
- printDefinedValues (cls, min_width=8)

  Print all defined (int) values of this enum-ish container, along with their text descriptions.

7.207.1 Detailed Description

Utility/convenience class for "enum-like" value container classes.

Provides a getInfo (class-)method that can be used to get a textual representation of contained values.

Example:

```
# default text is the (first) matching numeric member name:
class SOME_ENUM(EnumInfo):
    ONE   = 1
    TWO   = 2
    THREE = 3

>> SOME_ENUM.ONE
1
>> SOME_ENUM.getInfo(2)
'TWO'
>> SOME_ENUM.getInfo(0xFF)
'<undefined: 255>'
```

Example:

```
# customized text representations can be supplied via a
# member __info_mapping__:
class SOME_ENUM(EnumInfo):
    ONE   = 1
    TWO   = 2
    THREE = 3
    __info_mapping__ = {
         ONE:     "number one",
         TWO:     "number two",
         THREE:   "number three",
    }
>> SOME_ENUM.ONE
1
>> SOME_ENUM.getInfo(2)
'number two'
>> SOME_ENUM.getInfo(0xFF)
'<undefined: 255>'
```

7.207.2 Member Function Documentation

7.207.2.1 getInfo()

```
ttk_base._baseutils.EnumInfo.getInfo (
              cls,
              value,
              fallback = "<undefined: %(value)s>" )
```

Get a textual info for the supplied value (if one is available).

Info will be taken from a member dictionary **info_mapping** with a fallback to the first member name that matches the supplied value.

Parameters

value	one of this class' "constants"
fallback	fallback text for unknown values

Returns

an info string.

Reimplemented in ttk_base.baseutils.EnumInfo.

7.207.2.2 getName()

```
ttk_base._baseutils.EnumInfo.getName (
              cls,
              value,
              fallback = "<undefined: %(value)s>" )
```

Get a textual representation for the supplied value from matching "constant" in class members.

Parameters

value	one of this class' "constants"
fallback	fallback text for unknown values

Returns

a name string.

Reimplemented in ttk_base.baseutils.EnumInfo.

7.207.2.3 printDefinedValues()

```
ttk_base._baseutils.EnumInfo.printDefinedValues (
              cls,
              min_width = 8 )
```

Print all defined (int) values of this enum-ish container, along with their text descriptions.

Parameters

min_width	minimum width used for key formatting

Reimplemented in ttk_base.baseutils.EnumInfo.

7.207.2.4 values()

```
ttk_base._baseutils.EnumInfo.values (
```

```
        cls )
```

Get all defined values as a list.

Reimplemented in ttk_base.baseutils.EnumInfo.

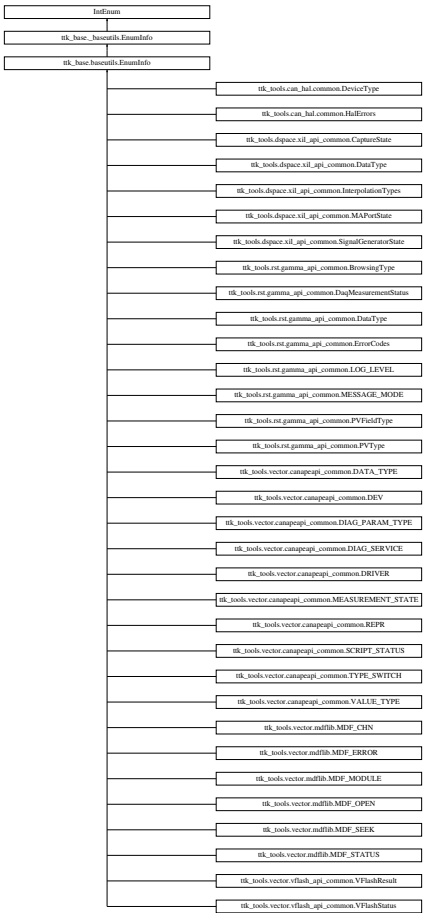
The documentation for this class was generated from the following file:

- ttk_base/_baseutils.py (v2.3.2)

7.208 ttk_base.baseutils.EnumInfo Class Reference

Utility/convenience class for "enum-like" value container classes.

Inheritance diagram for ttk_base.baseutils.EnumInfo:



Public Member Functions

- getInfo (cls, value, fallback="<undefined: %(value)s>")

  Get a textual info for the supplied value (if one is available).
- getName (cls, value, fallback="<undefined: %(value)s>")

  Get a textual representation for the supplied value from matching "constant" in class members.
- values (cls)

  Get all defined values as a list.
- printDefinedValues (cls, min_width=8)

  Print all defined (int) values of this enum-ish container, along with their text descriptions.

7.208.1 Detailed Description

Utility/convenience class for "enum-like" value container classes.

Provides a getInfo (class-)method that can be used to get a textual representation of contained values.

Example:

```
# default text is the (first) matching numeric member name:
class SOME_ENUM(EnumInfo):
    ONE   = 1
    TWO   = 2
    THREE = 3

»> SOME_ENUM.ONE
1
»> SOME_ENUM.getInfo(2)
'TWO'
»> SOME_ENUM.getInfo(0xFF)
'<undefined: 255>'
```

Example:

```
# customized text representations can be supplied via a
# member __info_mapping__:
class SOME_ENUM(EnumInfo):
    ONE   = 1
    TWO   = 2
    THREE = 3
    __info_mapping__ = {
         ONE:     "number one",
         TWO:     "number two",
         THREE:   "number three",
    }
»> SOME_ENUM.ONE
1
»> SOME_ENUM.getInfo(2)
'number two'
»> SOME_ENUM.getInfo(0xFF)
'<undefined: 255>'
```

7.208.2 Member Function Documentation

7.208.2.1 getInfo()

```
ttk_base.baseutils.EnumInfo.getInfo (
               cls,
               value,
               fallback = "<undefined: %(value)s>" )
```

Get a textual info for the supplied value (if one is available).

Info will be taken from a member dictionary **info_mapping** with a fallback to the first member name that matches the supplied value.

Parameters

value	one of this class' "constants"
fallback	fallback text for unknown values

Returns

an info string.

Reimplemented from ttk_base._baseutils.EnumInfo.

7.208.2.2 getName()

```
ttk_base.baseutils.EnumInfo.getName (
            cls,
            value,
            fallback = "<undefined: %(value)s>" )
```

Get a textual representation for the supplied value from matching "constant" in class members.

Parameters

value	one of this class' "constants"
fallback	fallback text for unknown values

Returns

a name string.

Reimplemented from ttk_base._baseutils.EnumInfo.

7.208.2.3 printDefinedValues()

```
ttk_base.baseutils.EnumInfo.printDefinedValues (
            cls,
            min_width = 8 )
```

Print all defined (int) values of this enum-ish container, along with their text descriptions.

Parameters

min_width	minimum width used for key formatting

Reimplemented from ttk_base._baseutils.EnumInfo.

7.208.2.4 values()

```
ttk_base.baseutils.EnumInfo.values (
            cls )
```

Get all defined values as a list.

Reimplemented from ttk_base._baseutils.EnumInfo.

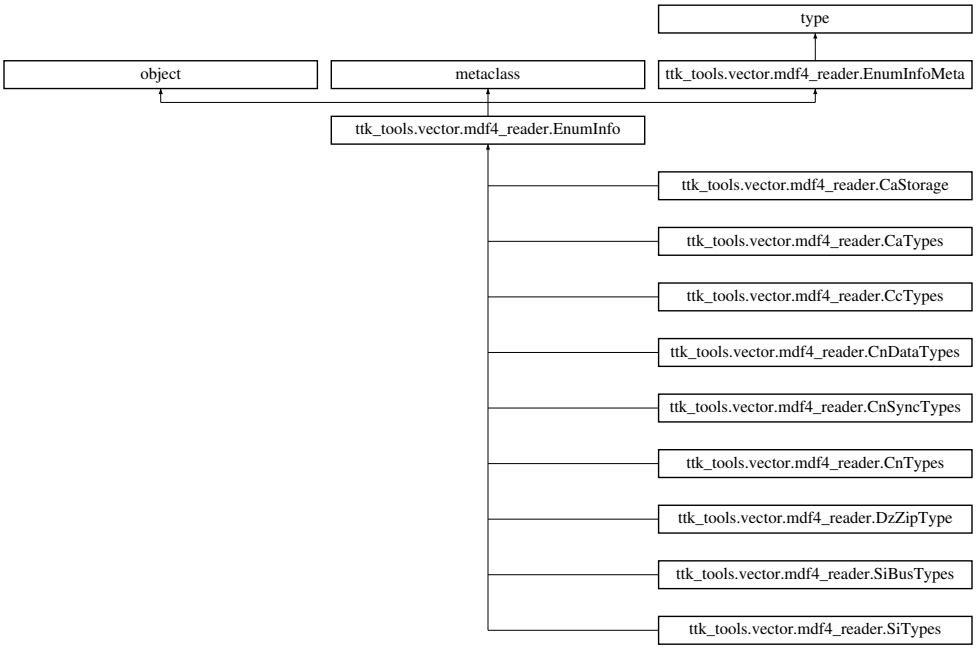
The documentation for this class was generated from the following file:

- ttk_base/baseutils.py (v2.3.1)

7.209 ttk_tools.vector.mdf4_reader.EnumInfo Class Reference

Placeholder for ttk_base.baseutils.EnumInfo.

Inheritance diagram for ttk_tools.vector.mdf4_reader.EnumInfo:



Public Member Functions

- getInfo (cls, value, fallback="<undefined: %(value)s>")

  Get a textual info for the supplied value (if one is available).
- getName (cls, value, fallback="<undefined: %(value)s>")

  Get a textual representation for the supplied value from matching "constant" in class members.

Static Protected Member Functions

- _stripPrefix (name, prefix)

  Strip a prefix from the supplied name.

7.209.1 Detailed Description

Placeholder for ttk_base.baseutils.EnumInfo.

7.209.2 Member Function Documentation

7.209.2.1 _stripPrefix()

```
ttk_tools.vector.mdf4_reader.EnumInfo._stripPrefix (
            name,
            prefix )  [static], [protected]
```

Strip a prefix from the supplied name.

Just for convenience.

Parameters

name	name
prefix	string with prefix to remove

Returns

name as a string.

7.209.2.2 getInfo()

```
ttk_tools.vector.mdf4_reader.EnumInfo.getInfo (
              cls,
              value,
              fallback = "<undefined: %(value)s>" )
```

Get a textual info for the supplied value (if one is available).

Info will be taken from a member dictionary **info_mapping** with a fallback to the first member name that matches the supplied value.

Parameters

value	one of this class' "constants"
fallback	fallback text for unknown values

Returns

an info string.

7.209.2.3 getName()

```
ttk_tools.vector.mdf4_reader.EnumInfo.getName (
              cls,
              value,
              fallback = "<undefined: %(value)s>" )
```

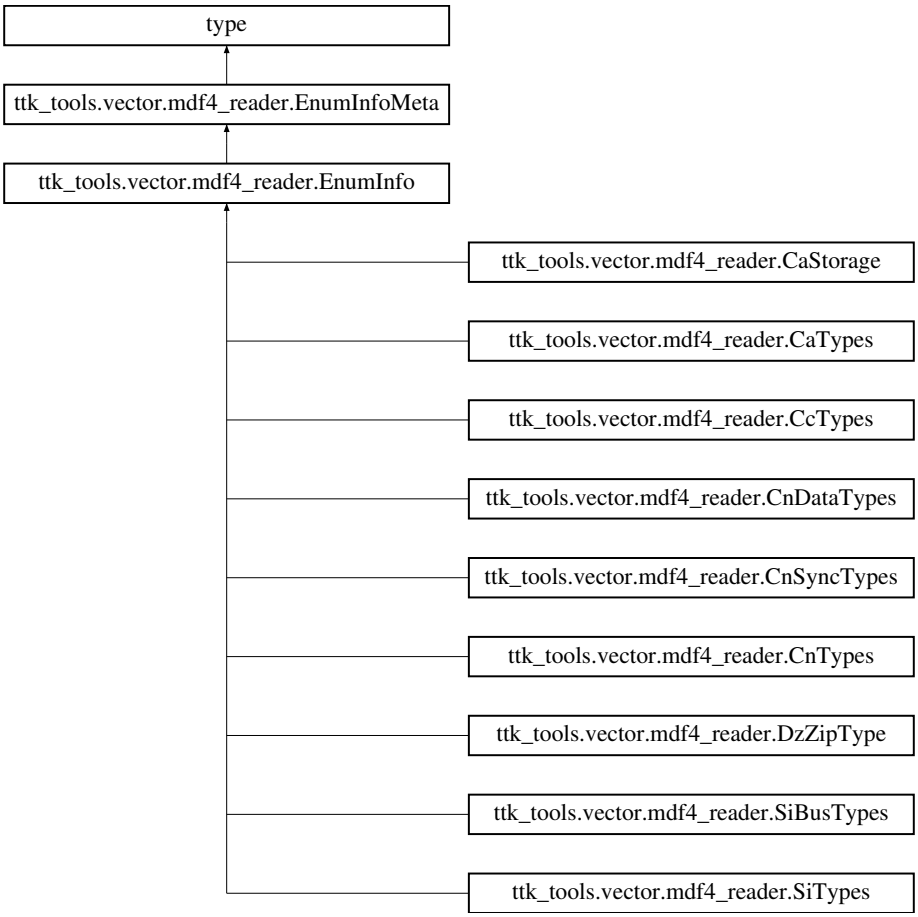
Get a textual representation for the supplied value from matching "constant" in class members.

The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.210 ttk_tools.vector.mdf4_reader.EnumInfoMeta Class Reference

Inheritance diagram for ttk_tools.vector.mdf4_reader.EnumInfoMeta:



Public Member Functions

- __getattr__ (self, name)

7.210.1 Member Function Documentation

7.210.1.1 __getattr__()

```
ttk_tools.vector.mdf4_reader.EnumInfoMeta.__getattr__ (
            self,
            name )
```

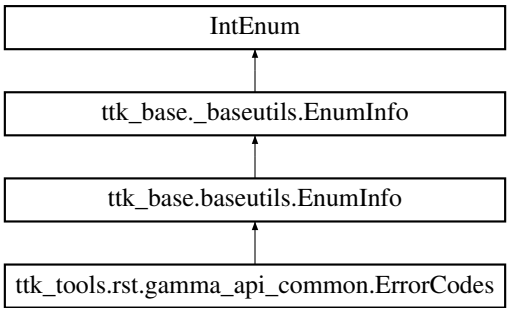
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.211 ttk_tools.rst.gamma_api_common.ErrorCodes Class Reference

Gamma error codes (default return codes of API functions).

Inheritance diagram for ttk_tools.rst.gamma_api_common.ErrorCodes:



Static Public Attributes

- int SUCCESS = 0
- int COM_INVALID_ARGUMENT = -1
- int COM_INTERNAL_ERROR = -2
- int COM_MALLOC_FAILED = -3
- int COM_NOT_IMPLEMENTED = -4
- int COM_STRING_TRUNCATED = -5
- int COM_IO_ERROR = -6
- int COM_VALUE_TRUNCATION = -7
- int COM_BAD_STATE = -8
- int COM_DEAD_CONNECTION = -9
- int COM_TIMED_OUT_RPC = -10
- int COM_SHM_INVALID = -11
- int COM_HIGH_LEVEL_INTERNAL_ERROR = -12
- int COM_SESSION_IS_DEAD = -13
- int ATTACH_MISSING = -100
- int ATTACH_ALREADY = -101
- int ATTACH_FAILED = -102
- int ATTACH_CORRUPTED = -103
- int DETACH_DONE_BUT_CON_DEAD = -104
- int PV_INVALID_SYNTAX = -200
- int PV_NAME_TOO_LONG = -201
- int PV_LOCK_FAILED = -202
- int PV_UNLOCK_FAILED = -203
- int PV_SIM_VALUE_SET_FAILED = -204
- int PV_SIM_VALUE_GET_FAILED = -205
- int PV_INVALID_DATA_TYPE = -206
- int PV_META_OP_NOT_AVAILABLE = -207
- int PV_BOUND_REJECTED = -208
- int PV_NO_COMMON_LOCK = -209
- int PV_DATA_TYPE_MISMATCH = -210
- int PV_CLOSE_DONE_BUT_CORRUPTION = -211
- int PV_FIFO_EMPTY = -212
- int PV_FIFO_FULL = -213
- int PV_TYPE_INVALID_OPERATION = -214

- int PV_FIFO_WOULD_BLOCK = -215
- int PV_SIG_ALREADY_ACTIVATED = -216
- int PV_SIG_ALREADY_DEACTIVATED = -217
- int SERVICE_SYSTEM_NOT_FOUND = -300
- int SERVICE_DUPLICATED_NAME = -301
- int SERVICE_OBJECT_NOT_EXIST = -302
- int SERVICE_NO_SUCH_HANDLE = -303
- int SERVICE_BAD_BROWSING_STATE = -304
- int SERVICE_OUT_OF_MEMORY = -305
- int SERVICE_INTERNAL_ERROR = -306
- int TASK_CREATE_FAILED = -400
- int TASK_WAIT_FAILED = -401
- int TASK_WAIT_TIMEOUT = -402
- int TASK_KILL_FAILED = -403
- int TASK_SCHED_FINISHED = -404
- int SCHEDULER_INVALID_STATE = -405
- int SCHEDULER_USER_INIT_FUNCTION_FAILED = -406
- int SCHEDULER_USER_RUN_FUNCTION_FAILED = -407
- int SCHEDULER_USER_DEINIT_FUNCTION_FAILED = -408
- int CLIENT_LIB_NOT_FOUND = -500
- int CLIENT_SYM_NOT_FOUND = -501
- int CLIENT_TIMED_OUT = -502
- int CLIENT_SOCKET_WAS_CLOSED_BY_SERVER = -503
- int CLIENT_SOCKET_RECEIVE_FAILED = -504
- int CLIENT_SOCKET_NO_SERVER_PORT = -505
- int CLIENT_SERVER_IS_BUSY = -506

7.211.1 Detailed Description

Gamma error codes (default return codes of API functions).

7.211.2 Member Data Documentation

7.211.2.1 ATTACH_ALREADY

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.ATTACH_ALREADY = -101  [static]
```

7.211.2.2 ATTACH_CORRUPTED

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.ATTACH_CORRUPTED = -103  [static]
```

7.211.2.3 ATTACH_FAILED

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.ATTACH_FAILED = -102  [static]
```

7.211.2.4 ATTACH_MISSING

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.ATTACH_MISSING = -100  [static]
```

7.211.2.5 CLIENT_LIB_NOT_FOUND

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.CLIENT_LIB_NOT_FOUND = -500  [static]
```

7.211.2.6 CLIENT_SERVER_IS_BUSY

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.CLIENT_SERVER_IS_BUSY = -506  [static]
```

7.211.2.7 CLIENT_SOCKET_NO_SERVER_PORT

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.CLIENT_SOCKET_NO_SERVER_PORT = -505  [static]
```

7.211.2.8 CLIENT_SOCKET_RECEIVE_FAILED

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.CLIENT_SOCKET_RECEIVE_FAILED = -504  [static]
```

7.211.2.9 CLIENT_SOCKET_WAS_CLOSED_BY_SERVER

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.CLIENT_SOCKET_WAS_CLOSED_BY_SERVER = -503  [static]
```

7.211.2.10 CLIENT_SYM_NOT_FOUND

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.CLIENT_SYM_NOT_FOUND = -501  [static]
```

7.211.2.11 CLIENT_TIMED_OUT

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.CLIENT_TIMED_OUT = -502  [static]
```

7.211.2.12 COM_BAD_STATE

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.COM_BAD_STATE = -8  [static]
```

7.211.2.13 COM_DEAD_CONNECTION

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.COM_DEAD_CONNECTION = -9  [static]
```

7.211.2.14 COM_HIGH_LEVEL_INTERNAL_ERROR

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.COM_HIGH_LEVEL_INTERNAL_ERROR = -12  [static]
```

7.211.2.15 COM_INTERNAL_ERROR

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.COM_INTERNAL_ERROR = -2  [static]
```

7.211.2.16 COM_INVALID_ARGUMENT

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.COM_INVALID_ARGUMENT = -1  [static]
```

7.211.2.17 COM_IO_ERROR

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.COM_IO_ERROR = -6  [static]
```

7.211.2.18 COM_MALLOC_FAILED

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.COM_MALLOC_FAILED = -3  [static]
```

7.211.2.19 COM_NOT_IMPLEMENTED

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.COM_NOT_IMPLEMENTED = -4  [static]
```

7.211.2.20 COM_SESSION_IS_DEAD

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.COM_SESSION_IS_DEAD = -13  [static]
```

7.211.2.21 COM_SHM_INVALID

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.COM_SHM_INVALID = -11  [static]
```

7.211.2.22 COM_STRING_TRUNCATED

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.COM_STRING_TRUNCATED = -5  [static]
```

7.211.2.23 COM_TIMED_OUT_RPC

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.COM_TIMED_OUT_RPC = -10  [static]
```

7.211.2.24 COM_VALUE_TRUNCATION

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.COM_VALUE_TRUNCATION = -7  [static]
```

7.211.2.25 DETACH_DONE_BUT_CON_DEAD

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.DETACH_DONE_BUT_CON_DEAD = -104  [static]
```

7.211.2.26 PV_BOUND_REJECTED

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.PV_BOUND_REJECTED = -208  [static]
```

7.211.2.27 PV_CLOSE_DONE_BUT_CORRUPTION

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.PV_CLOSE_DONE_BUT_CORRUPTION = -211  [static]
```

7.211.2.28 PV_DATA_TYPE_MISMATCH

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.PV_DATA_TYPE_MISMATCH = -210  [static]
```

7.211.2.29 PV_FIFO_EMPTY

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.PV_FIFO_EMPTY = -212  [static]
```

7.211.2.30 PV_FIFO_FULL

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.PV_FIFO_FULL = -213  [static]
```

7.211.2.31 PV_FIFO_WOULD_BLOCK

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.PV_FIFO_WOULD_BLOCK = -215  [static]
```

7.211.2.32 PV_INVALID_DATA_TYPE

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.PV_INVALID_DATA_TYPE = -206  [static]
```

7.211.2.33 PV_INVALID_SYNTAX

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.PV_INVALID_SYNTAX = -200  [static]
```

7.211.2.34 PV_LOCK_FAILED

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.PV_LOCK_FAILED = -202  [static]
```

7.211.2.35 PV_META_OP_NOT_AVAILABLE

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.PV_META_OP_NOT_AVAILABLE = -207  [static]
```

7.211.2.36 PV_NAME_TOO_LONG

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.PV_NAME_TOO_LONG = -201  [static]
```

7.211.2.37 PV_NO_COMMON_LOCK

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.PV_NO_COMMON_LOCK = -209  [static]
```

7.211.2.38 PV_SIG_ALREADY_ACTIVATED

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.PV_SIG_ALREADY_ACTIVATED = -216  [static]
```

7.211.2.39 PV_SIG_ALREADY_DEACTIVATED

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.PV_SIG_ALREADY_DEACTIVATED = -217  [static]
```

7.211.2.40 PV_SIM_VALUE_GET_FAILED

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.PV_SIM_VALUE_GET_FAILED = -205  [static]
```

7.211.2.41 PV_SIM_VALUE_SET_FAILED

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.PV_SIM_VALUE_SET_FAILED = -204  [static]
```

7.211.2.42 PV_TYPE_INVALID_OPERATION

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.PV_TYPE_INVALID_OPERATION = -214  [static]
```

7.211.2.43 PV_UNLOCK_FAILED

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.PV_UNLOCK_FAILED = -203  [static]
```

7.211.2.44 SCHEDULER_INVALID_STATE

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.SCHEDULER_INVALID_STATE = -405  [static]
```

7.211.2.45 SCHEDULER_USER_DEINIT_FUNCTION_FAILED

int ttk_tools.rst.gamma_api_common.ErrorCodes.SCHEDULER_USER_DEINIT_FUNCTION_FAILED = -408 [static]

7.211.2.46 SCHEDULER_USER_INIT_FUNCTION_FAILED

int ttk_tools.rst.gamma_api_common.ErrorCodes.SCHEDULER_USER_INIT_FUNCTION_FAILED = -406 [static]

7.211.2.47 SCHEDULER_USER_RUN_FUNCTION_FAILED

int ttk_tools.rst.gamma_api_common.ErrorCodes.SCHEDULER_USER_RUN_FUNCTION_FAILED = -407 [static]

7.211.2.48 SERVICE_BAD_BROWSING_STATE

int ttk_tools.rst.gamma_api_common.ErrorCodes.SERVICE_BAD_BROWSING_STATE = -304 [static]

7.211.2.49 SERVICE_DUPLICATED_NAME

int ttk_tools.rst.gamma_api_common.ErrorCodes.SERVICE_DUPLICATED_NAME = -301 [static]

7.211.2.50 SERVICE_INTERNAL_ERROR

int ttk_tools.rst.gamma_api_common.ErrorCodes.SERVICE_INTERNAL_ERROR = -306 [static]

7.211.2.51 SERVICE_NO_SUCH_HANDLE

int ttk_tools.rst.gamma_api_common.ErrorCodes.SERVICE_NO_SUCH_HANDLE = -303 [static]

7.211.2.52 SERVICE_OBJECT_NOT_EXIST

int ttk_tools.rst.gamma_api_common.ErrorCodes.SERVICE_OBJECT_NOT_EXIST = -302 [static]

7.211.2.53 SERVICE_OUT_OF_MEMORY

int ttk_tools.rst.gamma_api_common.ErrorCodes.SERVICE_OUT_OF_MEMORY = -305 [static]

7.211.2.54 SERVICE_SYSTEM_NOT_FOUND

int ttk_tools.rst.gamma_api_common.ErrorCodes.SERVICE_SYSTEM_NOT_FOUND = -300 [static]

7.211.2.55 SUCCESS

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.SUCCESS = 0  [static]
```

7.211.2.56 TASK_CREATE_FAILED

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.TASK_CREATE_FAILED = -400  [static]
```

7.211.2.57 TASK_KILL_FAILED

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.TASK_KILL_FAILED = -403  [static]
```

7.211.2.58 TASK_SCHED_FINISHED

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.TASK_SCHED_FINISHED = -404  [static]
```

7.211.2.59 TASK_WAIT_FAILED

```
int ttk_tools.rst.gamma_api_common.ErrorCodes.TASK_WAIT_FAILED = -401  [static]
```

7.211.2.60 TASK_WAIT_TIMEOUT

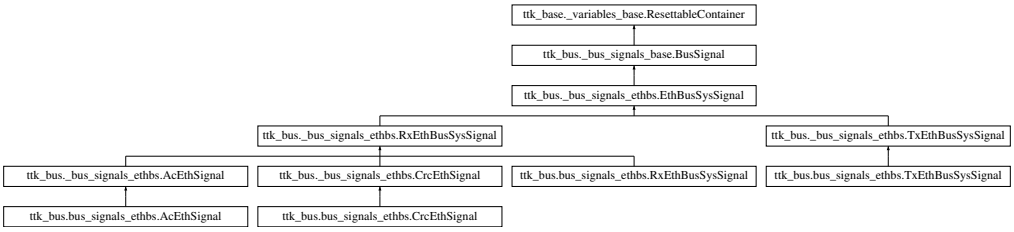
```
int ttk_tools.rst.gamma_api_common.ErrorCodes.TASK_WAIT_TIMEOUT = -402  [static]
```

The documentation for this class was generated from the following file:

- ttk_tools/rst/gamma_api_common.py (v1.5)

7.212 ttk_bus._bus_signals_ethbs.EthBusSysSignal Class Reference

Inheritance diagram for ttk_bus._bus_signals_ethbs.EthBusSysSignal:



Public Member Functions

- __init__ (self, context, signal_path, event=None, cycle_time=None, debounce_time=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  Ethernet signal initialization (BusSystems flavour).

Public Attributes

- event

  Reference to parent EthEvent instance.
- base_path

  base path below BusSystems, used for deriving internal HIL variables

7.212.1 Constructor & Destructor Documentation

7.212.1.1 __init__()

```
ttk_bus._bus_signals_ethbs.EthBusSysSignal.__init__ (
              self,
              context,
              signal_path,
              event = None,
              cycle_time = None,
              debounce_time = None,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Ethernet signal initialization (BusSystems flavour).

Parameters

context	parent context (an RTE application or XilTestbench instance to resolve signal/variable paths)
signal_path	signal path below BusSystems (typically: use .../foobar/Value)
event	reference to associated EthEvent instance
cycle_time	(opt) cycle time in [ms], overrides settings in EthEvent
debounce_time	(opt) minimum debounce delay time between events in [ms], overrides settings in EthEvent
unit	(opt) unit of physical signal value (as string, mainly used for report entries)
alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x cycle_time
timeout	(opt) detection time for timeout errors [ms], defaults to 2x cycle_time
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x cycle_time

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

Reimplemented in ttk_bus._bus_signals_ethbs.CrcEthSignal, ttk_bus._bus_signals_ethbs.AcEthSignal, ttk_bus.bus_signals_ethbs.Cr
ttk_bus.bus_signals_ethbs.AcEthSignal, ttk_bus._bus_signals_ethbs.RxEthBusSysSignal, and ttk_bus.bus_signals_ethbs.RxEthBusS

7.212.2 Member Data Documentation

7.212.2.1 base_path

`ttk_bus._bus_signals_ethbs.EthBusSysSignal.base_path`

base path below BusSystems, used for deriving internal HIL variables

7.212.2.2 event

`ttk_bus._bus_signals_ethbs.EthBusSysSignal.event`

Reference to parent EthEvent instance.

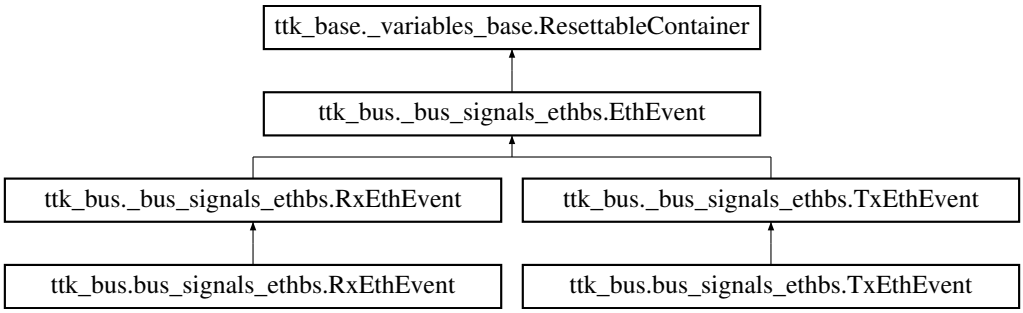
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_ethbs.py (v2.5)

7.213 ttk_bus._bus_signals_ethbs.EthEvent Class Reference

Ethernet Event (or Field) specific data and methods.

Inheritance diagram for ttk_bus._bus_signals_ethbs.EthEvent:



Public Member Functions

- __init__ (self, context, cycle_time=0, debounce_time=0, alias='', descr='', change_delay=None, timeout=None, recovery_time=None)

  Ethernet event (or field) data.
- __str__ (self)

  Return this event's alias name.

Public Attributes

- alias
  alias name of event
- descr
  description of event
- context
  context (reference to assigned RTE application instance)
- cycle_time
  cycle time of event in [ms]
- debounce_time
  debounce time of event in [ms]
- change_delay
  delay to wait after a value change [ms]
- timeout
  detection time for timeout errors [ms]
- recovery_time
  time for recovery/"wiedergut" checks [ms]

7.213.1 Detailed Description

Ethernet Event (or Field) specific data and methods.

7.213.2 Constructor & Destructor Documentation

7.213.2.1 __init__()

```
ttk_bus._bus_signals_ethbs.EthEvent.__init__ (
              self,
              context,
              cycle_time = 0,
              debounce_time = 0,
              alias = '',
              descr = '',
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Ethernet event (or field) data.

Parameters

context	parent context (an RTE application or XilTestbench instance to resolve signal/variable paths)
cycle_time	cycle time in [ms]. On-demand events have a cycle time of 0 (similar to fibex value)
debounce_time	minimum debounce delay time between events in [ms]
alias	Event alias/name
descr	(opt) Event description (may be used in report entries)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_base._variables_base.ResettableContainer.

Reimplemented in ttk_bus.bus_signals_ethbs.TxEthEvent, ttk_bus._bus_signals_ethbs.RxEthEvent, and ttk_bus.bus_signals_ethbs.RxEthEvent.

7.213.3 Member Function Documentation

7.213.3.1 __str__()

```
ttk_bus._bus_signals_ethbs.EthEvent.__str__ (
            self )
```

Return this event's alias name.

7.213.4 Member Data Documentation

7.213.4.1 alias

```
ttk_bus._bus_signals_ethbs.EthEvent.alias
```

alias name of event

7.213.4.2 change_delay

```
ttk_bus._bus_signals_ethbs.EthEvent.change_delay
```

delay to wait after a value change [ms]

7.213.4.3 context

```
ttk_bus._bus_signals_ethbs.EthEvent.context
```

context (reference to assigned RTE application instance)

7.213.4.4 cycle_time

```
ttk_bus._bus_signals_ethbs.EthEvent.cycle_time
```

cycle time of event in [ms]

7.213.4.5 debounce_time

```
ttk_bus._bus_signals_ethbs.EthEvent.debounce_time
```

debounce time of event in [ms]

7.213.4.6 descr

ttk_bus._bus_signals_ethbs.EthEvent.descr

description of event

7.213.4.7 recovery_time

ttk_bus._bus_signals_ethbs.EthEvent.recovery_time

time for recovery/"wiedergut" checks [ms]

7.213.4.8 timeout

ttk_bus._bus_signals_ethbs.EthEvent.timeout

detection time for timeout errors [ms]

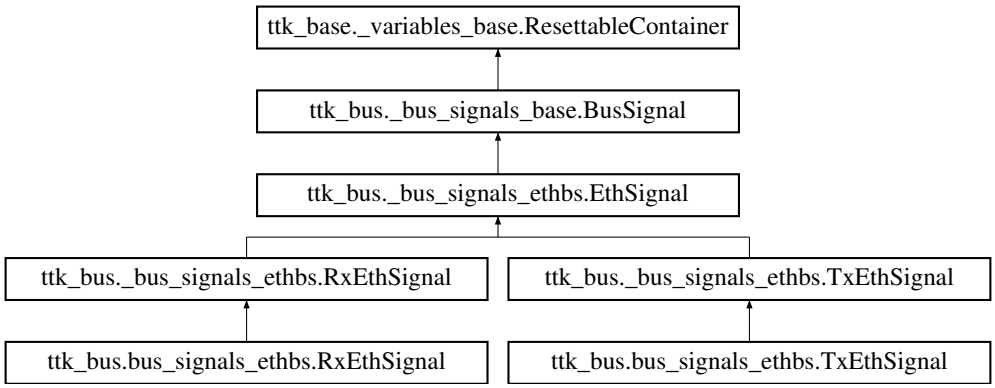
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_ethbs.py (v2.5)

7.214 ttk_bus._bus_signals_ethbs.EthSignal Class Reference

Base wrapper for a bus "signal" with HIL-side Ethernet Signals variable and relevant associated values.

Inheritance diagram for ttk_bus._bus_signals_ethbs.EthSignal:



Public Member Functions

- __init__ (self, context, signal_path, event=None, cycle_time=None, debounce_time=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None, **kwargs)

  Ethernet signal initialization (Model Path flavour).

Public Attributes

- event

  Reference to parent EthEvent instance.

7.214.1 Detailed Description

Base wrapper for a bus "signal" with HIL-side Ethernet Signals variable and relevant associated values.

Note

This class expects a "Model Root/" path structure.

7.214.2 Constructor & Destructor Documentation

7.214.2.1 __init__()

```
ttk_bus._bus_signals_ethbs.EthSignal.__init__ (
              self,
              context,
              signal_path,
              event = None,
              cycle_time = None,
              debounce_time = None,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              change_delay = None,
              timeout = None,
              recovery_time = None,
             ** kwargs )
```

Ethernet signal initialization (Model Path flavour).

Parameters

context	parent context (an RTE application or XilTestbench instance to resolve signal/variable paths)
signal_path	signal path below ModelRoot
event	reference to associated EthEvent instance
cycle_time	(opt) cycle time in [ms], overrides settings in EthEvent
debounce_time	(opt) minimum debounce delay time between events in [ms], overrides settings in EthEvent
unit	(opt) unit of physical signal value (as string, mainly used for report entries)
alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

Reimplemented in ttk_bus._bus_signals_ethbs.RxEthSignal, and ttk_bus.bus_signals_ethbs.RxEthSignal.

7.214.3 Member Data Documentation

7.214.3.1 event

```
ttk_bus._bus_signals_ethbs.EthSignal.event
```

Reference to parent EthEvent instance.

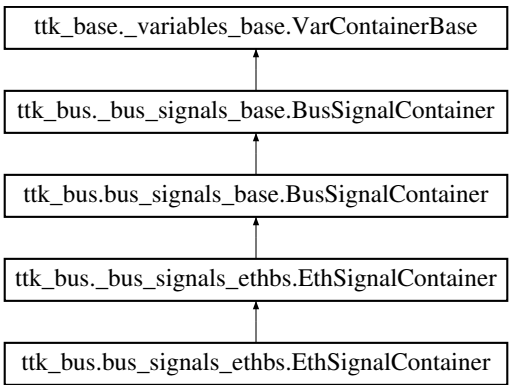
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_ethbs.py (v2.5)

7.215 ttk_bus._bus_signals_ethbs.EthSignalContainer Class Reference

A container for Ethernet Signals.

Inheritance diagram for ttk_bus._bus_signals_ethbs.EthSignalContainer:



Public Member Functions

- iterSignals (self, event)

  Iterate over all signals that reference the specified EthEvent.

7.215.1 Detailed Description

A container for Ethernet Signals.

7.215.2 Member Function Documentation

7.215.2.1 iterSignals()

```
ttk_bus._bus_signals_ethbs.EthSignalContainer.iterSignals (
            self,
            event )
```

Iterate over all signals that reference the specified EthEvent.

Parameters

event	an EthEvent instance

Usage:

```
for sig in eth.iterSignals(eth.evt_FRR_ACME_DASS_AWA_ESAssist)
    print sig
```

Returns

an iterator object.

Reimplemented in ttk_bus.bus_signals_ethbs.EthSignalContainer.

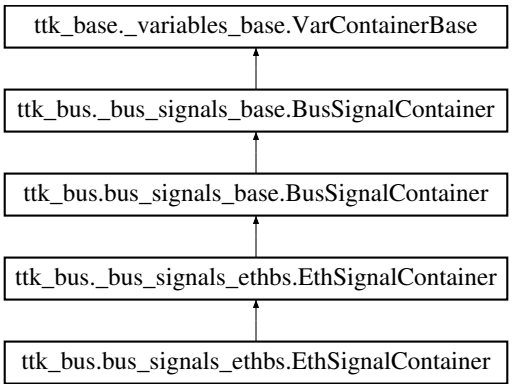
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_ethbs.py (v2.5)

7.216 ttk_bus.bus_signals_ethbs.EthSignalContainer Class Reference

A container for Ethernet Signals.

Inheritance diagram for ttk_bus.bus_signals_ethbs.EthSignalContainer:



Public Member Functions

- __init__ (self, context)
  Override to add signals.
- resetAll (self, verbosity=1, recursion_depth=4)
  Call reset() on all contained BusSignals that support it.
- iterSignals (self, event)
  Iterate over all signals that reference the specified EthEvent.

7.216.1 Detailed Description

A container for Ethernet Signals.

7.216.2 Constructor & Destructor Documentation

7.216.2.1 __init__()

```
ttk_bus.bus_signals_ethbs.EthSignalContainer.__init__ (
               self,
               context )
```

Override to add signals.

Parameters

context	context used for contained BusSignals

Reimplemented from ttk_bus.bus_signals_base.BusSignalContainer.

7.216.3 Member Function Documentation

7.216.3.1 iterSignals()

```
ttk_bus.bus_signals_ethbs.EthSignalContainer.iterSignals (
               self,
               event )
```

Iterate over all signals that reference the specified EthEvent.

Parameters

event	an EthEvent instance

Usage:

```
for sig in eth.iterSignals(eth.evt_FRR_ACME_DASS_AWA_ESAssist)
    print sig
```

Returns

an iterator object.

Reimplemented from ttk_bus._bus_signals_ethbs.EthSignalContainer.

7.216.3.2 resetAll()

```
ttk_bus.bus_signals_ethbs.EthSignalContainer.resetAll (
               self,
               verbosity = 1,
               recursion_depth = 4 )
```

Call reset() on all contained BusSignals that support it.

Parameters

verbosity	verbosity of log output: 0: silent except for errors 1: only minimal status info (begin/end) 2: also show processed variable names
recursion_depth	how "deep" to recurse into lists/dictionary structures when looking for variables to reset.

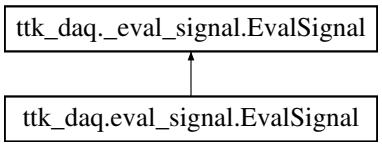
Reimplemented from ttk_bus.bus_signals_base.BusSignalContainer.

The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_ethbs.py (v2.5)

7.217 ttk_daq._eval_signal.EvalSignal Class Reference

Inheritance diagram for ttk_daq._eval_signal.EvalSignal:



Public Member Functions

- __init__ (self, *args)

  Initialize an EvalSignal with existing data.
- clearAll (self)

  Clear all current settings (evaluation range, current position) to start at the beginning of available data.
- seek (self, time_stamp, target=seek_closest, strict=True)

  Seek the data point at (or near) the supplied time stamp within the currently active evaluation range.
- seekStart (self)

  Seek data point at start position of currently active evaluation range.
- seekEnd (self)

  Seek data point at end position of currently active evaluation range.
- seekMiddle (self)

  Seek a data point closest to the middle of currently active evaluation range.
- markEvalRangeStart (self)

  Mark/set the current entry as new start of the evaluation range.
- markEvalRangeEnd (self)

  Mark/set the current entry as new end of the evaluation range.
- clearEvalRange (self)

  Clear any currently limited evaluation range to include all available data.
- clearEvalRangeStart (self)

  Clear evaluation range to start at beginning of available data.
- clearEvalRangeEnd (self)

  Clear evaluation range to include all entries until the end of available data.

- setEvalRange (self, start_time, stop_time)

  Limit evaluation to a defined time range.
- getEvalRangeData (self)

  Get a copy of signal data contained in the current evaluation range.
- getValue (self)

  Get the current value for the current/last found entry.
- getData (self)

  Alias for getValue().
- getTime (self)

  Get the current time stamp for the current/last found entry.
- isValid (self)

  Check if the current data/time/index position is valid.
- valid (self)

  True if current eval range and cursor position is valid, otherwise False.
- range_valid (self)

  True if current eval range is valid, otherwise False.
- __str__ (self)

  Get an information string about the current cursor position with value and timestamp (if available).
- find (self, operator="==", value=0, forward=True, abs_tol=0, rel_tol=0, skip_current_value=False)

  Find the next (or previous) value in the current data that matches the specified condition.
- findChanged (self, forward=True, threshold=None)

  Find the next (or previous) value that differs sufficiently from the current value.
- findNext (self, operator="==", value=0, abs_tol=0, rel_tol=0)

  Find next entry matching the condition.
- findPrev (self, operator="==", value=0, abs_tol=0, rel_tol=0)

  Find previous entry matching the condition.
- findNextChanged (self, threshold=None)

  Find next entry that changed more than the supplied `threshold` from the current value.
- findPrevChanged (self, threshold=None)

  Find previous entry that changed more than the supplied `threshold` from the current value.

Public Attributes

- time

  list of timestamp values
- data

  list of data values
- start_index

  start index of current evaluation range
- stop_index

  stop index of current evaluation range
- cur_index

  current index position of cursor, will be modified by seek/find methods

Static Public Attributes

- str seek_after = "after"

  seek() behavior, find entry at or directly after the supplied time value
- str seek_before = "before"

  seek() behavior, find entry at or directly before the supplied time value
- str seek_closest = "closest"

  seek() behavior, find entry closest to the supplied time value.

Protected Member Functions

- _genericFind (self, operator, value, index_range)

  Generic/basic find operations.
- _getFallbackThreshold (self, threshold)

  Get a fallback value if supplied threshold is None based on data type in self.data:

7.217.1 Constructor & Destructor Documentation

7.217.1.1 __init__()

```
ttk_daq._eval_signal.EvalSignal.__init__ (
              self,
             * args )
```

Initialize an EvalSignal with existing data.

Parameters

**args*	either a single signal dictionary with
	`{ "time": [<time1>, <time2>, ..., <timeN>], "data": [<data1>, <data2>, ..., <dataN>], }`
	or two separate lists with time and data entries:
	`[<time1>, ..., <timeN>], [<data1>, ..., <dataN>]`

Attention

Time values have to be monotonically increasing

Example: Using a signal dict

```
sig = EvalSignal(some_signal)
sig = EvalSignal(mat_data["signal_1"])
```

Example: Using separate time and data lists

```
sig = EvalSignal(time_stamps, data_points)
sig = EvalSignal(
    [0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7],
    [  0,   1,   1,   2,   3,   5,   8,  13],
)
```

Reimplemented in ttk_daq.eval_signal.EvalSignal.

7.217.2 Member Function Documentation

7.217.2.1 __str__()

```
ttk_daq._eval_signal.EvalSignal.__str__ (
              self )
```

Get an information string about the current cursor position with value and timestamp (if available).

Returns

a string formatted like "1.23 @ 0.456s" or "None" if cursor is out of range (position is invalid)

7.217.2.2 _genericFind()

```
ttk_daq._eval_signal.EvalSignal._genericFind (
              self,
              operator,
              value,
              index_range )  [protected]
```

Generic/basic find operations.

Will be used for most checks during find(), except for

- checks w/ defined tolerances and
- float data interpreted as integer

self.cur_index will be set to the index of the first matching item, or None if no match was found.

Parameters

operator	operator to use as string, one of <, <=, ==, !=, >= or >
value	target value to find
index_range	iterator with indexes into self.data to be checked

7.217.2.3 _getFallbackThreshold()

```
ttk_daq._eval_signal.EvalSignal._getFallbackThreshold (
              self,
              threshold )  [protected]
```

Get a fallback value if supplied threshold is None based on data type in self.data:

- 0 for integer data
- 0.001 for float data

Note

This avoids forcing checks of integer data to float-comparisons, which would cause issues for integers that do not fit into 64-bit doubles without loss of precision (i.e. values $> 2**53$). Pure integer comparisons have no practical size limitation.

7.217.2.4 clearAll()

```
ttk_daq._eval_signal.EvalSignal.clearAll (
              self )
```

Clear all current settings (evaluation range, current position) to start at the beginning of available data.

Reimplemented in ttk_daq.eval_signal.EvalSignal.

7.217.2.5 clearEvalRange()

```
ttk_daq._eval_signal.EvalSignal.clearEvalRange (
              self )
```

Clear any currently limited evaluation range to include all available data.

Current position will remain unchanged.

Reimplemented in ttk_daq.eval_signal.EvalSignal.

7.217.2.6 clearEvalRangeEnd()

```
ttk_daq._eval_signal.EvalSignal.clearEvalRangeEnd (
              self )
```

Clear evaluation range to include all entries until the end of available data.

Current position will remain unchanged.

Reimplemented in ttk_daq.eval_signal.EvalSignal.

7.217.2.7 clearEvalRangeStart()

```
ttk_daq._eval_signal.EvalSignal.clearEvalRangeStart (
              self )
```

Clear evaluation range to start at beginning of available data.

Current position will remain unchanged.

Reimplemented in ttk_daq.eval_signal.EvalSignal.

7.217.2.8 find()

```
ttk_daq._eval_signal.EvalSignal.find (
              self,
              operator = "==",
              value = 0,
              forward = True,
              abs_tol = 0,
              rel_tol = 0,
              skip_current_value = False )
```

Find the next (or previous) value in the current data that matches the specified condition.

Note

If the current value already matches the condition, it will be "found" right away unless `skip_current` is set to True.

If data consists of non-numeric entries, the contained objects will have to support relational operators and be comparable to the supplied value

Parameters

operator	comparison operator, one of ==, !=, >=, >, <, <= or a callable/function as custom operator, see Example
value	value to compare against
forward	direction of checks. True for forwards (current => end), False for backwards (current => start)
abs_tol	absolute tolerance value to use for numeric equality checks (==, !=)
rel_tol	relative tolerance factor to use for numeric equality checks (==, !=)
skip_current_value	True: the current value will be skipped (and thus not be "found"), which should be the expected behavior when seeking a "next" or "previous" value. See findNext(), findPrev()

Note

float-as-int data If data contains only float values that were originally integers (some interfaces only return double/float values, regardless of the original data type), then equality checks (==, !=) with both tolerances set to 0 will try to interpret data as integer values during comparison.
If encountered data values are too far off to be considered integer, a ValueError will be raised.

All other relational checks (>=, >, <, <=) will behave normally (tolerances will be disregarded), so e.g. 0.9999999 will be treated as < 1.0 and find(">=", 1) will not match it.

An invalid "current" position will also result in an invalid "find" result.
This means that you can specify a sequence of multiple find() operations and only check the last result in order to determine that the complete sequence was valid.

Attention

If float values are used for tolerances, the comparisons will be forced to use float values. Python internally uses double/float64 (IEEE-754 double-precision) as data type for "float", which will limit maximum precision.

If signal data contains 64-bit integer values, the implicit cast to float will limit the possible integer range to 52 bits.

In order to avoid this, the `threshold/abs_tol` parameter can be set to 0 (or any other integer value as desired). This way, an integer comparison will be performed which supports arbitrary bit lengths.

Example:

```
# e.g. signal should (within the available measurement time)
# assume state 2, then raise to or above state 3 and remain
# below state 5 until the end of the measurement:
status_sig.seekStart()
status_sig.findNext("==", 2)
status_sig.findNext(">=", 3)
valid = status_sig.findNext("<", 5) is not None
print("status valid:", valid)
```

Example:

```
# a status signal should assume status 2, followed by status 7.
# There should be no other states in between.
status_sig.seekStart()
first_state_2_ts       = status_sig.findNext("==", 2)
state_7_ts             = status_sig.findNext("==", 7)
last_before_state_7_ts = status_sig.findPrev("!=", 7)
status_sig.setEvalRange(first_state_2_ts, last_before_state_7_ts)
```

```
print("no other status between 2 and 7:", status_sig.findNext("!=", 2) is None)

# or with less time stamp fiddling:
status_sig.clearAll()
print("first status 2 at time", status_sig.findNext("==", 2))
status_sig.markEvalRangeStart()

print("first following status 7 at time", status_sig.findNext("==", 7))
# since findPrev finds the previous match before the current "7",
# we can simply use:
print("no other status between 2 and 7:", status_sig.findPrev("!=", 2) is None)
```

Example:

```
# a rpm signal should raise above 2000rpm and never drop below
# 1800rpm until the motor is stopped
rpm_sig.setEvalRange(motor_running_ts, motor_stopped_ts)
rpm_reached_ts = status_sig.findNext(">=", 2000)
rpm_valid = status_sig.findNext("<=", 1800)
print("rpm reached:", rpm_reached_ts is not None)
print("rpm remained valid:", rpm_valid is not None)

print("delay (motor enabled => rpm reached):", end=" ")
if None not in (motor_running_ts, rpm_reached_ts):
    print(f"{rpm_reached_ts - motor_running_ts :.3f}s")
else:
    print("not found")
```

Example: Custom Operator

```
# A function can be used as operator to get customized behavior,
# for example:
def op(current_value, target_value):
    # Custom find() operator.
    # Parameters:
    #    current_value - current data value to compare
    #    target_value  - value to compare against (from find()-parameter value)
    # Returns True if value was found, otherwise False
    return current_value & 0xFFFF > target_value & 0xFFFF

sig.findNext(op)
```

Returns

a time stamp for the found value or None if no matching value has been found (or if the the current data start position was unspecified to begin with).

Reimplemented in ttk_daq.eval_signal.EvalSignal.

7.217.2.9 findChanged()

```
ttk_daq._eval_signal.EvalSignal.findChanged (
            self,
            forward = True,
            threshold = None )
```

Find the next (or previous) value that differs sufficiently from the current value.

Parameters

forward	check direction. True for forwards (find next changed value), False for backwards (find previous changed value)
threshold	threshold, absolute, to determine if a value has changed. None: automatic; uses 0.001 for float data values or 0 for integer data values

Returns

the time stamp of the found value or None if no changed value has been found (or if the the current value was unspecified).

Reimplemented in ttk_daq.eval_signal.EvalSignal.

7.217.2.10 findNext()

```
ttk_daq._eval_signal.EvalSignal.findNext (
              self,
              operator = "==",
              value = 0,
              abs_tol = 0,
              rel_tol = 0 )
```

Find next entry matching the condition.

The current entry will be ignored, so multiple calls to findNext() will return the time stamp of the next following matching entry.

Returns

the time stamp of the found value or None if no changed value has been found. See find() for details.

Reimplemented in ttk_daq.eval_signal.EvalSignal.

7.217.2.11 findNextChanged()

```
ttk_daq._eval_signal.EvalSignal.findNextChanged (
              self,
              threshold = None )
```

Find next entry that changed more than the supplied `threshold` from the current value.

Parameters

threshold	threshold to determine if a value has changed. None: automatic; uses 0.001 for float and 0 for integer data values

Returns

the time stamp of the found value or None if no changed value has been found. See findChanged() for details.

Reimplemented in ttk_daq.eval_signal.EvalSignal.

7.217.2.12 findPrev()

```
ttk_daq._eval_signal.EvalSignal.findPrev (
              self,
```

```
                operator = "==",
                value = 0,
                abs_tol = 0,
                rel_tol = 0 )
```

Find previous entry matching the condition.

The current entry will be ignored, so multiple calls to findPrev() will return the time stamp of the preceding matching entry.

Returns

the time stamp of the found value or None if no changed value has been found. See find() for details.

Reimplemented in ttk_daq.eval_signal.EvalSignal.

7.217.2.13 findPrevChanged()

```
ttk_daq._eval_signal.EvalSignal.findPrevChanged (
                self,
                threshold = None )
```

Find previous entry that changed more than the supplied `threshold` from the current value.

Parameters

threshold	threshold to determine if a value has changed. None: automatic; uses 0.001 for float and 0 for integer data values

Returns

the time stamp of the found value or None if no changed value has been found. See findChanged() for details.

Reimplemented in ttk_daq.eval_signal.EvalSignal.

7.217.2.14 getData()

```
ttk_daq._eval_signal.EvalSignal.getData (
                self )
```

Alias for getValue().

Internal lists are named "data" and "time", so a function getData to get a value from data might also be expected.

Returns

a data value or None if current index position is invalid.

Reimplemented in ttk_daq.eval_signal.EvalSignal.

7.217.2.15 getEvalRangeData()

```
ttk_daq._eval_signal.EvalSignal.getEvalRangeData (
              self )
```

Get a copy of signal data contained in the current evaluation range.

First entry will be the first data point at range start, last entry will be the last data point at range end.

Returns

a dictionary with "time" and "data" lists: {
"time": [t0, t1, t2, ..., tn],
"data": [d0, d1, d2, ..., dn],
}
or None if the current evaluation range is invalid (that is, start or end timestamp are undefined)

Reimplemented in ttk_daq.eval_signal.EvalSignal.

7.217.2.16 getTime()

```
ttk_daq._eval_signal.EvalSignal.getTime (
              self )
```

Get the current time stamp for the current/last found entry.

Returns

a time value or None if current position is invalid.

Reimplemented in ttk_daq.eval_signal.EvalSignal.

7.217.2.17 getValue()

```
ttk_daq._eval_signal.EvalSignal.getValue (
              self )
```

Get the current value for the current/last found entry.

Returns

a data value or None if current index position is invalid.

Reimplemented in ttk_daq.eval_signal.EvalSignal.

7.217.2.18 isValid()

```
ttk_daq._eval_signal.EvalSignal.isValid (
              self )
```

Check if the current data/time/index position is valid.

Returns

True if cursor position is valid, otherwise False.

Reimplemented in ttk_daq.eval_signal.EvalSignal.

7.217.2.19 markEvalRangeEnd()

```
ttk_daq._eval_signal.EvalSignal.markEvalRangeEnd (
              self )
```

Mark/set the current entry as new end of the evaluation range.

Returns

True if the new range is valid, otherwise False.

Reimplemented in ttk_daq.eval_signal.EvalSignal.

7.217.2.20 markEvalRangeStart()

```
ttk_daq._eval_signal.EvalSignal.markEvalRangeStart (
              self )
```

Mark/set the current entry as new start of the evaluation range.

Returns

True if the new range is valid, otherwise False.

Reimplemented in ttk_daq.eval_signal.EvalSignal.

7.217.2.21 range_valid()

```
ttk_daq._eval_signal.EvalSignal.range_valid (
              self )
```

True if current eval range is valid, otherwise False.

7.217.2.22 seek()

```
ttk_daq._eval_signal.EvalSignal.seek (
              self,
              time_stamp,
              target = seek_closest,
              strict = True )
```

Seek the data point at (or near) the supplied time stamp within the currently active evaluation range.

Note that time values are expected to be stored in monotonically increasing order.

Parameters

time_stamp	time stamp value to seek.
target	target time stamp to seek, one of • "after" position at or directly after the supplied time (seek from beginning, match first value >= `time_stamp`) • "before": position at or directly before the supplied time (seek from end, match first value <= `time_stamp`) • "closest": position at or closest to `time_stamp` (in either direction)
strict	True: `time_stamp` has to fit inside the available time range (min time <= `time_stamp` <= max time). False: saturate on boundary value, if possible (see note/example)

Note

strict=False If `time_stamp` is outside the available time/evaluation range, seek may return one of the boundary values (start or end of eval range), depending on `target`.

Example:

```
sig = EvalSignal({
    "data": [  0,   1,   2,   3,   4,   5,   6,   7,   8,   9],
    "time": [0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9],
})

# seek returns the data value at the matched time
sig.seek(0.42, target="closest") == 4
sig.seek(0.42, target="before")  == 4
sig.seek(0.42, target="after")   == 5

sig.seek(0.56, target="closest") == 6
sig.seek(0.56, target="before")  == 5
sig.seek(0.56, target="after")   == 6

# out-of-range w/ strict=False:
sig.seek(1024, target="closest", strict=False) == 9
sig.seek(1024, target="before",  strict=False) == 9
sig.seek(1024, target="after",   strict=False) is None # no entry exists after time 1024

sig.seek(-512, target="closest", strict=False) == 0
sig.seek(-512, target="before",  strict=False) is None # no entry exists before time -512
sig.seek(-512, target="after",   strict=False) == 0
```

Returns

the value at the specified time or None if `time_stamp` is invalid / out of (evaluation) range.

Reimplemented in ttk_daq.eval_signal.EvalSignal.

7.217.2.23 seekEnd()

```
ttk_daq._eval_signal.EvalSignal.seekEnd (
            self )
```

Seek data point at end position of currently active evaluation range.

Returns

the value at end of range or None if current evaluation range is is invalid.

Reimplemented in ttk_daq.eval_signal.EvalSignal.

7.217.2.24 seekMiddle()

```
ttk_daq._eval_signal.EvalSignal.seekMiddle (
              self )
```

Seek a data point closest to the middle of currently active evaluation range.

Returns

the value at middle of range or None if current evaluation range is is invalid.

Reimplemented in ttk_daq.eval_signal.EvalSignal.

7.217.2.25 seekStart()

```
ttk_daq._eval_signal.EvalSignal.seekStart (
              self )
```

Seek data point at start position of currently active evaluation range.

Returns

the value at start of range or None if current evaluation range is is invalid.

Reimplemented in ttk_daq.eval_signal.EvalSignal.

7.217.2.26 setEvalRange()

```
ttk_daq._eval_signal.EvalSignal.setEvalRange (
              self,
              start_time,
              stop_time )
```

Limit evaluation to a defined time range.

Current position will be set to start of the new evaluation range.

Parameters

start_time	first time of evaluation range
stop_time	last time of evaluation range

Returns

True if the new range has been successfully set, otherwise False

Reimplemented in ttk_daq.eval_signal.EvalSignal.

7.217.2.27 valid()

```
ttk_daq._eval_signal.EvalSignal.valid (
              self )
```

True if current eval range and cursor position is valid, otherwise False.

7.217.3 Member Data Documentation

7.217.3.1 cur_index

```
ttk_daq._eval_signal.EvalSignal.cur_index
```

current index position of cursor, will be modified by seek/find methods

7.217.3.2 data

```
ttk_daq._eval_signal.EvalSignal.data
```

list of data values

7.217.3.3 seek_after

```
str ttk_daq._eval_signal.EvalSignal.seek_after = "after"  [static]
```

seek() behavior, find entry at or directly after the supplied time value

7.217.3.4 seek_before

```
str ttk_daq._eval_signal.EvalSignal.seek_before = "before"  [static]
```

seek() behavior, find entry at or directly before the supplied time value

7.217.3.5 seek_closest

```
str ttk_daq._eval_signal.EvalSignal.seek_closest = "closest"  [static]
```

seek() behavior, find entry closest to the supplied time value.

7.217.3.6 start_index

```
ttk_daq._eval_signal.EvalSignal.start_index
```

start index of current evaluation range

7.217.3.7 stop_index

`ttk_daq._eval_signal.EvalSignal.stop_index`

stop index of current evaluation range

7.217.3.8 time

`ttk_daq._eval_signal.EvalSignal.time`

list of timestamp values

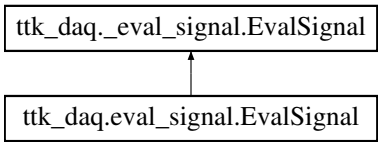
The documentation for this class was generated from the following file:

- ttk_daq/_eval_signal.py (v3.1.0)

7.218 ttk_daq.eval_signal.EvalSignal Class Reference

Class for evaluation of recorded signal data.

Inheritance diagram for ttk_daq.eval_signal.EvalSignal:



Public Member Functions

- __init__ (self, *args)
  Initialize an EvalSignal with existing data.
- clearAll (self)
  Clear all current settings (evaluation range, current position) to start at the beginning of available data.
- seek (self, time_stamp, target=seek_closest, strict=True)
  Seek the data point at (or near) the supplied time stamp within the currently active evaluation range.
- seekStart (self)
  Seek data point at start position of currently active evaluation range.
- seekEnd (self)
  Seek data point at end position of currently active evaluation range.
- seekMiddle (self)
  Seek a data point closest to the middle of currently active evaluation range.
- markEvalRangeStart (self)
  Mark/set the current entry as new start of the evaluation range.
- markEvalRangeEnd (self)
  Mark/set the current entry as new end of the evaluation range.
- clearEvalRange (self)
  Clear any currently limited evaluation range to include all available data.

- clearEvalRangeStart (self)

  Clear evaluation range to start at beginning of available data.
- clearEvalRangeEnd (self)

  Clear evaluation range to include all entries until the end of available data.
- setEvalRange (self, start_time, stop_time)

  Limit evaluation to a defined time range.
- getEvalRangeData (self)

  Get a copy of signal data contained in the current evaluation range.
- getValue (self)

  Get the current value for the current/last found entry.
- getData (self)

  Alias for getValue().
- getTime (self)

  Get the current time stamp for the current/last found entry.
- isValid (self)

  Check if the current data/time/index position is valid.
- find (self, operator="==", value=0, forward=True, abs_tol=0, rel_tol=0, skip_current_value=False)

  Find the next (or previous) value in the current data that matches the specified condition.
- findChanged (self, forward=True, threshold=None)

  Find the next (or previous) value that differs sufficiently from the current value.
- findNext (self, operator="==", value=0, abs_tol=0, rel_tol=0)

  Find next entry matching the condition.
- findPrev (self, operator="==", value=0, abs_tol=0, rel_tol=0)

  Find previous entry matching the condition.
- findNextChanged (self, threshold=None)

  Find next entry that changed more than the supplied `threshold` from the current value.
- findPrevChanged (self, threshold=None)

  Find previous entry that changed more than the supplied `threshold` from the current value.

Static Public Attributes

- str seek_after = "after"

  seek() behavior, find entry at or directly after the supplied time value
- str seek_before = "before"

  seek() behavior, find entry at or directly before the supplied time value
- str seek_closest = "closest"

  seek() behavior, find entry closest to the supplied time value.

7.218.1 Detailed Description

Class for evaluation of recorded signal data.

A signal can be searched similar to an open file handle with functions to seek in time or data domains.

Note

When searching for a certain data value, data points will be compared individually (values will **not** be interpolated between data points).
The first suitable/matching entry will be used/returned instead.

7.218.2 Constructor & Destructor Documentation

7.218.2.1 __init__()

```
ttk_daq.eval_signal.EvalSignal.__init__ (
              self,
             * args )
```

Initialize an EvalSignal with existing data.

Parameters

**args*	either a single signal dictionary with
	`{ "time": [<time1>, <time2>, ..., <timeN>], "data": [<data1>, <data2>, ..., <dataN>], }`
	or two separate lists with time and data entries:
	`[<time1>, ..., <timeN>], [<data1>, ..., <dataN>]`

Attention

Time values have to be monotonically increasing

Example: Using a signal dict

```
sig = EvalSignal(some_signal)
sig = EvalSignal(mat_data["signal_1"])
```

Example: Using separate time and data lists

```
sig = EvalSignal(time_stamps, data_points)
sig = EvalSignal(
    [0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7],
    [  0,   1,   1,   2,   3,   5,   8,  13],
)
```

Reimplemented from ttk_daq._eval_signal.EvalSignal.

7.218.3 Member Function Documentation

7.218.3.1 clearAll()

```
ttk_daq.eval_signal.EvalSignal.clearAll (
              self )
```

Clear all current settings (evaluation range, current position) to start at the beginning of available data.

Reimplemented from ttk_daq._eval_signal.EvalSignal.

7.218.3.2 clearEvalRange()

```
ttk_daq.eval_signal.EvalSignal.clearEvalRange (
              self )
```

Clear any currently limited evaluation range to include all available data.

Current position will remain unchanged.

Reimplemented from ttk_daq._eval_signal.EvalSignal.

7.218.3.3 clearEvalRangeEnd()

```
ttk_daq.eval_signal.EvalSignal.clearEvalRangeEnd (
              self )
```

Clear evaluation range to include all entries until the end of available data.

Current position will remain unchanged.

Reimplemented from ttk_daq._eval_signal.EvalSignal.

7.218.3.4 clearEvalRangeStart()

```
ttk_daq.eval_signal.EvalSignal.clearEvalRangeStart (
              self )
```

Clear evaluation range to start at beginning of available data.

Current position will remain unchanged.

Reimplemented from ttk_daq._eval_signal.EvalSignal.

7.218.3.5 find()

```
ttk_daq.eval_signal.EvalSignal.find (
              self,
              operator = "==",
              value = 0,
              forward = True,
              abs_tol = 0,
              rel_tol = 0,
              skip_current_value = False )
```

Find the next (or previous) value in the current data that matches the specified condition.

Note

If the current value already matches the condition, it will be "found" right away unless `skip_current` is set to True.

If data consists of non-numeric entries, the contained objects will have to support relational operators and be comparable to the supplied value

Parameters

operator	comparison operator, one of ==, !=, >=, >, <, <= or a callable/function as custom operator, see Example
value	value to compare against
forward	direction of checks. True for forwards (current => end), False for backwards (current => start)
abs_tol	absolute tolerance value to use for numeric equality checks (==, !=)
rel_tol	relative tolerance factor to use for numeric equality checks (==, !=)
skip_current_value	True: the current value will be skipped (and thus not be "found"), which should be the expected behavior when seeking a "next" or "previous" value. See findNext(), findPrev()

Note

float-as-int data If data contains only float values that were originally integers (some interfaces only return double/float values, regardless of the original data type), then equality checks (==, !=) with both tolerances set to 0 will try to interpret data as integer values during comparison.
If encountered data values are too far off to be considered integer, a ValueError will be raised.

All other relational checks (>=, >, <, <=) will behave normally (tolerances will be disregarded), so e.g. 0.9999999 will be treated as < 1.0 and find(">=", 1) will not match it.

An invalid "current" position will also result in an invalid "find" result.
This means that you can specify a sequence of multiple find() operations and only check the last result in order to determine that the complete sequence was valid.

Attention

If float values are used for tolerances, the comparisons will be forced to use float values. Python internally uses double/float64 (IEEE-754 double-precision) as data type for "float", which will limit maximum precision.

If signal data contains 64-bit integer values, the implicit cast to float will limit the possible integer range to 52 bits.

In order to avoid this, the `threshold/abs_tol` parameter can be set to 0 (or any other integer value as desired). This way, an integer comparison will be performed which supports arbitrary bit lengths.

Example:

```
# e.g. signal should (within the available measurement time)
# assume state 2, then raise to or above state 3 and remain
# below state 5 until the end of the measurement:
status_sig.seekStart()
status_sig.findNext("==", 2)
status_sig.findNext(">=", 3)
valid = status_sig.findNext("<", 5) is not None
print("status valid:", valid)
```

Example:

```
# a status signal should assume status 2, followed by status 7.
# There should be no other states in between.
status_sig.seekStart()
first_state_2_ts       = status_sig.findNext("==", 2)
state_7_ts             = status_sig.findNext("==", 7)
last_before_state_7_ts = status_sig.findPrev("!=", 7)
status_sig.setEvalRange(first_state_2_ts, last_before_state_7_ts)
```

```
print("no other status between 2 and 7:", status_sig.findNext("!=", 2) is None)

# or with less time stamp fiddling:
status_sig.clearAll()
print("first status 2 at time", status_sig.findNext("==", 2))
status_sig.markEvalRangeStart()

print("first following status 7 at time", status_sig.findNext("==", 7))
# since findPrev finds the previous match before the current "7",
# we can simply use:
print("no other status between 2 and 7:", status_sig.findPrev("!=", 2) is None)
```

Example:

```
# a rpm signal should raise above 2000rpm and never drop below
# 1800rpm until the motor is stopped
rpm_sig.setEvalRange(motor_running_ts, motor_stopped_ts)
rpm_reached_ts = status_sig.findNext(">=", 2000)
rpm_valid = status_sig.findNext("<=", 1800)
print("rpm reached:", rpm_reached_ts is not None)
print("rpm remained valid:", rpm_valid is not None)

print("delay (motor enabled => rpm reached):", end=" ")
if None not in (motor_running_ts, rpm_reached_ts):
    print(f"{rpm_reached_ts - motor_running_ts :.3f}s")
else:
    print("not found")
```

Example: Custom Operator

```
# A function can be used as operator to get customized behavior,
# for example:
def op(current_value, target_value):
    # Custom find() operator.
    # Parameters:
    #    current_value - current data value to compare
    #    target_value  - value to compare against (from find()-parameter value)
    # Returns True if value was found, otherwise False
    return current_value & 0xFFFF > target_value & 0xFFFF

sig.findNext(op)
```

Returns

a time stamp for the found value or None if no matching value has been found (or if the the current data start position was unspecified to begin with).

Reimplemented from ttk_daq._eval_signal.EvalSignal.

7.218.3.6 findChanged()

```
ttk_daq.eval_signal.EvalSignal.findChanged (
              self,
              forward = True,
              threshold = None )
```

Find the next (or previous) value that differs sufficiently from the current value.

Parameters

forward	check direction. True for forwards (find next changed value), False for backwards (find previous changed value)
threshold	threshold, absolute, to determine if a value has changed. None: automatic; uses 0.001 for float data values or 0 for integer data values

Returns

the time stamp of the found value or None if no changed value has been found (or if the the current value was unspecified).

Reimplemented from ttk_daq._eval_signal.EvalSignal.

7.218.3.7 findNext()

```
ttk_daq.eval_signal.EvalSignal.findNext (
              self,
              operator = "==",
              value = 0,
              abs_tol = 0,
              rel_tol = 0 )
```

Find next entry matching the condition.

The current entry will be ignored, so multiple calls to findNext() will return the time stamp of the next following matching entry.

Returns

the time stamp of the found value or None if no changed value has been found. See find() for details.

Reimplemented from ttk_daq._eval_signal.EvalSignal.

7.218.3.8 findNextChanged()

```
ttk_daq.eval_signal.EvalSignal.findNextChanged (
              self,
              threshold = None )
```

Find next entry that changed more than the supplied `threshold` from the current value.

Parameters

threshold	threshold to determine if a value has changed. None: automatic; uses 0.001 for float and 0 for integer data values

Returns

the time stamp of the found value or None if no changed value has been found. See findChanged() for details.

Reimplemented from ttk_daq._eval_signal.EvalSignal.

7.218.3.9 findPrev()

```
ttk_daq.eval_signal.EvalSignal.findPrev (
              self,
```

```
                operator = "==",
                value = 0,
                abs_tol = 0,
                rel_tol = 0 )
```

Find previous entry matching the condition.

The current entry will be ignored, so multiple calls to findPrev() will return the time stamp of the preceding matching entry.

Returns

the time stamp of the found value or None if no changed value has been found. See find() for details.

Reimplemented from ttk_daq._eval_signal.EvalSignal.

7.218.3.10 findPrevChanged()

```
ttk_daq.eval_signal.EvalSignal.findPrevChanged (
                self,
                threshold = None )
```

Find previous entry that changed more than the supplied `threshold` from the current value.

Parameters

threshold	threshold to determine if a value has changed. None: automatic; uses 0.001 for float and 0 for integer data values

Returns

the time stamp of the found value or None if no changed value has been found. See findChanged() for details.

Reimplemented from ttk_daq._eval_signal.EvalSignal.

7.218.3.11 getData()

```
ttk_daq.eval_signal.EvalSignal.getData (
                self )
```

Alias for getValue().

Internal lists are named "data" and "time", so a function getData to get a value from data might also be expected.

Returns

a data value or None if current index position is invalid.

Reimplemented from ttk_daq._eval_signal.EvalSignal.

7.218.3.12 getEvalRangeData()

```
ttk_daq.eval_signal.EvalSignal.getEvalRangeData (
              self )
```

Get a copy of signal data contained in the current evaluation range.

First entry will be the first data point at range start, last entry will be the last data point at range end.

Returns

a dictionary with "time" and "data" lists: {
"time": [t0, t1, t2, ..., tn],
"data": [d0, d1, d2, ..., dn],
}
or None if the current evaluation range is invalid (that is, start or end timestamp are undefined)

Reimplemented from ttk_daq._eval_signal.EvalSignal.

7.218.3.13 getTime()

```
ttk_daq.eval_signal.EvalSignal.getTime (
              self )
```

Get the current time stamp for the current/last found entry.

Returns

a time value or None if current position is invalid.

Reimplemented from ttk_daq._eval_signal.EvalSignal.

7.218.3.14 getValue()

```
ttk_daq.eval_signal.EvalSignal.getValue (
              self )
```

Get the current value for the current/last found entry.

Returns

a data value or None if current index position is invalid.

Reimplemented from ttk_daq._eval_signal.EvalSignal.

7.218.3.15 isValid()

```
ttk_daq.eval_signal.EvalSignal.isValid (
              self )
```

Check if the current data/time/index position is valid.

Returns

True if cursor position is valid, otherwise False.

Reimplemented from ttk_daq._eval_signal.EvalSignal.

7.218.3.16 markEvalRangeEnd()

```
ttk_daq.eval_signal.EvalSignal.markEvalRangeEnd (
              self )
```

Mark/set the current entry as new end of the evaluation range.

Returns

True if the new range is valid, otherwise False.

Reimplemented from ttk_daq._eval_signal.EvalSignal.

7.218.3.17 markEvalRangeStart()

```
ttk_daq.eval_signal.EvalSignal.markEvalRangeStart (
              self )
```

Mark/set the current entry as new start of the evaluation range.

Returns

True if the new range is valid, otherwise False.

Reimplemented from ttk_daq._eval_signal.EvalSignal.

7.218.3.18 seek()

```
ttk_daq.eval_signal.EvalSignal.seek (
              self,
              time_stamp,
              target = seek_closest,
              strict = True )
```

Seek the data point at (or near) the supplied time stamp within the currently active evaluation range.

Note that time values are expected to be stored in monotonically increasing order.

Parameters

time_stamp	time stamp value to seek.
target	target time stamp to seek, one of • "after" position at or directly after the supplied time (seek from beginning, match first value >= `time_stamp`) • "before": position at or directly before the supplied time (seek from end, match first value <= `time_stamp`) • "closest": position at or closest to `time_stamp` (in either direction)
strict	True: `time_stamp` has to fit inside the available time range (min time <= `time_stamp` <= max time). False: saturate on boundary value, if possible (see note/example)

Note

strict=False If `time_stamp` is outside the available time/evaluation range, seek may return one of the boundary values (start or end of eval range), depending on `target`.

Example:

```
sig = EvalSignal({
    "data": [  0,   1,   2,   3,   4,   5,   6,   7,   8,   9],
    "time": [0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9],
})

# seek returns the data value at the matched time
sig.seek(0.42, target="closest") == 4
sig.seek(0.42, target="before")  == 4
sig.seek(0.42, target="after")   == 5

sig.seek(0.56, target="closest") == 6
sig.seek(0.56, target="before")  == 5
sig.seek(0.56, target="after")   == 6

# out-of-range w/ strict=False:
sig.seek(1024, target="closest", strict=False) == 9
sig.seek(1024, target="before",  strict=False) == 9
sig.seek(1024, target="after",   strict=False) is None # no entry exists after time 1024

sig.seek(-512, target="closest", strict=False) == 0
sig.seek(-512, target="before",  strict=False) is None # no entry exists before time -512
sig.seek(-512, target="after",   strict=False) == 0
```

Returns

the value at the specified time or None if `time_stamp` is invalid / out of (evaluation) range.

Reimplemented from ttk_daq._eval_signal.EvalSignal.

7.218.3.19 seekEnd()

```
ttk_daq.eval_signal.EvalSignal.seekEnd (
              self )
```

Seek data point at end position of currently active evaluation range.

Returns

the value at end of range or None if current evaluation range is is invalid.

Reimplemented from ttk_daq._eval_signal.EvalSignal.

7.218.3.20 seekMiddle()

```
ttk_daq.eval_signal.EvalSignal.seekMiddle (
              self )
```

Seek a data point closest to the middle of currently active evaluation range.

Returns

the value at middle of range or None if current evaluation range is is invalid.

Reimplemented from ttk_daq._eval_signal.EvalSignal.

7.218.3.21 seekStart()

```
ttk_daq.eval_signal.EvalSignal.seekStart (
              self )
```

Seek data point at start position of currently active evaluation range.

Returns

the value at start of range or None if current evaluation range is is invalid.

Reimplemented from ttk_daq._eval_signal.EvalSignal.

7.218.3.22 setEvalRange()

```
ttk_daq.eval_signal.EvalSignal.setEvalRange (
              self,
              start_time,
              stop_time )
```

Limit evaluation to a defined time range.

Current position will be set to start of the new evaluation range.

Parameters

start_time	first time of evaluation range
stop_time	last time of evaluation range

Returns

True if the new range has been successfully set, otherwise False

Reimplemented from ttk_daq._eval_signal.EvalSignal.

7.218.4 Member Data Documentation

7.218.4.1 seek_after

```
str ttk_daq.eval_signal.EvalSignal.seek_after = "after"  [static]
```

seek() behavior, find entry at or directly after the supplied time value

7.218.4.2 seek_before

```
str ttk_daq.eval_signal.EvalSignal.seek_before = "before"  [static]
```

seek() behavior, find entry at or directly before the supplied time value

7.218.4.3 seek_closest

```
str ttk_daq.eval_signal.EvalSignal.seek_closest = "closest"  [static]
```

seek() behavior, find entry closest to the supplied time value.

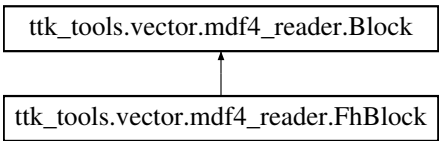
The documentation for this class was generated from the following file:

- ttk_daq/eval_signal.py (v3.1.0)

7.219 ttk_tools.vector.mdf4_reader.FhBlock Class Reference

File History Block.

Inheritance diagram for ttk_tools.vector.mdf4_reader.FhBlock:



Public Member Functions

- __init__ (self, file_handle, pos=None)

  Read FHBLOCK and its associated MDBLOCK from file_handle, starting at the current or supplied position.
- getTime (self)

  Get a time tuple for the entry's time stamp.
- __str__ (self)
- __repr__ (self)

Public Attributes

- fh_fh_next

  Link to next FHBLOCK (can be NIL if list finished)
- fh_md_comment

  Link to next FHBLOCK (can be NIL if list finished)
- fh_time_ns
- fh_tz_offset_min
- fh_dst_offset_min
- fh_time_flags
- comment
- tool_id
- tool_vendor
- tool_version
- user_name

7.219.1 Detailed Description

File History Block.

The FHBLOCK describes who/which tool generated or changed the MDF file. Each FHBLOCK contains a single change log entry.

Note

The first FHBLOCK referenced by the HDBLOCK must contain information about the tool which created the MDF file. Starting from this FHBLOCK a linked list of FHBLOCKs can be maintained with a chronological change history, i.e. any other application that changes the file must append a new FHBLOCK to the list.
A zero-based index value is used to reference blocks within this linked list, thus the ordering of the blocks must not be changed.

7.219.2 Constructor & Destructor Documentation

7.219.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.FhBlock.__init__ (
            self,
            file_handle,
            pos = None )
```

Read FHBLOCK and its associated MDBLOCK from file_handle, starting at the current or supplied position.

Note

File pointer position will be modified

Reimplemented from ttk_tools.vector.mdf4_reader.Block.

7.219.3 Member Function Documentation

7.219.3.1 __repr__()

```
ttk_tools.vector.mdf4_reader.FhBlock.__repr__ (
            self )
```

7.219.3.2 __str__()

```
ttk_tools.vector.mdf4_reader.FhBlock.__str__ (
            self )
```

7.219.3.3 getTime()

```
ttk_tools.vector.mdf4_reader.FhBlock.getTime (
            self )
```

Get a time tuple for the entry's time stamp.

Compare time.gmtime().

Returns

a time.gmtime()-like time tuple or None if time stamp data is inconsistent.

7.219.4 Member Data Documentation

7.219.4.1 comment

```
ttk_tools.vector.mdf4_reader.FhBlock.comment
```

7.219.4.2 fh_dst_offset_min

```
ttk_tools.vector.mdf4_reader.FhBlock.fh_dst_offset_min
```

7.219.4.3 fh_fh_next

```
ttk_tools.vector.mdf4_reader.FhBlock.fh_fh_next
```

Link to next FHBLOCK (can be NIL if list finished)

7.219.4.4 fh_md_comment

```
ttk_tools.vector.mdf4_reader.FhBlock.fh_md_comment
```

Link to next FHBLOCK (can be NIL if list finished)

Link to MDBLOCK containing comment about the creation or modification of the MDF file

7.219.4.5 fh_time_flags

ttk_tools.vector.mdf4_reader.FhBlock.fh_time_flags

7.219.4.6 fh_time_ns

ttk_tools.vector.mdf4_reader.FhBlock.fh_time_ns

7.219.4.7 fh_tz_offset_min

ttk_tools.vector.mdf4_reader.FhBlock.fh_tz_offset_min

7.219.4.8 tool_id

ttk_tools.vector.mdf4_reader.FhBlock.tool_id

7.219.4.9 tool_vendor

ttk_tools.vector.mdf4_reader.FhBlock.tool_vendor

7.219.4.10 tool_version

ttk_tools.vector.mdf4_reader.FhBlock.tool_version

7.219.4.11 user_name

ttk_tools.vector.mdf4_reader.FhBlock.user_name

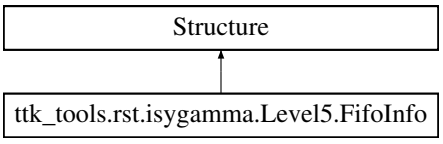
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.220 ttk_tools.rst.isygamma.Level5.FifoInfo Class Reference

typedef struct gc_FifoInfo

Inheritance diagram for ttk_tools.rst.isygamma.Level5.FifoInfo:



Static Protected Attributes

- list _fields_

7.220.1 Detailed Description

typedef struct gc_FifoInfo

7.220.2 Member Data Documentation

7.220.2.1 _fields_

```
list ttk_tools.rst.isygamma.Level5.FifoInfo._fields_  [static], [protected]
```

Initial value:

```
=  [
            ("address",      Addresses.Level5), # address of the FIFO
            ("name",         c_char * 1024),    # unique name of the FIFO
            ("type",         c_int),            # data type of elements within the FIFO (DataType Enum)
            ("typeLength",   c_uint32),         # maximum length of a complex data type.
            ("length",       c_uint32),         # length of the FIFO.
            ("isPersistent", c_bool),           #
            ("isUsingTimestamps", c_bool),      #
        ]
```

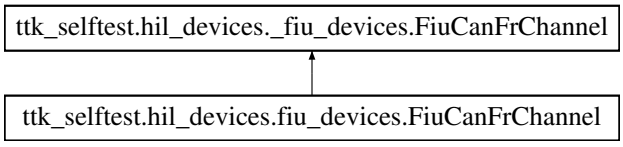
The documentation for this class was generated from the following file:

- ttk_tools/rst/isygamma.py (v1.6)

7.221 ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel Class Reference

Base FIU CAN FR channel container.

Inheritance diagram for ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel:



Public Member Functions

- __init__ (self, context, base_path, num_fiu_can_fr_card, num_fiu_can_fr_channel, wait_time_ms=5, alias="")

  FIU CAN FR Channel initialization.
- setHighOpenLoad (self)

  Set Open load on the high bus signal.
- setLowOpenLoad (self)

  Set Open load on the low bus signal.
- setHighEI1 (self)

  Connect EI1 to the high bus signal.
- setLowEI1 (self)

  Connect EI1 to the low bus signal.
- setHighEI2 (self)

  Connect EI2 to the high bus signal.
- setLowEI2 (self)

  Connect EI1 to the low bus signal.
- resetChannel (self)

  Reset FIU_CAN_FR channel to default.
- resetSignals (self)

  Reset the signals of the FIU_CAN_FR channel to default without triggering the CAN message.

Public Attributes

- fiu_msg_alias
- fiu_vars_alias
- wait_time_ms
- alias
- msg
- high_open_load
- low_open_load
- high_ei_1
- low_ei_1
- high_ei_2
- low_ei_2

7.221.1 Detailed Description

Base FIU CAN FR channel container.

7.221.2 Constructor & Destructor Documentation

7.221.2.1 __init__()

```
ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel.__init__ (
              self,
              context,
              base_path,
              num_fiu_can_fr_card,
              num_fiu_can_fr_channel,
              wait_time_ms = 5,
              alias = "" )
```

FIU CAN FR Channel initialization.

Parameters

context	parent context
base_path	base path of the iFIU_CAN_FR in the model (all required variables will be derived from this path)
num_fiu_can_fr_card	ID of the iFIU_CAN_FR
num_fiu_can_fr_channel	channel number of the specified iFIU_CAN_FR
wait_time_ms	(opt) waiting time in milliseconds after state of the FIU channel has been set defaults to 5ms
alias	(opt) alias of the FIU_CAN_FR channel defaults to number and channel of the iFIU_CAN_FR

Reimplemented in ttk_selftest.hil_devices.fiu_devices.FiuCanFrChannel.

7.221.3 Member Function Documentation

7.221.3.1 resetChannel()

```
ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel.resetChannel (
              self )
```

Reset FIU_CAN_FR channel to default.

No Error will be active.

Reimplemented in ttk_selftest.hil_devices.fiu_devices.FiuCanFrChannel.

7.221.3.2 resetSignals()

```
ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel.resetSignals (
              self )
```

Reset the signals of the FIU_CAN_FR channel to default without triggering the CAN message.

Reimplemented in ttk_selftest.hil_devices.fiu_devices.FiuCanFrChannel.

7.221.3.3 setHighEI1()

```
ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel.setHighEI1 (
              self )
```

Connect EI1 to the high bus signal.

Reimplemented in ttk_selftest.hil_devices.fiu_devices.FiuCanFrChannel.

7.221.3.4 setHighEI2()

```
ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel.setHighEI2 (
              self )
```

Connect EI2 to the high bus signal.

Reimplemented in ttk_selftest.hil_devices.fiu_devices.FiuCanFrChannel.

7.221.3.5 setHighOpenLoad()

```
ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel.setHighOpenLoad (
              self )
```

Set Open load on the high bus signal.

Reimplemented in ttk_selftest.hil_devices.fiu_devices.FiuCanFrChannel.

7.221.3.6 setLowEI1()

```
ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel.setLowEI1 (
              self )
```

Connect EI1 to the low bus signal.

Reimplemented in ttk_selftest.hil_devices.fiu_devices.FiuCanFrChannel.

7.221.3.7 setLowEI2()

```
ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel.setLowEI2 (
              self )
```

Connect EI1 to the low bus signal.

Reimplemented in ttk_selftest.hil_devices.fiu_devices.FiuCanFrChannel.

7.221.3.8 setLowOpenLoad()

```
ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel.setLowOpenLoad (
              self )
```

Set Open load on the low bus signal.

Reimplemented in ttk_selftest.hil_devices.fiu_devices.FiuCanFrChannel.

7.221.4 Member Data Documentation

7.221.4.1 alias

```
ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel.alias
```

7.221.4.2 fiu_msg_alias

ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel.fiu_msg_alias

7.221.4.3 fiu_vars_alias

ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel.fiu_vars_alias

7.221.4.4 high_ei_1

ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel.high_ei_1

7.221.4.5 high_ei_2

ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel.high_ei_2

7.221.4.6 high_open_load

ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel.high_open_load

7.221.4.7 low_ei_1

ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel.low_ei_1

7.221.4.8 low_ei_2

ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel.low_ei_2

7.221.4.9 low_open_load

ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel.low_open_load

7.221.4.10 msg

ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel.msg

7.221.4.11 wait_time_ms

ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel.wait_time_ms

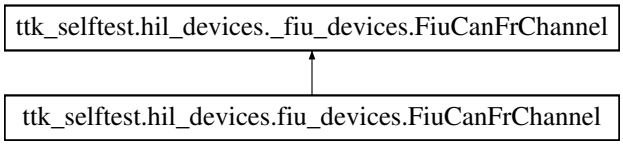
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/_fiu_devices.py (v2.0.0)

7.222 ttk_selftest.hil_devices.fiu_devices.FiuCanFrChannel Class Reference

Base FIU CAN FR channel container.

Inheritance diagram for ttk_selftest.hil_devices.fiu_devices.FiuCanFrChannel:



Public Member Functions

- __init__ (self, context, base_path, num_fiu_can_fr_card, num_fiu_can_fr_channel, wait_time_ms=5, alias="")
  FIU CAN FR Channel initialization.
- setHighOpenLoad (self)
  Set Open load on the high bus signal.
- setLowOpenLoad (self)
  Set Open load on the low bus signal.
- setHighEI1 (self)
  Connect EI1 to the high bus signal.
- setLowEI1 (self)
  Connect EI1 to the low bus signal.
- setHighEI2 (self)
  Connect EI2 to the high bus signal.
- setLowEI2 (self)
  Connect EI1 to the low bus signal.
- resetChannel (self)
  Reset FIU_CAN_FR channel to default.
- resetSignals (self)
  Reset the signals of the FIU_CAN_FR channel to default without triggering the CAN message.

7.222.1 Detailed Description

Base FIU CAN FR channel container.

7.222.2 Constructor & Destructor Documentation

7.222.2.1 __init__()

```
ttk_selftest.hil_devices.fiu_devices.FiuCanFrChannel.__init__ (
              self,
              context,
              base_path,
              num_fiu_can_fr_card,
              num_fiu_can_fr_channel,
              wait_time_ms = 5,
              alias = "" )
```

FIU CAN FR Channel initialization.

Parameters

context	parent context
base_path	base path of the iFIU_CAN_FR in the model (all required variables will be derived from this path)
num_fiu_can_fr_card	ID of the iFIU_CAN_FR
num_fiu_can_fr_channel	channel number of the specified iFIU_CAN_FR
wait_time_ms	(opt) waiting time in milliseconds after state of the FIU channel has been set defaults to 5ms
alias	(opt) alias of the FIU_CAN_FR channel defaults to number and channel of the iFIU_CAN_FR

Reimplemented from ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel.

7.222.3 Member Function Documentation

7.222.3.1 resetChannel()

```
ttk_selftest.hil_devices.fiu_devices.FiuCanFrChannel.resetChannel (
              self )
```

Reset FIU_CAN_FR channel to default.

No Error will be active.

Reimplemented from ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel.

7.222.3.2 resetSignals()

```
ttk_selftest.hil_devices.fiu_devices.FiuCanFrChannel.resetSignals (
              self )
```

Reset the signals of the FIU_CAN_FR channel to default without triggering the CAN message.

Reimplemented from ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel.

7.222.3.3 setHighEI1()

```
ttk_selftest.hil_devices.fiu_devices.FiuCanFrChannel.setHighEI1 (
              self )
```

Connect EI1 to the high bus signal.

Reimplemented from ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel.

7.222.3.4 setHighEI2()

```
ttk_selftest.hil_devices.fiu_devices.FiuCanFrChannel.setHighEI2 (
              self )
```

Connect EI2 to the high bus signal.

Reimplemented from ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel.

7.222.3.5 setHighOpenLoad()

```
ttk_selftest.hil_devices.fiu_devices.FiuCanFrChannel.setHighOpenLoad (
              self )
```

Set Open load on the high bus signal.

Reimplemented from ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel.

7.222.3.6 setLowEI1()

```
ttk_selftest.hil_devices.fiu_devices.FiuCanFrChannel.setLowEI1 (
              self )
```

Connect EI1 to the low bus signal.

Reimplemented from ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel.

7.222.3.7 setLowEI2()

```
ttk_selftest.hil_devices.fiu_devices.FiuCanFrChannel.setLowEI2 (
              self )
```

Connect EI1 to the low bus signal.

Reimplemented from ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel.

7.222.3.8 setLowOpenLoad()

```
ttk_selftest.hil_devices.fiu_devices.FiuCanFrChannel.setLowOpenLoad (
              self )
```

Set Open load on the low bus signal.

Reimplemented from ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel.

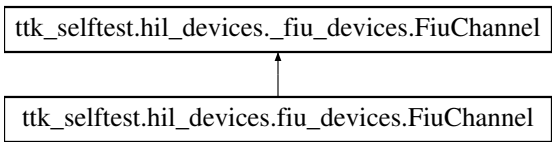
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/fiu_devices.py (v2.0.1)

7.223 ttk_selftest.hil_devices._fiu_devices.FiuChannel Class Reference

Base FIU channel container.

Inheritance diagram for ttk_selftest.hil_devices._fiu_devices.FiuChannel:



Public Member Functions

- __init__ (self, context, base_path, num_fiu_card, num_fiu_channel, wait_time_ms=5, alias="")
  FIU Channel initialization.
- setOpenLoad (self)
  Set Open load state on the FIU channel.
- resetOpenLoad (self)
  Reset Open load state on the FIU channel.
- setEI1 (self)
  Set connection to EI1 on the FIU channel.
- resetEI1 (self)
  Reset connection to EI1 on the FIU channel.
- setEI2 (self)
  Set connection to EI2 on the FIU channel.
- resetEI2 (self)
  Reset connection to EI2 on the FIU channel.
- resetChannel (self)
  Reset FIU channel to default.
- resetSignals (self)
  Reset the signals of the FIU channel to default without triggering the CAN message.

Public Attributes

- fiu_msg_alias
- fiu_vars_alias
- fiu_channel_name
- wait_time_ms
- alias
- msg
- open_load
- ei_1
- ei_2

7.223.1 Detailed Description

Base FIU channel container.

7.223.2 Constructor & Destructor Documentation

7.223.2.1 __init__()

```
ttk_selftest.hil_devices._fiu_devices.FiuChannel.__init__ (
               self,
               context,
               base_path,
               num_fiu_card,
               num_fiu_channel,
               wait_time_ms = 5,
               alias = "" )
```

FIU Channel initialization.

Parameters

context	parent context
base_path	base path of the iFIUCard in the model (all required variables will be derived from this path)
num_fiu_card	ID of the iFIUCard
num_fiu_channel	channel number of the specified iFIUCard
wait_time_ms	(opt) waiting time in milliseconds after state of the FIU channel has been set defaults to 5ms
alias	(opt) alias of the FIU channel defaults to number and channel of the FIU

Reimplemented in ttk_selftest.hil_devices.fiu_devices.FiuChannel.

7.223.3 Member Function Documentation

7.223.3.1 resetChannel()

```
ttk_selftest.hil_devices._fiu_devices.FiuChannel.resetChannel (
               self )
```

Reset FIU channel to default.

No Error will be active.

Reimplemented in ttk_selftest.hil_devices.fiu_devices.FiuChannel.

7.223.3.2 resetEI1()

```
ttk_selftest.hil_devices._fiu_devices.FiuChannel.resetEI1 (
               self )
```

Reset connection to EI1 on the FIU channel.

Reimplemented in ttk_selftest.hil_devices.fiu_devices.FiuChannel.

7.223.3.3 resetEI2()

```
ttk_selftest.hil_devices._fiu_devices.FiuChannel.resetEI2 (
              self )
```

Reset connection to EI2 on the FIU channel.

Reimplemented in ttk_selftest.hil_devices.fiu_devices.FiuChannel.

7.223.3.4 resetOpenLoad()

```
ttk_selftest.hil_devices._fiu_devices.FiuChannel.resetOpenLoad (
              self )
```

Reset Open load state on the FIU channel.

Reimplemented in ttk_selftest.hil_devices.fiu_devices.FiuChannel.

7.223.3.5 resetSignals()

```
ttk_selftest.hil_devices._fiu_devices.FiuChannel.resetSignals (
              self )
```

Reset the signals of the FIU channel to default without triggering the CAN message.

Reimplemented in ttk_selftest.hil_devices.fiu_devices.FiuChannel.

7.223.3.6 setEI1()

```
ttk_selftest.hil_devices._fiu_devices.FiuChannel.setEI1 (
              self )
```

Set connection to EI1 on the FIU channel.

Reimplemented in ttk_selftest.hil_devices.fiu_devices.FiuChannel.

7.223.3.7 setEI2()

```
ttk_selftest.hil_devices._fiu_devices.FiuChannel.setEI2 (
              self )
```

Set connection to EI2 on the FIU channel.

Reimplemented in ttk_selftest.hil_devices.fiu_devices.FiuChannel.

7.223.3.8 setOpenLoad()

```
ttk_selftest.hil_devices._fiu_devices.FiuChannel.setOpenLoad (
              self )
```

Set Open load state on the FIU channel.

Reimplemented in ttk_selftest.hil_devices.fiu_devices.FiuChannel.

7.223.4 Member Data Documentation

7.223.4.1 alias

ttk_selftest.hil_devices._fiu_devices.FiuChannel.alias

7.223.4.2 ei_1

ttk_selftest.hil_devices._fiu_devices.FiuChannel.ei_1

7.223.4.3 ei_2

ttk_selftest.hil_devices._fiu_devices.FiuChannel.ei_2

7.223.4.4 fiu_channel_name

ttk_selftest.hil_devices._fiu_devices.FiuChannel.fiu_channel_name

7.223.4.5 fiu_msg_alias

ttk_selftest.hil_devices._fiu_devices.FiuChannel.fiu_msg_alias

7.223.4.6 fiu_vars_alias

ttk_selftest.hil_devices._fiu_devices.FiuChannel.fiu_vars_alias

7.223.4.7 msg

ttk_selftest.hil_devices._fiu_devices.FiuChannel.msg

7.223.4.8 open_load

ttk_selftest.hil_devices._fiu_devices.FiuChannel.open_load

7.223.4.9 wait_time_ms

ttk_selftest.hil_devices._fiu_devices.FiuChannel.wait_time_ms

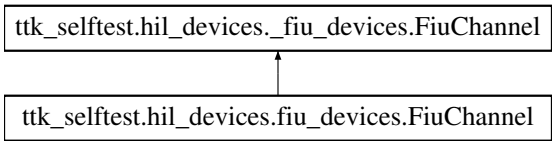
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/_fiu_devices.py (v2.0.0)

7.224 ttk_selftest.hil_devices.fiu_devices.FiuChannel Class Reference

Base FIU channel container.

Inheritance diagram for ttk_selftest.hil_devices.fiu_devices.FiuChannel:



Public Member Functions

- __init__ (self, context, base_path, num_fiu_card, num_fiu_channel, wait_time_ms=5, alias="")
  FIU Channel initialization.
- setOpenLoad (self)
  Set Open load state on the FIU channel.
- resetOpenLoad (self)
  Reset Open load state on the FIU channel.
- setEI1 (self)
  Set connection to EI1 on the FIU channel.
- resetEI1 (self)
  Reset connection to EI1 on the FIU channel.
- setEI2 (self)
  Set connection to EI2 on the FIU channel.
- resetEI2 (self)
  Reset connection to EI2 on the FIU channel.
- resetChannel (self)
  Reset FIU channel to default.
- resetSignals (self)
  Reset the signals of the FIU channel to default without triggering the CAN message.

7.224.1 Detailed Description

Base FIU channel container.

7.224.2 Constructor & Destructor Documentation

7.224.2.1 __init__()

```
ttk_selftest.hil_devices.fiu_devices.FiuChannel.__init__ (
              self,
              context,
              base_path,
              num_fiu_card,
              num_fiu_channel,
              wait_time_ms = 5,
              alias = "" )
```

FIU Channel initialization.

Parameters

context	parent context
base_path	base path of the iFIUCard in the model (all required variables will be derived from this path)
num_fiu_card	ID of the iFIUCard
num_fiu_channel	channel number of the specified iFIUCard
wait_time_ms	(opt) waiting time in milliseconds after state of the FIU channel has been set defaults to 5ms
alias	(opt) alias of the FIU channel defaults to number and channel of the FIU

Reimplemented from ttk_selftest.hil_devices._fiu_devices.FiuChannel.

7.224.3 Member Function Documentation

7.224.3.1 resetChannel()

```
ttk_selftest.hil_devices.fiu_devices.FiuChannel.resetChannel (
              self )
```

Reset FIU channel to default.

No Error will be active.

Reimplemented from ttk_selftest.hil_devices._fiu_devices.FiuChannel.

7.224.3.2 resetEI1()

```
ttk_selftest.hil_devices.fiu_devices.FiuChannel.resetEI1 (
              self )
```

Reset connection to EI1 on the FIU channel.

Reimplemented from ttk_selftest.hil_devices._fiu_devices.FiuChannel.

7.224.3.3 resetEI2()

```
ttk_selftest.hil_devices.fiu_devices.FiuChannel.resetEI2 (
              self )
```

Reset connection to EI2 on the FIU channel.

Reimplemented from ttk_selftest.hil_devices._fiu_devices.FiuChannel.

7.224.3.4 resetOpenLoad()

```
ttk_selftest.hil_devices.fiu_devices.FiuChannel.resetOpenLoad (
              self )
```

Reset Open load state on the FIU channel.

Reimplemented from ttk_selftest.hil_devices._fiu_devices.FiuChannel.

7.224.3.5 resetSignals()

```
ttk_selftest.hil_devices.fiu_devices.FiuChannel.resetSignals (
              self )
```

Reset the signals of the FIU channel to default without triggering the CAN message.

Reimplemented from ttk_selftest.hil_devices._fiu_devices.FiuChannel.

7.224.3.6 setEI1()

```
ttk_selftest.hil_devices.fiu_devices.FiuChannel.setEI1 (
              self )
```

Set connection to EI1 on the FIU channel.

Reimplemented from ttk_selftest.hil_devices._fiu_devices.FiuChannel.

7.224.3.7 setEI2()

```
ttk_selftest.hil_devices.fiu_devices.FiuChannel.setEI2 (
              self )
```

Set connection to EI2 on the FIU channel.

Reimplemented from ttk_selftest.hil_devices._fiu_devices.FiuChannel.

7.224.3.8 setOpenLoad()

```
ttk_selftest.hil_devices.fiu_devices.FiuChannel.setOpenLoad (
              self )
```

Set Open load state on the FIU channel.

Reimplemented from ttk_selftest.hil_devices._fiu_devices.FiuChannel.

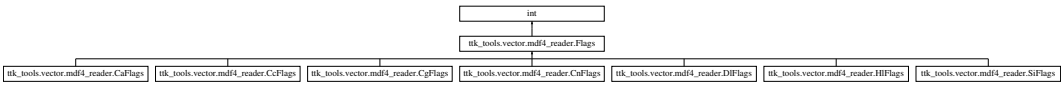
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/fiu_devices.py (v2.0.1)

7.225 ttk_tools.vector.mdf4_reader.Flags Class Reference

An integer/long as container for decoded bit values.

Inheritance diagram for ttk_tools.vector.mdf4_reader.Flags:



Public Member Functions

- __new__ (cls, flags, bit_width=8)
- __init__ (self, flags, bit_width=8)
- bit (self, n)

  Get boolean value of bit at offset n.
- __str__ (self)
- __repr__ (self)

Protected Member Functions

- _getNumNibbles (self)

  Get number of required nibbles to fit values with the set bit width.

Protected Attributes

- _bit_width

7.225.1 Detailed Description

An integer/long as container for decoded bit values.

Only intended for read-only access, changes in named flags will not modify original "flags" value. Bits that have no member assigned are considered to be "dont-care".

7.225.2 Constructor & Destructor Documentation

7.225.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.Flags.__init__ (
              self,
              flags,
              bit_width = 8 )
```

Reimplemented in ttk_tools.vector.mdf4_reader.CgFlags, ttk_tools.vector.mdf4_reader.CnFlags, ttk_tools.vector.mdf4_reader.CaFlags, ttk_tools.vector.mdf4_reader.DlFlags, ttk_tools.vector.mdf4_reader.HlFlags, ttk_tools.vector.mdf4_reader.SiFlags, and ttk_tools.vector.mdf4_reader.CcFlags.

7.225.3 Member Function Documentation

7.225.3.1 __new__()

```
ttk_tools.vector.mdf4_reader.Flags.__new__ (
              cls,
              flags,
              bit_width = 8 )
```

7.225.3.2 __repr__()

```
ttk_tools.vector.mdf4_reader.Flags.__repr__ (
              self )
```

7.225.3.3 __str__()

```
ttk_tools.vector.mdf4_reader.Flags.__str__ (
              self )
```

7.225.3.4 _getNumNibbles()

```
ttk_tools.vector.mdf4_reader.Flags._getNumNibbles (
              self )  [protected]
```

Get number of required nibbles to fit values with the set bit width.

7.225.3.5 bit()

```
ttk_tools.vector.mdf4_reader.Flags.bit (
              self,
              n )
```

Get boolean value of bit at offset n.

7.225.4 Member Data Documentation

7.225.4.1 _bit_width

```
ttk_tools.vector.mdf4_reader.Flags._bit_width  [protected]
```

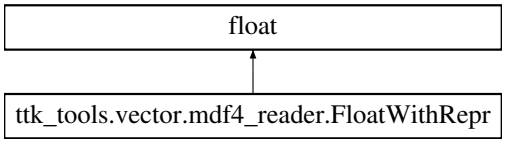
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.226 ttk_tools.vector.mdf4_reader.FloatWithRepr Class Reference

A float value that has an associated (textual) representation.

Inheritance diagram for ttk_tools.vector.mdf4_reader.FloatWithRepr:



Public Member Functions

- __new__ (cls, value, *args, **kwargs)
- __init__ (self, value, text=None)
- __repr__ (self)

Public Attributes

- text

7.226.1 Detailed Description

A float value that has an associated (textual) representation.

A representation of None marks an undefined representation (used for conversion types that may contain undefined values).

7.226.2 Constructor & Destructor Documentation

7.226.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.FloatWithRepr.__init__ (
              self,
              value,
              text = None )
```

Parameters

value	integer value
text	string with textual representation. If text is None, the value will be considered as "undefined/invalid"

7.226.3 Member Function Documentation

7.226.3.1 __new__()

```
ttk_tools.vector.mdf4_reader.FloatWithRepr.__new__ (
              cls,
              value,
             * args,
             ** kwargs )
```

7.226.3.2 __repr__()

```
ttk_tools.vector.mdf4_reader.FloatWithRepr.__repr__ (
              self )
```

7.226.4 Member Data Documentation

7.226.4.1 text

`ttk_tools.vector.mdf4_reader.FloatWithRepr.text`

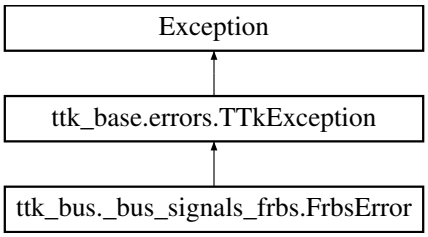
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.227 ttk_bus._bus_signals_frbs.FrbsError Class Reference

Basic Exception for FRBS (FlexRay block set) errors.

Inheritance diagram for ttk_bus._bus_signals_frbs.FrbsError:



7.227.1 Detailed Description

Basic Exception for FRBS (FlexRay block set) errors.

PDU-specific data and methods.

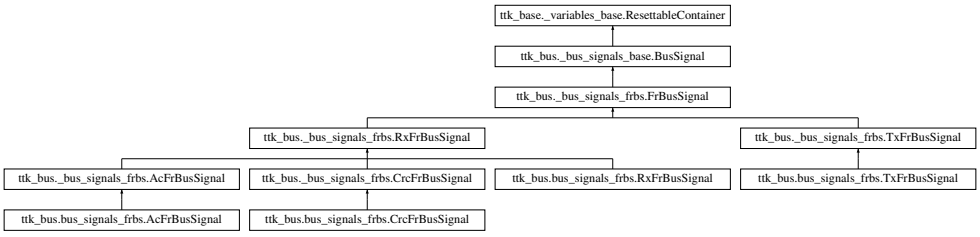
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_frbs.py (v2.5)

7.228 ttk_bus._bus_signals_frbs.FrBusSignal Class Reference

Base wrapper for a bus signal with HIL-side BusSignals variable and relevant associated values.

Inheritance diagram for ttk_bus._bus_signals_frbs.FrBusSignal:



Public Member Functions

- __init__ (self, context, base_path, pdu=None, cycle_time=None, debounce_time=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  FlexRay Bus signal initialization.

Public Attributes

- value_path

  model path of value variable
- base_path

  base model path (base path for deriving sub-element paths)
- pdu

  reference to parent FrPdu

7.228.1 Detailed Description

Base wrapper for a bus signal with HIL-side BusSignals variable and relevant associated values.

7.228.2 Constructor & Destructor Documentation

7.228.2.1 __init__()

```
ttk_bus._bus_signals_frbs.FrBusSignal.__init__ (
                self,
                context,
                base_path,
                pdu = None,
                cycle_time = None,
                debounce_time = None,
                unit = "",
                alias = "",
                descr = "",
                lookup = None,
                change_delay = None,
                timeout = None,
                recovery_time = None )
```

FlexRay Bus signal initialization.

Parameters

context	parent context (an RTE application instance to resolve signal/variable paths)
base_path	base path to signal below BusSystems (use path to `/Value` or `/Physical Value` as appropriate)
pdu	reference to associated FrPdu instance
cycle_time	(opt) cycle time in [ms], overrides settings in PDU
debounce_time	(opt) minimum debounce delay time in [ms], overrides settings in PDU
unit	(opt) unit of physical signal value (as string, mainly used for report entries)
alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)

Parameters

descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

Reimplemented in ttk_bus._bus_signals_frbs.CrcFrBusSignal, ttk_bus._bus_signals_frbs.AcFrBusSignal, ttk_bus.bus_signals_frbs.Cr
ttk_bus.bus_signals_frbs.AcFrBusSignal, ttk_bus._bus_signals_frbs.TxFrBusSignal, ttk_bus._bus_signals_frbs.RxFrBusSignal,
ttk_bus.bus_signals_frbs.TxFrBusSignal, and ttk_bus.bus_signals_frbs.RxFrBusSignal.

7.228.3 Member Data Documentation

7.228.3.1 base_path

`ttk_bus._bus_signals_frbs.FrBusSignal.base_path`

base model path (base path for deriving sub-element paths)

7.228.3.2 pdu

`ttk_bus._bus_signals_frbs.FrBusSignal.pdu`

reference to parent FrPdu

7.228.3.3 value_path

`ttk_bus._bus_signals_frbs.FrBusSignal.value_path`

model path of value variable

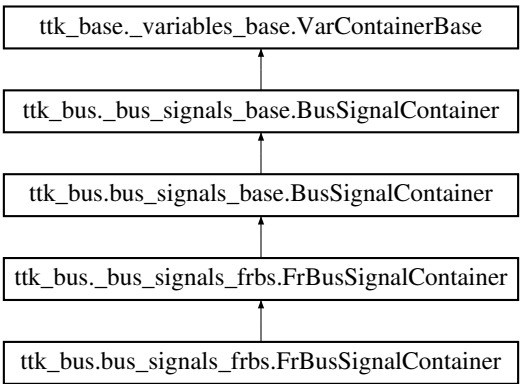
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_frbs.py (v2.5)

7.229 ttk_bus._bus_signals_frbs.FrBusSignalContainer Class Reference

A BusSignalContainer for FlexRay Bus Signals.

Inheritance diagram for ttk_bus._bus_signals_frbs.FrBusSignalContainer:



Public Member Functions

- iterSignals (self, pdu)

  Iterate over all signals that reference the specified PDU.

7.229.1 Detailed Description

A BusSignalContainer for FlexRay Bus Signals.

7.229.2 Member Function Documentation

7.229.2.1 iterSignals()

```
ttk_bus._bus_signals_frbs.FrBusSignalContainer.iterSignals (
            self,
            pdu )
```

Iterate over all signals that reference the specified PDU.

Parameters

pdu	a PDU instance

Usage:

```
for sig in bus.iterSignals(bus.tx_pdu_FNORD):
    print sig
```

Returns

an iterator object.

Reimplemented in ttk_bus.bus_signals_frbs.FrBusSignalContainer.

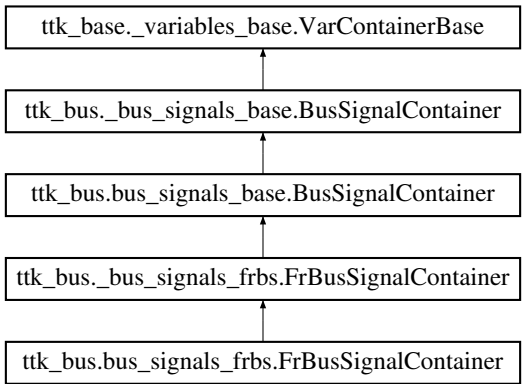
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_frbs.py (v2.5)

7.230 ttk_bus.bus_signals_frbs.FrBusSignalContainer Class Reference

A BusSignalContainer for FlexRay Bus Signals.

Inheritance diagram for ttk_bus.bus_signals_frbs.FrBusSignalContainer:



Public Member Functions

- __init__ (self, context)
  Override to add signals.
- resetAll (self, verbosity=1, recursion_depth=4)
  Call reset() on all contained BusSignals that support it.
- iterSignals (self, pdu)
  Iterate over all signals that reference the specified PDU.

7.230.1 Detailed Description

A BusSignalContainer for FlexRay Bus Signals.

7.230.2 Constructor & Destructor Documentation

7.230.2.1 __init__()

```
ttk_bus.bus_signals_frbs.FrBusSignalContainer.__init__ (
            self,
            context )
```

Override to add signals.

Parameters

context	context used for contained BusSignals

Reimplemented from ttk_bus.bus_signals_base.BusSignalContainer.

7.230.3 Member Function Documentation

7.230.3.1 iterSignals()

```
ttk_bus.bus_signals_frbs.FrBusSignalContainer.iterSignals (
              self,
              pdu )
```

Iterate over all signals that reference the specified PDU.

Parameters

pdu	a PDU instance

Usage:

```
for sig in bus.iterSignals(bus.tx_pdu_FNORD):
    print sig
```

Returns

an iterator object.

Reimplemented from ttk_bus._bus_signals_frbs.FrBusSignalContainer.

7.230.3.2 resetAll()

```
ttk_bus.bus_signals_frbs.FrBusSignalContainer.resetAll (
              self,
              verbosity = 1,
              recursion_depth = 4 )
```

Call reset() on all contained BusSignals that support it.

Parameters

verbosity	verbosity of log output: 0: silent except for errors 1: only minimal status info (begin/end) 2: also show processed variable names
recursion_depth	how "deep" to recurse into lists/dictionary structures when looking for variables to reset.

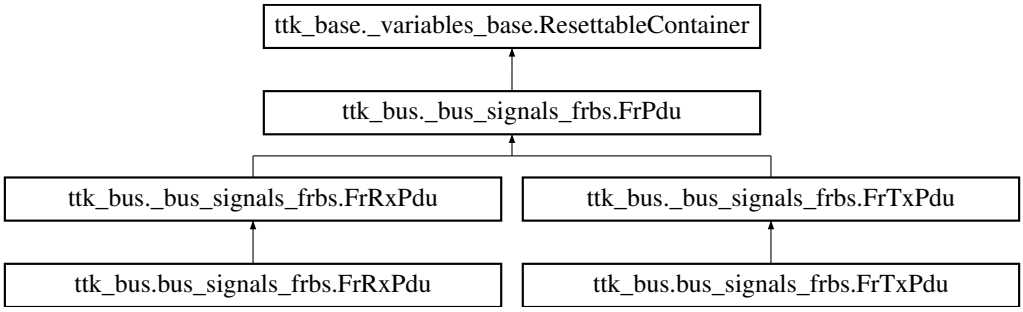
Reimplemented from ttk_bus.bus_signals_base.BusSignalContainer.

The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_frbs.py (v1.8)

7.231 ttk_bus._bus_signals_frbs.FrPdu Class Reference

Inheritance diagram for ttk_bus._bus_signals_frbs.FrPdu:



Public Member Functions

- __init__ (self, context, cycle_time=0, debounce_time=0, bus_systems_path=None, alias='', descr='', change_delay=None, timeout=None, recovery_time=None)

  FlexRay PDU data.
- __str__ (self)

  Returns the PDU's alias/name.

Public Attributes

- alias

  alias name of PDU
- descr

  description of PDU
- context

  context (reference to assigned RTE application instance)
- bus_systems_path

  base path to PDU below BusSystems
- cycle_time

  cycle time of PDU in [ms]
- debounce_time

  minimum debounce delay for event-triggered PDUs in [ms]
- change_delay

  delay to wait after a value change [ms]
- timeout

  detection time for timeout errors [ms]
- recovery_time

  time for recovery/"wiedergut" checks [ms]

Protected Member Functions

- _getRelativeHilVar (self, relative_model_path, **addkwargs)

  Get a HilVar based on a model path relative to the current bus_systems_path.

7.231.1 Constructor & Destructor Documentation

7.231.1.1 __init__()

```
ttk_bus._bus_signals_frbs.FrPdu.__init__ (
              self,
              context,
              cycle_time = 0,
              debounce_time = 0,
              bus_systems_path = None,
              alias = '',
              descr = '',
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

FlexRay PDU data.

Parameters

context	parent context (an RTE application instance to resolve signal/variable paths)
cycle_time	cycle time in [ms], 0 for event-triggered PDUs
debounce_time	minimum debounce delay time between events in [ms] for event-triggered PDUs
bus_systems_path	base path to PDU below BusSystems
alias	PDU alias/name
descr	(opt) PDU description (may be used in report entries)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_base._variables_base.ResettableContainer.

Reimplemented in ttk_bus._bus_signals_frbs.FrTxPdu, ttk_bus.bus_signals_frbs.FrTxPdu, ttk_bus._bus_signals_frbs.FrRxPdu, and ttk_bus.bus_signals_frbs.FrRxPdu.

7.231.2 Member Function Documentation

7.231.2.1 __str__()

```
ttk_bus._bus_signals_frbs.FrPdu.__str__ (
              self )
```

Returns the PDU's alias/name.

7.231.2.2 _getRelativeHilVar()

```
ttk_bus._bus_signals_frbs.FrPdu._getRelativeHilVar (
              self,
              relative_model_path,
             ** addkwargs )  [protected]
```

Get a HilVar based on a model path relative to the current bus_systems_path.

Parameters

relative_model_path	partial model path below self.bus_systems_path
***addkwargs*	additional keyword arguments for HilVar

Returns

a HilVar instance.

7.231.3 Member Data Documentation

7.231.3.1 alias

```
ttk_bus._bus_signals_frbs.FrPdu.alias
```

alias name of PDU

7.231.3.2 bus_systems_path

```
ttk_bus._bus_signals_frbs.FrPdu.bus_systems_path
```

base path to PDU below BusSystems

7.231.3.3 change_delay

```
ttk_bus._bus_signals_frbs.FrPdu.change_delay
```

delay to wait after a value change [ms]

7.231.3.4 context

```
ttk_bus._bus_signals_frbs.FrPdu.context
```

context (reference to assigned RTE application instance)

7.231.3.5 cycle_time

```
ttk_bus._bus_signals_frbs.FrPdu.cycle_time
```

cycle time of PDU in [ms]

7.231.3.6 debounce_time

`ttk_bus._bus_signals_frbs.FrPdu.debounce_time`

minimum debounce delay for event-triggered PDUs in [ms]

7.231.3.7 descr

`ttk_bus._bus_signals_frbs.FrPdu.descr`

description of PDU

7.231.3.8 recovery_time

`ttk_bus._bus_signals_frbs.FrPdu.recovery_time`

time for recovery/"wiedergut" checks [ms]

7.231.3.9 timeout

`ttk_bus._bus_signals_frbs.FrPdu.timeout`

detection time for timeout errors [ms]

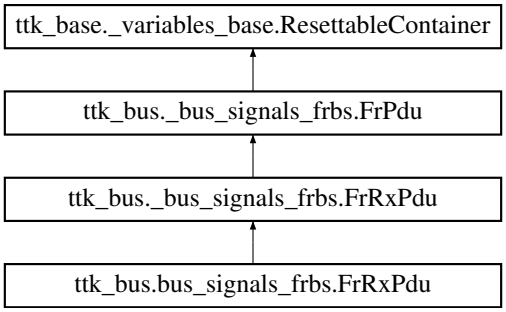
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_frbs.py (v2.5)

7.232 ttk_bus._bus_signals_frbs.FrRxPdu Class Reference

FlexRay ECU-Rx (HIL-Tx => ECU-Rx) PDU data.

Inheritance diagram for ttk_bus._bus_signals_frbs.FrRxPdu:



Public Member Functions

- __init__ (self, context, cycle_time=0, debounce_time=0, enable_path=None, bus_systems_path=None, alias='', descr='', change_delay=None, timeout=None, recovery_time=None, disable_mode=None)

  FlexRay ECU-Rx (HIL-Tx) PDU data.
- initUpdateBitControl (self)

  Initialize variables for Update Bit control.
- initTransmissionModeControl (self)

  Initialize variables for Transmission Mode control.
- initTxEnableControl (self)

  Initialize variables for Tx Enable control.
- enableHilTx (self)

  *(Re-)enable sending of PDU, depending on selected disable_mode (see **init**).*
- disableHilTx (self)

  *Disable sending of PDU, depending on selected disable_mode (see **init**).*
- kickout (self, delay_ms=None)

  Trigger sending of the PDU.

Public Attributes

- tx_enable_var

  HIL-Tx-enable/disable variable (for disable_mode: en) (a "HilVar" instance)
- tx_enable_switch

  HIL-Tx-enable/disable source-switch (for disable_mode: en) (a "HilVar" instance)
- update_bit_enable_var

  Update Bit handling variable (for disable_mode: ub) (a "HilVar" instance)
- update_bit_switch

  Update Bit handling source-switch (for disable_mode: ub) (a "HilVar" instance)
- tm_var

  Transmission Mode handling variable (for disable_mode: tm) (a "HilVar" instance)
- tm_switch

  Transmission Mode handling source-switch (for disable_mode: tm) (a "HilVar" instance)
- manual_enable_var

  Manual enable switch variable (when specified via enable_path) (a "HilVar" instance)
- kickout_var

  PDU Kickout/Trigger variable
  (a "HilVar" instance)
- kickout_switch

  PDU Kickout/Trigger source-switch
  (a "HilVar" instance)

Static Public Attributes

- str cfg_disable_mode = "tm"

  Configuration setting for new instances: Disable mode to use if no specific mode has been supplied in constructor.
- dict cfg_custom_defaults

  Configuration setting for new instances: Custom "hard" default values to use for internal real time system variables (for example: switch settings), so a reset() will always reset those variables to defined values (and not just those that were accessed).

Protected Member Functions

- _getDefault (self, name)

  Get a preconfigured default value for the supplied (member variable) name.
- _enableHilTxTM (self, enable=True)

  Enable PDU transmission via Transmission Mode (this overrides current setting in Simulink Model, use FrRxPdu.↩ reset() to restore initial behavior).
- _enableHilTxUB (self, enable=True)

  Enable PDU transmission via Update Bit override (this overrides current setting in Simulink Model, use FrRxPdu.↩ reset() to restore initial behavior).
- _enableHilTxEN (self, enable=True)

  Enable PDU transmission via an available Tx-/Hw- or SwEnable switch variable (this overrides current setting in Simulink Model, use FrRxPdu.reset() to restore initial behavior).

Protected Attributes

- _disable_mode
- _default_values

7.232.1 Detailed Description

FlexRay ECU-Rx (HIL-Tx => ECU-Rx) PDU data.

Extends FrPdu with methods for enableHilTx(), disableHilTx() and kickout() (trigger).

7.232.2 Constructor & Destructor Documentation

7.232.2.1 __init__()

```
ttk_bus._bus_signals_frbs.FrRxPdu.__init__ (
              self,
              context,
              cycle_time = 0,
              debounce_time = 0,
              enable_path = None,
              bus_systems_path = None,
              alias = '',
              descr = '',
              change_delay = None,
              timeout = None,
              recovery_time = None,
              disable_mode = None )
```

FlexRay ECU-Rx (HIL-Tx) PDU data.

Parameters

context	parent context (an RTE application instance to resolve signal/variable paths)
cycle_time	cycle time in [ms], 0 for event-triggered PDUs
debounce_time	minimum debounce delay time between events in [ms]

Parameters

enable_path	model path to manually added tx enable switch (if available). This will override any disable switches in BusSystems
bus_systems_path	base path to PDU below Bus Systems
alias	PDU alias/name
descr	(opt) PDU description (may be used in report entries)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)
disable_mode	(opt) mode used for disabling PDU Hil-Tx: • "tm": use the user defined transmission mode (99), which is strictly event/trigger-based • "ub": use Update Bit configuration (availability depends on FR configuration) • "en": use an available enable switch (Tx, HW or SW Enable) • None: configured cfg_disable_mode will be used.

Reimplemented from ttk_bus._bus_signals_frbs.FrPdu.

Reimplemented in ttk_bus.bus_signals_frbs.FrRxPdu.

7.232.3 Member Function Documentation

7.232.3.1 _enableHilTxEN()

```
ttk_bus._bus_signals_frbs.FrRxPdu._enableHilTxEN (
              self,
              enable = True )  [protected]
```

Enable PDU transmission via an available Tx-/Hw- or SwEnable switch variable (this overrides current setting in Simulink Model, use FrRxPdu.reset() to restore initial behavior).

Parameters

enable	True: enable transmission, False: disable transmission

7.232.3.2 _enableHilTxTM()

```
ttk_bus._bus_signals_frbs.FrRxPdu._enableHilTxTM (
```

```
                self,
                enable = True )  [protected]
```

Enable PDU transmission via Transmission Mode (this overrides current setting in Simulink Model, use FrRxPdu.↩ reset() to restore initial behavior).

Parameters

enable	True: enable transmission, False: disable transmission

7.232.3.3 _enableHilTxUB()

```
ttk_bus._bus_signals_frbs.FrRxPdu._enableHilTxUB (
                self,
                enable = True )  [protected]
```

Enable PDU transmission via Update Bit override (this overrides current setting in Simulink Model, use FrRxPdu.↩ reset() to restore initial behavior).

Parameters

enable	True: enable transmission, False: disable transmission

7.232.3.4 _getDefault()

```
ttk_bus._bus_signals_frbs.FrRxPdu._getDefault (
                self,
                name )  [protected]
```

Get a preconfigured default value for the supplied (member variable) name.

Empty or un-configured defaults will be returned as None, which enables normal HilVar behavior for default/reset values.

Parameters

name	name (as string) of member variable

Returns

a default value to use in HilVar constructor

7.232.3.5 disableHilTx()

```
ttk_bus._bus_signals_frbs.FrRxPdu.disableHilTx (
                self )
```

Disable sending of PDU, depending on selected disable_mode (see **init**).

This overrides current settings from Simulink Model, use FrRxPdu.reset() to restore initial behavior.

Reimplemented in ttk_bus.bus_signals_frbs.FrRxPdu.

7.232.3.6 enableHilTx()

```
ttk_bus._bus_signals_frbs.FrRxPdu.enableHilTx (
              self )
```

(Re-)enable sending of PDU, depending on selected disable_mode (see **init**).

This overrides current settings from Simulink Model, use FrRxPdu.reset() to restore initial behavior.

Reimplemented in ttk_bus.bus_signals_frbs.FrRxPdu.

7.232.3.7 initTransmissionModeControl()

```
ttk_bus._bus_signals_frbs.FrRxPdu.initTransmissionModeControl (
              self )
```

Initialize variables for Transmission Mode control.

Will be called automatically depending on selected disable_mode during **init**.
See FrRxPdu.tm_switch, FrRxPdu.tm_var

Reimplemented in ttk_bus.bus_signals_frbs.FrRxPdu.

7.232.3.8 initTxEnableControl()

```
ttk_bus._bus_signals_frbs.FrRxPdu.initTxEnableControl (
              self )
```

Initialize variables for Tx Enable control.

Will be called automatically depending on selected disable_mode during **init**.
See FrRxPdu.tx_enable_switch, FrRxPdu.tx_enable_var

Reimplemented in ttk_bus.bus_signals_frbs.FrRxPdu.

7.232.3.9 initUpdateBitControl()

```
ttk_bus._bus_signals_frbs.FrRxPdu.initUpdateBitControl (
              self )
```

Initialize variables for Update Bit control.

Will be called automatically depending on selected disable_mode during **init**.
See FrRxPdu.update_bit_switch, FrRxPdu.update_bit_enable_var

Reimplemented in ttk_bus.bus_signals_frbs.FrRxPdu.

7.232.3.10 kickout()

```
ttk_bus._bus_signals_frbs.FrRxPdu.kickout (
              self,
              delay_ms = None )
```

Trigger sending of the PDU.

Parameters

delay_ms	[ms] Delay to wait before clearing kickout/trigger variable again (it *should* auto-reset if trigger variable is located below BusSystems, but anyway).

Default Delay:

Delay defaults to 1/2 self.debounce_time (if debounce_time is explicitly set) or alternatively to 1/2 self.change_delay.
Default delay will never be shorter than 5 ms.

Reimplemented in ttk_bus.bus_signals_frbs.FrRxPdu.

7.232.4 Member Data Documentation

7.232.4.1 _default_values

`ttk_bus._bus_signals_frbs.FrRxPdu._default_values [protected]`

7.232.4.2 _disable_mode

`ttk_bus._bus_signals_frbs.FrRxPdu._disable_mode [protected]`

7.232.4.3 cfg_custom_defaults

`dict ttk_bus._bus_signals_frbs.FrRxPdu.cfg_custom_defaults [static]`

Initial value:

```
= {
        "tx_enable_var":         None,
        "tx_enable_switch":      None,
        "update_bit_enable_var": None,
        "update_bit_switch":     None,
        "tm_var":                None,
        "tm_switch":             None,
        "manual_enable_var":     None,
        "kickout_var":           None,
        "kickout_switch":        None,
    }
```

Configuration setting for new instances: Custom "hard" default values to use for internal real time system variables (for example: switch settings), so a reset() will always reset those variables to defined values (and not just those that were accessed).

This might be helpful if "other" tests fail to properly clean up the test environment after they are done.

Mapping: <variable name as string>: <default value>

A value of None (or a missing variable name) enables normal/automatic reset behavior.

Typical default values:

- source switches usually default to "source: simulink model"

  - 0: source: model/trc
  - 1: source: bus systems

- tx-enable vars default to "enabled"
  - 0: tx disabled
  - 1: tx enabled
- Transmission Mode depends on PDU setup, usually the default is "TM True"
  - 0: TM False
  - 1: TM True
  - 99: TM user defined
- kickout/trigger variables: Triggers below BusSystems will (should...) auto-reset to 0 once the PDU has been triggered.
- Update Bit behavior defaults to "automatic"
  - 0: use automatic Update Bit
  - 1: use static Update Bit (from associated /Value)

7.232.4.4 cfg_disable_mode

```
str ttk_bus._bus_signals_frbs.FrRxPdu.cfg_disable_mode = "tm"  [static]
```

Configuration setting for new instances: Disable mode to use if no specific mode has been supplied in constructor.

Mode used for disabling PDU Hil-Tx:

- "tm" - use user defined transmission mode (99), which is strictly event/trigger-based
- "ub" - use Update Bit configuration (availability depends on FR configuration)
- "en" - use an available enable switch (Tx, HW or SW Enable)

7.232.4.5 kickout_switch

```
ttk_bus._bus_signals_frbs.FrRxPdu.kickout_switch
```

PDU Kickout/Trigger source-switch
(a "HilVar" instance)

7.232.4.6 kickout_var

```
ttk_bus._bus_signals_frbs.FrRxPdu.kickout_var
```

PDU Kickout/Trigger variable
(a "HilVar" instance)

7.232.4.7 manual_enable_var

`ttk_bus._bus_signals_frbs.FrRxPdu.manual_enable_var`

Manual enable switch variable (when specified via enable_path) (a "HilVar" instance)

7.232.4.8 tm_switch

`ttk_bus._bus_signals_frbs.FrRxPdu.tm_switch`

Transmission Mode handling source-switch (for disable_mode: tm) (a "HilVar" instance)

7.232.4.9 tm_var

`ttk_bus._bus_signals_frbs.FrRxPdu.tm_var`

Transmission Mode handling variable (for disable_mode: tm) (a "HilVar" instance)

7.232.4.10 tx_enable_switch

`ttk_bus._bus_signals_frbs.FrRxPdu.tx_enable_switch`

HIL-Tx-enable/disable source-switch (for disable_mode: en) (a "HilVar" instance)

7.232.4.11 tx_enable_var

`ttk_bus._bus_signals_frbs.FrRxPdu.tx_enable_var`

HIL-Tx-enable/disable variable (for disable_mode: en) (a "HilVar" instance)

7.232.4.12 update_bit_enable_var

`ttk_bus._bus_signals_frbs.FrRxPdu.update_bit_enable_var`

Update Bit handling variable (for disable_mode: ub) (a "HilVar" instance)

7.232.4.13 update_bit_switch

`ttk_bus._bus_signals_frbs.FrRxPdu.update_bit_switch`

Update Bit handling source-switch (for disable_mode: ub) (a "HilVar" instance)

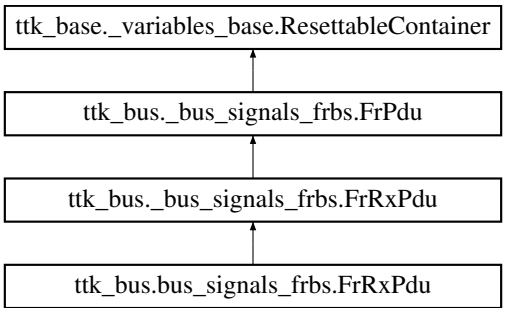
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_frbs.py (v2.5)

7.233 ttk_bus.bus_signals_frbs.FrRxPdu Class Reference

FlexRay ECU-Rx (HIL-Tx => ECU-Rx) PDU data.

Inheritance diagram for ttk_bus.bus_signals_frbs.FrRxPdu:



Public Member Functions

- __init__ (self, context, cycle_time=0, debounce_time=0, enable_path=None, bus_systems_path=None, alias='', descr='', change_delay=None, timeout=None, recovery_time=None, disable_mode=None)

  FlexRay ECU-Rx (HIL-Tx) PDU data.
- initUpdateBitControl (self)

  Initialize variables for Update Bit control.
- initTransmissionModeControl (self)

  Initialize variables for Transmission Mode control.
- initTxEnableControl (self)

  Initialize variables for Tx Enable control.
- enableHilTx (self)

  *(Re-)enable sending of PDU, depending on selected disable_mode (see **init**).*
- disableHilTx (self)

  *Disable sending of PDU, depending on selected disable_mode (see **init**).*
- kickout (self, delay_ms=None)

  Trigger sending of the PDU.
- reset (self, *args, **kwargs)

  Call reset() on all contained variables/signals that support it.

Public Attributes

- tx_enable_var

  HIL-Tx-enable/disable variable (for disable_mode: en) (a "HilVar" instance)
- tx_enable_switch

  HIL-Tx-enable/disable source-switch (for disable_mode: en)
  (a "HilVar" instance)
- update_bit_enable_var

  Update Bit handling variable (for disable_mode: ub)
  (a "HilVar" instance)
- update_bit_switch

  Update Bit handling source-switch (for disable_mode: ub)
  (a "HilVar" instance)
- tm_var

Transmission Mode handling variable (for disable_mode: tm)
(a "HilVar" instance)

- tm_switch

  Transmission Mode handling source-switch (for disable_mode: tm)
  (a "HilVar" instance)
- manual_enable_var

  Manual enable switch variable (when specified via enable_path)
  (a "HilVar" instance)
- kickout_var

  PDU Kickout/Trigger variable
  (a "HilVar" instance)
- kickout_switch

  PDU Kickout/Trigger source-switch
  (a "HilVar" instance)
- alias

  PDU alias name.
- descr

  PDU description text.
- context

  context (reference to assigned RTE application instance)
- bus_systems_path

  base path to PDU below BusSystems
- cycle_time

  cycle time of PDU in [ms]
- debounce_time

  minimum debounce delay for event-triggered PDUs in [ms]
- change_delay

  delay to wait after a value change [ms]
- timeout

  detection time for timeout errors [ms]
- recovery_time

  time for recovery/"wiedergut" checks [ms]

Static Public Attributes

- str cfg_disable_mode = "tm"

  Configuration setting for new instances: Disable mode to use if no specific mode has been supplied in constructor.
- dict cfg_custom_defaults

  Configuration setting for new instances: Custom "hard" default values to use for internal real time system variables (for example: switch settings), so a reset() will always reset those variables to defined values (and not just those that were accessed).

7.233.1 Detailed Description

FlexRay ECU-Rx (HIL-Tx => ECU-Rx) PDU data.

Extends FrPdu with methods for enableHilTx(), disableHilTx() and kickout() (trigger).

7.233.2 Constructor & Destructor Documentation

7.233.2.1 __init__()

```
ttk_bus.bus_signals_frbs.FrRxPdu.__init__ (
                self,
                context,
                cycle_time = 0,
                debounce_time = 0,
                enable_path = None,
                bus_systems_path = None,
                alias = '',
                descr = '',
                change_delay = None,
                timeout = None,
                recovery_time = None,
                disable_mode = None )
```

FlexRay ECU-Rx (HIL-Tx) PDU data.

Parameters

context	parent context (an RTE application instance to resolve signal/variable paths)
cycle_time	cycle time in [ms], 0 for event-triggered PDUs
debounce_time	minimum debounce delay time between events in [ms]
enable_path	model path to manually added tx enable switch (if available). This will override any disable switches in BusSystems
bus_systems_path	base path to PDU below Bus Systems
alias	PDU alias/name
descr	(opt) PDU description (may be used in report entries)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)
disable_mode	(opt) mode used for disabling PDU Hil-Tx: • "tm": use the user defined transmission mode (99), which is strictly event/trigger-based • "ub": use Update Bit configuration (availability depends on FR configuration) • "en": use an available enable switch (Tx, HW or SW Enable) • None: configured cfg_disable_mode will be used.

Reimplemented from ttk_bus._bus_signals_frbs.FrRxPdu.

7.233.3 Member Function Documentation

7.233.3.1 disableHilTx()

```
ttk_bus.bus_signals_frbs.FrRxPdu.disableHilTx (
              self )
```

Disable sending of PDU, depending on selected disable_mode (see **init**).

This overrides current settings from Simulink Model, use FrRxPdu.reset() to restore initial behavior.

Reimplemented from ttk_bus._bus_signals_frbs.FrRxPdu.

7.233.3.2 enableHilTx()

```
ttk_bus.bus_signals_frbs.FrRxPdu.enableHilTx (
              self )
```

(Re-)enable sending of PDU, depending on selected disable_mode (see **init**).

This overrides current settings from Simulink Model, use FrRxPdu.reset() to restore initial behavior.

Reimplemented from ttk_bus._bus_signals_frbs.FrRxPdu.

7.233.3.3 initTransmissionModeControl()

```
ttk_bus.bus_signals_frbs.FrRxPdu.initTransmissionModeControl (
              self )
```

Initialize variables for Transmission Mode control.

Will be called automatically depending on selected disable_mode during **init**.
See FrRxPdu.tm_switch, FrRxPdu.tm_var

Reimplemented from ttk_bus._bus_signals_frbs.FrRxPdu.

7.233.3.4 initTxEnableControl()

```
ttk_bus.bus_signals_frbs.FrRxPdu.initTxEnableControl (
              self )
```

Initialize variables for Tx Enable control.

Will be called automatically depending on selected disable_mode during **init**.
See FrRxPdu.tx_enable_switch, FrRxPdu.tx_enable_var

Reimplemented from ttk_bus._bus_signals_frbs.FrRxPdu.

7.233.3.5 initUpdateBitControl()

```
ttk_bus.bus_signals_frbs.FrRxPdu.initUpdateBitControl (
              self )
```

Initialize variables for Update Bit control.

Will be called automatically depending on selected disable_mode during **init**.
See FrRxPdu.update_bit_switch, FrRxPdu.update_bit_enable_var

Reimplemented from ttk_bus._bus_signals_frbs.FrRxPdu.

7.233.3.6 kickout()

```
ttk_bus.bus_signals_frbs.FrRxPdu.kickout (
              self,
              delay_ms = None )
```

Trigger sending of the PDU.

Parameters

delay_ms	[ms] Delay to wait before clearing kickout/trigger variable again (it *should* auto-reset if trigger variable is located below BusSystems, but anyway).

Default Delay:

Delay defaults to 1/2 self.debounce_time (if debounce_time is explicitly set) or alternatively to 1/2 self.↩ change_delay.
Default delay will never be shorter than 5 ms.

Reimplemented from ttk_bus._bus_signals_frbs.FrRxPdu.

7.233.3.7 reset()

```
ttk_bus.bus_signals_frbs.FrRxPdu.reset (
              self,
             * args,
             ** kwargs )
```

Call reset() on all contained variables/signals that support it.

Simple, non-recursive approach.

Parameters

∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.ResettableContainer.

7.233.4 Member Data Documentation

7.233.4.1 alias

```
ttk_bus.bus_signals_frbs.FrRxPdu.alias
```

PDU alias name.

7.233.4.2 bus_systems_path

```
ttk_bus.bus_signals_frbs.FrRxPdu.bus_systems_path
```

base path to PDU below BusSystems

7.233.4.3 cfg_custom_defaults

```
dict ttk_bus.bus_signals_frbs.FrRxPdu.cfg_custom_defaults  [static]
```

Initial value:

```
=  {
        "tx_enable_var":         None,
        "tx_enable_switch":      None,
        "update_bit_enable_var": None,
        "update_bit_switch":     None,
        "tm_var":                None,
        "tm_switch":             None,
        "manual_enable_var":     None,
        "kickout_var":           None,
        "kickout_switch":        None,
    }
```

Configuration setting for new instances: Custom "hard" default values to use for internal real time system variables (for example: switch settings), so a reset() will always reset those variables to defined values (and not just those that were accessed).

This might be helpful if "other" tests fail to properly clean up the test environment after they are done.

Mapping: <variable name as string>: <default value>

A value of None (or a missing variable name) enables normal/automatic reset behavior.

Typical default values:

- source switches usually default to "source: simulink model"
  - 0: source: model/trc
  - 1: source: bus systems
- tx-enable vars default to "enabled"
  - 0: tx disabled
  - 1: tx enabled
- Transmission Mode depends on PDU setup, usually the default is "TM True"
  - 0: TM False
  - 1: TM True
  - 99: TM user defined
- kickout/trigger variables: Triggers below BusSystems will (should...) auto-reset to 0 once the PDU has been triggered.
- Update Bit behavior defaults to "automatic"
  - 0: use automatic Update Bit
  - 1: use static Update Bit (from associated /Value)

7.233.4.4 cfg_disable_mode

```
str ttk_bus.bus_signals_frbs.FrRxPdu.cfg_disable_mode = "tm"  [static]
```

Configuration setting for new instances: Disable mode to use if no specific mode has been supplied in constructor.

Mode used for disabling PDU Hil-Tx:

- "tm" - use user defined transmission mode (99), which is strictly event/trigger-based
- "ub" - use Update Bit configuration (availability depends on FR configuration)
- "en" - use an available enable switch (Tx, HW or SW Enable)

7.233.4.5 change_delay

```
ttk_bus.bus_signals_frbs.FrRxPdu.change_delay
```

delay to wait after a value change [ms]

7.233.4.6 context

```
ttk_bus.bus_signals_frbs.FrRxPdu.context
```

context (reference to assigned RTE application instance)

7.233.4.7 cycle_time

```
ttk_bus.bus_signals_frbs.FrRxPdu.cycle_time
```

cycle time of PDU in [ms]

7.233.4.8 debounce_time

```
ttk_bus.bus_signals_frbs.FrRxPdu.debounce_time
```

minimum debounce delay for event-triggered PDUs in [ms]

7.233.4.9 descr

```
ttk_bus.bus_signals_frbs.FrRxPdu.descr
```

PDU description text.

7.233.4.10 kickout_switch

ttk_bus.bus_signals_frbs.FrRxPdu.kickout_switch

PDU Kickout/Trigger source-switch
(a "HilVar" instance)

7.233.4.11 kickout_var

ttk_bus.bus_signals_frbs.FrRxPdu.kickout_var

PDU Kickout/Trigger variable
(a "HilVar" instance)

7.233.4.12 manual_enable_var

ttk_bus.bus_signals_frbs.FrRxPdu.manual_enable_var

Manual enable switch variable (when specified via enable_path)
(a "HilVar" instance)

7.233.4.13 recovery_time

ttk_bus.bus_signals_frbs.FrRxPdu.recovery_time

time for recovery/"wiedergut" checks [ms]

7.233.4.14 timeout

ttk_bus.bus_signals_frbs.FrRxPdu.timeout

detection time for timeout errors [ms]

7.233.4.15 tm_switch

ttk_bus.bus_signals_frbs.FrRxPdu.tm_switch

Transmission Mode handling source-switch (for disable_mode: tm)
(a "HilVar" instance)

7.233.4.16 tm_var

ttk_bus.bus_signals_frbs.FrRxPdu.tm_var

Transmission Mode handling variable (for disable_mode: tm)
(a "HilVar" instance)

7.233.4.17 tx_enable_switch

`ttk_bus.bus_signals_frbs.FrRxPdu.tx_enable_switch`

HIL-Tx-enable/disable source-switch (for disable_mode: en)
(a "HilVar" instance)

7.233.4.18 tx_enable_var

`ttk_bus.bus_signals_frbs.FrRxPdu.tx_enable_var`

HIL-Tx-enable/disable variable (for disable_mode: en) (a "HilVar" instance)

7.233.4.19 update_bit_enable_var

`ttk_bus.bus_signals_frbs.FrRxPdu.update_bit_enable_var`

Update Bit handling variable (for disable_mode: ub)
(a "HilVar" instance)

7.233.4.20 update_bit_switch

`ttk_bus.bus_signals_frbs.FrRxPdu.update_bit_switch`

Update Bit handling source-switch (for disable_mode: ub)
(a "HilVar" instance)

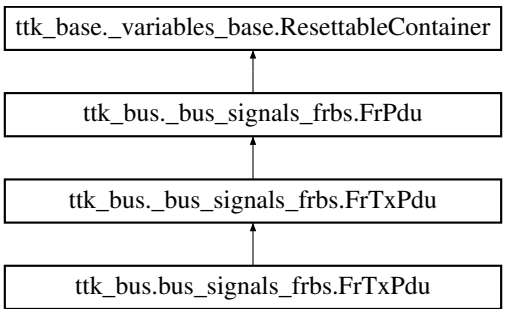
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_frbs.py (v1.8)

7.234 ttk_bus._bus_signals_frbs.FrTxPdu Class Reference

FlexRay ECU-Tx (ECU-Tx => HIL-Rx) PDU data.

Inheritance diagram for ttk_bus._bus_signals_frbs.FrTxPdu:



Public Member Functions

- __init__ (self, context, cycle_time=0, debounce_time=0, bus_systems_path=None, alias='', descr='', change_delay=None, timeout=None, recovery_time=None)

  FlexRay ECU-Tx (HIL-Rx) PDU data.

Public Attributes

- error_status_var

  HIL-Rx PDU Error Status variable (a "HilVar" instance).
- data_receive_counter_var

  HIL-Rx Data Receiver Counter variable (a "HilVar" instance).
- nullframe_var

  HIL-Rx Nullframe variable (a "HilVar" instance).

7.234.1 Detailed Description

FlexRay ECU-Tx (ECU-Tx => HIL-Rx) PDU data.

Extends FrPdu with access to status variables `error_status_var` and `data_receive_counter_var`.

7.234.2 Constructor & Destructor Documentation

7.234.2.1 __init__()

```
ttk_bus._bus_signals_frbs.FrTxPdu.__init__ (
                self,
                context,
                cycle_time = 0,
                debounce_time = 0,
                bus_systems_path = None,
                alias = '',
                descr = '',
                change_delay = None,
                timeout = None,
                recovery_time = None )
```

FlexRay ECU-Tx (HIL-Rx) PDU data.

Parameters

context	parent context (an RTE application instance to resolve signal/variable paths)
cycle_time	cycle time in [ms], 0 for event-triggered PDUs
debounce_time	minimum debounce delay time between events in [ms] for event-triggered PDUs
bus_systems_path	base path to PDU below Bus Systems
alias	PDU alias/name
descr	(opt) PDU description (may be used in report entries)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_frbs.FrPdu.

Reimplemented in ttk_bus.bus_signals_frbs.FrTxPdu.

7.234.3 Member Data Documentation

7.234.3.1 data_receive_counter_var

`ttk_bus._bus_signals_frbs.FrTxPdu.data_receive_counter_var`

HIL-Rx Data Receiver Counter variable (a "HilVar" instance).

"Counts the number of received PDUs. Null frames are ignored."

7.234.3.2 error_status_var

`ttk_bus._bus_signals_frbs.FrTxPdu.error_status_var`

HIL-Rx PDU Error Status variable (a "HilVar" instance).

"Indicates various statuses. Each status has its own bit,
 so parallel statuses are possible."

- Bit 0: Communication controller (CC) FlexRay bus synchronization error
  - 0: CC is synchronized
  - 1: CC not synchronized
- Bit 1: Buffer access error
  - 0: no buffer access error
  - 1: error while accessing buffer
- Bit 2: Boundary violation in FlexRay slot
  - 0: no boundary violation detected
  - 1: boundary violation detected
- Bit 3: Syntax error in FlexRay slot
  - 0: no syntax error detected
  - 1: syntax error detected
- Bit 4: Content error in FlexRay slot
  - 0: no content error detected
  - 1: content error detected
- Bit 5: Empty slot (FlexRay static frames only)
  - 0: data received in static slot
  - 1: slot is empty, no data received in static slot
- Bit 6: Software CRC calculation error
  - 0: CRC from last received frame is correct (or CRC calculation deactivated).
  - 1: CRC from last received frame is not correct.
- Bits 7 to 15: unused

7.234.3.3 nullframe_var

`ttk_bus._bus_signals_frbs.FrTxPdu.nullframe_var`

HIL-Rx Nullframe variable (a "HilVar" instance).

"Indicates whether the received frame was a null frame (only for static PDUs)"

- 0: last received frame was not a null frame
- 1: last received frame was a null frame

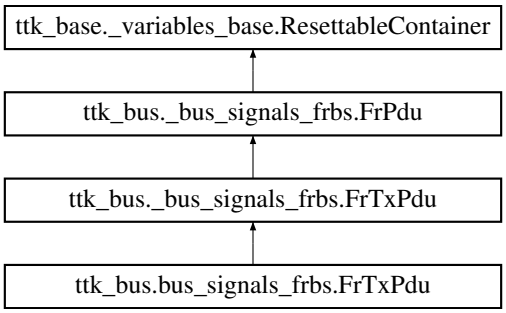
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_frbs.py (v2.5)

7.235 ttk_bus.bus_signals_frbs.FrTxPdu Class Reference

FlexRay ECU-Tx (ECU-Tx => HIL-Rx) PDU data.

Inheritance diagram for ttk_bus.bus_signals_frbs.FrTxPdu:



Public Member Functions

- __init__ (self, context, cycle_time=0, debounce_time=0, bus_systems_path=None, alias='', descr='', change_delay=None, timeout=None, recovery_time=None)

  FlexRay ECU-Tx (HIL-Rx) PDU data.
- reset (self, *args, **kwargs)

  Call reset() on all contained variables/signals that support it.

Public Attributes

- error_status_var
  HIL-Rx PDU Error Status variable (a "HilVar" instance).
- data_receive_counter_var
  HIL-Rx Data Receiver Counter variable (a "HilVar" instance).
- nullframe_var
  HIL-Rx Nullframe variable (a "HilVar" instance).
- alias
  PDU alias name.
- descr
  PDU description text.
- context
  context (reference to assigned RTE application instance)
- bus_systems_path
  base path to PDU below BusSystems
- cycle_time
  cycle time of PDU in [ms]
- debounce_time
  minimum debounce delay for event-triggered PDUs in [ms]
- change_delay
  delay to wait after a value change [ms]
- timeout
  detection time for timeout errors [ms]
- recovery_time
  time for recovery/"wiedergut" checks [ms]

7.235.1 Detailed Description

FlexRay ECU-Tx (ECU-Tx => HIL-Rx) PDU data.

Extends FrPdu with access to status variables `error_status_var` and `data_receive_counter_var`.

7.235.2 Constructor & Destructor Documentation

7.235.2.1 __init__()

```
ttk_bus.bus_signals_frbs.FrTxPdu.__init__ (
              self,
              context,
              cycle_time = 0,
              debounce_time = 0,
              bus_systems_path = None,
              alias = '',
              descr = '',
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

FlexRay ECU-Tx (HIL-Rx) PDU data.

Parameters

context	parent context (an RTE application instance to resolve signal/variable paths)
cycle_time	cycle time in [ms], 0 for event-triggered PDUs
debounce_time	minimum debounce delay time between events in [ms] for event-triggered PDUs
bus_systems_path	base path to PDU below Bus Systems
alias	PDU alias/name
descr	(opt) PDU description (may be used in report entries)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_frbs.FrTxPdu.

7.235.3 Member Function Documentation

7.235.3.1 reset()

```
ttk_bus.bus_signals_frbs.FrTxPdu.reset (
              self,
             * args,
             ** kwargs )
```

Call reset() on all contained variables/signals that support it.

Simple, non-recursive approach.

Parameters

∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.ResettableContainer.

7.235.4 Member Data Documentation

7.235.4.1 alias

```
ttk_bus.bus_signals_frbs.FrTxPdu.alias
```

PDU alias name.

7.235.4.2 bus_systems_path

```
ttk_bus.bus_signals_frbs.FrTxPdu.bus_systems_path
```

base path to PDU below BusSystems

7.235.4.3 change_delay

ttk_bus.bus_signals_frbs.FrTxPdu.change_delay

delay to wait after a value change [ms]

7.235.4.4 context

ttk_bus.bus_signals_frbs.FrTxPdu.context

context (reference to assigned RTE application instance)

7.235.4.5 cycle_time

ttk_bus.bus_signals_frbs.FrTxPdu.cycle_time

cycle time of PDU in [ms]

7.235.4.6 data_receive_counter_var

ttk_bus.bus_signals_frbs.FrTxPdu.data_receive_counter_var

HIL-Rx Data Receiver Counter variable (a "HilVar" instance).

> "Counts the number of received PDUs. Null frames are ignored."

7.235.4.7 debounce_time

ttk_bus.bus_signals_frbs.FrTxPdu.debounce_time

minimum debounce delay for event-triggered PDUs in [ms]

7.235.4.8 descr

ttk_bus.bus_signals_frbs.FrTxPdu.descr

PDU description text.

7.235.4.9 error_status_var

`ttk_bus.bus_signals_frbs.FrTxPdu.error_status_var`

HIL-Rx PDU Error Status variable (a "HilVar" instance).

"Indicates various statuses. Each status has its own bit,
 so parallel statuses are possible."

- Bit 0: Communication controller (CC) FlexRay bus synchronization error
  - 0: CC is synchronized
  - 1: CC not synchronized
- Bit 1: Buffer access error
  - 0: no buffer access error
  - 1: error while accessing buffer
- Bit 2: Boundary violation in FlexRay slot
  - 0: no boundary violation detected
  - 1: boundary violation detected
- Bit 3: Syntax error in FlexRay slot
  - 0: no syntax error detected
  - 1: syntax error detected
- Bit 4: Content error in FlexRay slot
  - 0: no content error detected
  - 1: content error detected
- Bit 5: Empty slot (FlexRay static frames only)
  - 0: data received in static slot
  - 1: slot is empty, no data received in static slot
- Bit 6: Software CRC calculation error
  - 0: CRC from last received frame is correct (or CRC calculation deactivated).
  - 1: CRC from last received frame is not correct.
- Bits 7 to 15: unused

7.235.4.10 nullframe_var

`ttk_bus.bus_signals_frbs.FrTxPdu.nullframe_var`

HIL-Rx Nullframe variable (a "HilVar" instance).

"Indicates whether the received frame was a null frame (only for static PDUs)"

- 0: last received frame was not a null frame
- 1: last received frame was a null frame

7.235.4.11 recovery_time

`ttk_bus.bus_signals_frbs.FrTxPdu.recovery_time`

time for recovery/"wiedergut" checks [ms]

7.235.4.12 timeout

`ttk_bus.bus_signals_frbs.FrTxPdu.timeout`

detection time for timeout errors [ms]

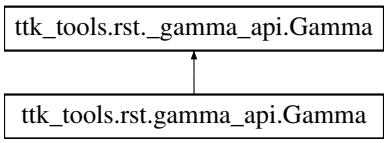
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_frbs.py (v1.8)

7.236 ttk_tools.rst._gamma_api.Gamma Class Reference

GammaService-connection (application).

Inheritance diagram for ttk_tools.rst._gamma_api.Gamma:



Public Member Functions

- __init__ (self, system_file, config_file, system_name_local=None, system_name_remote=None, io_↩ load=False, message_mode=MESSAGE_MODE.CONSOLE, log_level=LOG_LEVEL.ERROR, log_file↩ _path="", keep_gamma_alive=True, port=DEFAULT_LISTEN_PORT, timeout_ms=DEFAULT_ATTACH_↩ TIMEOUT)

  Initialize a GammaV connection.
- __del__ (self)

  Destructor/Cleanup.
- getGammaVersioninfo (self)

  Get version info of current Gamma V service.
- getReference (self)

  Get a gaapi reference for variable access.
- isGammaServiceAlive (self, verbose=True)

  Check if GammaV-Service is alive.
- detachFromService (self)

  Detach from GammaV-service.
- attachToService (self, port=None, timeout_ms=None)

  Attach to GammaV-Service.
- startScheduler (self, scheduler)

  Start a specific scheduler.
- stopScheduler (self, scheduler)

  Stop a scheduler.
- infoScheduler (self, scheduler)

  Get info about a specific scheduler.
- statSchedulerSlot (self, slot)

  Get timing info about a scheduler slot.
- getDAQ (self)

  Get a DAQ/"Capture" instance configured for this Gamma connection.

Static Public Member Functions

- getPVs ()

  Get PV addresses available on the currently connected system(s).

Protected Member Functions

- _callGammaCmd (self, *args)

  Call a simple "gacmd" command.

Protected Attributes

- _keep_gamma_alive
- _system_name_local
- _system_name_remote
- _system_file
- _config_file
- _port

  check if gamma service is running
- _timeout_ms

  check if gamma service is running
- _gamma_process
- _show_gamma_calls

  True: print gamma process calls to stdout.

7.236.1 Detailed Description

GammaService-connection (application).

7.236.2 Constructor & Destructor Documentation

7.236.2.1 __init__()

```
ttk_tools.rst._gamma_api.Gamma.__init__ (
              self,
              system_file,
              config_file,
              system_name_local = None,
              system_name_remote = None,
              io_load = False,
              message_mode = MESSAGE_MODE.CONSOLE,
              log_level = LOG_LEVEL.ERROR,
              log_file_path = "",
              keep_gamma_alive = True,
              port = DEFAULT_LISTEN_PORT,
              timeout_ms = DEFAULT_ATTACH_TIMEOUT )
```

Initialize a GammaV connection.

Parameters

system_file	system configuration file
config_file	service configuration file
system_name_local	system name for the local system (optional - only needed if system_file is not used)
system_name_remote	system name for a remote system (optional - needed if an already started gamma service should be checked and used)
io_load	True: load IO Plugins + IO mapping
message_mode	0: console (default) 1: file 2: syslog 3: console and file 4: console and syslog See gamma_api_common.MESSAGE_MODE
log_level	1: error (default) 2: warn 3: info See gamma_api_common.LOG_LEVEL
log_file_path	path to write the log file of the gamma service to (an empty path deactivates file logging)
keep_gamma_alive	True: leave the gamma service alive on shutdown False: Kill running gamma service on shutdown Leaving the service running/alive is the safe (and default) option.
port	Listen port of the gamma service (see gamma service configuration file)
timeout_ms	timeout to wait for the gamma service during attach

Example:

```
gamma_instance = Gamma(
    system_file = 'D:/Projekte/Kartentester/Datenmodell/system.xml',
    config_file = 'D:/Projekte/Kartentester/Datenmodell/config_local.xml',
    system_name_local = 'iSyst',
    message_mode      = MESSAGE_MODE.CONSOLE,
    log_level         = LOG_LEVEL.ERROR,
    io_load           = False
)
```

Reimplemented in ttk_tools.rst.gamma_api.Gamma.

7.236.2.2 __del__()

```
ttk_tools.rst._gamma_api.Gamma.__del__ (
              self )
```

Destructor/Cleanup.

7.236.3 Member Function Documentation

7.236.3.1 _callGammaCmd()

```
ttk_tools.rst._gamma_api.Gamma._callGammaCmd (
              self,
             * args )  [protected]
```

Call a simple "gacmd" command.

Parameters

**args*	command line arguments for GAMMA_CMD_PROCESS_NAME

Note

If an argument is of a list type, it will be expanded (mainly to make adding ids/addresses more convenient)

Example:

```
_callGammaCmd(a, [b, c], d)
# will be flattened/expanded to
_callGammaCmd(a, b, c, d)
```

Returns

the command's stdout as a string on success, None if the gamma command could not be resolved or a numeric (non-zero) return/exit code in case of errors.

7.236.3.2 attachToService()

```
ttk_tools.rst._gamma_api.Gamma.attachToService (
              self,
              port = None,
              timeout_ms = None )
```

Attach to GammaV-Service.

If no parameters are set, default settings (as supplied in constructor) will be used.

Parameters

port	override for Listen port of the gamma service
timeout_ms	override for timeout to wait for attach

Reimplemented in ttk_tools.rst.gamma_api.Gamma.

7.236.3.3 detachFromService()

```
ttk_tools.rst._gamma_api.Gamma.detachFromService (
              self )
```

Detach from GammaV-service.

Reimplemented in ttk_tools.rst.gamma_api.Gamma.

7.236.3.4 getDAQ()

```
ttk_tools.rst._gamma_api.Gamma.getDAQ (
              self )
```

Get a DAQ/"Capture" instance configured for this Gamma connection.

Reimplemented in ttk_tools.rst.gamma_api.Gamma.

7.236.3.5 getGammaVersioninfo()

```
ttk_tools.rst._gamma_api.Gamma.getGammaVersioninfo (
              self )
```

Get version info of current Gamma V service.

Returns

a version info string or None if access failed.

Reimplemented in ttk_tools.rst.gamma_api.Gamma.

7.236.3.6 getPVs()

```
ttk_tools.rst._gamma_api.Gamma.getPVs ( )  [static]
```

Get PV addresses available on the currently connected system(s).

Example: PV info dicts

```
# Each info dict contains entries similar to this:
pv_info = {
    "address":     'iSyst:DAQ.Memory.Group.DIO_IN_00', # full PV address as string (again)
    "name":        'DIO_IN_00',          # just the PV (array) name
    "data_type":   'UINT_8',             # data type as string, see gacommon.DataType
    "num_elements": 1,                   # number of elements of the PV (array)
}
```

Note

For data type BLOB there will be an additional entry "length" that contains the BLOB's length.

The PV (or actually: PV-Array) name will be the unique name below the "group" level (i.e. address level 4), and may still contain further dots for sub-structured PVs

Returns

a dict with { <pv_address>: {pv info} } mappings.

Reimplemented in ttk_tools.rst.gamma_api.Gamma.

7.236.3.7 getReference()

```
ttk_tools.rst._gamma_api.Gamma.getReference (
              self )
```

Get a gaapi reference for variable access.

Returns

a reference to the current gaapi instance.

Reimplemented in ttk_tools.rst.gamma_api.Gamma.

7.236.3.8 infoScheduler()

```
ttk_tools.rst._gamma_api.Gamma.infoScheduler (
              self,
              scheduler )
```

Get info about a specific scheduler.

Parameters

scheduler	id/address of scheduler to get info on, either as "name" or "index". See startScheduler()

Usage:

```
result = gamma_instance.infoScheduler('SchedulerMatlab')
print result
```

Returns

status string of gacmd executable, None if gacmd was not available or a numeric (non-zero) exit code if the gacmd call failed.

Reimplemented in ttk_tools.rst.gamma_api.Gamma.

7.236.3.9 isGammaServiceAlive()

```
ttk_tools.rst._gamma_api.Gamma.isGammaServiceAlive (
              self,
              verbose = True )
```

Check if GammaV-Service is alive.

Returns

True if service is alive, otherwise False.

Reimplemented in ttk_tools.rst.gamma_api.Gamma.

7.236.3.10 startScheduler()

```
ttk_tools.rst._gamma_api.Gamma.startScheduler (
              self,
              scheduler )
```

Start a specific scheduler.

Parameters

scheduler	id/address of scheduler to start, either as "name" or "index". Has to be supplied as string, as both variants may contain a system name separated by a colon and a assortment of dots. Compare Gamma V Specification and Overview, Chapter 10.2 Structure and Naming Conventions

Usage:

```
result = gamma_instance.startScheduler(':SchedulerMatlab')
# or 'SchedulerMatlab'       # a missing system name will be added as default system
# or 'iSyst:SchedulerMatlab' # w/ explicit system name
# or ':0'                    # default system, first scheduler
# or '0:0'                   # first system, first scheduler
print result
```

Returns

status string of gacmd executable (which is probably empty in this use-case), None if gacmd was not available or a numeric (non-zero) exit code if the gacmd call failed.

Reimplemented in ttk_tools.rst.gamma_api.Gamma.

7.236.3.11 statSchedulerSlot()

```
ttk_tools.rst._gamma_api.Gamma.statSchedulerSlot (
              self,
              slot )
```

Get timing info about a scheduler slot.

Parameters

slot	id/address of scheduler slot, either as "name" or "index". See startScheduler

Returns

status string of gacmd executable, None if gacmd was not available or a numeric (non-zero) exit code if the gacmd call failed.

Reimplemented in ttk_tools.rst.gamma_api.Gamma.

7.236.3.12 stopScheduler()

```
ttk_tools.rst._gamma_api.Gamma.stopScheduler (
              self,
              scheduler )
```

Stop a scheduler.

Parameters

scheduler	id/address of scheduler to stop, either as "name" or "index". See startScheduler()

Usage:

```
result = gamma_instance.stopScheduler('SchedulerMatlab')
print result
```

Returns

status string of gacmd executable (which is probably empty in this use-case), None if gacmd was not available or a numeric (non-zero) exit code if the gacmd call failed.

Reimplemented in ttk_tools.rst.gamma_api.Gamma.

7.236.4 Member Data Documentation

7.236.4.1 _config_file

ttk_tools.rst._gamma_api.Gamma._config_file [protected]

7.236.4.2 _gamma_process

ttk_tools.rst._gamma_api.Gamma._gamma_process [protected]

7.236.4.3 _keep_gamma_alive

ttk_tools.rst._gamma_api.Gamma._keep_gamma_alive [protected]

7.236.4.4 _port

ttk_tools.rst._gamma_api.Gamma._port [protected]

check if gamma service is running

7.236.4.5 _show_gamma_calls

ttk_tools.rst._gamma_api.Gamma._show_gamma_calls [protected]

True: print gamma process calls to stdout.

7.236.4.6 _system_file

ttk_tools.rst._gamma_api.Gamma._system_file [protected]

7.236.4.7 _system_name_local

ttk_tools.rst._gamma_api.Gamma._system_name_local [protected]

7.236.4.8 _system_name_remote

ttk_tools.rst._gamma_api.Gamma._system_name_remote [protected]

7.236.4.9 _timeout_ms

```
ttk_tools.rst._gamma_api.Gamma._timeout_ms  [protected]
```

check if gamma service is running

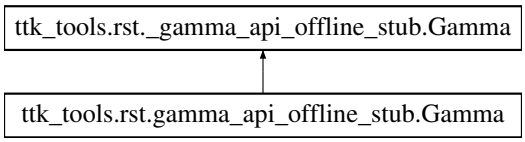
The documentation for this class was generated from the following file:

- ttk_tools/rst/_gamma_api.py (v3.4.1)

7.237 ttk_tools.rst._gamma_api_offline_stub.Gamma Class Reference

GammaService-connection (application) stub.

Inheritance diagram for ttk_tools.rst._gamma_api_offline_stub.Gamma:



Public Member Functions

- __init__ (self, system_file, config_file, system_name_local=None, system_name_remote=None, io_↩ load=False, message_mode=MESSAGE_MODE.CONSOLE, log_level=LOG_LEVEL.ERROR, log_file↩ _path="", keep_gamma_alive=True, port=DEFAULT_LISTEN_PORT, timeout_ms=DEFAULT_ATTACH_↩ TIMEOUT)

  Initialize a GammaV connection.
- getGammaVersioninfo (self)

  Get version info of current Gamma V service.
- getReference (self)

  Get a gaapi reference for variable access.
- isGammaServiceAlive (self, verbose=True)

  Check if GammaV-Service is alive.
- detachFromService (self)

  Detach from GammaV-service.
- attachToService (self, port=None, timeout_ms=None)

  Attach to GammaV-Service.
- startScheduler (self, scheduler)

  Start a specific scheduler.
- stopScheduler (self, scheduler)

  Stop a scheduler.
- infoScheduler (self, scheduler)

  Get info about a specific scheduler.
- statSchedulerSlot (self, slot)

  Get timing info about a scheduler slot.
- getDAQ (self)

  Get a DAQ/"Capture" instance configured for this Gamma connection.

Static Public Member Functions

- getPVs ()

  Get PV addresses available on the currently connected system(s).

Protected Attributes

- _keep_gamma_alive
- _system_name_local
- _system_name_remote
- _system_file
- _config_file
- _port
- _timeout_ms
- _gamma_process
- _show_gamma_calls

  True: print gamma process calls to stdout.

7.237.1 Detailed Description

GammaService-connection (application) stub.

7.237.2 Constructor & Destructor Documentation

7.237.2.1 __init__()

```
ttk_tools.rst._gamma_api_offline_stub.Gamma.__init__ (
              self,
              system_file,
              config_file,
              system_name_local = None,
              system_name_remote = None,
              io_load = False,
              message_mode = MESSAGE_MODE.CONSOLE,
              log_level = LOG_LEVEL.ERROR,
              log_file_path = "",
              keep_gamma_alive = True,
              port = DEFAULT_LISTEN_PORT,
              timeout_ms = DEFAULT_ATTACH_TIMEOUT )
```

Initialize a GammaV connection.

Parameters

system_file	system configuration file
config_file	service configuration file
system_name_local	system name for the local system (optional - only needed if system_file is not used)
system_name_remote	system name for a remote system (optional - needed if an already started gamma service should be checked and used)
io_load	True: load IO Plugins + IO mapping

Parameters

message_mode	0: console (default) 1: file 2: syslog 3: console and file 4: console and syslog See gamma_api_common.MESSAGE_MODE
log_level	1: error (default) 2: warn 3: info See gamma_api_common.LOG_LEVEL
log_file_path	path to write the log file of the gamma service to (an empty path deactivates file logging)
keep_gamma_alive	True: leave the gamma service alive on shutdown False: Kill running gamma service on shutdown Leaving the service running/alive is the safe (and default) option.
port	listen port of the gamma service (see gamma service configuration file)
timeout_ms	timeout to wait for the gamma service during attach

Example:

```
gamma_instance = Gamma(
    system_file = 'D:/Projekte/Kartentester/Datenmodell/system.xml',
    config_file = 'D:/Projekte/Kartentester/Datenmodell/config_local.xml',
    system_name_local = 'iSyst',
    message_mode      = MESSAGE_MODE.CONSOLE,
    log_level         = LOG_LEVEL.ERROR,
    io_load           = False
)
```

Reimplemented in ttk_tools.rst.gamma_api_offline_stub.Gamma.

7.237.3 Member Function Documentation

7.237.3.1 attachToService()

```
ttk_tools.rst._gamma_api_offline_stub.Gamma.attachToService (
              self,
              port = None,
              timeout_ms = None )
```

Attach to GammaV-Service.

If no parameters are set, default settings (as supplied in constructor) will be used.

Parameters

port	override for Listen port of the gamma service
timeout_ms	override for timeout to wait for attach

Reimplemented in ttk_tools.rst.gamma_api_offline_stub.Gamma.

7.237.3.2 detachFromService()

```
ttk_tools.rst._gamma_api_offline_stub.Gamma.detachFromService (
              self )
```

Detach from GammaV-service.

Reimplemented in ttk_tools.rst.gamma_api_offline_stub.Gamma.

7.237.3.3 getDAQ()

```
ttk_tools.rst._gamma_api_offline_stub.Gamma.getDAQ (
              self )
```

Get a DAQ/"Capture" instance configured for this Gamma connection.

Reimplemented in ttk_tools.rst.gamma_api_offline_stub.Gamma.

7.237.3.4 getGammaVersioninfo()

```
ttk_tools.rst._gamma_api_offline_stub.Gamma.getGammaVersioninfo (
              self )
```

Get version info of current Gamma V service.

Returns

a version info string or None if access failed.

Reimplemented in ttk_tools.rst.gamma_api_offline_stub.Gamma.

7.237.3.5 getPVs()

```
ttk_tools.rst._gamma_api_offline_stub.Gamma.getPVs ( )  [static]
```

Get PV addresses available on the currently connected system(s).

Example: PV info dicts

```
# Each info dict contains entries similar to this:
pv_info = {
    "address":     'iSyst:DAQ.Memory.Group.DIO_IN_00', # full PV address as string (again)
    "name":        'DIO_IN_00',          # just the PV (array) name
    "data_type":   'UINT_8',             # data type as string, see gacommon.DataType
    "num_elements": 1,                   # number of elements of the PV (array)
}
```

Note

For data type BLOB there will be an additional entry "length" that contains the BLOB's length.

The PV (or actually: PV-Array) name will be the unique name below the "group" level (i.e. address level 4), and may still contain further dots for sub-structured PVs

Returns

a dict with { <pv_address>: {pv info} } mappings.

Reimplemented in ttk_tools.rst.gamma_api_offline_stub.Gamma.

7.237.3.6 getReference()

```
ttk_tools.rst._gamma_api_offline_stub.Gamma.getReference (
              self )
```

Get a gaapi reference for variable access.

Returns

a reference to the current gaapi module.

Reimplemented in ttk_tools.rst.gamma_api_offline_stub.Gamma.

7.237.3.7 infoScheduler()

```
ttk_tools.rst._gamma_api_offline_stub.Gamma.infoScheduler (
              self,
              scheduler )
```

Get info about a specific scheduler.

Parameters

scheduler	id/address of scheduler to get info on, either as "name" or "index". See startScheduler()

Usage:

```
result = gamma_instance.infoScheduler('SchedulerMatlab')
print result
```

Returns

status string of gacmd executable, None if gacmd was not available or a numeric (non-zero) exit code if the gacmd call failed.

Reimplemented in ttk_tools.rst.gamma_api_offline_stub.Gamma.

7.237.3.8 isGammaServiceAlive()

```
ttk_tools.rst._gamma_api_offline_stub.Gamma.isGammaServiceAlive (
              self,
              verbose = True )
```

Check if GammaV-Service is alive.

Parameters

verbose	True: print service status

Returns

True if service is alive, otherwise False.

Reimplemented in ttk_tools.rst.gamma_api_offline_stub.Gamma.

7.237.3.9 startScheduler()

```
ttk_tools.rst._gamma_api_offline_stub.Gamma.startScheduler (
              self,
              scheduler )
```

Start a specific scheduler.

Parameters

scheduler	id/address of scheduler to start, either as "name" or "index". Has to be supplied as string, as both variants may contain a system name separated by a colon and a assortment of dots. Compare Gamma V Specification and Overview, Chapter 10.2 Structure and Naming Conventions

Usage:

```
result = gamma_instance.startScheduler(':SchedulerMatlab')
# or 'SchedulerMatlab'       # a missing system name will be added as default system
# or 'iSyst:SchedulerMatlab' # w/ explicit system name
# or ':0'                    # default system, first scheduler
# or '0:0'                   # first system, first scheduler
print result
```

Returns

status string of gacmd executable (which is probably empty in this use-case), None if gacmd was not available or a numeric (non-zero) exit code if the gacmd call failed.

Reimplemented in ttk_tools.rst.gamma_api_offline_stub.Gamma.

7.237.3.10 statSchedulerSlot()

```
ttk_tools.rst._gamma_api_offline_stub.Gamma.statSchedulerSlot (
              self,
              slot )
```

Get timing info about a scheduler slot.

Parameters

slot	id/address of scheduler slot, either as "name" or "index". See startScheduler

Returns

status string of gacmd executable, None if gacmd was not available or a numeric (non-zero) exit code if the gacmd call failed.

Reimplemented in ttk_tools.rst.gamma_api_offline_stub.Gamma.

7.237.3.11 stopScheduler()

```
ttk_tools.rst._gamma_api_offline_stub.Gamma.stopScheduler (
            self,
            scheduler )
```

Stop a scheduler.

Parameters

scheduler	id/address of scheduler to stop, either as "name" or "index". See startScheduler()

Usage:

```
result = gamma_instance.stopScheduler('SchedulerMatlab')
print result
```

Returns

status string of gacmd executable (which is probably empty in this use-case), None if gacmd was not available or a numeric (non-zero) exit code if the gacmd call failed.

Reimplemented in ttk_tools.rst.gamma_api_offline_stub.Gamma.

7.237.4 Member Data Documentation

7.237.4.1 _config_file

```
ttk_tools.rst._gamma_api_offline_stub.Gamma._config_file  [protected]
```

7.237.4.2 _gamma_process

```
ttk_tools.rst._gamma_api_offline_stub.Gamma._gamma_process  [protected]
```

7.237.4.3 _keep_gamma_alive

```
ttk_tools.rst._gamma_api_offline_stub.Gamma._keep_gamma_alive  [protected]
```

7.237.4.4 _port

```
ttk_tools.rst._gamma_api_offline_stub.Gamma._port  [protected]
```

7.237.4.5 _show_gamma_calls

ttk_tools.rst._gamma_api_offline_stub.Gamma._show_gamma_calls [protected]

True: print gamma process calls to stdout.

7.237.4.6 _system_file

ttk_tools.rst._gamma_api_offline_stub.Gamma._system_file [protected]

7.237.4.7 _system_name_local

ttk_tools.rst._gamma_api_offline_stub.Gamma._system_name_local [protected]

7.237.4.8 _system_name_remote

ttk_tools.rst._gamma_api_offline_stub.Gamma._system_name_remote [protected]

7.237.4.9 _timeout_ms

ttk_tools.rst._gamma_api_offline_stub.Gamma._timeout_ms [protected]

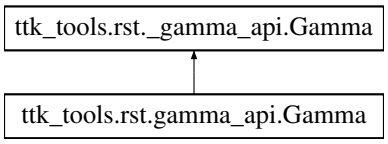
The documentation for this class was generated from the following file:

- ttk_tools/rst/_gamma_api_offline_stub.py (v1.7)

7.238 ttk_tools.rst.gamma_api.Gamma Class Reference

GammaService-connection (application).

Inheritance diagram for ttk_tools.rst.gamma_api.Gamma:



Public Member Functions

- __init__ (self, system_file, config_file, system_name_local=None, system_name_remote=None, io_load=False, message_mode=MESSAGE_MODE.CONSOLE, log_level=LOG_LEVEL.ERROR, log_file_path="", keep_gamma_alive=True, port=DEFAULT_LISTEN_PORT, timeout_ms=DEFAULT_ATTACH_TIMEOUT)

  Initialize a GammaV connection.
- getGammaVersioninfo (self)

  Get version info of current Gamma V service.
- getReference (self)

  Get a gaapi reference for variable access.
- isGammaServiceAlive (self, verbose=True)

  Check if GammaV-Service is alive.
- detachFromService (self)

  Detach from GammaV-service.
- attachToService (self, port=None, timeout_ms=None)

  Attach to GammaV-Service.
- startScheduler (self, scheduler)

  Start a specific scheduler.
- stopScheduler (self, scheduler)

  Stop a scheduler.
- infoScheduler (self, scheduler)

  Get info about a specific scheduler.
- statSchedulerSlot (self, slot)

  Get timing info about a scheduler slot.
- getDAQ (self)

  Get a DAQ/"Capture" instance configured for this Gamma connection.

Static Public Member Functions

- getPVs ()

  Get PV addresses available on the currently connected system(s).

7.238.1 Detailed Description

GammaService-connection (application).

7.238.2 Constructor & Destructor Documentation

7.238.2.1 __init__()

```
ttk_tools.rst.gamma_api.Gamma.__init__ (
              self,
              system_file,
              config_file,
              system_name_local = None,
              system_name_remote = None,
              io_load = False,
              message_mode = MESSAGE_MODE.CONSOLE,
              log_level = LOG_LEVEL.ERROR,
              log_file_path = "",
              keep_gamma_alive = True,
              port = DEFAULT_LISTEN_PORT,
              timeout_ms = DEFAULT_ATTACH_TIMEOUT )
```

Initialize a GammaV connection.

Parameters

system_file	system configuration file
config_file	service configuration file
system_name_local	system name for the local system (optional - only needed if system_file is not used)
system_name_remote	system name for a remote system (optional - needed if an already started gamma service should be checked and used)
io_load	True: load IO Plugins + IO mapping
message_mode	0: console (default) 1: file 2: syslog 3: console and file 4: console and syslog See gamma_api_common.MESSAGE_MODE
log_level	1: error (default) 2: warn 3: info See gamma_api_common.LOG_LEVEL
log_file_path	path to write the log file of the gamma service to (an empty path deactivates file logging)
keep_gamma_alive	True: leave the gamma service alive on shutdown False: Kill running gamma service on shutdown Leaving the service running/alive is the safe (and default) option.
port	listen port of the gamma service (see gamma service configuration file)
timeout_ms	timeout to wait for the gamma service during attach

Example:

```
gamma_instance = Gamma(
    system_file = 'D:/Projekte/Kartentester/Datenmodell/system.xml',
    config_file = 'D:/Projekte/Kartentester/Datenmodell/config_local.xml',
    system_name_local = 'iSyst',
    message_mode      = MESSAGE_MODE.CONSOLE,
    log_level         = LOG_LEVEL.ERROR,
    io_load           = False
)
```

Reimplemented from ttk_tools.rst._gamma_api.Gamma.

7.238.3 Member Function Documentation

7.238.3.1 attachToService()

```
ttk_tools.rst.gamma_api.Gamma.attachToService (
            self,
            port = None,
            timeout_ms = None )
```

Attach to GammaV-Service.

If no parameters are set, default settings (as supplied in constructor) will be used.

Parameters

port	override for Listen port of the gamma service
timeout_ms	override for timeout to wait for attach

Reimplemented from ttk_tools.rst._gamma_api.Gamma.

7.238.3.2 detachFromService()

```
ttk_tools.rst.gamma_api.Gamma.detachFromService (
              self )
```

Detach from GammaV-service.

Reimplemented from ttk_tools.rst._gamma_api.Gamma.

7.238.3.3 getDAQ()

```
ttk_tools.rst.gamma_api.Gamma.getDAQ (
              self )
```

Get a DAQ/"Capture" instance configured for this Gamma connection.

Reimplemented from ttk_tools.rst._gamma_api.Gamma.

7.238.3.4 getGammaVersioninfo()

```
ttk_tools.rst.gamma_api.Gamma.getGammaVersioninfo (
              self )
```

Get version info of current Gamma V service.

Returns

a version info string or None if access failed.

Reimplemented from ttk_tools.rst._gamma_api.Gamma.

7.238.3.5 getPVs()

```
ttk_tools.rst.gamma_api.Gamma.getPVs ( )  [static]
```

Get PV addresses available on the currently connected system(s).

Example: PV info dicts

```
# Each info dict contains entries similar to this:
pv_info = {
    "address":     'iSyst:DAQ.Memory.Group.DIO_IN_00', # full PV address as string (again)
    "name":        'DIO_IN_00',          # just the PV (array) name
    "data_type":   'UINT_8',             # data type as string, see gacommon.DataType
    "num_elements": 1,                   # number of elements of the PV (array)
}
```

Note

For data type BLOB there will be an additional entry "length" that contains the BLOB's length.

The PV (or actually: PV-Array) name will be the unique name below the "group" level (i.e. address level 4), and may still contain further dots for sub-structured PVs

Returns

a dict with { <pv_address>: {pv info} } mappings.

Reimplemented from ttk_tools.rst._gamma_api.Gamma.

7.238.3.6 getReference()

```
ttk_tools.rst.gamma_api.Gamma.getReference (
              self )
```

Get a gaapi reference for variable access.

Returns

a reference to the current gaapi module.

Reimplemented from ttk_tools.rst._gamma_api.Gamma.

7.238.3.7 infoScheduler()

```
ttk_tools.rst.gamma_api.Gamma.infoScheduler (
              self,
              scheduler )
```

Get info about a specific scheduler.

Parameters

scheduler	id/address of scheduler to get info on, either as "name" or "index". See startScheduler()

Usage:

```
result = gamma_instance.infoScheduler('SchedulerMatlab')
print result
```

Returns

status string of gacmd executable, None if gacmd was not available or a numeric (non-zero) exit code if the gacmd call failed.

Reimplemented from ttk_tools.rst._gamma_api.Gamma.

7.238.3.8 isGammaServiceAlive()

```
ttk_tools.rst.gamma_api.Gamma.isGammaServiceAlive (
              self,
              verbose = True )
```

Check if GammaV-Service is alive.

Parameters

verbose	True: print service status

Returns

True if service is alive, otherwise False.

Reimplemented from ttk_tools.rst._gamma_api.Gamma.

7.238.3.9 startScheduler()

```
ttk_tools.rst.gamma_api.Gamma.startScheduler (
              self,
              scheduler )
```

Start a specific scheduler.

Parameters

scheduler	id/address of scheduler to start, either as "name" or "index". Has to be supplied as string, as both variants may contain a system name separated by a colon and a assortment of dots. Compare Gamma V Specification and Overview, Chapter 10.2 Structure and Naming Conventions

Usage:

```
result = gamma_instance.startScheduler(':SchedulerMatlab')
# or 'SchedulerMatlab'       # a missing system name will be added as default system
# or 'iSyst:SchedulerMatlab' # w/ explicit system name
# or ':0'                    # default system, first scheduler
# or '0:0'                   # first system, first scheduler
print result
```

Returns

status string of gacmd executable (which is probably empty in this use-case), None if gacmd was not available or a numeric (non-zero) exit code if the gacmd call failed.

Reimplemented from ttk_tools.rst._gamma_api.Gamma.

7.238.3.10 statSchedulerSlot()

```
ttk_tools.rst.gamma_api.Gamma.statSchedulerSlot (
              self,
              slot )
```

Get timing info about a scheduler slot.

Parameters

slot	id/address of scheduler slot, either as "name" or "index". See startScheduler

Returns

status string of gacmd executable, None if gacmd was not available or a numeric (non-zero) exit code if the gacmd call failed.

Reimplemented from ttk_tools.rst._gamma_api.Gamma.

7.238.3.11 stopScheduler()

```
ttk_tools.rst.gamma_api.Gamma.stopScheduler (
            self,
            scheduler )
```

Stop a scheduler.

Parameters

scheduler	id/address of scheduler to stop, either as "name" or "index". See startScheduler()

Usage:

```
result = gamma_instance.stopScheduler('SchedulerMatlab')
print result
```

Returns

status string of gacmd executable (which is probably empty in this use-case), None if gacmd was not available or a numeric (non-zero) exit code if the gacmd call failed.

Reimplemented from ttk_tools.rst._gamma_api.Gamma.

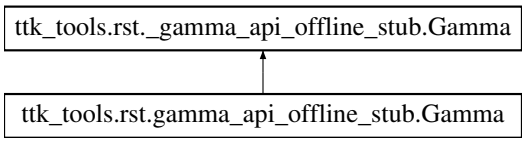
The documentation for this class was generated from the following file:

- ttk_tools/rst/gamma_api.py (v3.0)

7.239 ttk_tools.rst.gamma_api_offline_stub.Gamma Class Reference

GammaService-connection (application) stub.

Inheritance diagram for ttk_tools.rst.gamma_api_offline_stub.Gamma:



Public Member Functions

- __init__ (self, system_file, config_file, system_name_local=None, system_name_remote=None, io_load=False, message_mode=MESSAGE_MODE.CONSOLE, log_level=LOG_LEVEL.ERROR, log_file_path="", keep_gamma_alive=True, port=DEFAULT_LISTEN_PORT, timeout_ms=DEFAULT_ATTACH_TIMEOUT)

  Initialize a GammaV connection.
- getGammaVersioninfo (self)

  Get version info of current Gamma V service.
- getReference (self)

  Get a gaapi reference for variable access.
- isGammaServiceAlive (self, verbose=True)

  Check if GammaV-Service is alive.
- detachFromService (self)

  Detach from GammaV-service.
- attachToService (self, port=None, timeout_ms=None)

  Attach to GammaV-Service.
- startScheduler (self, scheduler)

  Start a specific scheduler.
- stopScheduler (self, scheduler)

  Stop a scheduler.
- infoScheduler (self, scheduler)

  Get info about a specific scheduler.
- statSchedulerSlot (self, slot)

  Get timing info about a scheduler slot.
- getDAQ (self)

  Get a DAQ/"Capture" instance configured for this Gamma connection.

Static Public Member Functions

- getPVs ()

  Get PV addresses available on the currently connected system(s).

7.239.1 Detailed Description

GammaService-connection (application) stub.

7.239.2 Constructor & Destructor Documentation

7.239.2.1 __init__()

```
ttk_tools.rst.gamma_api_offline_stub.Gamma.__init__ (
              self,
              system_file,
              config_file,
              system_name_local = None,
              system_name_remote = None,
              io_load = False,
              message_mode = MESSAGE_MODE.CONSOLE,
              log_level = LOG_LEVEL.ERROR,
              log_file_path = "",
              keep_gamma_alive = True,
              port = DEFAULT_LISTEN_PORT,
              timeout_ms = DEFAULT_ATTACH_TIMEOUT )
```

Initialize a GammaV connection.

Parameters

system_file	system configuration file
config_file	service configuration file
system_name_local	system name for the local system (optional - only needed if system_file is not used)
system_name_remote	system name for a remote system (optional - needed if an already started gamma service should be checked and used)
io_load	True: load IO Plugins + IO mapping
message_mode	0: console (default) 1: file 2: syslog 3: console and file 4: console and syslog See gamma_api_common.MESSAGE_MODE
log_level	1: error (default) 2: warn 3: info See gamma_api_common.LOG_LEVEL
log_file_path	path to write the log file of the gamma service to (an empty path deactivates file logging)
keep_gamma_alive	True: leave the gamma service alive on shutdown False: Kill running gamma service on shutdown Leaving the service running/alive is the safe (and default) option.
port	listen port of the gamma service (see gamma service configuration file)
timeout_ms	timeout to wait for the gamma service during attach

Example:

```
gamma_instance = Gamma(
    system_file = 'D:/Projekte/Kartentester/Datenmodell/system.xml',
    config_file = 'D:/Projekte/Kartentester/Datenmodell/config_local.xml',
    system_name_local = 'iSyst',
    message_mode      = MESSAGE_MODE.CONSOLE,
    log_level         = LOG_LEVEL.ERROR,
    io_load           = False
)
```

Reimplemented from ttk_tools.rst._gamma_api_offline_stub.Gamma.

7.239.3 Member Function Documentation

7.239.3.1 attachToService()

```
ttk_tools.rst.gamma_api_offline_stub.Gamma.attachToService (
            self,
            port = None,
            timeout_ms = None )
```

Attach to GammaV-Service.

If no parameters are set, default settings (as supplied in constructor) will be used.

Parameters

port	override for Listen port of the gamma service
timeout_ms	override for timeout to wait for attach

Reimplemented from ttk_tools.rst._gamma_api_offline_stub.Gamma.

7.239.3.2 detachFromService()

```
ttk_tools.rst.gamma_api_offline_stub.Gamma.detachFromService (
            self )
```

Detach from GammaV-service.

Reimplemented from ttk_tools.rst._gamma_api_offline_stub.Gamma.

7.239.3.3 getDAQ()

```
ttk_tools.rst.gamma_api_offline_stub.Gamma.getDAQ (
            self )
```

Get a DAQ/"Capture" instance configured for this Gamma connection.

Reimplemented from ttk_tools.rst._gamma_api_offline_stub.Gamma.

7.239.3.4 getGammaVersioninfo()

```
ttk_tools.rst.gamma_api_offline_stub.Gamma.getGammaVersioninfo (
            self )
```

Get version info of current Gamma V service.

Returns

a version info string or None if access failed.

Reimplemented from ttk_tools.rst._gamma_api_offline_stub.Gamma.

7.239.3.5 getPVs()

```
ttk_tools.rst.gamma_api_offline_stub.Gamma.getPVs ( )  [static]
```

Get PV addresses available on the currently connected system(s).

Example: PV info dicts

```
# Each info dict contains entries similar to this:
pv_info = {
    "address":     'iSyst:DAQ.Memory.Group.DIO_IN_00', # full PV address as string (again)
    "name":        'DIO_IN_00',          # just the PV (array) name
    "data_type":   'UINT_8',             # data type as string, see gacommon.DataType
    "num_elements": 1,                   # number of elements of the PV (array)
}
```

Note

For data type BLOB there will be an additional entry "length" that contains the BLOB's length.

The PV (or actually: PV-Array) name will be the unique name below the "group" level (i.e. address level 4), and may still contain further dots for sub-structured PVs

Returns

a dict with { <pv_address>: {pv info} } mappings.

Reimplemented from ttk_tools.rst._gamma_api_offline_stub.Gamma.

7.239.3.6 getReference()

```
ttk_tools.rst.gamma_api_offline_stub.Gamma.getReference (
              self )
```

Get a gaapi reference for variable access.

Returns

a reference to the current gaapi module.

Reimplemented from ttk_tools.rst._gamma_api_offline_stub.Gamma.

7.239.3.7 infoScheduler()

```
ttk_tools.rst.gamma_api_offline_stub.Gamma.infoScheduler (
              self,
              scheduler )
```

Get info about a specific scheduler.

Parameters

scheduler	id/address of scheduler to get info on, either as "name" or "index". See startScheduler()

Usage:

```
result = gamma_instance.infoScheduler('SchedulerMatlab')
print result
```

Returns

status string of gacmd executable, None if gacmd was not available or a numeric (non-zero) exit code if the gacmd call failed.

Reimplemented from ttk_tools.rst._gamma_api_offline_stub.Gamma.

7.239.3.8 isGammaServiceAlive()

```
ttk_tools.rst.gamma_api_offline_stub.Gamma.isGammaServiceAlive (
              self,
              verbose = True )
```

Check if GammaV-Service is alive.

Parameters

verbose	True: print service status

Returns

True if service is alive, otherwise False.

Reimplemented from ttk_tools.rst._gamma_api_offline_stub.Gamma.

7.239.3.9 startScheduler()

```
ttk_tools.rst.gamma_api_offline_stub.Gamma.startScheduler (
              self,
              scheduler )
```

Start a specific scheduler.

Parameters

scheduler	id/address of scheduler to start, either as "name" or "index". Has to be supplied as string, as both variants may contain a system name separated by a colon and a assortment of dots. Compare Gamma V Specification and Overview, Chapter 10.2 Structure and Naming Conventions

Usage:

```
result = gamma_instance.startScheduler(':SchedulerMatlab')
# or 'SchedulerMatlab'       # a missing system name will be added as default system
# or 'iSyst:SchedulerMatlab' # w/ explicit system name
# or ':0'                    # default system, first scheduler
# or '0:0'                   # first system, first scheduler
print result
```

Returns

status string of gacmd executable (which is probably empty in this use-case), None if gacmd was not available or a numeric (non-zero) exit code if the gacmd call failed.

Reimplemented from ttk_tools.rst._gamma_api_offline_stub.Gamma.

7.239.3.10 statSchedulerSlot()

```
ttk_tools.rst.gamma_api_offline_stub.Gamma.statSchedulerSlot (
              self,
              slot )
```

Get timing info about a scheduler slot.

Parameters

slot	id/address of scheduler slot, either as "name" or "index". See startScheduler

Returns

status string of gacmd executable, None if gacmd was not available or a numeric (non-zero) exit code if the gacmd call failed.

Reimplemented from ttk_tools.rst._gamma_api_offline_stub.Gamma.

7.239.3.11 stopScheduler()

```
ttk_tools.rst.gamma_api_offline_stub.Gamma.stopScheduler (
            self,
            scheduler )
```

Stop a scheduler.

Parameters

scheduler	id/address of scheduler to stop, either as "name" or "index". See startScheduler()

Usage:

```
result = gamma_instance.stopScheduler('SchedulerMatlab')
print result
```

Returns

status string of gacmd executable (which is probably empty in this use-case), None if gacmd was not available or a numeric (non-zero) exit code if the gacmd call failed.

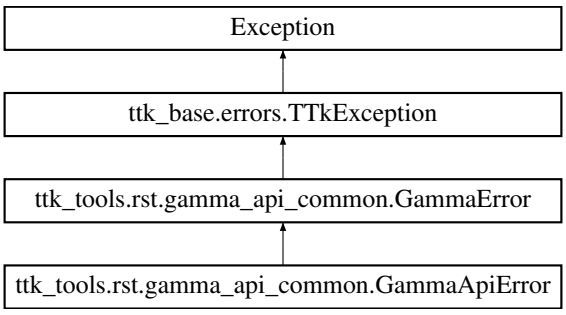
Reimplemented from ttk_tools.rst._gamma_api_offline_stub.Gamma.

The documentation for this class was generated from the following file:

- ttk_tools/rst/gamma_api_offline_stub.py (v1.5)

7.240 ttk_tools.rst.gamma_api_common.GammaApiError Class Reference

Inheritance diagram for ttk_tools.rst.gamma_api_common.GammaApiError:



Public Member Functions

- __init__ (self, message, error_code=None)

Public Attributes

- error_code

7.240.1 Constructor & Destructor Documentation

7.240.1.1 __init__()

```
ttk_tools.rst.gamma_api_common.GammaApiError.__init__ (
              self,
              message,
              error_code = None )
```

Parameters

message	error message
error_code	error code returned by API function, see ErrorCodes

7.240.2 Member Data Documentation

7.240.2.1 error_code

```
ttk_tools.rst.gamma_api_common.GammaApiError.error_code
```

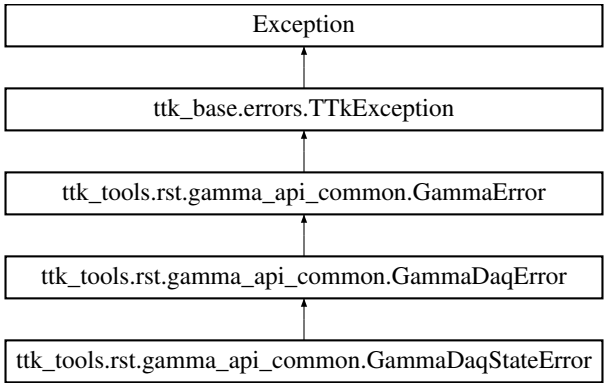
The documentation for this class was generated from the following file:

- ttk_tools/rst/gamma_api_common.py (v1.5)

7.241 ttk_tools.rst.gamma_api_common.GammaDaqError Class Reference

Base Error for GammaV DAQ / data logger functionality.

Inheritance diagram for ttk_tools.rst.gamma_api_common.GammaDaqError:



7.241.1 Detailed Description

Base Error for GammaV DAQ / data logger functionality.

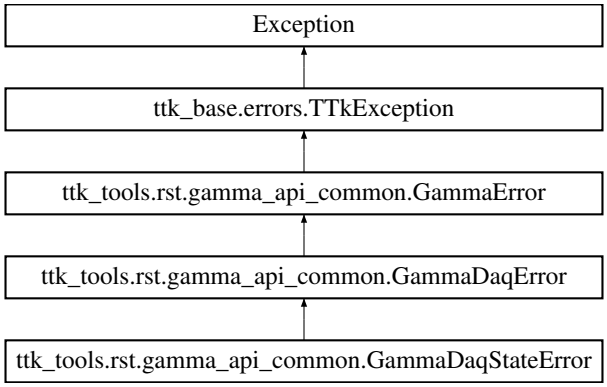
Gamma V DAQ state machine did not reach the expected state

The documentation for this class was generated from the following file:

- ttk_tools/rst/gamma_api_common.py (v1.5)

7.242 ttk_tools.rst.gamma_api_common.GammaDaqStateError Class Reference

Inheritance diagram for ttk_tools.rst.gamma_api_common.GammaDaqStateError:



Public Member Functions

- __init__ (self, message, expected_state, current_state)

Public Attributes

- expected_state
- current_state

7.242.1 Constructor & Destructor Documentation

7.242.1.1 __init__()

```
ttk_tools.rst.gamma_api_common.GammaDaqStateError.__init__ (
              self,
              message,
              expected_state,
              current_state )
```

Parameters

message	error message
expected_state	state that was expected (or potentially a list of expected states)
current_state	state that was actually reached (cause of error)

7.242.2 Member Data Documentation

7.242.2.1 current_state

`ttk_tools.rst.gamma_api_common.GammaDaqStateError.current_state`

7.242.2.2 expected_state

`ttk_tools.rst.gamma_api_common.GammaDaqStateError.expected_state`

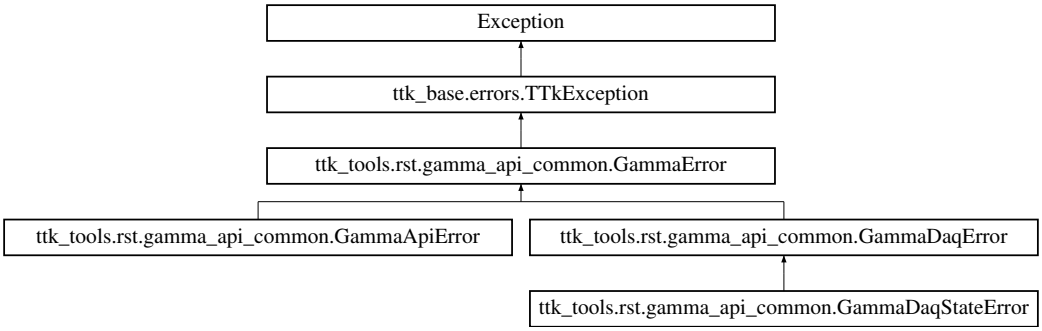
The documentation for this class was generated from the following file:

- ttk_tools/rst/gamma_api_common.py (v1.5)

7.243 ttk_tools.rst.gamma_api_common.GammaError Class Reference

Base Error for GammaV connection.

Inheritance diagram for ttk_tools.rst.gamma_api_common.GammaError:



7.243.1 Detailed Description

Base Error for GammaV connection.

Errors during C API access

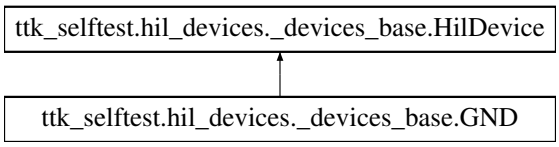
The documentation for this class was generated from the following file:

- ttk_tools/rst/gamma_api_common.py (v1.5)

7.244 ttk_selftest.hil_devices._devices_base.GND Class Reference

Base ground signal container for a testable ground signal and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices._devices_base.GND:



Public Member Functions

- __init__ (self, fiu_channel, preconditions=None, wait_time_ms=None, alias="", descr="")
  GND channel initialization.
- info (self)
  Get an info string for the ground device: alias, FIU channel plus description, if available.

7.244.1 Detailed Description

Base ground signal container for a testable ground signal and relevant associated variables/values.

7.244.2 Constructor & Destructor Documentation

7.244.2.1 __init__()

```
ttk_selftest.hil_devices._devices_base.GND.__init__ (
              self,
              fiu_channel,
              preconditions = None,
              wait_time_ms = None,
              alias = "",
              descr = "" )
```

GND channel initialization.

Parameters

fiu_channel	fiu channel of the ground signal
preconditions	(opt) preconditions that have to be set before the device can be checked. defaults to None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The functions specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
    # For special cases, the preconditions may also contain function calls
    # (which might be useful to insert additional delays or similar):
    preconditions = [
        { "device": Voltage, "value": 1},
        { "func":   time.sleep,
          "args":   [1.0], # positional arguments
          "kwargs": {},    # keyword arguments
        },
        {"device": ClampSwitch, "value": 1},
        # or a simple call like this:
        { "func":  someCustomFunction,
        },
        ...
    ]
wait_time_ms  - (opt) waiting time in milliseconds between
      ground stimulation and ground check
      defaults to 100ms
alias         - (opt) ground alias/name
      defaults is None
descr         - (opt) ground description (e.g. used in report entries)
```

Reimplemented from ttk_selftest.hil_devices._devices_base.HilDevice.

7.244.3 Member Function Documentation

7.244.3.1 info()

```
ttk_selftest.hil_devices._devices_base.GND.info (
              self )
```

Get an info string for the ground device: alias, FIU channel plus description, if available.

Returns

a formatted string.

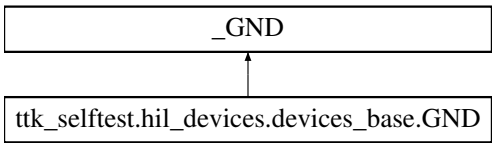
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/_devices_base.py (v2.2.0)

7.245 ttk_selftest.hil_devices.devices_base.GND Class Reference

Base ground signal container for a testable ground signal and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices.devices_base.GND:



Public Member Functions

- __init__ (self, fiu_channel, preconditions=None, wait_time_ms=None, alias="", descr="")

  GND channel initialization.
- info (self)

  Get an info string for the ground device: alias, FIU channel plus description, if available.

7.245.1 Detailed Description

Base ground signal container for a testable ground signal and relevant associated variables/values.

7.245.2 Constructor & Destructor Documentation

7.245.2.1 __init__()

```
ttk_selftest.hil_devices.devices_base.GND.__init__ (
              self,
              fiu_channel,
              preconditions = None,
              wait_time_ms = None,
              alias = "",
              descr = "" )
```

GND channel initialization.

Parameters

fiu_channel	fiu channel of the ground signal
preconditions	(opt) preconditions that have to be set before the device can be checked. defaults to None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The functions specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
# For special cases, the preconditions may also contain function calls
# (which might be useful to insert additional delays or similar):
preconditions = [
    { "device": Voltage, "value": 1},
    { "func":   time.sleep,
      "args":   [1.0], # positional arguments
      "kwargs": {},    # keyword arguments
    },
    {"device": ClampSwitch, "value": 1},
```

```
            # or a simple call like this:
            { "func":  someCustomFunction,
            },
            ...
        ]
    wait_time_ms  - (opt) waiting time in milliseconds between
          ground stimulation and ground check
          defaults to 100ms
    alias         - (opt) ground alias/name
          defaults is None
    descr         - (opt) ground description (e.g. used in report entries)
```

7.245.3 Member Function Documentation

7.245.3.1 info()

```
ttk_selftest.hil_devices.devices_base.GND.info (
              self )
```

Get an info string for the ground device: alias, FIU channel plus description, if available.

Returns

a formatted string.

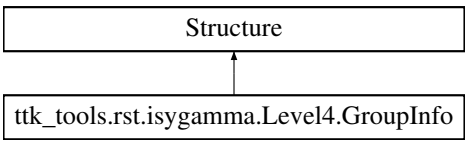
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/devices_base.py (v2.0.0)

7.246 ttk_tools.rst.isygamma.Level4.GroupInfo Class Reference

typedef struct gc_GroupInfo

Inheritance diagram for ttk_tools.rst.isygamma.Level4.GroupInfo:



Static Protected Attributes

- list _fields_

7.246.1 Detailed Description

typedef struct gc_GroupInfo

7.246.2 Member Data Documentation

7.246.2.1 _fields_

```
list ttk_tools.rst.isygamma.Level4.GroupInfo._fields_  [static], [protected]
```

Initial value:

```
=  [
            ("address",          Addresses.Level4), # address of the group
            ("name",             c_char * 1024),    # unique name of the group
            ("netActionAddress", Addresses.Level2), # address of optional network action
            ("sigActionAddress", Addresses.Level2), # address of optional signaling action
            ("logActionAddress", Addresses.Level2), # address of optional logging action
            ("isPersistent",     c_bool),           # lock persistent?
            ("isLocked",         c_bool),           # group currently locked?
            ("offset",           POINTER(c_int))    # offset to the location of the lock in associated
      memory
        ]
```

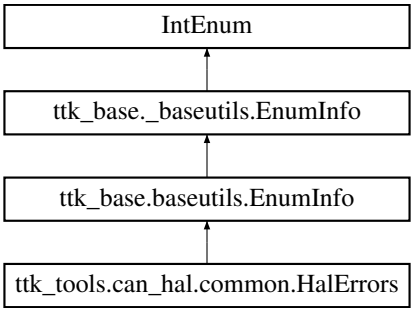
The documentation for this class was generated from the following file:

- ttk_tools/rst/isygamma.py (v1.6)

7.247 ttk_tools.can_hal.common.HalErrors Class Reference

HAL Error Codes (HalErrors.h, enum tHalError).

Inheritance diagram for ttk_tools.can_hal.common.HalErrors:



Static Public Attributes

- int HalOk = 0x0000
- int HalDriverNotAvailable = 0x0001
- int HalNoFreeHandles = 0x0002
- int HalNoFreeDevicListElement = 0x0003
- int HalUnknownSublayer = 0x0004
- int HalTooManyAppsConnected = 0x0005
- int HalWrongMessageTypeForDevice = 0x0006
- int HalHdlNotInitialized = 0x0007
- int CanHalDriverNotAvailable = 0x0101
- int CanHalNoFreeHandles = 0x0102
- int CanHalNoFreeDeviceListElement = 0x0103
- int CanHalNoDriverAvailable = 0x0104
- int CanHalRxNoData = 0x0105

- int CanHalNoPeakDriver = 0x0120
- int CanHalPeakNoFunction = 0x0121
- int CanHalPeakHwInitError = 0x0122
- int CanHalPeakWrongMessageStatus = 0x0123
- int CanHalPeakReadError = 0x0124
- int CanHalPeakWriteError = 0x0125
- int CanHalPeakNoMessage = 0x0126
- int CanHalPeakDeviceIdReadError = 0x0127
- int CanHalPeakContrNumReadError = 0x0128
- int CanHalNoVectorDriver = 0x0140
- int CanHalVectorNoFunction = 0x0141
- int CanHalVectorInitDriverError = 0x0142
- int CanHalVectorNoDeviceFound = 0x0143
- int CanHalVectorDeviceInitError = 0x0144
- int CanHalVectorNoFreeHandle = 0x0145
- int CanHalVectorWriteError = 0x0146
- int CanHalVectorReadError = 0x0147
- int CanHalVectorNoMessage = 0x0148
- int CanHalVectorGetStateError = 0x0149
- int CanHalNoSystecDriver = 0x0160
- int CanHalSystecNoFunction = 0x0161
- int CanHalSystecInitDriverError = 0x0162
- int CanHalSystecNoDeviceFound = 0x0163
- int CanHalSystecDeviceEnumError = 0x0164
- int CanHalSystecDeviceInitError = 0x0165
- int CanHalSystecHwInitError = 0x0166
- int CanHalSystecReadError = 0x0167
- int CanHalSystecWriteError = 0x0168
- int CanHalSystecNoMessage = 0x0169
- int CanHalSystecNoFreeHandle = 0x016A

7.247.1 Detailed Description

HAL Error Codes (HalErrors.h, enum tHalError).

7.247.2 Member Data Documentation

7.247.2.1 CanHalDriverNotAvailable

```
int ttk_tools.can_hal.common.HalErrors.CanHalDriverNotAvailable = 0x0101  [static]
```

7.247.2.2 CanHalNoDriverAvailable

```
int ttk_tools.can_hal.common.HalErrors.CanHalNoDriverAvailable = 0x0104  [static]
```

7.247.2.3 CanHalNoFreeDeviceListElement

```
int ttk_tools.can_hal.common.HalErrors.CanHalNoFreeDeviceListElement = 0x0103  [static]
```

7.247.2.4 CanHalNoFreeHandles

```
int ttk_tools.can_hal.common.HalErrors.CanHalNoFreeHandles = 0x0102  [static]
```

7.247.2.5 CanHalNoPeakDriver

```
int ttk_tools.can_hal.common.HalErrors.CanHalNoPeakDriver = 0x0120  [static]
```

7.247.2.6 CanHalNoSystecDriver

```
int ttk_tools.can_hal.common.HalErrors.CanHalNoSystecDriver = 0x0160  [static]
```

7.247.2.7 CanHalNoVectorDriver

```
int ttk_tools.can_hal.common.HalErrors.CanHalNoVectorDriver = 0x0140  [static]
```

7.247.2.8 CanHalPeakContrNumReadError

```
int ttk_tools.can_hal.common.HalErrors.CanHalPeakContrNumReadError = 0x0128  [static]
```

7.247.2.9 CanHalPeakDeviceIdReadError

```
int ttk_tools.can_hal.common.HalErrors.CanHalPeakDeviceIdReadError = 0x0127  [static]
```

7.247.2.10 CanHalPeakHwInitError

```
int ttk_tools.can_hal.common.HalErrors.CanHalPeakHwInitError = 0x0122  [static]
```

7.247.2.11 CanHalPeakNoFunction

```
int ttk_tools.can_hal.common.HalErrors.CanHalPeakNoFunction = 0x0121  [static]
```

7.247.2.12 CanHalPeakNoMessage

```
int ttk_tools.can_hal.common.HalErrors.CanHalPeakNoMessage = 0x0126  [static]
```

7.247.2.13 CanHalPeakReadError

```
int ttk_tools.can_hal.common.HalErrors.CanHalPeakReadError = 0x0124  [static]
```

7.247.2.14 CanHalPeakWriteError

```
int ttk_tools.can_hal.common.HalErrors.CanHalPeakWriteError = 0x0125  [static]
```

7.247.2.15 CanHalPeakWrongMessageStatus

```
int ttk_tools.can_hal.common.HalErrors.CanHalPeakWrongMessageStatus = 0x0123  [static]
```

7.247.2.16 CanHalRxNoData

```
int ttk_tools.can_hal.common.HalErrors.CanHalRxNoData = 0x0105  [static]
```

7.247.2.17 CanHalSystecDeviceEnumError

```
int ttk_tools.can_hal.common.HalErrors.CanHalSystecDeviceEnumError = 0x0164  [static]
```

7.247.2.18 CanHalSystecDeviceInitError

```
int ttk_tools.can_hal.common.HalErrors.CanHalSystecDeviceInitError = 0x0165  [static]
```

7.247.2.19 CanHalSystecHwInitError

```
int ttk_tools.can_hal.common.HalErrors.CanHalSystecHwInitError = 0x0166  [static]
```

7.247.2.20 CanHalSystecInitDriverError

```
int ttk_tools.can_hal.common.HalErrors.CanHalSystecInitDriverError = 0x0162  [static]
```

7.247.2.21 CanHalSystecNoDeviceFound

```
int ttk_tools.can_hal.common.HalErrors.CanHalSystecNoDeviceFound = 0x0163  [static]
```

7.247.2.22 CanHalSystecNoFreeHandle

```
int ttk_tools.can_hal.common.HalErrors.CanHalSystecNoFreeHandle = 0x016A  [static]
```

7.247.2.23 CanHalSystecNoFunction

```
int ttk_tools.can_hal.common.HalErrors.CanHalSystecNoFunction = 0x0161  [static]
```

7.247.2.24 CanHalSystecNoMessage

```
int ttk_tools.can_hal.common.HalErrors.CanHalSystecNoMessage = 0x0169  [static]
```

7.247.2.25 CanHalSystecReadError

```
int ttk_tools.can_hal.common.HalErrors.CanHalSystecReadError = 0x0167  [static]
```

7.247.2.26 CanHalSystecWriteError

```
int ttk_tools.can_hal.common.HalErrors.CanHalSystecWriteError = 0x0168  [static]
```

7.247.2.27 CanHalVectorDeviceInitError

```
int ttk_tools.can_hal.common.HalErrors.CanHalVectorDeviceInitError = 0x0144  [static]
```

7.247.2.28 CanHalVectorGetStateError

```
int ttk_tools.can_hal.common.HalErrors.CanHalVectorGetStateError = 0x0149  [static]
```

7.247.2.29 CanHalVectorInitDriverError

```
int ttk_tools.can_hal.common.HalErrors.CanHalVectorInitDriverError = 0x0142  [static]
```

7.247.2.30 CanHalVectorNoDeviceFound

```
int ttk_tools.can_hal.common.HalErrors.CanHalVectorNoDeviceFound = 0x0143  [static]
```

7.247.2.31 CanHalVectorNoFreeHandle

```
int ttk_tools.can_hal.common.HalErrors.CanHalVectorNoFreeHandle = 0x0145  [static]
```

7.247.2.32 CanHalVectorNoFunction

```
int ttk_tools.can_hal.common.HalErrors.CanHalVectorNoFunction = 0x0141  [static]
```

7.247.2.33 CanHalVectorNoMessage

```
int ttk_tools.can_hal.common.HalErrors.CanHalVectorNoMessage = 0x0148  [static]
```

7.247.2.34 CanHalVectorReadError

```
int ttk_tools.can_hal.common.HalErrors.CanHalVectorReadError = 0x0147  [static]
```

7.247.2.35 CanHalVectorWriteError

```
int ttk_tools.can_hal.common.HalErrors.CanHalVectorWriteError = 0x0146  [static]
```

7.247.2.36 HalDriverNotAvailable

```
int ttk_tools.can_hal.common.HalErrors.HalDriverNotAvailable = 0x0001  [static]
```

7.247.2.37 HalHdlNotInitialized

```
int ttk_tools.can_hal.common.HalErrors.HalHdlNotInitialized = 0x0007  [static]
```

7.247.2.38 HalNoFreeDevicListElement

```
int ttk_tools.can_hal.common.HalErrors.HalNoFreeDevicListElement = 0x0003  [static]
```

7.247.2.39 HalNoFreeHandles

```
int ttk_tools.can_hal.common.HalErrors.HalNoFreeHandles = 0x0002  [static]
```

7.247.2.40 HalOk

```
int ttk_tools.can_hal.common.HalErrors.HalOk = 0x0000  [static]
```

7.247.2.41 HalTooManyAppsConnected

```
int ttk_tools.can_hal.common.HalErrors.HalTooManyAppsConnected = 0x0005  [static]
```

7.247.2.42 HalUnknownSublayer

```
int ttk_tools.can_hal.common.HalErrors.HalUnknownSublayer = 0x0004  [static]
```

7.247.2.43 HalWrongMessageTypeForDevice

```
int ttk_tools.can_hal.common.HalErrors.HalWrongMessageTypeForDevice = 0x0006  [static]
```

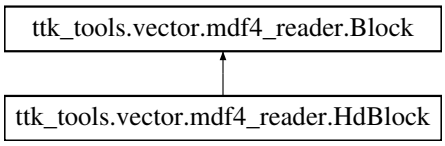
The documentation for this class was generated from the following file:

- ttk_tools/can_hal/common.py (v1.2)

7.248 ttk_tools.vector.mdf4_reader.HdBlock Class Reference

Header Block, unique.

Inheritance diagram for ttk_tools.vector.mdf4_reader.HdBlock:



Public Member Functions

- __init__ (self, file_handle, pos=None)

  Read HDBLOCK and its associated MDBLOCK (if linked) from file_handle, starting at the current or supplied position.
- getTime (self)

  Get a time tuple for the entry's start time stamp.
- __str__ (self)

Public Attributes

- hd_dg_first

  Pointer to the first data group block (DGBLOCK) (can be NIL)
- hd_fh_first

  Pointer to the first data group block (DGBLOCK) (can be NIL)
- hd_ch_first

  Pointer to the first data group block (DGBLOCK) (can be NIL)
- hd_at_first

  Pointer to the first data group block (DGBLOCK) (can be NIL)
- hd_ev_first

  Pointer to the first data group block (DGBLOCK) (can be NIL)
- hd_start_time_ns
- hd_tz_offset_min
- hd_dst_offset_min
- hd_time_flags
- hd_time_class
- hd_flags
- hd_reserved
- hd_start_angle_rad
- hd_start_distance_m
- comment
- file_history
- data_groups

7.248.1 Detailed Description

Header Block, unique.

The HDBLOCK always begins at file position 64 (i.e. right after IDBLOCK) It contains general information about the contents of the measured data file and is the root for the block hierarchy.

7.248.2 Constructor & Destructor Documentation

7.248.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.HdBlock.__init__ (
              self,
              file_handle,
              pos = None )
```

Read HDBLOCK and its associated MDBLOCK (if linked) from file_handle, starting at the current or supplied position.

Note

File pointer position will be modified

Reimplemented from ttk_tools.vector.mdf4_reader.Block.

7.248.3 Member Function Documentation

7.248.3.1 __str__()

```
ttk_tools.vector.mdf4_reader.HdBlock.__str__ (
              self )
```

7.248.3.2 getTime()

```
ttk_tools.vector.mdf4_reader.HdBlock.getTime (
              self )
```

Get a time tuple for the entry's start time stamp.

Returns

a time.gmtime()-like time tuple or None if time stamp data is inconsistent.

7.248.4 Member Data Documentation

7.248.4.1 comment

```
ttk_tools.vector.mdf4_reader.HdBlock.comment
```

7.248.4.2 data_groups

```
ttk_tools.vector.mdf4_reader.HdBlock.data_groups
```

7.248.4.3 file_history

ttk_tools.vector.mdf4_reader.HdBlock.file_history

7.248.4.4 hd_at_first

ttk_tools.vector.mdf4_reader.HdBlock.hd_at_first

Pointer to the first data group block (DGBLOCK) (can be NIL)

Pointer to first file history block (FHBLOCK) There must be at least one FHBLOCK with information about the application which created the MDF file.

Pointer to first channel hierarchy block (CHBLOCK) (can be NIL).

Pointer to first attachment block (ATBLOCK) (can be NIL)

7.248.4.5 hd_ch_first

ttk_tools.vector.mdf4_reader.HdBlock.hd_ch_first

Pointer to the first data group block (DGBLOCK) (can be NIL)

Pointer to first file history block (FHBLOCK) There must be at least one FHBLOCK with information about the application which created the MDF file.

Pointer to first channel hierarchy block (CHBLOCK) (can be NIL).

7.248.4.6 hd_dg_first

ttk_tools.vector.mdf4_reader.HdBlock.hd_dg_first

Pointer to the first data group block (DGBLOCK) (can be NIL)

7.248.4.7 hd_dst_offset_min

ttk_tools.vector.mdf4_reader.HdBlock.hd_dst_offset_min

7.248.4.8 hd_ev_first

ttk_tools.vector.mdf4_reader.HdBlock.hd_ev_first

Pointer to the first data group block (DGBLOCK) (can be NIL)

Pointer to first file history block (FHBLOCK) There must be at least one FHBLOCK with information about the application which created the MDF file.

Pointer to first channel hierarchy block (CHBLOCK) (can be NIL).

Pointer to first attachment block (ATBLOCK) (can be NIL)

Pointer to first event block (EVBLOCK) (can be NIL)

7.248.4.9 hd_fh_first

ttk_tools.vector.mdf4_reader.HdBlock.hd_fh_first

Pointer to the first data group block (DGBLOCK) (can be NIL)

Pointer to first file history block (FHBLOCK) There must be at least one FHBLOCK with information about the application which created the MDF file.

7.248.4.10 hd_flags

ttk_tools.vector.mdf4_reader.HdBlock.hd_flags

7.248.4.11 hd_reserved

ttk_tools.vector.mdf4_reader.HdBlock.hd_reserved

7.248.4.12 hd_start_angle_rad

ttk_tools.vector.mdf4_reader.HdBlock.hd_start_angle_rad

7.248.4.13 hd_start_distance_m

ttk_tools.vector.mdf4_reader.HdBlock.hd_start_distance_m

7.248.4.14 hd_start_time_ns

ttk_tools.vector.mdf4_reader.HdBlock.hd_start_time_ns

7.248.4.15 hd_time_class

ttk_tools.vector.mdf4_reader.HdBlock.hd_time_class

7.248.4.16 hd_time_flags

ttk_tools.vector.mdf4_reader.HdBlock.hd_time_flags

7.248.4.17 hd_tz_offset_min

ttk_tools.vector.mdf4_reader.HdBlock.hd_tz_offset_min

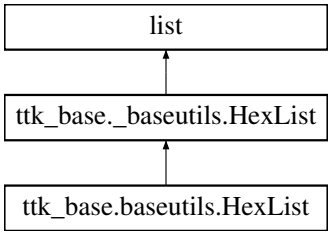
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.249 ttk_base._baseutils.HexList Class Reference

A list that returns a representation in hexadecimal notation (for easier reading).

Inheritance diagram for ttk_base._baseutils.HexList:



Public Member Functions

- __getitem__ (self, index)

  Get item at index.
- __str__ (self)
- __repr__ (self)

  Get list contents in hexadecimal formatting (if possible).

Protected Member Functions

- _getTweakedItem (cls, item)

  Interpret supplied item as a natural number, if possible/sensible.

7.249.1 Detailed Description

A list that returns a representation in hexadecimal notation (for easier reading).

This - obviously - makes the most sense for numeric values, especially bytes.

Note

- Getting an indexed item will return the numeric byte value, even if "bytes" have been added as (single) characters.
- Float values that are "close enough" (see FLOAT_CONVERSION_LIMIT) will be rounded to the nearest natural number before creating a hex representation
- if an entry cannot be interpreted, it will be represented as-is
- Only representation and indexed access is modified, the original list contents will not be modified

Examples:

```
HexList([0, 1, 2, 3, 4, 252, 253, 254, 255])
»» [0x00, 0x01, 0x02, 0x03, 0x04, 0xFC, 0xFD, 0xFE, 0xFF]

HexList("\x00\x01\x02\x03\x04\xFC\xFD\xFE\xFF")
»» [0x00, 0x01, 0x02, 0x03, 0x04, 0xFC, 0xFD, 0xFE, 0xFF]

HexList([1023, 1024, 0xFFFF, "\xAA", 0, -1, -0xFFFFFFFF])
»» [0x3FF, 0x400, 0xFFFF, 0xAA, 0x00, -0x01, -0xFFFFFFFF]

HexList([1, 2, "some string", u"ö", 1.000001, 2.99999, 1.123, None])
»» [0x01, 0x02, 'some string', 0xF6, 0x01, 0x03, 1.123, None]

list(HexList([1, 2, "three", u"ö", 1.123, 1.0, None]))
»» [1, 2, 'three', u'\xf6', 1.123, 1.0, None]

HexList([1, 2, "three", u"ö", 1.123, 1.0, None])[:]
»» [1, 2, 'three', u'\xf6', 1.123, 1.0, None]
```

7.249.2 Member Function Documentation

7.249.2.1 __getitem__()

```
ttk_base._baseutils.HexList.__getitem__ (
              self,
              index )
```

Get item at index.

Note

If the item is a single character, its ordinal number will be returned.

Parameters

index	index of item to get

Returns

the specified item as a numeric value (if possible).

Reimplemented in ttk_base.baseutils.HexList.

7.249.2.2 __repr__()

```
ttk_base._baseutils.HexList.__repr__ (
              self )
```

Get list contents in hexadecimal formatting (if possible).

Reimplemented in ttk_base.baseutils.HexList.

7.249.2.3 __str__()

```
ttk_base._baseutils.HexList.__str__ (
              self )
```

7.249.2.4 _getTweakedItem()

```
ttk_base._baseutils.HexList._getTweakedItem (
              cls,
              item )  [protected]
```

Interpret supplied item as a natural number, if possible/sensible.

Parameters

item	item to tweak

Returns

an int/long value or the item as-is.

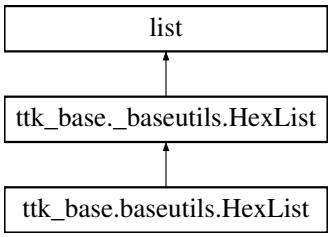
The documentation for this class was generated from the following file:

- ttk_base/_baseutils.py (v2.3.2)

7.250 ttk_base.baseutils.HexList Class Reference

A list that returns a representation in hexadecimal notation (for easier reading).

Inheritance diagram for ttk_base.baseutils.HexList:



Public Member Functions

- __getitem__ (self, index)
  Get item at index.
- __repr__ (self)
  Get list contents in hexadecimal formatting (if possible).

7.250.1 Detailed Description

A list that returns a representation in hexadecimal notation (for easier reading).

This - obviously - makes the most sense for numeric values, especially bytes.

Note

- Getting an indexed item will return the numeric byte value, even if "bytes" have been added as (single) characters.
- Float values that are "close enough" (see FLOAT_CONVERSION_LIMIT) will be rounded to the nearest natural number before creating a hex representation
- if an entry cannot be interpreted, it will be represented as-is
- Only representation and indexed access is modified, the original list contents will not be modified

Examples:

```
HexList([0, 1, 2, 3, 4, 252, 253, 254, 255])
»> [0x00, 0x01, 0x02, 0x03, 0x04, 0xFC, 0xFD, 0xFE, 0xFF]

HexList("\x00\x01\x02\x03\x04\xFC\xFD\xFE\xFF")
»> [0x00, 0x01, 0x02, 0x03, 0x04, 0xFC, 0xFD, 0xFE, 0xFF]

HexList([1023, 1024, 0xFFFF, "\xAA", 0, -1, -0xFFFFFFFF])
»> [0x3FF, 0x400, 0xFFFF, 0xAA, 0x00, -0x01, -0xFFFFFFFF]

HexList([1, 2, "some string", u"ö", 1.000001, 2.99999, 1.123, None])
»> [0x01, 0x02, 'some string', 0xF6, 0x01, 0x03, 1.123, None]

list(HexList([1, 2, "three", u"ö", 1.123, 1.0, None]))
»> [1, 2, 'three', u'\xf6', 1.123, 1.0, None]

HexList([1, 2, "three", u"ö", 1.123, 1.0, None])[:]
»> [1, 2, 'three', u'\xf6', 1.123, 1.0, None]
```

7.250.2 Member Function Documentation

7.250.2.1 __getitem__()

```
ttk_base.baseutils.HexList.__getitem__ (
              self,
              index )
```

Get item at index.

Note

If the item is a single character, its ordinal number will be returned.

Parameters

index	index of item to get

Returns

the specified item as a numeric value (if possible).

Reimplemented from ttk_base._baseutils.HexList.

7.250.2.2 __repr__()

```
ttk_base.baseutils.HexList.__repr__ (
              self )
```

Get list contents in hexadecimal formatting (if possible).

Reimplemented from ttk_base._baseutils.HexList.

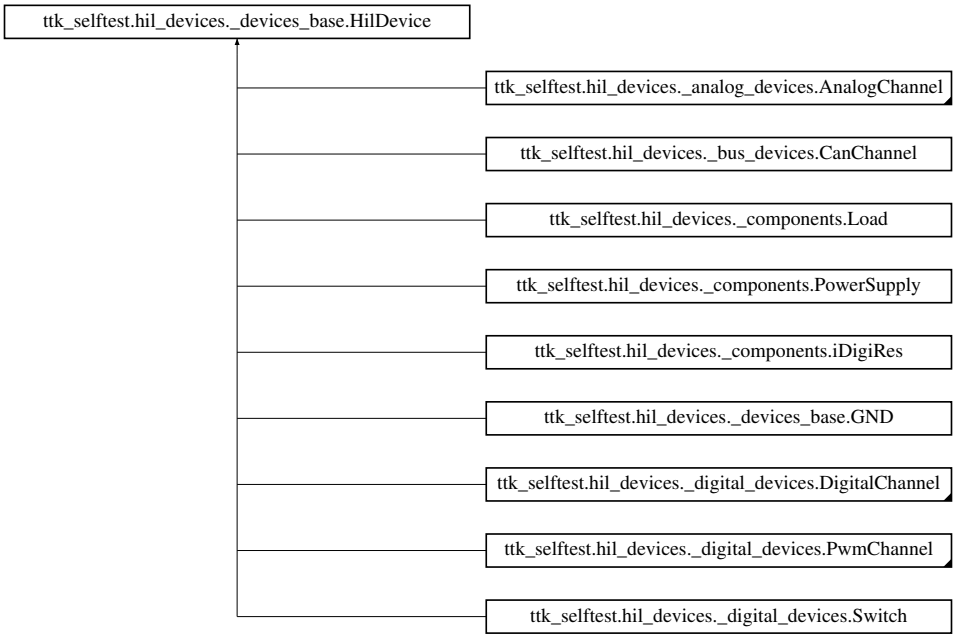
The documentation for this class was generated from the following file:

- ttk_base/baseutils.py (v2.3.1)

7.251 ttk_selftest.hil_devices._devices_base.HilDevice Class Reference

Base HIL device container for a testable HIL device and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices._devices_base.HilDevice:



Public Member Functions

- __init__ (self, fiu_channel_1, fiu_channel_2=None, preconditions=None, wait_time_ms=None, alias="", descr="")

  HIL device initialization.

Public Attributes

- fiu_channel_1

  First FIU channel.
- fiu_channel_2

  Second FIU channel.
- preconditions
- wait_time_ms

  Waiting time in [ms] between channel stimulation and channel check, defaults to 100ms.
- alias

  Device alias.
- descr

  Device description.
- verdict

  Test verdict.

7.251.1 Detailed Description

Base HIL device container for a testable HIL device and relevant associated variables/values.

7.251.2 Constructor & Destructor Documentation

7.251.2.1 __init__()

```
ttk_selftest.hil_devices._devices_base.HilDevice.__init__ (
                self,
                fiu_channel_1,
                fiu_channel_2 = None,
                preconditions = None,
                wait_time_ms = None,
                alias = "",
                descr = "" )
```

HIL device initialization.

Parameters

fiu_channel_1	first fiu channel of the device
fiu_channel_2	(opt) second fiu channel of the device default to None
preconditions	(opt) preconditions that have to be set before the device can be checked. defaults to None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The functions specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
    # For special cases, the preconditions may also contain function calls
    # (which might be useful to insert additional delays or similar):
    preconditions = [
        { "device": Voltage, "value": 1},
        { "func":   time.sleep,
          "args":   [1.0], # positional arguments
          "kwargs": {},    # keyword arguments
        },
        {"device": ClampSwitch, "value": 1},
        # or a simple call like this:
        { "func":  someCustomFunction,
        },
        ...
    ]
wait_time_ms  - (opt) waiting time in milliseconds between
      channel stimulation and channel check
      defaults to 100ms
alias         - (opt) device alias/name, defaults to the
      last (non-default) entry in identifier
      (i.e. signal model path)
descr         - (opt) device description
```

Reimplemented in ttk_selftest.hil_devices._digital_devices.PwmChannel, ttk_selftest.hil_devices._digital_devices.DigitalChannel, ttk_selftest.hil_devices._components.iDigiRes, ttk_selftest.hil_devices._digital_devices.PwmOutput, ttk_selftest.hil_devices._digital_d

ttk_selftest.hil_devices._digital_devices.DigitalOutput, ttk_selftest.hil_devices._digital_devices.DigitalInput, ttk_selftest.hil_devices._a
ttk_selftest.hil_devices._analog_devices.AnalogOutput, ttk_selftest.hil_devices._analog_devices.AnalogInput,
ttk_selftest.hil_devices._bus_devices.CanChannel, ttk_selftest.hil_devices._components.Load, ttk_selftest.hil_devices._digital_device
ttk_selftest.hil_devices._components.PowerSupply, and ttk_selftest.hil_devices._devices_base.GND.

7.251.3 Member Data Documentation

7.251.3.1 alias

`ttk_selftest.hil_devices._devices_base.HilDevice.alias`

Device alias.

7.251.3.2 descr

`ttk_selftest.hil_devices._devices_base.HilDevice.descr`

Device description.

7.251.3.3 fiu_channel_1

`ttk_selftest.hil_devices._devices_base.HilDevice.fiu_channel_1`

First FIU channel.

7.251.3.4 fiu_channel_2

`ttk_selftest.hil_devices._devices_base.HilDevice.fiu_channel_2`

Second FIU channel.

7.251.3.5 preconditions

`ttk_selftest.hil_devices._devices_base.HilDevice.preconditions`

7.251.3.6 verdict

`ttk_selftest.hil_devices._devices_base.HilDevice.verdict`

Test verdict.

The verdict contains the test result information of the device

7.251.3.7 wait_time_ms

```
ttk_selftest.hil_devices._devices_base.HilDevice.wait_time_ms
```

Waiting time in [ms] between channel stimulation and channel check, defaults to 100ms.

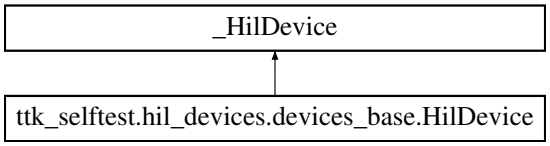
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/_devices_base.py (v2.2.0)

7.252 ttk_selftest.hil_devices.devices_base.HilDevice Class Reference

Base HIL device container for a testable HIL device and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices.devices_base.HilDevice:



Public Member Functions

- __init__ (self, fiu_channel_1, fiu_channel_2=None, preconditions=None, wait_time_ms=None, alias="", descr="")

  HIL device initialization.

7.252.1 Detailed Description

Base HIL device container for a testable HIL device and relevant associated variables/values.

7.252.2 Constructor & Destructor Documentation

7.252.2.1 __init__()

```
ttk_selftest.hil_devices.devices_base.HilDevice.__init__ (
              self,
              fiu_channel_1,
              fiu_channel_2 = None,
              preconditions = None,
              wait_time_ms = None,
              alias = "",
              descr = "" )
```

HIL device initialization.

Parameters

fiu_channel_1	first fiu channel of the device
fiu_channel_2	(opt) second fiu channel of the device default to None
preconditions	(opt) preconditions that have to be set before the device can be checked. defaults to None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The functions specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
    # For special cases, the preconditions may also contain function calls
    # (which might be useful to insert additional delays or similar):
    preconditions = [
        { "device": Voltage, "value": 1},
        { "func":   time.sleep,
          "args":   [1.0], # positional arguments
          "kwargs": {},    # keyword arguments
        },
        {"device": ClampSwitch, "value": 1},
        # or a simple call like this:
        { "func":  someCustomFunction,
        },
        ...
    ]
wait_time_ms  - (opt) waiting time in milliseconds between
channel stimulation and channel check
defaults to 100ms
alias         - (opt) device alias/name, defaults to the
last (non-default) entry in identifier
(i.e. signal model path)
descr         - (opt) device description
```

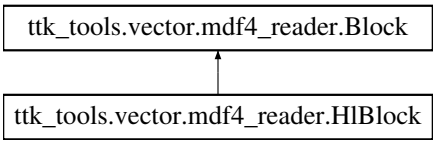
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/devices_base.py (v2.0.0)

7.253 ttk_tools.vector.mdf4_reader.HlBlock Class Reference

Header List Block.

Inheritance diagram for ttk_tools.vector.mdf4_reader.HlBlock:



Public Member Functions

- __init__ (self, file_handle, pos=None)

  Read HLBLOCK from file_handle, starting at the current or supplied position.

Public Attributes

- hl_dl_first

  Pointer to next data list block (DLBLOCK) (can be NIL).
- hl_flags
- hl_zip_type

7.253.1 Detailed Description

Header List Block.

A HLBLOCK represents the "header" of a list of data blocks, i.e. the start of a linked list of DLBLOCKs. It contains information about the DLBLOCKs and the contained data blocks.

The HLBLOCK indicates that the list of data blocks referenced by its descendant DLBLOCKs contains zipped data blocks (DZBLOCKs). It also states the compression algorithm used by the DZBLOCKs in the list.

The HLBLOCK type has been introduced in MDF version 4.1.0 and must not be used in previous MDF versions. Applications implemented for an earlier MDF 4.0.x version will not be able to recognize this block type.

7.253.2 Constructor & Destructor Documentation

7.253.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.HlBlock.__init__ (
              self,
              file_handle,
              pos = None )
```

Read HLBLOCK from file_handle, starting at the current or supplied position.

Reimplemented from ttk_tools.vector.mdf4_reader.Block.

7.253.3 Member Data Documentation

7.253.3.1 hl_dl_first

```
ttk_tools.vector.mdf4_reader.HlBlock.hl_dl_first
```

Pointer to next data list block (DLBLOCK) (can be NIL).

7.253.3.2 hl_flags

```
ttk_tools.vector.mdf4_reader.HlBlock.hl_flags
```

7.253.3.3 hl_zip_type

```
ttk_tools.vector.mdf4_reader.HlBlock.hl_zip_type
```

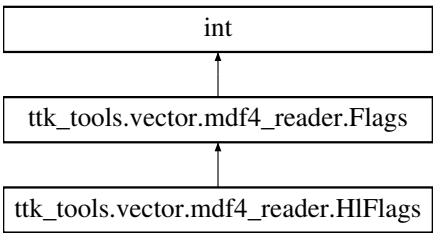
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.254 ttk_tools.vector.mdf4_reader.HlFlags Class Reference

Container for decoded uint8 hl_flags.

Inheritance diagram for ttk_tools.vector.mdf4_reader.HlFlags:



Public Member Functions

- __init__ (self, flags)

Public Attributes

- equal_length
  Bit 0: Equal length flag.

7.254.1 Detailed Description

Container for decoded uint8 hl_flags.

Only intended for read-only access, changes in named flags will not modify original "flags" value.

7.254.2 Constructor & Destructor Documentation

7.254.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.HlFlags.__init__ (
            self,
            flags )
```

Reimplemented from ttk_tools.vector.mdf4_reader.Flags.

7.254.3 Member Data Documentation

7.254.3.1 equal_length

`ttk_tools.vector.mdf4_reader.HlFlags.equal_length`

Bit 0: Equal length flag.

For the referenced DLBLOCK (and thus for each DLBLOCK in the linked list), the value of the "equal length" flag (bit 0 in dl_flags) must be equal to this flag.

The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.255 ttk_tools.isyst.ida_testware.IdaRecorder Class Reference

Data acquisition using an iDA Recorder application.

Public Member Functions

- __init__ (self, addr="127.0.0.1", port=50055)
- setup (self, variables)

  Setup recorder for data acquisition (configure what variables to measure).
- start (self)

  Start recorder data acquisition with configured variables.
- stop (self)

  Stop a currently running data acquisition.
- getData (self)

  Get recorded data and return it as a daq data structure.
- close (self)

  Close current connection.
- __enter__ (self)

  Start daq/recorder as a managed context.
- __exit__ (self, exc_type, exc_value, tb)

  End context manager.

Public Attributes

- mwrec

Static Public Attributes

- channel_options = None

  gRPC connection options.
- int verbosity = 1

  level of log verbosity: 0: mostly silent 1: status messages 2: more elaborate status messages

Protected Member Functions

- _log (self, level, *args, **kwargs)

7.255.1 Detailed Description

Data acquisition using an iDA Recorder application.

Example:

```
rec = IdaRecorder(addr="127.0.0.1")
# usually gets set up somewhere in TestEnv
# e.g. rec = testenv.getHilRecorder()

rec.setup([
    "scheduler.Primary.runtime",
    "tasklevel.Recorder_runtime",
    "some.other.variable",
    hil.motor_temp         # or directly use HIL Var instances
])

rec.start()
time.sleep(5.0) # or do something else while recording runs
rec.stop()

daq_data = rec.getData()

# usually gets closed in testenv.breakdown()
rec.close()
```

7.255.2 Constructor & Destructor Documentation

7.255.2.1 __init__()

```
ttk_tools.isyst.ida_testware.IdaRecorder.__init__ (
              self,
              addr = "127.0.0.1",
              port = 50055 )
```

Parameters

addr	connection target address string (IP), e.g. "192.168.2.123".
port	connection target port, defaults to 50055 for MwRec

7.255.3 Member Function Documentation

7.255.3.1 __enter__()

```
ttk_tools.isyst.ida_testware.IdaRecorder.__enter__ (
              self )
```

Start daq/recorder as a managed context.

Example:

```
with IdaRecorder(addr="127.0.0.1") as daq:
    daq.setup([
        "tasklevel.Recorder_runtime",
        "Analog_In.CVmon",
        "Analog_In.CCmon",
    ])
    daq.start()
    time.sleep(2.0) # or do something else while recording runs
    daq.stop()
    daq_data = daq.getData()
```

7.255.3.2 __exit__()

```
ttk_tools.isyst.ida_testware.IdaRecorder.__exit__ (
              self,
              exc_type,
              exc_value,
              tb )
```

End context manager.

Note

Exiting the context manager will not stop a currently running recording.

7.255.3.3 _log()

```
ttk_tools.isyst.ida_testware.IdaRecorder._log (
              self,
              level,
             * args,
             ** kwargs )  [protected]
```

7.255.3.4 close()

```
ttk_tools.isyst.ida_testware.IdaRecorder.close (
              self )
```

Close current connection.

Note that this will not stop a currently running recording.

7.255.3.5 getData()

```
ttk_tools.isyst.ida_testware.IdaRecorder.getData (
              self )
```

Get recorded data and return it as a daq data structure.

Returns

a daq data structure with {
<label>: {
"time": [<time_s>, <time_s>, ...],
"data": [<value>, <value>, ...],
"type": "<datatype-string>"
},
} entries.

7.255.3.6 setup()

```
ttk_tools.isyst.ida_testware.IdaRecorder.setup (
              self,
              variables )
```

Setup recorder for data acquisition (configure what variables to measure).

Basic variant for symmetry with other implementations

Note

setup() may only be performed while data acquisition is not already running.

Parameters

variables	list/iterable of gamma variables to measure

Usage:

```
# direct setup with variable names
daq.setup([
    "some_var.foo",
    "some_other_var"
])
# or, when using a VarContainer:
daq.setup([hil.some_var, hil.some_other_var])
```

7.255.3.7 start()

```
ttk_tools.isyst.ida_testware.IdaRecorder.start (
              self )
```

Start recorder data acquisition with configured variables.

Returns

a RecCommandStatus (usually success).

7.255.3.8 stop()

```
ttk_tools.isyst.ida_testware.IdaRecorder.stop (
              self )
```

Stop a currently running data acquisition.

Returns

a RecCommandStatus (usually success).

7.255.4 Member Data Documentation

7.255.4.1 channel_options

```
ttk_tools.isyst.ida_testware.IdaRecorder.channel_options = None  [static]
```

gRPC connection options.

If set, define entries as a list of (name, value) tuples, e.g. [('grpc.max_receive_message_length', 1024 ∗ 1024 ∗ 8),]

7.255.4.2 mwrec

```
ttk_tools.isyst.ida_testware.IdaRecorder.mwrec
```

7.255.4.3 verbosity

```
int ttk_tools.isyst.ida_testware.IdaRecorder.verbosity = 1  [static]
```

level of log verbosity: 0: mostly silent 1: status messages 2: more elaborate status messages

The documentation for this class was generated from the following file:

- ttk_tools/isyst/ida_testware.py (v1.2.1)

7.256 ttk_tools.isyst.ida_testware_offline_stub.IdaRecorder Class Reference

Data acquisition using an iDA Recorder application.

Public Member Functions

- __init__ (self, addr="127.0.0.1", port=50055)
- setup (self, variables)

  Setup recorder for data acquisition (configure what variables to measure).
- start (self)

  Start recorder data acquisition with configured variables.
- stop (self)

  Stop a currently running data acquisition.
- getData (self)

  Get recorded data and return it as a daq data structure.
- close (self)

  Close current connection.
- __enter__ (self)

  Start daq/recorder as a managed context.
- __exit__ (self, exc_type, exc_value, tb)

  End context manager.

Public Attributes

- mwrec

Static Public Attributes

- channel_options = None

  gRPC connection options.
- int verbosity = 1

  level of log verbosity: 0: mostly silent 1: status messages 2: more elaborate status messages

Protected Member Functions

- _log (self, level, ∗args, ∗∗kwargs)

7.256.1 Detailed Description

Data acquisition using an iDA Recorder application.

Example:

```
rec = IdaRecorder(addr="127.0.0.1")
# usually gets set up somewhere in TestEnv
# e.g. rec = testenv.getHilRecorder()

rec.setup([
    "scheduler.Primary.runtime",
    "tasklevel.Recorder_runtime",
    "some.other.variable",
    hil.motor_temp         # or directly use HIL Var instances
])

rec.start()
time.sleep(5.0) # or do something else while recording runs
rec.stop()

daq_data = rec.getData()

# usually gets closed in testenv.breakdown()
rec.close()
```

7.256.2 Constructor & Destructor Documentation

7.256.2.1 __init__()

```
ttk_tools.isyst.ida_testware_offline_stub.IdaRecorder.__init__ (
            self,
            addr = "127.0.0.1",
            port = 50055 )
```

Parameters

addr	connection target address string (IP), e.g. "192.168.2.123".
port	connection target port, defaults to 50055 for MwRec

7.256.3 Member Function Documentation

7.256.3.1 __enter__()

```
ttk_tools.isyst.ida_testware_offline_stub.IdaRecorder.__enter__ (
              self )
```

Start daq/recorder as a managed context.

Example:

```
with IdaRecorder(addr="127.0.0.1") as daq:
    daq.setup([
        "tasklevel.Recorder_runtime",
        "Analog_In.CVmon",
        "Analog_In.CCmon",
    ])
    daq.start()
    time.sleep(2.0) # or do something else while recording runs
    daq.stop()
    daq_data = daq.getData()
```

7.256.3.2 __exit__()

```
ttk_tools.isyst.ida_testware_offline_stub.IdaRecorder.__exit__ (
              self,
              exc_type,
              exc_value,
              tb )
```

End context manager.

Note

Exiting the context manager will not stop a currently running recording.

7.256.3.3 _log()

```
ttk_tools.isyst.ida_testware_offline_stub.IdaRecorder._log (
              self,
              level,
             * args,
             ** kwargs )  [protected]
```

7.256.3.4 close()

```
ttk_tools.isyst.ida_testware_offline_stub.IdaRecorder.close (
              self )
```

Close current connection.

7.256.3.5 getData()

```
ttk_tools.isyst.ida_testware_offline_stub.IdaRecorder.getData (
              self )
```

Get recorded data and return it as a daq data structure.

Returns

a daq data structure with {
<label>: {
"time": [<time_s>, <time_s>, ...],
"data": [<value>, <value>, ...],
"type": "<datatype-string>"
},
} entries.

7.256.3.6 setup()

```
ttk_tools.isyst.ida_testware_offline_stub.IdaRecorder.setup (
              self,
              variables )
```

Setup recorder for data acquisition (configure what variables to measure).

Basic variant for symmetry with other implementations

Note

setup() may only be performed while data acquisition is not already running.

Parameters

variables	list/iterable of gamma variables to measure

Usage:

```
# direct setup with variable names
daq.setup([
    "some_var.foo",
    "some_other_var"
])
# or, when using a VarContainer:
daq.setup([hil.some_var, hil.some_other_var])
```

7.256.3.7 start()

```
ttk_tools.isyst.ida_testware_offline_stub.IdaRecorder.start (
              self )
```

Start recorder data acquisition with configured variables.

Returns

a RecCommandStatus (usually success).

7.256.3.8 stop()

```
ttk_tools.isyst.ida_testware_offline_stub.IdaRecorder.stop (
              self )
```

Stop a currently running data acquisition.

Returns

a RecCommandStatus (usually success).

7.256.4 Member Data Documentation

7.256.4.1 channel_options

```
ttk_tools.isyst.ida_testware_offline_stub.IdaRecorder.channel_options = None  [static]
```

gRPC connection options.

If set, define entries as a list of (name, value) tuples, e.g. [('grpc.max_receive_message_length', 1024 ∗ 1024 ∗ 8),]

7.256.4.2 mwrec

```
ttk_tools.isyst.ida_testware_offline_stub.IdaRecorder.mwrec
```

7.256.4.3 verbosity

```
int ttk_tools.isyst.ida_testware_offline_stub.IdaRecorder.verbosity = 1  [static]
```

level of log verbosity: 0: mostly silent 1: status messages 2: more elaborate status messages

The documentation for this class was generated from the following file:

- ttk_tools/isyst/ida_testware_offline_stub.py (v1.0.1)

7.257 ttk_tools.isyst.ida_testware.IdaTestware Class Reference

Access to iDA Testware MwApp interface for reading/writing variables.

Public Member Functions

- __init__ (self, addr="127.0.0.1", port=50051)
- system_name (self)

  Name from active system configuration.
- service_version (self)

  Version of running iDA Testware service.
- instance_key (self)

  Key of the active iDA Testware service, changes each time the service is (re-) started.
- getMwVar (self, var_name)

  Factory method for creating an iDA Testware MwVar proxy instances.
- resetVars (self, var_ids=None, reset_all=False, complete_arrays=False)

  Reset one or more variables to their original values.
- getGroups (self)

  Get configured variable groups (and Var-IDs in those groups).
- getVarNames (self)

  Get a all configured variable names.
- close (self)

  Close current connection.
- __enter__ (self)

  Start connection as managed context.
- __exit__ (self, exc_type, exc_value, tb)

  End context manager.

Public Attributes

- mwapp

Static Public Attributes

- channel_options = None

  gRPC connection options.

7.257.1 Detailed Description

Access to iDA Testware MwApp interface for reading/writing variables.

7.257.2 Constructor & Destructor Documentation

7.257.2.1 __init__()

```
ttk_tools.isyst.ida_testware.IdaTestware.__init__ (
            self,
            addr = "127.0.0.1",
            port = 50051 )
```

Parameters

addr	connection target address string (IP), e.g. "192.168.2.123".
port	connection target port, defaults to 50051 for MwApp

7.257.3 Member Function Documentation

7.257.3.1 __enter__()

```
ttk_tools.isyst.ida_testware.IdaTestware.__enter__ (
            self )
```

Start connection as managed context.

7.257.3.2 __exit__()

```
ttk_tools.isyst.ida_testware.IdaTestware.__exit__ (
            self,
            exc_type,
            exc_value,
            tb )
```

End context manager.

7.257.3.3 close()

```
ttk_tools.isyst.ida_testware.IdaTestware.close (
            self )
```

Close current connection.

7.257.3.4 getGroups()

```
ttk_tools.isyst.ida_testware.IdaTestware.getGroups (
            self )
```

Get configured variable groups (and Var-IDs in those groups).

For array-Variables, only the head element's Var-ID will be included.

Returns

a dict with <group_id>: { "group_name": "<group_name>", "group_id": <group_id>, "var_ids": [<var_id1>, <var_id2>, ... <var_idn>], "var_names": [<var_name1>, <var_name2>, ... <var_namen>], }

7.257.3.5 getMwVar()

```
ttk_tools.isyst.ida_testware.IdaTestware.getMwVar (
            self,
            var_name )
```

Factory method for creating an iDA Testware MwVar proxy instances.

Mainly intended for convenience, so the variable abstraction (see ttk_base.isyst_ida) does not have to import additional packages.

Parameters

var_name	variable name on iDA Testware mwservice

MwVar Proxy Instances:

MwVar instances proxy access for a specific variable to the main MwApp interface.

Properties:

- mwvar.name - name of variable
- mwvar.var_id - internal Var-ID on service
- mwvar.data_type - data type of variable
- mwvar.size - size; 1: scalar, > 1: array variable
- mwvar.unit - value unit string
- mwvar.fmt - value format hint string

Methods:

- mwvar.getValue() => value
- mwvar.setValue(value)
- mwvar.updateValue(value) => prev_value
- mwvar.resetValue()

Returns

an initialized MwVar instance

7.257.3.6 getVarNames()

```
ttk_tools.isyst.ida_testware.IdaTestware.getVarNames (
              self )
```

Get a all configured variable names.

Meta data has been retrieved during init, so this will just return all cached names.

Note

Each element of array variables will be included separately

Returns

an iterator of variable names.

7.257.3.7 instance_key()

```
ttk_tools.isyst.ida_testware.IdaTestware.instance_key (
              self )
```

Key of the active iDA Testware service, changes each time the service is (re-) started.

7.257.3.8 resetVars()

```
ttk_tools.isyst.ida_testware.IdaTestware.resetVars (
              self,
              var_ids = None,
              reset_all = False,
              complete_arrays = False )
```

Reset one or more variables to their original values.

This will directly reset variables in the data model to their default values from system configuration.

Parameters

var_ids	list of Var-IDs to reset
reset_all	True: reset all variables, var_ids will be ignored False: reset only those variables listed in var_ids
complete_arrays	True: reset all array elements if head element (index [0]) is in var_ids False: only array elements in var_ids will be reset (does not matter with reset_all=True)

7.257.3.9 service_version()

```
ttk_tools.isyst.ida_testware.IdaTestware.service_version (
              self )
```

Version of running iDA Testware service.

7.257.3.10 system_name()

```
ttk_tools.isyst.ida_testware.IdaTestware.system_name (
              self )
```

Name from active system configuration.

7.257.4 Member Data Documentation

7.257.4.1 channel_options

```
ttk_tools.isyst.ida_testware.IdaTestware.channel_options = None  [static]
```

gRPC connection options.

If set, define as a list of (name, value) tuples, e.g. [('grpc.max_receive_message_length', 1024 ∗ 1024 ∗ 8),]

7.257.4.2 mwapp

`ttk_tools.isyst.ida_testware.IdaTestware.mwapp`

The documentation for this class was generated from the following file:

- ttk_tools/isyst/ida_testware.py (v1.2.1)

7.258 ttk_tools.isyst.ida_testware_offline_stub.IdaTestware Class Reference

Access to iDA Testware MwApp interface for reading/writing variables.

Public Member Functions

- __init__ (self, addr="127.0.0.1", port=50051)
- system_name (self)

  Name from active system configuration.
- service_version (self)

  Version of running iDA Testware service.
- instance_key (self)

  Key of the active iDA Testware service, changes each time the service is (re-) started.
- getMwVar (self, var_name)

  Factory method for creating an iDA Testware MwVar proxy instances.
- resetVars (self, var_ids=None, reset_all=False, complete_arrays=False)

  Reset one or more variables to their original values.
- getGroups (self)

  Get configured variable groups (and Var-IDs in those groups).
- getVarNames (self)

  Get a all configured variable names.
- close (self)

  Close current connection.
- __enter__ (self)

  Start connection as managed context.
- __exit__ (self, exc_type, exc_value, tb)

  End context manager.

Public Attributes

- mwapp

Static Public Attributes

- channel_options = None

  gRPC connection options.

7.258.1 Detailed Description

Access to iDA Testware MwApp interface for reading/writing variables.

7.258.2 Constructor & Destructor Documentation

7.258.2.1 __init__()

```
ttk_tools.isyst.ida_testware_offline_stub.IdaTestware.__init__ (
              self,
              addr = "127.0.0.1",
              port = 50051 )
```

Parameters

addr	connection target address string (IP), e.g. "192.168.2.123".
port	connection target port, defaults to 50051 for MwApp

7.258.3 Member Function Documentation

7.258.3.1 __enter__()

```
ttk_tools.isyst.ida_testware_offline_stub.IdaTestware.__enter__ (
              self )
```

Start connection as managed context.

7.258.3.2 __exit__()

```
ttk_tools.isyst.ida_testware_offline_stub.IdaTestware.__exit__ (
              self,
              exc_type,
              exc_value,
              tb )
```

End context manager.

7.258.3.3 close()

```
ttk_tools.isyst.ida_testware_offline_stub.IdaTestware.close (
              self )
```

Close current connection.

7.258.3.4 getGroups()

```
ttk_tools.isyst.ida_testware_offline_stub.IdaTestware.getGroups (
              self )
```

Get configured variable groups (and Var-IDs in those groups).

For array-Variables, only the head element's Var-ID will be included.

Returns

a dict with <group_id>: { "group_name": "<group_name>", "group_id": <group_id>, "var_ids": [<var_id1>, <var_id2>, ... <var_idn>], "var_names": [<var_name1>, <var_name2>, ... <var_namen>], }

7.258.3.5 getMwVar()

```
ttk_tools.isyst.ida_testware_offline_stub.IdaTestware.getMwVar (
              self,
              var_name )
```

Factory method for creating an iDA Testware MwVar proxy instances.

Mainly intended for convenience, so the variable abstraction (see ttk_base.isyst_ida) does not have to import additional packages.

Parameters

var_name	variable name on iDA Testware mwservice

MwVar Proxy Instances:

MwVar instances proxy access for a specific variable to the main MwApp interface.

Properties:

- mwvar.name - name of variable
- mwvar.var_id - internal Var-ID on service
- mwvar.data_type - data type of variable
- mwvar.size - size; 1: scalar, > 1: array variable
- mwvar.unit - value unit string
- mwvar.fmt - value format hint string

Methods:

- mwvar.getValue() => value
- mwvar.setValue(value)
- mwvar.updateValue(value) => prev_value
- mwvar.resetValue()

Returns

an initialized MwVar instance

7.258.3.6 getVarNames()

```
ttk_tools.isyst.ida_testware_offline_stub.IdaTestware.getVarNames (
              self )
```

Get a all configured variable names.

Meta data has been retrieved during init, so this will just return all cached names.

Note

Each element of array variables will be included separately

Returns

an iterator of variable names.

7.258.3.7 instance_key()

```
ttk_tools.isyst.ida_testware_offline_stub.IdaTestware.instance_key (
              self )
```

Key of the active iDA Testware service, changes each time the service is (re-) started.

7.258.3.8 resetVars()

```
ttk_tools.isyst.ida_testware_offline_stub.IdaTestware.resetVars (
              self,
              var_ids = None,
              reset_all = False,
              complete_arrays = False )
```

Reset one or more variables to their original values.

This will directly reset variables in the data model to their default values from system configuration.

Parameters

var_ids	list of Var-IDs to reset
reset_all	True: reset all variables, var_ids will be ignored False: reset only those variables listed in var_ids
complete_arrays	True: reset all array elements if head element (index [0]) is in var_ids False: only array elements in var_ids will be reset (does not matter with reset_all=True)

7.258.3.9 service_version()

```
ttk_tools.isyst.ida_testware_offline_stub.IdaTestware.service_version (
              self )
```

Version of running iDA Testware service.

7.258.3.10 system_name()

```
ttk_tools.isyst.ida_testware_offline_stub.IdaTestware.system_name (
              self )
```

Name from active system configuration.

7.258.4 Member Data Documentation

7.258.4.1 channel_options

```
ttk_tools.isyst.ida_testware_offline_stub.IdaTestware.channel_options = None  [static]
```

gRPC connection options.

If set, define as a list of (name, value) tuples, e.g. [('grpc.max_receive_message_length', 1024 ∗ 1024 ∗ 8),]

7.258.4.2 mwapp

```
ttk_tools.isyst.ida_testware_offline_stub.IdaTestware.mwapp
```

The documentation for this class was generated from the following file:

- ttk_tools/isyst/ida_testware_offline_stub.py (v1.0.1)

7.259 ttk_tools.vector.mdf4_reader.IdBlock Class Reference

File Identification Block.

Public Member Functions

- __init__ (self, data)

Public Attributes

- id_file
- id_vers
- id_prog
- id_ver
- id_unfin_flags
- id_custom_unfin_flags

7.259.1 Detailed Description

File Identification Block.

The IDBLOCK always begins at file position 0 and has a constant length of 64 Bytes (for compatibility with older MDF formats).

7.259.2 Constructor & Destructor Documentation

7.259.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.IdBlock.__init__ (
            self,
            data )
```

7.259.3 Member Data Documentation

7.259.3.1 id_custom_unfin_flags

```
ttk_tools.vector.mdf4_reader.IdBlock.id_custom_unfin_flags
```

7.259.3.2 id_file

```
ttk_tools.vector.mdf4_reader.IdBlock.id_file
```

7.259.3.3 id_prog

```
ttk_tools.vector.mdf4_reader.IdBlock.id_prog
```

7.259.3.4 id_unfin_flags

```
ttk_tools.vector.mdf4_reader.IdBlock.id_unfin_flags
```

7.259.3.5 id_ver

```
ttk_tools.vector.mdf4_reader.IdBlock.id_ver
```

7.259.3.6 id_vers

```
ttk_tools.vector.mdf4_reader.IdBlock.id_vers
```

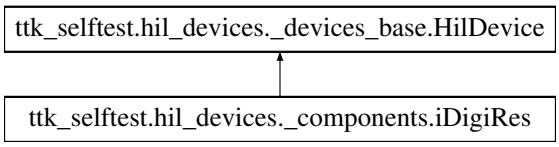
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.260 ttk_selftest.hil_devices._components.iDigiRes Class Reference

Base iDigiRes container for a testable iDigiRes device and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices._components.iDigiRes:



Public Member Functions

- __init__ (self, context, base_path, base_resistance, num_resistances, hs_channel=None, ls_channel=None, resistors=None, min_power=0.25, resistor_powers=None, preconditions=None, wait_time_ms=100, alias="", descr="")

  iDigiRes initialization.
- info (self)

  Get an info string for this iDigiResl: alias, FIU channel(s), base resistance, number of resistance and minimum power plus description, if available.

Public Attributes

- base_path
- hs_channel
- ls_channel
- base_resistance
- num_resistances
- min_power
- resistance_selftest_var
- enable_selftest_var
- resistors
- resistor_powers

7.260.1 Detailed Description

Base iDigiRes container for a testable iDigiRes device and relevant associated variables/values.

7.260.2 Constructor & Destructor Documentation

7.260.2.1 __init__()

```
ttk_selftest.hil_devices._components.iDigiRes.__init__ (
                self,
                context,
                base_path,
                base_resistance,
                num_resistances,
                hs_channel = None,
                ls_channel = None,
                resistors = None,
                min_power = 0.25,
                resistor_powers = None,
                preconditions = None,
                wait_time_ms = 100,
                alias = "",
                descr = "" )
```

iDigiRes initialization.

Parameters

context	parent context
base_path	base path to the iDigiRes in the model
base_resistance	base resistance value in Ohm that is fitted on the iDigiRes
num_resistances	number of resistances fitted on the iDigiRes
hs_channel	(opt) highside (first) fiu channel of the iDigiRes defaults to None
ls_channel	(opt) lowside (second) fiu channel of the iDigiRes defaults to None
resistors	(opt) list of resistors fitted on the iDigiRes Length of the list must be equal to the specified number of resistances. The values have to be specified in increasing order, beginning with the smallest and ending with the largest resistor. If the parameter is not specified, the resistors will be calculated according to base_resistance and num_resistances.
min_power	(opt) minimum power value of all resistances defaults to 0.25 (W)
resistor_powers	(opt) list of resistor powers Length of the list must be equal to the specified number of resistances. The values have to be ordered according to the power of R, 2R, 4R... If the parameter is not specified, the values of all resistors will be set to the parameter min_power.
preconditions	(opt) preconditions that have to be set before the device can be checked. defaults to None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The functions specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
        # For special cases, the preconditions may also contain function calls
        # (which might be useful to insert additional delays or similar):
        preconditions = [
            { "device": Voltage, "value": 1},
            { "func":   time.sleep,
              "args":   [1.0], # positional arguments
              "kwargs": {},    # keyword arguments
            },
            {"device": ClampSwitch, "value": 1},
            # or a simple call like this:
            { "func":  someCustomFunction,
            },
            ...
        ]
    wait_time_ms      - (opt) waiting time in milliseconds between
    channel stimulation and channel check
    defaults to 100ms
    alias             - (opt) channel alias/name, defaults to the
    base path of the iDigiRes in the model
    descr             - (opt) channel description (e.g. used in report entries)
    defaults to ""
```

Reimplemented from ttk_selftest.hil_devices._devices_base.HilDevice.

7.260.3 Member Function Documentation

7.260.3.1 info()

```
ttk_selftest.hil_devices._components.iDigiRes.info (
              self )
```

Get an info string for this iDigiResl: alias, FIU channel(s), base resistance, number of resistance and minimum power plus description, if available.

Returns

a formatted string.

7.260.4 Member Data Documentation

7.260.4.1 base_path

```
ttk_selftest.hil_devices._components.iDigiRes.base_path
```

7.260.4.2 base_resistance

```
ttk_selftest.hil_devices._components.iDigiRes.base_resistance
```

7.260.4.3 enable_selftest_var

```
ttk_selftest.hil_devices._components.iDigiRes.enable_selftest_var
```

7.260.4.4 hs_channel

ttk_selftest.hil_devices._components.iDigiRes.hs_channel

7.260.4.5 ls_channel

ttk_selftest.hil_devices._components.iDigiRes.ls_channel

7.260.4.6 min_power

ttk_selftest.hil_devices._components.iDigiRes.min_power

7.260.4.7 num_resistances

ttk_selftest.hil_devices._components.iDigiRes.num_resistances

7.260.4.8 resistance_selftest_var

ttk_selftest.hil_devices._components.iDigiRes.resistance_selftest_var

7.260.4.9 resistor_powers

ttk_selftest.hil_devices._components.iDigiRes.resistor_powers

7.260.4.10 resistors

ttk_selftest.hil_devices._components.iDigiRes.resistors

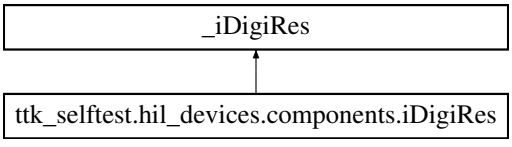
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/_components.py (v2.0.0)

7.261 ttk_selftest.hil_devices.components.iDigiRes Class Reference

Base iDigiRes container for a testable iDigiRes device and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices.components.iDigiRes:



Public Member Functions

- __init__ (self, context, base_path, base_resistance, num_resistances, hs_channel=None, ls_channel=None, resistors=None, min_power=0.25, resistor_powers=None, preconditions=None, wait_time_ms=100, alias="", descr="")

  iDigiRes initialization.
- info (self)

  Get an info string for this iDigiResl: alias, FIU channel(s), base resistance, number of resistance and minimum power plus description, if available.

7.261.1 Detailed Description

Base iDigiRes container for a testable iDigiRes device and relevant associated variables/values.

7.261.2 Constructor & Destructor Documentation

7.261.2.1 __init__()

```
ttk_selftest.hil_devices.components.iDigiRes.__init__ (
                self,
                context,
                base_path,
                base_resistance,
                num_resistances,
                hs_channel = None,
                ls_channel = None,
                resistors = None,
                min_power = 0.25,
                resistor_powers = None,
                preconditions = None,
                wait_time_ms = 100,
                alias = "",
                descr = "" )
```

iDigiRes initialization.

Parameters

context	parent context
base_path	base path to the iDigiRes in the model
base_resistance	base resistance value in Ohm that is fitted on the iDigiRes
num_resistances	number of resistances fitted on the iDigiRes
hs_channel	(opt) highside (first) fiu channel of the iDigiRes defaults to None
ls_channel	(opt) lowside (second) fiu channel of the iDigiRes defaults to None
resistors	(opt) list of resistors fitted on the iDigiRes Length of the list must be equal to the specified number of resistances. The values have to be specified in increasing order, beginning with the smallest and ending with the largest resistor. If the parameter is not specified, the resistors will be calculated according to base_resistance and num_resistances.
min_power	(opt) minimum power value of all resistances defaults to 0.25 (W)
resistor_powers	(opt) list of resistor powers Length of the list must be equal to the specified number of resistances. The values have to be ordered according to the power of R, 2R, 4R... If the parameter is not specified, the values of all resistors will be set to the parameter min_power.

Parameters

preconditions	(opt) preconditions that have to be set before the device can be checked. defaults to None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The functions specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
    # For special cases, the preconditions may also contain function calls
    # (which might be useful to insert additional delays or similar):
    preconditions = [
        { "device": Voltage, "value": 1},
        { "func":   time.sleep,
            "args":   [1.0], # positional arguments
            "kwargs": {},    # keyword arguments
        },
        {"device": ClampSwitch, "value": 1},
        # or a simple call like this:
        { "func":  someCustomFunction,
        },
        ...
    ]
wait_time_ms      - (opt) waiting time in milliseconds between
channel stimulation and channel check
defaults to 100ms
alias             - (opt) channel alias/name, defaults to the
base path of the iDigiRes in the model
descr             - (opt) channel description (e.g. used in report entries)
defaults to ""
```

7.261.3 Member Function Documentation

7.261.3.1 info()

```
ttk_selftest.hil_devices.components.iDigiRes.info (
              self )
```

Get an info string for this iDigiResl: alias, FIU channel(s), base resistance, number of resistance and minimum power plus description, if available.

Returns

a formatted string.

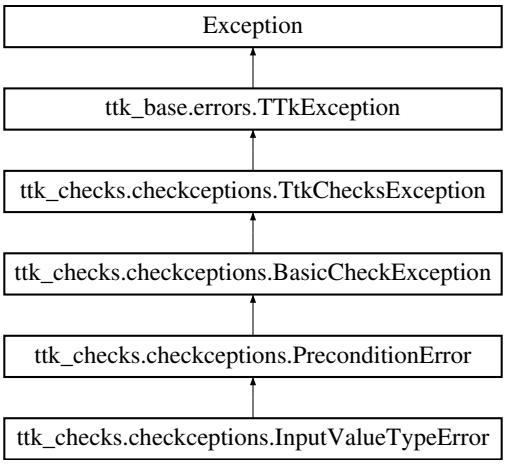
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/components.py (v2.0.0)

7.262 ttk_checks.checkceptions.InputValueTypeError Class Reference

A supplied value has a wrong format/type and cannot be converted.

Inheritance diagram for ttk_checks.checkceptions.InputValueTypeError:



Public Member Functions

- __init__ (self, msg='', log_msg=True)
  TtkChecksException init.

7.262.1 Detailed Description

A supplied value has a wrong format/type and cannot be converted.

7.262.2 Constructor & Destructor Documentation

7.262.2.1 __init__()

```
ttk_checks.checkceptions.InputValueTypeError.__init__ (
              self,
              msg = '',
              log_msg = True )
```

TtkChecksException init.

Parameters

msg	exception message text
log_msg	if True, the exception message will also be logged / printed to stdout

Reimplemented from ttk_checks.checkceptions.PreconditionError.

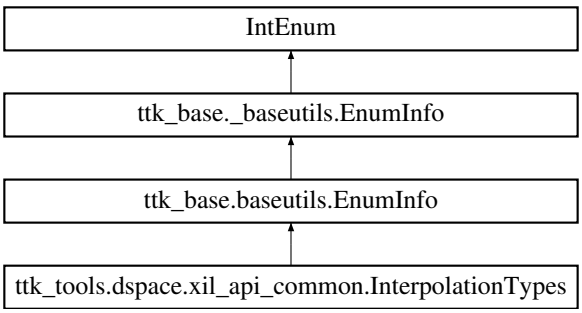
The documentation for this class was generated from the following file:

- ttk_checks/checkceptions.py (v2.0)

7.263 ttk_tools.dspace.xil_api_common.InterpolationTypes Class Reference

Data Interpolation Types.

Inheritance diagram for ttk_tools.dspace.xil_api_common.InterpolationTypes:



Static Public Attributes

- int eFORWARD = 0

  Next data point will be used immediately (staircase forward)
- int eBACKWARD = 1

  Actual data point will be used until next data point (staircase backward)
- int eLINEAR = 2

  Linear interpolation between data points.

7.263.1 Detailed Description

Data Interpolation Types.

See ASAM.XIL.Interfaces.Testbench.Common.Signal.Enum.InterpolationTypes

7.263.2 Member Data Documentation

7.263.2.1 eBACKWARD

```
int ttk_tools.dspace.xil_api_common.InterpolationTypes.eBACKWARD = 1  [static]
```

Actual data point will be used until next data point (staircase backward)

7.263.2.2 eFORWARD

```
int ttk_tools.dspace.xil_api_common.InterpolationTypes.eFORWARD = 0  [static]
```

Next data point will be used immediately (staircase forward)

7.263.2.3 eLINEAR

```
int ttk_tools.dspace.xil_api_common.InterpolationTypes.eLINEAR = 2  [static]
```

Linear interpolation between data points.

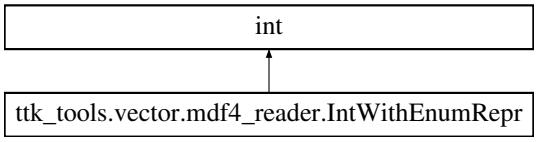
The documentation for this class was generated from the following file:

- ttk_tools/dspace/xil_api_common.py (v1.5.0)

7.264 ttk_tools.vector.mdf4_reader.IntWithEnumRepr Class Reference

An integer value that shows an EnumInfo repr of itself (if a match is found)

Inheritance diagram for ttk_tools.vector.mdf4_reader.IntWithEnumRepr:



Public Member Functions

- __new__ (cls, value, ∗args, ∗∗kwargs)
- __init__ (self, value, enum_class)
- __repr__ (self)
- text (self)

  Textual representation of this value, or None if no text is available.

Protected Attributes

- _enum

7.264.1 Detailed Description

An integer value that shows an EnumInfo repr of itself (if a match is found)

7.264.2 Constructor & Destructor Documentation

7.264.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.IntWithEnumRepr.__init__ (
              self,
              value,
              enum_class )
```

Parameters

value	integer value
enum_class	either a EnumInfo-derived class or a plain dict instance with {value: text} mappings

7.264.3 Member Function Documentation

7.264.3.1 __new__()

```
ttk_tools.vector.mdf4_reader.IntWithEnumRepr.__new__ (
              cls,
              value,
             * args,
             ** kwargs )
```

7.264.3.2 __repr__()

```
ttk_tools.vector.mdf4_reader.IntWithEnumRepr.__repr__ (
              self )
```

7.264.3.3 text()

```
ttk_tools.vector.mdf4_reader.IntWithEnumRepr.text (
              self )
```

Textual representation of this value, or None if no text is available.

7.264.4 Member Data Documentation

7.264.4.1 _enum

```
ttk_tools.vector.mdf4_reader.IntWithEnumRepr._enum  [protected]
```

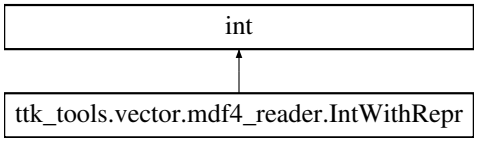
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.265 ttk_tools.vector.mdf4_reader.IntWithRepr Class Reference

An integer value that has an associated (textual) representation.

Inheritance diagram for ttk_tools.vector.mdf4_reader.IntWithRepr:



Public Member Functions

- __new__ (cls, value, *args, **kwargs)
- __init__ (self, value, text)
- __repr__ (self)

Public Attributes

- text

7.265.1 Detailed Description

An integer value that has an associated (textual) representation.

A representation of None marks an undefined representation (used for conversion types that may contain undefined values).

7.265.2 Constructor & Destructor Documentation

7.265.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.IntWithRepr.__init__ (
              self,
              value,
              text )
```

Parameters

value	integer value
text	string with textual representation. If text is None, the value will be considered as "undefined/invalid"

7.265.3 Member Function Documentation

7.265.3.1 __new__()

```
ttk_tools.vector.mdf4_reader.IntWithRepr.__new__ (
              cls,
              value,
             * args,
             ** kwargs )
```

7.265.3.2 __repr__()

```
ttk_tools.vector.mdf4_reader.IntWithRepr.__repr__ (
              self )
```

7.265.4 Member Data Documentation

7.265.4.1 text

```
ttk_tools.vector.mdf4_reader.IntWithRepr.text
```

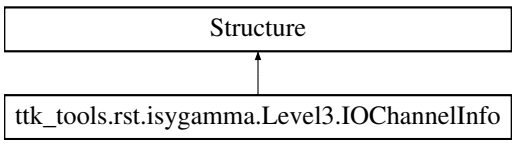
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.266 ttk_tools.rst.isygamma.Level3.IOChannelInfo Class Reference

typedef struct gc_IOChannelInfo

Inheritance diagram for ttk_tools.rst.isygamma.Level3.IOChannelInfo:



Static Protected Attributes

- list _fields_

7.266.1 Detailed Description

typedef struct gc_IOChannelInfo

7.266.2 Member Data Documentation

7.266.2.1 _fields_

```
list ttk_tools.rst.isygamma.Level3.IOChannelInfo._fields_  [static], [protected]
```

Initial value:

```
=  [
            ("address",        Addresses.Level3),# address of the I/O channel
            ("type",           c_int),           # data type of the I/O channel. See gc_DataType
            ("isInputChannel", c_bool),          # true: I/O channel used for input, false: used for output
            ("state",          c_int),           # current operation state of the channel.  See gc_IOStates
            ("description",    c_char * 256),    # additional description of the I/O channel.
        ]
```

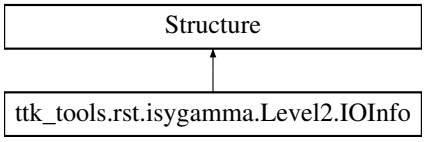
The documentation for this class was generated from the following file:

- ttk_tools/rst/isygamma.py (v1.6)

7.267 ttk_tools.rst.isygamma.Level2.IOInfo Class Reference

typedef struct gc_IOInfo

Inheritance diagram for ttk_tools.rst.isygamma.Level2.IOInfo:



Static Protected Attributes

- list _fields_

7.267.1 Detailed Description

typedef struct gc_IOInfo

7.267.2 Member Data Documentation

7.267.2.1 _fields_

```
list ttk_tools.rst.isygamma.Level2.IOInfo._fields_  [static], [protected]
```

Initial value:

```
=  [
            ("address",          Addresses.Level2), # address of the I/O
            ("noInputChannels",  c_uint32),         # number of input channels
            ("noOutputChannels", c_uint32),         # number of output channels
            ("state",            c_int),            # current operational state of the HW. See gc_IOStates
            ("vendor",           c_char * 256),     # name of hardware vendor
            ("description",      c_char * 256),     # additional description
            ("library",          c_char * 256),     # name of the Gamma plug-in library
            ("startParameters",  c_char * 256),     # start parameters for the plug-in
        ]
```

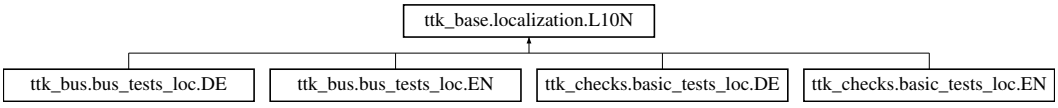
The documentation for this class was generated from the following file:

- ttk_tools/rst/isygamma.py (v1.6)

7.268 ttk_base.localization.L10N Class Reference

Base class for localization handling.

Inheritance diagram for ttk_base.localization.L10N:



Public Member Functions

- __init__ (self, debug_missing_entries=False)
- __del__ (self)
- makeText (self)

  Alias for getText.
- getText (self, text, n=None, default=None)

  Get a translated version of the supplied text.
- setDebugLog (self, log_file_path=None)

  Set a file for logging missing localization keys.

Static Public Member Functions

- getPluralForm (forms, n, fallback="")

  Get appropriate plural form from tuple forms for a given number n.

Protected Attributes

- _debug

  True: echo missing keys.
- _debug_log

  handle to debug log file
- _debug_already_logged

  lookup for already logged "missing" entries

Static Protected Attributes

- dict _localization = {}

  Localization Entries.

7.268.1 Detailed Description

Base class for localization handling.

7.268.2 Constructor & Destructor Documentation

7.268.2.1 __init__()

```
ttk_base.localization.L10N.__init__ (
            self,
            debug_missing_entries = False )
```

Parameters

debug_missing_entries	True: missing entries will be logged to stdout. If setDebugLog has been called, they will also be appended to the given log file.

7.268.2.2 __del__()

```
ttk_base.localization.L10N.__del__ (
              self )
```

7.268.3 Member Function Documentation

7.268.3.1 getPluralForm()

```
ttk_base.localization.L10N.getPluralForm (
              forms,
              n,
              fallback = "" )  [static]
```

Get appropriate plural form from tuple forms for a given number n.

This default implementation works for English and German.

Parameters

forms	a tuple (or list) of available plural forms, e.g. ("Ergebnis", "Ergebnisse")
n	value for which to select an appropriate plural form
fallback	fallback to use if no plural forms are available. This will usually be set to the lookup key.

Returns

a plural form string.

7.268.3.2 getText()

```
ttk_base.localization.L10N.getText (
              self,
              text,
              n = None,
              default = None )
```

Get a translated version of the supplied text.

Parameters

text	text to localize, also used as default value if the separate default parameter has not been specified (or explicitly set to None).
n	number for determining singular/plural forms. If left empty or no plural forms are available, the single/singular form will be returned.
default	use as explicit default value for longer texts (e.g. copyright statements), where parameter "text" serves as localization label.

Returns

a localized string.

7.268.3.3 makeText()

```
ttk_base.localization.L10N.makeText (
              self )
```

Alias for getText.

7.268.3.4 setDebugLog()

```
ttk_base.localization.L10N.setDebugLog (
              self,
              log_file_path = None )
```

Set a file for logging missing localization keys.

Parameters

log_file_path	path to log file. None: close a previously opened debug log

7.268.4 Member Data Documentation

7.268.4.1 _debug

```
ttk_base.localization.L10N._debug  [protected]
```

True: echo missing keys.

7.268.4.2 _debug_already_logged

```
ttk_base.localization.L10N._debug_already_logged  [protected]
```

lookup for already logged "missing" entries

7.268.4.3 _debug_log

```
ttk_base.localization.L10N._debug_log  [protected]
```

handle to debug log file

7.268.4.4 _localization

```
dict ttk_base.localization.L10N._localization = {}  [static], [protected]
```

Localization Entries.

Empty in base class, but derived classes will define their l10n lookups here. Contains entries like:

Simple entry (default text serves as l10n key)

"text": "translated text",

e.g.

"localized entry": "lokalisierter Eintrag",

Entry with singular and plural forms, see getPluralForm()

"base text": ("singular text", "plural text"),

e.g.

"result(s)": ("result", "results"),

Blank entries, value will default to l10n key.

Setting entries to None can be used to use default texts while

providing a way to check if all expected keys are defined,

see showL10nDifferences().

"this is also the default text": None,

Entries with non-default text l10n keys

Note: angular brackets in key names are not required

and only used by convention

"<some key>": "an overly long text like some licensing blurb that " "does not work very well as a l10n key [...]"

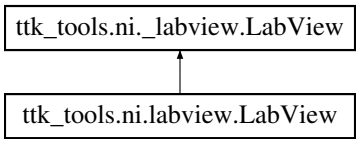
The documentation for this class was generated from the following file:

- ttk_base/localization.py (v1.6)

7.269 ttk_tools.ni._labview.LabView Class Reference

Access to LabView VI controls/variables via COM interface.

Inheritance diagram for ttk_tools.ni._labview.LabView:



Public Member Functions

- __init__ (self, com_obj_name, base_path)
- close (self)

  Close/quit current LabView connection.
- getVi (self, vi_name)

  Get a reference to the named virtual instrument.
- getViVar (self, identifier)

  Get current value of a VI's variable/control.
- setViVar (self, identifier, value)

  Set value to a VI's variable/control.

Public Attributes

- lv_application
- lv_base_path

Static Protected Member Functions

- _splitViVarNames (identifier)

  Split identifier into VI-name and var-name components.

Protected Attributes

- _vi_cache

7.269.1 Detailed Description

Access to LabView VI controls/variables via COM interface.

Note

The LabView App needs to be build with COM/ActiveX server support (and registered as a COM server) to enable remote access.

7.269.2 Constructor & Destructor Documentation

7.269.2.1 __init__()

```
ttk_tools.ni._labview.LabView.__init__ (
              self,
              com_obj_name,
              base_path )
```

Parameters

com_obj_name	name of the COM server application object, e.g. "FooApp1.Application"
base_path	base path to LabView-compiled COM server executable, e.g. 'C:/utilities/Foo_Ver1__By_Labview/Foo.exe'

Reimplemented in ttk_tools.ni.labview.LabView.

7.269.3 Member Function Documentation

7.269.3.1 _splitViVarNames()

```
ttk_tools.ni._labview.LabView._splitViVarNames (
              identifier )  [static], [protected]
```

Split identifier into VI-name and var-name components.

Parameters

identifier	identifier of variable with prefixed VI name, separated by a forward slash e.g. "Bar.vi/variable_name"

Returns

two elements: [vi_name, var_name]

7.269.3.2 close()

```
ttk_tools.ni._labview.LabView.close (
              self )
```

Close/quit current LabView connection.

Reimplemented in ttk_tools.ni.labview.LabView.

7.269.3.3 getVi()

```
ttk_tools.ni._labview.LabView.getVi (
              self,
              vi_name )
```

Get a reference to the named virtual instrument.

Parameters

vi_name	name of VI to get

Returns

a COM reference to the VI or None if COM application is not available.

Reimplemented in ttk_tools.ni.labview.LabView.

7.269.3.4 getViVar()

```
ttk_tools.ni._labview.LabView.getViVar (
              self,
              identifier )
```

Get current value of a VI's variable/control.

Parameters

identifier	identifier of variable with prefixed VI name, separated by a slash e.g. "Bar.vi/variable_name"

Returns

the current value

Reimplemented in ttk_tools.ni.labview.LabView.

7.269.3.5 setViVar()

```
ttk_tools.ni._labview.LabView.setViVar (
              self,
              identifier,
              value )
```

Set value to a VI's variable/control.

Parameters

identifier	identifier of variable with prefixed VI name, separated by a slash e.g. "Bar.vi/variable_name"
value	value to set

Reimplemented in ttk_tools.ni.labview.LabView.

7.269.4 Member Data Documentation

7.269.4.1 _vi_cache

```
ttk_tools.ni._labview.LabView._vi_cache  [protected]
```

7.269.4.2 lv_application

```
ttk_tools.ni._labview.LabView.lv_application
```

7.269.4.3 lv_base_path

```
ttk_tools.ni._labview.LabView.lv_base_path
```

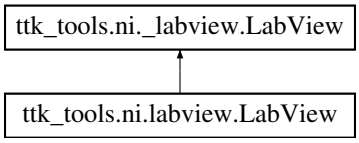
The documentation for this class was generated from the following file:

- ttk_tools/ni/_labview.py (v1.2.1)

7.270 ttk_tools.ni.labview.LabView Class Reference

Access to LabView VI controls/variables via COM interface.

Inheritance diagram for ttk_tools.ni.labview.LabView:



Public Member Functions

- __init__ (self, com_obj_name, base_path)
- close (self)

  Close/quit current LabView connection.
- getVi (self, vi_name)

  Get a reference to the named virtual instrument.
- getViVar (self, identifier)

  Get current value of a VI's variable/control.
- setViVar (self, identifier, value)

  Set value to a VI's variable/control.

7.270.1 Detailed Description

Access to LabView VI controls/variables via COM interface.

Note

The LabView App needs to be build with COM/ActiveX server support (and registered as a COM server) to enable remote access.

7.270.2 Constructor & Destructor Documentation

7.270.2.1 __init__()

```
ttk_tools.ni.labview.LabView.__init__ (
            self,
            com_obj_name,
            base_path )
```

Parameters

com_obj_name	name of the COM server application object, e.g. "FooApp1.Application"
base_path	base path to LabView-compiled COM server executable, e.g. 'C:/utilities/Foo_Ver1__By_Labview/Foo.exe'

Reimplemented from ttk_tools.ni._labview.LabView.

7.270.3 Member Function Documentation

7.270.3.1 close()

```
ttk_tools.ni.labview.LabView.close (
              self )
```

Close/quit current LabView connection.

Reimplemented from ttk_tools.ni._labview.LabView.

7.270.3.2 getVi()

```
ttk_tools.ni.labview.LabView.getVi (
              self,
              vi_name )
```

Get a reference to the named virtual instrument.

Parameters

vi_name	name of VI to get

Returns

a COM reference to the VI or None if COM application is not available.

Reimplemented from ttk_tools.ni._labview.LabView.

7.270.3.3 getViVar()

```
ttk_tools.ni.labview.LabView.getViVar (
              self,
              identifier )
```

Get current value of a VI's variable/control.

Parameters

identifier	identifier of variable with prefixed VI name, separated by a slash e.g. "Bar.vi/variable_name"

Returns

the current value

Reimplemented from ttk_tools.ni._labview.LabView.

7.270.3.4 setViVar()

```
ttk_tools.ni.labview.LabView.setViVar (
               self,
               identifier,
               value )
```

Set value to a VI's variable/control.

Parameters

identifier	identifier of variable with prefixed VI name, separated by a slash e.g. "Bar.vi/variable_name"
value	value to set

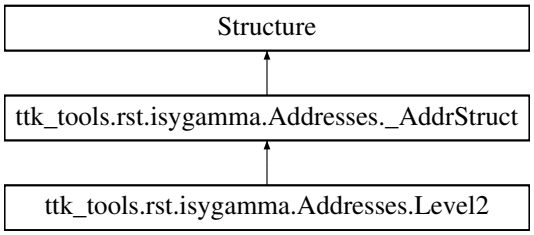
Reimplemented from ttk_tools.ni._labview.LabView.

The documentation for this class was generated from the following file:

- ttk_tools/ni/labview.py (v1.2.1)

7.271 ttk_tools.rst.isygamma.Addresses.Level2 Class Reference

Inheritance diagram for ttk_tools.rst.isygamma.Addresses.Level2:



Static Protected Attributes

- list _fields_ = [('system', c_int32), ('index1', c_uint32)]

7.271.1 Member Data Documentation

7.271.1.1 _fields_

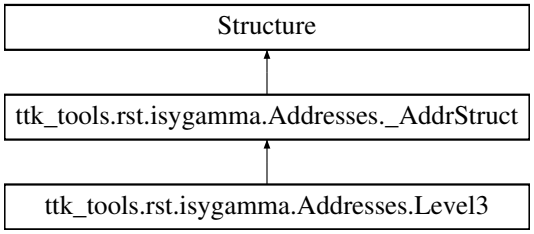
```
list ttk_tools.rst.isygamma.Addresses.Level2._fields_ = [('system', c_int32), ('index1', c_↩
uint32)]  [static], [protected]
```

The documentation for this class was generated from the following file:

- ttk_tools/rst/isygamma.py (v1.6)

7.272 ttk_tools.rst.isygamma.Addresses.Level3 Class Reference

Inheritance diagram for ttk_tools.rst.isygamma.Addresses.Level3:



Static Protected Attributes

- list _fields_ = [('system', c_int32), ('index1', c_uint32), ('index2', c_uint32)]

7.272.1 Member Data Documentation

7.272.1.1 _fields_

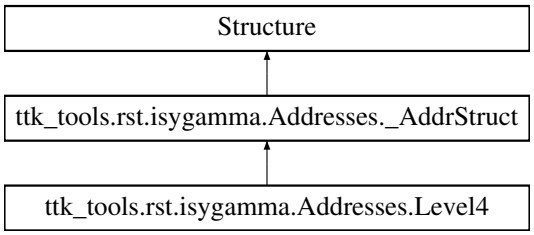
```
list ttk_tools.rst.isygamma.Addresses.Level3._fields_ = [('system', c_int32), ('index1', c_↩
uint32), ('index2', c_uint32)]  [static], [protected]
```

The documentation for this class was generated from the following file:

- ttk_tools/rst/isygamma.py (v1.6)

7.273 ttk_tools.rst.isygamma.Addresses.Level4 Class Reference

Inheritance diagram for ttk_tools.rst.isygamma.Addresses.Level4:



Static Protected Attributes

- list _fields_ = [('system', c_int32), ('index1', c_uint32), ('index2', c_uint32), ('index3', c_uint32)]

7.273.1 Member Data Documentation

7.273.1.1 _fields_

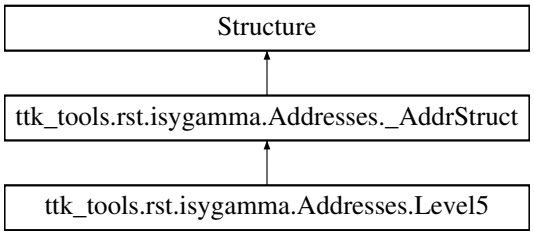
```
list ttk_tools.rst.isygamma.Addresses.Level4._fields_ = [('system', c_int32), ('index1', c_↩
uint32), ('index2', c_uint32), ('index3', c_uint32)]  [static], [protected]
```

The documentation for this class was generated from the following file:

- ttk_tools/rst/isygamma.py (v1.6)

7.274 ttk_tools.rst.isygamma.Addresses.Level5 Class Reference

Inheritance diagram for ttk_tools.rst.isygamma.Addresses.Level5:



Static Protected Attributes

- list _fields_ = [('system', c_int32), ('index1', c_uint32), ('index2', c_uint32), ('index3', c_uint32), ('index4', c_↩ uint32)]

7.274.1 Member Data Documentation

7.274.1.1 _fields_

```
list ttk_tools.rst.isygamma.Addresses.Level5._fields_ = [('system', c_int32), ('index1', c_↩
uint32), ('index2', c_uint32), ('index3', c_uint32), ('index4', c_uint32)]  [static], [protected]
```

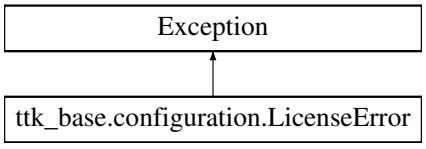
The documentation for this class was generated from the following file:

- ttk_tools/rst/isygamma.py (v1.6)

7.275 ttk_base.configuration.LicenseError Class Reference

Invalid license has been encountered.

Inheritance diagram for ttk_base.configuration.LicenseError:



Public Member Functions

- __init__ (self, msg="Invalid license")

7.275.1 Detailed Description

Invalid license has been encountered.

7.275.2 Constructor & Destructor Documentation

7.275.2.1 __init__()

```
ttk_base.configuration.LicenseError.__init__ (
              self,
              msg = "Invalid license" )
```

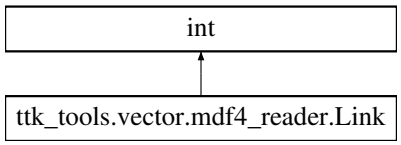
The documentation for this class was generated from the following file:

- ttk_base/configuration.py (v3.8.1)

7.276 ttk_tools.vector.mdf4_reader.Link Class Reference

LINK file position/offset as a long value with custom representation.

Inheritance diagram for ttk_tools.vector.mdf4_reader.Link:



Public Member Functions

- __new__ (cls, file_handle, *args, **kwargs)

  Get a single LINK position from file_handle (reads next 8 bytes and interprets it as LINK).
- __repr__ (self)

  just to get a (arguably) "nicer" representation

7.276.1 Detailed Description

LINK file position/offset as a long value with custom representation.

7.276.2 Member Function Documentation

7.276.2.1 __new__()

```
ttk_tools.vector.mdf4_reader.Link.__new__ (
              cls,
              file_handle,
             * args,
             ** kwargs )
```

Get a single LINK position from file_handle (reads next 8 bytes and interprets it as LINK).

A LINK is a sint64 and is used as byte position within the file. If a LINK is NIL (i.e. 0), this means the LINK cannot be de-referenced (no block is attached). A link must be a multiple of 8.

Parameters

file_handle	open file handle from which to read link data
∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

7.276.2.2 __repr__()

```
ttk_tools.vector.mdf4_reader.Link.__repr__ (
              self )
```

just to get a (arguably) "nicer" representation

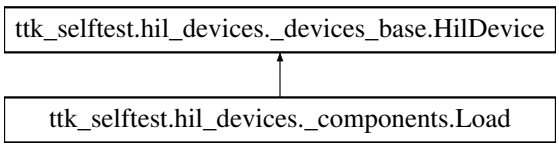
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.277 ttk_selftest.hil_devices._components.Load Class Reference

Base load container for a testable load and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices._components.Load:



Public Member Functions

- __init__ (self, context, hs_channel, ls_channel, max_voltage, max_current, load_resistance=None, base_path_v_mon=None, gnd_v_mon=None, abs_tol_v=0.1, rel_tol_v=0.1, base_path_hs_c_mon=None, base_path_ls_c_mon=None, max_voltage_c=None, abs_tol_c=0.1, rel_tol_c=0.1, avg_c=0, base_path_hs_pwm_mon=None, base_path_ls_pwm_mon=None, max_pwm_freq=100000, min_pwm_freq=0, base_path_hs_dig_mon=None, base_path_ls_dig_mon=None, max_on_rate=100, preconditions=None, wait_time_ms=100, alias="", model="", descr="")

  Load initialization.
- info (self)

  Get an info string for the load: alias, FIU channels, load resistance, maximum voltage, maximum current plus description, if available.

Public Attributes

- v_mon
- hs_c_mon
- ls_c_mon
- hs_pwm_mon
- ls_pwm_mon
- hs_dig_mon
- ls_dig_mon
- min_voltage
- hs_channel
- ls_channel
- resistance
- max_voltage
- max_current
- max_voltage_c
- max_on_rate
- max_pwm_freq
- min_pwm_freq
- abs_tol_v
- rel_tol_v
- abs_tol_c
- rel_tol_c
- average_c
- model
- alias

7.277.1 Detailed Description

Base load container for a testable load and relevant associated variables/values.

7.277.2 Constructor & Destructor Documentation

7.277.2.1 __init__()

```
ttk_selftest.hil_devices._components.Load.__init__ (
              self,
              context,
              hs_channel,
              ls_channel,
              max_voltage,
              max_current,
              load_resistance = None,
              base_path_v_mon = None,
              gnd_v_mon = None,
              abs_tol_v = 0.1,
              rel_tol_v = 0.1,
              base_path_hs_c_mon = None,
              base_path_ls_c_mon = None,
              max_voltage_c = None,
              abs_tol_c = 0.1,
              rel_tol_c = 0.1,
```

```
avg_c = 0,
base_path_hs_pwm_mon = None,
base_path_ls_pwm_mon = None,
max_pwm_freq = 100000,
min_pwm_freq = 0,
base_path_hs_dig_mon = None,
base_path_ls_dig_mon = None,
max_on_rate = 100,
preconditions = None,
wait_time_ms = 100,
alias = "",
model = "",
descr = "" )
```

Load initialization.

Parameters

context	parent context
hs_channel	fiu channel of the load highside
ls_channel	fiu channel of the load lowside can be either GND device or FiuChannel
max_voltage	maximum voltage in V of the load
max_current	maximum current in A of the load
load_resistance	(opt) expected load resistance in Ohm if constant load resistance can be specified defaults to None
base_path_v_mon	(opt) base path to analog channel subsystem used for voltage monitoring defaults to None
gnd_v_mon	(opt) ground of the voltage monitoring defaults to lowside fiu channel of the load
abs_tol_v	(opt) absolute tolerance for voltage monitoring defaults to 0.1 (equals 100mV)
rel_tol_v	(opt) relative tolerance for voltage monitoring defaults to 0.1 (equals 10%)
base_path_hs_c_mon	(opt) base path to analog channel subsystem used for highside current monitoring defaults to None
base_path_ls_c_mon	(opt) base path to analog channel subsystem used for lowside current monitoring defaults to None
max_voltage_c	(opt) maximum voltage during test of current monitoring defaults to maximum voltage of the load
abs_tol_c	(opt) absolute tolerance for current monitoring defaults to 0.1 (equals 100mA)
rel_tol_c	(opt) relative tolerance for current monitoring defaults to 0.1 (equals 10%)
avg_c	(opt) averaging for current reference measurement this parameter will be used to configure the averaging of the current sensor on the iSystemReferenceCard defaults to 0
base_path_hs_pwm_mon	(opt) base path to highside pwm channel subsystem used for highside pwm monitoring defaults to None
base_path_ls_pwm_mon	(opt) base path to lowside pwm channel subsystem used for lowside pwm monitoring defaults to None
max_pwm_freq	(opt) maximum frequency in Hz of the PWM channel limited by measurement hardware to 100kHz defaults to 100kHz
min_pwm_freq	(opt) minimum frequency in Hz of the PWM channel defaults to 0Hz
base_path_hs_dig_mon	(opt) base path to highside digital channel subsystem used for highside state monitoring defaults to None
base_path_ls_dig_mon	(opt) base path to lowside digital channel subsystem used for lowside state monitoring defaults to None
max_on_rate	(opt) maximum "on"-rate of the load in percent defaults to 100 (%)

Parameters

preconditions	(opt) preconditions that have to be set before the device can be checked. defaults to None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The functions specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
    # For special cases, the preconditions may also contain function calls
    # (which might be useful to insert additional delays or similar):
    preconditions = [
        { "device": Voltage, "value": 1},
        { "func":   time.sleep,
          "args":   [1.0], # positional arguments
          "kwargs": {},    # keyword arguments
        },
        {"device": ClampSwitch, "value": 1},
        # or a simple call like this:
        { "func":  someCustomFunction,
        },
        ...
    ]
wait_time_ms         - (opt) waiting time in milliseconds between
stimulation and check
defaults to 100ms
alias                - (opt) load alias/name
defaults to ""
model                - (opt) load model
defaults to ""
descr                - (opt) load description
(e.g. used in report entries)
defaults to ""
```

Reimplemented from ttk_selftest.hil_devices._devices_base.HilDevice.

7.277.3 Member Function Documentation

7.277.3.1 info()

```
ttk_selftest.hil_devices._components.Load.info (
              self )
```

Get an info string for the load: alias, FIU channels, load resistance, maximum voltage, maximum current plus description, if available.

Returns

a formatted string.

7.277.4 Member Data Documentation

7.277.4.1 abs_tol_c

ttk_selftest.hil_devices._components.Load.abs_tol_c

7.277.4.2 abs_tol_v

ttk_selftest.hil_devices._components.Load.abs_tol_v

7.277.4.3 alias

ttk_selftest.hil_devices._components.Load.alias

7.277.4.4 average_c

ttk_selftest.hil_devices._components.Load.average_c

7.277.4.5 hs_c_mon

ttk_selftest.hil_devices._components.Load.hs_c_mon

7.277.4.6 hs_channel

ttk_selftest.hil_devices._components.Load.hs_channel

7.277.4.7 hs_dig_mon

ttk_selftest.hil_devices._components.Load.hs_dig_mon

7.277.4.8 hs_pwm_mon

ttk_selftest.hil_devices._components.Load.hs_pwm_mon

7.277.4.9 ls_c_mon

ttk_selftest.hil_devices._components.Load.ls_c_mon

7.277.4.10 ls_channel

ttk_selftest.hil_devices._components.Load.ls_channel

7.277.4.11 ls_dig_mon

ttk_selftest.hil_devices._components.Load.ls_dig_mon

7.277.4.12 ls_pwm_mon

ttk_selftest.hil_devices._components.Load.ls_pwm_mon

7.277.4.13 max_current

ttk_selftest.hil_devices._components.Load.max_current

7.277.4.14 max_on_rate

ttk_selftest.hil_devices._components.Load.max_on_rate

7.277.4.15 max_pwm_freq

ttk_selftest.hil_devices._components.Load.max_pwm_freq

7.277.4.16 max_voltage

ttk_selftest.hil_devices._components.Load.max_voltage

7.277.4.17 max_voltage_c

ttk_selftest.hil_devices._components.Load.max_voltage_c

7.277.4.18 min_pwm_freq

ttk_selftest.hil_devices._components.Load.min_pwm_freq

7.277.4.19 min_voltage

ttk_selftest.hil_devices._components.Load.min_voltage

7.277.4.20 model

ttk_selftest.hil_devices._components.Load.model

7.277.4.21 rel_tol_c

ttk_selftest.hil_devices._components.Load.rel_tol_c

7.277.4.22 rel_tol_v

ttk_selftest.hil_devices._components.Load.rel_tol_v

7.277.4.23 resistance

ttk_selftest.hil_devices._components.Load.resistance

7.277.4.24 v_mon

ttk_selftest.hil_devices._components.Load.v_mon

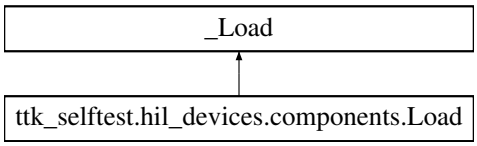
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/_components.py (v2.0.0)

7.278 ttk_selftest.hil_devices.components.Load Class Reference

Base load container for a testable load and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices.components.Load:



Public Member Functions

- __init__ (self, context, hs_channel, ls_channel, max_voltage, max_current, load_resistance=None, base_path_v_mon=None, gnd_v_mon=None, abs_tol_v=0.1, rel_tol_v=0.1, base_path_hs_c_mon=None, base_path_ls_c_mon=None, max_voltage_c=None, abs_tol_c=0.1, rel_tol_c=0.1, avg_c=0, base_path_hs_pwm_mon=None, base_path_ls_pwm_mon=None, max_pwm_freq=100000, min_pwm_freq=0, base_path_hs_dig_mon=None, base_path_ls_dig_mon=None, max_on_rate=100, preconditions=None, wait_time_ms=100, alias="", model="", descr="")

  Load initialization.
- info (self)

  Get an info string for the load: alias, FIU channels, load resistance, maximum voltage, maximum current plus description, if available.

7.278.1 Detailed Description

Base load container for a testable load and relevant associated variables/values.

7.278.2 Constructor & Destructor Documentation

7.278.2.1 __init__()

```
ttk_selftest.hil_devices.components.Load.__init__ (
                self,
                context,
                hs_channel,
                ls_channel,
                max_voltage,
                max_current,
                load_resistance = None,
                base_path_v_mon = None,
                gnd_v_mon = None,
                abs_tol_v = 0.1,
                rel_tol_v = 0.1,
                base_path_hs_c_mon = None,
                base_path_ls_c_mon = None,
                max_voltage_c = None,
                abs_tol_c = 0.1,
                rel_tol_c = 0.1,
                avg_c = 0,
                base_path_hs_pwm_mon = None,
                base_path_ls_pwm_mon = None,
                max_pwm_freq = 100000,
                min_pwm_freq = 0,
                base_path_hs_dig_mon = None,
                base_path_ls_dig_mon = None,
                max_on_rate = 100,
                preconditions = None,
                wait_time_ms = 100,
                alias = "",
                model = "",
                descr = "" )
```

Load initialization.

Parameters

context	parent context
hs_channel	fiu channel of the load highside
ls_channel	fiu channel of the load lowside can be either GND device or FiuChannel
max_voltage	maximum voltage in V of the load
max_current	maximum current in A of the load
load_resistance	(opt) expected load resistance in Ohm if constant load resistance can be specified defaults to None
base_path_v_mon	(opt) base path to analog channel subsystem used for voltage monitoring defaults to None
gnd_v_mon	(opt) ground of the voltage monitoring defaults to lowside fiu channel of the load
abs_tol_v	(opt) absolute tolerance for voltage monitoring defaults to 0.1 (equals 100mV)

Parameters

rel_tol_v	(opt) relative tolerance for voltage monitoring defaults to 0.1 (equals 10%)
base_path_hs_c_mon	(opt) base path to analog channel subsystem used for highside current monitoring defaults to None
base_path_ls_c_mon	(opt) base path to analog channel subsystem used for lowside current monitoring defaults to None
max_voltage_c	(opt) maximum voltage during test of current monitoring defaults to maximum voltage of the load
abs_tol_c	(opt) absolute tolerance for current monitoring defaults to 0.1 (equals 100mA)
rel_tol_c	(opt) relative tolerance for current monitoring defaults to 0.1 (equals 10%)
avg_c	(opt) averaging for current reference measurement this parameter will be used to configure the averaging of the current sensor on the iSystemReferenceCard defaults to 0
base_path_hs_pwm_mon	(opt) base path to highside pwm channel subsystem used for highside pwm monitoring defaults to None
base_path_ls_pwm_mon	(opt) base path to lowside pwm channel subsystem used for lowside pwm monitoring defaults to None
max_pwm_freq	(opt) maximum frequency in Hz of the PWM channel limited by measurement hardware to 100kHz defaults to 100kHz
min_pwm_freq	(opt) minimum frequency in Hz of the PWM channel defaults to 0Hz
base_path_hs_dig_mon	(opt) base path to highside digital channel subsystem used for highside state monitoring defaults to None
base_path_ls_dig_mon	(opt) base path to lowside digital channel subsystem used for lowside state monitoring defaults to None
max_on_rate	(opt) maximum "on"-rate of the load in percent defaults to 100 (%)
preconditions	(opt) preconditions that have to be set before the device can be checked. defaults to None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The functions specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
    # For special cases, the preconditions may also contain function calls
    # (which might be useful to insert additional delays or similar):
    preconditions = [
        { "device": Voltage, "value": 1},
        { "func":   time.sleep,
            "args":   [1.0], # positional arguments
            "kwargs": {},    # keyword arguments
        },
        {"device": ClampSwitch, "value": 1},
        # or a simple call like this:
        { "func":  someCustomFunction,
        },
        ...
    ]
```

```
wait_time_ms         - (opt) waiting time in milliseconds between
stimulation and check
defaults to 100ms
alias                - (opt) load alias/name
defaults to ""
model                - (opt) load model
defaults to ""
descr                - (opt) load description
(e.g. used in report entries)
defaults to ""
```

7.278.3 Member Function Documentation

7.278.3.1 info()

```
ttk_selftest.hil_devices.components.Load.info (
               self )
```

Get an info string for the load: alias, FIU channels, load resistance, maximum voltage, maximum current plus description, if available.

Returns

a formatted string.

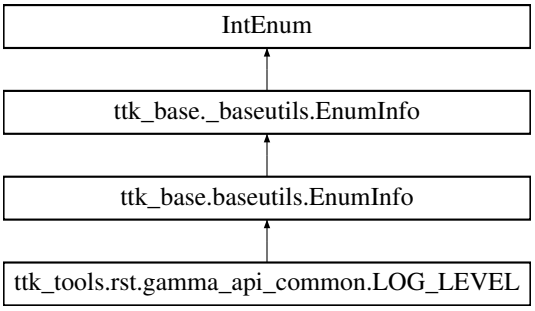
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/components.py (v2.0.0)

7.279 ttk_tools.rst.gamma_api_common.LOG_LEVEL Class Reference

Log levels.

Inheritance diagram for ttk_tools.rst.gamma_api_common.LOG_LEVEL:



Static Public Attributes

- int ERROR = 1
- int WARN = 2
- int INFO = 3

7.279.1 Detailed Description

Log levels.

7.279.2 Member Data Documentation

7.279.2.1 ERROR

```
int ttk_tools.rst.gamma_api_common.LOG_LEVEL.ERROR = 1  [static]
```

7.279.2.2 INFO

```
int ttk_tools.rst.gamma_api_common.LOG_LEVEL.INFO = 3  [static]
```

7.279.2.3 WARN

```
int ttk_tools.rst.gamma_api_common.LOG_LEVEL.WARN = 2  [static]
```

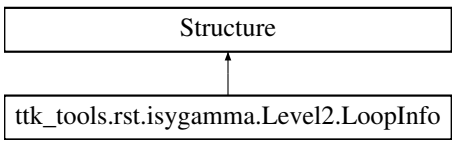
The documentation for this class was generated from the following file:

- ttk_tools/rst/gamma_api_common.py (v1.5)

7.280 ttk_tools.rst.isygamma.Level2.LoopInfo Class Reference

typedef struct gc_LoopInfo

Inheritance diagram for ttk_tools.rst.isygamma.Level2.LoopInfo:



7.280.1 Detailed Description

typedef struct gc_LoopInfo

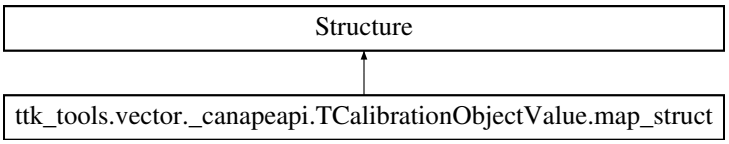
The documentation for this class was generated from the following file:

- ttk_tools/rst/isygamma.py (v1.6)

7.281 ttk_tools.vector._canapeapi.TCalibrationObjectValue.map_struct Class Reference

3D table data.

Inheritance diagram for ttk_tools.vector._canapeapi.TCalibrationObjectValue.map_struct:



Static Protected Attributes

- int _pack_ = 1
- list _fields_

7.281.1 Detailed Description

3D table data.

7.281.2 Member Data Documentation

7.281.2.1 _fields_

```
list ttk_tools.vector._canapeapi.TCalibrationObjectValue.map_struct._fields_  [static], [protected]
```

Initial value:

```
=  [
            ("type",        c_int),
            ("xDimension",  c_short),
            ("yDimension",  c_short),
            ("xAxis",       POINTER(c_double)),
            ("yAxis",       POINTER(c_double)),
            ("values",      POINTER(c_double)),
        ]
```

7.281.2.2 _pack_

```
int ttk_tools.vector._canapeapi.TCalibrationObjectValue.map_struct._pack_ = 1  [static], [protected]
```

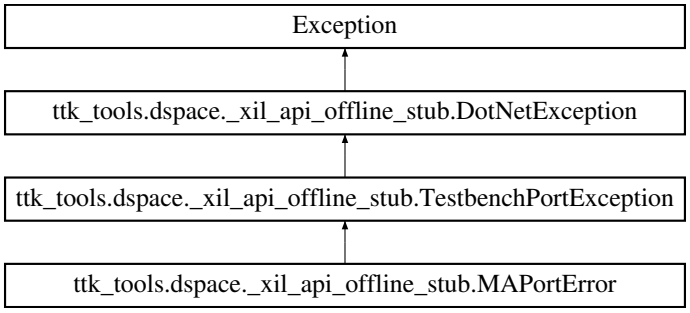
The documentation for this class was generated from the following file:

- ttk_tools/vector/_canapeapi.py (v7.3.3)

7.282 ttk_tools.dspace._xil_api_offline_stub.MAPortError Class Reference

XIL Model-Access-Port Error Stub.

Inheritance diagram for ttk_tools.dspace._xil_api_offline_stub.MAPortError:



7.282.1 Detailed Description

XIL Model-Access-Port Error Stub.

Non-stubbed hierarchy:

Exception -> (.NET) System.Exception -> TestbenchPortException -> MAPortError

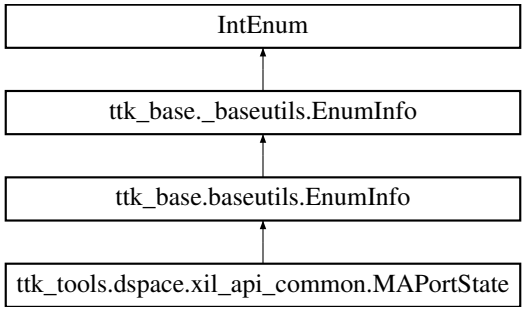
The documentation for this class was generated from the following file:

- ttk_tools/dspace/_xil_api_offline_stub.py (v2.1.2)

7.283 ttk_tools.dspace.xil_api_common.MAPortState Class Reference

Model Access Port States.

Inheritance diagram for ttk_tools.dspace.xil_api_common.MAPortState:



Static Public Attributes

- int eSIMULATION_STOPPED = 0x0000
- int eSIMULATION_RUNNING = 0x0001
- int eDISCONNECTED = 0x0002
- int eSIMULATION_PAUSED = 0x0003

7.283.1 Detailed Description

Model Access Port States.

See ASAM.XIL.Interfaces.Testbench.MAPort.Enum.MAPortState

7.283.2 Member Data Documentation

7.283.2.1 eDISCONNECTED

```
int ttk_tools.dspace.xil_api_common.MAPortState.eDISCONNECTED = 0x0002  [static]
```

7.283.2.2 eSIMULATION_PAUSED

```
int ttk_tools.dspace.xil_api_common.MAPortState.eSIMULATION_PAUSED = 0x0003  [static]
```

7.283.2.3 eSIMULATION_RUNNING

```
int ttk_tools.dspace.xil_api_common.MAPortState.eSIMULATION_RUNNING = 0x0001  [static]
```

7.283.2.4 eSIMULATION_STOPPED

```
int ttk_tools.dspace.xil_api_common.MAPortState.eSIMULATION_STOPPED = 0x0000  [static]
```

The documentation for this class was generated from the following file:

- ttk_tools/dspace/xil_api_common.py (v1.5.0)

7.284 ttk_tools.vector.mdf4_reader.Mdf4Reader Class Reference

Reader for MDF4 files; read data will be stored to member vars.

Public Member Functions

- __init__ (self, mdf_file, verbosity=20)
- getData (self)

  Get a DAQ data structure containing all decoded signals.
- showFileSummary (self, file_path=None)

  Show a summary of data groups / channel groups / channels present in the current MDF4-file.

Public Attributes

- verbosity
- idblock

  "The IDBLOCK always begins at file position 0 and has a constant length of 64 Bytes.
- mdf_version

  version of MDF file as a tuple
- hdblock

  First block following the ID block has to be the (unique) header (HD) block.

Protected Member Functions

- _decodeData (self)

  Decode data from each data group's payload data and sort it into the respective channels (to the <channel>.data member).
- _checkChannelNames (self)

  only for testing consistency of supplied sample data...
- _showFileHistory (self)

  Show File History Block entries.
- _showDataStructure (self)

  Show summary of parsed data structure.
- _showDataChannelSummary (self)

  Show summary of decoded data per data group, channel group and channel.

7.284.1 Detailed Description

Reader for MDF4 files; read data will be stored to member vars.

7.284.2 Constructor & Destructor Documentation

7.284.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.Mdf4Reader.__init__ (
              self,
              mdf_file,
              verbosity = 20 )
```

Parameters

mdf_file	file handle to .mf4 file (opened for reading, binary)
verbosity	verbosity of log output. 0: (mostly) silent larger number increase verbosity

7.284.3 Member Function Documentation

7.284.3.1 _checkChannelNames()

```
ttk_tools.vector.mdf4_reader.Mdf4Reader._checkChannelNames (
              self )  [protected]
```

only for testing consistency of supplied sample data...

7.284.3.2 _decodeData()

```
ttk_tools.vector.mdf4_reader.Mdf4Reader._decodeData (
              self )  [protected]
```

Decode data from each data group's payload data and sort it into the respective channels (to the <channel>.data member).

7.284.3.3 _showDataChannelSummary()

```
ttk_tools.vector.mdf4_reader.Mdf4Reader._showDataChannelSummary (
              self )  [protected]
```

Show summary of decoded data per data group, channel group and channel.

7.284.3.4 _showDataStructure()

```
ttk_tools.vector.mdf4_reader.Mdf4Reader._showDataStructure (
              self )  [protected]
```

Show summary of parsed data structure.

7.284.3.5 _showFileHistory()

```
ttk_tools.vector.mdf4_reader.Mdf4Reader._showFileHistory (
              self )  [protected]
```

Show File History Block entries.

7.284.3.6 getData()

```
ttk_tools.vector.mdf4_reader.Mdf4Reader.getData (
              self )
```

Get a DAQ data structure containing all decoded signals.

Example: DAQ data:

```
# Basic DAQ data:
daq_data = {
    <signal name>: {
        "data": [...],  # data values
        "time": [...]   # time values
    }
}

# For time-based recordings, the time-axis values will
# be stored in entry "time", but other sources exist:
#   * "time"     for time-based measurements
#   * "angle"    for angle-based measurements
#   * "distance" for distance-based measurements
# (and "other" for anything else)

# additional elements will be added from MDF meta data. e.g.

daq_data = {
    "signal name": {
        "data":         [...],          # list of recorded data values
        "data_comment": "",             # optional data comment
        "data_idx":     "0.0.2",        # data channel index in MDF
        "data_name":    "signal name",  # data name/identifier
        "data_unit":    "",             # unit of signal, if supplied
        "group":        "1ms",          # data group name
        "time":         [...],          # time axis values (for time-synchronized signals)
        "time_comment": "",             # optional sync axis comment
        "time_name":    "1ms",          # name/identifier of time axis
        "time_unit":    "",             # unit of sync axis, if supplied
    },
}
```

Note

If a signal of the same name should appear multiple times in the MDF file (e.g. same name in different groups), further occurrences will be added with a prepended (axis) data group name: "foo", "20ms.foo", "100ms.foo" etc.

Returns

a DAQ data dict with (typically) time and data value lists for each recorded signal/variable. See Example/Info for more details.

7.284.3.7 showFileSummary()

```
ttk_tools.vector.mdf4_reader.Mdf4Reader.showFileSummary (
              self,
              file_path = None )
```

Show a summary of data groups / channel groups / channels present in the current MDF4-file.

7.284.4 Member Data Documentation

7.284.4.1 hdblock

`ttk_tools.vector.mdf4_reader.Mdf4Reader.hdblock`

First block following the ID block has to be the (unique) header (HD) block.

HdBlock is the document's root block, all linked blocks will be resolved by their parent blocks (as needed)

7.284.4.2 idblock

`ttk_tools.vector.mdf4_reader.Mdf4Reader.idblock`

"The IDBLOCK always begins at file position 0 and has a constant length of 64 Bytes.

It contains information to identify the file."

7.284.4.3 mdf_version

`ttk_tools.vector.mdf4_reader.Mdf4Reader.mdf_version`

version of MDF file as a tuple

7.284.4.4 verbosity

`ttk_tools.vector.mdf4_reader.Mdf4Reader.verbosity`

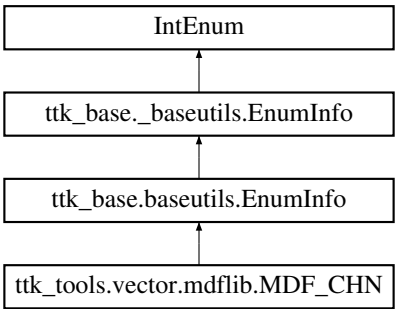
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.285 ttk_tools.vector.mdflib.MDF_CHN Class Reference

MDF Channel Types (`MDF_CHN_*`-constants)

Inheritance diagram for ttk_tools.vector.mdflib.MDF_CHN:



Static Public Attributes

- int VALUE = 0
- int TIME = 1

7.285.1 Detailed Description

MDF Channel Types (MDF_CHN_*-constants)

7.285.2 Member Data Documentation

7.285.2.1 TIME

```
int ttk_tools.vector.mdflib.MDF_CHN.TIME = 1  [static]
```

7.285.2.2 VALUE

```
int ttk_tools.vector.mdflib.MDF_CHN.VALUE = 0  [static]
```

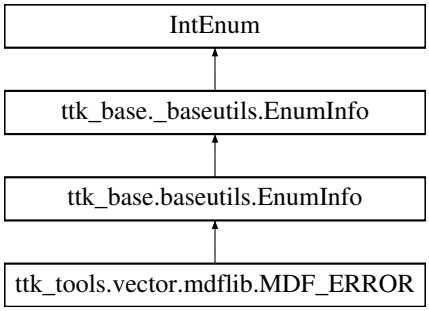
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdflib.py (v3.0.1)

7.286 ttk_tools.vector.mdflib.MDF_ERROR Class Reference

MDF Error Values (E_MDF_*-constants)

Inheritance diagram for ttk_tools.vector.mdflib.MDF_ERROR:



Static Public Attributes

- int OK = 0
- int FILE_NOTFND = 1000

  file not found

- int GROUP_NOTFND = 1001

  group not found
- int CHANNEL_NOTFND = 1002

  channel not found
- int NO_TIME = 1003

  time channel inexistent
- int FORM_NOTFND = 1004

  conversion rule not found
- int FORM_UNKNOWN = 1005

  conversion formula unknown
- int TRIGGER_NOTFND = 1006

  trigger not found
- int TRIGGER_UNKNOWN = 1007

  trigger not known
- int TIMECHANNEL_NOTFND = 1008

  time channel not found
- int MODULE_NOTFND = 1009

  module description not found
- int UNSORTED = 1010

  unsorted
- int FORM_NOT_TAB = 1011

  conversion formula non-tabular
- int NOT_INITIALIZED = -1000

  library not initialized
- int BAD_FILEHANDLE = -1001

  invalid file file_handle
- int BAD_GROUPHANDLE = -1002

  invalid group file_handle
- int BAD_CHANNELHANDLE = -1003

  invalid channel file_handle
- int BAD_OPENMODE = -1004

  invalid open mode
- int BAD_SEEKMODE = -1005

  invalid positioning mode
- int BAD_CHANNELTYPE = -1006

  invalid channel type
- int BAD_CHANNELVALNBIT = -1007

  invalid number of data bits
- int BAD_CHANNELVALTYPE = -1008

  invalid value type
- int GROUP_CREATE = -1009

  group cannot be created

7.286.1 Detailed Description

MDF Error Values (E_MDF_*-constants)

7.286.2 Member Data Documentation

7.286.2.1 BAD_CHANNELHANDLE

```
int ttk_tools.vector.mdflib.MDF_ERROR.BAD_CHANNELHANDLE = -1003  [static]
```

invalid channel file_handle

7.286.2.2 BAD_CHANNELTYPE

```
int ttk_tools.vector.mdflib.MDF_ERROR.BAD_CHANNELTYPE = -1006  [static]
```

invalid channel type

7.286.2.3 BAD_CHANNELVALNBIT

```
int ttk_tools.vector.mdflib.MDF_ERROR.BAD_CHANNELVALNBIT = -1007  [static]
```

invalid number of data bits

7.286.2.4 BAD_CHANNELVALTYPE

```
int ttk_tools.vector.mdflib.MDF_ERROR.BAD_CHANNELVALTYPE = -1008  [static]
```

invalid value type

7.286.2.5 BAD_FILEHANDLE

```
int ttk_tools.vector.mdflib.MDF_ERROR.BAD_FILEHANDLE = -1001  [static]
```

invalid file file_handle

7.286.2.6 BAD_GROUPHANDLE

```
int ttk_tools.vector.mdflib.MDF_ERROR.BAD_GROUPHANDLE = -1002  [static]
```

invalid group file_handle

7.286.2.7 BAD_OPENMODE

```
int ttk_tools.vector.mdflib.MDF_ERROR.BAD_OPENMODE = -1004  [static]
```

invalid open mode

7.286.2.8 BAD_SEEKMODE

```
int ttk_tools.vector.mdflib.MDF_ERROR.BAD_SEEKMODE = -1005  [static]
```

invalid positioning mode

7.286.2.9 CHANNEL_NOTFND

```
int ttk_tools.vector.mdflib.MDF_ERROR.CHANNEL_NOTFND = 1002  [static]
```

channel not found

7.286.2.10 FILE_NOTFND

```
int ttk_tools.vector.mdflib.MDF_ERROR.FILE_NOTFND = 1000  [static]
```

file not found

7.286.2.11 FORM_NOT_TAB

```
int ttk_tools.vector.mdflib.MDF_ERROR.FORM_NOT_TAB = 1011  [static]
```

conversion formula non-tabular

7.286.2.12 FORM_NOTFND

```
int ttk_tools.vector.mdflib.MDF_ERROR.FORM_NOTFND = 1004  [static]
```

conversion rule not found

7.286.2.13 FORM_UNKNOWN

```
int ttk_tools.vector.mdflib.MDF_ERROR.FORM_UNKNOWN = 1005  [static]
```

conversion formula unknown

7.286.2.14 GROUP_CREATE

```
int ttk_tools.vector.mdflib.MDF_ERROR.GROUP_CREATE = -1009  [static]
```

group cannot be created

7.286.2.15 GROUP_NOTFND

```
int ttk_tools.vector.mdflib.MDF_ERROR.GROUP_NOTFND = 1001  [static]
```

group not found

7.286.2.16 MODULE_NOTFND

```
int ttk_tools.vector.mdflib.MDF_ERROR.MODULE_NOTFND = 1009  [static]
```

module description not found

7.286.2.17 NO_TIME

```
int ttk_tools.vector.mdflib.MDF_ERROR.NO_TIME = 1003  [static]
```

time channel inexistent

7.286.2.18 NOT_INITIALIZED

```
int ttk_tools.vector.mdflib.MDF_ERROR.NOT_INITIALIZED = -1000  [static]
```

library not initialized

7.286.2.19 OK

```
int ttk_tools.vector.mdflib.MDF_ERROR.OK = 0  [static]
```

7.286.2.20 TIMECHANNEL_NOTFND

```
int ttk_tools.vector.mdflib.MDF_ERROR.TIMECHANNEL_NOTFND = 1008  [static]
```

time channel not found

7.286.2.21 TRIGGER_NOTFND

```
int ttk_tools.vector.mdflib.MDF_ERROR.TRIGGER_NOTFND = 1006  [static]
```

trigger not found

7.286.2.22 TRIGGER_UNKNOWN

```
int ttk_tools.vector.mdflib.MDF_ERROR.TRIGGER_UNKNOWN = 1007  [static]
```

trigger not known

7.286.2.23 UNSORTED

```
int ttk_tools.vector.mdflib.MDF_ERROR.UNSORTED = 1010  [static]
```

unsorted

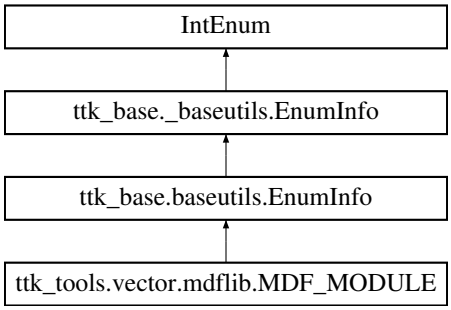
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdflib.py (v3.0.1)

7.287 ttk_tools.vector.mdflib.MDF_MODULE Class Reference

MDF Module/Block Types (`MDF_MODULE_*`-constants)

Inheritance diagram for ttk_tools.vector.mdflib.MDF_MODULE:



Static Public Attributes

- int NON = 0
- int VNA = 16
- int VNV = 17
- int AIM = 1
- int VA1 = 14
- int DIM = 2
- int ETK = 7
- int DM2 = 9
- int VP1 = 12
- int VP2 = 13
- int SSC = 18
- int CAN = 3
- int VNC = 15
- int VCN = 19
- int ACN = 6
- int V24 = 4
- int LA1 = 5
- int LA2 = 8
- int THS = 10
- int ADS = 11

7.287.1 Detailed Description

MDF Module/Block Types (MDF_MODULE_*-constants)

7.287.2 Member Data Documentation

7.287.2.1 ACN

```
int ttk_tools.vector.mdflib.MDF_MODULE.ACN = 6  [static]
```

7.287.2.2 ADS

```
int ttk_tools.vector.mdflib.MDF_MODULE.ADS = 11  [static]
```

7.287.2.3 AIM

```
int ttk_tools.vector.mdflib.MDF_MODULE.AIM = 1  [static]
```

7.287.2.4 CAN

```
int ttk_tools.vector.mdflib.MDF_MODULE.CAN = 3  [static]
```

7.287.2.5 DIM

```
int ttk_tools.vector.mdflib.MDF_MODULE.DIM = 2  [static]
```

7.287.2.6 DM2

```
int ttk_tools.vector.mdflib.MDF_MODULE.DM2 = 9  [static]
```

7.287.2.7 ETK

```
int ttk_tools.vector.mdflib.MDF_MODULE.ETK = 7  [static]
```

7.287.2.8 LA1

```
int ttk_tools.vector.mdflib.MDF_MODULE.LA1 = 5  [static]
```

7.287.2.9 LA2

```
int ttk_tools.vector.mdflib.MDF_MODULE.LA2 = 8  [static]
```

7.287.2.10 NON

```
int ttk_tools.vector.mdflib.MDF_MODULE.NON = 0  [static]
```

7.287.2.11 SSC

```
int ttk_tools.vector.mdflib.MDF_MODULE.SSC = 18  [static]
```

7.287.2.12 THS

```
int ttk_tools.vector.mdflib.MDF_MODULE.THS = 10  [static]
```

7.287.2.13 V24

```
int ttk_tools.vector.mdflib.MDF_MODULE.V24 = 4  [static]
```

7.287.2.14 VA1

```
int ttk_tools.vector.mdflib.MDF_MODULE.VA1 = 14  [static]
```

7.287.2.15 VCN

```
int ttk_tools.vector.mdflib.MDF_MODULE.VCN = 19  [static]
```

7.287.2.16 VNA

```
int ttk_tools.vector.mdflib.MDF_MODULE.VNA = 16  [static]
```

7.287.2.17 VNC

```
int ttk_tools.vector.mdflib.MDF_MODULE.VNC = 15  [static]
```

7.287.2.18 VNV

```
int ttk_tools.vector.mdflib.MDF_MODULE.VNV = 17  [static]
```

7.287.2.19 VP1

```
int ttk_tools.vector.mdflib.MDF_MODULE.VP1 = 12  [static]
```

7.287.2.20 VP2

```
int ttk_tools.vector.mdflib.MDF_MODULE.VP2 = 13  [static]
```

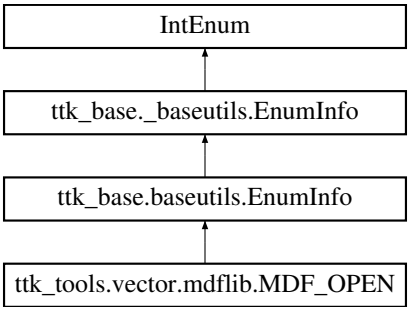
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdflib.py (v3.0.1)

7.288 ttk_tools.vector.mdflib.MDF_OPEN Class Reference

Values for `MDF_OPEN_*`-constants.

Inheritance diagram for ttk_tools.vector.mdflib.MDF_OPEN:



Static Public Attributes

- int READ = 0
- int MODIFY = 1
- int CREATE = 2

7.288.1 Detailed Description

Values for `MDF_OPEN_*`-constants.

7.288.2 Member Data Documentation

7.288.2.1 CREATE

```
int ttk_tools.vector.mdflib.MDF_OPEN.CREATE = 2  [static]
```

7.288.2.2 MODIFY

```
int ttk_tools.vector.mdflib.MDF_OPEN.MODIFY = 1  [static]
```

7.288.2.3 READ

```
int ttk_tools.vector.mdflib.MDF_OPEN.READ = 0  [static]
```

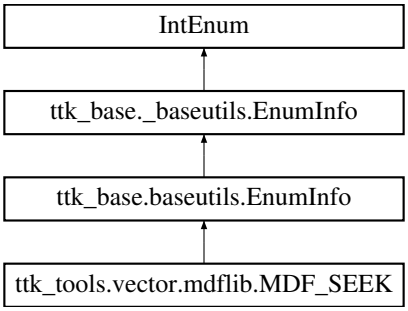
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdflib.py (v3.0.1)

7.289 ttk_tools.vector.mdflib.MDF_SEEK Class Reference

Values for `MDF_SEEK_*`-constants.

Inheritance diagram for ttk_tools.vector.mdflib.MDF_SEEK:



Static Public Attributes

- int ABSOLUTE = 0x00
- int RELATIVE = 0x10
- int NEAREST = 0x00
- int EQ = 0x01
- int LT = 0x02
- int LE = 0x03
- int GT = 0x04
- int GE = 0x05

7.289.1 Detailed Description

Values for `MDF_SEEK_*`-constants.

7.289.2 Member Data Documentation

7.289.2.1 ABSOLUTE

```
int ttk_tools.vector.mdflib.MDF_SEEK.ABSOLUTE = 0x00  [static]
```

7.289.2.2 EQ

```
int ttk_tools.vector.mdflib.MDF_SEEK.EQ = 0x01  [static]
```

7.289.2.3 GE

```
int ttk_tools.vector.mdflib.MDF_SEEK.GE = 0x05  [static]
```

7.289.2.4 GT

```
int ttk_tools.vector.mdflib.MDF_SEEK.GT = 0x04  [static]
```

7.289.2.5 LE

```
int ttk_tools.vector.mdflib.MDF_SEEK.LE = 0x03  [static]
```

7.289.2.6 LT

```
int ttk_tools.vector.mdflib.MDF_SEEK.LT = 0x02  [static]
```

7.289.2.7 NEAREST

```
int ttk_tools.vector.mdflib.MDF_SEEK.NEAREST = 0x00  [static]
```

7.289.2.8 RELATIVE

```
int ttk_tools.vector.mdflib.MDF_SEEK.RELATIVE = 0x10  [static]
```

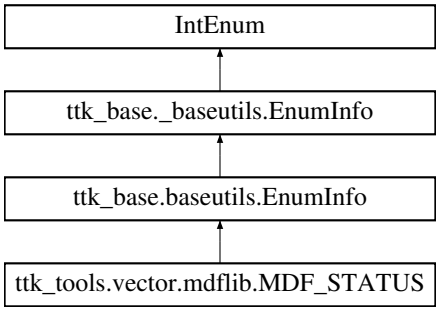
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdflib.py (v3.0.1)

7.290 ttk_tools.vector.mdflib.MDF_STATUS Class Reference

MDF status values (`MDF_*`-constants).

Inheritance diagram for ttk_tools.vector.mdflib.MDF_STATUS:



Static Public Attributes

- int OK = 0
  ok, no error
- int EOG = 1
  group end, no further records
- int RECORD_NOTFND = 2
  record not found
- int NAME_NOTFND = 3
  channel name not found
- int CHANNEL_NOTFND = 4
  channel not found
- int ALREADY_SORTED = 5
  file contents already sorted
- int RANGE_UNKNOWN = 6
  range unknown

7.290.1 Detailed Description

MDF status values (`MDF_*`-constants).

7.290.2 Member Data Documentation

7.290.2.1 ALREADY_SORTED

```
int ttk_tools.vector.mdflib.MDF_STATUS.ALREADY_SORTED = 5  [static]
```

file contents already sorted

7.290.2.2 CHANNEL_NOTFND

```
int ttk_tools.vector.mdflib.MDF_STATUS.CHANNEL_NOTFND = 4  [static]
```

channel not found

7.290.2.3 EOG

```
int ttk_tools.vector.mdflib.MDF_STATUS.EOG = 1  [static]
```

group end, no further records

7.290.2.4 NAME_NOTFND

```
int ttk_tools.vector.mdflib.MDF_STATUS.NAME_NOTFND = 3  [static]
```

channel name not found

7.290.2.5 OK

```
int ttk_tools.vector.mdflib.MDF_STATUS.OK = 0  [static]
```

ok, no error

7.290.2.6 RANGE_UNKNOWN

```
int ttk_tools.vector.mdflib.MDF_STATUS.RANGE_UNKNOWN = 6  [static]
```

range unknown

7.290.2.7 RECORD_NOTFND

```
int ttk_tools.vector.mdflib.MDF_STATUS.RECORD_NOTFND = 2  [static]
```

record not found

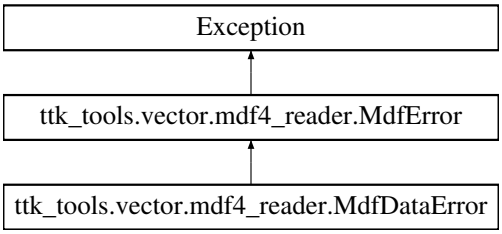
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdflib.py (v3.0.1)

7.291 ttk_tools.vector.mdf4_reader.MdfDataError Class Reference

Feature not (yet) supported)

Inheritance diagram for ttk_tools.vector.mdf4_reader.MdfDataError:



7.291.1 Detailed Description

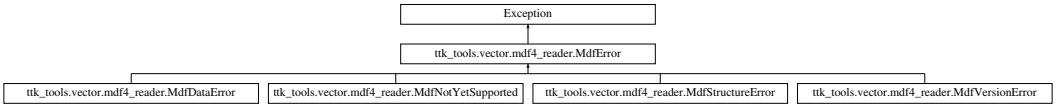
Feature not (yet) supported)

The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.292 ttk_tools.vector.mdf4_reader.MdfError Class Reference

Inheritance diagram for ttk_tools.vector.mdf4_reader.MdfError:



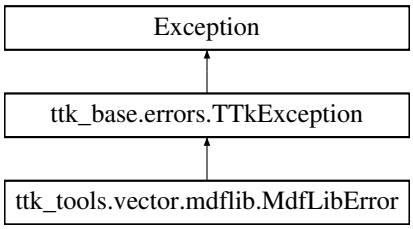
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.293 ttk_tools.vector.mdflib.MdfLibError Class Reference

Base Error for MdfLib access.

Inheritance diagram for ttk_tools.vector.mdflib.MdfLibError:



Public Member Functions

- __init__ (self, msg='', error_code=None, log_msg=False)

Public Attributes

- error_code

  API error code, if supplied/available.

7.293.1 Detailed Description

Base Error for MdfLib access.

Parameters

msg	error message
error_code	API error code (if available)
log_msg	True: log error message to stdout

7.293.2 Constructor & Destructor Documentation

7.293.2.1 __init__()

```
ttk_tools.vector.mdflib.MdfLibError.__init__ (
              self,
              msg = '',
              error_code = None,
              log_msg = False )
```

7.293.3 Member Data Documentation

7.293.3.1 error_code

```
ttk_tools.vector.mdflib.MdfLibError.error_code
```

API error code, if supplied/available.

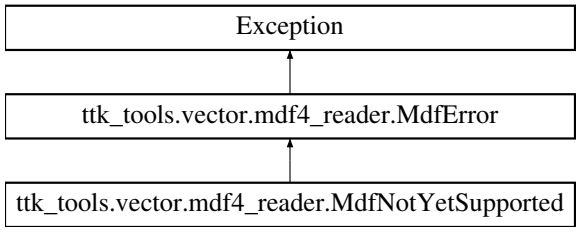
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdflib.py (v3.0.1)

7.294 ttk_tools.vector.mdf4_reader.MdfNotYetSupported Class Reference

Meta class for EnumInfo; adds a mechanism for custom value lookups.

Inheritance diagram for ttk_tools.vector.mdf4_reader.MdfNotYetSupported:



7.294.1 Detailed Description

Meta class for EnumInfo; adds a mechanism for custom value lookups.

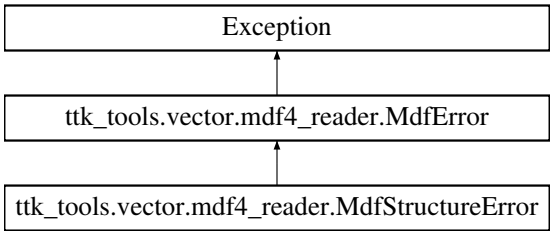
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.295 ttk_tools.vector.mdf4_reader.MdfStructureError Class Reference

Data or data types are inconsistent.

Inheritance diagram for ttk_tools.vector.mdf4_reader.MdfStructureError:



7.295.1 Detailed Description

Data or data types are inconsistent.

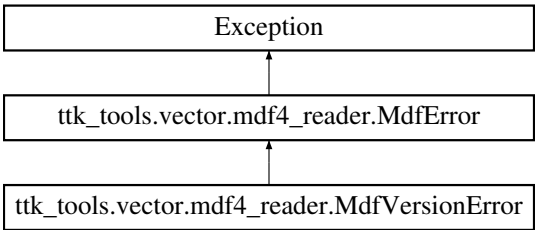
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.296 ttk_tools.vector.mdf4_reader.MdfVersionError Class Reference

Detected MDF version is unsupported.

Inheritance diagram for ttk_tools.vector.mdf4_reader.MdfVersionError:



7.296.1 Detailed Description

Detected MDF version is unsupported.

A sanity check for MDF structure failed

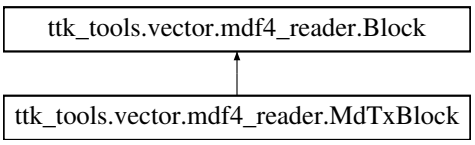
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.297 ttk_tools.vector.mdf4_reader.MdTxBlock Class Reference

Text block with optional meta data.

Inheritance diagram for ttk_tools.vector.mdf4_reader.MdTxBlock:



Public Member Functions

- __init__ (self, file_handle, pos=None)
  Read a TXBLOCK or MDBLOCK from file_handle, starting at the current or supplied position.
- __str__ (self)
- __repr__ (self)

Public Attributes

- xml
- dom
- text

7.297.1 Detailed Description

Text block with optional meta data.

Handled Blocks:

- Meta Data Block.
  The MDBLOCK contains information encoded as XML. For example, this can be comments for the measured data file, file history information or the identification of a channel. This information is ruled by the parent block and follows specific XML schemas definitions.

- Text Block.
  The TXBLOCK is very similar to a MDBLOCK but only contains a plain string encoded in UTF-8. The text length results from the block size.

7.297.2 Constructor & Destructor Documentation

7.297.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.MdTxBlock.__init__ (
              self,
              file_handle,
              pos = None )
```

Read a TXBLOCK or MDBLOCK from file_handle, starting at the current or supplied position.

self.text will contain either:

- TXBLOCK's text data or
- MDBLOCK's TX-element text

self.xml will contain

- None for TXBLOCKs and
- the raw XML string data for MDBLOCKs

self.dom will contain

- None for TXBLOCKs and
- XML DOM (as parsed by minidom) for MDBLOCKs

The DOM can be used to extract further meta data as needed.

Note

File pointer position will be modified (moved to end of block)

Reimplemented from ttk_tools.vector.mdf4_reader.Block.

7.297.3 Member Function Documentation

7.297.3.1 __repr__()

```
ttk_tools.vector.mdf4_reader.MdTxBlock.__repr__ (
              self )
```

7.297.3.2 __str__()

```
ttk_tools.vector.mdf4_reader.MdTxBlock.__str__ (
              self )
```

7.297.4 Member Data Documentation

7.297.4.1 dom

`ttk_tools.vector.mdf4_reader.MdTxBlock.dom`

7.297.4.2 text

`ttk_tools.vector.mdf4_reader.MdTxBlock.text`

7.297.4.3 xml

`ttk_tools.vector.mdf4_reader.MdTxBlock.xml`

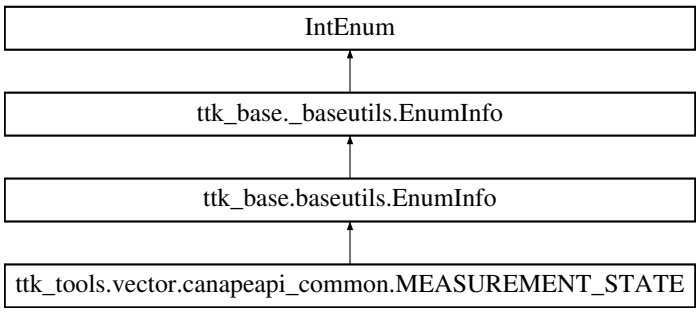
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.298 ttk_tools.vector.canapeapi_common.MEASUREMENT_STATE Class Reference

Measurement state constants (`eT_MEASUREMENT_*`) from enum `tMeasurementState` (new in CANape 9).

Inheritance diagram for ttk_tools.vector.canapeapi_common.MEASUREMENT_STATE:



Static Public Attributes

- int STOPPED = 0

  no measurement active
- int INIT = 1

  measurement started, but measurement thread not yet running
- int STOP_ON_START = 2

  measurement will be stopped via function in prStart
- int EXIT = 3

  measurement stop requested, but not yet stopped
- int THREAD_RUNNING = 4

  measurement started, measurement thread is running
- int RUNNING = 5

  measurement loop is running

7.298.1 Detailed Description

Measurement state constants (`eT_MEASUREMENT_*`) from enum `tMeasurementState` (new in CANape 9).

7.298.2 Member Data Documentation

7.298.2.1 EXIT

```
int ttk_tools.vector.canapeapi_common.MEASUREMENT_STATE.EXIT = 3  [static]
```

measurement stop requested, but not yet stopped

7.298.2.2 INIT

```
int ttk_tools.vector.canapeapi_common.MEASUREMENT_STATE.INIT = 1  [static]
```

measurement started, but measurement thread not yet running

7.298.2.3 RUNNING

```
int ttk_tools.vector.canapeapi_common.MEASUREMENT_STATE.RUNNING = 5  [static]
```

measurement loop is running

7.298.2.4 STOP_ON_START

```
int ttk_tools.vector.canapeapi_common.MEASUREMENT_STATE.STOP_ON_START = 2  [static]
```

measurement will be stopped via function in prStart

7.298.2.5 STOPPED

```
int ttk_tools.vector.canapeapi_common.MEASUREMENT_STATE.STOPPED = 0  [static]
```

no measurement active

7.298.2.6 THREAD_RUNNING

```
int ttk_tools.vector.canapeapi_common.MEASUREMENT_STATE.THREAD_RUNNING = 4  [static]
```

measurement started, measurement thread is running

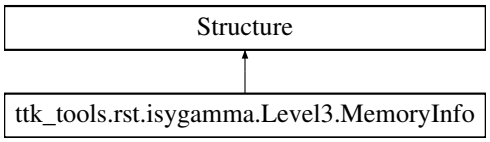
The documentation for this class was generated from the following file:

- ttk_tools/vector/canapeapi_common.py (v3.1.2)

7.299 ttk_tools.rst.isygamma.Level3.MemoryInfo Class Reference

typedef struct gc_MemoryInfo

Inheritance diagram for ttk_tools.rst.isygamma.Level3.MemoryInfo:



Static Protected Attributes

- list _fields_

7.299.1 Detailed Description

typedef struct gc_MemoryInfo

7.299.2 Member Data Documentation

7.299.2.1 _fields_

```
list ttk_tools.rst.isygamma.Level3.MemoryInfo._fields_  [static], [protected]
```

Initial value:

```
=  [
            ("address",         Addresses.Level3),# address of the memory
            ("isPersistent",    c_bool),          # is memory persistent?
            ("size",            c_uint32),        # maximum size the memory can have
            ("permission",      c_uint32),        # access permissions (see gc_PermissionSet)
            ("user",            c_char *  512),   # name of the user who owns the memory
            ("group",           c_char *  512),   # group name of the user who owns the memory
            ("name",            c_char *  512),   # name of memory
            ("modelName",       c_char * 1024),   # name of memory in model
        ]
```

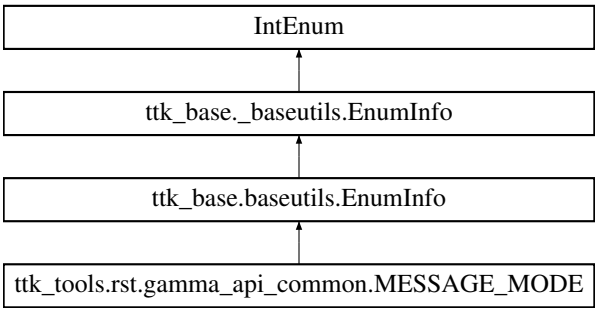
The documentation for this class was generated from the following file:

- ttk_tools/rst/isygamma.py (v1.6)

7.300 ttk_tools.rst.gamma_api_common.MESSAGE_MODE Class Reference

Message modes.

Inheritance diagram for ttk_tools.rst.gamma_api_common.MESSAGE_MODE:



Static Public Attributes

- int CONSOLE = 0
- int FILE = 1
- int SYSLOG = 2
- int CONSOLE_FILE = 3
- int CONSOLE_SYSLOG = 4

7.300.1 Detailed Description

Message modes.

7.300.2 Member Data Documentation

7.300.2.1 CONSOLE

```
int ttk_tools.rst.gamma_api_common.MESSAGE_MODE.CONSOLE = 0  [static]
```

7.300.2.2 CONSOLE_FILE

```
int ttk_tools.rst.gamma_api_common.MESSAGE_MODE.CONSOLE_FILE = 3  [static]
```

7.300.2.3 CONSOLE_SYSLOG

```
int ttk_tools.rst.gamma_api_common.MESSAGE_MODE.CONSOLE_SYSLOG = 4  [static]
```

7.300.2.4 FILE

```
int ttk_tools.rst.gamma_api_common.MESSAGE_MODE.FILE = 1  [static]
```

7.300.2.5 SYSLOG

```
int ttk_tools.rst.gamma_api_common.MESSAGE_MODE.SYSLOG = 2  [static]
```

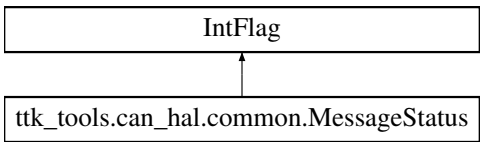
The documentation for this class was generated from the following file:

- ttk_tools/rst/gamma_api_common.py (v1.5)

7.301 ttk_tools.can_hal.common.MessageStatus Class Reference

CAN Message Status (CanHal.h, tCanHalMessageStatus).

Inheritance diagram for ttk_tools.can_hal.common.MessageStatus:



Static Public Attributes

- int NoStatus = 0b0000
- int RxMessage = 0b0001
- int TxMessage = 0b0010
- int RemoteRequest = 0b0100
- int ExtendedID = 0b1000

7.301.1 Detailed Description

CAN Message Status (CanHal.h, tCanHalMessageStatus).

7.301.2 Member Data Documentation

7.301.2.1 ExtendedID

```
int ttk_tools.can_hal.common.MessageStatus.ExtendedID = 0b1000  [static]
```

7.301.2.2 NoStatus

```
int ttk_tools.can_hal.common.MessageStatus.NoStatus = 0b0000  [static]
```

7.301.2.3 RemoteRequest

```
int ttk_tools.can_hal.common.MessageStatus.RemoteRequest = 0b0100  [static]
```

7.301.2.4 RxMessage

```
int ttk_tools.can_hal.common.MessageStatus.RxMessage = 0b0001  [static]
```

7.301.2.5 TxMessage

```
int ttk_tools.can_hal.common.MessageStatus.TxMessage = 0b0010  [static]
```

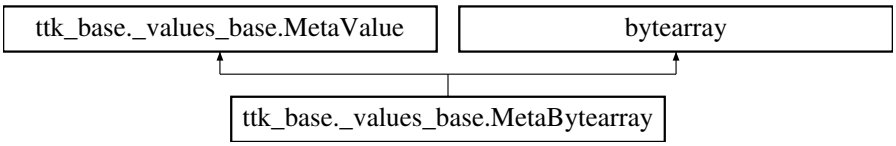
The documentation for this class was generated from the following file:

- ttk_tools/can_hal/common.py (v1.2)

7.302 ttk_base._values_base.MetaBytearray Class Reference

Bytearray values with additional meta info.

Inheritance diagram for ttk_base._values_base.MetaBytearray:



Public Member Functions

- __repr__ (self)

7.302.1 Detailed Description

Bytearray values with additional meta info.

7.302.2 Member Function Documentation

7.302.2.1 __repr__()

```
ttk_base._values_base.MetaBytearray.__repr__ (
              self )
```

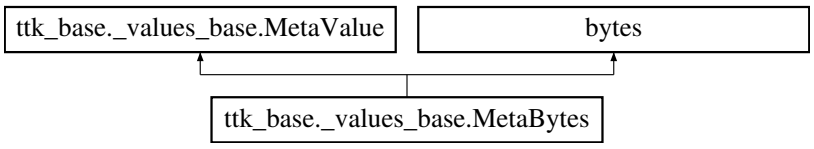
The documentation for this class was generated from the following file:

- ttk_base/_values_base.py (v2.6.0)

7.303 ttk_base._values_base.MetaBytes Class Reference

Raw Byte string values with additional meta info.

Inheritance diagram for ttk_base._values_base.MetaBytes:



Public Member Functions

- __new__ (cls, value, *args, **kwargs)

7.303.1 Detailed Description

Raw Byte string values with additional meta info.

7.303.2 Member Function Documentation

7.303.2.1 __new__()

```
ttk_base._values_base.MetaBytes.__new__ (
              cls,
              value,
             * args,
             ** kwargs )
```

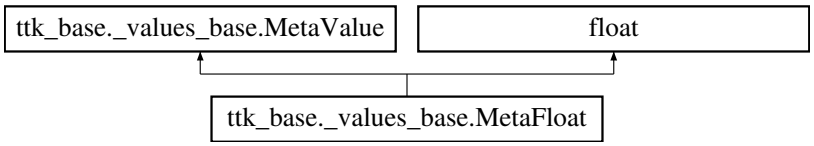
The documentation for this class was generated from the following file:

- ttk_base/_values_base.py (v2.6.0)

7.304 ttk_base._values_base.MetaFloat Class Reference

Float values with additional meta info.

Inheritance diagram for ttk_base._values_base.MetaFloat:



Public Member Functions

- __new__ (cls, value, *args, **kwargs)

7.304.1 Detailed Description

Float values with additional meta info.

7.304.2 Member Function Documentation

7.304.2.1 __new__()

```
ttk_base._values_base.MetaFloat.__new__ (
              cls,
              value,
             * args,
             ** kwargs )
```

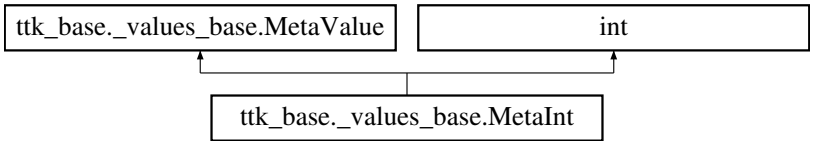
The documentation for this class was generated from the following file:

- ttk_base/_values_base.py (v2.6.0)

7.305 ttk_base._values_base.MetaInt Class Reference

Integer values with additional meta info.

Inheritance diagram for ttk_base._values_base.MetaInt:



Public Member Functions

- __new__ (cls, value, ∗args, ∗∗kwargs)

7.305.1 Detailed Description

Integer values with additional meta info.

7.305.2 Member Function Documentation

7.305.2.1 __new__()

```
ttk_base._values_base.MetaInt.__new__ (
              cls,
              value,
             * args,
             ** kwargs )
```

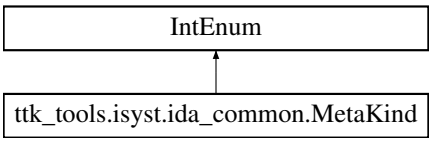
The documentation for this class was generated from the following file:

- ttk_base/_values_base.py (v2.6.0)

7.306 ttk_tools.isyst.ida_common.MetaKind Class Reference

MetaKind as defined in protobuf message MetaId.

Inheritance diagram for ttk_tools.isyst.ida_common.MetaKind:



Public Member Functions

- info (self)

Static Public Attributes

- int name = 0
- int size = 1
- int data_type = 2
- int unit = 3
- int display_format = 4
- int min_limit = 5
- int max_limit = 6
- int default_value = 7
- int description = 8
- int display_name = 9
- int system_name = 100
- int instance_key = 101
- int var_count = 102
- int version = 103

7.306.1 Detailed Description

MetaKind as defined in protobuf message MetaId.

7.306.2 Member Function Documentation

7.306.2.1 info()

```
ttk_tools.isyst.ida_common.MetaKind.info (
              self )
```

7.306.3 Member Data Documentation

7.306.3.1 data_type

```
int ttk_tools.isyst.ida_common.MetaKind.data_type = 2  [static]
```

7.306.3.2 default_value

```
int ttk_tools.isyst.ida_common.MetaKind.default_value = 7  [static]
```

7.306.3.3 description

```
int ttk_tools.isyst.ida_common.MetaKind.description = 8  [static]
```

7.306.3.4 display_format

```
int ttk_tools.isyst.ida_common.MetaKind.display_format = 4  [static]
```

7.306.3.5 display_name

```
int ttk_tools.isyst.ida_common.MetaKind.display_name = 9  [static]
```

7.306.3.6 instance_key

```
int ttk_tools.isyst.ida_common.MetaKind.instance_key = 101  [static]
```

7.306.3.7 max_limit

```
int ttk_tools.isyst.ida_common.MetaKind.max_limit = 6  [static]
```

7.306.3.8 min_limit

```
int ttk_tools.isyst.ida_common.MetaKind.min_limit = 5  [static]
```

7.306.3.9 name

```
int ttk_tools.isyst.ida_common.MetaKind.name = 0  [static]
```

7.306.3.10 size

```
int ttk_tools.isyst.ida_common.MetaKind.size = 1  [static]
```

7.306.3.11 system_name

```
int ttk_tools.isyst.ida_common.MetaKind.system_name = 100  [static]
```

7.306.3.12 unit

```
int ttk_tools.isyst.ida_common.MetaKind.unit = 3  [static]
```

7.306.3.13 var_count

```
int ttk_tools.isyst.ida_common.MetaKind.var_count = 102  [static]
```

7.306.3.14 version

```
int ttk_tools.isyst.ida_common.MetaKind.version = 103  [static]
```

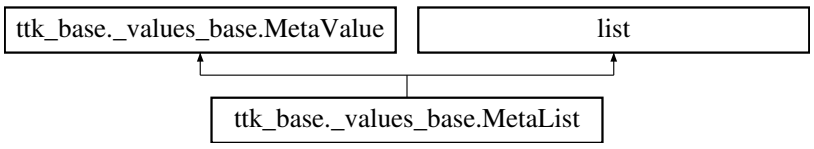
The documentation for this class was generated from the following file:

- ttk_tools/isyst/ida_common.py (v1.2.0)

7.307 ttk_base._values_base.MetaList Class Reference

A List with additional meta info.

Inheritance diagram for ttk_base._values_base.MetaList:



7.307.1 Detailed Description

A List with additional meta info.

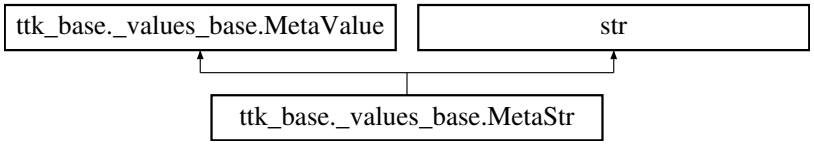
The documentation for this class was generated from the following file:

- ttk_base/_values_base.py (v2.6.0)

7.308 ttk_base._values_base.MetaStr Class Reference

String values with additional meta info.

Inheritance diagram for ttk_base._values_base.MetaStr:



Public Member Functions

- __new__ (cls, value, ∗args, ∗∗kwargs)

7.308.1 Detailed Description

String values with additional meta info.

7.308.2 Member Function Documentation

7.308.2.1 __new__()

```
ttk_base._values_base.MetaStr.__new__ (
              cls,
              value,
             * args,
             ** kwargs )
```

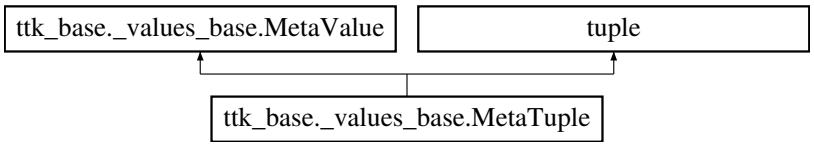
The documentation for this class was generated from the following file:

- ttk_base/_values_base.py (v2.6.0)

7.309 ttk_base._values_base.MetaTuple Class Reference

A tuple with additional meta info.

Inheritance diagram for ttk_base._values_base.MetaTuple:



Public Member Functions

- __new__ (cls, iterable, ∗args, ∗∗kwargs)

7.309.1 Detailed Description

A tuple with additional meta info.

7.309.2 Member Function Documentation

7.309.2.1 __new__()

```
ttk_base._values_base.MetaTuple.__new__ (
              cls,
              iterable,
             * args,
             ** kwargs )
```

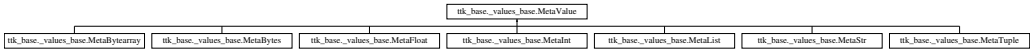
The documentation for this class was generated from the following file:

- ttk_base/_values_base.py (v2.6.0)

7.310 ttk_base._values_base.MetaValue Class Reference

Base class for meta attributes.

Inheritance diagram for ttk_base._values_base.MetaValue:



Public Member Functions

- __init__ (self, value, unit="", alias="", descr="", lookup=None, fmt=None)

  Construct a meta-value.
- lookup (self)

  Property `lookup`, an alias to `state_lookup`.
- last_value (self)

  Property `last_value`.
- info (self)

  Get an info string (alias and current value) for this value.
- hasStateLookup (self)

  Return True if a state/value lookup table is defined.
- getStateDescr (self, value=None)

  Get a state description text for the given value from the defined lookup table (if any).
- hasFormattedRepr (self)

  Return True if a formatting for value representation is defined.
- getFormattedValue (self, value=None)

  Get a formatted representation for the given value using the defined format string (see self.fmt).

Public Attributes

- unit
  a physical unit like V, mA or rpm
- alias
  alias/short name of this value
- descr
  description text
- state_lookup
  lookup table value => text
- fmt
  format string for custom value representation

7.310.1 Detailed Description

Base class for meta attributes.

Use this in conjunction with the "meta wrappers" below. Compare / see VarBase interface.

7.310.2 Constructor & Destructor Documentation

7.310.2.1 __init__()

```
ttk_base._values_base.MetaValue.__init__ (
              self,
              value,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              fmt = None )
```

Construct a meta-value.

Parameters

value	base value to wrap and add meta info to
unit	value's unit
alias	an alias name for the wrapped value
descr	description text for value
lookup	lookup table for {value => "state name"} mappings
fmt	format string to use for value representation

7.310.3 Member Function Documentation

7.310.3.1 getFormattedValue()

```
ttk_base._values_base.MetaValue.getFormattedValue (
```

```
                self,
                value = None )
```

Get a formatted representation for the given value using the defined format string (see self.fmt).

An empty (or non-defined) format string will use the value's default string representation.

Parameters

value	value for which to get a string representation If None, the meta-wrapped value will be used instead.

Returns

a formatted string representation.

7.310.3.2 getStateDescr()

```
ttk_base._values_base.MetaValue.getStateDescr (
                self,
                value = None )
```

Get a state description text for the given value from the defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the meta-wrapped value will be used instead.

Returns

the state text if a match was found, otherwise None.

7.310.3.3 hasFormattedRepr()

```
ttk_base._values_base.MetaValue.hasFormattedRepr (
                self )
```

Return True if a formatting for value representation is defined.

This can be used to decide if a custom formatting is available or if another variable's formatting should be applied

7.310.3.4 hasStateLookup()

```
ttk_base._values_base.MetaValue.hasStateLookup (
                self )
```

Return True if a state/value lookup table is defined.

7.310.3.5 info()

```
ttk_base._values_base.MetaValue.info (
              self )
```

Get an info string (alias and current value) for this value.

7.310.3.6 last_value()

```
ttk_base._values_base.MetaValue.last_value (
              self )
```

Property `last_value`.

As a MetaValue has no set() operation, its "last value" should just be its current and only value.

7.310.3.7 lookup()

```
ttk_base._values_base.MetaValue.lookup (
              self )
```

Property `lookup`, an alias to `state_lookup`.

As the supplied lookup values are internally stored as `state_lookup` (which maps value => text) this provides an alias to the values as originally supplied in the `lookup` parameter during **init**

Note

Meta values have no `value_lookup` which would map text => value, See _variables_base.VarBase

Example:

```
»> value = meta(1, lookup={0: "off", 1: "on", 2: "huh?"})
»> value.lookup
{0: 'off', 1: 'on', 2: 'huh?'}
»> value.state_lookup
{0: 'off', 1: 'on', 2: 'huh?'}
```

7.310.4 Member Data Documentation

7.310.4.1 alias

```
ttk_base._values_base.MetaValue.alias
```

alias/short name of this value

7.310.4.2 descr

```
ttk_base._values_base.MetaValue.descr
```

description text

7.310.4.3 fmt

`ttk_base._values_base.MetaValue.fmt`

format string for custom value representation

7.310.4.4 state_lookup

`ttk_base._values_base.MetaValue.state_lookup`

lookup table value => text

7.310.4.5 unit

`ttk_base._values_base.MetaValue.unit`

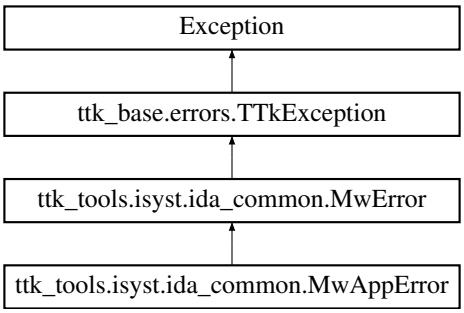
a physical unit like V, mA or rpm

The documentation for this class was generated from the following file:

- ttk_base/_values_base.py (v2.6.0)

7.311 ttk_tools.isyst.ida_common.MwAppError Class Reference

Inheritance diagram for ttk_tools.isyst.ida_common.MwAppError:



Public Member Functions

- __init__ (self, message, status)

Public Attributes

- status
- message

7.311.1 Constructor & Destructor Documentation

7.311.1.1 __init__()

```
ttk_tools.isyst.ida_common.MwAppError.__init__ (
              self,
              message,
              status )
```

Parameters

message	error message
status	MwAppStatus value

7.311.2 Member Data Documentation

7.311.2.1 message

`ttk_tools.isyst.ida_common.MwAppError.message`

7.311.2.2 status

`ttk_tools.isyst.ida_common.MwAppError.status`

The documentation for this class was generated from the following file:

- ttk_tools/isyst/ida_common.py (v1.2.0)

7.312 ttk_tools.isyst.mwapp_client.MwAppRpc Class Reference

Simple wrapper for MwApp RPCs.

Public Member Functions

- __init__ (self, target, channel_options=None)
- close (self)

  Close current gRPC channel.
- getMwVar (self, var_name)

  Factory method for creating MwVar proxy instances.
- getCachedVarId (self, var_name)

  Get Var-ID for a variable's (display) name from cached meta data.
- getValue (self, var_id)

  Get current value of MW-Variable.
- setValue (self, var_id, value, data_type=None)

  Set new value of a MW-Variable.
- updateValue (self, var_id, value, data_type=None)

  Update value of a MW-Variable: set new value, and return the previous value before the change.
- resetValue (self, var_ids=None, reset_all=False, complete_arrays=False)

  Reset one or more variables to their original values.
- getVarId (self, var_name)

  Get Var-ID for a MW-Variable name.
- getMeta (self, var_id=None, meta_id=MetaKind.system_name)

  Get a meta value.
- getGroups (self)

Get configured variable groups (and Var-IDs in those groups).

- bulkGetValue (self, var_ids)

  Get values for multiple variables.
- bulkGetMeta (self, var_meta_ids)

  Get meta data for multiple variables or system meta entries.
- bulkSetValue (self, var_type_values)

  Set values for multiple variables.
- __enter__ (self)

  Start MwAppRpc as managed context.
- __exit__ (self, exc_type, exc_value, tb)

  End context manager.

Public Attributes

- target
- channel
- stub
- system_name
- system_version
- instance_key
- var_meta_cache
- var_id_cache
- last_app_status

Protected Member Functions

- _connectionCheck (self)

  Check if connection is active.

Static Protected Member Functions

- _getOneOfValue (grpc_msg)

  Get used/filled element of OneOf-field "value".
- _setMwRpcValue (mw_rpc_value=None, data_type=MwDataType.UNDEFINED, value=0)

  Set member of MwRpcValue corresponding to `data_type` to the supplied `value`.

7.312.1 Detailed Description

Simple wrapper for MwApp RPCs.

7.312.2 Constructor & Destructor Documentation

7.312.2.1 __init__()

```
ttk_tools.isyst.mwapp_client.MwAppRpc.__init__ (
              self,
              target,
              channel_options = None )
```

Parameters

target	connection target string (address:port), e.g. "127.0.0.1:50051". If port is missing, default port 50051 will be used.
channel_options	gRPC connection options.

7.312.3 Member Function Documentation

7.312.3.1 __enter__()

```
ttk_tools.isyst.mwapp_client.MwAppRpc.__enter__ (
              self )
```

Start MwAppRpc as managed context.

7.312.3.2 __exit__()

```
ttk_tools.isyst.mwapp_client.MwAppRpc.__exit__ (
              self,
              exc_type,
              exc_value,
              tb )
```

End context manager.

7.312.3.3 _connectionCheck()

```
ttk_tools.isyst.mwapp_client.MwAppRpc._connectionCheck (
              self )  [protected]
```

Check if connection is active.

7.312.3.4 _getOneOfValue()

```
ttk_tools.isyst.mwapp_client.MwAppRpc._getOneOfValue (
              grpc_msg )  [static], [protected]
```

Get used/filled element of OneOf-field "value".

Returns

the field's value or None if OneOf is unset (or not present in `grpc_msg`).

7.312.3.5 _setMwRpcValue()

```
ttk_tools.isyst.mwapp_client.MwAppRpc._setMwRpcValue (
              mw_rpc_value = None,
              data_type = MwDataType.UNDEFINED,
              value = 0 )  [static], [protected]
```

Set member of MwRpcValue corresponding to `data_type` to the supplied `value`.

Note

Float values and int types When setting an integer type, float values that are close to integer values (less than 0.0001 difference) will be converted to the nearest integer.
See ttk_base._baseutils._getInteger or
see ttk_checks.checkutils.getInteger for a similar approach.

Parameters

mw_rpc_value	a MwRpcValue instance to set value to • or - None to create a new instance
data_type	MwDataType to use
value	value to set

Raises a ValueError if `value` is does not fit or is not suitable to set for `data_type`.

Returns

a MwRpcValue

7.312.3.6 bulkGetMeta()

```
ttk_tools.isyst.mwapp_client.MwAppRpc.bulkGetMeta (
              self,
              var_meta_ids )
```

Get meta data for multiple variables or system meta entries.

Note

Only used during connection check to retrieve system meta info. The variable meta data cache (_VarMeta↩ Cache) directly uses the RPC stub.

Parameters

var_meta_ids	a list of (var_id, meta_id) tuples. For system meta entries, var_id can be None.

Returns

a list of values, one for each supplied entry Value will be None if access failed (or meta value was undefined)

7.312.3.7 bulkGetValue()

```
ttk_tools.isyst.mwapp_client.MwAppRpc.bulkGetValue (
              self,
              var_ids )
```

Get values for multiple variables.

Parameters

var_ids	list of Var-ID

Returns

a list of values, one for each Var-ID . If a value cannot be accessed, its entry will be set to None.

7.312.3.8 bulkSetValue()

```
ttk_tools.isyst.mwapp_client.MwAppRpc.bulkSetValue (
              self,
              var_type_values )
```

Set values for multiple variables.

Parameters

var_type_values	list of tuples (var_id, data_type, new_value) var_id: int, unsigned data_type: MwDataType of var_id new_value: value suitable for data_type

Returns

a tuple: overall_errors - error count, should be 0 if everything was ok status_values - list of MwAppStatus values, one for each entry

7.312.3.9 close()

```
ttk_tools.isyst.mwapp_client.MwAppRpc.close (
              self )
```

Close current gRPC channel.

7.312.3.10 getCachedVarId()

```
ttk_tools.isyst.mwapp_client.MwAppRpc.getCachedVarId (
              self,
              var_name )
```

Get Var-ID for a variable's (display) name from cached meta data.

Parameters

var_name	display name of MW-Variable, potentially including an array index suffix

Returns

a numeric Var-ID, or None if no matching variable was found.

7.312.3.11 getGroups()

```
ttk_tools.isyst.mwapp_client.MwAppRpc.getGroups (
              self )
```

Get configured variable groups (and Var-IDs in those groups).

For array-Variables, only the head element's Var-ID will be included.

Note

Included var_names will be "display names", so the head element of array variables will show with index [0], e.g. "foo[0]" instead of the base name "foo".

Returns

a dict with <group_id>: { "group_name": "<group_name>", "group_id": <group_id>, "var_ids": [<var_id1>, <var_id2>, ... <var_idn>] "var_names": [<name_id1>, <name_id2>, ... <name_idn>], }

7.312.3.12 getMeta()

```
ttk_tools.isyst.mwapp_client.MwAppRpc.getMeta (
              self,
              var_id = None,
              meta_id = MetaKind.system_name )
```

Get a meta value.

Direct access via RPC. See MetaKind for available meta entries.

Parameters

var_id	Var-ID of variable to get meta data on. Will be ignored for system-level meta entries (can be set to None for this)
meta_id	MetaKind value specifying the meta entry to get.

Returns

a value (numeric or bytes/string, depending on meta id/type) Value may be None if meta entry is undefined. Raises a MwAppError if response status indicates an error.

7.312.3.13 getMwVar()

```
ttk_tools.isyst.mwapp_client.MwAppRpc.getMwVar (
              self,
              var_name )
```

Factory method for creating MwVar proxy instances.

This is mainly intended for convenience, so the variable abstraction does not have to import additional packages.

Parameters

var_name	variable name on mwservice

Returns

an initialized MwVar instance

7.312.3.14 getValue()

```
ttk_tools.isyst.mwapp_client.MwAppRpc.getValue (
              self,
              var_id )
```

Get current value of MW-Variable.

Returns

a value (numeric or bytes/string, depending on MW-Var data type)

7.312.3.15 getVarId()

```
ttk_tools.isyst.mwapp_client.MwAppRpc.getVarId (
              self,
              var_name )
```

Get Var-ID for a MW-Variable name.

Direct access via RPC.

Returns

a numeric value or None if no ID for the given name exists.

7.312.3.16 resetValue()

```
ttk_tools.isyst.mwapp_client.MwAppRpc.resetValue (
              self,
              var_ids = None,
              reset_all = False,
              complete_arrays = False )
```

Reset one or more variables to their original values.

Parameters

var_ids	list of Var-IDs to reset
reset_all	True: reset all variables, var_ids will be ignored False: reset only those variables listed in var_ids
complete_arrays	True: reset all array elements if head element (index [0]) is in var_ids False: only array elements in var_ids will be reset (does not matter with reset_all=True)

7.312.3.17 setValue()

```
ttk_tools.isyst.mwapp_client.MwAppRpc.setValue (
              self,
              var_id,
              value,
              data_type = None )
```

Set new value of a MW-Variable.

Parameters

var_id	ID of variable
value	value to set, type depends on data type of variable
data_type	data type (MwDataType) to be set in set message's MwRpcValue. This has to match the variable's data type If None, the data type will be retrieved from meta data

7.312.3.18 updateValue()

```
ttk_tools.isyst.mwapp_client.MwAppRpc.updateValue (
              self,
              var_id,
              value,
              data_type = None )
```

Update value of a MW-Variable: set new value, and return the previous value before the change.

Parameters

var_id	ID of variable
value	value to set, type depends on data type of variable
data_type	data type (MwDataType) to be set in set message's MwRpcValue. This has to match the variable's data type If None, the data type will be retrieved from meta data

Returns

the variable's value before the update.

7.312.4 Member Data Documentation

7.312.4.1 channel

```
ttk_tools.isyst.mwapp_client.MwAppRpc.channel
```

7.312.4.2 instance_key

```
ttk_tools.isyst.mwapp_client.MwAppRpc.instance_key
```

7.312.4.3 last_app_status

ttk_tools.isyst.mwapp_client.MwAppRpc.last_app_status

7.312.4.4 stub

ttk_tools.isyst.mwapp_client.MwAppRpc.stub

7.312.4.5 system_name

ttk_tools.isyst.mwapp_client.MwAppRpc.system_name

7.312.4.6 system_version

ttk_tools.isyst.mwapp_client.MwAppRpc.system_version

7.312.4.7 target

ttk_tools.isyst.mwapp_client.MwAppRpc.target

7.312.4.8 var_id_cache

ttk_tools.isyst.mwapp_client.MwAppRpc.var_id_cache

7.312.4.9 var_meta_cache

ttk_tools.isyst.mwapp_client.MwAppRpc.var_meta_cache

The documentation for this class was generated from the following file:

- ttk_tools/isyst/mwapp_client.py (v1.2.4)

7.313 ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc Class Reference

Simple wrapper for MwApp RPCs (offline stub/mock implementation)

Public Member Functions

- __init__ (self, target, channel_options=None)
- close (self)

  Close current gRPC channel.
- getMwVar (self, var_name)

  Factory method for creating MwVar proxy instances.
- getCachedVarId (self, var_name)

  Get Var-ID for a variable's (display) name from cached meta data.
- getValue (self, var_id)

  Get current value of MW-Variable.
- setValue (self, var_id, value, data_type=None)

  Set new value of a MW-Variable.
- updateValue (self, var_id, value, data_type=None)

  Update value of a MW-Variable: set new value, and return the previous value before the change.
- resetValue (self, var_ids=None, reset_all=False, complete_arrays=False)

  Reset one or more variables to their original values.
- getVarId (self, var_name)

  Get Var-ID for a MW-Variable name.
- getMeta (self, var_id=None, meta_id=MetaKind.system_name)

  Get a meta value.
- getGroups (self)

  Get configured variable groups (and Var-IDs in those groups).
- bulkGetValue (self, var_ids)

  Get values for multiple variables.
- bulkGetMeta (self, var_meta_ids)

  Get meta data for multiple variables or system meta entries.
- bulkSetValue (self, var_type_values)

  Set values for multiple variables.
- __enter__ (self)

  Start MwAppRpc as managed context.
- __exit__ (self, exc_type, exc_value, tb)

  End context manager.

Public Attributes

- target
- channel
- stub
- system_name
- system_version
- instance_key
- var_meta_cache
- var_id_cache

Protected Member Functions

- _connectionCheck (self)

  Check if "connection" is active (which it will always be in stub).
- _initCacheEntry (self, var_name)

  Initialize `var_meta_cache` and `var_id_cache` entries for the supplied `var_name`.

Protected Attributes

- _data_context
- _data_provider

7.313.1 Detailed Description

Simple wrapper for MwApp RPCs (offline stub/mock implementation)

7.313.2 Constructor & Destructor Documentation

7.313.2.1 __init__()

```
ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc.__init__ (
              self,
              target,
              channel_options = None )
```

Parameters

target	connection target string (address:port), e.g. "127.0.0.1:50051". If port is missing, default port 50051 will be used.
channel_options	gRPC connection options. Not used in stub

Note

Offline stub will use address part of target as context for the offline data provider.

7.313.3 Member Function Documentation

7.313.3.1 __enter__()

```
ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc.__enter__ (
              self )
```

Start MwAppRpc as managed context.

7.313.3.2 __exit__()

```
ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc.__exit__ (
              self,
              exc_type,
              exc_value,
              tb )
```

End context manager.

7.313.3.3 _connectionCheck()

```
ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc._connectionCheck (
              self )  [protected]
```

Check if "connection" is active (which it will always be in stub).

7.313.3.4 _initCacheEntry()

```
ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc._initCacheEntry (
              self,
              var_name )  [protected]
```

Initialize `var_meta_cache` and `var_id_cache` entries for the supplied `var_name`.

- Meta data will be derived from any existing default value set in offline data provider.
- A Var-ID will be assigned to the `var_name`

7.313.3.5 bulkGetMeta()

```
ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc.bulkGetMeta (
              self,
              var_meta_ids )
```

Get meta data for multiple variables or system meta entries.

Parameters

var_meta_ids	a list of (var_id, meta_id) tuples. For system meta entries, var_id can be None.

Returns

a list of values, one for each supplied entry Value will be None if access failed (or meta value was undefined)

7.313.3.6 bulkGetValue()

```
ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc.bulkGetValue (
              self,
              var_ids )
```

Get values for multiple variables.

Parameters

var_ids	list of Var-ID

Returns

a list of values, one for each Var-ID . If a value cannot be accessed, its entry will be set to None.

7.313.3.7 bulkSetValue()

```
ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc.bulkSetValue (
              self,
              var_type_values )
```

Set values for multiple variables.

Parameters

var_type_values	list of tuples (var_id, data_type, new_value) var_id: int, unsigned data_type: MwDataType of var_id new_value: value suitable for data_type

Returns

a tuple: overall_errors - error count, should be 0 if everything was ok status_values - list of MwAppStatus values, one for each entry

7.313.3.8 close()

```
ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc.close (
              self )
```

Close current gRPC channel.

7.313.3.9 getCachedVarId()

```
ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc.getCachedVarId (
              self,
              var_name )
```

Get Var-ID for a variable's (display) name from cached meta data.

Parameters

var_name	display name of MW-Variable, potentially including an array index suffix

Returns

a numeric Var-ID, or None if no matching variable was found.

7.313.3.10 getGroups()

```
ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc.getGroups (
              self )
```

Get configured variable groups (and Var-IDs in those groups).

For array-Variables, only the head element's Var-ID will be included.

Returns

a dict with <group_id>: { "group_name": "<group_name>", "group_id": <group_id>, "var_ids": [<var_id1>, <var_id2>, ... <var_idn>] }

7.313.3.11 getMeta()

```
ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc.getMeta (
              self,
              var_id = None,
              meta_id = MetaKind.system_name )
```

Get a meta value.

Direct access via RPC. See MetaKind for available meta entries.

Parameters

var_id	Var-ID of variable to get meta data on. Will be ignored for system-level meta entries (can be set to None for this)
meta_id	MetaKind value specifying the meta entry to get.

Returns

a value (numeric or bytes/string, depending on meta id/type) Value may be None if meta entry is undefined. Raises a MwAppError if response status indicates an error.

7.313.3.12 getMwVar()

```
ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc.getMwVar (
              self,
              var_name )
```

Factory method for creating MwVar proxy instances.

This is mainly intended for convenience, so the variable abstraction does not have to import additional packages.

Parameters

var_name	variable name on mwservice

Returns

an initialized MwVar instance

7.313.3.13 getValue()

```
ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc.getValue (
              self,
              var_id )
```

Get current value of MW-Variable.

Returns

a value (numeric or bytes/string, depending on MW-Var data type)

7.313.3.14 getVarId()

```
ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc.getVarId (
              self,
              var_name )
```

Get Var-ID for a MW-Variable name.

Direct access via RPC.

Returns

a numeric value or None if no ID for the given name exists.

7.313.3.15 resetValue()

```
ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc.resetValue (
              self,
              var_ids = None,
              reset_all = False,
              complete_arrays = False )
```

Reset one or more variables to their original values.

Parameters

var_ids	list of Var-IDs to reset
reset_all	True: reset all variables, var_ids will be ignored False: reset only those variables listed in var_ids
complete_arrays	True: reset all array elements if head element (index [0]) is in var_ids False: only array elements in var_ids will be reset (does not matter with reset_all=True)

7.313.3.16 setValue()

```
ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc.setValue (
              self,
```

```
                var_id,
                value,
                data_type = None )
```

Set new value of a MW-Variable.

Parameters

var_id	ID of variable
value	value to set, type depends on data type of variable
data_type	data type (MwDataType) to be set in set message's MwRpcValue. This has to match the variable's data type If None, the data type will be retrieved from meta data

7.313.3.17 updateValue()

```
ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc.updateValue (
                self,
                var_id,
                value,
                data_type = None )
```

Update value of a MW-Variable: set new value, and return the previous value before the change.

Parameters

var_id	ID of variable
value	value to set, type depends on data type of variable
data_type	data type (MwDataType) to be set in set message's MwRpcValue. This has to match the variable's data type If None, the data type will be retrieved from meta data

Returns

the variable's value before the update.

7.313.4 Member Data Documentation

7.313.4.1 _data_context

```
ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc._data_context  [protected]
```

7.313.4.2 _data_provider

```
ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc._data_provider  [protected]
```

7.313.4.3 channel

```
ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc.channel
```

7.313.4.4 instance_key

ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc.instance_key

7.313.4.5 stub

ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc.stub

7.313.4.6 system_name

ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc.system_name

7.313.4.7 system_version

ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc.system_version

7.313.4.8 target

ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc.target

7.313.4.9 var_id_cache

ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc.var_id_cache

7.313.4.10 var_meta_cache

ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc.var_meta_cache

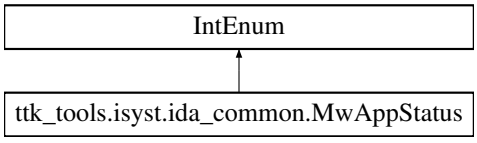
The documentation for this class was generated from the following file:

- ttk_tools/isyst/mwapp_client_offline_stub.py (v1.3.0)

7.314 ttk_tools.isyst.ida_common.MwAppStatus Class Reference

Application RPC status info (internal success/error).

Inheritance diagram for ttk_tools.isyst.ida_common.MwAppStatus:



Public Member Functions

- info (self)

Static Public Attributes

- int OK = 0x0000
- int NOT_FOUND = 0x0100
- int BAD_TYPE = 0x0200
- int UNDEFINED = 0x0300
- int OUT_OF_RANGE = 0x1000
- int NOT_AVAILABLE = 0x1100
- int UNEXPECTED_ERROR = 0xFFFF

7.314.1 Detailed Description

Application RPC status info (internal success/error).

7.314.2 Member Function Documentation

7.314.2.1 info()

```
ttk_tools.isyst.ida_common.MwAppStatus.info (
              self )
```

7.314.3 Member Data Documentation

7.314.3.1 BAD_TYPE

```
int ttk_tools.isyst.ida_common.MwAppStatus.BAD_TYPE = 0x0200  [static]
```

7.314.3.2 NOT_AVAILABLE

```
int ttk_tools.isyst.ida_common.MwAppStatus.NOT_AVAILABLE = 0x1100  [static]
```

7.314.3.3 NOT_FOUND

```
int ttk_tools.isyst.ida_common.MwAppStatus.NOT_FOUND = 0x0100  [static]
```

7.314.3.4 OK

```
int ttk_tools.isyst.ida_common.MwAppStatus.OK = 0x0000  [static]
```

7.314.3.5 OUT_OF_RANGE

```
int ttk_tools.isyst.ida_common.MwAppStatus.OUT_OF_RANGE = 0x1000  [static]
```

7.314.3.6 UNDEFINED

```
int ttk_tools.isyst.ida_common.MwAppStatus.UNDEFINED = 0x0300  [static]
```

7.314.3.7 UNEXPECTED_ERROR

```
int ttk_tools.isyst.ida_common.MwAppStatus.UNEXPECTED_ERROR = 0xFFFF  [static]
```

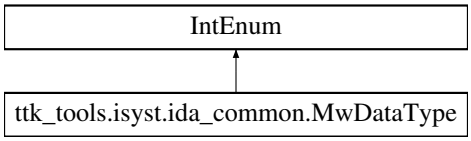
The documentation for this class was generated from the following file:

- ttk_tools/isyst/ida_common.py (v1.2.0)

7.315 ttk_tools.isyst.ida_common.MwDataType Class Reference

Data types for middleware values, compare mwvalue.h: dataType_t.

Inheritance diagram for ttk_tools.isyst.ida_common.MwDataType:



Public Member Functions

- isNumeric (self)
- isINT (self)
- isSINT (self)
- isUINT (self)
- isFLOAT (self)
- isBlobby (self)
- info (self)

  Get an info string for the supplied MwDataType.

Static Public Attributes

- int UNDEFINED = 0x00
- int UINT8 = 0x01
- int UINT16 = 0x02
- int UINT32 = 0x03
- int UINT64 = 0x04
- int SINT8 = 0x11
- int SINT16 = 0x12
- int SINT32 = 0x13
- int SINT64 = 0x14
- int FLOAT32 = 0x20
- int FLOAT64 = 0x21
- int CHAR = 0x30
- int BLOB = 0x40

7.315.1 Detailed Description

Data types for middleware values, compare mwvalue.h: dataType_t.

Note

Utility methods may be called as

- class methods with a value to interpret or
- on on an instance without a parameter

Example: Utility methods

```
»> MwDataType.FLOAT64.info()
"FLOAT64 [0x21]"

»> MwDataType.info(0x03)
"UINT32 [0x03]"

MwDataType.info(0x4711)
"<undefined> [0x4711]"
```

7.315.2 Member Function Documentation

7.315.2.1 info()

```
ttk_tools.isyst.ida_common.MwDataType.info (
              self )
```

Get an info string for the supplied MwDataType.

Undefined values will show as <undefined> and not raise an exception.

7.315.2.2 isBlobby()

```
ttk_tools.isyst.ida_common.MwDataType.isBlobby (
              self )
```

7.315.2.3 isFLOAT()

```
ttk_tools.isyst.ida_common.MwDataType.isFLOAT (
              self )
```

7.315.2.4 isINT()

```
ttk_tools.isyst.ida_common.MwDataType.isINT (
              self )
```

7.315.2.5 isNumeric()

```
ttk_tools.isyst.ida_common.MwDataType.isNumeric (
              self )
```

7.315.2.6 isSINT()

```
ttk_tools.isyst.ida_common.MwDataType.isSINT (
              self )
```

7.315.2.7 isUINT()

```
ttk_tools.isyst.ida_common.MwDataType.isUINT (
              self )
```

7.315.3 Member Data Documentation

7.315.3.1 BLOB

```
int ttk_tools.isyst.ida_common.MwDataType.BLOB = 0x40  [static]
```

7.315.3.2 CHAR

```
int ttk_tools.isyst.ida_common.MwDataType.CHAR = 0x30  [static]
```

7.315.3.3 FLOAT32

```
int ttk_tools.isyst.ida_common.MwDataType.FLOAT32 = 0x20  [static]
```

7.315.3.4 FLOAT64

```
int ttk_tools.isyst.ida_common.MwDataType.FLOAT64 = 0x21  [static]
```

7.315.3.5 SINT16

```
int ttk_tools.isyst.ida_common.MwDataType.SINT16 = 0x12  [static]
```

7.315.3.6 SINT32

```
int ttk_tools.isyst.ida_common.MwDataType.SINT32 = 0x13  [static]
```

7.315.3.7 SINT64

```
int ttk_tools.isyst.ida_common.MwDataType.SINT64 = 0x14  [static]
```

7.315.3.8 SINT8

```
int ttk_tools.isyst.ida_common.MwDataType.SINT8 = 0x11  [static]
```

7.315.3.9 UINT16

```
int ttk_tools.isyst.ida_common.MwDataType.UINT16 = 0x02  [static]
```

7.315.3.10 UINT32

```
int ttk_tools.isyst.ida_common.MwDataType.UINT32 = 0x03  [static]
```

7.315.3.11 UINT64

```
int ttk_tools.isyst.ida_common.MwDataType.UINT64 = 0x04  [static]
```

7.315.3.12 UINT8

```
int ttk_tools.isyst.ida_common.MwDataType.UINT8 = 0x01  [static]
```

7.315.3.13 UNDEFINED

```
int ttk_tools.isyst.ida_common.MwDataType.UNDEFINED = 0x00  [static]
```

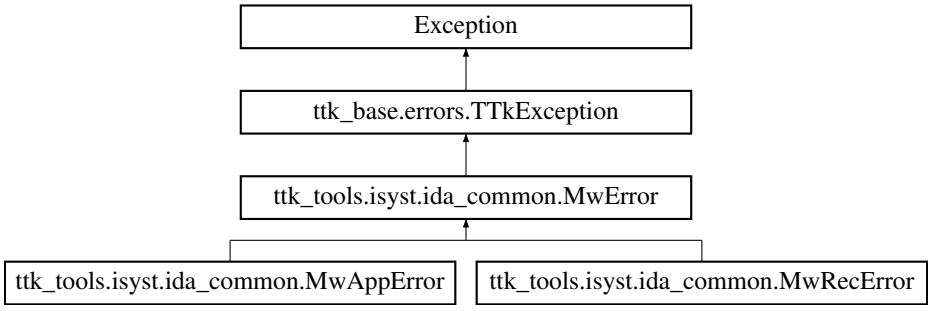
The documentation for this class was generated from the following file:

- ttk_tools/isyst/ida_common.py (v1.2.0)

7.316 ttk_tools.isyst.ida_common.MwError Class Reference

Generic iDA Testware middleware interface error.

Inheritance diagram for ttk_tools.isyst.ida_common.MwError:



7.316.1 Detailed Description

Generic iDA Testware middleware interface error.

Errors during Application RPC processing (i.e. errors detected server-side).

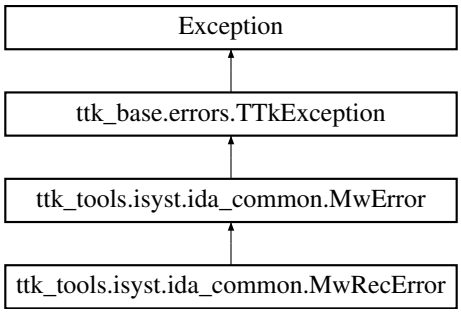
The documentation for this class was generated from the following file:

- ttk_tools/isyst/ida_common.py (v1.2.0)

7.317 ttk_tools.isyst.ida_common.MwRecError Class Reference

Errors during data recording.

Inheritance diagram for ttk_tools.isyst.ida_common.MwRecError:



Public Member Functions

- __init__ (self, message, status=None, info=None)

Public Attributes

- status
- message
- info

7.317.1 Detailed Description

Errors during data recording.

7.317.2 Constructor & Destructor Documentation

7.317.2.1 __init__()

```
ttk_tools.isyst.ida_common.MwRecError.__init__ (
              self,
              message,
              status = None,
              info = None )
```

Parameters

message	error message
status	RecCommandStatus value (if available)
info	additional error info (usually names of failed variables for status bad_config)

7.317.3 Member Data Documentation

7.317.3.1 info

`ttk_tools.isyst.ida_common.MwRecError.info`

7.317.3.2 message

`ttk_tools.isyst.ida_common.MwRecError.message`

7.317.3.3 status

`ttk_tools.isyst.ida_common.MwRecError.status`

The documentation for this class was generated from the following file:

- ttk_tools/isyst/ida_common.py (v1.2.0)

7.318 ttk_tools.isyst.mwrec_client.MwRecRpc Class Reference

Simple wrapper for MwRec RPCs.

Public Member Functions

- __init__ (self, target, channel_options=None)
- startRecording (self)
  Start a configured recording.
- stopRecording (self)
  Stop an active (running) recording.
- setupVars (self, var_names=None, var_ids=None)
  Setup data acquisition; configure which variables to record.
- setup (self, variables)
  Setup data acquisition (configure what variables to measure).
- getState (self, verbose=False)
  Get current state machine / recorder state.
- getIdaqDataIO (self)
  Get recorded data in idaq format.
- getData (self)
  Get recorded data and return it as a daq data structure.
- waitForState (self, rec_state=RecState.stopped, timeout_s=2, step_s=0.100)
  Wait for Recorder to enter the desired state (or a timeout to occur).
- close (self)
  Close current gRPC channel.
- __enter__ (self)
  Start MwRecRpc as managed context.
- __exit__ (self, exc_type, exc_value, tb)
  End context manager.

Public Attributes

- target
- channel
- stub

Static Public Attributes

- int verbosity = 1

Protected Member Functions

- _printStatus (self, prefix, status_response)

  Print a formatted command status response for verbosity >= 2.

7.318.1 Detailed Description

Simple wrapper for MwRec RPCs.

7.318.2 Constructor & Destructor Documentation

7.318.2.1 __init__()

```
ttk_tools.isyst.mwrec_client.MwRecRpc.__init__ (
              self,
              target,
              channel_options = None )
```

Parameters

target	connection target string (IP and port), e.g. "127.0.0.1:50055". If port is missing, default port 50055 will be used.
channel_options	gRPC connection options.

7.318.3 Member Function Documentation

7.318.3.1 __enter__()

```
ttk_tools.isyst.mwrec_client.MwRecRpc.__enter__ (
              self )
```

Start MwRecRpc as managed context.

7.318.3.2 __exit__()

```
ttk_tools.isyst.mwrec_client.MwRecRpc.__exit__ (
              self,
              exc_type,
              exc_value,
              tb )
```

End context manager.

7.318.3.3 _printStatus()

```
ttk_tools.isyst.mwrec_client.MwRecRpc._printStatus (
              self,
              prefix,
              status_response )  [protected]
```

Print a formatted command status response for verbosity >= 2.

Parameters

prefix	prefix to show, usually the rpc name
status_response	MwRec Status response message

7.318.3.4 close()

```
ttk_tools.isyst.mwrec_client.MwRecRpc.close (
              self )
```

Close current gRPC channel.

7.318.3.5 getData()

```
ttk_tools.isyst.mwrec_client.MwRecRpc.getData (
              self )
```

Get recorded data and return it as a daq data structure.

Can only be (successfully) called in RecState.stopped.

Returns

a daq data structure with {
<label>: {
"time": [<time_s>, <time_s>, ...],
"data": [<value>, <value>, ...],
"type": "<datatype-string>"
},
} entries.

7.318.3.6 getIdaqDataIO()

```
ttk_tools.isyst.mwrec_client.MwRecRpc.getIdaqDataIO (
              self )
```

Get recorded data in idaq format.

Can only be (successfully) called in RecState.stopped. In states other than stopped, an empty response will be returned.

Returns

a BytesIO instance containing data in idaq format. File position will be rewound to the start.

7.318.3.7 getState()

```
ttk_tools.isyst.mwrec_client.MwRecRpc.getState (
              self,
              verbose = False )
```

Get current state machine / recorder state.

Parameters; verbose - True: Print status to console

Returns

a RecState

7.318.3.8 setup()

```
ttk_tools.isyst.mwrec_client.MwRecRpc.setup (
              self,
              variables )
```

Setup data acquisition (configure what variables to measure).

Basic variant of setupVars for symmetry with other implementations

Note

setup() may only be performed while data acquisition is not already running.

Parameters

variables	list/iterable of gamma variables to measure

Usage:

```
        # direct setup with variable names
        daq.setup([
            "some_var.foo",
            "some_other_var"
        ])
        # or, when using a VarContainer:
        daq.setup([hil.some_var, hil.some_other_var])

    Raises an MwRecError if setup was unsuccessful.
```

7.318.3.9 setupVars()

```
ttk_tools.isyst.mwrec_client.MwRecRpc.setupVars (
               self,
               var_names = None,
               var_ids = None )
```

Setup data acquisition; configure which variables to record.

If both variable names and IDs are supplied, the recording will contain all referenced variables.

Note

setupVars() may only be performed while data acquisition is not already running.

Array Variables Data acquisition will only record scalar variables. To record an array variable, all its array elements have to be added separately.

Parameters

var_names	list of variable names to record
var_ids	list of Var-IDs to record (optional, in case IDs are available)

Raises an MwRecError if setup was unsuccessful.

7.318.3.10 startRecording()

```
ttk_tools.isyst.mwrec_client.MwRecRpc.startRecording (
               self )
```

Start a configured recording.

Returns

a RecCommandStatus

7.318.3.11 stopRecording()

```
ttk_tools.isyst.mwrec_client.MwRecRpc.stopRecording (
               self )
```

Stop an active (running) recording.

Returns

a RecCommandStatus

7.318.3.12 waitForState()

```
ttk_tools.isyst.mwrec_client.MwRecRpc.waitForState (
              self,
              rec_state = RecState.stopped,
              timeout_s = 2,
              step_s = 0.100 )
```

Wait for Recorder to enter the desired state (or a timeout to occur).

Parameters

rec_state	state to wait for
timeout_s	timeout in [s] for state to be reached
step_s	step time between checks in [s]

Returns

the current recorder state, which will be rec_state if the state has been reached, or the last state at time of timeout.

7.318.4 Member Data Documentation

7.318.4.1 channel

```
ttk_tools.isyst.mwrec_client.MwRecRpc.channel
```

7.318.4.2 stub

```
ttk_tools.isyst.mwrec_client.MwRecRpc.stub
```

7.318.4.3 target

```
ttk_tools.isyst.mwrec_client.MwRecRpc.target
```

7.318.4.4 verbosity

```
int ttk_tools.isyst.mwrec_client.MwRecRpc.verbosity = 1  [static]
```

The documentation for this class was generated from the following file:

- ttk_tools/isyst/mwrec_client.py (v1.0.2)

7.319 ttk_tools.isyst.mwrec_client_offline_stub.MwRecRpc Class Reference

Simple wrapper for MwRec RPCs (offline stub/mock implementation)

Public Member Functions

- __init__ (self, target, channel_options=None)
- startRecording (self)

  Start a configured recording.
- stopRecording (self)

  Stop an active (running) recording.
- setupVars (self, var_names=None, var_ids=None)

  Setup data acquisition; configure which variables to record.
- setup (self, variables)

  Setup data acquisition (configure what variables to measure).
- getState (self, verbose=False)

  Get current state machine / recorder state.
- getIdaqDataIO (self)

  Get recorded data in idaq format.
- getData (self)

  Get recorded data and return it as a daq data structure.
- waitForState (self, rec_state=RecState.stopped, timeout_s=2, step_s=0.100)

  Wait for Recorder to enter the desired state (or a timeout to occur).
- close (self)

  Close current gRPC channel.
- __enter__ (self)

  Start MwRecRpc as managed context.
- __exit__ (self, exc_type, exc_value, tb)

  End context manager.

Public Attributes

- target
- channel
- stub

Static Public Attributes

- int verbosity = 1

Protected Member Functions

- _printStatus (self, prefix, status)

  Print a formatted command status response for verbosity >= 2.
- _getDummyData (self)

  create dummy data between _start_time and _stop_time

Protected Attributes

- _data_context
- _data_provider
- _polling_rate_ms
- _variables
- _start_time
- _stop_time
- _recorder_state
- _stert_time

7.319.1 Detailed Description

Simple wrapper for MwRec RPCs (offline stub/mock implementation)

7.319.2 Constructor & Destructor Documentation

7.319.2.1 __init__()

```
ttk_tools.isyst.mwrec_client_offline_stub.MwRecRpc.__init__ (
              self,
              target,
              channel_options = None )
```

Parameters

target	connection target string (IP and port), e.g. "127.0.0.1:50055". If port is missing, default port 50055 will be used.
channel_options	gRPC connection options.

Note

Offline stub will use address part of target as context for the offline data provider.

7.319.3 Member Function Documentation

7.319.3.1 __enter__()

```
ttk_tools.isyst.mwrec_client_offline_stub.MwRecRpc.__enter__ (
              self )
```

Start MwRecRpc as managed context.

7.319.3.2 __exit__()

```
ttk_tools.isyst.mwrec_client_offline_stub.MwRecRpc.__exit__ (
              self,
              exc_type,
              exc_value,
              tb )
```

End context manager.

7.319.3.3 _getDummyData()

```
ttk_tools.isyst.mwrec_client_offline_stub.MwRecRpc._getDummyData (
              self )  [protected]
```

create dummy data between _start_time and _stop_time

7.319.3.4 _printStatus()

```
ttk_tools.isyst.mwrec_client_offline_stub.MwRecRpc._printStatus (
              self,
              prefix,
              status )  [protected]
```

Print a formatted command status response for verbosity >= 2.

Parameters

prefix	prefix to show, usually the rpc name
status	RecCommandStatus status

7.319.3.5 close()

```
ttk_tools.isyst.mwrec_client_offline_stub.MwRecRpc.close (
              self )
```

Close current gRPC channel.

7.319.3.6 getData()

```
ttk_tools.isyst.mwrec_client_offline_stub.MwRecRpc.getData (
              self )
```

Get recorded data and return it as a daq data structure.

Can only be (successfully) called in RecState.stopped.

Returns

a daq data structure with {
<label>: {
"time": [<time_s>, <time_s>, ...],
"data": [<value>, <value>, ...],
"type": "<datatype-string>"
},
} entries.

7.319.3.7 getIdaqDataIO()

```
ttk_tools.isyst.mwrec_client_offline_stub.MwRecRpc.getIdaqDataIO (
              self )
```

Get recorded data in idaq format.

Can only be (successfully) called in RecState.stopped. In states other than stopped, an empty response will be returned.

Returns

a BytesIO instance containing data in idaq format. File position will be rewound to the start.

7.319.3.8 getState()

```
ttk_tools.isyst.mwrec_client_offline_stub.MwRecRpc.getState (
              self,
              verbose = False )
```

Get current state machine / recorder state.

Parameters; verbose - True: Print status to console

Returns

a RecState

7.319.3.9 setup()

```
ttk_tools.isyst.mwrec_client_offline_stub.MwRecRpc.setup (
              self,
              variables )
```

Setup data acquisition (configure what variables to measure).

Basic variant of setupVars for symmetry with other implementations

Note

setup() may only be performed while data acquisition is not already running.

Parameters

variables	list/iterable of gamma variables to measure

Usage:

```
        # direct setup with variable names
        daq.setup([
            "some_var.foo",
            "some_other_var"
        ])
        # or, when using a VarContainer:
        daq.setup([hil.some_var, hil.some_other_var])

    Raises an MwRecError if setup was unsuccessful.
```

7.319.3.10 setupVars()

```
ttk_tools.isyst.mwrec_client_offline_stub.MwRecRpc.setupVars (
              self,
              var_names = None,
              var_ids = None )
```

Setup data acquisition; configure which variables to record.

If both variable names and IDs are supplied, the recording will contain all referenced variables.

Note

setupVars() may only be performed while data acquisition is not already running.

Array Variables Data acquisition will only record scalar variables. To record an array variable, all its array elements have to be added separately.

Variables added via var_ids will be ignored in offline stub (as offline data provider references variables by name)

Parameters

var_names	list of variable names to record
var_ids	list of Var-IDs to record (optional, in case IDs are available)

Raises an MwRecError if setup was unsuccessful.

7.319.3.11 startRecording()

```
ttk_tools.isyst.mwrec_client_offline_stub.MwRecRpc.startRecording (
              self )
```

Start a configured recording.

Returns

a RecCommandStatus

7.319.3.12 stopRecording()

```
ttk_tools.isyst.mwrec_client_offline_stub.MwRecRpc.stopRecording (
              self )
```

Stop an active (running) recording.

Returns

a RecCommandStatus

7.319.3.13 waitForState()

```
ttk_tools.isyst.mwrec_client_offline_stub.MwRecRpc.waitForState (
              self,
              rec_state = RecState.stopped,
              timeout_s = 2,
              step_s = 0.100 )
```

Wait for Recorder to enter the desired state (or a timeout to occur).

Parameters

rec_state	state to wait for
timeout_s	timeout in [s] for state to be reached
step_s	step time between checks in [s]

Returns

the current recorder state, which will be rec_state if the state has been reached, or the last state at time of timeout.

7.319.4 Member Data Documentation

7.319.4.1 _data_context

```
ttk_tools.isyst.mwrec_client_offline_stub.MwRecRpc._data_context  [protected]
```

7.319.4.2 _data_provider

```
ttk_tools.isyst.mwrec_client_offline_stub.MwRecRpc._data_provider  [protected]
```

7.319.4.3 _polling_rate_ms

```
ttk_tools.isyst.mwrec_client_offline_stub.MwRecRpc._polling_rate_ms  [protected]
```

7.319.4.4 _recorder_state

ttk_tools.isyst.mwrec_client_offline_stub.MwRecRpc._recorder_state [protected]

7.319.4.5 _start_time

ttk_tools.isyst.mwrec_client_offline_stub.MwRecRpc._start_time [protected]

7.319.4.6 _stert_time

ttk_tools.isyst.mwrec_client_offline_stub.MwRecRpc._stert_time [protected]

7.319.4.7 _stop_time

ttk_tools.isyst.mwrec_client_offline_stub.MwRecRpc._stop_time [protected]

7.319.4.8 _variables

ttk_tools.isyst.mwrec_client_offline_stub.MwRecRpc._variables [protected]

7.319.4.9 channel

ttk_tools.isyst.mwrec_client_offline_stub.MwRecRpc.channel

7.319.4.10 stub

ttk_tools.isyst.mwrec_client_offline_stub.MwRecRpc.stub

7.319.4.11 target

ttk_tools.isyst.mwrec_client_offline_stub.MwRecRpc.target

7.319.4.12 verbosity

int ttk_tools.isyst.mwrec_client_offline_stub.MwRecRpc.verbosity = 1 [static]

The documentation for this class was generated from the following file:

- ttk_tools/isyst/mwrec_client_offline_stub.py (v1.0.0)

7.320 ttk_tools.isyst.mwapp_client.MwVar Class Reference

Container for a single MW-Variable.

Public Member Functions

- __init__ (self, var_name, mwapp)
- getValue (self)
- setValue (self, value)
- updateValue (self, value)
- resetValue (self)

Public Attributes

- name
- var_id
- data_type
- size
- unit
- fmt

Protected Attributes

- _mwapp

7.320.1 Detailed Description

Container for a single MW-Variable.

Proxies access to MwAppRpc.

7.320.2 Constructor & Destructor Documentation

7.320.2.1 __init__()

```
ttk_tools.isyst.mwapp_client.MwVar.__init__ (
              self,
              var_name,
              mwapp )
```

7.320.3 Member Function Documentation

7.320.3.1 getValue()

```
ttk_tools.isyst.mwapp_client.MwVar.getValue (
              self )
```

7.320.3.2 resetValue()

```
ttk_tools.isyst.mwapp_client.MwVar.resetValue (
              self )
```

7.320.3.3 setValue()

```
ttk_tools.isyst.mwapp_client.MwVar.setValue (
              self,
              value )
```

7.320.3.4 updateValue()

```
ttk_tools.isyst.mwapp_client.MwVar.updateValue (
              self,
              value )
```

7.320.4 Member Data Documentation

7.320.4.1 _mwapp

```
ttk_tools.isyst.mwapp_client.MwVar._mwapp  [protected]
```

7.320.4.2 data_type

```
ttk_tools.isyst.mwapp_client.MwVar.data_type
```

7.320.4.3 fmt

```
ttk_tools.isyst.mwapp_client.MwVar.fmt
```

7.320.4.4 name

```
ttk_tools.isyst.mwapp_client.MwVar.name
```

7.320.4.5 size

```
ttk_tools.isyst.mwapp_client.MwVar.size
```

7.320.4.6 unit

```
ttk_tools.isyst.mwapp_client.MwVar.unit
```

7.320.4.7 var_id

ttk_tools.isyst.mwapp_client.MwVar.var_id

The documentation for this class was generated from the following file:

- ttk_tools/isyst/mwapp_client.py (v1.2.4)

7.321 ttk_tools.isyst.mwapp_client_offline_stub.MwVar Class Reference

Container for a single MW-Variable.

Public Member Functions

- __init__ (self, var_name, mwapp)
- getValue (self)
- setValue (self, value)
- updateValue (self, value)
- resetValue (self)

Public Attributes

- name
- var_id
- data_type
- size
- unit
- fmt

Protected Attributes

- _mwapp

7.321.1 Detailed Description

Container for a single MW-Variable.

Proxies access to MwAppRpc.

7.321.2 Constructor & Destructor Documentation

7.321.2.1 __init__()

```
ttk_tools.isyst.mwapp_client_offline_stub.MwVar.__init__ (
            self,
            var_name,
            mwapp )
```

7.321.3 Member Function Documentation

7.321.3.1 getValue()

```
ttk_tools.isyst.mwapp_client_offline_stub.MwVar.getValue (
              self )
```

7.321.3.2 resetValue()

```
ttk_tools.isyst.mwapp_client_offline_stub.MwVar.resetValue (
              self )
```

7.321.3.3 setValue()

```
ttk_tools.isyst.mwapp_client_offline_stub.MwVar.setValue (
              self,
              value )
```

7.321.3.4 updateValue()

```
ttk_tools.isyst.mwapp_client_offline_stub.MwVar.updateValue (
              self,
              value )
```

7.321.4 Member Data Documentation

7.321.4.1 _mwapp

```
ttk_tools.isyst.mwapp_client_offline_stub.MwVar._mwapp  [protected]
```

7.321.4.2 data_type

```
ttk_tools.isyst.mwapp_client_offline_stub.MwVar.data_type
```

7.321.4.3 fmt

```
ttk_tools.isyst.mwapp_client_offline_stub.MwVar.fmt
```

7.321.4.4 name

```
ttk_tools.isyst.mwapp_client_offline_stub.MwVar.name
```

7.321.4.5 size

ttk_tools.isyst.mwapp_client_offline_stub.MwVar.size

7.321.4.6 unit

ttk_tools.isyst.mwapp_client_offline_stub.MwVar.unit

7.321.4.7 var_id

ttk_tools.isyst.mwapp_client_offline_stub.MwVar.var_id

The documentation for this class was generated from the following file:

- ttk_tools/isyst/mwapp_client_offline_stub.py (v1.3.0)

7.322 ttk_selftest._selftest_tests.NegWrapper Class Reference

Class used to get values multiplied by -1.

Public Member Functions

- __init__ (self, var)
- get (self)
- __getattr__ (self, name)
- __str__ (self)

Public Attributes

- var

7.322.1 Detailed Description

Class used to get values multiplied by -1.

Necessary for current measurement to compensate the current direction.

7.322.2 Constructor & Destructor Documentation

7.322.2.1 __init__()

```
ttk_selftest._selftest_tests.NegWrapper.__init__ (
              self,
              var )
```

7.322.3 Member Function Documentation

7.322.3.1 __getattr__()

```
ttk_selftest._selftest_tests.NegWrapper.__getattr__ (
            self,
            name )
```

7.322.3.2 __str__()

```
ttk_selftest._selftest_tests.NegWrapper.__str__ (
            self )
```

7.322.3.3 get()

```
ttk_selftest._selftest_tests.NegWrapper.get (
            self )
```

7.322.4 Member Data Documentation

7.322.4.1 var

```
ttk_selftest._selftest_tests.NegWrapper.var
```

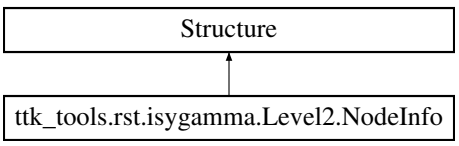
The documentation for this class was generated from the following file:

- ttk_selftest/_selftest_tests.py (v2.2.1)

7.323 ttk_tools.rst.isygamma.Level2.NodeInfo Class Reference

typedef struct gc_NodeInfo

Inheritance diagram for ttk_tools.rst.isygamma.Level2.NodeInfo:



Static Protected Attributes

- list _fields_

7.323.1 Detailed Description

typedef struct gc_NodeInfo

7.323.2 Member Data Documentation

7.323.2.1 _fields_

```
list ttk_tools.rst.isygamma.Level2.NodeInfo._fields_  [static], [protected]
```

Initial value:

```
=  [
            ("address",         Addresses.Level2),  # address of the node
            ("name",            c_char * 1024),     # unique name of the node
            ("isPersistent",    c_bool)             # is node persistent?
        ]
```

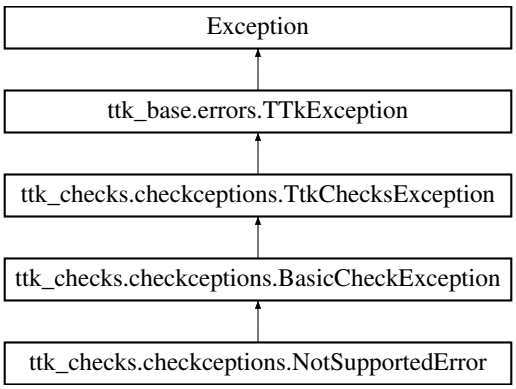
The documentation for this class was generated from the following file:

- ttk_tools/rst/isygamma.py (v1.6)

7.324 ttk_checks.checkceptions.NotSupportedError Class Reference

Exception: Feature or input is not or not yet supported.

Inheritance diagram for ttk_checks.checkceptions.NotSupportedError:



7.324.1 Detailed Description

Exception: Feature or input is not or not yet supported.

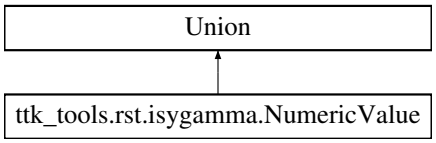
An exception that gets raised in case an unexpected error occurred, e.g. during calculations/checks where all input parameters have already been validated and no further trouble is expected.

The documentation for this class was generated from the following file:

- ttk_checks/checkceptions.py (v2.0)

7.325 ttk_tools.rst.isygamma.NumericValue Class Reference

Inheritance diagram for ttk_tools.rst.isygamma.NumericValue:



Public Member Functions

- getMemberForDataType (cls, data_type)

  Get member name matching a DataType enum value.
- getValueForDataType (self, data_type)

  Get member value for a DataType enum value.

Static Protected Attributes

- list _fields_
- dict _data_type_mapping

7.325.1 Member Function Documentation

7.325.1.1 getMemberForDataType()

```
ttk_tools.rst.isygamma.NumericValue.getMemberForDataType (
              cls,
              data_type )
```

Get member name matching a DataType enum value.

Returns

a member's name (e.g. "uint16") or None for unsupported types.

7.325.1.2 getValueForDataType()

```
ttk_tools.rst.isygamma.NumericValue.getValueForDataType (
              self,
              data_type )
```

Get member value for a DataType enum value.

Returns

the current value or None for unsupported types.

7.325.2 Member Data Documentation

7.325.2.1 _data_type_mapping

```
dict ttk_tools.rst.isygamma.NumericValue._data_type_mapping  [static], [protected]
```

Initial value:

```
=  {
        DataType.SINT_8:  "sint8",
        DataType.UINT_8:  "uint8",
        DataType.SINT_16: "sint16",
        DataType.UINT_16: "uint16",
        DataType.SINT_32: "sint32",
        DataType.UINT_32: "uint32",
        DataType.SINT_64: "sint64",
        DataType.UINT_64: "uint64",
        DataType.IEEE_32: "ieee32",
        DataType.IEEE_64: "ieee64",
    }
```

7.325.2.2 _fields_

```
list ttk_tools.rst.isygamma.NumericValue._fields_  [static], [protected]
```

Initial value:

```
=  [
        ('sint8',  c_int8   ),
        ('uint8',  c_uint8  ),
        ('sint16', c_int16  ),
        ('uint16', c_uint16 ),
        ('sint32', c_int32  ),
        ('uint32', c_uint32 ),
        ('sint64', c_int64  ),
        ('uint64', c_uint64 ),
        ('ieee32', c_float  ),
        ('ieee64', c_double )
    ]
```

The documentation for this class was generated from the following file:

- ttk_tools/rst/isygamma.py (v1.6)

7.326 ttk_tools.offline_data_provider.OfflineDataProvider Class Reference

Provider of dummy data for use in offline debugging.

Public Member Functions

- __init__ (self, verbosity=0)
- loadData (self, file_path)

  Load offline data from a file.
- storeData (self, file_path=None)

  Store current offline data to a file.
- setVerbosity (self, level=0)

  Set verbosity level of log output.
- addContext (self, name, default_values=None)

  Add and init a new context.
- getData (self, context, variable)

  Get a value for the supplied variable within context.
- setData (self, context, variable, value)

  Get a value for the supplied variable within context.

Public Attributes

- offline_data

  dictionary with offline data as {identifier: value} entries
- verbosity

  verbosity of status output
- loaded_file_path

7.326.1 Detailed Description

Provider of dummy data for use in offline debugging.

7.326.2 Constructor & Destructor Documentation

7.326.2.1 __init__()

```
ttk_tools.offline_data_provider.OfflineDataProvider.__init__ (
              self,
              verbosity = 0 )
```

Parameters

verbosity	verbosity of status output, see setVerbosity()

7.326.3 Member Function Documentation

7.326.3.1 addContext()

```
ttk_tools.offline_data_provider.OfflineDataProvider.addContext (
              self,
              name,
              default_values = None )
```

Add and init a new context.

If context already exists, it will stay unchanged.

Parameters

name	name of new context
default_values	dictionary with default variable => value mappings

7.326.3.2 getData()

```
ttk_tools.offline_data_provider.OfflineDataProvider.getData (
              self,
```

```
                context,
                variable )
```

Get a value for the supplied variable within context.

Parameters

context	data context (e.g. board name, device name, ...)
variable	variable name

Returns

the variable's value, which defaults to DEFAULT_VALUE (0.0) if variable did not yet exist.

7.326.3.3 loadData()

```
ttk_tools.offline_data_provider.OfflineDataProvider.loadData (
                self,
                file_path )
```

Load offline data from a file.

Note

Not yet implemented

Parameters

file_path	path to data file

7.326.3.4 setData()

```
ttk_tools.offline_data_provider.OfflineDataProvider.setData (
                self,
                context,
                variable,
                value )
```

Get a value for the supplied variable within context.

Parameters

context	data context (e.g. board name, device name, ...)
variable	variable name
value	value to set

Returns

the variable's value, which defaults to DEFAULT_VALUE (0.0) if variable did not yet exist.

7.326.3.5 setVerbosity()

```
ttk_tools.offline_data_provider.OfflineDataProvider.setVerbosity (
              self,
              level = 0 )
```

Set verbosity level of log output.

Parameters

level	0: silent, 1: somewhat chatty

7.326.3.6 storeData()

```
ttk_tools.offline_data_provider.OfflineDataProvider.storeData (
              self,
              file_path = None )
```

Store current offline data to a file.

Note

Not yet implemented

Parameters

file_path	path to data file

7.326.4 Member Data Documentation

7.326.4.1 loaded_file_path

```
ttk_tools.offline_data_provider.OfflineDataProvider.loaded_file_path
```

7.326.4.2 offline_data

```
ttk_tools.offline_data_provider.OfflineDataProvider.offline_data
```

dictionary with offline data as {identifier: value} entries

7.326.4.3 verbosity

`ttk_tools.offline_data_provider.OfflineDataProvider.verbosity`

verbosity of status output

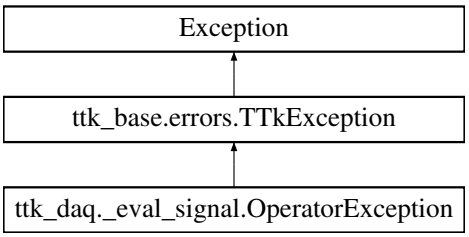
The documentation for this class was generated from the following file:

- ttk_tools/offline_data_provider.py (v1.5)

7.327 ttk_daq._eval_signal.OperatorException Class Reference

Errors during custom find operator function calls.

Inheritance diagram for ttk_daq._eval_signal.OperatorException:



7.327.1 Detailed Description

Errors during custom find operator function calls.

Class for evaluation of recorded signal data. A signal can be searched similar to an open file handle with functions to seek in time or data domains.

Note

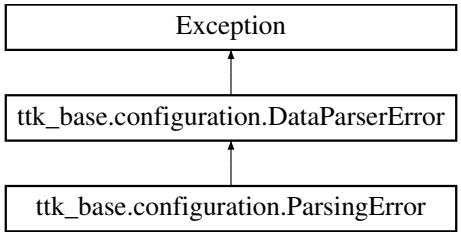
When searching for a certain data value, data points will be compared individually (values will **not** be interpolated between data points).
The first suitable/matching entry will be used/returned instead.

The documentation for this class was generated from the following file:

- ttk_daq/_eval_signal.py (v3.1.0)

7.328 ttk_base.configuration.ParsingError Class Reference

Inheritance diagram for ttk_base.configuration.ParsingError:



Public Member Functions

- __init__ (self, filename='<???>')
- append (self, lineno, line)
- __str__ (self)

Public Attributes

- filename

  file name where error occurred
- errors

  list of error messages as (lineno, line) tuples

Protected Attributes

- _msg

  keep message as member, so we can later append further error info

7.328.1 Constructor & Destructor Documentation

7.328.1.1 __init__()

```
ttk_base.configuration.ParsingError.__init__ (
            self,
            filename = '<???>' )
```

7.328.2 Member Function Documentation

7.328.2.1 __str__()

```
ttk_base.configuration.ParsingError.__str__ (
            self )
```

7.328.2.2 append()

```
ttk_base.configuration.ParsingError.append (
            self,
            lineno,
            line )
```

7.328.3 Member Data Documentation

7.328.3.1 _msg

```
ttk_base.configuration.ParsingError._msg  [protected]
```

keep message as member, so we can later append further error info

7.328.3.2 errors

`ttk_base.configuration.ParsingError.errors`

list of error messages as (lineno, line) tuples

7.328.3.3 filename

`ttk_base.configuration.ParsingError.filename`

file name where error occurred

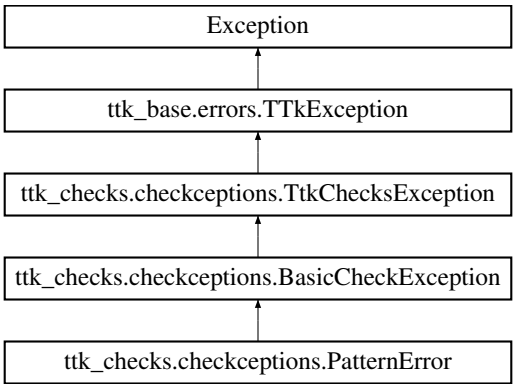
The documentation for this class was generated from the following file:

- ttk_base/configuration.py (v3.8.1)

7.329 ttk_checks.checkceptions.PatternError Class Reference

Exception: A supplied pattern is malformed and cannot be interpreted.

Inheritance diagram for ttk_checks.checkceptions.PatternError:



Public Member Functions

- __init__ (self, pattern='', msg='', log_msg=True)
  TtkChecksException init.

7.329.1 Detailed Description

Exception: A supplied pattern is malformed and cannot be interpreted.

7.329.2 Constructor & Destructor Documentation

7.329.2.1 __init__()

```
ttk_checks.checkceptions.PatternError.__init__ (
              self,
              msg = '',
              log_msg = '',
              log_msg = True )
```

TtkChecksException init.

Parameters

msg	exception message text
log_msg	if True, the exception message will also be logged / printed to stdout

Reimplemented from ttk_checks.checkceptions.TtkChecksException.

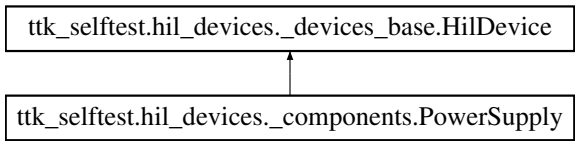
The documentation for this class was generated from the following file:

- ttk_checks/checkceptions.py (v2.0)

7.330 ttk_selftest.hil_devices._components.PowerSupply Class Reference

Base power supply container for a testable power supply and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices._components.PowerSupply:



Public Member Functions

- __init__ (self, context, power_channel, gnd_channel, base_path_cv_prog, base_path_cc_prog, base_path_cv_mon, base_path_cc_mon, max_voltage, max_current, abs_tol_v=0.05, rel_tol_v=0.01, abs_tol_c=0.05, rel_tol_c=0.01, preconditions=None, wait_time_ms=100, load_wait_time_ms=100, alias="", model="", descr="")

  Power Supply initialization.
- info (self)

  Get an info string for the Power Supply: alias, model, FIU channels, minimum and maximum value plus description, if available.
- setVoltage (self, voltage)

  Function to set the target voltage of the power supply.
- setCurrent (self, current)

  Function to set the current limit of the power supply.

Public Attributes

- max_voltage
- max_current
- load_wait_time_ms
- model
- gnd
- cv_prog
- cc_prog
- cv_mon
- cc_mon
- verdict_cvprog_to_ref
- verdict_cvprog_to_cv_mon
- verdict_cvmon_to_ref
- verdict_voltage
- verdict_ccprog_to_ref
- verdict_ccmon_to_ref
- verdict_current
- alias

7.330.1 Detailed Description

Base power supply container for a testable power supply and relevant associated variables/values.

7.330.2 Constructor & Destructor Documentation

7.330.2.1 __init__()

```
ttk_selftest.hil_devices._components.PowerSupply.__init__ (
              self,
              context,
              power_channel,
              gnd_channel,
              base_path_cv_prog,
              base_path_cc_prog,
              base_path_cv_mon,
              base_path_cc_mon,
              max_voltage,
              max_current,
              abs_tol_v = 0.05,
              rel_tol_v = 0.01,
              abs_tol_c = 0.05,
              rel_tol_c = 0.01,
              preconditions = None,
              wait_time_ms = 100,
              load_wait_time_ms = 100,
              alias = "",
              model = "",
              descr = "" )
```

Power Supply initialization.

Parameters

context	parent context
power_channel	fiu channel of the positive output of the power supply
gnd_channel	ground device of the power supply equals negative output of the power supply must be a GND device
base_path_cv_prog	base path to analog channel subsystem used for voltage programming
base_path_cc_prog	base path to analog channel subsystem used for current programming
base_path_cv_mon	base path to analog channel subsystem used for voltage monitoring
base_path_cc_mon	base path to analog channel subsystem used for current monitoring
max_voltage	maximum output voltage of the power supply
max_current	maximum output current of the power supply
abs_tol_v	absolute tolerance for voltage programming/ monitoring, defaults to 0.05 (V)
rel_tol_v	relative tolerance for voltage programming/ monitoring, defaults to 0.01 = 1%
abs_tol_c	absolute tolerance for current programming/ monitoring, defaults to 0.05 (A)
rel_tol_c	relative tolerance for current programming/ monitoring, defaults to 0.01 = 1%
preconditions	(opt) preconditions that have to be set before the device can be checked. defaults to None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The functions specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
    # For special cases, the preconditions may also contain function calls
    # (which might be useful to insert additional delays or similar):
    preconditions = [
        { "device": Voltage, "value": 1},
        { "func":   time.sleep,
            "args":   [1.0], # positional arguments
            "kwargs": {},    # keyword arguments
        },
        {"device": ClampSwitch, "value": 1},
        # or a simple call like this:
        { "func":  someCustomFunction,
        },
        ...
    ]
wait_time_ms      - (opt) waiting time in milliseconds between
power supply stimulation and check
defaults to 100ms
load_wait_time_ms - (opt) waiting time after load stimulation
necessary to unload inductive loads
defaults to 100ms
alias             - (opt) power supply alias/name
defaults to ""
model             - (opt) power supply model
defaults to ""
descr             - (opt) power supply description (e.g. used in report entries)
defaults to ""
```

Reimplemented from ttk_selftest.hil_devices._devices_base.HilDevice.

7.330.3 Member Function Documentation

7.330.3.1 info()

```
ttk_selftest.hil_devices._components.PowerSupply.info (
              self )
```

Get an info string for the Power Supply: alias, model, FIU channels, minimum and maximum value plus description, if available.

Returns

a formatted string.

7.330.3.2 setCurrent()

```
ttk_selftest.hil_devices._components.PowerSupply.setCurrent (
              self,
              current )
```

Function to set the current limit of the power supply.

Parameters

current	current limit in A to be set

Returns

the original current limit before the set operation occurred.

7.330.3.3 setVoltage()

```
ttk_selftest.hil_devices._components.PowerSupply.setVoltage (
              self,
              voltage )
```

Function to set the target voltage of the power supply.

Parameters

voltage	target voltage in V to be set

Returns

the original voltage before the set operation occurred.

7.330.4 Member Data Documentation

7.330.4.1 alias

ttk_selftest.hil_devices._components.PowerSupply.alias

7.330.4.2 cc_mon

ttk_selftest.hil_devices._components.PowerSupply.cc_mon

7.330.4.3 cc_prog

ttk_selftest.hil_devices._components.PowerSupply.cc_prog

7.330.4.4 cv_mon

ttk_selftest.hil_devices._components.PowerSupply.cv_mon

7.330.4.5 cv_prog

ttk_selftest.hil_devices._components.PowerSupply.cv_prog

7.330.4.6 gnd

ttk_selftest.hil_devices._components.PowerSupply.gnd

7.330.4.7 load_wait_time_ms

ttk_selftest.hil_devices._components.PowerSupply.load_wait_time_ms

7.330.4.8 max_current

ttk_selftest.hil_devices._components.PowerSupply.max_current

7.330.4.9 max_voltage

ttk_selftest.hil_devices._components.PowerSupply.max_voltage

7.330.4.10 model

ttk_selftest.hil_devices._components.PowerSupply.model

7.330.4.11 verdict_ccmon_to_ref

ttk_selftest.hil_devices._components.PowerSupply.verdict_ccmon_to_ref

7.330.4.12 verdict_ccprog_to_ref

ttk_selftest.hil_devices._components.PowerSupply.verdict_ccprog_to_ref

7.330.4.13 verdict_current

ttk_selftest.hil_devices._components.PowerSupply.verdict_current

7.330.4.14 verdict_cvmon_to_ref

ttk_selftest.hil_devices._components.PowerSupply.verdict_cvmon_to_ref

7.330.4.15 verdict_cvprog_to_cv_mon

ttk_selftest.hil_devices._components.PowerSupply.verdict_cvprog_to_cv_mon

7.330.4.16 verdict_cvprog_to_ref

ttk_selftest.hil_devices._components.PowerSupply.verdict_cvprog_to_ref

7.330.4.17 verdict_voltage

ttk_selftest.hil_devices._components.PowerSupply.verdict_voltage

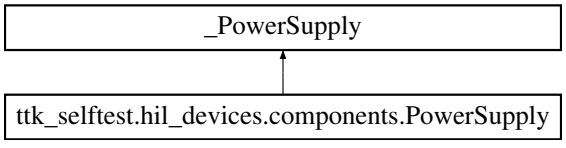
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/_components.py (v2.0.0)

7.331 ttk_selftest.hil_devices.components.PowerSupply Class Reference

Base power supply container for a testable power supply and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices.components.PowerSupply:



Public Member Functions

- __init__ (self, context, power_channel, gnd_channel, base_path_cv_prog, base_path_cc_prog, base_path_cv_mon, base_path_cc_mon, max_voltage, max_current, abs_tol_v=0.05, rel_tol_v=0.01, abs_tol_c=0.05, rel_tol_c=0.01, preconditions=None, wait_time_ms=100, load_wait_time_ms=100, alias="", model="", descr="")

  Power Supply initialization.
- info (self)

  Get an info string for the Power Supply: alias, model, FIU channels, minimum and maximum value plus description, if available.
- setVoltage (self, voltage)

  Function to set the target voltage of the power supply.
- setCurrent (self, current)

  Function to set the current limit of the power supply.

7.331.1 Detailed Description

Base power supply container for a testable power supply and relevant associated variables/values.

7.331.2 Constructor & Destructor Documentation

7.331.2.1 __init__()

```
ttk_selftest.hil_devices.components.PowerSupply.__init__ (
              self,
              context,
              power_channel,
              gnd_channel,
              base_path_cv_prog,
              base_path_cc_prog,
              base_path_cv_mon,
              base_path_cc_mon,
              max_voltage,
              max_current,
              abs_tol_v = 0.05,
              rel_tol_v = 0.01,
              abs_tol_c = 0.05,
              rel_tol_c = 0.01,
              preconditions = None,
              wait_time_ms = 100,
              load_wait_time_ms = 100,
              alias = "",
              model = "",
              descr = "" )
```

Power Supply initialization.

Parameters

context	parent context
power_channel	fiu channel of the positive output of the power supply
gnd_channel	ground device of the power supply equals negative output of the power supply must be a GND device

Parameters

base_path_cv_prog	base path to analog channel subsystem used for voltage programming
base_path_cc_prog	base path to analog channel subsystem used for current programming
base_path_cv_mon	base path to analog channel subsystem used for voltage monitoring
base_path_cc_mon	base path to analog channel subsystem used for current monitoring
max_voltage	maximum output voltage of the power supply
max_current	maximum output current of the power supply
abs_tol_v	absolute tolerance for voltage programming/ monitoring, defaults to 0.05 (V)
rel_tol_v	relative tolerance for voltage programming/ monitoring, defaults to 0.01 = 1%
abs_tol_c	absolute tolerance for current programming/ monitoring, defaults to 0.05 (A)
rel_tol_c	relative tolerance for current programming/ monitoring, defaults to 0.01 = 1%
preconditions	(opt) preconditions that have to be set before the device can be checked. defaults to None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The functions specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
    # For special cases, the preconditions may also contain function calls
    # (which might be useful to insert additional delays or similar):
    preconditions = [
        { "device": Voltage, "value": 1},
        { "func":   time.sleep,
          "args":   [1.0], # positional arguments
          "kwargs": {},    # keyword arguments
        },
        {"device": ClampSwitch, "value": 1},
        # or a simple call like this:
        { "func":  someCustomFunction,
        },
        ...
    ]
wait_time_ms      - (opt) waiting time in milliseconds between
power supply stimulation and check
defaults to 100ms
load_wait_time_ms - (opt) waiting time after load stimulation
necessary to unload inductive loads
defaults to 100ms
alias             - (opt) power supply alias/name
defaults to ""
model             - (opt) power supply model
defaults to ""
descr             - (opt) power supply description (e.g. used in report entries)
defaults to ""
```

7.331.3 Member Function Documentation

7.331.3.1 info()

```
ttk_selftest.hil_devices.components.PowerSupply.info (
               self )
```

Get an info string for the Power Supply: alias, model, FIU channels, minimum and maximum value plus description, if available.

Returns

a formatted string.

7.331.3.2 setCurrent()

```
ttk_selftest.hil_devices.components.PowerSupply.setCurrent (
              self,
              current )
```

Function to set the current limit of the power supply.

Parameters

current	current limit in A to be set

Returns

the original current limit before the set operation occurred.

7.331.3.3 setVoltage()

```
ttk_selftest.hil_devices.components.PowerSupply.setVoltage (
              self,
              voltage )
```

Function to set the target voltage of the power supply.

Parameters

voltage	target voltage in V to be set

Returns

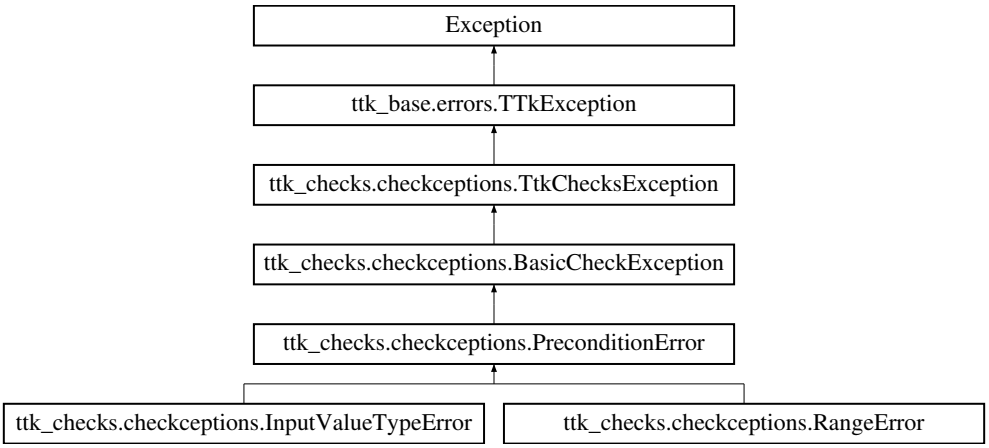
the original voltage before the set operation occurred.

The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/components.py (v2.0.0)

7.332 ttk_checks.checkceptions.PreconditionError Class Reference

Inheritance diagram for ttk_checks.checkceptions.PreconditionError:



Public Member Functions

- __init__ (self, msg='', log_msg=True)

  TtkChecksException init.

7.332.1 Constructor & Destructor Documentation

7.332.1.1 __init__()

```
ttk_checks.checkceptions.PreconditionError.__init__ (
              self,
              msg = '',
              log_msg = True )
```

TtkChecksException init.

Parameters

msg	exception message text
log_msg	if True, the exception message will also be logged / printed to stdout

Reimplemented from ttk_checks.checkceptions.TtkChecksException.

Reimplemented in ttk_checks.checkceptions.InputValueTypeError, and ttk_checks.checkceptions.RangeError.

The documentation for this class was generated from the following file:

- ttk_checks/checkceptions.py (v2.0)

7.333 ttk_tools.rst.isygamma.PV Class Reference

Gamma PV.

Public Member Functions

- __init__ (self, gaapi, address, properties=PVProperty.ACC_NON_BLOCKING)

  Process Variable access proxy.
- __del__ (self)
- __enter__ (self)

  Context handler: Enter.
- __exit__ (self, _exc_type, _exc_value, _tb)

  Context handler: Exit.
- __str__ (self)
- value (self)

  Get current value of PV.
- value (self, value)

  Set new value of PV.
- address (self)

  PV address as supplied in init.
- data_type (self)

  Data Type of PV.
- pv_type (self)

  Type/kind of PV.
- length (self)

  Get length / size info of PV for types STRING and BLOB.
- getField (self, field=PVFieldType.ORIG)

  Get field value of a numeric PV (string/blob types do not support fields).
- setField (self, value, field=PVFieldType.ORIG)

  Set field value of a numeric PV.

Protected Attributes

- _gaapi

7.333.1 Detailed Description

Gamma PV.

7.333.2 Constructor & Destructor Documentation

7.333.2.1 __init__()

```
ttk_tools.rst.isygamma.PV.__init__ (
              self,
              gaapi,
              address,
              properties = PVProperty.ACC_NON_BLOCKING )
```

Process Variable access proxy.

Usually invoked via _GaAPI.PV factory method (in gaapi instance).

Parameters

gaapi	reference to _GaAPI instance
address	PV "address" as string
properties	PV properties to set, see PVProperty

7.333.2.2 __del__()

```
ttk_tools.rst.isygamma.PV.__del__ (
              self )
```

7.333.3 Member Function Documentation

7.333.3.1 __enter__()

```
ttk_tools.rst.isygamma.PV.__enter__ (
              self )
```

Context handler: Enter.

7.333.3.2 __exit__()

```
ttk_tools.rst.isygamma.PV.__exit__ (
              self,
              _exc_type,
              _exc_value,
              _tb )
```

Context handler: Exit.

7.333.3.3 __str__()

```
ttk_tools.rst.isygamma.PV.__str__ (
              self )
```

7.333.3.4 address()

```
ttk_tools.rst.isygamma.PV.address (
              self )
```

PV address as supplied in init.

7.333.3.5 data_type()

```
ttk_tools.rst.isygamma.PV.data_type (
              self )
```

Data Type of PV.

7.333.3.6 getField()

```
ttk_tools.rst.isygamma.PV.getField (
              self,
              field = PVFieldType.ORIG )
```

Get field value of a numeric PV (string/blob types do not support fields).

Parameters

field	field type, see PVFieldType for available fields.

Returns

the field's value (if available/accessible) or None if PV is not a numeric type.

7.333.3.7 length()

```
ttk_tools.rst.isygamma.PV.length (
              self )
```

Get length / size info of PV for types STRING and BLOB.

Returns

a named tuple LengthInfo(cur_len, max_len) (or None, None for non-blob types)

7.333.3.8 pv_type()

```
ttk_tools.rst.isygamma.PV.pv_type (
              self )
```

Type/kind of PV.

7.333.3.9 setField()

```
ttk_tools.rst.isygamma.PV.setField (
              self,
              value,
              field = PVFieldType.ORIG )
```

Set field value of a numeric PV.

Parameters

field	field type, see PVFieldType for available fields.
value	value to set, has to match PV's data type.

7.333.3.10 value() [1/2]

```
ttk_tools.rst.isygamma.PV.value (
              self )
```

Get current value of PV.

7.333.3.11 value() [2/2]

```
ttk_tools.rst.isygamma.PV.value (
              self,
              value )
```

Set new value of PV.

7.333.4 Member Data Documentation

7.333.4.1 _gaapi

```
ttk_tools.rst.isygamma.PV._gaapi  [protected]
```

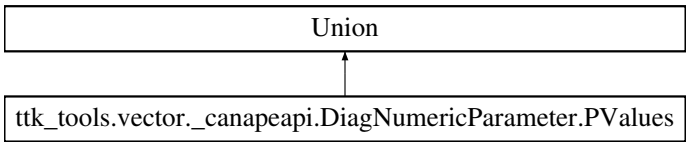
The documentation for this class was generated from the following file:

- ttk_tools/rst/isygamma.py (v1.6)

7.334 ttk_tools.vector._canapeapi.DiagNumericParameter.PValues Class Reference

Different value representations.

Inheritance diagram for ttk_tools.vector._canapeapi.DiagNumericParameter.PValues:



Static Protected Attributes

- int _pack_ = 1
- list _fields_

7.334.1 Detailed Description

Different value representations.

7.334.2 Member Data Documentation

7.334.2.1 _fields_

list ttk_tools.vector._canapeapi.DiagNumericParameter.PValues._fields_ [static], [protected]

Initial value:

```
=  [
            ("IVal",   c_int),    # 32 bits
            ("UIVal",  c_uint),   # 32 bits
            ("FVal",   c_float),  # 32 bits
            ("DVal",   c_double), # 64 bits
            ("Raw",    c_uint64)  # not available in original union,
        ]
```

7.334.2.2 _pack_

int ttk_tools.vector._canapeapi.DiagNumericParameter.PValues._pack_ = 1 [static], [protected]

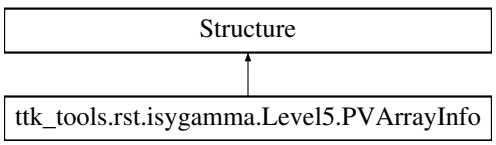
The documentation for this class was generated from the following file:

- ttk_tools/vector/_canapeapi.py (v7.3.3)

7.335 ttk_tools.rst.isygamma.Level5.PVArrayInfo Class Reference

typedef struct gc_PVArrayInfo

Inheritance diagram for ttk_tools.rst.isygamma.Level5.PVArrayInfo:



Static Protected Attributes

- list _fields_

7.335.1 Detailed Description

typedef struct gc_PVArrayInfo

7.335.2 Member Data Documentation

7.335.2.1 _fields_

list ttk_tools.rst.isygamma.Level5.PVArrayInfo._fields_ [static], [protected]

Initial value:

```
= [
            ("address",      Addresses.Level5), # address of the variable/PVArray
            ("name",         c_char * 1024),    # unique name of the PVArray
            ("type",         c_int),            # data type of PVs within the PV array (DataType Enum)
            ("noElements",   c_uint32),         # number of elements of the PV array
            ("length",       c_uint32),         # length of the gc_Blob type, if it is one
            ("isPersistent", c_bool),           #
        ]
```

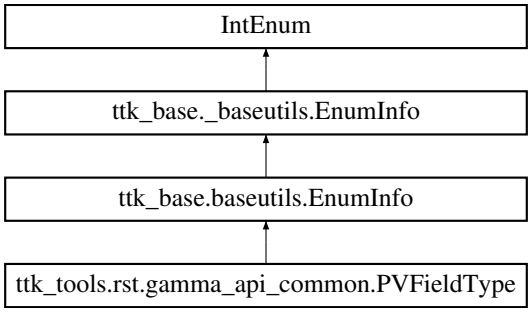
The documentation for this class was generated from the following file:

- ttk_tools/rst/isygamma.py (v1.6)

7.336 ttk_tools.rst.gamma_api_common.PVFieldType Class Reference

gc_PVFieldType - Field types of numeric PVs (gc_PV_FIELD_∗ entries).

Inheritance diagram for ttk_tools.rst.gamma_api_common.PVFieldType:



Static Public Attributes

- int VALUE = 1
- int ORIG = 2
- int SIM = 3
- int MIN = 4
- int MAX = 5

7.336.1 Detailed Description

gc_PVFieldType - Field types of numeric PVs (gc_PV_FIELD_∗ entries).

7.336.2 Member Data Documentation

7.336.2.1 MAX

```
int ttk_tools.rst.gamma_api_common.PVFieldType.MAX = 5  [static]
```

7.336.2.2 MIN

```
int ttk_tools.rst.gamma_api_common.PVFieldType.MIN = 4  [static]
```

7.336.2.3 ORIG

```
int ttk_tools.rst.gamma_api_common.PVFieldType.ORIG = 2  [static]
```

7.336.2.4 SIM

```
int ttk_tools.rst.gamma_api_common.PVFieldType.SIM = 3  [static]
```

7.336.2.5 VALUE

```
int ttk_tools.rst.gamma_api_common.PVFieldType.VALUE = 1  [static]
```

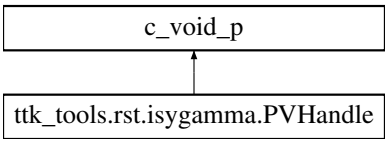
The documentation for this class was generated from the following file:

- ttk_tools/rst/gamma_api_common.py (v1.5)

7.337 ttk_tools.rst.isygamma.PVHandle Class Reference

Opaque PV handle.

Inheritance diagram for ttk_tools.rst.isygamma.PVHandle:



Public Member Functions

- __repr__ (self)

7.337.1 Detailed Description

Opaque PV handle.

7.337.2 Member Function Documentation

7.337.2.1 __repr__()

```
ttk_tools.rst.isygamma.PVHandle.__repr__ (
            self )
```

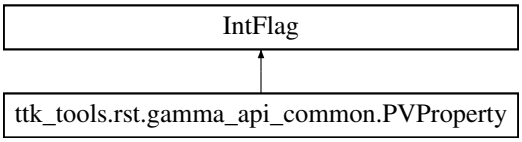
The documentation for this class was generated from the following file:

- ttk_tools/rst/isygamma.py (v1.6)

7.338 ttk_tools.rst.gamma_api_common.PVProperty Class Reference

Properties for PV operation/access mode.

Inheritance diagram for ttk_tools.rst.gamma_api_common.PVProperty:



Static Public Attributes

- int ACC_BLOCKING = 0b00000000_00000001
- int ACC_NON_BLOCKING = 0b00000000_00000010
- int LOOKUP_LOCAL = 0b00000000_00000100
- int LOOKUP_GLOBAL = 0b00000000_00001000
- int PUT_FULL_OVERWRITE = 0b00000000_00010000
- int PUT_FULL_BLOCK = 0b00000000_00100000
- int PUT_FULL_ERROR = 0b00000000_01000000
- int GET_EMPTY_BLOCK = 0b00000000_10000000
- int GET_EMPTY_ERROR = 0b00000001_00000000

7.338.1 Detailed Description

Properties for PV operation/access mode.

7.338.2 Member Data Documentation

7.338.2.1 ACC_BLOCKING

```
int ttk_tools.rst.gamma_api_common.PVProperty.ACC_BLOCKING = 0b00000000_00000001  [static]
```

7.338.2.2 ACC_NON_BLOCKING

```
int ttk_tools.rst.gamma_api_common.PVProperty.ACC_NON_BLOCKING = 0b00000000_00000010  [static]
```

7.338.2.3 GET_EMPTY_BLOCK

```
int ttk_tools.rst.gamma_api_common.PVProperty.GET_EMPTY_BLOCK = 0b00000000_10000000  [static]
```

7.338.2.4 GET_EMPTY_ERROR

```
int ttk_tools.rst.gamma_api_common.PVProperty.GET_EMPTY_ERROR = 0b00000001_00000000  [static]
```

7.338.2.5 LOOKUP_GLOBAL

```
int ttk_tools.rst.gamma_api_common.PVProperty.LOOKUP_GLOBAL = 0b00000000_00001000  [static]
```

7.338.2.6 LOOKUP_LOCAL

```
int ttk_tools.rst.gamma_api_common.PVProperty.LOOKUP_LOCAL = 0b00000000_00000100  [static]
```

7.338.2.7 PUT_FULL_BLOCK

```
int ttk_tools.rst.gamma_api_common.PVProperty.PUT_FULL_BLOCK = 0b00000000_00100000  [static]
```

7.338.2.8 PUT_FULL_ERROR

```
int ttk_tools.rst.gamma_api_common.PVProperty.PUT_FULL_ERROR = 0b00000000_01000000  [static]
```

7.338.2.9 PUT_FULL_OVERWRITE

```
int ttk_tools.rst.gamma_api_common.PVProperty.PUT_FULL_OVERWRITE = 0b00000000_00010000  [static]
```

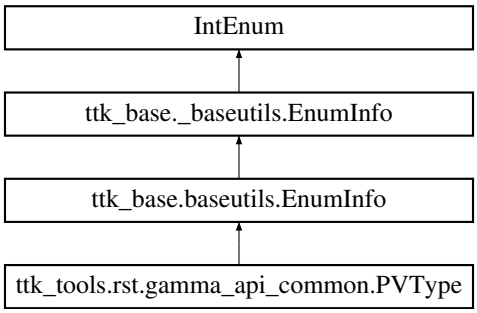
The documentation for this class was generated from the following file:

- ttk_tools/rst/gamma_api_common.py (v1.5)

7.339 ttk_tools.rst.gamma_api_common.PVType Class Reference

ga_PVType

Inheritance diagram for ttk_tools.rst.gamma_api_common.PVType:



Static Public Attributes

- int SINGLE_VALUE = 0x00
- int FIFO = 0x01
- int INVALID = 0xFF

7.339.1 Detailed Description

ga_PVType

7.339.2 Member Data Documentation

7.339.2.1 FIFO

```
int ttk_tools.rst.gamma_api_common.PVType.FIFO = 0x01  [static]
```

7.339.2.2 INVALID

```
int ttk_tools.rst.gamma_api_common.PVType.INVALID = 0xFF  [static]
```

7.339.2.3 SINGLE_VALUE

```
int ttk_tools.rst.gamma_api_common.PVType.SINGLE_VALUE = 0x00  [static]
```

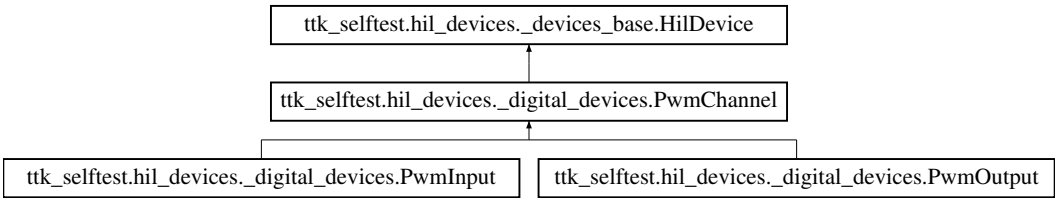
The documentation for this class was generated from the following file:

- ttk_tools/rst/gamma_api_common.py (v1.5)

7.340 ttk_selftest.hil_devices._digital_devices.PwmChannel Class Reference

Base PWM channel container for a testable PWM channel and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices._digital_devices.PwmChannel:



Public Member Functions

- __init__ (self, base_path, sig_channel, max_freq=None, abs_tol_freq=0.1, rel_tol_freq=0.05, abs_tol_dc=5, threshold_voltage=None, min_freq=None, preconditions=None, wait_time_ms=None, alias="", descr="")

  PWM channel initialization.
- info (self)

  Get an info string for this PWM channel: alias, FIU channel, threshold voltage, minimum and maximum frequency plus description, if available.

Public Attributes

- base_path
- dc_selftest_var
- freq_selfetst_var
- threshold_voltage
- max_freq
- min_freq
- abs_tol_freq
- rel_tol_freq
- abs_tol_dc

7.340.1 Detailed Description

Base PWM channel container for a testable PWM channel and relevant associated variables/values.

7.340.2 Constructor & Destructor Documentation

7.340.2.1 __init__()

```
ttk_selftest.hil_devices._digital_devices.PwmChannel.__init__ (
                self,
                base_path,
                sig_channel,
                max_freq = None,
                abs_tol_freq = 0.1,
                rel_tol_freq = 0.05,
                abs_tol_dc = 5,
                threshold_voltage = None,
                min_freq = None,
                preconditions = None,
                wait_time_ms = None,
                alias = "",
                descr = "" )
```

PWM channel initialization.

Parameters

base_path	base path to PWM channel subsystem
sig_channel	fiu channel of the PWM channel
max_freq	(opt) maximum frequency of the PWM channel defaults to "Inf"
abs_tol_freq	(opt) absolute frequency tolerance defaults to 0.1Hz
rel_tol_freq	(opt) relative frequency tolerance defaults to 0.05 = 5%
abs_tol_dc	(opt) absolute duty cycle tolerance defaults to 5%
threshold_voltage	(opt) voltage threshold in V for high/low detection defaults to 2.5V (due to the threshold voltage of the iSystemReferenceCard iSRC)
min_freq	(opt) minimum frequency in Hz of the PWM channel defaults to 0Hz
preconditions	(opt) preconditions that have to be set before the device can be checked. default is None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The function specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
# For special cases, the preconditions may also contain function calls
# (which might be useful to insert additional delays or similar):
preconditions = [
    { "device": Voltage, "value": 1},
    { "func":   time.sleep,
```

```
            "args":   [1.0], # positional arguments
            "kwargs": {},    # keyword arguments
          },
          {"device": ClampSwitch, "value": 1},
          # or a simple call like this:
          { "func":  someCustomFunction,
          },
          ...
      ]
    wait_time_ms      - (opt) waiting time in milliseconds between
    channel stimulation and channel check
    defaults to 100ms
    alias             - (opt) channel alias/name, defaults to the
    last (non-default) entry in identifier
    (i.e. signal model path)
    descr             - (opt) channel description (e.g. used in report entries)
```

Reimplemented from ttk_selftest.hil_devices._devices_base.HilDevice.

Reimplemented in ttk_selftest.hil_devices._digital_devices.PwmOutput, and ttk_selftest.hil_devices._digital_devices.PwmInput.

7.340.3 Member Function Documentation

7.340.3.1 info()

```
ttk_selftest.hil_devices._digital_devices.PwmChannel.info (
              self )
```

Get an info string for this PWM channel: alias, FIU channel, threshold voltage, minimum and maximum frequency plus description, if available.

Returns

a formatted string.

7.340.4 Member Data Documentation

7.340.4.1 abs_tol_dc

```
ttk_selftest.hil_devices._digital_devices.PwmChannel.abs_tol_dc
```

7.340.4.2 abs_tol_freq

```
ttk_selftest.hil_devices._digital_devices.PwmChannel.abs_tol_freq
```

7.340.4.3 base_path

```
ttk_selftest.hil_devices._digital_devices.PwmChannel.base_path
```

7.340.4.4 dc_selftest_var

```
ttk_selftest.hil_devices._digital_devices.PwmChannel.dc_selftest_var
```

7.340.4.5 freq_selfetst_var

ttk_selftest.hil_devices._digital_devices.PwmChannel.freq_selfetst_var

7.340.4.6 max_freq

ttk_selftest.hil_devices._digital_devices.PwmChannel.max_freq

7.340.4.7 min_freq

ttk_selftest.hil_devices._digital_devices.PwmChannel.min_freq

7.340.4.8 rel_tol_freq

ttk_selftest.hil_devices._digital_devices.PwmChannel.rel_tol_freq

7.340.4.9 threshold_voltage

ttk_selftest.hil_devices._digital_devices.PwmChannel.threshold_voltage

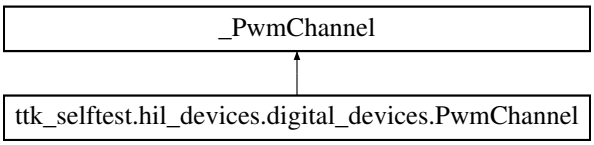
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/_digital_devices.py (v2.0.0)

7.341 ttk_selftest.hil_devices.digital_devices.PwmChannel Class Reference

Base PWM channel container for a testable PWM channel and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices.digital_devices.PwmChannel:



Public Member Functions

- __init__ (self, base_path, sig_channel, max_freq=None, abs_tol_freq=0.1, rel_tol_freq=0.05, abs_tol_dc=5, threshold_voltage=None, min_freq=None, preconditions=None, wait_time_ms=None, alias="", descr="")

  PWM channel initialization.
- info (self)

  Get an info string for this PWM channel: alias, FIU channel, threshold voltage, minimum and maximum frequency plus description, if available.

7.341.1 Detailed Description

Base PWM channel container for a testable PWM channel and relevant associated variables/values.

7.341.2 Constructor & Destructor Documentation

7.341.2.1 __init__()

```
ttk_selftest.hil_devices.digital_devices.PwmChannel.__init__ (
               self,
               base_path,
               sig_channel,
               max_freq = None,
               abs_tol_freq = 0.1,
               rel_tol_freq = 0.05,
               abs_tol_dc = 5,
               threshold_voltage = None,
               min_freq = None,
               preconditions = None,
               wait_time_ms = None,
               alias = "",
               descr = "" )
```

PWM channel initialization.

Parameters

base_path	base path to PWM channel subsystem
sig_channel	fiu channel of the PWM channel
max_freq	(opt) maximum frequency of the PWM channel defaults to "Inf"
abs_tol_freq	(opt) absolute frequency tolerance defaults to 0.1Hz
rel_tol_freq	(opt) relative frequency tolerance defaults to 0.05 = 5%
abs_tol_dc	(opt) absolute duty cycle tolerance defaults to 5%
threshold_voltage	(opt) voltage threshold in V for high/low detection defaults to 2.5V (due to the threshold voltage of the iSystemReferenceCard iSRC)
min_freq	(opt) minimum frequency in Hz of the PWM channel defaults to 0Hz
preconditions	(opt) preconditions that have to be set before the device can be checked. default is None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The function specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
    # For special cases, the preconditions may also contain function calls
    # (which might be useful to insert additional delays or similar):
    preconditions = [
        { "device": Voltage, "value": 1},
        { "func":   time.sleep,
          "args":   [1.0], # positional arguments
          "kwargs": {},    # keyword arguments
        },
        {"device": ClampSwitch, "value": 1},
        # or a simple call like this:
        { "func":  someCustomFunction,
        },
        ...
    ]
  wait_time_ms      - (opt) waiting time in milliseconds between
  channel stimulation and channel check
  defaults to 100ms
  alias             - (opt) channel alias/name, defaults to the
  last (non-default) entry in identifier
  (i.e. signal model path)
  descr             - (opt) channel description (e.g. used in report entries)
```

7.341.3 Member Function Documentation

7.341.3.1 info()

```
ttk_selftest.hil_devices.digital_devices.PwmChannel.info (
              self )
```

Get an info string for this PWM channel: alias, FIU channel, threshold voltage, minimum and maximum frequency plus description, if available.

Returns

a formatted string.

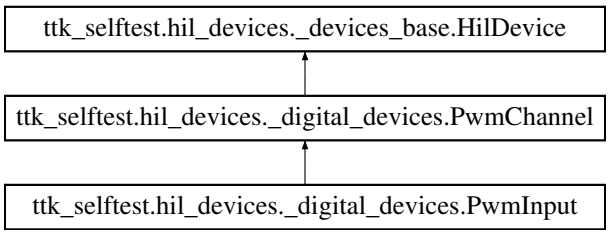
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/digital_devices.py (v2.0.0)

7.342 ttk_selftest.hil_devices._digital_devices.PwmInput Class Reference

Base PWM input container for a testable PWM input and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices._digital_devices.PwmInput:



Public Member Functions

- __init__ (self, context, base_path, sig_channel, max_voltage, max_freq=None, abs_tol_freq=0.↩ 1, rel_tol_freq=0.05, abs_tol_dc=5, threshold_voltage=None, min_freq=None, preconditions=None, wait_time_ms=None, alias="", descr="")

  PWM input initialization.

Public Attributes

- max_voltage
- wait_time_ms
- dc_selftest_var
- freq_selftest_var

7.342.1 Detailed Description

Base PWM input container for a testable PWM input and relevant associated variables/values.

7.342.2 Constructor & Destructor Documentation

7.342.2.1 __init__()

```
ttk_selftest.hil_devices._digital_devices.PwmInput.__init__ (
                self,
                context,
                base_path,
                sig_channel,
                max_voltage,
                max_freq = None,
                abs_tol_freq = 0.1,
                rel_tol_freq = 0.05,
                abs_tol_dc = 5,
                threshold_voltage = None,
                min_freq = None,
                preconditions = None,
                wait_time_ms = None,
                alias = "",
                descr = "" )
```

PWM input initialization.

Parameters

context	parent context
base_path	base path to PWM channel subsystem
sig_channel	fiu channel of the PWM channel
max_voltage	maximum allowed physical voltage in V of the channel
max_freq	(opt) maximum frequency of the PWM channel defaults to "Inf"
abs_tol_freq	(opt) absolute frequency tolerance defaults to 0.1Hz
rel_tol_freq	(opt) relative frequency tolerance defaults to 0.05 = 5%
abs_tol_dc	(opt) absolute duty cycle tolerance defaults to 5%

Parameters

threshold_voltage	(opt) voltage threshold in V for high/low detection defaults to 2.5V
min_freq	(opt) minimum frequency in Hz of the PWM channel defaults to 0Hz
preconditions	(opt) preconditions that have to be set before the device can be checked. default is None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The function specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
  # For special cases, the preconditions may also contain function calls
  # (which might be useful to insert additional delays or similar):
  preconditions = [
      { "device": Voltage, "value": 1},
      { "func":   time.sleep,
        "args":   [1.0], # positional arguments
        "kwargs": {},    # keyword arguments
      },
      {"device": ClampSwitch, "value": 1},
      # or a simple call like this:
      { "func":  someCustomFunction,
      },
      ...
  ]
wait_time_ms      - (opt) waiting time in milliseconds between
channel stimulation and channel check
defaults to 100ms
alias             - (opt) channel alias/name, defaults to the
last (non-default) entry in identifier
(i.e. signal model path)
descr             - (opt) channel description (e.g. used in report entries)
```

Reimplemented from ttk_selftest.hil_devices._digital_devices.PwmChannel.

7.342.3 Member Data Documentation

7.342.3.1 dc_selftest_var

ttk_selftest.hil_devices._digital_devices.PwmInput.dc_selftest_var

7.342.3.2 freq_selftest_var

ttk_selftest.hil_devices._digital_devices.PwmInput.freq_selftest_var

7.342.3.3 max_voltage

ttk_selftest.hil_devices._digital_devices.PwmInput.max_voltage

7.342.3.4 wait_time_ms

ttk_selftest.hil_devices._digital_devices.PwmInput.wait_time_ms

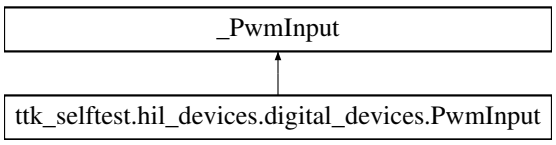
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/_digital_devices.py (v2.0.0)

7.343 ttk_selftest.hil_devices.digital_devices.PwmInput Class Reference

Base PWM input container for a testable PWM input and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices.digital_devices.PwmInput:



Public Member Functions

- __init__ (self, context, base_path, sig_channel, max_voltage, max_freq=None, abs_tol_freq=0.↩ 1, rel_tol_freq=0.05, abs_tol_dc=5, threshold_voltage=None, min_freq=None, preconditions=None, wait_time_ms=None, alias="", descr="")

  PWM input initialization.

7.343.1 Detailed Description

Base PWM input container for a testable PWM input and relevant associated variables/values.

7.343.2 Constructor & Destructor Documentation

7.343.2.1 __init__()

```
ttk_selftest.hil_devices.digital_devices.PwmInput.__init__ (
              self,
              context,
              base_path,
              sig_channel,
              max_voltage,
              max_freq = None,
              abs_tol_freq = 0.1,
              rel_tol_freq = 0.05,
              abs_tol_dc = 5,
              threshold_voltage = None,
              min_freq = None,
              preconditions = None,
              wait_time_ms = None,
              alias = "",
              descr = "" )
```

PWM input initialization.

Parameters

context	parent context
base_path	base path to PWM channel subsystem
sig_channel	fiu channel of the PWM channel
max_voltage	maximum allowed physical voltage in V of the channel
max_freq	(opt) maximum frequency of the PWM channel defaults to "Inf"
abs_tol_freq	(opt) absolute frequency tolerance defaults to 0.1Hz
rel_tol_freq	(opt) relative frequency tolerance defaults to 0.05 = 5%
abs_tol_dc	(opt) absolute duty cycle tolerance defaults to 5%
threshold_voltage	(opt) voltage threshold in V for high/low detection defaults to 2.5V
min_freq	(opt) minimum frequency in Hz of the PWM channel defaults to 0Hz
preconditions	(opt) preconditions that have to be set before the device can be checked. default is None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The function specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
  # For special cases, the preconditions may also contain function calls
  # (which might be useful to insert additional delays or similar):
  preconditions = [
      { "device": Voltage, "value": 1},
      { "func":   time.sleep,
        "args":   [1.0], # positional arguments
        "kwargs": {},    # keyword arguments
      },
      {"device": ClampSwitch, "value": 1},
      # or a simple call like this:
      { "func":  someCustomFunction,
      },
      ...
  ]
wait_time_ms      - (opt) waiting time in milliseconds between
channel stimulation and channel check
defaults to 100ms
alias             - (opt) channel alias/name, defaults to the
last (non-default) entry in identifier
(i.e. signal model path)
descr             - (opt) channel description (e.g. used in report entries)
```

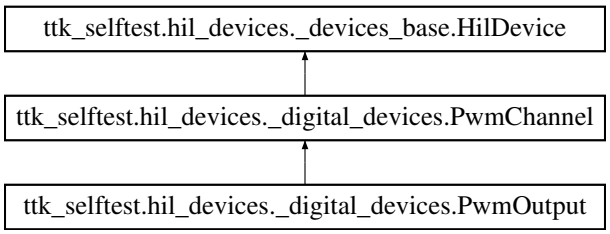
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/digital_devices.py (v2.0.0)

7.344 ttk_selftest.hil_devices._digital_devices.PwmOutput Class Reference

Base PWM output container for a testable PWM output and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices._digital_devices.PwmOutput:



Public Member Functions

- __init__ (self, context, base_path, sig_channel, max_freq=None, abs_tol_freq=0.1, rel_tol_freq=0.01, abs_tol_dc=5, threshold_voltage=None, min_freq=None, preconditions=None, wait_time_ms=None, alias="", descr="")

  PWM output initialization.

Public Attributes

- dc_selftest_var
- freq_selftest_var

7.344.1 Detailed Description

Base PWM output container for a testable PWM output and relevant associated variables/values.

7.344.2 Constructor & Destructor Documentation

7.344.2.1 __init__()

```
ttk_selftest.hil_devices._digital_devices.PwmOutput.__init__ (
              self,
              context,
              base_path,
              sig_channel,
              max_freq = None,
              abs_tol_freq = 0.1,
              rel_tol_freq = 0.01,
              abs_tol_dc = 5,
              threshold_voltage = None,
              min_freq = None,
              preconditions = None,
              wait_time_ms = None,
              alias = "",
              descr = "" )
```

PWM output initialization.

Parameters

context	parent context
base_path	base path to PWM channel subsystem
sig_channel	fiu channel of the PWM channel
max_freq	(opt) maximum frequency of the PWM channel defaults to "Inf"
abs_tol_freq	(opt) absolute frequency tolerance defaults to 0.1Hz
rel_tol_freq	(opt) relative frequency tolerance defaults to 0.01 = 1%
abs_tol_dc	(opt) absolute duty cycle tolerance defaults to 5%
threshold_voltage	(opt) voltage threshold in V for high/low detection defaults to 2.5V
min_freq	(opt) minimum frequency in Hz of the PWM channel defaults to 0Hz
preconditions	(opt) preconditions that have to be set before the device can be checked. default is None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The function specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
  # For special cases, the preconditions may also contain function calls
  # (which might be useful to insert additional delays or similar):
  preconditions = [
      { "device": Voltage, "value": 1},
      { "func":   time.sleep,
        "args":   [1.0], # positional arguments
        "kwargs": {},    # keyword arguments
      },
      {"device": ClampSwitch, "value": 1},
      # or a simple call like this:
      { "func":  someCustomFunction,
      },
      ...
  ]
wait_time_ms      - (opt) waiting time in milliseconds between
channel stimulation and channel check
defaults to 100ms
alias             - (opt) channel alias/name, defaults to the
last (non-default) entry in identifier
(i.e. signal model path)
descr             - (opt) channel description (e.g. used in report entries)
```

Reimplemented from ttk_selftest.hil_devices._digital_devices.PwmChannel.

7.344.3 Member Data Documentation

7.344.3.1 dc_selftest_var

ttk_selftest.hil_devices._digital_devices.PwmOutput.dc_selftest_var

7.344.3.2 freq_selftest_var

```
ttk_selftest.hil_devices._digital_devices.PwmOutput.freq_selftest_var
```

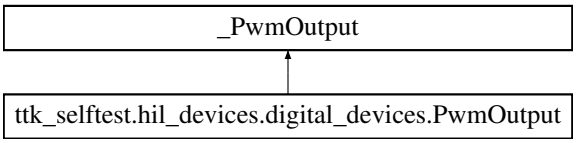
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/_digital_devices.py (v2.0.0)

7.345 ttk_selftest.hil_devices.digital_devices.PwmOutput Class Reference

Base PWM output container for a testable PWM output and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices.digital_devices.PwmOutput:



Public Member Functions

- __init__ (self, context, base_path, sig_channel, max_freq=None, abs_tol_freq=0.1, rel_tol_freq=0.01, abs_tol_dc=5, threshold_voltage=None, min_freq=None, preconditions=None, wait_time_ms=None, alias="", descr="")

  PWM output initialization.

7.345.1 Detailed Description

Base PWM output container for a testable PWM output and relevant associated variables/values.

7.345.2 Constructor & Destructor Documentation

7.345.2.1 __init__()

```
ttk_selftest.hil_devices.digital_devices.PwmOutput.__init__ (
              self,
              context,
              base_path,
              sig_channel,
              max_freq = None,
              abs_tol_freq = 0.1,
              rel_tol_freq = 0.01,
              abs_tol_dc = 5,
              threshold_voltage = None,
              min_freq = None,
              preconditions = None,
              wait_time_ms = None,
              alias = "",
              descr = "" )
```

PWM output initialization.

Parameters

context	parent context
base_path	base path to PWM channel subsystem
sig_channel	fiu channel of the PWM channel
max_freq	(opt) maximum frequency of the PWM channel defaults to "Inf"
abs_tol_freq	(opt) absolute frequency tolerance defaults to 0.1Hz
rel_tol_freq	(opt) relative frequency tolerance defaults to 0.01 = 1%
abs_tol_dc	(opt) absolute duty cycle tolerance defaults to 5%
threshold_voltage	(opt) voltage threshold in V for high/low detection defaults to 2.5V
min_freq	(opt) minimum frequency in Hz of the PWM channel defaults to 0Hz
preconditions	(opt) preconditions that have to be set before the device can be checked. default is None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The function specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
  # For special cases, the preconditions may also contain function calls
  # (which might be useful to insert additional delays or similar):
  preconditions = [
      { "device": Voltage, "value": 1},
      { "func":   time.sleep,
        "args":   [1.0], # positional arguments
        "kwargs": {},    # keyword arguments
      },
      {"device": ClampSwitch, "value": 1},
      # or a simple call like this:
      { "func":  someCustomFunction,
      },
      ...
  ]
wait_time_ms      - (opt) waiting time in milliseconds between
channel stimulation and channel check
defaults to 100ms
alias             - (opt) channel alias/name, defaults to the
last (non-default) entry in identifier
(i.e. signal model path)
descr             - (opt) channel description (e.g. used in report entries)
```

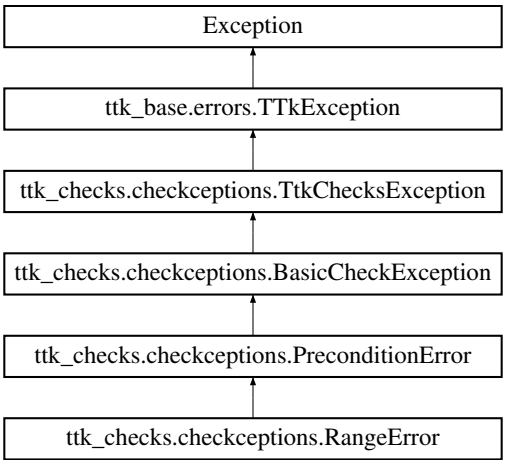
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/digital_devices.py (v2.0.0)

7.346 ttk_checks.checkceptions.RangeError Class Reference

A (supplied) value is outside the supported range (but has a suitable data type otherwise).

Inheritance diagram for ttk_checks.checkceptions.RangeError:



Public Member Functions

- __init__ (self, msg='', log_msg=True)
  TtkChecksException init.

7.346.1 Detailed Description

A (supplied) value is outside the supported range (but has a suitable data type otherwise).

7.346.2 Constructor & Destructor Documentation

7.346.2.1 __init__()

```
ttk_checks.checkceptions.RangeError.__init__ (
              self,
              msg = '',
              log_msg = True )
```

TtkChecksException init.

Parameters

msg	exception message text
log_msg	if True, the exception message will also be logged / printed to stdout

Reimplemented from ttk_checks.checkceptions.PreconditionError.

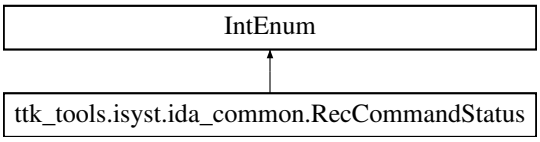
The documentation for this class was generated from the following file:

- ttk_checks/checkceptions.py (v2.0)

7.347 ttk_tools.isyst.ida_common.RecCommandStatus Class Reference

Recorder command RPC status info (internal success/error).

Inheritance diagram for ttk_tools.isyst.ida_common.RecCommandStatus:



Public Member Functions

- info (self)

Static Public Attributes

- int success = 0
- int bad_config = 1
- int wrong_state = 2
- int already_in_state = 3

7.347.1 Detailed Description

Recorder command RPC status info (internal success/error).

7.347.2 Member Function Documentation

7.347.2.1 info()

```
ttk_tools.isyst.ida_common.RecCommandStatus.info (
              self )
```

7.347.3 Member Data Documentation

7.347.3.1 already_in_state

```
int ttk_tools.isyst.ida_common.RecCommandStatus.already_in_state = 3  [static]
```

7.347.3.2 bad_config

```
int ttk_tools.isyst.ida_common.RecCommandStatus.bad_config = 1  [static]
```

7.347.3.3 success

```
int ttk_tools.isyst.ida_common.RecCommandStatus.success = 0  [static]
```

7.347.3.4 wrong_state

```
int ttk_tools.isyst.ida_common.RecCommandStatus.wrong_state = 2  [static]
```

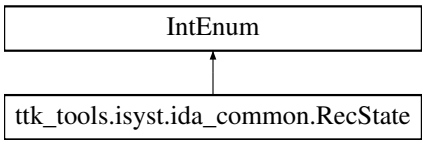
The documentation for this class was generated from the following file:

- ttk_tools/isyst/ida_common.py (v1.2.0)

7.348 ttk_tools.isyst.ida_common.RecState Class Reference

Recorder state machine state.

Inheritance diagram for ttk_tools.isyst.ida_common.RecState:



Public Member Functions

- info (self)

Static Public Attributes

- int undefined = 0
- int not_configured = 1
- int configuring = 2
- int ready = 3
- int start_recording = 4
- int recording = 5
- int stop_recording = 6
- int stopped = 7
- int sending_data = 8

7.348.1 Detailed Description

Recorder state machine state.

7.348.2 Member Function Documentation

7.348.2.1 info()

```
ttk_tools.isyst.ida_common.RecState.info (
              self )
```

7.348.3 Member Data Documentation

7.348.3.1 configuring

```
int ttk_tools.isyst.ida_common.RecState.configuring = 2  [static]
```

7.348.3.2 not_configured

```
int ttk_tools.isyst.ida_common.RecState.not_configured = 1  [static]
```

7.348.3.3 ready

```
int ttk_tools.isyst.ida_common.RecState.ready = 3  [static]
```

7.348.3.4 recording

```
int ttk_tools.isyst.ida_common.RecState.recording = 5  [static]
```

7.348.3.5 sending_data

```
int ttk_tools.isyst.ida_common.RecState.sending_data = 8  [static]
```

7.348.3.6 start_recording

```
int ttk_tools.isyst.ida_common.RecState.start_recording = 4  [static]
```

7.348.3.7 stop_recording

```
int ttk_tools.isyst.ida_common.RecState.stop_recording = 6  [static]
```

7.348.3.8 stopped

```
int ttk_tools.isyst.ida_common.RecState.stopped = 7  [static]
```

7.348.3.9 undefined

```
int ttk_tools.isyst.ida_common.RecState.undefined = 0  [static]
```

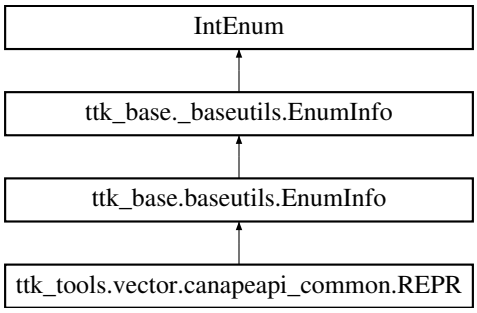
The documentation for this class was generated from the following file:

- ttk_tools/isyst/ida_common.py (v1.2.0)

7.349 ttk_tools.vector.canapeapi_common.REPR Class Reference

Measurement data format representation constants from enum `TFormat` (original constants have no specific prefix)

Inheritance diagram for ttk_tools.vector.canapeapi_common.REPR:



Static Public Attributes

- int HEX = 0

  ECU-internal data format.
- int PHYS = 1

  physical value representation

7.349.1 Detailed Description

Measurement data format representation constants from enum `TFormat` (original constants have no specific prefix)

7.349.2 Member Data Documentation

7.349.2.1 HEX

```
int ttk_tools.vector.canapeapi_common.REPR.HEX = 0  [static]
```

ECU-internal data format.

7.349.2.2 PHYS

```
int ttk_tools.vector.canapeapi_common.REPR.PHYS = 1  [static]
```

physical value representation

The documentation for this class was generated from the following file:

- ttk_tools/vector/canapeapi_common.py (v3.1.2)

7.350 ttk_base._variables_base.ResettableContainer Class Reference

Simple container class that provides a non-recursive reset() method.

Inheritance diagram for ttk_base._variables_base.ResettableContainer:



Public Member Functions

- __init__ (self, resettable=True, reset_nested_containers=False)
- reset (self, *args, **kwargs)

  Call reset() on all contained members (variables/signals) that support it.

Public Attributes

- resettable
- reset_nested_containers

7.350.1 Detailed Description

Simple container class that provides a non-recursive reset() method.

Intended as mix-in for PDUs/Messages/Signals or other collections of associated variables without the need for nested data structures.

7.350.2 Constructor & Destructor Documentation

7.350.2.1 __init__()

```
ttk_base._variables_base.ResettableContainer.__init__ (
              self,
              resettable = True,
              reset_nested_containers = False )
```

Parameters

resettable	True: this ResettableContainer can be reset False: container should not be reset (for special cases)
reset_nested_containers	True: reset() on this container will also reset any found ResettableContainers False: only reset members that are not instances of ResettableContainers themselves

Note

reset_nested_containers use-case Typical use-case are BusSignals and BusMessages. Both are derived from ResettableContainer (as they may contain a set of related Var instances).
However, a BusSignal references its parent BusMessage so resetting a BusSignal would also reset the Bus↩ Message. This might - for example - re-enable a manually disabled message, which is usually not the intended behavior.

Reimplemented in ttk_bus._bus_signals_cangamma.CanBusMessage, ttk_bus._bus_signals_canida.CanBusMessage, ttk_bus._bus_signals_cangamma.TxCanBusMessage, ttk_bus._bus_signals_canida.TxCanBusMessage, ttk_bus._bus_signals_canid ttk_bus.bus_signals_cangamma.TxCanBusMessage, ttk_bus.bus_signals_canida.TxCanBusMessage, ttk_bus.bus_signals_canida.F ttk_bus._bus_signals_frbs.CrcFrBusSignal, ttk_bus._bus_signals_frbs.AcFrBusSignal, ttk_bus.bus_signals_frbs.CrcFrBusSignal, ttk_bus.bus_signals_frbs.AcFrBusSignal, ttk_bus._bus_signals_ethbs.CrcEthSignal, ttk_bus._bus_signals_ethbs.AcEthSignal, ttk_bus.bus_signals_ethbs.CrcEthSignal, ttk_bus.bus_signals_ethbs.AcEthSignal, ttk_bus._bus_signals_ethbs.RxEthBusSysSignal, ttk_bus.bus_signals_ethbs.RxEthBusSysSignal, ttk_bus._bus_signals_frbs.FrBusSignal, ttk_bus._bus_signals_frbs.TxFrBusSignal, ttk_bus._bus_signals_frbs.RxFrBusSignal, ttk_bus.bus_signals_frbs.TxFrBusSignal, ttk_bus.bus_signals_frbs.RxFrBusSignal, ttk_bus._bus_signals_canmmbs.CanBusMessage, ttk_bus._bus_signals_canmmbs.TxCanBusMessage, ttk_bus.bus_signals_canmm ttk_bus.bus_signals_ethbs.TxEthEvent, ttk_bus._bus_signals_ethbs.EthEvent, ttk_bus._bus_signals_frbs.FrPdu, ttk_bus._bus_signals_frbs.FrTxPdu, ttk_bus.bus_signals_frbs.FrTxPdu, ttk_bus._bus_signals_frbs.FrRxPdu, ttk_bus.bus_signals_frbs.FrRxPdu, ttk_bus._bus_signals_cangamma.RxCanBusMessage, ttk_bus.bus_signals_cangamma.RxCanBu ttk_bus._bus_signals_ethbs.RxEthEvent, ttk_bus.bus_signals_ethbs.RxEthEvent, ttk_bus._bus_signals_canmmbs.RxCanBusMessag ttk_bus.bus_signals_canmmbs.RxCanBusMessage, ttk_bus._bus_signals_base.CrcBusSignal, ttk_bus._bus_signals_base.AcBusSig ttk_bus.bus_signals_base.CrcBusSignal, ttk_bus.bus_signals_base.AcBusSignal, ttk_bus._bus_signals_base.BusSignal, ttk_bus.bus_signals_base.TxBusSignal, ttk_bus.bus_signals_base.RxBusSignal, ttk_bus._bus_signals_ethbs.EthBusSysSignal, ttk_bus._bus_signals_ethbs.EthSignal, ttk_bus._bus_signals_ethbs.RxEthSignal, ttk_bus.bus_signals_ethbs.RxEthSignal, ttk_bus._bus_signals_canmmbs.CrcCanBusSignal, ttk_bus._bus_signals_canmmbs.AcCanBusSignal, ttk_bus.bus_signals_canmmbs ttk_bus.bus_signals_canmmbs.AcCanBusSignal, ttk_bus._bus_signals_canmmbs.CanBusSignal, ttk_bus._bus_signals_canmmbs.Rx ttk_bus.bus_signals_canmmbs.TxCanBusSignal, and ttk_bus.bus_signals_canmmbs.RxCanBusSignal.

7.350.3 Member Function Documentation

7.350.3.1 reset()

```
ttk_base._variables_base.ResettableContainer.reset (
              self,
             * args,
             ** kwargs )
```

Call reset() on all contained members (variables/signals) that support it.

Simple, non-recursive approach.

Note

Each member will only be reset once, even if there are multiple references to the same member.

Parameters

**args*	unused, just as precaution to tolerate optional parameters
***kwargs*	unused, just as precaution to tolerate optional parameters

Reimplemented in ttk_bus.bus_signals_base.TxBusSignal, ttk_bus.bus_signals_base.RxBusSignal, ttk_bus.bus_signals_base.CrcBu
ttk_bus.bus_signals_base.AcBusSignal, ttk_bus.bus_signals_cangamma.TxCanBusMessage, ttk_bus.bus_signals_cangamma.RxCa
ttk_bus.bus_signals_canida.TxCanBusMessage, ttk_bus.bus_signals_canida.RxCanBusMessage, ttk_bus.bus_signals_canmmbs.Tx
ttk_bus.bus_signals_canmmbs.RxCanBusMessage, ttk_bus.bus_signals_canmmbs.TxCanBusSignal, ttk_bus.bus_signals_canmmbs
ttk_bus.bus_signals_canmmbs.CrcCanBusSignal, ttk_bus.bus_signals_canmmbs.AcCanBusSignal, ttk_bus.bus_signals_ethbs.RxEth
ttk_bus.bus_signals_ethbs.TxEthEvent, ttk_bus.bus_signals_ethbs.RxEthSignal, ttk_bus.bus_signals_ethbs.TxEthSignal,
ttk_bus.bus_signals_ethbs.RxEthBusSysSignal, ttk_bus.bus_signals_ethbs.TxEthBusSysSignal, ttk_bus.bus_signals_ethbs.CrcEthSi
ttk_bus.bus_signals_ethbs.AcEthSignal, ttk_bus.bus_signals_frbs.FrTxPdu, ttk_bus.bus_signals_frbs.FrRxPdu,
ttk_bus.bus_signals_frbs.TxFrBusSignal, ttk_bus.bus_signals_frbs.RxFrBusSignal, ttk_bus.bus_signals_frbs.CrcFrBusSignal,
and ttk_bus.bus_signals_frbs.AcFrBusSignal.

7.350.4 Member Data Documentation

7.350.4.1 reset_nested_containers

`ttk_base._variables_base.ResettableContainer.reset_nested_containers`

7.350.4.2 resettable

`ttk_base._variables_base.ResettableContainer.resettable`

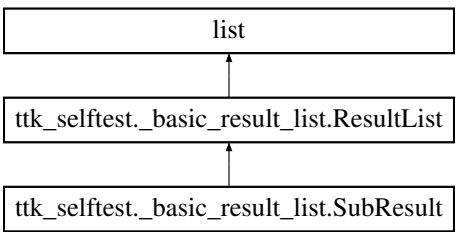
The documentation for this class was generated from the following file:

- ttk_base/_variables_base.py (v2.8.1)

7.351 ttk_selftest._basic_result_list.ResultList Class Reference

Add logging, some sanity checks and helper functions to a plain list class in order to make it more convenient for storing test result entries.

Inheritance diagram for ttk_selftest._basic_result_list.ResultList:



Public Member Functions

- __init__ (self)
- __iadd__ (self, other)
- append (self, other)
- extend (self, other_iterable)
- add (self, *items)

  Add a single result, last item has to contain test verdict.
- addHeader (self, text)
- addSubHeader (self, text)
- addInfo (self, *msg)
- addError (self, msg, ex=None, show_tb=True)

  Add an ERROR result.
- addBoolean (self, *items)

  Add a boolean result.
- addCheckBitMask (self, value, mask, operator="AND", bit_length=None, pass_on_true=True, descr="")

  Check result of the operation given in `operator`.
- addCheckBitPattern (self, value, pattern, invert=False, descr="")

  Check that a numeric value matches a certain pattern.
- addCheckBit (self, value, offset, status=1, descr="")

  Check if the bit given by the value and offset is set or cleared depending on the status.
- addCheckTolerance (self, current_value, rated_value, rel_pos=0, abs_pos=0, rel_neg=None, abs_neg=None, invert=False, descr="")

  Check if the current value is within a specified range of the rated value.
- addCheckAbsTol (self, current_value, rated_value, abs_tol=0, invert=False, descr="")

  Check if the current value is within a specified range of the rated value.
- addCheckRelTol (self, current_value, rated_value, rel_tol=0.0, invert=False, descr="")

  Check if the current value is within a specified range of the rated value.
- addCheckStatus (self, current_status, nominal_status, equal=True, descr="")

  Check if the current status value is identical to the nominal status value.
- addCheckRange (self, value, min_value, max_value, invert=False, descr="")

  Check if value is in defined range (`min_value <= value <= max_value`).
- addCompare (self, left_value, operator, right_value, abs_tol=0, descr="")

  Verify the conditional relationship between two operands based on the given operator.
- addCompareCE (self, current_value, operator, expected_value, abs_tol=0, descr="")

  Verify the conditional relationship between current and expected value based on the given operator.
- addContains (self, defined_values, current_value, mode="all", descr="")

  Check if current_value is an element (or sub set) of defined_values.
- addTestVars (self, set_var_list=None, check_var_list=None, min_wait_time_ms=100, descr='')

  Check the values of multiple variables/signals as a result of setting multiple other variables/signals.
- setTestcaseId (self, testcase_id=None, *testcase_ids)

  Set the currently active Testcase-ID (or IDs) to be assigned to all following result entries.
- clearTestcaseId (self)

  Clear the current Testcase-ID, following results will not get IDs assigned.
- enableEcho (self, enable=True)

  en-/disable echoing of new entries to stdout.
- getCombinedResult (self, divider=None, trim_descrs=None, failed_prefix=None, compact_descrs=None)

  Get a single result entry containing the joined description texts and the resulting verdict of all results currently present in this ResultList instance.

Public Attributes

- header_format_verdicts

  For which verdicts to use "header" formatting in log (default: empty verdicts)
- header_block_width

  width of "header" format blocks in log output
- echo_enabled

  en-/disable echoing of new entries, see enableEcho()
- divider

  getCombinedResult: default string to place between joined description texts.
- failed_prefix

  getCombinedResult: Prefix to add to descriptions of failed test steps.
- trim_descrs

  getCombinedResult: True: whitespace will be trimmed/stripped from beginning and end of each description to make output more compact.
- compact_descrs

  getCombinedResult: True: "compact" descriptions by removing all lines from the beginning that are already present in the directly preceding description.
- current_testcase_id

  Currently active testcase ID (or potentially multiple Testcase-IDs).

Protected Member Functions

- _getResultingVerdict (self, verdicts)

  Get the resulting verdict of the supplied entries according to the configured verdict group settings.

Protected Attributes

- _fail_verdicts
- _pass_verdicts
- _info_verdicts

7.351.1 Detailed Description

Add logging, some sanity checks and helper functions to a plain list class in order to make it more convenient for storing test result entries.

7.351.2 Constructor & Destructor Documentation

7.351.2.1 __init__()

```
ttk_selftest._basic_result_list.ResultList.__init__ (
              self )
```

Reimplemented in ttk_selftest._basic_result_list.SubResult.

7.351.3 Member Function Documentation

7.351.3.1 __iadd__()

```
ttk_selftest._basic_result_list.ResultList.__iadd__ (
              self,
              other )
```

7.351.3.2 _getResultingVerdict()

```
ttk_selftest._basic_result_list.ResultList._getResultingVerdict (
              self,
              verdicts )  [protected]
```

Get the resulting verdict of the supplied entries according to the configured verdict group settings.

Fail > Pass > Info > (others). See ResultList._fail_verdicts, ._pass_verdicts and ._info_verdicts.

Parameters

verdicts	list/tuple/set of verdicts to process

Returns

the resulting verdict string.

7.351.3.3 add()

```
ttk_selftest._basic_result_list.ResultList.add (
              self,
             * items )
```

Add a single result, last item has to contain test verdict.

Example:

```
# result.add vs. default list.append
result.append(["argument is a list", "with result entry items", "VERDICT"])
result.add("items", "as separate arguments", "VERDICT")
```

7.351.3.4 addBoolean()

```
ttk_selftest._basic_result_list.ResultList.addBoolean (
              self,
             * items )
```

Add a boolean result.

Last item will be interpreted as bool (True=>PASSED, False=>FAILED). If less than two items a present, verdict will be ERROR.

7.351.3.5 addCheckAbsTol()

```
ttk_selftest._basic_result_list.ResultList.addCheckAbsTol (
               self,
               current_value,
               rated_value,
               abs_tol = 0,
               invert = False,
               descr = "" )
```

Check if the current value is within a specified range of the rated value.

The valid range is defined by an absolute tolerance as a symmetric range around the `rated_value`. Result will be added to this ResultList.

Parameters

current_value	value to be checked
rated_value	expected value
abs_tol	absolute tolerance value, valid range will be `rated_value` +/- `abs_tol`
invert	True: inverted, check current value != rated value (+/- tolerance) False: default, check current value == rated value (+/- tolerance)
descr	user specific description shown before the single test result

7.351.3.6 addCheckBit()

```
ttk_selftest._basic_result_list.ResultList.addCheckBit (
               self,
               value,
               offset,
               status = 1,
               descr = "" )
```

Check if the bit given by the value and offset is set or cleared depending on the status.

Result will be added to this ResultList.

Parameters

value	Numeric value where the bit has to be checked
offset	Offset of the bit inside of value (based on 0)
status	The expected value of the bit (default 1)
descr	user specific description printed on the start of the single test result

7.351.3.7 addCheckBitMask()

```
ttk_selftest._basic_result_list.ResultList.addCheckBitMask (
               self,
               value,
               mask,
```

```
                operator = "AND",
                bit_length = None,
                pass_on_true = True,
                descr = "" )
```

Check result of the operation given in `operator`.

Result will be added to this ResultList. Depending on parameter `pass_on_true`, the test will be PASSED if the result is True (i.e. 'greater 0') or False (i.e. 'equal 0').

Parameters

value	value to be checked
mask	mask to be used for operation in operator
operator	operator for check operation, one of ("AND", "OR", "XOR", "NAND", "NOR", "NXOR") see *basic_checks.CHECKBITMASK∗*-constants.
bit_length	bit length for value and mask
pass_on_true	if True, test will be PASSED if result is 'greater 0', otherwise it will be PASSED if the result is 'equal 0'
descr	user specific description shown at the start of the result text

7.351.3.8 addCheckBitPattern()

```
ttk_selftest._basic_result_list.ResultList.addCheckBitPattern (
                self,
                value,
                pattern,
                invert = False,
                descr = "" )
```

Check that a numeric value matches a certain pattern.

Result will be added to this ResultList.

Parameters

value	value to check. The value must match the full pattern in order to pass. Values may be numeric (int/long), string representations of binary/hexadecimal values ("0xABCD", "0b1010" etc.) or a list of bytes.
pattern	a string describing the desired bit pattern. • a pattern must always describe the full value (i.e. all bits from start to end). • a pattern starting with "0x" will be interpreted as hexadecimal • "0b" at the beginning denotes a binary pattern • "X" marks "don't care" sections (may appear anywhere in pattern)
invert	True: inverted check: pattern must not match False: default check: pattern has to match
descr	user specific description shown at the start of the result text

Pattern String Examples:

- `0xABC123` - a hexadecimal 24-bit pattern
- `123ABC` - without a prefix, pattern defaults to hexadecimal
- `0b0101101` - a binary 6-bit pattern
- `0xAABXX2` - X designates "don't care" (4-bit-nibbles in hex patterns)
- `0b1XX001` - "don't care" on a bit level in binary patterns
- `0xAA[8]BB` - longer "don't care"-sections can be written as a repetition number in square brackets. This example pattern is 48 bits long with 32 don't care bits in the middle.
- `0b10[8]00` - as above but in a binary pattern: 12 bits with 8 don't care bits in the middle.
- `0xAA BB CC` - whitespace will be ignored for pattern check, but plain spaces will be retained in formatted intermediate values
- `0xab Ab aB` - patterns are not case sensitive

7.351.3.9 addCheckRange()

```
ttk_selftest._basic_result_list.ResultList.addCheckRange (
              self,
              value,
              min_value,
              max_value,
              invert = False,
              descr = "" )
```

Check if value is in defined range (`min_value <= value <= max_value`).

Result will be added to this ResultList.

Parameters

value	value to be checked
min_value	lower Limit
max_value	upper Limit
invert	True: inverted, check that value is outside of range False: default, check that value is inside of range
descr	user specific description printed on the start of the single test result

7.351.3.10 addCheckRelTol()

```
ttk_selftest._basic_result_list.ResultList.addCheckRelTol (
              self,
              current_value,
              rated_value,
              rel_tol = 0.0,
              invert = False,
              descr = "" )
```

Check if the current value is within a specified range of the rated value.

The valid range is defined by relative tolerance factor (as a symmetric range around the `rated_value`). Result will be added to this ResultList.

Info: Relative Tolerances

Relative tolerance values are specified as factors in relation to the supplied `rated_value`.
A range of `0.0 .. 1.0` maps to `0 .. 100 %`

Note

`checkRelTol` will treat `rel_tol` values >= 1.0 as unintended and mark the returned result as ERROR. Use the main test function `checkTolerance` if checks with tolerance values >= 100% should ever be needed.

Parameters

current_value	value to be checked
rated_value	expected value
rel_tol	relative tolerance as factor, in range 0.0..1.0
invert	True: inverted, check current value != rated value (+/- tolerance) False: default: check current value == rated value (+/- tolerance)
descr	user specific description shown before the single test result

7.351.3.11 addCheckStatus()

```
ttk_selftest._basic_result_list.ResultList.addCheckStatus (
              self,
              current_status,
              nominal_status,
              equal = True,
              descr = "" )
```

Check if the current status value is identical to the nominal status value.

Both statuses will be interpreted as int/long values for the comparison. Result will be added to this ResultList.

Note

If one or both values cannot be represented as integer numbers, the test will fail with verdicts.ERROR.

Parameters

current_status	status to be checked
nominal_status	expected status to be checked against
equal	True: check current status == nominal status, False: check current status != nominal status
descr	user specific description printed on the start of the single test result

7.351.3.12 addCheckTolerance()

```
ttk_selftest._basic_result_list.ResultList.addCheckTolerance (
              self,
```

```
            current_value,
            rated_value,
            rel_pos = 0,
            abs_pos = 0,
            rel_neg = None,
            abs_neg = None,
            invert = False,
            descr = "" )
```

Check if the current value is within a specified range of the rated value.

The range is defined by an absolute and a relative tolerance. Both tolerance types will be added to get the effective tolerance. Result will be added to this ResultList.

Note

If a *symmetric* tolerance range is desired, only `rel_pos` and/or `abs_pos` need to be set.

Info: Relative Tolerances

Relative tolerance values are specified as factors in relation to the supplied `rated_value`.
A range of `0.0 .. 1.0` maps to `0 .. 100 %`

Using both relative and absolute tolerances:

Internally, both tolerances are added to determine the effective boundary values:

```
lower_boundary = abs_neg + rel_neg * abs(rated_value) + rated_value
upper_boundary = abs_pos + rel_pos * abs(rated_value) + rated_value
```

`current_value` will pass the check if its value lies between the calculated boundary values:

```
lower_boundary <= current_value <= upper_boundary
```

Parameters

current_value	value to be checked
rated_value	expected value
rel_pos	(optional) relative positive tolerance as factor, typically in range 0.0..1.0
abs_pos	(optional) absolute positive tolerance
rel_neg	(optional) relative negative tolerance as factor, typically in range 0.0..-1.0
abs_neg	(optional) absolute negative tolerance
invert	True: inverted, check current value != rated value (+/- tolerance) False: default: check current value == rated value (+/- tolerance)
descr	user specific description printed on the start of the single test result

7.351.3.13 addCompare()

```
ttk_selftest._basic_result_list.ResultList.addCompare (
            self,
```

```
                left_value,
                operator,
                right_value,
                abs_tol = 0,
                descr = "" )
```

Verify the conditional relationship between two operands based on the given operator.

Result will be added to this ResultList.

Parameters

left_value	left operand of the relational expression
operator	relational operator given as a string, see below
right_value	right operand of the relational expression
abs_tol	absolute tolerance for equality/non-equality checks of numeric types
descr	user specific description printed on the start of the single test result

Info: Tolerance

A a tolerance > 0 is mandatory for equality / non-equality checks involving float values.
See checkTolerance() for more in-depth options like relative and non-symmetric tolerances.

Relational operator strings:

- "==" : equal to
- "<" : less than
- ">" : greater than
- "<=" : less than or equal to
- ">=" : greater than or equal to
- "!=" : not equal to

7.351.3.14 addCompareCE()

```
ttk_selftest._basic_result_list.ResultList.addCompareCE (
                self,
                current_value,
                operator,
                expected_value,
                abs_tol = 0,
                descr = "" )
```

Verify the conditional relationship between current and expected value based on the given operator.

Result will be added to this ResultList.

This is a compare variant with an alternative output formatting to explicitly label input values as "current" and "expected". See also basic_tests.compare.

Parameters

current_value	left operand of the relational expression
operator	relational operator given as a string, see below
expected_value	right operand of the relational expression
abs_tol	absolute tolerance for equality/non-equality checks of numeric types
descr	user specific description printed on the start of the single test result

Info: Tolerance

A tolerance > 0 is mandatory for equality / non-equality checks involving float values.

Relational operator strings:

- "==" : equal to
- "<" : less than
- ">" : greater than
- "<=" : less than or equal to
- ">=" : greater than or equal to
- "!=" : not equal to

7.351.3.15 addContains()

```
ttk_selftest._basic_result_list.ResultList.addContains (
              self,
              defined_values,
              current_value,
              mode = "all",
              descr = "" )
```

Check if current_value is an element (or sub set) of defined_values.

Result will be added to this ResultList.

Parameters

defined_values	List of scalars to be searched in
current_value	the function checks if this scalar or list is contained in 'defined_values'
mode	search mode (see below)
descr	user specific description printed on the start of the single test result

Available Modes:

- "all": each value of 'current_value' should be in `defined_values`
- "any": at least one element of 'current_value' should be in `defined_values`
- "none"/"not": no element of 'current_value' should be in `defined_values`

7.351.3.16 addError()

```
ttk_selftest._basic_result_list.ResultList.addError (
              self,
              msg,
              ex = None,
              show_tb = True )
```

Add an ERROR result.

Parameters

msg	error message
ex	optional Exception instance to add to message
show_tb	True: if ex has been supplied, print traceback to console

7.351.3.17 addHeader()

```
ttk_selftest._basic_result_list.ResultList.addHeader (
              self,
              text )
```

7.351.3.18 addInfo()

```
ttk_selftest._basic_result_list.ResultList.addInfo (
              self,
             * msg )
```

7.351.3.19 addSubHeader()

```
ttk_selftest._basic_result_list.ResultList.addSubHeader (
              self,
              text )
```

7.351.3.20 addTestVars()

```
ttk_selftest._basic_result_list.ResultList.addTestVars (
              self,
              set_var_list = None,
              check_var_list = None,
              min_wait_time_ms = 100,
              descr = '' )
```

Check the values of multiple variables/signals as a result of setting multiple other variables/signals.

Between setting and checking, a configurable time will be waited to deal with propagation delay. This wait time defaults to the minimum practicable wait time of all involved bus signals. Result will be added to this ResultList.

See ttk_checks.adv_tests.testVars for details.

Parameters

set_var_list	Variables to set: a list of dictionaries with variable and value
check_var_list	Variables to check: a list of dictionaries with variable, expected value and tolerance
min_wait_time_ms	minimum wait time between setting variables and performing checks
descr	user specific description shown at the start of the result text

7.351.3.21 append()

```
ttk_selftest._basic_result_list.ResultList.append (
              self,
              other )
```

7.351.3.22 clearTestcaseId()

```
ttk_selftest._basic_result_list.ResultList.clearTestcaseId (
              self )
```

Clear the current Testcase-ID, following results will not get IDs assigned.

7.351.3.23 enableEcho()

```
ttk_selftest._basic_result_list.ResultList.enableEcho (
              self,
              enable = True )
```

en-/disable echoing of new entries to stdout.

7.351.3.24 extend()

```
ttk_selftest._basic_result_list.ResultList.extend (
              self,
              other_iterable )
```

7.351.3.25 getCombinedResult()

```
ttk_selftest._basic_result_list.ResultList.getCombinedResult (
              self,
              divider = None,
              trim_descrs = None,
              failed_prefix = None,
              compact_descrs = None )
```

Get a single result entry containing the joined description texts and the resulting verdict of all results currently present in this ResultList instance.

Info:

If results contain "special" entries (testcase-ids, images, ...) the combined result will contain all unique testcase-ids, and all encountered objects/images, so the result might end up with multiple images and testcase-ids. Any existing delta timestamps will be ignored, though.

Parameters

divider	string to place between joined description texts. At least a single newline is recommended, or descriptions will tend to flow into each other. None: use default divider from self.divider
trim_descrs	if True, whitespace will be trimmed/stripped from beginning and end of each description to make the output more compact. None: use default setting from self.trim_descr
failed_prefix	a string to prefix to descriptions of failed test steps to make them more evident in the combined description text. Set it to an empty string if prefixing is not desired. A list of multiple prefixes may also be used: First prefix will then be used to mark failed entries, all others will be replaced if already present in description lines None: use default setting from self.failed_prefix
compact_descrs	if True, descriptions will be "compacted" by removing all lines from the beginning that are already present in the directly preceding description. None: use default setting from self.compact_descrs

Returns

a single result entry [<description>, <resulting verdict>]. In more complex cases, the result entry might look more similar to [<description>, <[[TC-ID]]>, <[[IMG]]>, ..., <resulting verdict>]

7.351.3.26 setTestcaseId()

```
ttk_selftest._basic_result_list.ResultList.setTestcaseId (
              self,
              testcase_id = None,
             * testcase_ids )
```

Set the currently active Testcase-ID (or IDs) to be assigned to all following result entries.

Set the ID to None (or call clearTestcaseId()) to disable automatic ID assignment.

Parameters

testcase_id	testcase id string to assign
∗*testcase_ids*	additional testcase ids to set

Note

First `testcase_id` kept as a separate argument for backwards compatibility to previous ResultList versions.

7.351.4 Member Data Documentation

7.351.4.1 _fail_verdicts

```
ttk_selftest._basic_result_list.ResultList._fail_verdicts  [protected]
```

7.351.4.2 _info_verdicts

ttk_selftest._basic_result_list.ResultList._info_verdicts [protected]

7.351.4.3 _pass_verdicts

ttk_selftest._basic_result_list.ResultList._pass_verdicts [protected]

7.351.4.4 compact_descrs

ttk_selftest._basic_result_list.ResultList.compact_descrs

getCombinedResult: True: "compact" descriptions by removing all lines from the beginning that are already present in the directly preceding description.

7.351.4.5 current_testcase_id

ttk_selftest._basic_result_list.ResultList.current_testcase_id

Currently active testcase ID (or potentially multiple Testcase-IDs).

None if no testcase ID is active, otherwise a list of IDs

7.351.4.6 divider

ttk_selftest._basic_result_list.ResultList.divider

getCombinedResult: default string to place between joined description texts.

7.351.4.7 echo_enabled

ttk_selftest._basic_result_list.ResultList.echo_enabled

en-/disable echoing of new entries, see enableEcho()

7.351.4.8 failed_prefix

ttk_selftest._basic_result_list.ResultList.failed_prefix

getCombinedResult: Prefix to add to descriptions of failed test steps.

7.351.4.9 header_block_width

ttk_selftest._basic_result_list.ResultList.header_block_width

width of "header" format blocks in log output

7.351.4.10 header_format_verdicts

`ttk_selftest._basic_result_list.ResultList.header_format_verdicts`

For which verdicts to use "header" formatting in log (default: empty verdicts)

7.351.4.11 trim_descrs

`ttk_selftest._basic_result_list.ResultList.trim_descrs`

getCombinedResult: True: whitespace will be trimmed/stripped from beginning and end of each description to make output more compact.

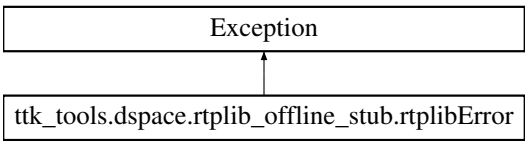
The documentation for this class was generated from the following file:

- ttk_selftest/_basic_result_list.py (v1.0.1)

7.352 ttk_tools.dspace.rtplib_offline_stub.rtplibError Class Reference

Exception of the module rtplib.

Inheritance diagram for ttk_tools.dspace.rtplib_offline_stub.rtplibError:



7.352.1 Detailed Description

Exception of the module rtplib.

Check if a Python object has the type Appl

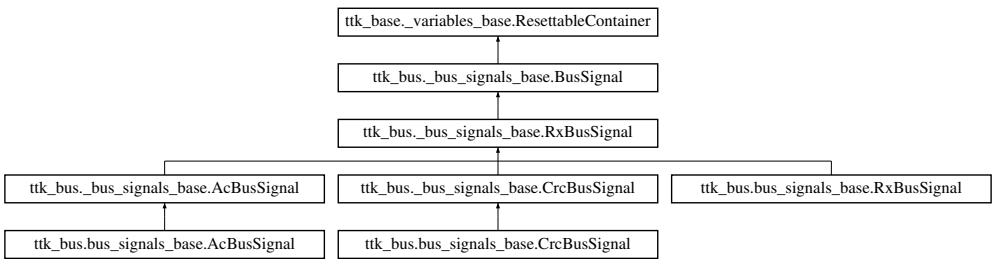
The documentation for this class was generated from the following file:

- ttk_tools/dspace/rtplib_offline_stub.py (v2.8)

7.353 ttk_bus._bus_signals_base.RxBusSignal Class Reference

A RX bus signal (HIL => ECU) that models a CRC value with additional error injections.

Inheritance diagram for ttk_bus._bus_signals_base.RxBusSignal:



7.353.1 Detailed Description

A RX bus signal (HIL => ECU) that models a CRC value with additional error injections.

This is the generic base class for CRC signals, blockset/implementation specific CRC signal classes will inherit from this.

Attention

This is just a base class, using it directly will not accomplish much. At the very least, setBasicCrcError and clearBasicCrcError need to be overridden depending on the target interface.

See bus_signals_frbs or bus_signals_canmmbs for existing implementations.

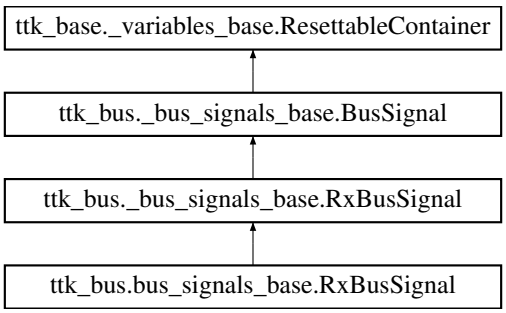
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_base.py (v3.0.1)

7.354 ttk_bus.bus_signals_base.RxBusSignal Class Reference

A signal sent from the test system to the DUT (ECU Rx <= HIL Tx).

Inheritance diagram for ttk_bus.bus_signals_base.RxBusSignal:



Public Member Functions

- __init__ (self, context, identifier, cycle_time=0, debounce_time=0, signal_var=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  Bus signal initialization.
- info (self)

  Get a current info string about the HIL signal variable.
- get (self)

  Get the current value of the HIL signal variable.
- set (self, value)

  Set a value to the HIL signal variable.
- getState (self)

  Get a textual state for the HIL signal variable's current data.
- setState (self, state)

  Set a textual state to the HIL signal variable.

- hasStateLookup (self)

  Return True if a state/value lookup table is defined for the HIL signal variable.
- getStateDescr (self, value=None, fallback=None)

  Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).
- hasFormattedRepr (self)

  Return True if a formatting for value representation is defined for the HIL signal variable.
- getFormattedValue (self, value=None)

  Get a formatted representation for the given value using the defined format string of the the HIL signal variable.
- reset (self, ∗args, ∗∗kwargs)

  Call reset() on all contained variables/signals that support it.

Public Attributes

- context

  context for "HilVar" variables
- signal_var

  Main bus signal variable on HIL-side.

7.354.1 Detailed Description

A signal sent from the test system to the DUT (ECU Rx <= HIL Tx).

7.354.2 Constructor & Destructor Documentation

7.354.2.1 __init__()

```
ttk_bus.bus_signals_base.RxBusSignal.__init__ (
              self,
              context,
              identifier,
              cycle_time = 0,
              debounce_time = 0,
              signal_var = None,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Bus signal initialization.

Parameters

context	parent context for instantiating HIL variables. This gets mainly used in derived classes
identifier	base signal identifier/name/path in the HIL model (typically this is read-only for Tx-signals and writable for Rx-signals). Derived classes typically use this identifier as starting point for deriving additional helper variables.
cycle_time	cycle time in [ms], 0 if event-triggered

Parameters

debounce_time	minimum debounce delay time between events in [ms]
signal_var	(opt) an already initialized instance of Var (or most likely one of its derived classes) for the main signal value. If not supplied, a DummyVar will be used/ created as fall back
unit	(opt) unit of physical signal value (as string, mainly used for report entries)
alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions, defaults to signal_var.unit.
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.354.3 Member Function Documentation

7.354.3.1 get()

```
ttk_bus.bus_signals_base.RxBusSignal.get (
              self )
```

Get the current value of the HIL signal variable.

Returns

the current value.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.354.3.2 getFormattedValue()

```
ttk_bus.bus_signals_base.RxBusSignal.getFormattedValue (
              self,
              value = None )
```

Get a formatted representation for the given value using the defined format string of the the HIL signal variable.

An empty (or non-defined) format string will use the value's default string representation.

Parameters

value	value for which to get a string representation. If None, the last value (read or written) will be used instead.

Returns

a formatted string representation.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.354.3.3 getState()

```
ttk_bus.bus_signals_base.RxBusSignal.getState (
              self )
```

Get a textual state for the HIL signal variable's current data.

This uses the lookup values supplied in constructor. Fallback (if no matching state was found) will be the unmapped numeric value.

Returns

current state's description.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.354.3.4 getStateDescr()

```
ttk_bus.bus_signals_base.RxBusSignal.getStateDescr (
              self,
              value = None,
              fallback = None )
```

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the last read value will be used instead.
fallback	fallback value to return if no match was found. Default: None

Returns

the state text if a match was found, otherwise the specified fallback value.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.354.3.5 hasFormattedRepr()

```
ttk_bus.bus_signals_base.RxBusSignal.hasFormattedRepr (
              self )
```

Return True if a formatting for value representation is defined for the HIL signal variable.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.354.3.6 hasStateLookup()

```
ttk_bus.bus_signals_base.RxBusSignal.hasStateLookup (
              self )
```

Return True if a state/value lookup table is defined for the HIL signal variable.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.354.3.7 info()

```
ttk_bus.bus_signals_base.RxBusSignal.info (
              self )
```

Get a current info string about the HIL signal variable.

Returns

an info string.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.354.3.8 reset()

```
ttk_bus.bus_signals_base.RxBusSignal.reset (
              self,
             * args,
             ** kwargs )
```

Call reset() on all contained variables/signals that support it.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.ResettableContainer.

7.354.3.9 set()

```
ttk_bus.bus_signals_base.RxBusSignal.set (
              self,
              value )
```

Set a value to the HIL signal variable.

Parameters

value	new value to be set

Returns

the original value before the set operation occurred.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.354.3.10 setState()

```
ttk_bus.bus_signals_base.RxBusSignal.setState (
              self,
              state )
```

Set a textual state to the HIL signal variable.

This uses the lookup values supplied in constructor.

Parameters

state	named state to set

Returns

the previous state's description.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.354.4 Member Data Documentation

7.354.4.1 context

```
ttk_bus.bus_signals_base.RxBusSignal.context
```

context for "HilVar" variables

7.354.4.2 signal_var

```
ttk_bus.bus_signals_base.RxBusSignal.signal_var
```

Main bus signal variable on HIL-side.

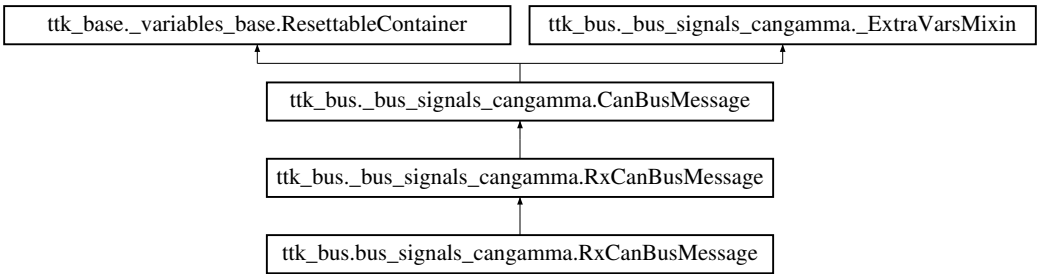
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_base.py (v3.1.0)

7.355 ttk_bus._bus_signals_cangamma.RxCanBusMessage Class Reference

CAN Message/PDU ECU-Rx (HIL-Tx => ECU-Rx).

Inheritance diagram for ttk_bus._bus_signals_cangamma.RxCanBusMessage:



Public Member Functions

- __init__ (self, context, enable_identifier=None, base_identifier=None, cycle_time=0, debounce_time=0, alias="", descr='', change_delay=None, timeout=None, recovery_time=None)

  Rx CAN Message/PDU data (HIL-Tx => ECU-Rx).
- enableHilTx (self)

  (Re-)enable sending of this Message/PDU.
- disableHilTx (self)

  Disable (cyclic) sending of this Message/PDU.
- kickout (self, delay_ms=None)

  Trigger sending of this CAN-Message/PDU.

Public Attributes

- cycle_time

Static Public Attributes

- dict cfg_custom_defaults

  Configuration setting for new instances: Custom "hard" default values to use for internal real time system variables, so a reset() will always reset those variables to defined values (and not just those variables that were set/accessed).
- bool cfg_init_cycle_time = True

  True: automatically initialize message's cycle time from the configured cycle_time value (sets self.cycle_time_var to self.cycle_time during init)
- manual_enable_var

  custom/manual HIL-Tx enable switch PV
- default
- kickout_var

  HIL-Tx-kickout/trigger variable.
- cycle_time_var

  Message/PDU (Hil-Tx) cycle time access.
- unit
- cycle_counter_var
- min_delay_var
- prev_cycle_time = self.cycle_time_var.set(self.cycle_time)

7.355.1 Detailed Description

CAN Message/PDU ECU-Rx (HIL-Tx => ECU-Rx).

Extends the basic CanBusMessage with methods for enableHilTx/disableHilTx and message kickout (trigger).

7.355.2 Constructor & Destructor Documentation

7.355.2.1 __init__()

```
ttk_bus._bus_signals_cangamma.RxCanBusMessage.__init__ (
                self,
                context,
                enable_identifier = None,
                base_identifier = None,
                cycle_time = 0,
                debounce_time = 0,
                alias = "",
                descr = '',
                change_delay = None,
                timeout = None,
                recovery_time = None )
```

Rx CAN Message/PDU data (HIL-Tx => ECU-Rx).

Parameters

context	parent gamma_api context
enable_identifier	identifier of a custom/manual tx enable switch PV. Keep at None to use default "enable" behavior. If defined, the enable variable will later be set to 1 to enable or to 0 to disable message transmission.
base_identifier	base identifier to PDU/message. Helper entries will be derived from this, e.g. kickout or status variables (though this is mostly relevant to messages sent to DUT, see RxCanBusMessage)
cycle_time	cycle time in [ms]. Value will be used to enable/disable sending of cyclic messages, see cycle_time_var. Event-triggered messages have a cycle time of 0 (similar to DBC value)
debounce_time	minimum debounce delay between events in [ms]
alias	Message alias/name
descr	(opt) Message description (may be used in report entries)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_cangamma.CanBusMessage.

Reimplemented in ttk_bus.bus_signals_cangamma.RxCanBusMessage.

7.355.3 Member Function Documentation

7.355.3.1 disableHilTx()

```
ttk_bus._bus_signals_cangamma.RxCanBusMessage.disableHilTx (
              self )
```

Disable (cyclic) sending of this Message/PDU.

Reimplemented in ttk_bus.bus_signals_cangamma.RxCanBusMessage.

7.355.3.2 enableHilTx()

```
ttk_bus._bus_signals_cangamma.RxCanBusMessage.enableHilTx (
              self )
```

(Re-)enable sending of this Message/PDU.

Reimplemented in ttk_bus.bus_signals_cangamma.RxCanBusMessage.

7.355.3.3 kickout()

```
ttk_bus._bus_signals_cangamma.RxCanBusMessage.kickout (
              self,
              delay_ms = None )
```

Trigger sending of this CAN-Message/PDU.

Parameters

delay_ms	[ms] Delay to wait before clearing kickout/trigger variable again.

Default Delay:

Delay defaults to 1/2 self.debounce_time (if explicitly set) or alternatively to 1/2 self.change_delay. Default delay will never be shorter than 5 ms.

Reimplemented in ttk_bus.bus_signals_cangamma.RxCanBusMessage.

7.355.4 Member Data Documentation

7.355.4.1 cfg_custom_defaults

```
dict ttk_bus._bus_signals_cangamma.RxCanBusMessage.cfg_custom_defaults  [static]
```

Initial value:

```
=  {
        "manual_enable_var":    None,
        "kickout_var":          None,
```

```
    }
```

Configuration setting for new instances: Custom "hard" default values to use for internal real time system variables, so a reset() will always reset those variables to defined values (and not just those variables that were set/accessed).

This might be helpful if "other" tests fail to properly clean up the test environment after they are done.

Mapping: <variable name as string>: <default value>

A value of None (or a missing variable name) enables normal/automatic reset behavior.

Typical default values:

- Enable variables are usually default to "enabled"
  - 0: disabled
  - 1: enabled
- Message/PDU (Hil-TX) cycle time access: Default value depends on configured cycle time (in [s]). Override standard behavior only if absolutely necessary.
- Kickout/trigger variables: Triggers should auto-reset to 0 once the PDU has been triggered. 0 is a safe default.

7.355.4.2 cfg_init_cycle_time

```
bool ttk_bus._bus_signals_cangamma.RxCanBusMessage.cfg_init_cycle_time = True  [static]
```

True: automatically initialize message's cycle time from the configured cycle_time value (sets self.cycle_time_var to self.cycle_time during init)

7.355.4.3 cycle_counter_var

```
ttk_bus._bus_signals_cangamma.RxCanBusMessage.cycle_counter_var  [static]
```

7.355.4.4 cycle_time

```
ttk_bus._bus_signals_cangamma.RxCanBusMessage.cycle_time
```

7.355.4.5 cycle_time_var

```
ttk_bus._bus_signals_cangamma.RxCanBusMessage.cycle_time_var  [static]
```

Message/PDU (Hil-Tx) cycle time access.

Cycle time in [ms]. Note: This maps to the .period element of the current message

7.355.4.6 default

```
ttk_bus._bus_signals_cangamma.RxCanBusMessage.default  [static]
```

7.355.4.7 kickout_var

```
ttk_bus._bus_signals_cangamma.RxCanBusMessage.kickout_var  [static]
```

HIL-Tx-kickout/trigger variable.

7.355.4.8 manual_enable_var

```
ttk_bus._bus_signals_cangamma.RxCanBusMessage.manual_enable_var  [static]
```

custom/manual HIL-Tx enable switch PV

7.355.4.9 min_delay_var

```
ttk_bus._bus_signals_cangamma.RxCanBusMessage.min_delay_var  [static]
```

7.355.4.10 prev_cycle_time

```
ttk_bus._bus_signals_cangamma.RxCanBusMessage.prev_cycle_time = self.cycle_time_var.set(self.↩
cycle_time)  [static]
```

7.355.4.11 unit

```
ttk_bus._bus_signals_cangamma.RxCanBusMessage.unit  [static]
```

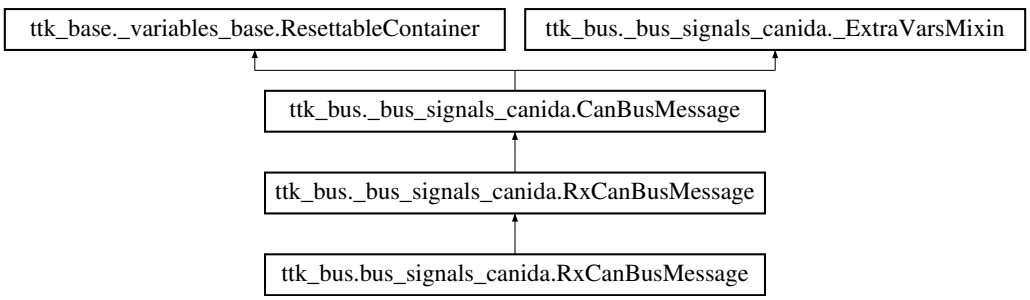
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_cangamma.py (v1.12)

7.356 ttk_bus._bus_signals_canida.RxCanBusMessage Class Reference

CAN Message/PDU ECU-Rx (HIL-Tx => ECU-Rx).

Inheritance diagram for ttk_bus._bus_signals_canida.RxCanBusMessage:



Public Member Functions

- __init__ (self, context, base_identifier=None, cycle_time=0, debounce_time=0, alias="", descr='', change_delay=None, timeout=None, recovery_time=None)

  Rx CAN Message/PDU data (HIL-Tx => ECU-Rx).
- enableHilTx (self)

  (Re-)enable sending of this Message/PDU.
- disableHilTx (self)

  Disable (cyclic) sending of this Message/PDU.
- kickout (self, delay_ms=None)

  Trigger sending of this CAN-Message/PDU.

Public Attributes

- cycle_time

Static Public Attributes

- dict cfg_custom_defaults

  Configuration setting for new instances: Custom "hard" default values to use for internal real time system variables, so a reset() will always reset those variables to defined values (and not just those variables that were set/accessed).
- bool cfg_init_cycle_time = True

  True: automatically initialize message's cycle time from the configured cycle_time value (sets self.cycle_time_var to self.cycle_time during init)
- kickout_var

  HIL-Tx-kickout/trigger variable.
- cycle_time_var

  Message/PDU (Hil-Tx) cycle time access.
- unit
- cycle_counter_var
- min_delay_var
- prev_cycle_time = self.cycle_time_var.set(self.cycle_time)

7.356.1 Detailed Description

CAN Message/PDU ECU-Rx (HIL-Tx => ECU-Rx).

Extends the basic CanBusMessage with methods for enableHilTx/disableHilTx and message kickout (trigger).

7.356.2 Constructor & Destructor Documentation

7.356.2.1 __init__()

```
ttk_bus._bus_signals_canida.RxCanBusMessage.__init__ (
              self,
              context,
              base_identifier = None,
              cycle_time = 0,
              debounce_time = 0,
              alias = "",
              descr = '',
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Rx CAN Message/PDU data (HIL-Tx => ECU-Rx).

Parameters

context	parent iDA context
base_identifier	base identifier to PDU/message. Helper entries will be derived from this, like kickout or status variables. e.g. ecu-rx (hil-tx) "CAN.can3_iFIU_iSRC.HIL_TX.N0FIU_SetError_Card1"
cycle_time	cycle time in [ms]. Value will be used to enable/disable sending of cyclic messages, see cycle_time_var. Event-triggered messages have a cycle time of 0 (similar to DBC value)
debounce_time	minimum debounce delay between events in [ms]
alias	Message alias/name
descr	(opt) Message description (may be used in report entries)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_canida.CanBusMessage.

Reimplemented in ttk_bus.bus_signals_canida.RxCanBusMessage.

7.356.3 Member Function Documentation

7.356.3.1 disableHilTx()

```
ttk_bus._bus_signals_canida.RxCanBusMessage.disableHilTx (
              self )
```

Disable (cyclic) sending of this Message/PDU.

Reimplemented in ttk_bus.bus_signals_canida.RxCanBusMessage.

7.356.3.2 enableHilTx()

```
ttk_bus._bus_signals_canida.RxCanBusMessage.enableHilTx (
              self )
```

(Re-)enable sending of this Message/PDU.

Reimplemented in ttk_bus.bus_signals_canida.RxCanBusMessage.

7.356.3.3 kickout()

```
ttk_bus._bus_signals_canida.RxCanBusMessage.kickout (
              self,
              delay_ms = None )
```

Trigger sending of this CAN-Message/PDU.

Parameters

delay_ms	[ms] Delay to wait before clearing kickout/trigger variable again.

Default Delay:

Delay defaults to 1/2 self.debounce_time (if explicitly set) or alternatively to 1/2 self.change_delay. Default delay will never be shorter than 5 ms.

Reimplemented in ttk_bus.bus_signals_canida.RxCanBusMessage.

7.356.4 Member Data Documentation

7.356.4.1 cfg_custom_defaults

```
dict ttk_bus._bus_signals_canida.RxCanBusMessage.cfg_custom_defaults  [static]
```

Initial value:

```
=  {
        "kickout_var": None,
    }
```

Configuration setting for new instances: Custom "hard" default values to use for internal real time system variables, so a reset() will always reset those variables to defined values (and not just those variables that were set/accessed).

This might be helpful if "other" tests fail to properly clean up the test environment after they are done.

Mapping: <variable name as string>: <default value>

A value of None (or a missing variable name) enables normal/automatic reset behavior.

Typical default values:

- Enable variables are usually default to "enabled"
  - 0: disabled
  - 1: enabled
- Message/PDU (Hil-TX) cycle time access: Default value depends on configured cycle time (in [s]). Override standard behavior only if absolutely necessary.
- Kickout/trigger variables: Triggers should auto-reset to 0 once the PDU has been triggered. 0 is a safe default.

7.356.4.2 cfg_init_cycle_time

```
bool ttk_bus._bus_signals_canida.RxCanBusMessage.cfg_init_cycle_time = True  [static]
```

True: automatically initialize message's cycle time from the configured cycle_time value (sets self.cycle_time_var to self.cycle_time during init)

7.356.4.3 cycle_counter_var

```
ttk_bus._bus_signals_canida.RxCanBusMessage.cycle_counter_var  [static]
```

7.356.4.4 cycle_time

```
ttk_bus._bus_signals_canida.RxCanBusMessage.cycle_time
```

7.356.4.5 cycle_time_var

```
ttk_bus._bus_signals_canida.RxCanBusMessage.cycle_time_var  [static]
```

Message/PDU (Hil-Tx) cycle time access.

Cycle time in [ms]. Note: This maps to the .period element of the current message

7.356.4.6 kickout_var

```
ttk_bus._bus_signals_canida.RxCanBusMessage.kickout_var  [static]
```

HIL-Tx-kickout/trigger variable.

7.356.4.7 min_delay_var

```
ttk_bus._bus_signals_canida.RxCanBusMessage.min_delay_var  [static]
```

7.356.4.8 prev_cycle_time

```
ttk_bus._bus_signals_canida.RxCanBusMessage.prev_cycle_time = self.cycle_time_var.set(self.↩
cycle_time)  [static]
```

7.356.4.9 unit

```
ttk_bus._bus_signals_canida.RxCanBusMessage.unit  [static]
```

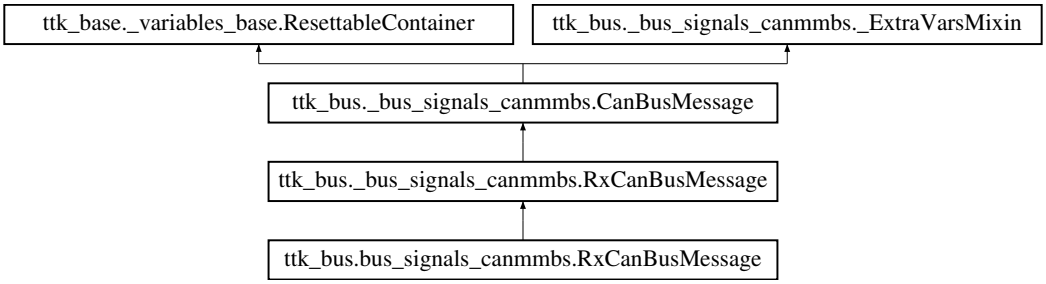
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_canida.py (v1.0.2)

7.357 ttk_bus._bus_signals_canmmbs.RxCanBusMessage Class Reference

CAN Message/PDU ECU-Rx (HIL-Tx => ECU-Rx).

Inheritance diagram for ttk_bus._bus_signals_canmmbs.RxCanBusMessage:



Public Member Functions

- __init__ (self, context, enable_path=None, bus_systems_path=None, cycle_time=0, debounce_time=0, alias="", descr='', change_delay=None, timeout=None, recovery_time=None)

  Rx CAN Message/PDU data (HIL-Tx => ECU-Rx).
- enableHilTx (self)

  (Re-)enable sending of this Message/PDU.
- disableHilTx (self)

  Disable sending of this Message/PDU.
- kickout (self, delay_ms=None)

  Trigger sending of this CAN-Message/PDU.

Static Public Attributes

- dict cfg_custom_defaults

  Configuration setting for new instances:
  Custom "hard" default values to use for internal real time system variables, so a reset() will always reset those variables to defined values (and not just those variables that were set/accessed).
- enable_var

  HIL-Tx-enable/disable variable.
- kickout_var

  Message/PDU Kickout/Trigger variable
- cycle_time_var

  Message/PDU (Hil-TX) cycle time access.

7.357.1 Detailed Description

CAN Message/PDU ECU-Rx (HIL-Tx => ECU-Rx).

Extends the basic CanBusMessage with methods for enableHilTx/disableHilTx and message kickout (trigger).

7.357.2 Constructor & Destructor Documentation

7.357.2.1 __init__()

```
ttk_bus._bus_signals_canmmbs.RxCanBusMessage.__init__ (
              self,
              context,
              enable_path = None,
              bus_systems_path = None,
              cycle_time = 0,
              debounce_time = 0,
              alias = "",
              descr = '',
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Rx CAN Message/PDU data (HIL-Tx => ECU-Rx).

Parameters

context	parent context (an RTE application instance to resolve signal/variable paths)
enable_path	manual path to a tx enable switch. Keep at None to use default enable switch below BusSystems (see parameter bus_systems_path). The enable variable will later be set to 1 to enable or to 0 to disable message transmission.
bus_systems_path	base path to PDU/message below BusSystems, use "BusSystems/CAN/<CAN_NAME>/<MSG_NAME>/TX/<MSG_NAME>" or "BusSystems/CAN/<CAN_NAME>/<MSG_NAME>/TX/<MSG_NAME>_Enable" Helper entries will be derived from this path, e.g. kickout or status variables.
cycle_time	cycle time in [ms]. Event-triggered messages have a cycle time of 0 (similar to DBC value)
debounce_time	minimum debounce delay between events in [ms]
alias	Message alias/name
descr	(opt) Message description (may be used in report entries)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_canmmbs.CanBusMessage.

Reimplemented in ttk_bus.bus_signals_canmmbs.RxCanBusMessage.

7.357.3 Member Function Documentation

7.357.3.1 disableHilTx()

```
ttk_bus._bus_signals_canmmbs.RxCanBusMessage.disableHilTx (
              self )
```

Disable sending of this Message/PDU.

Reimplemented in ttk_bus.bus_signals_canmmbs.RxCanBusMessage.

7.357.3.2 enableHilTx()

```
ttk_bus._bus_signals_canmmbs.RxCanBusMessage.enableHilTx (
              self )
```

(Re-)enable sending of this Message/PDU.

Reimplemented in ttk_bus.bus_signals_canmmbs.RxCanBusMessage.

7.357.3.3 kickout()

```
ttk_bus._bus_signals_canmmbs.RxCanBusMessage.kickout (
              self,
              delay_ms = None )
```

Trigger sending of this CAN-Message/PDU.

Parameters

delay_ms	[ms] Delay to wait before clearing kickout/trigger variable again (it *should* auto-reset if trigger variable is located below BusSystems, but anyway).

Default Delay:

Delay defaults to 1/2 self.debounce_time (if explicitly set) or alternatively to 1/2 self.change_delay. Default delay will never be shorter than 5 ms.

Reimplemented in ttk_bus.bus_signals_canmmbs.RxCanBusMessage.

7.357.4 Member Data Documentation

7.357.4.1 cfg_custom_defaults

```
dict ttk_bus._bus_signals_canmmbs.RxCanBusMessage.cfg_custom_defaults  [static]
```

Initial value:

```
=  {
        "enable_var":       None,
        "kickout_var":      None,
        "cycle_time_var":   None,
    }
```

Configuration setting for new instances:
Custom "hard" default values to use for internal real time system variables, so a reset() will always reset those variables to defined values (and not just those variables that were set/accessed).

This might be helpful if "other" tests fail to properly clean up the test environment after they are done.

Mapping: <variable name as string>: <default value>

A value of None (or a missing variable name) enables normal/automatic reset behavior.

Typical default values:

- Enable variables are usually default to "enabled"
  - 0: disabled
  - 1: enabled
- Message/PDU (Hil-TX) cycle time access:
  Default value depends on configured cycle time (in [s]).
  Override standard behavior only if absolutely necessary.
- Kickout/trigger variables:
  Triggers below BusSystems should auto-reset to 0 once the PDU has been triggered. 0 is a safe default.

7.357.4.2 cycle_time_var

`ttk_bus._bus_signals_canmmbs.RxCanBusMessage.cycle_time_var [static]`

Message/PDU (Hil-TX) cycle time access.

7.357.4.3 enable_var

`ttk_bus._bus_signals_canmmbs.RxCanBusMessage.enable_var [static]`

HIL-Tx-enable/disable variable.

7.357.4.4 kickout_var

`ttk_bus._bus_signals_canmmbs.RxCanBusMessage.kickout_var [static]`

Message/PDU Kickout/Trigger variable

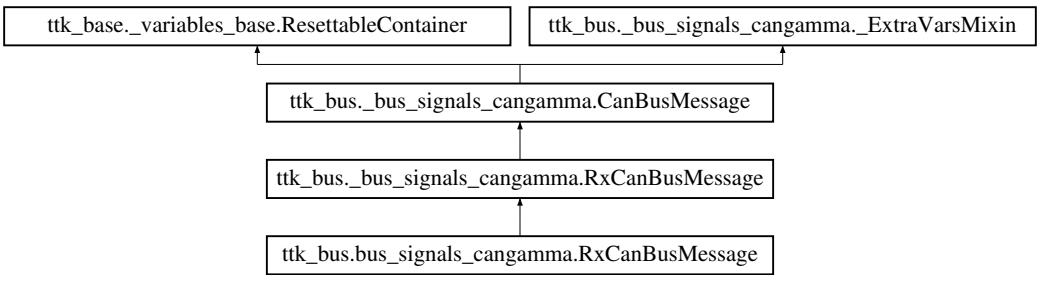
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_canmmbs.py (v2.3)

7.358 ttk_bus.bus_signals_cangamma.RxCanBusMessage Class Reference

CAN Message/PDU ECU-Rx (HIL-Tx => ECU-Rx).

Inheritance diagram for ttk_bus.bus_signals_cangamma.RxCanBusMessage:



Public Member Functions

- __init__ (self, context, enable_identifier=None, base_identifier=None, cycle_time=0, debounce_time=0, alias="", descr='', change_delay=None, timeout=None, recovery_time=None)

  Rx CAN Message/PDU data (HIL-Tx => ECU-Rx).
- enableHilTx (self)

  (Re-)enable sending of this Message/PDU.
- disableHilTx (self)

  Disable (cyclic) sending of this Message/PDU.
- kickout (self, delay_ms=None)

  Trigger sending of this CAN-Message/PDU.
- reset (self, *args, **kwargs)

  Call reset() on all contained variables/signals that support it.

Public Attributes

- enable_var

  HIL-Tx-enable/disable variable (only if set via enable_identifier)
- kickout_var

  HIL-Tx-kickout/trigger variable.
- cycle_time_var

  Message/PDU (Hil-Tx) cycle time access.
- cycle_counter_var
- min_delay_var

  minimum delay between triggered messages in [ms], compare debounce delay in .debounce_time
- context

  Parent context (a gaapi reference to resolve signal/variable paths)
- alias

  Message alias name.
- descr

  Message description text.
- base_identifier

  Base BusSystems path to Message/PDU helper variables below BusSystems e.g.
- cycle_time

  cycle time of Message/PDU in [ms]
- debounce_time

  minimum debounce delay for event-triggered Messages/PDUs in [ms]
- change_delay

  delay to wait after a value change [ms]
- timeout

  detection time for timeout errors [ms]
- recovery_time

  time for recovery/"wiedergut" checks [ms]

Static Public Attributes

- dict cfg_custom_defaults

  Configuration setting for new instances: Custom "hard" default values to use for internal real time system variables, so a reset() will always reset those variables to defined values (and not just those variables that were set/accessed).
- bool cfg_init_cycle_time = True

  True: automatically initialize message's cycle time from the configured cycle_time value (sets self.cycle_time_var to self.cycle_time during init)

7.358.1 Detailed Description

CAN Message/PDU ECU-Rx (HIL-Tx => ECU-Rx).

Extends the basic CanBusMessage with methods for enableHilTx/disableHilTx and message kickout (trigger).

7.358.2 Constructor & Destructor Documentation

7.358.2.1 __init__()

```
ttk_bus.bus_signals_cangamma.RxCanBusMessage.__init__ (
               self,
               context,
               enable_identifier = None,
               base_identifier = None,
               cycle_time = 0,
               debounce_time = 0,
               alias = "",
               descr = '',
               change_delay = None,
               timeout = None,
               recovery_time = None )
```

Rx CAN Message/PDU data (HIL-Tx => ECU-Rx).

Parameters

context	parent gamma_api context
enable_identifier	identifier of a custom/manual tx enable switch PV. Keep at None to use default "enable" behavior. If defined, the enable variable will later be set to 1 to enable or to 0 to disable message transmission.
base_identifier	base identifier to PDU/message. Helper entries will be derived from this, e.g. kickout or status variables (though this is mostly relevant to messages sent to DUT, see RxCanBusMessage) e.g. "BusSystems.<CAN_NAME>.HIL_TX.<MSG_NAME>"
cycle_time	cycle time in [ms]. Value will be used to enable/disable sending of cyclic messages, see cycle_time_var. Event-triggered messages have a cycle time of 0 (similar to DBC value)
debounce_time	minimum debounce delay between events in [ms]
alias	Message alias/name
descr	(opt) Message description (may be used in report entries)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_cangamma.RxCanBusMessage.

7.358.3 Member Function Documentation

7.358.3.1 disableHilTx()

```
ttk_bus.bus_signals_cangamma.RxCanBusMessage.disableHilTx (
              self )
```

Disable (cyclic) sending of this Message/PDU.

Reimplemented from ttk_bus._bus_signals_cangamma.RxCanBusMessage.

7.358.3.2 enableHilTx()

```
ttk_bus.bus_signals_cangamma.RxCanBusMessage.enableHilTx (
              self )
```

(Re-)enable sending of this Message/PDU.

Reimplemented from ttk_bus._bus_signals_cangamma.RxCanBusMessage.

7.358.3.3 kickout()

```
ttk_bus.bus_signals_cangamma.RxCanBusMessage.kickout (
              self,
              delay_ms = None )
```

Trigger sending of this CAN-Message/PDU.

Parameters

delay_ms	[ms] Delay to wait before clearing kickout/trigger variable again.

Default Delay:

Delay defaults to 1/2 self.debounce_time (if explicitly set) or alternatively to 1/2 self.change_delay. Default delay will never be shorter than 5 ms.

Reimplemented from ttk_bus._bus_signals_cangamma.RxCanBusMessage.

7.358.3.4 reset()

```
ttk_bus.bus_signals_cangamma.RxCanBusMessage.reset (
              self,
             * args,
             ** kwargs )
```

Call reset() on all contained variables/signals that support it.

Simple, non-recursive approach.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.ResettableContainer.

7.358.4 Member Data Documentation

7.358.4.1 alias

`ttk_bus.bus_signals_cangamma.RxCanBusMessage.alias`

Message alias name.

7.358.4.2 base_identifier

`ttk_bus.bus_signals_cangamma.RxCanBusMessage.base_identifier`

Base BusSystems path to Message/PDU helper variables below BusSystems e.g.

"BusSystems.<CAN_NAME>.HIL_TX.<MSG_NAME>"

7.358.4.3 cfg_custom_defaults

`dict ttk_bus.bus_signals_cangamma.RxCanBusMessage.cfg_custom_defaults [static]`

Initial value:

```
=  {
        "manual_enable_var":    None,
        "kickout_var":          None,
    }
```

Configuration setting for new instances: Custom "hard" default values to use for internal real time system variables, so a reset() will always reset those variables to defined values (and not just those variables that were set/accessed).

This might be helpful if "other" tests fail to properly clean up the test environment after they are done.

Mapping: <variable name as string>: <default value>

A value of None (or a missing variable name) enables normal/automatic reset behavior.

Typical default values:

- Enable variables are usually default to "enabled"
  - 0: disabled
  - 1: enabled
- Message/PDU (Hil-TX) cycle time access: Default value depends on configured cycle time (in [s]). Override standard behavior only if absolutely necessary.
- Kickout/trigger variables: Triggers should auto-reset to 0 once the PDU has been triggered. 0 is a safe default.

7.358.4.4 cfg_init_cycle_time

```
bool ttk_bus.bus_signals_cangamma.RxCanBusMessage.cfg_init_cycle_time = True  [static]
```

True: automatically initialize message's cycle time from the configured cycle_time value (sets self.cycle_time_var to self.cycle_time during init)

7.358.4.5 change_delay

```
ttk_bus.bus_signals_cangamma.RxCanBusMessage.change_delay
```

delay to wait after a value change [ms]

7.358.4.6 context

```
ttk_bus.bus_signals_cangamma.RxCanBusMessage.context
```

Parent context (a gaapi reference to resolve signal/variable paths)

7.358.4.7 cycle_counter_var

```
ttk_bus.bus_signals_cangamma.RxCanBusMessage.cycle_counter_var
```

7.358.4.8 cycle_time

```
ttk_bus.bus_signals_cangamma.RxCanBusMessage.cycle_time
```

cycle time of Message/PDU in [ms]

7.358.4.9 cycle_time_var

```
ttk_bus.bus_signals_cangamma.RxCanBusMessage.cycle_time_var
```

Message/PDU (Hil-Tx) cycle time access.

Cycle time in [ms]. Note: This maps to the .period element of the current message

7.358.4.10 debounce_time

```
ttk_bus.bus_signals_cangamma.RxCanBusMessage.debounce_time
```

minimum debounce delay for event-triggered Messages/PDUs in [ms]

7.358.4.11 descr

ttk_bus.bus_signals_cangamma.RxCanBusMessage.descr

Message description text.

7.358.4.12 enable_var

ttk_bus.bus_signals_cangamma.RxCanBusMessage.enable_var

HIL-Tx-enable/disable variable (only if set via enable_identifier)

7.358.4.13 kickout_var

ttk_bus.bus_signals_cangamma.RxCanBusMessage.kickout_var

HIL-Tx-kickout/trigger variable.

7.358.4.14 min_delay_var

ttk_bus.bus_signals_cangamma.RxCanBusMessage.min_delay_var

minimum delay between triggered messages in [ms], compare debounce delay in .debounce_time

7.358.4.15 recovery_time

ttk_bus.bus_signals_cangamma.RxCanBusMessage.recovery_time

time for recovery/"wiedergut" checks [ms]

7.358.4.16 timeout

ttk_bus.bus_signals_cangamma.RxCanBusMessage.timeout

detection time for timeout errors [ms]

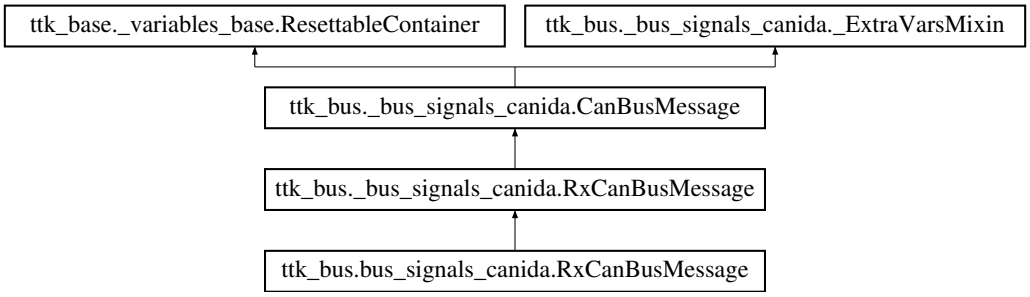
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_cangamma.py (v1.3)

7.359 ttk_bus.bus_signals_canida.RxCanBusMessage Class Reference

CAN Message/PDU ECU-Rx (HIL-Tx => ECU-Rx).

Inheritance diagram for ttk_bus.bus_signals_canida.RxCanBusMessage:



Public Member Functions

- __init__ (self, context, base_identifier=None, cycle_time=0, debounce_time=0, alias="", descr='', change_delay=None, timeout=None, recovery_time=None)

  Rx CAN Message/PDU data (HIL-Tx => ECU-Rx).
- enableHilTx (self)

  (Re-)enable sending of this Message/PDU.
- disableHilTx (self)

  Disable (cyclic) sending of this Message/PDU.
- kickout (self, delay_ms=None)

  Trigger sending of this CAN-Message/PDU.
- reset (self, *args, **kwargs)

  Call reset() on all contained variables/signals that support it.

Public Attributes

- enable_var

  HIL-Tx-enable/disable variable (only if set via enable_identifier)
- kickout_var

  HIL-Tx-kickout/trigger variable.
- cycle_time_var

  Message/PDU (Hil-Tx) cycle time access.
- min_delay_var

  minimum delay between triggered messages in [ms], compare debounce delay in .debounce_time
- context

  Parent context (a gaapi reference to resolve signal/variable paths)
- alias

  Message alias name.
- descr

  Message description text.
- base_identifier

  Base BusSystems path to Message/PDU helper variables below BusSystems e.g.
- cycle_time

  cycle time of Message/PDU in [ms]

- debounce_time

  minimum debounce delay for event-triggered Messages/PDUs in [ms]
- change_delay

  delay to wait after a value change [ms]
- timeout

  detection time for timeout errors [ms]
- recovery_time

  time for recovery/"wiedergut" checks [ms]

Static Public Attributes

- dict cfg_custom_defaults

  Configuration setting for new instances: Custom "hard" default values to use for internal real time system variables, so a reset() will always reset those variables to defined values (and not just those variables that were set/accessed).
- bool cfg_init_cycle_time = True

  True: automatically initialize message's cycle time from the configured cycle_time value (sets self.cycle_time_var to self.cycle_time during init)

7.359.1 Detailed Description

CAN Message/PDU ECU-Rx (HIL-Tx => ECU-Rx).

Extends the basic CanBusMessage with methods for enableHilTx/disableHilTx and message kickout (trigger).

7.359.2 Constructor & Destructor Documentation

7.359.2.1 __init__()

```
ttk_bus.bus_signals_canida.RxCanBusMessage.__init__ (
              self,
              context,
              base_identifier = None,
              cycle_time = 0,
              debounce_time = 0,
              alias = "",
              descr = '',
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Rx CAN Message/PDU data (HIL-Tx => ECU-Rx).

Parameters

context	parent iDA context
base_identifier	base identifier to PDU/message. Helper entries will be derived from this, like kickout or status variables. e.g. ecu-rx (hil-tx) "CAN.can3_iFIU_iSRC.HIL_TX.N0FIU_SetError_Card1"
cycle_time	cycle time in [ms]. Value will be used to enable/disable sending of cyclic messages, see cycle_time_var. Event-triggered messages have a cycle time of 0 (similar to DBC value)

Parameters

debounce_time	minimum debounce delay between events in [ms]
alias	Message alias/name
descr	(opt) Message description (may be used in report entries)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_canida.RxCanBusMessage.

7.359.3 Member Function Documentation

7.359.3.1 disableHilTx()

```
ttk_bus.bus_signals_canida.RxCanBusMessage.disableHilTx (
              self )
```

Disable (cyclic) sending of this Message/PDU.

Reimplemented from ttk_bus._bus_signals_canida.RxCanBusMessage.

7.359.3.2 enableHilTx()

```
ttk_bus.bus_signals_canida.RxCanBusMessage.enableHilTx (
              self )
```

(Re-)enable sending of this Message/PDU.

Reimplemented from ttk_bus._bus_signals_canida.RxCanBusMessage.

7.359.3.3 kickout()

```
ttk_bus.bus_signals_canida.RxCanBusMessage.kickout (
              self,
              delay_ms = None )
```

Trigger sending of this CAN-Message/PDU.

Parameters

delay_ms	[ms] Delay to wait before clearing kickout/trigger variable again.

Default Delay:

Delay defaults to 1/2 self.debounce_time (if explicitly set) or alternatively to 1/2 self.change_delay. Default delay will never be shorter than 5 ms.

Reimplemented from ttk_bus._bus_signals_canida.RxCanBusMessage.

7.359.3.4 reset()

```
ttk_bus.bus_signals_canida.RxCanBusMessage.reset (
              self,
             * args,
             ** kwargs )
```

Call reset() on all contained variables/signals that support it.

Simple, non-recursive approach.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.ResettableContainer.

7.359.4 Member Data Documentation

7.359.4.1 alias

```
ttk_bus.bus_signals_canida.RxCanBusMessage.alias
```

Message alias name.

7.359.4.2 base_identifier

```
ttk_bus.bus_signals_canida.RxCanBusMessage.base_identifier
```

Base BusSystems path to Message/PDU helper variables below BusSystems e.g.

"BusSystems.<CAN_NAME>.HIL_TX.<MSG_NAME>"

7.359.4.3 cfg_custom_defaults

```
dict ttk_bus.bus_signals_canida.RxCanBusMessage.cfg_custom_defaults  [static]
```

Initial value:

```
=  {
        "manual_enable_var":    None,
        "kickout_var":          None,
    }
```

Configuration setting for new instances: Custom "hard" default values to use for internal real time system variables, so a reset() will always reset those variables to defined values (and not just those variables that were set/accessed).

This might be helpful if "other" tests fail to properly clean up the test environment after they are done.

Mapping: <variable name as string>: <default value>

A value of None (or a missing variable name) enables normal/automatic reset behavior.

Typical default values:

- Enable variables are usually default to "enabled"
  - 0: disabled
  - 1: enabled
- Message/PDU (Hil-TX) cycle time access: Default value depends on configured cycle time (in [s]). Override standard behavior only if absolutely necessary.
- Kickout/trigger variables: Triggers should auto-reset to 0 once the PDU has been triggered. 0 is a safe default.

7.359.4.4 cfg_init_cycle_time

```
bool ttk_bus.bus_signals_canida.RxCanBusMessage.cfg_init_cycle_time = True  [static]
```

True: automatically initialize message's cycle time from the configured cycle_time value (sets self.cycle_time_var to self.cycle_time during init)

7.359.4.5 change_delay

```
ttk_bus.bus_signals_canida.RxCanBusMessage.change_delay
```

delay to wait after a value change [ms]

7.359.4.6 context

```
ttk_bus.bus_signals_canida.RxCanBusMessage.context
```

Parent context (a gaapi reference to resolve signal/variable paths)

7.359.4.7 cycle_time

```
ttk_bus.bus_signals_canida.RxCanBusMessage.cycle_time
```

cycle time of Message/PDU in [ms]

7.359.4.8 cycle_time_var

```
ttk_bus.bus_signals_canida.RxCanBusMessage.cycle_time_var
```

Message/PDU (Hil-Tx) cycle time access.

Cycle time in [ms]. Note: This maps to the .period element of the current message

7.359.4.9 debounce_time

```
ttk_bus.bus_signals_canida.RxCanBusMessage.debounce_time
```

minimum debounce delay for event-triggered Messages/PDUs in [ms]

7.359.4.10 descr

ttk_bus.bus_signals_canida.RxCanBusMessage.descr

Message description text.

7.359.4.11 enable_var

ttk_bus.bus_signals_canida.RxCanBusMessage.enable_var

HIL-Tx-enable/disable variable (only if set via enable_identifier)

7.359.4.12 kickout_var

ttk_bus.bus_signals_canida.RxCanBusMessage.kickout_var

HIL-Tx-kickout/trigger variable.

7.359.4.13 min_delay_var

ttk_bus.bus_signals_canida.RxCanBusMessage.min_delay_var

minimum delay between triggered messages in [ms], compare debounce delay in .debounce_time

7.359.4.14 recovery_time

ttk_bus.bus_signals_canida.RxCanBusMessage.recovery_time

time for recovery/"wiedergut" checks [ms]

7.359.4.15 timeout

ttk_bus.bus_signals_canida.RxCanBusMessage.timeout

detection time for timeout errors [ms]

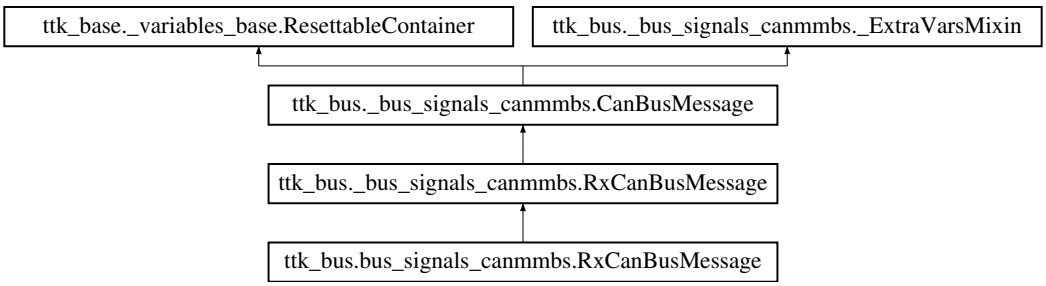
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_canida.py (v1.0.0)

7.360 ttk_bus.bus_signals_canmmbs.RxCanBusMessage Class Reference

CAN Message/PDU ECU-Rx (HIL-Tx => ECU-Rx).

Inheritance diagram for ttk_bus.bus_signals_canmmbs.RxCanBusMessage:



Public Member Functions

- __init__ (self, context, enable_path=None, bus_systems_path=None, cycle_time=0, debounce_time=0, alias="", descr='', change_delay=None, timeout=None, recovery_time=None)

  Rx CAN Message/PDU data (HIL-Tx => ECU-Rx).
- enableHilTx (self)

  (Re-)enable sending of this Message/PDU.
- disableHilTx (self)

  Disable sending of this Message/PDU.
- kickout (self, delay_ms=None)

  Trigger sending of this CAN-Message/PDU.
- reset (self, *args, **kwargs)

  Call reset() on all contained variables/signals that support it.

Public Attributes

- enable_var

  HIL-Tx-enable/disable variable (a HilVar instance)
- kickout_var

  HIL-Tx-kickout variable (a HilVar instance)
- cycle_time_var

  HIL-Tx-cycle_time variable (currently active cycle time in [s]) (a HilVar instance)
- status_var

  HIL-Tx-status variable (a HilVar instance)
- context

  Parent context (an RTE application instance to resolve signal/variable paths)
- alias

  Message alias name.
- descr

  Message description text.
- bus_systems_path

  Base BusSystems path to Message/PDU helper variables below BusSystems e.g.
- cycle_time

cycle time of Message/PDU in [ms]

- debounce_time

  minimum debounce delay for event-triggered Messages/PDUs in [ms]
- change_delay

  delay to wait after a value change [ms]
- timeout

  detection time for timeout errors [ms]
- recovery_time

  time for recovery/"wiedergut" checks [ms]

Static Public Attributes

- dict cfg_custom_defaults

  Configuration setting for new instances:
  Custom "hard" default values to use for internal real time system variables, so a reset() will always reset those variables to defined values (and not just those variables that were set/accessed).

7.360.1 Detailed Description

CAN Message/PDU ECU-Rx (HIL-Tx => ECU-Rx).

Extends the basic CanBusMessage with methods for enableHilTx/disableHilTx and message kickout (trigger).

7.360.2 Constructor & Destructor Documentation

7.360.2.1 __init__()

```
ttk_bus.bus_signals_canmmbs.RxCanBusMessage.__init__ (
              self,
              context,
              enable_path = None,
              bus_systems_path = None,
              cycle_time = 0,
              debounce_time = 0,
              alias = "",
              descr = '',
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Rx CAN Message/PDU data (HIL-Tx => ECU-Rx).

Parameters

context	parent context (an RTE application instance to resolve signal/variable paths)
enable_path	manual path to a tx enable switch. Keep at None to use default enable switch below BusSystems (see parameter bus_systems_path). The enable variable will later be set to 1 to enable or to 0 to disable message transmission.
bus_systems_path	base path to PDU/message below BusSystems, use "BusSystems/CAN/<CAN_NAME>/<MSG_NAME>/TX/<MSG_NAME>" or "BusSystems/CAN/<CAN_NAME>/<MSG_NAME>/TX/<MSG_NAME>_Enable" Helper entries will be derived from this path, e.g. kickout or status variables.

Parameters

cycle_time	cycle time in [ms]. Event-triggered messages have a cycle time of 0 (similar to DBC value)
debounce_time	minimum debounce delay between events in [ms]
alias	Message alias/name
descr	(opt) Message description (may be used in report entries)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_canmmbs.RxCanBusMessage.

7.360.3 Member Function Documentation

7.360.3.1 disableHilTx()

```
ttk_bus.bus_signals_canmmbs.RxCanBusMessage.disableHilTx (
            self )
```

Disable sending of this Message/PDU.

Reimplemented from ttk_bus._bus_signals_canmmbs.RxCanBusMessage.

7.360.3.2 enableHilTx()

```
ttk_bus.bus_signals_canmmbs.RxCanBusMessage.enableHilTx (
            self )
```

(Re-)enable sending of this Message/PDU.

Reimplemented from ttk_bus._bus_signals_canmmbs.RxCanBusMessage.

7.360.3.3 kickout()

```
ttk_bus.bus_signals_canmmbs.RxCanBusMessage.kickout (
            self,
            delay_ms = None )
```

Trigger sending of this CAN-Message/PDU.

Parameters

delay_ms	[ms] Delay to wait before clearing kickout/trigger variable again (it *should* auto-reset if trigger variable is located below BusSystems, but anyway).

Default Delay:

Delay defaults to 1/2 self.debounce_time (if explicitly set) or alternatively to 1/2 self.change_delay. Default delay will never be shorter than 5 ms.

Reimplemented from ttk_bus._bus_signals_canmmbs.RxCanBusMessage.

7.360.3.4 reset()

```
ttk_bus.bus_signals_canmmbs.RxCanBusMessage.reset (
              self,
             * args,
             ** kwargs )
```

Call reset() on all contained variables/signals that support it.

Simple, non-recursive approach.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.ResettableContainer.

7.360.4 Member Data Documentation

7.360.4.1 alias

```
ttk_bus.bus_signals_canmmbs.RxCanBusMessage.alias
```

Message alias name.

7.360.4.2 bus_systems_path

```
ttk_bus.bus_signals_canmmbs.RxCanBusMessage.bus_systems_path
```

Base BusSystems path to Message/PDU helper variables below BusSystems e.g.

"BusSystems/CAN/<CAN_NAME>/<MSG_NAME>/TX/<MSG_NAME>"

7.360.4.3 cfg_custom_defaults

```
dict ttk_bus.bus_signals_canmmbs.RxCanBusMessage.cfg_custom_defaults  [static]
```

Initial value:

```
=  {
        "enable_var":       None,
        "kickout_var":      None,
        "cycle_time_var":   None,
    }
```

Configuration setting for new instances:
Custom "hard" default values to use for internal real time system variables, so a reset() will always reset those variables to defined values (and not just those variables that were set/accessed).

This might be helpful if "other" tests fail to properly clean up the test environment after they are done.

Mapping: <variable name as string>: <default value>

A value of None (or a missing variable name) enables normal/automatic reset behavior.

Typical default values:

- Enable variables are usually default to "enabled"
  - 0: disabled
  - 1: enabled
- Message/PDU (Hil-TX) cycle time access:
  Default value depends on configured cycle time (in [s]).
  Override standard behavior only if absolutely necessary.
- Kickout/trigger variables:
  Triggers below BusSystems should auto-reset to 0 once the PDU has been triggered. 0 is a safe default.

7.360.4.4 change_delay

```
ttk_bus.bus_signals_canmmbs.RxCanBusMessage.change_delay
```

delay to wait after a value change [ms]

7.360.4.5 context

```
ttk_bus.bus_signals_canmmbs.RxCanBusMessage.context
```

Parent context (an RTE application instance to resolve signal/variable paths)

7.360.4.6 cycle_time

```
ttk_bus.bus_signals_canmmbs.RxCanBusMessage.cycle_time
```

cycle time of Message/PDU in [ms]

7.360.4.7 cycle_time_var

ttk_bus.bus_signals_canmmbs.RxCanBusMessage.cycle_time_var

HIL-Tx-cycle_time variable (currently active cycle time in [s]) (a HilVar instance)

7.360.4.8 debounce_time

ttk_bus.bus_signals_canmmbs.RxCanBusMessage.debounce_time

minimum debounce delay for event-triggered Messages/PDUs in [ms]

7.360.4.9 descr

ttk_bus.bus_signals_canmmbs.RxCanBusMessage.descr

Message description text.

7.360.4.10 enable_var

ttk_bus.bus_signals_canmmbs.RxCanBusMessage.enable_var

HIL-Tx-enable/disable variable (a HilVar instance)

7.360.4.11 kickout_var

ttk_bus.bus_signals_canmmbs.RxCanBusMessage.kickout_var

HIL-Tx-kickout variable (a HilVar instance)

7.360.4.12 recovery_time

ttk_bus.bus_signals_canmmbs.RxCanBusMessage.recovery_time

time for recovery/"wiedergut" checks [ms]

7.360.4.13 status_var

ttk_bus.bus_signals_canmmbs.RxCanBusMessage.status_var

HIL-Tx-status variable (a HilVar instance)

7.360.4.14 timeout

`ttk_bus.bus_signals_canmmbs.RxCanBusMessage.timeout`

detection time for timeout errors [ms]

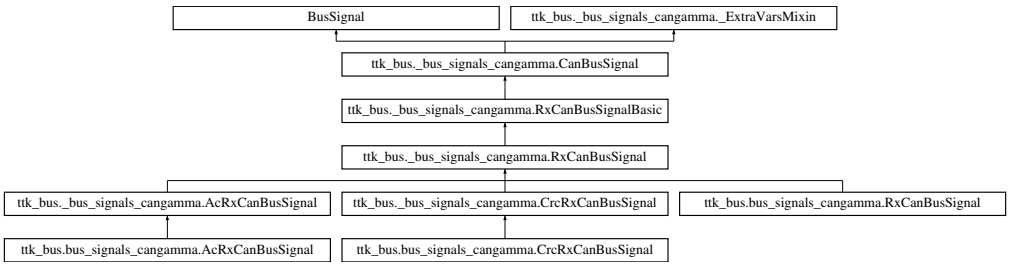
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_canmmbs.py (v1.8)

7.361 ttk_bus._bus_signals_cangamma.RxCanBusSignal Class Reference

A CAN signal sent from the test system to the DUT (ECU Rx <= HIL Tx).

Inheritance diagram for ttk_bus._bus_signals_cangamma.RxCanBusSignal:



Public Member Functions

- __init__ (self, context, signal_identifier, message=None, cycle_time=None, debounce_time=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  Rx bus signal initialization (ECU Rx <= HIL Tx).
- set (self, value)

  Set a value to the static override variable (`.static_value_phy_var`) and switch mode to use this value (mode 0x01).
- setState (self, state)

  Set a textual state to the static override variable (`.static_value_phy_var`) and switch mode to use this value (mode 0x01).

Static Public Attributes

- dict cfg_custom_defaults

  Configuration setting for new instances: Custom "hard" default value(s) to use for internal real time system variables (e.g.
- bool cfg_value_workaround = False

  [temporary] set True to also write to `.signal_var` (and not just `.static_value_phy_var`) as a workaround for missing raw/phys conversion in realtime model
- mode_var

  Mode switch variable for signal manipulation.
- lookup

- fmt
- dyn_count_var

  Dynamic Countdown Value, see modes 0x03 and beyond.
- dyn_value_phy_var

  Dynamic Value: value that is active while dyn_count_var decrements.
- dyn_pattern_var

  Dynamic Pattern Value, see modes 0x15/0x16/0x23.
- static_value_phy_var

  physical static override value (for mode 0x01)
- unit
- static_value_raw_var

  raw static override value (for mode 0x02)

7.361.1 Detailed Description

A CAN signal sent from the test system to the DUT (ECU Rx <= HIL Tx).

Standard implementation, accesses converted physical or raw values and provides options for error injections.

Use this for signal entries generated with iSyDBC mode "standard" . See _bus_signals_cangamma.RxCanBus↩SignalBasic.

7.361.2 Constructor & Destructor Documentation

7.361.2.1 __init__()

```
ttk_bus._bus_signals_cangamma.RxCanBusSignal.__init__ (
              self,
              context,
              signal_identifier,
              message = None,
              cycle_time = None,
              debounce_time = None,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Rx bus signal initialization (ECU Rx <= HIL Tx).

Parameters

context	parent gamma_api context
signal_identifier	identifier for signal value PV
message	reference to associated CanBusMessage instance
cycle_time	(opt) cycle time in [ms], overrides settings in Message
debounce_time	(opt) minimum debounce delay between events in [ms], overrides settings in Message
unit	(opt) unit of physical signal value (as string, mainly used for report entries)

Parameters

alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

Info: signal_identifier

HIL-Tx/ECU-Rx signals will have a PV element `.value` which contains the value as sent to CAN-plugin (typically in "raw" format).
For signal manipulation the PV elements `.static_value_phy` or `.static_value_raw` (and an appropriate mode setting) will be used.

Attention

When using the default PV `signal_identifier` ending in `.value`, reading the signal value via `.get()` will return the most recently written value in *raw* format

Reimplemented from ttk_bus._bus_signals_cangamma.RxCanBusSignalBasic.

Reimplemented in ttk_bus._bus_signals_cangamma.CrcRxCanBusSignal, ttk_bus._bus_signals_cangamma.AcRxCanBusSignal, ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal, ttk_bus.bus_signals_cangamma.AcRxCanBusSignal, and ttk_bus.bus_signals_cangamma.RxCanBusSignal.

7.361.3 Member Function Documentation

7.361.3.1 set()

```
ttk_bus._bus_signals_cangamma.RxCanBusSignal.set (
              self,
              value )
```

Set a value to the static override variable (`.static_value_phy_var`) and switch mode to use this value (mode 0x01).

Use `.reset()` to clear override (or manually clear the mode setting).

Parameters

value	value to set (physical)

Returns

the previous value from static_value_phy.

Reimplemented in ttk_bus.bus_signals_cangamma.RxCanBusSignal, ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal, and ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.

7.361.3.2 setState()

```
ttk_bus._bus_signals_cangamma.RxCanBusSignal.setState (
              self,
              state )
```

Set a textual state to the static override variable (`.static_value_phy_var`) and switch mode to use this value (mode 0x01).

This uses the lookup values supplied in constructor.

Parameters

state	named state to set

Returns

the previous value from static_value_phy.

Reimplemented in ttk_bus.bus_signals_cangamma.RxCanBusSignal, ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal, and ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.

7.361.4 Member Data Documentation

7.361.4.1 cfg_custom_defaults

```
dict ttk_bus._bus_signals_cangamma.RxCanBusSignal.cfg_custom_defaults  [static]
```

Initial value:

```
=  {
        "mode_var":          None,
        "dyn_count_var":     None,
        "dyn_value_phy_var": None,
    }
```

Configuration setting for new instances: Custom "hard" default value(s) to use for internal real time system variables (e.g.

switch settings), so a reset() will always reset those variables to defined values (and not just those variables that have actually been accessed). This might be helpful if "other" tests fail to properly clean up the test environment after they are done.

Mapping: <variable name as string>: <default value>

A value of None (or a missing variable name) enables normal/automatic reset behavior.

Typical default values:

- Switch variable (signal manipulation)
  0: default value from model input, see .mode_var for details.
- Dynamic Countdown Value: auto-decrements until 0 is reached if a dynamic mode is active
- Dynamic Value: value that is active while dyn_count_var decrements.

7.361.4.2 cfg_value_workaround

```
bool ttk_bus._bus_signals_cangamma.RxCanBusSignal.cfg_value_workaround = False  [static]
```

[temporary] set True to also write to `.signal_var` (and not just `.static_value_phy_var`) as a workaround for missing raw/phys conversion in realtime model

7.361.4.3 dyn_count_var

```
ttk_bus._bus_signals_cangamma.RxCanBusSignal.dyn_count_var  [static]
```

Dynamic Countdown Value, see modes 0x03 and beyond.

7.361.4.4 dyn_pattern_var

```
ttk_bus._bus_signals_cangamma.RxCanBusSignal.dyn_pattern_var  [static]
```

Dynamic Pattern Value, see modes 0x15/0x16/0x23.

7.361.4.5 dyn_value_phy_var

```
ttk_bus._bus_signals_cangamma.RxCanBusSignal.dyn_value_phy_var  [static]
```

Dynamic Value: value that is active while dyn_count_var decrements.

7.361.4.6 fmt

```
ttk_bus._bus_signals_cangamma.RxCanBusSignal.fmt  [static]
```

7.361.4.7 lookup

```
ttk_bus._bus_signals_cangamma.RxCanBusSignal.lookup  [static]
```

7.361.4.8 mode_var

ttk_bus._bus_signals_cangamma.RxCanBusSignal.mode_var [static]

Mode switch variable for signal manipulation.

- 0x00: simulink model inport value
- 0x01: <signal>.static_value_phy (manual override w/ physical value)
- 0x02: <signal>.static_value_raw (manual override w/ raw value)
- 0x03: <signal>.dyn_value_phy, while <signal>.dyn_count decrements down to 0, then model value (mode 0x00)
- 0x04: <signal>.dyn_value_phy, while <signal>.dyn_count decrements down to 0, then <signal>.static_value_phy (mode 0x01)
- 0x05: <signal>.dyn_value_phy, while <signal>.dyn_count decrements down to 0, then <signal>.static_value_raw (mode 0x02)

For alive counter signals (see dedicated sub-class):

- 0x10: alive-counter, running/counting
- 0x11: alive-counter, stopped (static value)
- 0x12: alive-counter, stopped while <signal>.dyn_count decrements, then continues counting/running from last count (mode 0x10)
- 0x13: alive-counter, stopped while <signal>.dyn_count decrements but continues counting in background (then mode 0x10)
- 0x14: alive-counter, add .offset to value,
- 0x15: alive-counter stopped/started according to .dyn_pattern
- 0x16: alive-counter stopped/started according to .dyn_pattern (continues counting while stopped)

For CRC signals (see dedicated sub-class):

- 0x20: CRC calculated with default/correct algorithm (.algo)
- 0x21: CRC calculated with algorithm set in .dyn_algo
- 0x22: CRC calculated with algorithm from .dyn_algo while .dyn_count decrements, then mode 0x20
- 0x23: CRC calculated with .algo/.dyn_algo according to .dyn_pattern

7.361.4.9 static_value_phy_var

ttk_bus._bus_signals_cangamma.RxCanBusSignal.static_value_phy_var [static]

physical static override value (for mode 0x01)

7.361.4.10 static_value_raw_var

```
ttk_bus._bus_signals_cangamma.RxCanBusSignal.static_value_raw_var  [static]
```

raw static override value (for mode 0x02)

7.361.4.11 unit

```
ttk_bus._bus_signals_cangamma.RxCanBusSignal.unit  [static]
```

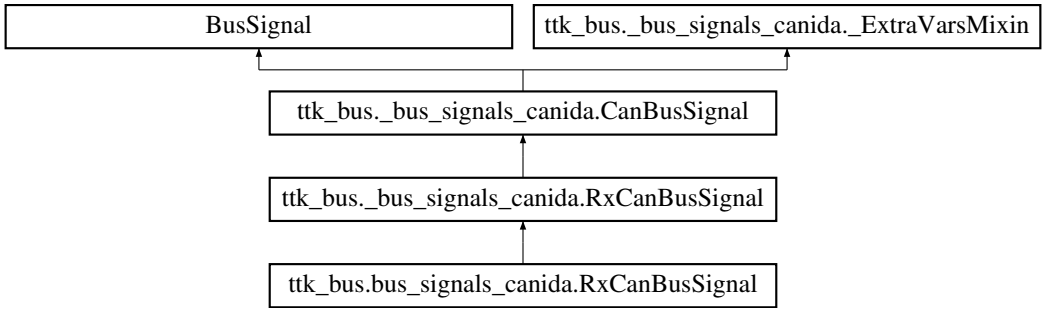
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_cangamma.py (v1.12)

7.362 ttk_bus._bus_signals_canida.RxCanBusSignal Class Reference

A basic CAN signal sent from the test system to the DUT (ECU Rx <= HIL Tx).

Inheritance diagram for ttk_bus._bus_signals_canida.RxCanBusSignal:



Public Member Functions

- __init__ (self, context, signal_identifier, message=None, cycle_time=None, debounce_time=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  Basic Rx bus signal initialization (ECU Rx <= HIL Tx).
- enableHilTx (self)

  Enable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).
- disableHilTx (self)

  Disable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).
- kickout (self)

  Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent CAN-Message/PDU).

7.362.1 Detailed Description

A basic CAN signal sent from the test system to the DUT (ECU Rx <= HIL Tx).

Basic implementation, directly accesses raw value as-is from CAN, without a converted physical value or error injections.

7.362.2 Constructor & Destructor Documentation

7.362.2.1 __init__()

```
ttk_bus._bus_signals_canida.RxCanBusSignal.__init__ (
              self,
              context,
              signal_identifier,
              message = None,
              cycle_time = None,
              debounce_time = None,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Basic Rx bus signal initialization (ECU Rx <= HIL Tx).

Note

`signal_identifier` var name Use `.value` entry (or optionally `.value_phy`/`.value_raw`, if available), e.g.
`"CAN.CanName.HIL_TX.MessageName.signals.SignalName.value"`
or the base signal identifier to default to use `.value`:
`"CAN.CanName.HIL_TX.MessageName.signals.SignalName"`

Parameters

context	parent iDA context
signal_identifier	identifier for signal value var (see note for samples)
message	reference to associated CanBusMessage instance
cycle_time	(opt) cycle time in [ms], overrides settings in Message
debounce_time	(opt) minimum debounce delay between events in [ms], overrides settings in Message
unit	(opt) unit of physical signal value (as string, mainly used for report entries)
alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

Minimal/Basic Signals:

ECU-Rx/HIL-Tx Signals generated by iSyDBC in minimal/basic mode will only contain PVs for:

- `<SignalName>.value` - raw value as sent to CAN-Plugin
- `<SignalName>.period` - tx cycle time / period in ms.

- `<SignalName>.timestamp` - Tx time stamp
- `<SignalName>.length` - message length

Reimplemented from ttk_bus._bus_signals_canida.CanBusSignal.

Reimplemented in ttk_bus.bus_signals_canida.RxCanBusSignal.

7.362.3 Member Function Documentation

7.362.3.1 disableHilTx()

```
ttk_bus._bus_signals_canida.RxCanBusSignal.disableHilTx (
              self )
```

Disable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).

Note

This will disable transmission of *all* signals of the parent Message/PDU.

Reimplemented in ttk_bus.bus_signals_canida.RxCanBusSignal.

7.362.3.2 enableHilTx()

```
ttk_bus._bus_signals_canida.RxCanBusSignal.enableHilTx (
              self )
```

Enable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).

Note

This will enable transmission of *all* signals of the parent Message/PDU.

Reimplemented in ttk_bus.bus_signals_canida.RxCanBusSignal.

7.362.3.3 kickout()

```
ttk_bus._bus_signals_canida.RxCanBusSignal.kickout (
              self )
```

Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent CAN-Message/PDU).

Note

All other signals of the parent Message/PDU will also be sent.

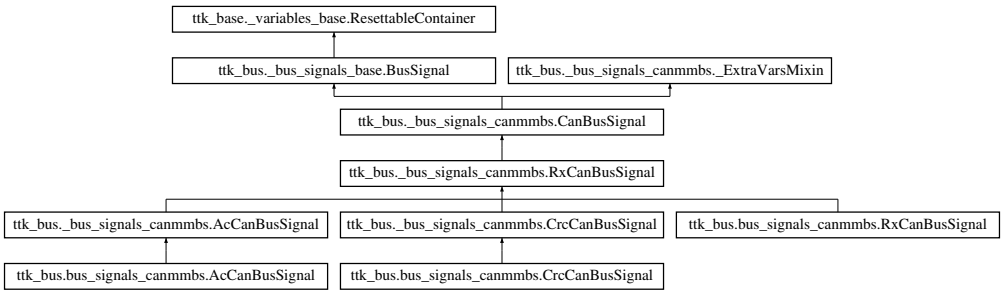
Reimplemented in ttk_bus.bus_signals_canida.RxCanBusSignal.

The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_canida.py (v1.0.2)

7.363 ttk_bus._bus_signals_canmmbs.RxCanBusSignal Class Reference

Inheritance diagram for ttk_bus._bus_signals_canmmbs.RxCanBusSignal:



Public Member Functions

- __init__ (self, context, signal_path, message=None, cycle_time=None, debounce_time=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  Rx bus signal initialization.
- switchToBusSystemsSource (self, bus_systems=True)

  Switch signal source to "BusSystems".
- set (self, value)

  Set a value to the HIL signal variable below BusSystems (and switch the signal source to "BusSystems" if necessary).
- setState (self, state)

  Set a textual state to the HIL signal variable below BusSystems (and switches the signal source to "BusSystems" if necessary).
- enableHilTx (self)

  Enable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).
- disableHilTx (self)

  Disable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).
- kickout (self)

  Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent CAN-Message/PDU).

Public Attributes

- switch_var

  Switch variable for signal manipulation.
- dyncount_var

  Dynamic Countdown Value.
- dynvalue_var

  Dynamic Value: value that is active while dyncount_var decrements.

Static Public Attributes

- bool cfg_init_signal_manipulation = False

  Configuration setting for new instances:
  Try to initialize (additional) variables that are only available if "dynamic values" input manipulation has been selected during RTICANMM configuration for a signal.
- dict cfg_custom_defaults

  Configuration setting for new instances:
  Custom "hard" default value(s) to use for internal real time system variables (e.g.

7.363.1 Constructor & Destructor Documentation

7.363.1.1 __init__()

```
ttk_bus._bus_signals_canmmbs.RxCanBusSignal.__init__ (
                self,
                context,
                signal_path,
                message = None,
                cycle_time = None,
                debounce_time = None,
                unit = "",
                alias = "",
                descr = "",
                lookup = None,
                change_delay = None,
                timeout = None,
                recovery_time = None )
```

Rx bus signal initialization.

Parameters

context	parent context (an RTE application instance to resolve signal/variable paths)
signal_path	base path to signal value below BusSystems
message	reference to associated CanBusMessage instance
cycle_time	(opt) cycle time in [ms], overrides settings in Message
debounce_time	(opt) minimum debounce delay between events in [ms], overrides settings in Message
unit	(opt) unit of physical signal value (as string, mainly used for report entries)
alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_canmmbs.CanBusSignal.

Reimplemented in ttk_bus._bus_signals_canmmbs.CrcCanBusSignal, ttk_bus._bus_signals_canmmbs.AcCanBusSignal, ttk_bus.bus_signals_canmmbs.CrcCanBusSignal, ttk_bus.bus_signals_canmmbs.AcCanBusSignal, and ttk_bus.bus_signals_canmm

7.363.2 Member Function Documentation

7.363.2.1 disableHilTx()

```
ttk_bus._bus_signals_canmmbs.RxCanBusSignal.disableHilTx (
                self )
```

Disable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).

Note

This will disable transmission of *all* signals of the parent Message/PDU.

Reimplemented in ttk_bus.bus_signals_canmmbs.RxCanBusSignal, ttk_bus.bus_signals_canmmbs.CrcCanBusSignal, and ttk_bus.bus_signals_canmmbs.AcCanBusSignal.

7.363.2.2 enableHilTx()

```
ttk_bus._bus_signals_canmmbs.RxCanBusSignal.enableHilTx (
              self )
```

Enable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).

Note

This will enable transmission of *all* signals of the parent Message/PDU.

Reimplemented in ttk_bus.bus_signals_canmmbs.RxCanBusSignal, ttk_bus.bus_signals_canmmbs.CrcCanBusSignal, and ttk_bus.bus_signals_canmmbs.AcCanBusSignal.

7.363.2.3 kickout()

```
ttk_bus._bus_signals_canmmbs.RxCanBusSignal.kickout (
              self )
```

Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent CAN-Message/PDU).

Note

All other signals of the parent Message/PDU will also be sent.

Reimplemented in ttk_bus.bus_signals_canmmbs.RxCanBusSignal, ttk_bus.bus_signals_canmmbs.CrcCanBusSignal, and ttk_bus.bus_signals_canmmbs.AcCanBusSignal.

7.363.2.4 set()

```
ttk_bus._bus_signals_canmmbs.RxCanBusSignal.set (
              self,
              value )
```

Set a value to the HIL signal variable below BusSystems (and switch the signal source to "BusSystems" if necessary).

Parameters

value	new value to set

Returns

the previous value from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

Reimplemented in ttk_bus.bus_signals_canmmbs.RxCanBusSignal, ttk_bus.bus_signals_canmmbs.CrcCanBusSignal, and ttk_bus.bus_signals_canmmbs.AcCanBusSignal.

7.363.2.5 setState()

```
ttk_bus._bus_signals_canmmbs.RxCanBusSignal.setState (
              self,
              state )
```

Set a textual state to the HIL signal variable below BusSystems (and switches the signal source to "BusSystems" if necessary).

This uses the lookup values supplied in constructor.

Parameters

state	named state to set

Returns

the previous value representation from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

Reimplemented in ttk_bus.bus_signals_canmmbs.RxCanBusSignal, ttk_bus.bus_signals_canmmbs.CrcCanBusSignal, and ttk_bus.bus_signals_canmmbs.AcCanBusSignal.

7.363.2.6 switchToBusSystemsSource()

```
ttk_bus._bus_signals_canmmbs.RxCanBusSignal.switchToBusSystemsSource (
              self,
              bus_systems = True )
```

Switch signal source to "BusSystems".

Mainly intended for internal use.

Parameters

bus_systems	True: use BusSystems as source False: use Simulink Model (Model Root/...) as source

7.363.3 Member Data Documentation

7.363.3.1 cfg_custom_defaults

```
dict ttk_bus._bus_signals_canmmbs.RxCanBusSignal.cfg_custom_defaults  [static]
```

Initial value:

```
=  {
        "switch_var":   None,
        "dyncount_var": None,
        "dynvalue_var": None,
    }
```

Configuration setting for new instances:
Custom "hard" default value(s) to use for internal real time system variables (e.g.

switch settings), so a reset() will always reset those variables to defined values (and not just those variables that have actually been accessed).
This might be helpful if "other" tests fail to properly clean up the test environment after they are done.

Mapping: <variable name as string>: <default value>

A value of None (or a missing variable name) enables normal/automatic reset behavior.

Typical default values:

- Switch variable (signal manipulation)
  - 0: model input (if configured/available)
  - 1: constant from Trc/BusSystems
  - 2: counter running (for counter signals, default switch setting for enabled counters)
  - 3: constant + counter (for counter signals)
  - 4: toggle value (for toggle signals)
  - 5: parity value (for parity signals)
  - 6: ~parity value (for parity signals)
  - 7: error value (if an error value is configured)
  - 8: dynamic value (if dynamic value manipulation is configured)
- Dynamic Countdown Value: Auto-decrements until 0 is reached if signal manipulation switch is set to 8 => 0 would be a safe default value
- Dynamic Value: value that is active while dyncount_var decrements.

7.363.3.2 cfg_init_signal_manipulation

```
bool ttk_bus._bus_signals_canmmbs.RxCanBusSignal.cfg_init_signal_manipulation = False  [static]
```

Configuration setting for new instances:
Try to initialize (additional) variables that are only available if "dynamic values" input manipulation has been selected during RTICANMM configuration for a signal.

Additional variables:

- dyncount_var
- dynvalue_var

7.363.3.3 dyncount_var

`ttk_bus._bus_signals_canmmbs.RxCanBusSignal.dyncount_var`

Dynamic Countdown Value.

Only available if signal manipulation is configured in CANMM and enabled in class config.
See self.cfg_init_signal_manipulation

7.363.3.4 dynvalue_var

`ttk_bus._bus_signals_canmmbs.RxCanBusSignal.dynvalue_var`

Dynamic Value: value that is active while dyncount_var decrements.

Only available if signal manipulation is configured in CANMM and enabled in class config.
See self.cfg_init_signal_manipulation

7.363.3.5 switch_var

`ttk_bus._bus_signals_canmmbs.RxCanBusSignal.switch_var`

Switch variable for signal manipulation.

Only available if more than one signal manipulation option is configured in CANMM.

Values:

- 0: model input (if configured/available)
- 1: constant from Trc/BusSystems
- 2: counter running (for counter signals, default switch setting for enabled counters)
- 3: constant + counter (for counter signals)
- 4: toggle value (for toggle signals)
- 5: parity value (for parity signals)
- 6: ~parity value (for parity signals)
- 7: error value (if an error value is configured)
- 8: dynamic value (if dynamic value manipulation is configured)

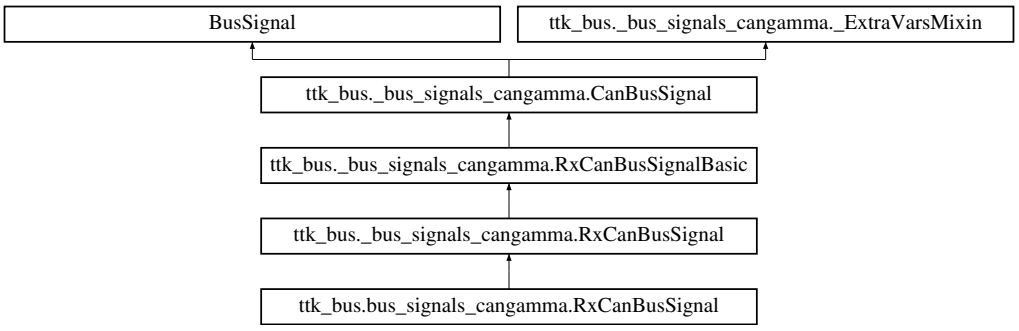
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_canmmbs.py (v2.3)

7.364 ttk_bus.bus_signals_cangamma.RxCanBusSignal Class Reference

A CAN signal sent from the test system to the DUT (ECU Rx <= HIL Tx).

Inheritance diagram for ttk_bus.bus_signals_cangamma.RxCanBusSignal:



Public Member Functions

- __init__ (self, context, signal_identifier, message=None, cycle_time=None, debounce_time=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  Rx bus signal initialization (ECU Rx <= HIL Tx).
- info (self)

  Get a current info string about the HIL signal variable, see `signal_var`.
- get (self)

  Get the current value of the HIL signal variable, see `signal_var`.
- set (self, value)

  Set a value to the static override variable (`.static_value_phy_var`) and switch mode to use this value (mode 0x01).
- getState (self)

  Get a textual state for the HIL signal variable's current data, see `signal_var`.
- setState (self, state)

  Set a textual state to the static override variable (`.static_value_phy_var`) and switch to use this value (mode 0x01).
- getStateDescr (self, value=None, fallback=None)

  Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).
- enableHilTx (self)

  Enable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).
- disableHilTx (self)

  Disable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).
- kickout (self)

  Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent CAN-Message/PDU).
- reset (self, *args, **kwargs)

  Reset all resettable variables of this bus signal.

Public Attributes

- mode_var

  Mode switch variable for signal manipulation.
- dyn_count_var

  Dynamic Countdown Value, see modes 0x03 and beyond.
- dyn_value_phy_var

  Dynamic Value: value that is active while dyn_count_var decrements.
- dyn_pattern_var

  Dynamic Pattern Value, see modes 0x15/0x16/0x23.
- static_value_phy_var

  physical static override value (for mode 0x01)
- static_value_raw_var

  raw static override value (for mode 0x02)
- base_identifier

  Signal PV address/identifier as base identifier for deriving sub-elements.
- message

  reference to associated (Rx-)CanBusMessage instance
- context

  parent context (a gaapi reference to resolve signal/variable PVs)
- signal_var

  signal value variable.
- unit

  unit of physical signal value (as string, mainly used for report entries)
- alias

  bus signal alias/name
- descr

  bus signal description text
- cycle_time

  cycle time of Message/PDU in [ms]
- debounce_time

  minimum debounce delay for event-triggered Messages/PDUs in [ms]
- change_delay

  delay to wait after a value change [ms]
- timeout

  detection time for timeout errors [ms]
- recovery_time

  time for recovery/"wiedergut" checks [ms]

Static Public Attributes

- dict cfg_custom_defaults

  Configuration setting for new instances: Custom "hard" default value(s) to use for internal real time system variables (e.g.
- bool cfg_value_workaround = False

  [temporary] set True to also write to `.signal_var` (and not just `.static_value_phy_var`) as a workaround for missing raw/phys conversion in realtime model

7.364.1 Detailed Description

A CAN signal sent from the test system to the DUT (ECU Rx <= HIL Tx).

Standard implementation, accesses converted physical or raw values and provides options for error injections.

Use this for signal entries generated with iSyDBC mode "standard" . See _bus_signals_cangamma.RxCanBus↩SignalBasic.

7.364.2 Constructor & Destructor Documentation

7.364.2.1 __init__()

```
ttk_bus.bus_signals_cangamma.RxCanBusSignal.__init__ (
                self,
                context,
                signal_identifier,
                message = None,
                cycle_time = None,
                debounce_time = None,
                unit = "",
                alias = "",
                descr = "",
                lookup = None,
                change_delay = None,
                timeout = None,
                recovery_time = None )
```

Rx bus signal initialization (ECU Rx <= HIL Tx).

Parameters

context	parent gamma_api context
signal_identifier	identifier for signal value PV
message	reference to associated CanBusMessage instance
cycle_time	(opt) cycle time in [ms], overrides settings in Message
debounce_time	(opt) minimum debounce delay between events in [ms], overrides settings in Message
unit	(opt) unit of physical signal value (as string, mainly used for report entries)
alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

Info: signal_identifier

HIL-Tx/ECU-Rx signals will have a PV element `.value` which contains the value as sent to CAN-plugin (typically in "raw" format).

For signal manipulation the PV elements `.static_value_phy` or `.static_value_raw` (and an appropriate mode setting) will be used.

Attention

When using the default PV `signal_identifier` ending in `.value`, reading the signal value via `.get()` will return the most recently written value in *raw* format

Reimplemented from ttk_bus._bus_signals_cangamma.RxCanBusSignal.

7.364.3 Member Function Documentation

7.364.3.1 disableHilTx()

```
ttk_bus.bus_signals_cangamma.RxCanBusSignal.disableHilTx (
              self )
```

Disable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).

Note

This will disable transmission of *all* signals of the parent Message/PDU.

Reimplemented from ttk_bus._bus_signals_cangamma.RxCanBusSignalBasic.

7.364.3.2 enableHilTx()

```
ttk_bus.bus_signals_cangamma.RxCanBusSignal.enableHilTx (
              self )
```

Enable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).

Note

This will enable transmission of *all* signals of the parent Message/PDU.

Reimplemented from ttk_bus._bus_signals_cangamma.RxCanBusSignalBasic.

7.364.3.3 get()

```
ttk_bus.bus_signals_cangamma.RxCanBusSignal.get (
              self )
```

Get the current value of the HIL signal variable, see `signal_var`.

Attention

For RxCanBusSignals, this will get the current (raw) value of the HIL-send-buffer, not the last seen signal value on CAN bus.

Returns

the current value from `signal_var`.

7.364.3.4 getState()

```
ttk_bus.bus_signals_cangamma.RxCanBusSignal.getState (
              self )
```

Get a textual state for the HIL signal variable's current data, see `signal_var`.

This uses the lookup values supplied in constructor. Fallback (if no matching state was found) will be the unmapped numeric value.

Attention

For RxCanBusSignals, this will get the current (raw) value of the HIL-send-buffer, not the last seen signal value on CAN bus.

Returns

the current state representation from `signal_var`.

7.364.3.5 getStateDescr()

```
ttk_bus.bus_signals_cangamma.RxCanBusSignal.getStateDescr (
              self,
              value = None,
              fallback = None )
```

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the last read value will be used instead.
fallback	fallback value to return if no match was found. Default: None

Returns

the state text if a match was found, otherwise the specified fallback value.

7.364.3.6 info()

```
ttk_bus.bus_signals_cangamma.RxCanBusSignal.info (
              self )
```

Get a current info string about the HIL signal variable, see `signal_var`.

Attention

For RxCanBusSignals, this will show info on the contents of the HIL-send-buffer, not the value from Model/↩ Simulink source nor the last seen signal value on CAN bus.

Returns

the current info string from `signal_var`.

7.364.3.7 kickout()

```
ttk_bus.bus_signals_cangamma.RxCanBusSignal.kickout (
              self )
```

Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent CAN-Message/PDU).

Note

All other signals of the parent Message/PDU will also be sent.

Reimplemented from ttk_bus._bus_signals_cangamma.RxCanBusSignalBasic.

7.364.3.8 reset()

```
ttk_bus.bus_signals_cangamma.RxCanBusSignal.reset (
              self,
             * args,
             ** kwargs )
```

Reset all resettable variables of this bus signal.

Note that this will also switch the variable source switch back to its original setting.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

7.364.3.9 set()

```
ttk_bus.bus_signals_cangamma.RxCanBusSignal.set (
              self,
              value )
```

Set a value to the static override variable (`.static_value_phy_var`) and switch mode to use this value (mode 0x01).

Use `.reset()` to clear override (or manually clear the mode setting).

Parameters

value	value to set (physical)

Returns

the previous value from static_value_phy.

Reimplemented from ttk_bus._bus_signals_cangamma.RxCanBusSignal.

7.364.3.10 setState()

```
ttk_bus.bus_signals_cangamma.RxCanBusSignal.setState (
              self,
              state )
```

Set a textual state to the static override variable (`.static_value_phy_var`) and switch to use this value (mode 0x01).

This uses the lookup values supplied in constructor.

Parameters

state	named state to set

Returns

the previous value from static_value_phy.

Reimplemented from ttk_bus._bus_signals_cangamma.RxCanBusSignal.

7.364.4 Member Data Documentation

7.364.4.1 alias

```
ttk_bus.bus_signals_cangamma.RxCanBusSignal.alias
```

bus signal alias/name

7.364.4.2 base_identifier

```
ttk_bus.bus_signals_cangamma.RxCanBusSignal.base_identifier
```

Signal PV address/identifier as base identifier for deriving sub-elements.

For CAN/Gamma signals, related variables follow the form `<base_identifier>.<suffix>`

7.364.4.3 cfg_custom_defaults

```
dict ttk_bus.bus_signals_cangamma.RxCanBusSignal.cfg_custom_defaults  [static]
```

Initial value:

```
=  {
        "mode_var":          None,
        "dyn_count_var":     None,
        "dyn_value_phy_var": None,
    }
```

Configuration setting for new instances: Custom "hard" default value(s) to use for internal real time system variables (e.g.

switch settings), so a reset() will always reset those variables to defined values (and not just those variables that have actually been accessed). This might be helpful if "other" tests fail to properly clean up the test environment after they are done.

Mapping: <variable name as string>: <default value>

A value of None (or a missing variable name) enables normal/automatic reset behavior.

Typical default values:

- Switch variable (signal manipulation)
  0: default value from model input, see .mode_var for details.
- Dynamic Countdown Value: auto-decrements until 0 is reached if a dynamic mode is active
- Dynamic Value: value that is active while dyn_count_var decrements.

7.364.4.4 cfg_value_workaround

```
bool ttk_bus.bus_signals_cangamma.RxCanBusSignal.cfg_value_workaround = False  [static]
```

[temporary] set True to also write to `.signal_var` (and not just `.static_value_phy_var`) as a workaround for missing raw/phys conversion in realtime model

7.364.4.5 change_delay

```
ttk_bus.bus_signals_cangamma.RxCanBusSignal.change_delay
```

delay to wait after a value change [ms]

7.364.4.6 context

```
ttk_bus.bus_signals_cangamma.RxCanBusSignal.context
```

parent context (a gaapi reference to resolve signal/variable PVs)

7.364.4.7 cycle_time

```
ttk_bus.bus_signals_cangamma.RxCanBusSignal.cycle_time
```

cycle time of Message/PDU in [ms]

7.364.4.8 debounce_time

```
ttk_bus.bus_signals_cangamma.RxCanBusSignal.debounce_time
```

minimum debounce delay for event-triggered Messages/PDUs in [ms]

7.364.4.9 descr

```
ttk_bus.bus_signals_cangamma.RxCanBusSignal.descr
```

bus signal description text

7.364.4.10 dyn_count_var

ttk_bus.bus_signals_cangamma.RxCanBusSignal.dyn_count_var

Dynamic Countdown Value, see modes 0x03 and beyond.

7.364.4.11 dyn_pattern_var

ttk_bus.bus_signals_cangamma.RxCanBusSignal.dyn_pattern_var

Dynamic Pattern Value, see modes 0x15/0x16/0x23.

7.364.4.12 dyn_value_phy_var

ttk_bus.bus_signals_cangamma.RxCanBusSignal.dyn_value_phy_var

Dynamic Value: value that is active while dyn_count_var decrements.

7.364.4.13 message

ttk_bus.bus_signals_cangamma.RxCanBusSignal.message

reference to associated (Rx-)CanBusMessage instance

7.364.4.14 mode_var

ttk_bus.bus_signals_cangamma.RxCanBusSignal.mode_var

Mode switch variable for signal manipulation.

- 0x00: simulink model inport value
- 0x01: <signal>.static_value_phy (manual override w/ physical value)
- 0x02: <signal>.static_value_raw (manual override w/ raw value)
- 0x03: <signal>.dyn_value_phy, while <signal>.dyn_count decrements down to 0, then model value (mode 0x00)
- 0x04: <signal>.dyn_value_phy, while <signal>.dyn_count decrements down to 0, then <signal>.static_value_phy (mode 0x01)
- 0x05: <signal>.dyn_value_phy, while <signal>.dyn_count decrements down to 0, then <signal>.static_value_raw (mode 0x02)

For alive counter signals (see dedicated sub-class):

- 0x10: alive-counter, running/counting

- 0x11: alive-counter, stopped (static value)
- 0x12: alive-counter, stopped while `<signal>.dyn_count` decrements, then continues counting/running from last count (mode 0x10)
- 0x13: alive-counter, stopped while `<signal>.dyn_count` decrements but continues counting in background (then mode 0x10)
- 0x14: alive-counter, add `.offset` to value,
- 0x15: alive-counter stopped/started according to `.dyn_pattern`
- 0x16: alive-counter stopped/started according to `.dyn_pattern` (continues counting while stopped)

For CRC signals (see dedicated sub-class):

- 0x20: CRC calculated with default/correct algorithm (`.algo`)
- 0x21: CRC calculated with algorithm set in `.dyn_algo`
- 0x22: CRC calculated with algorithm from `.dyn_algo` while `.dyn_count` decrements, then mode 0x20
- 0x23: CRC calculated with `.algo`/`.dyn_algo` according to `.dyn_pattern`

7.364.4.15 recovery_time

`ttk_bus.bus_signals_cangamma.RxCanBusSignal.recovery_time`

time for recovery/"wiedergut" checks [ms]

7.364.4.16 signal_var

`ttk_bus.bus_signals_cangamma.RxCanBusSignal.signal_var`

signal value variable.

get/info methods of this RxCanBusSignal will access this variable.

7.364.4.17 static_value_phy_var

`ttk_bus.bus_signals_cangamma.RxCanBusSignal.static_value_phy_var`

physical static override value (for mode 0x01)

7.364.4.18 static_value_raw_var

`ttk_bus.bus_signals_cangamma.RxCanBusSignal.static_value_raw_var`

raw static override value (for mode 0x02)

7.364.4.19 timeout

`ttk_bus.bus_signals_cangamma.RxCanBusSignal.timeout`

detection time for timeout errors [ms]

7.364.4.20 unit

`ttk_bus.bus_signals_cangamma.RxCanBusSignal.unit`

unit of physical signal value (as string, mainly used for report entries)

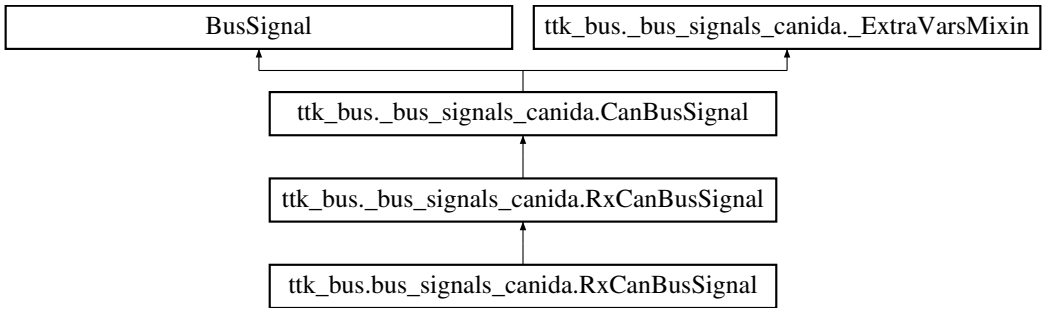
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_cangamma.py (v1.3)

7.365 ttk_bus.bus_signals_canida.RxCanBusSignal Class Reference

A basic CAN signal sent from the test system to the DUT (ECU Rx <= HIL Tx).

Inheritance diagram for ttk_bus.bus_signals_canida.RxCanBusSignal:



Public Member Functions

- __init__ (self, context, signal_identifier, message=None, cycle_time=None, debounce_time=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  Basic Rx bus signal initialization (ECU Rx <= HIL Tx).
- info (self)

  Get a current info string about the HIL signal variable, see `signal_var`.
- get (self)

  Get the current value of the HIL signal variable, see `signal_var`.
- set (self, value)

  Set a value to the HIL signal variable (`.signal_var`).
- getState (self)

  Get a textual state for the HIL signal variable's current data, see `value_var`.
- setState (self, state)

  Set a textual state to the HIL signal variable (`.value_var`).
- getStateDescr (self, value=None, fallback=None)

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

- reset (self, ∗args, ∗∗kwargs)

  Reset all resettable variables of this bus signal.
- enableHilTx (self)

  Enable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).
- disableHilTx (self)

  Disable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).
- kickout (self)

  Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent CAN-Message/PDU).

Public Attributes

- base_identifier

  Signal PV address/identifier as base identifier for deriving sub-elements.
- message

  reference to associated (Rx-)CanBusMessage instance
- context

  parent context (a gaapi reference to resolve signal/variable PVs)
- signal_var

  signal value variable.
- unit

  unit of physical signal value (as string, mainly used for report entries)
- alias

  bus signal alias/name
- descr

  bus signal description text
- cycle_time

  cycle time of Message/PDU in [ms]
- debounce_time

  minimum debounce delay for event-triggered Messages/PDUs in [ms]
- change_delay

  delay to wait after a value change [ms]
- timeout

  detection time for timeout errors [ms]
- recovery_time

  time for recovery/"wiedergut" checks [ms]

7.365.1 Detailed Description

A basic CAN signal sent from the test system to the DUT (ECU Rx <= HIL Tx).

Basic implementation, directly accesses raw value as-is from CAN, without a converted physical value or error injections.

7.365.2 Constructor & Destructor Documentation

7.365.2.1 __init__()

```
ttk_bus.bus_signals_canida.RxCanBusSignal.__init__ (
              self,
              context,
              signal_identifier,
              message = None,
              cycle_time = None,
              debounce_time = None,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Basic Rx bus signal initialization (ECU Rx <= HIL Tx).

Note

`signal_identifier` var name Use `.value` entry (or optionally `.value_phy`/`.value_raw`, if available), e.g.
`"CAN.CanName.HIL_TX.MessageName.signals.SignalName.value"`
or the base signal identifier to default to use `.value`:
`"CAN.CanName.HIL_TX.MessageName.signals.SignalName"`

Parameters

context	parent iDA context
signal_identifier	identifier for signal value var (see note for samples)
message	reference to associated CanBusMessage instance
cycle_time	(opt) cycle time in [ms], overrides settings in Message
debounce_time	(opt) minimum debounce delay between events in [ms], overrides settings in Message
unit	(opt) unit of physical signal value (as string, mainly used for report entries)
alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

Minimal/Basic Signals:

ECU-Rx/HIL-Tx Signals generated by iSyDBC in minimal/basic mode will only contain PVs for:

- `<SignalName>.value` - raw value as sent to CAN-Plugin
- `<SignalName>.period` - tx cycle time / period in ms.

- `<SignalName>.timestamp` - Tx time stamp
- `<SignalName>.length` - message length

Reimplemented from ttk_bus._bus_signals_canida.RxCanBusSignal.

7.365.3 Member Function Documentation

7.365.3.1 disableHilTx()

```
ttk_bus.bus_signals_canida.RxCanBusSignal.disableHilTx (
              self )
```

Disable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).

Note

This will disable transmission of *all* signals of the parent Message/PDU.

Reimplemented from ttk_bus._bus_signals_canida.RxCanBusSignal.

7.365.3.2 enableHilTx()

```
ttk_bus.bus_signals_canida.RxCanBusSignal.enableHilTx (
              self )
```

Enable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).

Note

This will enable transmission of *all* signals of the parent Message/PDU.

Reimplemented from ttk_bus._bus_signals_canida.RxCanBusSignal.

7.365.3.3 get()

```
ttk_bus.bus_signals_canida.RxCanBusSignal.get (
              self )
```

Get the current value of the HIL signal variable, see `signal_var`.

Attention

For RxCanBusSignals, this will get the current (raw) value of the HIL-send-buffer, not the last seen signal value on CAN bus.

Returns

the current value from `signal_var`.

7.365.3.4 getState()

```
ttk_bus.bus_signals_canida.RxCanBusSignal.getState (
              self )
```

Get a textual state for the HIL signal variable's current data, see `value_var`.

This uses the lookup values supplied in constructor. Fallback (if no matching state was found) will be the unmapped numeric value.

Attention

For RxCanBusSignals, this will get the current (raw) value of the HIL-send-buffer, not the last seen signal value on CAN bus.

Returns

the current state representation from `signal_var`.

7.365.3.5 getStateDescr()

```
ttk_bus.bus_signals_canida.RxCanBusSignal.getStateDescr (
              self,
              value = None,
              fallback = None )
```

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the last read value will be used instead.
fallback	fallback value to return if no match was found. Default: None

Returns

the state text if a match was found, otherwise the specified fallback value.

7.365.3.6 info()

```
ttk_bus.bus_signals_canida.RxCanBusSignal.info (
              self )
```

Get a current info string about the HIL signal variable, see `signal_var`.

Attention

For RxCanBusSignals, this will show info on the contents of the HIL-send-buffer, not the value from Model/↩ Simulink source nor the last seen signal value on CAN bus.

Returns

the current info string from `signal_var`.

7.365.3.7 kickout()

```
ttk_bus.bus_signals_canida.RxCanBusSignal.kickout (
              self )
```

Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent CAN-Message/PDU).

Note

All other signals of the parent Message/PDU will also be sent.

Reimplemented from ttk_bus._bus_signals_canida.RxCanBusSignal.

7.365.3.8 reset()

```
ttk_bus.bus_signals_canida.RxCanBusSignal.reset (
              self,
             * args,
             ** kwargs )
```

Reset all resettable variables of this bus signal.

Note that this will also switch the variable source switch back to its original setting.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

7.365.3.9 set()

```
ttk_bus.bus_signals_canida.RxCanBusSignal.set (
              self,
              value )
```

Set a value to the HIL signal variable (`.signal_var`).

Parameters

value	value to set (physical)

Returns

the previous value from `signal_var`.

7.365.3.10 setState()

```
ttk_bus.bus_signals_canida.RxCanBusSignal.setState (
              self,
              state )
```

Set a textual state to the HIL signal variable (`.value_var`).

This uses the lookup values supplied in constructor.

Parameters

state	named state to set

Returns

the previous value from `signal_var`.

7.365.4 Member Data Documentation

7.365.4.1 alias

`ttk_bus.bus_signals_canida.RxCanBusSignal.alias`

bus signal alias/name

7.365.4.2 base_identifier

`ttk_bus.bus_signals_canida.RxCanBusSignal.base_identifier`

Signal PV address/identifier as base identifier for deriving sub-elements.

For CAN/Gamma signals, related variables follow the form `<base_identifier>.<suffix>`

7.365.4.3 change_delay

`ttk_bus.bus_signals_canida.RxCanBusSignal.change_delay`

delay to wait after a value change [ms]

7.365.4.4 context

`ttk_bus.bus_signals_canida.RxCanBusSignal.context`

parent context (a gaapi reference to resolve signal/variable PVs)

7.365.4.5 cycle_time

`ttk_bus.bus_signals_canida.RxCanBusSignal.cycle_time`

cycle time of Message/PDU in [ms]

7.365.4.6 debounce_time

ttk_bus.bus_signals_canida.RxCanBusSignal.debounce_time

minimum debounce delay for event-triggered Messages/PDUs in [ms]

7.365.4.7 descr

ttk_bus.bus_signals_canida.RxCanBusSignal.descr

bus signal description text

7.365.4.8 message

ttk_bus.bus_signals_canida.RxCanBusSignal.message

reference to associated (Rx-)CanBusMessage instance

7.365.4.9 recovery_time

ttk_bus.bus_signals_canida.RxCanBusSignal.recovery_time

time for recovery/"wiedergut" checks [ms]

7.365.4.10 signal_var

ttk_bus.bus_signals_canida.RxCanBusSignal.signal_var

signal value variable.

get/info methods of this RxCanBusSignal will access this variable.

7.365.4.11 timeout

ttk_bus.bus_signals_canida.RxCanBusSignal.timeout

detection time for timeout errors [ms]

7.365.4.12 unit

ttk_bus.bus_signals_canida.RxCanBusSignal.unit

unit of physical signal value (as string, mainly used for report entries)

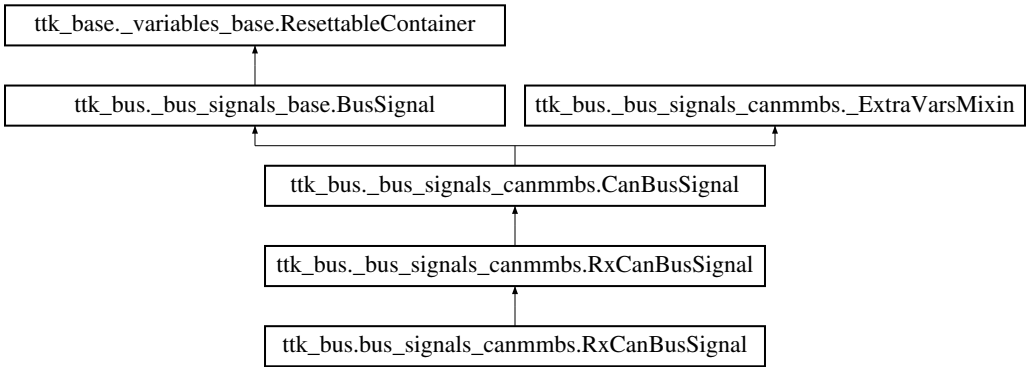
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_canida.py (v1.0.0)

7.366 ttk_bus.bus_signals_canmmbs.RxCanBusSignal Class Reference

A CAN signal sent from the test system to the DUT (ECU Rx <= HIL Tx).

Inheritance diagram for ttk_bus.bus_signals_canmmbs.RxCanBusSignal:



Public Member Functions

- __init__ (self, context, signal_path, message=None, cycle_time=None, debounce_time=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  Rx bus signal initialization.
- info (self)

  Get a current info string about the HIL signal variable, see signal_var.
- get (self)

  Get the current value of the HIL signal variable, see signal_var.
- set (self, value)

  Set a value to the HIL signal variable below BusSystems (and switch the signal source to "BusSystems" if necessary).
- getState (self)

  Get a textual state for the HIL signal variable's current data, see signal_var.
- setState (self, state)

  Set a textual state to the HIL signal variable below BusSystems (and switches the signal source to "BusSystems" if necessary).
- getStateDescr (self, value=None, fallback=None)

  Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).
- enableHilTx (self)

  Enable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).
- disableHilTx (self)

  Disable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).
- kickout (self)

  Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent CAN-Message/PDU).
- reset (self, *args, **kwargs)

  Reset all resettable variables of this bus signal.

Public Attributes

- switch_var
  Switch variable for signal manipulation.
- dyncount_var
  Dynamic Countdown Value.
- dynvalue_var
  Dynamic Value: value that is active while dyncount_var decrements.
- base_path
  Signal model path as base path for deriving sub-element paths.
- message
  reference to associated (Rx-)CanBusMessage instance
- signal_var
  signal value variable.
- context
  parent context (an RTE application instance to resolve signal/variable paths)
- unit
  unit of physical signal value (as string, mainly used for report entries)
- alias
  bus signal alias/name
- descr
  bus signal description text
- cycle_time
  cycle time of Message/PDU in [ms]
- debounce_time
  minimum debounce delay for event-triggered Messages/PDUs in [ms]
- change_delay
  delay to wait after a value change [ms]
- timeout
  detection time for timeout errors [ms]
- recovery_time
  time for recovery/"wiedergut" checks [ms]

Static Public Attributes

- bool cfg_init_signal_manipulation = False
  Configuration setting for new instances:
  Try to initialize (additional) variables that are only available if "dynamic values" input manipulation has been selected during RTICANMM configuration for a signal.
- dict cfg_custom_defaults
  Configuration setting for new instances:
  Custom "hard" default value(s) to use for internal real time system variables (e.g.

7.366.1 Detailed Description

A CAN signal sent from the test system to the DUT (ECU Rx <= HIL Tx).

Note

RxCanBusSignal requires at least an available input manipulation via TRC/BusSystems (which is the default setting in CANMM, but can be disabled).

See _bus_signals_canmmbs.CanBusSignal.

7.366.2 Constructor & Destructor Documentation

7.366.2.1 __init__()

```
ttk_bus.bus_signals_canmmbs.RxCanBusSignal.__init__ (
                self,
                context,
                signal_path,
                message = None,
                cycle_time = None,
                debounce_time = None,
                unit = "",
                alias = "",
                descr = "",
                lookup = None,
                change_delay = None,
                timeout = None,
                recovery_time = None )
```

Rx bus signal initialization.

Parameters

context	parent context (an RTE application instance to resolve signal/variable paths)
signal_path	base path to signal value below BusSystems
message	reference to associated CanBusMessage instance
cycle_time	(opt) cycle time in [ms], overrides settings in Message
debounce_time	(opt) minimum debounce delay between events in [ms], overrides settings in Message
unit	(opt) unit of physical signal value (as string, mainly used for report entries)
alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_canmmbs.RxCanBusSignal.

7.366.3 Member Function Documentation

7.366.3.1 disableHilTx()

```
ttk_bus.bus_signals_canmmbs.RxCanBusSignal.disableHilTx (
                self )
```

Disable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).

Note

This will disable transmission of *all* signals of the parent Message/PDU.

Reimplemented from ttk_bus._bus_signals_canmmbs.RxCanBusSignal.

7.366.3.2 enableHilTx()

```
ttk_bus.bus_signals_canmmbs.RxCanBusSignal.enableHilTx (
              self )
```

Enable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).

Note

This will enable transmission of *all* signals of the parent Message/PDU.

Reimplemented from ttk_bus._bus_signals_canmmbs.RxCanBusSignal.

7.366.3.3 get()

```
ttk_bus.bus_signals_canmmbs.RxCanBusSignal.get (
              self )
```

Get the current value of the HIL signal variable, see signal_var.

Attention

For RxCanBusSignals, this will get the current value of the HIL-send-buffer (below BusSystems), not the last seen signal value on CAN bus.

Returns

the current value from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.366.3.4 getState()

```
ttk_bus.bus_signals_canmmbs.RxCanBusSignal.getState (
              self )
```

Get a textual state for the HIL signal variable's current data, see signal_var.

This uses the lookup values supplied in constructor. Fallback (if no matching state was found) will be the unmapped numeric value.

Attention

For RxCanBusSignals, this will get the current value of the HIL-send-buffer (below BusSystems), not the last seen signal value on CAN bus.

Returns

the current state representation from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.366.3.5 getStateDescr()

```
ttk_bus.bus_signals_canmmbs.RxCanBusSignal.getStateDescr (
               self,
               value = None,
               fallback = None )
```

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the last read value will be used instead.
fallback	fallback value to return if no match was found. Default: None

Returns

the state text if a match was found, otherwise the specified fallback value.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.366.3.6 info()

```
ttk_bus.bus_signals_canmmbs.RxCanBusSignal.info (
               self )
```

Get a current info string about the HIL signal variable, see signal_var.

Attention

For RxCanBusSignals, this will show info on the contents of the HIL-send-buffer (below BusSystems), not the value from Model/Simulink source nor the last seen signal value on CAN bus.

Returns

the current info string from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.366.3.7 kickout()

```
ttk_bus.bus_signals_canmmbs.RxCanBusSignal.kickout (
               self )
```

Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent CAN-Message/PDU).

Note

All other signals of the parent Message/PDU will also be sent.

Reimplemented from ttk_bus._bus_signals_canmmbs.RxCanBusSignal.

7.366.3.8 reset()

```
ttk_bus.bus_signals_canmmbs.RxCanBusSignal.reset (
               self,
              * args,
              ** kwargs )
```

Reset all resettable variables of this bus signal.

Note that this will also switch the variable source switch back to its original setting.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.ResettableContainer.

7.366.3.9 set()

```
ttk_bus.bus_signals_canmmbs.RxCanBusSignal.set (
              self,
              value )
```

Set a value to the HIL signal variable below BusSystems (and switch the signal source to "BusSystems" if necessary).

Parameters

value	new value to set

Returns

the previous value from signal_var.

Reimplemented from ttk_bus._bus_signals_canmmbs.RxCanBusSignal.

7.366.3.10 setState()

```
ttk_bus.bus_signals_canmmbs.RxCanBusSignal.setState (
              self,
              state )
```

Set a textual state to the HIL signal variable below BusSystems (and switches the signal source to "BusSystems" if necessary).

This uses the lookup values supplied in constructor.

Parameters

state	named state to set

Returns

the previous value representation from signal_var.

Reimplemented from ttk_bus._bus_signals_canmmbs.RxCanBusSignal.

7.366.4 Member Data Documentation

7.366.4.1 alias

ttk_bus.bus_signals_canmmbs.RxCanBusSignal.alias

bus signal alias/name

7.366.4.2 base_path

ttk_bus.bus_signals_canmmbs.RxCanBusSignal.base_path

Signal model path as base path for deriving sub-element paths.

For CANMM signals, related variables follow the form <path_to_signal>_suffix

7.366.4.3 cfg_custom_defaults

dict ttk_bus.bus_signals_canmmbs.RxCanBusSignal.cfg_custom_defaults [static]

Initial value:

```
=  {
        "switch_var":   None,
        "dyncount_var": None,
        "dynvalue_var": None,
    }
```

Configuration setting for new instances:
Custom "hard" default value(s) to use for internal real time system variables (e.g.

switch settings), so a reset() will always reset those variables to defined values (and not just those variables that have actually been accessed).
This might be helpful if "other" tests fail to properly clean up the test environment after they are done.

Mapping: <variable name as string>: <default value>

A value of None (or a missing variable name) enables normal/automatic reset behavior.

Typical default values:

- Switch variable (signal manipulation)
  - 0: model input (if configured/available)
  - 1: constant from Trc/BusSystems
  - 2: counter running (for counter signals, default switch setting for enabled counters)
  - 3: constant + counter (for counter signals)
  - 4: toggle value (for toggle signals)
  - 5: parity value (for parity signals)
  - 6: ∼parity value (for parity signals)
  - 7: error value (if an error value is configured)
  - 8: dynamic value (if dynamic value manipulation is configured)
- Dynamic Countdown Value: Auto-decrements until 0 is reached if signal manipulation switch is set to 8 => 0 would be a safe default value
- Dynamic Value: value that is active while dyncount_var decrements.

7.366.4.4 cfg_init_signal_manipulation

```
bool ttk_bus.bus_signals_canmmbs.RxCanBusSignal.cfg_init_signal_manipulation = False  [static]
```

Configuration setting for new instances:
Try to initialize (additional) variables that are only available if "dynamic values" input manipulation has been selected during RTICANMM configuration for a signal.

Additional variables:

- dyncount_var
- dynvalue_var

7.366.4.5 change_delay

```
ttk_bus.bus_signals_canmmbs.RxCanBusSignal.change_delay
```

delay to wait after a value change [ms]

7.366.4.6 context

```
ttk_bus.bus_signals_canmmbs.RxCanBusSignal.context
```

parent context (an RTE application instance to resolve signal/variable paths)

7.366.4.7 cycle_time

```
ttk_bus.bus_signals_canmmbs.RxCanBusSignal.cycle_time
```

cycle time of Message/PDU in [ms]

7.366.4.8 debounce_time

```
ttk_bus.bus_signals_canmmbs.RxCanBusSignal.debounce_time
```

minimum debounce delay for event-triggered Messages/PDUs in [ms]

7.366.4.9 descr

```
ttk_bus.bus_signals_canmmbs.RxCanBusSignal.descr
```

bus signal description text

7.366.4.10 dyncount_var

ttk_bus.bus_signals_canmmbs.RxCanBusSignal.dyncount_var

Dynamic Countdown Value.

Only available if signal manipulation is configured in CANMM and enabled in class config.
See self.cfg_init_signal_manipulation

7.366.4.11 dynvalue_var

ttk_bus.bus_signals_canmmbs.RxCanBusSignal.dynvalue_var

Dynamic Value: value that is active while dyncount_var decrements.

Only available if signal manipulation is configured in CANMM and enabled in class config.
See self.cfg_init_signal_manipulation

7.366.4.12 message

ttk_bus.bus_signals_canmmbs.RxCanBusSignal.message

reference to associated (Rx-)CanBusMessage instance

7.366.4.13 recovery_time

ttk_bus.bus_signals_canmmbs.RxCanBusSignal.recovery_time

time for recovery/"wiedergut" checks [ms]

7.366.4.14 signal_var

ttk_bus.bus_signals_canmmbs.RxCanBusSignal.signal_var

signal value variable.

get/info methods of this RxCanBusSignal will access this variable. (a HilVar instance)

7.366.4.15 switch_var

`ttk_bus.bus_signals_canmmbs.RxCanBusSignal.switch_var`

Switch variable for signal manipulation.

Only available if more than one signal manipulation option is configured in CANMM.

Values:

- 0: model input (if configured/available)
- 1: constant from Trc/BusSystems
- 2: counter running (for counter signals, default switch setting for enabled counters)
- 3: constant + counter (for counter signals)
- 4: toggle value (for toggle signals)
- 5: parity value (for parity signals)
- 6: ∼parity value (for parity signals)
- 7: error value (if an error value is configured)
- 8: dynamic value (if dynamic value manipulation is configured)

7.366.4.16 timeout

`ttk_bus.bus_signals_canmmbs.RxCanBusSignal.timeout`

detection time for timeout errors [ms]

7.366.4.17 unit

`ttk_bus.bus_signals_canmmbs.RxCanBusSignal.unit`

unit of physical signal value (as string, mainly used for report entries)

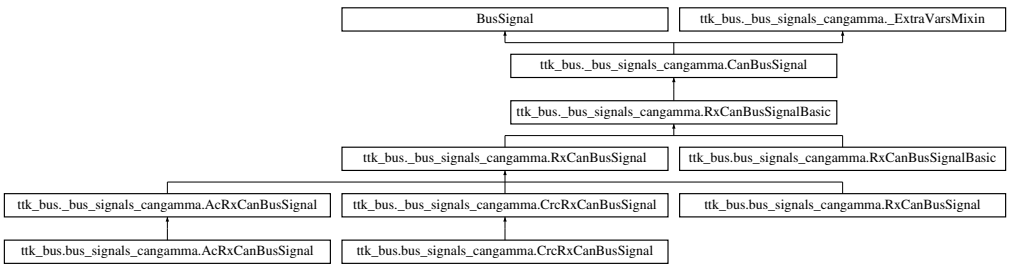
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_canmmbs.py (v1.8)

7.367 ttk_bus._bus_signals_cangamma.RxCanBusSignalBasic Class Reference

A basic CAN signal sent from the test system to the DUT (ECU Rx <= HIL Tx).

Inheritance diagram for ttk_bus._bus_signals_cangamma.RxCanBusSignalBasic:



Public Member Functions

- __init__ (self, context, signal_identifier, message=None, cycle_time=None, debounce_time=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  Basic Rx bus signal initialization (ECU Rx <= HIL Tx).
- enableHilTx (self)

  Enable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).
- disableHilTx (self)

  Disable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).
- kickout (self)

  Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent CAN-Message/PDU).

7.367.1 Detailed Description

A basic CAN signal sent from the test system to the DUT (ECU Rx <= HIL Tx).

Basic implementation, directly accesses raw value as-is from CAN, without a converted physical value or error injections.

Use this for signal entries generated with iSyDBC mode "minimal" . See _bus_signals_cangamma.CanBusSignal.

7.367.2 Constructor & Destructor Documentation

7.367.2.1 __init__()

```
ttk_bus._bus_signals_cangamma.RxCanBusSignalBasic.__init__ (
                self,
                context,
                signal_identifier,
                message = None,
                cycle_time = None,
                debounce_time = None,
                unit = "",
                alias = "",
                descr = "",
                lookup = None,
                change_delay = None,
                timeout = None,
                recovery_time = None )
```

Basic Rx bus signal initialization (ECU Rx <= HIL Tx).

Parameters

context	parent gamma_api context
signal_identifier	identifier for signal value PV. Use `.value` entry, e.g. `"BusSystems.can.HIL_TX.MessageName.signals.SignalName.value"`
message	reference to associated CanBusMessage instance
cycle_time	(opt) cycle time in [ms], overrides settings in Message
debounce_time	(opt) minimum debounce delay between events in [ms], overrides settings in Message
unit	(opt) unit of physical signal value (as string, mainly used for report entries)

Parameters

alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

Minimal/Basic Signals:

ECU-Rx/HIL-Tx Signals generated by iSyDBC in minimal/basic mode will only contain PVs for:

- `<SignalName>.value` - raw value as sent to CAN-Plugin
- `<SignalName>.period` - tx cycle time / period in ms.
- `<SignalName>.timestamp` - Tx time stamp
- `<SignalName>.length` - message length

Reimplemented from ttk_bus._bus_signals_cangamma.CanBusSignal.

Reimplemented in ttk_bus._bus_signals_cangamma.CrcRxCanBusSignal, ttk_bus._bus_signals_cangamma.AcRxCanBusSignal, ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal, ttk_bus.bus_signals_cangamma.AcRxCanBusSignal, ttk_bus._bus_signals_cangamma.RxCanBusSignal, ttk_bus.bus_signals_cangamma.RxCanBusSignalBasic, and ttk_bus.bus_signals_cangamma.RxCanBusSignal.

7.367.3 Member Function Documentation

7.367.3.1 disableHilTx()

```
ttk_bus._bus_signals_cangamma.RxCanBusSignalBasic.disableHilTx (
              self )
```

Disable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).

Note

This will disable transmission of *all* signals of the parent Message/PDU.

Reimplemented in ttk_bus.bus_signals_cangamma.RxCanBusSignalBasic, ttk_bus.bus_signals_cangamma.RxCanBusSignal, ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal, and ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.

7.367.3.2 enableHilTx()

```
ttk_bus._bus_signals_cangamma.RxCanBusSignalBasic.enableHilTx (
              self )
```

Enable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).

Note

This will enable transmission of *all* signals of the parent Message/PDU.

Reimplemented in ttk_bus.bus_signals_cangamma.RxCanBusSignalBasic, ttk_bus.bus_signals_cangamma.RxCanBusSignal, ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal, and ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.

7.367.3.3 kickout()

```
ttk_bus._bus_signals_cangamma.RxCanBusSignalBasic.kickout (
              self )
```

Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent CAN-Message/PDU).

Note

All other signals of the parent Message/PDU will also be sent.

Reimplemented in ttk_bus.bus_signals_cangamma.RxCanBusSignalBasic, ttk_bus.bus_signals_cangamma.RxCanBusSignal, ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal, and ttk_bus.bus_signals_cangamma.AcRxCanBusSignal.

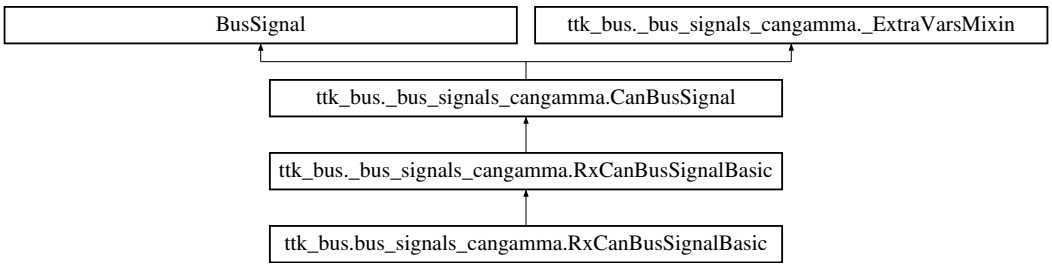
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_cangamma.py (v1.12)

7.368 ttk_bus.bus_signals_cangamma.RxCanBusSignalBasic Class Reference

A basic CAN signal sent from the test system to the DUT (ECU Rx <= HIL Tx).

Inheritance diagram for ttk_bus.bus_signals_cangamma.RxCanBusSignalBasic:



Public Member Functions

- __init__ (self, context, signal_identifier, message=None, cycle_time=None, debounce_time=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  Basic Rx bus signal initialization (ECU Rx <= HIL Tx).
- info (self)

  Get a current info string about the HIL signal variable, see `signal_var`
- get (self)

  Get the current value of the HIL signal variable, see `signal_var`.
- set (self, value)

  Set a value to the HIL signal variable (`.signal_var`*).*
- getState (self)

  Get a textual state for the HIL signal variable's current data, see `value_var`.
- setState (self, state)

  Set a textual state to the HIL signal variable (`.signal_var`*).*

- getStateDescr (self, value=None, fallback=None)

  Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).
- enableHilTx (self)

  Enable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).
- disableHilTx (self)

  Disable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).
- kickout (self)

  Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent CAN-Message/PDU).
- reset (self, ∗args, ∗∗kwargs)

  Reset all resettable variables of this bus signal.

7.368.1 Detailed Description

A basic CAN signal sent from the test system to the DUT (ECU Rx <= HIL Tx).

Basic implementation, directly accesses raw value as-is from CAN, without a converted physical value or error injections.

Use this for signal entries generated with iSyDBC mode "minimal" .

7.368.2 Constructor & Destructor Documentation

7.368.2.1 __init__()

```
ttk_bus.bus_signals_cangamma.RxCanBusSignalBasic.__init__ (
                self,
                context,
                signal_identifier,
                message = None,
                cycle_time = None,
                debounce_time = None,
                unit = "",
                alias = "",
                descr = "",
                lookup = None,
                change_delay = None,
                timeout = None,
                recovery_time = None )
```

Basic Rx bus signal initialization (ECU Rx <= HIL Tx).

Parameters

context	parent gamma_api context
signal_identifier	identifier for signal value PV. Use `.value` entry, e.g. `"BusSystems.can.HIL_TX.MessageName.signals.SignalName.value"`
message	reference to associated CanBusMessage instance
cycle_time	(opt) cycle time in [ms], overrides settings in Message
debounce_time	(opt) minimum debounce delay between events in [ms], overrides settings in Message
unit	(opt) unit of physical signal value (as string, mainly used for report entries)

Parameters

alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

Minimal/Basic Signals:

ECU-Rx/HIL-Tx Signals generated by iSyDBC in minimal/basic mode will only contain PVs for:

- `<SignalName>.value` - raw value as sent to CAN-Plugin
- `<SignalName>.period` - tx cycle time / period in ms.
- `<SignalName>.timestamp` - Tx time stamp
- `<SignalName>.length` - message length

Reimplemented from ttk_bus._bus_signals_cangamma.RxCanBusSignalBasic.

7.368.3 Member Function Documentation

7.368.3.1 disableHilTx()

```
ttk_bus.bus_signals_cangamma.RxCanBusSignalBasic.disableHilTx (
              self )
```

Disable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).

Note

This will disable transmission of *all* signals of the parent Message/PDU.

Reimplemented from ttk_bus._bus_signals_cangamma.RxCanBusSignalBasic.

7.368.3.2 enableHilTx()

```
ttk_bus.bus_signals_cangamma.RxCanBusSignalBasic.enableHilTx (
              self )
```

Enable transmission of signal HIL => DUT (if the assigned CAN-Message/PDU supports it).

Note

This will enable transmission of *all* signals of the parent Message/PDU.

Reimplemented from ttk_bus._bus_signals_cangamma.RxCanBusSignalBasic.

7.368.3.3 get()

```
ttk_bus.bus_signals_cangamma.RxCanBusSignalBasic.get (
              self )
```

Get the current value of the HIL signal variable, see `signal_var`.

Attention

For RxCanBusSignals, this will get the current (raw) value of the HIL-send-buffer, not the last seen signal value on CAN bus.

Returns

the current value from `signal_var`.

7.368.3.4 getState()

```
ttk_bus.bus_signals_cangamma.RxCanBusSignalBasic.getState (
              self )
```

Get a textual state for the HIL signal variable's current data, see `value_var`.

This uses the lookup values supplied in constructor. Fallback (if no matching state was found) will be the unmapped numeric value.

Attention

For RxCanBusSignals, this will get the current (raw) value of the HIL-send-buffer, not the last seen signal value on CAN bus.

Returns

the current state representation from `signal_var`.

7.368.3.5 getStateDescr()

```
ttk_bus.bus_signals_cangamma.RxCanBusSignalBasic.getStateDescr (
              self,
              value = None,
              fallback = None )
```

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the last read value will be used instead.
fallback	fallback value to return if no match was found. Default: None

Returns

the state text if a match was found, otherwise the specified fallback value.

7.368.3.6 info()

```
ttk_bus.bus_signals_cangamma.RxCanBusSignalBasic.info (
              self )
```

Get a current info string about the HIL signal variable, see `signal_var`

Attention

For RxCanBusSignals, this will show info on the contents of the HIL-send-buffer, not the value from Model/↩ Simulink source nor the last seen signal value on CAN bus.

Returns

the current info string from `signal_var`.

7.368.3.7 kickout()

```
ttk_bus.bus_signals_cangamma.RxCanBusSignalBasic.kickout (
              self )
```

Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent CAN-Message/PDU).

Note

All other signals of the parent Message/PDU will also be sent.

Reimplemented from ttk_bus._bus_signals_cangamma.RxCanBusSignalBasic.

7.368.3.8 reset()

```
ttk_bus.bus_signals_cangamma.RxCanBusSignalBasic.reset (
              self,
             * args,
             ** kwargs )
```

Reset all resettable variables of this bus signal.

Note that this will also switch the variable source switch back to its original setting.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

7.368.3.9 set()

```
ttk_bus.bus_signals_cangamma.RxCanBusSignalBasic.set (
              self,
              value )
```

Set a value to the HIL signal variable (`.signal_var`).

Parameters

value	value to set (physical)

Returns

the previous value from `signal_var`.

7.368.3.10 setState()

```
ttk_bus.bus_signals_cangamma.RxCanBusSignalBasic.setState (
              self,
              state )
```

Set a textual state to the HIL signal variable (`.signal_var`).

This uses the lookup values supplied in constructor.

Parameters

state	named state to set

Returns

the previous value from `signal_var`.

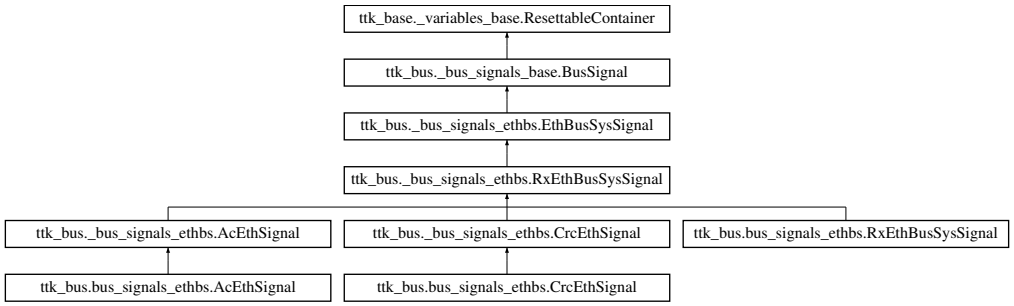
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_cangamma.py (v1.3)

7.369 ttk_bus._bus_signals_ethbs.RxEthBusSysSignal Class Reference

An Ethernet event signal/parameter/sub-parameter sent from test system to DUT (HIL PSI Tx => ECU CSI Rx).

Inheritance diagram for ttk_bus._bus_signals_ethbs.RxEthBusSysSignal:



Public Member Functions

- __init__ (self, context, base_path, event=None, cycle_time=None, debounce_time=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  Ethernet Rx signal initialization (BusSystems flavour).
- switchToBusSystemsSource (self, bus_systems=True)

  Switch signal source to variables below "BusSystems".
- set (self, value)

  Set a value to the HIL signal variable.
- setState (self, state)

  Set a textual state to the HIL signal variable.
- enableHilTx (self)

  Enable transmission of signal HIL => DUT (if the assigned EthEvent supports it).
- disableHilTx (self)

  Disable transmission of signal HIL => DUT (if the assigned EthEvent supports it).
- kickout (self)

  Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent EthEvent).

Public Attributes

- switch_path

  model path to switch variable
- switch_var

  Source switch variable.
- context

Static Public Attributes

- int fallback_change_delay = 100

  Fallback value in [ms] for event-triggered signals w/o a defined or derived change delay to permit signal propagation.

7.369.1 Detailed Description

An Ethernet event signal/parameter/sub-parameter sent from test system to DUT (HIL PSI Tx => ECU CSI Rx).

Note

This class expects a BusSystems path structure.

7.369.2 Constructor & Destructor Documentation

7.369.2.1 __init__()

```
ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.__init__ (
                self,
                context,
                base_path,
                event = None,
                cycle_time = None,
                debounce_time = None,
                unit = "",
                alias = "",
                descr = "",
                lookup = None,
                change_delay = None,
                timeout = None,
                recovery_time = None )
```

Ethernet Rx signal initialization (BusSystems flavour).

Parameters

context	parent context (an RTE application or XilTestbench instance to resolve signal/variable paths)
base_path	signal path below BusSystems (typically: use .../foobar/Value)
event	reference to associated EthEvent instance
cycle_time	(opt) cycle time in [ms], overrides settings in EthEvent
debounce_time	(opt) minimum debounce delay time between events in [ms], overrides settings in EthEvent
unit	(opt) unit of physical signal value (as string, mainly used for report entries)
alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x cycle_time
timeout	(opt) detection time for timeout errors [ms], defaults to 2x cycle_time
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x cycle_time

Reimplemented from ttk_bus._bus_signals_ethbs.EthBusSysSignal.

Reimplemented in ttk_bus._bus_signals_ethbs.CrcEthSignal, ttk_bus._bus_signals_ethbs.AcEthSignal, ttk_bus.bus_signals_ethbs.Cr
ttk_bus.bus_signals_ethbs.AcEthSignal, and ttk_bus.bus_signals_ethbs.RxEthBusSysSignal.

7.369.3 Member Function Documentation

7.369.3.1 disableHilTx()

```
ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.disableHilTx (
                self )
```

Disable transmission of signal HIL => DUT (if the assigned EthEvent supports it).

Note

This disables transmission of all signals in the parent Event.

Reimplemented in ttk_bus.bus_signals_ethbs.RxEthBusSysSignal, ttk_bus.bus_signals_ethbs.CrcEthSignal, and ttk_bus.bus_signals_ethbs.AcEthSignal.

7.369.3.2 enableHilTx()

```
ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.enableHilTx (
              self )
```

Enable transmission of signal HIL => DUT (if the assigned EthEvent supports it).

Note

This enables transmission of all signals in the parent Event.

Reimplemented in ttk_bus.bus_signals_ethbs.RxEthBusSysSignal, ttk_bus.bus_signals_ethbs.CrcEthSignal, and ttk_bus.bus_signals_ethbs.AcEthSignal.

7.369.3.3 kickout()

```
ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.kickout (
              self )
```

Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent EthEvent).

Note

All other signals of the parent EthEvent also will be sent.

Reimplemented in ttk_bus.bus_signals_ethbs.RxEthBusSysSignal, ttk_bus.bus_signals_ethbs.CrcEthSignal, and ttk_bus.bus_signals_ethbs.AcEthSignal.

7.369.3.4 set()

```
ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.set (
              self,
              value )
```

Set a value to the HIL signal variable.

Parameters

value	value to be set

Returns

the original value before the set operation occurred.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

Reimplemented in ttk_bus.bus_signals_ethbs.RxEthBusSysSignal, ttk_bus.bus_signals_ethbs.CrcEthSignal, and ttk_bus.bus_signals_ethbs.AcEthSignal.

7.369.3.5 setState()

```
ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.setState (
              self,
              state )
```

Set a textual state to the HIL signal variable.

This uses the lookup values supplied in constructor.

Parameters

state	named state to set

Returns

the previous state's description.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

Reimplemented in ttk_bus.bus_signals_ethbs.RxEthBusSysSignal, ttk_bus.bus_signals_ethbs.CrcEthSignal, and ttk_bus.bus_signals_ethbs.AcEthSignal.

7.369.3.6 switchToBusSystemsSource()

```
ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.switchToBusSystemsSource (
              self,
              bus_systems = True )
```

Switch signal source to variables below "BusSystems".

This is mainly intended for internal use.

Parameters

bus_systems	True: use "BusSystems" paths as source False: use "Model Root" paths as source

Reimplemented in ttk_bus.bus_signals_ethbs.RxEthBusSysSignal.

7.369.4 Member Data Documentation

7.369.4.1 context

`ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.context`

7.369.4.2 fallback_change_delay

`int ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.fallback_change_delay = 100  [static]`

Fallback value in [ms] for event-triggered signals w/o a defined or derived change delay to permit signal propagation.

7.369.4.3 switch_path

`ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.switch_path`

model path to switch variable

7.369.4.4 switch_var

`ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.switch_var`

Source switch variable.

- 0: Simulink Model
- 1: BusSystems

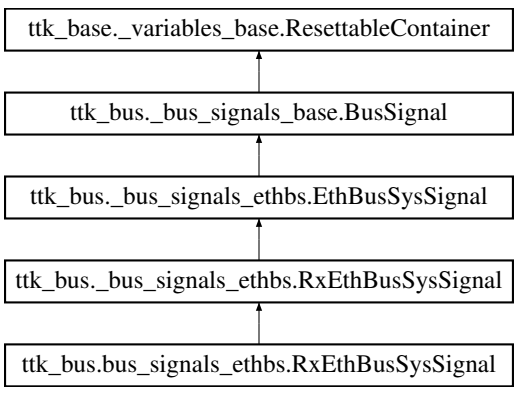
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_ethbs.py (v2.5)

7.370 ttk_bus.bus_signals_ethbs.RxEthBusSysSignal Class Reference

An Ethernet event signal/parameter/sub-parameter sent from test system to DUT (HIL PSI Tx => ECU CSI Rx).

Inheritance diagram for ttk_bus.bus_signals_ethbs.RxEthBusSysSignal:



Public Member Functions

- __init__ (self, context, base_path, event=None, cycle_time=None, debounce_time=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  Ethernet Rx signal initialization (BusSystems flavour).
- switchToBusSystemsSource (self, bus_systems=True)

  Switch signal source to variables below "BusSystems".
- enableHilTx (self)

  Enable transmission of signal HIL => DUT (if the assigned EthEvent supports it).
- disableHilTx (self)

  Disable transmission of signal HIL => DUT (if the assigned EthEvent supports it).
- kickout (self)

  Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent EthEvent).
- get (self)

  Get the current value of the HIL signal variable (typically not that useful for ECU-Rx/HIL-Tx signals, though).
- set (self, value)

  Set a value to the HIL signal variable.
- getState (self)

  Get a textual state for the HIL signal variable's current data (typically not that useful for ECU-Rx/HIL-Tx signals, though).
- setState (self, state)

  Set a textual state to the HIL signal variable.
- getStateDescr (self, value=None, fallback=None)

  Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).
- info (self)

  Get a current info string about the HIL signal variable.
- reset (self, *args, **kwargs)

  Reset all resettable variables of this bus signal.

Public Attributes

- switch_path

  Source Switch path (a "HilVar" instance)
- switch_var

  Source Switch variable (a HilVar instance)
- context

  parent context (an RTE application instance to resolve signal/variable paths)
- signal_path

  base path to signal below BusSystems
- event

  reference to associated EthEvent instance
- unit

  unit of physical signal value (as string, mainly used for report entries)
- alias

  bus signal alias name
- descr

  signal description text
- cycle_time

  cycle time of signal in [ms]
- debounce_time

  minimum debounce delay for triggered signals in [ms]

- change_delay

  delay to wait after a value change [ms]
- timeout

  detection time for timeout errors [ms]
- recovery_time

  time for recovery/"wiedergut" checks [ms]

7.370.1 Detailed Description

An Ethernet event signal/parameter/sub-parameter sent from test system to DUT (HIL PSI Tx => ECU CSI Rx).

Note

This class expects a BusSystems path structure.

7.370.2 Constructor & Destructor Documentation

7.370.2.1 __init__()

```
ttk_bus.bus_signals_ethbs.RxEthBusSysSignal.__init__ (
               self,
               context,
               base_path,
               event = None,
               cycle_time = None,
               debounce_time = None,
               unit = "",
               alias = "",
               descr = "",
               lookup = None,
               change_delay = None,
               timeout = None,
               recovery_time = None )
```

Ethernet Rx signal initialization (BusSystems flavour).

Parameters

context	parent context (an RTE application or XilTestbench instance to resolve signal/variable paths)
base_path	signal path below BusSystems (typically: use .../foobar/Value)
event	reference to associated EthEvent instance
cycle_time	(opt) cycle time in [ms], overrides settings in EthEvent
debounce_time	(opt) minimum debounce delay time between events in [ms], overrides settings in EthEvent
unit	(opt) unit of physical signal value (as string, mainly used for report entries)
alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x cycle_time
timeout	(opt) detection time for timeout errors [ms], defaults to 2x cycle_time
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x cycle_time

Reimplemented from ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.

7.370.3 Member Function Documentation

7.370.3.1 disableHilTx()

```
ttk_bus.bus_signals_ethbs.RxEthBusSysSignal.disableHilTx (
              self )
```

Disable transmission of signal HIL => DUT (if the assigned EthEvent supports it).

Note

This disables transmission of all signals in the parent Event.

Reimplemented from ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.

7.370.3.2 enableHilTx()

```
ttk_bus.bus_signals_ethbs.RxEthBusSysSignal.enableHilTx (
              self )
```

Enable transmission of signal HIL => DUT (if the assigned EthEvent supports it).

Note

This enables transmission of all signals in the parent Event.

Reimplemented from ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.

7.370.3.3 get()

```
ttk_bus.bus_signals_ethbs.RxEthBusSysSignal.get (
              self )
```

Get the current value of the HIL signal variable (typically not that useful for ECU-Rx/HIL-Tx signals, though).

Returns

the current value from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.370.3.4 getState()

```
ttk_bus.bus_signals_ethbs.RxEthBusSysSignal.getState (
              self )
```

Get a textual state for the HIL signal variable's current data (typically not that useful for ECU-Rx/HIL-Tx signals, though).

Returns

the current state's description.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.370.3.5 getStateDescr()

```
ttk_bus.bus_signals_ethbs.RxEthBusSysSignal.getStateDescr (
              self,
              value = None,
              fallback = None )
```

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the last read value will be used instead.
fallback	fallback value to return if no match was found. Default: None

Returns

the state text if a match was found, otherwise the specified fallback value.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.370.3.6 info()

```
ttk_bus.bus_signals_ethbs.RxEthBusSysSignal.info (
              self )
```

Get a current info string about the HIL signal variable.

Returns

an info string.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.370.3.7 kickout()

```
ttk_bus.bus_signals_ethbs.RxEthBusSysSignal.kickout (
              self )
```

Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent EthEvent).

Note

All other signals of the parent EthEvent also will be sent.

Reimplemented from ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.

7.370.3.8 reset()

```
ttk_bus.bus_signals_ethbs.RxEthBusSysSignal.reset (
              self,
             * args,
             ** kwargs )
```

Reset all resettable variables of this bus signal.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.ResettableContainer.

7.370.3.9 set()

```
ttk_bus.bus_signals_ethbs.RxEthBusSysSignal.set (
              self,
              value )
```

Set a value to the HIL signal variable.

Parameters

value	value to be set

Returns

the original value before the set operation occurred.

Reimplemented from ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.

7.370.3.10 setState()

```
ttk_bus.bus_signals_ethbs.RxEthBusSysSignal.setState (
              self,
              state )
```

Set a textual state to the HIL signal variable.

This uses the lookup values supplied in constructor.

Parameters

state	named state to set

Returns

the previous state's description.

Reimplemented from ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.

7.370.3.11 switchToBusSystemsSource()

```
ttk_bus.bus_signals_ethbs.RxEthBusSysSignal.switchToBusSystemsSource (
              self,
              bus_systems = True )
```

Switch signal source to variables below "BusSystems".

This is mainly intended for internal use.

Parameters

bus_systems	True: use "BusSystems" paths as source False: use "Model Root" paths as source

Reimplemented from ttk_bus._bus_signals_ethbs.RxEthBusSysSignal.

7.370.4 Member Data Documentation

7.370.4.1 alias

```
ttk_bus.bus_signals_ethbs.RxEthBusSysSignal.alias
```

bus signal alias name

7.370.4.2 change_delay

```
ttk_bus.bus_signals_ethbs.RxEthBusSysSignal.change_delay
```

delay to wait after a value change [ms]

7.370.4.3 context

ttk_bus.bus_signals_ethbs.RxEthBusSysSignal.context

parent context (an RTE application instance to resolve signal/variable paths)

7.370.4.4 cycle_time

ttk_bus.bus_signals_ethbs.RxEthBusSysSignal.cycle_time

cycle time of signal in [ms]

7.370.4.5 debounce_time

ttk_bus.bus_signals_ethbs.RxEthBusSysSignal.debounce_time

minimum debounce delay for triggered signals in [ms]

7.370.4.6 descr

ttk_bus.bus_signals_ethbs.RxEthBusSysSignal.descr

signal description text

7.370.4.7 event

ttk_bus.bus_signals_ethbs.RxEthBusSysSignal.event

reference to associated EthEvent instance

7.370.4.8 recovery_time

ttk_bus.bus_signals_ethbs.RxEthBusSysSignal.recovery_time

time for recovery/"wiedergut" checks [ms]

7.370.4.9 signal_path

ttk_bus.bus_signals_ethbs.RxEthBusSysSignal.signal_path

base path to signal below BusSystems

7.370.4.10 switch_path

ttk_bus.bus_signals_ethbs.RxEthBusSysSignal.switch_path

Source Switch path (a "HilVar" instance)

7.370.4.11 switch_var

`ttk_bus.bus_signals_ethbs.RxEthBusSysSignal.switch_var`

Source Switch variable (a HilVar instance)

- 0: value source is Simulink Model
- 1: value source is BusSystems

7.370.4.12 timeout

`ttk_bus.bus_signals_ethbs.RxEthBusSysSignal.timeout`

detection time for timeout errors [ms]

7.370.4.13 unit

`ttk_bus.bus_signals_ethbs.RxEthBusSysSignal.unit`

unit of physical signal value (as string, mainly used for report entries)

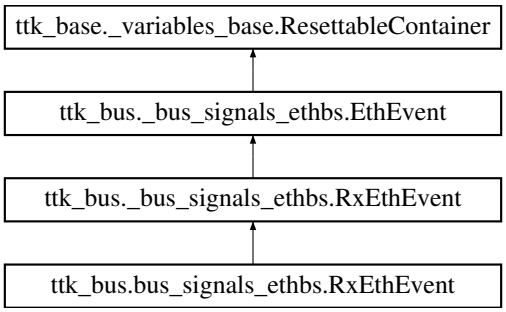
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_ethbs.py (v2.5)

7.371 ttk_bus._bus_signals_ethbs.RxEthEvent Class Reference

Ethernet-Event-specific data and methods - Rx (HIL-Tx => ECU-Rx).

Inheritance diagram for ttk_bus._bus_signals_ethbs.RxEthEvent:



Public Member Functions

- __init__ (self, context, enable_path, trigger_path, cycle_time=0, debounce_time=0, alias='', descr='', change_delay=None, timeout=None, recovery_time=None)

  Ethernet Rx-Event (or field) data.
- enableHilTx (self)

  (Re-)enable sending of Event.
- disableHilTx (self)

  Disable sending of Event.
- kickout (self, delay_ms=None)

  Trigger sending of the Ethernet Event / Field.

Public Attributes

- enable_var

  HIL-Tx-enable/disable variable.
- kickout_var

  HIL-Tx-kickout variable.

7.371.1 Detailed Description

Ethernet-Event-specific data and methods - Rx (HIL-Tx => ECU-Rx).

Extends EthEvent with methods for enableHilTx/disableHilTx and kickout (trigger).

Note

This class expects a "Model Root/" path structure.

Also note that BusSystems-based implementations (see below) may use Model-Root-based EthEvents without problems.

7.371.2 Constructor & Destructor Documentation

7.371.2.1 __init__()

```
ttk_bus._bus_signals_ethbs.RxEthEvent.__init__ (
              self,
              context,
              enable_path,
              trigger_path,
              cycle_time = 0,
              debounce_time = 0,
              alias = '',
              descr = '',
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Ethernet Rx-Event (or field) data.

Parameters

context	parent context (an RTE application or XilTestbench instance to resolve signal/variable paths)
enable_path	model path to tx enable switch (if available)
trigger_path	model path to tx trigger (if available)
cycle_time	cycle time in [ms]. On-demand events have a cycle time of 0 (similar to fibex value)
debounce_time	minimum debounce delay time between events in [ms]
alias	Event alias/name
descr	(opt) Event description (may be used in report entries)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_ethbs.EthEvent.

Reimplemented in ttk_bus.bus_signals_ethbs.RxEthEvent.

7.371.3 Member Function Documentation

7.371.3.1 disableHilTx()

```
ttk_bus._bus_signals_ethbs.RxEthEvent.disableHilTx (
              self )
```

Disable sending of Event.

Reimplemented in ttk_bus.bus_signals_ethbs.RxEthEvent.

7.371.3.2 enableHilTx()

```
ttk_bus._bus_signals_ethbs.RxEthEvent.enableHilTx (
              self )
```

(Re-)enable sending of Event.

Reimplemented in ttk_bus.bus_signals_ethbs.RxEthEvent.

7.371.3.3 kickout()

```
ttk_bus._bus_signals_ethbs.RxEthEvent.kickout (
              self,
              delay_ms = None )
```

Trigger sending of the Ethernet Event / Field.

Parameters

delay_ms	[ms] Delay to wait before clearing kickout/trigger variable again (it *should* auto-reset, but anyway). Defaults to self.debounce_time (if explicitly set) or alternatively to self.change_delay

Reimplemented in ttk_bus.bus_signals_ethbs.RxEthEvent.

7.371.4 Member Data Documentation

7.371.4.1 enable_var

`ttk_bus._bus_signals_ethbs.RxEthEvent.enable_var`

HIL-Tx-enable/disable variable.

7.371.4.2 kickout_var

`ttk_bus._bus_signals_ethbs.RxEthEvent.kickout_var`

HIL-Tx-kickout variable.

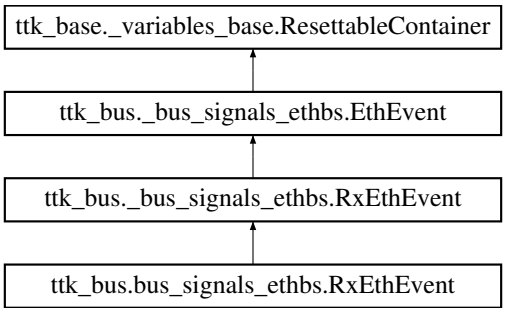
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_ethbs.py (v2.5)

7.372 ttk_bus.bus_signals_ethbs.RxEthEvent Class Reference

Ethernet-Event-specific data and methods - Rx (HIL-Tx => ECU-Rx).

Inheritance diagram for ttk_bus.bus_signals_ethbs.RxEthEvent:



Public Member Functions

- __init__ (self, context, enable_path, trigger_path, cycle_time=0, debounce_time=0, alias='', descr='', change_delay=None, timeout=None, recovery_time=None)

  Ethernet Rx-Event (or field) data.
- enableHilTx (self)

  (Re-)enable sending of Event.
- disableHilTx (self)

  Disable sending of Event.
- kickout (self, delay_ms=None)

  Trigger sending of the Ethernet Event / Field.
- reset (self, *args, **kwargs)

  Call reset() on all contained variables/signals that support it.

Public Attributes

- enable_var
  HIL-Tx enable/disable variable (a "HilVar" instance)
- kickout_var
  HIL-Tx kickout/trigger variable (a "HilVar" instance)
- context
  parent context (an RTE application instance to resolve signal/variable paths)
- alias
  alias name of event
- descr
  description text of event
- enable_path
  model path to tx enable switch (if available).
- cycle_time
  cycle time of event in [ms]
- debounce_time
  minimum debounce delay for triggered events in [ms]
- change_delay
  delay to wait after a value change [ms]
- timeout
  detection time for timeout errors [ms]
- recovery_time
  time for recovery/"wiedergut" checks [ms]

7.372.1 Detailed Description

Ethernet-Event-specific data and methods - Rx (HIL-Tx => ECU-Rx).

Extends EthEvent with methods for enableHilTx/disableHilTx and kickout (trigger).

Note

This class expects a "Model Root/" path structure.

Also note that BusSystems-based implementations (see below) may use Model-Root-based EthEvents without problems.

7.372.2 Constructor & Destructor Documentation

7.372.2.1 __init__()

```
ttk_bus.bus_signals_ethbs.RxEthEvent.__init__ (
              self,
              context,
              enable_path,
              trigger_path,
              cycle_time = 0,
              debounce_time = 0,
              alias = '',
              descr = '',
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Ethernet Rx-Event (or field) data.

Parameters

context	parent context (an RTE application or XilTestbench instance to resolve signal/variable paths)
enable_path	model path to tx enable switch (if available)
trigger_path	model path to tx trigger (if available)
cycle_time	cycle time in [ms]. On-demand events have a cycle time of 0 (similar to fibex value)
debounce_time	minimum debounce delay time between events in [ms]
alias	Event alias/name
descr	(opt) Event description (may be used in report entries)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_ethbs.RxEthEvent.

7.372.3 Member Function Documentation

7.372.3.1 disableHilTx()

```
ttk_bus.bus_signals_ethbs.RxEthEvent.disableHilTx (
              self )
```

Disable sending of Event.

Reimplemented from ttk_bus._bus_signals_ethbs.RxEthEvent.

7.372.3.2 enableHilTx()

```
ttk_bus.bus_signals_ethbs.RxEthEvent.enableHilTx (
              self )
```

(Re-)enable sending of Event.

Reimplemented from ttk_bus._bus_signals_ethbs.RxEthEvent.

7.372.3.3 kickout()

```
ttk_bus.bus_signals_ethbs.RxEthEvent.kickout (
              self,
              delay_ms = None )
```

Trigger sending of the Ethernet Event / Field.

Parameters

delay_ms	[ms] Delay to wait before clearing kickout/trigger variable again (it *should* auto-reset, but anyway). Defaults to self.debounce_time (if explicitly set) or alternatively to self.change_delay

Reimplemented from ttk_bus._bus_signals_ethbs.RxEthEvent.

7.372.3.4 reset()

```
ttk_bus.bus_signals_ethbs.RxEthEvent.reset (
              self,
             * args,
             ** kwargs )
```

Call reset() on all contained variables/signals that support it.

Simple, non-recursive approach.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.ResettableContainer.

7.372.4 Member Data Documentation

7.372.4.1 alias

```
ttk_bus.bus_signals_ethbs.RxEthEvent.alias
```

alias name of event

7.372.4.2 change_delay

```
ttk_bus.bus_signals_ethbs.RxEthEvent.change_delay
```

delay to wait after a value change [ms]

7.372.4.3 context

```
ttk_bus.bus_signals_ethbs.RxEthEvent.context
```

parent context (an RTE application instance to resolve signal/variable paths)

7.372.4.4 cycle_time

```
ttk_bus.bus_signals_ethbs.RxEthEvent.cycle_time
```

cycle time of event in [ms]

7.372.4.5 debounce_time

ttk_bus.bus_signals_ethbs.RxEthEvent.debounce_time

minimum debounce delay for triggered events in [ms]

7.372.4.6 descr

ttk_bus.bus_signals_ethbs.RxEthEvent.descr

description text of event

7.372.4.7 enable_path

ttk_bus.bus_signals_ethbs.RxEthEvent.enable_path

model path to tx enable switch (if available).

Other relevant entries will be derived from this path (e.g. kickout, status, ...)

7.372.4.8 enable_var

ttk_bus.bus_signals_ethbs.RxEthEvent.enable_var

HIL-Tx enable/disable variable (a "HilVar" instance)

7.372.4.9 kickout_var

ttk_bus.bus_signals_ethbs.RxEthEvent.kickout_var

HIL-Tx kickout/trigger variable (a "HilVar" instance)

7.372.4.10 recovery_time

ttk_bus.bus_signals_ethbs.RxEthEvent.recovery_time

time for recovery/"wiedergut" checks [ms]

7.372.4.11 timeout

ttk_bus.bus_signals_ethbs.RxEthEvent.timeout

detection time for timeout errors [ms]

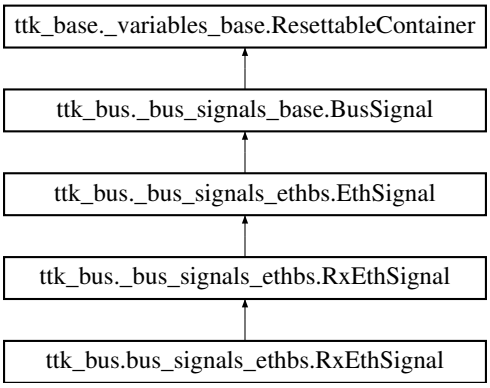
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_ethbs.py (v2.5)

7.373 ttk_bus._bus_signals_ethbs.RxEthSignal Class Reference

An Ethernet event signal/parameter/sub-parameter sent from test system to DUT (ECU CSI Rx <= HIL PSI Tx).

Inheritance diagram for ttk_bus._bus_signals_ethbs.RxEthSignal:



Public Member Functions

- __init__ (self, context, signal_path, event=None, cycle_time=None, debounce_time=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None, **kwargs)

  Ethernet Rx signal initialization ("Model Root/"-path flavour).
- enableHilTx (self)

  Enable transmission of signal HIL => DUT (if the assigned EthEvent supports it).
- disableHilTx (self)

  Disable transmission of signal HIL => DUT (if the assigned EthEvent supports it).
- kickout (self)

  Trigger sending of the parent EthEvent.

Static Public Attributes

- int fallback_change_delay = 100

  Fallback value in [ms] for event-triggered signals w/o a defined or derived change delay to permit signal propagation.

7.373.1 Detailed Description

An Ethernet event signal/parameter/sub-parameter sent from test system to DUT (ECU CSI Rx <= HIL PSI Tx).

Note

This class expects a "Model Root/" path structure.

7.373.2 Constructor & Destructor Documentation

7.373.2.1 __init__()

```
ttk_bus._bus_signals_ethbs.RxEthSignal.__init__ (
              self,
              context,
              signal_path,
              event = None,
              cycle_time = None,
              debounce_time = None,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              change_delay = None,
              timeout = None,
              recovery_time = None,
             ** kwargs )
```

Ethernet Rx signal initialization ("Model Root/"-path flavour).

Parameters

context	parent context (an RTE application or XilTestbench instance to resolve signal/variable paths)
signal_path	signal path below ModelRoot
event	reference to associated EthEvent instance
cycle_time	(opt) cycle time in [ms], overrides settings in EthEvent
debounce_time	(opt) minimum debounce delay time between events in [ms], overrides settings in EthEvent
unit	(opt) unit of physical signal value (as string, mainly used for report entries)
alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_bus._bus_signals_ethbs.EthSignal.

Reimplemented in ttk_bus.bus_signals_ethbs.RxEthSignal.

7.373.3 Member Function Documentation

7.373.3.1 disableHilTx()

```
ttk_bus._bus_signals_ethbs.RxEthSignal.disableHilTx (
              self )
```

Disable transmission of signal HIL => DUT (if the assigned EthEvent supports it).

Note

This disables transmission of all signals in the parent Event.

Reimplemented in ttk_bus.bus_signals_ethbs.RxEthSignal.

7.373.3.2 enableHilTx()

```
ttk_bus._bus_signals_ethbs.RxEthSignal.enableHilTx (
              self )
```

Enable transmission of signal HIL => DUT (if the assigned EthEvent supports it).

Note

This enables transmission of all signals in the parent Event.

Reimplemented in ttk_bus.bus_signals_ethbs.RxEthSignal.

7.373.3.3 kickout()

```
ttk_bus._bus_signals_ethbs.RxEthSignal.kickout (
              self )
```

Trigger sending of the parent EthEvent.

Reimplemented in ttk_bus.bus_signals_ethbs.RxEthSignal.

7.373.4 Member Data Documentation

7.373.4.1 fallback_change_delay

```
int ttk_bus._bus_signals_ethbs.RxEthSignal.fallback_change_delay = 100  [static]
```

Fallback value in [ms] for event-triggered signals w/o a defined or derived change delay to permit signal propagation.

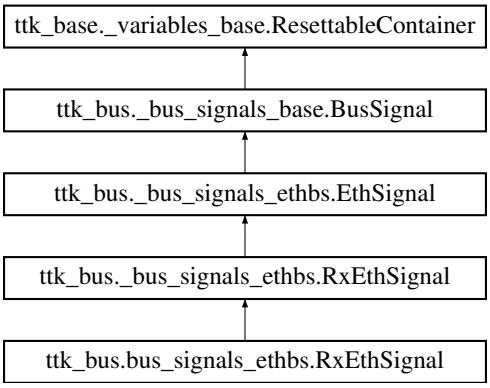
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_ethbs.py (v2.5)

7.374 ttk_bus.bus_signals_ethbs.RxEthSignal Class Reference

An Ethernet event signal/parameter/sub-parameter sent from test system to DUT (ECU CSI Rx <= HIL PSI Tx).

Inheritance diagram for ttk_bus.bus_signals_ethbs.RxEthSignal:



Public Member Functions

- __init__ (self, context, signal_path, event=None, cycle_time=None, debounce_time=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None, **kwargs)

  Ethernet Rx signal initialization ("Model Root/"-path flavour).
- enableHilTx (self)

  Enable transmission of signal HIL => DUT (if the assigned EthEvent supports it).
- disableHilTx (self)

  Disable transmission of signal HIL => DUT (if the assigned EthEvent supports it).
- kickout (self)

  Trigger sending of the Ethernet Event / Field.
- get (self)

  Get the current value of the HIL signal variable (typically not that useful for ECU-Rx/HIL-Tx signals, though).
- set (self, value)

  Set a value to the HIL signal variable.
- getState (self)

  Get a textual state for the HIL signal variable's current data.
- setState (self, state)

  Set a textual state to the HIL signal variable.
- getStateDescr (self, value=None, fallback=None)

  Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).
- info (self)

  Get a current info string about the HIL signal variable.
- reset (self, *args, **kwargs)

  Reset all resettable variables of this bus signal.

Public Attributes

- context
  parent context (an RTE application instance to resolve signal/variable paths)
- signal_path
  Base path to signal below BusSystems.
- event
  reference to associated EthEvent instance
- unit
  unit of physical signal value (as string, mainly used for report entries)
- alias
  bus signal alias name
- descr
  signal description text
- cycle_time
  cycle time of signal in [ms]
- debounce_time
  minimum debounce delay for triggered signals in [ms]
- change_delay
  delay to wait after a value change [ms]
- timeout
  detection time for timeout errors [ms]
- recovery_time
  time for recovery/"wiedergut" checks [ms]

7.374.1 Detailed Description

An Ethernet event signal/parameter/sub-parameter sent from test system to DUT (ECU CSI Rx <= HIL PSI Tx).

Note

This class expects a "Model Root/" path structure.

7.374.2 Constructor & Destructor Documentation

7.374.2.1 __init__()

```
ttk_bus.bus_signals_ethbs.RxEthSignal.__init__ (
              self,
              context,
              signal_path,
              event = None,
              cycle_time = None,
              debounce_time = None,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              change_delay = None,
              timeout = None,
              recovery_time = None,
             ** kwargs )
```

Ethernet Rx signal initialization ("Model Root/"-path flavour).

Parameters

context	parent context (an RTE application or XilTestbench instance to resolve signal/variable paths)
signal_path	signal path below ModelRoot
event	reference to associated EthEvent instance
cycle_time	(opt) cycle time in [ms], overrides settings in EthEvent
debounce_time	(opt) minimum debounce delay time between events in [ms], overrides settings in EthEvent
unit	(opt) unit of physical signal value (as string, mainly used for report entries)
alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_bus._bus_signals_ethbs.RxEthSignal.

7.374.3 Member Function Documentation

7.374.3.1 disableHilTx()

```
ttk_bus.bus_signals_ethbs.RxEthSignal.disableHilTx (
              self )
```

Disable transmission of signal HIL => DUT (if the assigned EthEvent supports it).

Note

This disables transmission of all signals in the parent Event.

Reimplemented from ttk_bus._bus_signals_ethbs.RxEthSignal.

7.374.3.2 enableHilTx()

```
ttk_bus.bus_signals_ethbs.RxEthSignal.enableHilTx (
              self )
```

Enable transmission of signal HIL => DUT (if the assigned EthEvent supports it).

Note

This enables transmission of all signals in the parent Event.

Reimplemented from ttk_bus._bus_signals_ethbs.RxEthSignal.

7.374.3.3 get()

```
ttk_bus.bus_signals_ethbs.RxEthSignal.get (
              self )
```

Get the current value of the HIL signal variable (typically not that useful for ECU-Rx/HIL-Tx signals, though).

Returns

the current value from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.374.3.4 getState()

```
ttk_bus.bus_signals_ethbs.RxEthSignal.getState (
              self )
```

Get a textual state for the HIL signal variable's current data.

(typically not that useful for ECU-Rx/HIL-Tx signals, though). This uses the lookup values supplied in constructor.

Returns

the current state representation from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.374.3.5 getStateDescr()

```
ttk_bus.bus_signals_ethbs.RxEthSignal.getStateDescr (
              self,
              value = None,
              fallback = None )
```

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the last read value will be used instead.
fallback	fallback value to return if no match was found. Default: None

Returns

the state text if a match was found, otherwise the specified fallback value.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.374.3.6 info()

```
ttk_bus.bus_signals_ethbs.RxEthSignal.info (
              self )
```

Get a current info string about the HIL signal variable.

Returns

an info string.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.374.3.7 kickout()

```
ttk_bus.bus_signals_ethbs.RxEthSignal.kickout (
              self )
```

Trigger sending of the Ethernet Event / Field.

Reimplemented from ttk_bus._bus_signals_ethbs.RxEthSignal.

7.374.3.8 reset()

```
ttk_bus.bus_signals_ethbs.RxEthSignal.reset (
              self,
             * args,
             ** kwargs )
```

Reset all resettable variables of this bus signal.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.ResettableContainer.

7.374.3.9 set()

```
ttk_bus.bus_signals_ethbs.RxEthSignal.set (
              self,
              value )
```

Set a value to the HIL signal variable.

Parameters

value	value to set

Returns

the previous value from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.374.3.10 setState()

```
ttk_bus.bus_signals_ethbs.RxEthSignal.setState (
            self,
            state )
```

Set a textual state to the HIL signal variable.

This uses the lookup values supplied in constructor.

Parameters

state	named state to set

Returns

the previous value representation from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.374.4 Member Data Documentation

7.374.4.1 alias

```
ttk_bus.bus_signals_ethbs.RxEthSignal.alias
```

bus signal alias name

7.374.4.2 change_delay

```
ttk_bus.bus_signals_ethbs.RxEthSignal.change_delay
```

delay to wait after a value change [ms]

7.374.4.3 context

```
ttk_bus.bus_signals_ethbs.RxEthSignal.context
```

parent context (an RTE application instance to resolve signal/variable paths)

7.374.4.4 cycle_time

ttk_bus.bus_signals_ethbs.RxEthSignal.cycle_time

cycle time of signal in [ms]

7.374.4.5 debounce_time

ttk_bus.bus_signals_ethbs.RxEthSignal.debounce_time

minimum debounce delay for triggered signals in [ms]

7.374.4.6 descr

ttk_bus.bus_signals_ethbs.RxEthSignal.descr

signal description text

7.374.4.7 event

ttk_bus.bus_signals_ethbs.RxEthSignal.event

reference to associated EthEvent instance

7.374.4.8 recovery_time

ttk_bus.bus_signals_ethbs.RxEthSignal.recovery_time

time for recovery/"wiedergut" checks [ms]

7.374.4.9 signal_path

ttk_bus.bus_signals_ethbs.RxEthSignal.signal_path

Base path to signal below BusSystems.

7.374.4.10 timeout

ttk_bus.bus_signals_ethbs.RxEthSignal.timeout

detection time for timeout errors [ms]

7.374.4.11 unit

`ttk_bus.bus_signals_ethbs.RxEthSignal.unit`

unit of physical signal value (as string, mainly used for report entries)

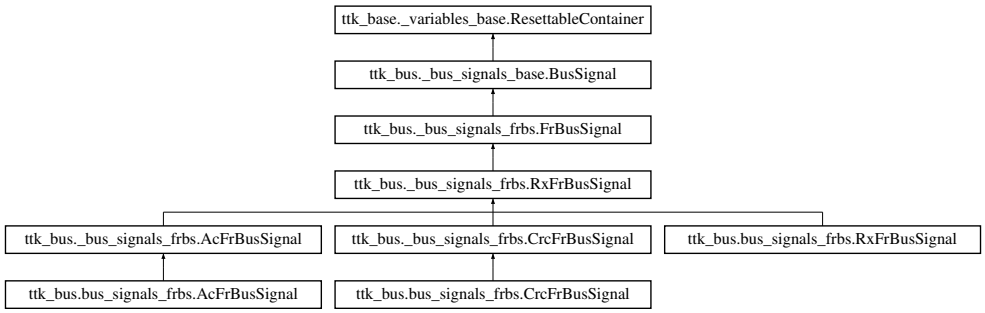
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_ethbs.py (v2.5)

7.375 ttk_bus._bus_signals_frbs.RxFrBusSignal Class Reference

A FlexRay signal sent from test system to the DUT (ECU Rx <= HIL Tx).

Inheritance diagram for ttk_bus._bus_signals_frbs.RxFrBusSignal:



Public Member Functions

- __init__ (self, context, base_path, pdu=None, cycle_time=None, debounce_time=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  Rx bus signal initialization.
- switchToBusSystemsSource (self, bus_systems=True)

  Switch signal source to "BusSystems".
- set (self, value)

  Set a value to the HIL signal variable below BusSystems (and switch the signal source to "BusSystems").
- setState (self, state)

  Set a textual state to the HIL signal variable below BusSystems (and switches the signal source to "BusSystems").
- enableHilTx (self)

  Enable transmission of signal HIL => DUT (if the assigned PDU supports it).
- disableHilTx (self)

  Disable transmission of signal HIL => DUT (if the assigned PDU supports it).
- kickout (self)

  Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent PDU).

Public Attributes

- switch_var

  Source Switch variable (a "HilVar" instance).

Static Public Attributes

- dict cfg_custom_defaults

  Configuration setting for new instances: Custom "hard" default value(s) to use for internal real time system variables (for example: switch settings), so a reset() will always reset those variables to defined values (and not just those variables that have actually been accessed).

Protected Attributes

- _default_values

7.375.1 Detailed Description

A FlexRay signal sent from test system to the DUT (ECU Rx <= HIL Tx).

7.375.2 Constructor & Destructor Documentation

7.375.2.1 __init__()

```
ttk_bus._bus_signals_frbs.RxFrBusSignal.__init__ (
                self,
                context,
                base_path,
                pdu = None,
                cycle_time = None,
                debounce_time = None,
                unit = "",
                alias = "",
                descr = "",
                lookup = None,
                change_delay = None,
                timeout = None,
                recovery_time = None )
```

Rx bus signal initialization.

Example: BusSystems Signal Path Structure:

```
# path to signal value variable
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/SIGNAL_NAME_OUT/Value
# or
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/SIGNAL_NAME_OUT/Physical Value
```

Parameters

context	parent context (an RTE application instance to resolve signal/variable paths)
base_path	base path to signal below BusSystems (use path to `/Value` or `/Physical Value` as appropriate)
pdu	reference to associated RxFrPdu instance
cycle_time	(opt) cycle time in [ms], overrides settings in PDU
debounce_time	(opt) minimum debounce delay time in [ms], overrides settings in PDU
unit	(opt) unit of physical signal value (as string, mainly used for report entries)

Parameters

alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_frbs.FrBusSignal.

Reimplemented in ttk_bus._bus_signals_frbs.CrcFrBusSignal, ttk_bus._bus_signals_frbs.AcFrBusSignal, ttk_bus.bus_signals_frbs.Cr
ttk_bus.bus_signals_frbs.AcFrBusSignal, and ttk_bus.bus_signals_frbs.RxFrBusSignal.

7.375.3 Member Function Documentation

7.375.3.1 disableHilTx()

```
ttk_bus._bus_signals_frbs.RxFrBusSignal.disableHilTx (
              self )
```

Disable transmission of signal HIL => DUT (if the assigned PDU supports it).

Note

This will disable transmission of all signals of the parent PDU.

Reimplemented in ttk_bus.bus_signals_frbs.RxFrBusSignal, ttk_bus.bus_signals_frbs.CrcFrBusSignal, and ttk_bus.bus_signals_frbs.AcFrBusSignal.

7.375.3.2 enableHilTx()

```
ttk_bus._bus_signals_frbs.RxFrBusSignal.enableHilTx (
              self )
```

Enable transmission of signal HIL => DUT (if the assigned PDU supports it).

Note

This will enable transmission of all signals of the parent PDU.

Reimplemented in ttk_bus.bus_signals_frbs.RxFrBusSignal, ttk_bus.bus_signals_frbs.CrcFrBusSignal, and ttk_bus.bus_signals_frbs.AcFrBusSignal.

7.375.3.3 kickout()

```
ttk_bus._bus_signals_frbs.RxFrBusSignal.kickout (
              self )
```

Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent PDU).

Note

This means all other signals of the parent PDU will be sent as well

Reimplemented in ttk_bus.bus_signals_frbs.RxFrBusSignal, ttk_bus.bus_signals_frbs.CrcFrBusSignal, and ttk_bus.bus_signals_frbs.AcFrBusSignal.

7.375.3.4 set()

```
ttk_bus._bus_signals_frbs.RxFrBusSignal.set (
              self,
              value )
```

Set a value to the HIL signal variable below BusSystems (and switch the signal source to "BusSystems").

Parameters

value	value to set

Returns

the previous value from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

Reimplemented in ttk_bus.bus_signals_frbs.RxFrBusSignal, ttk_bus.bus_signals_frbs.CrcFrBusSignal, and ttk_bus.bus_signals_frbs.AcFrBusSignal.

7.375.3.5 setState()

```
ttk_bus._bus_signals_frbs.RxFrBusSignal.setState (
              self,
              state )
```

Set a textual state to the HIL signal variable below BusSystems (and switches the signal source to "BusSystems").

This uses the lookup values supplied in constructor.

Parameters

state	named state to set

Returns

the previous state representation from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

Reimplemented in ttk_bus.bus_signals_frbs.RxFrBusSignal, ttk_bus.bus_signals_frbs.CrcFrBusSignal, and ttk_bus.bus_signals_frbs.AcFrBusSignal.

7.375.3.6 switchToBusSystemsSource()

```
ttk_bus._bus_signals_frbs.RxFrBusSignal.switchToBusSystemsSource (
              self,
              bus_systems = True )
```

Switch signal source to "BusSystems".

Mainly intended for internal use.

Parameters

bus_systems	True: use BusSystems as source False: use Simulink Model (Model Root/...) as source

7.375.4 Member Data Documentation

7.375.4.1 _default_values

```
ttk_bus._bus_signals_frbs.RxFrBusSignal._default_values  [protected]
```

7.375.4.2 cfg_custom_defaults

```
dict ttk_bus._bus_signals_frbs.RxFrBusSignal.cfg_custom_defaults  [static]
```

Initial value:

```
=  {
        "switch_var": None,
    }
```

Configuration setting for new instances: Custom "hard" default value(s) to use for internal real time system variables (for example: switch settings), so a reset() will always reset those variables to defined values (and not just those variables that have actually been accessed).

This might be helpful if "other" tests fail to properly clean up the test environment after they are done.

Mapping: <variable name as string>: <default value>

A value of None (or a missing variable name) enables normal/automatic reset behavior.

Typical default values:

- source switches usually default to "source: simulink model"
  - 0: source: simulink model/trc
  - 1: source: bus systems

7.375.4.3 switch_var

`ttk_bus._bus_signals_frbs.RxFrBusSignal.switch_var`

Source Switch variable (a "HilVar" instance).

- 0: value source is Simulink Model
- 1: value source is BusSystems

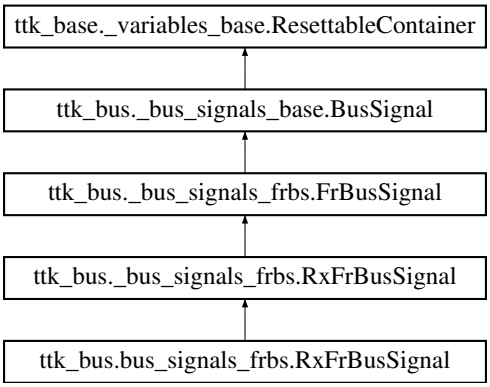
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_frbs.py (v2.5)

7.376 ttk_bus.bus_signals_frbs.RxFrBusSignal Class Reference

A FlexRay signal sent from test system to the DUT (ECU Rx <= HIL Tx).

Inheritance diagram for ttk_bus.bus_signals_frbs.RxFrBusSignal:



Public Member Functions

- __init__ (self, context, base_path, pdu=None, cycle_time=None, debounce_time=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  Rx bus signal initialization.
- info (self)

  Get a current info string about the HIL signal variable, see signal_var.
- get (self)

  Get the current value of the HIL signal variable, see signal_var.
- set (self, value)

  Set a value to the HIL signal variable below BusSystems (and switch the signal source to "BusSystems").
- getState (self)

  Get a textual state for the HIL signal variable's current data.
- setState (self, state)

  Set a textual state to the HIL signal variable below BusSystems (and switches the signal source to "BusSystems").
- getStateDescr (self, value=None, fallback=None)

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

- enableHilTx (self)

  Enable transmission of signal HIL => DUT (if the assigned PDU supports it).

- disableHilTx (self)

  Disable transmission of signal HIL => DUT (if the assigned PDU supports it).

- kickout (self)

  Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent PDU).

- reset (self, *args, **kwargs)

  Reset all resettable variables of this bus signal.

Public Attributes

- switch_var

  Source Switch variable (a "HilVar" instance).

- value_path

  model path of value variable

- base_path

  base model path (base path for deriving sub-element paths)

- pdu

  reference to parent FrRxPdu

- context

  context (reference to assigned RTE application instance)

- signal_var

  main bus signal variable on HIL-side, (a "HilVar" instance)

- alias

  signal alias name (from signal_var)

- unit

  signal unit (from signal_var)

- descr

  signal description text

- cycle_time

  cycle time of signal in [ms] (that is, of the associated message or PDU)

- debounce_time

  minimum debounce delay for event-triggered signals in [ms]

- change_delay

  delay to wait after a value change [ms]

- timeout

  detection time for timeout errors [ms]

- recovery_time

  time for recovery/"wiedergut" checks [ms]

Static Public Attributes

- dict cfg_custom_defaults

  Configuration setting for new instances: Custom "hard" default value(s) to use for internal real time system variables (for example: switch settings), so a reset() will always reset those variables to defined values (and not just those variables that have actually been accessed).

7.376.1 Detailed Description

A FlexRay signal sent from test system to the DUT (ECU Rx <= HIL Tx).

7.376.2 Constructor & Destructor Documentation

7.376.2.1 __init__()

```
ttk_bus.bus_signals_frbs.RxFrBusSignal.__init__ (
                self,
                context,
                base_path,
                pdu = None,
                cycle_time = None,
                debounce_time = None,
                unit = "",
                alias = "",
                descr = "",
                lookup = None,
                change_delay = None,
                timeout = None,
                recovery_time = None )
```

Rx bus signal initialization.

Example: BusSystems Signal Path Structure:

```
# path to signal value variable
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/SIGNAL_NAME_OUT/Value
# or
BusSystems/FlexRay/Sending/ECU_NAME/Channel A/PDU_NAME/Signals/SIGNAL_NAME_OUT/Physical Value
```

Parameters

context	parent context (an RTE application instance to resolve signal/variable paths)
base_path	base path to signal below BusSystems (use path to `/Value` or `/Physical Value` as appropriate)
pdu	reference to associated RxFrPdu instance
cycle_time	(opt) cycle time in [ms], overrides settings in PDU
debounce_time	(opt) minimum debounce delay time in [ms], overrides settings in PDU
unit	(opt) unit of physical signal value (as string, mainly used for report entries)
alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_frbs.RxFrBusSignal.

7.376.3 Member Function Documentation

7.376.3.1 disableHilTx()

```
ttk_bus.bus_signals_frbs.RxFrBusSignal.disableHilTx (
              self )
```

Disable transmission of signal HIL => DUT (if the assigned PDU supports it).

Note

This will disable transmission of all signals of the parent PDU.

Reimplemented from ttk_bus._bus_signals_frbs.RxFrBusSignal.

7.376.3.2 enableHilTx()

```
ttk_bus.bus_signals_frbs.RxFrBusSignal.enableHilTx (
              self )
```

Enable transmission of signal HIL => DUT (if the assigned PDU supports it).

Note

This will enable transmission of all signals of the parent PDU.

Reimplemented from ttk_bus._bus_signals_frbs.RxFrBusSignal.

7.376.3.3 get()

```
ttk_bus.bus_signals_frbs.RxFrBusSignal.get (
              self )
```

Get the current value of the HIL signal variable, see signal_var.

Attention

Please note that for RxFrBusSignals, this will get the current value of the HIL-send-buffer (below BusSystems), not the value from Model/Simulink source nor the last seen signal value on FlexRay bus.

Returns

the current value from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.376.3.4 getState()

```
ttk_bus.bus_signals_frbs.RxFrBusSignal.getState (
              self )
```

Get a textual state for the HIL signal variable's current data.

This uses the lookup values supplied in constructor, see signal_var. Fallback (if no matching state was found) will be the unmapped numeric value.

Attention

Please note that for RxFrBusSignals, this will get the current value of the HIL-send-buffer (below BusSystems), not the value from Model/Simulink source nor the last seen signal value on FlexRay bus.

Returns

the current state representation from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.376.3.5 getStateDescr()

```
ttk_bus.bus_signals_frbs.RxFrBusSignal.getStateDescr (
              self,
              value = None,
              fallback = None )
```

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the last read value will be used instead.
fallback	fallback value to return if no match was found. Default: None

Returns

the state text if a match was found, otherwise the specified fallback value.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.376.3.6 info()

```
ttk_bus.bus_signals_frbs.RxFrBusSignal.info (
              self )
```

Get a current info string about the HIL signal variable, see signal_var.

Attention

Please note that for RxFrBusSignals, this will show info on the contents of the HIL-send-buffer (below Bus↩ Systems), not the value from Model/Simulink source nor the last seen signal value on FlexRay bus.

Returns

the current info string from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.376.3.7 kickout()

```
ttk_bus.bus_signals_frbs.RxFrBusSignal.kickout (
              self )
```

Trigger transmission ("kickout") of signal HIL => DUT (by triggering the parent PDU).

Note

This means all other signals of the parent PDU will be sent as well

Reimplemented from ttk_bus._bus_signals_frbs.RxFrBusSignal.

7.376.3.8 reset()

```
ttk_bus.bus_signals_frbs.RxFrBusSignal.reset (
              self,
             * args,
             ** kwargs )
```

Reset all resettable variables of this bus signal.

Note that this will also switch the variable source switch back from BusSystems to (default) Simulink/Model.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.ResettableContainer.

7.376.3.9 set()

```
ttk_bus.bus_signals_frbs.RxFrBusSignal.set (
              self,
              value )
```

Set a value to the HIL signal variable below BusSystems (and switch the signal source to "BusSystems").

Parameters

value	value to set

Returns

the previous value from signal_var.

Reimplemented from ttk_bus._bus_signals_frbs.RxFrBusSignal.

7.376.3.10 setState()

```
ttk_bus.bus_signals_frbs.RxFrBusSignal.setState (
              self,
              state )
```

Set a textual state to the HIL signal variable below BusSystems (and switches the signal source to "BusSystems").

This uses the lookup values supplied in constructor.

Parameters

state	named state to set

Returns

the previous state representation from signal_var.

Reimplemented from ttk_bus._bus_signals_frbs.RxFrBusSignal.

7.376.4 Member Data Documentation

7.376.4.1 alias

```
ttk_bus.bus_signals_frbs.RxFrBusSignal.alias
```

signal alias name (from signal_var)

7.376.4.2 base_path

```
ttk_bus.bus_signals_frbs.RxFrBusSignal.base_path
```

base model path (base path for deriving sub-element paths)

7.376.4.3 cfg_custom_defaults

```
dict ttk_bus.bus_signals_frbs.RxFrBusSignal.cfg_custom_defaults  [static]
```

Initial value:

```
=  {
        "switch_var": None,
    }
```

Configuration setting for new instances: Custom "hard" default value(s) to use for internal real time system variables (for example: switch settings), so a reset() will always reset those variables to defined values (and not just those variables that have actually been accessed).

This might be helpful if "other" tests fail to properly clean up the test environment after they are done.

Mapping: <variable name as string>: <default value>

A value of None (or a missing variable name) enables normal/automatic reset behavior.

Typical default values:

- source switches usually default to "source: simulink model"
  - 0: source: simulink model/trc
  - 1: source: bus systems

7.376.4.4 change_delay

```
ttk_bus.bus_signals_frbs.RxFrBusSignal.change_delay
```

delay to wait after a value change [ms]

7.376.4.5 context

```
ttk_bus.bus_signals_frbs.RxFrBusSignal.context
```

context (reference to assigned RTE application instance)

7.376.4.6 cycle_time

```
ttk_bus.bus_signals_frbs.RxFrBusSignal.cycle_time
```

cycle time of signal in [ms] (that is, of the associated message or PDU)

7.376.4.7 debounce_time

```
ttk_bus.bus_signals_frbs.RxFrBusSignal.debounce_time
```

minimum debounce delay for event-triggered signals in [ms]

7.376.4.8 descr

ttk_bus.bus_signals_frbs.RxFrBusSignal.descr

signal description text

7.376.4.9 pdu

ttk_bus.bus_signals_frbs.RxFrBusSignal.pdu

reference to parent FrRxPdu

7.376.4.10 recovery_time

ttk_bus.bus_signals_frbs.RxFrBusSignal.recovery_time

time for recovery/"wiedergut" checks [ms]

7.376.4.11 signal_var

ttk_bus.bus_signals_frbs.RxFrBusSignal.signal_var

main bus signal variable on HIL-side, (a "HilVar" instance)

7.376.4.12 switch_var

ttk_bus.bus_signals_frbs.RxFrBusSignal.switch_var

Source Switch variable (a "HilVar" instance).

- 0: value source is Simulink Model
- 1: value source is BusSystems

7.376.4.13 timeout

ttk_bus.bus_signals_frbs.RxFrBusSignal.timeout

detection time for timeout errors [ms]

7.376.4.14 unit

ttk_bus.bus_signals_frbs.RxFrBusSignal.unit

signal unit (from signal_var)

7.376.4.15 value_path

```
ttk_bus.bus_signals_frbs.RxFrBusSignal.value_path
```

model path of value variable

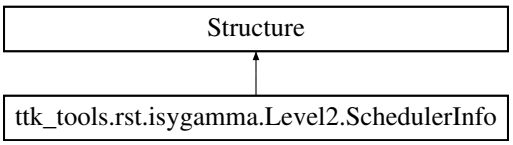
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_frbs.py (v1.8)

7.377 ttk_tools.rst.isygamma.Level2.SchedulerInfo Class Reference

typedef struct gc_SchedulerInfo

Inheritance diagram for ttk_tools.rst.isygamma.Level2.SchedulerInfo:



Static Protected Attributes

- list _fields_

7.377.1 Detailed Description

typedef struct gc_SchedulerInfo

7.377.2 Member Data Documentation

7.377.2.1 _fields_

```
list ttk_tools.rst.isygamma.Level2.SchedulerInfo._fields_  [static], [protected]
```

Initial value:

```
=  [
            ("address",         Addresses.Level2),  # address of the scheduler
            ("name",            c_char * 1024),     # unique name of the scheduler
            ("isPersistent",    c_bool),            # persistence state of the scheduler
            ("numSlots",        c_uint32),          # number of slots to be scheduled
            ("topology",        POINTER(c_uint32)), # topology as an NxN array. See gc_SchedulingTopology
            ("cycleTicks",          c_uint32),      # configuration of the scheduler
            ("timeoutUs",           c_uint32),      # scheduling timeout
            ("expectedMinUs",       c_uint32),      # minimum expected cycle duration
            ("expectedMaxUs",       c_uint32),      # maximum expected cycle duration
            ("abortOnNumOverruns",  c_uint32),      # abort margin for overruns.  0: disabled
            ("abortOnNumUnderruns", c_uint32),      # abort margin for underruns. 0: disabled
            ("minUs",               c_uint32),      # statistics of the last execution
            ("maxUs",               c_uint32),      # maximum frame duration
            ("meanUs",              c_uint32),      # mean frame duration
            ("stdDevUs",            c_uint32),      # standard deviation of frame duration
            ("numFrames",           c_uint32),      # number of executed frames
            ("overrunFrames",       c_uint32),      # number of frame duration overruns
            ("underrunFrames",      c_uint32),      # number of frame duration underruns
            ("state",               c_int),         # scheduler state. See gc_SchedulerState
        ]
```

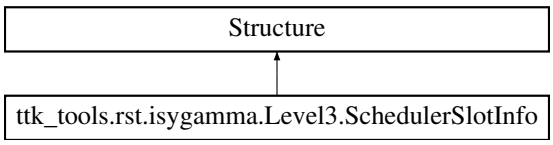
The documentation for this class was generated from the following file:

- ttk_tools/rst/isygamma.py (v1.6)

7.378 ttk_tools.rst.isygamma.Level3.SchedulerSlotInfo Class Reference

typedef struct gc_SchedulerSlotInfo

Inheritance diagram for ttk_tools.rst.isygamma.Level3.SchedulerSlotInfo:



Static Protected Attributes

- list _fields_

7.378.1 Detailed Description

typedef struct gc_SchedulerSlotInfo

7.378.2 Member Data Documentation

7.378.2.1 _fields_

```
list ttk_tools.rst.isygamma.Level3.SchedulerSlotInfo._fields_  [static], [protected]
```

Initial value:

```
=  [
            ("address",     Addresses.Level3),   # address of the scheduler slot
            ("name",           c_char * 1024),   # unique name of the scheduler slot
            ("isAssigned",          c_bool),     # true: slot is used
            ("timeoutUs",           c_uint32),   # timeout
            ("expectedMinUs",       c_uint32),   # minimum expected slot duration.
            ("expectedMaxUs",       c_uint32),   # maximum expected cycle duration.
            ("abortOnNumOverruns",  c_uint32),   # abort margin for overruns.  Zero: disabled.
            ("abortOnNumUnderruns", c_uint32),   # abort margin for underruns. Zero: disabled.
            ("state",               c_int),      # status. See gc_SchedulerState
            ("initState",           c_int32),    # return code of initialization function call.
            ("runState",            c_int32),    # return code of run function call.
            ("deinitState",         c_int32),    # return code of cleanup function call.
            ("minUs",               c_int32),    # minimum slot duration
            ("maxUs",               c_int32),    # maximum slot duration
            ("meanUs",              c_int32),    # mean slot duration
            ("stdDevUs",            c_int32),    # standard deviation of frame duration
            ("numFrames",           c_int32),    # number of executed frames
            ("overrunFrames",       c_int32),    # number of slot duration overruns
            ("underrunFrames",      c_int32),    # number of slot duration underruns
        ]
```

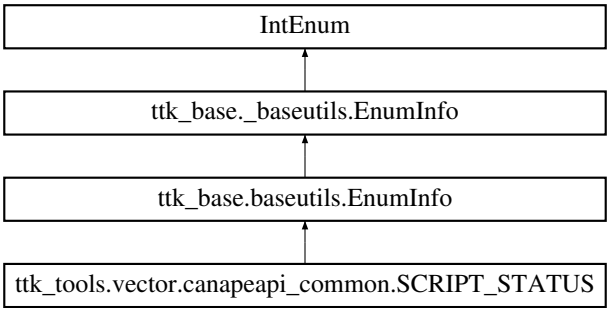
The documentation for this class was generated from the following file:

- ttk_tools/rst/isygamma.py (v1.6)

7.379 ttk_tools.vector.canapeapi_common.SCRIPT_STATUS Class Reference

Script execution status values from enum `TScriptStatus`.

Inheritance diagram for ttk_tools.vector.canapeapi_common.SCRIPT_STATUS:



Static Public Attributes

- int Ready = 1
- int Starting = 2
- int Running = 3
- int Sleeping = 4
- int Suspended = 5
- int Terminated = 6
- int FinishedReturn = 7
- int FinishedCancel = 8
- int Failure = 9
- int Timeout = 10

7.379.1 Detailed Description

Script execution status values from enum `TScriptStatus`.

(added in CANape 9).

7.379.2 Member Data Documentation

7.379.2.1 Failure

```
int ttk_tools.vector.canapeapi_common.SCRIPT_STATUS.Failure = 9  [static]
```

7.379.2.2 FinishedCancel

```
int ttk_tools.vector.canapeapi_common.SCRIPT_STATUS.FinishedCancel = 8  [static]
```

7.379.2.3 FinishedReturn

```
int ttk_tools.vector.canapeapi_common.SCRIPT_STATUS.FinishedReturn = 7  [static]
```

7.379.2.4 Ready

```
int ttk_tools.vector.canapeapi_common.SCRIPT_STATUS.Ready = 1  [static]
```

7.379.2.5 Running

```
int ttk_tools.vector.canapeapi_common.SCRIPT_STATUS.Running = 3  [static]
```

7.379.2.6 Sleeping

```
int ttk_tools.vector.canapeapi_common.SCRIPT_STATUS.Sleeping = 4  [static]
```

7.379.2.7 Starting

```
int ttk_tools.vector.canapeapi_common.SCRIPT_STATUS.Starting = 2  [static]
```

7.379.2.8 Suspended

```
int ttk_tools.vector.canapeapi_common.SCRIPT_STATUS.Suspended = 5  [static]
```

7.379.2.9 Terminated

```
int ttk_tools.vector.canapeapi_common.SCRIPT_STATUS.Terminated = 6  [static]
```

7.379.2.10 Timeout

```
int ttk_tools.vector.canapeapi_common.SCRIPT_STATUS.Timeout = 10  [static]
```

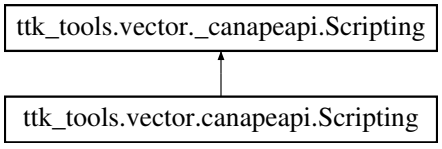
The documentation for this class was generated from the following file:

- ttk_tools/vector/canapeapi_common.py (v3.1.2)

7.380 ttk_tools.vector._canapeapi.Scripting Class Reference

Access to extended scripting functions added in CANape 9.

Inheritance diagram for ttk_tools.vector._canapeapi.Scripting:



Public Member Functions

- __init__ (self, device, verbosity=2)
- getStatus (self)

  Get execution status for the currently defined script/command.
- create (self, script_command, is_file_name=False)

  Create a new script handle for a CAPL script or script command via CANapeAPI.
- start (self, command_line=None)

  Start execution of a currently defined script.
- stop (self)

  Stop a currently running script.
- getReturnValue (self)

  Get return value of finished script (set via "return 4711;" in script).
- getResult (self)

  Get string result value of finished script.
- release (self)

  Release a currently defined script.
- waitUntilFinished (self, timeout_s=10)

  Wait until a currently running script is finished.
- run (self, script_command, is_file_name=False, command_line=None, timeout_s=10)

  Run a script command (or CAPL script) via CANapeAPI.

Public Attributes

- module_handle

  Module handle to be separately provided as (optional) "current_device" to Asap3StartScript during start().
- verbosity

Protected Member Functions

- _log (self, level, *msgs)

Protected Attributes

- _device
- _asap3
- _script_handle
- _api_log_level
- _logApiCallStart
- _logApiCallEnd

7.380.1 Detailed Description

Access to extended scripting functions added in CANape 9.

Example:

```
# separate calls
scripting = Scripting(cal_device)
scripting.create('Write("Log output"); SetScriptResult("Foobar Baz"); return 1234;')
scripting.start()
scripting.waitUntilFinished()
result = scripting.getResult()
retval = scripting.getReturnValue()
scripting.release()

assert result == "Foobar Baz"
assert retval == 1234.0
```

Example:

```
# single call, similar to classic runScript
# (but method blocks until script is finished)
retval = scripting.run('Write("Log output, again"); return 1234;')
assert retval == 1234.0
```

7.380.2 Constructor & Destructor Documentation

7.380.2.1 __init__()

```
ttk_tools.vector._canapeapi.Scripting.__init__ (
              self,
              device,
              verbosity = 2 )
```

Parameters

device	CANape device/module (BaseDevice or child instances)
verbosity	verbosity of messages logged to console: 0: only errors 1: + status info

Reimplemented in ttk_tools.vector.canapeapi.Scripting.

7.380.3 Member Function Documentation

7.380.3.1 _log()

```
ttk_tools.vector._canapeapi.Scripting._log (
              self,
              level,
             * msgs )  [protected]
```

7.380.3.2 create()

```
ttk_tools.vector._canapeapi.Scripting.create (
              self,
              script_command,
              is_file_name = False )
```

Create a new script handle for a CAPL script or script command via CANapeAPI.

Parameters

script_command	CAPL command or name of CAPL script file to execute. A CAPL command can also be the name of a custom "project function" that has been defined in the current workspace.
is_file_name	True if `script_command` contains the name of a script file to be run, False if it directly contains a CAPL command.

Reimplemented in ttk_tools.vector.canapeapi.Scripting.

7.380.3.3 getResult()

```
ttk_tools.vector._canapeapi.Scripting.getResult (
              self )
```

Get string result value of finished script.

Note

To get a result here, the function "SetScriptResult" has to be called in the executed script.

Returns

a string or None if no script was available.

Reimplemented in ttk_tools.vector.canapeapi.Scripting.

7.380.3.4 getReturnValue()

```
ttk_tools.vector._canapeapi.Scripting.getReturnValue (
              self )
```

Get return value of finished script (set via "return 4711;" in script).

Returns

a float (double) value or None if no script was available. Value may be NaN if no return value has been provided.

Reimplemented in ttk_tools.vector.canapeapi.Scripting.

7.380.3.5 getStatus()

```
ttk_tools.vector._canapeapi.Scripting.getStatus (
              self )
```

Get execution status for the currently defined script/command.

Returns

a SCRIPT_STATUS value or None if API access failed.

Reimplemented in ttk_tools.vector.canapeapi.Scripting.

7.380.3.6 release()

```
ttk_tools.vector._canapeapi.Scripting.release (
              self )
```

Release a currently defined script.

Reimplemented in ttk_tools.vector.canapeapi.Scripting.

7.380.3.7 run()

```
ttk_tools.vector._canapeapi.Scripting.run (
              self,
              script_command,
              is_file_name = False,
              command_line = None,
              timeout_s = 10 )
```

Run a script command (or CAPL script) via CANapeAPI.

Essentially just calls

- create
- start
- waitUntilFinished
- getReturnValue and
- release
  in sequence.

Parameters

script_command	CAPL command or name of CAPL script file to execute. A CAPL command can also be the name of a custom "project function" that has been defined in the current workspace.
is_file_name	True if `script_command` contains the name of a script file to be run, False if it directly contains a CAPL command.
command_line	optional "command line" string to hand to start()
timeout_s	seconds to wait for script execution to finish

Example:

```
# multiple commands can be added in sequence
scripting.run('Speak("Hello World"); Write("Done."); return 123;')
```

Returns

the script's return value (see getReturnValue)

Reimplemented in ttk_tools.vector.canapeapi.Scripting.

7.380.3.8 start()

```
ttk_tools.vector._canapeapi.Scripting.start (
              self,
              command_line = None )
```

Start execution of a currently defined script.

Parameters

command_line	optional "command line" string to hand to Asap3StartScript.

Reimplemented in ttk_tools.vector.canapeapi.Scripting.

7.380.3.9 stop()

```
ttk_tools.vector._canapeapi.Scripting.stop (
              self )
```

Stop a currently running script.

Returns

True if script was stopped, False if stop failed and None if there was no current script handle to stop.

Reimplemented in ttk_tools.vector.canapeapi.Scripting.

7.380.3.10 waitUntilFinished()

```
ttk_tools.vector._canapeapi.Scripting.waitUntilFinished (
              self,
              timeout_s = 10 )
```

Wait until a currently running script is finished.

Parameters

timeout↩ _s	timeout in [s] before script will be stopped.

Reimplemented in ttk_tools.vector.canapeapi.Scripting.

7.380.4 Member Data Documentation

7.380.4.1 _api_log_level

```
ttk_tools.vector._canapeapi.Scripting._api_log_level  [protected]
```

7.380.4.2 _asap3

```
ttk_tools.vector._canapeapi.Scripting._asap3  [protected]
```

7.380.4.3 _device

```
ttk_tools.vector._canapeapi.Scripting._device  [protected]
```

7.380.4.4 _logApiCallEnd

```
ttk_tools.vector._canapeapi.Scripting._logApiCallEnd  [protected]
```

7.380.4.5 _logApiCallStart

```
ttk_tools.vector._canapeapi.Scripting._logApiCallStart  [protected]
```

7.380.4.6 _script_handle

```
ttk_tools.vector._canapeapi.Scripting._script_handle  [protected]
```

7.380.4.7 module_handle

```
ttk_tools.vector._canapeapi.Scripting.module_handle
```

Module handle to be separately provided as (optional) "current_device" to Asap3StartScript during start().

7.380.4.8 verbosity

`ttk_tools.vector._canapeapi.Scripting.verbosity`

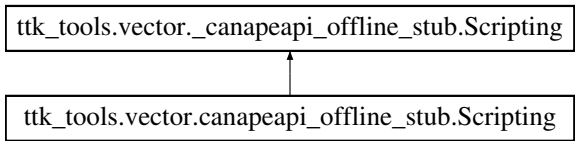
The documentation for this class was generated from the following file:

- ttk_tools/vector/_canapeapi.py (v7.3.3)

7.381 ttk_tools.vector._canapeapi_offline_stub.Scripting Class Reference

Access to extended scripting functions.

Inheritance diagram for ttk_tools.vector._canapeapi_offline_stub.Scripting:



Public Member Functions

- __init__ (self, device, verbosity=2)
- getStatus (self)

  Get execution status for the currently defined script/command.
- create (self, script_command, is_file_name=False)

  Create a new script handle for a CAPL script or script command via CANapeAPI.
- start (self, command_line=None)

  Start execution of a currently defined script.
- stop (self)

  Stop a currently running script.
- getReturnValue (self)

  Get return value of finished script (set via "return 4711;" in script).
- getResult (self)

  Get string result value of finished script.
- release (self)

  Release a currently defined script.
- waitUntilFinished (self, timeout_s=10)

  Wait until a currently running script is finished.
- run (self, script_command, is_file_name=False, command_line=None, timeout_s=10)

  Run a script command (or CAPL script) via CANapeAPI.

Public Attributes

- module_handle

  Module handle to be separately provided as (optional) "current_device" to Asap3StartScript during start().
- verbosity

Protected Member Functions

- _log (self, level, *msgs)

Protected Attributes

- _device
- _asap3
- _script_handle
- _script_status
- _retval
- _result

7.381.1 Detailed Description

Access to extended scripting functions.

Example:

```
# separate calls
scripting = Scripting(cal_device)
scripting.create('Write("Log output"); SetScriptResult("Foobar Baz"); return 1234;')
scripting.start()
scripting.waitUntilFinished()
result = scripting.getResult()
retval = scripting.getReturnValue()
scripting.release()

assert result == "Foobar Baz"
assert retval == 1234.0
```

Example:

```
# single call, similar to classic runScript
# (but method blocks until script is finished)
retval = scripting.run('Write("Log output, again"); return 1234;')
assert retval == 1234.0
```

7.381.2 Constructor & Destructor Documentation

7.381.2.1 __init__()

```
ttk_tools.vector._canapeapi_offline_stub.Scripting.__init__ (
            self,
            device,
            verbosity = 2 )
```

Parameters

device	CANape device/module (BaseDevice or child instances)
verbosity	verbosity of messages logged to console: 0: only errors 1: + status info

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.Scripting.

7.381.3 Member Function Documentation

7.381.3.1 _log()

```
ttk_tools.vector._canapeapi_offline_stub.Scripting._log (
              self,
              level,
             * msgs )  [protected]
```

7.381.3.2 create()

```
ttk_tools.vector._canapeapi_offline_stub.Scripting.create (
              self,
              script_command,
              is_file_name = False )
```

Create a new script handle for a CAPL script or script command via CANapeAPI.

Parameters

script_command	CAPL command or name of CAPL script file to execute. A CAPL command can also be the name of a custom "project function" that has been defined in the current workspace.
is_file_name	True if `script_command` contains the name of a script file to be run, False if it directly contains a CAPL command.

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.Scripting.

7.381.3.3 getResult()

```
ttk_tools.vector._canapeapi_offline_stub.Scripting.getResult (
              self )
```

Get string result value of finished script.

Note

To get a result here, the function "SetScriptResult" has to be called in the executed script.

Returns

a string or None if no script was available.

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.Scripting.

7.381.3.4 getReturnValue()

```
ttk_tools.vector._canapeapi_offline_stub.Scripting.getReturnValue (
              self )
```

Get return value of finished script (set via "return 4711;" in script).

Returns

a float (double) value or None if no script was available. Value may be NaN if no return value has been provided.

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.Scripting.

7.381.3.5 getStatus()

```
ttk_tools.vector._canapeapi_offline_stub.Scripting.getStatus (
              self )
```

Get execution status for the currently defined script/command.

Returns

a SCRIPT_STATUS value or None if API access failed.

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.Scripting.

7.381.3.6 release()

```
ttk_tools.vector._canapeapi_offline_stub.Scripting.release (
              self )
```

Release a currently defined script.

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.Scripting.

7.381.3.7 run()

```
ttk_tools.vector._canapeapi_offline_stub.Scripting.run (
              self,
              script_command,
              is_file_name = False,
              command_line = None,
              timeout_s = 10 )
```

Run a script command (or CAPL script) via CANapeAPI.

Essentially just calls

- create
- start
- waitUntilFinished
- getReturnValue and
- release
  in sequence.

Parameters

script_command	CAPL command or name of CAPL script file to execute. A CAPL command can also be the name of a custom "project function" that has been defined in the current workspace.
is_file_name	True if `script_command` contains the name of a script file to be run, False if it directly contains a CAPL command.
command_line	optional "command line" string to hand to start()
timeout_s	seconds to wait for script execution to finish

Example:

```
# multiple commands can be added in sequence
scripting.run('Speak("Hello World"); Write("Done."); return 123;')
```

Returns

the script's return value (see getReturnValue)

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.Scripting.

7.381.3.8 start()

```
ttk_tools.vector._canapeapi_offline_stub.Scripting.start (
              self,
              command_line = None )
```

Start execution of a currently defined script.

Parameters

command_line	optional "command line" string to hand to Asap3StartScript.

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.Scripting.

7.381.3.9 stop()

```
ttk_tools.vector._canapeapi_offline_stub.Scripting.stop (
              self )
```

Stop a currently running script.

Returns

True if script was stopped, False if stop failed and None if there was no current script handle to stop.

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.Scripting.

7.381.3.10 waitUntilFinished()

```
ttk_tools.vector._canapeapi_offline_stub.Scripting.waitUntilFinished (
              self,
              timeout_s = 10 )
```

Wait until a currently running script is finished.

Parameters

timeout_s	timeout in [s] before script will be stopped.

Reimplemented in ttk_tools.vector.canapeapi_offline_stub.Scripting.

7.381.4 Member Data Documentation

7.381.4.1 _asap3

`ttk_tools.vector._canapeapi_offline_stub.Scripting._asap3 [protected]`

7.381.4.2 _device

`ttk_tools.vector._canapeapi_offline_stub.Scripting._device [protected]`

7.381.4.3 _result

`ttk_tools.vector._canapeapi_offline_stub.Scripting._result [protected]`

7.381.4.4 _retval

`ttk_tools.vector._canapeapi_offline_stub.Scripting._retval [protected]`

7.381.4.5 _script_handle

`ttk_tools.vector._canapeapi_offline_stub.Scripting._script_handle [protected]`

7.381.4.6 _script_status

`ttk_tools.vector._canapeapi_offline_stub.Scripting._script_status [protected]`

7.381.4.7 module_handle

`ttk_tools.vector._canapeapi_offline_stub.Scripting.module_handle`

Module handle to be separately provided as (optional) "current_device" to Asap3StartScript during start().

7.381.4.8 verbosity

`ttk_tools.vector._canapeapi_offline_stub.Scripting.verbosity`

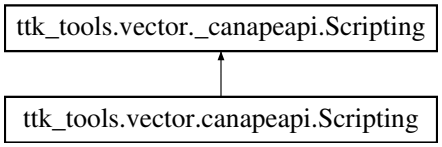
The documentation for this class was generated from the following file:

- ttk_tools/vector/_canapeapi_offline_stub.py (v4.7.1)

7.382 ttk_tools.vector.canapeapi.Scripting Class Reference

Access to extended scripting functions added in CANape 9.

Inheritance diagram for ttk_tools.vector.canapeapi.Scripting:



Public Member Functions

- __init__ (self, device, verbosity=2)
- getStatus (self)

  Get execution status for the currently defined script/command.
- create (self, script_command, is_file_name=False)

  Create a new script handle for a CAPL script or script command via CANapeAPI.
- start (self, command_line=None)

  Start execution of a currently defined script.
- stop (self)

  Stop a currently running script.
- getReturnValue (self)

  Get return value of finished script (set via "return 4711;" in script).
- getResult (self)

  Get string result value of finished script.
- release (self)

  Release a currently defined script.
- waitUntilFinished (self, timeout_s=10)

  Wait until a currently running script is finished.
- run (self, script_command, is_file_name=False, command_line=None, timeout_s=10)

  Run a script command (or CAPL script) via CANapeAPI.

7.382.1 Detailed Description

Access to extended scripting functions added in CANape 9.

Example:

```
# separate calls
scripting = Scripting(cal_device)
scripting.create('Write("Log output"); SetScriptResult("Foobar Baz"); return 1234;')
scripting.start()
scripting.waitUntilFinished()
result = scripting.getResult()
retval = scripting.getReturnValue()
scripting.release()

assert result == "Foobar Baz"
assert retval == 1234.0
```

Example:

```
# single call, similar to classic runScript
# (but method blocks until script is finished)
retval = scripting.run('Write("Log output, again"); return 1234;')
assert retval == 1234.0
```

7.382.2 Constructor & Destructor Documentation

7.382.2.1 __init__()

```
ttk_tools.vector.canapeapi.Scripting.__init__ (
              self,
              device,
              verbosity = 2 )
```

Parameters

device	CANape device/module (BaseDevice or child instances)
verbosity	verbosity of messages logged to console: 0: only errors 1: + status info

Reimplemented from ttk_tools.vector._canapeapi.Scripting.

7.382.3 Member Function Documentation

7.382.3.1 create()

```
ttk_tools.vector.canapeapi.Scripting.create (
              self,
              script_command,
              is_file_name = False )
```

Create a new script handle for a CAPL script or script command via CANapeAPI.

Parameters

script_command	CAPL command or name of CAPL script file to execute. A CAPL command can also be the name of a custom "project function" that has been defined in the current workspace.
is_file_name	True if `script_command` contains the name of a script file to be run, False if it directly contains a CAPL command.

Reimplemented from ttk_tools.vector._canapeapi.Scripting.

7.382.3.2 getResult()

```
ttk_tools.vector.canapeapi.Scripting.getResult (
              self )
```

Get string result value of finished script.

Note

To get a result here, the function "SetScriptResult" has to be called in the executed script.

Returns

a string or None if no script was available.

Reimplemented from ttk_tools.vector._canapeapi.Scripting.

7.382.3.3 getReturnValue()

```
ttk_tools.vector.canapeapi.Scripting.getReturnValue (
              self )
```

Get return value of finished script (set via "return 4711;" in script).

Returns

a float (double) value or None if no script was available. Value may be NaN if no return value has been provided.

Reimplemented from ttk_tools.vector._canapeapi.Scripting.

7.382.3.4 getStatus()

```
ttk_tools.vector.canapeapi.Scripting.getStatus (
              self )
```

Get execution status for the currently defined script/command.

Returns

a SCRIPT_STATUS value or None if API access failed.

Reimplemented from ttk_tools.vector._canapeapi.Scripting.

7.382.3.5 release()

```
ttk_tools.vector.canapeapi.Scripting.release (
              self )
```

Release a currently defined script.

Reimplemented from ttk_tools.vector._canapeapi.Scripting.

7.382.3.6 run()

```
ttk_tools.vector.canapeapi.Scripting.run (
              self,
              script_command,
              is_file_name = False,
              command_line = None,
              timeout_s = 10 )
```

Run a script command (or CAPL script) via CANapeAPI.

Essentially just calls

- create
- start
- waitUntilFinished
- getReturnValue and
- release
  in sequence.

Parameters

script_command	CAPL command or name of CAPL script file to execute. A CAPL command can also be the name of a custom "project function" that has been defined in the current workspace.
is_file_name	True if `script_command` contains the name of a script file to be run, False if it directly contains a CAPL command.
command_line	optional "command line" string to hand to start()
timeout_s	seconds to wait for script execution to finish

Example:

```
# multiple commands can be added in sequence
scripting.run('Speak("Hello World"); Write("Done."); return 123;')
```

Returns

the script's return value (see getReturnValue)

Reimplemented from ttk_tools.vector._canapeapi.Scripting.

7.382.3.7 start()

```
ttk_tools.vector.canapeapi.Scripting.start (
              self,
              command_line = None )
```

Start execution of a currently defined script.

Parameters

command_line	optional "command line" string to hand to Asap3StartScript.

Reimplemented from ttk_tools.vector._canapeapi.Scripting.

7.382.3.8 stop()

```
ttk_tools.vector.canapeapi.Scripting.stop (
              self )
```

Stop a currently running script.

Returns

True if script was stopped, False if stop failed and None if there was no current script handle to stop.

Reimplemented from ttk_tools.vector._canapeapi.Scripting.

7.382.3.9 waitUntilFinished()

```
ttk_tools.vector.canapeapi.Scripting.waitUntilFinished (
              self,
              timeout_s = 10 )
```

Wait until a currently running script is finished.

Parameters

timeout_s	timeout in [s] before script will be stopped.

Reimplemented from ttk_tools.vector._canapeapi.Scripting.

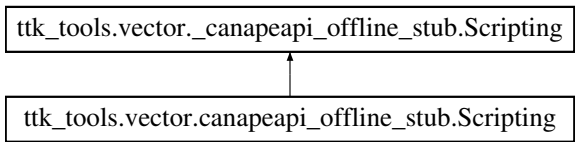
The documentation for this class was generated from the following file:

- ttk_tools/vector/canapeapi.py (v3.1.1)

7.383 ttk_tools.vector.canapeapi_offline_stub.Scripting Class Reference

Access to extended scripting functions added in CANape 9.

Inheritance diagram for ttk_tools.vector.canapeapi_offline_stub.Scripting:



Public Member Functions

- __init__ (self, device, verbosity=2)
- getStatus (self)

  Get execution status for the currently defined script/command.
- create (self, script_command, is_file_name=False)

  Create a new script handle for a CAPL script or script command via CANapeAPI.
- start (self, command_line=None)

  Start execution of a currently defined script.
- stop (self)

  Stop a currently running script.
- getReturnValue (self)

  Get return value of finished script (set via "return 4711;" in script).
- getResult (self)

  Get string result value of finished script.
- release (self)

  Release a currently defined script.
- waitUntilFinished (self, timeout_s=10)

  Wait until a currently running script is finished.
- run (self, script_command, is_file_name=False, command_line=None, timeout_s=10)

  Run a script command (or CAPL script) via CANapeAPI.

7.383.1 Detailed Description

Access to extended scripting functions added in CANape 9.

Example:

```
# separate calls
scripting = Scripting(cal_device)
scripting.create('Write("Log output"); SetScriptResult("Foobar Baz"); return 1234;')
scripting.start()
scripting.waitUntilFinished()
result = scripting.getResult()
retval = scripting.getReturnValue()
scripting.release()

assert result == "Foobar Baz"
assert retval == 1234.0
```

Example:

```
# single call, similar to classic runScript
# (but method blocks until script is finished)
retval = scripting.run('Write("Log output, again"); return 1234;')
assert retval == 1234.0
```

7.383.2 Constructor & Destructor Documentation

7.383.2.1 __init__()

```
ttk_tools.vector.canapeapi_offline_stub.Scripting.__init__ (
              self,
              device,
              verbosity = 2 )
```

Parameters

device	CANape device/module (BaseDevice or child instances)
verbosity	verbosity of messages logged to console: 0: only errors 1: + status info

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.Scripting.

7.383.3 Member Function Documentation

7.383.3.1 create()

```
ttk_tools.vector.canapeapi_offline_stub.Scripting.create (
              self,
              script_command,
              is_file_name = False )
```

Create a new script handle for a CAPL script or script command via CANapeAPI.

Parameters

script_command	CAPL command or name of CAPL script file to execute. A CAPL command can also be the name of a custom "project function" that has been defined in the current workspace.
is_file_name	True if `script_command` contains the name of a script file to be run, False if it directly contains a CAPL command.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.Scripting.

7.383.3.2 getResult()

```
ttk_tools.vector.canapeapi_offline_stub.Scripting.getResult (
              self )
```

Get string result value of finished script.

Note

To get a result here, the function "SetScriptResult" has to be called in the executed script.

Returns

a string or None if no script was available.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.Scripting.

7.383.3.3 getReturnValue()

```
ttk_tools.vector.canapeapi_offline_stub.Scripting.getReturnValue (
              self )
```

Get return value of finished script (set via "return 4711;" in script).

Returns

a float (double) value or None if no script was available. Value may be NaN if no return value has been provided.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.Scripting.

7.383.3.4 getStatus()

```
ttk_tools.vector.canapeapi_offline_stub.Scripting.getStatus (
              self )
```

Get execution status for the currently defined script/command.

Returns

a SCRIPT_STATUS value or None if API access failed.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.Scripting.

7.383.3.5 release()

```
ttk_tools.vector.canapeapi_offline_stub.Scripting.release (
              self )
```

Release a currently defined script.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.Scripting.

7.383.3.6 run()

```
ttk_tools.vector.canapeapi_offline_stub.Scripting.run (
              self,
              script_command,
              is_file_name = False,
              command_line = None,
              timeout_s = 10 )
```

Run a script command (or CAPL script) via CANapeAPI.

Essentially just calls

- create
- start
- waitUntilFinished
- getReturnValue and
- release
  in sequence.

Parameters

script_command	CAPL command or name of CAPL script file to execute. A CAPL command can also be the name of a custom "project function" that has been defined in the current workspace.
is_file_name	True if `script_command` contains the name of a script file to be run, False if it directly contains a CAPL command.
command_line	optional "command line" string to hand to start()
timeout_s	seconds to wait for script execution to finish

Example:

```
# multiple commands can be added in sequence
scripting.run('Speak("Hello World"); Write("Done."); return 123;')
```

Returns

the script's return value (see getReturnValue)

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.Scripting.

7.383.3.7 start()

```
ttk_tools.vector.canapeapi_offline_stub.Scripting.start (
              self,
              command_line = None )
```

Start execution of a currently defined script.

Parameters

command_line	optional "command line" string to hand to Asap3StartScript.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.Scripting.

7.383.3.8 stop()

```
ttk_tools.vector.canapeapi_offline_stub.Scripting.stop (
              self )
```

Stop a currently running script.

Returns

True if script was stopped, False if stop failed and None if there was no current script handle to stop.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.Scripting.

7.383.3.9 waitUntilFinished()

```
ttk_tools.vector.canapeapi_offline_stub.Scripting.waitUntilFinished (
              self,
              timeout_s = 10 )
```

Wait until a currently running script is finished.

Parameters

timeout_s	timeout in [s] before script will be stopped.

Reimplemented from ttk_tools.vector._canapeapi_offline_stub.Scripting.

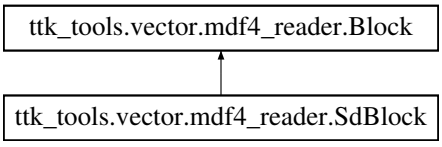
The documentation for this class was generated from the following file:

- ttk_tools/vector/canapeapi_offline_stub.py (v4.1.0)

7.384 ttk_tools.vector.mdf4_reader.SdBlock Class Reference

Signal Data Block.

Inheritance diagram for ttk_tools.vector.mdf4_reader.SdBlock:



Public Member Functions

- __init__ (self, file_handle, pos=None)

  Read SDBLOCK from file_handle, starting at the current or supplied position.

Protected Attributes

- _data

7.384.1 Detailed Description

Signal Data Block.

The data section of an SDBLOCK contains a sequence of signal values of variable length.

A variable length signal data (VLSD) value starts with a 32-bit uint value (LE/Intel Byte order) which specifies the length of the value data, i.e. the number of contiguous bytes following the 32-bit integer value. These bytes represent the actual data of the signal value. The next VLSD value must start directly after the previous one without any gap

An SDBLOCK can only contain signal values of its parent CNBLOCK, which must be a "variable length data channel" (cn_type = 1). Channel data type and conversion rule (or MIME content-type string) of the parent CNBLOCK apply to the VLSD value in the SDBLOCK.

The parent CNBLOCK also specifies a value in the (fixed length) records of its DTBLOCK. Independent of the channel data type and the conversion rule, the record value always must be interpreted as an unsigned integer value (LE/Intel Byte order) with the number of bits specified in the CNBLOCK (usually 32 or 64 bits). This value specifies the position of the VLSD value, i.e. the start offset of the 32-bit length value within the data section of the SDBLOCK (counted from the beginning of the data section of the SDBLOCK). In other words, the fixed length record for the channel group of the "variable length data channel" simply contains a reference to the VLSD signal value in the SDBLOCK of this channel. Thus each VLSD value implicitly is synchronized with the other values in the fixed length record.

7.384.2 Constructor & Destructor Documentation

7.384.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.SdBlock.__init__ (
              self,
              file_handle,
              pos = None )
```

Read SDBLOCK from file_handle, starting at the current or supplied position.

Also handles SDBLOCKs with compressed data (DZBLOCKs).

Reimplemented from ttk_tools.vector.mdf4_reader.Block.

7.384.3 Member Data Documentation

7.384.3.1 _data

`ttk_tools.vector.mdf4_reader.SdBlock._data [protected]`

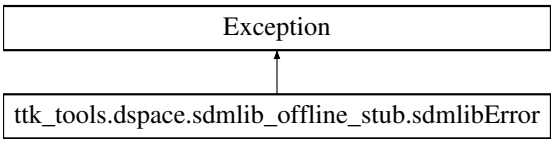
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.385 ttk_tools.dspace.sdmlib_offline_stub.sdmlibError Class Reference

sdmlib-exceptions

Inheritance diagram for ttk_tools.dspace.sdmlib_offline_stub.sdmlibError:



7.385.1 Detailed Description

sdmlib-exceptions

Each channel of the signal generator has to be declared as an object of the SIGNAL class. The signal curves can be constructed as definitions of signal segments with periodical and non-periodical signal forms using the methods of the SIGNAL class. The target variable of a signal must be a model parameter and not an output value, because the signal generator writes the calculated signal values at the beginning of each sampling period to the target variables. See DSTAPythonRef.pdf

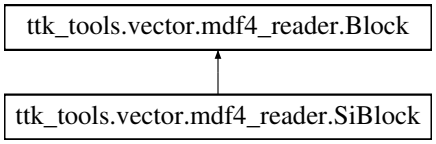
The documentation for this class was generated from the following file:

- ttk_tools/dspace/sdmlib_offline_stub.py (v1.4)

7.386 ttk_tools.vector.mdf4_reader.SiBlock Class Reference

Source Information Block.

Inheritance diagram for ttk_tools.vector.mdf4_reader.SiBlock:



Public Member Functions

- __init__ (self, file_handle, pos=None)

  Read a SIBLOCK from file_handle, starting at the current or supplied position.
- __str__ (self)
- __repr__ (self)

Public Attributes

- si_tx_name

  Pointer to TXBLOCK with name (identification) of source (can be NIL).
- si_tx_path

  Pointer to TXBLOCK with name (identification) of source (can be NIL).
- si_md_comment

  Pointer to TXBLOCK with name (identification) of source (can be NIL).
- si_type
- si_bus_type
- si_flags
- name

  source name, may be None
- path

  tool specific path, may be None
- comment

  comment
- xml
- alt_names

  Alternative names and description for source name (si_tx_name).
- alt_paths

  Alternative names and description for path string (si_tx_path).
- alt_bus_names

  Alternative names and description of the bus (e.g.
- protocol_name

  Name of protocol (e.g.

7.386.1 Detailed Description

Source Information Block.

A SIBLOCK describes the source of an acquisition mode or of a signal. The source information is also used to ensure a unique identification of a channel

Note that several channels may have the same source and thus might share the same SIBLOCK. Also the source of an acquisition mode and of a channel can be equal.

7.386.2 Constructor & Destructor Documentation

7.386.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.SiBlock.__init__ (
              self,
              file_handle,
              pos = None )
```

Read a SIBLOCK from file_handle, starting at the current or supplied position.

Reimplemented from ttk_tools.vector.mdf4_reader.Block.

7.386.3 Member Function Documentation

7.386.3.1 __repr__()

```
ttk_tools.vector.mdf4_reader.SiBlock.__repr__ (
              self )
```

7.386.3.2 __str__()

```
ttk_tools.vector.mdf4_reader.SiBlock.__str__ (
              self )
```

7.386.4 Member Data Documentation

7.386.4.1 alt_bus_names

```
ttk_tools.vector.mdf4_reader.SiBlock.alt_bus_names
```

Alternative names and description of the bus (e.g.

CAN1). The actual bus name is given by attribute "name" of <bus> tag. (optional)

7.386.4.2 alt_names

```
ttk_tools.vector.mdf4_reader.SiBlock.alt_names
```

Alternative names and description for source name (si_tx_name).

(optional)

7.386.4.3 alt_paths

```
ttk_tools.vector.mdf4_reader.SiBlock.alt_paths
```

Alternative names and description for path string (si_tx_path).

(optional)

7.386.4.4 comment

```
ttk_tools.vector.mdf4_reader.SiBlock.comment
```

comment

7.386.4.5 name

ttk_tools.vector.mdf4_reader.SiBlock.name

source name, may be None

7.386.4.6 path

ttk_tools.vector.mdf4_reader.SiBlock.path

tool specific path, may be None

7.386.4.7 protocol_name

ttk_tools.vector.mdf4_reader.SiBlock.protocol_name

Name of protocol (e.g.

XCP) (optional)

7.386.4.8 si_bus_type

ttk_tools.vector.mdf4_reader.SiBlock.si_bus_type

7.386.4.9 si_flags

ttk_tools.vector.mdf4_reader.SiBlock.si_flags

7.386.4.10 si_md_comment

ttk_tools.vector.mdf4_reader.SiBlock.si_md_comment

Pointer to TXBLOCK with name (identification) of source (can be NIL).

The source name must follow naming rules in section 5.4.2.

Pointer to TXBLOCK with (tool-specific) path of source (can be NIL). Each tool may generate a different path string. The only purpose is to ensure uniqueness.

As a recommendation, the path should be a human readable string containing additional information about the source. However, the path string should not be used to store this information in order to retrieve it later by parsing the string. Instead, additional source information should be stored in generic or custom XML fields in the comment MDBLOCK si_md_comment.

Pointer to source comment and additional information (TXBLOCK or MDBLOCK) (can be NIL)

7.386.4.11 si_tx_name

ttk_tools.vector.mdf4_reader.SiBlock.si_tx_name

Pointer to TXBLOCK with name (identification) of source (can be NIL).

The source name must follow naming rules in section 5.4.2.

7.386.4.12 si_tx_path

ttk_tools.vector.mdf4_reader.SiBlock.si_tx_path

Pointer to TXBLOCK with name (identification) of source (can be NIL).

The source name must follow naming rules in section 5.4.2.

Pointer to TXBLOCK with (tool-specific) path of source (can be NIL). Each tool may generate a different path string. The only purpose is to ensure uniqueness.

As a recommendation, the path should be a human readable string containing additional information about the source. However, the path string should not be used to store this information in order to retrieve it later by parsing the string. Instead, additional source information should be stored in generic or custom XML fields in the comment MDBLOCK si_md_comment.

7.386.4.13 si_type

ttk_tools.vector.mdf4_reader.SiBlock.si_type

7.386.4.14 xml

ttk_tools.vector.mdf4_reader.SiBlock.xml

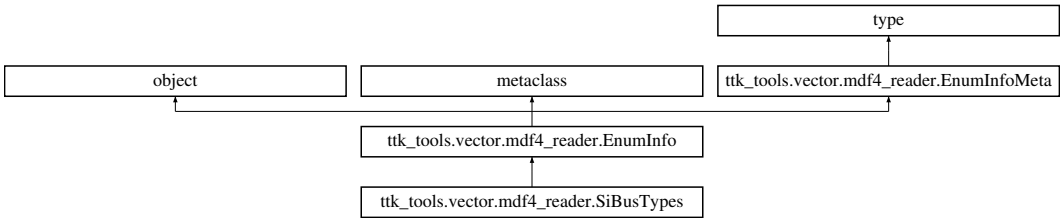
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.387 ttk_tools.vector.mdf4_reader.SiBusTypes Class Reference

Source bus type (si_bus_type), additional classification of used bus (should be 0 for si_type >= 3).

Inheritance diagram for ttk_tools.vector.mdf4_reader.SiBusTypes:



Static Public Attributes

- int NONE = 0
- int OTHER = 1
- int CAN = 2
- int LIN = 3
- int MOST = 4
- int FLEXRAY = 5
- int K_LINE = 6
- int ETHERNET = 7
- int USB = 8

Protected Member Functions

- _customNameLookup (cls, value)
- _customValueLookup (cls, name)

7.387.1 Detailed Description

Source bus type (si_bus_type), additional classification of used bus (should be 0 for si_type >= 3).

Vendor defined bus types can be added starting with value 128.

7.387.2 Member Function Documentation

7.387.2.1 _customNameLookup()

```
ttk_tools.vector.mdf4_reader.SiBusTypes._customNameLookup (
              cls,
              value )  [protected]
```

7.387.2.2 _customValueLookup()

```
ttk_tools.vector.mdf4_reader.SiBusTypes._customValueLookup (
              cls,
              name )  [protected]
```

7.387.3 Member Data Documentation

7.387.3.1 CAN

```
int ttk_tools.vector.mdf4_reader.SiBusTypes.CAN = 2  [static]
```

7.387.3.2 ETHERNET

```
int ttk_tools.vector.mdf4_reader.SiBusTypes.ETHERNET = 7  [static]
```

7.387.3.3 FLEXRAY

```
int ttk_tools.vector.mdf4_reader.SiBusTypes.FLEXRAY = 5  [static]
```

7.387.3.4 K_LINE

```
int ttk_tools.vector.mdf4_reader.SiBusTypes.K_LINE = 6  [static]
```

7.387.3.5 LIN

```
int ttk_tools.vector.mdf4_reader.SiBusTypes.LIN = 3  [static]
```

7.387.3.6 MOST

```
int ttk_tools.vector.mdf4_reader.SiBusTypes.MOST = 4  [static]
```

7.387.3.7 NONE

```
int ttk_tools.vector.mdf4_reader.SiBusTypes.NONE = 0  [static]
```

7.387.3.8 OTHER

```
int ttk_tools.vector.mdf4_reader.SiBusTypes.OTHER = 1  [static]
```

7.387.3.9 USB

```
int ttk_tools.vector.mdf4_reader.SiBusTypes.USB = 8  [static]
```

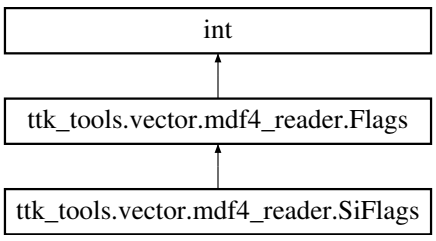
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.388 ttk_tools.vector.mdf4_reader.SiFlags Class Reference

Container for decoded uint8 si_flags.

Inheritance diagram for ttk_tools.vector.mdf4_reader.SiFlags:



Public Member Functions

- __init__ (self, flags)

Public Attributes

- simulated

  Bit 0: simulated source.

7.388.1 Detailed Description

Container for decoded uint8 si_flags.

Only intended for read-only access, changes in named flags will not modify original "flags" value.

7.388.2 Constructor & Destructor Documentation

7.388.2.1 __init__()

```
ttk_tools.vector.mdf4_reader.SiFlags.__init__ (
            self,
            flags )
```

Reimplemented from ttk_tools.vector.mdf4_reader.Flags.

7.388.3 Member Data Documentation

7.388.3.1 simulated

```
ttk_tools.vector.mdf4_reader.SiFlags.simulated
```

Bit 0: simulated source.

Source is only a simulation (can be hardware or software simulated) Cannot be set for si_type = 4 (TOOL).

The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.389 ttk_tools.dspace.sdmlib_offline_stub.SIGNAL Class Reference

Public Member Functions

- __init__ (self, name=None)

  Create a stimulus signal instance.
- CONST (self, value)

  Generate a constant value for a signal.
- EXP (self, offset, limit, tau)

  Generate an exponential signal according to the equation
  f(t) = offset + (limit - offset) ∗ (1 - e∗∗(-t/tau))
- IDLE (self)

  Stop signal generation temporarily and allow modification from the outside.
- NOISE_GAUSS (self, mean=0, variance=1, fc1=None, fc2=None, damp=0.707, seed=1234)

  Generate a signal with a Gaussian-like probability density function.
- PULSE (self, low, high, frequency)

  Generate a square wave pulse with a 50% duty cycle.
- RAMP (self, offset, slope, limit)

  Generate a ramp signal according to the equation:
  f(t) = offset + slope ∗ t
  slope > 0; f(t) > limit => f(t) = limit
  slope < 0; f(t) < limit => f(t) = limit
- SAW (self, offset, size, frequency)

  Generate a sawtooth waveform.
- SIN (self, offset, size, frequency)

  Generate a sine wave signal according to the equation:
  f(t) = offset + amplitude ∗ sin(2pi ∗ frequency ∗ t)
- SQR (self, offset, gain, limit)

  Generate a quadratic polynomial signal according to the equation:
  f(t) = offset + gain ∗ t∗∗2
  gain > 0; f(t) > limit => f(t) = limit
  gain < 0; f(t) < limit => f(t) = limit
- TRI (self, offset, size, frequency)

  Generate a triangular waveform.
- ADD (self, value)

  Add a constant value or another signal to the signal.
- MULT (self, value)

  Multiply the signal a constant value or another signal.
- TABLE_HOST (self, matfile, times='times', values='values', interpolation='INTERPOLATION')

  Replay measured data in real time.

7.389.1 Constructor & Destructor Documentation

7.389.1.1 __init__()

```
ttk_tools.dspace.sdmlib_offline_stub.SIGNAL.__init__ (
              self,
              name = None )
```

Create a stimulus signal instance.

Parameters

name	Name of the real-time variable to be used as Stimulus Engine channel.

7.389.2 Member Function Documentation

7.389.2.1 ADD()

```
ttk_tools.dspace.sdmlib_offline_stub.SIGNAL.ADD (
              self,
              value )
```

Add a constant value or another signal to the signal.

Parameters

value	a constant value or a SIGNAL object

7.389.2.2 CONST()

```
ttk_tools.dspace.sdmlib_offline_stub.SIGNAL.CONST (
              self,
              value )
```

Generate a constant value for a signal.

CONST is the abbreviation for a RAMP with a slope of zero. Float and Integer variables are supported.

Parameters

value	constant value of the signal

7.389.2.3 EXP()

```
ttk_tools.dspace.sdmlib_offline_stub.SIGNAL.EXP (
              self,
              offset,
              limit,
              tau )
```

Generate an exponential signal according to the equation
f(t) = offset + (limit - offset) ∗ (1 - e∗∗(-t/tau))

Parameters

offset	bottom value of the signal (float)
limit	value to which the function converges (float)
tau	time constant (float)

7.389.2.4 IDLE()

```
ttk_tools.dspace.sdmlib_offline_stub.SIGNAL.IDLE (
              self )
```

Stop signal generation temporarily and allow modification from the outside.

7.389.2.5 MULT()

```
ttk_tools.dspace.sdmlib_offline_stub.SIGNAL.MULT (
              self,
              value )
```

Multiply the signal a constant value or another signal.

Parameters

value	a constant value or a SIGNAL object

7.389.2.6 NOISE_GAUSS()

```
ttk_tools.dspace.sdmlib_offline_stub.SIGNAL.NOISE_GAUSS (
              self,
              mean = 0,
              variance = 1,
              fc1 = None,
              fc2 = None,
              damp = 0.707,
              seed = 1234 )
```

Generate a signal with a Gaussian-like probability density function.

Parameters

mean	mean value of the signal. Defaults to 0.0.
variance	variance value of the signal. Defaults to 1.0.
fc1	lower cut off frequency of the connected filter. Defaults to None
fc2	upper cut off frequency of the connected filter. Defaults to None
damp	damping of the connected filter. Defaults to 0.707.
seed	Seed of the random generator (int). Defaults to 1234.

7.389.2.7 PULSE()

```
ttk_tools.dspace.sdmlib_offline_stub.SIGNAL.PULSE (
              self,
              low,
              high,
              frequency )
```

Generate a square wave pulse with a 50% duty cycle.

For positive amplitudes, it starts with a positive block pulse. Float and Integer variables are supported.

Parameters

low	low value of the pulse train
high	high value of the pulse train
frequency	frequency of the pulse train [Hz]

7.389.2.8 RAMP()

```
ttk_tools.dspace.sdmlib_offline_stub.SIGNAL.RAMP (
              self,
              offset,
              slope,
              limit )
```

Generate a ramp signal according to the equation:
f(t) = offset + slope $*$ t
slope $>$ 0; f(t) $>$ limit => f(t) = limit
slope $<$ 0; f(t) $<$ limit => f(t) = limit

Parameters

offset	initial value of the signal
slope	signal change per second
limit	saturation value

7.389.2.9 SAW()

```
ttk_tools.dspace.sdmlib_offline_stub.SIGNAL.SAW (
              self,
              offset,
              size,
              frequency )
```

Generate a sawtooth waveform.

For positive amplitudes, the triangular waveform starts with a positive block ramp.

Parameters

offset	bottom value of the signal
size	peak to peak value
frequency	frequency of the signal [Hz]

7.389.2.10 SIN()

```
ttk_tools.dspace.sdmlib_offline_stub.SIGNAL.SIN (
              self,
              offset,
              size,
              frequency )
```

Generate a sine wave signal according to the equation:
f(t) = offset + amplitude $*$ sin(2pi $*$ frequency $*$ t)

Parameters

offset	initial value of the signal
size	amplitude of the sine wave
frequency	frequency of the sine wave [Hz]

7.389.2.11 SQR()

```
ttk_tools.dspace.sdmlib_offline_stub.SIGNAL.SQR (
              self,
              offset,
              gain,
              limit )
```

Generate a quadratic polynomial signal according to the equation:
f(t) = offset + gain $*$ t$**$2
gain $>$ 0; f(t) $>$ limit => f(t) = limit
gain $<$ 0; f(t) $<$ limit => f(t) = limit

Parameters

offset	initial value of the signal
gain	coefficient of the quadratic term
limit	saturation value

7.389.2.12 TABLE_HOST()

```
ttk_tools.dspace.sdmlib_offline_stub.SIGNAL.TABLE_HOST (
              self,
              matfile,
              times = 'times',
              values = 'values',
              interpolation = 'INTERPOLATION' )
```

Replay measured data in real time.

The measured data must be stored in a binary MATLAB file format. This MAT file must contain an array with time values and an array with data values. The time values can have varying distances. In the default case, the signal generation algorithm interpolates the signal values between consecutive interpolation points in real time.

Parameters

matfile	file path to a binary file containing time stamps and measured data values
times	string identifier of the time stamp array. Defaults to identifier "times".
values	string identifier of the measured data values. Defaults to identifier "values".
interpolation	interpretation of the data values, which can be interpolated ("INTERPOLATION") or not interpolated ("NO_INTERPOLATION"). Defaults to "INTERPOLATION".

7.389.2.13 TRI()

```
ttk_tools.dspace.sdmlib_offline_stub.SIGNAL.TRI (
                self,
                offset,
                size,
                frequency )
```

Generate a triangular waveform.

For positive amplitudes, the triangular waveform starts with a positive peak.

Parameters

offset	bottom value of the signal
size	peak to peak value
frequency	frequency of the signal [Hz]

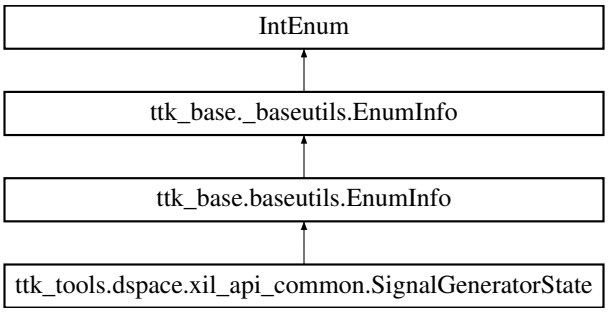
The documentation for this class was generated from the following file:

- ttk_tools/dspace/sdmlib_offline_stub.py (v1.4)

7.390 ttk_tools.dspace.xil_api_common.SignalGeneratorState Class Reference

Signal Generator States.

Inheritance diagram for ttk_tools.dspace.xil_api_common.SignalGeneratorState:



Static Public Attributes

- int eIN_CONFIGURATION = 0x00
  not yet loaded to target (or destroyed/unloaded)
- int eREADY = 0x01
  loaded to target
- int eFINISHED = 0x02
  stimulus finished running
- int ePAUSED = 0x03
  paused while running
- int eRUNNING = 0x04
  stimulus started and running
- int eSTOPPED = 0x05
  running stopped

7.390.1 Detailed Description

Signal Generator States.

See ASAM.XIL.Interfaces.Testbench.Common.SignalGenerator.Enum.SignalGeneratorState

7.390.2 Member Data Documentation

7.390.2.1 eFINISHED

```
int ttk_tools.dspace.xil_api_common.SignalGeneratorState.eFINISHED = 0x02  [static]
```

stimulus finished running

7.390.2.2 eIN_CONFIGURATION

```
int ttk_tools.dspace.xil_api_common.SignalGeneratorState.eIN_CONFIGURATION = 0x00  [static]
```

not yet loaded to target (or destroyed/unloaded)

7.390.2.3 ePAUSED

```
int ttk_tools.dspace.xil_api_common.SignalGeneratorState.ePAUSED = 0x03  [static]
```

paused while running

7.390.2.4 eREADY

```
int ttk_tools.dspace.xil_api_common.SignalGeneratorState.eREADY = 0x01  [static]
```

loaded to target

7.390.2.5 eRUNNING

```
int ttk_tools.dspace.xil_api_common.SignalGeneratorState.eRUNNING = 0x04  [static]
```

stimulus started and running

7.390.2.6 eSTOPPED

```
int ttk_tools.dspace.xil_api_common.SignalGeneratorState.eSTOPPED = 0x05  [static]
```

running stopped

The documentation for this class was generated from the following file:

- ttk_tools/dspace/xil_api_common.py (v1.5.0)

7.391 ttk_tools.siemens.simit_shm.SimitShm Class Reference

Wrapper for SIMIT SHM access.

Public Member Functions

- __init__ (self, shm_name="SIMIT_SHM")
- exit (self)

  Exit SHM and close connection.
- __del__ (self)
- __str__ (self)
- cycle_counter (self)

  current cycle count from shm header.
- write (self, pos, data)

  Write "raw" bytes to shm.
- read (self, pos, size, use_mutex=True)

  Read bytes from shm.
- getSimitSignal (self, name)

  Get a SimitSignal instance for a signal `name` from the SHM.
- getSignalNames (self, direction=None)

  Get a list of available signal names.

Static Public Member Functions

- getSimitSignalAddr (address, direction=None, data_type=DataType.byte)

  Get a "SIMIT-Style" address as used in exported signal lists.

Public Attributes

- mmap_file
- cycle_time_ms

  SHM cycle/update time in [ms] (from SHM header)
- mutex_name

  Name of SHM Mutex (from SHM header)
- mutex_handle

  Handle to SHM Mutex.
- mutex_timeout_ms

  Timeout for SHM Mutex access in [ms].
- name_encoding

Static Public Attributes

- int verbosity = 1

  verbosity level of log output.
- str name_encoding = "iso-8859-1"

Protected Member Functions

- _log (self, level, ∗args, ∗∗kwargs)

  log/print supplied message if level is below configured `verbosity`.

Protected Attributes

- _signals
- _shm_name
- _shm_version
- _shm_size
- _header_size

7.391.1 Detailed Description

Wrapper for SIMIT SHM access.

Handles/extracts default SHM headers and provides signal proxies. Requires SHM to be configured with "↩ Signalbeschreibung im Header" to extract signals definitions (and name of access mutex)

7.391.2 Constructor & Destructor Documentation

7.391.2.1 __init__()

```
ttk_tools.siemens.simit_shm.SimitShm.__init__ (
            self,
            shm_name = "SIMIT_SHM" )
```

7.391.2.2 __del__()

```
ttk_tools.siemens.simit_shm.SimitShm.__del__ (
              self )
```

7.391.3 Member Function Documentation

7.391.3.1 __str__()

```
ttk_tools.siemens.simit_shm.SimitShm.__str__ (
              self )
```

7.391.3.2 _log()

```
ttk_tools.siemens.simit_shm.SimitShm._log (
              self,
              level,
             * args,
             ** kwargs )  [protected]
```

log/print supplied message if level is below configured `verbosity`.

Parameters

level	numeric log level for output
∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

7.391.3.3 cycle_counter()

```
ttk_tools.siemens.simit_shm.SimitShm.cycle_counter (
              self )
```

current cycle count from shm header.

7.391.3.4 exit()

```
ttk_tools.siemens.simit_shm.SimitShm.exit (
              self )
```

Exit SHM and close connection.

7.391.3.5 getSignalNames()

```
ttk_tools.siemens.simit_shm.SimitShm.getSignalNames (
              self,
              direction = None )
```

Get a list of available signal names.

Parameters

direction	None: return both input and output signals SimitDirection.shm_to_simit: only return "output" signals SimitDirection.simit_to_shm: only return "input" signals

Returns

a list of name strings.

7.391.3.6 getSimitSignal()

```
ttk_tools.siemens.simit_shm.SimitShm.getSimitSignal (
              self,
              name )
```

Get a SimitSignal instance for a signal `name` from the SHM.

Raises a KeyError if SHN does not contain a matching signal.

7.391.3.7 getSimitSignalAddr()

```
ttk_tools.siemens.simit_shm.SimitShm.getSimitSignalAddr (
              address,
              direction = None,
              data_type = DataType.byte )  [static]
```

Get a "SIMIT-Style" address as used in exported signal lists.

7.391.3.8 read()

```
ttk_tools.siemens.simit_shm.SimitShm.read (
              self,
              pos,
              size,
              use_mutex = True )
```

Read bytes from shm.

Parameters

pos	start position / address to start reading from
size	number of bytes to read
use_mutex	True: use mutex for access control, False: do not try acquire mutex (for special cases)

Returns

a "raw" bytes value Raises a RuntimeError if mutex could not be acquired within `mutex_timeout_ms`

7.391.3.9 write()

```
ttk_tools.siemens.simit_shm.SimitShm.write (
              self,
              pos,
              data )
```

Write "raw" bytes to shm.

Write will be secured by the shm mutex.

Parameters

pos	start position / address to write to
data	bytes to write Raises a RuntimeError if mutex could not be acquired within `mutex_timeout_ms`

7.391.4 Member Data Documentation

7.391.4.1 _header_size

```
ttk_tools.siemens.simit_shm.SimitShm._header_size  [protected]
```

7.391.4.2 _shm_name

```
ttk_tools.siemens.simit_shm.SimitShm._shm_name  [protected]
```

7.391.4.3 _shm_size

```
ttk_tools.siemens.simit_shm.SimitShm._shm_size  [protected]
```

7.391.4.4 _shm_version

```
ttk_tools.siemens.simit_shm.SimitShm._shm_version  [protected]
```

7.391.4.5 _signals

```
ttk_tools.siemens.simit_shm.SimitShm._signals  [protected]
```

7.391.4.6 cycle_time_ms

```
ttk_tools.siemens.simit_shm.SimitShm.cycle_time_ms
```

SHM cycle/update time in [ms] (from SHM header)

7.391.4.7 mmap_file

ttk_tools.siemens.simit_shm.SimitShm.mmap_file

7.391.4.8 mutex_handle

ttk_tools.siemens.simit_shm.SimitShm.mutex_handle

Handle to SHM Mutex.

7.391.4.9 mutex_name

ttk_tools.siemens.simit_shm.SimitShm.mutex_name

Name of SHM Mutex (from SHM header)

7.391.4.10 mutex_timeout_ms

ttk_tools.siemens.simit_shm.SimitShm.mutex_timeout_ms

Timeout for SHM Mutex access in [ms].

An Exception will be raised during read/write operations if Mutex cannot be acquired within this time frame.

7.391.4.11 name_encoding [1/2]

str ttk_tools.siemens.simit_shm.SimitShm.name_encoding = "iso-8859-1" [static]

7.391.4.12 name_encoding [2/2]

ttk_tools.siemens.simit_shm.SimitShm.name_encoding

7.391.4.13 verbosity

int ttk_tools.siemens.simit_shm.SimitShm.verbosity = 1 [static]

verbosity level of log output.

0: mostly silent
1: info output
2: additional info

The documentation for this class was generated from the following file:

- ttk_tools/siemens/simit_shm.py (v1.4.0)

7.392 ttk_tools.siemens.simit_shm.SimitSignal Class Reference

A SIMIT signal configured in SHM Coupling.

Public Member Functions

- __init__ (self, shm, name, address, data_type, direction=Direction.simit_to_shm)
- read (self)

  Read current signal value.
- write (self, value)

  Write value to signal.
- __str__ (self)

Public Attributes

- name

  signal name
- address

  address, absolute to SHM start
- data_type

  data type of signal
- direction

  direction of signal, shm=>simit (r/w) or simit=>shm (r)

Protected Attributes

- _shm

  SimitShm instance.
- _struct

  struct definition for binary value packing/unpacking
- _bit_mask

  bit mask for bool types

7.392.1 Detailed Description

A SIMIT signal configured in SHM Coupling.

7.392.2 Constructor & Destructor Documentation

7.392.2.1 __init__()

```
ttk_tools.siemens.simit_shm.SimitSignal.__init__ (
              self,
              shm,
              name,
              address,
              data_type,
              direction = Direction.simit_to_shm )
```

Parameters

shm	SimitShm instance used for SHM access
name	signal name
address	address, absolute to SHM start (addresses in signal definition are relative to end-of-header)
data_type	Simit DataType of signal
direction	Simit Direction of signal (input or output)

7.392.3 Member Function Documentation

7.392.3.1 __str__()

```
ttk_tools.siemens.simit_shm.SimitSignal.__str__ (
              self )
```

7.392.3.2 read()

```
ttk_tools.siemens.simit_shm.SimitSignal.read (
              self )
```

Read current signal value.

Note

Data types The returned Python type depends on the signals's configured SIMIT-SHM data type:

- int for BYTE/INT/DWORD/DINT-types
- float for REAL
- bool for BOOL

Returns

the unpacked/decoded value

7.392.3.3 write()

```
ttk_tools.siemens.simit_shm.SimitSignal.write (
              self,
              value )
```

Write value to signal.

Parameters

value	new value to write

Returns

the previous value.

7.392.4 Member Data Documentation

7.392.4.1 _bit_mask

`ttk_tools.siemens.simit_shm.SimitSignal._bit_mask [protected]`

bit mask for bool types

7.392.4.2 _shm

`ttk_tools.siemens.simit_shm.SimitSignal._shm [protected]`

SimitShm instance.

7.392.4.3 _struct

`ttk_tools.siemens.simit_shm.SimitSignal._struct [protected]`

struct definition for binary value packing/unpacking

7.392.4.4 address

`ttk_tools.siemens.simit_shm.SimitSignal.address`

address, absolute to SHM start

7.392.4.5 data_type

`ttk_tools.siemens.simit_shm.SimitSignal.data_type`

data type of signal

7.392.4.6 direction

`ttk_tools.siemens.simit_shm.SimitSignal.direction`

direction of signal, shm=>simit (r/w) or simit=>shm (r)

7.392.4.7 name

```
ttk_tools.siemens.simit_shm.SimitSignal.name
```

signal name

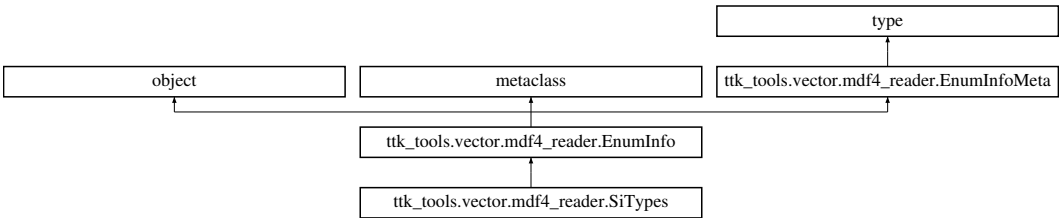
The documentation for this class was generated from the following file:

- ttk_tools/siemens/simit_shm.py (v1.4.0)

7.393 ttk_tools.vector.mdf4_reader.SiTypes Class Reference

Source type (si_type), additional source classification.

Inheritance diagram for ttk_tools.vector.mdf4_reader.SiTypes:



Static Public Attributes

- int OTHER = 0
- int ECU = 1
- int BUS = 2
- int IO = 3
- int TOOL = 4
- int USER = 5

7.393.1 Detailed Description

Source type (si_type), additional source classification.

7.393.2 Member Data Documentation

7.393.2.1 BUS

```
int ttk_tools.vector.mdf4_reader.SiTypes.BUS = 2  [static]
```

7.393.2.2 ECU

```
int ttk_tools.vector.mdf4_reader.SiTypes.ECU = 1  [static]
```

7.393.2.3 IO

```
int ttk_tools.vector.mdf4_reader.SiTypes.IO = 3  [static]
```

7.393.2.4 OTHER

```
int ttk_tools.vector.mdf4_reader.SiTypes.OTHER = 0  [static]
```

7.393.2.5 TOOL

```
int ttk_tools.vector.mdf4_reader.SiTypes.TOOL = 4  [static]
```

7.393.2.6 USER

```
int ttk_tools.vector.mdf4_reader.SiTypes.USER = 5  [static]
```

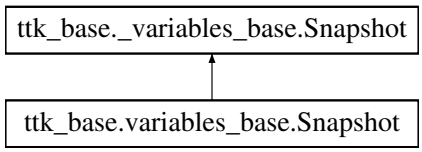
The documentation for this class was generated from the following file:

- ttk_tools/vector/mdf4_reader.py (v1.1.0)

7.394 ttk_base._variables_base.Snapshot Class Reference

Store and restore a "snapshot" of variables' values.

Inheritance diagram for ttk_base._variables_base.Snapshot:



Public Member Functions

- __init__ (self, *variables)
  Create a snapshot of the supplied variables.
- add (self, variable)
  Add a variable to the snapshot and store its value.
- remove (self, variable)
  Remove a variable from the snapshot discarding any stored value.
- recapture (self)
  Take a fresh snapshot of all contained variables.
- restore (self, verbosity=1)
  Restore all contained variables to the states/values they had when the snapshot was taken.

Protected Attributes

- _snapshots

7.394.1 Detailed Description

Store and restore a "snapshot" of variables' values.

Example:

```
snapshot = Snapshot(cal.foo, cal.bar, hil.baz)
cal.foo.set(1234)
hil.baz.set(0.05)
# [...]
snapshot.restore()
```

7.394.2 Constructor & Destructor Documentation

7.394.2.1 __init__()

```
ttk_base._variables_base.Snapshot.__init__ (
              self,
             * variables )
```

Create a snapshot of the supplied variables.

Note

Multiple snapshot instances can be used to quickly switch between snapshots.

Parameters

∗variables	either a list of variables or each variable as individual argument

Reimplemented in ttk_base.variables_base.Snapshot.

7.394.3 Member Function Documentation

7.394.3.1 add()

```
ttk_base._variables_base.Snapshot.add (
              self,
              variable )
```

Add a variable to the snapshot and store its value.

If a variable is already in the snapshot, its snapshot-value will be updated.

Parameters

variable	variable to add to snapshot; has to support a get()-method.

Reimplemented in ttk_base.variables_base.Snapshot.

7.394.3.2 recapture()

```
ttk_base._variables_base.Snapshot.recapture (
              self )
```

Take a fresh snapshot of all contained variables.

Reimplemented in ttk_base.variables_base.Snapshot.

7.394.3.3 remove()

```
ttk_base._variables_base.Snapshot.remove (
              self,
              variable )
```

Remove a variable from the snapshot discarding any stored value.

Parameters

variable	variable to remove

Returns

True if variable has been removed, otherwise False.

Reimplemented in ttk_base.variables_base.Snapshot.

7.394.3.4 restore()

```
ttk_base._variables_base.Snapshot.restore (
              self,
              verbosity = 1 )
```

Restore all contained variables to the states/values they had when the snapshot was taken.

Note

The order in which values will be restored is not specified.

Parameters

verbosity	debug messages: 0: only warnings, 1: more chatty

Reimplemented in ttk_base.variables_base.Snapshot.

7.394.4 Member Data Documentation

7.394.4.1 _snapshots

`ttk_base._variables_base.Snapshot._snapshots [protected]`

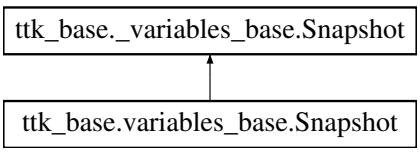
The documentation for this class was generated from the following file:

- ttk_base/_variables_base.py (v2.8.1)

7.395 ttk_base.variables_base.Snapshot Class Reference

Store and restore a "snapshot" of multiple variables' values.

Inheritance diagram for ttk_base.variables_base.Snapshot:



Public Member Functions

- __init__ (self, *variables)
  Create a snapshot of the supplied variables.
- add (self, variable)
  Add a variable to the snapshot and store its value.
- remove (self, variable)
  Remove a variable from the snapshot discarding any stored value.
- recapture (self)
  Take a fresh snapshot of all contained variables.
- restore (self, verbosity=1)
  Restore all contained variables to the states/values they had when the snapshot was.

7.395.1 Detailed Description

Store and restore a "snapshot" of multiple variables' values.

Example:

```
snapshot = Snapshot(cal.foo, cal.bar, hil.baz)
cal.foo.set(1234)
hil.baz.set(0.05)
# [...]
snapshot.restore()
```

7.395.2 Constructor & Destructor Documentation

7.395.2.1 __init__()

```
ttk_base.variables_base.Snapshot.__init__ (
              self,
             * variables )
```

Create a snapshot of the supplied variables.

Note that you can use multiple snapshot instances to quickly switch between snapshots.

Parameters

variables	either a list of variables or each variable as individual argument

Reimplemented from ttk_base._variables_base.Snapshot.

7.395.3 Member Function Documentation

7.395.3.1 add()

```
ttk_base.variables_base.Snapshot.add (
              self,
              variable )
```

Add a variable to the snapshot and store its value.

If a variable is already in the snapshot, its snapshot-value will be updated.

Parameters

variable	variable to add to snapshot; has to support a get()-method.

Reimplemented from ttk_base._variables_base.Snapshot.

7.395.3.2 recapture()

```
ttk_base.variables_base.Snapshot.recapture (
```

```
                self )
```

Take a fresh snapshot of all contained variables.

Reimplemented from ttk_base._variables_base.Snapshot.

7.395.3.3 remove()

```
ttk_base.variables_base.Snapshot.remove (
                self,
                variable )
```

Remove a variable from the snapshot discarding any stored value.

Parameters

variable	variable to remove

Returns

True if variable has been removed, otherwise False.

Reimplemented from ttk_base._variables_base.Snapshot.

7.395.3.4 restore()

```
ttk_base.variables_base.Snapshot.restore (
                self,
                verbosity = 1 )
```

Restore all contained variables to the states/values they had when the snapshot was.

Note

The order in which values will be restored is **not** specified.

Parameters

verbosity	debug messages: 0: only warnings 1: more chatty

Reimplemented from ttk_base._variables_base.Snapshot.

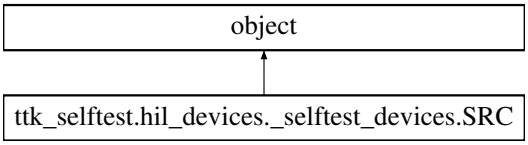
The documentation for this class was generated from the following file:

- ttk_base/variables_base.py (v2.2)

7.396 ttk_selftest.hil_devices._selftest_devices.SRC Class Reference

Base iSystemReferenceCard (iSRC) container.

Inheritance diagram for ttk_selftest.hil_devices._selftest_devices.SRC:



Public Member Functions

- __init__ (self, context, base_path, node_id, psu, alias="", descr="")

  iSystemReferenceCard initialization.
- setModeId (self, mode_id)

  Function to set mode of iSRC according to supplied mode id.
- setMode (self, mode)

  Function to set mode of iSRC.
- resetMode (self)

  Function to set iSRC in standby mode.
- hardReset (self)

  Function to perform a hard reset of the iSRC.
- reqData (self, cycle_time=0)

  Function to request data from iSRC and check the reception of the response message.
- setAlert (self, ∗∗kwargs)

  Function to set the alert limit of iSRC for voltage and current measurements.
- resetAlert (self)

  Function to reset the alert limit of iSRC for voltage and current measurements to their default values.
- setTimeLimit (self, current_limit, time_limit)

  Function to configure the timebased limit of the iSRC.
- setSensorConfig (self, ∗∗kwargs)

  Function to set the sensor configuration of iSRC for voltage and current measurements.
- resetSensorConfig (self)

  Function to reset the sensor configuration of iSRC for voltage and current measurements to their default values.
- setVoltage (self, voltage)

  Function to set the output voltage of the iSRC.
- setPWM (self, frequency, duty_cycle)

  Function to set the PWM output of the iSRC.
- getCurrentCheckVars (self)

  Function to get the current measurement variable and timestamp variable of the associated CAN message.
- getVoltageCheckVars (self)

  Function to get the voltage measurement variable and timestamp variable of the associated CAN message.
- getPwmCheckVars (self)

  Function to get the PWM measurement variables and timestamp variable of the associated CAN message.
- getCanCheckVars (self)

  Function to get the variables for CAN bus test, which are.
- getVoltageReferenceModes (self)

  Function to get the voltage reference modes.

- getResistance (self, measurement_mode="Highside", current_limit=1, voltage=None)

  Function to determine the wire/connection resistance.
- getStatus (self)

  Function to get an updated status of the iSRC.
- configureCAN (self, bitrate, termination)

  Function to configure the second CAN channel of the iSRC.

Public Attributes

- context
- base_path
- node_id
- psu
- response_timeout
- protection_factor
- alias
- descr
- ref_voltage_low
- ref_voltage_mid
- ref_voltage_high
- min_voltage
- max_voltage
- current_limit_low
- current_limit_mid
- current_limit_high
- averaging_mode
- bus_vol_conv_time
- shunt_vol_conv_time
- max_output_voltage
- max_pwm_frequency
- lut_status
- lut_bitrate
- lut_termination
- lut_averaging_mode
- lut_bus_shunt_vol_conv_time
- msg_bootup
- hw_major_version
- hw_minor_version
- sw_major_version
- sw_minor_version
- sw_patch_version
- configuration_status
- msg_meas_high
- meas_current_high
- meas_voltage_high
- msg_meas_mid
- meas_current_mid
- meas_voltage_mid
- msg_meas_low
- meas_current_low
- meas_voltage_low
- msg_meas_voltage
- meas_voltage

- msg_status
- status
- timestamp_last_status
- msg_pwm
- duty_cycle
- frequency
- pwm_in_status
- msg_can
- can_msg_id
- can_msg_length
- standby

  Define modes of iSRC (see DBC file) ##################################.
- cur_meas_qu
- ls_cur_meas_high
- ls_cur_meas_mid
- ls_cur_meas_low
- hs_cur_meas_high
- hs_cur_meas_mid
- hs_cur_meas_low
- vol_meas_gnd
- vol_meas_diff
- ref_high
- ref_mid
- ref_low
- pwm_generation
- pwm_measurement
- can_test
- voltage_generation
- hardreset
- hs_cur_meas_modes
- ls_cur_meas_modes
- vol_ref_modes
- msg_mode
- mode
- mode_dict
- msg_reqData
- sig_reqData
- msg_alert_limit
- alert_voltage
- alert_current_high
- alert_current_mid
- alert_current_low
- msg_time_limit
- time_current_limit
- time_limit
- msg_sensor_config
- cfg_averaging_mode
- cfg_bus_vol_conv_time
- cfg_shunt_vol_conv_time
- msg_set_voltage
- voltage_output
- msg_set_pwm
- duty_cycle_output
- frequency_output

- msg_configure_can
- can_bitrate
- can_termination
- verdict
- mode_active

7.396.1 Detailed Description

Base iSystemReferenceCard (iSRC) container.

7.396.2 Constructor & Destructor Documentation

7.396.2.1 __init__()

```
ttk_selftest.hil_devices._selftest_devices.SRC.__init__ (
                self,
                context,
                base_path,
                node_id,
                psu,
                alias = "",
                descr = "" )
```

iSystemReferenceCard initialization.

Parameters

context	parent context
base_path	base path of CAN signals to communicate with iSRC
node_id	node id of the iSRC
psu	power supply unit, which is connected to the iSRC
alias	(opt) device alias/name, defaults to "iSRC"
descr	(opt) device description defaults to ""

7.396.3 Member Function Documentation

7.396.3.1 configureCAN()

```
ttk_selftest.hil_devices._selftest_devices.SRC.configureCAN (
                self,
                bitrate,
                termination )
```

Function to configure the second CAN channel of the iSRC.

Parameters

bitrate	target bitrate in kBit/s
termination	flag to enable(True) or disable(False) the 120R termination of the second CAN channel of the iSRC

Returns

a single [<info string>, <verdict>] test result entry.

7.396.3.2 getCanCheckVars()

```
ttk_selftest.hil_devices._selftest_devices.SRC.getCanCheckVars (
              self )
```

Function to get the variables for CAN bus test, which are.

- message ID of received CAN message
- message length of received CAN message
- timestamp of iSRCs response message

```
None
```

Returns

a list [can_message_id, can_message_length, timestamp_var] consisting of the variables of the received message and the timestamp variable of the iSRC.

7.396.3.3 getCurrentCheckVars()

```
ttk_selftest.hil_devices._selftest_devices.SRC.getCurrentCheckVars (
              self )
```

Function to get the current measurement variable and timestamp variable of the associated CAN message.

```
None
```

Returns

a list [current_var, timestamp_var] consisting of the current measurement and timestamp variable of the iSRC if current measurement is possible for the recent mode. Otherwise both variables will be set None.

7.396.3.4 getPwmCheckVars()

```
ttk_selftest.hil_devices._selftest_devices.SRC.getPwmCheckVars (
              self )
```

Function to get the PWM measurement variables and timestamp variable of the associated CAN message.

```
None
```

Returns

a list [freq_var, dc_var, status_var, timestamp_var] consisting of the PWM measurement and timestamp variable of the iSRC.

7.396.3.5 getResistance()

```
ttk_selftest.hil_devices._selftest_devices.SRC.getResistance (
              self,
              measurement_mode = "Highside",
              current_limit = 1,
              voltage = None )
```

Function to determine the wire/connection resistance.

The resistance will be calculated as devision of voltage and current measured by the iSRC. For this, the PSU connected to the iSRC will be used as supply. Depending on the desired measurement mode (see parameter), the iSRC will be set to Highside measurement or Lowside measurement.

Precondition

1. In Highside measurement mode: The wire/connection to be measured has to be switched between EI1 and EI2. Moreover, the PSU ground has to be connected to EI2. If the wire/connection to be measured already has a connection to PSU ground, the connection to EI1 is sufficient.
2. In Lowside measurement mode: The wire/connection to be measurend has to be switched between EI2 and EI1. Moreover, the PSU output has to be connected to EI2. If the wire/connection to be measured already has a connection to PSU output, the connection to EI1 is sufficient. As the supply voltage cannot be measured in Lowside measurement mode, it has to be provided as parameter.

Note

The calculated resistance may be influenced by connections of the test connection to the ground of the PSU in the HIL system. On one hand, this may leed to wrong resistance values. On the other hand, this can be used to measure connection resistance to the PSU ground.

Parameters

measurement_mode	(opt) mode of the iSRC measurement that will be used to determine the resistance, can be "Highside" or "Lowside" defaults to "Highside"
current_limit	(opt) maximum allowed current while checking the resistance in A, defaults to 1A
voltage	(opt) voltage in V used for resistance calculation Parameter has to be given for Lowside measurement mode, defaults to measurement of the iSRC

Returns

a list [info, resistance] where "info" is a list of test result entrys and "resistance" is the wire/connection resistance in Ohm or None in case of missing response from iSRC

7.396.3.6 getStatus()

```
ttk_selftest.hil_devices._selftest_devices.SRC.getStatus (
              self )
```

Function to get an updated status of the iSRC.

```
None
```

Returns

a single [<status string>, <verdict>] test result entry or None if no updated status message has been received.

7.396.3.7 getVoltageCheckVars()

```
ttk_selftest.hil_devices._selftest_devices.SRC.getVoltageCheckVars (
              self )
```

Function to get the voltage measurement variable and timestamp variable of the associated CAN message.

None

Returns

a list [voltage_var, timestamp_var] consisting of the voltage measurement and timestamp variable of the iSRC if voltage measurement is possible for the recent mode. Otherwise None will be returned.

7.396.3.8 getVoltageReferenceModes()

```
ttk_selftest.hil_devices._selftest_devices.SRC.getVoltageReferenceModes (
              self )
```

Function to get the voltage reference modes.

None

Returns

a list of all voltage reference (in order of ascending reference voltages)

7.396.3.9 hardReset()

```
ttk_selftest.hil_devices._selftest_devices.SRC.hardReset (
              self )
```

Function to perform a hard reset of the iSRC.

None

Returns

a single [<info string>, <verdict>] test result entry.

7.396.3.10 reqData()

```
ttk_selftest.hil_devices._selftest_devices.SRC.reqData (
              self,
              cycle_time = 0 )
```

Function to request data from iSRC and check the reception of the response message.

Parameters

cycle_time	(opt) cycle time in ms of measurement data provided by iSRC on CAN bus default is one snapshot of measurement data

Returns

a single [<info string>, <verdict>] test result entry.

7.396.3.11 resetAlert()

```
ttk_selftest.hil_devices._selftest_devices.SRC.resetAlert (
              self )
```

Function to reset the alert limit of iSRC for voltage and current measurements to their default values.

```
None
```

Returns

a single [<info string>, <verdict>] test result entry.

7.396.3.12 resetMode()

```
ttk_selftest.hil_devices._selftest_devices.SRC.resetMode (
              self )
```

Function to set iSRC in standby mode.

None

Returns

an info string.

7.396.3.13 resetSensorConfig()

```
ttk_selftest.hil_devices._selftest_devices.SRC.resetSensorConfig (
              self )
```

Function to reset the sensor configuration of iSRC for voltage and current measurements to their default values.

```
None
```

Returns

a single [<info string>, <verdict>] test result entry.

7.396.3.14 setAlert()

```
ttk_selftest.hil_devices._selftest_devices.SRC.setAlert (
              self,
             ** kwargs )
```

Function to set the alert limit of iSRC for voltage and current measurements.

Parameters

kwargs	key words can be any of max_voltage, current_limit_low, current_limit_mid or current_limit_high. The corresponding value specifies the new limitation to be set for the associated measurement. Unspecified limitations will be left unchanged. Unit are:
max_voltage	V
current_limit_low	A
current_limit_mid	A current_limit_high: A
∗∗*kwargs*	remaining keyword arguments (catch-all)

Returns

a single [<info string>, <verdict>] test result entry.

7.396.3.15 setMode()

```
ttk_selftest.hil_devices._selftest_devices.SRC.setMode (
              self,
              mode )
```

Function to set mode of iSRC.

Parameters

mode	target mode of iSRC specified as instance of class _SrcMode

Returns

an info string.

7.396.3.16 setModeId()

```
ttk_selftest.hil_devices._selftest_devices.SRC.setModeId (
              self,
              mode_id )
```

Function to set mode of iSRC according to supplied mode id.

Parameters

mode_id	target mode id of iSRC

Returns

an info string.

7.396.3.17 setPWM()

```
ttk_selftest.hil_devices._selftest_devices.SRC.setPWM (
              self,
              frequency,
              duty_cycle )
```

Function to set the PWM output of the iSRC.

Parameters

frequency	target frequency in Hz
duty_cycle	target duty cycle in % High

Returns

a single [<info string>, <verdict>] test result entry.

7.396.3.18 setSensorConfig()

```
ttk_selftest.hil_devices._selftest_devices.SRC.setSensorConfig (
              self,
             ** kwargs )
```

Function to set the sensor configuration of iSRC for voltage and current measurements.

Parameters

kwargs	key words can be any of averaging_mode, bus_vol_conv_time or shunt_vol_conv_time. The supplied value must be valid and within the specified lookup tables.
∗∗*kwargs*	remaining keyword arguments (catch-all)

Returns

a single [<info string>, <verdict>] test result entry.

7.396.3.19 setTimeLimit()

```
ttk_selftest.hil_devices._selftest_devices.SRC.setTimeLimit (
              self,
              current_limit,
              time_limit )
```

Function to configure the timebased limit of the iSRC.

The iSRC will interrupt the connection if the specified current will be exceeded for the specified time.

Parameters

current_limit	current limit in mA
time_limit	time limit in ms

Returns

a single [<info string>, <verdict>] test result entry.

7.396.3.20 setVoltage()

```
ttk_selftest.hil_devices._selftest_devices.SRC.setVoltage (
              self,
              voltage )
```

Function to set the output voltage of the iSRC.

Parameters

voltage	target voltage in mV

Returns

a single [<info string>, <verdict>] test result entry.

7.396.4 Member Data Documentation

7.396.4.1 alert_current_high

```
ttk_selftest.hil_devices._selftest_devices.SRC.alert_current_high
```

7.396.4.2 alert_current_low

```
ttk_selftest.hil_devices._selftest_devices.SRC.alert_current_low
```

7.396.4.3 alert_current_mid

```
ttk_selftest.hil_devices._selftest_devices.SRC.alert_current_mid
```

7.396.4.4 alert_voltage

```
ttk_selftest.hil_devices._selftest_devices.SRC.alert_voltage
```

7.396.4.5 alias

ttk_selftest.hil_devices._selftest_devices.SRC.alias

7.396.4.6 averaging_mode

ttk_selftest.hil_devices._selftest_devices.SRC.averaging_mode

7.396.4.7 base_path

ttk_selftest.hil_devices._selftest_devices.SRC.base_path

7.396.4.8 bus_vol_conv_time

ttk_selftest.hil_devices._selftest_devices.SRC.bus_vol_conv_time

7.396.4.9 can_bitrate

ttk_selftest.hil_devices._selftest_devices.SRC.can_bitrate

7.396.4.10 can_msg_id

ttk_selftest.hil_devices._selftest_devices.SRC.can_msg_id

7.396.4.11 can_msg_length

ttk_selftest.hil_devices._selftest_devices.SRC.can_msg_length

7.396.4.12 can_termination

ttk_selftest.hil_devices._selftest_devices.SRC.can_termination

7.396.4.13 can_test

ttk_selftest.hil_devices._selftest_devices.SRC.can_test

7.396.4.14 cfg_averaging_mode

ttk_selftest.hil_devices._selftest_devices.SRC.cfg_averaging_mode

7.396.4.15 cfg_bus_vol_conv_time

ttk_selftest.hil_devices._selftest_devices.SRC.cfg_bus_vol_conv_time

7.396.4.16 cfg_shunt_vol_conv_time

ttk_selftest.hil_devices._selftest_devices.SRC.cfg_shunt_vol_conv_time

7.396.4.17 configuration_status

ttk_selftest.hil_devices._selftest_devices.SRC.configuration_status

7.396.4.18 context

ttk_selftest.hil_devices._selftest_devices.SRC.context

7.396.4.19 cur_meas_qu

ttk_selftest.hil_devices._selftest_devices.SRC.cur_meas_qu

7.396.4.20 current_limit_high

ttk_selftest.hil_devices._selftest_devices.SRC.current_limit_high

7.396.4.21 current_limit_low

ttk_selftest.hil_devices._selftest_devices.SRC.current_limit_low

7.396.4.22 current_limit_mid

ttk_selftest.hil_devices._selftest_devices.SRC.current_limit_mid

7.396.4.23 descr

ttk_selftest.hil_devices._selftest_devices.SRC.descr

7.396.4.24 duty_cycle

ttk_selftest.hil_devices._selftest_devices.SRC.duty_cycle

7.396.4.25 duty_cycle_output

ttk_selftest.hil_devices._selftest_devices.SRC.duty_cycle_output

7.396.4.26 frequency

ttk_selftest.hil_devices._selftest_devices.SRC.frequency

7.396.4.27 frequency_output

ttk_selftest.hil_devices._selftest_devices.SRC.frequency_output

7.396.4.28 hardreset

ttk_selftest.hil_devices._selftest_devices.SRC.hardreset

7.396.4.29 hs_cur_meas_high

ttk_selftest.hil_devices._selftest_devices.SRC.hs_cur_meas_high

7.396.4.30 hs_cur_meas_low

ttk_selftest.hil_devices._selftest_devices.SRC.hs_cur_meas_low

7.396.4.31 hs_cur_meas_mid

ttk_selftest.hil_devices._selftest_devices.SRC.hs_cur_meas_mid

7.396.4.32 hs_cur_meas_modes

ttk_selftest.hil_devices._selftest_devices.SRC.hs_cur_meas_modes

7.396.4.33 hw_major_version

ttk_selftest.hil_devices._selftest_devices.SRC.hw_major_version

7.396.4.34 hw_minor_version

ttk_selftest.hil_devices._selftest_devices.SRC.hw_minor_version

7.396.4.35 ls_cur_meas_high

ttk_selftest.hil_devices._selftest_devices.SRC.ls_cur_meas_high

7.396.4.36 ls_cur_meas_low

ttk_selftest.hil_devices._selftest_devices.SRC.ls_cur_meas_low

7.396.4.37 ls_cur_meas_mid

ttk_selftest.hil_devices._selftest_devices.SRC.ls_cur_meas_mid

7.396.4.38 ls_cur_meas_modes

ttk_selftest.hil_devices._selftest_devices.SRC.ls_cur_meas_modes

7.396.4.39 lut_averaging_mode

ttk_selftest.hil_devices._selftest_devices.SRC.lut_averaging_mode

7.396.4.40 lut_bitrate

ttk_selftest.hil_devices._selftest_devices.SRC.lut_bitrate

7.396.4.41 lut_bus_shunt_vol_conv_time

ttk_selftest.hil_devices._selftest_devices.SRC.lut_bus_shunt_vol_conv_time

7.396.4.42 lut_status

ttk_selftest.hil_devices._selftest_devices.SRC.lut_status

7.396.4.43 lut_termination

ttk_selftest.hil_devices._selftest_devices.SRC.lut_termination

7.396.4.44 max_output_voltage

ttk_selftest.hil_devices._selftest_devices.SRC.max_output_voltage

7.396.4.45 max_pwm_frequency

ttk_selftest.hil_devices._selftest_devices.SRC.max_pwm_frequency

7.396.4.46 max_voltage

ttk_selftest.hil_devices._selftest_devices.SRC.max_voltage

7.396.4.47 meas_current_high

ttk_selftest.hil_devices._selftest_devices.SRC.meas_current_high

7.396.4.48 meas_current_low

ttk_selftest.hil_devices._selftest_devices.SRC.meas_current_low

7.396.4.49 meas_current_mid

ttk_selftest.hil_devices._selftest_devices.SRC.meas_current_mid

7.396.4.50 meas_voltage

ttk_selftest.hil_devices._selftest_devices.SRC.meas_voltage

7.396.4.51 meas_voltage_high

ttk_selftest.hil_devices._selftest_devices.SRC.meas_voltage_high

7.396.4.52 meas_voltage_low

ttk_selftest.hil_devices._selftest_devices.SRC.meas_voltage_low

7.396.4.53 meas_voltage_mid

ttk_selftest.hil_devices._selftest_devices.SRC.meas_voltage_mid

7.396.4.54 min_voltage

ttk_selftest.hil_devices._selftest_devices.SRC.min_voltage

7.396.4.55 mode

ttk_selftest.hil_devices._selftest_devices.SRC.mode

7.396.4.56 mode_active

ttk_selftest.hil_devices._selftest_devices.SRC.mode_active

7.396.4.57 mode_dict

ttk_selftest.hil_devices._selftest_devices.SRC.mode_dict

7.396.4.58 msg_alert_limit

ttk_selftest.hil_devices._selftest_devices.SRC.msg_alert_limit

7.396.4.59 msg_bootup

ttk_selftest.hil_devices._selftest_devices.SRC.msg_bootup

7.396.4.60 msg_can

ttk_selftest.hil_devices._selftest_devices.SRC.msg_can

7.396.4.61 msg_configure_can

ttk_selftest.hil_devices._selftest_devices.SRC.msg_configure_can

7.396.4.62 msg_meas_high

ttk_selftest.hil_devices._selftest_devices.SRC.msg_meas_high

7.396.4.63 msg_meas_low

ttk_selftest.hil_devices._selftest_devices.SRC.msg_meas_low

7.396.4.64 msg_meas_mid

ttk_selftest.hil_devices._selftest_devices.SRC.msg_meas_mid

7.396.4.65 msg_meas_voltage

ttk_selftest.hil_devices._selftest_devices.SRC.msg_meas_voltage

7.396.4.66 msg_mode

ttk_selftest.hil_devices._selftest_devices.SRC.msg_mode

7.396.4.67 msg_pwm

ttk_selftest.hil_devices._selftest_devices.SRC.msg_pwm

7.396.4.68 msg_reqData

ttk_selftest.hil_devices._selftest_devices.SRC.msg_reqData

7.396.4.69 msg_sensor_config

ttk_selftest.hil_devices._selftest_devices.SRC.msg_sensor_config

7.396.4.70 msg_set_pwm

ttk_selftest.hil_devices._selftest_devices.SRC.msg_set_pwm

7.396.4.71 msg_set_voltage

ttk_selftest.hil_devices._selftest_devices.SRC.msg_set_voltage

7.396.4.72 msg_status

ttk_selftest.hil_devices._selftest_devices.SRC.msg_status

7.396.4.73 msg_time_limit

ttk_selftest.hil_devices._selftest_devices.SRC.msg_time_limit

7.396.4.74 node_id

ttk_selftest.hil_devices._selftest_devices.SRC.node_id

7.396.4.75 protection_factor

ttk_selftest.hil_devices._selftest_devices.SRC.protection_factor

7.396.4.76 psu

ttk_selftest.hil_devices._selftest_devices.SRC.psu

7.396.4.77 pwm_generation

ttk_selftest.hil_devices._selftest_devices.SRC.pwm_generation

7.396.4.78 pwm_in_status

ttk_selftest.hil_devices._selftest_devices.SRC.pwm_in_status

7.396.4.79 pwm_measurement

ttk_selftest.hil_devices._selftest_devices.SRC.pwm_measurement

7.396.4.80 ref_high

ttk_selftest.hil_devices._selftest_devices.SRC.ref_high

7.396.4.81 ref_low

ttk_selftest.hil_devices._selftest_devices.SRC.ref_low

7.396.4.82 ref_mid

ttk_selftest.hil_devices._selftest_devices.SRC.ref_mid

7.396.4.83 ref_voltage_high

ttk_selftest.hil_devices._selftest_devices.SRC.ref_voltage_high

7.396.4.84 ref_voltage_low

ttk_selftest.hil_devices._selftest_devices.SRC.ref_voltage_low

7.396.4.85 ref_voltage_mid

ttk_selftest.hil_devices._selftest_devices.SRC.ref_voltage_mid

7.396.4.86 response_timeout

ttk_selftest.hil_devices._selftest_devices.SRC.response_timeout

7.396.4.87 shunt_vol_conv_time

ttk_selftest.hil_devices._selftest_devices.SRC.shunt_vol_conv_time

7.396.4.88 sig_reqData

ttk_selftest.hil_devices._selftest_devices.SRC.sig_reqData

7.396.4.89 standby

ttk_selftest.hil_devices._selftest_devices.SRC.standby

Define modes of iSRC (see DBC file) ##############################.

7.396.4.90 status

ttk_selftest.hil_devices._selftest_devices.SRC.status

7.396.4.91 sw_major_version

ttk_selftest.hil_devices._selftest_devices.SRC.sw_major_version

7.396.4.92 sw_minor_version

ttk_selftest.hil_devices._selftest_devices.SRC.sw_minor_version

7.396.4.93 sw_patch_version

ttk_selftest.hil_devices._selftest_devices.SRC.sw_patch_version

7.396.4.94 time_current_limit

ttk_selftest.hil_devices._selftest_devices.SRC.time_current_limit

7.396.4.95 time_limit

ttk_selftest.hil_devices._selftest_devices.SRC.time_limit

7.396.4.96 timestamp_last_status

ttk_selftest.hil_devices._selftest_devices.SRC.timestamp_last_status

7.396.4.97 verdict

ttk_selftest.hil_devices._selftest_devices.SRC.verdict

7.396.4.98 vol_meas_diff

ttk_selftest.hil_devices._selftest_devices.SRC.vol_meas_diff

7.396.4.99 vol_meas_gnd

ttk_selftest.hil_devices._selftest_devices.SRC.vol_meas_gnd

7.396.4.100 vol_ref_modes

ttk_selftest.hil_devices._selftest_devices.SRC.vol_ref_modes

7.396.4.101 voltage_generation

ttk_selftest.hil_devices._selftest_devices.SRC.voltage_generation

7.396.4.102 voltage_output

ttk_selftest.hil_devices._selftest_devices.SRC.voltage_output

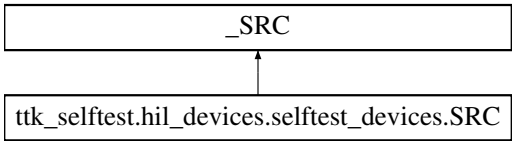
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/_selftest_devices.py (v2.1.1)

7.397 ttk_selftest.hil_devices.selftest_devices.SRC Class Reference

Base iSystemReferenceCard (iSRC) container.

Inheritance diagram for ttk_selftest.hil_devices.selftest_devices.SRC:



Public Member Functions

- __init__ (self, context, base_path, node_id, psu, alias="", descr="")

  iSystemReferenceCard initialization.
- setModeId (self, mode_id)

  Function to set mode of iSRC according to supplied mode id.
- setMode (self, mode)

  Function to set mode of iSRC.
- resetMode (self)

  Function to set iSRC in standby mode.
- hardReset (self)

  Function to perform a hard reset of the iSRC.
- reqData (self, cycle_time=0)

  Function to request data from iSRC and check the reception of the response message.
- setAlert (self, ∗∗kwargs)

  Function to set the alert limit of iSRC for voltage and current measurements.
- resetAlert (self)

  Function to reset the alert limit of iSRC for voltage and current measurements to their default values.
- setTimeLimit (self, current_limit, time_limit)

  Function to configure the timebased limit of the iSRC.
- setSensorConfig (self, ∗∗kwargs)

  Function to set the sensor configuration of iSRC for voltage and current measurements.
- resetSensorConfig (self)

  Function to reset the sensor configuration of iSRC for voltage and current measurements to their default values.
- setVoltage (self, voltage)

  Function to set the output voltage of the iSRC.
- setPWM (self, frequency, duty_cycle)

  Function to set the PWM output of the iSRC.
- getCurrentCheckVars (self)

  Function to get the current measurement variable and timestamp variable of the associated CAN message.
- getVoltageCheckVars (self)

  Function to get the voltage measurement variable and timestamp variable of the associated CAN message.
- getPwmCheckVars (self)

  Function to get the PWM measurement variables and timestamp variable of the associated CAN message.
- getCanCheckVars (self)

  Function to get the variables for CAN bus test, which are.
- getVoltageReferenceModes (self)

  Function to get the voltage reference modes.
- getResistance (self, measurement_mode="Highside", current_limit=1, voltage=None)

  Function to determine the wire/connection resistance.
- getStatus (self)

  Function to get an updated status of the iSRC.
- configureCAN (self, bitrate, termination)

  Function to configure the second CAN channel of the iSRC.

7.397.1 Detailed Description

Base iSystemReferenceCard (iSRC) container.

7.397.2 Constructor & Destructor Documentation

7.397.2.1 __init__()

```
ttk_selftest.hil_devices.selftest_devices.SRC.__init__ (
              self,
              context,
              base_path,
              node_id,
              psu,
              alias = "",
              descr = "" )
```

iSystemReferenceCard initialization.

Parameters

context	parent context
base_path	base path of CAN signals to communicate with iSRC
node_id	node id of the iSRC
psu	power supply unit, which is connected to the iSRC
alias	(opt) device alias/name, defaults to "iSRC"
descr	(opt) device description defaults to ""

7.397.3 Member Function Documentation

7.397.3.1 configureCAN()

```
ttk_selftest.hil_devices.selftest_devices.SRC.configureCAN (
              self,
              bitrate,
              termination )
```

Function to configure the second CAN channel of the iSRC.

Parameters

bitrate	target bitrate in kBit/s
termination	flag to enable(True) or disable(False) the 120R termination of the second CAN channel of the iSRC

Returns

a single [<info string>, <verdict>] test result entry.

7.397.3.2 getCanCheckVars()

```
ttk_selftest.hil_devices.selftest_devices.SRC.getCanCheckVars (
              self )
```

Function to get the variables for CAN bus test, which are.

- message ID of received CAN message
- message length of received CAN message
- timestamp of iSRCs response message

```
None
```

Returns

a list [can_message_id, can_message_length, timestamp_var] consisting of the variables of the received message and the timestamp variable of the iSRC.

7.397.3.3 getCurrentCheckVars()

```
ttk_selftest.hil_devices.selftest_devices.SRC.getCurrentCheckVars (
              self )
```

Function to get the current measurement variable and timestamp variable of the associated CAN message.

```
None
```

Returns

a list [current_var, timestamp_var] consisting of the current measurement and timestamp variable of the iSRC if current measurement is possible for the recent mode. Otherwise both variables will be set None.

7.397.3.4 getPwmCheckVars()

```
ttk_selftest.hil_devices.selftest_devices.SRC.getPwmCheckVars (
              self )
```

Function to get the PWM measurement variables and timestamp variable of the associated CAN message.

```
None
```

Returns

a list [freq_var, dc_var, status_var, timestamp_var] consisting of the PWM measurement and timestamp variable of the iSRC.

7.397.3.5 getResistance()

```
ttk_selftest.hil_devices.selftest_devices.SRC.getResistance (
              self,
              measurement_mode = "Highside",
              current_limit = 1,
              voltage = None )
```

Function to determine the wire/connection resistance.

The resistance will be calculated as devision of voltage and current measured by the iSRC. For this, the PSU connected to the iSRC will be used as supply. Depending on the desired measurement mode (see parameter), the iSRC will be set to Highside measurement or Lowside measurement.

Precondition

1. In Highside measurement mode: The wire/connection to be measured has to be switched between EI1 and EI2. Moreover, the PSU ground has to be connected to EI2. If the wire/connection to be measured already has a connection to PSU ground, the connection to EI1 is sufficient.
2. In Lowside measurement mode: The wire/connection to be measurend has to be switched between EI2 and EI1. Moreover, the PSU output has to be connected to EI2. If the wire/connection to be measured already has a connection to PSU output, the connection to EI1 is sufficient. As the supply voltage cannot be measured in Lowside measurement mode, it has to be provided as parameter.

Note

The calculated resistance may be influenced by connections of the test connection to the ground of the PSU in the HIL system. On one hand, this may leed to wrong resistance values. On the other hand, this can be used to measure connection resistance to the PSU ground.

Parameters

measurement_mode	(opt) mode of the iSRC measurement that will be used to determine the resistance, can be "Highside" or "Lowside" defaults to "Highside"
current_limit	(opt) maximum allowed current while checking the resistance in A, defaults to 1A
voltage	(opt) voltage in V used for resistance calculation Parameter has to be given for Lowside measurement mode, defaults to measurement of the iSRC

Returns

a list [info, resistance] where "info" is a list of test result entrys and "resistance" is the wire/connection resistance in Ohm or None in case of missing response from iSRC

7.397.3.6 getStatus()

```
ttk_selftest.hil_devices.selftest_devices.SRC.getStatus (
              self )
```

Function to get an updated status of the iSRC.

```
None
```

Returns

a single [<status string>, <verdict>] test result entry or None if no updated status message has been received.

7.397.3.7 getVoltageCheckVars()

```
ttk_selftest.hil_devices.selftest_devices.SRC.getVoltageCheckVars (
              self )
```

Function to get the voltage measurement variable and timestamp variable of the associated CAN message.

```
None
```

Returns

a list [voltage_var, timestamp_var] consisting of the voltage measurement and timestamp variable of the iSRC if voltage measurement is possible for the recent mode. Otherwise None will be returned.

7.397.3.8 getVoltageReferenceModes()

```
ttk_selftest.hil_devices.selftest_devices.SRC.getVoltageReferenceModes (
              self )
```

Function to get the voltage reference modes.

```
None
```

Returns

a list of all voltage reference (in order of ascending reference voltages)

7.397.3.9 hardReset()

```
ttk_selftest.hil_devices.selftest_devices.SRC.hardReset (
              self )
```

Function to perform a hard reset of the iSRC.

None

Returns

a single [<info string>, <verdict>] test result entry.

7.397.3.10 reqData()

```
ttk_selftest.hil_devices.selftest_devices.SRC.reqData (
              self,
              cycle_time = 0 )
```

Function to request data from iSRC and check the reception of the response message.

Parameters

cycle_time	(opt) cycle time in ms of measurement data provided by iSRC on CAN bus default is one snapshot of measurement data

Returns

a single [<info string>, <verdict>] test result entry.

7.397.3.11 resetAlert()

```
ttk_selftest.hil_devices.selftest_devices.SRC.resetAlert (
              self )
```

Function to reset the alert limit of iSRC for voltage and current measurements to their default values.

```
None
```

Returns

a single [<info string>, <verdict>] test result entry.

7.397.3.12 resetMode()

```
ttk_selftest.hil_devices.selftest_devices.SRC.resetMode (
              self )
```

Function to set iSRC in standby mode.

None

Returns

an info string.

7.397.3.13 resetSensorConfig()

```
ttk_selftest.hil_devices.selftest_devices.SRC.resetSensorConfig (
              self )
```

Function to reset the sensor configuration of iSRC for voltage and current measurements to their default values.

```
None
```

Returns

a single [<info string>, <verdict>] test result entry.

7.397.3.14 setAlert()

```
ttk_selftest.hil_devices.selftest_devices.SRC.setAlert (
              self,
             ** kwargs )
```

Function to set the alert limit of iSRC for voltage and current measurements.

Parameters

kwargs	key words can be any of max_voltage, current_limit_low, current_limit_mid or current_limit_high. The corresponding value specifies the new limitation to be set for the associated measurement. Unspecified limitations will be left unchanged. Unit are:
max_voltage	V
current_limit_low	A
current_limit_mid	A current_limit_high: A
∗∗*kwargs*	remaining keyword arguments (catch-all)

Returns

a single [<info string>, <verdict>] test result entry.

7.397.3.15 setMode()

```
ttk_selftest.hil_devices.selftest_devices.SRC.setMode (
              self,
              mode )
```

Function to set mode of iSRC.

Parameters

mode	target mode of iSRC specified as instance of class _SrcMode

Returns

an info string.

7.397.3.16 setModeId()

```
ttk_selftest.hil_devices.selftest_devices.SRC.setModeId (
              self,
              mode_id )
```

Function to set mode of iSRC according to supplied mode id.

Parameters

mode↩ _id	target mode id of iSRC

Returns

an info string.

7.397.3.17 setPWM()

```
ttk_selftest.hil_devices.selftest_devices.SRC.setPWM (
              self,
              frequency,
              duty_cycle )
```

Function to set the PWM output of the iSRC.

Parameters

frequency	target frequency in Hz
duty_cycle	target duty cycle in % High

Returns

a single [<info string>, <verdict>] test result entry.

7.397.3.18 setSensorConfig()

```
ttk_selftest.hil_devices.selftest_devices.SRC.setSensorConfig (
              self,
             ** kwargs )
```

Function to set the sensor configuration of iSRC for voltage and current measurements.

Parameters

kwargs	key words can be any of averaging_mode, bus_vol_conv_time or shunt_vol_conv_time. The supplied value must be valid and within the specified lookup tables.
∗∗*kwargs*	remaining keyword arguments (catch-all)

Returns

a single [<info string>, <verdict>] test result entry.

7.397.3.19 setTimeLimit()

```
ttk_selftest.hil_devices.selftest_devices.SRC.setTimeLimit (
              self,
              current_limit,
              time_limit )
```

Function to configure the timebased limit of the iSRC.

The iSRC will interrupt the connection if the specified current will be exceeded for the specified time.

Parameters

current_limit	current limit in mA
time_limit	time limit in ms

Returns

a single [<info string>, <verdict>] test result entry.

7.397.3.20 setVoltage()

```
ttk_selftest.hil_devices.selftest_devices.SRC.setVoltage (
              self,
              voltage )
```

Function to set the output voltage of the iSRC.

Parameters

voltage	target voltage in mV

Returns

a single [<info string>, <verdict>] test result entry.

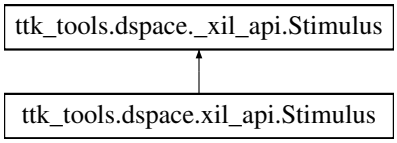
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/selftest_devices.py (v2.0.0)

7.398 ttk_tools.dspace._xil_api.Stimulus Class Reference

Stimulus / Signal Generator.

Inheritance diagram for ttk_tools.dspace._xil_api.Stimulus:



Public Member Functions

- __init__ (self, xil_testbench, appl_name=None)

  Stimulus / Signal Generator.
- __del__ (self)
- loadStimulusFile (self, file_path, assignments=None)

  Load signal generator settings from an STI or STZ file.
- addSignalDescription (self, stim_sig_descr)

  Add a `StimulusSignalDescription` instance to this Stimulus.
- loadToTarget (self)

  Load stimulus to target application.
- destroyOnTarget (self)

  Destroy (well, unload) a currently loaded stimulus on/from target application.
- start (self)

  Start a loaded stimulus.
- pause (self)

  Pause a running stimulus.
- stop (self)

  Stop a running (or paused) stimulus.
- getState (self, verbosity=1)

  Get a numeric representation of the current stimulus state.
- getStateText (self, state=None)

  Get a string description for a numeric stimulus state.
- isRunning (self)

  Return True if signal generation is currently running.
- isDone (self)

  Return True if signal generation is done (finished or stopped).
- getElapsedTime (self)

  Get elapsed time in seconds.
- waitUntilDone (self, timeout, step_size=0.500, auto_destroy=True, verbosity=2)

  Wait until a running stimulus is done (that is, has reached states `eSTOPPED` or `eFINISHED`, see isDone()).

Public Attributes

- parent

  reference to parent testbench
- testbench

  reference to testbench api
- signal_descriptions

  container for <label>: <stimulus_signal_description> mappings (separate container so we can more easily perform sanity checks, actual assignment will happen right before loadToTarget)
- signal_generator

  XIL SignalGenerator instance.

Protected Member Functions

- _isInState (self, *states)

7.398.1 Detailed Description

Stimulus / Signal Generator.

7.398.2 Constructor & Destructor Documentation

7.398.2.1 __init__()

```
ttk_tools.dspace._xil_api.Stimulus.__init__ (
              self,
              xil_testbench,
              appl_name = None )
```

Stimulus / Signal Generator.

Parameters

xil_testbench	parent XilTestbench instance
appl_name	target application name for multi core systems (e.g. "masterAppl"), None for single core setups.

Reimplemented in ttk_tools.dspace.xil_api.Stimulus.

7.398.2.2 __del__()

```
ttk_tools.dspace._xil_api.Stimulus.__del__ (
              self )
```

7.398.3 Member Function Documentation

7.398.3.1 _isInState()

```
ttk_tools.dspace._xil_api.Stimulus._isInState (
              self,
             * states )  [protected]
```

7.398.3.2 addSignalDescription()

```
ttk_tools.dspace._xil_api.Stimulus.addSignalDescription (
              self,
              stim_sig_descr )
```

Add a `StimulusSignalDescription` instance to this Stimulus.

Parameters

stim_sig_descr	StimulusSignalDescription instance to add

Reimplemented in ttk_tools.dspace.xil_api.Stimulus.

7.398.3.3 destroyOnTarget()

```
ttk_tools.dspace._xil_api.Stimulus.destroyOnTarget (
              self )
```

Destroy (well, unload) a currently loaded stimulus on/from target application.

Reimplemented in ttk_tools.dspace.xil_api.Stimulus.

7.398.3.4 getElapsedTime()

```
ttk_tools.dspace._xil_api.Stimulus.getElapsedTime (
              self )
```

Get elapsed time in seconds.

Reimplemented in ttk_tools.dspace.xil_api.Stimulus.

7.398.3.5 getState()

```
ttk_tools.dspace._xil_api.Stimulus.getState (
              self,
              verbosity = 1 )
```

Get a numeric representation of the current stimulus state.

Parameters

verbosity	0: silent, 1: print current state as info to log console

Returns

a numeric value. See `SignalGeneratorState` enum.

Reimplemented in ttk_tools.dspace.xil_api.Stimulus.

7.398.3.6 getStateText()

```
ttk_tools.dspace._xil_api.Stimulus.getStateText (
              self,
              state = None )
```

Get a string description for a numeric stimulus state.

Parameters

state	numeric state to get a description for None: use current state

Returns

a string.

Reimplemented in ttk_tools.dspace.xil_api.Stimulus.

7.398.3.7 isDone()

```
ttk_tools.dspace._xil_api.Stimulus.isDone (
              self )
```

Return True if signal generation is done (finished or stopped).

Reimplemented in ttk_tools.dspace.xil_api.Stimulus.

7.398.3.8 isRunning()

```
ttk_tools.dspace._xil_api.Stimulus.isRunning (
              self )
```

Return True if signal generation is currently running.

Reimplemented in ttk_tools.dspace.xil_api.Stimulus.

7.398.3.9 loadStimulusFile()

```
ttk_tools.dspace._xil_api.Stimulus.loadStimulusFile (
              self,
              file_path,
              assignments = None )
```

Load signal generator settings from an STI or STZ file.

Note

The STI/STZ file might not contain signal mappings, so signals may still have to be assigned manually.

If you add signal descriptions via addSignalDescription(), they will replace/overwrite the loaded stimulus.

Parameters

file_path	path to STI/STZ file
assignments	optional dictionary with <name>: <signal path> assignments.

Reimplemented in ttk_tools.dspace.xil_api.Stimulus.

7.398.3.10 loadToTarget()

```
ttk_tools.dspace._xil_api.Stimulus.loadToTarget (
              self )
```

Load stimulus to target application.

Reimplemented in ttk_tools.dspace.xil_api.Stimulus.

7.398.3.11 pause()

```
ttk_tools.dspace._xil_api.Stimulus.pause (
              self )
```

Pause a running stimulus.

Reimplemented in ttk_tools.dspace.xil_api.Stimulus.

7.398.3.12 start()

```
ttk_tools.dspace._xil_api.Stimulus.start (
              self )
```

Start a loaded stimulus.

Reimplemented in ttk_tools.dspace.xil_api.Stimulus.

7.398.3.13 stop()

```
ttk_tools.dspace._xil_api.Stimulus.stop (
              self )
```

Stop a running (or paused) stimulus.

Reimplemented in ttk_tools.dspace.xil_api.Stimulus.

7.398.3.14 waitUntilDone()

```
ttk_tools.dspace._xil_api.Stimulus.waitUntilDone (
              self,
              timeout,
              step_size = 0.500,
              auto_destroy = True,
              verbosity = 2 )
```

Wait until a running stimulus is done (that is, has reached states `eSTOPPED` or `eFINISHED`, see isDone()).

Parameters

timeout	timeout in [s]; stimulus will be stopped after timeout has been reached.
step_size	step_size in [s] to wait between state checks
auto_destroy	automatically unload stimulus from target after it is done. See destroyOnTarget()
verbosity	verbosity of status messages: 0: (mostly) silent 1: waiting ... done messages plus verbose exception info 2: + stimulus running / unloading / done info

Reimplemented in ttk_tools.dspace.xil_api.Stimulus.

7.398.4 Member Data Documentation

7.398.4.1 parent

`ttk_tools.dspace._xil_api.Stimulus.parent`

reference to parent testbench

7.398.4.2 signal_descriptions

`ttk_tools.dspace._xil_api.Stimulus.signal_descriptions`

container for <label>: <stimulus_signal_description> mappings (separate container so we can more easily perform sanity checks, actual assignment will happen right before loadToTarget)

7.398.4.3 signal_generator

`ttk_tools.dspace._xil_api.Stimulus.signal_generator`

XIL SignalGenerator instance.

7.398.4.4 testbench

`ttk_tools.dspace._xil_api.Stimulus.testbench`

reference to testbench api

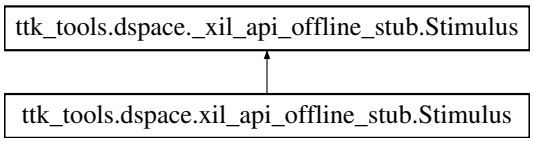
The documentation for this class was generated from the following file:

- ttk_tools/dspace/_xil_api.py (v2.3.1)

7.399 ttk_tools.dspace._xil_api_offline_stub.Stimulus Class Reference

Offline stub for Stimulus / Signal Generator.

Inheritance diagram for ttk_tools.dspace._xil_api_offline_stub.Stimulus:



Public Member Functions

- __init__ (self, xil_testbench, appl_name=None)
  Stimulus / Signal Generator stub.
- loadStimulusFile (self, file_path, assignments=None)
  Load signal generator settings from an STI or STZ file.
- addSignalDescription (self, stim_sig_descr)
  Add a `StimulusSignalDescription` instance to this Stimulus.
- loadToTarget (self)
  Load stimulus to target application.
- destroyOnTarget (self)
  Destroy (well, unload) a currently loaded stimulus on/from target application.
- start (self)
  Start a loaded stimulus.
- pause (self)
  Pause a running stimulus.
- stop (self)
  Stop a running (or paused) stimulus.
- getState (self, verbosity=1)
  Get a numeric representation of the current stimulus state.
- getStateText (self, state=None)
  Get a string description for a numeric stimulus state.
- isRunning (self)
  Return True if signal generation is currently running.
- isDone (self)
  Return True if signal generation is done (finished or stopped).
- getElapsedTime (self)
  Get elapsed time in seconds.
- waitUntilDone (self, timeout, step_size=0.500, auto_destroy=True, verbosity=2)
  Wait until a running stimulus is done (that is, has reached states `eSTOPPED` or `eFINISHED`, see isDone()).

Public Attributes

- parent
  reference to parent testbench
- testbench
  reference to testbench api
- signal_generator
  [XIL-API] SignalGenerator instance
- signal_descriptions
  container for `<label>: <stimulus_signal_description>` mappings.

Protected Member Functions

- _update (self)

  Stub function: update virtual signal generator state based on "expected" run time.

Protected Attributes

- _state
- _run_time
- _start_time
- _stop_time
- _max_duration

7.399.1 Detailed Description

Offline stub for Stimulus / Signal Generator.

7.399.2 Constructor & Destructor Documentation

7.399.2.1 __init__()

```
ttk_tools.dspace._xil_api_offline_stub.Stimulus.__init__ (
              self,
              xil_testbench,
              appl_name = None )
```

Stimulus / Signal Generator stub.

Parameters

xil_testbench	parent XilTestbench instance
appl_name	target application name for multi core systems (e.g. "masterAppl"), None for single core setups.

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.Stimulus.

7.399.3 Member Function Documentation

7.399.3.1 _update()

```
ttk_tools.dspace._xil_api_offline_stub.Stimulus._update (
              self )  [protected]
```

Stub function: update virtual signal generator state based on "expected" run time.

7.399.3.2 addSignalDescription()

```
ttk_tools.dspace._xil_api_offline_stub.Stimulus.addSignalDescription (
              self,
              stim_sig_descr )
```

Add a `StimulusSignalDescription` instance to this Stimulus.

Parameters

stim_sig_descr	StimulusSignalDescription instance to add

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.Stimulus.

7.399.3.3 destroyOnTarget()

```
ttk_tools.dspace._xil_api_offline_stub.Stimulus.destroyOnTarget (
              self )
```

Destroy (well, unload) a currently loaded stimulus on/from target application.

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.Stimulus.

7.399.3.4 getElapsedTime()

```
ttk_tools.dspace._xil_api_offline_stub.Stimulus.getElapsedTime (
              self )
```

Get elapsed time in seconds.

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.Stimulus.

7.399.3.5 getState()

```
ttk_tools.dspace._xil_api_offline_stub.Stimulus.getState (
              self,
              verbosity = 1 )
```

Get a numeric representation of the current stimulus state.

Parameters

verbosity	0: silent, 1: print current state as info to log console

Returns

a numeric value. See `SignalGeneratorState` enum.

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.Stimulus.

7.399.3.6 getStateText()

```
ttk_tools.dspace._xil_api_offline_stub.Stimulus.getStateText (
              self,
              state = None )
```

Get a string description for a numeric stimulus state.

Parameters

state	numeric state to get a description for None: use current state

Returns

a string.

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.Stimulus.

7.399.3.7 isDone()

```
ttk_tools.dspace._xil_api_offline_stub.Stimulus.isDone (
              self )
```

Return True if signal generation is done (finished or stopped).

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.Stimulus.

7.399.3.8 isRunning()

```
ttk_tools.dspace._xil_api_offline_stub.Stimulus.isRunning (
              self )
```

Return True if signal generation is currently running.

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.Stimulus.

7.399.3.9 loadStimulusFile()

```
ttk_tools.dspace._xil_api_offline_stub.Stimulus.loadStimulusFile (
              self,
              file_path,
              assignments = None )
```

Load signal generator settings from an STI or STZ file.

Note

The STI/STZ file might not contain signal mappings, so signals may still have to be assigned manually.

If you add signal descriptions via addSignalDescription(), they will replace/overwrite the loaded stimulus.

Parameters

file_path	path to STI/STZ file
assignments	optional dictionary with <name>: <signal path> assignments.

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.Stimulus.

7.399.3.10 loadToTarget()

```
ttk_tools.dspace._xil_api_offline_stub.Stimulus.loadToTarget (
              self )
```

Load stimulus to target application.

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.Stimulus.

7.399.3.11 pause()

```
ttk_tools.dspace._xil_api_offline_stub.Stimulus.pause (
              self )
```

Pause a running stimulus.

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.Stimulus.

7.399.3.12 start()

```
ttk_tools.dspace._xil_api_offline_stub.Stimulus.start (
              self )
```

Start a loaded stimulus.

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.Stimulus.

7.399.3.13 stop()

```
ttk_tools.dspace._xil_api_offline_stub.Stimulus.stop (
              self )
```

Stop a running (or paused) stimulus.

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.Stimulus.

7.399.3.14 waitUntilDone()

```
ttk_tools.dspace._xil_api_offline_stub.Stimulus.waitUntilDone (
              self,
              timeout,
              step_size = 0.500,
              auto_destroy = True,
              verbosity = 2 )
```

Wait until a running stimulus is done (that is, has reached states `eSTOPPED` or `eFINISHED`, see isDone()).

Parameters

timeout	timeout in [s]; stimulus will be stopped after timeout has been reached.
step_size	step_size in [s] to wait between state checks
auto_destroy	automatically unload stimulus from target after it is done. See destroyOnTarget()
verbosity	verbosity of status messages: 0: (mostly) silent 1: waiting ... done messages plus verbose exception info 2: + stimulus running / unloading / done info

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.Stimulus.

7.399.4 Member Data Documentation

7.399.4.1 _max_duration

```
ttk_tools.dspace._xil_api_offline_stub.Stimulus._max_duration  [protected]
```

7.399.4.2 _run_time

```
ttk_tools.dspace._xil_api_offline_stub.Stimulus._run_time  [protected]
```

7.399.4.3 _start_time

```
ttk_tools.dspace._xil_api_offline_stub.Stimulus._start_time  [protected]
```

7.399.4.4 _state

```
ttk_tools.dspace._xil_api_offline_stub.Stimulus._state  [protected]
```

7.399.4.5 _stop_time

```
ttk_tools.dspace._xil_api_offline_stub.Stimulus._stop_time  [protected]
```

7.399.4.6 parent

```
ttk_tools.dspace._xil_api_offline_stub.Stimulus.parent
```

reference to parent testbench

7.399.4.7 signal_descriptions

ttk_tools.dspace._xil_api_offline_stub.Stimulus.signal_descriptions

container for <label>: <stimulus_signal_description> mappings.

A separate container so we can more easily perform sanity checks, actual assignment will happen right before self.loadToTarget()

7.399.4.8 signal_generator

ttk_tools.dspace._xil_api_offline_stub.Stimulus.signal_generator

[XIL-API] SignalGenerator instance

7.399.4.9 testbench

ttk_tools.dspace._xil_api_offline_stub.Stimulus.testbench

reference to testbench api

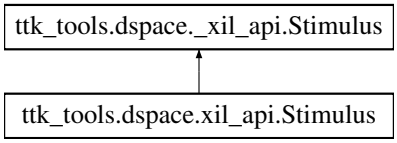
The documentation for this class was generated from the following file:

- ttk_tools/dspace/_xil_api_offline_stub.py (v2.1.2)

7.400 ttk_tools.dspace.xil_api.Stimulus Class Reference

Stimulus / Signal Generator.

Inheritance diagram for ttk_tools.dspace.xil_api.Stimulus:



Public Member Functions

- __init__ (self, xil_testbench, appl_name=None)

  Stimulus / Signal Generator.
- loadStimulusFile (self, file_path, assignments=None)

  Load signal generator settings from an STI or STZ file.
- addSignalDescription (self, stim_sig_descr)

  Add a `StimulusSignalDescription` instance to this Stimulus.
- loadToTarget (self)

  Load stimulus to target application.
- destroyOnTarget (self)

  Destroy (well, unload) a currently loaded stimulus on/from target application.
- start (self)

  Start a loaded stimulus.
- pause (self)

  Pause a running stimulus.
- stop (self)

  Stop a running (or paused) stimulus.
- getState (self, verbosity=1)

  Get a numeric representation of the current stimulus state.
- getStateText (self, state=None)

  Get a string description for a numeric stimulus state.
- isRunning (self)

  Return True if signal generation is currently running.
- isDone (self)

  Return True if signal generation is done (finished or stopped).
- getElapsedTime (self)

  Get elapsed time in seconds.
- waitUntilDone (self, timeout, step_size=0.500, auto_destroy=True, verbosity=2)

  Wait until a running stimulus is done (that is, has reached states `eSTOPPED` or `eFINISHED`, see isDone()).

Public Attributes

- parent

  reference to parent testbench
- testbench

  reference to testbench api
- signal_generator

  [XIL-API] SignalGenerator instance

7.400.1 Detailed Description

Stimulus / Signal Generator.

7.400.2 Constructor & Destructor Documentation

7.400.2.1 __init__()

```
ttk_tools.dspace.xil_api.Stimulus.__init__ (
            self,
            xil_testbench,
            appl_name = None )
```

Stimulus / Signal Generator.

Parameters

xil_testbench	parent XilTestbench instance
appl_name	target application name for multi core systems (e.g. "masterAppl"), None for single core setups.

Reimplemented from ttk_tools.dspace._xil_api.Stimulus.

7.400.3 Member Function Documentation

7.400.3.1 addSignalDescription()

```
ttk_tools.dspace.xil_api.Stimulus.addSignalDescription (
              self,
              stim_sig_descr )
```

Add a `StimulusSignalDescription` instance to this Stimulus.

Parameters

stim_sig_descr	StimulusSignalDescription instance to add

Reimplemented from ttk_tools.dspace._xil_api.Stimulus.

7.400.3.2 destroyOnTarget()

```
ttk_tools.dspace.xil_api.Stimulus.destroyOnTarget (
              self )
```

Destroy (well, unload) a currently loaded stimulus on/from target application.

Reimplemented from ttk_tools.dspace._xil_api.Stimulus.

7.400.3.3 getElapsedTime()

```
ttk_tools.dspace.xil_api.Stimulus.getElapsedTime (
              self )
```

Get elapsed time in seconds.

Reimplemented from ttk_tools.dspace._xil_api.Stimulus.

7.400.3.4 getState()

```
ttk_tools.dspace.xil_api.Stimulus.getState (
              self,
              verbosity = 1 )
```

Get a numeric representation of the current stimulus state.

Parameters

verbosity	0: silent, 1: print current state as info to log console

Returns

a numeric value. See `SignalGeneratorState` enum.

Reimplemented from ttk_tools.dspace._xil_api.Stimulus.

7.400.3.5 getStateText()

```
ttk_tools.dspace.xil_api.Stimulus.getStateText (
              self,
              state = None )
```

Get a string description for a numeric stimulus state.

Parameters

state	numeric state to get a description for None: use current state

Returns

a string.

Reimplemented from ttk_tools.dspace._xil_api.Stimulus.

7.400.3.6 isDone()

```
ttk_tools.dspace.xil_api.Stimulus.isDone (
              self )
```

Return True if signal generation is done (finished or stopped).

Reimplemented from ttk_tools.dspace._xil_api.Stimulus.

7.400.3.7 isRunning()

```
ttk_tools.dspace.xil_api.Stimulus.isRunning (
              self )
```

Return True if signal generation is currently running.

Reimplemented from ttk_tools.dspace._xil_api.Stimulus.

7.400.3.8 loadStimulusFile()

```
ttk_tools.dspace.xil_api.Stimulus.loadStimulusFile (
              self,
              file_path,
              assignments = None )
```

Load signal generator settings from an STI or STZ file.

Note

The STI/STZ file might not contain signal mappings, so signals may still have to be assigned manually.

If you add signal descriptions via addSignalDescription(), they will replace/overwrite the loaded stimulus.

Parameters

file_path	path to STI/STZ file
assignments	optional dictionary with <name>: <signal path> assignments.

Reimplemented from ttk_tools.dspace._xil_api.Stimulus.

7.400.3.9 loadToTarget()

```
ttk_tools.dspace.xil_api.Stimulus.loadToTarget (
              self )
```

Load stimulus to target application.

Reimplemented from ttk_tools.dspace._xil_api.Stimulus.

7.400.3.10 pause()

```
ttk_tools.dspace.xil_api.Stimulus.pause (
              self )
```

Pause a running stimulus.

Reimplemented from ttk_tools.dspace._xil_api.Stimulus.

7.400.3.11 start()

```
ttk_tools.dspace.xil_api.Stimulus.start (
              self )
```

Start a loaded stimulus.

Reimplemented from ttk_tools.dspace._xil_api.Stimulus.

7.400.3.12 stop()

```
ttk_tools.dspace.xil_api.Stimulus.stop (
              self )
```

Stop a running (or paused) stimulus.

Reimplemented from ttk_tools.dspace._xil_api.Stimulus.

7.400.3.13 waitUntilDone()

```
ttk_tools.dspace.xil_api.Stimulus.waitUntilDone (
              self,
              timeout,
              step_size = 0.500,
              auto_destroy = True,
              verbosity = 2 )
```

Wait until a running stimulus is done (that is, has reached states `eSTOPPED` or `eFINISHED`, see isDone()).

Parameters

timeout	timeout in [s]; stimulus will be stopped after timeout has been reached.
step_size	step_size in [s] to wait between state checks
auto_destroy	automatically unload stimulus from target after it is done. See destroyOnTarget()
verbosity	verbosity of status messages: 0: (mostly) silent 1: waiting ... done messages plus verbose exception info 2: + stimulus running / unloading / done info

Reimplemented from ttk_tools.dspace._xil_api.Stimulus.

7.400.4 Member Data Documentation

7.400.4.1 parent

```
ttk_tools.dspace.xil_api.Stimulus.parent
```

reference to parent testbench

7.400.4.2 signal_generator

```
ttk_tools.dspace.xil_api.Stimulus.signal_generator
```

[XIL-API] SignalGenerator instance

7.400.4.3 testbench

`ttk_tools.dspace.xil_api.Stimulus.testbench`

reference to testbench api

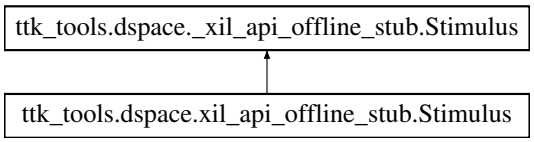
The documentation for this class was generated from the following file:

- ttk_tools/dspace/xil_api.py (v2.0.1)

7.401 ttk_tools.dspace.xil_api_offline_stub.Stimulus Class Reference

Offline stub for Stimulus / Signal Generator.

Inheritance diagram for ttk_tools.dspace.xil_api_offline_stub.Stimulus:



Public Member Functions

- __init__ (self, xil_testbench, appl_name=None)
  Stimulus / Signal Generator stub.
- loadStimulusFile (self, file_path, assignments=None)
  Load signal generator settings from an STI or STZ file.
- addSignalDescription (self, stim_sig_descr)
  Add a `StimulusSignalDescription` instance to this Stimulus.
- loadToTarget (self)
  Load stimulus to target application.
- destroyOnTarget (self)
  Destroy (well, unload) a currently loaded stimulus on/from target application.
- start (self)
  Start a loaded stimulus.
- pause (self)
  Pause a running stimulus.
- stop (self)
  Stop a running (or paused) stimulus.
- getState (self, verbosity=1)
  Get a numeric representation of the current stimulus state.
- getStateText (self, state=None)
  Get a string description for a numeric stimulus state.
- isRunning (self)
  Return True if signal generation is currently running.
- isDone (self)
  Return True if signal generation is done (finished or stopped).
- getElapsedTime (self)
  Get elapsed time in seconds.
- waitUntilDone (self, timeout, step_size=0.500, auto_destroy=True, verbosity=2)
  Wait until a running stimulus is done (that is, has reached states `eSTOPPED` or `eFINISHED`, see isDone()).

Public Attributes

- parent
  reference to parent testbench
- testbench
  reference to testbench api
- signal_generator
  [XIL-API] SignalGenerator instance

7.401.1 Detailed Description

Offline stub for Stimulus / Signal Generator.

7.401.2 Constructor & Destructor Documentation

7.401.2.1 __init__()

```
ttk_tools.dspace.xil_api_offline_stub.Stimulus.__init__ (
              self,
              xil_testbench,
              appl_name = None )
```

Stimulus / Signal Generator stub.

Parameters

xil_testbench	parent XilTestbench instance
appl_name	target application name for multi core systems (e.g. "masterAppl"), None for single core setups.

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.Stimulus.

7.401.3 Member Function Documentation

7.401.3.1 addSignalDescription()

```
ttk_tools.dspace.xil_api_offline_stub.Stimulus.addSignalDescription (
              self,
              stim_sig_descr )
```

Add a `StimulusSignalDescription` instance to this Stimulus.

Parameters

stim_sig_descr	StimulusSignalDescription instance to add

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.Stimulus.

7.401.3.2 destroyOnTarget()

```
ttk_tools.dspace.xil_api_offline_stub.Stimulus.destroyOnTarget (
              self )
```

Destroy (well, unload) a currently loaded stimulus on/from target application.

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.Stimulus.

7.401.3.3 getElapsedTime()

```
ttk_tools.dspace.xil_api_offline_stub.Stimulus.getElapsedTime (
              self )
```

Get elapsed time in seconds.

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.Stimulus.

7.401.3.4 getState()

```
ttk_tools.dspace.xil_api_offline_stub.Stimulus.getState (
              self,
              verbosity = 1 )
```

Get a numeric representation of the current stimulus state.

Parameters

verbosity	0: silent, 1: print current state as info to log console

Returns

a numeric value. See `SignalGeneratorState` enum.

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.Stimulus.

7.401.3.5 getStateText()

```
ttk_tools.dspace.xil_api_offline_stub.Stimulus.getStateText (
              self,
              state = None )
```

Get a string description for a numeric stimulus state.

Parameters

state	numeric state to get a description for None: use current state

Returns

a string.

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.Stimulus.

7.401.3.6 isDone()

```
ttk_tools.dspace.xil_api_offline_stub.Stimulus.isDone (
              self )
```

Return True if signal generation is done (finished or stopped).

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.Stimulus.

7.401.3.7 isRunning()

```
ttk_tools.dspace.xil_api_offline_stub.Stimulus.isRunning (
              self )
```

Return True if signal generation is currently running.

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.Stimulus.

7.401.3.8 loadStimulusFile()

```
ttk_tools.dspace.xil_api_offline_stub.Stimulus.loadStimulusFile (
              self,
              file_path,
              assignments = None )
```

Load signal generator settings from an STI or STZ file.

Note

The STI/STZ file might not contain signal mappings, so signals may still have to be assigned manually.

If you add signal descriptions via addSignalDescription(), they will replace/overwrite the loaded stimulus.

Parameters

file_path	path to STI/STZ file
assignments	optional dictionary with <name>: <signal path> assignments.

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.Stimulus.

7.401.3.9 loadToTarget()

```
ttk_tools.dspace.xil_api_offline_stub.Stimulus.loadToTarget (
              self )
```

Load stimulus to target application.

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.Stimulus.

7.401.3.10 pause()

```
ttk_tools.dspace.xil_api_offline_stub.Stimulus.pause (
              self )
```

Pause a running stimulus.

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.Stimulus.

7.401.3.11 start()

```
ttk_tools.dspace.xil_api_offline_stub.Stimulus.start (
              self )
```

Start a loaded stimulus.

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.Stimulus.

7.401.3.12 stop()

```
ttk_tools.dspace.xil_api_offline_stub.Stimulus.stop (
              self )
```

Stop a running (or paused) stimulus.

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.Stimulus.

7.401.3.13 waitUntilDone()

```
ttk_tools.dspace.xil_api_offline_stub.Stimulus.waitUntilDone (
              self,
              timeout,
              step_size = 0.500,
              auto_destroy = True,
              verbosity = 2 )
```

Wait until a running stimulus is done (that is, has reached states `eSTOPPED` or `eFINISHED`, see isDone()).

Parameters

timeout	timeout in [s]; stimulus will be stopped after timeout has been reached.
step_size	step_size in [s] to wait between state checks
auto_destroy	automatically unload stimulus from target after it is done. See destroyOnTarget()
verbosity	verbosity of status messages: 0: (mostly) silent 1: waiting ... done messages plus verbose exception info 2: + stimulus running / unloading / done info

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.Stimulus.

7.401.4 Member Data Documentation

7.401.4.1 parent

`ttk_tools.dspace.xil_api_offline_stub.Stimulus.parent`

reference to parent testbench

7.401.4.2 signal_generator

`ttk_tools.dspace.xil_api_offline_stub.Stimulus.signal_generator`

[XIL-API] SignalGenerator instance

7.401.4.3 testbench

`ttk_tools.dspace.xil_api_offline_stub.Stimulus.testbench`

reference to testbench api

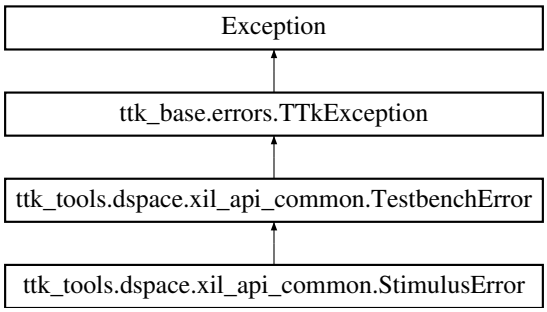
The documentation for this class was generated from the following file:

- ttk_tools/dspace/xil_api_offline_stub.py (v2.0.1)

7.402 ttk_tools.dspace.xil_api_common.StimulusError Class Reference

Errors during Stimulus setup/execution.

Inheritance diagram for ttk_tools.dspace.xil_api_common.StimulusError:



7.402.1 Detailed Description

Errors during Stimulus setup/execution.

Write a simple MAPort XML configuration file for dSPACE implementations to the specified path.

Parameters

sdf_file_path	path to system description file (∗.sdf). Note that this file does not have to exist to write the XML
platform_name	name of target platform hardware
xml_file_path	file path to write configuration to. If None, a default file named "MAPortConfig.xml" will be created in same folder as the SDF-file.

Note

`sdf_file_path` can be specified as an absolute path or as relative to the location of the XML file (i.e. `xml_file_path`).

Example:

```
writeMAPortConfigXML(
    sdf_file_path = "D:/HIL-Projekte/Demo/Modell/demo.sdf",
    platform_name = "DS1006"
)
# => configuration will be written to
# D:/HIL-Projekte/Demo/Modell/MAPortConfig.xml
```

Example:

```
maport_config_path = os.path.join(
    _base_project_path, "ControlDesk", "MAPortConfig.xml"
)
sdf_file_path = os.path.join(
    # "scalexio" is the project folder below <hil_project>/ControlDesk/
    "scalexio", "Variable Descriptions",
    "Inbetriebnahme.sdf", "Inbetriebnahme.sdf" # yep, folder and file are named the same
)
xil_api_common.writeMAPortConfigXML(
    sdf_file_path = sdf_file_path, # a relative path
    platform_name = "SCALEXIO",    # see platform properties in ControlDesk
    xml_file_path = maport_config_path
)
```

Returns

the path to the written MAPort configuration file.

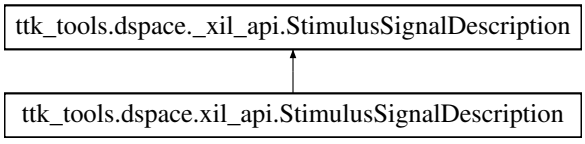
The documentation for this class was generated from the following file:

- ttk_tools/dspace/xil_api_common.py (v1.5.0)

7.403 ttk_tools.dspace._xil_api.StimulusSignalDescription Class Reference

Signal value description for stimulus / signal generator.

Inheritance diagram for ttk_tools.dspace._xil_api.StimulusSignalDescription:



Public Member Functions

- __init__ (self, xil_testbench, name, model_path)

  Stimulus signal description.
- name (self)

  Property name: signal name/alias/label (read only).
- addConstSegment (self, duration, value)

  Add a `ConstSegment`, a segment with, well, a constant value.
- addRampSegment (self, duration, start_value, stop_value)

  Add a `RampSegment`, a segment with a linear ramp moving from start to stop value during the specified time.
- addIdleSegment (self, duration)

  Add an `IdleSegment`.
- addNoiseSegment (self, duration, mean, sigma, seed=None)

  Add a `NoiseSegment`, a segment with noise having a Gaussian/normal distribution.
- addRampSlopeSegment (self, duration, offset, slope)

  Add a `RampSlopeSegment`, a segment with a ramp according to a linear equation.
- addSineSegment (self, duration, offset, amplitude, period, phase=0)

  Add a `SineSegment`, a segment with a sine waveform.
- addSawSegment (self, duration, offset, amplitude, period, phase=0, duty_cycle=0.5)

  Add a `SawSegment`, a segment of the `sig_descr` with a saw tooth (or triangle) wave form.
- addPulseSegment (self, duration, offset, amplitude, period, phase=0, duty_cycle=0.5)

  Add a `PulseSegment`, a segment with periodic rectangular pulses (that is, a PWM `sig_descr` with a fixed duty cycle).
- addExpSegment (self, duration, start_value, stop_value, tau)

  Add a `ExpSegment`, a segment following an exponential curve.
- addSignalValueSegment (self, time_values, data_values, interpolation="backward")

  Add a `SignalValueSegment`, a segment which directly uses numerical data.
- addDataFileSegment (self, file_path, data_label, time_label=None, start_time=None, duration=None, interpolation="backward")

  Add a `DataFileSegment`, a segment which uses numerical data stored in a file.

Public Attributes

- testbench

  reference to testbench api
- sig_factory

  reference to testbench's signal factory
- sig_descr

  [XIL-API] SegmentSignalDescription.
- model_path

  Model Path for this signal.

Protected Member Functions

- _getConstSymbol (self, value)

  Get a ConstSymbol.
- _getSignalValue (self, timestamps, data)

  Get a SignalValue from supplied timestamps and data values.
- _getInterpolationMode (self, text)

  Get a suitable InterpolationType for the supplied text description.

7.403.1 Detailed Description

Signal value description for stimulus / signal generator.

Describes signal value changes over time.

7.403.2 Constructor & Destructor Documentation

7.403.2.1 __init__()

```
ttk_tools.dspace._xil_api.StimulusSignalDescription.__init__ (
              self,
              xil_testbench,
              name,
              model_path )
```

Stimulus signal description.

Parameters

xil_testbench	parent XilTestbench instance
name	`sig_descr` description name/label/alias
model_path	path to signal to be stimulated

Reimplemented in ttk_tools.dspace.xil_api.StimulusSignalDescription.

7.403.3 Member Function Documentation

7.403.3.1 _getConstSymbol()

```
ttk_tools.dspace._xil_api.StimulusSignalDescription._getConstSymbol (
              self,
              value )  [protected]
```

Get a ConstSymbol.

Parameters

value	value to use for ConstSymbol

Returns

a configured ConstSymbol instance.

7.403.3.2 _getInterpolationMode()

```
ttk_tools.dspace._xil_api.StimulusSignalDescription._getInterpolationMode (
              self,
              text )  [protected]
```

Get a suitable InterpolationType for the supplied text description.

TODO: or use separate / cleaned up "enum"-like "constants"?

7.403.3.3 _getSignalValue()

```
ttk_tools.dspace._xil_api.StimulusSignalDescription._getSignalValue (
            self,
            timestamps,
            data )  [protected]
```

Get a SignalValue from supplied timestamps and data values.

Parameters

timestamps	list of time stamps
data	list of data points (int/long or float values, type of first data point determines data type)

Returns

a configured SignalValue instance.

7.403.3.4 addConstSegment()

```
ttk_tools.dspace._xil_api.StimulusSignalDescription.addConstSegment (
            self,
            duration,
            value )
```

Add a `ConstSegment`, a segment with, well, a constant value.

Info:

f(t) = value

Parameters

duration	duration of segment in [s]
value	`sig_descr` value/amplitude

Reimplemented in ttk_tools.dspace.xil_api.StimulusSignalDescription.

7.403.3.5 addDataFileSegment()

```
ttk_tools.dspace._xil_api.StimulusSignalDescription.addDataFileSegment (
            self,
            file_path,
```

```
                data_label,
                time_label = None,
                start_time = None,
                duration = None,
                interpolation = "backward" )
```

Add a `DataFileSegment`, a segment which uses numerical data stored in a file.

The data file is typically a measurement data file which contains some measured signals and one or more time axes. The `DataFileSegment` holds a link to the data file and does not store or serialize the used numerical data.

Limitations:

dSPACE XIL API Implementation Guide May 2015:
The `eFORWARD` and the `eLINEAR` interpolation types are not supported by the `SignalValueSegment` and the `DataFileSegment`. All methods which access the `InterpolationType` property implicitly use `eBACKWARD`. An exception is not thrown.

If you use a `DataFileSegment`, only the following file formats are supported:

- MAT file level 5
- MDF 4.x

To create a level 5 MAT file in MATLAB use -v6 as input argument of MATLAB's save function.

Parameters

file_path	path to data file in one of the supported formats
data_label	name/label of data vector/sig_descr to use
time_label	name/label of time vector/raster to use (not needed for MDF files, leave as None)
start_time	start time stamp in [s] None: start at beginning of available data
duration	maximum duration the segment in [s] None: run until end of file
interpolation	interpolation method • "backward": current data point will be used until next data point (staircase backward) • "linear": Linear interpolation between data points • "forward": next data point will be used immediately (staircase forward)

Reimplemented in ttk_tools.dspace.xil_api.StimulusSignalDescription.

7.403.3.6 addExpSegment()

```
ttk_tools.dspace._xil_api.StimulusSignalDescription.addExpSegment (
                self,
```

```
                duration,
                start_value,
                stop_value,
                tau )
```

Add a `ExpSegment`, a segment following an exponential curve.

Info:

f(t) = amplitude * (1 - e**(t/tau)) + offset
with
offset = start_value
amplitude = stop_value - start_value

Parameters

duration	duration of segment in [s]
start_value	start value (offset of `sig_descr`)
stop_value	stop value
tau	time constant of exponential curve in [s]

Reimplemented in ttk_tools.dspace.xil_api.StimulusSignalDescription.

7.403.3.7 addIdleSegment()

```
ttk_tools.dspace._xil_api.StimulusSignalDescription.addIdleSegment (
                self,
                duration )
```

Add an `IdleSegment`.

This sets `sig_descr` generation into idle-mode for the given duration. During idle time, the `sig_descr` generator will not write to the corresponding model variable.

Parameters

duration	duration of segment in [s]

Reimplemented in ttk_tools.dspace.xil_api.StimulusSignalDescription.

7.403.3.8 addNoiseSegment()

```
ttk_tools.dspace._xil_api.StimulusSignalDescription.addNoiseSegment (
                self,
                duration,
                mean,
                sigma,
                seed = None )
```

Add a `NoiseSegment`, a segment with noise having a Gaussian/normal distribution.

Parameters

duration	duration of segment in [s]
mean	mean value of Gaussian noise
sigma	standard deviation of amplitude against mean value
seed	seed/start value for random number generator. None: uses current time (epoch) as seed value

Reimplemented in ttk_tools.dspace.xil_api.StimulusSignalDescription.

7.403.3.9 addPulseSegment()

```
ttk_tools.dspace._xil_api.StimulusSignalDescription.addPulseSegment (
              self,
              duration,
              offset,
              amplitude,
              period,
              phase = 0,
              duty_cycle = 0.5 )
```

Add a `PulseSegment`, a segment with periodic rectangular pulses (that is, a PWM `sig_descr` with a fixed duty cycle).

Parameters

duration	duration of segment in [s]
offset	offset value of waveform
amplitude	amplitude of waveform, maximum value = offset + amplitude, minimum value = offset
period	cycle time of the waveform in [s]
phase	initial phase shift as positive or negative factor of the period [-1.0 <= phase <= +1.0] e.g. 0.25 equals a 90° phase shift, -0.33 equals a -120° phase shift
duty_cycle	ratio of high time to period as a positive factor [0.0 <= duty_cycle <= 1.0], DC 0.0: constant value (low @ offset) DC 0.5: symmetric rectangular `sig_descr` DC 1.0: constant value (high @ offset + amplitude)

Reimplemented in ttk_tools.dspace.xil_api.StimulusSignalDescription.

7.403.3.10 addRampSegment()

```
ttk_tools.dspace._xil_api.StimulusSignalDescription.addRampSegment (
              self,
              duration,
              start_value,
              stop_value )
```

Add a `RampSegment`, a segment with a linear ramp moving from start to stop value during the specified time.

Info:

f(t) = (stop_value - start_value) / duration ∗ t + start_value

Parameters

duration	duration of segment in [s]
start_value	`sig_descr` value at beginning of ramp
stop_value	`sig_descr` value at end of ramp

Reimplemented in ttk_tools.dspace.xil_api.StimulusSignalDescription.

7.403.3.11 addRampSlopeSegment()

```
ttk_tools.dspace._xil_api.StimulusSignalDescription.addRampSlopeSegment (
              self,
              duration,
              offset,
              slope )
```

Add a `RampSlopeSegment`, a segment with a ramp according to a linear equation.

Info:

f(t) = slope ∗ t + offset

Parameters

duration	duration of segment in [s]
offset	offset value of ramp
slope	slope of ramp

Reimplemented in ttk_tools.dspace.xil_api.StimulusSignalDescription.

7.403.3.12 addSawSegment()

```
ttk_tools.dspace._xil_api.StimulusSignalDescription.addSawSegment (
              self,
              duration,
              offset,
              amplitude,
              period,
              phase = 0,
              duty_cycle = 0.5 )
```

Add a `SawSegment`, a segment of the `sig_descr` with a saw tooth (or triangle) wave form.

Parameters

duration	duration of segment in [s]

Parameters

offset	offset value of saw tooth waveform
amplitude	amplitude of saw tooth waveform maximum value = offset + amplitude, minimum value = offset
period	cycle time of the waveform in [s]
phase	initial phase shift as positive or negative factor of the period [-1.0 <= phase <= +1.0] e.g. 0.25 equals a 90° phase shift, -0.33 equals a -120° phase shift
duty_cycle	ratio of raise time to period as a positive factor [0.0 <= duty_cycle <= 1.0], use factor 0.5 to get a triangle wave form.

Reimplemented in ttk_tools.dspace.xil_api.StimulusSignalDescription.

7.403.3.13 addSignalValueSegment()

```
ttk_tools.dspace._xil_api.StimulusSignalDescription.addSignalValueSegment (
              self,
              time_values,
              data_values,
              interpolation = "backward" )
```

Add a `SignalValueSegment`, a segment which directly uses numerical data.

The amplitude results from data points in numerical data and the given interpolation type.
Duration of the segment is implicitly derived from the time vector.

Note

Normally this segment is used to replay measured data.

Limitations:

dSPACE XIL API Implementation Guide May 2015:
The `eFORWARD` and the `eLINEAR` interpolation types are not supported by the `SignalValueSegment` and the `DataFileSegment`.
All methods which access the InterpolationType property implicitly use `eBACKWARD`. An exception is not thrown.

The `SignalValueSegment` only supports values of `FloatVectorValue` data type.

Parameters

time_values	a list of time stamps
data_values	a list of data points
interpolation	interpolation method: • "backward": current data point will be used until next data point (staircase backward) • "linear": Linear interpolation between data points • "forward": next data point will be used immediately (staircase forward)

Reimplemented in ttk_tools.dspace.xil_api.StimulusSignalDescription.

7.403.3.14 addSineSegment()

```
ttk_tools.dspace._xil_api.StimulusSignalDescription.addSineSegment (
              self,
              duration,
              offset,
              amplitude,
              period,
              phase = 0 )
```

Add a `SineSegment`, a segment with a sine waveform.

Parameters

duration	duration of segment in [s]
offset	offset value of the sine waveform
amplitude	amplitude of sine waveform, maximum value = offset + amplitude, minimum value = offset - amplitude
period	cycle time of the sine waveform in [s]
phase	initial phase shift as positive or negative factor of the period [-1.0 <= phase <= +1.0] e.g. 0.25 equals a 90° phase shift, -0.33 equals a -120° phase shift

Reimplemented in ttk_tools.dspace.xil_api.StimulusSignalDescription.

7.403.3.15 name()

```
ttk_tools.dspace._xil_api.StimulusSignalDescription.name (
              self )
```

Property name: signal name/alias/label (read only).

7.403.4 Member Data Documentation

7.403.4.1 model_path

```
ttk_tools.dspace._xil_api.StimulusSignalDescription.model_path
```

Model Path for this signal.

7.403.4.2 sig_descr

```
ttk_tools.dspace._xil_api.StimulusSignalDescription.sig_descr
```

[XIL-API] SegmentSignalDescription.

A SegmentSignalDescription is used to define a signal waveform based on a chronologically ordered sequence of different segments.

7.403.4.3 sig_factory

`ttk_tools.dspace._xil_api.StimulusSignalDescription.sig_factory`

reference to testbench's signal factory

7.403.4.4 testbench

`ttk_tools.dspace._xil_api.StimulusSignalDescription.testbench`

reference to testbench api

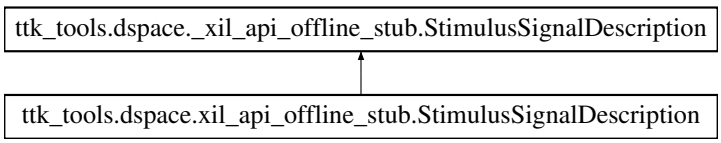
The documentation for this class was generated from the following file:

- ttk_tools/dspace/_xil_api.py (v2.3.1)

7.404 ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription Class Reference

Stub for signal value description for stimulus / signal generator.

Inheritance diagram for ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription:



Public Member Functions

- __init__ (self, xil_testbench, name, model_path)

  Stimulus signal description stub.
- name (self)

  Property name: signal name/alias/label (read only).
- addConstSegment (self, duration, value)
- addRampSegment (self, duration, start_value, stop_value)
- addIdleSegment (self, duration)

  Add an `IdleSegment`.
- addNoiseSegment (self, duration, mean, sigma, seed=None)
- addRampSlopeSegment (self, duration, offset, slope)
- addSineSegment (self, duration, offset, amplitude, period, phase=0)
- addSawSegment (self, duration, offset, amplitude, period, phase=0, duty_cycle=0.5)
- addPulseSegment (self, duration, offset, amplitude, period, phase=0, duty_cycle=0.5)
- addExpSegment (self, duration, start_value, stop_value, tau)
- addSignalValueSegment (self, time_values, data_values, interpolation="backward")
- addDataFileSegment (self, file_path, data_label, time_label=None, start_time=None, duration=None, interpolation="backward")

Public Attributes

- testbench
  reference to testbench api
- sig_descr
  [XIL-API] SegmentSignalDescription.
- model_path
  Model Path for this signal.

Protected Attributes

- _name
- _total_duration

7.404.1 Detailed Description

Stub for signal value description for stimulus / signal generator.

Describes signal value changes over time.

7.404.2 Constructor & Destructor Documentation

7.404.2.1 __init__()

```
ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription.__init__ (
              self,
              xil_testbench,
              name,
              model_path )
```

Stimulus signal description stub.

Parameters

xil_testbench	parent XilTestbench instance
name	`sig_descr` description name/label/alias
model_path	path to signal to be stimulated

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription.

7.404.3 Member Function Documentation

7.404.3.1 addConstSegment()

```
ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription.addConstSegment (
              self,
              duration,
              value )
```

```
 Add a `ConstSegment`, a segment with, well, a constant value.

    Info:
        f(t) = value

    Parameters:
        duration - duration of segment in [s]
        value    - `sig_descr` value/amplitude
```

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription.

7.404.3.2 addDataFileSegment()

```
ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription.addDataFileSegment (
             self,
             file_path,
             data_label,
             time_label = None,
             start_time = None,
             duration = None,
             interpolation = "backward" )


 Add a `DataFileSegment`, a segment which uses numerical data stored
    in a file.

    The data file is typically a measurement data file which contains
    some measured signals and one or more time axes. The `DataFileSegment`
    holds a link to the data file and does not store or serialize the
    used numerical data.

    Info: Limitations:
        dSPACE XIL API Implementation Guide May 2015:
        The `eFORWARD` and the `eLINEAR` interpolation types are not
        supported by the `SignalValueSegment` and the `DataFileSegment`.
        All methods which access the `InterpolationType` property
        implicitly use `eBACKWARD`. An exception is not thrown.

        If you use a `DataFileSegment`, only the following file formats
        are supported:
        * MAT file level 5
        * MDF 4.x

        To create a level 5 MAT file in MATLAB use -v6 as input
        argument of MATLAB's save function.

    Parameters:
        file_path     - path to data file in one of the supported formats
        data_label    - name/label of data vector/sig_descr to use
        time_label    - name/label of time vector/raster to use
                        (not needed for MDF files, leave as None)

        start_time    - start time stamp in [s]
                        None: start at beginning of available data
        duration      - maximum duration the segment in [s]
                        None: run until end of file
        interpolation - interpolation method
                        * "backward": current data point will be used
                                      until next data point (staircase backward)
                        * "linear":   Linear interpolation between data points
                        * "forward":  next data point will be used immediately
                                      (staircase forward)
```

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription.

7.404.3.3 addExpSegment()

```
ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription.addExpSegment (
              self,
              duration,
              start_value,
              stop_value,
              tau )


 Add an 'ExpSegment', a segment following an exponential curve.

    Info:
        f(t) = amplitude * (1 - e**(t/tau)) + offset
        with
            offset    = start_value
            amplitude = stop_value - start_value

    Parameters:
        duration    -  duration of segment in [s]
        start_value -  start value (offset of 'sig_descr')
        stop_value  -  stop value
        tau         -  time constant of exponential curve in [s]
```

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription.

7.404.3.4 addIdleSegment()

```
ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription.addIdleSegment (
              self,
              duration )
```

Add an `IdleSegment`.

This sets `sig_descr` generation into idle-mode for the given duration. During idle time, the `sig_descr` generator will not write to the corresponding model variable.

Parameters

duration	duration of segment in [s]

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription.

7.404.3.5 addNoiseSegment()

```
ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription.addNoiseSegment (
              self,
              duration,
              mean,
              sigma,
              seed = None )


 Add a 'NoiseSegment', a segment with noise having a Gaussian/normal
    distribution.
```

```
    Parameters:
        duration - duration of segment in [s]
        mean     - mean value of Gaussian noise
        sigma    - standard deviation of amplitude against mean value
        seed     - seed/start value for random number generator.
                   None: uses current time (epoch) as seed value
```

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription.

7.404.3.6 addPulseSegment()

```
ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription.addPulseSegment (
              self,
              duration,
              offset,
              amplitude,
              period,
              phase = 0,
              duty_cycle = 0.5 )

 Add a `PulseSegment`, a segment with periodic rectangular pulses
    (that is, a PWM `sig_descr` with a fixed duty cycle).

    Parameters:
        duration    -  duration of segment in [s]
        offset      -  offset value of waveform
        amplitude   -  amplitude of waveform,
                       maximum value = offset + amplitude,
                       minimum value = offset
        period      -  cycle time of the waveform in [s]
        phase       -  initial phase shift as positive or negative
                       factor of the period [-1.0 <= phase <= +1.0]
                       e.g. 0.25 equals a 90° phase shift,
                       -0.33 equals a -120° phase shift
        duty_cycle  -  ratio of high time to period as a positive factor
                       [0.0 <= duty_cycle <= 1.0],
                       DC 0.0: constant value (low @ offset)
                       DC 0.5: symmetric rectangular `sig_descr`
                       DC 1.0: constant value (high @ offset + amplitude)
```

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription.

7.404.3.7 addRampSegment()

```
ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription.addRampSegment (
              self,
              duration,
              start_value,
              stop_value )

 Add a `RampSegment`, a segment with a linear ramp moving from start
    to stop value during the specified time.

    Info:
        f(t) = (stop_value - start_value) / duration * t + start_value

    Parameters:
        duration    -  duration of segment in [s]
        start_value -  `sig_descr` value at beginning of ramp
        stop_value  -  `sig_descr` value at end of ramp
```

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription.

7.404.3.8 addRampSlopeSegment()

```
ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription.addRampSlopeSegment (
              self,
              duration,
              offset,
              slope )


 Add a `RampSlopeSegment`, a segment with a ramp according to a
    linear equation.

    Info:
        f(t) = slope * t + offset

    Parameters:
        duration    -  duration of segment in [s]
        offset      -  offset value of ramp
        slope       -  slope of ramp
```

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription.

7.404.3.9 addSawSegment()

```
ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription.addSawSegment (
              self,
              duration,
              offset,
              amplitude,
              period,
              phase = 0,
              duty_cycle = 0.5 )


 Add a `SawSegment`, a segment of the `sig_descr` with a saw tooth
    (or triangle) wave form.

    Parameters:
        duration    -  duration of segment in [s]
        offset      -  offset value of saw tooth waveform
        amplitude   -  amplitude of saw tooth waveform
                       maximum value = offset + amplitude,
                       minimum value = offset
        period      -  cycle time of the waveform in [s]
        phase       -  initial phase shift as positive or negative
                       factor of the period [-1.0 <= phase <= +1.0]
                       e.g. 0.25 equals a 90° phase shift,
                       -0.33 equals a -120° phase shift
        duty_cycle  -  ratio of raise time to period as a positive factor
                       [0.0 <= duty_cycle <= 1.0], use factor 0.5 to
                       get a triangle wave form.
```

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription.

7.404.3.10 addSignalValueSegment()

```
ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription.addSignalValueSegment (
              self,
              time_values,
              data_values,
              interpolation = "backward" )


 Add a `SignalValueSegment`, a segment which directly uses numerical
    data. The amplitude results from data points in numerical data and
    the given interpolation type.
    Duration of the segment is implicitly derived from the time vector.

    Note:
        Normally this segment is used to replay measured data.

    Info: Limitations:
        dSPACE XIL API Implementation Guide May 2015:
        The `eFORWARD` and the `eLINEAR` interpolation types are not
        supported by the `SignalValueSegment` and the `DataFileSegment`.
        All methods which access the InterpolationType property
        implicitly use `eBACKWARD`. An exception is not thrown.

        The `SignalValueSegment` only supports values of `FloatVectorValue`
        data type.

    Parameters:
        time_values   - a list of time stamps
        data_values   - a list of data points
        interpolation - interpolation method:
                        * "backward": current data point will be used
                                      until next data point (staircase backward)
                        * "linear":   Linear interpolation between data points
                        * "forward":  next data point will be used immediately
                                      (staircase forward)
```

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription.

7.404.3.11 addSineSegment()

```
ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription.addSineSegment (
              self,
              duration,
              offset,
              amplitude,
              period,
              phase = 0 )


 Add a `SineSegment`, a segment with a sine waveform.

    Parameters:
        duration    -  duration of segment in [s]
        offset      -  offset value of the sine waveform
        amplitude   -  amplitude of sine waveform,
                       maximum value = offset + amplitude,
                       minimum value = offset - amplitude
        period      -  cycle time of the sine waveform in [s]
        phase       -  initial phase shift as positive or negative
                       factor of the period [-1.0 <= phase <= +1.0]
                       e.g. 0.25 equals a 90° phase shift,
                       -0.33 equals a -120° phase shift
```

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription.

7.404.3.12 name()

```
ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription.name (
              self )
```

Property name: signal name/alias/label (read only).

7.404.4 Member Data Documentation

7.404.4.1 _name

```
ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription._name  [protected]
```

7.404.4.2 _total_duration

```
ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription._total_duration  [protected]
```

7.404.4.3 model_path

```
ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription.model_path
```

Model Path for this signal.

7.404.4.4 sig_descr

```
ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription.sig_descr
```

[XIL-API] SegmentSignalDescription.

A SegmentSignalDescription is used to define a signal waveform based on a chronologically ordered sequence of different segments.

7.404.4.5 testbench

```
ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription.testbench
```

reference to testbench api

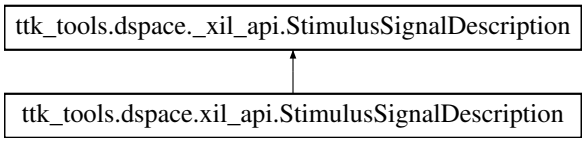
The documentation for this class was generated from the following file:

- ttk_tools/dspace/_xil_api_offline_stub.py (v2.1.2)

7.405 ttk_tools.dspace.xil_api.StimulusSignalDescription Class Reference

Signal value description for stimulus / signal generator.

Inheritance diagram for ttk_tools.dspace.xil_api.StimulusSignalDescription:



Public Member Functions

- __init__ (self, xil_testbench, name, model_path)

  Stimulus signal description.
- addConstSegment (self, duration, value)

  Add a `ConstSegment`, a segment with, well, a constant value.
- addRampSegment (self, duration, start_value, stop_value)

  Add a `RampSegment`, a segment with a linear ramp moving from start to stop value during the specified time.
- addIdleSegment (self, duration)

  Add an `IdleSegment`.
- addNoiseSegment (self, duration, mean, sigma, seed=None)

  Add a `NoiseSegment`, a segment with noise having a Gaussian/normal distribution.
- addRampSlopeSegment (self, duration, offset, slope)

  Add a `RampSlopeSegment`, a segment with a ramp according to a linear equation.
- addSineSegment (self, duration, offset, amplitude, period, phase=0)

  Add a `SineSegment`, a segment with a sine waveform.
- addSawSegment (self, duration, offset, amplitude, period, phase=0, duty_cycle=0.5)

  Add a `SawSegment`, a segment of the `sig_descr` with a saw tooth (or triangle) wave form.
- addPulseSegment (self, duration, offset, amplitude, period, phase=0, duty_cycle=0.5)

  Add a `PulseSegment`, a segment with periodic rectangular pulses (that is, a PWM `sig_descr` with a fixed duty cycle).
- addExpSegment (self, duration, start_value, stop_value, tau)

  Add a `ExpSegment`, a segment following an exponential curve.
- addSignalValueSegment (self, time_values, data_values, interpolation="backward")

  Add a `SignalValueSegment`, a segment which directly uses numerical data.
- addDataFileSegment (self, file_path, data_label, time_label=None, start_time=None, duration=None, interpolation="backward")

  Add a `DataFileSegment`, a segment which uses numerical data stored in a file.

Public Attributes

- sig_descr

  [XIL-API] SegmentSignalDescription.
- model_path

  Model Path for this signal.

7.405.1 Detailed Description

Signal value description for stimulus / signal generator.

Describes signal value changes over time.

7.405.2 Constructor & Destructor Documentation

7.405.2.1 __init__()

```
ttk_tools.dspace.xil_api.StimulusSignalDescription.__init__ (
              self,
              xil_testbench,
              name,
              model_path )
```

Stimulus signal description.

Parameters

xil_testbench	parent XilTestbench instance
name	`sig_descr` description name/label/alias
model_path	path to signal to be stimulated

Reimplemented from ttk_tools.dspace._xil_api.StimulusSignalDescription.

7.405.3 Member Function Documentation

7.405.3.1 addConstSegment()

```
ttk_tools.dspace.xil_api.StimulusSignalDescription.addConstSegment (
              self,
              duration,
              value )
```

Add a `ConstSegment`, a segment with, well, a constant value.

Info:

f(t) = value

Parameters

duration	duration of segment in [s]
value	`sig_descr` value/amplitude

Reimplemented from ttk_tools.dspace._xil_api.StimulusSignalDescription.

7.405.3.2 addDataFileSegment()

```
ttk_tools.dspace.xil_api.StimulusSignalDescription.addDataFileSegment (
                self,
                file_path,
                data_label,
                time_label = None,
                start_time = None,
                duration = None,
                interpolation = "backward" )
```

Add a `DataFileSegment`, a segment which uses numerical data stored in a file.

The data file is typically a measurement data file which contains some measured signals and one or more time axes. The `DataFileSegment` holds a link to the data file and does not store or serialize the used numerical data.

Limitations:

dSPACE XIL API Implementation Guide May 2015:
The `eFORWARD` and the `eLINEAR` interpolation types are not supported by the `SignalValueSegment` and the `DataFileSegment`. All methods which access the `InterpolationType` property implicitly use `eBACKWARD`. An exception is not thrown.

If you use a `DataFileSegment`, only the following file formats are supported:

- MAT file level 5
- MDF 4.x

To create a level 5 MAT file in MATLAB use -v6 as input argument of MATLAB's save function.

Parameters

file_path	path to data file in one of the supported formats
data_label	name/label of data vector/sig_descr to use
time_label	name/label of time vector/raster to use (not needed for MDF files, leave as None)
start_time	start time stamp in [s] None: start at beginning of available data
duration	maximum duration the segment in [s] None: run until end of file
interpolation	interpolation method • "backward": current data point will be used until next data point (staircase backward) • "linear": Linear interpolation between data points • "forward": next data point will be used immediately (staircase forward)

Reimplemented from ttk_tools.dspace._xil_api.StimulusSignalDescription.

7.405.3.3 addExpSegment()

```
ttk_tools.dspace.xil_api.StimulusSignalDescription.addExpSegment (
              self,
              duration,
              start_value,
              stop_value,
              tau )
```

Add a `ExpSegment`, a segment following an exponential curve.

Info:

f(t) = amplitude * (1 - e**(t/tau)) + offset
with
offset = start_value
amplitude = stop_value - start_value

Parameters

duration	duration of segment in [s]
start_value	start value (offset of `sig_descr`)
stop_value	stop value
tau	time constant of exponential curve in [s]

Reimplemented from ttk_tools.dspace._xil_api.StimulusSignalDescription.

7.405.3.4 addIdleSegment()

```
ttk_tools.dspace.xil_api.StimulusSignalDescription.addIdleSegment (
              self,
              duration )
```

Add an `IdleSegment`.

This sets `sig_descr` generation into idle-mode for the given duration. During idle time, the `sig_descr` generator will not write to the corresponding model variable.

Parameters

duration	duration of segment in [s]

Reimplemented from ttk_tools.dspace._xil_api.StimulusSignalDescription.

7.405.3.5 addNoiseSegment()

```
ttk_tools.dspace.xil_api.StimulusSignalDescription.addNoiseSegment (
              self,
```

```
                duration,
                mean,
                sigma,
                seed = None )
```

Add a `NoiseSegment`, a segment with noise having a Gaussian/normal distribution.

Parameters

duration	duration of segment in [s]
mean	mean value of Gaussian noise
sigma	standard deviation of amplitude against mean value
seed	seed/start value for random number generator. None: uses current time (epoch) as seed value

Reimplemented from ttk_tools.dspace._xil_api.StimulusSignalDescription.

7.405.3.6 addPulseSegment()

```
ttk_tools.dspace.xil_api.StimulusSignalDescription.addPulseSegment (
                self,
                duration,
                offset,
                amplitude,
                period,
                phase = 0,
                duty_cycle = 0.5 )
```

Add a `PulseSegment`, a segment with periodic rectangular pulses (that is, a PWM `sig_descr` with a fixed duty cycle).

Parameters

duration	duration of segment in [s]
offset	offset value of waveform
amplitude	amplitude of waveform, maximum value = offset + amplitude, minimum value = offset
period	cycle time of the waveform in [s]
phase	initial phase shift as positive or negative factor of the period [-1.0 <= phase <= +1.0] e.g. 0.25 equals a 90° phase shift, -0.33 equals a -120° phase shift
duty_cycle	ratio of high time to period as a positive factor [0.0 <= duty_cycle <= 1.0], DC 0.0: constant value (low @ offset) DC 0.5: symmetric rectangular `sig_descr` DC 1.0: constant value (high @ offset + amplitude)

Reimplemented from ttk_tools.dspace._xil_api.StimulusSignalDescription.

7.405.3.7 addRampSegment()

```
ttk_tools.dspace.xil_api.StimulusSignalDescription.addRampSegment (
```

```
                self,
                duration,
                start_value,
                stop_value )
```

Add a `RampSegment`, a segment with a linear ramp moving from start to stop value during the specified time.

Info:

f(t) = (stop_value - start_value) / duration ∗ t + start_value

Parameters

duration	duration of segment in [s]
start_value	`sig_descr` value at beginning of ramp
stop_value	`sig_descr` value at end of ramp

Reimplemented from ttk_tools.dspace._xil_api.StimulusSignalDescription.

7.405.3.8 addRampSlopeSegment()

```
ttk_tools.dspace.xil_api.StimulusSignalDescription.addRampSlopeSegment (
                self,
                duration,
                offset,
                slope )
```

Add a `RampSlopeSegment`, a segment with a ramp according to a linear equation.

Info:

f(t) = slope ∗ t + offset

Parameters

duration	duration of segment in [s]
offset	offset value of ramp
slope	slope of ramp

Reimplemented from ttk_tools.dspace._xil_api.StimulusSignalDescription.

7.405.3.9 addSawSegment()

```
ttk_tools.dspace.xil_api.StimulusSignalDescription.addSawSegment (
                self,
                duration,
                offset,
                amplitude,
                period,
```

```
                phase = 0,
                duty_cycle = 0.5 )
```

Add a `SawSegment`, a segment of the `sig_descr` with a saw tooth (or triangle) wave form.

Parameters

duration	duration of segment in [s]
offset	offset value of saw tooth waveform
amplitude	amplitude of saw tooth waveform maximum value = offset + amplitude, minimum value = offset
period	cycle time of the waveform in [s]
phase	initial phase shift as positive or negative factor of the period [-1.0 <= phase <= +1.0] e.g. 0.25 equals a 90° phase shift, -0.33 equals a -120° phase shift
duty_cycle	ratio of raise time to period as a positive factor [0.0 <= duty_cycle <= 1.0], use factor 0.5 to get a triangle wave form.

Reimplemented from ttk_tools.dspace._xil_api.StimulusSignalDescription.

7.405.3.10 addSignalValueSegment()

```
ttk_tools.dspace.xil_api.StimulusSignalDescription.addSignalValueSegment (
                self,
                time_values,
                data_values,
                interpolation = "backward" )
```

Add a `SignalValueSegment`, a segment which directly uses numerical data.

The amplitude results from data points in numerical data and the given interpolation type.
Duration of the segment is implicitly derived from the time vector.

Note

Normally this segment is used to replay measured data.

Limitations:

dSPACE XIL API Implementation Guide May 2015:
The `eFORWARD` and the `eLINEAR` interpolation types are not supported by the `SignalValueSegment` and the `DataFileSegment`.
All methods which access the InterpolationType property implicitly use `eBACKWARD`. An exception is not thrown.

The `SignalValueSegment` only supports values of `FloatVectorValue` data type.

Parameters

time_values	a list of time stamps
data_values	a list of data points
interpolation	interpolation method: • "backward": current data point will be used until next data point (staircase backward) • "linear": Linear interpolation between data points • "forward": next data point will be used immediately (staircase forward)

Reimplemented from ttk_tools.dspace._xil_api.StimulusSignalDescription.

7.405.3.11 addSineSegment()

```
ttk_tools.dspace.xil_api.StimulusSignalDescription.addSineSegment (
              self,
              duration,
              offset,
              amplitude,
              period,
              phase = 0 )
```

Add a `SineSegment`, a segment with a sine waveform.

Parameters

duration	duration of segment in [s]
offset	offset value of the sine waveform
amplitude	amplitude of sine waveform, maximum value = offset + amplitude, minimum value = offset - amplitude
period	cycle time of the sine waveform in [s]
phase	initial phase shift as positive or negative factor of the period [-1.0 <= phase <= +1.0] e.g. 0.25 equals a 90° phase shift, -0.33 equals a -120° phase shift

Reimplemented from ttk_tools.dspace._xil_api.StimulusSignalDescription.

7.405.4 Member Data Documentation

7.405.4.1 model_path

```
ttk_tools.dspace.xil_api.StimulusSignalDescription.model_path
```

Model Path for this signal.

7.405.4.2 sig_descr

`ttk_tools.dspace.xil_api.StimulusSignalDescription.sig_descr`

[XIL-API] SegmentSignalDescription.

A SegmentSignalDescription is used to define a signal waveform based on a chronologically ordered sequence of different segments.

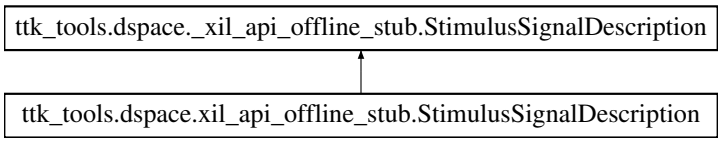
The documentation for this class was generated from the following file:

- ttk_tools/dspace/xil_api.py (v2.0.1)

7.406 ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription Class Reference

Stub for signal value description for stimulus / signal generator.

Inheritance diagram for ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription:



Public Member Functions

- __init__ (self, xil_testbench, name, model_path)

  Stimulus signal description stub.
- addConstSegment (self, duration, value)

  Add a `ConstSegment`, a segment with, well, a constant value.
- addRampSegment (self, duration, start_value, stop_value)

  Add a `RampSegment`, a segment with a linear ramp moving from start to stop value during the specified time.
- addIdleSegment (self, duration)

  Add an `IdleSegment`.
- addNoiseSegment (self, duration, mean, sigma, seed=None)

  Add a `NoiseSegment`, a segment with noise having a Gaussian/normal distribution.
- addRampSlopeSegment (self, duration, offset, slope)

  Add a `RampSlopeSegment`, a segment with a ramp according to a linear equation.
- addSineSegment (self, duration, offset, amplitude, period, phase=0)

  Add a `SineSegment`, a segment with a sine waveform.
- addSawSegment (self, duration, offset, amplitude, period, phase=0, duty_cycle=0.5)

  Add a `SawSegment`, a segment of the `sig_descr` with a saw tooth (or triangle) wave form.
- addPulseSegment (self, duration, offset, amplitude, period, phase=0, duty_cycle=0.5)

  Add a `PulseSegment`, a segment with periodic rectangular pulses (that is, a PWM `sig_descr` with a fixed duty cycle).
- addExpSegment (self, duration, start_value, stop_value, tau)

  Add an `ExpSegment`, a segment following an exponential curve.
- addSignalValueSegment (self, time_values, data_values, interpolation="backward")

  Add a `SignalValueSegment`, a segment which directly uses numerical data.
- addDataFileSegment (self, file_path, data_label, time_label=None, start_time=None, duration=None, interpolation="backward")

  Add a `DataFileSegment`, a segment which uses numerical data stored in a file.

Public Attributes

- testbench

  reference to testbench api
- sig_descr

  [XIL-API] SegmentSignalDescription.
- model_path

  Model Path for this signal.

7.406.1 Detailed Description

Stub for signal value description for stimulus / signal generator.

Describes signal value changes over time.

7.406.2 Constructor & Destructor Documentation

7.406.2.1 __init__()

```
ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription.__init__ (
              self,
              xil_testbench,
              name,
              model_path )
```

Stimulus signal description stub.

Parameters

xil_testbench	parent XilTestbench instance
name	`sig_descr` description name/label/alias
model_path	path to signal to be stimulated

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription.

7.406.3 Member Function Documentation

7.406.3.1 addConstSegment()

```
ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription.addConstSegment (
              self,
              duration,
              value )
```

Add a `ConstSegment`, a segment with, well, a constant value.

Info:

f(t) = value

Parameters

duration	duration of segment in [s]
value	`sig_descr` value/amplitude

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription.

7.406.3.2 addDataFileSegment()

```
ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription.addDataFileSegment (
              self,
              file_path,
              data_label,
              time_label = None,
              start_time = None,
              duration = None,
              interpolation = "backward" )
```

Add a `DataFileSegment`, a segment which uses numerical data stored in a file.

The data file is typically a measurement data file which contains some measured signals and one or more time axes. The `DataFileSegment` holds a link to the data file and does not store or serialize the used numerical data.

Limitations:

dSPACE XIL API Implementation Guide May 2015:
The `eFORWARD` and the `eLINEAR` interpolation types are not supported by the `SignalValueSegment` and the `DataFileSegment`. All methods which access the `InterpolationType` property implicitly use `eBACKWARD`. An exception is not thrown.

If you use a `DataFileSegment`, only the following file formats are supported:

- MAT file level 5
- MDF 4.x

To create a level 5 MAT file in MATLAB use -v6 as input argument of MATLAB's save function.

Parameters

file_path	path to data file in one of the supported formats
data_label	name/label of data vector/sig_descr to use
time_label	name/label of time vector/raster to use (not needed for MDF files, leave as None)
start_time	start time stamp in [s] None: start at beginning of available data
duration	maximum duration the segment in [s] None: run until end of file
interpolation	interpolation method • "backward": current data point will be used until next data point (staircase backward) • "linear": Linear interpolation between data points • "forward": next data point will be used immediately (staircase forward)

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription.

7.406.3.3 addExpSegment()

```
ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription.addExpSegment (
              self,
              duration,
              start_value,
              stop_value,
              tau )
```

Add an `ExpSegment`, a segment following an exponential curve.

Info:

f(t) = amplitude * (1 - e**(t/tau)) + offset
with
offset = start_value
amplitude = stop_value - start_value

Parameters

duration	duration of segment in [s]
start_value	start value (offset of `sig_descr`)
stop_value	stop value
tau	time constant of exponential curve in [s]

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription.

7.406.3.4 addIdleSegment()

```
ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription.addIdleSegment (
              self,
              duration )
```

Add an `IdleSegment`.

This sets `sig_descr` generation into idle-mode for the given duration. During idle time, the `sig_descr` generator will not write to the corresponding model variable.

Parameters

duration	duration of segment in [s]

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription.

7.406.3.5 addNoiseSegment()

```
ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription.addNoiseSegment (
```

```
                self,
                duration,
                mean,
                sigma,
                seed = None )
```

Add a `NoiseSegment`, a segment with noise having a Gaussian/normal distribution.

Parameters

duration	duration of segment in [s]
mean	mean value of Gaussian noise
sigma	standard deviation of amplitude against mean value
seed	seed/start value for random number generator. None: uses current time (epoch) as seed value

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription.

7.406.3.6 addPulseSegment()

```
ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription.addPulseSegment (
                self,
                duration,
                offset,
                amplitude,
                period,
                phase = 0,
                duty_cycle = 0.5 )
```

Add a `PulseSegment`, a segment with periodic rectangular pulses (that is, a PWM `sig_descr` with a fixed duty cycle).

Parameters

duration	duration of segment in [s]
offset	offset value of waveform
amplitude	amplitude of waveform, maximum value = offset + amplitude, minimum value = offset
period	cycle time of the waveform in [s]
phase	initial phase shift as positive or negative factor of the period [-1.0 <= phase <= +1.0] e.g. 0.25 equals a 90° phase shift, -0.33 equals a -120° phase shift
duty_cycle	ratio of high time to period as a positive factor [0.0 <= duty_cycle <= 1.0], DC 0.0: constant value (low @ offset) DC 0.5: symmetric rectangular `sig_descr` DC 1.0: constant value (high @ offset + amplitude)

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription.

7.406.3.7 addRampSegment()

```
ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription.addRampSegment (
              self,
              duration,
              start_value,
              stop_value )
```

Add a `RampSegment`, a segment with a linear ramp moving from start to stop value during the specified time.

Info:

f(t) = (stop_value - start_value) / duration ∗ t + start_value

Parameters

duration	duration of segment in [s]
start_value	`sig_descr` value at beginning of ramp
stop_value	`sig_descr` value at end of ramp

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription.

7.406.3.8 addRampSlopeSegment()

```
ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription.addRampSlopeSegment (
              self,
              duration,
              offset,
              slope )
```

Add a `RampSlopeSegment`, a segment with a ramp according to a linear equation.

Info:

f(t) = slope ∗ t + offset

Parameters

duration	duration of segment in [s]
offset	offset value of ramp
slope	slope of ramp

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription.

7.406.3.9 addSawSegment()

```
ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription.addSawSegment (
              self,
```

```
                duration,
                offset,
                amplitude,
                period,
                phase = 0,
                duty_cycle = 0.5 )
```

Add a `SawSegment`, a segment of the `sig_descr` with a saw tooth (or triangle) wave form.

Parameters

duration	duration of segment in [s]
offset	offset value of saw tooth waveform
amplitude	amplitude of saw tooth waveform maximum value = offset + amplitude, minimum value = offset
period	cycle time of the waveform in [s]
phase	initial phase shift as positive or negative factor of the period [-1.0 <= phase <= +1.0] e.g. 0.25 equals a 90° phase shift, -0.33 equals a -120° phase shift
duty_cycle	ratio of raise time to period as a positive factor [0.0 <= duty_cycle <= 1.0], use factor 0.5 to get a triangle wave form.

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription.

7.406.3.10 addSignalValueSegment()

```
ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription.addSignalValueSegment (
                self,
                time_values,
                data_values,
                interpolation = "backward" )
```

Add a `SignalValueSegment`, a segment which directly uses numerical data.

The amplitude results from data points in numerical data and the given interpolation type.
Duration of the segment is implicitly derived from the time vector.

Note

Normally this segment is used to replay measured data.

Limitations:

dSPACE XIL API Implementation Guide May 2015:
The `eFORWARD` and the `eLINEAR` interpolation types are not supported by the `SignalValueSegment` and the `DataFileSegment`.
All methods which access the InterpolationType property implicitly use `eBACKWARD`. An exception is not thrown.

The `SignalValueSegment` only supports values of `FloatVectorValue` data type.

Parameters

time_values	a list of time stamps
data_values	a list of data points
interpolation	interpolation method: • "backward": current data point will be used until next data point (staircase backward) • "linear": Linear interpolation between data points • "forward": next data point will be used immediately (staircase forward)

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription.

7.406.3.11 addSineSegment()

```
ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription.addSineSegment (
              self,
              duration,
              offset,
              amplitude,
              period,
              phase = 0 )
```

Add a `SineSegment`, a segment with a sine waveform.

Parameters

duration	duration of segment in [s]
offset	offset value of the sine waveform
amplitude	amplitude of sine waveform, maximum value = offset + amplitude, minimum value = offset - amplitude
period	cycle time of the sine waveform in [s]
phase	initial phase shift as positive or negative factor of the period [-1.0 <= phase <= +1.0] e.g. 0.25 equals a 90° phase shift, -0.33 equals a -120° phase shift

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription.

7.406.4 Member Data Documentation

7.406.4.1 model_path

```
ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription.model_path
```

Model Path for this signal.

7.406.4.2 sig_descr

`ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription.sig_descr`

[XIL-API] SegmentSignalDescription.

A SegmentSignalDescription is used to define a signal waveform based on a chronologically ordered sequence of different segments.

7.406.4.3 testbench

`ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription.testbench`

reference to testbench api

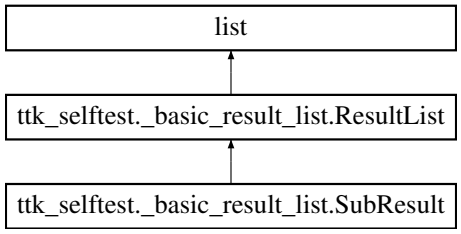
The documentation for this class was generated from the following file:

- ttk_tools/dspace/xil_api_offline_stub.py (v2.0.1)

7.407 ttk_selftest._basic_result_list.SubResult Class Reference

Context manager for Sub-Results.

Inheritance diagram for ttk_selftest._basic_result_list.SubResult:



Public Member Functions

- __init__ (self, parent_result_list=None, divider=None)
- __enter__ (self)

  Start context manager.
- __exit__ (self, exc_type, exc_value, tb)

  End context manager.

Public Attributes

- echo_enabled
- divider

  string to place between joined description texts.
- failed_prefix

  Prefix to add to descriptions of failed test steps.
- trim_descrs

  True: whitespace will be trimmed/stripped from beginning and end of each description to make output more compact.
- compact_descrs

  True: "compact" descriptions by removing all lines from the beginning that are already present in the directly preceding description.

Protected Attributes

- _parent

  parent list to which combined sub-result will be appended

7.407.1 Detailed Description

Context manager for Sub-Results.

Upon leaving the context, the combined sub-results will be added to the parent ResultList (compare ResultList.↩ getCombinedResult). Otherwise (e.g. outside of a managed context) this will behave just like a ResultList with some adapted default settings.

Note

Exceptions occurring within the context will be caught and added as an ERROR result.

Usage:

```
testresults = ResultList()
# [...]
with SubResult(testresults) as subresult:
    subresult.append(["Some result", "PASSED"])
    subresult.append(["Another result", "PASSED"])
```

7.407.2 Constructor & Destructor Documentation

7.407.2.1 __init__()

```
ttk_selftest._basic_result_list.SubResult.__init__ (
            self,
            parent_result_list = None,
            divider = None )
```

Parameters

parent_result_list	a parent ResultList (or basic list) instance to which the finished combined sub-result will be added
divider	divider string to use to combine sub-results None: use divider from parent_result_list

Reimplemented from ttk_selftest._basic_result_list.ResultList.

7.407.3 Member Function Documentation

7.407.3.1 __enter__()

```
ttk_selftest._basic_result_list.SubResult.__enter__ (
            self )
```

Start context manager.

Example:

```
with SubResult(testresult) as subresult:
    subresult.append(["Hello World!", "PASSED"])
```

7.407.3.2 __exit__()

```
ttk_selftest._basic_result_list.SubResult.__exit__ (
            self,
            exc_type,
            exc_value,
            tb )
```

End context manager.

Exceptions within the context will be added as an ERROR result.

7.407.4 Member Data Documentation

7.407.4.1 _parent

```
ttk_selftest._basic_result_list.SubResult._parent  [protected]
```

parent list to which combined sub-result will be appended

7.407.4.2 compact_descrs

```
ttk_selftest._basic_result_list.SubResult.compact_descrs
```

True: "compact" descriptions by removing all lines from the beginning that are already present in the directly preceding description.

See ResultList.getCombinedResult

7.407.4.3 divider

```
ttk_selftest._basic_result_list.SubResult.divider
```

string to place between joined description texts.

See ResultList.getCombinedResult

7.407.4.4 echo_enabled

```
ttk_selftest._basic_result_list.SubResult.echo_enabled
```

7.407.4.5 failed_prefix

`ttk_selftest._basic_result_list.SubResult.failed_prefix`

Prefix to add to descriptions of failed test steps.

See ResultList.getCombinedResult

7.407.4.6 trim_descrs

`ttk_selftest._basic_result_list.SubResult.trim_descrs`

True: whitespace will be trimmed/stripped from beginning and end of each description to make output more compact.

See ResultList.getCombinedResult

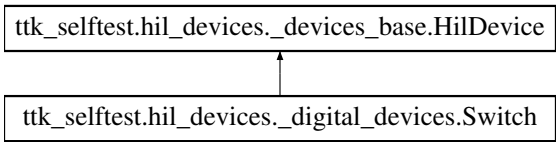
The documentation for this class was generated from the following file:

- ttk_selftest/_basic_result_list.py (v1.0.1)

7.408 ttk_selftest.hil_devices._digital_devices.Switch Class Reference

Base switch container for a testable switch and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices._digital_devices.Switch:



Public Member Functions

- __init__ (self, context, identifier, on_channel, off_channel=None, preconditions=None, wait_time_ms=100, alias="", descr="")

  Switch initialization.
- set (self, value, ∗args, ∗∗kwargs)

  Function to set the switch according to its type.
- setState (self, state)

  Function to set the switch according to its type.
- reset (self)

  Function to reset the switch according to its type to its initial value.

Public Attributes

- switch_var

7.408.1 Detailed Description

Base switch container for a testable switch and relevant associated variables/values.

7.408.2 Constructor & Destructor Documentation

7.408.2.1 __init__()

```
ttk_selftest.hil_devices._digital_devices.Switch.__init__ (
                self,
                context,
                identifier,
                on_channel,
                off_channel = None,
                preconditions = None,
                wait_time_ms = 100,
                alias = "",
                descr = "" )
```

Switch initialization.

Parameters

context	parent context
identifier	variable identifier: model path, CAN signal or C2DOutput should be C2DOuput, DigOut or CAN signal of digital output any time
on_channel	fiu channel for on-state of the switch device
off_channel	(opt) fiu channel for off-state of the switch device
preconditions	(opt) preconditions that have to be set before the device can be checked. default is None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The function specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
    # For special cases, the preconditions may also contain function calls
    # (which might be useful to insert additional delays or similar):
    preconditions = [
        { "device": Voltage, "value": 1},
        { "func":   time.sleep,
          "args":   [1.0], # positional arguments
          "kwargs": {},    # keyword arguments
        },
        {"device": ClampSwitch, "value": 1},
          # or a simple call like this:
          { "func":  someCustomFunction,
```

```
            },
              ...
        ]
    wait_time_ms  - (opt) waiting time in milliseconds between
    channel stimulation and channel check
    defaults to 100ms
    alias         - (opt) channel alias/name, defaults to the
    last (non-default) entry in identifier
    (i.e. signal model path)
    descr         - (opt) channel description (e.g. used in report entries)
```

Reimplemented from ttk_selftest.hil_devices._devices_base.HilDevice.

7.408.3 Member Function Documentation

7.408.3.1 reset()

```
ttk_selftest.hil_devices._digital_devices.Switch.reset (
              self )
```

Function to reset the switch according to its type to its initial value.

```
None
```

Returns

the original value (which may be None if the variable has not been set before)

7.408.3.2 set()

```
ttk_selftest.hil_devices._digital_devices.Switch.set (
              self,
              value,
             * args,
             ** kwargs )
```

Function to set the switch according to its type.

Parameters

value	value to set
∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Note

Any additional parameters will be forwarded to the write implementation of the variable instance

Returns

the original value before the set operation occurred.

7.408.3.3 setState()

```
ttk_selftest.hil_devices._digital_devices.Switch.setState (
              self,
              state )
```

Function to set the switch according to its type.

Parameters

state	named state to set

Returns

the previously set state (or the raw value if no matching state was available).

7.408.4 Member Data Documentation

7.408.4.1 switch_var

```
ttk_selftest.hil_devices._digital_devices.Switch.switch_var
```

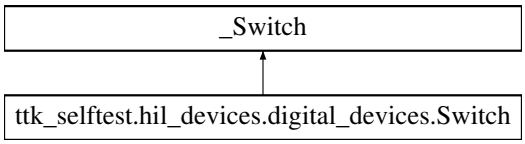
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/_digital_devices.py (v2.0.0)

7.409 ttk_selftest.hil_devices.digital_devices.Switch Class Reference

Base switch container for a testable switch and relevant associated variables/values.

Inheritance diagram for ttk_selftest.hil_devices.digital_devices.Switch:



Public Member Functions

- __init__ (self, context, identifier, on_channel, off_channel=None, preconditions=None, wait_time_ms=100, alias="", descr="")

  Switch initialization.
- set (self, value, ∗args, ∗∗kwargs)

  Function to set the switch according to its type.
- setState (self, state)

  Function to set the switch according to its type.
- reset (self)

  Function to reset the switch according to its type to its initial value.

7.409.1 Detailed Description

Base switch container for a testable switch and relevant associated variables/values.

7.409.2 Constructor & Destructor Documentation

7.409.2.1 __init__()

```
ttk_selftest.hil_devices.digital_devices.Switch.__init__ (
                self,
                context,
                identifier,
                on_channel,
                off_channel = None,
                preconditions = None,
                wait_time_ms = 100,
                alias = "",
                descr = "" )
```

Switch initialization.

Parameters

context	parent context
identifier	variable identifier: model path, CAN signal or C2DOutput should be C2DOuput, DigOut or CAN signal of digital output any time
on_channel	fiu channel for on-state of the switch device
off_channel	(opt) fiu channel for off-state of the switch device
preconditions	(opt) preconditions that have to be set before the device can be checked. default is None preconditions have to be specified as a list of dictionaries where each list element specifies one precondition. Each dictionary consists of the keys "device" and "value", where "value" specifies the value to be set for the "device". Alternatively, functions can be called. In this case, the keys "func", "set_func" or "reset_func" and optional arguments have to be part of the dictionary. The function specified with "func" will be called when the preconditions are set and reset, the functions specified with "set_func" will only be called when preconditions are set, the functions specified with "reset_func" will only be called when preconditions are reset.

Example: Set device Voltage to 13V and ClampSwitch to 1

```
preconditions = [
    {"device": Voltage, "value": 13},
    {"device": ClampSwitch, "value": 1},
]
```

Example: function calls

```
    # For special cases, the preconditions may also contain function calls
    # (which might be useful to insert additional delays or similar):
    preconditions = [
        { "device": Voltage, "value": 1},
        { "func":   time.sleep,
          "args":   [1.0], # positional arguments
          "kwargs": {},    # keyword arguments
        },
        {"device": ClampSwitch, "value": 1},
          # or a simple call like this:
          { "func":  someCustomFunction,
```

```
            },
              ...
        ]
    wait_time_ms  - (opt) waiting time in milliseconds between
    channel stimulation and channel check
    defaults to 100ms
    alias         - (opt) channel alias/name, defaults to the
    last (non-default) entry in identifier
    (i.e. signal model path)
    descr         - (opt) channel description (e.g. used in report entries)
```

7.409.3 Member Function Documentation

7.409.3.1 reset()

```
ttk_selftest.hil_devices.digital_devices.Switch.reset (
              self )
```

Function to reset the switch according to its type to its initial value.

```
None
```

Returns

the original value (which may be None if the variable has not been set before)

7.409.3.2 set()

```
ttk_selftest.hil_devices.digital_devices.Switch.set (
              self,
              value,
             * args,
             ** kwargs )
```

Function to set the switch according to its type.

Parameters

value	value to set
∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Note

Any additional parameters will be forwarded to the write implementation of the variable instance

Returns

the original value before the set operation occurred.

7.409.3.3 setState()

```
ttk_selftest.hil_devices.digital_devices.Switch.setState (
               self,
               state )
```

Function to set the switch according to its type.

Parameters

state	named state to set

Returns

the previously set state (or the raw value if no matching state was available).

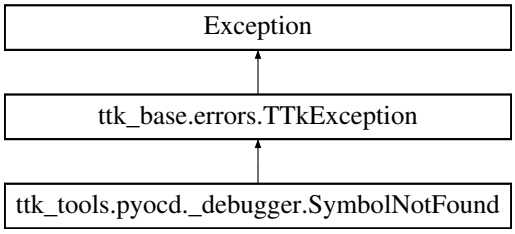
The documentation for this class was generated from the following file:

- ttk_selftest/hil_devices/digital_devices.py (v2.0.0)

7.410 ttk_tools.pyocd._debugger.SymbolNotFound Class Reference

No matching symbol was found in ELF file.

Inheritance diagram for ttk_tools.pyocd._debugger.SymbolNotFound:



7.410.1 Detailed Description

No matching symbol was found in ELF file.

Wrapper for a PyOCD session and utility methods. Debugger session will be connected to running system in "attach" mode.

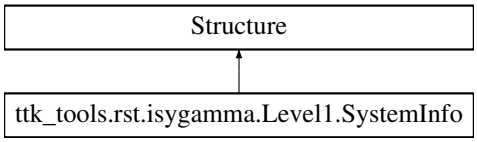
The documentation for this class was generated from the following file:

- ttk_tools/pyocd/_debugger.py (v1.2.0)

7.411 ttk_tools.rst.isygamma.Level1.SystemInfo Class Reference

typedef struct gc_SystemInfo

Inheritance diagram for ttk_tools.rst.isygamma.Level1.SystemInfo:



Static Protected Attributes

- list _fields_

7.411.1 Detailed Description

typedef struct gc_SystemInfo

7.411.2 Member Data Documentation

7.411.2.1 _fields_

```
list ttk_tools.rst.isygamma.Level1.SystemInfo._fields_  [static], [protected]
```

Initial value:

```
=  [
            ("address",             Addresses.Level1),  # address of the system
            ("projectID",           c_uint32 * 4),      # project ID
            ("name",                c_char * 256),      # name of the system
            ("clockPrecisionUsec",  c_uint32),          # configured clock precision in us
            ("startedAt",           c_uint64)           # seconds since epoch (1970-01-01T00:00:00)
        ]
```

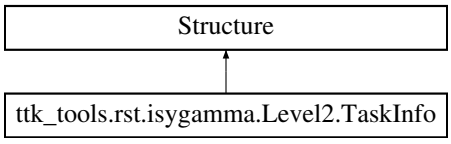
The documentation for this class was generated from the following file:

- ttk_tools/rst/isygamma.py (v1.6)

7.412 ttk_tools.rst.isygamma.Level2.TaskInfo Class Reference

typedef struct gc_TaskInfo

Inheritance diagram for ttk_tools.rst.isygamma.Level2.TaskInfo:



Static Protected Attributes

- list _fields_

7.412.1 Detailed Description

typedef struct gc_TaskInfo

7.412.2 Member Data Documentation

7.412.2.1 _fields_

```
list ttk_tools.rst.isygamma.Level2.TaskInfo._fields_  [static], [protected]
```

Initial value:

```
=  [
            ("address",         Addresses.Level2),  # address of the task
            ("name",            c_char * 1024),     # unique name of the task
            ("pid",             c_uint32),          # pid of the task
            ("signalNumber",    c_uint32),          # signal number
            ("memory",          Level3.MemoryInfo), # memory that contains task-specific information
        ]
```

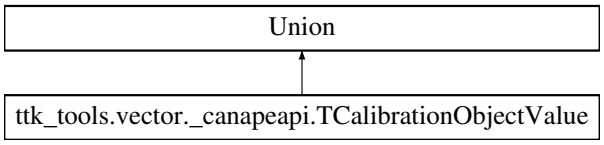
The documentation for this class was generated from the following file:

- ttk_tools/rst/isygamma.py (v1.6)

7.413 ttk_tools.vector._canapeapi.TCalibrationObjectValue Class Reference

Calibration Object Value data structure as used in CANapeAPI.

Inheritance diagram for ttk_tools.vector._canapeapi.TCalibrationObjectValue:



Classes

- class ascii_struct

  ASCII/string data.
- class axis_struct

  axis definitions for maps or curves.
- class curve_struct

  2D table data.
- class map_struct

  3D table data.
- class valblk_struct

  Value Block data.
- class value_struct

  plain scalar values

Static Protected Attributes

- _pack_
- list _fields_

7.413.1 Detailed Description

Calibration Object Value data structure as used in CANapeAPI.

7.413.2 Member Data Documentation

7.413.2.1 _fields_

```
list ttk_tools.vector._canapeapi.TCalibrationObjectValue._fields_  [static], [protected]
```

Initial value:

```
=  [
        ("type",    c_int),
        ("value",   value_struct),
        ("axis",    axis_struct),
        ("ascii",   ascii_struct),
        ("curve",   curve_struct),
        ("map",     map_struct),
        ("valblk",  valblk_struct), # added in CANape 16 (or at least after CANape 14)

    ]
```

7.413.2.2 _pack_

```
ttk_tools.vector._canapeapi.TCalibrationObjectValue._pack_  [static], [protected]
```

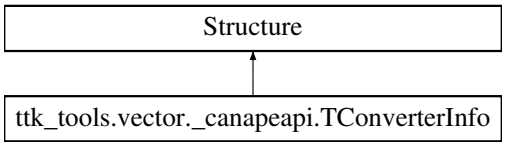
The documentation for this class was generated from the following file:

- ttk_tools/vector/_canapeapi.py (v7.3.3)

7.414 ttk_tools.vector._canapeapi.TConverterInfo Class Reference

Basic information about available MDF converters.

Inheritance diagram for ttk_tools.vector._canapeapi.TConverterInfo:



Static Protected Attributes

- int _pack_ = 1
- list _fields_

7.414.1 Detailed Description

Basic information about available MDF converters.

7.414.2 Member Data Documentation

7.414.2.1 _fields_

```
list ttk_tools.vector._canapeapi.TConverterInfo._fields_  [static], [protected]
```

Initial value:

```
=  [ # win32con.MAX_PATH == 260
        ('Comment', c_char * 260),  # comment text
        ('Name',    c_char * 260),  # Converter name
        ('ID',      c_char * 260),  # Identification id
    ]
```

7.414.2.2 _pack_

```
int ttk_tools.vector._canapeapi.TConverterInfo._pack_ = 1  [static], [protected]
```

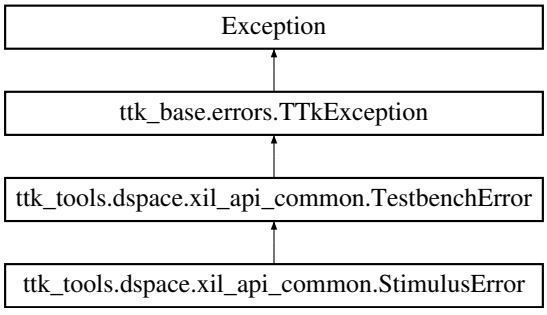
The documentation for this class was generated from the following file:

- ttk_tools/vector/_canapeapi.py (v7.3.3)

7.415 ttk_tools.dspace.xil_api_common.TestbenchError Class Reference

Base class for "other" testbench errors that are not handled by XIL TestbenchPortExceptions.

Inheritance diagram for ttk_tools.dspace.xil_api_common.TestbenchError:



Public Member Functions

- __init__ (self, msg, source=None, orig_exception=None)

Public Attributes

- source
- orig_exception

7.415.1 Detailed Description

Base class for "other" testbench errors that are not handled by XIL TestbenchPortExceptions.

7.415.2 Constructor & Destructor Documentation

7.415.2.1 __init__()

```
ttk_tools.dspace.xil_api_common.TestbenchError.__init__ (
              self,
              msg,
              source = None,
              orig_exception = None )
```

7.415.3 Member Data Documentation

7.415.3.1 orig_exception

```
ttk_tools.dspace.xil_api_common.TestbenchError.orig_exception
```

7.415.3.2 source

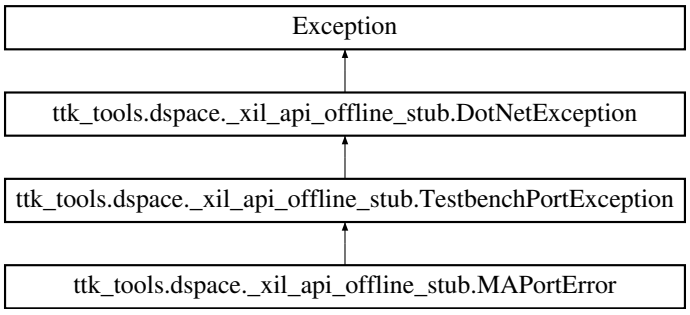
```
ttk_tools.dspace.xil_api_common.TestbenchError.source
```

The documentation for this class was generated from the following file:

- ttk_tools/dspace/xil_api_common.py (v1.5.0)

7.416 ttk_tools.dspace._xil_api_offline_stub.TestbenchPortException Class Reference

Inheritance diagram for ttk_tools.dspace._xil_api_offline_stub.TestbenchPortException:



Public Member Functions

- __init__ (self, msg)

Public Attributes

- CodeDescription
- VendorCodeDescription

7.416.1 Constructor & Destructor Documentation

7.416.1.1 __init__()

```
ttk_tools.dspace._xil_api_offline_stub.TestbenchPortException.__init__ (
              self,
              msg )
```

7.416.2 Member Data Documentation

7.416.2.1 CodeDescription

```
ttk_tools.dspace._xil_api_offline_stub.TestbenchPortException.CodeDescription
```

7.416.2.2 VendorCodeDescription

```
ttk_tools.dspace._xil_api_offline_stub.TestbenchPortException.VendorCodeDescription
```

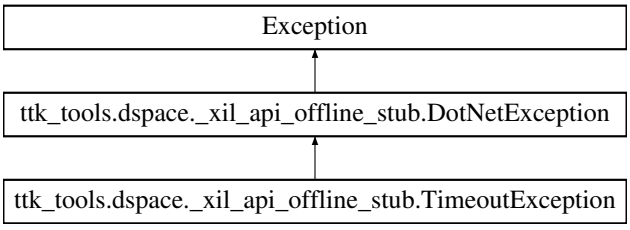
The documentation for this class was generated from the following file:

- ttk_tools/dspace/_xil_api_offline_stub.py (v2.1.2)

7.417 ttk_tools.dspace._xil_api_offline_stub.TimeoutException Class Reference

XIL TestbenchPortException Stub.

Inheritance diagram for ttk_tools.dspace._xil_api_offline_stub.TimeoutException:



7.417.1 Detailed Description

XIL TestbenchPortException Stub.

Non-stubbed hierarchy:

Exception -> (.NET) System.Exception -> TestbenchPortException

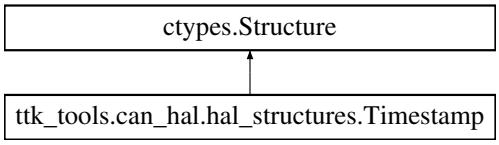
The documentation for this class was generated from the following file:

- ttk_tools/dspace/_xil_api_offline_stub.py (v2.1.2)

7.418 ttk_tools.can_hal.hal_structures.Timestamp Class Reference

Message time stamps (CanHal.h, tCanHalTimestamp), values in µs.

Inheritance diagram for ttk_tools.can_hal.hal_structures.Timestamp:



Static Protected Attributes

- list _fields_

7.418.1 Detailed Description

Message time stamps (CanHal.h, tCanHalTimestamp), values in µs.

7.418.2 Member Data Documentation

7.418.2.1 _fields_

```
list ttk_tools.can_hal.hal_structures.Timestamp._fields_  [static], [protected]
```

Initial value:

```
=  [
        ("HardwareTimeStamp", ctypes.c_uint64), # UINT64 m_ullHardwareTimeStamp
        ("SystemTimeStamp",   ctypes.c_uint64)  # UINT64 m_ullSystemTimeStamp
    ]
```

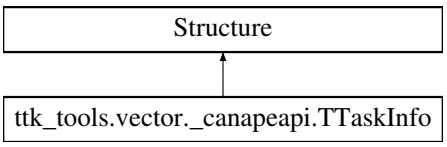
The documentation for this class was generated from the following file:

- ttk_tools/can_hal/hal_structures.py (v1.0.0)

7.419 ttk_tools.vector._canapeapi.TTaskInfo Class Reference

ECU Task data structure as used in CANapeAPI.

Inheritance diagram for ttk_tools.vector._canapeapi.TTaskInfo:



Static Protected Attributes

- int _pack_ = 1
- list _fields_

7.419.1 Detailed Description

ECU Task data structure as used in CANapeAPI.

7.419.2 Member Data Documentation

7.419.2.1 _fields_

```
list ttk_tools.vector._canapeapi.TTaskInfo._fields_  [static], [protected]
```

Initial value:

```
=  [
        ("description", c_char_p), # char *description;       // description text
        ("taskId",      c_ushort), # unsigned short taskId;   // CANape-internal task id number
        ("taskCycle",   c_ulong),  # unsigned long  taskCycle // cycle rate in msec;
        # taskCycle (from CANapAPI.h):
        #   "Cycle rate in msec 0 if not a cyclic task or unknown.         #   In case of modes polling or
      cycle this parameter have no sense."
    ]
```

7.419.2.2 _pack_

```
int ttk_tools.vector._canapeapi.TTaskInfo._pack_ = 1  [static], [protected]
```

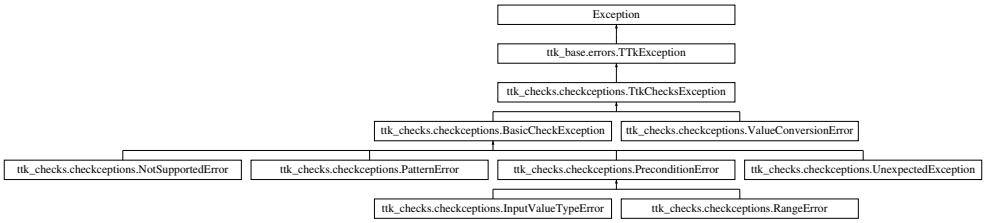
The documentation for this class was generated from the following file:

- ttk_tools/vector/_canapeapi.py (v7.3.3)

7.420 ttk_checks.checkceptions.TtkChecksException Class Reference

Base Exception class for ttk_checks package.

Inheritance diagram for ttk_checks.checkceptions.TtkChecksException:



Public Member Functions

- __init__ (self, msg='', log_msg=True)
  TtkChecksException init.

7.420.1 Detailed Description

Base Exception class for ttk_checks package.

7.420.2 Constructor & Destructor Documentation

7.420.2.1 __init__()

```
ttk_checks.checkceptions.TtkChecksException.__init__ (
              self,
              msg = '',
              log_msg = True )
```

TtkChecksException init.

Parameters

msg	exception message text
log_msg	if True, the exception message will also be logged / printed to stdout

Reimplemented in ttk_checks.checkceptions.PreconditionError, ttk_checks.checkceptions.InputValueTypeError, ttk_checks.checkceptions.RangeError, ttk_checks.checkceptions.UnexpectedException, and ttk_checks.checkceptions.PatternError.

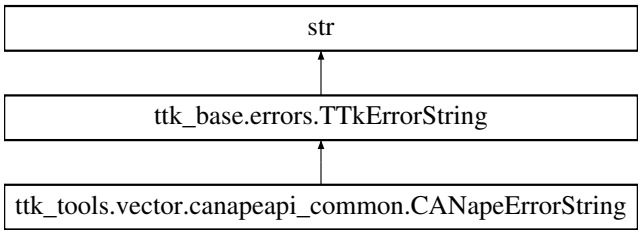
The documentation for this class was generated from the following file:

- ttk_checks/checkceptions.py (v2.0)

7.421 ttk_base.errors.TTkErrorString Class Reference

A wrapper for error message strings so they can be distinguished from "normal" string values (like contents of ASCII-type variables in the case of calibration devices)

Inheritance diagram for ttk_base.errors.TTkErrorString:



7.421.1 Detailed Description

A wrapper for error message strings so they can be distinguished from "normal" string values (like contents of ASCII-type variables in the case of calibration devices)

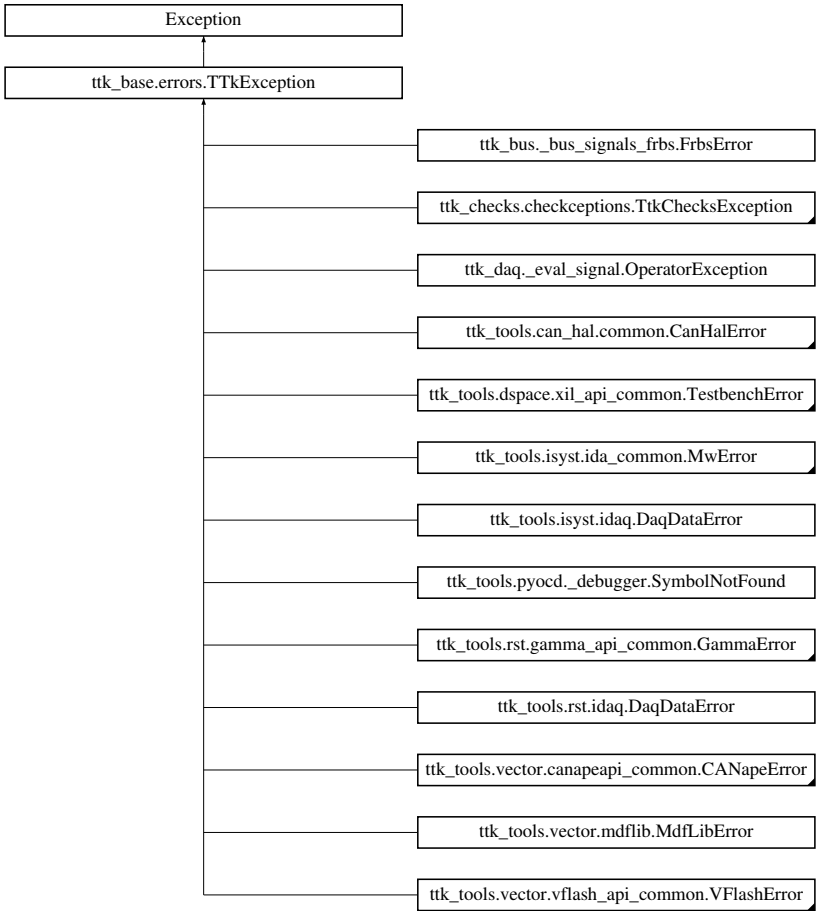
Base Exception for TestToolkit packages.

The documentation for this class was generated from the following file:

- ttk_base/errors.py (v1.2)

7.422 ttk_base.errors.TTkException Class Reference

Inheritance diagram for ttk_base.errors.TTkException:



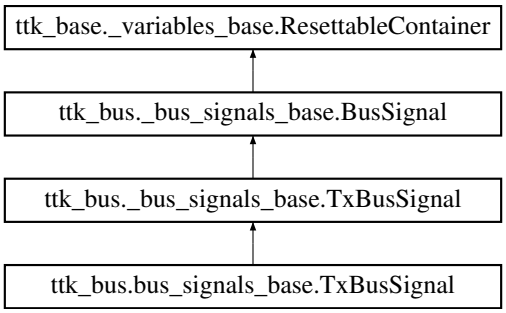
The documentation for this class was generated from the following file:

- ttk_base/errors.py (v1.2)

7.423 ttk_bus._bus_signals_base.TxBusSignal Class Reference

A signal sent from the DUT to the test system (ECU Tx => HIL Rx).

Inheritance diagram for ttk_bus._bus_signals_base.TxBusSignal:



7.423.1 Detailed Description

A signal sent from the DUT to the test system (ECU Tx => HIL Rx).

A signal sent from the test system to the DUT (ECU Rx <= HIL Tx).

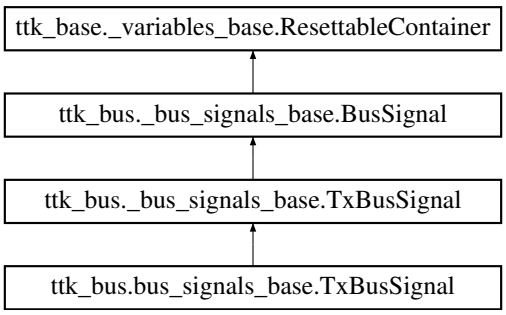
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_base.py (v3.0.1)

7.424 ttk_bus.bus_signals_base.TxBusSignal Class Reference

A signal sent from the DUT to the test system (ECU Tx => HIL Rx).

Inheritance diagram for ttk_bus.bus_signals_base.TxBusSignal:



Public Member Functions

- __init__ (self, context, identifier, cycle_time=0, debounce_time=0, signal_var=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  Bus signal initialization.
- info (self)

  Get a current info string about the HIL signal variable.
- get (self)

  Get the current value of the HIL signal variable.
- set (self, value)

  Set a value to the HIL signal variable.
- getState (self)

  Get a textual state for the HIL signal variable's current data.
- setState (self, state)

  Set a textual state to the HIL signal variable.
- hasStateLookup (self)

  Return True if a state/value lookup table is defined for the HIL signal variable.
- getStateDescr (self, value=None, fallback=None)

  Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).
- hasFormattedRepr (self)

  Return True if a formatting for value representation is defined for the HIL signal variable.
- getFormattedValue (self, value=None)

  Get a formatted representation for the given value using the defined format string of the the HIL signal variable.
- reset (self, *args, **kwargs)

  Call reset() on all contained variables/signals that support it.

Public Attributes

- context

  context for "HilVar" variables
- signal_var

  Main bus signal variable on HIL-side.

7.424.1 Detailed Description

A signal sent from the DUT to the test system (ECU Tx => HIL Rx).

7.424.2 Constructor & Destructor Documentation

7.424.2.1 __init__()

```
ttk_bus.bus_signals_base.TxBusSignal.__init__ (
              self,
              context,
              identifier,
              cycle_time = 0,
              debounce_time = 0,
              signal_var = None,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Bus signal initialization.

Parameters

context	parent context for instantiating HIL variables. This gets mainly used in derived classes
identifier	base signal identifier/name/path in the HIL model (typically this is read-only for Tx-signals and writable for Rx-signals). Derived classes typically use this identifier as starting point for deriving additional helper variables.
cycle_time	cycle time in [ms], 0 if event-triggered
debounce_time	minimum debounce delay time between events in [ms]
signal_var	(opt) an already initialized instance of Var (or most likely one of its derived classes) for the main signal value. If not supplied, a DummyVar will be used/ created as fall back
unit	(opt) unit of physical signal value (as string, mainly used for report entries), defaults to signal_var.unit.
alias	(opt) bus signal alias/name, defaults to signal_var.unit.
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions, defaults to signal_var.unit.
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.424.3 Member Function Documentation

7.424.3.1 get()

```
ttk_bus.bus_signals_base.TxBusSignal.get (
              self )
```

Get the current value of the HIL signal variable.

Returns

the current value.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.424.3.2 getFormattedValue()

```
ttk_bus.bus_signals_base.TxBusSignal.getFormattedValue (
              self,
              value = None )
```

Get a formatted representation for the given value using the defined format string of the the HIL signal variable.

An empty (or non-defined) format string will use the value's default string representation.

Parameters

value	value for which to get a string representation. If None, the last value (read or written) will be used instead.

Returns

a formatted string representation.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.424.3.3 getState()

```
ttk_bus.bus_signals_base.TxBusSignal.getState (
              self )
```

Get a textual state for the HIL signal variable's current data.

This uses the lookup values supplied in constructor. Fallback (if no matching state was found) will be the unmapped numeric value.

Returns

the current state's description.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.424.3.4 getStateDescr()

```
ttk_bus.bus_signals_base.TxBusSignal.getStateDescr (
              self,
              value = None,
              fallback = None )
```

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the last read value will be used instead.
fallback	fallback value to return if no match was found. Default: None

Returns

the state text if a match was found, otherwise the specified fallback value.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.424.3.5 hasFormattedRepr()

```
ttk_bus.bus_signals_base.TxBusSignal.hasFormattedRepr (
              self )
```

Return True if a formatting for value representation is defined for the HIL signal variable.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.424.3.6 hasStateLookup()

```
ttk_bus.bus_signals_base.TxBusSignal.hasStateLookup (
              self )
```

Return True if a state/value lookup table is defined for the HIL signal variable.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.424.3.7 info()

```
ttk_bus.bus_signals_base.TxBusSignal.info (
              self )
```

Get a current info string about the HIL signal variable.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.424.3.8 reset()

```
ttk_bus.bus_signals_base.TxBusSignal.reset (
              self,
             * args,
             ** kwargs )
```

Call reset() on all contained variables/signals that support it.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.ResettableContainer.

7.424.3.9 set()

```
ttk_bus.bus_signals_base.TxBusSignal.set (
              self,
              value )
```

Set a value to the HIL signal variable.

Parameters

value	new value to set

Returns

the previous value.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.424.3.10 setState()

```
ttk_bus.bus_signals_base.TxBusSignal.setState (
              self,
              state )
```

Set a textual state to the HIL signal variable.

This uses the lookup values supplied in constructor.

Parameters

state	named state to set

Returns

the previous state's description.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.424.4 Member Data Documentation

7.424.4.1 context

```
ttk_bus.bus_signals_base.TxBusSignal.context
```

context for "HilVar" variables

7.424.4.2 signal_var

`ttk_bus.bus_signals_base.TxBusSignal.signal_var`

Main bus signal variable on HIL-side.

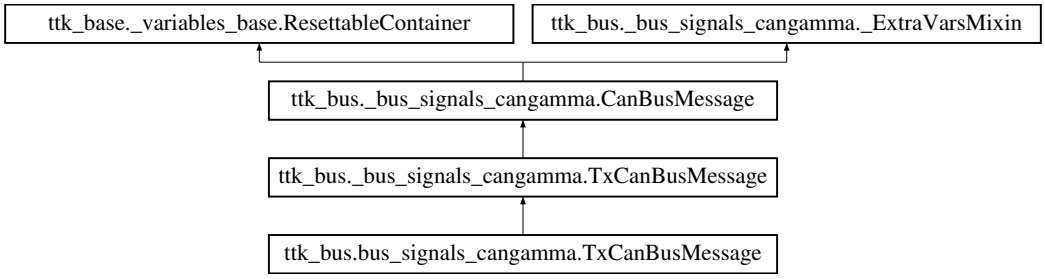
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_base.py (v3.1.0)

7.425 ttk_bus._bus_signals_cangamma.TxCanBusMessage Class Reference

CAN Message/PDU ECU-Tx (ECU-Tx => HIL-Rx).

Inheritance diagram for ttk_bus._bus_signals_cangamma.TxCanBusMessage:



Public Member Functions

- __init__ (self, context, base_identifier=None, cycle_time=0, debounce_time=0, alias="", descr='', change_delay=None, timeout=None, recovery_time=None)

  Tx CAN Message data (ECU-Tx => HIL-Rx).

7.425.1 Detailed Description

CAN Message/PDU ECU-Tx (ECU-Tx => HIL-Rx).

7.425.2 Constructor & Destructor Documentation

7.425.2.1 __init__()

```
ttk_bus._bus_signals_cangamma.TxCanBusMessage.__init__ (
              self,
              context,
              base_identifier = None,
              cycle_time = 0,
              debounce_time = 0,
              alias = "",
              descr = '',
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Tx CAN Message data (ECU-Tx => HIL-Rx).

Parameters

context	parent gamma_api context
base_identifier	base identifier to PDU/message. Helper entries will be derived from this, e.g. kickout or status variables (though this is mostly relevant to messages sent to DUT, see RxCanBusMessage)
cycle_time	cycle time in [ms]. Event-triggered messages have a cycle time of 0 (similar to DBC value)
debounce_time	minimum debounce delay between events in [ms]
alias	Message alias/name
descr	(opt) Message description (may be used in report entries)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_cangamma.CanBusMessage.

Reimplemented in ttk_bus.bus_signals_cangamma.TxCanBusMessage.

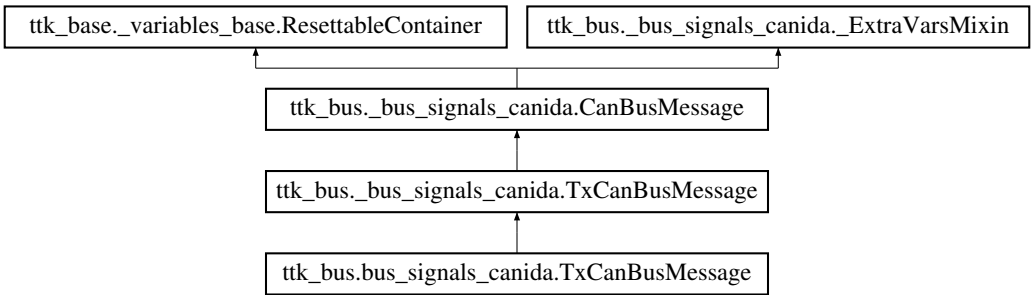
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_cangamma.py (v1.12)

7.426 ttk_bus._bus_signals_canida.TxCanBusMessage Class Reference

CAN Message/PDU ECU-Tx (ECU-Tx => HIL-Rx).

Inheritance diagram for ttk_bus._bus_signals_canida.TxCanBusMessage:



Public Member Functions

- __init__ (self, context, base_identifier=None, cycle_time=0, debounce_time=0, alias="", descr='', change_delay=None, timeout=None, recovery_time=None)

  Tx CAN Message data (ECU-Tx => HIL-Rx).

7.426.1 Detailed Description

CAN Message/PDU ECU-Tx (ECU-Tx => HIL-Rx).

7.426.2 Constructor & Destructor Documentation

7.426.2.1 __init__()

```
ttk_bus._bus_signals_canida.TxCanBusMessage.__init__ (
                self,
                context,
                base_identifier = None,
                cycle_time = 0,
                debounce_time = 0,
                alias = "",
                descr = '',
                change_delay = None,
                timeout = None,
                recovery_time = None )
```

Tx CAN Message data (ECU-Tx => HIL-Rx).

Parameters

context	parent iDA context
base_identifier	base identifier to PDU/message. Helper entries will be derived from this, e.g. kickout or status variables (though this is mostly relevant to messages sent to DUT, see RxCanBusMessage) e.g. ecu-rx (hil-tx): "CAN.can3_iFIU_iSRC.HIL_RX.N0FIU_Status_Card19"
cycle_time	cycle time in [ms]. Event-triggered messages have a cycle time of 0 (similar to DBC value)
debounce_time	minimum debounce delay between events in [ms]
alias	Message alias/name
descr	(opt) Message description (may be used in report entries)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_canida.CanBusMessage.

Reimplemented in ttk_bus.bus_signals_canida.TxCanBusMessage.

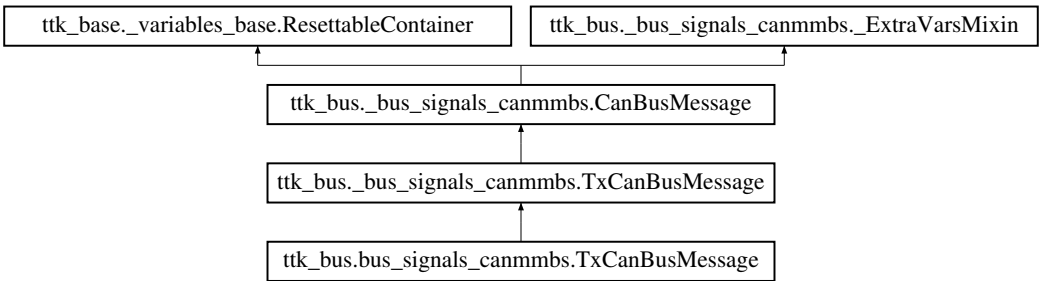
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_canida.py (v1.0.2)

7.427 ttk_bus._bus_signals_canmmbs.TxCanBusMessage Class Reference

CAN Message/PDU ECU-Tx (ECU-Tx => HIL-Rx).

Inheritance diagram for ttk_bus._bus_signals_canmmbs.TxCanBusMessage:



Public Member Functions

- __init__ (self, context, bus_systems_path=None, cycle_time=0, debounce_time=0, alias="", descr='', change_delay=None, timeout=None, recovery_time=None)

  Tx CAN Message data (ECU-Tx => HIL-Rx).

Public Attributes

- deltatime_var

  Message Rx delta time.
- time_var

  Message Rx time (in seconds)

7.427.1 Detailed Description

CAN Message/PDU ECU-Tx (ECU-Tx => HIL-Rx).

Extends the basic CanBusMessage with access to time_var (rx time of message) and deltatime_var (rx delta time of message)

7.427.2 Constructor & Destructor Documentation

7.427.2.1 __init__()

```
ttk_bus._bus_signals_canmmbs.TxCanBusMessage.__init__ (
              self,
              context,
              bus_systems_path = None,
              cycle_time = 0,
              debounce_time = 0,
              alias = "",
              descr = '',
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Tx CAN Message data (ECU-Tx => HIL-Rx).

Parameters

context	parent context (an RTE application instance to resolve signal/variable paths)
bus_systems_path	base path to PDU/message below BusSystems, use `BusSystems/CAN/<CAN_NAME>/<MSG_NAME>/RX/<MSG_NAME>` Helper entries will be derived from this path, e.g. kickout or status variables (though this is more relevant to messages sent to DUT, see RxCanBusMessage)
cycle_time	cycle time in [ms]. Event-triggered messages have a cycle time of 0 (similar to DBC value)
debounce_time	minimum debounce delay between events in [ms]
alias	Message alias/name
descr	(opt) Message description (may be used in report entries)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_canmmbs.CanBusMessage.

Reimplemented in ttk_bus.bus_signals_canmmbs.TxCanBusMessage.

7.427.3 Member Data Documentation

7.427.3.1 deltatime_var

`ttk_bus._bus_signals_canmmbs.TxCanBusMessage.deltatime_var`

Message Rx delta time.

Time between current and previous received message in seconds.

7.427.3.2 time_var

`ttk_bus._bus_signals_canmmbs.TxCanBusMessage.time_var`

Message Rx time (in seconds)

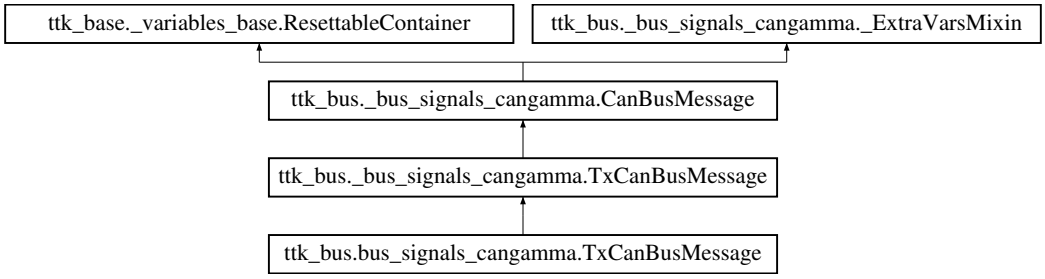
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_canmmbs.py (v2.3)

7.428 ttk_bus.bus_signals_cangamma.TxCanBusMessage Class Reference

CAN Message/PDU ECU-Tx (ECU-Tx => HIL-Rx).

Inheritance diagram for ttk_bus.bus_signals_cangamma.TxCanBusMessage:



Public Member Functions

- __init__ (self, context, base_identifier=None, cycle_time=0, debounce_time=0, alias="", descr='', change_delay=None, timeout=None, recovery_time=None)

  Tx CAN Message data (ECU-Tx => HIL-Rx).
- reset (self, *args, **kwargs)

  Call reset() on all contained variables/signals that support it.

Public Attributes

- time_var

  Message Rx time variable (in seconds)
- length_var

  Message payload data length.
- context

  Parent context (a gaapi reference to resolve signal/variable paths)
- alias

  Message alias name.
- descr

  Message description text.
- base_identifier

  Base BusSystems path to Message/PDU helper variables e.g.
- cycle_time

  cycle time of Message/PDU in [ms]
- debounce_time

  minimum debounce delay for event-triggered Messages/PDUs in [ms]
- change_delay

  delay to wait after a value change [ms]
- timeout

  detection time for timeout errors [ms]
- recovery_time

  time for recovery/"wiedergut" checks [ms]

7.428.1 Detailed Description

CAN Message/PDU ECU-Tx (ECU-Tx => HIL-Rx).

7.428.2 Constructor & Destructor Documentation

7.428.2.1 __init__()

```
ttk_bus.bus_signals_cangamma.TxCanBusMessage.__init__ (
               self,
               context,
               base_identifier = None,
               cycle_time = 0,
               debounce_time = 0,
               alias = "",
               descr = '',
               change_delay = None,
               timeout = None,
               recovery_time = None )
```

Tx CAN Message data (ECU-Tx => HIL-Rx).

Parameters

context	parent gamma_api context
base_identifier	base identifier to PDU/message. Helper entries will be derived from this, e.g. kickout or status variables (though this is mostly relevant to messages sent to DUT, see RxCanBusMessage) e.g. "BusSystems.<CAN_NAME>.HIL_RX.<MSG_NAME>"
cycle_time	cycle time in [ms]. Event-triggered messages have a cycle time of 0 (similar to DBC value)
debounce_time	minimum debounce delay between events in [ms]
alias	Message alias/name
descr	(opt) Message description (may be used in report entries)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_cangamma.TxCanBusMessage.

7.428.3 Member Function Documentation

7.428.3.1 reset()

```
ttk_bus.bus_signals_cangamma.TxCanBusMessage.reset (
               self,
              * args,
              ** kwargs )
```

Call reset() on all contained variables/signals that support it.

Simple, non-recursive approach.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.ResettableContainer.

7.428.4 Member Data Documentation

7.428.4.1 alias

`ttk_bus.bus_signals_cangamma.TxCanBusMessage.alias`

Message alias name.

7.428.4.2 base_identifier

`ttk_bus.bus_signals_cangamma.TxCanBusMessage.base_identifier`

Base BusSystems path to Message/PDU helper variables e.g.

"BusSystems.<CAN_NAME>.HIL_RX.<MSG_NAME>"

7.428.4.3 change_delay

`ttk_bus.bus_signals_cangamma.TxCanBusMessage.change_delay`

delay to wait after a value change [ms]

7.428.4.4 context

`ttk_bus.bus_signals_cangamma.TxCanBusMessage.context`

Parent context (a gaapi reference to resolve signal/variable paths)

7.428.4.5 cycle_time

`ttk_bus.bus_signals_cangamma.TxCanBusMessage.cycle_time`

cycle time of Message/PDU in [ms]

7.428.4.6 debounce_time

`ttk_bus.bus_signals_cangamma.TxCanBusMessage.debounce_time`

minimum debounce delay for event-triggered Messages/PDUs in [ms]

7.428.4.7 descr

`ttk_bus.bus_signals_cangamma.TxCanBusMessage.descr`

Message description text.

7.428.4.8 length_var

`ttk_bus.bus_signals_cangamma.TxCanBusMessage.length_var`

Message payload data length.

7.428.4.9 recovery_time

`ttk_bus.bus_signals_cangamma.TxCanBusMessage.recovery_time`

time for recovery/"wiedergut" checks [ms]

7.428.4.10 time_var

`ttk_bus.bus_signals_cangamma.TxCanBusMessage.time_var`

Message Rx time variable (in seconds)

7.428.4.11 timeout

`ttk_bus.bus_signals_cangamma.TxCanBusMessage.timeout`

detection time for timeout errors [ms]

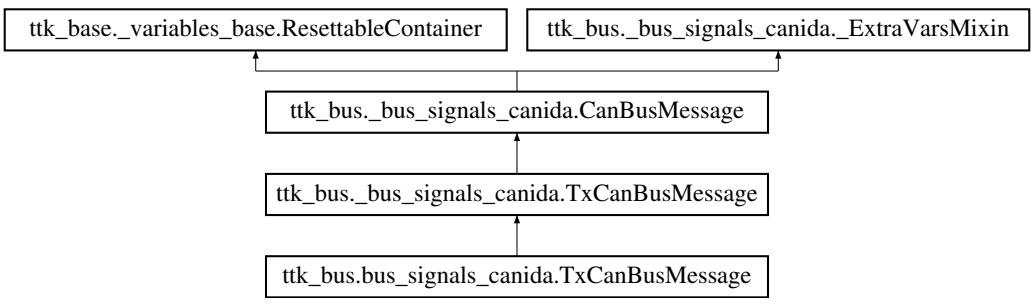
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_cangamma.py (v1.3)

7.429 ttk_bus.bus_signals_canida.TxCanBusMessage Class Reference

CAN Message/PDU ECU-Tx (ECU-Tx => HIL-Rx).

Inheritance diagram for ttk_bus.bus_signals_canida.TxCanBusMessage:



Public Member Functions

- __init__ (self, context, base_identifier=None, cycle_time=0, debounce_time=0, alias="", descr='', change_delay=None, timeout=None, recovery_time=None)

  Tx CAN Message data (ECU-Tx => HIL-Rx).
- reset (self, *args, **kwargs)

  Call reset() on all contained variables/signals that support it.

Public Attributes

- time_var

  Message Rx time variable (in seconds)
- length_var

  Message payload data length.
- context

  Parent context (a gaapi reference to resolve signal/variable paths)
- alias

  Message alias name.
- descr

  Message description text.
- base_identifier

  Base Bus path to Message/PDU helper variables e.g.
- cycle_time

  cycle time of Message/PDU in [ms]
- debounce_time

  minimum debounce delay for event-triggered Messages/PDUs in [ms]
- change_delay

  delay to wait after a value change [ms]
- timeout

  detection time for timeout errors [ms]
- recovery_time

  time for recovery/"wiedergut" checks [ms]

7.429.1 Detailed Description

CAN Message/PDU ECU-Tx (ECU-Tx => HIL-Rx).

7.429.2 Constructor & Destructor Documentation

7.429.2.1 __init__()

```
ttk_bus.bus_signals_canida.TxCanBusMessage.__init__ (
              self,
              context,
              base_identifier = None,
              cycle_time = 0,
              debounce_time = 0,
              alias = "",
              descr = '',
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Tx CAN Message data (ECU-Tx => HIL-Rx).

Parameters

context	parent iDA context
base_identifier	base identifier to PDU/message. Helper entries will be derived from this, e.g. kickout or status variables (though this is mostly relevant to messages sent to DUT, see RxCanBusMessage) e.g. ecu-rx (hil-tx): "CAN.can3_iFIU_iSRC.HIL_RX.N0FIU_Status_Card19"
cycle_time	cycle time in [ms]. Event-triggered messages have a cycle time of 0 (similar to DBC value)
debounce_time	minimum debounce delay between events in [ms]
alias	Message alias/name
descr	(opt) Message description (may be used in report entries)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_canida.TxCanBusMessage.

7.429.3 Member Function Documentation

7.429.3.1 reset()

```
ttk_bus.bus_signals_canida.TxCanBusMessage.reset (
              self,
             * args,
             ** kwargs )
```

Call reset() on all contained variables/signals that support it.

Simple, non-recursive approach.

Parameters

∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.ResettableContainer.

7.429.4 Member Data Documentation

7.429.4.1 alias

```
ttk_bus.bus_signals_canida.TxCanBusMessage.alias
```

Message alias name.

7.429.4.2 base_identifier

ttk_bus.bus_signals_canida.TxCanBusMessage.base_identifier

Base Bus path to Message/PDU helper variables e.g.

"CAN.<CAN_NAME>.HIL_RX.<MSG_NAME>"

7.429.4.3 change_delay

ttk_bus.bus_signals_canida.TxCanBusMessage.change_delay

delay to wait after a value change [ms]

7.429.4.4 context

ttk_bus.bus_signals_canida.TxCanBusMessage.context

Parent context (a gaapi reference to resolve signal/variable paths)

7.429.4.5 cycle_time

ttk_bus.bus_signals_canida.TxCanBusMessage.cycle_time

cycle time of Message/PDU in [ms]

7.429.4.6 debounce_time

ttk_bus.bus_signals_canida.TxCanBusMessage.debounce_time

minimum debounce delay for event-triggered Messages/PDUs in [ms]

7.429.4.7 descr

ttk_bus.bus_signals_canida.TxCanBusMessage.descr

Message description text.

7.429.4.8 length_var

ttk_bus.bus_signals_canida.TxCanBusMessage.length_var

Message payload data length.

7.429.4.9 recovery_time

`ttk_bus.bus_signals_canida.TxCanBusMessage.recovery_time`

time for recovery/"wiedergut" checks [ms]

7.429.4.10 time_var

`ttk_bus.bus_signals_canida.TxCanBusMessage.time_var`

Message Rx time variable (in seconds)

7.429.4.11 timeout

`ttk_bus.bus_signals_canida.TxCanBusMessage.timeout`

detection time for timeout errors [ms]

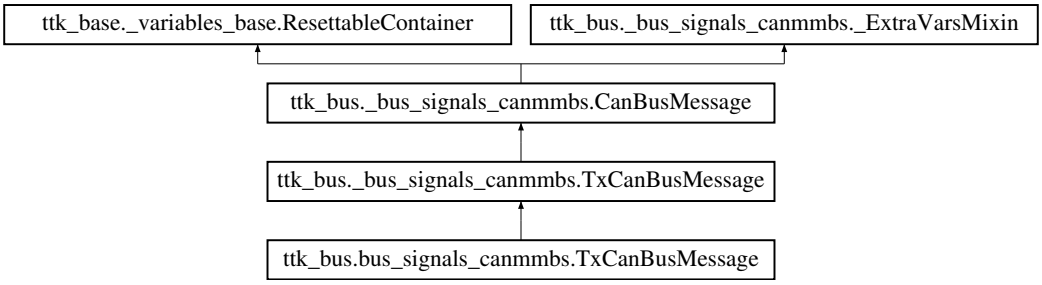
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_canida.py (v1.0.0)

7.430 ttk_bus.bus_signals_canmmbs.TxCanBusMessage Class Reference

CAN Message/PDU ECU-Tx (ECU-Tx => HIL-Rx).

Inheritance diagram for ttk_bus.bus_signals_canmmbs.TxCanBusMessage:



Public Member Functions

- __init__ (self, context, bus_systems_path=None, cycle_time=0, debounce_time=0, alias="", descr='', change_delay=None, timeout=None, recovery_time=None)

  Tx CAN Message data (ECU-Tx => HIL-Rx).
- reset (self, *args, **kwargs)

  Call reset() on all contained variables/signals that support it.

Public Attributes

- deltatime_var
  Message Rx delta time variable.
- time_var
  Message Rx time variable (in seconds) (a HilVar instance)
- status_var
  HIL-Rx-status variable (a HilVar instance)
- context
  Parent context (an RTE application instance to resolve signal/variable paths)
- alias
  Message alias name.
- descr
  Message description text.
- bus_systems_path
  Base BusSystems path to Message/PDU helper variables below BusSystems e.g.
- cycle_time
  cycle time of Message/PDU in [ms]
- debounce_time
  minimum debounce delay for event-triggered Messages/PDUs in [ms]
- change_delay
  delay to wait after a value change [ms]
- timeout
  detection time for timeout errors [ms]
- recovery_time
  time for recovery/"wiedergut" checks [ms]

7.430.1 Detailed Description

CAN Message/PDU ECU-Tx (ECU-Tx => HIL-Rx).

Extends the basic CanBusMessage with access to time_var (rx time of message) and deltatime_var (rx delta time of message)

7.430.2 Constructor & Destructor Documentation

7.430.2.1 __init__()

```
ttk_bus.bus_signals_canmmbs.TxCanBusMessage.__init__ (
              self,
              context,
              bus_systems_path = None,
              cycle_time = 0,
              debounce_time = 0,
              alias = "",
              descr = '',
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Tx CAN Message data (ECU-Tx => HIL-Rx).

Parameters

context	parent context (an RTE application instance to resolve signal/variable paths)
bus_systems_path	base path to PDU/message below BusSystems, use `BusSystems/CAN/<CAN↩ _NAME>/<MSG_NAME>/RX/<MSG_NAME>` Helper entries will be derived from this path, e.g. kickout or status variables (though this is more relevant to messages sent to DUT, see RxCanBusMessage)
cycle_time	cycle time in [ms]. Event-triggered messages have a cycle time of 0 (similar to DBC value)
debounce_time	minimum debounce delay between events in [ms]
alias	Message alias/name
descr	(opt) Message description (may be used in report entries)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_canmmbs.TxCanBusMessage.

7.430.3 Member Function Documentation

7.430.3.1 reset()

```
ttk_bus.bus_signals_canmmbs.TxCanBusMessage.reset (
              self,
             * args,
             ** kwargs )
```

Call reset() on all contained variables/signals that support it.

Simple, non-recursive approach.

Parameters

∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.ResettableContainer.

7.430.4 Member Data Documentation

7.430.4.1 alias

```
ttk_bus.bus_signals_canmmbs.TxCanBusMessage.alias
```

Message alias name.

7.430.4.2 bus_systems_path

ttk_bus.bus_signals_canmmbs.TxCanBusMessage.bus_systems_path

Base BusSystems path to Message/PDU helper variables below BusSystems e.g.

"BusSystems/CAN/<CAN_NAME>/<MSG_NAME>/RX/<MSG_NAME>"

7.430.4.3 change_delay

ttk_bus.bus_signals_canmmbs.TxCanBusMessage.change_delay

delay to wait after a value change [ms]

7.430.4.4 context

ttk_bus.bus_signals_canmmbs.TxCanBusMessage.context

Parent context (an RTE application instance to resolve signal/variable paths)

7.430.4.5 cycle_time

ttk_bus.bus_signals_canmmbs.TxCanBusMessage.cycle_time

cycle time of Message/PDU in [ms]

7.430.4.6 debounce_time

ttk_bus.bus_signals_canmmbs.TxCanBusMessage.debounce_time

minimum debounce delay for event-triggered Messages/PDUs in [ms]

7.430.4.7 deltatime_var

ttk_bus.bus_signals_canmmbs.TxCanBusMessage.deltatime_var

Message Rx delta time variable.

Time between current and previous received message in seconds (a HilVar instance)

7.430.4.8 descr

ttk_bus.bus_signals_canmmbs.TxCanBusMessage.descr

Message description text.

7.430.4.9 recovery_time

`ttk_bus.bus_signals_canmmbs.TxCanBusMessage.recovery_time`

time for recovery/"wiedergut" checks [ms]

7.430.4.10 status_var

`ttk_bus.bus_signals_canmmbs.TxCanBusMessage.status_var`

HIL-Rx-status variable (a HilVar instance)

7.430.4.11 time_var

`ttk_bus.bus_signals_canmmbs.TxCanBusMessage.time_var`

Message Rx time variable (in seconds) (a HilVar instance)

7.430.4.12 timeout

`ttk_bus.bus_signals_canmmbs.TxCanBusMessage.timeout`

detection time for timeout errors [ms]

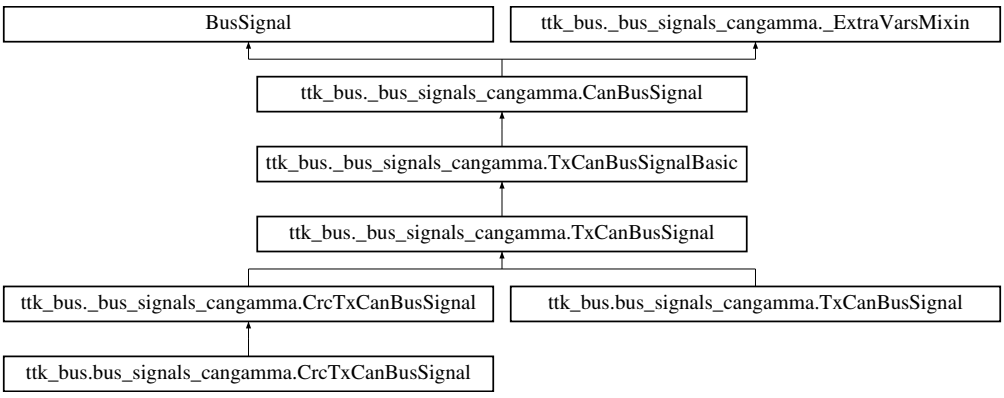
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_canmmbs.py (v1.8)

7.431 ttk_bus._bus_signals_cangamma.TxCanBusSignal Class Reference

A CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).

Inheritance diagram for ttk_bus._bus_signals_cangamma.TxCanBusSignal:



Public Member Functions

- __init__ (self, context, signal_identifier, message=None, cycle_time=None, debounce_time=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  Tx bus signal initialization (ECU Tx => HIL Rx).

Public Attributes

- signal_raw_var

  Raw value of signal.
- signal_phy_var

  Physical value of signal.

7.431.1 Detailed Description

A CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).

Standard implementation, accesses converted physical or raw values.

Use this for signal entries generated with iSyDBC mode "standard" . See _bus_signals_cangamma.TxCanBus↩SignalBasic.

7.431.2 Constructor & Destructor Documentation

7.431.2.1 __init__()

```
ttk_bus._bus_signals_cangamma.TxCanBusSignal.__init__ (
                self,
                context,
                signal_identifier,
                message = None,
                cycle_time = None,
                debounce_time = None,
                unit = "",
                alias = "",
                descr = "",
                lookup = None,
                change_delay = None,
                timeout = None,
                recovery_time = None )
```

Tx bus signal initialization (ECU Tx => HIL Rx).

Parameters

context	parent gamma_api context
signal_identifier	identifier for signal value PV
message	reference to associated TxCanBusMessage instance
cycle_time	(opt) cycle time in [ms], overrides settings in message
debounce_time	(opt) minimum debounce delay between events in [ms], overrides settings in Message
unit	(opt) unit of physical signal value (as string, mainly used for report entries)

Parameters

alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

signal_identifier:

HIL-Rx/ECU-Tx signals will have the following "value" PV elements:

- `.value_phy` - physical representation of value
- `.value_raw` - raw representation of value
- `.value` - value as received from CAN-Plugin (typically "raw")

Usually, the "phy" variant will be used, e.g. something like `signal_identifier="BusSystems.↩ can.HIL_TX.MessageName.signals.SignalName.value_phy"`

Reimplemented from ttk_bus._bus_signals_cangamma.TxCanBusSignalBasic.

Reimplemented in ttk_bus._bus_signals_cangamma.CrcTxCanBusSignal, ttk_bus.bus_signals_cangamma.CrcTxCanBusSignal, and ttk_bus.bus_signals_cangamma.TxCanBusSignal.

7.431.3 Member Data Documentation

7.431.3.1 signal_phy_var

`ttk_bus._bus_signals_cangamma.TxCanBusSignal.signal_phy_var`

Physical value of signal.

Note that the default self.signal_var usually maps to member "value_phy", (depending on supplied signal_identifier).

7.431.3.2 signal_raw_var

`ttk_bus._bus_signals_cangamma.TxCanBusSignal.signal_raw_var`

Raw value of signal.

Note that the default self.signal_var usually maps to member "value_phy", (depending on supplied signal_identifier).

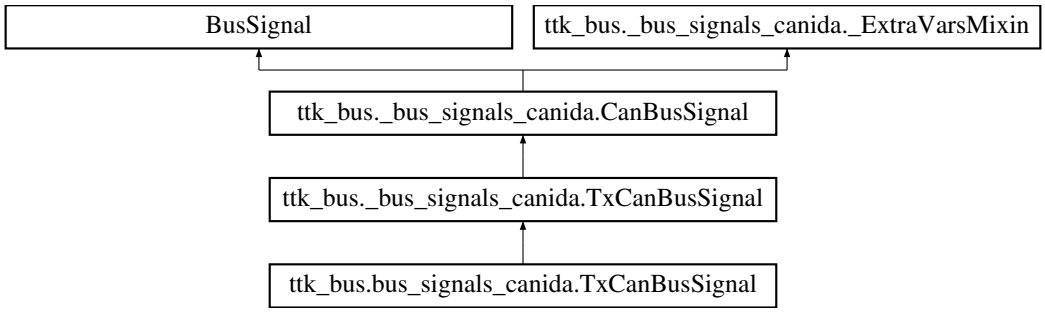
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_cangamma.py (v1.12)

7.432 ttk_bus._bus_signals_canida.TxCanBusSignal Class Reference

A basic CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).

Inheritance diagram for ttk_bus._bus_signals_canida.TxCanBusSignal:



Public Member Functions

- __init__ (self, context, signal_identifier, message=None, cycle_time=None, debounce_time=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  Basic Tx bus signal initialization (ECU Tx => HIL Rx).

7.432.1 Detailed Description

A basic CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).

Basic implementation, directly accesses raw .value as-is from CAN-Plugin, without a converted physical value or error injections.

7.432.2 Constructor & Destructor Documentation

7.432.2.1 __init__()

```
ttk_bus._bus_signals_canida.TxCanBusSignal.__init__ (
              self,
              context,
              signal_identifier,
              message = None,
              cycle_time = None,
              debounce_time = None,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Basic Tx bus signal initialization (ECU Tx => HIL Rx).

Note

`signal_identifier` var name Use `.value` entry (or optionally `.value_phy`/`.value_raw`, if available), e.g. `"CAN.CanName.HIL_RX.MessageName.signals.SignalName.value"`
or just the base signal identifier to default to use `.value`:
`"CAN.CanName.HIL_RX.MessageName.signals.SignalName"`

Parameters

context	parent iDA context
signal_identifier	identifier for signal value var (see note)
message	reference to associated TxCanBusMessage instance
cycle_time	(opt) cycle time in [ms], overrides settings in message
debounce_time	(opt) minimum debounce delay between events in [ms], overrides settings in Message
unit	(opt) unit of physical signal value (as string, mainly used for report entries)
alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

Minimal/Basic Signals:

ECU-Tx/HIL-Rx Signals will only contain variables for:

- `<SignalName>.value` - raw value as received from CAN-Plugin
- `<SignalName>.timestamp` - Rx time stamp
- `<SignalName>.length` - message length

Reimplemented from ttk_bus._bus_signals_canida.CanBusSignal.

Reimplemented in ttk_bus.bus_signals_canida.TxCanBusSignal.

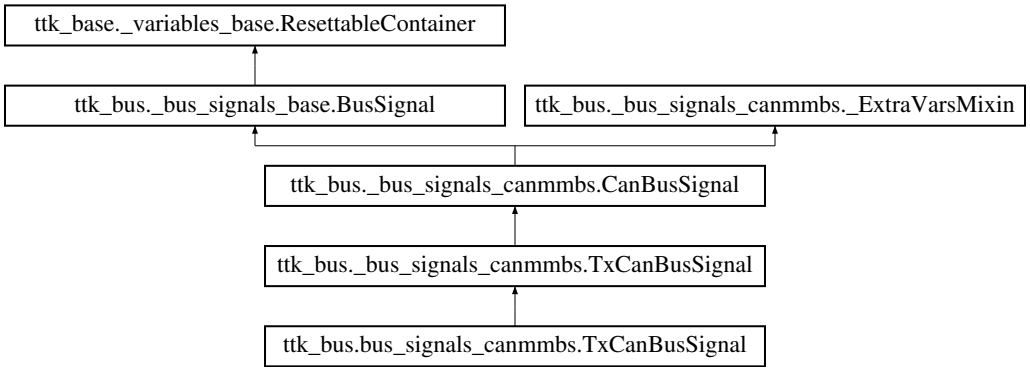
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_canida.py (v1.0.2)

7.433 ttk_bus._bus_signals_canmmbs.TxCanBusSignal Class Reference

A CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).

Inheritance diagram for ttk_bus._bus_signals_canmmbs.TxCanBusSignal:



7.433.1 Detailed Description

A CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).

See _bus_signals_canmmbs.CanBusSignal.

Note

Right now, this class does not add any further functionality

A CAN signal sent from the test system to the DUT (ECU Rx <= HIL Tx).

Note

RxCanBusSignal requires at least an available input manipulation via TRC/BusSystems (which is the default setting in CANMM, but can be disabled).

See _bus_signals_canmmbs.CanBusSignal.

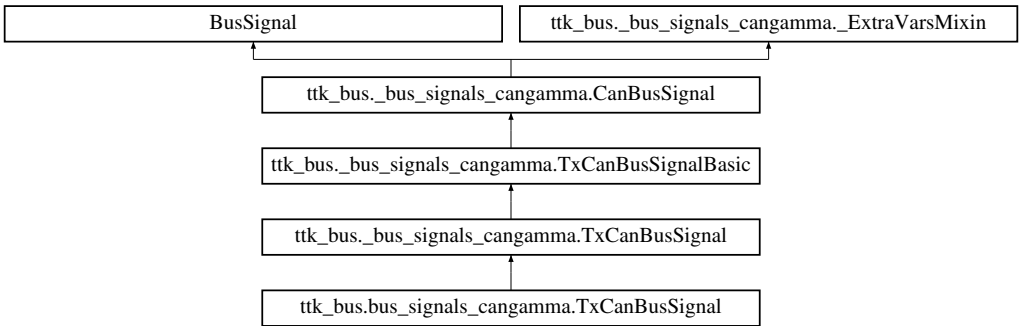
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_canmmbs.py (v2.3)

7.434 ttk_bus.bus_signals_cangamma.TxCanBusSignal Class Reference

A CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).

Inheritance diagram for ttk_bus.bus_signals_cangamma.TxCanBusSignal:



Public Member Functions

- __init__ (self, context, signal_identifier, message=None, cycle_time=None, debounce_time=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  Tx bus signal initialization (ECU Tx => HIL Rx).
- info (self)

  Get a current info string about the HIL signal variable, see `signal_var`
- get (self)

  Get the current value of the HIL signal variable, see `signal_var`
- set (self, value)

  Set a value to the HIL signal variable, see `signal_var`
- getState (self)

  Get a textual state for the HIL signal variable's current data, see `signal_var`.
- setState (self, state)

  Set a textual state to the HIL signal variable, see `signal_var`.
- getStateDescr (self, value=None, fallback=None)

  Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).
- reset (self, *args, **kwargs)

  Reset all resettable variables of this bus signal.

Public Attributes

- base_identifier
  Signal PV address/identifier as base identifier for deriving sub-elements.
- message
  reference to associated (Tx-)CanBusMessage instance
- context
  parent context (a gaapi reference to resolve signal/variable paths)
- signal_var
  Signal value variable, default representation.
- signal_raw_var
  Raw signal value variable.
- signal_phy_var
  Physical signal value variable.
- unit
  unit of physical signal value (as string, mainly used for report entries)
- alias
  bus signal alias name
- descr
  bus signal description text
- cycle_time
  cycle time of Message/PDU in [ms]
- debounce_time
  minimum debounce delay for event-triggered Messages/PDUs in [ms]
- change_delay
  delay to wait after a value change [ms]
- timeout
  detection time for timeout errors [ms]
- recovery_time
  time for recovery/"wiedergut" checks [ms]

7.434.1 Detailed Description

A CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).

Standard implementation, accesses converted physical or raw values.

Use this for signal entries generated with iSyDBC mode "standard" .

7.434.2 Constructor & Destructor Documentation

7.434.2.1 __init__()

```
ttk_bus.bus_signals_cangamma.TxCanBusSignal.__init__ (
              self,
              context,
              signal_identifier,
              message = None,
              cycle_time = None,
              debounce_time = None,
```

```
            unit = "",
            alias = "",
            descr = "",
            lookup = None,
            change_delay = None,
            timeout = None,
            recovery_time = None )
```

Tx bus signal initialization (ECU Tx => HIL Rx).

Parameters

context	parent gamma_api context
signal_identifier	identifier for signal value PV
message	reference to associated TxCanBusMessage instance
cycle_time	(opt) cycle time in [ms], overrides settings in message
debounce_time	(opt) minimum debounce delay between events in [ms], overrides settings in Message
unit	(opt) unit of physical signal value (as string, mainly used for report entries)
alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

signal_identifier:

HIL-Rx/ECU-Tx signals will have the following "value" PV elements:

- `.value_phy` - physical representation of value
- `.value_raw` - raw representation of value
- `.value` - value as received from CAN-Plugin (typically "raw")

Usually, the "phy" variant will be used, e.g. something like `signal_identifier="BusSystems.↩ can.HIL_TX.MessageName.signals.SignalName.value_phy"`

Reimplemented from ttk_bus._bus_signals_cangamma.TxCanBusSignal.

7.434.3 Member Function Documentation

7.434.3.1 get()

```
ttk_bus.bus_signals_cangamma.TxCanBusSignal.get (
            self )
```

Get the current value of the HIL signal variable, see `signal_var`

Returns

the current value from `signal_var`.

7.434.3.2 getState()

```
ttk_bus.bus_signals_cangamma.TxCanBusSignal.getState (
              self )
```

Get a textual state for the HIL signal variable's current data, see `signal_var`.

This uses the lookup values supplied in constructor. Fallback (if no matching state was found) will be the unmapped numeric value.

Returns

the current state representation from `signal_var`.

7.434.3.3 getStateDescr()

```
ttk_bus.bus_signals_cangamma.TxCanBusSignal.getStateDescr (
              self,
              value = None,
              fallback = None )
```

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the last read value will be used instead.
fallback	fallback value to return if no match was found. Default: None

Returns

the state text if a match was found, otherwise the specified fallback value.

7.434.3.4 info()

```
ttk_bus.bus_signals_cangamma.TxCanBusSignal.info (
              self )
```

Get a current info string about the HIL signal variable, see `signal_var`

Returns

the current info string from `signal_var`.

7.434.3.5 reset()

```
ttk_bus.bus_signals_cangamma.TxCanBusSignal.reset (
              self,
             * args,
             ** kwargs )
```

Reset all resettable variables of this bus signal.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

7.434.3.6 set()

```
ttk_bus.bus_signals_cangamma.TxCanBusSignal.set (
              self,
              value )
```

Set a value to the HIL signal variable, see `signal_var`

Attention

For TxCanBusSignals, this will only overwrite the current contents of the HIl receive buffer of the signal (which is typically not too useful as it will be overwritten on next update with data from CAN plugin)

Parameters

value	value to set

Returns

the previous value from `signal_var`.

7.434.3.7 setState()

```
ttk_bus.bus_signals_cangamma.TxCanBusSignal.setState (
              self,
              state )
```

Set a textual state to the HIL signal variable, see `signal_var`.

This uses the lookup values supplied in constructor.

Attention

For TxFrBusSignals this will only overwrite the current contents of the HIl receive buffer of the signal (which typically is not too useful).

Parameters

state	named state to set

Returns

the previous state representation from `signal_var`.

7.434.4 Member Data Documentation

7.434.4.1 alias

`ttk_bus.bus_signals_cangamma.TxCanBusSignal.alias`

bus signal alias name

7.434.4.2 base_identifier

`ttk_bus.bus_signals_cangamma.TxCanBusSignal.base_identifier`

Signal PV address/identifier as base identifier for deriving sub-elements.

For CAN/Gamma signals, related variables follow the form `<base_identifier>.<suffix>`

7.434.4.3 change_delay

`ttk_bus.bus_signals_cangamma.TxCanBusSignal.change_delay`

delay to wait after a value change [ms]

7.434.4.4 context

`ttk_bus.bus_signals_cangamma.TxCanBusSignal.context`

parent context (a gaapi reference to resolve signal/variable paths)

7.434.4.5 cycle_time

`ttk_bus.bus_signals_cangamma.TxCanBusSignal.cycle_time`

cycle time of Message/PDU in [ms]

7.434.4.6 debounce_time

`ttk_bus.bus_signals_cangamma.TxCanBusSignal.debounce_time`

minimum debounce delay for event-triggered Messages/PDUs in [ms]

7.434.4.7 descr

ttk_bus.bus_signals_cangamma.TxCanBusSignal.descr

bus signal description text

7.434.4.8 message

ttk_bus.bus_signals_cangamma.TxCanBusSignal.message

reference to associated (Tx-)CanBusMessage instance

7.434.4.9 recovery_time

ttk_bus.bus_signals_cangamma.TxCanBusSignal.recovery_time

time for recovery/"wiedergut" checks [ms]

7.434.4.10 signal_phy_var

ttk_bus.bus_signals_cangamma.TxCanBusSignal.signal_phy_var

Physical signal value variable.

7.434.4.11 signal_raw_var

ttk_bus.bus_signals_cangamma.TxCanBusSignal.signal_raw_var

Raw signal value variable.

7.434.4.12 signal_var

ttk_bus.bus_signals_cangamma.TxCanBusSignal.signal_var

Signal value variable, default representation.

get/info methods of this TxCanBusSignal will access this variable.
Note that the default self.signal_var usually maps to member "value_phy", (depending on supplied signal_identifier).

7.434.4.13 timeout

ttk_bus.bus_signals_cangamma.TxCanBusSignal.timeout

detection time for timeout errors [ms]

7.434.4.14 unit

`ttk_bus.bus_signals_cangamma.TxCanBusSignal.unit`

unit of physical signal value (as string, mainly used for report entries)

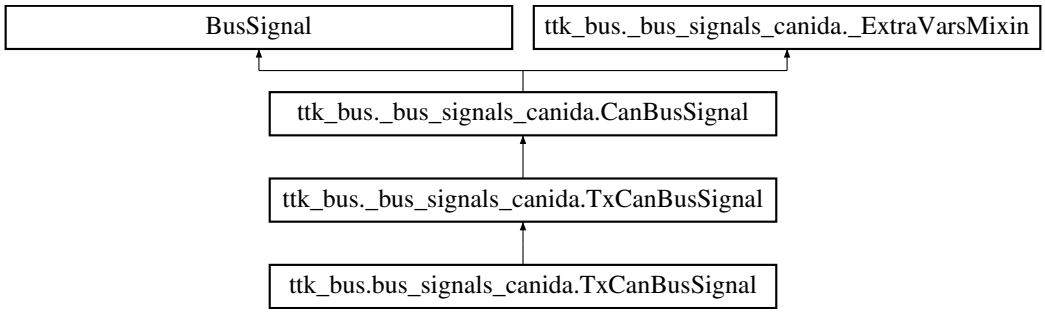
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_cangamma.py (v1.3)

7.435 ttk_bus.bus_signals_canida.TxCanBusSignal Class Reference

A basic CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).

Inheritance diagram for ttk_bus.bus_signals_canida.TxCanBusSignal:



Public Member Functions

- __init__ (self, context, signal_identifier, message=None, cycle_time=None, debounce_time=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  Basic Tx bus signal initialization (ECU Tx => HIL Rx).
- info (self)

  Get a current info string about the HIL signal variable, see `signal_var`
- get (self)

  Get the current value of the HIL signal variable, see `signal_var`
- set (self, value)

  Set a value to the HIL signal variable, see `signal_var`
- getState (self)

  Get a textual state for the HIL signal variable's current data, see `signal_var`.
- setState (self, state)

  Set a textual state to the HIL signal variable, see `signal_var`.
- getStateDescr (self, value=None, fallback=None)

  Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).
- reset (self, *args, **kwargs)

  Reset all resettable variables of this bus signal.

Public Attributes

- base_identifier
  Signal PV address/identifier as base identifier for deriving sub-elements.
- message
  reference to associated (Tx-)CanBusMessage instance
- context
  parent context (a gaapi reference to resolve signal/variable paths)
- signal_var
  Signal value variable, default representation.
- unit
  unit of physical signal value (as string, mainly used for report entries)
- alias
  bus signal alias name
- descr
  bus signal description text
- cycle_time
  cycle time of Message/PDU in [ms]
- debounce_time
  minimum debounce delay for event-triggered Messages/PDUs in [ms]
- change_delay
  delay to wait after a value change [ms]
- timeout
  detection time for timeout errors [ms]
- recovery_time
  time for recovery/"wiedergut" checks [ms]

7.435.1 Detailed Description

A basic CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).

Basic implementation, directly accesses raw .value as-is from CAN-Plugin, without a converted physical value or error injections.

7.435.2 Constructor & Destructor Documentation

7.435.2.1 __init__()

```
ttk_bus.bus_signals_canida.TxCanBusSignal.__init__ (
              self,
              context,
              signal_identifier,
              message = None,
              cycle_time = None,
              debounce_time = None,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Basic Tx bus signal initialization (ECU Tx => HIL Rx).

Note

`signal_identifier` var name Use `.value` entry (or optionally `.value_phy`/`.value_raw`, if available), e.g. `"CAN.CanName.HIL_RX.MessageName.signals.SignalName.value"`
or just the base signal identifier to default to use `.value`:
`"CAN.CanName.HIL_RX.MessageName.signals.SignalName"`

Parameters

context	parent iDA context
signal_identifier	identifier for signal value var (see note)
message	reference to associated TxCanBusMessage instance
cycle_time	(opt) cycle time in [ms], overrides settings in message
debounce_time	(opt) minimum debounce delay between events in [ms], overrides settings in Message
unit	(opt) unit of physical signal value (as string, mainly used for report entries)
alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

Minimal/Basic Signals:

ECU-Tx/HIL-Rx Signals will only contain variables for:

- `<SignalName>.value` - raw value as received from CAN-Plugin
- `<SignalName>.timestamp` - Rx time stamp
- `<SignalName>.length` - message length

Reimplemented from ttk_bus._bus_signals_canida.TxCanBusSignal.

7.435.3 Member Function Documentation

7.435.3.1 get()

```
ttk_bus.bus_signals_canida.TxCanBusSignal.get (
              self )
```

Get the current value of the HIL signal variable, see `signal_var`

Returns

the current value from `signal_var`.

7.435.3.2 getState()

```
ttk_bus.bus_signals_canida.TxCanBusSignal.getState (
              self )
```

Get a textual state for the HIL signal variable's current data, see `signal_var`.

This uses the lookup values supplied in constructor. Fallback (if no matching state was found) will be the unmapped numeric value.

Returns

the current state representation from `signal_var`.

7.435.3.3 getStateDescr()

```
ttk_bus.bus_signals_canida.TxCanBusSignal.getStateDescr (
              self,
              value = None,
              fallback = None )
```

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the last read value will be used instead.
fallback	fallback value to return if no match was found. Default: None

Returns

the state text if a match was found, otherwise the specified fallback value.

7.435.3.4 info()

```
ttk_bus.bus_signals_canida.TxCanBusSignal.info (
              self )
```

Get a current info string about the HIL signal variable, see `signal_var`

Returns

the current info string from `signal_var`.

7.435.3.5 reset()

```
ttk_bus.bus_signals_canida.TxCanBusSignal.reset (
              self,
             * args,
             ** kwargs )
```

Reset all resettable variables of this bus signal.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

7.435.3.6 set()

```
ttk_bus.bus_signals_canida.TxCanBusSignal.set (
              self,
              value )
```

Set a value to the HIL signal variable, see `signal_var`

Attention

For TxCanBusSignals, this will only overwrite the current contents of the HIl receive buffer of the signal (which is typically not too useful as it will be overwritten on next update with data from CAN plugin)

Parameters

value	value to set

Returns

the previous value from `signal_var`.

7.435.3.7 setState()

```
ttk_bus.bus_signals_canida.TxCanBusSignal.setState (
              self,
              state )
```

Set a textual state to the HIL signal variable, see `signal_var`.

This uses the lookup values supplied in constructor.

Attention

For TxFrBusSignals this will only overwrite the current contents of the HIl receive buffer of the signal (which typically is not too useful).

Parameters

state	named state to set

Returns

the previous state representation from `signal_var`.

7.435.4 Member Data Documentation

7.435.4.1 alias

`ttk_bus.bus_signals_canida.TxCanBusSignal.alias`

bus signal alias name

7.435.4.2 base_identifier

`ttk_bus.bus_signals_canida.TxCanBusSignal.base_identifier`

Signal PV address/identifier as base identifier for deriving sub-elements.

For CAN/Gamma signals, related variables follow the form `<base_identifier>.<suffix>`

7.435.4.3 change_delay

`ttk_bus.bus_signals_canida.TxCanBusSignal.change_delay`

delay to wait after a value change [ms]

7.435.4.4 context

`ttk_bus.bus_signals_canida.TxCanBusSignal.context`

parent context (a gaapi reference to resolve signal/variable paths)

7.435.4.5 cycle_time

`ttk_bus.bus_signals_canida.TxCanBusSignal.cycle_time`

cycle time of Message/PDU in [ms]

7.435.4.6 debounce_time

`ttk_bus.bus_signals_canida.TxCanBusSignal.debounce_time`

minimum debounce delay for event-triggered Messages/PDUs in [ms]

7.435.4.7 descr

ttk_bus.bus_signals_canida.TxCanBusSignal.descr

bus signal description text

7.435.4.8 message

ttk_bus.bus_signals_canida.TxCanBusSignal.message

reference to associated (Tx-)CanBusMessage instance

7.435.4.9 recovery_time

ttk_bus.bus_signals_canida.TxCanBusSignal.recovery_time

time for recovery/"wiedergut" checks [ms]

7.435.4.10 signal_var

ttk_bus.bus_signals_canida.TxCanBusSignal.signal_var

Signal value variable, default representation.

get/info methods of this TxCanBusSignal will access this variable.
Note that the default self.signal_var usually maps to member "value" or "value_phy", (depending on supplied signal_identifier).

7.435.4.11 timeout

ttk_bus.bus_signals_canida.TxCanBusSignal.timeout

detection time for timeout errors [ms]

7.435.4.12 unit

ttk_bus.bus_signals_canida.TxCanBusSignal.unit

unit of physical signal value (as string, mainly used for report entries)

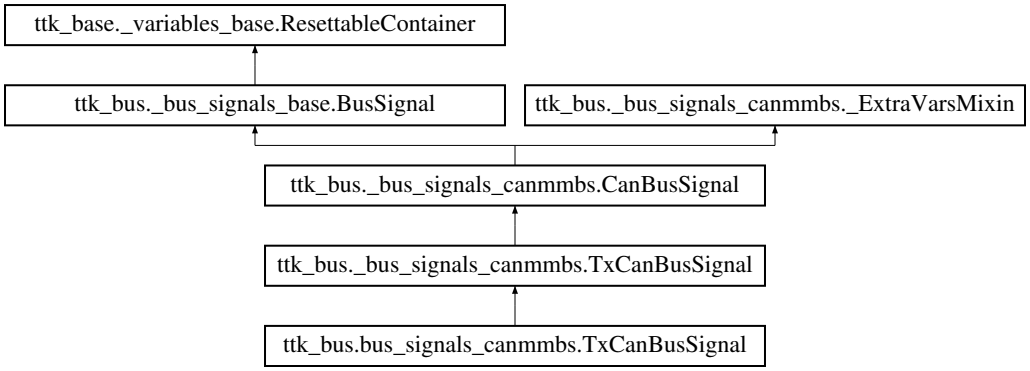
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_canida.py (v1.0.0)

7.436 ttk_bus.bus_signals_canmmbs.TxCanBusSignal Class Reference

A CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).

Inheritance diagram for ttk_bus.bus_signals_canmmbs.TxCanBusSignal:



Public Member Functions

- __init__ (self, context, signal_path, message=None, cycle_time=None, debounce_time=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  Tx bus signal initialization.
- info (self)

  Get a current info string about the HIL signal variable, see signal_var.
- get (self)

  Get the current value of the HIL signal variable, see signal_var.
- set (self, value)

  Set a value to the HIL signal variable, see signal_var.
- getState (self)

  Get a textual state for the HIL signal variable's current data, see signal_var.
- setState (self, state)

  Set a textual state to the HIL signal variable, see signal_var.
- getStateDescr (self, value=None, fallback=None)

  Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).
- reset (self, *args, **kwargs)

  Reset all resettable variables of this bus signal.

Public Attributes

- base_path

  Signal model path as base path for deriving sub-element paths.
- message

  reference to associated (Tx-)CanBusMessage instance
- signal_var

  signal value variable.
- context

  parent context (an RTE application instance to resolve signal/variable paths)
- unit

  unit of physical signal value (as string, mainly used for report entries)

- alias

  bus signal alias name
- descr

  bus signal description text
- cycle_time

  cycle time of Message/PDU in [ms]
- debounce_time

  minimum debounce delay for event-triggered Messages/PDUs in [ms]
- change_delay

  delay to wait after a value change [ms]
- timeout

  detection time for timeout errors [ms]
- recovery_time

  time for recovery/"wiedergut" checks [ms]

7.436.1 Detailed Description

A CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).

See _bus_signals_canmmbs.CanBusSignal.

7.436.2 Constructor & Destructor Documentation

7.436.2.1 __init__()

```
ttk_bus.bus_signals_canmmbs.TxCanBusSignal.__init__ (
                self,
                context,
                signal_path,
                message = None,
                cycle_time = None,
                debounce_time = None,
                unit = "",
                alias = "",
                descr = "",
                lookup = None,
                change_delay = None,
                timeout = None,
                recovery_time = None )
```

Tx bus signal initialization.

Parameters

context	parent context (an RTE application instance to resolve signal/variable paths)
signal_path	base path to signal value below BusSystems
message	reference to associated CanBusMessage instance
cycle_time	(opt) cycle time in [ms], overrides settings in Message
debounce_time	(opt) minimum debounce delay between events in [ms], overrides settings in Message
unit	(opt) unit of physical signal value (as string, mainly used for report entries)

Parameters

alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_canmmbs.CanBusSignal.

7.436.3 Member Function Documentation

7.436.3.1 get()

```
ttk_bus.bus_signals_canmmbs.TxCanBusSignal.get (
              self )
```

Get the current value of the HIL signal variable, see signal_var.

Returns

the current value from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.436.3.2 getState()

```
ttk_bus.bus_signals_canmmbs.TxCanBusSignal.getState (
              self )
```

Get a textual state for the HIL signal variable's current data, see signal_var.

This uses the lookup values supplied in constructor. Fallback (if no matching state was found) will be the unmapped numeric value.

Returns

the current state representation from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.436.3.3 getStateDescr()

```
ttk_bus.bus_signals_canmmbs.TxCanBusSignal.getStateDescr (
              self,
              value = None,
              fallback = None )
```

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the last read value will be used instead.
fallback	fallback value to return if no match was found. Default: None

Returns

the state text if a match was found, otherwise the specified fallback value.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.436.3.4 info()

```
ttk_bus.bus_signals_canmmbs.TxCanBusSignal.info (
              self )
```

Get a current info string about the HIL signal variable, see signal_var.

Returns

the current info string from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.436.3.5 reset()

```
ttk_bus.bus_signals_canmmbs.TxCanBusSignal.reset (
              self,
             * args,
             ** kwargs )
```

Reset all resettable variables of this bus signal.

Parameters

∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.ResettableContainer.

7.436.3.6 set()

```
ttk_bus.bus_signals_canmmbs.TxCanBusSignal.set (
              self,
              value )
```

Set a value to the HIL signal variable, see signal_var.

Attention

For TxCanBusSignals this will only overwrite the current contents of the HIl receive buffer of the signal (which is typically not too useful).

Parameters

value	value to set

Returns

the previous value from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.436.3.7 setState()

```
ttk_bus.bus_signals_canmmbs.TxCanBusSignal.setState (
              self,
              state )
```

Set a textual state to the HIL signal variable, see signal_var.

This uses the lookup values supplied in constructor.

Attention

For TxFrBusSignals this will only overwrite the current contents of the HIl receive buffer of the signal (which typically is not too useful).

Parameters

state	named state to set

Returns

the previous state representation from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.436.4 Member Data Documentation

7.436.4.1 alias

```
ttk_bus.bus_signals_canmmbs.TxCanBusSignal.alias
```

bus signal alias name

7.436.4.2 base_path

ttk_bus.bus_signals_canmmbs.TxCanBusSignal.base_path

Signal model path as base path for deriving sub-element paths.

For CANMM signals, related variables follow the form <path_to_signal>_suffix

7.436.4.3 change_delay

ttk_bus.bus_signals_canmmbs.TxCanBusSignal.change_delay

delay to wait after a value change [ms]

7.436.4.4 context

ttk_bus.bus_signals_canmmbs.TxCanBusSignal.context

parent context (an RTE application instance to resolve signal/variable paths)

7.436.4.5 cycle_time

ttk_bus.bus_signals_canmmbs.TxCanBusSignal.cycle_time

cycle time of Message/PDU in [ms]

7.436.4.6 debounce_time

ttk_bus.bus_signals_canmmbs.TxCanBusSignal.debounce_time

minimum debounce delay for event-triggered Messages/PDUs in [ms]

7.436.4.7 descr

ttk_bus.bus_signals_canmmbs.TxCanBusSignal.descr

bus signal description text

7.436.4.8 message

ttk_bus.bus_signals_canmmbs.TxCanBusSignal.message

reference to associated (Tx-)CanBusMessage instance

7.436.4.9 recovery_time

`ttk_bus.bus_signals_canmmbs.TxCanBusSignal.recovery_time`

time for recovery/"wiedergut" checks [ms]

7.436.4.10 signal_var

`ttk_bus.bus_signals_canmmbs.TxCanBusSignal.signal_var`

signal value variable.

get/info methods of this TxCanBusSignal will access this variable. (a HilVar instance)

7.436.4.11 timeout

`ttk_bus.bus_signals_canmmbs.TxCanBusSignal.timeout`

detection time for timeout errors [ms]

7.436.4.12 unit

`ttk_bus.bus_signals_canmmbs.TxCanBusSignal.unit`

unit of physical signal value (as string, mainly used for report entries)

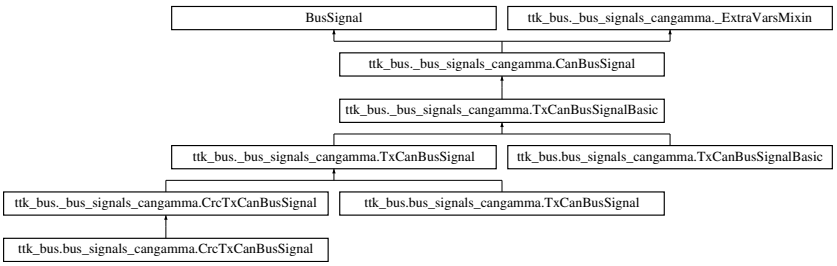
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_canmmbs.py (v1.8)

7.437 ttk_bus._bus_signals_cangamma.TxCanBusSignalBasic Class Reference

A basic CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).

Inheritance diagram for ttk_bus._bus_signals_cangamma.TxCanBusSignalBasic:



Public Member Functions

- __init__ (self, context, signal_identifier, message=None, cycle_time=None, debounce_time=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  Basic Tx bus signal initialization (ECU Tx => HIL Rx).

7.437.1 Detailed Description

A basic CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).

Basic implementation, directly accesses raw value as-is from CAN, without a converted physical value or error injections.

Use this for signal entries generated with iSyDBC mode "minimal" . See _bus_signals_cangamma.CanBusSignal.

7.437.2 Constructor & Destructor Documentation

7.437.2.1 __init__()

```
ttk_bus._bus_signals_cangamma.TxCanBusSignalBasic.__init__ (
                self,
                context,
                signal_identifier,
                message = None,
                cycle_time = None,
                debounce_time = None,
                unit = "",
                alias = "",
                descr = "",
                lookup = None,
                change_delay = None,
                timeout = None,
                recovery_time = None )
```

Basic Tx bus signal initialization (ECU Tx => HIL Rx).

Parameters

context	parent gamma_api context
signal_identifier	identifier for signal value PV. Use `.value` entry, e.g. `"BusSystems.can.HIL_TX.MessageName.signals.SignalName.value"`
message	reference to associated TxCanBusMessage instance
cycle_time	(opt) cycle time in [ms], overrides settings in message
debounce_time	(opt) minimum debounce delay between events in [ms], overrides settings in Message
unit	(opt) unit of physical signal value (as string, mainly used for report entries)
alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

Minimal/Basic Signals:

ECU-Tx/HIL-Rx Signals generated by iSyDBC in minimal/basic mode will only contain PVs for:

- `<SignalName>.value` - raw value as received from CAN-Plugin
- `<SignalName>.timestamp` - Rx time stamp
- `<SignalName>.length` - message length

Reimplemented from ttk_bus._bus_signals_cangamma.CanBusSignal.

Reimplemented in ttk_bus._bus_signals_cangamma.CrcTxCanBusSignal, ttk_bus.bus_signals_cangamma.CrcTxCanBusSignal, ttk_bus._bus_signals_cangamma.TxCanBusSignal, ttk_bus.bus_signals_cangamma.TxCanBusSignalBasic, and ttk_bus.bus_signals_cangamma.TxCanBusSignal.

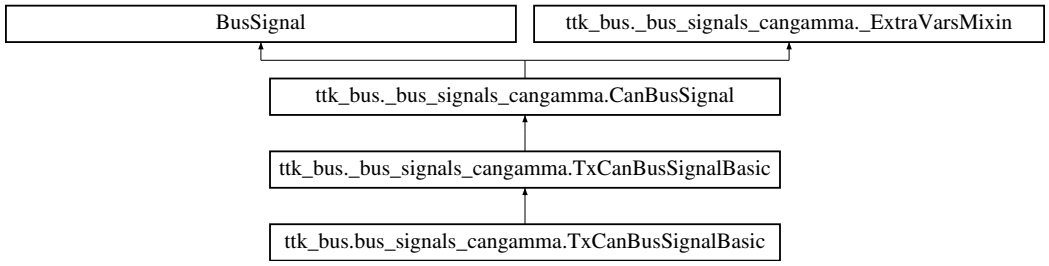
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_cangamma.py (v1.12)

7.438 ttk_bus.bus_signals_cangamma.TxCanBusSignalBasic Class Reference

A basic CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).

Inheritance diagram for ttk_bus.bus_signals_cangamma.TxCanBusSignalBasic:



Public Member Functions

- __init__ (self, context, signal_identifier, message=None, cycle_time=None, debounce_time=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  Basic Tx bus signal initialization (ECU Tx => HIL Rx).
- info (self)

  Get a current info string about the HIL signal variable, see `signal_var`
- get (self)

  Get the current value of the HIL signal variable, see `signal_var`
- set (self, value)

  Set a value to the HIL signal variable, see `signal_var`
- getState (self)

  Get a textual state for the HIL signal variable's current data, see `signal_var`.
- setState (self, state)

  Set a textual state to the HIL signal variable, see `signal_var`.
- getStateDescr (self, value=None, fallback=None)

  Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).
- reset (self, *args, **kwargs)

  Reset all resettable variables of this bus signal.

Public Attributes

- base_identifier
  Signal PV address/identifier as base identifier for deriving sub-elements.
- message
  reference to associated (Tx-)CanBusMessage instance
- context
  parent context (a gaapi reference to resolve signal/variable paths)
- signal_var
  Signal value variable, default representation.
- unit
  unit of physical signal value (as string, mainly used for report entries)
- alias
  bus signal alias name
- descr
  bus signal description text
- cycle_time
  cycle time of Message/PDU in [ms]
- debounce_time
  minimum debounce delay for event-triggered Messages/PDUs in [ms]
- change_delay
  delay to wait after a value change [ms]
- timeout
  detection time for timeout errors [ms]
- recovery_time
  time for recovery/"wiedergut" checks [ms]

7.438.1 Detailed Description

A basic CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).

Basic implementation, directly accesses raw value as-is from CAN, without a converted physical value or error injections.

Use this for signal entries generated with iSyDBC mode "minimal" .

7.438.2 Constructor & Destructor Documentation

7.438.2.1 __init__()

```
ttk_bus.bus_signals_cangamma.TxCanBusSignalBasic.__init__ (
              self,
              context,
              signal_identifier,
              message = None,
              cycle_time = None,
              debounce_time = None,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Basic Tx bus signal initialization (ECU Tx => HIL Rx).

Parameters

context	parent gamma_api context
signal_identifier	identifier for signal value PV. Use `.value` entry, e.g. `"BusSystems.can.HIL_TX.MessageName.signals.SignalName.value"`
message	reference to associated TxCanBusMessage instance
cycle_time	(opt) cycle time in [ms], overrides settings in message
debounce_time	(opt) minimum debounce delay between events in [ms], overrides settings in Message
unit	(opt) unit of physical signal value (as string, mainly used for report entries)
alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

Minimal/Basic Signals:

ECU-Tx/HIL-Rx Signals generated by iSyDBC in minimal/basic mode will only contain PVs for:

- `<SignalName>.value` - raw value as received from CAN-Plugin
- `<SignalName>.timestamp` - Rx time stamp
- `<SignalName>.length` - message length

Reimplemented from ttk_bus._bus_signals_cangamma.TxCanBusSignalBasic.

7.438.3 Member Function Documentation

7.438.3.1 get()

```
ttk_bus.bus_signals_cangamma.TxCanBusSignalBasic.get (
              self )
```

Get the current value of the HIL signal variable, see `signal_var`

Returns

the current value from `signal_var`.

7.438.3.2 getState()

```
ttk_bus.bus_signals_cangamma.TxCanBusSignalBasic.getState (
              self )
```

Get a textual state for the HIL signal variable's current data, see `signal_var`.

This uses the lookup values supplied in constructor. Fallback (if no matching state was found) will be the unmapped numeric value.

Returns

the current state representation from `signal_var`.

7.438.3.3 getStateDescr()

```
ttk_bus.bus_signals_cangamma.TxCanBusSignalBasic.getStateDescr (
              self,
              value = None,
              fallback = None )
```

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the last read value will be used instead.
fallback	fallback value to return if no match was found. Default: None

Returns

the state text if a match was found, otherwise the specified fallback value.

7.438.3.4 info()

```
ttk_bus.bus_signals_cangamma.TxCanBusSignalBasic.info (
              self )
```

Get a current info string about the HIL signal variable, see `signal_var`

Returns

the current info string from `signal_var`.

7.438.3.5 reset()

```
ttk_bus.bus_signals_cangamma.TxCanBusSignalBasic.reset (
              self,
             * args,
             ** kwargs )
```

Reset all resettable variables of this bus signal.

Parameters

∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

7.438.3.6 set()

```
ttk_bus.bus_signals_cangamma.TxCanBusSignalBasic.set (
              self,
              value )
```

Set a value to the HIL signal variable, see `signal_var`

Attention

For TxCanBusSignals, this will only overwrite the current contents of the HIl receive buffer of the signal (which is typically not too useful as it will be overwritten on next update with data from CAN plugin)

Parameters

value	value to set

Returns

the previous value from `signal_var`.

7.438.3.7 setState()

```
ttk_bus.bus_signals_cangamma.TxCanBusSignalBasic.setState (
              self,
              state )
```

Set a textual state to the HIL signal variable, see `signal_var`.

This uses the lookup values supplied in constructor.

Attention

For TxFrBusSignals this will only overwrite the current contents of the HIl receive buffer of the signal (which typically is not too useful).

Parameters

state	named state to set

Returns

the previous state representation from `signal_var`.

7.438.4 Member Data Documentation

7.438.4.1 alias

```
ttk_bus.bus_signals_cangamma.TxCanBusSignalBasic.alias
```

bus signal alias name

7.438.4.2 base_identifier

ttk_bus.bus_signals_cangamma.TxCanBusSignalBasic.base_identifier

Signal PV address/identifier as base identifier for deriving sub-elements.

For CAN/Gamma signals, related variables follow the form <base_identifier>.<suffix>

7.438.4.3 change_delay

ttk_bus.bus_signals_cangamma.TxCanBusSignalBasic.change_delay

delay to wait after a value change [ms]

7.438.4.4 context

ttk_bus.bus_signals_cangamma.TxCanBusSignalBasic.context

parent context (a gaapi reference to resolve signal/variable paths)

7.438.4.5 cycle_time

ttk_bus.bus_signals_cangamma.TxCanBusSignalBasic.cycle_time

cycle time of Message/PDU in [ms]

7.438.4.6 debounce_time

ttk_bus.bus_signals_cangamma.TxCanBusSignalBasic.debounce_time

minimum debounce delay for event-triggered Messages/PDUs in [ms]

7.438.4.7 descr

ttk_bus.bus_signals_cangamma.TxCanBusSignalBasic.descr

bus signal description text

7.438.4.8 message

ttk_bus.bus_signals_cangamma.TxCanBusSignalBasic.message

reference to associated (Tx-)CanBusMessage instance

7.438.4.9 recovery_time

`ttk_bus.bus_signals_cangamma.TxCanBusSignalBasic.recovery_time`

time for recovery/"wiedergut" checks [ms]

7.438.4.10 signal_var

`ttk_bus.bus_signals_cangamma.TxCanBusSignalBasic.signal_var`

Signal value variable, default representation.

get/info methods of this TxCanBusSignal will access this variable.
Note that the default self.signal_var usually maps to member "value_phy", (depending on supplied signal_identifier).

7.438.4.11 timeout

`ttk_bus.bus_signals_cangamma.TxCanBusSignalBasic.timeout`

detection time for timeout errors [ms]

7.438.4.12 unit

`ttk_bus.bus_signals_cangamma.TxCanBusSignalBasic.unit`

unit of physical signal value (as string, mainly used for report entries)

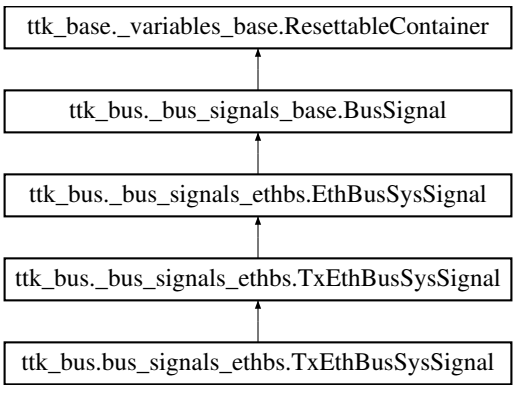
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_cangamma.py (v1.3)

7.439 ttk_bus._bus_signals_ethbs.TxEthBusSysSignal Class Reference

An Ethernet event signal/parameter/sub-parameter sent from DUT to test system (ECU PSI Tx => HIL CSI Rx).

Inheritance diagram for ttk_bus._bus_signals_ethbs.TxEthBusSysSignal:



7.439.1 Detailed Description

An Ethernet event signal/parameter/sub-parameter sent from DUT to test system (ECU PSI Tx => HIL CSI Rx).

Note

This class expects a BusSystems path structure.

CRC Tx-Event/Signal status (ECU PSI Tx => HIL CSI Rx) and associated switch variables for dSPACE Ethernet blockset.

Note

This class expects a BusSystems path structure.

This behaves essentially as a basic "HilVar" containing the CRC status plus additional sub-variables.

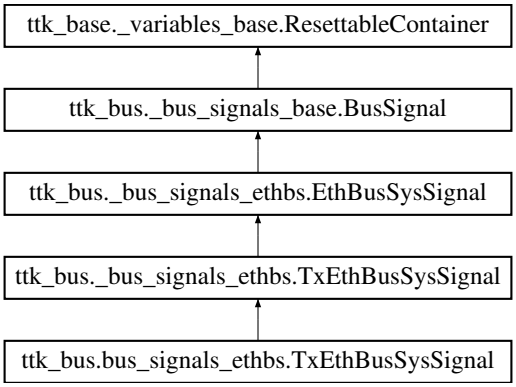
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_ethbs.py (v2.5)

7.440 ttk_bus.bus_signals_ethbs.TxEthBusSysSignal Class Reference

An Ethernet event signal/parameter/sub-parameter sent from DUT to test system (ECU PSI Tx => HIL CSI Rx).

Inheritance diagram for ttk_bus.bus_signals_ethbs.TxEthBusSysSignal:



Public Member Functions

- get (self)
  Get the current value of the HIL signal variable.
- set (self, value)
  Set a value to the HIL signal variable (typically not that useful for ECU-Tx/HIL-Rx signals, though).
- getState (self)
  Get a textual state for the HIL signal variable's current data.
- setState (self, state)
  Set a textual state to the HIL signal variable.
- getStateDescr (self, value=None, fallback=None)
  Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).
- info (self)
  Get a current info string about the HIL signal variable.
- reset (self, *args, **kwargs)
  Reset all resettable variables of this bus signal.

7.440.1 Detailed Description

An Ethernet event signal/parameter/sub-parameter sent from DUT to test system (ECU PSI Tx => HIL CSI Rx).

Note

This class expects a BusSystems path structure.

7.440.2 Member Function Documentation

7.440.2.1 get()

```
ttk_bus.bus_signals_ethbs.TxEthBusSysSignal.get (
              self )
```

Get the current value of the HIL signal variable.

Returns

the current value from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.440.2.2 getState()

```
ttk_bus.bus_signals_ethbs.TxEthBusSysSignal.getState (
              self )
```

Get a textual state for the HIL signal variable's current data.

Returns

the current state's description.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.440.2.3 getStateDescr()

```
ttk_bus.bus_signals_ethbs.TxEthBusSysSignal.getStateDescr (
              self,
              value = None,
              fallback = None )
```

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the last read value will be used instead.
fallback	fallback value to return if no match was found. Default: None

Returns

the state text if a match was found, otherwise the specified fallback value.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.440.2.4 info()

```
ttk_bus.bus_signals_ethbs.TxEthBusSysSignal.info (
              self )
```

Get a current info string about the HIL signal variable.

Returns

an info string.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.440.2.5 reset()

```
ttk_bus.bus_signals_ethbs.TxEthBusSysSignal.reset (
              self,
             * args,
             ** kwargs )
```

Reset all resettable variables of this bus signal.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.ResettableContainer.

7.440.2.6 set()

```
ttk_bus.bus_signals_ethbs.TxEthBusSysSignal.set (
              self,
              value )
```

Set a value to the HIL signal variable (typically not that useful for ECU-Tx/HIL-Rx signals, though).

Parameters

value	value to set

Returns

the previous value from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.440.2.7 setState()

```
ttk_bus.bus_signals_ethbs.TxEthBusSysSignal.setState (
              self,
              state )
```

Set a textual state to the HIL signal variable.

This uses the lookup values supplied in constructor (typically not that useful for ECU-Tx/HIL-Rx signals, though).

Parameters

state	named state to set

Returns

the previous state's description.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

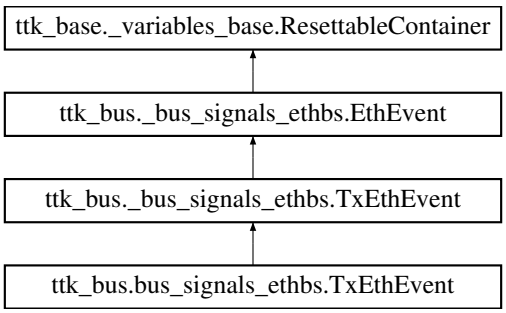
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_ethbs.py (v2.5)

7.441 ttk_bus._bus_signals_ethbs.TxEthEvent Class Reference

Ethernet-Event-specific data and methods - Tx (ECU-Tx => HIL-Rx).

Inheritance diagram for ttk_bus._bus_signals_ethbs.TxEthEvent:



7.441.1 Detailed Description

Ethernet-Event-specific data and methods - Tx (ECU-Tx => HIL-Rx).

Note

This is currently just a container for meta entries, as there are (so far) no HIL variables needed for Tx events.

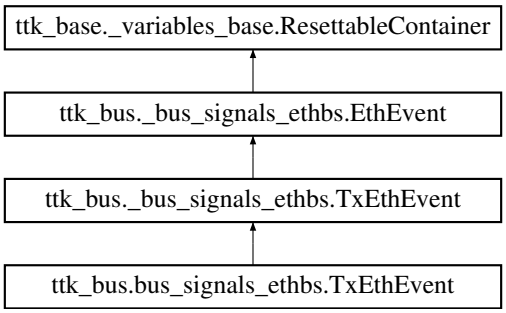
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_ethbs.py (v2.5)

7.442 ttk_bus.bus_signals_ethbs.TxEthEvent Class Reference

Ethernet-Event-specific data and methods - Tx (ECU-Tx => HIL-Rx).

Inheritance diagram for ttk_bus.bus_signals_ethbs.TxEthEvent:



Public Member Functions

- __init__ (self, context, cycle_time=0, debounce_time=0, alias="", descr="", change_delay=None, timeout=None, recovery_time=None)

  Ethernet Tx-Event (or field) data.
- reset (self, *args, **kwargs)

  Call reset() on all contained variables/signals that support it.

Public Attributes

- context

  parent context (an RTE application instance to resolve signal/variable paths)
- alias

  event alias name
- descr

  event description text
- cycle_time

  cycle time of signal in [ms]
- debounce_time

  minimum debounce delay for triggered signals in [ms]
- change_delay

  delay to wait after a value change [ms]
- timeout

  detection time for timeout errors [ms]
- recovery_time

  time for recovery/"wiedergut" checks [ms]

7.442.1 Detailed Description

Ethernet-Event-specific data and methods - Tx (ECU-Tx => HIL-Rx).

Note

This is currently just a container for meta entries, as there are (so far) no HIL variables needed for Tx events.

7.442.2 Constructor & Destructor Documentation

7.442.2.1 __init__()

```
ttk_bus.bus_signals_ethbs.TxEthEvent.__init__ (
              self,
              context,
              cycle_time = 0,
              debounce_time = 0,
              alias = "",
              descr = "",
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Ethernet Tx-Event (or field) data.

Note

This is currently just a container for meta entries, as there are (so far) no HIL variable needed for Tx events.

Parameters

context	parent context, an RTE application or XilTestbench instance to resolve variable paths (if any)
cycle_time	(opt) cycle time in [ms]
debounce_time	(opt) minimum debounce delay time between events in [ms]
alias	(opt) event alias/name
descr	(opt) event description (e.g. used in report entries)
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms] defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_ethbs.EthEvent.

7.442.3 Member Function Documentation

7.442.3.1 reset()

```
ttk_bus.bus_signals_ethbs.TxEthEvent.reset (
              self,
```

```
        * args,
        ** kwargs )
```

Call reset() on all contained variables/signals that support it.

Simple, non-recursive approach.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.ResettableContainer.

7.442.4 Member Data Documentation

7.442.4.1 alias

`ttk_bus.bus_signals_ethbs.TxEthEvent.alias`

event alias name

7.442.4.2 change_delay

`ttk_bus.bus_signals_ethbs.TxEthEvent.change_delay`

delay to wait after a value change [ms]

7.442.4.3 context

`ttk_bus.bus_signals_ethbs.TxEthEvent.context`

parent context (an RTE application instance to resolve signal/variable paths)

7.442.4.4 cycle_time

`ttk_bus.bus_signals_ethbs.TxEthEvent.cycle_time`

cycle time of signal in [ms]

7.442.4.5 debounce_time

`ttk_bus.bus_signals_ethbs.TxEthEvent.debounce_time`

minimum debounce delay for triggered signals in [ms]

7.442.4.6 descr

`ttk_bus.bus_signals_ethbs.TxEthEvent.descr`

event description text

7.442.4.7 recovery_time

`ttk_bus.bus_signals_ethbs.TxEthEvent.recovery_time`

time for recovery/"wiedergut" checks [ms]

7.442.4.8 timeout

`ttk_bus.bus_signals_ethbs.TxEthEvent.timeout`

detection time for timeout errors [ms]

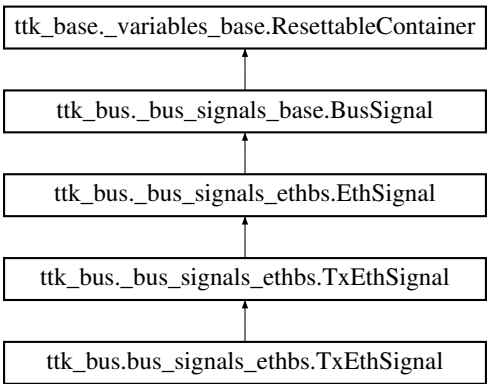
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_ethbs.py (v2.5)

7.443 ttk_bus._bus_signals_ethbs.TxEthSignal Class Reference

An Ethernet event signal/parameter/sub-parameter sent from DUT to test system (ECU PSI Tx => HIL CSI Rx).

Inheritance diagram for ttk_bus._bus_signals_ethbs.TxEthSignal:



7.443.1 Detailed Description

An Ethernet event signal/parameter/sub-parameter sent from DUT to test system (ECU PSI Tx => HIL CSI Rx).

Note

This class expects a "Model Root/" path structure.

Base wrapper for an Ethernet event with HIL-side BusSignals variables and relevant associated values. Derives info from paths below BusSystems.

Note

This class expects a BusSystems path structure.

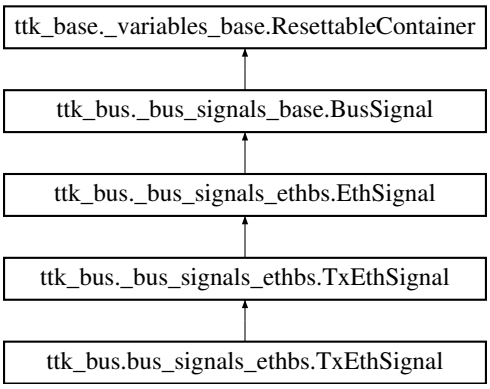
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_ethbs.py (v2.5)

7.444 ttk_bus.bus_signals_ethbs.TxEthSignal Class Reference

An Ethernet event signal/parameter/sub-parameter sent from DUT to test system (ECU PSI Tx => HIL CSI Rx).

Inheritance diagram for ttk_bus.bus_signals_ethbs.TxEthSignal:



Public Member Functions

- get (self)
  Get the current value of the HIL signal variable.
- set (self, value)
  Set a value to the HIL signal variable (typically not that useful for ECU-Tx/HIL-Rx signals, though).
- getState (self)
  Get a textual state for the HIL signal variable's current data.
- setState (self, state)
  Set a textual state to the HIL signal variable.
- getStateDescr (self, value=None, fallback=None)
  Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).
- info (self)
  Get a current info string about the HIL signal variable.
- reset (self, *args, **kwargs)
  Reset all resettable variables of this bus signal.

7.444.1 Detailed Description

An Ethernet event signal/parameter/sub-parameter sent from DUT to test system (ECU PSI Tx => HIL CSI Rx).

Note

This class expects a "Model Root/" path structure.

7.444.2 Member Function Documentation

7.444.2.1 get()

```
ttk_bus.bus_signals_ethbs.TxEthSignal.get (
              self )
```

Get the current value of the HIL signal variable.

Returns

the current value from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.444.2.2 getState()

```
ttk_bus.bus_signals_ethbs.TxEthSignal.getState (
              self )
```

Get a textual state for the HIL signal variable's current data.

Returns

the current state representation from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.444.2.3 getStateDescr()

```
ttk_bus.bus_signals_ethbs.TxEthSignal.getStateDescr (
              self,
              value = None,
              fallback = None )
```

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the last read value will be used instead.
fallback	fallback value to return if no match was found. Default: None

Returns

the state text if a match was found, otherwise the specified fallback value.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.444.2.4 info()

```
ttk_bus.bus_signals_ethbs.TxEthSignal.info (
              self )
```

Get a current info string about the HIL signal variable.

Returns

an info string.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.444.2.5 reset()

```
ttk_bus.bus_signals_ethbs.TxEthSignal.reset (
              self,
             * args,
             ** kwargs )
```

Reset all resettable variables of this bus signal.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.ResettableContainer.

7.444.2.6 set()

```
ttk_bus.bus_signals_ethbs.TxEthSignal.set (
              self,
              value )
```

Set a value to the HIL signal variable (typically not that useful for ECU-Tx/HIL-Rx signals, though).

Parameters

value	value to set

Returns

the previous value from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.444.2.7 setState()

```
ttk_bus.bus_signals_ethbs.TxEthSignal.setState (
              self,
              state )
```

Set a textual state to the HIL signal variable.

This uses the lookup values supplied in constructor (typically not that useful for ECU-Tx/HIL-Rx signals, though).

Parameters

state	named state to set

Returns

the previous value representation from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

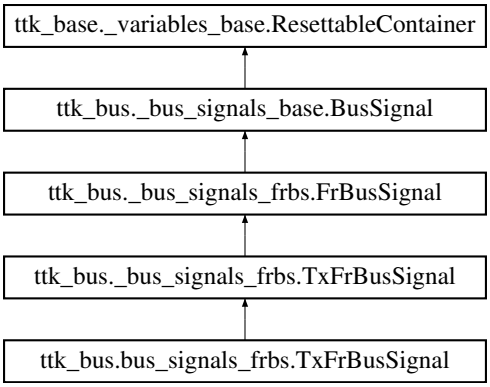
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_ethbs.py (v2.5)

7.445 ttk_bus._bus_signals_frbs.TxFrBusSignal Class Reference

A FLexRay signal sent from DUT to the test system (ECU Tx => HIL Rx).

Inheritance diagram for ttk_bus._bus_signals_frbs.TxFrBusSignal:



Public Member Functions

- __init__ (self, context, base_path, pdu=None, cycle_time=None, debounce_time=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  Tx bus signal initialization.

Public Attributes

- status_var

  HIL-Rx-Status information (a "HilVar" instance).
- context

7.445.1 Detailed Description

A FLexRay signal sent from DUT to the test system (ECU Tx => HIL Rx).

This class mainly adds a member variable `status_var` that contains the HIL-Rx status of the signal (useful for `NOT-VALID` values, as the received signal will remain at the last valid value).

7.445.2 Constructor & Destructor Documentation

7.445.2.1 __init__()

```
ttk_bus._bus_signals_frbs.TxFrBusSignal.__init__ (
              self,
              context,
              base_path,
              pdu = None,
              cycle_time = None,
              debounce_time = None,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Tx bus signal initialization.

Example: BusSystems Path Structure:

```
# path to signal value variable
BusSystems/FlexRay/Monitoring/ECU_NAME/Channel A/PDU_NAME/Signals/SIGNAL_NAME_IN/Value
# or
BusSystems/FlexRay/Monitoring/ECU_NAME/Channel A/PDU_NAME/Signals/SIGNAL_NAME_IN/Physical Value
```

Parameters

context	parent context (an RTE application instance to resolve signal/variable paths)
base_path	base path to signal below BusSystems (use path to `/Value` or `/Physical Value` as appropriate)

Parameters

pdu	reference to associated TxFrPdu instance
cycle_time	(opt) cycle time in [ms], overrides settings in PDU
debounce_time	(opt) minimum debounce delay time in [ms], overrides settings in PDU
unit	(opt) unit of physical signal value (as string, mainly used for report entries)
alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_frbs.FrBusSignal.

Reimplemented in ttk_bus.bus_signals_frbs.TxFrBusSignal.

7.445.3 Member Data Documentation

7.445.3.1 context

`ttk_bus._bus_signals_frbs.TxFrBusSignal.context`

7.445.3.2 status_var

`ttk_bus._bus_signals_frbs.TxFrBusSignal.status_var`

HIL-Rx-Status information (a "HilVar" instance).

> "Status information of the received or monitored signal. Only available if signal is selected for 'Signal RX Status Access' feature in FlexRay Configuration Tool" (compare FlexRayConfigFeatures.pdf)

The values have the following meanings:

- 0: No error
- 1: Access error
- 2: Signal is not received
- 4: Signal is invalid
- 8: CRC is incorrect

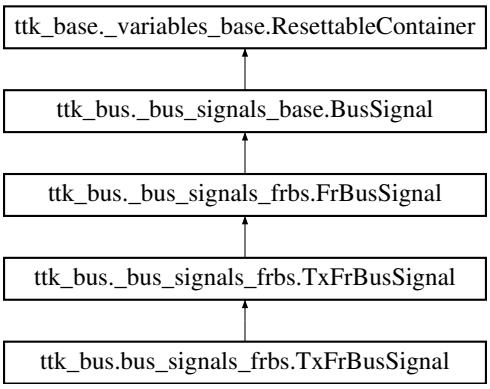
The documentation for this class was generated from the following file:

- ttk_bus/_bus_signals_frbs.py (v2.5)

7.446 ttk_bus.bus_signals_frbs.TxFrBusSignal Class Reference

A FLexRay signal sent from DUT to the test system (ECU Tx => HIL Rx).

Inheritance diagram for ttk_bus.bus_signals_frbs.TxFrBusSignal:



Public Member Functions

- __init__ (self, context, base_path, pdu=None, cycle_time=None, debounce_time=None, unit="", alias="", descr="", lookup=None, change_delay=None, timeout=None, recovery_time=None)

  Tx bus signal initialization.
- info (self)

  Get a current info string about the HIL signal variable, see signal_var.
- get (self)

  Get the current value of the HIL signal variable, see signal_var.
- set (self, value)

  Set a value to the HIL signal variable, see signal_var.
- getState (self)

  Get a textual state for the HIL signal variable's current data.
- setState (self, state)

  Set a textual state to the HIL signal variable, see signal_var.
- getStateDescr (self, value=None, fallback=None)

  Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).
- reset (self, *args, **kwargs)

  Reset all resettable members/variables of this bus signal.

Public Attributes

- status_var

  HIL-Rx-Status information (a "HilVar" instance).
- value_path

  model path of value variable
- base_path

  base model path (base path used for deriving sub-element paths)
- pdu

  reference to parent FrRxPdu
- context

context (reference to assigned RTE application instance)

- signal_var

  main bus signal variable on HIL-side (a "HilVar" instance)
- alias

  signal alias name (from signal_var)
- unit

  signal unit (from signal_var)
- descr

  signal description text
- cycle_time

  cycle time of signal in [ms] (that is, of the associated message or PDU)
- debounce_time

  minimum debounce delay for event-triggered signals in [ms]
- change_delay

  delay to wait after a value change [ms]
- timeout

  detection time for timeout errors [ms]
- recovery_time

  time for recovery/"wiedergut" checks [ms]

7.446.1 Detailed Description

A FLexRay signal sent from DUT to the test system (ECU Tx => HIL Rx).

This class mainly adds a member variable `status_var` that contains the HIL-Rx status of the signal (useful for `NOT-VALID` values, as the received signal will remain at the last valid value).

7.446.2 Constructor & Destructor Documentation

7.446.2.1 __init__()

```
ttk_bus.bus_signals_frbs.TxFrBusSignal.__init__ (
              self,
              context,
              base_path,
              pdu = None,
              cycle_time = None,
              debounce_time = None,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              change_delay = None,
              timeout = None,
              recovery_time = None )
```

Tx bus signal initialization.

Example: BusSystems Path Structure:

```
# path to signal value variable
BusSystems/FlexRay/Monitoring/ECU_NAME/Channel A/PDU_NAME/Signals/SIGNAL_NAME_IN/Value
# or
BusSystems/FlexRay/Monitoring/ECU_NAME/Channel A/PDU_NAME/Signals/SIGNAL_NAME_IN/Physical Value
```

Parameters

context	parent context (an RTE application instance to resolve signal/variable paths)
base_path	base path to signal below BusSystems (use path to `/Value` or `/Physical Value` as appropriate)
pdu	reference to associated TxFrPdu instance
cycle_time	(opt) cycle time in [ms], overrides settings in PDU
debounce_time	(opt) minimum debounce delay time in [ms], overrides settings in PDU
unit	(opt) unit of physical signal value (as string, mainly used for report entries)
alias	(opt) bus signal alias/name, defaults to the last (non-default) entry in identifier (i.e. signal model path)
descr	(opt) signal description (e.g. used in report entries)
lookup	(opt) a lookup-table for mapping discrete signal states to textual descriptions
change_delay	(opt) delay to wait after a value change [ms], defaults to 2x max(cycle_time, debounce_time)
timeout	(opt) detection time for timeout errors [ms], defaults to 2x max(cycle_time, debounce_time)
recovery_time	(opt) time for recovery/"wiedergut" checks [ms], defaults to 2x max(cycle_time, debounce_time)

Reimplemented from ttk_bus._bus_signals_frbs.TxFrBusSignal.

7.446.3 Member Function Documentation

7.446.3.1 get()

```
ttk_bus.bus_signals_frbs.TxFrBusSignal.get (
              self )
```

Get the current value of the HIL signal variable, see signal_var.

Returns

the current value from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.446.3.2 getState()

```
ttk_bus.bus_signals_frbs.TxFrBusSignal.getState (
              self )
```

Get a textual state for the HIL signal variable's current data.

This uses the lookup values supplied in constructor, see signal_var. Fallback (if no matching state was found) will be the unmapped numeric value.

Returns

the current state representation from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.446.3.3 getStateDescr()

```
ttk_bus.bus_signals_frbs.TxFrBusSignal.getStateDescr (
               self,
               value = None,
               fallback = None )
```

Get a state description text for the given value from the HIL signal variable's defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the last read value will be used instead.
fallback	fallback value to return if no match was found. Default: None

Returns

the state text if a match was found, otherwise the specified fallback value.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.446.3.4 info()

```
ttk_bus.bus_signals_frbs.TxFrBusSignal.info (
               self )
```

Get a current info string about the HIL signal variable, see signal_var.

Returns

the current info string from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.446.3.5 reset()

```
ttk_bus.bus_signals_frbs.TxFrBusSignal.reset (
               self,
              * args,
              ** kwargs )
```

Reset all resettable members/variables of this bus signal.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Reimplemented from ttk_base._variables_base.ResettableContainer.

7.446.3.6 set()

```
ttk_bus.bus_signals_frbs.TxFrBusSignal.set (
              self,
              value )
```

Set a value to the HIL signal variable, see signal_var.

Note

For TxFrBusSignals this will only overwrite the current contents of the HIl receive buffer of the signal (which is not too useful).

Parameters

value	new value to set

Returns

the previous value from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.446.3.7 setState()

```
ttk_bus.bus_signals_frbs.TxFrBusSignal.setState (
              self,
              state )
```

Set a textual state to the HIL signal variable, see signal_var.

This uses the lookup values supplied in constructor.

Note

For TxFrBusSignals this will only overwrite the current contents of the HIl receive buffer of the signal (which is not too useful).

Parameters

state	named state to set

Returns

the previous state description from signal_var.

Reimplemented from ttk_bus._bus_signals_base.BusSignal.

7.446.4 Member Data Documentation

7.446.4.1 alias

ttk_bus.bus_signals_frbs.TxFrBusSignal.alias

signal alias name (from signal_var)

7.446.4.2 base_path

ttk_bus.bus_signals_frbs.TxFrBusSignal.base_path

base model path (base path used for deriving sub-element paths)

7.446.4.3 change_delay

ttk_bus.bus_signals_frbs.TxFrBusSignal.change_delay

delay to wait after a value change [ms]

7.446.4.4 context

ttk_bus.bus_signals_frbs.TxFrBusSignal.context

context (reference to assigned RTE application instance)

7.446.4.5 cycle_time

ttk_bus.bus_signals_frbs.TxFrBusSignal.cycle_time

cycle time of signal in [ms] (that is, of the associated message or PDU)

7.446.4.6 debounce_time

ttk_bus.bus_signals_frbs.TxFrBusSignal.debounce_time

minimum debounce delay for event-triggered signals in [ms]

7.446.4.7 descr

ttk_bus.bus_signals_frbs.TxFrBusSignal.descr

signal description text

7.446.4.8 pdu

ttk_bus.bus_signals_frbs.TxFrBusSignal.pdu

reference to parent FrRxPdu

7.446.4.9 recovery_time

ttk_bus.bus_signals_frbs.TxFrBusSignal.recovery_time

time for recovery/"wiedergut" checks [ms]

7.446.4.10 signal_var

ttk_bus.bus_signals_frbs.TxFrBusSignal.signal_var

main bus signal variable on HIL-side (a "HilVar" instance)

7.446.4.11 status_var

ttk_bus.bus_signals_frbs.TxFrBusSignal.status_var

HIL-Rx-Status information (a "HilVar" instance).

> "Status information of the received or monitored signal. Only available if signal is selected for 'Signal RX Status Access' feature in FlexRay Configuration Tool"
> (compare FlexRayConfigFeatures.pdf)

The values have the following meanings:

- 0: No error
- 1: Access error
- 2: Signal is not received
- 4: Signal is invalid
- 8: CRC is incorrect

7.446.4.12 timeout

ttk_bus.bus_signals_frbs.TxFrBusSignal.timeout

detection time for timeout errors [ms]

7.446.4.13 unit

`ttk_bus.bus_signals_frbs.TxFrBusSignal.unit`

signal unit (from signal_var)

7.446.4.14 value_path

`ttk_bus.bus_signals_frbs.TxFrBusSignal.value_path`

model path of value variable

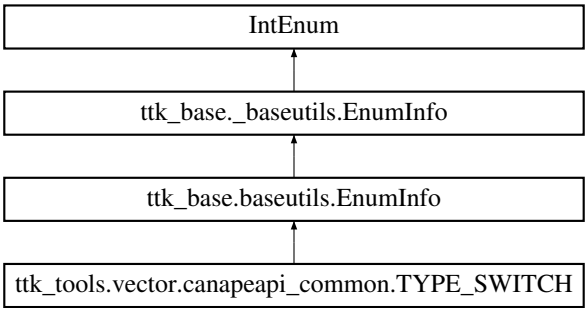
The documentation for this class was generated from the following file:

- ttk_bus/bus_signals_frbs.py (v1.8)

7.447 ttk_tools.vector.canapeapi_common.TYPE_SWITCH Class Reference

Device online switch state constants (`TYPE_SWITCH_*`) from enum `TAsap3ECUState`.

Inheritance diagram for ttk_tools.vector.canapeapi_common.TYPE_SWITCH:



Static Public Attributes

- int ONLINE = 0
- int OFFLINE = 1
- int DEACTIVATED = 3

7.447.1 Detailed Description

Device online switch state constants (`TYPE_SWITCH_*`) from enum `TAsap3ECUState`.

7.447.2 Member Data Documentation

7.447.2.1 DEACTIVATED

`int ttk_tools.vector.canapeapi_common.TYPE_SWITCH.DEACTIVATED = 3 [static]`

7.447.2.2 OFFLINE

```
int ttk_tools.vector.canapeapi_common.TYPE_SWITCH.OFFLINE = 1  [static]
```

7.447.2.3 ONLINE

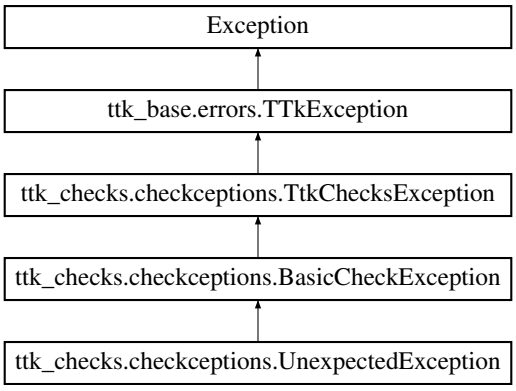
```
int ttk_tools.vector.canapeapi_common.TYPE_SWITCH.ONLINE = 0  [static]
```

The documentation for this class was generated from the following file:

- ttk_tools/vector/canapeapi_common.py (v3.1.2)

7.448 ttk_checks.checkceptions.UnexpectedException Class Reference

Inheritance diagram for ttk_checks.checkceptions.UnexpectedException:



Public Member Functions

- __init__ (self, msg='', original_exception=None, log_msg=True)

Public Attributes

- original_exception

7.448.1 Constructor & Destructor Documentation

7.448.1.1 __init__()

```
ttk_checks.checkceptions.UnexpectedException.__init__ (
              self,
              msg = '',
              original_exception = None,
              log_msg = True )
```

Parameters

msg	exception message text
original_exception	original exception, if any. Its message will be appended to the message text
log_msg	if True, the exception message will also be logged / printed to stdout

Reimplemented from ttk_checks.checkceptions.TtkChecksException.

7.448.2 Member Data Documentation

7.448.2.1 original_exception

`ttk_checks.checkceptions.UnexpectedException.original_exception`

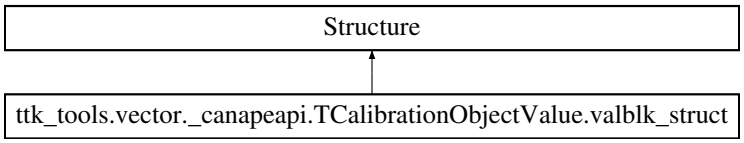
The documentation for this class was generated from the following file:

- ttk_checks/checkceptions.py (v2.0)

7.449 ttk_tools.vector._canapeapi.TCalibrationObjectValue.valblk_struct Class Reference

Value Block data.

Inheritance diagram for ttk_tools.vector._canapeapi.TCalibrationObjectValue.valblk_struct:



Static Protected Attributes

- int _pack_ = 1
- list _fields_

7.449.1 Detailed Description

Value Block data.

Number of elements = xDimension + yDimension.

7.449.2 Member Data Documentation

7.449.2.1 _fields_

```
list ttk_tools.vector._canapeapi.TCalibrationObjectValue.valblk_struct._fields_  [static],
[protected]
```

Initial value:

```
=  [
            ("type",        c_int),
            ("xDimension",  c_short),
            ("yDimension",  c_short),

            ("values",      POINTER(c_double)),
        ]
```

7.449.2.2 _pack_

```
int ttk_tools.vector._canapeapi.TCalibrationObjectValue.valblk_struct._pack_ = 1  [static],
[protected]
```

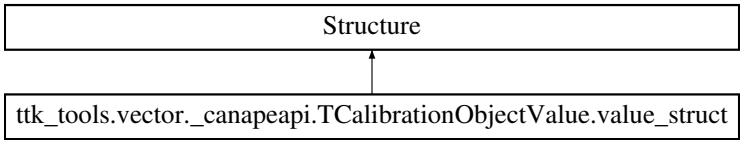
The documentation for this class was generated from the following file:

- ttk_tools/vector/_canapeapi.py (v7.3.3)

7.450 ttk_tools.vector._canapeapi.TCalibrationObjectValue.value_struct Class Reference

plain scalar values

Inheritance diagram for ttk_tools.vector._canapeapi.TCalibrationObjectValue.value_struct:



Static Protected Attributes

- int _pack_ = 1
- list _fields_

7.450.1 Detailed Description

plain scalar values

7.450.2 Member Data Documentation

7.450.2.1 _fields_

```
list ttk_tools.vector._canapeapi.TCalibrationObjectValue.value_struct._fields_  [static],
[protected]
```

Initial value:

```
=  [
            ("type",        c_int),
            ("value",       c_double),
        ]
```

7.450.2.2 _pack_

```
int ttk_tools.vector._canapeapi.TCalibrationObjectValue.value_struct._pack_ = 1  [static],
[protected]
```

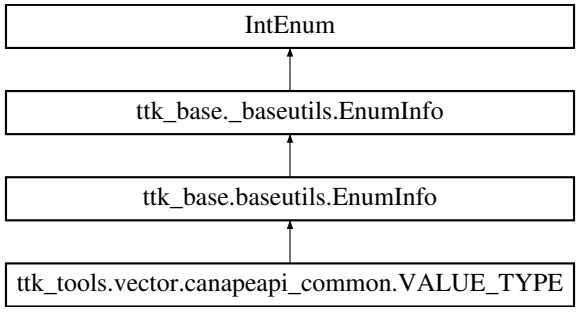
The documentation for this class was generated from the following file:

- ttk_tools/vector/_canapeapi.py (v7.3.3)

7.451 ttk_tools.vector.canapeapi_common.VALUE_TYPE Class Reference

Value type constants from enum `TValueType` (original constants have no specific prefix)

Inheritance diagram for ttk_tools.vector.canapeapi_common.VALUE_TYPE:



Static Public Attributes

- int VALUE = 0
  scalar value
- int CURVE = 1
  curve, 2D
- int MAP = 2
  map, 3D
- int AXIS = 3
  axis points
- int ASCII = 4
  string value
- int VAL_BLK = 5
  value block, 2D

7.451.1 Detailed Description

Value type constants from enum `TValueType` (original constants have no specific prefix)

7.451.2 Member Data Documentation

7.451.2.1 ASCII

```
int ttk_tools.vector.canapeapi_common.VALUE_TYPE.ASCII = 4  [static]
```

string value

7.451.2.2 AXIS

```
int ttk_tools.vector.canapeapi_common.VALUE_TYPE.AXIS = 3  [static]
```

axis points

7.451.2.3 CURVE

```
int ttk_tools.vector.canapeapi_common.VALUE_TYPE.CURVE = 1  [static]
```

curve, 2D

7.451.2.4 MAP

```
int ttk_tools.vector.canapeapi_common.VALUE_TYPE.MAP = 2  [static]
```

map, 3D

7.451.2.5 VAL_BLK

```
int ttk_tools.vector.canapeapi_common.VALUE_TYPE.VAL_BLK = 5  [static]
```

value block, 2D

7.451.2.6 VALUE

```
int ttk_tools.vector.canapeapi_common.VALUE_TYPE.VALUE = 0  [static]
```

scalar value

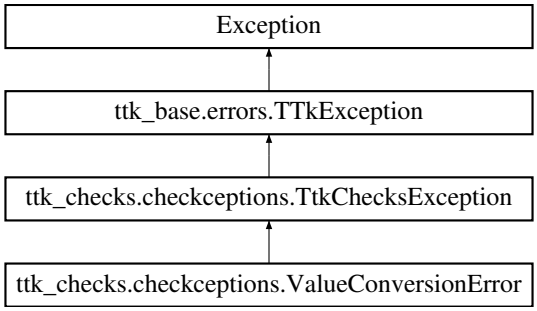
The documentation for this class was generated from the following file:

- ttk_tools/vector/canapeapi_common.py (v3.1.2)

7.452 ttk_checks.checkceptions.ValueConversionError Class Reference

A required precondition is not given.

Inheritance diagram for ttk_checks.checkceptions.ValueConversionError:



7.452.1 Detailed Description

A required precondition is not given.

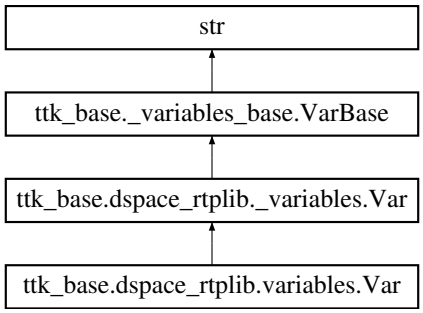
The documentation for this class was generated from the following file:

- ttk_checks/checkceptions.py (v2.0)

7.453 ttk_base.dspace_rtplib._variables.Var Class Reference

Simple wrapper for dSPACE (rtplib) "HIL" application variables.

Inheritance diagram for ttk_base.dspace_rtplib._variables.Var:



Public Member Functions

- __init__ (self, context, identifier, unit="", alias="", descr="", lookup=None, default=None, resettable=True, fmt=None)

  Rtplib-Variable initialization.
- get (self, *args, **kwargs)

  Get the current value of the application / model variable.
- set (self, value, *args, **kwargs)

  Set a value to the application / model variable.

Public Attributes

- variable

  reference to actual rtplib Appl.Variable instance
- alias

  alias/short name of this variable (potentially derived from model path)
- last_value

  last value of get/set operation
- orig_value

  original/reset value

7.453.1 Detailed Description

Simple wrapper for dSPACE (rtplib) "HIL" application variables.

Uses rtplib.Appl (or rtplib2.Appl) for accessing dSPACE real time variables.

7.453.2 Constructor & Destructor Documentation

7.453.2.1 __init__()

```
ttk_base.dspace_rtplib._variables.Var.__init__ (
              self,
              context,
              identifier,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              default = None,
              resettable = True,
              fmt = None )
```

Rtplib-Variable initialization.

Parameters

context	parent context (rtplib.Appl instance)
identifier	variable identifier: model path
unit	(optional) unit string for the variable
alias	(optional) alias name, defaults to last (non- default) entry in the identifier model path
descr	description for the variable
lookup	lookup dictionary for states {value => "state name"}
default	manually defined default value for the variable. Will be used as "reset" value instead of the original value before the first set operation if set to anything but None.
resettable	enable/disable reset() for this variable. If set to False, a call to reset() will be ignored.
fmt	format string or keyword to use for value representation, e.g. "%.2f" or "hex"

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_base.dspace_rtplib.variables.Var.

7.453.3 Member Function Documentation

7.453.3.1 get()

```
ttk_base.dspace_rtplib._variables.Var.get (
              self,
             * args,
             ** kwargs )
```

Get the current value of the application / model variable.

Note

Any specified parameters will be handed on to the underlying "read" operation, which may or may not be able to handle them. Use with care and only in very special cases.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Returns

the current value.

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_base.dspace_rtplib.variables.Var.

7.453.3.2 set()

```
ttk_base.dspace_rtplib._variables.Var.set (
              self,
              value,
             * args,
             ** kwargs )
```

Set a value to the application / model variable.

Parameters

value	value to set
**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Note

Any specified additional parameters will be handed on to the underlying "read" operation, which may or may not be able to handle them. Use with care and only in very special cases.

Returns

the original value before the set operation occurred.

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_base.dspace_rtplib.variables.Var.

7.453.4 Member Data Documentation

7.453.4.1 alias

`ttk_base.dspace_rtplib._variables.Var.alias`

alias/short name of this variable (potentially derived from model path)

7.453.4.2 last_value

`ttk_base.dspace_rtplib._variables.Var.last_value`

last value of get/set operation

7.453.4.3 orig_value

`ttk_base.dspace_rtplib._variables.Var.orig_value`

original/reset value

7.453.4.4 variable

`ttk_base.dspace_rtplib._variables.Var.variable`

reference to actual rtplib Appl.Variable instance

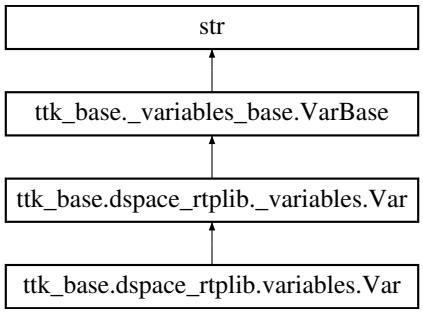
The documentation for this class was generated from the following file:

- ttk_base/dspace_rtplib/_variables.py (v1.4)

7.454 ttk_base.dspace_rtplib.variables.Var Class Reference

Simple wrapper for dSPACE (rtplib) "HIL" application variables.

Inheritance diagram for ttk_base.dspace_rtplib.variables.Var:



Public Member Functions

- __init__ (self, context, identifier, unit="", alias="", descr="", lookup=None, default=None, resettable=True, fmt=None)

  Rtplib Variable initialization.
- get (self, ∗args, ∗∗kwargs)

  Get the current value of the application / model variable.
- set (self, value, ∗args, ∗∗kwargs)

  Set a value to the application / model variable.
- getState (self)

  Get a textual state for the variable's current data.
- setState (self, state)

  Set a textual state to the variable.
- reset (self, verbosity=1)

  Reset the variable to its initial value (default value specified in constructor or value before the first set() call).
- info (self)

  Get an info string for this variable (alias and current value plus unit and lookup value, if available).

7.454.1 Detailed Description

Simple wrapper for dSPACE (rtplib) "HIL" application variables.

Uses rtplib.Appl (or rtplib2.Appl) for accessing dSPACE real time variables.

7.454.2 Constructor & Destructor Documentation

7.454.2.1 __init__()

```
ttk_base.dspace_rtplib.variables.Var.__init__ (
              self,
              context,
              identifier,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              default = None,
              resettable = True,
              fmt = None )
```

Rtplib Variable initialization.

Parameters

context	parent context (rtplib.Appl instance)
identifier	variable identifier: model path
unit	(optional) unit string for the variable
alias	(optional) alias name, defaults to last (non- default) entry in the identifier model path
descr	description for the variable
lookup	lookup dictionary for states {value => "state name"}
default	manually defined default value for the variable. Will be used as "reset" value instead of the original value before the first set operation if set to anything but None.
resettable	enable/disable reset() for this variable. If set to False, a call to reset() will be ignored.
fmt	format string or keyword to use for value representation, e.g. "%.2f" or "hex"

Reimplemented from ttk_base.dspace_rtplib._variables.Var.

7.454.3 Member Function Documentation

7.454.3.1 get()

```
ttk_base.dspace_rtplib.variables.Var.get (
              self,
             * args,
             ** kwargs )
```

Get the current value of the application / model variable.

Note

Any specified parameters will be handed on to the underlying "read" operation, which may or may not be able to handle them. Use with care and only in very special cases.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Returns

the current value.

Reimplemented from ttk_base.dspace_rtplib._variables.Var.

7.454.3.2 getState()

```
ttk_base.dspace_rtplib.variables.Var.getState (
              self )
```

Get a textual state for the variable's current data.

This uses the lookup values supplied in constructor.

Returns

a textual state description (if available) or the unmapped value (if no matching state was found).

Reimplemented from ttk_base._variables_base.VarBase.

7.454.3.3 info()

```
ttk_base.dspace_rtplib.variables.Var.info (
              self )
```

Get an info string for this variable (alias and current value plus unit and lookup value, if available).

Example:

```
>> print voltage_var.info()
Hil-Var Name: 12 V
>> print status_var.info()
Status Var Name: 1 (valid)
```

Returns

a formatted string.

Reimplemented from ttk_base._variables_base.VarBase.

7.454.3.4 reset()

```
ttk_base.dspace_rtplib.variables.Var.reset (
              self,
              verbosity = 1 )
```

Reset the variable to its initial value (default value specified in constructor or value before the first set() call).

Parameters

verbosity	verbosity of log output, 0: silent, 1: show calls to non-resettable variables 2: show all reset calls

Returns

the original value (which may be None if the variable has not been set before)

Reimplemented from ttk_base._variables_base.VarBase.

7.454.3.5 set()

```
ttk_base.dspace_rtplib.variables.Var.set (
              self,
              value,
             * args,
             ** kwargs )
```

Set a value to the application / model variable.

Parameters

value	value to set
∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Note

Any specified additional parameters will be handed on to the underlying "read" operation, which may or may not be able to handle them. Use with care and only in very special cases.

Returns

the original value before the set operation occurred.

Reimplemented from ttk_base.dspace_rtplib._variables.Var.

7.454.3.6 setState()

```
ttk_base.dspace_rtplib.variables.Var.setState (
              self,
              state )
```

Set a textual state to the variable.

This uses the lookup values supplied in constructor. Fallback (if no matching state text was found) will be the unmapped input state.

Parameters

state	named state to set

Returns

the previously set state (if possible, otherwise the previous unmapped value, see getState)

Reimplemented from ttk_base._variables_base.VarBase.

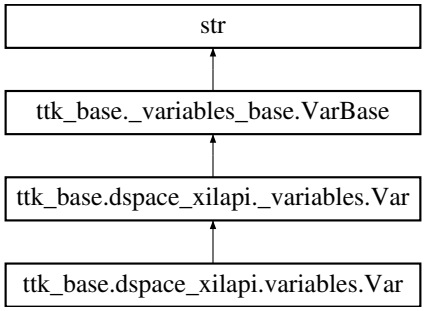
The documentation for this class was generated from the following file:

- ttk_base/dspace_rtplib/variables.py (v1.2)

7.455 ttk_base.dspace_xilapi._variables.Var Class Reference

Simple wrapper for dSPACE (XIL-API) "HIL" application variables.

Inheritance diagram for ttk_base.dspace_xilapi._variables.Var:



Public Member Functions

- __init__ (self, context, identifier, unit="", alias="", descr="", lookup=None, default=None, resettable=True, fmt=None)

  XIL-API/Testbench Variable initialization.
- get (self, *args, **kwargs)

  Get the current value of the application / model variable.
- set (self, value, *args, **kwargs)

  Set a value to the application / model variable.

Public Attributes

- context

  reference to XilTestbench instance
- alias

  alias/short name of this variable (potentially derived from model path)
- last_value

  last value of get/set operation
- orig_value

  original/reset value

7.455.1 Detailed Description

Simple wrapper for dSPACE (XIL-API) "HIL" application variables.

Uses a ttk_tools.dspace.xil_api.XilTestbench instance for variable access via a XIL MAPort.

7.455.2 Constructor & Destructor Documentation

7.455.2.1 __init__()

```
ttk_base.dspace_xilapi._variables.Var.__init__ (
              self,
              context,
              identifier,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              default = None,
              resettable = True,
              fmt = None )
```

XIL-API/Testbench Variable initialization.

Parameters

context	parent context (here: a XilTestbench instance)
identifier	variable identifier: model path
unit	(optional) unit string for the variable
alias	(optional) alias name, defaults to last (non- default) entry in the identifier model path
descr	description for the variable
lookup	lookup dictionary for states {value => "state name"}
default	manually defined default value for the variable. Will be used as "reset" value instead of the original value before the first set operation if set to anything but None.
resettable	enable/disable reset() for this variable. If set to False, a call to reset() will be ignored.
fmt	format string or keyword to use for value representation, e.g. "%.2f" or "hex"

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_base.dspace_xilapi.variables.Var.

7.455.3 Member Function Documentation

7.455.3.1 get()

```
ttk_base.dspace_xilapi._variables.Var.get (
              self,
             * args,
             ** kwargs )
```

Get the current value of the application / model variable.

Note

Any specified parameters will be discarded (the XIL-Testbench read implementation does not handle any additional parameters)

Parameters

∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Returns

the current value.

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_base.dspace_xilapi.variables.Var.

7.455.3.2 set()

```
ttk_base.dspace_xilapi._variables.Var.set (
              self,
              value,
             * args,
             ** kwargs )
```

Set a value to the application / model variable.

Parameters

value	value to set
∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Note

Any additional parameters will be discarded (the XIL-Testbench write implementation does not support any additional parameters)

Returns

the original value before the set operation occurred.

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_base.dspace_xilapi.variables.Var.

7.455.4 Member Data Documentation

7.455.4.1 alias

```
ttk_base.dspace_xilapi._variables.Var.alias
```

alias/short name of this variable (potentially derived from model path)

7.455.4.2 context

`ttk_base.dspace_xilapi._variables.Var.context`

reference to XilTestbench instance

7.455.4.3 last_value

`ttk_base.dspace_xilapi._variables.Var.last_value`

last value of get/set operation

7.455.4.4 orig_value

`ttk_base.dspace_xilapi._variables.Var.orig_value`

original/reset value

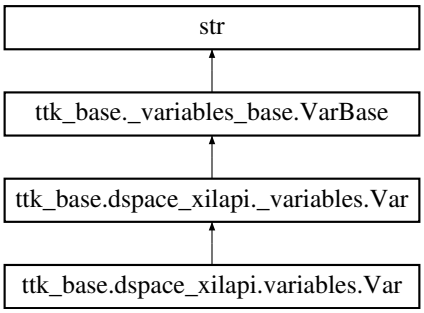
The documentation for this class was generated from the following file:

- ttk_base/dspace_xilapi/_variables.py (v1.4)

7.456 ttk_base.dspace_xilapi.variables.Var Class Reference

Simple wrapper for dSPACE (XIL-API) "HIL" application variables.

Inheritance diagram for ttk_base.dspace_xilapi.variables.Var:



Public Member Functions

- __init__ (self, context, identifier, unit="", alias="", descr="", lookup=None, default=None, resettable=True, fmt=None)
  XIL-API/Testbench Variable initialization.
- get (self, *args, **kwargs)
  Get the current value of the application / model variable.
- set (self, value, *args, **kwargs)
  Set a value to the application / model variable.
- getState (self)
  Get a textual state for the variable's current data.
- setState (self, state)
  Set a textual state to the variable.
- reset (self, verbosity=1)
  Reset the variable to its initial value (default value specified in constructor or value before the first set() call).
- info (self)
  Get an info string for this variable (alias and current value plus unit and lookup value, if available).

7.456.1 Detailed Description

Simple wrapper for dSPACE (XIL-API) "HIL" application variables.

Uses a ttk_tools.dspace.xil_api.XilTestbench instance for variable access via a XIL MAPort.

7.456.2 Constructor & Destructor Documentation

7.456.2.1 __init__()

```
ttk_base.dspace_xilapi.variables.Var.__init__ (
              self,
              context,
              identifier,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              default = None,
              resettable = True,
              fmt = None )
```

XIL-API/Testbench Variable initialization.

Parameters

context	parent context (here: a XilTestbench instance)
identifier	variable identifier: model path
unit	(optional) unit string for the variable
alias	(optional) alias name, defaults to last (non- default) entry in the identifier model path
descr	description for the variable
lookup	lookup dictionary for states {value => "state name"}
default	manually defined default value for the variable. Will be used as "reset" value instead of the original value before the first set operation if set to anything but None.
resettable	enable/disable reset() for this variable. If set to False, a call to reset() will be ignored.
fmt	format string or keyword to use for value representation, e.g. "%.2f" or "hex"

Reimplemented from ttk_base.dspace_xilapi._variables.Var.

7.456.3 Member Function Documentation

7.456.3.1 get()

```
ttk_base.dspace_xilapi.variables.Var.get (
             self,
            * args,
            ** kwargs )
```

Get the current value of the application / model variable.

Note

Any specified parameters will be discarded (the XIL-Testbench read implementation does not handle any additional parameters)

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Returns

the current value.

Reimplemented from ttk_base.dspace_xilapi._variables.Var.

7.456.3.2 getState()

```
ttk_base.dspace_xilapi.variables.Var.getState (
              self )
```

Get a textual state for the variable's current data.

This uses the lookup values supplied in constructor.

Returns

a textual state description (if available) or the unmapped value (if no matching state was found).

Reimplemented from ttk_base._variables_base.VarBase.

7.456.3.3 info()

```
ttk_base.dspace_xilapi.variables.Var.info (
              self )
```

Get an info string for this variable (alias and current value plus unit and lookup value, if available).

Example:

```
>> print voltage_var.info()
Xil-Var Name: 12 V
>> print status_var.info()
Status Var Name: 1 (valid)
```

Returns

a formatted string.

Reimplemented from ttk_base._variables_base.VarBase.

7.456.3.4 reset()

```
ttk_base.dspace_xilapi.variables.Var.reset (
              self,
              verbosity = 1 )
```

Reset the variable to its initial value (default value specified in constructor or value before the first set() call).

Parameters

verbosity	verbosity of log output. 0: silent, 1: show calls to non-resettable variables 2: show all reset calls

Returns

the original value (which may be None if the variable has not been set before)

Reimplemented from ttk_base._variables_base.VarBase.

7.456.3.5 set()

```
ttk_base.dspace_xilapi.variables.Var.set (
              self,
              value,
             * args,
             ** kwargs )
```

Set a value to the application / model variable.

Parameters

value	value to set
∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Note

Any additional parameters will be discarded (the XIL-Testbench write implementation does not support any additional parameters)

Returns

the original value before the set operation occurred.

Reimplemented from ttk_base.dspace_xilapi._variables.Var.

7.456.3.6 setState()

```
ttk_base.dspace_xilapi.variables.Var.setState (
              self,
              state )
```

Set a textual state to the variable.

This uses the lookup values supplied in constructor. Fallback (if no matching state text was found) will be the unmapped input state.

Parameters

state	named state to set

Returns

the previously set state (if possible, otherwise the previous unmapped value, see getState)

Reimplemented from ttk_base._variables_base.VarBase.

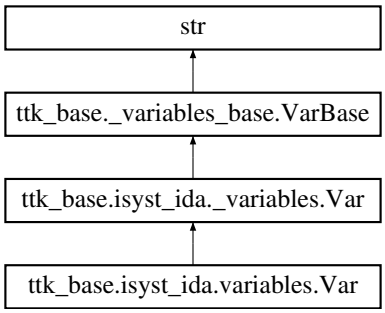
The documentation for this class was generated from the following file:

- ttk_base/dspace_xilapi/variables.py (v1.2)

7.457 ttk_base.isyst_ida._variables.Var Class Reference

Simple wrapper for iDA Testware "HIL" application variables.

Inheritance diagram for ttk_base.isyst_ida._variables.Var:



Public Member Functions

- __init__ (self, context, identifier, unit="", alias="", descr="", lookup=None, default=None, resettable=True, fmt=None)

  iDA-Variable initialization.
- get (self, *args, **kwargs)

  Get the current value of the application / model variable.
- set (self, value, *args, **kwargs)

  Set a value to the application / model variable.

Public Attributes

- context

  reference to MwAppRpc instance
- mwvar

  MwVar proxy that handles communication with the service for this variable.
- alias

  alias/short name of this variable
- unit

  unit of this variable.
- fmt

  display format hint of this variable.
- last_value

  last value of get/set operation
- orig_value

  original/reset value

Static Public Attributes

- bool use_getset_fallback = False
- bool use_reset_fallback = False

7.457.1 Detailed Description

Simple wrapper for iDA Testware "HIL" application variables.

7.457.2 Constructor & Destructor Documentation

7.457.2.1 __init__()

```
ttk_base.isyst_ida._variables.Var.__init__ (
              self,
              context,
              identifier,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              default = None,
              resettable = True,
              fmt = None )
```

iDA-Variable initialization.

Parameters

context	parent context (here: an IdaTestware or MwAppRpc instance)
identifier	variable identifier/name
unit	(optional) unit string for the variable
alias	(optional) alias name, defaults to last (non- default) entry in the identifier model path
descr	description for the variable
lookup	lookup dictionary for states {value => "state name"}
default	manually defined default value for the variable. Will be used as "reset" value instead of the original value before the first set operation if set to anything but None.
resettable	enable/disable reset() for this variable. If set to False, a call to reset() will be ignored.
fmt	format string or keyword to use for value representation, e.g. "%.2f" or "hex"

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_base.isyst_ida.variables.Var.

7.457.3 Member Function Documentation

7.457.3.1 get()

```
ttk_base.isyst_ida._variables.Var.get (
              self,
```

```
            * args,
            ** kwargs )
```

Get the current value of the application / model variable.

Note

Any specified parameters will be discarded (iDA's read implementation does not handle any additional parameters)

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Returns

the current value.

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_base.isyst_ida.variables.Var.

7.457.3.2 set()

```
ttk_base.isyst_ida._variables.Var.set (
              self,
              value,
             * args,
             ** kwargs )
```

Set a value to the application / model variable.

Parameters

value	value to set
**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Note

Any additional parameters will be discarded (iDA's write implementation does not support any additional parameters)

Returns

the original value before the set operation occurred.

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_base.isyst_ida.variables.Var.

7.457.4 Member Data Documentation

7.457.4.1 alias

ttk_base.isyst_ida._variables.Var.alias

alias/short name of this variable

7.457.4.2 context

ttk_base.isyst_ida._variables.Var.context

reference to MwAppRpc instance

7.457.4.3 fmt

ttk_base.isyst_ida._variables.Var.fmt

display format hint of this variable.

Potentially provided by mwservice configuration

7.457.4.4 last_value

ttk_base.isyst_ida._variables.Var.last_value

last value of get/set operation

7.457.4.5 mwvar

ttk_base.isyst_ida._variables.Var.mwvar

MwVar proxy that handles communication with the service for this variable.

7.457.4.6 orig_value

ttk_base.isyst_ida._variables.Var.orig_value

original/reset value

7.457.4.7 unit

ttk_base.isyst_ida._variables.Var.unit

unit of this variable.

Potentially provided by mwservice configuration

7.457.4.8 use_getset_fallback

```
bool ttk_base.isyst_ida._variables.Var.use_getset_fallback = False  [static]
```

7.457.4.9 use_reset_fallback

```
bool ttk_base.isyst_ida._variables.Var.use_reset_fallback = False  [static]
```

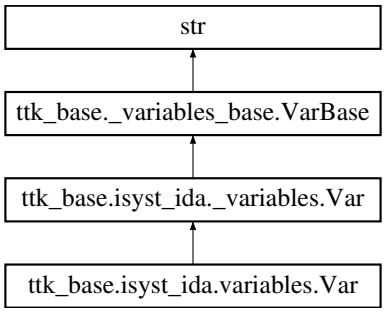
The documentation for this class was generated from the following file:

- ttk_base/isyst_ida/_variables.py (v1.2.0)

7.458 ttk_base.isyst_ida.variables.Var Class Reference

Simple wrapper for iDA Testware "HIL" application variables.

Inheritance diagram for ttk_base.isyst_ida.variables.Var:



Public Member Functions

- __init__ (self, context, identifier, unit="", alias="", descr="", lookup=None, default=None, resettable=True, fmt=None)

  iDA-Variable initialization.
- get (self, *args, **kwargs)

  Get the current value of the application / model variable.
- set (self, value, *args, **kwargs)

  Set a value to the application / model variable.
- getState (self)

  Get a textual state for the variable's current data.
- setState (self, state)

  Set a textual state to the variable.
- reset (self, verbosity=1)

  Reset the variable to its initial value (default value specified in constructor or value before the first set() call).
- info (self)

  Get an info string for this variable (alias and current value plus unit and lookup value, if available).

7.458.1 Detailed Description

Simple wrapper for iDA Testware "HIL" application variables.

7.458.2 Constructor & Destructor Documentation

7.458.2.1 __init__()

```
ttk_base.isyst_ida.variables.Var.__init__ (
              self,
              context,
              identifier,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              default = None,
              resettable = True,
              fmt = None )
```

iDA-Variable initialization.

Parameters

context	parent context (here: an IdaTestware or MwAppRpc instance)
identifier	variable identifier: model path
unit	(optional) unit string for the variable
alias	(optional) alias name, defaults to last (non- default) entry in the identifier model path
descr	description for the variable
lookup	lookup dictionary for states {value => "state name"}
default	manually defined default value for the variable. Will be used as "reset" value instead of the original value before the first set operation if set to anything but None.
resettable	enable/disable reset() for this variable. If set to False, a call to reset() will be ignored.
fmt	format string or keyword to use for value representation, e.g. "%.2f" or "hex"

Reimplemented from ttk_base.isyst_ida._variables.Var.

7.458.3 Member Function Documentation

7.458.3.1 get()

```
ttk_base.isyst_ida.variables.Var.get (
             self,
            * args,
            ** kwargs )
```

Get the current value of the application / model variable.

Note

Any specified parameters will be discarded (iDA's read implementation does not handle any additional parameters)

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Returns

the current value.

Reimplemented from ttk_base.isyst_ida._variables.Var.

7.458.3.2 getState()

```
ttk_base.isyst_ida.variables.Var.getState (
              self )
```

Get a textual state for the variable's current data.

This uses the lookup values supplied in constructor.

Returns

a textual state description (if available) or the unmapped value (if no matching state was found).

Reimplemented from ttk_base._variables_base.VarBase.

7.458.3.3 info()

```
ttk_base.isyst_ida.variables.Var.info (
              self )
```

Get an info string for this variable (alias and current value plus unit and lookup value, if available).

Example:

```
>> print voltage_var.info()
MW-Var Name: 12 V
>> print status_var.info()
Status Var Name: 1 (valid)
```

Returns

a formatted string.

Reimplemented from ttk_base._variables_base.VarBase.

7.458.3.4 reset()

```
ttk_base.isyst_ida.variables.Var.reset (
              self,
              verbosity = 1 )
```

Reset the variable to its initial value (default value specified in constructor or value before the first set() call).

Parameters

verbosity	verbosity of log output. 0: silent, 1: show calls to non-resettable variables 2: show all reset calls

Returns

the original value (which may be None if the variable has not been set before)

Reimplemented from ttk_base._variables_base.VarBase.

7.458.3.5 set()

```
ttk_base.isyst_ida.variables.Var.set (
              self,
              value,
             * args,
             ** kwargs )
```

Set a value to the application / model variable.

Parameters

value	value to set
∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Note

Any additional parameters will be discarded (iDA's write implementation does not support any additional parameters)

Returns

the original value before the set operation occurred.

Reimplemented from ttk_base.isyst_ida._variables.Var.

7.458.3.6 setState()

```
ttk_base.isyst_ida.variables.Var.setState (
              self,
              state )
```

Set a textual state to the variable.

This uses the lookup values supplied in constructor. Fallback (if no matching state text was found) will be the unmapped input state.

Parameters

state	named state to set

Returns

the previously set state (if possible, otherwise the previous unmapped value, see getState)

Reimplemented from ttk_base._variables_base.VarBase.

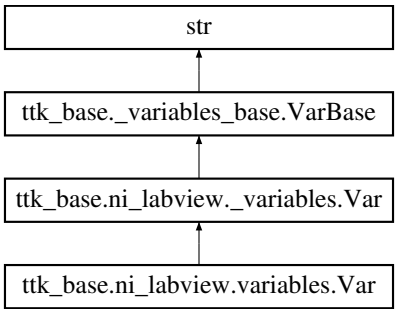
The documentation for this class was generated from the following file:

- ttk_base/isyst_ida/variables.py (v1.1.1)

7.459 ttk_base.ni_labview._variables.Var Class Reference

Simple wrapper for NI LabView "HIL" variables/controls.

Inheritance diagram for ttk_base.ni_labview._variables.Var:



Public Member Functions

- __init__ (self, context, identifier, unit="", alias="", descr="", lookup=None, default=None, resettable=True, fmt=None)
- get (self, *args, **kwargs)

  Get the current value of the VI control / variable.
- set (self, value, *args, **kwargs)

  Set a value to the VI control / variable.

Public Attributes

- context

  reference to LabView COM-wrapper instance
- alias
- last_value

  last value of get/set operation
- orig_value

  original/reset value

7.459.1 Detailed Description

Simple wrapper for NI LabView "HIL" variables/controls.

Uses a ttk_tools.ni.labview.LabView instance for variable access via COM.

7.459.2 Constructor & Destructor Documentation

7.459.2.1 __init__()

```
ttk_base.ni_labview._variables.Var.__init__ (
              self,
              context,
              identifier,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              default = None,
              resettable = True,
              fmt = None )
```

Parameters

context	parent context (here: a LabView instance)
identifier	variable identifier: "vi_name.vi/variable_name"
unit	(optional) unit string for the variable
alias	(optional) alias name, defaults to last (non- default) entry in the identifier model path
descr	description for the variable
lookup	lookup dictionary for states {value => "state name"}
default	manually defined default value for the variable. Will be used as "reset" value instead of the original value before the first set operation if set to anything but None.
resettable	enable/disable reset() for this variable. If set to False, a call to reset() will be ignored.
fmt	format string or keyword to use for value representation, e.g. "%.2f" or "hex"

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_base.ni_labview.variables.Var.

7.459.3 Member Function Documentation

7.459.3.1 get()

```
ttk_base.ni_labview._variables.Var.get (
              self,
             * args,
             ** kwargs )
```

Get the current value of the VI control / variable.

Note

Any specified parameters will be discarded (the XIL-Testbench read implementation does not handle any additional parameters)

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Returns

the current value.

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_base.ni_labview.variables.Var.

7.459.3.2 set()

```
ttk_base.ni_labview._variables.Var.set (
              self,
              value,
             * args,
             ** kwargs )
```

Set a value to the VI control / variable.

Parameters

value	value to set
**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Note

Any additional parameters will be discarded (the XIL-Testbench write implementation does not support any additional parameters)

Returns

the original value before the set operation occurred.

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_base.ni_labview.variables.Var.

7.459.4 Member Data Documentation

7.459.4.1 alias

```
ttk_base.ni_labview._variables.Var.alias
```

7.459.4.2 context

`ttk_base.ni_labview._variables.Var.context`

reference to LabView COM-wrapper instance

7.459.4.3 last_value

`ttk_base.ni_labview._variables.Var.last_value`

last value of get/set operation

7.459.4.4 orig_value

`ttk_base.ni_labview._variables.Var.orig_value`

original/reset value

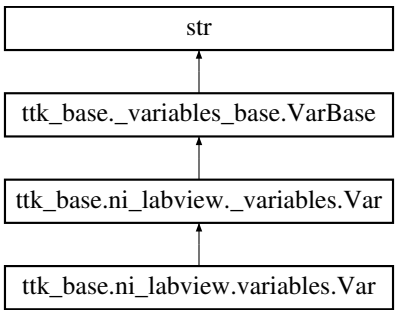
The documentation for this class was generated from the following file:

- ttk_base/ni_labview/_variables.py (v1.1)

7.460 ttk_base.ni_labview.variables.Var Class Reference

Simple wrapper for NI LabView "HIL" variables/controls.

Inheritance diagram for ttk_base.ni_labview.variables.Var:



Public Member Functions

- __init__ (self, context, identifier, unit="", alias="", descr="", lookup=None, default=None, resettable=True, fmt=None)
- get (self, *args, **kwargs)

  Get the current value of the VI control / variable.
- set (self, value, *args, **kwargs)

  Set a value to the VI control / variable.
- getState (self)

  Get a textual state for the variable's current data.
- setState (self, state)

  Set a textual state to the variable.
- reset (self, verbosity=1)

  Reset the variable to its initial value (default value specified in constructor or value before the first set() call).
- info (self)

  Get an info string for this variable (alias and current value plus unit and lookup value, if available).

7.460.1 Detailed Description

Simple wrapper for NI LabView "HIL" variables/controls.

Uses a ttk_tools.ni.labview.LabView instance for variable access via COM.

7.460.2 Constructor & Destructor Documentation

7.460.2.1 __init__()

```
ttk_base.ni_labview.variables.Var.__init__ (
              self,
              context,
              identifier,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              default = None,
              resettable = True,
              fmt = None )
```

Parameters

context	parent context (here: a LabView instance)
identifier	variable identifier: "vi_name.vi/variable_name"
unit	(optional) unit string for the variable
alias	(optional) alias name, defaults to last (non- default) entry in the identifier model path
descr	description for the variable
lookup	lookup dictionary for states {value => "state name"}
default	manually defined default value for the variable. Will be used as "reset" value instead of the original value before the first set operation if set to anything but None.
resettable	enable/disable reset() for this variable. If set to False, a call to reset() will be ignored.
fmt	format string or keyword to use for value representation, e.g. "%.2f" or "hex"

Reimplemented from ttk_base.ni_labview._variables.Var.

7.460.3 Member Function Documentation

7.460.3.1 get()

```
ttk_base.ni_labview.variables.Var.get (
              self,
             * args,
             ** kwargs )
```

Get the current value of the VI control / variable.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Returns

the current value.

Reimplemented from ttk_base.ni_labview._variables.Var.

7.460.3.2 getState()

```
ttk_base.ni_labview.variables.Var.getState (
              self )
```

Get a textual state for the variable's current data.

This uses the lookup values supplied in constructor.

Returns

a textual state description (if available) or the unmapped value (if no matching state was found).

Reimplemented from ttk_base._variables_base.VarBase.

7.460.3.3 info()

```
ttk_base.ni_labview.variables.Var.info (
              self )
```

Get an info string for this variable (alias and current value plus unit and lookup value, if available).

Example:

```
>> print voltage_var.info()
Var Name: 12 V
>> print status_var.info()
Status Var Name: 1 (valid)
```

Returns

a formatted string.

Reimplemented from ttk_base._variables_base.VarBase.

7.460.3.4 reset()

```
ttk_base.ni_labview.variables.Var.reset (
              self,
              verbosity = 1 )
```

Reset the variable to its initial value (default value specified in constructor or value before the first set() call).

Parameters

verbosity	verbosity of log output. 0: silent, 1: show calls to non-resettable variables 2: show all reset calls

Returns

the original value (which may be None if the variable has not been set before)

Reimplemented from ttk_base._variables_base.VarBase.

7.460.3.5 set()

```
ttk_base.ni_labview.variables.Var.set (
              self,
              value,
             * args,
             ** kwargs )
```

Set a value to the VI control / variable.

Parameters

value	value to set
∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Returns

the original value before the set operation occurred.

Reimplemented from ttk_base.ni_labview._variables.Var.

7.460.3.6 setState()

```
ttk_base.ni_labview.variables.Var.setState (
              self,
              state )
```

Set a textual state to the variable.

This uses the lookup values supplied in constructor. Fallback (if no matching state text was found) will be the unmapped input state.

Parameters

state	named state to set

Returns

the previously set state (if possible, otherwise the previous unmapped value, see getState)

Reimplemented from ttk_base._variables_base.VarBase.

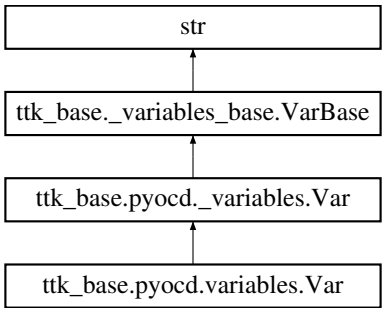
The documentation for this class was generated from the following file:

- ttk_base/ni_labview/variables.py (v1.1)

7.461 ttk_base.pyocd._variables.Var Class Reference

Simple wrapper for PyOCD variables in a attached debugging session.

Inheritance diagram for ttk_base.pyocd._variables.Var:



Public Member Functions

- __init__ (self, context, identifier, addr_offset=0, data_type="uint8", unit="", alias="", descr="", lookup=None, default=None, resettable=True, fmt=None)

  Debugger variable initialization.
- get (self, *args, **kwargs)

  Get the current value of the application / model variable.
- set (self, value, *args, **kwargs)

  Set a value to the application variable.

Public Attributes

- context

  reference to Debugger instance
- alias

  alias/short name of this variable
- data_type

  data type of variable
- base_address

  address of variable in target memory
- address
- last_value

  last value of get/set operation
- orig_value

  original/reset value

Protected Member Functions

- _setAccessors (self)

  Select/cache appropriate read/write functions for later use.

Static Protected Member Functions

- _normalizeDataType (name)

  Normalize data type name to a standard form.

Protected Attributes

- _byte_size

  size of variable in Bytes (will be updated when selecting suitable read/write implementation for data type, see _set↩ Accessors)
- _readValue
- _writeValue

7.461.1 Detailed Description

Simple wrapper for PyOCD variables in a attached debugging session.

7.461.2 Constructor & Destructor Documentation

7.461.2.1 __init__()

```
ttk_base.pyocd._variables.Var.__init__ (
              self,
              context,
              identifier,
              addr_offset = 0,
              data_type = "uint8",
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              default = None,
              resettable = True,
              fmt = None )
```

Debugger variable initialization.

Note

Array elements: Using an identifier with index, e.g. foobar[2] will calculate the required address offset to the selected index based on the variable's data type.
This only works for basic arrays and data types, not for nested data structures.

Parameters

context	parent context (here: a ttk_tools.pyocd.debugger.Debugger instance)
identifier	variable identifier: symbol name from ELF file
addr_offset	offset in Bytes to address of identifier symbol
data_type	data type for decoding/encoding memory contents
unit	(optional) unit string for the variable
alias	(optional) alias name, defaults to last (non- default) entry in the identifier model path
descr	description for the variable
lookup	lookup dictionary for states {value => "state name"}
default	manually defined default value for the variable. Will be used as "reset" value instead of the original value before the first set operation if set to anything but None.
resettable	enable/disable reset() for this variable. If set to False, a call to reset() will be ignored.
fmt	format string or keyword to use for value representation, e.g. "%.2f" or "hex"

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_base.pyocd.variables.Var.

7.461.3 Member Function Documentation

7.461.3.1 _normalizeDataType()

```
ttk_base.pyocd._variables.Var._normalizeDataType (
              name )  [static], [protected]
```

Normalize data type name to a standard form.

Note

Data Types:

- signed: sint8, sint16, sint32, or s8, s16, s32, or int8_t
- unsigned: uint8, uint16, uint32, or u8, u16, u32, or uint8_t
- float: float32, float64, or f32, f64 or float, double

Returns

a data type name string

7.461.3.2 _setAccessors()

```
ttk_base.pyocd._variables.Var._setAccessors (
              self )  [protected]
```

Select/cache appropriate read/write functions for later use.

7.461.3.3 get()

```
ttk_base.pyocd._variables.Var.get (
              self,
             * args,
             ** kwargs )
```

Get the current value of the application / model variable.

Note

Any specified parameters will be discarded read implementation does not handle any additional parameters)

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Returns

the current value.

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_base.pyocd.variables.Var.

7.461.3.4 set()

```
ttk_base.pyocd._variables.Var.set (
              self,
              value,
             * args,
             ** kwargs )
```

Set a value to the application variable.

Parameters

value	value to set
**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Note

Any additional parameters will be discarded (write implementation does not support any additional parameters)

Returns

the original value before the set operation occurred.

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_base.pyocd.variables.Var.

7.461.4 Member Data Documentation

7.461.4.1 _byte_size

ttk_base.pyocd._variables.Var._byte_size [protected]

size of variable in Bytes (will be updated when selecting suitable read/write implementation for data type, see _↩ setAccessors)

7.461.4.2 _readValue

ttk_base.pyocd._variables.Var._readValue [protected]

7.461.4.3 _writeValue

ttk_base.pyocd._variables.Var._writeValue [protected]

7.461.4.4 address

ttk_base.pyocd._variables.Var.address

7.461.4.5 alias

ttk_base.pyocd._variables.Var.alias

alias/short name of this variable

7.461.4.6 base_address

ttk_base.pyocd._variables.Var.base_address

address of variable in target memory

7.461.4.7 context

ttk_base.pyocd._variables.Var.context

reference to Debugger instance

7.461.4.8 data_type

ttk_base.pyocd._variables.Var.data_type

data type of variable

7.461.4.9 last_value

`ttk_base.pyocd._variables.Var.last_value`

last value of get/set operation

7.461.4.10 orig_value

`ttk_base.pyocd._variables.Var.orig_value`

original/reset value

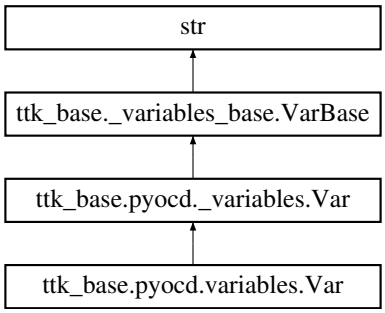
The documentation for this class was generated from the following file:

- ttk_base/pyocd/_variables.py (v1.0.0)

7.462 ttk_base.pyocd.variables.Var Class Reference

Simple wrapper for PyOCD variables in a attached debugging session.

Inheritance diagram for ttk_base.pyocd.variables.Var:



Public Member Functions

- __init__ (self, context, identifier, addr_offset=0, data_type="uint8", unit="", alias="", descr="", lookup=None, default=None, resettable=True, fmt=None)

  Debugger variable initialization.
- get (self, ∗args, ∗∗kwargs)

  Get the current value of the application / model variable.
- set (self, value, ∗args, ∗∗kwargs)

  Set a value to the application / model variable.
- getState (self)

  Get a textual state for the variable's current data.
- setState (self, state)

  Set a textual state to the variable.
- reset (self, verbosity=1)

  Reset the variable to its initial value (default value specified in constructor or value before the first set() call).
- info (self)

  Get an info string for this variable (alias and current value plus unit and lookup value, if available).

7.462.1 Detailed Description

Simple wrapper for PyOCD variables in a attached debugging session.

7.462.2 Constructor & Destructor Documentation

7.462.2.1 __init__()

```
ttk_base.pyocd.variables.Var.__init__ (
                self,
                context,
                identifier,
                addr_offset = 0,
                data_type = "uint8",
                unit = "",
                alias = "",
                descr = "",
                lookup = None,
                default = None,
                resettable = True,
                fmt = None )
```

Debugger variable initialization.

Note

Array elements: Using an identifier with index, e.g. foobar[2] will calculate the required address offset to the selected index based on the variable's data type.
This only works for basic arrays and data typs, not for nested data structures.

Data Types:

- unsigned: uint8, uint16, uint32 (, uint64) | u8, u16, u32 (u64)
- signed: sint8, sint16, sint32 (, sint64) | s8, s16, s32 (s64)
- float: float32 (, float64) | f32 (f64)

64-Bit types cannot be read directly via the debug interface with STM32 controllers

Parameters

context	parent context (here: a ttk_tools.pyocd.debugger.Debugger instance)
identifier	variable identifier: symbol name from ELF file
addr_offset	offset in Bytes to address of identifier symbol
data_type	data type for decoding/encoding memory contents
unit	(optional) unit string for the variable
alias	(optional) alias name, defaults to last (non- default) entry in the identifier model path
descr	description for the variable
lookup	lookup dictionary for states {value => "state name"}
default	manually defined default value for the variable. Will be used as "reset" value instead of the original value before the first set operation if set to anything but None.
resettable	enable/disable reset() for this variable. If set to False, a call to reset() will be ignored.
fmt	format string or keyword to use for value representation, e.g. "%.2f" or "hex"

Reimplemented from ttk_base.pyocd._variables.Var.

7.462.3 Member Function Documentation

7.462.3.1 get()

```
ttk_base.pyocd.variables.Var.get (
              self,
             * args,
             ** kwargs )
```

Get the current value of the application / model variable.

Note

Any specified parameters will be discarded read implementation does not handle any additional parameters)

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Returns

the current value.

Reimplemented from ttk_base.pyocd._variables.Var.

7.462.3.2 getState()

```
ttk_base.pyocd.variables.Var.getState (
              self )
```

Get a textual state for the variable's current data.

This uses the lookup values supplied in constructor.

Returns

a textual state description (if available) or the unmapped value (if no matching state was found).

Reimplemented from ttk_base._variables_base.VarBase.

7.462.3.3 info()

```
ttk_base.pyocd.variables.Var.info (
              self )
```

Get an info string for this variable (alias and current value plus unit and lookup value, if available).

Example:

```
>> print voltage_var.info()
MW-Var Name: 12 V
>> print status_var.info()
Status Var Name: 1 (valid)
```

Returns

a formatted string.

Reimplemented from ttk_base._variables_base.VarBase.

7.462.3.4 reset()

```
ttk_base.pyocd.variables.Var.reset (
              self,
              verbosity = 1 )
```

Reset the variable to its initial value (default value specified in constructor or value before the first set() call).

Parameters

verbosity	verbosity of log output. 0: silent, 1: show calls to non-resettable variables 2: show all reset calls

Returns

the original value (which may be None if the variable has not been set before)

Reimplemented from ttk_base._variables_base.VarBase.

7.462.3.5 set()

```
ttk_base.pyocd.variables.Var.set (
              self,
              value,
             * args,
             ** kwargs )
```

Set a value to the application / model variable.

Parameters

value	value to set
∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Note

Any additional parameters will be discarded (write implementation does not support any additional parameters)

Returns

the original value before the set operation occurred.

Reimplemented from ttk_base.pyocd._variables.Var.

7.462.3.6 setState()

```
ttk_base.pyocd.variables.Var.setState (
              self,
              state )
```

Set a textual state to the variable.

This uses the lookup values supplied in constructor. Fallback (if no matching state text was found) will be the unmapped input state.

Parameters

state	named state to set

Returns

the previously set state (if possible, otherwise the previous unmapped value, see getState)

Reimplemented from ttk_base._variables_base.VarBase.

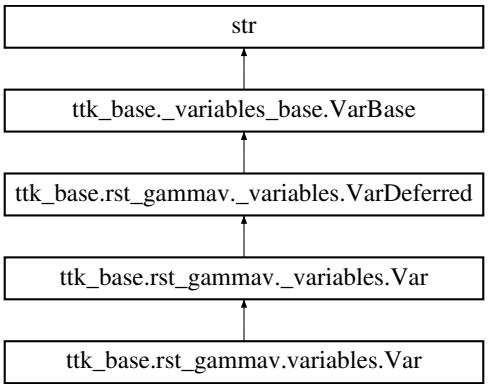
The documentation for this class was generated from the following file:

- ttk_base/pyocd/variables.py (v1.0.0)

7.463 ttk_base.rst_gammav._variables.Var Class Reference

Simple wrapper for Gamma V variables.

Inheritance diagram for ttk_base.rst_gammav._variables.Var:



Public Member Functions

- __init__ (self, context, identifier, unit="", alias="", descr="", lookup=None, default=None, resettable=True, fmt=None)

  Variable initialization.

7.463.1 Detailed Description

Simple wrapper for Gamma V variables.

Directly initializes the Gamma process variable during init, which might be a bit time consuming if many variables need to get initialized. See GammaVarDeferred

7.463.2 Constructor & Destructor Documentation

7.463.2.1 __init__()

```
ttk_base.rst_gammav._variables.Var.__init__ (
              self,
              context,
              identifier,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              default = None,
              resettable = True,
              fmt = None )
```

Variable initialization.

Parameters

context	parent context (a gaapi reference)
identifier	variable identifier: model path
unit	(optional) unit string for the variable
alias	(optional) alias name, defaults to last (non- default) entry in the identifier model path
descr	description for the variable
lookup	lookup dictionary for states {value => "state name"}
default	manually defined default value for the variable. Will be used as "reset" value instead of the original value before the first set operation if set to anything but None.
resettable	enable/disable reset() for this variable. If set to False, a call to reset() will be ignored.
fmt	format string or keyword to use for value representation, e.g. "% 2f" or "hex"

Reimplemented from ttk_base.rst_gammav._variables.VarDeferred.

Reimplemented in ttk_base.rst_gammav.variables.Var.

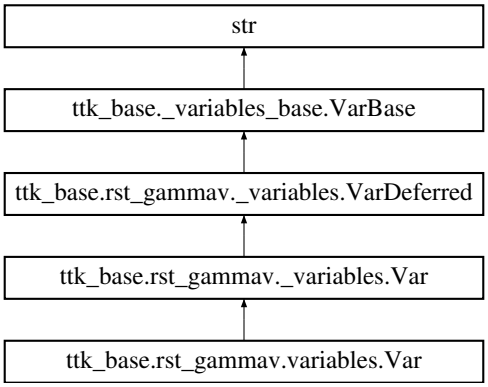
The documentation for this class was generated from the following file:

- ttk_base/rst_gammav/_variables.py (v1.4)

7.464 ttk_base.rst_gammav.variables.Var Class Reference

Simple wrapper for Gamma V variables.

Inheritance diagram for ttk_base.rst_gammav.variables.Var:



Public Member Functions

- __init__ (self, context, identifier, unit="", alias="", descr="", lookup=None, default=None, resettable=True, fmt=None)

  Gamma-Variable initialization.
- get (self, ∗args, ∗∗kwargs)

  Get the current value of the process variable.
- set (self, value, ∗args, ∗∗kwargs)

  Set a value to the process variable.
- getState (self)

  Get a textual state for the variable's current data.
- setState (self, state)

  Set a textual state to the variable.
- reset (self, verbosity=1)

  Reset the variable to its initial value (default value specified in constructor or value before the first set() call).
- info (self)

  Get an info string for this variable (alias and current value plus unit and lookup value, if available).

7.464.1 Detailed Description

Simple wrapper for Gamma V variables.

7.464.2 Constructor & Destructor Documentation

7.464.2.1 __init__()

```
ttk_base.rst_gammav.variables.Var.__init__ (
              self,
              context,
              identifier,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              default = None,
              resettable = True,
              fmt = None )
```

Gamma-Variable initialization.

Parameters

context	parent context (gaapi reference)
identifier	process variable (PV) identifier
unit	(optional) unit string for the variable
alias	(optional) alias name, defaults to last (non- default) entry in the identifier model path
descr	description for the variable
lookup	lookup dictionary for states {value => "state name"}
default	manually defined default value for the variable. Will be used as "reset" value instead of the original value before the first set operation if set to anything but None.
resettable	enable/disable reset() for this variable. If set to False, a call to reset() will be ignored.
fmt	format string or keyword to use for value representation, e.g. "%.2f" or "hex"

Reimplemented from ttk_base.rst_gammav._variables.Var.

7.464.3 Member Function Documentation

7.464.3.1 get()

```
ttk_base.rst_gammav.variables.Var.get (
              self,
             * args,
             ** kwargs )
```

Get the current value of the process variable.

Note

Any specified parameters will be discarded (the XIL-Testbench read implementation does not handle any additional parameters)

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Returns

the current value.

Reimplemented from ttk_base.rst_gammav._variables.VarDeferred.

7.464.3.2 getState()

```
ttk_base.rst_gammav.variables.Var.getState (
              self )
```

Get a textual state for the variable's current data.

This uses the lookup values supplied in constructor.

Returns

a textual state description (if available) or the unmapped value (if no matching state was found).

Reimplemented from ttk_base._variables_base.VarBase.

7.464.3.3 info()

```
ttk_base.rst_gammav.variables.Var.info (
              self )
```

Get an info string for this variable (alias and current value plus unit and lookup value, if available).

Example:

```
>> print voltage_var.info()
Gamma-Var Name: 12 V
>> print status_var.info()
Status Var Name: 1 (valid)
```

Returns

a formatted string.

Reimplemented from ttk_base._variables_base.VarBase.

7.464.3.4 reset()

```
ttk_base.rst_gammav.variables.Var.reset (
              self,
              verbosity = 1 )
```

Reset the variable to its initial value (default value specified in constructor or value before the first set() call).

Parameters

verbosity	verbosity of log output, 0: silent, 1: show calls to non-resettable variables 2: show all reset calls

Returns

the original value (which may be None if the variable has not been set before)

Reimplemented from ttk_base._variables_base.VarBase.

7.464.3.5 set()

```
ttk_base.rst_gammav.variables.Var.set (
              self,
              value,
             * args,
             ** kwargs )
```

Set a value to the process variable.

Parameters

value	value to set
∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Note

Any additional parameters will be ignored, as Gamma process variable access does not handle optional parameters.

Returns

the original value before the set operation occurred.

Reimplemented from ttk_base.rst_gammav._variables.VarDeferred.

7.464.3.6 setState()

```
ttk_base.rst_gammav.variables.Var.setState (
              self,
              state )
```

Set a textual state to the variable.

This uses the lookup values supplied in constructor. Fallback (if no matching state text was found) will be the unmapped input state.

Parameters

state	named state to set

Returns

the previously set state (if possible, otherwise the previous unmapped value, see getState)

Reimplemented from ttk_base._variables_base.VarBase.

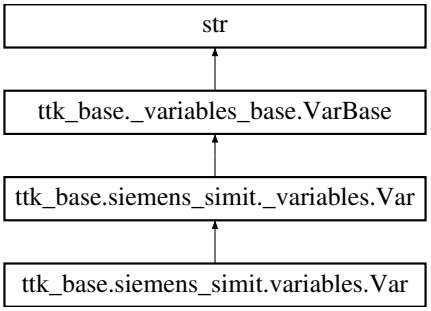
The documentation for this class was generated from the following file:

- ttk_base/rst_gammav/variables.py (v1.2)

7.465 ttk_base.siemens_simit._variables.Var Class Reference

Simple wrapper for SIMIT SHM Coupling variables.

Inheritance diagram for ttk_base.siemens_simit._variables.Var:



Public Member Functions

- __init__ (self, context, identifier, unit="", alias="", descr="", lookup=None, default=None, resettable=True, fmt=None)

  SIMIT-SHM-Variable initialization.
- get (self, *args, **kwargs)

  Get the current value of the application / model variable.
- set (self, value, *args, **kwargs)

  Set a value to the application / model variable.

Public Attributes

- simit_signal

  proxy that handles data access
- alias

  alias/short name of this variable
- last_value

  last value of get/set operation
- orig_value

  original/reset value

7.465.1 Detailed Description

Simple wrapper for SIMIT SHM Coupling variables.

7.465.2 Constructor & Destructor Documentation

7.465.2.1 __init__()

```
ttk_base.siemens_simit._variables.Var.__init__ (
              self,
              context,
              identifier,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              default = None,
              resettable = True,
              fmt = None )
```

SIMIT-SHM-Variable initialization.

Parameters

context	parent context (here: a SimitShm instance, see ttk_tools.siemens.simit_shm)
identifier	variable/signal identifier/name
unit	(optional) unit string for the variable
alias	(optional) alias name, defaults to last (non- default) entry in the identifier model path
descr	description for the variable
lookup	lookup dictionary for states {value => "state name"}
default	manually defined default value for the variable. Will be used as "reset" value instead of the original value before the first set operation if set to anything but None.
resettable	enable/disable reset() for this variable. If set to False, a call to reset() will be ignored.
fmt	format string or keyword to use for value representation, e.g. "%.2f" or "hex"

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_base.siemens_simit.variables.Var.

7.465.3 Member Function Documentation

7.465.3.1 get()

```
ttk_base.siemens_simit._variables.Var.get (
             self,
            * args,
            ** kwargs )
```

Get the current value of the application / model variable.

Note

Any specified parameters will be discarded (the SimitSignal read implementation does not handle any additional parameters)

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Returns

the current value.

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_base.siemens_simit.variables.Var.

7.465.3.2 set()

```
ttk_base.siemens_simit._variables.Var.set (
              self,
              value,
             * args,
             ** kwargs )
```

Set a value to the application / model variable.

Parameters

value	value to set
**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Note

Any additional parameters will be discarded (the SimitSignal write implementation does not support any additional parameters)

Returns

the original value before the set operation occurred.

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_base.siemens_simit.variables.Var.

7.465.4 Member Data Documentation

7.465.4.1 alias

```
ttk_base.siemens_simit._variables.Var.alias
```

alias/short name of this variable

7.465.4.2 last_value

`ttk_base.siemens_simit._variables.Var.last_value`

last value of get/set operation

7.465.4.3 orig_value

`ttk_base.siemens_simit._variables.Var.orig_value`

original/reset value

7.465.4.4 simit_signal

`ttk_base.siemens_simit._variables.Var.simit_signal`

proxy that handles data access

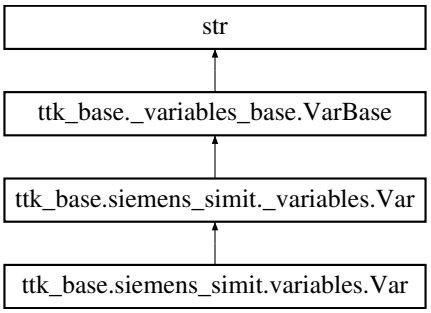
The documentation for this class was generated from the following file:

- ttk_base/siemens_simit/_variables.py (v1.0.0)

7.466 ttk_base.siemens_simit.variables.Var Class Reference

Simple wrapper for SIMIT SHM Coupling variables.

Inheritance diagram for ttk_base.siemens_simit.variables.Var:



Public Member Functions

- __init__ (self, context, identifier, unit="", alias="", descr="", lookup=None, default=None, resettable=True, fmt=None)

  SIMIT-SHM-Variable initialization.
- get (self, *args, **kwargs)

  Get the current value of the application / model variable.
- set (self, value, *args, **kwargs)

  Set a value to the application / model variable.
- getState (self)

  Get a textual state for the variable's current data.
- setState (self, state)

  Set a textual state to the variable.
- reset (self, verbosity=1)

  Reset the variable to its initial value (default value specified in constructor or value before the first set() call).
- info (self)

  Get an info string for this variable (alias and current value plus unit and lookup value, if available).

7.466.1 Detailed Description

Simple wrapper for SIMIT SHM Coupling variables.

7.466.2 Constructor & Destructor Documentation

7.466.2.1 __init__()

```
ttk_base.siemens_simit.variables.Var.__init__ (
              self,
              context,
              identifier,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              default = None,
              resettable = True,
              fmt = None )
```

SIMIT-SHM-Variable initialization.

Parameters

context	parent context (here: a SimitShm instance, see ttk_tools.siemens.simit_shm)
identifier	variable/signal identifier/name
unit	(optional) unit string for the variable
alias	(optional) alias name, defaults to last (non- default) entry in the identifier model path
descr	description for the variable
lookup	lookup dictionary for states {value => "state name"}
default	manually defined default value for the variable. Will be used as "reset" value instead of the original value before the first set operation if set to anything but None.
resettable	enable/disable reset() for this variable. If set to False, a call to reset() will be ignored.
fmt	format string or keyword to use for value representation, e.g. "%.2f" or "hex"

Reimplemented from ttk_base.siemens_simit._variables.Var.

7.466.3 Member Function Documentation

7.466.3.1 get()

```
ttk_base.siemens_simit.variables.Var.get (
              self,
             * args,
             ** kwargs )
```

Get the current value of the application / model variable.

Note

Any specified parameters will be discarded (the SimitSignal read implementation does not handle any additional parameters)

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Returns

the current value.

Reimplemented from ttk_base.siemens_simit._variables.Var.

7.466.3.2 getState()

```
ttk_base.siemens_simit.variables.Var.getState (
              self )
```

Get a textual state for the variable's current data.

This uses the lookup values supplied in constructor.

Returns

a textual state description (if available) or the unmapped value (if no matching state was found).

Reimplemented from ttk_base._variables_base.VarBase.

7.466.3.3 info()

```
ttk_base.siemens_simit.variables.Var.info (
              self )
```

Get an info string for this variable (alias and current value plus unit and lookup value, if available).

Example:

```
>> print voltage_var.info()
MW-Var Name: 12 V
>> print status_var.info()
Status Var Name: 1 (valid)
```

Returns

a formatted string.

Reimplemented from ttk_base._variables_base.VarBase.

7.466.3.4 reset()

```
ttk_base.siemens_simit.variables.Var.reset (
              self,
              verbosity = 1 )
```

Reset the variable to its initial value (default value specified in constructor or value before the first set() call).

Parameters

verbosity	verbosity of log output. 0: silent, 1: show calls to non-resettable variables 2: show all reset calls

Returns

the original value (which may be None if the variable has not been set before)

Reimplemented from ttk_base._variables_base.VarBase.

7.466.3.5 set()

```
ttk_base.siemens_simit.variables.Var.set (
              self,
              value,
             * args,
             ** kwargs )
```

Set a value to the application / model variable.

Parameters

value	value to set
∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Note

Any additional parameters will be discarded (the SimitSignal write implementation does not support any additional parameters)

Returns

the original value before the set operation occurred.

Reimplemented from ttk_base.siemens_simit._variables.Var.

7.466.3.6 setState()

```
ttk_base.siemens_simit.variables.Var.setState (
              self,
              state )
```

Set a textual state to the variable.

This uses the lookup values supplied in constructor. Fallback (if no matching state text was found) will be the unmapped input state.

Parameters

state	named state to set

Returns

the previously set state (if possible, otherwise the previous unmapped value, see getState)

Reimplemented from ttk_base._variables_base.VarBase.

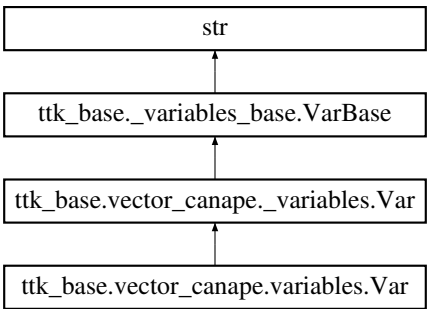
The documentation for this class was generated from the following file:

- ttk_base/siemens_simit/variables.py (v1.0.0)

7.467 ttk_base.vector_canape._variables.Var Class Reference

Simple wrapper for Vector CANape calibration variables.

Inheritance diagram for ttk_base.vector_canape._variables.Var:



Public Member Functions

- __init__ (self, context, identifier, unit="", alias="", descr="", lookup=None, default=None, resettable=True, fmt=None, raw_mode=False)

  CANape-Variable initialization.
- get (self, *args, **kwargs)

  Get the current value of the calibration variable.
- set (self, value, *args, **kwargs)

  Set a value to the calibration variable.

Public Attributes

- context

  reference to parent context (usually a Cal VarContainer)
- alias

  alias/short name of this variable (defaults to identifier)
- raw_mode

  True: use raw/internal representation of values
  False: use physical representation of values (default)
- last_value

  last value of get/set operation
- orig_value

  original/reset value

Protected Member Functions

- _getCurrentReprMode (self)

  Get PHYS/HEX representation constant for canapeapi calls depending on currently active representation mode.

7.467.1 Detailed Description

Simple wrapper for Vector CANape calibration variables.

Uses ttk_tools.vector.canapeapi for accessing CANape application.

7.467.2 Constructor & Destructor Documentation

7.467.2.1 __init__()

```
ttk_base.vector_canape._variables.Var.__init__ (
              self,
              context,
              identifier,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              default = None,
              resettable = True,
              fmt = None,
              raw_mode = False )
```

CANape-Variable initialization.

Parameters

context	parent context that handles tool references This is usually a reference to a variables container for CANape "Cal"-Vars See ttk_base.vector_canape.variables.VarContainer
identifier	variable identifier/name in the calibration tool
unit	(optional) unit string for the variable
alias	(optional) alias name, defaults to varname
descr	description for the variable
lookup	lookup dictionary for states {value => "state name"}
default	manually defined default value for the variable. Will be used as "reset" value instead of the original value before the first set operation if set to anything but None.
resettable	enable/disable reset() for this variable. If set to False, a call to reset() will be ignored.
fmt	format string or keyword to use for value representation, e.g. "%.2f" or "hex"
raw_mode	True: use raw/internal representation of values False: use physical representation of values (default)

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_base.vector_canape.variables.Var.

7.467.3 Member Function Documentation

7.467.3.1 _getCurrentReprMode()

```
ttk_base.vector_canape._variables.Var._getCurrentReprMode (
              self )  [protected]
```

Get PHYS/HEX representation constant for canapeapi calls depending on currently active representation mode.

7.467.3.2 get()

```
ttk_base.vector_canape._variables.Var.get (
              self,
             * args,
             ** kwargs )
```

Get the current value of the calibration variable.

Note

Any specified parameters will be handed on to the underlying cal-tool "read" operation, which may or may not be able to handle them. Use with care and only in very special cases.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Returns

the current value.

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_base.vector_canape.variables.Var.

7.467.3.3 set()

```
ttk_base.vector_canape._variables.Var.set (
              self,
              value,
             * args,
             ** kwargs )
```

Set a value to the calibration variable.

Parameters

value	value to set
**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Note

Any additional parameters will be handed on to the underlying cal-tool "write" operation, which may or may not be able to handle them. Use with care and only in very special cases.

Returns

the original value before the set operation occurred.

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_base.vector_canape.variables.Var.

7.467.4 Member Data Documentation

7.467.4.1 alias

`ttk_base.vector_canape._variables.Var.alias`

alias/short name of this variable (defaults to identifier)

7.467.4.2 context

`ttk_base.vector_canape._variables.Var.context`

reference to parent context (usually a Cal VarContainer)

7.467.4.3 last_value

`ttk_base.vector_canape._variables.Var.last_value`

last value of get/set operation

7.467.4.4 orig_value

`ttk_base.vector_canape._variables.Var.orig_value`

original/reset value

7.467.4.5 raw_mode

`ttk_base.vector_canape._variables.Var.raw_mode`

True: use raw/internal representation of values
False: use physical representation of values (default)

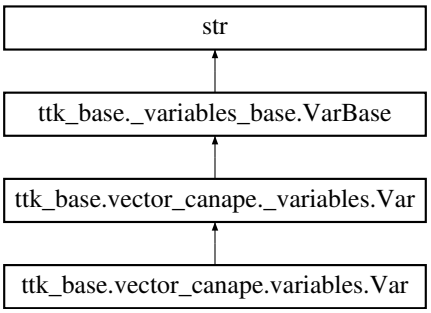
The documentation for this class was generated from the following file:

- ttk_base/vector_canape/_variables.py (v1.4)

7.468 ttk_base.vector_canape.variables.Var Class Reference

Simple wrapper for Vector CANape calibration variables.

Inheritance diagram for ttk_base.vector_canape.variables.Var:



Public Member Functions

- __init__ (self, context, identifier, unit="", alias="", descr="", lookup=None, default=None, resettable=True, fmt=None, raw_mode=False)

  CANape-CAL-Variable initialization.
- get (self, *args, **kwargs)

  Get the current value of the calibration variable.
- set (self, value, *args, **kwargs)

  Set a value to the calibration variable.
- getState (self)

  Get a textual state for the variable's current data.
- setState (self, state)

  Set a textual state to the variable.
- reset (self, verbosity=1)

  Reset the variable to its initial value (default value specified in constructor or value before the first set() call).
- info (self)

  Get an info string for this variable (alias and current value plus unit and lookup value, if available).

7.468.1 Detailed Description

Simple wrapper for Vector CANape calibration variables.

Uses ttk_tools.vector_canape.canapeapi for accessing CANape application.

7.468.2 Constructor & Destructor Documentation

7.468.2.1 __init__()

```
ttk_base.vector_canape.variables.Var.__init__ (
                self,
                context,
                identifier,
                unit = "",
                alias = "",
                descr = "",
                lookup = None,
                default = None,
                resettable = True,
                fmt = None,
                raw_mode = False )
```

CANape-CAL-Variable initialization.

Parameters

context	parent context that handles tool references This is usually a reference to a variables container for CANape "Cal"-Vars See ttk_base.vector_canape.variables.VarContainer
identifier	variable identifier/name in the calibration tool
unit	(optional) unit string for the variable
alias	(optional) alias name, defaults to varname
descr	description for the variable
lookup	lookup dictionary for states {value => "state name"}
default	manually defined default value for the variable. Will be used as "reset" value instead of the original value before the first set operation if set to anything but None.
resettable	enable/disable reset() for this variable. If set to False, a call to reset() will be ignored.
fmt	format string or keyword to use for value representation, e.g. "%.2f" or "hex"
raw_mode	True: use raw/internal representation of values False: use physical representation of values (default)

Reimplemented from ttk_base.vector_canape._variables.Var.

7.468.3 Member Function Documentation

7.468.3.1 get()

```
ttk_base.vector_canape.variables.Var.get (
               self,
              * args,
              ** kwargs )
```

Get the current value of the calibration variable.

Note

Any specified parameters will be handed on to the underlying "read" operation, which may or may not be able to handle them. Use with care and only in very special cases.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Returns

the current value.

Reimplemented from ttk_base.vector_canape._variables.Var.

7.468.3.2 getState()

```
ttk_base.vector_canape.variables.Var.getState (
              self )
```

Get a textual state for the variable's current data.

This uses the lookup values supplied in constructor.

Returns

a textual state description (if available) or the unmapped value (if no matching state was found).

Reimplemented from ttk_base._variables_base.VarBase.

7.468.3.3 info()

```
ttk_base.vector_canape.variables.Var.info (
              self )
```

Get an info string for this variable (alias and current value plus unit and lookup value, if available).

Example:

```
>> print voltage_var.info()
Cal-Var Name: 12 V
>> print status_var.info()
Status Var Name: 1 (valid)
```

Returns

a formatted string.

Reimplemented from ttk_base._variables_base.VarBase.

7.468.3.4 reset()

```
ttk_base.vector_canape.variables.Var.reset (
              self,
              verbosity = 1 )
```

Reset the variable to its initial value (default value specified in constructor or value before the first set() call).

Parameters

verbosity	verbosity of log output, 0: silent, 1: show calls to non-resettable variables 2: show all reset calls

Returns

the original value (which may be None if the variable has not been set before)

Reimplemented from ttk_base._variables_base.VarBase.

7.468.3.5 set()

```
ttk_base.vector_canape.variables.Var.set (
              self,
              value,
             * args,
             ** kwargs )
```

Set a value to the calibration variable.

Parameters

value	value to set
∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Note

Any additional parameters will be handed on to the underlying cal-tool "write" operation, which may or may not be able to handle them. Use with care and only in very special cases.

Returns

the original value before the set operation occurred.

Reimplemented from ttk_base.vector_canape._variables.Var.

7.468.3.6 setState()

```
ttk_base.vector_canape.variables.Var.setState (
              self,
              state )
```

Set a textual state to the variable.

This uses the lookup values supplied in constructor. Fallback (if no matching state text was found) will be the unmapped input state.

Parameters

state	named state to set

Returns

the previously set state (if possible, otherwise the previous unmapped value, see getState)

Reimplemented from ttk_base._variables_base.VarBase.

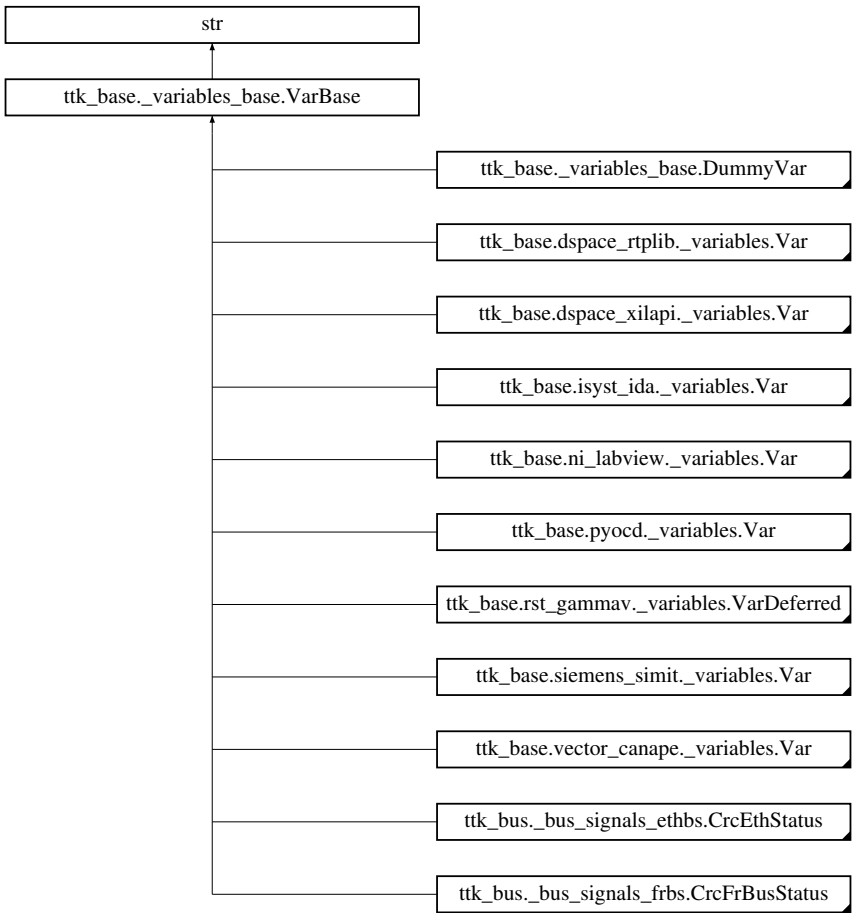
The documentation for this class was generated from the following file:

- ttk_base/vector_canape/variables.py (v1.2)

7.469 ttk_base._variables_base.VarBase Class Reference

Base wrapper for a single variable.

Inheritance diagram for ttk_base._variables_base.VarBase:



Public Member Functions

- __new__ (cls, context, identifier, *args, **kwargs)

  Create a custom instance of the immutable str type.
- __init__ (self, context, identifier, unit="", alias="", descr="", lookup=None, default=None, resettable=True, fmt=None)

  Variable initialization.
- lookup (self)

  Property `lookup`, an alias to `state_lookup`.
- get (self, *args, **kwargs)

  Get the current value of the variable.
- set (self, value, *args, **kwargs)

  Set a value to the variable.
- getState (self)

  Get a textual state for the variable's current data.
- setState (self, state)

  Set a textual state to the variable.
- reset (self, verbosity=1)

  Reset the variable to its initial value (default value specified in constructor or value before the first set() call).
- info (self)

  Get an info string for this variable: alias and current value plus unit and lookup value, if available.
- hasStateLookup (self)

  Return True if a state/value lookup table is defined.
- getStateDescr (self, value=None, fallback=None)

  Get a state description text for the given value from the defined lookup table (if any).
- hasFormattedRepr (self)

  Return True if a formatting for value representation is defined.
- getFormattedValue (self, value=None)

  Get a formatted representation for the given value using the defined format string (see self.fmt).

Public Attributes

- orig_value

  original/reset value
- last_value

  last value of get/set operation
- state_lookup

  lookup table value => text
- value_lookup

  lookup table text => value
- resettable

  status if this variable is resettable
- fmt

  format string for custom value representation
- context

  reference to context of variable (depends on variable type/source/origin in specific implementations)
- unit

  physical unit like V, mA or rpm (only for report output, no conversion gets performed)
- alias

  alias/short name of this variable
- descr

  description text
- lookup

7.469.1 Detailed Description

Base wrapper for a single variable.

7.469.2 Constructor & Destructor Documentation

7.469.2.1 __init__()

```
ttk_base._variables_base.VarBase.__init__ (
            self,
            context,
            identifier,
            unit = "",
            alias = "",
            descr = "",
            lookup = None,
            default = None,
            resettable = True,
            fmt = None )
```

Variable initialization.

Parameters

context	parent context that handles tool references This is usually a reference to a variables container or to a tool/api instance.
identifier	variable identifier/name in the calibration tool
unit	(optional) unit string for the variable
alias	(optional) alias name, defaults to varname
descr	description for the variable
lookup	lookup dictionary for states {value => "state name"}
default	manually defined default value for the variable. Will be used as "reset" value instead of the original value before the first set operation if set to anything but None.
resettable	enable/disable reset() for this variable. If set to False, a call to reset() will be ignored.
fmt	format string or keyword to use for custom value representation, e.g. "%.2f" or "hex"

Reimplemented in ttk_bus._bus_signals_ethbs.CrcEthStatus, ttk_bus._bus_signals_frbs.CrcFrBusStatus, ttk_bus.bus_signals_ethbs.CrcEthStatus, ttk_bus.bus_signals_frbs.CrcFrBusStatus, ttk_base.pyocd._variables.Var, ttk_base.pyocd.variables.Var, ttk_base._variables_base.DummyVar, ttk_base.dspace_rtplib._variables.Var, ttk_base.dspace_rtplib.variables.Var, ttk_base.dspace_xilapi._variables.Var, ttk_base.dspace_xilapi.variables.Var, ttk_base.isyst_ida._variables.Var, ttk_base.isyst_ida.variables.Var, ttk_base.ni_labview._variables.Var, ttk_base.ni_labview.variables.V ttk_base.rst_gammav._variables.VarDeferred, ttk_base.rst_gammav._variables.Var, ttk_base.rst_gammav.variables.Var, ttk_base.rst_gammav.variables.VarDeferred, ttk_base.siemens_simit._variables.Var, ttk_base.siemens_simit.variables.Var, ttk_base.variables_base.DummyVar, ttk_base.vector_canape._variables.Var, and ttk_base.vector_canape.variables.Var.

7.469.3 Member Function Documentation

7.469.3.1 __new__()

```
ttk_base._variables_base.VarBase.__new__ (
            cls,
```

```
                context,
                identifier,
               * args,
               ** kwargs )
```

Create a custom instance of the immutable str type.

We derive our wrapper class from str, so the default behaviour will be that of a string of the variable identifier (model path, var name, ...) (just as it was used in previous test scripts).

Parameters

context	parent context, see **init**
identifier	variable identifier, see **init**
∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Reimplemented in ttk_bus._bus_signals_ethbs.CrcEthStatus.

7.469.3.2 get()

```
ttk_base._variables_base.VarBase.get (
               self,
               * args,
               ** kwargs )
```

Get the current value of the variable.

Member `last_value` will also be assigned the current value.

Override this in specific implementations.

Note

When implementing a custom get method, member self.last_value has to be set to the "gotten" value.

Parameters

∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Reimplemented in ttk_bus._bus_signals_ethbs.CrcEthStatus, ttk_bus._bus_signals_frbs.CrcFrBusStatus, ttk_bus.bus_signals_ethbs.CrcEthStatus, ttk_bus.bus_signals_frbs.CrcFrBusStatus, ttk_base._variables_base.DummyVar, ttk_base.dspace_rtplib._variables.Var, ttk_base.dspace_rtplib.variables.Var, ttk_base.dspace_xilapi._variables.Var, ttk_base.dspace_xilapi.variables.Var, ttk_base.isyst_ida._variables.Var, ttk_base.isyst_ida.variables.Var, ttk_base.ni_labview._variabl ttk_base.ni_labview.variables.Var, ttk_base.pyocd._variables.Var, ttk_base.pyocd.variables.Var, ttk_base.rst_gammav._variables.VarD ttk_base.rst_gammav.variables.Var, ttk_base.rst_gammav.variables.VarDeferred, ttk_base.siemens_simit._variables.Var, ttk_base.siemens_simit.variables.Var, ttk_base.variables_base.DummyVar, ttk_base.vector_canape._variables.Var, and ttk_base.vector_canape.variables.Var.

7.469.3.3 getFormattedValue()

```
ttk_base._variables_base.VarBase.getFormattedValue (
              self,
              value = None )
```

Get a formatted representation for the given value using the defined format string (see self.fmt).

An empty (or non-defined) format string will use the value's default string representation.

Parameters

value	value for which to get a string representation. If None, the last value (read or written) will be used instead (see self.last_value).

Returns

a formatted string representation.

7.469.3.4 getState()

```
ttk_base._variables_base.VarBase.getState (
              self )
```

Get a textual state for the variable's current data.

This uses the lookup values supplied in constructor.

Fallback (if no matching state was found) will be the unmapped numeric value.

Returns

a matching status representation from the lookup table, or the "raw" value if no match was found.

Reimplemented in ttk_base.dspace_rtplib.variables.Var, ttk_base.dspace_xilapi.variables.Var, ttk_base.isyst_ida.variables.Var, ttk_base.ni_labview.variables.Var, ttk_base.pyocd.variables.Var, ttk_base.rst_gammav.variables.Var, ttk_base.rst_gammav.variables.V ttk_base.siemens_simit.variables.Var, ttk_base.variables_base.DummyVar, ttk_base.vector_canape.variables.Var, ttk_bus.bus_signals_ethbs.CrcEthStatus, and ttk_bus.bus_signals_frbs.CrcFrBusStatus.

7.469.3.5 getStateDescr()

```
ttk_base._variables_base.VarBase.getStateDescr (
              self,
              value = None,
              fallback = None )
```

Get a state description text for the given value from the defined lookup table (if any).

Parameters

value	value for which to get a state description. If None, the last read value will be used instead.
fallback	fallback value to return if no match was found. Default: None

Returns

the state text if a match was found, otherwise the specified fallback value.

7.469.3.6 hasFormattedRepr()

```
ttk_base._variables_base.VarBase.hasFormattedRepr (
              self )
```

Return True if a formatting for value representation is defined.

Can be used to decide if a custom formatting is available, i.e. if this variable can be used as a formatting source for other variables in the same evaluation context.

Note

See ttk_base.baseutils.isCustomRepr for a way to test if an existing representation has a "custom" format or is a default representation.
Per default, meta data updates will prefer "custom" formatted over default formatted strings. See ttk_base.baseutils.getCustomRepr for a function that will create string "marked" as "customized".

7.469.3.7 hasStateLookup()

```
ttk_base._variables_base.VarBase.hasStateLookup (
              self )
```

Return True if a state/value lookup table is defined.

7.469.3.8 info()

```
ttk_base._variables_base.VarBase.info (
              self )
```

Get an info string for this variable: alias and current value plus unit and lookup value, if available.

Note

Current value's representation will be formatted, if a value format (parameter "fmt") has been specified.

Example:

```
>> print voltage_var.info()
Var Name: 12 V
>> print status_var.info()
Status Var Name: 0x01 (valid)
```

Returns

a formatted string.

Reimplemented in ttk_base.dspace_rtplib.variables.Var, ttk_base.dspace_xilapi.variables.Var, ttk_base.isyst_ida.variables.Var, ttk_base.ni_labview.variables.Var, ttk_base.pyocd.variables.Var, ttk_base.rst_gammav.variables.Var, ttk_base.rst_gammav.variables.V ttk_base.siemens_simit.variables.Var, ttk_base.variables_base.DummyVar, and ttk_base.vector_canape.variables.Var.

7.469.3.9 lookup()

```
ttk_base._variables_base.VarBase.lookup (
            self )
```

Property `lookup`, an alias to `state_lookup`.

As the supplied lookup values are internally stored as two separate lookup tables `state_lookup` (which maps value => text) and `value_lookup` (which maps text => value), this provides an alias to the values as originally supplied in the `lookup` parameter during **init**

Example:

```
>> var = VarBase(context=None, identifier="example", lookup={0: "off", 1: "on", 2: "huh?"})
>> var.lookup
{0: 'off', 1: 'on', 2: 'huh?'}
>> var.state_lookup
{0: 'off', 1: 'on', 2: 'huh?'}
>> var.value_lookup
{'off':0, 'on': 1, 'huh?': 2}
```

7.469.3.10 reset()

```
ttk_base._variables_base.VarBase.reset (
            self,
            verbosity = 1 )
```

Reset the variable to its initial value (default value specified in constructor or value before the first set() call).

Parameters

verbosity	verbosity of log output, 0: silent, 1: show calls to non-resettable variables 2: show all reset calls

Returns

the original value (which may be None if the variable has not been set before)

Reimplemented in ttk_bus._bus_signals_ethbs.CrcEthStatus, ttk_bus._bus_signals_frbs.CrcFrBusStatus, ttk_bus.bus_signals_ethbs.CrcEthStatus, ttk_bus.bus_signals_frbs.CrcFrBusStatus, ttk_base.dspace_rtplib.variables.Var, ttk_base.dspace_xilapi.variables.Var, ttk_base.isyst_ida.variables.Var, ttk_base.ni_labview.variables.Var, ttk_base.pyocd.variables.Va ttk_base.rst_gammav.variables.Var, ttk_base.rst_gammav.variables.VarDeferred, ttk_base.siemens_simit.variables.Var, ttk_base.variables_base.DummyVar, and ttk_base.vector_canape.variables.Var.

7.469.3.11 set()

```
ttk_base._variables_base.VarBase.set (
            self,
            value,
```

```
        * args,
        ** kwargs )
```

Set a value to the variable.

Member `last_value` will also be assigned the set value.

Override this in specific implementations.

Parameters

value	value to set
∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Note

When implementing a custom set method, member self.last_value has to be set to the "set" value.

Returns

the original value before the set operation occurred.

Reimplemented in ttk_base._variables_base.DummyVar, ttk_base.dspace_rtplib._variables.Var, ttk_base.dspace_rtplib.variables.Var, ttk_base.dspace_xilapi._variables.Var, ttk_base.dspace_xilapi.variables.Var, ttk_base.isyst_ida._variables.Var, ttk_base.isyst_ida.variables.Var, ttk_base.ni_labview._variables.Var, ttk_base.ni_labview.variables.Var, ttk_base.pyocd._variables.Var, ttk_base.pyocd.variables.Var, ttk_base.rst_gammav._variables.VarDeferred, ttk_base.rst_gammav.variables.Var, ttk_base.rst_gammav.variables.VarDeferred, ttk_base.siemens_simit._variables.Var, ttk_base.siemens_simit.variables.Var, ttk_base.variables_base.DummyVar, ttk_base.vector_canape._variables.Var, and ttk_base.vector_canape.variables.Var.

7.469.3.12 setState()

```
ttk_base._variables_base.VarBase.setState (
              self,
              state )
```

Set a textual state to the variable.

This uses the lookup values supplied in constructor.

Fallback (if no matching state text was found) will be the unmapped input state (which will most likely fail to set)

Parameters

state	named state to set

Returns

the previously set state (or the raw value if no matching state was available).

Reimplemented in ttk_base.dspace_rtplib.variables.Var, ttk_base.dspace_xilapi.variables.Var, ttk_base.isyst_ida.variables.Var,

ttk_base.ni_labview.variables.Var, ttk_base.pyocd.variables.Var, ttk_base.rst_gammav.variables.Var, ttk_base.rst_gammav.variables.V
ttk_base.siemens_simit.variables.Var, ttk_base.variables_base.DummyVar, and ttk_base.vector_canape.variables.Var.

7.469.4 Member Data Documentation

7.469.4.1 alias

`ttk_base._variables_base.VarBase.alias`

alias/short name of this variable

7.469.4.2 context

`ttk_base._variables_base.VarBase.context`

reference to context of variable (depends on variable type/source/origin in specific implementations)

7.469.4.3 descr

`ttk_base._variables_base.VarBase.descr`

description text

7.469.4.4 fmt

`ttk_base._variables_base.VarBase.fmt`

format string for custom value representation

7.469.4.5 last_value

`ttk_base._variables_base.VarBase.last_value`

last value of get/set operation

7.469.4.6 lookup

`ttk_base._variables_base.VarBase.lookup`

7.469.4.7 orig_value

`ttk_base._variables_base.VarBase.orig_value`

original/reset value

7.469.4.8 resettable

ttk_base._variables_base.VarBase.resettable

status if this variable is resettable

7.469.4.9 state_lookup

ttk_base._variables_base.VarBase.state_lookup

lookup table value => text

7.469.4.10 unit

ttk_base._variables_base.VarBase.unit

physical unit like V, mA or rpm (only for report output, no conversion gets performed)

7.469.4.11 value_lookup

ttk_base._variables_base.VarBase.value_lookup

lookup table text => value

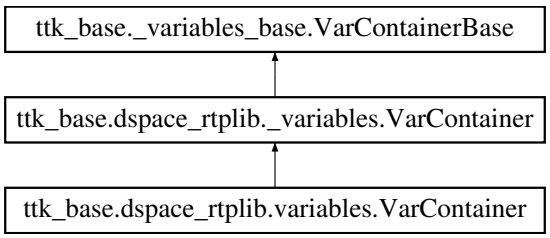
The documentation for this class was generated from the following file:

- ttk_base/_variables_base.py (v2.8.1)

7.470 ttk_base.dspace_rtplib._variables.VarContainer Class Reference

Base class for dSPACE (rtplib) HIL Var containers.

Inheritance diagram for ttk_base.dspace_rtplib._variables.VarContainer:



Public Member Functions

- __init__ (self, application)

7.470.1 Detailed Description

Base class for dSPACE (rtplib) HIL Var containers.

7.470.2 Constructor & Destructor Documentation

7.470.2.1 __init__()

```
ttk_base.dspace_rtplib._variables.VarContainer.__init__ (
            self,
            application )
```

Parameters

application	rtplib.Appl instance (will be used in constructors of rtplib-based HIL variables)

Reimplemented in ttk_base.dspace_rtplib.variables.VarContainer.

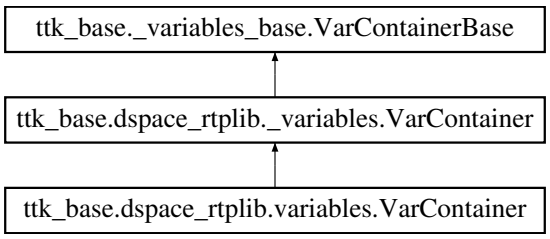
The documentation for this class was generated from the following file:

- ttk_base/dspace_rtplib/_variables.py (v1.4)

7.471 ttk_base.dspace_rtplib.variables.VarContainer Class Reference

Base class for rtplib-based "HIL" variable containers.

Inheritance diagram for ttk_base.dspace_rtplib.variables.VarContainer:



Public Member Functions

- __init__ (self, application)
- resetAll (self, verbosity=1, recursion_depth=4)

  Call reset() on all contained variables that support it.

7.471.1 Detailed Description

Base class for rtplib-based "HIL" variable containers.

7.471.2 Constructor & Destructor Documentation

7.471.2.1 __init__()

```
ttk_base.dspace_rtplib.variables.VarContainer.__init__ (
            self,
            application )
```

Parameters

application	rtplib.Appl instance (will be used in constructors of rtplib-based HIL variables)

Reimplemented from ttk_base.dspace_rtplib._variables.VarContainer.

7.471.3 Member Function Documentation

7.471.3.1 resetAll()

```
ttk_base.dspace_rtplib.variables.VarContainer.resetAll (
               self,
               verbosity = 1,
               recursion_depth = 4 )
```

Call reset() on all contained variables that support it.

Info:

Variables will be reset in an arbitrary order (first found, first reset). It might therefore be preferable to mark all variables used in DUT's startup/shutdown sequences (like power supply settings, KL30, KL15, ...) as not "resettable" in order to get a more predictable behavior.

Parameters

verbosity	verbosity of debug/log messages. 0: silent, 1: plain info, 2: +var names, 3: overkill
recursion_depth	how "deep" to recurse into lists/dictionary structures when looking for variables to reset.

Reimplemented from ttk_base._variables_base.VarContainerBase.

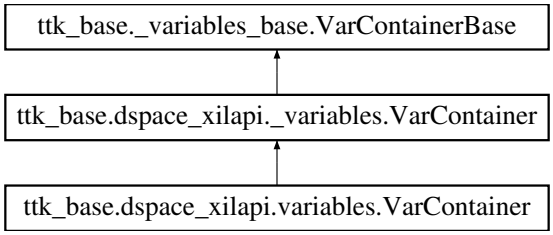
The documentation for this class was generated from the following file:

- ttk_base/dspace_rtplib/variables.py (v1.2)

7.472 ttk_base.dspace_xilapi._variables.VarContainer Class Reference

Base class for dSPACE (XIL API) "HIL" variable containers.

Inheritance diagram for ttk_base.dspace_xilapi._variables.VarContainer:



Public Member Functions

- __init__ (self, xiltb)

7.472.1 Detailed Description

Base class for dSPACE (XIL API) "HIL" variable containers.

7.472.2 Constructor & Destructor Documentation

7.472.2.1 __init__()

```
ttk_base.dspace_xilapi._variables.VarContainer.__init__ (
              self,
              xiltb )
```

Parameters

xiltb	XilTestbench instance (will be used in constructors of XIL-API-based HIL variables)

Reimplemented in ttk_base.dspace_xilapi.variables.VarContainer.

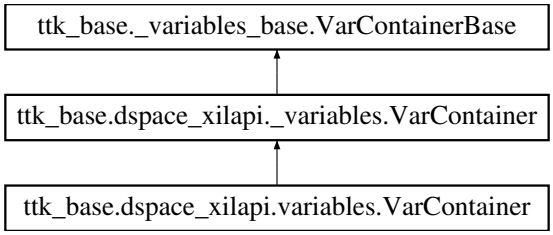
The documentation for this class was generated from the following file:

- ttk_base/dspace_xilapi/_variables.py (v1.4)

7.473 ttk_base.dspace_xilapi.variables.VarContainer Class Reference

Base class for dSPACE (XIL API) "HIL" variable containers.

Inheritance diagram for ttk_base.dspace_xilapi.variables.VarContainer:



Public Member Functions

- __init__ (self, xiltb)
- resetAll (self, verbosity=1, recursion_depth=4)

  Call reset() on all contained variables that support it.

7.473.1 Detailed Description

Base class for dSPACE (XIL API) "HIL" variable containers.

7.473.2 Constructor & Destructor Documentation

7.473.2.1 __init__()

```
ttk_base.dspace_xilapi.variables.VarContainer.__init__ (
              self,
              xiltb )
```

Parameters

xiltb	XilTestbench instance (will be used in constructors of XIL-API-based HIL variables)

Reimplemented from ttk_base.dspace_xilapi._variables.VarContainer.

7.473.3 Member Function Documentation

7.473.3.1 resetAll()

```
ttk_base.dspace_xilapi.variables.VarContainer.resetAll (
              self,
              verbosity = 1,
              recursion_depth = 4 )
```

Call reset() on all contained variables that support it.

Info:

Variables will be reset in an arbitrary order (first found, first reset). It might therefore be preferable to mark all variables used in DUT's startup/shutdown sequences (like power supply settings, KL30, KL15, ...) as not "resettable" in order to get a more predictable behavior.

Parameters

verbosity	verbosity of debug/log messages. 0: silent, 1: plain info, 2: +var names, 3: overkill
recursion_depth	how "deep" to recurse into lists/dictionary structures when looking for variables to reset.

Reimplemented from ttk_base._variables_base.VarContainerBase.

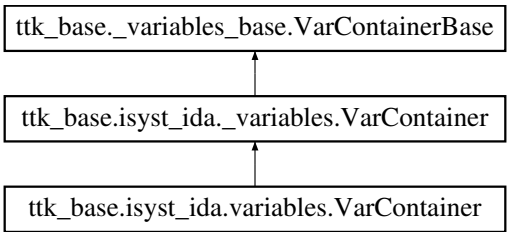
The documentation for this class was generated from the following file:

- ttk_base/dspace_xilapi/variables.py (v1.2)

7.474 ttk_base.isyst_ida._variables.VarContainer Class Reference

Base class for iDA Testware "HIL" variable containers.

Inheritance diagram for ttk_base.isyst_ida._variables.VarContainer:



Public Member Functions

- __init__ (self, ida)

7.474.1 Detailed Description

Base class for iDA Testware "HIL" variable containers.

7.474.2 Constructor & Destructor Documentation

7.474.2.1 __init__()

```
ttk_base.isyst_ida._variables.VarContainer.__init__ (
              self,
              ida )
```

Parameters

ida	IdaTestware (or MwAppRpc) instance (will be used in constructors of iDA-based HIL variables)

Reimplemented in ttk_base.isyst_ida.variables.VarContainer.

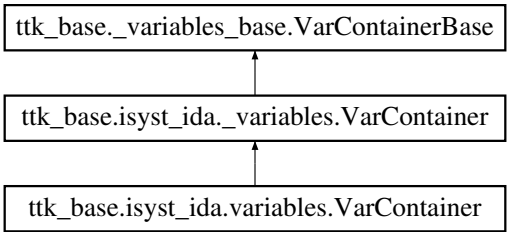
The documentation for this class was generated from the following file:

- ttk_base/isyst_ida/_variables.py (v1.2.0)

7.475 ttk_base.isyst_ida.variables.VarContainer Class Reference

Base class for iDA Testware "HIL" variable containers.

Inheritance diagram for ttk_base.isyst_ida.variables.VarContainer:



Public Member Functions

- __init__ (self, ida)
- resetAll (self, verbosity=1, recursion_depth=4)

  Call reset() on all contained variables that support it.

7.475.1 Detailed Description

Base class for iDA Testware "HIL" variable containers.

7.475.2 Constructor & Destructor Documentation

7.475.2.1 __init__()

```
ttk_base.isyst_ida.variables.VarContainer.__init__ (
              self,
              ida )
```

Parameters

ida	IdaTestware (or MwAppRpc) instance (will be used in constructors of iDA-based HIL variables)

Reimplemented from ttk_base.isyst_ida._variables.VarContainer.

7.475.3 Member Function Documentation

7.475.3.1 resetAll()

```
ttk_base.isyst_ida.variables.VarContainer.resetAll (
              self,
              verbosity = 1,
              recursion_depth = 4 )
```

Call reset() on all contained variables that support it.

Info:

Variables will be reset in an arbitrary order (first found, first reset). It might therefore be preferable to mark all variables used in DUT's startup/shutdown sequences (like power supply settings, KL30, KL15, ...) as not "resettable" in order to get a more predictable behavior.

Parameters

verbosity	verbosity of debug/log messages. 0: silent, 1: plain info, 2: +var names, 3: overkill
recursion_depth	how "deep" to recurse into lists/dictionary structures when looking for variables to reset.

Reimplemented from ttk_base._variables_base.VarContainerBase.

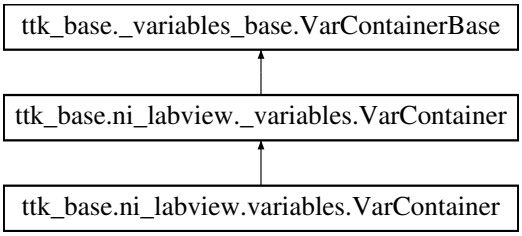
The documentation for this class was generated from the following file:

- ttk_base/isyst_ida/variables.py (v1.1.1)

7.476 ttk_base.ni_labview._variables.VarContainer Class Reference

Base class for LabView "HIL" variable containers.

Inheritance diagram for ttk_base.ni_labview._variables.VarContainer:



Public Member Functions

- __init__ (self, lv)

7.476.1 Detailed Description

Base class for LabView "HIL" variable containers.

7.476.2 Constructor & Destructor Documentation

7.476.2.1 __init__()

```
ttk_base.ni_labview._variables.VarContainer.__init__ (
              self,
              lv )
```

Parameters

lv	a ttk_tools.ni.labview.LabView instance connected to a suitable COM server "LabView App"

Reimplemented in ttk_base.ni_labview.variables.VarContainer.

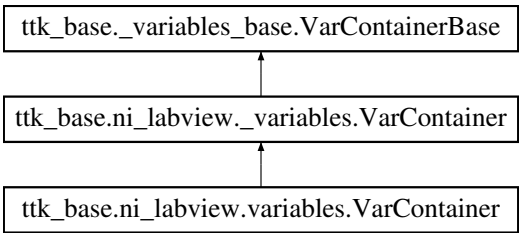
The documentation for this class was generated from the following file:

- ttk_base/ni_labview/_variables.py (v1.1)

7.477 ttk_base.ni_labview.variables.VarContainer Class Reference

Base class for LabView "HIL" variable containers.

Inheritance diagram for ttk_base.ni_labview.variables.VarContainer:



Public Member Functions

- __init__ (self, lv)
- resetAll (self, verbosity=1, recursion_depth=4)

  Call reset() on all contained variables that support it.

7.477.1 Detailed Description

Base class for LabView "HIL" variable containers.

7.477.2 Constructor & Destructor Documentation

7.477.2.1 __init__()

```
ttk_base.ni_labview.variables.VarContainer.__init__ (
              self,
              lv )
```

Parameters

lv	a ttk_tools.ni.labview.LabView instance connected to a suitable COM server "LabView App"

Reimplemented from ttk_base.ni_labview._variables.VarContainer.

7.477.3 Member Function Documentation

7.477.3.1 resetAll()

```
ttk_base.ni_labview.variables.VarContainer.resetAll (
              self,
              verbosity = 1,
              recursion_depth = 4 )
```

Call reset() on all contained variables that support it.

Info:

Variables will be reset in an arbitrary order (first found, first reset). It might therefore be preferable to mark all variables used in DUT's startup/shutdown sequences (like power supply settings, KL30, KL15, ...) as not "resettable" in order to get a more predictable behavior.

Parameters

verbosity	verbosity of debug/log messages. 0: silent, 1: plain info, 2: +var names, 3: overkill
recursion_depth	how "deep" to recurse into lists/dictionary structures when looking for variables to reset.

Reimplemented from ttk_base._variables_base.VarContainerBase.

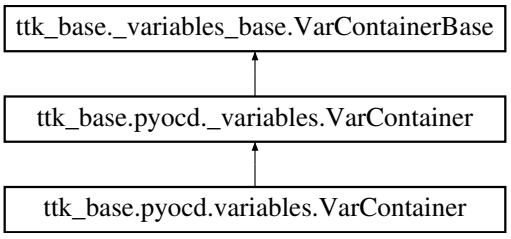
The documentation for this class was generated from the following file:

- ttk_base/ni_labview/variables.py (v1.1)

7.478 ttk_base.pyocd._variables.VarContainer Class Reference

Base class for variable containers.

Inheritance diagram for ttk_base.pyocd._variables.VarContainer:



Public Member Functions

- __init__ (self, debugger)

7.478.1 Detailed Description

Base class for variable containers.

7.478.2 Constructor & Destructor Documentation

7.478.2.1 __init__()

```
ttk_base.pyocd._variables.VarContainer.__init__ (
              self,
              debugger )
```

Parameters

debugger	PyOCD debugger wrapper instance (will be used in constructors of debugger-based variables)

Reimplemented in ttk_base.pyocd.variables.VarContainer.

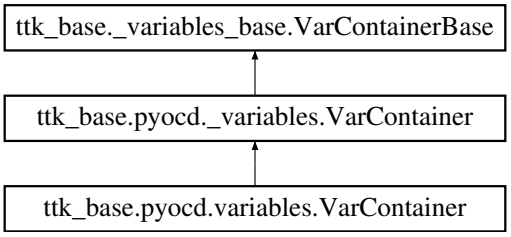
The documentation for this class was generated from the following file:

- ttk_base/pyocd/_variables.py (v1.0.0)

7.479 ttk_base.pyocd.variables.VarContainer Class Reference

Base class for variable containers.

Inheritance diagram for ttk_base.pyocd.variables.VarContainer:



Public Member Functions

- __init__ (self, debugger)
- resetAll (self, verbosity=1, recursion_depth=4)

  Call reset() on all contained variables that support it.

7.479.1 Detailed Description

Base class for variable containers.

7.479.2 Constructor & Destructor Documentation

7.479.2.1 __init__()

```
ttk_base.pyocd.variables.VarContainer.__init__ (
            self,
            debugger )
```

Parameters

debugger	PyOCD debugger wrapper instance (will be used in constructors of debugger-based variables)

Reimplemented from ttk_base.pyocd._variables.VarContainer.

7.479.3 Member Function Documentation

7.479.3.1 resetAll()

```
ttk_base.pyocd.variables.VarContainer.resetAll (
              self,
              verbosity = 1,
              recursion_depth = 4 )
```

Call reset() on all contained variables that support it.

Info:

Variables will be reset in an arbitrary order (first found, first reset). It might therefore be preferable to mark all variables used in DUT's startup/shutdown sequences (like power supply settings, KL30, KL15, ...) as not "resettable" in order to get a more predictable behavior.

Parameters

verbosity	verbosity of debug/log messages. 0: silent, 1: plain info, 2: +var names, 3: overkill
recursion_depth	how "deep" to recurse into lists/dictionary structures when looking for variables to reset.

Reimplemented from ttk_base._variables_base.VarContainerBase.

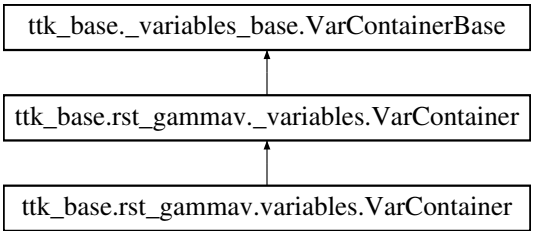
The documentation for this class was generated from the following file:

- ttk_base/pyocd/variables.py (v1.0.0)

7.480 ttk_base.rst_gammav._variables.VarContainer Class Reference

Base class for Gamma V "GammaVar" containers.

Inheritance diagram for ttk_base.rst_gammav._variables.VarContainer:



Public Member Functions

- __init__ (self, gamma_api)

7.480.1 Detailed Description

Base class for Gamma V "GammaVar" containers.

7.480.2 Constructor & Destructor Documentation

7.480.2.1 __init__()

```
ttk_base.rst_gammav._variables.VarContainer.__init__ (
            self,
            gamma_api )
```

Parameters

gamma_api	reference to gamma api (will be used in constructors of gamma-based HIL variables)

Reimplemented in ttk_base.rst_gammav.variables.VarContainer.

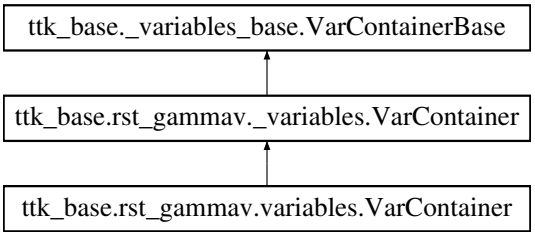
The documentation for this class was generated from the following file:

- ttk_base/rst_gammav/_variables.py (v1.4)

7.481 ttk_base.rst_gammav.variables.VarContainer Class Reference

Base class for Gamma-V-based "HIL" variable containers.

Inheritance diagram for ttk_base.rst_gammav.variables.VarContainer:



Public Member Functions

- __init__ (self, gamma_api)
- resetAll (self, verbosity=1, recursion_depth=4)

  Call reset() on all contained variables that support it.

7.481.1 Detailed Description

Base class for Gamma-V-based "HIL" variable containers.

7.481.2 Constructor & Destructor Documentation

7.481.2.1 __init__()

```
ttk_base.rst_gammav.variables.VarContainer.__init__ (
              self,
              gamma_api )
```

Parameters

gamma_api	reference to gamma api (will be used in constructors of Gamma-based HIL variables)

Reimplemented from ttk_base.rst_gammav._variables.VarContainer.

7.481.3 Member Function Documentation

7.481.3.1 resetAll()

```
ttk_base.rst_gammav.variables.VarContainer.resetAll (
              self,
              verbosity = 1,
              recursion_depth = 4 )
```

Call reset() on all contained variables that support it.

Info:

Variables will be reset in an arbitrary order (first found, first reset).
It might therefore be preferable to mark all variables used in DUT's startup/shutdown sequences (like power supply settings, KL30, KL15, ...) as not "resettable" in order to get a more predictable behavior.

Parameters

verbosity	verbosity of debug/log messages. 0: silent, 1: plain info, 2: +var names, 3: overkill
recursion_depth	how "deep" to recurse into lists/dictionary structures when looking for variables to reset.

Reimplemented from ttk_base._variables_base.VarContainerBase.

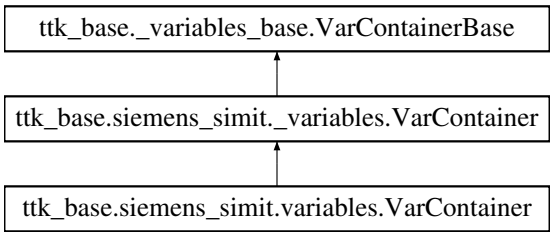
The documentation for this class was generated from the following file:

- ttk_base/rst_gammav/variables.py (v1.2)

7.482 ttk_base.siemens_simit._variables.VarContainer Class Reference

Base class for SIMIT SHM coupling signal variable containers.

Inheritance diagram for ttk_base.siemens_simit._variables.VarContainer:



Public Member Functions

- __init__ (self, simshm)
- __getattr__ (self, name)

  Get a SimitSignal-based variable for non-defined entries.

Protected Attributes

- _simshm

7.482.1 Detailed Description

Base class for SIMIT SHM coupling signal variable containers.

7.482.2 Constructor & Destructor Documentation

7.482.2.1 __init__()

```
ttk_base.siemens_simit._variables.VarContainer.__init__ (
              self,
              simshm )
```

Parameters

simshm	SimitShm instance (will be used in constructors of SimitSignal-based variables)

Reimplemented in ttk_base.siemens_simit.variables.VarContainer.

7.482.3 Member Function Documentation

7.482.3.1 __getattr__()

```
ttk_base.siemens_simit._variables.VarContainer.__getattr__ (
              self,
              name )
```

Get a SimitSignal-based variable for non-defined entries.

Creates and assigns a default siemens_simit._variable.Var for the supplied name.

This will only work well for SHM coupling signal/symbol names that are also valid Python names.

Adding Var members beforehand/statically allows to define custom meta data and assign custom member names for more free-form signal/symbol name

Note

Dynamically created Vars will be instances of the base implementation's Var, not of the interface wrapper to avoid circular imports. The only time this might matter is during isinstance checks, as the interface is an instance of the implementation, but not vice versa.

Parameters

name	attribute/member name to lookup

Returns

a Var instance for the supplied name

7.482.4 Member Data Documentation

7.482.4.1 _simshm

`ttk_base.siemens_simit._variables.VarContainer._simshm [protected]`

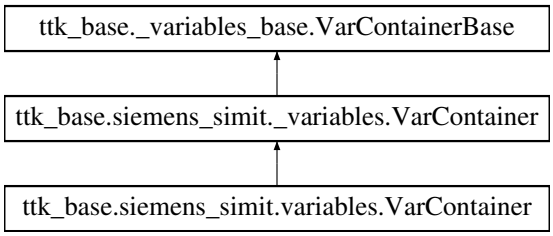
The documentation for this class was generated from the following file:

- ttk_base/siemens_simit/_variables.py (v1.0.0)

7.483 ttk_base.siemens_simit.variables.VarContainer Class Reference

Base class for SIMIT SHM coupling signal variable containers.

Inheritance diagram for ttk_base.siemens_simit.variables.VarContainer:



Public Member Functions

- __init__ (self, simshm)
- resetAll (self, verbosity=1, recursion_depth=4)

  Call reset() on all contained variables that support it.

7.483.1 Detailed Description

Base class for SIMIT SHM coupling signal variable containers.

7.483.2 Constructor & Destructor Documentation

7.483.2.1 __init__()

```
ttk_base.siemens_simit.variables.VarContainer.__init__ (
              self,
              simshm )
```

Parameters

simshm	SimitShm instance (will be used in constructors of SimitSignal-based variables)

Reimplemented from ttk_base.siemens_simit._variables.VarContainer.

7.483.3 Member Function Documentation

7.483.3.1 resetAll()

```
ttk_base.siemens_simit.variables.VarContainer.resetAll (
              self,
              verbosity = 1,
              recursion_depth = 4 )
```

Call reset() on all contained variables that support it.

Info:

Variables will be reset in an arbitrary order (first found, first reset). It might therefore be preferable to mark all variables used in DUT's startup/shutdown sequences (like power supply settings, KL30, KL15, ...) as not "resettable" in order to get a more predictable behavior.

Parameters

verbosity	verbosity of debug/log messages. 0: silent, 1: plain info, 2: +var names, 3: overkill
recursion_depth	how "deep" to recurse into lists/dictionary structures when looking for variables to reset.

Reimplemented from ttk_base._variables_base.VarContainerBase.

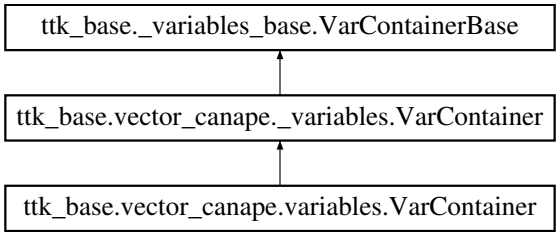
The documentation for this class was generated from the following file:

- ttk_base/siemens_simit/variables.py (v1.0.0)

7.484 ttk_base.vector_canape._variables.VarContainer Class Reference

Base class for CalVar containers.

Inheritance diagram for ttk_base.vector_canape._variables.VarContainer:



Public Member Functions

- __init__ (self, tool_ref=None)
- setToolRef (self, tool_ref)

  Set a calibration tool reference for cal var access as weak reference that will not prevent tool shutdown.
- getToolRef (self)

  Get a reference to the calibration tool.

Protected Attributes

- _tool_ref

7.484.1 Detailed Description

Base class for CalVar containers.

7.484.2 Constructor & Destructor Documentation

7.484.2.1 __init__()

```
ttk_base.vector_canape._variables.VarContainer.__init__ (
              self,
              tool_ref = None )
```

Parameters

tool_ref	calibration tool reference, see setToolRef

Reimplemented in ttk_base.vector_canape.variables.VarContainer.

7.484.3 Member Function Documentation

7.484.3.1 getToolRef()

```
ttk_base.vector_canape._variables.VarContainer.getToolRef (
              self )
```

Get a reference to the calibration tool.

Returns

a reference or None if the tool has been closed/deleted.

7.484.3.2 setToolRef()

```
ttk_base.vector_canape._variables.VarContainer.setToolRef (
              self,
              tool_ref )
```

Set a calibration tool reference for cal var access as weak reference that will not prevent tool shutdown.

Reimplemented in ttk_base.vector_canape.variables.VarContainer.

7.484.4 Member Data Documentation

7.484.4.1 _tool_ref

```
ttk_base.vector_canape._variables.VarContainer._tool_ref  [protected]
```

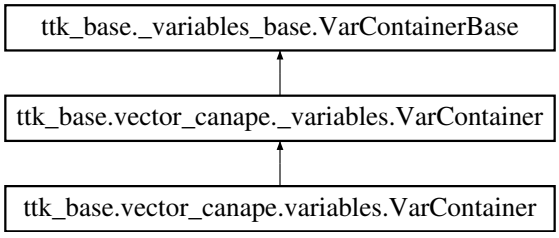
The documentation for this class was generated from the following file:

- ttk_base/vector_canape/_variables.py (v1.4)

7.485 ttk_base.vector_canape.variables.VarContainer Class Reference

Base class for CalVar containers.

Inheritance diagram for ttk_base.vector_canape.variables.VarContainer:



Public Member Functions

- __init__ (self, tool_ref=None)
- setToolRef (self, tool_ref)

  Set a calibration tool reference for cal var access.
- resetAll (self, verbosity=1, recursion_depth=4)

  Call reset() on all contained variables that support it.

7.485.1 Detailed Description

Base class for CalVar containers.

7.485.2 Constructor & Destructor Documentation

7.485.2.1 __init__()

```
ttk_base.vector_canape.variables.VarContainer.__init__ (
              self,
              tool_ref = None )
```

Parameters

tool_ref	calibration tool reference, see setToolRef

Reimplemented from ttk_base.vector_canape._variables.VarContainer.

7.485.3 Member Function Documentation

7.485.3.1 resetAll()

```
ttk_base.vector_canape.variables.VarContainer.resetAll (
              self,
              verbosity = 1,
              recursion_depth = 4 )
```

Call reset() on all contained variables that support it.

Info:

Variables will be reset in an arbitrary order (first found, first reset).

Parameters

verbosity	verbosity of debug/log messages. 0: silent, 1: plain info, 2: +var names, 3: overkill
recursion_depth	how "deep" to recurse into lists/dictionary structures when looking for variables to reset.

Reimplemented from ttk_base._variables_base.VarContainerBase.

7.485.3.2 setToolRef()

```
ttk_base.vector_canape.variables.VarContainer.setToolRef (
              self,
              tool_ref )
```

Set a calibration tool reference for cal var access.

Reimplemented from ttk_base.vector_canape._variables.VarContainer.

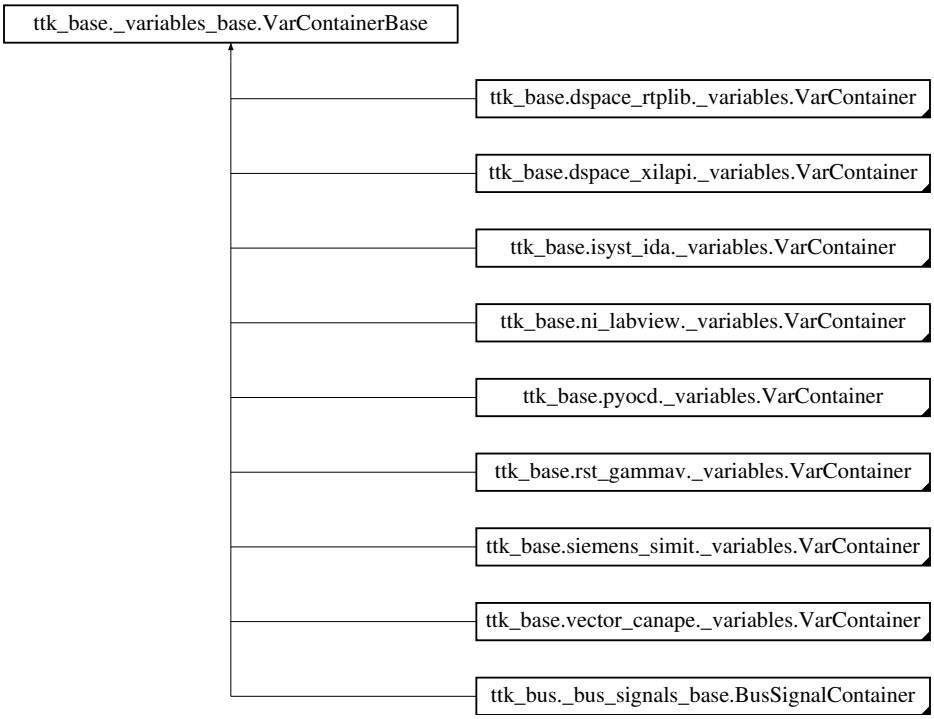
The documentation for this class was generated from the following file:

- ttk_base/vector_canape/variables.py (v1.2)

7.486 ttk_base._variables_base.VarContainerBase Class Reference

Base class for variables containers.

Inheritance diagram for ttk_base._variables_base.VarContainerBase:



Public Member Functions

- resetAll (self, verbosity=1, recursion_depth=4)

  Call reset() on all contained variables that support it.

7.486.1 Detailed Description

Base class for variables containers.

7.486.2 Member Function Documentation

7.486.2.1 resetAll()

```
ttk_base._variables_base.VarContainerBase.resetAll (
            self,
            verbosity = 1,
            recursion_depth = 4 )
```

Call reset() on all contained variables that support it.

Traverses list/dict/set structures, but does not examine arbitrary contained objects. Provide a reset() method if you want a specific custom member to be reset by this method.

Info:

Variables will be reset in an arbitrary order (first found, first reset).
It might therefore be preferable to mark all variables used in DUT's startup/shutdown sequences (like power supply settings, KL30, KL15, ...) as not "resettable" in order to get a more predictable behavior.

Note

Each member/variable will only be reset once, even if there are multiple references to the same member.

Example: Sample output at verbosity levels

```
# 1: plain info
Resetting... ...done.

# 2: plain info + "progress dots"
Resetting... . . . . . . . . ...done.

# 3: plain info + variable names/aliases
(Resetting... foo bar baz qux ...done.)
```

Parameters

verbosity	verbosity of debug/log messages. 0: silent 1: plain info 2: + "progress dots" (.: reset, -: skipped) 3: + var names (<name>: reset, -: skipped) 4: overkill
recursion_depth	how "deep" to recurse into lists/dictionary structures (and sets) when looking for variables to reset.

Reimplemented in ttk_base.dspace_rtplib.variables.VarContainer, ttk_base.dspace_xilapi.variables.VarContainer, ttk_base.isyst_ida.variables.VarContainer, ttk_base.ni_labview.variables.VarContainer, ttk_base.pyocd.variables.VarContainer, ttk_base.rst_gammav.variables.VarContainer, ttk_base.siemens_simit.variables.VarContainer, ttk_base.vector_canape.variables.VarC

ttk_bus.bus_signals_base.BusSignalContainer, ttk_bus.bus_signals_cangamma.CanBusSignalContainer, ttk_bus.bus_signals_canid
ttk_bus.bus_signals_canmmbs.CanBusSignalContainer, ttk_bus.bus_signals_ethbs.EthSignalContainer, and
ttk_bus.bus_signals_frbs.FrBusSignalContainer.

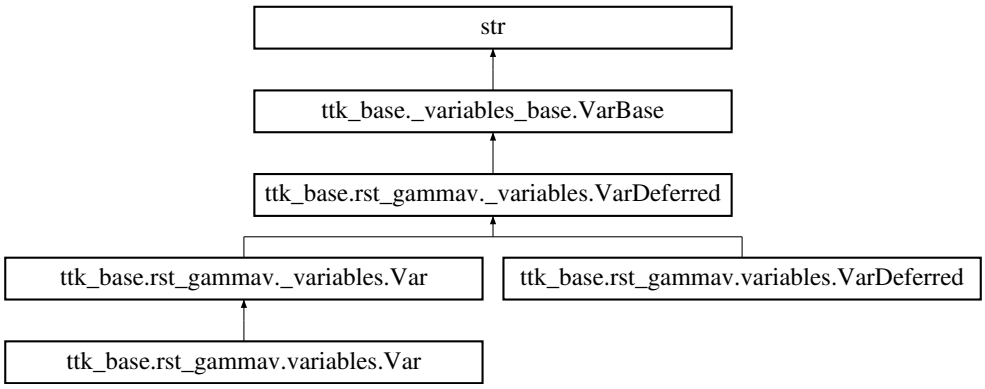
The documentation for this class was generated from the following file:

- ttk_base/_variables_base.py (v2.8.1)

7.487 ttk_base.rst_gammav._variables.VarDeferred Class Reference

Simple wrapper for Gamma V process variables with deferred initialization to speed up loading of containers with many GammaVars.

Inheritance diagram for ttk_base.rst_gammav._variables.VarDeferred:



Public Member Functions

- __init__ (self, context, identifier, unit="", alias="", descr="", lookup=None, default=None, resettable=True, fmt=None)

  Gamma-Variable initialization.
- get (self, *args, **kwargs)

  Get the current value of the process variable.
- set (self, value, *args, **kwargs)

  Set a value to the process variable.

Public Attributes

- context

  reference to gaapi
- variable

  reference to actual process variable
- alias

  alias/short name of this variable (potentially derived from PV address)
- last_value

  last value of get/set operation
- orig_value

  original/reset value

Protected Member Functions

- _deferredInit (self)

  Deferred part of constructor.

Protected Attributes

- _init_complete

7.487.1 Detailed Description

Simple wrapper for Gamma V process variables with deferred initialization to speed up loading of containers with many GammaVars.

Note that the actual process variable initialization will take place at the beginning of the first get/set-operation (and all operations that build on them, like getState/setState/reset...).

7.487.2 Constructor & Destructor Documentation

7.487.2.1 __init__()

```
ttk_base.rst_gammav._variables.VarDeferred.__init__ (
              self,
              context,
              identifier,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              default = None,
              resettable = True,
              fmt = None )
```

Gamma-Variable initialization.

Parameters

context	parent context (a gaapi reference)
identifier	process variable (PV) identifier
unit	(optional) unit string for the variable
alias	(optional) alias name, defaults to last (non- default) entry in the identifier model path
descr	description for the variable
lookup	lookup dictionary for states {value => "state name"}
default	manually defined default value for the variable. Will be used as "reset" value instead of the original value before the first set operation if set to anything but None.
resettable	enable/disable reset() for this variable. If set to False, a call to reset() will be ignored.
fmt	format string or keyword to use for value representation, e.g. "%.2f" or "hex"

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_base.rst_gammav._variables.Var, ttk_base.rst_gammav.variables.Var, and ttk_base.rst_gammav.variables.VarD

7.487.3 Member Function Documentation

7.487.3.1 _deferredInit()

```
ttk_base.rst_gammav._variables.VarDeferred._deferredInit (
              self )  [protected]
```

Deferred part of constructor.

Will be called prior to access of the underlying process variable.

7.487.3.2 get()

```
ttk_base.rst_gammav._variables.VarDeferred.get (
              self,
             * args,
             ** kwargs )
```

Get the current value of the process variable.

Note

Any additional parameters will be ignored, as Gamma process variable access does not handle optional parameters.

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Returns

the current value.

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_base.rst_gammav.variables.Var, and ttk_base.rst_gammav.variables.VarDeferred.

7.487.3.3 set()

```
ttk_base.rst_gammav._variables.VarDeferred.set (
              self,
              value,
             * args,
             ** kwargs )
```

Set a value to the process variable.

Parameters

value	value to set
∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Note

Any additional parameters will be ignored, as Gamma process variable access does not handle optional parameters.

Returns

the original value before the set operation occurred.

Reimplemented from ttk_base._variables_base.VarBase.

Reimplemented in ttk_base.rst_gammav.variables.Var, and ttk_base.rst_gammav.variables.VarDeferred.

7.487.4 Member Data Documentation

7.487.4.1 _init_complete

`ttk_base.rst_gammav._variables.VarDeferred._init_complete [protected]`

7.487.4.2 alias

`ttk_base.rst_gammav._variables.VarDeferred.alias`

alias/short name of this variable (potentially derived from PV address)

7.487.4.3 context

`ttk_base.rst_gammav._variables.VarDeferred.context`

reference to gaapi

7.487.4.4 last_value

`ttk_base.rst_gammav._variables.VarDeferred.last_value`

last value of get/set operation

7.487.4.5 orig_value

`ttk_base.rst_gammav._variables.VarDeferred.orig_value`

original/reset value

7.487.4.6 variable

`ttk_base.rst_gammav._variables.VarDeferred.variable`

reference to actual process variable

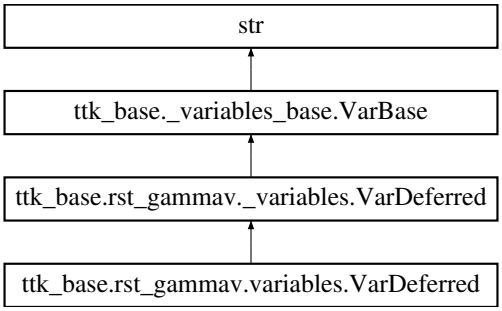
The documentation for this class was generated from the following file:

- ttk_base/rst_gammav/_variables.py (v1.4)

7.488 ttk_base.rst_gammav.variables.VarDeferred Class Reference

Simple wrapper for Gamma V variables with deferred initialization to speed up loading of containers with many GammaVars.

Inheritance diagram for ttk_base.rst_gammav.variables.VarDeferred:



Public Member Functions

- __init__ (self, context, identifier, unit="", alias="", descr="", lookup=None, default=None, resettable=True, fmt=None)

  Variable initialization.
- get (self, *args, **kwargs)

  Get the current value of the process variable.
- set (self, value, *args, **kwargs)

  Set a value to the process variable.
- getState (self)

  Get a textual state for the variable's current data.
- setState (self, state)

  Set a textual state to the variable.
- reset (self, verbosity=1)

  Reset the variable to its initial value (default value specified in constructor or value before the first set() call).
- info (self)

  Get an info string for this variable (alias and current value plus unit and lookup value, if available).

7.488.1 Detailed Description

Simple wrapper for Gamma V variables with deferred initialization to speed up loading of containers with many GammaVars.

Note that the actual process variable initialization will take place at the beginning of the first get/set-operation (and all that build on them, getState/setState/reset...)

7.488.2 Constructor & Destructor Documentation

7.488.2.1 __init__()

```
ttk_base.rst_gammav.variables.VarDeferred.__init__ (
              self,
              context,
              identifier,
              unit = "",
              alias = "",
              descr = "",
              lookup = None,
              default = None,
              resettable = True,
              fmt = None )
```

Variable initialization.

Parameters

context	parent context (gaapi reference)
identifier	variable identifier: model path
unit	(optional) unit string for the variable
alias	(optional) alias name, defaults to last (non- default) entry in the identifier model path
descr	description for the variable
lookup	lookup dictionary for states {value => "state name"}
default	manually defined default value for the variable. Will be used as "reset" value instead of the original value before the first set operation if set to anything but None.
resettable	enable/disable reset() for this variable. If set to False, a call to reset() will be ignored.
fmt	format string or keyword to use for value representation, e.g. "%.2f" or "hex"

Reimplemented from ttk_base.rst_gammav._variables.VarDeferred.

7.488.3 Member Function Documentation

7.488.3.1 get()

```
ttk_base.rst_gammav.variables.VarDeferred.get (
              self,
             * args,
             ** kwargs )
```

Get the current value of the process variable.

Note

Any specified parameters will be discarded (the XIL-Testbench read implementation does not handle any additional parameters)

Parameters

**args*	remaining positional arguments (catch-all)
***kwargs*	remaining keyword arguments (catch-all)

Returns

the current value.

Reimplemented from ttk_base.rst_gammav._variables.VarDeferred.

7.488.3.2 getState()

```
ttk_base.rst_gammav.variables.VarDeferred.getState (
              self )
```

Get a textual state for the variable's current data.

This uses the lookup values supplied in constructor.

Returns

a textual state description (if available) or the unmapped value (if no matching state was found).

Reimplemented from ttk_base._variables_base.VarBase.

7.488.3.3 info()

```
ttk_base.rst_gammav.variables.VarDeferred.info (
              self )
```

Get an info string for this variable (alias and current value plus unit and lookup value, if available).

Returns

a formatted string.

Reimplemented from ttk_base._variables_base.VarBase.

7.488.3.4 reset()

```
ttk_base.rst_gammav.variables.VarDeferred.reset (
              self,
              verbosity = 1 )
```

Reset the variable to its initial value (default value specified in constructor or value before the first set() call).

Parameters

verbosity	verbosity of log output, 0: silent, 1: show calls to non-resettable variables 2: show all reset calls

Returns

the original value (which may be None if the variable has not been set before)

Reimplemented from ttk_base._variables_base.VarBase.

7.488.3.5 set()

```
ttk_base.rst_gammav.variables.VarDeferred.set (
              self,
              value,
             * args,
             ** kwargs )
```

Set a value to the process variable.

Parameters

value	value to set
∗*args*	remaining positional arguments (catch-all)
∗∗*kwargs*	remaining keyword arguments (catch-all)

Note

Any additional parameters will be ignored, as Gamma process variable access does not handle optional parameters.

Returns

the original value before the set operation occurred.

Reimplemented from ttk_base.rst_gammav._variables.VarDeferred.

7.488.3.6 setState()

```
ttk_base.rst_gammav.variables.VarDeferred.setState (
              self,
              state )
```

Set a textual state to the variable.

This uses the lookup values supplied in constructor. Fallback (if no matching state text was found) will be the unmapped input state.

Parameters

state	named state to set

Returns

the previously set state (if possible, otherwise the previous unmapped value, see getState)

Reimplemented from ttk_base._variables_base.VarBase.

The documentation for this class was generated from the following file:

- ttk_base/rst_gammav/variables.py (v1.2)

7.489 ttk_tools.isyst.mwapp_client._VarMetaCache.VarMeta Class Reference

Simple container for iDA-Variable meta data.

Public Member Functions

- __init__ (self, var_id)
- name (self)
- __repr__ (self)

Public Attributes

- id

  var id
- display_name

  unique var name, including index-suffix (if any)
- data_type

  data type of variable
- size

  dimension/size/length of variable (array length, blob size).
- unit
- fmt
- descr
- default_value
- min_limit
- max_limit

7.489.1 Detailed Description

Simple container for iDA-Variable meta data.

7.489.2 Constructor & Destructor Documentation

7.489.2.1 __init__()

```
ttk_tools.isyst.mwapp_client._VarMetaCache.VarMeta.__init__ (
              self,
              var_id )
```

7.489.3 Member Function Documentation

7.489.3.1 __repr__()

```
ttk_tools.isyst.mwapp_client._VarMetaCache.VarMeta.__repr__ (
              self )
```

7.489.3.2 name()

```
ttk_tools.isyst.mwapp_client._VarMetaCache.VarMeta.name (
              self )
```

7.489.4 Member Data Documentation

7.489.4.1 data_type

```
ttk_tools.isyst.mwapp_client._VarMetaCache.VarMeta.data_type
```

data type of variable

7.489.4.2 default_value

```
ttk_tools.isyst.mwapp_client._VarMetaCache.VarMeta.default_value
```

7.489.4.3 descr

```
ttk_tools.isyst.mwapp_client._VarMetaCache.VarMeta.descr
```

7.489.4.4 display_name

```
ttk_tools.isyst.mwapp_client._VarMetaCache.VarMeta.display_name
```

unique var name, including index-suffix (if any)

7.489.4.5 fmt

ttk_tools.isyst.mwapp_client._VarMetaCache.VarMeta.fmt

7.489.4.6 id

ttk_tools.isyst.mwapp_client._VarMetaCache.VarMeta.id

var id

7.489.4.7 max_limit

ttk_tools.isyst.mwapp_client._VarMetaCache.VarMeta.max_limit

7.489.4.8 min_limit

ttk_tools.isyst.mwapp_client._VarMetaCache.VarMeta.min_limit

7.489.4.9 size

ttk_tools.isyst.mwapp_client._VarMetaCache.VarMeta.size

dimension/size/length of variable (array length, blob size).

For numeric values, `size` will be

- 1 for scalar variables,
- $>$ 1 for head elements of array variables
- 0 for child elements of array variables

For UTF-8-Strings (data type CHAR), `size` will be the maximum available length in Bytes, which will usually be less than the string length in characters (UTF-8 uses a variable number of bytes, depending on the encoded character/glyph). Read strings may be shorter than the maximum storage length.

For BLOBs (data type BLOB), `size` will simply contain the available size in Bytes. Read BLOBs will always return a sequence of `size` Bytes

7.489.4.10 unit

ttk_tools.isyst.mwapp_client._VarMetaCache.VarMeta.unit

The documentation for this class was generated from the following file:

- ttk_tools/isyst/mwapp_client.py (v1.2.4)

7.490 ttk_tools.isyst.mwapp_client_offline_stub._VarMetaCache.VarMeta Class Reference

Simple container for iDA-Variable meta data.

Public Member Functions

- __init__ (self, var_id)
- name (self)
- __repr__ (self)

Public Attributes

- id

  var id
- display_name

  unique var name, including index-suffix (if any)
- data_type

  data type of variable
- size

  dimension/size/length of variable (array length, blob size).
- unit
- fmt
- descr
- default_value
- min_limit
- max_limit

7.490.1 Detailed Description

Simple container for iDA-Variable meta data.

7.490.2 Constructor & Destructor Documentation

7.490.2.1 __init__()

```
ttk_tools.isyst.mwapp_client_offline_stub._VarMetaCache.VarMeta.__init__ (
              self,
              var_id )
```

7.490.3 Member Function Documentation

7.490.3.1 __repr__()

```
ttk_tools.isyst.mwapp_client_offline_stub._VarMetaCache.VarMeta.__repr__ (
              self )
```

7.490.3.2 name()

```
ttk_tools.isyst.mwapp_client_offline_stub._VarMetaCache.VarMeta.name (
            self )
```

7.490.4 Member Data Documentation

7.490.4.1 data_type

```
ttk_tools.isyst.mwapp_client_offline_stub._VarMetaCache.VarMeta.data_type
```

data type of variable

7.490.4.2 default_value

```
ttk_tools.isyst.mwapp_client_offline_stub._VarMetaCache.VarMeta.default_value
```

7.490.4.3 descr

```
ttk_tools.isyst.mwapp_client_offline_stub._VarMetaCache.VarMeta.descr
```

7.490.4.4 display_name

```
ttk_tools.isyst.mwapp_client_offline_stub._VarMetaCache.VarMeta.display_name
```

unique var name, including index-suffix (if any)

7.490.4.5 fmt

```
ttk_tools.isyst.mwapp_client_offline_stub._VarMetaCache.VarMeta.fmt
```

7.490.4.6 id

```
ttk_tools.isyst.mwapp_client_offline_stub._VarMetaCache.VarMeta.id
```

var id

7.490.4.7 max_limit

```
ttk_tools.isyst.mwapp_client_offline_stub._VarMetaCache.VarMeta.max_limit
```

7.490.4.8 min_limit

ttk_tools.isyst.mwapp_client_offline_stub._VarMetaCache.VarMeta.min_limit

7.490.4.9 size

ttk_tools.isyst.mwapp_client_offline_stub._VarMetaCache.VarMeta.size

dimension/size/length of variable (array length, blob size).

7.490.4.10 unit

ttk_tools.isyst.mwapp_client_offline_stub._VarMetaCache.VarMeta.unit

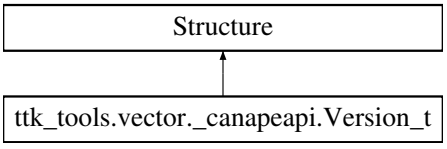
The documentation for this class was generated from the following file:

- ttk_tools/isyst/mwapp_client_offline_stub.py (v1.3.0)

7.491 ttk_tools.vector._canapeapi.Version_t Class Reference

CANape API version info structure.

Inheritance diagram for ttk_tools.vector._canapeapi.Version_t:



Static Protected Attributes

- int _pack_ = 1
- list _fields_

7.491.1 Detailed Description

CANape API version info structure.

7.491.2 Member Data Documentation

7.491.2.1 _fields_

list ttk_tools.vector._canapeapi.Version_t._fields_ [static], [protected]

Initial value:

```
=  [
        ('dllMainVersion',  c_int),
        ('dllSubVersion',   c_int),
        ('dllRelease',      c_int),
        ('osVersion',       c_char * 50),
        ('osRelease',       c_int),
    ]
```

7.491.2.2 _pack_

```
int ttk_tools.vector._canapeapi.Version_t._pack_ = 1  [static], [protected]
```

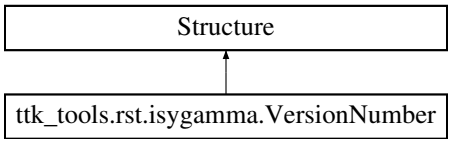
The documentation for this class was generated from the following file:

- ttk_tools/vector/_canapeapi.py (v7.3.3)

7.492 ttk_tools.rst.isygamma.VersionNumber Class Reference

gc_VersionNumber

Inheritance diagram for ttk_tools.rst.isygamma.VersionNumber:



Public Member Functions

- __str__ (self)
- getVersionInfo (self)

  Get version number as a VersionInfo named tuple.

Public Attributes

- major
- minor
- release
- build

Static Protected Attributes

- list _fields_

7.492.1 Detailed Description

gc_VersionNumber

7.492.2 Member Function Documentation

7.492.2.1 __str__()

```
ttk_tools.rst.isygamma.VersionNumber.__str__ (
             self )
```

7.492.2.2 getVersionInfo()

```
ttk_tools.rst.isygamma.VersionNumber.getVersionInfo (
              self )
```

Get version number as a VersionInfo named tuple.

7.492.3 Member Data Documentation

7.492.3.1 _fields_

```
list ttk_tools.rst.isygamma.VersionNumber._fields_  [static], [protected]
```

Initial value:

```
=  [
        ('major',   c_uint32 ),
        ('minor',   c_uint32 ),
        ('release', c_uint32 ),
        ('build',   c_uint32 )
    ]
```

7.492.3.2 build

```
ttk_tools.rst.isygamma.VersionNumber.build
```

7.492.3.3 major

```
ttk_tools.rst.isygamma.VersionNumber.major
```

7.492.3.4 minor

```
ttk_tools.rst.isygamma.VersionNumber.minor
```

7.492.3.5 release

```
ttk_tools.rst.isygamma.VersionNumber.release
```

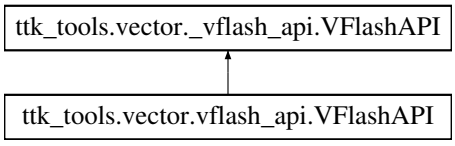
The documentation for this class was generated from the following file:

- ttk_tools/rst/isygamma.py (v1.6)

7.493 ttk_tools.vector._vflash_api.VFlashAPI Class Reference

Python Wrapper for accessing vFlash API functions.

Inheritance diagram for ttk_tools.vector._vflash_api.VFlashAPI:



Public Member Functions

- __init__ (self, dll_path=None)
- flash (self, flash_file, testresult=None, activate_network=False, flash_timeout=FLASH_TIMEOUT)

  Start flashing the software with vFlash C-API,.

Static Public Member Functions

- cbReportProgress (progress_in_percent, remaining_time_in_s)

  Callback function for C-API.
- cbReportStatus (flash_status)

  Status callback function for C-API.

Public Attributes

- project_handle

  handle to loaded VFlash project (typedef void VFLASHHANDLE)*
- c_func_CallbackProgress
- c_func_CallbackStatus
- dll_path

  detected/used path to VFlashAutomation DLL
- api_handle

  Handle to loaded DLL.
- max_len_error_text

Static Public Attributes

- bool cbvar_reprogramming_in_progress = False
- bool cbvar_reprogramming_success = False
- cbvar_reprogramming_status = None
- int cbvar_progress_in_percent = 0
- int cbvar_remaining_time_in_s = 0
- prototype_CallbackProgress
- prototype_CallbackStatus
- int max_len_error_text = 1024

  maximum size for string buffer for vFlashGetLastError.

Protected Member Functions

- _vFlashInitialize (self)

  Initialize vFlashAPI.
- _vFlashLoadProject (self, flash_file)

  Load flash project.
- _vFlashActivateNetwork (self)

  Activates FlexRay Network Management.
- _vFlashStart (self)

  Start flashing.
- _vFlashStop (self)

  Stop an active flash process.

- _vFlashUnloadProject (self)

  Unload the flash project.
- _vFlashDeinitialize (self)

  Deinitialize vFlash library.
- _vFlashGetLastError (self)

  Read the last error from the C-API.
- _isFlashing (self)

  Check if flash process is currently active.

7.493.1 Detailed Description

Python Wrapper for accessing vFlash API functions.

7.493.2 Constructor & Destructor Documentation

7.493.2.1 __init__()

```
ttk_tools.vector._vflash_api.VFlashAPI.__init__ (
              self,
              dll_path = None )
```

Parameters

dll_path	Path to vFlash C-API DLL file. None: Detect suitable DLL from system PATH

Note

DLL Use path to "VFlashAutomation.dll" for 32-bit Python and path to "VFlashAutomation64.dll" for 64-bit Python.

Both DLLs should be available in the "Bin/" sub folder of the vFlash installation (or alternatively in the "Bin/↩ Automation/" sub folder)

If `dll_path` is set to a folder path, VFlashAPI will try to load a suitable DLL from this folder.

If `dll_path` is None, VFlashAPI will try to locate a suitable DLL from system PATH entries (default vFlash installer adds the vFlash Bin/Automation/ folder to PATH)

Usage:

```
# with DLL auto-detect:
vflash_api = VFlashAPI()
```

Reimplemented in ttk_tools.vector.vflash_api.VFlashAPI.

7.493.3 Member Function Documentation

7.493.3.1 _isFlashing()

```
ttk_tools.vector._vflash_api.VFlashAPI._isFlashing (
              self )  [protected]
```

Check if flash process is currently active.

Returns

True if flashing is active, otherwise False.

7.493.3.2 _vFlashActivateNetwork()

```
ttk_tools.vector._vflash_api.VFlashAPI._vFlashActivateNetwork (
              self )  [protected]
```

Activates FlexRay Network Management.

Note

New in vFlash version 2.5, not necessary in earlier versions.

Info: vFlashActivateNetwork

"Synchronously activates the FlexRay Network Management for a FlexRay network. In case no Network Management is required (vFlash does not have to simulate a Coldstart Node), the function call completes successfully, but with actually no operation." - vFlashAutomation.chm

Raises a VFlashResultError if API call was not successful.

Returns

a VFlashResult value from vFlash C-API call.

7.493.3.3 _vFlashDeinitialize()

```
ttk_tools.vector._vflash_api.VFlashAPI._vFlashDeinitialize (
              self )  [protected]
```

Deinitialize vFlash library.

After executing any other command the library has to be deinitialized.

Raises a VFlashResultError if API call was not successful.

Returns

a VFlashResult value from vFlash C-API call.

7.493.3.4 _vFlashGetLastError()

```
ttk_tools.vector._vflash_api.VFlashAPI._vFlashGetLastError (
              self )  [protected]
```

Read the last error from the C-API.

Returns

the error text retrieved from the C-API

7.493.3.5 _vFlashInitialize()

```
ttk_tools.vector._vflash_api.VFlashAPI._vFlashInitialize (
              self )  [protected]
```

Initialize vFlashAPI.

Returns

a VFlashResult value from vFlash C-API call.

7.493.3.6 _vFlashLoadProject()

```
ttk_tools.vector._vflash_api.VFlashAPI._vFlashLoadProject (
              self,
              flash_file )  [protected]
```

Load flash project.

Parameters

flash_file	Path to the flash file

Usage:

```
flash_result = _vFlashLoadProject(
    flash_file = 'D:\\Hil_Projekte\\flashfile.odx'
)
```

Returns

a VFlashResult value from vFlash C-API call.

7.493.3.7 _vFlashStart()

```
ttk_tools.vector._vflash_api.VFlashAPI._vFlashStart (
              self )  [protected]
```

Start flashing.

Raises a VFlashResultError if API call was not successful.

Returns

a VFlashResult value from vFlash C-API call.

7.493.3.8 _vFlashStop()

```
ttk_tools.vector._vflash_api.VFlashAPI._vFlashStop (
              self )  [protected]
```

Stop an active flash process.

Returns

a VFlashResult value from vFlash C-API call.

7.493.3.9 _vFlashUnloadProject()

```
ttk_tools.vector._vflash_api.VFlashAPI._vFlashUnloadProject (
              self )  [protected]
```

Unload the flash project.

Raises a VFlashResultError if API call was not successful.

Returns

a VFlashResult value from vFlash C-API call.

7.493.3.10 cbReportProgress()

```
ttk_tools.vector._vflash_api.VFlashAPI.cbReportProgress (
              progress_in_percent,
              remaining_time_in_s )  [static]
```

Callback function for C-API.

Required C-signature for callback:

typedef void(∗ CallbackProgress)(unsigned int progressInPercent,unsigned int remainingTimeInS)

Attention

Function signature must not be changed.
Do not call any other vFlash API function within this callback.

Parameters

progress_in_percent	flash progress in percent (int)
remaining_time_in_s	remaining flash time in seconds (int)

7.493.3.11 cbReportStatus()

```
ttk_tools.vector._vflash_api.VFlashAPI.cbReportStatus (
              flash_status )  [static]
```

Status callback function for C-API.

Gets called at the end of a flash cycle.

Required C-signature for callback:

typedef void(∗ CallbackStatus)(VFlashStatus flashStatus)

Attention

Function signature must not be changed.
Do not call any other vFlash API function within this callback.

Parameters

flash_status	flash status int value, see VFlashStatus

7.493.3.12 flash()

```
ttk_tools.vector._vflash_api.VFlashAPI.flash (
              self,
              flash_file,
              testresult = None,
              activate_network = False,
              flash_timeout = FLASH_TIMEOUT )
```

Start flashing the software with vFlash C-API,.

Steps of the state machine:

1 - vFlash Initialization 2 - Load Project 3 - Activate Network (optional), availability depends on DLL version 4 - Flash software 5 - Unload Project 6 - Deinitialize vFlash Library

Parameters

flash_file	Path to the flash file
testresult	List where all the testresults are stored
activate_network	Only necessary when an active FlexRay node should be simulated. Available since vFlash version 2.5
flash_timeout	Timeout of flash procedure in seconds standard: 1000 seconds, see FLASH_TIMEOUT

Example:

```
flash(
    flash_file = 'D:\\Hil_Projekte\\flashfile.odx',
```

```
        testresult = testresult,
        activate_network = False
    )
```

Returns

a testresult list

Reimplemented in ttk_tools.vector.vflash_api.VFlashAPI.

7.493.4 Member Data Documentation

7.493.4.1 api_handle

```
ttk_tools.vector._vflash_api.VFlashAPI.api_handle
```

Handle to loaded DLL.

7.493.4.2 c_func_CallbackProgress

```
ttk_tools.vector._vflash_api.VFlashAPI.c_func_CallbackProgress
```

7.493.4.3 c_func_CallbackStatus

```
ttk_tools.vector._vflash_api.VFlashAPI.c_func_CallbackStatus
```

7.493.4.4 cbvar_progress_in_percent

```
int ttk_tools.vector._vflash_api.VFlashAPI.cbvar_progress_in_percent = 0  [static]
```

7.493.4.5 cbvar_remaining_time_in_s

```
int ttk_tools.vector._vflash_api.VFlashAPI.cbvar_remaining_time_in_s = 0  [static]
```

7.493.4.6 cbvar_reprogramming_in_progress

```
bool ttk_tools.vector._vflash_api.VFlashAPI.cbvar_reprogramming_in_progress = False  [static]
```

7.493.4.7 cbvar_reprogramming_status

```
ttk_tools.vector._vflash_api.VFlashAPI.cbvar_reprogramming_status = None  [static]
```

7.493.4.8 cbvar_reprogramming_success

```
bool ttk_tools.vector._vflash_api.VFlashAPI.cbvar_reprogramming_success = False  [static]
```

7.493.4.9 dll_path

```
ttk_tools.vector._vflash_api.VFlashAPI.dll_path
```

detected/used path to VFlashAutomation DLL

7.493.4.10 max_len_error_text [1/2]

```
int ttk_tools.vector._vflash_api.VFlashAPI.max_len_error_text = 1024  [static]
```

maximum size for string buffer for vFlashGetLastError.

7.493.4.11 max_len_error_text [2/2]

```
ttk_tools.vector._vflash_api.VFlashAPI.max_len_error_text
```

7.493.4.12 project_handle

```
ttk_tools.vector._vflash_api.VFlashAPI.project_handle
```

handle to loaded VFlash project (typedef void* VFLASHHANDLE)

7.493.4.13 prototype_CallbackProgress

```
ttk_tools.vector._vflash_api.VFlashAPI.prototype_CallbackProgress  [static]
```

Initial value:

```
=  ctypes.CFUNCTYPE(
        ctypes.c_void_p, ctypes.c_uint, ctypes.c_uint
    )
```

7.493.4.14 prototype_CallbackStatus

```
ttk_tools.vector._vflash_api.VFlashAPI.prototype_CallbackStatus  [static]
```

Initial value:

```
=  ctypes.CFUNCTYPE(
        ctypes.c_void_p, ctypes.c_uint
    )
```

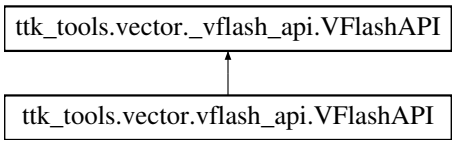
The documentation for this class was generated from the following file:

- ttk_tools/vector/_vflash_api.py (v1.6.1)

7.494 ttk_tools.vector.vflash_api.VFlashAPI Class Reference

Python Wrapper for accessing vFlash API functions.

Inheritance diagram for ttk_tools.vector.vflash_api.VFlashAPI:



Public Member Functions

- __init__ (self, dll_path=None)
- flash (self, flash_file, testresult=None, activate_network=False, flash_timeout=FLASH_TIMEOUT)

  Start flashing the software with vFlash C-API,.

7.494.1 Detailed Description

Python Wrapper for accessing vFlash API functions.

7.494.2 Constructor & Destructor Documentation

7.494.2.1 __init__()

```
ttk_tools.vector.vflash_api.VFlashAPI.__init__ (
            self,
            dll_path = None )
```

Parameters

dll_path	Path to vFlash C-API DLL file. None: Detect suitable DLL from system PATH

Note

DLL Use path to "VFlashAutomation.dll" for 32-bit Python and path to "VFlashAutomation64.dll" for 64-bit Python.

Both DLLs should be available in the "Bin/" sub folder of the vFlash installation (or alternatively in the "Bin/↩ Automation/" sub folder)

If `dll_path` is set to a folder path, VFlashAPI will try to load a suitable DLL from this folder.

If `dll_path` is None, VFlashAPI will try to locate a suitable DLL from system PATH entries (default vFlash installer adds the vFlash Bin/Automation/ folder to PATH)

Usage:

```
# with DLL auto-detect:
vflash_api = VFlashAPI()
```

Reimplemented from ttk_tools.vector._vflash_api.VFlashAPI.

7.494.3 Member Function Documentation

7.494.3.1 flash()

```
ttk_tools.vector.vflash_api.VFlashAPI.flash (
              self,
              flash_file,
              testresult = None,
              activate_network = False,
              flash_timeout = FLASH_TIMEOUT )
```

Start flashing the software with vFlash C-API,.

Steps of the state machine:

1 - vFlash Initialization 2 - Load Project 3 - Activate Network (optional), availability depends on DLL version 4 - Flash software 5 - Unload Project 6 - Deinitialize vFlash Library

Parameters

flash_file	Path to the flash file
testresult	List where all the testresults are stored
activate_network	Only necessary when an active FlexRay node should be simulated. Available since vFlash version 2.5
flash_timeout	Timeout of flash procedure in seconds standard: 1000 seconds, see FLASH_TIMEOUT

Example:

```
flash(
    flash_file = 'D:\\Hil_Projekte\\flashfile.odx',
    testresult = testresult,
    activate_network = False
)
```

Returns

a testresult list

Reimplemented from ttk_tools.vector._vflash_api.VFlashAPI.

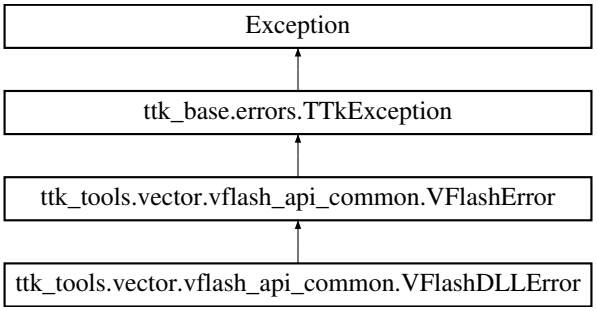
The documentation for this class was generated from the following file:

- ttk_tools/vector/vflash_api.py (v1.6.1)

7.495 ttk_tools.vector.vflash_api_common.VFlashDLLError Class Reference

Flashing ended with an error state.

Inheritance diagram for ttk_tools.vector.vflash_api_common.VFlashDLLError:



7.495.1 Detailed Description

Flashing ended with an error state.

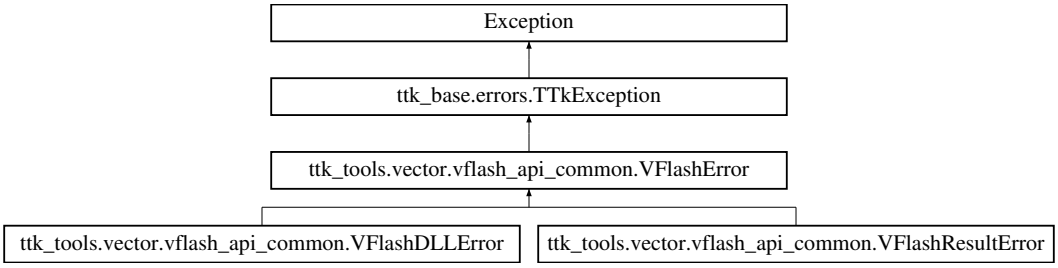
The documentation for this class was generated from the following file:

- ttk_tools/vector/vflash_api_common.py (v1.4)

7.496 ttk_tools.vector.vflash_api_common.VFlashError Class Reference

Base Error for vFlash API.

Inheritance diagram for ttk_tools.vector.vflash_api_common.VFlashError:



7.496.1 Detailed Description

Base Error for vFlash API.

DLL access failed

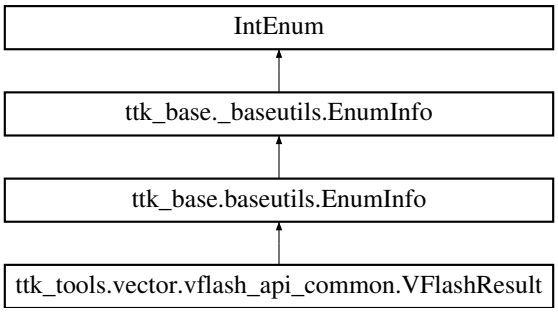
The documentation for this class was generated from the following file:

- ttk_tools/vector/vflash_api_common.py (v1.4)

7.497 ttk_tools.vector.vflash_api_common.VFlashResult Class Reference

Flash programming result values.

Inheritance diagram for ttk_tools.vector.vflash_api_common.VFlashResult:



Static Public Attributes

- int FR_SUCCESS = 0
- int FR_INITIALIZATION_ERROR = 10
- int FR_DEINITIALIZATION_ERROR = 11
- int FR_INVALID_PROJECT_HANDLE = 12
- int FR_INVALID_LICENSE = 13
- int FR_STOP_FAILED = 20
- int FR_PROJECT_IN_PROCESSING = 21
- int FR_NO_PROJECT_LOADED = 30
- int FR_PROJECT_PATH_INVALID = 31
- int FR_INCONSISTENT_PROJECT = 32
- int FR_INVALID_CALLBACKS = 33
- int FR_PROCESSING_FLASHWARE_ERROR = 34
- int FR_CHANNEL_ALREADY_IN_USE = 35
- int FR_NETWORK_ACTIVATION_FAILED = 36
- int FR_NO_CAN_DRIVER_AVAILABLE = 40
- int FR_INVALID_COMMAND_ORDER = 50
- int FR_FILE_NOT_FOUND = 51
- int FR_PROJECT_LOAD_FAILED = 52
- int FR_PROJECT_UNLOAD_FAILED = 53
- int FR_MAX_PROJECT_NUMBER_EXCEEDED = 54
- int FR_UNKNOWN_ERROR = 100

7.497.1 Detailed Description

Flash programming result values.

7.497.2 Member Data Documentation

7.497.2.1 FR_CHANNEL_ALREADY_IN_USE

```
int ttk_tools.vector.vflash_api_common.VFlashResult.FR_CHANNEL_ALREADY_IN_USE = 35  [static]
```

7.497.2.2 FR_DEINITIALIZATION_ERROR

```
int ttk_tools.vector.vflash_api_common.VFlashResult.FR_DEINITIALIZATION_ERROR = 11  [static]
```

7.497.2.3 FR_FILE_NOT_FOUND

```
int ttk_tools.vector.vflash_api_common.VFlashResult.FR_FILE_NOT_FOUND = 51  [static]
```

7.497.2.4 FR_INCONSISTENT_PROJECT

```
int ttk_tools.vector.vflash_api_common.VFlashResult.FR_INCONSISTENT_PROJECT = 32  [static]
```

7.497.2.5 FR_INITIALIZATION_ERROR

```
int ttk_tools.vector.vflash_api_common.VFlashResult.FR_INITIALIZATION_ERROR = 10  [static]
```

7.497.2.6 FR_INVALID_CALLBACKS

```
int ttk_tools.vector.vflash_api_common.VFlashResult.FR_INVALID_CALLBACKS = 33  [static]
```

7.497.2.7 FR_INVALID_COMMAND_ORDER

```
int ttk_tools.vector.vflash_api_common.VFlashResult.FR_INVALID_COMMAND_ORDER = 50  [static]
```

7.497.2.8 FR_INVALID_LICENSE

```
int ttk_tools.vector.vflash_api_common.VFlashResult.FR_INVALID_LICENSE = 13  [static]
```

7.497.2.9 FR_INVALID_PROJECT_HANDLE

```
int ttk_tools.vector.vflash_api_common.VFlashResult.FR_INVALID_PROJECT_HANDLE = 12  [static]
```

7.497.2.10 FR_MAX_PROJECT_NUMBER_EXCEEDED

```
int ttk_tools.vector.vflash_api_common.VFlashResult.FR_MAX_PROJECT_NUMBER_EXCEEDED = 54  [static]
```

7.497.2.11 FR_NETWORK_ACTIVATION_FAILED

```
int ttk_tools.vector.vflash_api_common.VFlashResult.FR_NETWORK_ACTIVATION_FAILED = 36  [static]
```

7.497.2.12 FR_NO_CAN_DRIVER_AVAILABLE

```
int ttk_tools.vector.vflash_api_common.VFlashResult.FR_NO_CAN_DRIVER_AVAILABLE = 40  [static]
```

7.497.2.13 FR_NO_PROJECT_LOADED

```
int ttk_tools.vector.vflash_api_common.VFlashResult.FR_NO_PROJECT_LOADED = 30  [static]
```

7.497.2.14 FR_PROCESSING_FLASHWARE_ERROR

```
int ttk_tools.vector.vflash_api_common.VFlashResult.FR_PROCESSING_FLASHWARE_ERROR = 34  [static]
```

7.497.2.15 FR_PROJECT_IN_PROCESSING

```
int ttk_tools.vector.vflash_api_common.VFlashResult.FR_PROJECT_IN_PROCESSING = 21  [static]
```

7.497.2.16 FR_PROJECT_LOAD_FAILED

```
int ttk_tools.vector.vflash_api_common.VFlashResult.FR_PROJECT_LOAD_FAILED = 52  [static]
```

7.497.2.17 FR_PROJECT_PATH_INVALID

```
int ttk_tools.vector.vflash_api_common.VFlashResult.FR_PROJECT_PATH_INVALID = 31  [static]
```

7.497.2.18 FR_PROJECT_UNLOAD_FAILED

```
int ttk_tools.vector.vflash_api_common.VFlashResult.FR_PROJECT_UNLOAD_FAILED = 53  [static]
```

7.497.2.19 FR_STOP_FAILED

```
int ttk_tools.vector.vflash_api_common.VFlashResult.FR_STOP_FAILED = 20  [static]
```

7.497.2.20 FR_SUCCESS

```
int ttk_tools.vector.vflash_api_common.VFlashResult.FR_SUCCESS = 0  [static]
```

7.497.2.21 FR_UNKNOWN_ERROR

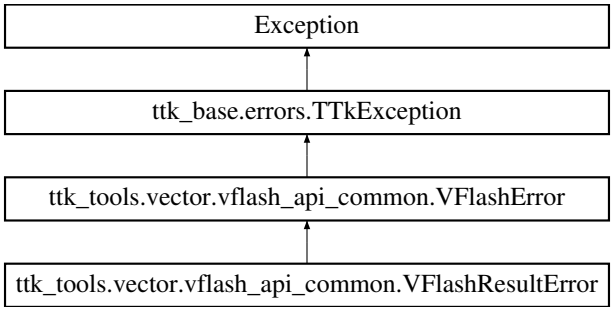
```
int ttk_tools.vector.vflash_api_common.VFlashResult.FR_UNKNOWN_ERROR = 100  [static]
```

The documentation for this class was generated from the following file:

- ttk_tools/vector/vflash_api_common.py (v1.4)

7.498 ttk_tools.vector.vflash_api_common.VFlashResultError Class Reference

Inheritance diagram for ttk_tools.vector.vflash_api_common.VFlashResultError:



Public Member Functions

- __init__ (self, flash_result, last_error)

7.498.1 Constructor & Destructor Documentation

7.498.1.1 __init__()

```
ttk_tools.vector.vflash_api_common.VFlashResultError.__init__ (
              self,
              flash_result,
              last_error )
```

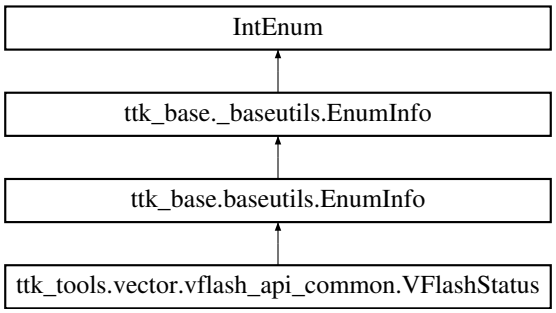
The documentation for this class was generated from the following file:

- ttk_tools/vector/vflash_api_common.py (v1.4)

7.499 ttk_tools.vector.vflash_api_common.VFlashStatus Class Reference

Flash programming status values.

Inheritance diagram for ttk_tools.vector.vflash_api_common.VFlashStatus:



Static Public Attributes

- int FS_SUCCESS = 0
- int FS_LICENSE_IS_NOT_AVAILABLE = 9
- int FS_ABORTED = 10
- int FS_UNKNOWN_ERROR = 11
- int FS_PROCESSING_FLASHWARE_FAILED = 15
- int FS_INITIALIZING_COMMUNICATION_FAILED = 16
- int FS_STOP_REPROGRAMMING_FAILED = 17
- int FS_PROJECT_DATA_INVALID = 18
- int FS_INITIALIZING_FLASH_LIBRARY_FAILED = 19
- int FS_SCRIPT_VERSION_MISMATCH = 20
- int FS_LOADING_PROJECT_FAILED = 21
- int FS_FLASHWARE_CHANGED = 22
- int FS_READING_EXPECTED_IDENTS_FAILED = 23
- int FS_FORCE_BOOT_MODE_FAILED = 24
- int FS_SWITCH_BAUD_RATE_FAILED = 25
- int FS_SEND_WAKEUP_PATTERN_FAILED = 26
- int FS_GENERAL_SCRIPT_FAILURE = 40
- int FS_SEED_KEY_HANDLING_FAILED = 41
- int FS_SECURITY_ACCESS_FAILED = 42
- int FS_COMMUNICATION_FAILED = 43
- int FS_DIAGNOSTIC_TRANSACTION_FAILED = 44
- int FS_SOFTWARE_INTEGRITY_CHECK_FAILED = 45
- int FS_NEGATIVE_RESPONSE_RECEIVED = 46
- int FS_ERASE_MEMORY_FAILED = 47
- int FS_SOFTWARE_AUTHENTICITY_CHECK_FAILED = 48
- int FS_SOFTWARE_COMPATIBILITY_CHECK_FAILED = 49
- int FS_FINGERPRINT_CHECK_FAILED = 50
- int FS_HARDWARE_COMPATIBILITY_CHECK_FAILED = 51
- int FS_PROGRAMMING_PRECONDITIONS_CHECK_FAILED = 52
- int FS_INVALID_PARAMETER_DETECTED = 53
- int FS_GENERAL_CUSTOM_ACTION_FAILURE = 60
- int FS_CUSTOM_ACTION_ATTRIBUTE_CONVERSION_FAILED = 61

7.499.1 Detailed Description

Flash programming status values.

7.499.2 Member Data Documentation

7.499.2.1 FS_ABORTED

```
int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_ABORTED = 10  [static]
```

7.499.2.2 FS_COMMUNICATION_FAILED

```
int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_COMMUNICATION_FAILED = 43  [static]
```

7.499.2.3 FS_CUSTOM_ACTION_ATTRIBUTE_CONVERSION_FAILED

```
int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_CUSTOM_ACTION_ATTRIBUTE_CONVERSION_↩
FAILED = 61  [static]
```

7.499.2.4 FS_DIAGNOSTIC_TRANSACTION_FAILED

```
int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_DIAGNOSTIC_TRANSACTION_FAILED = 44
[static]
```

7.499.2.5 FS_ERASE_MEMORY_FAILED

```
int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_ERASE_MEMORY_FAILED = 47  [static]
```

7.499.2.6 FS_FINGERPRINT_CHECK_FAILED

```
int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_FINGERPRINT_CHECK_FAILED = 50  [static]
```

7.499.2.7 FS_FLASHWARE_CHANGED

```
int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_FLASHWARE_CHANGED = 22  [static]
```

7.499.2.8 FS_FORCE_BOOT_MODE_FAILED

```
int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_FORCE_BOOT_MODE_FAILED = 24  [static]
```

7.499.2.9 FS_GENERAL_CUSTOM_ACTION_FAILURE

```
int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_GENERAL_CUSTOM_ACTION_FAILURE = 60
[static]
```

7.499.2.10 FS_GENERAL_SCRIPT_FAILURE

```
int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_GENERAL_SCRIPT_FAILURE = 40  [static]
```

7.499.2.11 FS_HARDWARE_COMPATIBILITY_CHECK_FAILED

```
int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_HARDWARE_COMPATIBILITY_CHECK_FAILED =
51  [static]
```

7.499.2.12 FS_INITIALIZING_COMMUNICATION_FAILED

int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_INITIALIZING_COMMUNICATION_FAILED = 16 [static]

7.499.2.13 FS_INITIALIZING_FLASH_LIBRARY_FAILED

int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_INITIALIZING_FLASH_LIBRARY_FAILED = 19 [static]

7.499.2.14 FS_INVALID_PARAMETER_DETECTED

int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_INVALID_PARAMETER_DETECTED = 53 [static]

7.499.2.15 FS_LICENSE_IS_NOT_AVAILABLE

int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_LICENSE_IS_NOT_AVAILABLE = 9 [static]

7.499.2.16 FS_LOADING_PROJECT_FAILED

int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_LOADING_PROJECT_FAILED = 21 [static]

7.499.2.17 FS_NEGATIVE_RESPONSE_RECEIVED

int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_NEGATIVE_RESPONSE_RECEIVED = 46 [static]

7.499.2.18 FS_PROCESSING_FLASHWARE_FAILED

int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_PROCESSING_FLASHWARE_FAILED = 15 [static]

7.499.2.19 FS_PROGRAMMING_PRECONDITIONS_CHECK_FAILED

int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_PROGRAMMING_PRECONDITIONS_CHECK_FAILED = 52 [static]

7.499.2.20 FS_PROJECT_DATA_INVALID

int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_PROJECT_DATA_INVALID = 18 [static]

7.499.2.21 FS_READING_EXPECTED_IDENTS_FAILED

int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_READING_EXPECTED_IDENTS_FAILED = 23 [static]

7.499.2.22 FS_SCRIPT_VERSION_MISMATCH

```
int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_SCRIPT_VERSION_MISMATCH = 20  [static]
```

7.499.2.23 FS_SECURITY_ACCESS_FAILED

```
int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_SECURITY_ACCESS_FAILED = 42  [static]
```

7.499.2.24 FS_SEED_KEY_HANDLING_FAILED

```
int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_SEED_KEY_HANDLING_FAILED = 41  [static]
```

7.499.2.25 FS_SEND_WAKEUP_PATTERN_FAILED

```
int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_SEND_WAKEUP_PATTERN_FAILED = 26  [static]
```

7.499.2.26 FS_SOFTWARE_AUTHENTICITY_CHECK_FAILED

```
int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_SOFTWARE_AUTHENTICITY_CHECK_FAILED = 48
[static]
```

7.499.2.27 FS_SOFTWARE_COMPATIBILITY_CHECK_FAILED

```
int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_SOFTWARE_COMPATIBILITY_CHECK_FAILED =
49  [static]
```

7.499.2.28 FS_SOFTWARE_INTEGRITY_CHECK_FAILED

```
int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_SOFTWARE_INTEGRITY_CHECK_FAILED = 45
[static]
```

7.499.2.29 FS_STOP_REPROGRAMMING_FAILED

```
int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_STOP_REPROGRAMMING_FAILED = 17  [static]
```

7.499.2.30 FS_SUCCESS

```
int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_SUCCESS = 0  [static]
```

7.499.2.31 FS_SWITCH_BAUD_RATE_FAILED

```
int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_SWITCH_BAUD_RATE_FAILED = 25  [static]
```

7.499.2.32 FS_UNKNOWN_ERROR

```
int ttk_tools.vector.vflash_api_common.VFlashStatus.FS_UNKNOWN_ERROR = 11  [static]
```

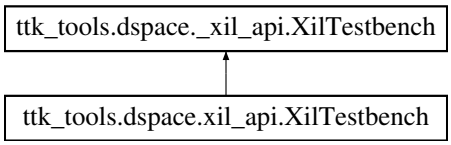
The documentation for this class was generated from the following file:

- ttk_tools/vector/vflash_api_common.py (v1.4)

7.500 ttk_tools.dspace._xil_api.XilTestbench Class Reference

XilTestbench access.

Inheritance diagram for ttk_tools.dspace._xil_api.XilTestbench:



Public Member Functions

- __init__ (self, config_file_path, force_config=False, port_name="MAPort", product_name="XIL API", product_version="2022-B")

  Setup a dSPACE XIL testbench along with a MAPort.
- __del__ (self)
- dispose (self)

  Dispose of resources allocated by this testbench instance.
- read (self, model_path)

  Read a value from a variable.
- write (self, model_path, value)

  Write a value to a variable.
- getCapture (self)

  Get a Capture instance configured to use this XilTestbench.
- getStimulus (self)

  Get a Stimulus instance configured to use this XilTestbench.

Public Attributes

- testbench

  [XIL-API] dSPACE-flavored testbench interface
- VENDOR_NAME
- ma_port

  [XIL-API] model access port

Static Public Attributes

- str VENDOR_NAME = "dSPACE GmbH"

  vendor name for dSPACE's XIL testbench implementation.
- write_value = None
- value = getIntValue(value)
- xil_type_info = DataType.getInfo(data_type)
- msg = str(ex)
- str addon_msg = ""
- ex_name = type(ex).__name__
- str ex_name = ".NET-System.Exception"
- formatted_ex_msg = str(ex).replace("\n", "\n>" + " " * (len(ex_name) + 5))
- Message
- source
- Source
- orig_exception

Protected Member Functions

- _read (self, model_path)

  Read a value from a variable.

Static Protected Attributes

- int _MAPORT_CONNECT_ATTEMPTS = 4

  number of attempts when trying to create/connect to an MAPort

7.500.1 Detailed Description

XilTestbench access.

7.500.2 Constructor & Destructor Documentation

7.500.2.1 __init__()

```
ttk_tools.dspace._xil_api.XilTestbench.__init__ (
              self,
              config_file_path,
              force_config = False,
              port_name = "MAPort",
              product_name = "XIL API",
              product_version = "2022-B" )
```

Setup a dSPACE XIL testbench along with a MAPort.

Parameters

config_file_path	path to XML configuration file
force_config	if True, an already loaded application will be replaced with the "new" configuration
port_name	name of MA Port (not really useful, though)
product_name	product name used in testbench factory
product_version	product version used in testbench factory. This usually matches the current release name.

Reimplemented in ttk_tools.dspace.xil_api.XilTestbench.

7.500.2.2 __del__()

```
ttk_tools.dspace._xil_api.XilTestbench.__del__ (
              self )
```

7.500.3 Member Function Documentation

7.500.3.1 _read()

```
ttk_tools.dspace._xil_api.XilTestbench._read (
              self,
              model_path )  [protected]
```

Read a value from a variable.

Parameters

model_path	path to variable, e.g. a Simulink-path

Returns

the "unpacked" value plus a (numeric) XIL DataType as a tuple (<value>, <data_↩ type>)

7.500.3.2 dispose()

```
ttk_tools.dspace._xil_api.XilTestbench.dispose (
              self )
```

Dispose of resources allocated by this testbench instance.

After this is called, the MAPort will no longer be available and further access will fail.

Reimplemented in ttk_tools.dspace.xil_api.XilTestbench.

7.500.3.3 getCapture()

```
ttk_tools.dspace._xil_api.XilTestbench.getCapture (
              self )
```

Get a Capture instance configured to use this XilTestbench.

Reimplemented in ttk_tools.dspace.xil_api.XilTestbench.

7.500.3.4 getStimulus()

```
ttk_tools.dspace._xil_api.XilTestbench.getStimulus (
              self )
```

Get a Stimulus instance configured to use this XilTestbench.

Reimplemented in ttk_tools.dspace.xil_api.XilTestbench.

7.500.3.5 read()

```
ttk_tools.dspace._xil_api.XilTestbench.read (
              self,
              model_path )
```

Read a value from a variable.

Parameters

model_path	path to variable, e.g. a Simulink-path

Returns

the current value of the variable

Reimplemented in ttk_tools.dspace.xil_api.XilTestbench.

7.500.3.6 write()

```
ttk_tools.dspace._xil_api.XilTestbench.write (
              self,
              model_path,
              value )
```

Write a value to a variable.

Parameters

model_path	path to variable, e.g. a Simulink-path
value	value to write

Returns

the previous value before the write operation.

Reimplemented in ttk_tools.dspace.xil_api.XilTestbench.

7.500.4 Member Data Documentation

7.500.4.1 _MAPORT_CONNECT_ATTEMPTS

```
int ttk_tools.dspace._xil_api.XilTestbench._MAPORT_CONNECT_ATTEMPTS = 4  [static], [protected]
```

number of attempts when trying to create/connect to an MAPort

7.500.4.2 addon_msg

```
str ttk_tools.dspace._xil_api.XilTestbench.addon_msg = ""  [static]
```

7.500.4.3 ex_name [1/2]

```
ttk_tools.dspace._xil_api.XilTestbench.ex_name = type(ex).__name__  [static]
```

7.500.4.4 ex_name [2/2]

```
str ttk_tools.dspace._xil_api.XilTestbench.ex_name = ".NET-System.Exception"  [static]
```

7.500.4.5 formatted_ex_msg

```
ttk_tools.dspace._xil_api.XilTestbench.formatted_ex_msg = str(ex).replace("\n", "\n>" + " " *
(len(ex_name) + 5))  [static]
```

7.500.4.6 ma_port

```
ttk_tools.dspace._xil_api.XilTestbench.ma_port
```

[XIL-API] model access port

7.500.4.7 Message

```
ttk_tools.dspace._xil_api.XilTestbench.Message  [static]
```

7.500.4.8 msg

```
ttk_tools.dspace._xil_api.XilTestbench.msg = str(ex)  [static]
```

7.500.4.9 orig_exception

```
ttk_tools.dspace._xil_api.XilTestbench.orig_exception  [static]
```

7.500.4.10 source

```
ttk_tools.dspace._xil_api.XilTestbench.source  [static]
```

7.500.4.11 Source

```
ttk_tools.dspace._xil_api.XilTestbench.Source  [static]
```

7.500.4.12 testbench

```
ttk_tools.dspace._xil_api.XilTestbench.testbench
```

[XIL-API] dSPACE-flavored testbench interface

7.500.4.13 value

```
ttk_tools.dspace._xil_api.XilTestbench.value = getIntValue(value)  [static]
```

7.500.4.14 VENDOR_NAME [1/2]

```
str ttk_tools.dspace._xil_api.XilTestbench.VENDOR_NAME = "dSPACE GmbH"  [static]
```

vendor name for dSPACE's XIL testbench implementation.

This should not really change (but neither should product name, which already changed between Releases 2014-B and 2015-A...)

7.500.4.15 VENDOR_NAME [2/2]

```
ttk_tools.dspace._xil_api.XilTestbench.VENDOR_NAME
```

7.500.4.16 write_value

```
ttk_tools.dspace._xil_api.XilTestbench.write_value = None  [static]
```

7.500.4.17 xil_type_info

```
ttk_tools.dspace._xil_api.XilTestbench.xil_type_info = DataType.getInfo(data_type)  [static]
```

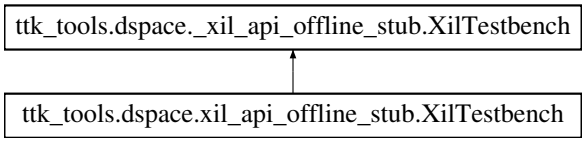
The documentation for this class was generated from the following file:

- ttk_tools/dspace/_xil_api.py (v2.3.1)

7.501 ttk_tools.dspace._xil_api_offline_stub.XilTestbench Class Reference

XilTestbench access offline stub.

Inheritance diagram for ttk_tools.dspace._xil_api_offline_stub.XilTestbench:



Public Member Functions

- __init__ (self, config_file_path, force_config=False, port_name="MAPort", product_name="XIL API", product_version="2022-B")

  Setup a (stubbed) dSPACE XIL testbench along with a (stubbed) MAPort.
- dispose (self)

  Dispose of resources allocated by this testbench instance.
- read (self, model_path)

  Read a value from a variable.
- write (self, model_path, value)

  Write a value to a variable.
- getCapture (self)

  Get a Capture instance configured to use this XilTestbench.
- getStimulus (self)

  Get a Stimulus instance configured to use this XilTestbench.

Public Attributes

- testbench

  [XIL-API] dSPACE-flavored testbench interface.
- ma_port

  [XIL-API] model access port.

Static Public Attributes

- str VENDOR_NAME = "dSPACE GmbH"

  vendor name for dSPACE's XIL testbench implementation.

7.501.1 Detailed Description

XilTestbench access offline stub.

7.501.2 Constructor & Destructor Documentation

7.501.2.1 __init__()

```
ttk_tools.dspace._xil_api_offline_stub.XilTestbench.__init__ (
              self,
              config_file_path,
              force_config = False,
              port_name = "MAPort",
              product_name = "XIL API",
              product_version = "2022-B" )
```

Setup a (stubbed) dSPACE XIL testbench along with a (stubbed) MAPort.

Parameters

config_file_path	path to XML configuration file
force_config	if True, an already loaded application will be replaced with the "new" configuration
port_name	name of MA Port (not really useful, though)
product_name	product name used in testbench factory
product_version	product version used in testbench factory. This usually matches the current release name.

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.XilTestbench.

7.501.3 Member Function Documentation

7.501.3.1 dispose()

```
ttk_tools.dspace._xil_api_offline_stub.XilTestbench.dispose (
              self )
```

Dispose of resources allocated by this testbench instance.

After this is called, the MAPort will no longer be available and further access will fail.

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.XilTestbench.

7.501.3.2 getCapture()

```
ttk_tools.dspace._xil_api_offline_stub.XilTestbench.getCapture (
              self )
```

Get a Capture instance configured to use this XilTestbench.

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.XilTestbench.

7.501.3.3 getStimulus()

```
ttk_tools.dspace._xil_api_offline_stub.XilTestbench.getStimulus (
              self )
```

Get a Stimulus instance configured to use this XilTestbench.

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.XilTestbench.

7.501.3.4 read()

```
ttk_tools.dspace._xil_api_offline_stub.XilTestbench.read (
              self,
              model_path )
```

Read a value from a variable.

Parameters

model_path	path to variable, e.g. a Simulink-path

Returns

the current value of the variable

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.XilTestbench.

7.501.3.5 write()

```
ttk_tools.dspace._xil_api_offline_stub.XilTestbench.write (
              self,
              model_path,
              value )
```

Write a value to a variable.

Parameters

model_path	path to variable, e.g. a Simulink-path
value	value to write

Returns

the previous value before the write operation.

Reimplemented in ttk_tools.dspace.xil_api_offline_stub.XilTestbench.

7.501.4 Member Data Documentation

7.501.4.1 ma_port

```
ttk_tools.dspace._xil_api_offline_stub.XilTestbench.ma_port
```

[XIL-API] model access port.

In a XilTestbench stub, this will only reference a minimal MAPort stub that reads/writes offline data (see ttk_tools.offline_data_provider)

7.501.4.2 testbench

```
ttk_tools.dspace._xil_api_offline_stub.XilTestbench.testbench
```

[XIL-API] dSPACE-flavored testbench interface.

In a XilTestbench stub, this will only reference a minimal testbench stub containing members for VendorName, ProductName and ProductVersion.

7.501.4.3 VENDOR_NAME

```
str ttk_tools.dspace._xil_api_offline_stub.XilTestbench.VENDOR_NAME = "dSPACE GmbH"  [static]
```

vendor name for dSPACE's XIL testbench implementation.

This should not really change (but neither should product name, which already changed between Releases 2014-B and 2015-A...)

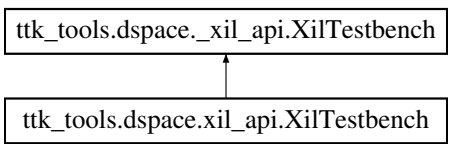
The documentation for this class was generated from the following file:

- ttk_tools/dspace/_xil_api_offline_stub.py (v2.1.2)

7.502 ttk_tools.dspace.xil_api.XilTestbench Class Reference

XilTestbench access.

Inheritance diagram for ttk_tools.dspace.xil_api.XilTestbench:



Public Member Functions

- __init__ (self, config_file_path, force_config=False, port_name="MAPort", product_name="XIL API", product_version="2022-B")

  Setup a dSPACE XIL testbench along with a MAPort.
- dispose (self)

  Dispose of resources allocated by this testbench instance.
- read (self, model_path)

  Read a value from a variable.
- write (self, model_path, value)

  Write a value to a variable.
- getCapture (self)

  Get a Capture instance configured to use this XilTestbench.
- getStimulus (self)

  Get a Stimulus instance configured to use this XilTestbench.

Public Attributes

- testbench

  [XIL-API] dSPACE-flavored testbench interface, will be initialized in parent constructor
- ma_port

  [XIL-API] model access port, will be initialized in parent constructor

Static Public Attributes

- str VENDOR_NAME = "dSPACE GmbH"

  vendor name for dSPACE's XIL testbench implementation.

7.502.1 Detailed Description

XilTestbench access.

7.502.2 Constructor & Destructor Documentation

7.502.2.1 __init__()

```
ttk_tools.dspace.xil_api.XilTestbench.__init__ (
              self,
              config_file_path,
              force_config = False,
              port_name = "MAPort",
              product_name = "XIL API",
              product_version = "2022-B" )
```

Setup a dSPACE XIL testbench along with a MAPort.

Parameters

config_file_path	path to XML configuration file
force_config	if True, an already loaded application will be replaced with the "new" configuration
port_name	name of MA Port (not really useful, though)
product_name	product name used in testbench factory
product_version	product version used in testbench factory. This usually matches the current release name.

Reimplemented from ttk_tools.dspace._xil_api.XilTestbench.

7.502.3 Member Function Documentation

7.502.3.1 dispose()

```
ttk_tools.dspace.xil_api.XilTestbench.dispose (
              self )
```

Dispose of resources allocated by this testbench instance.

After this is called, the MAPort will no longer be available and further access will fail.

Reimplemented from ttk_tools.dspace._xil_api.XilTestbench.

7.502.3.2 getCapture()

```
ttk_tools.dspace.xil_api.XilTestbench.getCapture (
              self )
```

Get a Capture instance configured to use this XilTestbench.

Reimplemented from ttk_tools.dspace._xil_api.XilTestbench.

7.502.3.3 getStimulus()

```
ttk_tools.dspace.xil_api.XilTestbench.getStimulus (
              self )
```

Get a Stimulus instance configured to use this XilTestbench.

Reimplemented from ttk_tools.dspace._xil_api.XilTestbench.

7.502.3.4 read()

```
ttk_tools.dspace.xil_api.XilTestbench.read (
              self,
              model_path )
```

Read a value from a variable.

Parameters

model_path	path to variable, e.g. a Simulink-path

Returns

the current value of the variable

Reimplemented from ttk_tools.dspace._xil_api.XilTestbench.

7.502.3.5 write()

```
ttk_tools.dspace.xil_api.XilTestbench.write (
              self,
              model_path,
              value )
```

Write a value to a variable.

Parameters

model_path	path to variable, e.g. a Simulink-path
value	value to write

Returns

the previous value before the write operation.

Reimplemented from ttk_tools.dspace._xil_api.XilTestbench.

7.502.4 Member Data Documentation

7.502.4.1 ma_port

```
ttk_tools.dspace.xil_api.XilTestbench.ma_port
```

[XIL-API] model access port, will be initialized in parent constructor

7.502.4.2 testbench

```
ttk_tools.dspace.xil_api.XilTestbench.testbench
```

[XIL-API] dSPACE-flavored testbench interface, will be initialized in parent constructor

7.502.4.3 VENDOR_NAME

```
str ttk_tools.dspace.xil_api.XilTestbench.VENDOR_NAME = "dSPACE GmbH"  [static]
```

vendor name for dSPACE's XIL testbench implementation.

This should not really change (but neither should product name, which already changed between Releases 2014-B and 2015-A...)

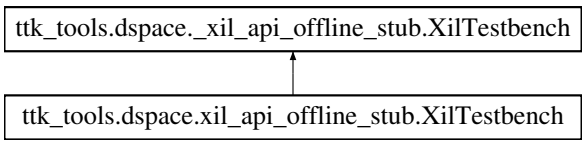
The documentation for this class was generated from the following file:

- ttk_tools/dspace/xil_api.py (v2.0.1)

7.503 ttk_tools.dspace.xil_api_offline_stub.XilTestbench Class Reference

XilTestbench access offline stub.

Inheritance diagram for ttk_tools.dspace.xil_api_offline_stub.XilTestbench:



Public Member Functions

- __init__ (self, config_file_path, force_config=False, port_name="MAPort", product_name="XIL API", product_version="2022-B")

  Setup a (stubbed) dSPACE XIL testbench along with a (stubbed) MAPort.
- dispose (self)

  Dispose of resources allocated by this testbench instance.
- read (self, model_path)

  Read a value from a variable.
- write (self, model_path, value)

  Write a value to a variable.
- getCapture (self)

  Get a Capture instance configured to use this XilTestbench.
- getStimulus (self)

  Get a Stimulus instance configured to use this XilTestbench.

Public Attributes

- testbench

  [XIL-API] dSPACE-flavored testbench interface.
- ma_port

  [XIL-API] model access port.

Static Public Attributes

- str VENDOR_NAME = "dSPACE GmbH"

  vendor name for dSPACE's XIL testbench implementation.

7.503.1 Detailed Description

XilTestbench access offline stub.

7.503.2 Constructor & Destructor Documentation

7.503.2.1 __init__()

```
ttk_tools.dspace.xil_api_offline_stub.XilTestbench.__init__ (
              self,
              config_file_path,
              force_config = False,
              port_name = "MAPort",
              product_name = "XIL API",
              product_version = "2022-B" )
```

Setup a (stubbed) dSPACE XIL testbench along with a (stubbed) MAPort.

Parameters

config_file_path	path to XML configuration file
force_config	if True, an already loaded application will be replaced with the "new" configuration
port_name	name of MA Port (not really useful, though)
product_name	product name used in testbench factory
product_version	product version used in testbench factory. This usually matches the current release name.

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.XilTestbench.

7.503.3 Member Function Documentation

7.503.3.1 dispose()

```
ttk_tools.dspace.xil_api_offline_stub.XilTestbench.dispose (
              self )
```

Dispose of resources allocated by this testbench instance.

After this is called, the MAPort will no longer be available and further access will fail.

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.XilTestbench.

7.503.3.2 getCapture()

```
ttk_tools.dspace.xil_api_offline_stub.XilTestbench.getCapture (
              self )
```

Get a Capture instance configured to use this XilTestbench.

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.XilTestbench.

7.503.3.3 getStimulus()

```
ttk_tools.dspace.xil_api_offline_stub.XilTestbench.getStimulus (
              self )
```

Get a Stimulus instance configured to use this XilTestbench.

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.XilTestbench.

7.503.3.4 read()

```
ttk_tools.dspace.xil_api_offline_stub.XilTestbench.read (
              self,
              model_path )
```

Read a value from a variable.

Parameters

model_path	path to variable, e.g. a Simulink-path

Returns

the current value of the variable

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.XilTestbench.

7.503.3.5 write()

```
ttk_tools.dspace.xil_api_offline_stub.XilTestbench.write (
              self,
              model_path,
              value )
```

Write a value to a variable.

Parameters

model_path	path to variable, e.g. a Simulink-path
value	value to write

Returns

the previous value before the write operation.

Reimplemented from ttk_tools.dspace._xil_api_offline_stub.XilTestbench.

7.503.4 Member Data Documentation

7.503.4.1 ma_port

`ttk_tools.dspace.xil_api_offline_stub.XilTestbench.ma_port`

[XIL-API] model access port.

In a XilTestbench stub, this will only reference a minimal MAPort stub that reads/writes offline data (see ttk_tools.offline_data_provider)

7.503.4.2 testbench

`ttk_tools.dspace.xil_api_offline_stub.XilTestbench.testbench`

[XIL-API] dSPACE-flavored testbench interface.

In a XilTestbench stub, this will only reference a minimal testbench stub containing members for VendorName, ProductName and ProductVersion.

7.503.4.3 VENDOR_NAME

`str ttk_tools.dspace.xil_api_offline_stub.XilTestbench.VENDOR_NAME = "dSPACE GmbH"  [static]`

vendor name for dSPACE's XIL testbench implementation.

This should not really change (but neither should product name, which already changed between Releases 2014-B and 2015-A...)

The documentation for this class was generated from the following file:

- ttk_tools/dspace/xil_api_offline_stub.py (v2.0.1)

Chapter 8

File Documentation

8.1 mainpage.txt File Reference

8.2 ttk_base/_baseutils.py File Reference

Classes

- class ttk_base._baseutils.HexList

  A list that returns a representation in hexadecimal notation (for easier reading).
- class ttk_base._baseutils.EnumInfo

  Utility/convenience class for "enum-like" value container classes.
- class ttk_base._baseutils._CustomRepr

  A utility class for "marking" customized (i.e.

Namespaces

- namespace ttk_base
- namespace ttk_base._baseutils

  General utility functions and classes.

Functions

- ttk_base._baseutils.getSpacedHexString (value, min_bytes=1, spacer=" ")

  Get a hex string representation of Bytes in the supplied (unsigned) integer value.
- ttk_base._baseutils._getInteger (value)
- ttk_base._baseutils._bin (value, prefix='0b', min_digits=0)

  Create a binary representation of the input value.
- ttk_base._baseutils._hex (value, prefix='0x', min_digits=0)

  Create a hexadecimal representation of the input value.
- ttk_base._baseutils._dec (value, prefix='0d', min_digits=0)

  Create a integer/decimal representation of the input value.
- ttk_base._baseutils._oct (value, prefix='0o', min_digits=0)

  Create an octal representation of the input value.
- ttk_base._baseutils._pct (factor, suffix=' %', decimal_places=None)

Create an "percentage" representation of the input factor value.

- ttk_base._baseutils._flt (value, decimal_places=None)

  Create a float representation of the input value.
- ttk_base._baseutils.getCustomRepr (value, fmt=None)

  Get a custom string representation of a value.
- ttk_base._baseutils.isCustomRepr (s)

  Check if s is formatted using a custom representation or just a default/fallback string representation (or anything else).
- ttk_base._baseutils.bstr (obj, encoding=DEFAULT_ENCODING)

  Get a byte string representation of an object, that is, like str() but returns a raw bytes string.
- ttk_base._baseutils.ustr (obj, encoding=DEFAULT_ENCODING)

  Get a unicode/text string representation of an object.
- ttk_base._baseutils.uni (s)

  Enforce unicode/text strings.
- ttk_base._baseutils.fsEncode (s)

  Get a representation of (unicode/ASCII) string s, using the current file system encoding.
- ttk_base._baseutils.fsDecode (s)

  Get a unicode representation of byte string s, using the current file system encoding (if available).
- ttk_base._baseutils.sleepWithFeedback (delay_ms, comment=None, step_ms=100, done_msg=None)

  Sleep for the specified delay and print some feedback/progress to console/log.

Variables

- str ttk_base._baseutils.DEFAULT_ENCODING = "UTF-8"

  Default encoding for encoding/decoding raw byte strings as text.
- str ttk_base._baseutils.BYTES_ENCODING = "ASCII"

  Encoding to use when adding raw byte strings to description text.
- float ttk_base._baseutils._FLOAT_CONVERSION_LIMIT = 0.0001

  maximum divergence to still accept a float value to be "intended" as int/long
- ttk_base._baseutils._FMT_KEYWORD_RE = re.compile(r"^\s*([a-z]+)\s*(\d*)\s*$")

  regex for format keyword matching, see getCustomRepr

8.3 ttk_base/_values_base.py File Reference

Classes

- class ttk_base._values_base.MetaValue

  Base class for meta attributes.
- class ttk_base._values_base.MetaFloat

  Float values with additional meta info.
- class ttk_base._values_base.MetaInt

  Integer values with additional meta info.
- class ttk_base._values_base.MetaStr

  String values with additional meta info.
- class ttk_base._values_base.MetaBytes

  Raw Byte string values with additional meta info.
- class ttk_base._values_base.MetaBytearray

  Bytearray values with additional meta info.
- class ttk_base._values_base.MetaTuple

  A tuple with additional meta info.
- class ttk_base._values_base.MetaList

  A List with additional meta info.

Namespaces

- namespace ttk_base
- namespace ttk_base._values_base

  Base classes for values of built-in types like long, int, float or string with additional meta information attached.

Functions

- ttk_base._values_base.meta (value, unit="", alias="", descr="", lookup=None, fmt=None)

  Factory function to create "meta" instances of the supplied value that have some additional meta attributes bolted on.

8.4 ttk_base/_variables_base.py File Reference

Classes

- class ttk_base._variables_base.VarBase

  Base wrapper for a single variable.
- class ttk_base._variables_base.DummyVar

  A minimal dummy test variable not connected to any external interface.
- class ttk_base._variables_base.VarContainerBase

  Base class for variables containers.
- class ttk_base._variables_base.ResettableContainer

  Simple container class that provides a non-recursive reset() method.
- class ttk_base._variables_base.Snapshot

  Store and restore a "snapshot" of variables' values.

Namespaces

- namespace ttk_base
- namespace ttk_base._variables_base

  Base classes for test variables (hil, cal, ...) to provide a unified interface and containers to group variables into sets.

Functions

- ttk_base._variables_base._getArg (args, kwargs, pos, key, fallback=None)

  Extract a specific positional or keyword argument from generic args/kwargs containers.
- ttk_base._variables_base._voidKwArgs (kwargs, ∗keys)

  Remove keyword arguments from the supplied dict (to make sure it will not interfere with other arguments we will directly supply).
- ttk_base._variables_base._getModelPathAlias (model_path)

  Get a fallback name/alias based on the supplied Simulink model path.

8.5 ttk_base/baseutils.py File Reference

Classes

- class ttk_base.baseutils.HexList

  A list that returns a representation in hexadecimal notation (for easier reading).
- class ttk_base.baseutils.EnumInfo

  Utility/convenience class for "enum-like" value container classes.

Namespaces

- namespace ttk_base
- namespace ttk_base.baseutils

  Interface wrapper for general utility functions and classes in ttk_base._baseutils.

Functions

- ttk_base.baseutils.getSpacedHexString (value, min_bytes=1, spacer=" ")

  Get a hex string representation of Bytes in the supplied (unsigned) integer value.
- ttk_base.baseutils.getCustomRepr (value, fmt=None)

  Get a custom string representation of a value.
- ttk_base.baseutils.isCustomRepr (s)

  Check if s is formatted using a custom representation or just a default/fallback string representation (or anything else).
- ttk_base.baseutils.bstr (obj, encoding=DEFAULT_ENCODING)

  Get a byte string representation of an object, that is, like str() but returns a raw bytes string.
- ttk_base.baseutils.ustr (obj, encoding=DEFAULT_ENCODING)

  Get a unicode/text string representation of an object.
- ttk_base.baseutils.uni (s)

  Enforce unicode/text strings.
- ttk_base.baseutils.fsEncode (s)

  Get a representation of (unicode/ASCII) string s, using the current file system encoding.
- ttk_base.baseutils.fsDecode (s)

  Get a unicode representation of byte string s, using the current file system encoding (if available).
- ttk_base.baseutils.sleepWithFeedback (delay_ms, comment=None, step_ms=100, done_msg=None)

  Sleep for the specified delay and print some feedback/progress to console/log.

8.6 ttk_base/configuration.py File Reference

Classes

- class ttk_base.configuration.DongleLibError

  Errors while loading Dongle DLLs.
- class ttk_base.configuration.DongleLibDependencyError

  Base class for data parser errors.
- class ttk_base.configuration.DataParserError

  Errors during parsing of a configuration file.
- class ttk_base.configuration.ParsingError
- class ttk_base.configuration.LicenseError

  Invalid license has been encountered.
- class ttk_base.configuration.DataParser

  A reduced "light" implementation of the original DataParser (which was in turn an alternate, less finnicky implementation of Python 2.2-era ConfigParser).
- class ttk_base.configuration.Configuration

  A simple class that reads/stores configuration info in ini files.

Namespaces

- namespace ttk_base
- namespace ttk_base.configuration

  Settings and configuration for TestToolkit packages.

Functions

- ttk_base.configuration.getExpectedDongleVersion ()

  Get a string representation of the expected Dongle-DLL version for this release (see EXPECTED_DONGLE_DLL↩ _VERSION)
- ttk_base.configuration.findFile (file_name, start_path=None, max_parent_levels=6)

  Find a matching file by traversing the folder tree upwards, starting at start_path.
- ttk_base.configuration.getSettings ()

  Get the current project settings (which may be empty if no configuration file is available).
- ttk_base.configuration._getTTkDebugLevel ()

  Get current TTk status output debug level.
- ttk_base.configuration.getL10nInstance (locale_id=None, available_locales=None)

  Get a L10N instance matching the supplied locale id.

Variables

- str ttk_base.configuration.DEFAULT_INI_ENCODING = "windows-1252"

  default encoding to use for configuration files (.ini)

8.7 ttk_base/__init__.py File Reference

Namespaces

- namespace ttk_base

Variables

- ttk_base.ttk_path
- str ttk_base.version = ".".join(str(v) for v in sys.version_info[:3])
- str ttk_base.arch = "x64" if sys.maxsize > 2**32 else "x86"
- str ttk_base.plat
- ttk_base.build_info_path = os.path.join(ttk_path, "ttk_build.info")
- ttk_base.encoding
- ttk_base.errors
- ttk_base.tracebacklimit

8.8 ttk_base/dspace_rtplib/__init__.py File Reference

Namespaces

- namespace ttk_base
- namespace ttk_base.dspace_rtplib

  Abstraction layer for accessing HIL variables using dSPACE RTPlib interfaces.

8.9 ttk_base/dspace_xilapi/__init__.py File Reference

Namespaces

- namespace ttk_base
- namespace ttk_base.dspace_xilapi

  Abstraction layer for accessing HIL variables using dSPACE XIL-API/Testbench interfaces.

8.10 ttk_base/isyst_ida/__init__.py File Reference

Namespaces

- namespace ttk_base
- namespace ttk_base.isyst_ida

  Abstraction layer for accessing HIL variables using iSyst iDA Testware.

8.11 ttk_base/ni_labview/__init__.py File Reference

Namespaces

- namespace ttk_base
- namespace ttk_base.ni_labview

  Abstraction layer for accessing National Instruments LabView variables/controls.

8.12 ttk_base/pyocd/__init__.py File Reference

Namespaces

- namespace ttk_base
- namespace ttk_base.pyocd

  Abstraction layer for accessing PyOCD debugger symbols/variables.

8.13 ttk_base/rst_gammav/__init__.py File Reference

Namespaces

- namespace ttk_base
- namespace ttk_base.rst_gammav

  Abstraction layer for accessing RST (RST Industrie Automation GmbH) Gamma V realtime middleware process variables.

8.14 ttk_base/siemens_simit/__init__.py File Reference

Namespaces

- namespace ttk_base
- namespace ttk_base.siemens_simit

  Abstraction layer for accessing variables on SIMIT SHM Coupling interfaces.

8.15 ttk_base/vector_canape/__init__.py File Reference

Namespaces

- namespace ttk_base
- namespace ttk_base.vector_canape

  Abstraction layer for accessing Vector CANape device variables.

8.16 ttk_bus/__init__.py File Reference

Namespaces

- namespace ttk_bus

  Test functions for bus/interface tests and abstraction layer for "bus signals".

8.17 ttk_checks/__init__.py File Reference

Namespaces

- namespace ttk_checks

  Check and test utility functions.

8.18 ttk_daq/__init__.py File Reference

Namespaces

- namespace ttk_daq

  Utilities for data acquisition/signal/measurement evaluation.

8.19 ttk_selftest/__init__.py File Reference

Namespaces

- namespace ttk_selftest

  Test functions for HIL system self test and abstraction layer for "HIL devices".

8.20 ttk_selftest/hil_devices/__init__.py File Reference

Namespaces

- namespace ttk_selftest

  Test functions for HIL system self test and abstraction layer for "HIL devices".
- namespace ttk_selftest.hil_devices

  Access to hil devices (dSPACE and GammaV).

8.21 ttk_tools/__init__.py File Reference

Namespaces

- namespace ttk_tools

  Modules for tool access.

8.22 ttk_tools/can_hal/__init__.py File Reference

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.can_hal

  Access to iSyst Hardware Abstraction Layer for CAN interfaces.

8.23 ttk_tools/dspace/__init__.py File Reference

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.dspace

  Access to dSPACE (dSPACE GmbH) tools and interfaces.

8.24 ttk_tools/isyst/__init__.py File Reference

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.isyst

  Access to iSyst Intelligente Systeme GmbH tools and interfaces.

8.25 ttk_tools/ni/__init__.py File Reference

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.ni

  Access to NI (National Instruments) tools and interfaces, primarily LabView.

8.26 ttk_tools/pyocd/__init__.py File Reference

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.pyocd

  Wrapper for using PyOCD (https://pyocd.io/) as a data source for HIL tests.

8.27 ttk_tools/rst/__init__.py File Reference

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.rst

  Access to RST (RST Industrie Automation GmbH) tools and interfaces.

8.28 ttk_tools/siemens/__init__.py File Reference

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.siemens

  Access to SIEMENS automation technology tools and interfaces.

8.29 ttk_tools/vector/__init__.py File Reference

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.vector

  Access to Vector (Vector Informatik GmbH) tools and interfaces.

8.30 ttk_base/dspace_rtplib/_variables.py File Reference

Classes

- class ttk_base.dspace_rtplib._variables.Var

  Simple wrapper for dSPACE (rtplib) "HIL" application variables.
- class ttk_base.dspace_rtplib._variables.VarContainer

  Base class for dSPACE (rtplib) HIL Var containers.

Namespaces

- namespace ttk_base
- namespace ttk_base.dspace_rtplib

  Abstraction layer for accessing HIL variables using dSPACE RTPlib interfaces.
- namespace ttk_base.dspace_rtplib._variables

  Base classes for dSPACE rtplib/rtplib2 hil variables to provide a unified interface and containers to group variables into sets.

8.31 ttk_base/dspace_xilapi/_variables.py File Reference

Classes

- class ttk_base.dspace_xilapi._variables.Var

  Simple wrapper for dSPACE (XIL-API) "HIL" application variables.
- class ttk_base.dspace_xilapi._variables.VarContainer

  Base class for dSPACE (XIL API) "HIL" variable containers.

Namespaces

- namespace ttk_base
- namespace ttk_base.dspace_xilapi

  Abstraction layer for accessing HIL variables using dSPACE XIL-API/Testbench interfaces.
- namespace ttk_base.dspace_xilapi._variables

  Base classes for dSPACE XIL-API-based hil variables to provide a unified interface and containers to group variables into sets.

8.32 ttk_base/isyst_ida/_variables.py File Reference

Classes

- class ttk_base.isyst_ida._variables.Var

  Simple wrapper for iDA Testware "HIL" application variables.
- class ttk_base.isyst_ida._variables.VarContainer

  Base class for iDA Testware "HIL" variable containers.

Namespaces

- namespace ttk_base
- namespace ttk_base.isyst_ida

  Abstraction layer for accessing HIL variables using iSyst iDA Testware.
- namespace ttk_base.isyst_ida._variables

  Base classes for iDA-Testware-based hil variables to provide a unified interface and containers to group variables into sets.

8.33 ttk_base/ni_labview/_variables.py File Reference

Classes

- class ttk_base.ni_labview._variables.Var

  Simple wrapper for NI LabView "HIL" variables/controls.
- class ttk_base.ni_labview._variables.VarContainer

  Base class for LabView "HIL" variable containers.

Namespaces

- namespace ttk_base
- namespace ttk_base.ni_labview

  Abstraction layer for accessing National Instruments LabView variables/controls.
- namespace ttk_base.ni_labview._variables

  Base classes for hil variables using an interface to NI LabView to provide a unified interface and containers to group variables into sets.

8.34 ttk_base/pyocd/_variables.py File Reference

Classes

- class ttk_base.pyocd._variables.Var

  Simple wrapper for PyOCD variables in a attached debugging session.
- class ttk_base.pyocd._variables.VarContainer

  Base class for variable containers.

Namespaces

- namespace ttk_base
- namespace ttk_base.pyocd

  Abstraction layer for accessing PyOCD debugger symbols/variables.
- namespace ttk_base.pyocd._variables

  Base classes for PyOCD-based hil variables to provide a unified interface and containers to group variables into sets.

8.35 ttk_base/rst_gammav/_variables.py File Reference

Classes

- class ttk_base.rst_gammav._variables.VarDeferred

  Simple wrapper for Gamma V process variables with deferred initialization to speed up loading of containers with many GammaVars.
- class ttk_base.rst_gammav._variables.Var

  Simple wrapper for Gamma V variables.
- class ttk_base.rst_gammav._variables.VarContainer

  Base class for Gamma V "GammaVar" containers.

Namespaces

- namespace ttk_base
- namespace ttk_base.rst_gammav

  Abstraction layer for accessing RST (RST Industrie Automation GmbH) Gamma V realtime middleware process variables.
- namespace ttk_base.rst_gammav._variables

  Base classes for RST Gamma V HIL variables to provide a unified interface and containers to group variables into sets.

8.36 ttk_base/siemens_simit/_variables.py File Reference

Classes

- class ttk_base.siemens_simit._variables.Var

  Simple wrapper for SIMIT SHM Coupling variables.
- class ttk_base.siemens_simit._variables.VarContainer

  Base class for SIMIT SHM coupling signal variable containers.

Namespaces

- namespace ttk_base
- namespace ttk_base.siemens_simit

  Abstraction layer for accessing variables on SIMIT SHM Coupling interfaces.
- namespace ttk_base.siemens_simit._variables

  Base classes for variables using SIMIT SHM Coupling signals to provide a unified interface and containers to group variables into sets.

8.37 ttk_base/vector_canape/_variables.py File Reference

Classes

- class ttk_base.vector_canape._variables.Var

  Simple wrapper for Vector CANape calibration variables.
- class ttk_base.vector_canape._variables.VarContainer

  Base class for CalVar containers.

Namespaces

- namespace ttk_base
- namespace ttk_base.vector_canape

  Abstraction layer for accessing Vector CANape device variables.
- namespace ttk_base.vector_canape._variables

  Base classes for cal variables using Vector CANape to provide a unified interface and containers to group variables into sets.

8.38 ttk_base/dspace_rtplib/variables.py File Reference

Classes

- class ttk_base.dspace_rtplib.variables.Var

  Simple wrapper for dSPACE (rtplib) "HIL" application variables.
- class ttk_base.dspace_rtplib.variables.VarContainer

  Base class for rtplib-based "HIL" variable containers.

Namespaces

- namespace ttk_base
- namespace ttk_base.dspace_rtplib

  Abstraction layer for accessing HIL variables using dSPACE RTPlib interfaces.
- namespace ttk_base.dspace_rtplib.variables

  Wrapper classes for dSPACE rtplib/rtplib2 hil variables to provide a unified interface and containers to group variables into sets.

8.39 ttk_base/dspace_xilapi/variables.py File Reference

Classes

- class ttk_base.dspace_xilapi.variables.Var

  Simple wrapper for dSPACE (XIL-API) "HIL" application variables.
- class ttk_base.dspace_xilapi.variables.VarContainer

  Base class for dSPACE (XIL API) "HIL" variable containers.

Namespaces

- namespace ttk_base
- namespace ttk_base.dspace_xilapi

  Abstraction layer for accessing HIL variables using dSPACE XIL-API/Testbench interfaces.
- namespace ttk_base.dspace_xilapi.variables

  Wrapper classes for dSPACE XIL-API / MAPort hil variables to provide a unified interface and containers to group variables into sets.

8.40 ttk_base/isyst_ida/variables.py File Reference

Classes

- class ttk_base.isyst_ida.variables.Var

  Simple wrapper for iDA Testware "HIL" application variables.
- class ttk_base.isyst_ida.variables.VarContainer

  Base class for iDA Testware "HIL" variable containers.

Namespaces

- namespace ttk_base
- namespace ttk_base.isyst_ida

  Abstraction layer for accessing HIL variables using iSyst iDA Testware.
- namespace ttk_base.isyst_ida.variables

  Wrapper classes for iDA-Testware-based HIL variables to provide a unified interface and containers to group variables into sets.

8.41 ttk_base/ni_labview/variables.py File Reference

Classes

- class ttk_base.ni_labview.variables.Var

  Simple wrapper for NI LabView "HIL" variables/controls.
- class ttk_base.ni_labview.variables.VarContainer

  Base class for LabView "HIL" variable containers.

Namespaces

- namespace ttk_base
- namespace ttk_base.ni_labview

  Abstraction layer for accessing National Instruments LabView variables/controls.
- namespace ttk_base.ni_labview.variables

  Wrapper classes for hil variables using an interface to NI LabView to provide a unified interface and containers to group variables into sets.

8.42 ttk_base/pyocd/variables.py File Reference

Classes

- class ttk_base.pyocd.variables.Var

  Simple wrapper for PyOCD variables in a attached debugging session.
- class ttk_base.pyocd.variables.VarContainer

  Base class for variable containers.

Namespaces

- namespace ttk_base
- namespace ttk_base.pyocd

  Abstraction layer for accessing PyOCD debugger symbols/variables.
- namespace ttk_base.pyocd.variables
- namespace ttk_base.pyocd._variables

  Base classes for PyOCD-based hil variables to provide a unified interface and containers to group variables into sets.

8.43 ttk_base/rst_gammav/variables.py File Reference

Classes

- class ttk_base.rst_gammav.variables.Var

  Simple wrapper for Gamma V variables.
- class ttk_base.rst_gammav.variables.VarDeferred

  Simple wrapper for Gamma V variables with deferred initialization to speed up loading of containers with many GammaVars.
- class ttk_base.rst_gammav.variables.VarContainer

  Base class for Gamma-V-based "HIL" variable containers.

Namespaces

- namespace ttk_base
- namespace ttk_base.rst_gammav

  Abstraction layer for accessing RST (RST Industrie Automation GmbH) Gamma V realtime middleware process variables.
- namespace ttk_base.rst_gammav.variables

  Wrapper classes for RST Gamma V hil variables to provide a unified interface and containers to group variables into sets.

8.44 ttk_base/siemens_simit/variables.py File Reference

Classes

- class ttk_base.siemens_simit.variables.Var

  Simple wrapper for SIMIT SHM Coupling variables.
- class ttk_base.siemens_simit.variables.VarContainer

  Base class for SIMIT SHM coupling signal variable containers.

Namespaces

- namespace ttk_base
- namespace ttk_base.siemens_simit

  Abstraction layer for accessing variables on SIMIT SHM Coupling interfaces.
- namespace ttk_base.siemens_simit.variables
- namespace ttk_base.isyst_ida.variables

  Wrapper classes for iDA-Testware-based HIL variables to provide a unified interface and containers to group variables into sets.

8.45 ttk_base/vector_canape/variables.py File Reference

Classes

- class ttk_base.vector_canape.variables.Var

  Simple wrapper for Vector CANape calibration variables.
- class ttk_base.vector_canape.variables.VarContainer

  Base class for CalVar containers.

Namespaces

- namespace ttk_base
- namespace ttk_base.vector_canape

  Abstraction layer for accessing Vector CANape device variables.
- namespace ttk_base.vector_canape.variables

  Wrapper classes for Vector CANape cal variables to provide a unified interface and containers to group variables into sets.

8.46 ttk_base/errors.py File Reference

Classes

- class ttk_base.errors.TTkErrorString

  A wrapper for error message strings so they can be distinguished from "normal" string values (like contents of ASCII-type variables in the case of calibration devices)
- class ttk_base.errors.TTkException

Namespaces

- namespace ttk_base
- namespace ttk_base.errors

  Base classes and definitions for TestToolkit exceptions/errors.

8.47 ttk_base/localization.py File Reference

Classes

- class ttk_base.localization.L10N

  Base class for localization handling.

Namespaces

- namespace ttk_base
- namespace ttk_base.localization

  Base class for localization.

Functions

- ttk_base.localization.showL10nDifferences (loc_a, loc_b)

  Show/print differences between localization keys in two L10N-derived instances.

8.48 ttk_base/values_base.py File Reference

Namespaces

- namespace ttk_base
- namespace ttk_base.values_base

  Interface Wrapper for values with additional meta information in ttk_base._values_base.

Functions

- ttk_base.values_base.meta (value, unit="", alias="", descr="", lookup=None, fmt=None)

  Factory function to create "meta" instances of the supplied value that have some additional meta attributes bolted on.

8.49 ttk_base/variables_base.py File Reference

Classes

- class ttk_base.variables_base.DummyVar

  A minimal dummy test variable not connected to any external interface.
- class ttk_base.variables_base.Snapshot

  Store and restore a "snapshot" of multiple variables' values.

Namespaces

- namespace ttk_base
- namespace ttk_base.variables_base

  Interface Wrapper for variables' base classes in ttk_base._variables_base.

8.50 ttk_bus/_bus_signals_base.py File Reference

Classes

- class ttk_bus._bus_signals_base.BusSignal

  Base signal container for a bus signal with a HIL-side model variable and relevant associated variables/values.
- class ttk_bus._bus_signals_base.TxBusSignal

  A signal sent from the DUT to the test system (ECU Tx => HIL Rx).
- class ttk_bus._bus_signals_base.RxBusSignal

  A RX bus signal (HIL => ECU) that models a CRC value with additional error injections.
- class ttk_bus._bus_signals_base.CrcBusSignal
- class ttk_bus._bus_signals_base.AcBusSignal

  A RX bus signal (HIL => ECU) that models an alive counter value with additional error injections.
- class ttk_bus._bus_signals_base.BusSignalContainer

  Base class for BusSignal containers.

Namespaces

- namespace ttk_bus

  Test functions for bus/interface tests and abstraction layer for "bus signals".
- namespace ttk_bus._bus_signals_base

  Container classes for bus/interface signal descriptions.

8.51 ttk_bus/_bus_signals_cangamma.py File Reference

Classes

- class ttk_bus._bus_signals_cangamma._ExtraVarsMixin

  Mixin to add a utility method to initialize "extra" helper Gamma-HilVars.
- class ttk_bus._bus_signals_cangamma.CanBusMessage

  CAN-Message/PDU-specific data and methods.
- class ttk_bus._bus_signals_cangamma.TxCanBusMessage

  CAN Message/PDU ECU-Tx (ECU-Tx => HIL-Rx).
- class ttk_bus._bus_signals_cangamma.RxCanBusMessage

  CAN Message/PDU ECU-Rx (HIL-Tx => ECU-Rx).
- class ttk_bus._bus_signals_cangamma.CanBusSignal

  Base wrapper for a bus signal with HIL-side BusSignals variable and relevant associated values.
- class ttk_bus._bus_signals_cangamma.TxCanBusSignalBasic

  A basic CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).
- class ttk_bus._bus_signals_cangamma.TxCanBusSignal

  A CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).
- class ttk_bus._bus_signals_cangamma.RxCanBusSignalBasic

  A basic CAN signal sent from the test system to the DUT (ECU Rx <= HIL Tx).
- class ttk_bus._bus_signals_cangamma.RxCanBusSignal

  A CAN signal sent from the test system to the DUT (ECU Rx <= HIL Tx).
- class ttk_bus._bus_signals_cangamma.CrcRxCanBusSignal

  A CRC RX bus signal (HIL-Tx => ECU-Rx)
- class ttk_bus._bus_signals_cangamma.CrcTxCanBusSignal

  A CRC TX bus signal (ECU-Tx => HIL-Rx), adds CRC validity info.
- class ttk_bus._bus_signals_cangamma.AcRxCanBusSignal

  Alive counter RX bus signal (HIL-Tx => ECU-Rx).
- class ttk_bus._bus_signals_cangamma.CanBusSignalContainer

  A BusSignalContainer for CAN Bus Signals.

Namespaces

- namespace ttk_bus

  Test functions for bus/interface tests and abstraction layer for "bus signals".
- namespace ttk_bus._bus_signals_cangamma

  Container classes for bus/interface signal descriptions.

Variables

- int ttk_bus._bus_signals_cangamma.FALLBACK_CHANGE_DELAY_MS = 100

  Fallback value used for change_delay in event-triggered RxCanBusSignals w/o a specified change_delay, to use as a somewhat sensible default delay to permit signal propagation.
- int ttk_bus._bus_signals_cangamma.FALLBACK_ERROR_DELAY_MS = 200

  Fallback: Somewhat sensible default error delay used in AcCanBusSignal and CrcCanBusSignal for signals/PDUs without any specified cycle_time / debounce_time)

8.52 ttk_bus/_bus_signals_canida.py File Reference

Classes

- class ttk_bus._bus_signals_canida._ExtraVarsMixin

  Mixin to add a utility method to initialize "extra" helper Gamma-HilVars.
- class ttk_bus._bus_signals_canida.CanBusMessage

  CAN-Message/PDU-specific data and methods.
- class ttk_bus._bus_signals_canida.TxCanBusMessage

  CAN Message/PDU ECU-Tx (ECU-Tx => HIL-Rx).
- class ttk_bus._bus_signals_canida.RxCanBusMessage

  CAN Message/PDU ECU-Rx (HIL-Tx => ECU-Rx).
- class ttk_bus._bus_signals_canida.CanBusSignal

  Base wrapper for a bus signal with HIL-side BusSignals variable and relevant associated values.
- class ttk_bus._bus_signals_canida.TxCanBusSignal

  A basic CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).
- class ttk_bus._bus_signals_canida.RxCanBusSignal

  A basic CAN signal sent from the test system to the DUT (ECU Rx <= HIL Tx).
- class ttk_bus._bus_signals_canida.CanBusSignalContainer

  A BusSignalContainer for CAN Bus Signals.

Namespaces

- namespace ttk_bus

  Test functions for bus/interface tests and abstraction layer for "bus signals".
- namespace ttk_bus._bus_signals_canida

  Container classes for bus/interface signal descriptions.

Variables

- int ttk_bus._bus_signals_canida.FALLBACK_CHANGE_DELAY_MS = 100

  Fallback value used for change_delay in event-triggered RxCanBusSignals (and RxBusMessages) w/o a specified change_delay (that is, both cycle time and change delay are 0), to use as a somewhat sensible default delay to permit signal propagation.
- int ttk_bus._bus_signals_canida.FALLBACK_ERROR_DELAY_MS = 200

  Fallback: Somewhat sensible default error delay used in AcCanBusSignal and CrcCanBusSignal for signals/PDUs without any specified cycle_time / debounce_time)

8.53 ttk_bus/_bus_signals_canmmbs.py File Reference

Classes

- class ttk_bus._bus_signals_canmmbs._ExtraVarsMixin

  Mixin to add a utility method to initialize "extra" helper CANMM-HilVars.
- class ttk_bus._bus_signals_canmmbs.CanBusMessage

  CAN-Message/PDU-specific data and methods.
- class ttk_bus._bus_signals_canmmbs.TxCanBusMessage

  CAN Message/PDU ECU-Tx (ECU-Tx => HIL-Rx).
- class ttk_bus._bus_signals_canmmbs.RxCanBusMessage

  CAN Message/PDU ECU-Rx (HIL-Tx => ECU-Rx).
- class ttk_bus._bus_signals_canmmbs.CanBusSignal

  Base wrapper for a bus signal with HIL-side BusSignals variable and relevant associated values.
- class ttk_bus._bus_signals_canmmbs.TxCanBusSignal

  A CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).
- class ttk_bus._bus_signals_canmmbs.RxCanBusSignal
- class ttk_bus._bus_signals_canmmbs.CrcCanBusSignal

  A CRC RX bus signal derived from RxCanBusSignal (HIL-Tx => ECU-Rx) for use with dSPACE CAN Multi-Message blockset.
- class ttk_bus._bus_signals_canmmbs.AcCanBusSignal

  Alive counter bus signal derived from RxCanBusSignal (HIL-Tx => ECU-Rx) for use with dSPACE CAN Multi-↩ Message blockset data structures.
- class ttk_bus._bus_signals_canmmbs.CanBusSignalContainer

  A BusSignalContainer for CAN Bus Signals.

Namespaces

- namespace ttk_bus

  Test functions for bus/interface tests and abstraction layer for "bus signals".
- namespace ttk_bus._bus_signals_canmmbs

  Container classes for bus/interface signal descriptions.

Variables

- int ttk_bus._bus_signals_canmmbs.FALLBACK_CHANGE_DELAY_MS = 100

  Fallback value used for change_delay in event-triggered RxCanBusSignals w/o a specified change_delay, to use as a somewhat sensible default delay to permit signal propagation.
- int ttk_bus._bus_signals_canmmbs.FALLBACK_ERROR_DELAY_MS = 200

  Fallback: Somewhat sensible default error delay used in AcCanBusSignal and CrcCanBusSignal for signals/PDUs without any specified cycle_time / debounce_time)

8.54 ttk_bus/_bus_signals_ethbs.py File Reference

Classes

- class ttk_bus._bus_signals_ethbs.EthEvent

  Ethernet Event (or Field) specific data and methods.
- class ttk_bus._bus_signals_ethbs.RxEthEvent

  Ethernet-Event-specific data and methods - Rx (HIL-Tx => ECU-Rx).
- class ttk_bus._bus_signals_ethbs.TxEthEvent

  Ethernet-Event-specific data and methods - Tx (ECU-Tx => HIL-Rx).
- class ttk_bus._bus_signals_ethbs.EthSignal

  Base wrapper for a bus "signal" with HIL-side Ethernet Signals variable and relevant associated values.
- class ttk_bus._bus_signals_ethbs.RxEthSignal

  An Ethernet event signal/parameter/sub-parameter sent from test system to DUT (ECU CSI Rx <= HIL PSI Tx).
- class ttk_bus._bus_signals_ethbs.TxEthSignal

  An Ethernet event signal/parameter/sub-parameter sent from DUT to test system (ECU PSI Tx => HIL CSI Rx).
- class ttk_bus._bus_signals_ethbs.EthBusSysSignal
- class ttk_bus._bus_signals_ethbs.RxEthBusSysSignal

  An Ethernet event signal/parameter/sub-parameter sent from test system to DUT (HIL PSI Tx => ECU CSI Rx).
- class ttk_bus._bus_signals_ethbs.TxEthBusSysSignal

  An Ethernet event signal/parameter/sub-parameter sent from DUT to test system (ECU PSI Tx => HIL CSI Rx).
- class ttk_bus._bus_signals_ethbs.CrcEthStatus
- class ttk_bus._bus_signals_ethbs.CrcEthSignal

  A CRC RX Ethernet event signal/parameter (HIL-Tx => ECU-Rx) for use with dSPACE Ethernet blockset.
- class ttk_bus._bus_signals_ethbs.AcEthSignal

  An alive counter Rx Ethernet signal/parameter (HIL PSI => ECU CSI) for use with dSPACE Ethernet blockset.
- class ttk_bus._bus_signals_ethbs.EthSignalContainer

  A container for Ethernet Signals.

Namespaces

- namespace ttk_bus

  Test functions for bus/interface tests and abstraction layer for "bus signals".
- namespace ttk_bus._bus_signals_ethbs

  Container classes for "bus"/interface signal descriptions.

8.55 ttk_bus/_bus_signals_frbs.py File Reference

Classes

- class ttk_bus._bus_signals_frbs.FrbsError

  Basic Exception for FRBS (FlexRay block set) errors.
- class ttk_bus._bus_signals_frbs.FrPdu
- class ttk_bus._bus_signals_frbs.FrTxPdu

  FlexRay ECU-Tx (ECU-Tx => HIL-Rx) PDU data.
- class ttk_bus._bus_signals_frbs.FrRxPdu

  FlexRay ECU-Rx (HIL-Tx => ECU-Rx) PDU data.
- class ttk_bus._bus_signals_frbs.FrBusSignal

Base wrapper for a bus signal with HIL-side BusSignals variable and relevant associated values.

- class ttk_bus._bus_signals_frbs.TxFrBusSignal

  A FLexRay signal sent from DUT to the test system (ECU Tx => HIL Rx).
- class ttk_bus._bus_signals_frbs.RxFrBusSignal

  A FlexRay signal sent from test system to the DUT (ECU Rx <= HIL Tx).
- class ttk_bus._bus_signals_frbs.CrcFrBusStatus

  CRC Tx-Signal status (ECU Tx => HIL Rx) and associated switch variables for dSPACE FlexRay blockset with PDU-based modeling.
- class ttk_bus._bus_signals_frbs.CrcFrBusSignal

  A CRC RX bus signal derived from RxFrBusSignal (HIL-Tx => ECU-Rx) for use with dSPACE FlexRay blockset with PDU-based structure.
- class ttk_bus._bus_signals_frbs.AcFrBusSignal

  An alive counter RX bus signal derived from RxFrBusSignal (HIL => ECU) for use with dSPACE FlexRay blockset PDU-based data structures.
- class ttk_bus._bus_signals_frbs.FrBusSignalContainer

  A BusSignalContainer for FlexRay Bus Signals.

Namespaces

- namespace ttk_bus

  Test functions for bus/interface tests and abstraction layer for "bus signals".
- namespace ttk_bus._bus_signals_frbs

  Container classes for bus/interface signal descriptions.

Variables

- int ttk_bus._bus_signals_frbs.FALLBACK_CHANGE_DELAY_MS = 100

  Fallback value used for change_delay in event-triggered RxFrBusSignals w/o a specified change_delay, to use as a somewhat sensible default delay to permit signal propagation.
- int ttk_bus._bus_signals_frbs.FALLBACK_ERROR_DELAY_MS = 200

  Fallback: Somewhat sensible default error delay used in AcFrBusSignal and CrcFrBusSignal for signals/PDUs without any specified cycle_time / debounce_time)

8.56 ttk_bus/_bus_test_utils.py File Reference

Namespaces

- namespace ttk_bus

  Test functions for bus/interface tests and abstraction layer for "bus signals".
- namespace ttk_bus._bus_test_utils

  Utility functions for bus tests.

Functions

- ttk_bus._bus_test_utils.getIntRangeValues (min_value=None, max_value=None, amount=5)

  Get a list of equally distributed (unique) values within min and max value boundaries.
- ttk_bus._bus_test_utils.getRangeValues (min_value=None, max_value=None, amount=5)

  Get a list of equally distributed values within min and max value boundaries.

8.57 ttk_bus/_bus_tests.py File Reference

Namespaces

- namespace ttk_bus

  Test functions for bus/interface tests and abstraction layer for "bus signals".
- namespace ttk_bus._bus_tests

  Test functions for bus/interface tests.

Functions

- ttk_bus._bus_tests._addToEntry (d, key, value)

  Add value to an existing key, create a new key as needed.
- ttk_bus._bus_tests.testRxSignal (bus_signal_data, set_value, set_value_descr='', check_var_list=None, max_retests=0, descr='', format=None)

  Check the behaviour of a single bus signal sent from HIL system and received by the ECU.
- ttk_bus._bus_tests.testTxSignal (set_var_list, bus_signal_data, check_value, check_value_descr="", rel_↩ pos=None, abs_pos=None, rel_neg=None, abs_neg=None, max_retests=0, descr='', format=None)

  Check the behaviour of a single bus signal sent from the ECU and received by the HIL system.
- ttk_bus._bus_tests.testSignals (set_var_list=None, check_var_list=None, min_wait_time_ms=0, descr='')

  Check the values of multiple variables/signals as a result of setting multiple other variables/signals.
- ttk_bus._bus_tests._testBasicSignalError (bus_signal_data, check_var_list_before=None, check_var↩ _list_after=None, manual_wait_cycles=None, manual_triggers_before=0, manual_triggers_between=0, manual_trigger_delay_ms=0, log_title="testBasicSignalError", signal_type="Generic", set_error_method_↩ name="set", clear_error_method_name="clear", descr='', format=None)

  Check ECU behaviour on errors at HIL-generated dynamic signals (like CRCs or Alive Counters) for messages sent from HIL system and received by the ECU.
- ttk_bus._bus_tests.testBasicCrcError (crc_signal_data, check_var_list_before=None, check_var_list_↩ after=None, manual_wait_cycles=None, manual_triggers_before=0, manual_triggers_between=0, manual↩ _trigger_delay_ms=0, descr='', format=None)

  Check ECU behaviour on CRC signal errors (for messages sent from HIL system and received by the ECU).
- ttk_bus._bus_tests.testBasicAcError (ac_signal_data, check_var_list_before=None, check_var_list_↩ after=None, manual_wait_cycles=None, manual_triggers_before=0, manual_triggers_between=0, manual↩ _trigger_delay_ms=0, descr='', format=None)

  Check ECU behaviour on alive counter signal errors (for messages sent from HIL system and received by the ECU).

8.58 ttk_bus/bus_signals_base.py File Reference

Classes

- class ttk_bus.bus_signals_base.TxBusSignal

  A signal sent from the DUT to the test system (ECU Tx => HIL Rx).
- class ttk_bus.bus_signals_base.RxBusSignal

  A signal sent from the test system to the DUT (ECU Rx <= HIL Tx).
- class ttk_bus.bus_signals_base.CrcBusSignal

  A RX bus signal (HIL => ECU) that models a CRC value with additional error injections.
- class ttk_bus.bus_signals_base.AcBusSignal

  A RX bus signal (HIL => ECU) that models an alive counter value with additional error injections.
- class ttk_bus.bus_signals_base.BusSignalContainer

  Base class for BusSignal containers.

Namespaces

- namespace ttk_bus

  Test functions for bus/interface tests and abstraction layer for "bus signals".
- namespace ttk_bus.bus_signals_base

  Interface wrapper for bus/interface signal classes in ttk_bus._bus_signals_base.

8.59 ttk_bus/bus_signals_cangamma.py File Reference

Classes

- class ttk_bus.bus_signals_cangamma.TxCanBusMessage

  CAN Message/PDU ECU-Tx (ECU-Tx => HIL-Rx).
- class ttk_bus.bus_signals_cangamma.RxCanBusMessage

  CAN Message/PDU ECU-Rx (HIL-Tx => ECU-Rx).
- class ttk_bus.bus_signals_cangamma.TxCanBusSignalBasic

  A basic CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).
- class ttk_bus.bus_signals_cangamma.TxCanBusSignal

  A CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).
- class ttk_bus.bus_signals_cangamma.RxCanBusSignalBasic

  A basic CAN signal sent from the test system to the DUT (ECU Rx <= HIL Tx).
- class ttk_bus.bus_signals_cangamma.RxCanBusSignal

  A CAN signal sent from the test system to the DUT (ECU Rx <= HIL Tx).
- class ttk_bus.bus_signals_cangamma.CrcRxCanBusSignal

  A CRC RX bus signal (HIL-Tx => ECU-Rx)
- class ttk_bus.bus_signals_cangamma.CrcTxCanBusSignal

  A CRC TX bus signal (ECU-Tx => HIL-Rx), adds CRC validity info.
- class ttk_bus.bus_signals_cangamma.AcRxCanBusSignal

  Alive counter RX bus signal (HIL-Tx => ECU-Rx).
- class ttk_bus.bus_signals_cangamma.CanBusSignalContainer

  A BusSignalContainer for CAN Bus Signals.

Namespaces

- namespace ttk_bus

  Test functions for bus/interface tests and abstraction layer for "bus signals".
- namespace ttk_bus.bus_signals_cangamma

  Interface wrapper bus/interface signal classes in ttk_bus._bus_signals_cangamma.

8.60 ttk_bus/bus_signals_canida.py File Reference

Classes

- class ttk_bus.bus_signals_canida.TxCanBusMessage

  CAN Message/PDU ECU-Tx (ECU-Tx => HIL-Rx).
- class ttk_bus.bus_signals_canida.RxCanBusMessage

  CAN Message/PDU ECU-Rx (HIL-Tx => ECU-Rx).
- class ttk_bus.bus_signals_canida.TxCanBusSignal

  A basic CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).
- class ttk_bus.bus_signals_canida.RxCanBusSignal

  A basic CAN signal sent from the test system to the DUT (ECU Rx <= HIL Tx).
- class ttk_bus.bus_signals_canida.CanBusSignalContainer

  A BusSignalContainer for CAN Bus Signals.

Namespaces

- namespace ttk_bus

  Test functions for bus/interface tests and abstraction layer for "bus signals".
- namespace ttk_bus.bus_signals_canida

  Interface wrapper bus/interface signal classes in ttk_bus._bus_signals_canida.

8.61 ttk_bus/bus_signals_canmmbs.py File Reference

Classes

- class ttk_bus.bus_signals_canmmbs.TxCanBusMessage

  CAN Message/PDU ECU-Tx (ECU-Tx => HIL-Rx).
- class ttk_bus.bus_signals_canmmbs.RxCanBusMessage

  CAN Message/PDU ECU-Rx (HIL-Tx => ECU-Rx).
- class ttk_bus.bus_signals_canmmbs.TxCanBusSignal

  A CAN signal sent from the DUT to the test system (ECU Tx => HIL Rx).
- class ttk_bus.bus_signals_canmmbs.RxCanBusSignal

  A CAN signal sent from the test system to the DUT (ECU Rx <= HIL Tx).
- class ttk_bus.bus_signals_canmmbs.CrcCanBusSignal

  A CRC RX bus signal derived from RxCanBusSignal (HIL-Tx => ECU-Rx) for use with dSPACE CAN Multi-Message blockset.
- class ttk_bus.bus_signals_canmmbs.AcCanBusSignal

  Alive counter bus signal derived from RxCanBusSignal (HIL-Tx => ECU-Rx) for use with dSPACE CAN Multi-Message blockset data structures.
- class ttk_bus.bus_signals_canmmbs.CanBusSignalContainer

  A BusSignalContainer for CAN Bus Signals.

Namespaces

- namespace ttk_bus

  Test functions for bus/interface tests and abstraction layer for "bus signals".
- namespace ttk_bus.bus_signals_canmmbs

  Interface wrapper bus/interface signal classes in ttk_bus._bus_signals_canmmbs.

8.62 ttk_bus/bus_signals_ethbs.py File Reference

Classes

- class ttk_bus.bus_signals_ethbs.RxEthEvent

  Ethernet-Event-specific data and methods - Rx (HIL-Tx => ECU-Rx).
- class ttk_bus.bus_signals_ethbs.TxEthEvent

  Ethernet-Event-specific data and methods - Tx (ECU-Tx => HIL-Rx).
- class ttk_bus.bus_signals_ethbs.RxEthSignal

  An Ethernet event signal/parameter/sub-parameter sent from test system to DUT (ECU CSI Rx <= HIL PSI Tx).
- class ttk_bus.bus_signals_ethbs.TxEthSignal

  An Ethernet event signal/parameter/sub-parameter sent from DUT to test system (ECU PSI Tx => HIL CSI Rx).
- class ttk_bus.bus_signals_ethbs.RxEthBusSysSignal

An Ethernet event signal/parameter/sub-parameter sent from test system to DUT (HIL PSI Tx => ECU CSI Rx).

- class ttk_bus.bus_signals_ethbs.TxEthBusSysSignal

  An Ethernet event signal/parameter/sub-parameter sent from DUT to test system (ECU PSI Tx => HIL CSI Rx).
- class ttk_bus.bus_signals_ethbs.CrcEthStatus

  CRC Tx-Event/Signal status (ECU PSI Tx => HIL CSI Rx) and associated switch variables for dSPACE Ethernet blockset.
- class ttk_bus.bus_signals_ethbs.CrcEthSignal

  A CRC RX Ethernet event signal/parameter (HIL-Tx => ECU-Rx) for use with dSPACE Ethernet blockset.
- class ttk_bus.bus_signals_ethbs.AcEthSignal

  An alive counter Rx Ethernet signal/parameter (HIL PSI => ECU CSI) for use with dSPACE Ethernet blockset.
- class ttk_bus.bus_signals_ethbs.EthSignalContainer

  A container for Ethernet Signals.

Namespaces

- namespace ttk_bus

  Test functions for bus/interface tests and abstraction layer for "bus signals".
- namespace ttk_bus.bus_signals_ethbs

  Interface wrapper for "bus"/interface signal classes in ttk_bus._bus_signals_ethbs.

8.63 ttk_bus/bus_signals_frbs.py File Reference

Classes

- class ttk_bus.bus_signals_frbs.FrTxPdu

  FlexRay ECU-Tx (ECU-Tx => HIL-Rx) PDU data.
- class ttk_bus.bus_signals_frbs.FrRxPdu

  FlexRay ECU-Rx (HIL-Tx => ECU-Rx) PDU data.
- class ttk_bus.bus_signals_frbs.TxFrBusSignal

  A FLexRay signal sent from DUT to the test system (ECU Tx => HIL Rx).
- class ttk_bus.bus_signals_frbs.RxFrBusSignal

  A FlexRay signal sent from test system to the DUT (ECU Rx <= HIL Tx).
- class ttk_bus.bus_signals_frbs.CrcFrBusStatus

  CRC Tx-Signal status (ECU Tx => HIL Rx) and associated switch variables for dSPACE FlexRay blockset with PDU-based modeling.
- class ttk_bus.bus_signals_frbs.CrcFrBusSignal

  A CRC RX bus signal derived from RxFrBusSignal (HIL-Tx => ECU-Rx) for use with dSPACE FlexRay blockset with PDU-based structure.
- class ttk_bus.bus_signals_frbs.AcFrBusSignal

  An alive counter RX bus signal derived from RxFrBusSignal (HIL => ECU) for use with dSPACE FlexRay blockset PDU-based data structures.
- class ttk_bus.bus_signals_frbs.FrBusSignalContainer

  A BusSignalContainer for FlexRay Bus Signals.

Namespaces

- namespace ttk_bus

  Test functions for bus/interface tests and abstraction layer for "bus signals".
- namespace ttk_bus.bus_signals_frbs

  Interface wrapper for bus/interface signal classes in ttk_bus._bus_signals_frbs.

8.64 ttk_bus/bus_test_utils.py File Reference

Namespaces

- namespace ttk_bus

  Test functions for bus/interface tests and abstraction layer for "bus signals".
- namespace ttk_bus.bus_test_utils

  Interface wrapper for bus test utility functions in ttk_bus._bus_test_utils.

Functions

- ttk_bus.bus_test_utils.getIntRangeValues (min_value=None, max_value=None, amount=5)

  Get a list of equally distributed (unique) values within min and max value boundaries.
- ttk_bus.bus_test_utils.getRangeValues (min_value=None, max_value=None, amount=5)

  Get a list of equally distributed values within min and max value boundaries.

8.65 ttk_bus/bus_tests.py File Reference

Namespaces

- namespace ttk_bus

  Test functions for bus/interface tests and abstraction layer for "bus signals".
- namespace ttk_bus.bus_tests

  Interface wrapper for bus/interface test functions (HIL <–bus-signal–> DUT) in ttk_bus._bus_tests.

Functions

- ttk_bus.bus_tests.testRxSignal (bus_signal_data, set_value, set_value_descr='', check_var_list=None, max_retests=0, descr='', format=None)

  Check the behaviour of a single bus signal sent from HIL system and received by the ECU.
- ttk_bus.bus_tests.testTxSignal (set_var_list, bus_signal_data, check_value, check_value_descr="", rel_↩ pos=None, abs_pos=None, rel_neg=None, abs_neg=None, max_retests=0, descr='', format=None)

  Check the behaviour of a single bus signal sent from the ECU and received by the HIL system.
- ttk_bus.bus_tests.testSignals (set_var_list=None, check_var_list=None, min_wait_time_ms=0, descr='')

  Check the values of multiple variables/signals as a result of setting multiple other variables/signals.
- ttk_bus.bus_tests.testBasicCrcError (crc_signal_data, check_var_list_before=None, check_var_list_↩ after=None, manual_wait_cycles=None, manual_triggers_before=0, manual_triggers_between=0, manual↩ _trigger_delay_ms=0, descr='', format=None)

  Check ECU behaviour on CRC signal errors (for messages sent from HIL system and received by the ECU).
- ttk_bus.bus_tests.testBasicAcError (ac_signal_data, check_var_list_before=None, check_var_list_↩ after=None, manual_wait_cycles=None, manual_triggers_before=0, manual_triggers_between=0, manual↩ _trigger_delay_ms=0, descr='', format=None)

  Check ECU behaviour on alive counter signal errors (for messages sent from HIL system and received by the ECU).

8.66 ttk_bus/bus_tests_loc.py File Reference

Classes

- class ttk_bus.bus_tests_loc.DE

  German localization.
- class ttk_bus.bus_tests_loc.EN

  English localization.

Namespaces

- namespace ttk_bus

  Test functions for bus/interface tests and abstraction layer for "bus signals".
- namespace ttk_bus.bus_tests_loc

  Localization look-up for bus test descriptions.

Functions

- ttk_bus.bus_tests_loc.getInstance (locale_id=None)

  Get a L10N instance matching the supplied locale id.

Variables

- dict ttk_bus.bus_tests_loc._available_locales
- ttk_bus.bus_tests_loc._loc = getInstance("fnord_debug")

8.67 ttk_checks/_adv_tests.py File Reference

Namespaces

- namespace ttk_checks

  Check and test utility functions.
- namespace ttk_checks._adv_tests

  Advanced test/evaluation functions for test automation that directly generate test report entries.

Functions

- ttk_checks._adv_tests._callFunction (var_list_entry)

  Call a function from the supplied entry.
- ttk_checks._adv_tests._getFuncEntryInfo (var_list_entry)

  Get a textual representation of the function in var_list_entry.
- ttk_checks._adv_tests._getToleranceFormat (abs_pos=0, rel_pos=0, abs_neg=None, rel_neg=None)

  Build a reasonably concise tolerance info format.
- ttk_checks._adv_tests._getFormattedValues (original_v, actual_v, lookup_src_var=None, fmt_src_var=None)

  Get a formatted string describing the supplied values.
- ttk_checks._adv_tests._needsTriggering (signal)

  Check whether a signal needs to be manually "triggered" for values to be passed on (if not, it should update automatically).
- ttk_checks._adv_tests._trigger (var)

  Trigger transmission of the given BusSignal/Var.
- ttk_checks._adv_tests._setVariables (set_var_list)

  Set preconditions: variables/signals.
- ttk_checks._adv_tests._checkVariables (check_var_list, show_int_note=True)

  Check a list of target values (hil/cal variables, signals).
- ttk_checks._adv_tests.testVars (set_var_list=None, check_var_list=None, min_wait_time_ms=100, descr='')

  Check the values of multiple variables/signals as a result of setting multiple other variables/signals.

8.68 ttk_checks/_basic_checks.py File Reference

Namespaces

- namespace ttk_checks

  Check and test utility functions.
- namespace ttk_checks._basic_checks

  Basic "low-level" check functions for use in test automation.

Functions

- ttk_checks._basic_checks._checkNumeric (label, value)

  Check if value is one of the numeric types (i.e.
- ttk_checks._basic_checks.checkBit (value, offset, status=1)

  Check if the bit given by the value and offset is set or cleared depending on the status.
- ttk_checks._basic_checks.checkBitMask (value, mask, operator="AND", bit_length=CHECKBITMASK_DEFAULT_BIT_LENGTH

  Check if the result of the operation given in `operator` is 'greater 0' or 'equal 0'.
- ttk_checks._basic_checks.checkBitPattern (value, pattern)

  Check that a numeric value matches a certain pattern.
- ttk_checks._basic_checks.checkRange (value, min_value, max_value)

  Check if value is within defined range (min_value <= value <= max_value).
- ttk_checks._basic_checks.checkTolerance (current_value, rated_value, rel_pos=0, abs_pos=0, rel_↩ neg=None, abs_neg=None)

  Check if the current value is within a specified range of the rated value.
- ttk_checks._basic_checks.contains (defined_values, current_value, mode="all")

  Check if current_value is an element or sub-set of defined_values.
- ttk_checks._basic_checks.compare (left_value, operator, right_value, abs_tol=0)

  Verify the conditional relationship between two operands based on the given operator.

Variables

- int ttk_checks._basic_checks.CHECKBITMASK_MAX_BIT_LENGTH = 128
- int ttk_checks._basic_checks.CHECKBITMASK_DEFAULT_BIT_LENGTH = 16
- tuple ttk_checks._basic_checks.CHECKBITMASK_OPERATORLIST_AND = ('AND', '&')
- tuple ttk_checks._basic_checks.CHECKBITMASK_OPERATORLIST_OR = ('OR', '|')
- tuple ttk_checks._basic_checks.CHECKBITMASK_OPERATORLIST_XOR = ('XOR', '^')
- tuple ttk_checks._basic_checks.CHECKBITMASK_OPERATORLIST_NAND = ('NAND', '!&')
- tuple ttk_checks._basic_checks.CHECKBITMASK_OPERATORLIST_NOR = ('NOR', '!|')
- tuple ttk_checks._basic_checks.CHECKBITMASK_OPERATORLIST_NXOR = ('NXOR', '!^', 'XNOR')
- tuple ttk_checks._basic_checks.CONTAIN_MODES = ('all', 'any', 'none', 'not')
- tuple ttk_checks._basic_checks.COMPARE_OPERATORS = ("=", "==", "<", ">", "<=", ">=", "!=", "<>")
- int ttk_checks._basic_checks.verbosity = 1

  level of debug/log message verbosity (0: mostly silent, >0: more chatty)

8.69 ttk_checks/_basic_tests.py File Reference

Namespaces

- namespace ttk_checks

  Check and test utility functions.
- namespace ttk_checks._basic_tests

  Basic test/evaluation functions for test automation that directly generate test report entries.

Functions

- ttk_checks._basic_tests._getErrorDescr (info, ex_info, *params)

  Get a formatted error description with added parameter information (for internal use in exception handlers).
- ttk_checks._basic_tests.checkBitMask (value, mask, operator="AND", bit_length=basic_checks.↩ CHECKBITMASK_DEFAULT_BIT_LENGTH, pass_on_true=True, descr="", format=None)

  Check result of the operation given in `operator`.
- ttk_checks._basic_tests.checkBitPattern (value, pattern, invert=False, descr="", format=None)

  Check that a numeric value matches a certain pattern.
- ttk_checks._basic_tests.checkBit (value, offset, status=1, descr="", format=None)

  Check if the bit given by the value and offset is set or cleared depending on the status.
- ttk_checks._basic_tests.checkTolerance (current_value, rated_value, rel_pos=0, abs_pos=0, rel_neg=None, abs_neg=None, invert=False, descr="", format=None)

  Check if the current value is within a specified range of the rated value.
- ttk_checks._basic_tests.checkAbsTol (current_value, rated_value, abs_tol=0, invert=False, descr="")

  Check if the current value is within a specified range of the rated value.
- ttk_checks._basic_tests.checkRelTol (current_value, rated_value, rel_tol=0.0, invert=False, descr="")

  Check if the current value is within a specified range of the rated value.
- ttk_checks._basic_tests.checkStatus (current_status, nominal_status, equal=True, descr="", format=None)

  Check if the current status value is identical to the nominal status value.
- ttk_checks._basic_tests.checkRange (value, min_value, max_value, invert=False, descr="", format=None)

  Check if value is in defined range (`min_value <= value <= max_value`).
- ttk_checks._basic_tests.compare (left_value, operator, right_value, abs_tol=0, descr="", format=None)

  Verify the conditional relationship between two operands based on the given operator.
- ttk_checks._basic_tests.compareCE (current_value, operator, expected_value, abs_tol=0, descr="")

  Verify the conditional relationship between current and expected value based on the given operator.
- ttk_checks._basic_tests.contains (defined_values, current_value, mode="all", descr="", format=None)

  Check if current_value is an element (or sub set) of defined_values.

8.70 ttk_checks/_checkutils.py File Reference

Classes

- class ttk_checks._checkutils.DefaultDict

  A dictionary that returns empty strings instead of raising KeyErrors.

Namespaces

- namespace ttk_checks

  Check and test utility functions.
- namespace ttk_checks._checkutils

  Utilities for check functions.

Functions

- ttk_checks._checkutils.buildDescription (descr_format, entries)

  Build a result description from a supplied format string and entries.
- ttk_checks._checkutils.getInteger (value)

  Cast a numeric value to an integer (int/long) number, but perform sanity checks before (relevant for float input values).
- ttk_checks._checkutils.resolveValue (value)

  Get a value of the input parameter if it supports a "get" method, otherwise return it as-is.
- ttk_checks._checkutils._getValueRepresentation (value, fallback_value=None, fallback_fmt_source=None)

  Get a string representation of the supplied value (preferably formatted if a format hint has been specified).
- ttk_checks._checkutils._getStateDescr (value, fallback_lookup_source=None)

  Get a state description of the supplied value if a value lookup is available.
- ttk_checks._checkutils.updateMetaInfo (meta, value, key_prefix="", meta_attrs=None, setdefaults=False, fallback_value=None)

  Extract meta info from `value` (if info is available) and update the supplied meta dictionary with those values.
- ttk_checks._checkutils.updateMetaStateDescr (meta, source_var, value, key)

  Update (or create) a meta entry with a state description for the supplied value, using the lookup-table from `source_var`.
- ttk_checks._checkutils.updateMetaFormattedValue (meta, source_var, value, key="fmt", fallback_value=None)

  Update (or create) a meta entry with a formatted value representation for the supplied value, using the formatting defined in `source_var`.
- ttk_checks._checkutils.updateMetaEntries (meta, *source_vars_info)

  All-in-one update of meta data.
- ttk_checks._checkutils.bin (value, prefix='0b', min_bit_width=0)

  Create a binary representation of the input value.
- ttk_checks._checkutils.getTracebackString (limit=None)

  Get current traceback info as a string (or an empty string if no exception is currently active).

Variables

- ttk_checks._checkutils._ = basic_tests_loc.getInstance().getText
- float ttk_checks._checkutils.FLOAT_CONVERSION_LIMIT = 0.0001

  maximum divergence to still accept a float value to be "intended" as int/long
- int ttk_checks._checkutils.FLOAT_CONVERSION_MAX_VALUE = 2**52

  IEEE-754 double-precision => 52 bits explicitly stored.

8.71 ttk_checks/_verdicts.py File Reference

Namespaces

- namespace ttk_checks

  Check and test utility functions.
- namespace ttk_checks._verdicts

  Defined test status verdicts.

Functions

- ttk_checks._verdicts.getResultingVerdict (verdicts)

  Get the resulting verdict for a sequence of verdicts.
- ttk_checks._verdicts.getInvertedVerdict (verdict)

  Invert verdict.

Variables

- str ttk_checks._verdicts.PASSED = "PASSED"

  Verdict: Test step successfully completed.
- str ttk_checks._verdicts.FAILED = "FAILED"

  Verdict: Test step failed.
- str ttk_checks._verdicts.ERROR = "ERROR"

  Verdict: Error occurred during test execution (in environment, tool, script)
- str ttk_checks._verdicts.RETEST_PASSED = "RETEST PASSED"

  Verdict: A manual re-test of a previously failed test step passed / was successful.
- list ttk_checks._verdicts.PASS_VERDICTS = [RETEST_PASSED, PASSED]

  All verdicts that evaluate to "Passed".
- list ttk_checks._verdicts.FAIL_VERDICTS = [FAILED, ERROR]

  All verdicts that evaluate to "Failed".

8.72 ttk_checks/adv_tests.py File Reference

Namespaces

- namespace ttk_checks

  Check and test utility functions.
- namespace ttk_checks.adv_tests

  Interface wrapper for advanced test/evaluation functions in ttk_checks._adv_tests.

Functions

- ttk_checks.adv_tests.testVars (set_var_list=None, check_var_list=None, min_wait_time_ms=100, descr='')

  Check the values of multiple variables/signals as a result of setting multiple other variables/signals.

8.73 ttk_checks/basic_checks.py File Reference

Namespaces

- namespace ttk_checks

  Check and test utility functions.
- namespace ttk_checks.basic_checks

  Interface wrapper for basic check functions ("low-level") in ttk_checks._basic_checks.

Functions

- ttk_checks.basic_checks.checkBit (value, offset, status=1)

  Check if the bit given by the value and offset is set or cleared depending on the status.
- ttk_checks.basic_checks.checkBitMask (value, mask, operator="AND", bit_length=CHECKBITMASK_↩ DEFAULT_BIT_LENGTH)

  Check if the result of the operation given in 'operator' is 'greater 0' or 'equal 0'.
- ttk_checks.basic_checks.checkBitPattern (value, pattern)

  Check that a numeric value matches a certain pattern.
- ttk_checks.basic_checks.checkRange (value, min_value, max_value)

  Checks if value is in defined range (min_value <= value <= max_value).
- ttk_checks.basic_checks.checkTolerance (current_value, rated_value, rel_pos=0, abs_pos=0, rel_neg=None, abs_neg=None)

  Check if the current value is within a specified range of the rated value.
- ttk_checks.basic_checks.contains (defined_values, current_value, mode="all")

  Checks if current_value is an element or sub-set of defined_values.
- ttk_checks.basic_checks.compare (left_value, operator, right_value, abs_tol=0)

  Verify the conditional relationship between two operands based on the given operator.

8.74 ttk_checks/basic_tests.py File Reference

Namespaces

- namespace ttk_checks

  Check and test utility functions.
- namespace ttk_checks.basic_tests

  Interface wrapper for basic test/evaluation functions that directly generate test report entries in ttk_checks._basic_tests.

Functions

- ttk_checks.basic_tests.checkBitMask (value, mask, operator="AND", bit_length=CHECKBITMASK_↩ DEFAULT_BIT_LENGTH, pass_on_true=True, descr="", format=None)

  Check result of the operation given in `operator`.
- ttk_checks.basic_tests.checkBitPattern (value, pattern, invert=False, descr="", format=None)

  Check that a numeric value matches a certain pattern.
- ttk_checks.basic_tests.checkBit (value, offset, status=1, descr="", format=None)

  Check if the bit given by the value and offset is set or cleared depending on the status.
- ttk_checks.basic_tests.checkTolerance (current_value, rated_value, rel_pos=0, abs_pos=0, rel_neg=None, abs_neg=None, invert=False, descr="", format=None)

Check if the current value is within a specified range of the rated value.

- ttk_checks.basic_tests.checkAbsTol (current_value, rated_value, abs_tol=0, invert=False, descr="")

  Check if the current value is within a specified range of the rated value.
- ttk_checks.basic_tests.checkRelTol (current_value, rated_value, rel_tol=0.0, invert=False, descr="")

  Check if the current value is within a specified range of the rated value.
- ttk_checks.basic_tests.checkStatus (current_status, nominal_status, equal=True, descr="", format=None)

  Check if the current status value is identical to the nominal status value.
- ttk_checks.basic_tests.checkRange (value, min_value, max_value, invert=False, descr="", format=None)

  Check if value is in defined range (`min_value <= value <= max_value`).
- ttk_checks.basic_tests.compare (left_value, operator, right_value, abs_tol=0, descr="", format=None)

  Verify the conditional relationship between two operands based on the given operator.
- ttk_checks.basic_tests.compareCE (current_value, operator, expected_value, abs_tol=0, descr="")

  Verify the conditional relationship between current and expected value based on the given operator.
- ttk_checks.basic_tests.contains (defined_values, current_value, mode="all", descr="", format=None)

  Check if current_value is an element (or sub set) of defined_values.

8.75 ttk_checks/basic_tests_loc.py File Reference

Classes

- class ttk_checks.basic_tests_loc.DE

  German localization.
- class ttk_checks.basic_tests_loc.EN

  English localization.

Namespaces

- namespace ttk_checks

  Check and test utility functions.
- namespace ttk_checks.basic_tests_loc

  Localization look-up for basic test descriptions.

Functions

- ttk_checks.basic_tests_loc.getInstance (locale_id=None)

  Get a L10N instance matching the supplied locale id.

Variables

- dict ttk_checks.basic_tests_loc._available_locales
- tuple ttk_checks.basic_tests_loc._cached_instance = (None, None)
- ttk_checks.basic_tests_loc._l10n = getInstance("fnord_debug")

8.76 ttk_checks/checkceptions.py File Reference

Classes

- class ttk_checks.checkceptions.TtkChecksException

  Base Exception class for ttk_checks package.
- class ttk_checks.checkceptions.BasicCheckException

  Base Exception for basic checks.
- class ttk_checks.checkceptions.ValueConversionError

  A required precondition is not given.
- class ttk_checks.checkceptions.PreconditionError
- class ttk_checks.checkceptions.InputValueTypeError

  A supplied value has a wrong format/type and cannot be converted.
- class ttk_checks.checkceptions.RangeError

  A (supplied) value is outside the supported range (but has a suitable data type otherwise).
- class ttk_checks.checkceptions.PatternError

  Exception: A supplied pattern is malformed and cannot be interpreted.
- class ttk_checks.checkceptions.NotSupportedError

  Exception: Feature or input is not or not yet supported.
- class ttk_checks.checkceptions.UnexpectedException

Namespaces

- namespace ttk_checks

  Check and test utility functions.
- namespace ttk_checks.checkceptions

  Custom exception classes for package ttk_checks.

8.77 ttk_checks/checkutils.py File Reference

Classes

- class ttk_checks.checkutils.DefaultDict

  A dictionary that returns empty strings instead of raising KeyErrors.

Namespaces

- namespace ttk_checks

  Check and test utility functions.
- namespace ttk_checks.checkutils

  Interface wrapper for check functions utilities in ttk_checks._checkutils.

Functions

- ttk_checks.checkutils.buildDescription (descr_format, entries)
- ttk_checks.checkutils.getInteger (value)

  Cast a numeric value to an integer (int/long) number, but perform sanity checks before (relevant for float input values).
- ttk_checks.checkutils.resolveValue (value)

  Get a value of the input parameter if it supports a "get" function, otherwise return it as-is.
- ttk_checks.checkutils.updateMetaInfo (meta, value, key_prefix="", meta_attrs=None, setdefaults=False, fallback_value=None)

  Extract meta info from `value` (if info is available) and update the supplied meta dictionary with those values.
- ttk_checks.checkutils.updateMetaStateDescr (meta, source_var, value, key)

  Update (or create) a meta entry with a state description for the supplied value, using the lookup-table from `source_var`.
- ttk_checks.checkutils.updateMetaFormattedValue (meta, source_var, value, key="fmt", fallback_value=None)

  Update (or create) a meta entry with a formatted value representation for the supplied value, using the formatting defined in `source_var`.
- ttk_checks.checkutils.updateMetaEntries (meta, *source_vars_info)

  All-in-one update of meta data.
- ttk_checks.checkutils.bin (value, prefix='0b', min_bit_width=0)

  Create a binary representation of the input value.
- ttk_checks.checkutils.getTracebackString (limit=None)

  Get current traceback info as a string (or an empty string if no exception is currently active).

8.78 ttk_checks/verdicts.py File Reference

Namespaces

- namespace ttk_checks

  Check and test utility functions.
- namespace ttk_checks.verdicts

  Interface wrapper for defined test status verdicts in ttk_checks._verdicts.

Functions

- ttk_checks.verdicts.getResultingVerdict (verdicts)

  Get the resulting verdict for a set or sequence of verdicts.
- ttk_checks.verdicts.getInvertedVerdict (verdict)

  Invert verdict.

8.79 ttk_daq/_daq_utils.py File Reference

Namespaces

- namespace ttk_daq

  Utilities for data acquisition/signal/measurement evaluation.
- namespace ttk_daq._daq_utils

  Utility functions for recorded signal data.

Functions

- ttk_daq._daq_utils.printDataInfo (values, indent_level=1, outer_n=2)

  Print a summary of the supplied values to stdout.
- ttk_daq._daq_utils._printDataInfo (values, indent_level=1, outer_n=2)

  alias for original protected function
- ttk_daq._daq_utils.getDaqDataStruct (capture_data)

  Get a "DAQ"-like data structure from a classic dSPACE rtplib capture.Fetch() dictionary.
- ttk_daq._daq_utils.roundDaqData (daq_data, digits=6)

  Round all data points in daq_data to the specified number of digits.
- ttk_daq._daq_utils.getOuterTimeBoundaries (daq_data)

  Get minimum and maximum time stamps for all measurements in daq_data to contain all measurements, even if some might not have data for all time stamps (e.g.
- ttk_daq._daq_utils.getInnerTimeBoundaries (daq_data)

  Get minimum and maximum time stamps for all measurements in daq_data, so that all measurements have samples between those boundaries (i.e.
- ttk_daq._daq_utils._getDerivedStepTime (daq_data, limit_data_points=1000)

  Get a suitable "average" delta time between data points in daq_data.
- ttk_daq._daq_utils.shiftTimeAxes (daq_data, offset)

  Shift all time axes values in daq_data by the specified offset.
- ttk_daq._daq_utils.appendDaqData (cur_daq_data, new_daq_data, only_append_to_existing=False, time↩ _between_s=None, verbosity=0)

  Append the measurement data in new_daq_data to the data in cur_daq_data.
- ttk_daq._daq_utils.trimDaqData (daq_data, verbosity=0)

  Trim measurements in a daq data structure to the area where all signals have data points (i.e.
- ttk_daq._daq_utils.getNormalizedDaqData (daq_data)

  Get a data dictionary with normalized/shifted time axes, so the earliest data point(s) will start at time 0.0.
- ttk_daq._daq_utils.linearInterpolate (x, x0, y0, x1, y1)

  Linear interpolation at point `x` between two points `(x0, y0)` and `(x1, y1)`.
- ttk_daq._daq_utils.tableLookup (x, x_values, y_values)

  Get an interpolated y-value for a given x-value in a x/y lookup table.
- ttk_daq._daq_utils.getInterpolatedValue (signal_data, time_stamp)

  Get a value at the specified time.

8.80 ttk_daq/_eval_signal.py File Reference

Classes

- class ttk_daq._eval_signal.OperatorException

  Errors during custom find operator function calls.
- class ttk_daq._eval_signal.EvalSignal

Namespaces

- namespace ttk_daq

  Utilities for data acquisition/signal/measurement evaluation.
- namespace ttk_daq._eval_signal

  Utility class for evaluation of recorded signal data.

8.81 ttk_daq/daq_utils.py File Reference

Namespaces

- namespace ttk_daq

  Utilities for data acquisition/signal/measurement evaluation.
- namespace ttk_daq.daq_utils

  Interface wrapper for recorded signal data utility functions in ttk_daq._daq_utils.

Functions

- ttk_daq.daq_utils.printDataInfo (values, indent_level=1, outer_n=2)

  Print a summary of the supplied values to stdout.
- ttk_daq.daq_utils.getDaqDataStruct (capture_data)

  Get a "DAQ"-like data structure from a classic dSPACE rtplib capture.Fetch() dictionary.
- ttk_daq.daq_utils.roundDaqData (daq_data, digits=6)

  Round all data points in daq_data to the specified number of digits.
- ttk_daq.daq_utils.getOuterTimeBoundaries (daq_data)

  Get minimum and maximum time stamps for all measurements in daq_data to contain all measurements, even if some might not have data for all time stamps (e.g.
- ttk_daq.daq_utils.getInnerTimeBoundaries (daq_data)

  Get minimum and maximum time stamps for all measurements in daq_data, so that all measurements have samples between those boundaries (i.e.
- ttk_daq.daq_utils.shiftTimeAxes (daq_data, offset)

  Shift all time axes values in daq_data by the specified offset.
- ttk_daq.daq_utils.appendDaqData (cur_daq_data, new_daq_data, only_append_to_existing=False, time_↩ between_s=None, verbosity=0)

  Append the measurement data in new_daq_data to the data in cur_daq_data.
- ttk_daq.daq_utils.trimDaqData (daq_data, verbosity=0)

  Trim measurements in a daq data structure to the area where all signals have data points (i.e.
- ttk_daq.daq_utils.getNormalizedDaqData (daq_data)

  Get a data dictionary with normalized/shifted time axes, so the earliest data point(s) will start at time 0.0.
- ttk_daq.daq_utils.linearInterpolate (x, x0, y0, x1, y1)

  Linear interpolation at point `x` between two points `(x0, y0)` and `(x1, y1)`.
- ttk_daq.daq_utils.tableLookup (x, x_values, y_values)

  Get an interpolated y-value for a given x-value in a x/y lookup table.
- ttk_daq.daq_utils.getInterpolatedValue (signal_data, time_stamp)

  Get a value at the specified time.

8.82 ttk_daq/data_sync.py File Reference

Namespaces

- namespace ttk_daq

  Utilities for data acquisition/signal/measurement evaluation.
- namespace ttk_daq.data_sync

  Examples/utilities for daq data synchronization.

Functions

- ttk_daq.data_sync.synchronize (daq_data_a, sync_signal_a, sync_value_a, daq_data_b, sync_signal_↩ b, sync_value_b, interpolate=True, offset=0)

  Synchronize timestamps of two sets of DAQ measurements (a and b) by using a synchronization signal value change recorded in both.

8.83 ttk_daq/eval_signal.py File Reference

Classes

- class ttk_daq.eval_signal.EvalSignal

  Class for evaluation of recorded signal data.

Namespaces

- namespace ttk_daq

  Utilities for data acquisition/signal/measurement evaluation.
- namespace ttk_daq.eval_signal

  Interface wrapper for recorded signal data evaluation utility class in ttk_daq._eval_signal.

8.84 ttk_selftest/_adjust_exception.py File Reference

Classes

- class ttk_selftest._adjust_exception.AdjustError

  Base class to raise adjustment Errors.

Namespaces

- namespace ttk_selftest

  Test functions for HIL system self test and abstraction layer for "HIL devices".
- namespace ttk_selftest._adjust_exception

  Exception class definitions.

8.85 ttk_selftest/_adjustment_calculation.py File Reference

Classes

- class ttk_selftest._adjustment_calculation.AdjustmentCalculation

  Class for adjustment one hil channel which is currently in state self test and adjustment.

Namespaces

- namespace ttk_selftest

  Test functions for HIL system self test and abstraction layer for "HIL devices".
- namespace ttk_selftest._adjustment_calculation

  Adjustment functions for self test at hil systems.

Variables

- str ttk_selftest._adjustment_calculation._NOTTESTED = "NOT TESTED"
- str ttk_selftest._adjustment_calculation._PASSED = "PASSED"
- str ttk_selftest._adjustment_calculation._FAILED = "FAILED"
- str ttk_selftest._adjustment_calculation._MAX = "max"
- str ttk_selftest._adjustment_calculation._MIN = "min"
- ttk_selftest._adjustment_calculation.delta = TestVar.DummyVar(context=None, identifier="", default=1.00)
- ttk_selftest._adjustment_calculation.bravo = TestVar.DummyVar(context=None, identifier="", default=0.00)
- ttk_selftest._adjustment_calculation.a = AdjustmentCalculation(hilvar_gain=delta, hilvar_offset=bravo)
- ttk_selftest._adjustment_calculation.expected
- ttk_selftest._adjustment_calculation.measured

8.86 ttk_selftest/_basic_result_list.py File Reference

Classes

- class ttk_selftest._basic_result_list.ResultList

  Add logging, some sanity checks and helper functions to a plain list class in order to make it more convenient for storing test result entries.
- class ttk_selftest._basic_result_list.SubResult

  Context manager for Sub-Results.

Namespaces

- namespace ttk_selftest

  Test functions for HIL system self test and abstraction layer for "HIL devices".
- namespace ttk_selftest._basic_result_list
- namespace _basic_result_list

  A basic result list implementation that contains only the most commonly used features; intended for internal use in TTk test functions while avoiding external dependencies to TestSupport libs.

Functions

- ttk_selftest._basic_result_list._parseDescription (description)

  Parse a description text to detect "special" keywords.
- ttk_selftest._basic_result_list._removeDoubledBeginnings (descriptions)

  Remove first lines of description texts that are already present in the (directly) preceding description.
- ttk_selftest._basic_result_list.printDescriptionBlock (descr, verdict='', min_indent=7)

  Print a formatted description and verdict (if available)

Variables

- str ttk_selftest._basic_result_list.TESTCASE_ID_MARKER = "[[TCID]]"

  default marker for testcase-IDs (pre-wrapped with brackets)
- str ttk_selftest._basic_result_list.COMMENT_MARKER = "[[COMMENT]]"

  default marker for comments (pre-wrapped with brackets)
- str ttk_selftest._basic_result_list.DEFAULT_DIVIDER = "\n----------\n"

  default divider to use with getCombinedResult
- str ttk_selftest._basic_result_list.DEFAULT_FAILED_PREFIX = "> "

  default prefix marker for failed sub result entries
- dict ttk_selftest._basic_result_list.COMMENT_KEYWORDS = {'COMMENT', 'CMNT'}
- dict ttk_selftest._basic_result_list.TESTCASE_ID_KEYWORDS = {'TCID', 'TC-ID', 'TC'}
- dict ttk_selftest._basic_result_list.OBJECT_KEYWORDS = {'IMG', 'OBJ'}
- dict ttk_selftest._basic_result_list.DELTATIME_KEYWORDS = {"DELTATIME"}
- ttk_selftest._basic_result_list._keyword_regex = re.compile(r'\[\[([-\w]+)\]\]\s*(.*?)\s*$', re.DOTALL|re.ASCII)

8.87 ttk_selftest/_selftest_tests.py File Reference

Classes

- class ttk_selftest._selftest_tests.NegWrapper

  Class used to get values multiplied by -1.

Namespaces

- namespace ttk_selftest

  Test functions for HIL system self test and abstraction layer for "HIL devices".
- namespace ttk_selftest._selftest_tests

  Test functions for self test of hil systems.

Functions

- ttk_selftest._selftest_tests.mean (it)
- ttk_selftest._selftest_tests._setFiu (fiu_channel, OpenLoad=True, EI1=True, EI2=False)

  Function to set the FIU channel.
- ttk_selftest._selftest_tests._resetFiu (fiu_channel)

  Function to reset the whole FIU channel.
- ttk_selftest._selftest_tests._setFiuCanFr (fiu_channel, High_OpenLoad=True, High_EI1=True, High_↩ EI2=False, Low_OpenLoad=True, Low_EI1=False, Low_EI2=True)

  Function to set the FIU_CAN_FR channel.
- ttk_selftest._selftest_tests._resetFiuCanFr (fiu_channel)

  Function to reset the whole FIU_CAN_FR channel.
- ttk_selftest._selftest_tests._getCommonVoltageRange (ch1, ch2, offset_voltage=0)

  Function to get common voltage range of two analog channels.
- ttk_selftest._selftest_tests._setPreconditions (preconditions)

  Function to set the preconditions for the test of a hil device.
- ttk_selftest._selftest_tests._resetPreconditions (preconditions)

  Function to reset the preconditions for the test of a hil device.
- ttk_selftest._selftest_tests._setPowerSupply (power_supply, voltage, current_limit)

Function to set the power supply output.

- ttk_selftest._selftest_tests._resetPowerSupply (power_supply)

  Function to reset the power supply output.
- ttk_selftest._selftest_tests._checkAndSetPowerSupply (testenv, power_supply, voltage, current_limit)

  Function to set the power supply output, if the test result of the power supply is PASSED.
- ttk_selftest._selftest_tests._resetAndWait (device, stimulation_var, min_wait_time_ms, timestamp_start=None)

  Function to reset the stimulation and pause the test execution.
- ttk_selftest._selftest_tests._addValuesToSelftestDataset (device, expected_value, measured_value)

  Function to add a measurement snapshot consisting of expected value and measured value to the dataset dictionary of the device.
- ttk_selftest._selftest_tests.testGND (testenv, gnd, threshold=10000, ignore_results=False)

  Function to test ground connections.
- ttk_selftest._selftest_tests.testAnalogOutput (testenv, channel, ref_channel, num_testvalues=5, gnd_offset=0.5)

  Function to test an analog output channel.
- ttk_selftest._selftest_tests.testAnalogVoltageInput (testenv, device, num_testvalues=5)

  Function to test an analog voltage input channel.
- ttk_selftest._selftest_tests.testAnalogCurrentInput (testenv, device, num_testvalues=5)

  Function to test an analog current input channel.
- ttk_selftest._selftest_tests.testPowerSupplyVoltage (testenv, power_supply, num_testvalues=5)

  Function to test power supply voltage control and measurement.
- ttk_selftest._selftest_tests.testPowerSupplyCurrent (testenv, power_supply, max_test_current=None, num_testvalues=5)

  Function to test power supply current control and measurement.
- ttk_selftest._selftest_tests.testPowerSupply (testenv, power_supply, max_test_current=None, num_testvalues=5)

  Function to test Power Supply voltage and current.
- ttk_selftest._selftest_tests.testPwmInput (testenv, device, num_testvalues=5)

  Function to test a PWM input channel.
- ttk_selftest._selftest_tests.testLoad (testenv, load, num_testvalues=5)

  Function to test a load.
- ttk_selftest._selftest_tests.testPwmOutput (testenv, channel, num_testvalues=5)

  Function to test an PWM output channel.
- ttk_selftest._selftest_tests.testCan (testenv, channel)

  Function to test a CAN channel.
- ttk_selftest._selftest_tests.testDigitalInput (testenv, device)

  Function to test a digital input channel.
- ttk_selftest._selftest_tests.testDigitalOutput (testenv, channel)

  Function to test a digital output channel.
- ttk_selftest._selftest_tests.testiDigiRes (testenv, idigires)

  Function to test an iDigiRes.
- ttk_selftest._selftest_tests.resetFiu (hil_devices)

  Function to reset the FIU.

8.88 ttk_selftest/_xml_handling.py File Reference

Namespaces

- namespace ttk_selftest

  Test functions for HIL system self test and abstraction layer for "HIL devices".
- namespace ttk_selftest._xml_handling

  Read and Write methods for self test at hil systems.

Functions

- ttk_selftest._xml_handling.writeDefaultXml (filepath, filename, prjname="ProjectName", hilinfo={}, boardinfo={})

  Write a default xml file in specified structure.

Variables

- dict ttk_selftest._xml_handling.a = {1: "0215-02686", 2: "0215-3386", 3: "0555-02686"}
- ttk_selftest._xml_handling.current_path = os.path.realpath(inspect.currentframe().f_code.co_filename)
- ttk_selftest._xml_handling.ttk_selftest_path = os.path.dirname(current_path)
- ttk_selftest._xml_handling.lib_path = os.path.dirname(ttk_selftest_path)
- ttk_selftest._xml_handling.python_path = os.path.dirname(lib_path)
- ttk_selftest._xml_handling.project_path = os.path.dirname(python_path)
- ttk_selftest._xml_handling.file_path = os.path.join(project_path, "Model", "_Standardmodel")
- str ttk_selftest._xml_handling.file_name = "FMC4008A_Test.xml"
- ttk_selftest._xml_handling.filepath
- ttk_selftest._xml_handling.filename
- ttk_selftest._xml_handling.prjname
- ttk_selftest._xml_handling.hilinfo
- ttk_selftest._xml_handling.boardinfo

8.89 ttk_selftest/adjustment_calculation.py File Reference

Classes

- class ttk_selftest.adjustment_calculation.AdjustmentCalculation

  Class for adjustment one hil channel which is currently in state self test and adjustment.

Namespaces

- namespace ttk_selftest

  Test functions for HIL system self test and abstraction layer for "HIL devices".
- namespace ttk_selftest.adjustment_calculation

  Interface wrapper for adjustment calculation class.

8.90 ttk_selftest/hil_devices/_analog_devices.py File Reference

Classes

- class ttk_selftest.hil_devices._analog_devices.AnalogChannel

  Base analog channel container for a testable analog channel and relevant associated variables/values.
- class ttk_selftest.hil_devices._analog_devices.AnalogOutput

  Base analog output container for a testable analog output and relevant associated variables/values.
- class ttk_selftest.hil_devices._analog_devices.AnalogInput

  Base analog input container for a testable analog input and relevant associated variables/values.

Namespaces

- namespace ttk_selftest

  Test functions for HIL system self test and abstraction layer for "HIL devices".
- namespace ttk_selftest.hil_devices

  Access to hil devices (dSPACE and GammaV).
- namespace ttk_selftest.hil_devices._analog_devices

  Container classes for analog device descriptions.

8.91 ttk_selftest/hil_devices/_bus_devices.py File Reference

Classes

- class ttk_selftest.hil_devices._bus_devices.CanChannel

  Base CAN channel container for a testable CAN channel and relevant associated variables/values.

Namespaces

- namespace ttk_selftest

  Test functions for HIL system self test and abstraction layer for "HIL devices".
- namespace ttk_selftest.hil_devices

  Access to hil devices (dSPACE and GammaV).
- namespace ttk_selftest.hil_devices._bus_devices

  Container classes for bus device descriptions.

8.92 ttk_selftest/hil_devices/_components.py File Reference

Classes

- class ttk_selftest.hil_devices._components.PowerSupply

  Base power supply container for a testable power supply and relevant associated variables/values.
- class ttk_selftest.hil_devices._components.Load

  Base load container for a testable load and relevant associated variables/values.
- class ttk_selftest.hil_devices._components.iDigiRes

  Base iDigiRes container for a testable iDigiRes device and relevant associated variables/values.

Namespaces

- namespace ttk_selftest

  Test functions for HIL system self test and abstraction layer for "HIL devices".
- namespace ttk_selftest.hil_devices

  Access to hil devices (dSPACE and GammaV).
- namespace ttk_selftest.hil_devices._components

  Container classes for several hil component descriptions.

8.93 ttk_selftest/hil_devices/_devices_base.py File Reference

Classes

- class ttk_selftest.hil_devices._devices_base.HilDevice

  Base HIL device container for a testable HIL device and relevant associated variables/values.
- class ttk_selftest.hil_devices._devices_base.GND

  Base ground signal container for a testable ground signal and relevant associated variables/values.
- class ttk_selftest.hil_devices._devices_base.DeviceContainer

  Base class for HIL device containers.

Namespaces

- namespace ttk_selftest

  Test functions for HIL system self test and abstraction layer for "HIL devices".
- namespace ttk_selftest.hil_devices

  Access to hil devices (dSPACE and GammaV).
- namespace ttk_selftest.hil_devices._devices_base

  Container classes for HIL device descriptions.

Functions

- ttk_selftest.hil_devices._devices_base.isGamma (context)

  Function to detect if the system context is a GammaV environment.
- ttk_selftest.hil_devices._devices_base.isDspace (context)

  Function to detect if the system context is a dSPACE environment.
- ttk_selftest.hil_devices._devices_base.isIdaTestware (context)

  Function to detect if the system context is an iDA Testware environment.
- ttk_selftest.hil_devices._devices_base.hilVarFactory (context, identifier, ∗args, ∗∗kwargs)

  Factory for "HilVars", decides actual implementation to use based on supplied context (i.e.
- ttk_selftest.hil_devices._devices_base._buildCanMessagePath (context, base_path, msg_name, direction)

  Function to build a CAN message path, decides actual implementation to use based on supplied context and message direction based on supplied direction.
- ttk_selftest.hil_devices._devices_base._buildCanSignalPath (context, base_path, msg_name, sig_name, direction)

  Function to build a CAN signal path, decides actual implementation to use based on supplied context and signal direction based on supplied direction.
- ttk_selftest.hil_devices._devices_base._readHilVarFactory (context, identifier, variable=None, ∗args, ∗∗kwargs)

  Factory for "ReadHilVars" (HilVars that will be read only e.g.
- ttk_selftest.hil_devices._devices_base._writeHilVarFactory (context, identifier, variable=None, ∗args, ∗∗kwargs)

  Factory for "WriteHilVars" (HilVars that will be read/write e.g.
- ttk_selftest.hil_devices._devices_base._rxCanBusMessageFactory (context, base_path, msg_name, ∗args, ∗∗kwargs)

  Factory for "RxCanBusMessages", decides actual implementation to use based on supplied context.
- ttk_selftest.hil_devices._devices_base._txCanBusMessageFactory (context, base_path, msg_name, ∗args, ∗∗kwargs)

  Factory for "TxCanBusMessages", decides actual implementation to use based on supplied context.
- ttk_selftest.hil_devices._devices_base._rxCanBusSignalFactory (context, base_path, msg_name, sig_name, ∗args, ∗∗kwargs)

  Factory for "RxCanBusSignals", decides actual implementation to use based on supplied context.
- ttk_selftest.hil_devices._devices_base._txCanBusSignalFactory (context, base_path, msg_name, sig_name, ∗args, ∗∗kwargs)

  Factory for "TxCanBusSignals", decides actual implementation to use based on supplied context.

Variables

- str ttk_selftest.hil_devices._devices_base.DSPACE_SEPARATOR = "/"
- ttk_selftest.hil_devices._devices_base.XilTestbench = OfflineXilTestbench
- str ttk_selftest.hil_devices._devices_base.GAMMA_SEPARATOR = "."
- ttk_selftest.hil_devices._devices_base.gaapi = _GaApiStub
- str ttk_selftest.hil_devices._devices_base.IDA_SEPARATOR = "."
- ttk_selftest.hil_devices._devices_base.IdaTestware = OfflineIdaTestware

8.94 ttk_selftest/hil_devices/_digital_devices.py File Reference

Classes

- class ttk_selftest.hil_devices._digital_devices.DigitalChannel

  Base digital channel container for a testable digital channel and relevant associated variables/values.
- class ttk_selftest.hil_devices._digital_devices.DigitalInput

  Base digital input container for a testable digital input and relevant associated variables/values.
- class ttk_selftest.hil_devices._digital_devices.DigitalOutput

  Base digital output container for a testable digital output and relevant associated variables/values.
- class ttk_selftest.hil_devices._digital_devices.PwmChannel

  Base PWM channel container for a testable PWM channel and relevant associated variables/values.
- class ttk_selftest.hil_devices._digital_devices.PwmInput

  Base PWM input container for a testable PWM input and relevant associated variables/values.
- class ttk_selftest.hil_devices._digital_devices.PwmOutput

  Base PWM output container for a testable PWM output and relevant associated variables/values.
- class ttk_selftest.hil_devices._digital_devices.Switch

  Base switch container for a testable switch and relevant associated variables/values.
- class ttk_selftest.hil_devices._digital_devices.C2DOutput

  Base iCAN2DIG output container for a testable iCAN2DIG output and relevant associated variables/values.

Namespaces

- namespace ttk_selftest

  Test functions for HIL system self test and abstraction layer for "HIL devices".
- namespace ttk_selftest.hil_devices

  Access to hil devices (dSPACE and GammaV).
- namespace ttk_selftest.hil_devices._digital_devices

  Container classes for digital device descriptions.

Functions

- ttk_selftest.hil_devices._digital_devices._isCanRxBusSignal (identifier)

  Function to identify CAN RX bus signals (e.g.

8.95 ttk_selftest/hil_devices/_fiu_devices.py File Reference

Classes

- class ttk_selftest.hil_devices._fiu_devices.FiuChannel

  Base FIU channel container.
- class ttk_selftest.hil_devices._fiu_devices.FiuCanFrChannel

  Base FIU CAN FR channel container.

Namespaces

- namespace ttk_selftest

  Test functions for HIL system self test and abstraction layer for "HIL devices".
- namespace ttk_selftest.hil_devices

  Access to hil devices (dSPACE and GammaV).
- namespace ttk_selftest.hil_devices._fiu_devices

  Container classes for fiu device descriptions.

Functions

- ttk_selftest.hil_devices._fiu_devices._checkValueAndRange (label, value, upper_limit)

  Function to check if value is one of the integer types (i.e.

8.96 ttk_selftest/hil_devices/_selftest_devices.py File Reference

Classes

- class ttk_selftest.hil_devices._selftest_devices._SrcMode

  Base class for modes of the iSRC.
- class ttk_selftest.hil_devices._selftest_devices.SRC

  Base iSystemReferenceCard (iSRC) container.

Namespaces

- namespace ttk_selftest

  Test functions for HIL system self test and abstraction layer for "HIL devices".
- namespace ttk_selftest.hil_devices

  Access to hil devices (dSPACE and GammaV).
- namespace ttk_selftest.hil_devices._selftest_devices

  Container classes for selftest device descriptions.

8.97 ttk_selftest/hil_devices/analog_devices.py File Reference

Classes

- class ttk_selftest.hil_devices.analog_devices.AnalogChannel

  Base analog channel container for a testable analog channel and relevant associated variables/values.
- class ttk_selftest.hil_devices.analog_devices.AnalogOutput

  Base analog output container for a testable analog output and relevant associated variables/values.
- class ttk_selftest.hil_devices.analog_devices.AnalogInput

  Base analog input container for a testable analog input and relevant associated variables/values.

Namespaces

- namespace ttk_selftest

  Test functions for HIL system self test and abstraction layer for "HIL devices".
- namespace ttk_selftest.hil_devices

  Access to hil devices (dSPACE and GammaV).
- namespace ttk_selftest.hil_devices.analog_devices

  Interface wrapper for analog device descriptions.

8.98 ttk_selftest/hil_devices/bus_devices.py File Reference

Classes

- class ttk_selftest.hil_devices.bus_devices.CanChannel

  Base CAN channel container for a testable CAN channel and relevant associated variables/values.

Namespaces

- namespace ttk_selftest

  Test functions for HIL system self test and abstraction layer for "HIL devices".
- namespace ttk_selftest.hil_devices

  Access to hil devices (dSPACE and GammaV).
- namespace ttk_selftest.hil_devices.bus_devices

  Interface wrapper for bus device descriptions.

8.99 ttk_selftest/hil_devices/components.py File Reference

Classes

- class ttk_selftest.hil_devices.components.PowerSupply

  Base power supply container for a testable power supply and relevant associated variables/values.
- class ttk_selftest.hil_devices.components.Load

  Base load container for a testable load and relevant associated variables/values.
- class ttk_selftest.hil_devices.components.iDigiRes

  Base iDigiRes container for a testable iDigiRes device and relevant associated variables/values.

Namespaces

- namespace ttk_selftest

  Test functions for HIL system self test and abstraction layer for "HIL devices".
- namespace ttk_selftest.hil_devices

  Access to hil devices (dSPACE and GammaV).
- namespace ttk_selftest.hil_devices.components

  Wrapper classes for several hil component descriptions.

8.100 ttk_selftest/hil_devices/devices_base.py File Reference

Classes

- class ttk_selftest.hil_devices.devices_base.HilDevice

  Base HIL device container for a testable HIL device and relevant associated variables/values.
- class ttk_selftest.hil_devices.devices_base.GND

  Base ground signal container for a testable ground signal and relevant associated variables/values.
- class ttk_selftest.hil_devices.devices_base.DeviceContainer

  Base class for HIL device containers.

Namespaces

- namespace ttk_selftest

  Test functions for HIL system self test and abstraction layer for "HIL devices".
- namespace ttk_selftest.hil_devices

  Access to hil devices (dSPACE and GammaV).
- namespace ttk_selftest.hil_devices.devices_base
- namespace ttk_selftest.devices_base

  Interface wrapper for HIL device descriptions.

8.101 ttk_selftest/hil_devices/digital_devices.py File Reference

Classes

- class ttk_selftest.hil_devices.digital_devices.DigitalChannel

  Base digital channel container for a testable digital channel and relevant associated variables/values.
- class ttk_selftest.hil_devices.digital_devices.DigitalInput

  Base digital input container for a testable digital input and relevant associated variables/values.
- class ttk_selftest.hil_devices.digital_devices.DigitalOutput

  Base digital output container for a testable digital output and relevant associated variables/values.
- class ttk_selftest.hil_devices.digital_devices.PwmChannel

  Base PWM channel container for a testable PWM channel and relevant associated variables/values.
- class ttk_selftest.hil_devices.digital_devices.PwmInput

  Base PWM input container for a testable PWM input and relevant associated variables/values.
- class ttk_selftest.hil_devices.digital_devices.PwmOutput

  Base PWM output container for a testable PWM output and relevant associated variables/values.
- class ttk_selftest.hil_devices.digital_devices.Switch

  Base switch container for a testable switch and relevant associated variables/values.
- class ttk_selftest.hil_devices.digital_devices.C2DOutput

  Base iCAN2DIG output container for a testable iCAN2DIG output and relevant associated variables/values.

Namespaces

- namespace ttk_selftest

  Test functions for HIL system self test and abstraction layer for "HIL devices".
- namespace ttk_selftest.hil_devices

  Access to hil devices (dSPACE and GammaV).
- namespace ttk_selftest.hil_devices.digital_devices

  Interface wrapper for digital device descriptions.

8.102 ttk_selftest/hil_devices/fiu_devices.py File Reference

Classes

- class ttk_selftest.hil_devices.fiu_devices.FiuChannel

  Base FIU channel container.
- class ttk_selftest.hil_devices.fiu_devices.FiuCanFrChannel

  Base FIU CAN FR channel container.

Namespaces

- namespace ttk_selftest

  Test functions for HIL system self test and abstraction layer for "HIL devices".
- namespace ttk_selftest.hil_devices

  Access to hil devices (dSPACE and GammaV).
- namespace ttk_selftest.hil_devices.fiu_devices

  Interface wrapper for fiu device descriptions.

8.103 ttk_selftest/hil_devices/selftest_devices.py File Reference

Classes

- class ttk_selftest.hil_devices.selftest_devices.SRC

  Base iSystemReferenceCard (iSRC) container.

Namespaces

- namespace ttk_selftest

  Test functions for HIL system self test and abstraction layer for "HIL devices".
- namespace ttk_selftest.hil_devices

  Access to hil devices (dSPACE and GammaV).
- namespace ttk_selftest.hil_devices.selftest_devices

  Interface wrapper for selftest device descriptions.

8.104 ttk_selftest/selftest_tests.py File Reference

Namespaces

- namespace ttk_selftest

  Test functions for HIL system self test and abstraction layer for "HIL devices".
- namespace ttk_selftest.selftest_tests

  Wrapper for self test test functions This serves as "interface" to the precompiled module in delivery to enable code-completion in PyDev.

Functions

- ttk_selftest.selftest_tests.testGND (testenv, gnd, threshold=10000, ignore_results=False)

  Function to test ground connections.
- ttk_selftest.selftest_tests.testAnalogOutput (testenv, channel, ref_channel, num_testvalues=5, gnd_↩ offset=0.5)

  Function to test an analog output channel.
- ttk_selftest.selftest_tests.testAnalogVoltageInput (testenv, device, num_testvalues=5)

  Function to test an analog voltage input channel.
- ttk_selftest.selftest_tests.testAnalogCurrentInput (testenv, device, num_testvalues=5)

  Function to test an analog current input channel.
- ttk_selftest.selftest_tests.testPowerSupplyVoltage (testenv, power_supply, num_testvalues=5)

  Function to test power supply voltage control and measurement.
- ttk_selftest.selftest_tests.testPowerSupplyCurrent (testenv, power_supply, max_test_current=None, num_↩ testvalues=5)

  Function to test power supply current control and measurement.
- ttk_selftest.selftest_tests.testPowerSupply (testenv, power_supply, max_test_current=None, num_↩ testvalues=5)

  Function to test Power Supply voltage and current.
- ttk_selftest.selftest_tests.testPwmInput (testenv, device, num_testvalues=5)

  Function to test a PWM input channel.
- ttk_selftest.selftest_tests.testLoad (testenv, load, num_testvalues=5)

  Function to test a load.
- ttk_selftest.selftest_tests.testPwmOutput (testenv, channel, num_testvalues=5)

  Function to test an PWM output channel.
- ttk_selftest.selftest_tests.testCan (testenv, channel)

  Function to test a CAN channel.
- ttk_selftest.selftest_tests.testDigitalInput (testenv, device)

  Function to test a digital input channel.
- ttk_selftest.selftest_tests.testDigitalOutput (testenv, channel)

  Function to test a digital output channel.
- ttk_selftest.selftest_tests.testiDigiRes (testenv, idigires)

  Function to test an iDigiRes.
- ttk_selftest.selftest_tests.resetFiu (hil_devices)

  Function to reset the FIU.

8.105 ttk_tools/_process_utils.py File Reference

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools._process_utils

  Utilities for handling win32 (and x64) processes.

Functions

- ttk_tools._process_utils.killProcess (target_names, verbosity=1)

  Kill all processes that match the given target_name(s) (using kernel32).
- ttk_tools._process_utils.getCurrentProcessNames (normalize_case=True)

  Get image names of currently active processes (that is, the executable file names).
- ttk_tools._process_utils.getCurrentProcessPaths (normalize_case=True, file_names_only=False)

  Get image names/paths of currently active processes (that is, names or paths of the executable files).
- ttk_tools._process_utils.isProcessActive (process_name, ignore_case=True)

  Check whether at least one process using the supplied executable name is currently running.
- ttk_tools._process_utils.killProcess2 (image_name)

  Kill all processes that match the given image_name using the windows "taskkill" command running in a sub process.
- ttk_tools._process_utils.getCurrentPIDs ()

  Get a list of PIDs of running processes, similar to the output of win32process.EnumProcesses() (which was not yet available in older Python versions).
- ttk_tools._process_utils.killByPID (pid)

  Kill the process with the supplied ID (as long as the user has sufficient access rights).

Variables

- int ttk_tools._process_utils.PROCESS_TERMINATE = 0x0001
- int ttk_tools._process_utils.PROCESS_VM_READ = 0x0010
- int ttk_tools._process_utils.PROCESS_QUERY_INFORMATION = 0x0400
- int ttk_tools._process_utils.PROCESS_QUERY_LIMITED_INFORMATION = 0x1000

8.106 ttk_tools/_win32_utils.py File Reference

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools._win32_utils

  Utility functions for use on Windows systems.

Functions

- ttk_tools._win32_utils.disableAppCrashDialog ()

  Disable displaying of application/program crash dialog for the current process ("XYZ has stopped working" / "XYZ funktioniert nicht mehr").
- ttk_tools._win32_utils.waitPumpMessages (delay)

  Wait for (approximately) delay [s] and pump windows messages/events
  while waiting, which is a "good thing" to do if a COM connection is active in the current process/thread.
- ttk_tools._win32_utils._getRegKeys (path)

  Get a `_winreg.HKEY_*`-constant matching the supplied `start_key`.
- ttk_tools._win32_utils.getRegPath (path, name, wow64_32_key=None, verbosity=1)

  Get a value from an entry in the windows registry.
- ttk_tools._win32_utils.setRegPath (path, name, value, value_type=None, wow64_32_key=None, verbosity=1)

  Set a value to an entry in the windows registry.

Variables

- ttk_tools._win32_utils._stop_event = win32event.CreateEvent(None, 0, 0, None)

  alibi-event to wait for with waitPumpMessages() Note: this event could be triggered via win32event.SetEvent(stop↩ _event) to stop a running waitPumpMessages() early

8.107 ttk_tools/can_hal/_hal.py File Reference

Classes

- class ttk_tools.can_hal._hal.CanHal

  Low(-ish)-Level interface to CAN_HAL_(32/64).dll.
- class ttk_tools.can_hal._hal.CanHalDevice

  CAN device connected to a CanHal instance.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.can_hal

  Access to iSyst Hardware Abstraction Layer for CAN interfaces.
- namespace ttk_tools.can_hal._hal

  Access to CAN HAL (via CAN_HAL_32.dll / CAN_HAL_64.dll).

Functions

- ttk_tools.can_hal._hal._errCheck (result, func, args)

  Default "errcheck" callable for API calls that return a tHalError value.
- ttk_tools.can_hal._hal._settypes (lib, name, argtypes=None, restype=None)

  Assign "restype", "errcheck" and "argtypes" attributes to DLL functions.

Variables

- ttk_tools.can_hal._hal._base_path = os.path.abspath(inspect.currentframe().f_code.co_filename)

8.108 ttk_tools/can_hal/_hal_offline_stub.py File Reference

Classes

- class ttk_tools.can_hal._hal_offline_stub.CanHal

  Low(-ish)-Level interface to CAN_HAL_(32/64).dll.
- class ttk_tools.can_hal._hal_offline_stub.CanHal._DeviceStub
- class ttk_tools.can_hal._hal_offline_stub.CanHalDevice

  CAN device connected to a CanHal instance.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.can_hal

  Access to iSyst Hardware Abstraction Layer for CAN interfaces.
- namespace ttk_tools.can_hal._hal_offline_stub

  Offline stub for CAN HAL access in ttk_tools.can_hal.hal.

Variables

- ttk_tools.can_hal._hal_offline_stub._base_path = os.path.abspath(inspect.currentframe().f_code.co_↩ filename)

8.109 ttk_tools/can_hal/common.py File Reference

Classes

- class ttk_tools.can_hal.common.CanHalError

  Base class for CAN HAL exceptions.
- class ttk_tools.can_hal.common.CanHalApiError
- class ttk_tools.can_hal.common.CanHalReadError

  Error during read access.
- class ttk_tools.can_hal.common.CanHalWriteError

  Device Type (CanHAl.h, tCanHalDeviceType).
- class ttk_tools.can_hal.common.DeviceType
- class ttk_tools.can_hal.common.CanBusState

  CAN bus states (CanHal.h, tCanHalCanBusState).
- class ttk_tools.can_hal.common.ConfigFlag

  Flags used during CAN configuration (CanHal.h, tCanHalCanConfig).
- class ttk_tools.can_hal.common.CanBitrate

  Connection speed settings (CanHal.h, tCanHalBautrate), bits per s.
- class ttk_tools.can_hal.common.HalErrors

  HAL Error Codes (HalErrors.h, enum tHalError).
- class ttk_tools.can_hal.common.MessageStatus

  CAN Message Status (CanHal.h, tCanHalMessageStatus).

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.can_hal

  Access to iSyst Hardware Abstraction Layer for CAN interfaces.
- namespace ttk_tools.can_hal.common

  Common CAN HAL definitions, classes and utility functions.

Variables

- ttk_tools.can_hal.common.ms = MessageStatus.RxMessage | MessageStatus.RemoteRequest
- int ttk_tools.can_hal.common.error_code = 292

8.110 ttk_tools/can_hal/hal.py File Reference

Classes

- class ttk_tools.can_hal.hal.CanMessage

  CAN message data (CanHal.h, tCanHalCanMessage).
- class ttk_tools.can_hal.hal.CanHal

  Low(-ish)-Level interface to CAN_HAL_(32/64).dll.
- class ttk_tools.can_hal.hal.CanHalDevice

  CAN device connected to a CanHal instance.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.can_hal

  Access to iSyst Hardware Abstraction Layer for CAN interfaces.
- namespace ttk_tools.can_hal.hal
- namespace ttk_tools.can_hal._hal

  Access to CAN HAL (via CAN_HAL_32.dll / CAN_HAL_64.dll).

8.111 ttk_tools/can_hal/hal_offline_stub.py File Reference

Classes

- class ttk_tools.can_hal.hal_offline_stub.CanMessage

  CAN message data (CanHal.h, tCanHalCanMessage).
- class ttk_tools.can_hal.hal_offline_stub.CanHal

  Low(-ish)-Level interface to CAN_HAL_(32/64).dll.
- class ttk_tools.can_hal.hal_offline_stub.CanHalDevice

  CAN device connected to a CanHal instance.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.can_hal

  Access to iSyst Hardware Abstraction Layer for CAN interfaces.
- namespace ttk_tools.can_hal.hal_offline_stub
- namespace ttk_tools.can_hal._hal

  Access to CAN HAL (via CAN_HAL_32.dll / CAN_HAL_64.dll).

8.112 ttk_tools/can_hal/hal_structures.py File Reference

Classes

- class ttk_tools.can_hal.hal_structures._CanId

  An int-wrapper that uses a hexadecimal representation.
- class ttk_tools.can_hal.hal_structures.CanHalVersion

  tCanHalVersion (CanHal.h)
- class ttk_tools.can_hal.hal_structures._EnumMapMixin

  Mix-in for ctypes.Structure.
- class ttk_tools.can_hal.hal_structures.CanDeviceConfig

  CAN device configuration (CanHal.h, tCanHalDeviceConfig).
- class ttk_tools.can_hal.hal_structures.DeviceListEntry

  A single device list entry (CanHal.h, tCanHalDeviceList).
- class ttk_tools.can_hal.hal_structures.Timestamp

  Message time stamps (CanHal.h, tCanHalTimestamp), values in µs.
- class ttk_tools.can_hal.hal_structures.CanMessage

  CAN message data (CanHal.h, tCanHalCanMessage).

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.can_hal

  Access to iSyst Hardware Abstraction Layer for CAN interfaces.
- namespace ttk_tools.can_hal.hal_structures

  C data structures for use with CAN_HAL DLL.

Variables

- ttk_tools.can_hal.hal_structures.CanDeviceHdl = ctypes.c_ulong

  Handle for connected devices: typedef unsigned long tHalCanDeviceHdl;.
- int ttk_tools.can_hal.hal_structures.CanPayload = ctypes.c_ubyte * 8

  data type for CAN payload data (8 bytes, just for convenience)

8.113 ttk_tools/dspace/_xil_api.py File Reference

Classes

- class ttk_tools.dspace._xil_api.XilTestbench
  XilTestbench access.
- class ttk_tools.dspace._xil_api.Capture
  Capture / Data Acquisition.
- class ttk_tools.dspace._xil_api.StimulusSignalDescription
  Signal value description for stimulus / signal generator.
- class ttk_tools.dspace._xil_api.Stimulus
  Stimulus / Signal Generator.

Namespaces

- namespace ttk_tools
  Modules for tool access.
- namespace ttk_tools.dspace
  Access to dSPACE (dSPACE GmbH) tools and interfaces.
- namespace ttk_tools.dspace._xil_api
  Interface to the dSPACE XIL-2.x-Testbench implementation.

Variables

- dict ttk_tools.dspace._xil_api._SCALAR_DATA_TYPES
- dict ttk_tools.dspace._xil_api._VECTOR_DATA_TYPES
- dict ttk_tools.dspace._xil_api._MATRIX_DATA_TYPES

8.114 ttk_tools/dspace/_xil_api_offline_stub.py File Reference

Classes

- class ttk_tools.dspace._xil_api_offline_stub.DotNetException
  Stub for .NET System.Exception.
- class ttk_tools.dspace._xil_api_offline_stub.TimeoutException
  XIL TestbenchPortException Stub.
- class ttk_tools.dspace._xil_api_offline_stub.TestbenchPortException
- class ttk_tools.dspace._xil_api_offline_stub.MAPortError
  XIL Model-Access-Port Error Stub.
- class ttk_tools.dspace._xil_api_offline_stub._TestbenchStub
  Minimal (very minimal) stub for XIL-API testbench implementation.
- class ttk_tools.dspace._xil_api_offline_stub._MaPortStub
  Minimal stub for a MAPort-ish implementation.
- class ttk_tools.dspace._xil_api_offline_stub.XilTestbench
  XilTestbench access offline stub.
- class ttk_tools.dspace._xil_api_offline_stub.Capture
  Capture / Data Acquisition Offline Stub.
- class ttk_tools.dspace._xil_api_offline_stub.StimulusSignalDescription
  Stub for signal value description for stimulus / signal generator.
- class ttk_tools.dspace._xil_api_offline_stub.Stimulus
  Offline stub for Stimulus / Signal Generator.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.dspace

  Access to dSPACE (dSPACE GmbH) tools and interfaces.
- namespace ttk_tools.dspace._xil_api_offline_stub

  A dummy/stub implementation of the default ttk_tools.dspace.xil_api for debugging without available hardware or license keys.

8.115 ttk_tools/dspace/dspace_utils.py File Reference

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.dspace

  Access to dSPACE (dSPACE GmbH) tools and interfaces.
- namespace ttk_tools.dspace.dspace_utils

  Utilities for classic dSPACE rtplib-based real-time platform access.

Functions

- ttk_tools.dspace.dspace_utils.hilVarFactory (context, identifier, ∗args, ∗∗kwargs)

  Factory for "HilVars", decides actual implementation to use based on supplied context (i.e.

Variables

- ttk_tools.dspace.dspace_utils.XilTestbench = OfflineXilTestbench

8.116 ttk_tools/dspace/rtplib_offline_stub.py File Reference

Classes

- class ttk_tools.dspace.rtplib_offline_stub.rtplibError

  Exception of the module rtplib.
- class ttk_tools.dspace.rtplib_offline_stub._RTPApplicationDummy

  Dummy implementation of `rtplib.Appl()` compatible objects for offline use.
- class ttk_tools.dspace.rtplib_offline_stub._RTPVariableDummy

  Dummy implementation of `rtplib.Appl().Variable()` compatible objects for offline use.
- class ttk_tools.dspace.rtplib_offline_stub._RTPCaptureDummy

  A dummy implementation of `rtplib.Capture()` compatible objects for offline use.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.dspace

  Access to dSPACE (dSPACE GmbH) tools and interfaces.
- namespace ttk_tools.dspace.rtplib_offline_stub

  A dummy/stub implementation of the dSPACE rtplib to enable offline debugging without available real time hardware or license keys.

Functions

- ttk_tools.dspace.rtplib_offline_stub.ApplCheck (Object)
- ttk_tools.dspace.rtplib_offline_stub.VariableCheck (Object)

  Check if an Python object has the type Variable.
- ttk_tools.dspace.rtplib_offline_stub.CaptureCheck (Object)

  Check if an Python object has the type Capture.
- ttk_tools.dspace.rtplib_offline_stub.Appl (VariableFile, SystemName, SystemType)

  Create an Appl class object.

Variables

- str ttk_tools.dspace.rtplib_offline_stub.rtplibVersion = "0.0.0.0"
- int ttk_tools.dspace.rtplib_offline_stub.CaptureNone = 0
- int ttk_tools.dspace.rtplib_offline_stub.CaptureTriggerArmed = 1
- int ttk_tools.dspace.rtplib_offline_stub.CaptureInProgress = 2
- int ttk_tools.dspace.rtplib_offline_stub.CaptureDone = 3
- int ttk_tools.dspace.rtplib_offline_stub.CaptureAborted = 4
- int ttk_tools.dspace.rtplib_offline_stub.CaptureWaitingForTriggerArmed = 5

8.117 ttk_tools/dspace/sdmlib_offline_stub.py File Reference

Classes

- class ttk_tools.dspace.sdmlib_offline_stub.Constants

  Constants-Enum.
- class ttk_tools.dspace.sdmlib_offline_stub.sdmlibError

  sdmlib-exceptions
- class ttk_tools.dspace.sdmlib_offline_stub.SIGNAL
- class ttk_tools.dspace.sdmlib_offline_stub.BUFFER

  Instances of the class BUFFER are used to hold values of Stimulus Engine channels for later use.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.dspace

  Access to dSPACE (dSPACE GmbH) tools and interfaces.
- namespace ttk_tools.dspace.sdmlib_offline_stub

  A dummy/stub implementation of the classic dSPACE sdmlib Stimulus Engine to enable offline debugging without available real-time hardware or license keys.

Functions

- ttk_tools.dspace.sdmlib_offline_stub.SG_INIT (TrcFile, BoardName=None, MPSubsystem=None)

  Initialize the global stimulus object with an application.
- ttk_tools.dspace.sdmlib_offline_stub.SG_CLOSE ()

  Shut down the Signal Generator if it is running in a separate thread.
- ttk_tools.dspace.sdmlib_offline_stub.SG_GET_STATE ()

  Get the current state of the stimulus engine.
- ttk_tools.dspace.sdmlib_offline_stub.SG_GET_TIME ()

  Get the current time of the Stimulus Engine.
- ttk_tools.dspace.sdmlib_offline_stub.SG_SET_STATE (NewState)

  Set the state of the stimulus engine.
- ttk_tools.dspace.sdmlib_offline_stub.SG_STOP ()

  Stop the Stimulus Engine on the real-time platform.
- ttk_tools.dspace.sdmlib_offline_stub.SG_RUN ()

  Start the Stimulus Engine on the real-time platform.
- ttk_tools.dspace.sdmlib_offline_stub.SG_PAUSE ()

  Pause the stimulus engine on the real-time platform.
- ttk_tools.dspace.sdmlib_offline_stub.ELSE ()

  Mark the beginning of the else branch of an if statement.
- ttk_tools.dspace.sdmlib_offline_stub.ENDIF ()

  Mark the end of an IF statement.
- ttk_tools.dspace.sdmlib_offline_stub.ENDLOOP ()

  Mark the end of a repeated code block.
- ttk_tools.dspace.sdmlib_offline_stub.EVAL (eval_time_length, save_per_period=30, save_time_step=None, mat_file_name=None)

  Calculate the specified signal forms offline on the host PC and save the results in a binary MATLAB-compatible file.
- ttk_tools.dspace.sdmlib_offline_stub.GOTO (Label)

  Go to the specified label.
- ttk_tools.dspace.sdmlib_offline_stub.IF (result, timeout=None)

  Define a condition which is evaluated by the Stimulus Engine.
- ttk_tools.dspace.sdmlib_offline_stub.LABEL (Label)

  Set a label for GOTO() calls.
- ttk_tools.dspace.sdmlib_offline_stub.LOAD_SEQ ()

  Download the instruction table.
- ttk_tools.dspace.sdmlib_offline_stub.LOOP (Number)

  Specify the number of repetitions of a repeated code block.
- ttk_tools.dspace.sdmlib_offline_stub.MAKE_SEQ (seq_file_name=None)

  Create the sequence control table (SEQ) for the signal generation.
- ttk_tools.dspace.sdmlib_offline_stub.MsgSleep (Interval)

  Suspends the execution of the current thread for at least the specified interval.
- ttk_tools.dspace.sdmlib_offline_stub.STOP ()

  This function is used to finish the stimulus description.
- ttk_tools.dspace.sdmlib_offline_stub.TH_START ()

  Start the data replay mechanism.
- ttk_tools.dspace.sdmlib_offline_stub.TH_STOP ()

  Stop the data replay mechanism.
- ttk_tools.dspace.sdmlib_offline_stub.TT (Delay)

  Specify the time delay between two Stimulus Engine actions.
- ttk_tools.dspace.sdmlib_offline_stub.UNLOAD_SEQ ()

  Unload the instruction table.
- ttk_tools.dspace.sdmlib_offline_stub.ASCIItoMat (ASCII_file, mat_file)

  Convert ASCII data to a binary MAT file.

Variables

- str ttk_tools.dspace.sdmlib_offline_stub._version_ = '0.0.0.0'

8.118 ttk_tools/dspace/xil_api.py File Reference

Classes

- class ttk_tools.dspace.xil_api.XilTestbench

  XilTestbench access.
- class ttk_tools.dspace.xil_api.Capture

  Capture / Data Acquisition.
- class ttk_tools.dspace.xil_api.StimulusSignalDescription

  Signal value description for stimulus / signal generator.
- class ttk_tools.dspace.xil_api.Stimulus

  Stimulus / Signal Generator.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.dspace

  Access to dSPACE (dSPACE GmbH) tools and interfaces.
- namespace ttk_tools.dspace.xil_api

  Wrapper for interface to the dSPACE XIL-2.x-Testbench implementation in ttk_tools.dspace._xil_api.

8.119 ttk_tools/dspace/xil_api_common.py File Reference

Classes

- class ttk_tools.dspace.xil_api_common.DataType

  Data Types.
- class ttk_tools.dspace.xil_api_common.InterpolationTypes

  Data Interpolation Types.
- class ttk_tools.dspace.xil_api_common.MAPortState

  Model Access Port States.
- class ttk_tools.dspace.xil_api_common.CaptureState

  Data Capture States.
- class ttk_tools.dspace.xil_api_common.SignalGeneratorState

  Signal Generator States.
- class ttk_tools.dspace.xil_api_common.TestbenchError

  Base class for "other" testbench errors that are not handled by XIL TestbenchPortExceptions.
- class ttk_tools.dspace.xil_api_common.StimulusError

  Errors during Stimulus setup/execution.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.dspace

  Access to dSPACE (dSPACE GmbH) tools and interfaces.
- namespace ttk_tools.dspace.xil_api_common

  Common shared definitions and error classes for xil_api.

Functions

- ttk_tools.dspace.xil_api_common.writeMAPortConfigXML (sdf_file_path, platform_name="DS1006", xml_↩ file_path=None)

8.120 ttk_tools/dspace/xil_api_offline_stub.py File Reference

Classes

- class ttk_tools.dspace.xil_api_offline_stub.XilTestbench

  XilTestbench access offline stub.
- class ttk_tools.dspace.xil_api_offline_stub.Capture

  Capture / Data Acquisition Offline Stub.
- class ttk_tools.dspace.xil_api_offline_stub.StimulusSignalDescription

  Stub for signal value description for stimulus / signal generator.
- class ttk_tools.dspace.xil_api_offline_stub.Stimulus

  Offline stub for Stimulus / Signal Generator.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.dspace

  Access to dSPACE (dSPACE GmbH) tools and interfaces.
- namespace ttk_tools.dspace.xil_api_offline_stub

  Wrapper for a dummy/stub implementation of the default ttk_tools.dspace.xil_api for debugging without available hardware or license keys.

8.121 ttk_tools/dspace/xil_net.py File Reference

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.dspace

  Access to dSPACE (dSPACE GmbH) tools and interfaces.
- namespace ttk_tools.dspace.xil_net

  Import helper module as workaround for current import hook compatibility issues between packages Cython and pythonnet.

Variables

- str ttk_tools.dspace.xil_net.default_xil_version = "2.2.0"

  fallback/default version
- ttk_tools.dspace.xil_net.available_xil_versions = _detectXilVersions(verbose=False)

  detected versions in ascending order (if any)
- ttk_tools.dspace.xil_net.preferred_xil_version

  used specified version (from current hil_project.ini)
- list ttk_tools.dspace.xil_net.candidates = [default_xil_version]
- list ttk_tools.dspace.xil_net.xil_version = candidates[-1]

8.122 ttk_tools/isyst/ida_common.py File Reference

Classes

- class ttk_tools.isyst.ida_common.MwError

  Generic iDA Testware middleware interface error.
- class ttk_tools.isyst.ida_common.MwAppError
- class ttk_tools.isyst.ida_common.MwRecError

  Errors during data recording.
- class ttk_tools.isyst.ida_common.MwAppStatus

  Application RPC status info (internal success/error).
- class ttk_tools.isyst.ida_common.MwDataType

  Data types for middleware values, compare mwvalue.h: dataType_t.
- class ttk_tools.isyst.ida_common.MetaKind

  MetaKind as defined in protobuf message MetaId.
- class ttk_tools.isyst.ida_common.RecCommandStatus

  Recorder command RPC status info (internal success/error).
- class ttk_tools.isyst.ida_common.RecState

  Recorder state machine state.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.isyst

  Access to iSyst Intelligente Systeme GmbH tools and interfaces.
- namespace ttk_tools.isyst.ida_common

  Common iDA Testware / middleware definitions, classes and utility functions.

Functions

- ttk_tools.isyst.ida_common._enumName (EnumClass, value)

  Get value's name of an enum, or "<undefined>" for undefined values.
- ttk_tools.isyst.ida_common._hexRepr (value, nibbles=0)

  Get a hexadecimal representation of value, fallback to default representation for non-integer values.

8.123 ttk_tools/isyst/ida_testware.py File Reference

Classes

- class ttk_tools.isyst.ida_testware.IdaTestware

  Access to iDA Testware MwApp interface for reading/writing variables.
- class ttk_tools.isyst.ida_testware.IdaRecorder

  Data acquisition using an iDA Recorder application.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.isyst

  Access to iSyst Intelligente Systeme GmbH tools and interfaces.
- namespace ttk_tools.isyst.ida_testware

  Access to iDA Testware interfaces.

8.124 ttk_tools/isyst/ida_testware_offline_stub.py File Reference

Classes

- class ttk_tools.isyst.ida_testware_offline_stub.IdaTestware

  Access to iDA Testware MwApp interface for reading/writing variables.
- class ttk_tools.isyst.ida_testware_offline_stub.IdaRecorder

  Data acquisition using an iDA Recorder application.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.isyst

  Access to iSyst Intelligente Systeme GmbH tools and interfaces.
- namespace ttk_tools.isyst.ida_testware_offline_stub
- namespace ttk_tools.isyst.ida_testware

  Access to iDA Testware interfaces.

8.125 ttk_tools/isyst/idaq.py File Reference

Classes

- class ttk_tools.isyst.idaq.DaqDataError

  Errors during DAQ data decoding.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.isyst

  Access to iSyst Intelligente Systeme GmbH tools and interfaces.
- namespace ttk_tools.isyst.idaq

  I/O for "idaq" data acquisition format.

Functions

- ttk_tools.isyst.idaq._bstr (s)
- ttk_tools.isyst.idaq._ustr (b)

  Get a text string using the default encoding.
- ttk_tools.isyst.idaq._getByteSizeString (size)

  Get a string representation of the supplied size in Bytes.
- ttk_tools.isyst.idaq._getSamplesPath ()

  Get path to a folder /samples on same level as a parent folder /src.
- ttk_tools.isyst.idaq.readDaqData (file_handle)

  Read DAQ data in "idaq" format.
- ttk_tools.isyst.idaq.writeDaqData (daq_data, file_handle, comment=None, verbosity=1)

  Write Read DAQ data in "idaq" format.
- ttk_tools.isyst.idaq.printDataInfo (values, indent_level=1, outer_n=2)

  Print a summary of the supplied values to stdout.

Variables

- str ttk_tools.isyst.idaq.FILE_FORMAT = "1.2"

  marker for supported file format version
- str ttk_tools.isyst.idaq.ENCODING = "latin-1"

  text encoding to use for non-binary parts (header, meta data)
- dict ttk_tools.isyst.idaq.data_type_to_format_mapping

  Mapping of data type names to Python struct format codes.

8.126 ttk_tools/rst/idaq.py File Reference

Classes

- class ttk_tools.rst.idaq.DaqDataError

  Errors during DAQ data decoding.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.rst

  Access to RST (RST Industrie Automation GmbH) tools and interfaces.
- namespace ttk_tools.rst.idaq

Functions

- ttk_tools.rst.idaq._bstr (s)
- ttk_tools.rst.idaq._ustr (b)

  Get a text string using the default CANape encoding.
- ttk_tools.rst.idaq._getByteSizeString (size)

  Get a string representation of the supplied size in Bytes.
- ttk_tools.rst.idaq._getSamplesPath ()

  Get path to a folder /samples on same level as a parent folder /src.
- ttk_tools.rst.idaq.readDaqData (file_handle)

  Read DAQ data in "idaq" format.
- ttk_tools.rst.idaq.writeDaqData (daq_data, file_handle, comment=None, verbosity=1)

  Write Read DAQ data in "idaq" format.

Variables

- str ttk_tools.rst.idaq.FILE_FORMAT = "1.2"

  marker for supported file format version
- str ttk_tools.rst.idaq.ENCODING = "latin-1"

  text encoding to use for non-binary parts (header, meta data)
- dict ttk_tools.rst.idaq.data_type_to_format_mapping

  Mapping of data type names to Python struct format codes.

8.127 ttk_tools/isyst/mwapp_client.py File Reference

Classes

- class ttk_tools.isyst.mwapp_client._VarMetaCache

  Cache for iDA-Variable meta data.
- class ttk_tools.isyst.mwapp_client._VarMetaCache.VarMeta

  Simple container for iDA-Variable meta data.
- class ttk_tools.isyst.mwapp_client.MwVar

  Container for a single MW-Variable.
- class ttk_tools.isyst.mwapp_client.MwAppRpc

  Simple wrapper for MwApp RPCs.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.isyst

  Access to iSyst Intelligente Systeme GmbH tools and interfaces.
- namespace ttk_tools.isyst.mwapp_client

  iDA Testware data access via gRPC "MwApp" interface

8.128 ttk_tools/isyst/mwapp_client_offline_stub.py File Reference

Classes

- class ttk_tools.isyst.mwapp_client_offline_stub._VarMetaCache
  Cache for iDA-variable meta data.
- class ttk_tools.isyst.mwapp_client_offline_stub._VarMetaCache.VarMeta
  Simple container for iDA-Variable meta data.
- class ttk_tools.isyst.mwapp_client_offline_stub.MwVar
  Container for a single MW-Variable.
- class ttk_tools.isyst.mwapp_client_offline_stub.MwAppRpc
  Simple wrapper for MwApp RPCs (offline stub/mock implementation)

Namespaces

- namespace ttk_tools
  Modules for tool access.
- namespace ttk_tools.isyst
  Access to iSyst Intelligente Systeme GmbH tools and interfaces.
- namespace ttk_tools.isyst.mwapp_client_offline_stub
  Offline stub for iDA Testware data access via gRPC "MwApp" interface.

Variables

- str ttk_tools.isyst.mwapp_client_offline_stub._STUB_VERSION = "1.3.0"

8.129 ttk_tools/isyst/mwrec_client.py File Reference

Classes

- class ttk_tools.isyst.mwrec_client.MwRecRpc
  Simple wrapper for MwRec RPCs.

Namespaces

- namespace ttk_tools
  Modules for tool access.
- namespace ttk_tools.isyst
  Access to iSyst Intelligente Systeme GmbH tools and interfaces.
- namespace ttk_tools.isyst.mwrec_client
  Access to iDA Recorder (data acquisition) via gRPC "MwRec" interface.

8.130 ttk_tools/isyst/mwrec_client_offline_stub.py File Reference

Classes

- class ttk_tools.isyst.mwrec_client_offline_stub.MwRecRpc
  Simple wrapper for MwRec RPCs (offline stub/mock implementation)

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.isyst

  Access to iSyst Intelligente Systeme GmbH tools and interfaces.
- namespace ttk_tools.isyst.mwrec_client_offline_stub

  Offline stub for access to iDA Recorder (data acquisition)

Variables

- str ttk_tools.isyst.mwrec_client_offline_stub._STUB_VERSION = "1.0.0"

8.131 ttk_tools/ni/_labview.py File Reference

Classes

- class ttk_tools.ni._labview.LabView

  Access to LabView VI controls/variables via COM interface.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.ni

  Access to NI (National Instruments) tools and interfaces, primarily LabView.
- namespace ttk_tools.ni._labview

  Access to LabView VIs via COM server.

8.132 ttk_tools/ni/labview.py File Reference

Classes

- class ttk_tools.ni.labview.LabView

  Access to LabView VI controls/variables via COM interface.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.ni

  Access to NI (National Instruments) tools and interfaces, primarily LabView.
- namespace ttk_tools.ni.labview

  Interface wrapper for LabView VI COM access in ttk_tools.ni._labview.

8.133 ttk_tools/offline_data_provider.py File Reference

Classes

- class ttk_tools.offline_data_provider.OfflineDataProvider

  Provider of dummy data for use in offline debugging.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.offline_data_provider

  Data provider for dummy/offline/mock implementations of tool access modules (like ccp/xcp or real time platform/model data).

Functions

- ttk_tools.offline_data_provider.getDefaultProvider ()

  Get global default provider.
- ttk_tools.offline_data_provider.setDefaultProvider (data_provider)

  Set a new global default provider.

Variables

- float ttk_tools.offline_data_provider.DEFAULT_VALUE = 0.0

  initial/default value for new "variables"

8.134 ttk_tools/process_utils.py File Reference

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.process_utils

  Interface to utilities for handling win32 (and x64) processes in ttk_tools._process_utils.

Functions

- ttk_tools.process_utils.killProcess (target_names, verbosity=1)

  Kill all processes that match the given target_name(s) (using kernel32).
- ttk_tools.process_utils.getCurrentProcessNames (normalize_case=True)

  Get image names of currently active processes (that is, the executable file names).
- ttk_tools.process_utils.getCurrentProcessPaths (normalize_case=True, file_names_only=False)

  Get image names/paths of currently active processes (that is, names or paths of the executable files).
- ttk_tools.process_utils.isProcessActive (process_name, ignore_case=True)

  Check whether at least one process using the supplied executable name is currently running.
- ttk_tools.process_utils.killProcess2 (image_name)

  Kill all processes that match the given image_name using the windows "taskkill" command running in a sub process.
- ttk_tools.process_utils.getCurrentPIDs ()

  Get a list of PIDs of running processes, similar to the output of win32process.EnumProcesses() (which was not yet available in older Python versions).
- ttk_tools.process_utils.killByPID (pid)

  Kill the process with the supplied ID (as long as the user has sufficient access rights).

8.135 ttk_tools/pyocd/_debugger.py File Reference

Classes

- class ttk_tools.pyocd._debugger.SymbolNotFound

  No matching symbol was found in ELF file.
- class ttk_tools.pyocd._debugger.Debugger

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.pyocd

  Wrapper for using PyOCD (https://pyocd.io/) as a data source for HIL tests.
- namespace ttk_tools.pyocd._debugger

  Access to PyOCD session, provides attaching to running controllers and read/write of variables via debug symbols.

Variables

- ttk_tools.pyocd._debugger.level

8.136 ttk_tools/pyocd/debugger.py File Reference

Classes

- class ttk_tools.pyocd.debugger.Debugger

  Wrapper for a PyOCD session and utility methods.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.pyocd

  Wrapper for using PyOCD (https://pyocd.io/) as a data source for HIL tests.
- namespace ttk_tools.pyocd.debugger

  Access to PyOCD session, provides attaching to running controllers and read/write of variables via debug symbols.

8.137 ttk_tools/rst/_gamma_api.py File Reference

Classes

- class ttk_tools.rst._gamma_api.Gamma

  GammaService-connection (application).
- class ttk_tools.rst._gamma_api.DataAcquisition

  Data Acquisition with GammaV Logger Task.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.rst

  Access to RST (RST Industrie Automation GmbH) tools and interfaces.
- namespace ttk_tools.rst._gamma_api

  Access to GammaV Service and related functionality.

Functions

- ttk_tools.rst._gamma_api._getIdArgs (ident)

  Get an identifier (or "address") argument as numeric index or as a name for use with gacmd calls.
- ttk_tools.rst._gamma_api.loadDaqDataFile (file_path)

  Read DAQ data from a file in "idaq" format.

Variables

- tuple ttk_tools.rst._gamma_api.is_win32 = (sys.platform == "win32")

  True if current platform is detected as Windows, otherwise False.
- tuple ttk_tools.rst._gamma_api.is_linux = (sys.platform.startswith("linux"))

  True if current platform is detected as Linux, otherwise False.
- str ttk_tools.rst._gamma_api.GAMMA_PROCESS_NAME = "gaservice.exe"

  name of gamma service executable (depends on used platform)
- str ttk_tools.rst._gamma_api.GAMMA_CMD_PROCESS_NAME = "gacmd.exe"

  name of gamma command executable (depends on used platform)
- int ttk_tools.rst._gamma_api._GAMMA_SERVICE_START_DELAY_MS = 2000

  delay in [ms] to wait after starting a "new" Gamma V Service
- int ttk_tools.rst._gamma_api._GAMMA_SERVICE_UPDATE_DELAY_MS = 2000

  delay in [ms] to wait after attaching to a "new" Gamma V Service (to give it time to update the variables list)
- bool ttk_tools.rst._gamma_api._DEFAULT_SHOW_GAMMA_CALLS = False

  default value for `Gamma._show_gamma_calls`, set to True to print gamma process calls to stdout for debugging purposes.

8.138 ttk_tools/rst/_gamma_api_offline_stub.py File Reference

Classes

- class ttk_tools.rst._gamma_api_offline_stub._PVStub

  Basic stub for class gaapi.PV (or rather, isygamma.PV).
- class ttk_tools.rst._gamma_api_offline_stub._GaApiStub

  Minimal stub for "gaapi" (which is normally a module)
- class ttk_tools.rst._gamma_api_offline_stub.Gamma

  GammaService-connection (application) stub.
- class ttk_tools.rst._gamma_api_offline_stub.DataAcquisition

  Data Acquisition with GammaV Logger Task.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.rst

  Access to RST (RST Industrie Automation GmbH) tools and interfaces.
- namespace ttk_tools.rst._gamma_api_offline_stub

  A stub implementation of ttk_tools.rst.gamma_api for debugging without a installed GammaV middleware.

Functions

- ttk_tools.rst._gamma_api_offline_stub.loadDaqDataFile (file_path)

  Read DAQ data from a file in "idaq" format.

Variables

- tuple ttk_tools.rst._gamma_api_offline_stub.is_win32 = (sys.platform == "win32")

  True if current platform is detected as Windows, otherwise False.
- tuple ttk_tools.rst._gamma_api_offline_stub.is_linux = (sys.platform.startswith("linux"))

  True if current platform is detected as Linux, otherwise False.
- str ttk_tools.rst._gamma_api_offline_stub.GAMMA_PROCESS_NAME = "gaservice.exe"

  name of gamma service executable (depends on used platform)
- str ttk_tools.rst._gamma_api_offline_stub.GAMMA_CMD_PROCESS_NAME = "gacmd.exe"

  name of gamma command executable (depends on used platform)
- str ttk_tools.rst._gamma_api_offline_stub._OFFLINE_DATA_CONTEXT = "gamma"
- ttk_tools.rst._gamma_api_offline_stub.gaapi = _GaApiStub()

8.139 ttk_tools/rst/gamma_api.py File Reference

Classes

- class ttk_tools.rst.gamma_api.Gamma

  GammaService-connection (application).
- class ttk_tools.rst.gamma_api.DataAcquisition

  Data Acquisition with GammaV Logger Task.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.rst

  Access to RST (RST Industrie Automation GmbH) tools and interfaces.
- namespace ttk_tools.rst.gamma_api

  Interface wrapper for GammaV Service Access in ttk_tools.rst._gamma_api.

Functions

- ttk_tools.rst.gamma_api.loadDaqDataFile (file_path)

  Read DAQ data from a file in "idaq" format.

8.140 ttk_tools/rst/gamma_api_common.py File Reference

Classes

- class ttk_tools.rst.gamma_api_common.GammaError

  Base Error for GammaV connection.
- class ttk_tools.rst.gamma_api_common.GammaApiError
- class ttk_tools.rst.gamma_api_common.GammaDaqError

  Base Error for GammaV DAQ / data logger functionality.
- class ttk_tools.rst.gamma_api_common.GammaDaqStateError
- class ttk_tools.rst.gamma_api_common.MESSAGE_MODE

  Message modes.
- class ttk_tools.rst.gamma_api_common.LOG_LEVEL

  Log levels.
- class ttk_tools.rst.gamma_api_common.DataType

  Gamma Data Types (gc_DataType).
- class ttk_tools.rst.gamma_api_common.PVType

  ga_PVType
- class ttk_tools.rst.gamma_api_common.PVProperty

  Properties for PV operation/access mode.
- class ttk_tools.rst.gamma_api_common.PVFieldType

  gc_PVFieldType - Field types of numeric PVs (gc_PV_FIELD_∗ entries).
- class ttk_tools.rst.gamma_api_common.ErrorCodes

  Gamma error codes (default return codes of API functions).
- class ttk_tools.rst.gamma_api_common.DaqMeasurementStatus

  State machine states for DAQ task.
- class ttk_tools.rst.gamma_api_common.BrowsingType

  Wrapper for enum gc_BrowsingType (BROWSING_TYPE_∗).

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.rst

  Access to RST (RST Industrie Automation GmbH) tools and interfaces.
- namespace ttk_tools.rst.gamma_api_common

  Common shared definitions and error classes for gamma_api.

Variables

- str ttk_tools.rst.gamma_api_common.DEFAULT_LOGFILE_NAME = "Gamma_Logfile.txt"

  default name for log files
- int ttk_tools.rst.gamma_api_common.DEFAULT_LISTEN_PORT = 0

  default listen port of the gamma service (see gamma service configuration file)
- int ttk_tools.rst.gamma_api_common.DEFAULT_ATTACH_TIMEOUT = 5000

  default timeout in [ms] to wait for the gamma service during attach

8.141 ttk_tools/rst/gamma_api_offline_stub.py File Reference

Classes

- class ttk_tools.rst.gamma_api_offline_stub.Gamma

  GammaService-connection (application) stub.
- class ttk_tools.rst.gamma_api_offline_stub.DataAcquisition

  Data Acquisition with GammaV Logger Task.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.rst

  Access to RST (RST Industrie Automation GmbH) tools and interfaces.
- namespace ttk_tools.rst.gamma_api_offline_stub

  Interface wrapper for a stub implementation of gamma_api in ttk_tools.rst._gamma_api_offline_stub.

Functions

- ttk_tools.rst.gamma_api_offline_stub.loadDaqDataFile (file_path)

  Read DAQ data from a file in "idaq" format.

8.142 ttk_tools/rst/isygamma.py File Reference

Classes

- class ttk_tools.rst.isygamma.PVHandle

  Opaque PV handle.
- class ttk_tools.rst.isygamma.NumericValue
- class ttk_tools.rst.isygamma.VersionNumber

  gc_VersionNumber
- class ttk_tools.rst.isygamma.Addresses

  Gamma addresses by nesting level.
- class ttk_tools.rst.isygamma.Addresses._AddrStruct
- class ttk_tools.rst.isygamma.Addresses.Level2
- class ttk_tools.rst.isygamma.Addresses.Level3
- class ttk_tools.rst.isygamma.Addresses.Level4
- class ttk_tools.rst.isygamma.Addresses.Level5
- class ttk_tools.rst.isygamma.Level5

  Address Level 5 elements (pv-array/fifo). More...
- class ttk_tools.rst.isygamma.Level5.PVArrayInfo

  typedef struct gc_PVArrayInfo
- class ttk_tools.rst.isygamma.Level5.FifoInfo

  typedef struct gc_FifoInfo
- class ttk_tools.rst.isygamma.Level4

  Address Level 4 elements (group). More...
- class ttk_tools.rst.isygamma.Level4.GroupInfo

typedef struct gc_GroupInfo

- class ttk_tools.rst.isygamma.Level3

  Address Level 3 elements (memory/io-channel/scheduler-slot). More...
- class ttk_tools.rst.isygamma.Level3.MemoryInfo

  typedef struct gc_MemoryInfo
- class ttk_tools.rst.isygamma.Level3.IOChannelInfo

  typedef struct gc_IOChannelInfo
- class ttk_tools.rst.isygamma.Level3.SchedulerSlotInfo

  typedef struct gc_SchedulerSlotInfo
- class ttk_tools.rst.isygamma.Level2

  Address Level 2 elements (node/io/task/action/scheduler) More...
- class ttk_tools.rst.isygamma.Level2.NodeInfo

  typedef struct gc_NodeInfo
- class ttk_tools.rst.isygamma.Level2.TaskInfo

  typedef struct gc_TaskInfo
- class ttk_tools.rst.isygamma.Level2.ActionInfo

  typedef struct gc_ActionInfo
- class ttk_tools.rst.isygamma.Level2.LoopInfo

  typedef struct gc_LoopInfo
- class ttk_tools.rst.isygamma.Level2.IOInfo

  typedef struct gc_IOInfo
- class ttk_tools.rst.isygamma.Level2.SchedulerInfo

  typedef struct gc_SchedulerInfo
- class ttk_tools.rst.isygamma.Level1

  Address Level 1 elements (system). More...
- class ttk_tools.rst.isygamma.Level1.SystemInfo

  typedef struct gc_SystemInfo
- class ttk_tools.rst.isygamma.BrowserInfo

  typedef struct gc_BrowserInfo.
- class ttk_tools.rst.isygamma.BrowserInfo.BrowsingInfoUnion

  container for the various "info" flavors.
- class ttk_tools.rst.isygamma._GaAPI

  Wrapper for gaapi dynamic library.
- class ttk_tools.rst.isygamma._GaCommon

  Wrapper for gacommon dynamic library.
- class ttk_tools.rst.isygamma.PV

  Gamma PV.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.rst

  Access to RST (RST Industrie Automation GmbH) tools and interfaces.
- namespace ttk_tools.rst.isygamma

  Access to Gamma V DLLs.

Functions

- ttk_tools.rst.isygamma._errCheck (result, func, args)

  Default "errcheck" callable for API calls that return a Gamma ErrorID.
- ttk_tools.rst.isygamma._settypes (lib, name, argtypes=None, restype=None)

  Assign "restype", "errcheck" and "argtypes" attributes to DLL functions.
- ttk_tools.rst.isygamma._bstr (s)

  Get a raw bytes string using the default encoding.
- ttk_tools.rst.isygamma._ustr (b)

  Get a text string using the default encoding.

Variables

- tuple ttk_tools.rst.isygamma.IS_WIN32 = (sys.platform == "win32")
- str ttk_tools.rst.isygamma.API_ENCODING = "latin-1"
- ttk_tools.rst.isygamma.LengthInfo = namedtuple('LengthInfo', ['cur_len', 'max_len'])
- ttk_tools.rst.isygamma.VersionInfo = namedtuple('VersionInfo', ['major', 'minor', 'release', 'build'])
- ttk_tools.rst.isygamma.gaapi = _GaAPI()

  instance of wrapped gaapi DLL
- ttk_tools.rst.isygamma.gacommon = _GaCommon()

  instance of gacommon DLL

8.142.1 Class Documentation

8.142.1.1 class ttk_tools::rst::isygamma::Level5

Address Level 5 elements (pv-array/fifo).

8.142.1.2 class ttk_tools::rst::isygamma::Level4

Address Level 4 elements (group).

8.142.1.3 class ttk_tools::rst::isygamma::Level3

Address Level 3 elements (memory/io-channel/scheduler-slot).

8.142.1.4 class ttk_tools::rst::isygamma::Level2

Address Level 2 elements (node/io/task/action/scheduler)

8.142.1.5 class ttk_tools::rst::isygamma::Level1

Address Level 1 elements (system).

8.143 ttk_tools/siemens/simit_common.py File Reference

Classes

- class ttk_tools.siemens.simit_common.DataType

  Data types from "Tabelle 2-3 Typkennung der Signale".
- class ttk_tools.siemens.simit_common.Direction

  "E/A-Kennung, d.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.siemens

  Access to SIEMENS automation technology tools and interfaces.
- namespace ttk_tools.siemens.simit_common

  Common definitions for SIMIT signal data access via SHM coupling.

8.144 ttk_tools/siemens/simit_shm.py File Reference

Classes

- class ttk_tools.siemens.simit_shm.SimitSignal

  A SIMIT signal configured in SHM Coupling.
- class ttk_tools.siemens.simit_shm.SimitShm

  Wrapper for SIMIT SHM access.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.siemens

  Access to SIEMENS automation technology tools and interfaces.
- namespace ttk_tools.siemens.simit_shm

  Access SIMIT signal data via SHM coupling.

Variables

- str ttk_tools.siemens.simit_shm.shm_name = "SIMIT_SHM"
- ttk_tools.siemens.simit_shm.shm = None

8.145 ttk_tools/vector/_canapeapi.py File Reference

Classes

- class ttk_tools.vector._canapeapi.Version_t
  CANape API version info structure.
- class ttk_tools.vector._canapeapi.Appversion
  CANape API application version info structure.
- class ttk_tools.vector._canapeapi.TCalibrationObjectValue
  Calibration Object Value data structure as used in CANapeAPI.
- class ttk_tools.vector._canapeapi.TCalibrationObjectValue.value_struct
  plain scalar values
- class ttk_tools.vector._canapeapi.TCalibrationObjectValue.axis_struct
  axis definitions for maps or curves.
- class ttk_tools.vector._canapeapi.TCalibrationObjectValue.ascii_struct
  ASCII/string data.
- class ttk_tools.vector._canapeapi.TCalibrationObjectValue.curve_struct
  2D table data.
- class ttk_tools.vector._canapeapi.TCalibrationObjectValue.map_struct
  3D table data.
- class ttk_tools.vector._canapeapi.TCalibrationObjectValue.valblk_struct
  Value Block data.
- class ttk_tools.vector._canapeapi.TConverterInfo
  Basic information about available MDF converters.
- class ttk_tools.vector._canapeapi.DiagNumericParameter
  Numeric values for diagnostic parameters.
- class ttk_tools.vector._canapeapi.DiagNumericParameter.PValues
  Different value representations.
- class ttk_tools.vector._canapeapi.TTaskInfo
  ECU Task data structure as used in CANapeAPI.
- class ttk_tools.vector._canapeapi.CANapeASAP3
  CANape ASAP3-connection (application).
- class ttk_tools.vector._canapeapi.DataAcquisition
  Data acquisition for multiple CANape devices.
- class ttk_tools.vector._canapeapi.BaseDevice
  Base class for CANape devices with generic/common/basic functionality.
- class ttk_tools.vector._canapeapi.CANapeDevice
  CANape device class (ASAP3-devices: CCP, XCP).
- class ttk_tools.vector._canapeapi.CANapeDiag
  Diagnostics device class (UDS, ...).
- class ttk_tools.vector._canapeapi.Scripting
  Access to extended scripting functions added in CANape 9.

Namespaces

- namespace ttk_tools
  Modules for tool access.
- namespace ttk_tools.vector
  Access to Vector (Vector Informatik GmbH) tools and interfaces.
- namespace ttk_tools.vector._canapeapi
  Access to CANape devices via CANapeAPI.

Functions

- ttk_tools.vector._canapeapi._bstr (s)

  Get a raw bytes string using the default CANape encoding.
- ttk_tools.vector._canapeapi._ustr (b)

  Get a text string using the default CANape encoding.
- ttk_tools.vector._canapeapi._printCurrentExceptionInfo (header=None)

  Print information about the currently active exception (mainly for debugging purposes).
- ttk_tools.vector._canapeapi._getCanapeRegPath ()

  Get the current installation path of CANape from the windows registry (HKLM\SOFTWARE\VECTOR\CANape).
- ttk_tools.vector._canapeapi._setCanapeRegPath (path)

  Select the active CANape installation by setting the path in the windows registry (HKLM\SOFTWARE\VECTOR\CANape).

Variables

- str ttk_tools.vector._canapeapi.CANAPE_REG_KEY = r"HKEY_LOCAL_MACHINE\SOFTWARE\↩ VECTOR\CANape"

  Registry key for CANape installations.

8.146 ttk_tools/vector/_canapeapi_offline_stub.py File Reference

Classes

- class ttk_tools.vector._canapeapi_offline_stub.CANapeOfflineStubNotSupported

  Error class for features not supported in the stub implementation.
- class ttk_tools.vector._canapeapi_offline_stub.CANapeASAP3

  Stub for CANape ASAP3-connection (application).
- class ttk_tools.vector._canapeapi_offline_stub.DataAcquisition

  Data acquisition (stub) for multiple CANape devices.
- class ttk_tools.vector._canapeapi_offline_stub.BaseDevice

  Base class (stub) for CANape devices with generic/common/basic functionality.
- class ttk_tools.vector._canapeapi_offline_stub.CANapeDevice

  CANape device class (ASAP3-devices: CCP, XCP).
- class ttk_tools.vector._canapeapi_offline_stub.CANapeDiag

  Diagnostics device stub class (UDS, ...).
- class ttk_tools.vector._canapeapi_offline_stub.Scripting

  Access to extended scripting functions.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.vector

  Access to Vector (Vector Informatik GmbH) tools and interfaces.
- namespace ttk_tools.vector._canapeapi_offline_stub

  A dummy/stub implementation of the default ttk_tools.vector.canapeapi for debugging without access to vector software, hardware or license keys.

Variables

- tuple ttk_tools.vector._canapeapi_offline_stub.STUB_VERSION = (4, 7, 1)

8.147 ttk_tools/vector/_vflash_api.py File Reference

Classes

- class ttk_tools.vector._vflash_api.VFlashAPI

  Python Wrapper for accessing vFlash API functions.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.vector

  Access to Vector (Vector Informatik GmbH) tools and interfaces.
- namespace ttk_tools.vector._vflash_api

  Access to vFlash API functions.

Functions

- ttk_tools.vector._vflash_api._bstr (s)

  Get a raw bytes string using the default encoding.

8.148 ttk_tools/vector/canapeapi.py File Reference

Classes

- class ttk_tools.vector.canapeapi.CANapeASAP3

  CANape ASAP3-connection (application).
- class ttk_tools.vector.canapeapi.DataAcquisition

  Data acquisition for multiple CANape devices.
- class ttk_tools.vector.canapeapi.CANapeDevice

  CANape device class (ASAP3-devices: CCP, XCP).
- class ttk_tools.vector.canapeapi.CANapeDiag

  Diagnostics device class (UDS, ...).
- class ttk_tools.vector.canapeapi.Scripting

  Access to extended scripting functions added in CANape 9.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.vector

  Access to Vector (Vector Informatik GmbH) tools and interfaces.
- namespace ttk_tools.vector.canapeapi

  Wrapper for Access to CANape devices via CANapeAPI.

8.149 ttk_tools/vector/canapeapi_common.py File Reference

Classes

- class ttk_tools.vector.canapeapi_common.REPR

  Measurement data format representation constants from enum `TFormat` (original constants have no specific prefix)
- class ttk_tools.vector.canapeapi_common.VALUE_TYPE

  Value type constants from enum `TValueType` (original constants have no specific prefix)
- class ttk_tools.vector.canapeapi_common.DATA_TYPE

  Data type constants (`TYPE_*`) from enum `TAsap3DataType`
- class ttk_tools.vector.canapeapi_common.DRIVER

  Driver type constants (`ASAP3_DRIVER_*`) from enum `tDriverType`.
- class ttk_tools.vector.canapeapi_common.TYPE_SWITCH

  Device online switch state constants (`TYPE_SWITCH_*`) from enum `TAsap3ECUState`.
- class ttk_tools.vector.canapeapi_common.MEASUREMENT_STATE

  Measurement state constants (`eT_MEASUREMENT_*`) from enum `tMeasurementState` (new in CANape 9).
- class ttk_tools.vector.canapeapi_common.DIAG_SERVICE

  Diagnostic service states (`DIAG_SERVICE_*`) from enum `eServiceStates`.
- class ttk_tools.vector.canapeapi_common.DIAG_PARAM_TYPE

  Diagnostic response parameter types (`Param*`) from enum `EnParamType`.
- class ttk_tools.vector.canapeapi_common.DEV

  Device channel constants (DEV_*) from CANapAPI.h.
- class ttk_tools.vector.canapeapi_common.SCRIPT_STATUS

  Script execution status values from enum `TScriptStatus`.
- class ttk_tools.vector.canapeapi_common.CANapeError

  Base error for CANape connection.
- class ttk_tools.vector.canapeapi_common.CANapeAccessError

  Cal-object access errors for CANape connection.
- class ttk_tools.vector.canapeapi_common.CANapeDaqError

  Data Acquisition errors for CANape connection.
- class ttk_tools.vector.canapeapi_common.CANapeDiagError

  Errors for CANape diagnostic connections.
- class ttk_tools.vector.canapeapi_common.CANapeDiagTimeoutError

  Timeout occurred during a diag request.
- class ttk_tools.vector.canapeapi_common.CANapeScriptError

  A wrapper for error message strings so they can be distinguished from "normal" string values (like contents of ASCII-type variables).
- class ttk_tools.vector.canapeapi_common.CANapeErrorString

  Container for diagnostic response values that may have multiple value representations.
- class ttk_tools.vector.canapeapi_common.DiagParameterValue
- class ttk_tools.vector.canapeapi_common.DiagParameterName

  Container for diagnostic response name that may contain sub-parameter names for "complex parameters".

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.vector

  Access to Vector (Vector Informatik GmbH) tools and interfaces.
- namespace ttk_tools.vector.canapeapi_common

  Common utility functions and classes for CANape access (used in canapeapi.py).

8.150 ttk_tools/vector/canapeapi_offline_stub.py File Reference

Classes

- class ttk_tools.vector.canapeapi_offline_stub.CANapeASAP3

  CANape ASAP3-connection (application).
- class ttk_tools.vector.canapeapi_offline_stub.DataAcquisition

  Data acquisition (stub) for multiple CANape devices.
- class ttk_tools.vector.canapeapi_offline_stub.CANapeDevice

  CANape device class (ASAP3-devices: CCP, XCP).
- class ttk_tools.vector.canapeapi_offline_stub.CANapeDiag

  Diagnostics device class (UDS, ...).
- class ttk_tools.vector.canapeapi_offline_stub.Scripting

  Access to extended scripting functions added in CANape 9.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.vector

  Access to Vector (Vector Informatik GmbH) tools and interfaces.
- namespace ttk_tools.vector.canapeapi_offline_stub

  Interface to a stub implementation of the default canapeapi.py for debugging without access to vector software, hardware or license keys.

8.151 ttk_tools/vector/mdf4_reader.py File Reference

Classes

- class ttk_tools.vector.mdf4_reader.MdfError
- class ttk_tools.vector.mdf4_reader.MdfVersionError

  Detected MDF version is unsupported.
- class ttk_tools.vector.mdf4_reader.MdfStructureError

  Data or data types are inconsistent.
- class ttk_tools.vector.mdf4_reader.MdfDataError

  Feature not (yet) supported)
- class ttk_tools.vector.mdf4_reader.MdfNotYetSupported

  Meta class for EnumInfo; adds a mechanism for custom value lookups.
- class ttk_tools.vector.mdf4_reader.EnumInfoMeta
- class ttk_tools.vector.mdf4_reader.EnumInfo

  Placeholder for ttk_base.baseutils.EnumInfo.
- class ttk_tools.vector.mdf4_reader.IntWithEnumRepr

  An integer value that shows an EnumInfo repr of itself (if a match is found)
- class ttk_tools.vector.mdf4_reader.IntWithRepr

  An integer value that has an associated (textual) representation.
- class ttk_tools.vector.mdf4_reader.FloatWithRepr

  A float value that has an associated (textual) representation.
- class ttk_tools.vector.mdf4_reader.Flags

  An integer/long as container for decoded bit values.

- class ttk_tools.vector.mdf4_reader.DataRecords

  Iterator for fixed and variable length signal data records, returns sliced records from raw byte payload data.
- class ttk_tools.vector.mdf4_reader.BlockHeader

  String encoding used in an MDF file is UTF-8 (applies to TXBLOCK and MDBLOCK data)
- class ttk_tools.vector.mdf4_reader.Block

  A generic block with header, links and data sections (for all blocks except for special case IDBLOCK)
- class ttk_tools.vector.mdf4_reader.Link

  LINK file position/offset as a long value with custom representation.
- class ttk_tools.vector.mdf4_reader.MdTxBlock

  Text block with optional meta data.
- class ttk_tools.vector.mdf4_reader.IdBlock

  File Identification Block.
- class ttk_tools.vector.mdf4_reader.HdBlock

  Header Block, unique.
- class ttk_tools.vector.mdf4_reader.FhBlock

  File History Block.
- class ttk_tools.vector.mdf4_reader.CgFlags

  Container for decoded cg_flags.
- class ttk_tools.vector.mdf4_reader.DgBlock

  Data Group Block.
- class ttk_tools.vector.mdf4_reader.CgBlock

  Channel Group Block.
- class ttk_tools.vector.mdf4_reader.CnFlags

  Container for decoded uint32 cn_flags.
- class ttk_tools.vector.mdf4_reader.CnTypes

  Channel type (cn_type) values.
- class ttk_tools.vector.mdf4_reader.CnSyncTypes

  Channel synchronization type (cn_sync_type) values.
- class ttk_tools.vector.mdf4_reader.CnDataTypes

  Channel data type (cn_data_type) values.
- class ttk_tools.vector.mdf4_reader.CaFlags

  Container for decoded uint32 ca_flags.
- class ttk_tools.vector.mdf4_reader.CaTypes

  Channel array type (ca_type) values: array semantics.
- class ttk_tools.vector.mdf4_reader.CaStorage

  Channel storage (ca_storage) values: how element values are stored.
- class ttk_tools.vector.mdf4_reader.CANopenDate

  Container for channel data type "CANopen date".
- class ttk_tools.vector.mdf4_reader.CANopenTime

  Container for channel data type "CANopen time".
- class ttk_tools.vector.mdf4_reader.CnBlock

  Channel Block.
- class ttk_tools.vector.mdf4_reader.CaBlock

  Channel Array Block.
- class ttk_tools.vector.mdf4_reader.DzZipType

  Zip algorithm type (dz_zip_type) used to compress data stored in dz_data.
- class ttk_tools.vector.mdf4_reader.DlFlags

  Container for decoded uint8 dl_flags.
- class ttk_tools.vector.mdf4_reader.HlFlags

  Container for decoded uint8 hl_flags.
- class ttk_tools.vector.mdf4_reader.DtBlock

Data Block.

- class ttk_tools.vector.mdf4_reader.SdBlock

  Signal Data Block.
- class ttk_tools.vector.mdf4_reader.DlBlock

  Data List Block.
- class ttk_tools.vector.mdf4_reader.HlBlock

  Header List Block.
- class ttk_tools.vector.mdf4_reader.SiFlags

  Container for decoded uint8 si_flags.
- class ttk_tools.vector.mdf4_reader.SiTypes

  Source type (si_type), additional source classification.
- class ttk_tools.vector.mdf4_reader.SiBusTypes

  Source bus type (si_bus_type), additional classification of used bus (should be 0 for si_type >= 3).
- class ttk_tools.vector.mdf4_reader.SiBlock

  Source Information Block.
- class ttk_tools.vector.mdf4_reader.CcFlags

  Container for decoded uint16 cc_flags.
- class ttk_tools.vector.mdf4_reader.CcTypes

  Conversion type / formula identifier (cc_type) values.
- class ttk_tools.vector.mdf4_reader.CcBlock

  Channel Conversion Block.
- class ttk_tools.vector.mdf4_reader.Mdf4Reader

  Reader for MDF4 files; read data will be stored to member vars.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.vector

  Access to Vector (Vector Informatik GmbH) tools and interfaces.
- namespace ttk_tools.vector.mdf4_reader

Functions

- ttk_tools.vector.mdf4_reader.printMembers (inst, header=None, indent="")
- ttk_tools.vector.mdf4_reader.parseXmlSnippet (xml)
- ttk_tools.vector.mdf4_reader.getNodeText (node)

  Get text content from a DOM node.
- ttk_tools.vector.mdf4_reader.getTagText (parent_node, tag_name="TX", joiner="\n")

  Get text from all elements with the given tag name.
- ttk_tools.vector.mdf4_reader.decodeZeroTerminatedString (raw_bytes, encoding="utf-8")

  Get a string representation from a zero-terminated byte string buffer.
- ttk_tools.vector.mdf4_reader.getLinkedBlocks (file_handle, block_class, first_pos, next_pos_attr_name)

  Get/resolve a linked list of MDF-Blocks where each block contains a LINK / start position of the following block.
- ttk_tools.vector.mdf4_reader._fixup2digitYear (yy_year)

  Get "full" year value for the supplied two-digit year that is closest to the current year.
- ttk_tools.vector.mdf4_reader.getUnzippedData (raw_data)

  Uncompress block data from a DZBLOCK to get the "original" data.
- ttk_tools.vector.mdf4_reader.getData (file_handle, pos=None)

  Get (potentially merged) data payload.

- ttk_tools.vector.mdf4_reader.getSignalData (file_handle, pos=None)

  Get (potentially merged) signal data payload (i.e.
- ttk_tools.vector.mdf4_reader.clearCache ()

  MDF Spec 4.1.0, Section 5.4.4 Linking of Blocks: "Note that there are block types for which there are no unique links.
- ttk_tools.vector.mdf4_reader.getCachedBlock (file_handle, pos, block_class, fallback=None, optional=True)

  Get a block from file handle (see Block) and cache it.
- ttk_tools.vector.mdf4_reader.getCachedText (file_handle, pos, fallback=None)

  Get contents of an (optional) text block (TX or MD) link (i.e.
- ttk_tools.vector.mdf4_reader.printLineHeader (text, indent="", width=80)

  Print a "header" line for log output.
- ttk_tools.vector.mdf4_reader.readMdf4 (file_path, verbosity=0)

  Read an MDF4 file.
- ttk_tools.vector.mdf4_reader._printDataInfo (values, indent_level=1, print_all=False, outer_n=3, label_min↩ _width=16)

  Print a summary of the supplied values to stdout.

Variables

- ttk_tools.vector.mdf4_reader._current_year = time.localtime()[0]
- ttk_tools.vector.mdf4_reader._current_century = _current_year - (_current_year % 100)
- dict ttk_tools.vector.mdf4_reader._block_cache = {}
- ttk_tools.vector.mdf4_reader._handle_ref = None
- ttk_tools.vector.mdf4_reader.examples_path = os.path.join("..", "samples")
- ttk_tools.vector.mdf4_reader.file_path = os.path.join(examples_path, "Simple", "Vector_CANape.MF4")

8.152 ttk_tools/vector/mdflib.py File Reference

Classes

- class ttk_tools.vector.mdflib.MdfLibError

  Base Error for MdfLib access.
- class ttk_tools.vector.mdflib.MDF_STATUS

  MDF status values (`MDF_*`-constants).
- class ttk_tools.vector.mdflib.MDF_ERROR

  MDF Error Values (`E_MDF_*`-constants)
- class ttk_tools.vector.mdflib.MDF_OPEN

  Values for `MDF_OPEN_*`-constants.
- class ttk_tools.vector.mdflib.MDF_SEEK

  Values for `MDF_SEEK_*`-constants.
- class ttk_tools.vector.mdflib.MDF_CHN

  MDF Channel Types (`MDF_CHN_*`-constants)
- class ttk_tools.vector.mdflib.MDF_MODULE

  MDF Module/Block Types (`MDF_MODULE_*`-constants)

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.vector

  Access to Vector (Vector Informatik GmbH) tools and interfaces.
- namespace ttk_tools.vector.mdflib

  MDF-file I/O via Vector MdfLib DLL (for MDF version 3).

Functions

- ttk_tools.vector.mdflib.setMdfLibPath (dll_path)

  Set the path to the MDFLib DLL.
- ttk_tools.vector.mdflib.getMdfLibHandle ()

  Get a handle to the MdfLib DLL.
- ttk_tools.vector.mdflib.getMdfError (error_code)

  Get a correct interpretation of MDFERROR return values, which are 16-bit signed int values.
- ttk_tools.vector.mdflib.getCurrentError (verbosity=1)

  Get info on the most recent error condition from the MDFLib DLL.
- ttk_tools.vector.mdflib._handleMdfErrorStatus (mdf_error, context="", ignore_non_severe_errors=False)

  Raise a MdfLibError if decoded MDF-Error indicates an error state.
- ttk_tools.vector.mdflib.readMdf (file_path, verbosity=1, signal_names=None)

  Read channel data from the supplied MDF-File.

Variables

- ttk_tools.vector.mdflib._lib_path = None
- ttk_tools.vector.mdflib._lib = None
- int ttk_tools.vector.mdflib._IS_X64 = sys.maxsize > 2**32

8.153 ttk_tools/vector/vflash_api.py File Reference

Classes

- class ttk_tools.vector.vflash_api.VFlashAPI

  Python Wrapper for accessing vFlash API functions.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.vector

  Access to Vector (Vector Informatik GmbH) tools and interfaces.
- namespace ttk_tools.vector.vflash_api

  Interface to vFlash API function access in ttk_tools.vector._vflash_api.

8.154 ttk_tools/vector/vflash_api_common.py File Reference

Classes

- class ttk_tools.vector.vflash_api_common.VFlashError

  Base Error for vFlash API.
- class ttk_tools.vector.vflash_api_common.VFlashDLLError

  Flashing ended with an error state.
- class ttk_tools.vector.vflash_api_common.VFlashResultError
- class ttk_tools.vector.vflash_api_common.VFlashStatus

  Flash programming status values.
- class ttk_tools.vector.vflash_api_common.VFlashResult

  Flash programming result values.

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.vector

  Access to Vector (Vector Informatik GmbH) tools and interfaces.
- namespace ttk_tools.vector.vflash_api_common

  Common definitions, utility functions and classes for vFlash API access.

Variables

- int ttk_tools.vector.vflash_api_common.FLASH_TIMEOUT = 2000

  timeout for flash process in [s]

8.155 ttk_tools/win32_utils.py File Reference

Namespaces

- namespace ttk_tools

  Modules for tool access.
- namespace ttk_tools.win32_utils

  Interface wrapper for Windows systems utility functions, see ttk_tools._win32_utils.

Functions

- ttk_tools.win32_utils.disableAppCrashDialog ()

  Disable displaying of application/program crash dialog for the current process ("XYZ has stopped working" / "XYZ funktioniert nicht mehr").
- ttk_tools.win32_utils.waitPumpMessages (delay)

  Wait for (approximately) delay [s] and pump windows messages/events
  while waiting, which is a "good thing" to do if a COM connection is active in the current process/thread.
- ttk_tools.win32_utils.getRegPath (path, name, wow64_32_key=None, verbosity=1)

  Get a value from an entry in the windows registry.
- ttk_tools.win32_utils.setRegPath (path, name, value, value_type=None, wow64_32_key=None, verbosity=1)

  Set a value to an entry in the windows registry.

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